EFFIGY MOUNDS NATIONAL MONUMENT

Rehabilitation Visitor Center
N170 EFMO 318539

Redesign Parking and Replace Curbs,
Walks, and Pavement
EFMO 224159

Final Construction Documents
PROJECT SPECIFICATIONS

NATIONAL PARK SERVICE (NPS)
DENVER SERVICE CENTER (DSC)

June 2026

Table of Contents

N170 EFMO 318539 and EFMO 224159 **No. of Pages**

Combined Project Title Page 1
Table of Contents 8

Division 01: General Requirements

01 11 00 Summary of Work 5
01 26 01 Contract Modification Procedures 6
01 27 00 Definition of Contract Line Items 4
01 31 00 Project Management and Coordination 11
01 32 16 Construction Schedule 10
01 32 33 Photo Documentation 3
01 33 23 Submittal Procedures 8
01 33 23A Submittal List 16
01 35 13.22 Archeological Protection 3
01 35 23 Safety Requirements 4
01 35 50 Natural & Cultural Resources Protection 16
01 40 00 Quality Requirements 50
01 42 00 Reference Standards 13
01 50 00 Temporary Facilities and Controls 18
01 57 19.11 Indoor Air Quality Management 6
01 57 19.12 Noise and Acoustics Management 3
01 57 23 Temporary Storm Water Pollution Prevention (SWPP) 7
01 67 00 Product Requirements 9
01 73 29 Cutting and Patching 4
01 73 40 Execution 7
01 74 19 Construction Waste Management and Disposal 9
01 74 19A Project Waste Management Plan Worksheet Sample 1
01 77 00 Closeout Procedures 5
01 77 00A Closeout and Operations & Maintenance Requirements 5
01 78 23 Operation and Maintenance (O&M) Data 10
01 79 00 Demonstration and Training 4
01 81 13 Sustainable Design Requirements – Non-LEED™ Projects 4
01 91 14 Total Building Commissioning 12

N170 EFMO 318539 **No. of Pages**

Project Title Page 1
Seals Page 7

Division 02: Existing Conditions

02 41 13 Selective Site Demolition 4
02 41 19 Selective Building Demolition 7
02 81 01 Off-Site Transportation and Disposal 4
02 82 13 Asbestos Abatement 22
02 83 33 Removal and Disposal of Material Containing Lead 12
02 84 16 Handling of Lamps Containing Mercury 6

Division 03: Concrete

03 30 00 Cast-In-Place Concrete 23

Division 04: Masonry

04 01 20.63 Brick Masonry Repair and Cleaning 9
04 01 20.64 Brick Masonry Repointing 7
04 22 00 Concrete Unit Masonry 10
04 26 13 Masonry Veneer 15

Division 05: Metals

05 12 00 Structural Steel Framing 10
05 31 00 Steel Decking 4
05 50 00 Metal Fabrications 11

Division 06: Wood, Plastics & Composites

06 10 00 Rough Carpentry 9
06 16 00 Sheathing 4
06 18 00 Glued-Laminated Construction 4
06 20 23 Interior Finish Carpentry 5
06 40 13 Exterior Architectural Woodwork 6
06 41 00 Architectural Wood Casework 9
06 61 16 Solid Surface Fabrications 7

Division 07: Thermal & Moisture Protection

07 14 16 Cold Fluid-Applied Waterproofing 7
07 18 00 Traffic Coatings 7
07 21 00 Thermal Insulation 8
07 21 19 Foamed-In-Place Insulation 3
07 26 00 Vapor Retarders 3
07 27 15 Nonbituminous Self-Adhering Sheet Air Barriers 8
07 27 26 Fluid-Applied Membrane Air Barriers 8
07 41 13.16 Standing-Seam Metal Roof Panels 11
07 42 13.13 Formed Metal Wall Panels 9
07 46 16 Aluminum Cladding 6
07 62 00 Sheet Metal Flashing and Trim 10
07 71 00 Roof Specialties 7
07 72 53 Snow Guards 4
07 84 13 Penetration Firestopping 6
07 84 43 Joint Firestopping 5
07 92 00 Joint Sealants 7

Division 8: Openings

08 11 13 Hollow Metal Doors and Frames 9
08 14 16 Flush Wood Doors 9
08 31 13 Access Doors and Frames 3
08 41 13 Aluminum-Framed Entrances and Storefronts 12
08 71 00 Door Hardware 44
08 80 00 Glazing 14

Division 09: Finishes

09 22 16 Non-Structural Metal Framing 3
09 29 00 Gypsum Board 8
09 51 23 Acoustical Tile Ceilings 5
09 65 19 Resilient Tile Rubber Flooring 9
09 67 23 Resinous Epoxy Flooring 6
09 68 13 Tile Carpeting 7
09 91 13 Exterior Painting 7
09 91 23 Interior Painting 10
09 93 00 Staining and Transparent Finishing 6

Division 10: Specialties

10 14 23.16 Room-Identification Panel Signage 5
10 21 13.19 Plastic Toilet Compartments 5
10 22 39 Folding Panel Partitions 10
10 28 00 Toilet Accessories 7
10 44 13 Fire Protection Cabinets 5
10 44 16 Fire Extinguishers 3
10 51 16 Wood Lockers 8
10 75 16 Ground-Set Flagpoles 5
10 81 19 Rodent and Bird Control 6

Division 11: Equipment

11 00 00 Requirements For Miscellaneous Equipment 3
11 30 13 Residential Appliances 5

Division 12: Furnishings

12 24 13 Roller Window Shades 6
12 32 16 Manufactured Thermally Fused-Laminate-Clad Casework 7
12 36 61 Simulated Stone Countertops 5

Division 14: Conveying Equipment

14 01 20.71 Elevator Rehabilitation 6

Division 21: Fire Suppression

21 00 00 Fire Suppression Basic Requirements 16
21 05 00 Common Work Results for Fire Suppression 16
21 08 00 Commissioning of Fire Suppression Systems 6
21 13 00 Fire Suppression Sprinkler Systems 9
21 13 19 Fire Suppression Preaction Sprinkler Systems 13
21 22 00 Clean-Agent Fire-Extinguishing Systems 19

Division 22: Plumbing

22 05 00 Common Work Results for Plumbing 8
22 05 17 Sleeves and Sleeve Seals for Plumbing Piping 5
22 05 18 Escutcheons for Plumbing Piping 2
22 05 19 Meters and Gages for Plumbing Piping 5
22 05 23 General-Duty Valves for Plumbing Piping 4

22 05 29 Hangers and Supports for Plumbing Piping and Equipment 10
22 05 53 Identification for Plumbing Piping and Equipment 4
22 07 19 Plumbing Piping Insulation 14
22 11 16 Domestic Water Piping 10
22 11 19 Domestic Water Piping Specialties 8
22 11 23 Domestic Water Pumps 4
22 13 16 Sanitary Waste and Vent Piping 10
22 13 19 Sanitary Waste Piping Specialties 6
22 33 00 Electric, Domestic-Water Heaters 5
22 4 213.13 Commercial Water Closets 4
22 4 213.16 Commercial Urinals 5
22 42 16.13 Commercial Lavatories 4
22 42 16.16 Commercial Sinks 5
22 47 13 Drinking Fountains 3

Division 23: Heating, Ventilating, and Air-Conditioning (HVAC)

23 05 00 Common Work Results for HVAC 10
23 05 17 Sleeves and Sleeve Seals for HVAC Piping 5
23 05 29 Hangers and Supports for HVAC Piping and Equipment 13
23 05 53 Identification for HVAC Piping and Equipment 4
23 05 93 Testing, Adjusting, and Balancing for HVAC 11
23 07 13 Duct Insulation 9
23 07 19 HVAC Piping Insulation 13
23 11 26 Facility Liquefied-Petroleum Gas Piping 9
23 23 00 Refrigerant Piping 8
23 31 13 Metal Ducts 12
23 33 00 Air Duct Accessories 9
23 34 23 HVAC Power Ventilators 5
23 37 13 Diffusers, Registers, and Grilles 2
23 72 23.13 Packaged Indoor Heat Wheel Energy Recovery Units 4
23 81 26 Split-System Air Conditioners 6
23 8 239 Propeller Unit Heaters 3
23 84 16.16 Indoor, Mechanical Dehumidification Units 3

Division 26: Electrical

26 05 00 Common Work Results for Electrical 2
26 05 19 Low-Voltage Electrical Power Conductors and Cables 4
26 05 26 Grounding and Bonding for Electrical Systems 5
26 05 29 Hangers and Supports for Electrical Systems 8

26 05 33 Raceway and Boxes for Electrical Systems 8
26 05 44 Sleeves and Sleeve Seals for Electrical Raceways and Cabling 5
26 05 53 Identification for Electrical Systems 8
26 05 73 Electrical System Studies 6
26 08 00 Commissioning of Electrical Systems 4
26 09 23 Lighting Control Devices 4
26 09 43 Network Lighting Controls 7
26 24 16 Panelboards 8
26 27 13 Electricity Metering 3
26 27 26 Wiring Devices 6
26 28 16 Enclosed Switches and Circuit Breakers 5
26 41 13 Lightning Protection for Structures 3
26 43 13 Transient-Voltage Suppression for Low-Voltage Electrical Power Circuits 4
26 51 00 Lighting Fixtures 8

Division 27: Communications

27 00 10 Supplemental Requirements for Communications 11
27 05 26 Grounding and Bonding for Communications Systems 8
27 05 28 Pathways for Communications Systems 14
27 05 36 Cable Trays for Communications Systems 7
27 05 44 Sleeves and Sleeve Seals for Communications Pathways and Cabling 4
27 05 53 Identification for Communications Systems 6
27 11 16 Communications Racks, Frames, and Enclosures 7
27 13 23 Communications Optical Fiber Backbone Cabling 9
27 15 13 Communications Copper Horizontal Cabling 11
27 41 00 Audio Video Systems 34

Division 28: Electrical Safety & Security

28 00 01 Electronic Safety Basic Requirements 13
28 00 10 Supplemental Requirements for Electronic Safety and Security 11
28 08 00 Commissioning of Fire Alarm 7
28 13 00 Access Control Software and Database Management 6
28 14 00 Access Control System Hardware 11
28 16 00 Intrusion Detection 14
28 20 00 Video Surveillance 9
28 46 00 Fire Detection and Alarm 22

Division 31: Earthwork

31 10 00 Site Clearing 3
31 20 00 Earth Moving 14
31 21 13 Radon Mitigation 3
31 23 16 Topsoil 4
31 25 00 Erosion and Sediment Control 2

Division 32: Exterior Improvements

32 13 20 Site Concrete 13
32 33 00 Site Furnishings 5
32 91 13 Soil Preparation 3
32 92 00 Turf and Grasses 6
32 93 00 Plants 8

Division 33: Utilities

33 14 16 Site Water Utility Distribution 8
33 40 00 Storm Drainage 6
33 46 00 Subdrainage 3

EFMO 224159 **No. of Pages**

Project Title Page 1
Seals Page 1

Division 02: Existing Conditions

02 41 13 Selective Site Demolition 4

Division 26: Electrical

26 05 00 Common Work Results for Electrical 2
26 05 19 Low-Voltage Electrical Power Conductors and Cables 4
26 05 26 Grounding and Bonding for Electrical Systems 5
26 05 29 Hangers and Supports for Electrical Systems 8
26 05 33 Raceway and Boxes for Electrical Systems 8
26 05 44 Sleeves and Sleeve Seals for Electrical Raceways and Cabling 5
26 05 53 Identification for Electrical Systems 8
26 51 00 Lighting Fixtures 6

Division 31: Earthwork

31 10 00 Site Clearing 3
31 20 00 Earth Moving 14
31 23 16 Topsoil 4
31 25 00 Erosion and Sediment Control 2

Division 32: Exterior Improvements

32 12 16 Asphalt Paving 5
32 13 20 Site Concrete 12
32 17 00 Paving Specialties 7
32 31 13 Gates 4
32 91 13 Soil Preparation 3
32 92 00 Turf and Grasses 8
32 93 00 Plants 9

Division 33: Utilities

33 40 00 Storm Drainage 5

END OF TABLE OF CONTENTS

EFFIGY MOUNDS NATIONAL MONUMENT

Rehabilitation Visitor Center
N170 EFMO 318539

Redesign Parking and Replace Curbs,
Walks, and Pavement
EFMO 224159

Division 01

Final Construction Documents

PROJECT SPECIFICATIONS

SECTION 01 11 00 – SUMMARY OF WORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes the following:

1. Work Covered by Contract Documents
2. Work Phases
3. Work Under Other Contracts
4. Government-Furnished Materials
5. Construction Contractor Use of Site
6. Public Use of Site
7. Occupancy Requirements for Buildings
8. Conduct of Operations
9. Work Restrictions
10. Special Construction Requirements
11. Soils Investigation Report
12. Additional Reports

1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Location: Effigy Mounds National Monument, Visitor Center complex, 151 Highway 76, Harpers Ferry, IA 52146-7519. Rehabilitation of the visitor center building, visitor center parking lot, and maintenance area drive and parking area.

B. The Work consists of:

1. The Work includes rehabilitation of the visitor center, including foundation improvements, and remodeling of the basement and main floor. Includes selective demolition, hazardous materials removal, new HVAC system, electrical system, domestic water, fire alarm, and fire suppression improvements. New windows and doors, interior and exterior building finishes and roofing, and addition of new insulation at foundation, exterior walls, and attic. Structural changes (relocating columns, structural improvements for code, new beams) and moving/modifying interior partitions, relocating restrooms (new fixtures – plumbing and electrical – new toilet partitions, wall paneling, epoxy floor, toilet accessories, etc.) and elevator tune up. Construction of a new entry plaza. Vehicle and pedestrian circulation improvements include parking lot drainage improvements, new parking lot layout with curb and gutters, sidewalks and asphalt. Parking lot island redesign will create visitor use areas with benches and picnic tables. Improvements to the park staff maintenance areas include delivery circulation, asphalt paving, pavement drainage and parking improvements.

a. Refer to Section 01 27 00 “Definition of Contract Line Items” for more information regarding additional work in options for the visitor center.

C. Project will be constructed under a single prime contract.

1.3 WORK PHASES

A. The Work shall be conducted in one phase. Total duration is approximately 640 calendar days. Recommended sequencing to construct the project is below:

1. Site Readiness: Mobilization, survey, temporary site protections, TESC
2. Visitor Center Rehabilitation: Includes structural, architectural, MEP systems, civil engineering, and landscaping.

    a. Abatement and Salvage: Selective demolition and hazardous materials abatement (exterior and interior). Salvage of materials.
    b. Structural Improvements: Structural upgrades including foundation work, beams and columns, SOG, and CMU reinforcement. Work is included on the basement level, main level, attic level and exterior.
    c. Exterior Shell: Includes demolition scope, new sheathing and WRB, new siding, paintings and coating and new storefront systems. Location of work occurs on the West, North, and South elevations, and Roof activities.
    d. Interior Improvements: Includes demolition scope, new sheathing and WRB, new siding, paintings and coating and new storefront systems. Location of work occurs on the West, North, and South elevations, and Roof activities.
    e. Civil: Includes salvage of existing site elements, utility improvements, exterior concrete work.

3. Parking Lot Improvements: Demolition of existing pavement, grading, base preparation, drainage and utility improvements, and new paving. Parking lot striping.
4. Visitor Center and Parking Lot site furnishings and landscaping.

1.4 CONSTRUCTION CONTRACTOR USE OF SITE

A. General: Construction-Contractor shall have full use of the site for construction operations during the construction period. Construction–Contractor's use of the site is limited only by the Government's right to perform work or to retain other construction-Contractors on portions of Project.

B. Storage of Materials: Confine storage of materials to staging area shown on plans.

C. Parking: Confine parking to staging area as shown on drawing C000.

D. Stockpiling: Confine stockpiling to staging area.

E. Preservation of Natural Features:

1. Prevent damage to natural surroundings. Restore damaged areas, repairing or replacing damaged trees and plants, at no additional expense to the Government.

2. Provide temporary barriers to protect existing trees and plants and root zones.
3. Do not remove, injure, or destroy trees or other plants without prior approval. Consult with Contracting Officer (CO) and remove agreed-on roots and branches that interfere with construction.
4. Do not fasten ropes, cables, or guys to existing trees.
5. Carefully supervise excavating, grading, filling, and other construction operations near trees to prevent damage.

F. Driveways and Entrances: Keep driveways and entrances serving premises clear and available to Government employees, and emergency vehicles at all times. Do not use for parking or storage of materials.

1. Schedule deliveries to minimize use of driveways and entrances.
2. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

G. Construction Camp: Establishment of a camp within park will not be permitted.

H. Hauling Restrictions: Comply with legal load restrictions in hauling of materials. Load restrictions on park roads are identical to state load restrictions with such additional regulations as may be imposed by the Park Superintendent. Information regarding rules and regulations for vehicular traffic on park roads may be obtained from the Office of the Park Superintendent. A special permit will not relieve Construction Contractor of liability for damage which may result from moving of equipment.

I. Bridge Restrictions: Identify jurisdictions, load restrictions, permit requirements, time and calendar restrictions as outlined.

J. Repairs as a result of damage to existing facilities during construction shall be at no additional cost to the Government.

1.5 PUBLIC USE OF SITE

A. The building, driveway and parking will be closed to the public during construction.

1.6 OCCUPANCY REQUIRMENTS FOR BUILIDINGS

A. Existing Buildings

1. Partial Government Occupancy: Government will occupy premises during entire construction period, with exception of areas under construction. Cooperate with Government during construction operations to minimize conflicts and facilitate Government usage. Perform Work so as not to interfere with Government’s operations. Maintain existing exits, unless otherwise indicated.

a. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from Contracting Officer.

b. Park data center is located in the visitor center and shall remain functional throughout construction. Park electrical service shall be used to patch in temporary power. Contractor shall provide plan to keep equipment dust free, accessible, temperature controlled, and functioning.
c. Park radio tower shall remain functional throughout construction. Contractor shall request any necessary downtime to the CO 48 hours prior to radio disconnect.

1.7 CONDUCT OF OPERATIONS

A. Construction Contractor shall conduct his operations in conformance with rules and regulations promulgated by the Secretary of the Interior for the National Park Service, and applicable park rules and regulations prescribed by Park Superintendent.

B. Work on Saturdays, Sundays, Federal holidays or at night may not be performed unless stated in the Work Restrictions below or without prior consent from the Contracting Officer. Submit requests 2 business/calendar days in advance of the work to the Contracting Officer for approval.

C. No signs or advertisements (except those specified herein) shall be displayed on the construction site or within the park unless approved by the Contracting Officer.

1.8 WORK RESTRICTIONS

A. On-Site Work Hours: Work shall be generally performed during normal business working hours of 7:00 a.m. to 5:00 p.m., Monday through Friday, except when otherwise indicated.

1. Weekend Hours: subject to Contracting Officer's approval.
2. Early Morning Hours: work before 7:00 am requires Contracting Officer's approval.

B. Existing Utilities

1. Existing Utilities: Notify Contracting Officer and utility companies of proposed locations and times for excavation.
2. Construction Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs, repair utility at no additional expense to the Government.
3. If damage occurs to an unknown utility, repair utility. An equitable adjustment will be made in accordance with the Changes clause of the contract.

C. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Government or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:

1. Notify Contracting Officer not less than **two** business days in advance of proposed utility interruptions.
2. Do not proceed with utility interruptions without Contracting Officer's written permission.
3. Hours and length of Utility Shutdowns: subject to Contracting Officer's approval.

D. Nonsmoking Building/Tobacco Use/Vaping: Smoking is not permitted within building or within 25 feet of entrances, operable windows, or outdoor air intakes.

1.9 SPECIAL CONSTRUCTION REQUIREMENTS

A. Project Management and Communication Software: Software administered by NPS shall be used to manage communication and document sharing during construction.

1. See Section 01 31 00 "Project Management and Coordination" for requirements on using NPS project management and communication software.

1.10 SOILS INVESTIGATION REPORT

A. The Report for Geotechnical Investigation, April 25, 2025, prepared by WSP is an appendix with this package (Appendix A).

B. In case of conflict between report and drawings or specifications, the drawings and specifications govern.

1.11 ADDITIONAL REPORTS

A. The report Hazardous Materials Inspection for Effigy Mounds National Monument Visitor's Center, Harpers Ferry, Iowa, January 31, 2025, prepared by Landmark Environmental, Inc. is an appendix with this package (Appendix B).

B. The report Bat and Nesting Bird Survey for the Effigy Mounds National Monument Visitor's Center in Allamakee County, Iowa, July 2025, prepared by SWCA is an appendix with this package (Appendix C).

C. An elevator handbook titled Elevator Modernization Work Coordination Handbook, prepared by TK Elevator (TKE) is an appendix with this package (Appendix D).

D. In case of conflict between report and drawings or specifications, drawings and specifications govern.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 11 00

SECTION 01 26 01 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of administrative and procedural requirements for contract modifications.

1.2 DEFINITIONS AND ALLOWANCES

A. Home Office Overhead: Costs incurred in support of all of a contractor's projects and not attributable to a specific job. The cost for home office overhead is only allowed as a percentage of all direct work excluding profit. The following items represent allowable home office overhead costs identified in Part 31 of the Federal Acquisition Regulation (FAR):

1. Rent
2. Utilities
3. Furnishings
4. Office equipment
5. Executive and management staff not exclusively assigned to the project
6. Support, accounting, and administrative staff
7. Preparation of cost proposals, estimating, and schedule analyses connected with Modifications
8. Estimating and preconstruction services
9. Mortgage costs
10. Real estate and corporate taxes
11. Automobile maintenance and travel costs for home office personnel
12. Home office insurances i.e. structure, automotive, umbrella, flood, etc.
13. Depreciation of equipment and other assets
14. Home office supplies (paper, staples, etc.)
15. Legal services
16. Accounting and data processing
17. Professional fees/registration

B. General Conditions (Field Office Overhead): Management and administrative costs incurred on site for the designated project. Costs associated with preparation of modifications will not be allowed. Costs for these items are to be included only in the general conditions of the modification estimate. Only in the case of a contract time extension are additional general conditions included in modifications. The following items, if applicable, are considered allowable costs for calculating General Conditions:

1. Project Manager (PM), Assistant Project Manager
2. Superintendent, Assistant Superintendent
3. Quality Control, Safety Officer, Environmental Manager, etc.
4. Engineers

5. Travel, lodging, and per diem (as established by Federal Travel Regulations)
6. Scheduling
7. Field Office Trailers and associated temporary utilities
8. Field office supplies

    a. Mailing and couriers
    b. Reproduction costs
    c. Storage
    d. Phones
    e. Computers
    f. Copiers

9. Personal vehicles i.e. Superintendent Pickup trucks

C. General Requirements: Costs directly associated with the project and are necessary to perform the actual work of the modification. These costs shall be shown as direct costs in the estimate. The following items, if applicable, are considered allowable costs for calculating General Requirements:

1. Hoisting
2. Material handling
3. Temporary fencing
4. Port-a-lets
5. Trash removal, dumpsters
6. Barricades
7. Small tools
8. Safety supplies
9. Scaffolding
10. Daily cleaning
11. Traffic control
12. Temporary signage
13. Temporary heating and power

D. Personnel Costs: Costs included in the modification must only be for General Conditions staff and workers actually present and working on project site. Modification costs for salaried workers are only allowed within the structure of a 40-hour week and no overtime or holiday pay will be allowed.

1. Worker Hourly Rates are costs directly associated with the individual worker and consist of the following:

    a. Base Rate: The hourly rate paid directly to the worker
    b. Labor Burden: Employer payments of all applicable burdens; includes insurance and taxes the business must pay on behalf of the worker to government entities and educational forums, such as:

        1) Social Security
        2) Medicare
        3) Workers Compensation – Policy and company calculation to be made available.

4) Federal Unemployment Tax Act (FUTA) - Cap Rate and percentage to be proportionally allocated over one year.
5) State Unemployment Tax Act (SUTA) - Cap Rate and percentage to be proportionally allocated over one year.
6) Union agreement costs - Other costs required under an enforceable collective bargaining agreement.

c. Fringe Benefits: Various non-wage compensations provided to employees such as:

1) Health Care Insurance Premiums
2) Cell Phone
3) Clothing
4) 401K and Pensions
5) Vehicle allowances
6) Gas allowance
7) Life insurance premiums
8) Disability insurance
9) Other Fringe Benefits required under an enforceable collective bargaining agreement

E. Bonuses or Deferred Compensation: No Bonus or Deferred Compensation will be allowed within any components of pricing including Home Office Overhead, General Conditions, General Requirements, Hourly Worker Rates, or the direct costs of work.

F. General Liability Insurance: An insurance policy that protects Contractor from claims resulting from bodily injury or property damage to a third party. Include as a separate line item within all modification proposals and provide a current insurance quote upon request.

G. Performance and Payment Bonds: A performance bond is a surety bond issued by an insurance company or bank to guarantee satisfactory completion of a project. The Payment Bond guarantees the Contractor will pay the labor and material costs incurred. Banks and Insurance companies charge a premium for individual project based on a sliding scale related to the size of the project. Include as a separate line item in modification proposals and provide current company bonding rates upon request.

H. Builder's Risk Insurance: Covers the contractor's loss due to fire, high winds, or other natural forces. Not reimbursed by the National Park Service (NPS) and shall not be included in modification proposals.

1.3 MODIFICATION PROPOSAL PRICING REQUIREMENTS

A. General:

1. Proposal shall be received in the format and within the time frame specified in the Request for Proposal (RFP) letter. Costs or delays resulting from failure of contractor to submit within the time frame specified will not be compensable.
2. Proposal shall be detailed with itemized lists of equipment, materials, labor, production rates, overhead, profit, and bond markup for each item. Labor costs must be itemized by craft and hourly rate, including Fringe Benefits and Labor Burden. If the costs of Fringe Benefits and Labor Burden are not itemized, it is assumed they are included in the hourly

rate shown, or contractor is not requesting reimbursement. Contractor may utilize the government provided Contractor Estimate Form, or their own form, provided that it contains the same information and level of detail as the Government's form.

3. Requests for extensions of contract time as a result of change must be justified with a Time Impact Analysis (TIA). Refer to Section 01 32 16 "Construction Schedule", for time impact analysis requirements. TIA and associated costs shall be received with the proposal by the date shown within the Request for Proposal letter. Contractor's failure to submit within the specified time frame will be construed as the Contractor waiving right for additional time and no time extension will be allowed.
4. All supporting documentation used to justify the proposed modification will be made available to the Contracting Officer (CO) upon request.
5. Contractor shall review and approve all subcontractor/supplier pricing in detail for proper format, scope, production rates, and pricing prior to submission to NPS. All delay costs associated with not reviewing and approving subcontractor/supplier pricing will be borne by the Contractor.
6. All pricing and production rates within the estimate must be based on fair and reasonable pricing and cannot include built-in contingency.

B. Labor:

1. Contractor shall estimate cost of labor by itemizing each craft involved, indicating worker hourly rate (base rate + labor burden + fringe benefits) for each and itemizing hours required for each craft directly engaged in modification work. Any work proposed requiring overtime work or premium pay shall be itemized separately. Rates shall be in accordance with the Davis-Bacon Act as incorporated herein. Labor Burden may include payroll taxes, Social Security, unemployment insurances, workers compensation insurance, Federal Insurance Contributions Act (FICA), FUTA, and other direct costs resulting from Federal, State or local laws.
2. Itemize labor costs for equipment operators separate from equipment costs.
3. Labor cost for foremen shall only be costs for related work required for the modification.

C. Materials:

1. Estimated cost for materials shall include quotes from multiple sources. Material prices shall include applicable fees and credits, including but not limited to, sales tax, freight and delivery charges, and tax rebates.
2. No markup shall be applied to any material provided by NPS.

D. Equipment:

1. Equipment used for the project must be appropriately sized for work being performed.
2. Do not include costs for "miscellaneous tools and equipment", in your proposal for a replacement value of $500 or less. Costs shown in excess of $500 shall be broken out separately.
3. Regardless of ownership, rates to be used in determining equipment rental costs shall be the lowest cost from one of the following sources:

    a. United States (U.S.) Army Corps of Engineers, Ownership and Operating Expense Schedule (use latest edition and applicable region)
    b. Construction Blue Book

c. Local equipment rental rates, documented by actual invoice charges, or itemized vendor quotes.

4. Estimated equipment rates shall include operating costs of all fuel, oil, lubrication, supplies, small tools, necessary attachments, ground engaging components, tires and tracks, routine repairs and maintenance (cost of major repair and overhaul is not allowed per Federal Acquisition Regulation (FAR) 31.105(d)(2)), depreciation, storage, insurance, and all incidentals. Mobilization, if applicable, may be included for equipment solely used on the modification work but must be listed separately.
5. Estimate full rate for equipment only for duration that equipment will be utilized to accomplish work of the modification.
6. Standby unit rates used in accordance with paragraph 1.3, D, 2, above. If the U.S. Army Corp of Engineers is utilized then their standby rates prevail. If Bluebook or local equipment pricing is accepted, then 1/2 of equipment costs minus any operating costs, major repair and overhaul will be accepted.
7. If equipment is in standby mode due solely to a documented NPS delay, established standby rate shall apply from the first day of the delay.
8. Equipment not used and on job site for up to five consecutive days may be classified at standby rates, provided the equipment is or has been used solely to perform work on the modification and will be necessary to complete additional modification work. Equipment still on the jobsite but not in use after five consecutive days will not be considered in the modification pricing.
9. Requests for compensation for equipment stand by time must be justified, documented and itemized separately.
10. The estimated timeframe (daily, weekly, monthly) for use of the equipment must reflect the lowest cost to the Government.

E. Establishment and Application of Overhead and Profit Percentages:

1. Utilize Profit Calculator to generate allowable maximum rate for Profit on self-performed work. Insert result below where indicated by the red X.XX%. Work collaboratively with Contracting Officer Representative (COR) to produce draft Profit Calculator for Design Development (DD) submission. Denver Service Center Contracting Services (DSC CS) division will approve final number. Submit completed Profit Calculator with DD submission.
2. Home Office Overhead and Profit (OH&P) shall be applied to direct costs only. Profit shall not be applied to overhead amounts; and overhead shall not be applied to profit. Home office overhead shall contain only allowable, allocable, and reasonable costs per the contract documents and FAR Part 31. Profit percentages are based on risk factors found in FAR Part 31which have been applied to the specific type of work included in this project. Negotiated rates shall not exceed the following percentages for OH&P for contractor self-performed work:

   Overhead……15%
   Profit……..……10.00%

3. Total aggregate limit of markup (OH&P) for Contractor and Subcontractors on modification work shall not exceed 25%. The NPS will not be responsible for allocation of percentages between contractor and subcontractors at any tier.
4. If Contractors form a partnership, partnership may only receive home office overhead and profit in same amount as an individual Contractor (refer to paragraph 1.3,E,1 above). It is the responsibility of the partners to decide on division of revenue.

5. Combined Increases and Decreases: On proposals involving both increases and decreases in the Contract Price, overhead and profit mark-ups are required on net increases and deducted on net decreases.
6. At no time can profit be calculated on Overhead or itself, it must be calculated on direct costs of work only.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 26 01

SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section explains in general, what is and is not included in a contract line item, and limits or cut-off points where one item ends and another begins.

B. If no contract line item exists for a portion of work, include costs in a related item.

C. Unit pricing for masonry repointing, masonry crack repair, and brick masonry replacement.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 LIST OF CONTRACT LINE ITEMS

A. Contract Line Item Number 0001: Visitor Center Base Scope (Legacy Restoration Fund (LRF) Funded).

1. This item consists of foundation and superstructure improvements, site restoration around the visitor center, the conduit for the low voltage cable, and foundation repair/waterproofing.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

B. Contract Line Item Number 0002: Temporary Visitor Center Trailer in Parking Lot (LRF Funded).

1. This item consists of an ABA compliant mobile trailer to serve as a temporary visitor center during construction, with 1 lockable office. A separate temporary portable ABA compliant restroom trailer for four (4) gender neutral visitor toilets, and temporary utilities to support each mobile unit.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

C. Contract Line Item Number 0003: Temporary Office Space Rental and Move (LRF Funded).

1. This item consists of establishing a temporary office space for visitor center staff to include rental of the space and moving costs from the Visitor Center to McGregor.
2. Measurement for payment will be lump sum.

3. Payment will be made at the contract lump sum price.

D. Contract Line Item Number 0004: Visitor Center Collections Move (LRF Funded).

1. This item consists of moving the visitor center collections to a temporary storage location during construction of the visitor center and move of collections back to the visitor center after construction is complete.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

E. Contract Line Item Number 0005: Parking Lot Replacement (Federal Highways Funded).

1. This item consists of selective demolition within the parking lot and maintenance area, parking lot construction to include stormwater improvements, pedestrian facilities, asphalt and concrete paving, curbs, gutters, and sidewalks, vehicular signage, and pedestrian lighting.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

F. Contract Line Item Number 0006 (Option A): Visitor Center HVAC & Fire Suppression (LRF Funded).

1. This item consists of removing existing HVAC and fire suppression, installing new HVAC and fire suppression and security/fire alarm, radon mitigation, and elevator tune up. If this CLIN is performed without the interior work, it assumes main ducting is in attic and ceiling removal for basement level fire alarm conduit is not repaired. Includes batt insulation, wood catwalk in attic, and reconfiguring the attic access hatch.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

G. Contract Line Item Number 0007 (Option B): Visitor Center Exterior Envelope Rehabilitation (LRF Funded).

1. This item consists of removing exterior siding and roofing, reframing new window openings, install new roofing, repair roof sheathing and roof accessories (snow guards and roof vents), rehabilitate existing brick, install new exterior weather barrier, insulation, aluminum cladding, metal panels, exterior windows and doors, soffit, exterior columns (including concrete plinths), and rehab existing railing. Includes finalizing site work, such as installing the flag pole and other site furnishings.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

H. Contract Line Item Number 0008 (Option C): Visitor Center Interior Accessibility Compliance, Plumbing, & Electrical (LRF Funded).

1. This item consists of performing hazardous material abatement. Removal of interior finishes and demo select interior walls for interior reconfiguration, install new walls at reconfigured restrooms and office spaces, install new transaction desk, light fixtures, electrical/data conduit and cabling, lobby monitors, plumbing and plumbing fixtures, finishes (epoxy, solid surface paneling, carpet, rubber flooring, rubber treads/risers), toilet

accessories, interior doors and windows, ACT where indicated, paint all walls, select ceilings, retained interior doors, and stain lobby wood ceiling.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

I. Contract Line Item Number 0009 (Option D): Visitor Center Low Voltage Fire Alarm Update (LRF Funded).

1. This item consists of approximately 800 LF of new cabling through the parking lot 2" conduit for low voltage VC fire alarm connection improvements. The new 2" conduit for the cabling is included in CLIN 1 and shown in the Parking Lot Replacement Base drawing set.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

J. Contract Line Item Number 0010 (Option E): Visitor Center Lightning Protection System (LRF Funded).

1. This item consists of the installation of a new lightning protection system at the visitor center.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

K. Contract Line Item Number 0011 (Option F): Visitor Center Meeting Room/Auditorium AV & Exhibit Area Renovation (NPS Funded).

1. This item consists of the installation of new audio-visual equipment to include display, headend equipment, wall mounted and floor connections for input from PC or laptop, video player, speakers, microphones, folding acoustic movable wall, steel structure supporting the movable wall, and window treatments.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

L. Contract Line Item Number 0012 (Option G): Visitor Center Expand Deck at NE Corner (Alternative Funds, TBD).

1. This item consists of expansion of the concrete expansion deck at the visitor center, with steel columns and framing, concrete footings and topping slab, including a new railing system to match existing with dual height viewing scope.
2. Measurement for payment will be lump sum.
3. Payment will be made at the contract lump sum price.

3.2 UNIT PRICING

A. Contractor shall provide proposal unit pricing masonry work as a part of CLIN 02 Base Bid work for additions and deletions from Work as authorized by change orders.

1. Unit Price 1: Masonry Repointing.

a. As defined in section 04 01 20.64. Unit price shall include all materials, equipment, labor, overhead and profit for a complete installation.
b. Measurement for payment shall be per linear foot.

2. Unit Price 2: Masonry Crack Repair.

a. As defined in section 04 01 20.63. Unit price shall include all materials, equipment, labor, overhead and profit for a complete installation.
b. Measurement for payment shall be per linear foot.

3. Unit Price 3: Brick Masonry Replacement.

a. As defined in section 04 26 13. Unit price shall include all materials, equipment, labor, overhead and profit for a complete installation.
b. Measurement for payment shall be per linear foot.

END OF SECTION 01 27 00

SECTION 01 31 00 – PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:

1. Definitions
2. Construction Coordination
3. Submittals
4. Coordination Drawings
5. Requests for Information (RFIs)
6. NPS Project Management and Communication Software
7. Project Meetings
8. Environmental Coordination
9. Permits

B. Related Requirements:

1. Section 01 32 16 "Construction Schedule" for preparing and submitting Construction Contractor's construction schedule.
2. Section 01 73 40 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
3. Section 01 77 00 "Closeout Procedures" for coordinating closeout of the Contract.
4. Section 01 91 14 "Total Building Commissioning" for coordinating the work with Government's Commissioning Authority.

1.2 DEFINITIONS

A. Agency with Jurisdiction

B. Construction Permits – Contractor Provided

C. Government Furnished Permits

1.3 CONSTRUCTION COORDINATION

A. Coordination: Coordinate construction operations included in different Sections of Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, which depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
2. Coordinate installation of different components with other Construction Contractors to ensure maximum accessibility for required maintenance, service, and repair.
3. Make provisions to accommodate items scheduled for later installation.
4. Where availability of space is limited, coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair of components, including mechanical and electrical.
5. Properly plan construction operations to include permit requirements. Allow enough time to execute permit provisions to maintain work schedule, site visits, inspections, and reporting deadlines.

B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.

C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other Construction Contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to:

1. Preparation of Construction Contractor Construction Schedule
2. Preparation of the Schedule of Values
3. Installation and removal of temporary facilities and controls
4. Delivery and processing of submittals
5. Progress meetings
6. Permit requirements
7. Pre-installation conferences
8. Project closeout activities
9. Commissioning activities

1.4 SUBMITTALS

A. Coordination Drawings: Prepare Coordination Drawings if limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.

1. Content: Project-specific information, drawn accurately to scale. Do not base Coordination Drawings on reproductions of Contract Documents or standard printed data. Include following information, as applicable:

    a. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems.
    b. Indicate dimensions shown on Contract Drawings and make specific note of dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternate sketches to Contracting Officer (CO) for resolution of such conflicts.
    c. Indicate space requirements for routine maintenance and anticipated replacement of components during the life of the installation.

d. Show location and size of access doors required for access to concealed dampers, valves, and controls.
e. Indicate required installation sequences.

2. Refer to individual Sections for Coordination Drawing requirements for Work in those Sections.
3. Follow NPS DSC Electronic Drawing Standards**.** See CAD/BIM (Computer Aided Design/Building Information Modeling) & Drafting Standards page > DSC Requirements section.

B. Coordination Drawing Organization: Organize coordination drawings as follows:

1. Floor Plans and Reflected Ceiling Plans: Show architectural, structural, mechanical, plumbing, fire-protection, fire-alarm, and electrical elements. Show locations of visible ceiling-mounted devices relative to acoustical ceiling grid. Supplement plan drawings with section drawings where required to adequately represent the Work.
2. Plenum Space: Indicate sub-framing for support of ceiling and wall systems, mechanical and electrical equipment, and related Work. Locate components within ceiling plenum to accommodate layout of light fixtures indicated on Drawings. Indicate areas of conflict between light fixtures and other components.
3. Mechanical Rooms: Provide coordination drawings for mechanical rooms showing plans and elevations of mechanical, plumbing, fire-protection, fire-alarm, and electrical equipment.
4. Structural Penetrations: Indicate penetrations and openings required for all disciplines.
5. Slab Edge and Embedded Items: Indicate slab edge locations and sizes and locations of embedded items for metal fabrications, sleeves, anchor bolts, bearing plates, angles, door floor closers, slab depressions for floor finishes, curbs and housekeeping pads, and similar items.
6. Mechanical and Plumbing Work:

    a. Sizes and bottom elevations of ductwork, piping, and conduit runs, including insulation, bracing, flanges, and support systems.
    b. Dimensions of major components, such as dampers, valves, diffusers, access doors, cleanouts and electrical distribution equipment.
    c. Fire-rated enclosures around ductwork.

7. Electrical Work:

    a. Runs of vertical and horizontal conduit 1-1/4 inches in diameter and larger.
    b. Light fixture, exit light, emergency battery pack, smoke detector, and other fire-alarm locations.
    c. Panel board, switch board, switchgear, transformer, busway, generator, and motor control center locations.
    d. Location of pull boxes and junction boxes, dimensioned from column center lines.

8. Fire-Protection System:

    a. Locations of standpipes, mains piping, branch lines, pipe drops, and sprinkler heads.

9. Review: Contracting Officer will review coordination drawings to confirm Work is being coordinated; details of coordination are Construction Contractor's responsibility. If Contracting Officer determines coordination drawings are not prepared in scope or detail, or are otherwise deficient, Contracting Officer will inform Construction Contractor, who shall make changes and resubmit.
10. Coordination Drawing Prints: Prepare coordination drawing prints according to requirements in Section 01 33 23 "Submittal Procedures."

C. Coordination Digital Data Files: Prepare coordination digital data files according to:

1. File Preparation Format: Same digital data software program, version, and operating system as original Drawings.
2. File Submittal Format: Submit or post coordination drawing files using Portable Document Format (PDF) file format.
3. Contracting Officer will furnish Construction Contractor one set of digital data files (AutoCad.dwg) of Drawings for use in preparing coordination digital data files.

    a. Contracting Officer makes no representations as to accuracy or completeness of digital data files as they relate to Drawings.
    b. Digital Data Software Program: Drawings are available in AutoCad.dwg.

D. Division 1 documents:

1. Letter designating Project Superintendent
2. Construction Schedule
3. A Comprehensive Schedule of Values
4. Accident Prevention Plan / Site Specific Health and Safety Plan
5. A List of Subcontractors for this project
6. Written statements from Subcontractors certifying compliance with applicable labor standard clauses.
7. Certificates of Insurance and Standard Form SF1413 for Contactor and all Subcontractors
8. Waste Management Plan
9. Quality Control Plan
10. Temporary Storm Water Pollution Prevention Plan (SWPP)
11. Indoor Air Quality (IAQ) Management Plan
12. Construction Contractor Commissioning Plan
13. Hazardous Materials Management Plan
14. Wildlife Exclusion Plan
15. Park Data Protection and Access Plan
16. List of Required Construction Permits. Include the following information for each permit:

    a. Name of Permit
    b. Agency(ies) with Jurisdiction issuing the permit
    c. Information required from Government to complete permit application

E. Provide items listed to Contracting Officer at least one week prior to Pre-Construction Conference, or as directed by Contracting Officer.

1. Failure to submit items may result in:

a. Construction conference may be cancelled, Notice to Proceed may not be issued, and Contracting Officer will consider other contractual remedies.
b. Work shall not commence until written Notice to Proceed has been issued and Certificates of Insurance and SF1413 documents are received for Construction Contractor and all Subcontractors.
c. Notice To Proceed (NTP) will not be issued until all submittals as requested above are received and accepted by the Contractor Officer (CO).

1.5 REQUESTS FOR INFORMATION (RFIs)

A. General: Immediately on discovery of the need for additional information or interpretation of Contract Documents, Construction Contractor shall prepare and submit an RFI utilizing form created on NPS project management and communication software.

1. Contracting Officer will not respond to RFIs submitted by other entities controlled by Construction Contractor.
2. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.

B. Content of RFI: Include detailed, legible description of item needing information or interpretation and the following:

1. RFI number, numbered sequentially
2. Date
3. RFI subject
4. Specification Section number and title and related paragraphs, as appropriate.
5. Drawing number and detail references, as appropriate.
6. Field dimensions and conditions, as appropriate.
7. Construction Contractor's suggested resolution: If suggested resolution impacts Contract Time or Contract Sum, Construction Contractor shall state impact in RFI.
8. Construction Contractor's signature
9. Requested date for response
10. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.

a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.

C. RFI Form: Complete and submit on NPS project management and communication software.

D. Contracting Officer’s Action: Contracting Officer will review each RFI, determine action required, and respond. Contracting Officer will determine critical nature of each RFI and issue response accordingly.

1. The following are not considered to be RFIs and will receive no action:

a. Requests for approval of submittals.
b. Requests for approval of substitutions.
c. Requests for approval of Construction Contractor's means and methods.
d. Requests for coordination information already indicated in Contract Documents.

e. Requests for adjustments in Contract Time or Contract Sum.
f. Requests for interpretation of Architect's actions on submittals.
g. Incomplete RFIs or inaccurately prepared RFIs.

2. Contracting Officer's action may include a request for additional information; time for response will date from time of receipt of additional information.
3. Contracting Officer's action on RFIs may result in need for a change to Contract Time or Contract Sum. All contract changes will be processed following terms and conditions of contract.

1.6 PROJECT MANAGEMENT AND COMMUNICATION SOFTWARE

A. Use to manage project communication and document sharing during contract period on:

1. Project directory
2. Project correspondence
3. Meeting agendas and minutes
4. Contract modifications forms and logs
5. RFI form and processing
6. Task and issue management
7. Photo documentation
8. Baseline schedule, schedule updates and calendar management
9. Submittal form and processing
10. Payment coordination documentation
11. Reminder and tracking functions
12. Archiving functions
13. Notification of submittal and RFI statuses and current responsible party
14. Permits and addendums

B. Some documents are not suitable to be shared using NPS project management and communication software. Documents containing Personal Identifying Information (PII) (i.e. certified payrolls) shall not be shared and shall be coordinated with project team as appropriate.

1.7 PROJECT MEETINGS

A. Preconstruction Conference: Before start of construction, Contracting Officer will arrange an on-site meeting with Construction Contractor. Meeting agenda will include the following as a minimum:

1. Roles & Responsibilities / Lines of Authority
2. Park rules and regulations
3. Cultural resource compliance requirements
4. Jobsite Safety
5. Resolution of comments on required Division 1 documents
6. Coordination of Subcontractors
7. Labor law application
8. Modifications
9. Payments to Construction Contractor
10. Payroll reports

11. Contract time
12. Liquidated damages
13. Construction Contractor Performance Evaluation
14. Display of Hotline posters
15. Notice to proceed
16. Correspondence procedures
17. NPS project management and communication software
18. Acceptance/rejection of work
19. Progress meetings
20. Submittal procedures
21. NPS Final Accessibility Inspection
22. Environmental requirements
23. Permit requirements
24. As-constructed drawings/operation and maintenance (O&M) manuals.
25. Saturday, Sunday, holiday and night work.
26. Reference materials
27. Value engineering
28. Schedule of Values

B. Progress Meetings: Contracting Officer will schedule weekly meetings with Construction Contractor.

1. Attendees: In addition to Government Representatives, each Construction Contractor, Subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented. Participants at meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Meeting agenda will include:

    a. Approval of minutes of previous meetings
    b. Submittal status
    c. Review of off-site fabrication and delivery schedules.
    d. Requests for information (RFI) and other issues.
    e. Modifications
    f. Work in progress and projected.

        1) Status of required inspections (Special Inspections, Accessibility, etc.)

    g. Inspections of work in progress and projected (Special inspections, Accessibility, etc.)
    h. Construction Schedule update (provide updated Critical Path Method (CPM)).
    i. Status of Project Record Drawings and O&M manuals.
    j. Other business relating to work.
    k. Permit requirements

C. Preinstallation Conferences: Conduct at Project site before each construction activity that requires coordination with other construction.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by installation and its coordination or integration with other materials and

installations that have preceded or will follow, shall attend meeting. Advise Contracting Officer of scheduled meeting dates.

2. Agenda: Review progress of other construction activities and preparations for particular activity under consideration, including requirements for:

    a. Contract Documents
    b. Options
    c. Related RFIs
    d. Related Change Orders
    e. Purchases
    f. Deliveries
    g. Submittals
    h. Review of mockups
    i. Possible conflicts
    j. Compatibility requirements
    k. Time schedules
    l. Weather limitations
    m. Manufacturer's written instructions
    n. Warranty requirements
    o. Compatibility of materials
    p. Acceptability of substrates
    q. Temporary facilities and controls
    r. Space and access limitations
    s. Regulations of agency(ies) with jurisdiction
    t. Testing and inspecting requirements
    u. Installation procedures
    v. Coordination with other work
    w. Required performance results
    x. Protection of adjacent work
    y. Protection of construction and personnel

3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
4. Reporting: Distribute minutes of meeting to each party present and to other parties requiring information.
5. Do not proceed with installation if conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of Work and reconvene conference at earliest feasible date.

1.8 ENVIRONMENTAL COORDINATION

A. Construction Contractor’s Environmental Manager: Designate on-site party responsible for overseeing Construction Contractor’s conformance to environmental goals for project and implementing procedures for environmental protection.

1. Qualifications: Minimum 3 years Construction experience on projects of similar size and scope; with environmental procedures similar to this project; familiar with environmental regulations applicable to construction operations.
2. Responsibilities: Responsibilities shall include:

a. Compliance with applicable Federal, State, and local environmental regulations, including maintaining required documentation.
b. Implementation of Waste Management Plan (WMP).
c. Implementation of Indoor Air Quality (IAQ) Management Plan.
d. Implementation of Storm Water Pollution Prevention Plan (SWPPP).
e. Implementation of Hazardous Materials Management Plan
f. Implementation of Wildlife Exclusion Plan
g. Present overview of environmental issues and summarize site specific procedures relating to management plans at Preconstruction conference.
h. Training for Construction Contractor personnel in accordance with position requirements.
i. Monitoring and documentation of environmental procedures.

B. Perform project quality control in accordance with requirements specified in Related Sections, including:

1. Quality Requirements
2. Regulatory Requirements
3. Indoor Air Quality (IAQ) Management
4. Noise and Acoustics Management
5. Temporary Storm Water Pollution Prevention Environmental Management
6. Construction Waste Management

C. Construction Contractor's Environmental Training Program: Construction Contractor shall provide environmental training for workers performing work on project site. Training shall include:

1. Overview of environmental issues related to building industry.
2. Overview of environmental issues related to Project.
3. Review of site-specific procedures and management plans:

a. Construction Waste Management
b. Indoor Air Quality (IAQ) Management
c. Noise and Acoustics Management
d. Temporary Storm Water Pollution Prevention

4. Pollution Prevention (P2) practices: Submit evidence of familiarity with P2 practices.
5. Compliance with environmental regulations: As specified in Regulatory Requirements. Submit Construction Contractor 40 CFR (Code of Federal Regulations) employee training records upon request of Contracting Officer.

D. Provide documentation for environmental procedures as specified herein and in accordance with approved Waste Management Plan, IAQ Management Plan, and Storm Water Pollution Prevention Plan.

1.9 PERMITS

A. General:

1. Permits and Responsibilities: Construction Contractor shall, without additional expense to the Government, be responsible for obtaining necessary licenses and permits, and for complying with Federal, State and municipal laws, codes, and regulations applicable to the performance of the work. Construction Contractor shall also be responsible for damages to persons or property that occur as a result of Construction Contractor's fault or negligence; and for materials delivered and work performed until completion and acceptance of the work.
2. For the purpose of this contract, Construction Contractor will not be considered an agent of the Government. Construction Contractor shall comply with appropriate Federal, State and local laws.

B. Potential Permits: Permits listed below were identified during the design process as likely to be required based on typical means and methods of construction. The list is provided to assist Construction Contractor in determining which permits will be required for contract's chosen means and methods. The list shall not be considered complete; it is the Construction Contractors' responsibility to determine means and methods and obtain required permits. Construction Contractor shall obtain all permits required to legally conduct work.

1. Electrical Utility Hookup – Allamakee-Clayton Electric Cooperative
2. Gas Utility Hookup – National Park Service
3. Sewer Hookup – National Park Service
4. New Water Meter – National Park Service
5. Construction trailer, Utilities hookup permit – National Park Service
6. Stormwater Pollution Prevention Permit – Iowa Department of Natural Resources
7. Dewatering Permit – Iowa Department of Natural Resources

C. Coordination with Agency(ies) with Jurisdiction Issuing Permits

1. Coordination: Contact the Agency(ies) with Jurisdiction as needed and sufficiently in advance to avoid delaying work: Coordinate meetings, reporting requirements, inspections, and other requirements.

D. Administrative Procedures:

1. Coordinate scheduling and timing of required administrative provisions of project permits with Agency(ies) with Jurisdiction, Construction Manager, and Park to avoid conflicts.
2. Supply needed information to Agency(ies) with Jurisdiction issuing permits, pay fees required and provide material needed to comply with permit's conditions and provisions.
3. Upload permits to NPS project management and communication software when permits are obtained.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 31 00

SECTION 01 32 16 – CONSTRUCTION SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of Construction Schedule requirements including:

1. Schedule of Values
2. Construction Schedule Requirements.
3. Construction Schedule Updates.
4. Time Impact Analysis.

B. Purpose: The Construction Schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the work by the Construction Contractor. It shall assist the Construction Contractor and Contracting Officer (CO) in monitoring the progress of the work, evaluating proposed changes, and processing Construction Contractor's monthly progress payments. It shall include the dates in the contract, phases, milestones, occupancies, holidays, weather consideration, a critical path, and the requirements of this section.

1.2 DEFINITIONS

A. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.

1. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times.
2. Predecessor Activity: An activity that precedes another activity in the network.
3. Successor Activity: An activity that follows another activity in the network.

B. Cost Loading: Allocation of the Schedule of Values for completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by the Contracting Officer.

C. Critical Path Method (CPM): Method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.

D. Critical Path: Longest connected chain of interdependent activities through the network schedule that establishes minimum overall Project duration and contains no float.

E. Float: Measure of leeway in starting and completing an activity.

1. Float: Not for the exclusive use or benefit of the Government or Construction Contractor but is jointly owned.
2. Free Float: Amount of time an activity can be delayed without adversely affecting the early start of the successor activity.
3. Total Float: Measure of leeway in starting or completing an activity without adversely affecting planned Project completion date.

F. Resource Loading: Allocation of manpower and equipment necessary for completion of an activity as scheduled.

G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

1.3 SUBMITTALS

A. Electronic Copies: Schedules and reports submitted shall be posted on the NPS project management and communication software as PDF (portable document format) files. The intent of the Government is to limit the number of printed reports to those determined by the project team as essential.

B. Schedule of Values: After contract award and before Pre-Construction conference, submit schedule of dollar values based on Contract Price Schedule.

C. Construction Baseline Schedule: After contract award and before Pre-Construction conference, submit electronic copies of baseline schedule, large enough to show entire schedule for entire construction period. Utilize Schedule of Values in preparation of Construction Baseline Schedule.

D. Critical Path Method (CPM) Reports: Concurrent with CPM schedule, submit electronic copies of the following computer-generated reports. For each activity, include activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.

1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
2. Logic Report: List of predecessor and successor tasks for activities sorted in ascending order by activity number and then early start date, or actual start date if known.
3. Total Float Report: List of activities sorted in ascending order of total float.

E. Construction Schedule Updates: On or before 7th day preceding progress payment request date, submit estimates of percent completion of each schedule activity and necessary supporting data.

F. Construction Schedule Revisions and Time Impact Analysis: For each Construction Schedule revision, submit electronic copies of a Time Impact Analysis. Incorporate a Fragmentary Network (Fragnet) into currently accepted Construction Schedule demonstrating proposal to incorporate a modification, change, delay, or Construction Contractor request.

1.4 QUALITY ASSURANCE

A. Construction Contractor shall meet with Contracting Officer on day of the preconstruction conference to go over:

1. Review software limitations, content and format for reports.
2. Verify availability of qualified personnel needed to develop and update schedule.
3. Discuss constraints, including phasing, work stages, area separations, interim milestones, and partial Government occupancy/substantial completions.
4. Review delivery dates for Government-furnished products.
5. Review schedule for work of separate Government contracts.
6. Review time required for review of submittals and re-submittals.
7. Review requirements for tests and inspections by independent testing and inspecting agencies.
8. Review time required for completion and startup procedures.
9. Review time required for obtaining and activating permits.
10. Review and finalize list of construction activities to be included in schedule.
11. Review baseline schedule comments, resolve issues and progress on incorporating them
12. Review procedures for updating schedule.
13. Discuss reporting requirements and establish protocol for naming and transmitting electronic schedules.

B. Construction Contractor's Schedule Representative: Before the preconstruction conference, designate an authorized representative to be responsible for preparing and maintaining the Construction Schedule. Submit resume outlining qualifications of Scheduler to Contracting Officer for acceptance. Scheduler shall have prepared and maintained at least 5 previous schedules of similar size and complexity similar to this Contract, demonstrating proficiency of using scheduling software. Authorized representative will be responsible for preparing the Baseline Schedule, required updates, revisions, Time Impact Analyses, and reports.

1.5 COORDINATION

A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate Construction Contractors.

B. Coordinate Construction Baseline Schedule with Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.

1. In developing Construction Baseline Schedule, ensure Subcontractor’s work at all tiers, and prime Construction Contractor’s work, is included and coordinated.
2. Secure time commitments for performing critical elements of work from parties involved.
3. Coordinate each construction activity in network with other activities and schedule in proper sequence.

PART 2 - PRODUCTS

2.1 SCHEDULE OF VALUES

A. Breakdown each lump-sum item into component work activities used in the schedule for which progress payments may be requested. Work activities broken out within schedule of values shall be integrated into and made a logical part of the construction baseline schedule. Total costs for the component work activities shall equal contract price for that lump-sum item. Contracting Officer may request data to verify accuracy of dollar values. Include mobilization, general condition costs, overhead and profit in the total dollar value of unit price items and in the component work activities for each lump-sum item. Do not include mobilization, general condition costs, overhead or profit as a separate item. SOV shall be developed in a way to clearly identify SOV items on the project schedule coordinating item numbers. SOV items hall be arranged in a sequence to match the order of activities in the schedule and shall reference the corresponding CLIN. SOV shall clearly communicate current billing vs. billing to-date to be coordinated with future contractor progress payment applications.

B. Do not break down unit price items. Use only the contract price for unit price items.

C. Total cost of all items shall equal the contract price. The Schedule of Values will form the basis for progress payments and the Construction Schedule.

2.2 CONSTRUCTION SCHEDULE REQUIREMENTS

A. Construction Baseline Schedule: Prepare Construction Baseline Schedule using a computerized, cost and resource-based**,** time-scaled Critical Path Method network analysis diagram for the Work.

1. Develop and finalize Construction Baseline Schedule so it can be accepted for use no later than 30 days after date established for the Notice of Award.

a. Failure to include any work item required for performance of this Contract shall not excuse Construction Contractor from completing work within applicable completion dates, regardless of Governments acceptance of schedule.

2. Establish procedures for monitoring and updating Construction Baseline Schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.

B. Construction Baseline Schedule Preparation: Prepare a list of all activities required to complete the Work. Using preliminary Critical Path Method network diagram, prepare a skeleton network to identify probable critical paths.

1. Activities: Indicate estimated duration, sequence requirements, and relationship of each activity in relation to other activities.
2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates shall be consistent with Contract milestone dates.

3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the Critical Path Method schedule within the limitations of the Contract Time.
4. Show sequence and interdependence of activities required for completion of work. Ensure work sequences are logical and Construction Baseline Schedule shows a coordinated plan of the work.
5. Resource loading of each activity shall include personnel by labor category and equipment type and capacity proposed to complete the activity in duration shown.
6. Consider seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time.
7. Time Frame: Proposed duration assigned to each activity shall be Construction Contractor's best estimate of time required to complete activity considering the scope and resources planned for activity.

    a. An early finish date may be shown but the late finish date shall be same date as last day of contract period. An early completion schedule shall contain:

        1) Insert an activity titled "Project Float" as a successor to last activity in early project completion schedule network.
        2) Add a milestone titled "Contract End Date" as a successor to the activity "Project Float".
        3) Add duration to the activity "Project Float" as required so the milestone "Contract End Date" equals the last day of Contract Period.

    b. Contract completion date shall not be changed by submission of a schedule that shows an early completion date.
    c. Construction Contractor shall limit use of lead or lag duration’s between schedule activities.
    d. Project Calendars: Develop and incorporate the following calendars:

        1) Administrative Calendar: Include calendar based on a 7-day week to be used on activities based on calendar days. Apply this calendar to administrative tasks or other tasks not affected by non-working days (Federal Holidays, weather, etc.).
        2) Project Calendar: Include calendar based on planned work week for the project. Include Federal Holidays, weekends, and non-workdays indicated in contract documents. Apply this calendar to activities not anticipated to be affected by weather. Be clear when identifying number of work days in work week.
        3) Weather Calendar: Utilize Project Calendar and show anticipated normal downtime related to weather as non-working time. Weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer. Apply this calendar to activities anticipated to be affected by weather.

    e. Activity Duration: Define so no activity is longer than 14 days, except for non-construction activities including mobilization, shop drawings and submittals, fabrication and delivery of materials and equipment.

f. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 calendar days, as separate activities in the schedule. Procurement cycle activities can include submittals, approvals, purchasing, fabrication, and delivery.

1) Steel beams
2) Mechanical system equipment
3) Parking lot and visitor center lighting fixtures

g. Submittal Review Time: Include review and re-submittal times indicated. Coordinate submittal review times in Construction Baseline Schedule.
h. Startup and Testing Time: Include not less than 30 days for startup and testing and commissioning activities.
i. Substantial Completion: Allow time for Government administrative procedures necessary for certification of Substantial Completion. (For more information, refer to Specification 01 77 00 "Closeout Procedures.")

8. Constraints: Include constraints and work restrictions indicated in Contract Documents and as follows in schedule and show how the sequence of Work is affected.

a. Phasing: Arrange list of activities on schedule by phase.
b. Work under More Than One Contract: Include a separate activity for each contract.
c. Work Restrictions: Show effect of the following on the schedule:

1) Coordination with existing construction
2) Limitations of continued occupancies
3) Uninterruptible services
4) Partial occupancy before Substantial Completion
5) Use of premises restrictions
6) Provisions for future construction
7) Seasonal variations
8) Environmental control
9) Permit provisions

d. Work Stages: Indicate important stages of construction for each major portion of the Work.

1) Subcontract awards
2) Submittals
3) Purchases
4) Mockups
5) Fabrication
6) Sample testing
7) Deliveries
8) Installation
9) Tests and inspections
10) Adjusting
11) Curing
12) Building flush-out.
13) Building commissioning activities.
14) Move-in of FF&E.

9. Milestones: Include milestones indicated in Contract Documents in schedule, including, but not limited to, Notice to Proceed, Substantial Completion, and the following interim milestones:

    a. Division One accepted by Contracting Officer
    b. Temporary facilities installed and hooked up/NPS move out/move in
    c. Final acceptance by Contracting Officer

C. Joint Review, Revision, and Acceptance:

1. Within seven calendar days of receiving Construction Contractor's proposed Construction Baseline Schedule, Contracting Officer shall review initial Construction Baseline Schedule.
2. Within seven calendar days after review, Construction Contractor shall revise and resubmit Construction Baseline Schedule in accordance with comments presented from review.
3. In the event the Construction Contractor fails to define any element of work, activity, or logic and the Contracting Officer review does not detect this omission or error, such omission or error, when discovered by Construction Contractor or Contracting Officer, shall be corrected by Construction Contractor within seven calendar days and shall not affect contract period.
4. Upon acceptance of the Construction Baseline Schedule, Contracting Officer saves schedule as a baseline and updates on a monthly basis. Construction schedule update will be used to evaluate Construction Contractor's monthly applications for payment based upon information developed at monthly Construction Schedule update meeting.

D. Cost Correlation: In the heading of the schedule, provide a cost correlation line, indicating planned and actual costs. On the line, show dollar volume of Work performed as of dates used to prepare payment requests.

1. Construction Contractor shall assign cost to construction activities on Construction Baseline Schedule. Costs shall not be assigned to submittal activities unless specified otherwise but may, with Contracting Officer’s approval, be assigned to fabrication and delivery activities. Costs shall be included for testing and commissioning activities, operation and maintenance manuals, punch list activities, Project Record Documents, and demonstration and training (if applicable).
2. Each activity cost shall reflect an accurate value based on the Contract Price Schedule.
3. Total cost assigned to activities shall equal total Contract Price.

E. Recovery Schedule: When periodic schedule update indicates Work is fourteen < 14 > or more calendar days behind current accepted schedule, a separate recovery schedule indicating means by which Construction Contractor intends to regain compliance with the schedule shall also be submitted. Indicate changes to working hours, working days, crew sizes, and equipment required to achieve compliance, and date by which recovery shall be accomplished.

F. Computer Software: Prepare schedules using a program developed specifically to manage construction schedules.

1. Use Microsoft Project or Primavera (Windows 7 and newer.)

PART 3 - EXECUTION

3.1 CONSTRUCTION SCHEDULE UPDATES

A. Progress Meeting Updates: Provide a 3 week look-ahead schedule (consisting of the current week and 3 additional weeks), derived from the currently accepted schedule, before each weekly progress meeting. Utilize look-ahead schedule to facilitate and take notes on discussions held during progress meeting.

B. Monthly Schedule Updates:

1. General: Update Construction Schedule on monthly basis to reflect construction progress and activities throughout entire contract period and until project substantial completion. The status date of each schedule update shall be the 7th day preceding the progress payment request date.
2. Procedure: Construction Contractor shall meet with Contracting Officer each month at Construction Schedule update meeting to review progress made through the status date of the Construction Schedule update, including dates activities were started or completed and percentage of work completed on each activity started or completed.
3. Reports: Concurrent schedule revisions, prepare tabulated reports showing:

    a. Identification of activities that have changed
    b. Changes in early and late start dates
    c. Changes in early and late finish dates
    d. Changes in activity durations in workdays
    e. Changes in the critical path
    f. Changes in total float or slack time
    g. Changes in the Contract Time

4. Narrative: Report shall include a brief description of actual progress made during update period; actual and potential delaying activities; impediments to progress; issues related to inclement weather; progress toward established milestones and project float. Report shall include a brief description of work anticipated to be performed in the next month. Minor revisions to the schedule should be identified for evaluation and acceptance or rejection.
5. As Work progresses, indicate Actual Completion percentage for each activity.
6. If schedule update shows a late finish date after contract completion date, include:

    a. Known delays
    b. Actions to get back on schedule
    c. Pending modifications
    d. Impediments or constraints affecting progress

7. Progress Payments: Monthly updating of the currently accepted Construction Schedule shall be an integral part of the process upon which progress payments will be made. If Construction Contractor fails to provide schedule updates or revisions, a portion of the monthly payment may be retained until corrections have been made.

C. Distribution: Distribute copies of accepted schedule to Contracting Officer, Contracting Officers Representative, Construction Management Representative, Subcontractors, testing and inspecting

agencies, and other parties identified by Construction Contractor with a need-to-know schedule responsibility.

1. Post copies in Project meeting rooms and temporary field offices.
2. When revisions are made, distribute updated schedules to same parties and post in same locations. Delete parties from distribution when they have completed their assigned portion of the Work.

D. Construction Schedule Revisions:

1. Required Revisions: If, as a result of the monthly schedule update, it appears the currently accepted Construction Schedule no longer represents actual prosecution and progress of the work, Contracting Officer will request, and Construction Contractor shall submit, a revision to the Construction Schedule. Construction Contractor may also request reasonable revisions to currently accepted Construction Schedule in event the Construction Contractor's planning for the work is revised. If Construction Contractor desires to make changes, Construction Contractor shall notify Contracting Officer in writing, stating reason for proposed revision. Accepted revisions shall be incorporated into currently accepted Construction Schedule for next monthly schedule update.
2. Procedure: If revision to currently accepted Construction Schedule is contemplated, Construction Contractor or Contracting Officer shall advise the other in writing at least seven calendar days prior to next monthly schedule update meeting, describing revision and reasons for the revision. Government-requested revisions will be presented in writing to the Construction Contractor, who shall respond in writing within seven calendar days.
3. Reports: Concurrent with making revisions to schedule, prepare tabulated reports showing:

    a. Identification of activities changed
    b. Changes in early and late start dates
    c. Changes in early and late finish dates
    d. Changes in activity durations in workdays
    e. Changes in critical path
    f. Changes in total float or slack time

3.2 TIME IMPACT ANALYSIS FOR CONTRACT MODIFICATIONS CHANGES DELAYS AND CONSTRUCTION CONTRACTOR REQUESTS:

A. Requirements: When contract modifications or changes are initiated, delays experienced, or Construction Contractor desires to revise currently accepted Construction Schedule, Construction Contractor shall submit to Contracting Officer a written time impact analysis illustrating the influence of modification, change, delay, or Construction Contractor request on contract time.

B. Time Extensions: Activity delays, resulting in a late completion date projection, shall not automatically mean an extension of contract time is warranted or due to Construction Contractor. It is possible a modification, change, or delay will not affect existing critical path activities or cause non-critical activities to become critical. A modification, change, or delay may result in absorbing a part of available total float that may exist within an activity chain of the Schedule, not causing any effect on contract time. Time extensions will be granted in accordance with terms of contract.

C. Extension of contract time will be granted only to the extent the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.

D. Procedure: Each time impact analysis shall be submitted within time period stated in a request for proposal, or time period designated under the clauses entitled Changes or Default. In cases where Construction Contractor does not submit a written request for extension of time and a time impact analysis within the designated time, it is mutually agreed that the particular modification, change, delay, or Construction Contractor request does not require an extension of the contract time. Upon acceptance, time impact analysis shall be incorporated into currently accepted Construction Schedule at next monthly schedule update.

1. Contract Modifications: Prepare time-impact analysis using fragnets to demonstrate effect of proposed change on overall Construction Schedule for each proposed contract modification concurrent with submission.

END OF SECTION 01 32 16

SECTION 01 32 33 – PHOTO DOCUMENTATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for:

1. Existing condition images
2. Periodic construction images

B. See Section 01 77 00 "Closeout Procedures" for a complete listing of closeout documents.

C. See Section 01 79 00 "Demonstration and Training" for submitting videotapes of demonstration of equipment and training of National Park Service (NPS) personnel.

1.2 SUBMITTALS

A. Existing Condition Images: Submit images electronically within [seven] <7> days of taking the image. Include:

1. Date, time and number (sequentially number all images) in filename.
2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.
4. Upload images to the NPS Project Management Software and organized into folders or albums for file management.
5. Images shall be sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 1600 by 1200, and 300 dpi (dots per inch).

B. Construction Images: Submit images electronically within [seven] <7> days of taking the image. Include:

1. Date, time and number (sequentially number all images) in filename.
2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.
4. Upload images to the NPS Project Management Software and organized into folders or albums for file management.
5. Images shall be sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 1600 by 1200, and 300 dpi (dots per inch).

C. Closeout: Submit complete set of digital image electronic files as a Project Record Document**.** Submit on Digital Video Disc (DVD).

1. Provide index as separate file on Disc. List each image as a file name with number, date, and time. Include description and or vantage point image was taken.
2. Submit images that have the same aspect ratio as the sensor, un-cropped.
3. Upload images to the NPS Project Management Software and organized into folders or albums for file management.
4. Images shall be sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 1600 by 1200, and 300 dpi (dots per inch).

PART 2 - PRODUCTS

2.1 FORMAT REQUIREMENTS

A. Images: Provide sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 1600 by 1200, and 300 dpi (dots per inch).

B. Upload images to project management software.

PART 3 - EXECUTION

3.1 CONSTRUCTION IMAGES

A. General: Take digital images using the maximum range of depth of field, in-focus, to clearly show the Work. No blurry or out-of-focus areas accepted.

1. Maintain index with each set of Construction images and identify the number, date, time, and description for each.
2. Maintain one set of images accessible in field office at Project site available for reference.

B. Existing Condition Images: Before mobilization and commencement of work activities, take color digital images of Project site and surrounding properties, including existing items to remain during construction, from different vantage points.

1. Flag construction limits before recording construction images.
2. Take eight separate images to show existing conditions adjacent to property before starting Work.
3. Take eight separate images of existing buildings either on or adjoining property to accurately record physical conditions at start of construction.

C. Periodic Construction Images: Take 12 color, digital images monthly, with cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last images were taken.

D. Additional Images: Contracting Officer may issue requests for additional images.

1. Three days advance, where feasible.
2. In emergency situations, take additional images within 24 hours of request.
3. Additional images include, but are not limited to:

    a. Immediate follow-up when on-site events result in construction damage or losses.
    b. Fabrication locations away from Project site.
    c. Substantial Completion of a major phase or component of Work.
    d. Extra record images at time of final acceptance.

END OF SECTION 01 32 33

SECTION 01 33 23 – SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.2 DEFINITIONS

A. Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.

B. Informational Submittals: Written information that does not require Government's responsive action. Submittals may be rejected for not complying with requirements.

C. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access files.

D. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

1.3 GENERAL SUBMITTAL PROCEDURES

A. General: Prepare and submit submittals required by individual Specification Sections and in some cases as requested in drawings. Types of submittals are indicated in individual specific sections.

1. Contracting Officer (CO) reserves right to require submittals in addition to those called for in individual sections.

B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Review for legibility, accuracy, completeness, and compliance with Contract Documents.

1. Coordinate submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
2. Coordinate transmittal of different types of submittals for related parts of Work so processing will not be delayed because of need for concurrent review coordination.

a. Contracting Officer reserves right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

C. Submittal List: Submittal list is attached to the end of this Specification Section. The intent is to provide an overall summary of submittal requirements. The requirements of individual Specification Sections and terms and conditions of the Contract still apply regardless of what is shown on submittal list.

D. Processing Time: Allow enough time for submittal review and re-submittals. Contract time extensions will not be authorized if submittals are not transmitted with enough time to permit processing, including re-submittals.

1. Action Submittals

a. Initial Review: Allow 15 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
b. Re-submittal Review: Allow 15 days for review of each re-submittal.

2. Informational submittals

a. Review: Allow 10 days for review of each submittal.

E. Approved Equal Submittal Requirements:

1. For each item proposed, submit supporting data, including:

a. Drawings and samples as appropriate.
b. Comparison of the salient characteristics of the proposed item with that specified.
c. Changes required in other elements of the work because of the substitution.
d. Name, address, and telephone number of vendor.
e. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.

2. A request for approval constitutes a representation that Construction Contractor:

a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
b. Will provide the same warranties for the proposed item as for the item specified.
c. Has determined that the proposed item is compatible with interfacing items.
d. Will coordinate installation of an approved item and make changes required in other elements of the work because of the substitution.
e. Waives claims for additional expenses that may be incurred as a result of the substitution.

F. Submittals: Identify and incorporate information in each submittal file as follows:

1. Transmittal Form (CM-16): Utilize NPS Transmittal Form for all submittals. No action will be taken on a submittal item unless accompanied by this Transmittal Form.

2. Physical samples: Complete Transmittal Form and deliver physical sample to the Contracting Officer (or designee) on site for processing. All comments and actions will be documented on the Transmittal Form.

G. Identification: Submittal number or other unique identifier, including revision identifier.

1. Submittal number shall use a sequential number (e.g. .001). Re-submittals shall include alphabetic suffix after another decimal point (e.g. .001.A).

H. Re-submittals: Make re-submittals using same process used with initial submittal.

1. Note date and content of previous submittal.
2. Note date and content of revision in the title block on the Transmittal Form (CM-16) and clearly indicate extent of revision.
3. Re-submit submittals until they are marked "Approved" or "Approved with notations".

I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, and others as necessary for performance of construction activities.

J. Use for Construction: Use only final submittals with mark indicating "Approved" or "Approved with notations". Ensure notations have been incorporated and, at a minimum, keep one copy of final approved submittal on site for use during construction.

1.4 CONSTRUCTION CONTRACTOR'S USE OF CAD/BIM FILES

A. General: At Construction Contractor's written request, copies of CAD/BIM (Computer Aided Design/Building Information Modeling) files will be provided to Construction Contractor for use in connection with Project, subject to:

1. Files provided as is; no format or other changes to files or changes to objects in the drawing will be done by the Government.

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
2. Mark each submittal to show which products and options are applicable.
3. As applicable, include:

a. Manufacturer's product specifications.
b. Manufacturer's installation instructions: When Contract Documents require compliance with manufacturer's printed instructions, provide one complete set of instructions to Contracting Officer and keep another complete set of instructions at the project site until substantial completion.

c. Manufacturer's catalog cuts: Submit only pertinent pages; mark each page of standard printed data to identify specific products proposed for use.
d. Wiring diagrams showing factory-installed wiring.
e. Printed performance curves.
f. Operational range diagrams.
g. Compliance with specified referenced standards.
h. Testing by recognized testing agency.

4. Submit product data in PDF (portable document format) file format before or concurrent with samples.

B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data, unless submittal of CAD/BIM Drawings is otherwise permitted.

1. Preparation: Fully illustrate requirements in Contract Documents. As applicable, include:

a. Dimensions
b. Identification of products
c. Fabrication and installation drawings
d. Roughing-in and setting diagrams
e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring
f. Shopwork manufacturing instructions
g. Templates and patterns
h. Schedules
i. Notation of coordination requirements
j. Notation of dimensions established by field measurement
k. Relationship to adjoining construction clearly indicated
l. Seal and signature of professional engineer if specified
m. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring

2. Submit shop drawings as PDF electronic file

C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.

1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
2. Complete and post the Transmittal Form (CM-16) on the NPS project management and communication software website for processing and documentation of action on submitted samples.
3. Identification: Attach label on unexposed side of Samples that includes:

a. Generic description of Sample
b. Product name and name of manufacturer
c. Sample source
d. Submittal Number and title of appropriate Specification Section

4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing full range of colors, textures, and patterns available.

    a. Number of Samples: Submit two full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Contracting Officer will return with options selected.

6. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.

    a. Number of Samples: Submit four sets of Samples. Contracting Officer will retain three Sample sets; remainder will be returned. Retain Sample set as a Project Record Sample.

D. Construction Materials: Construction Contractor is encouraged to submit products made out of recycled or environmentally responsible material. Every effort will be made by NPS to approve these materials.

E. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."

F. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."

G. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

H. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. As applicable, include:

1. Statement on condition of substrates and their acceptability for installation of product.
2. Summary of installation procedures being followed, compliance with requirements and, if not, what corrective action was taken.
3. Results of operational and other tests and a statement of whether observed performance complies with the requirements.

2.2 INFORMATIONAL SUBMITTALS

A. General: Prepare and submit Informational Submittals required by individual Specification Sections.

1. Post informational submittals as PDF electronic files directly to the NPS project management and communication software.
2. Certificates and Certifications: Provide a notarized statement with signature of entity responsible for preparing certification. Certificates and certifications shall be signed by officer or other individual authorized to sign documents on behalf of that entity.
3. Informational submittals that do not comply with requirements specified in Contract Documents will be rejected and one copy will be returned.

B. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."

C. Construction Contractors Construction Schedule: Comply with requirements specified in Section 01 32 16 "Construction Schedule."

D. Accident Prevention Plan: Comply with requirements specified in Section 01 35 23 "Safety Requirements."

E. Schedule of Values: Comply with requirements specified in Section 01 32 16 "Construction Schedule."

F. Waste Recycling Plan: Comply with requirements specified in Section 01 74 19 "Construction Waste Management and Disposal."

G. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."

H. Storm Water Pollution Prevention Plan: Comply with requirements specified in Section 01 57 23 "Temporary Storm Water Pollution Prevention" and storm water permit requirements identified in Section 01 31 00 "Project Management and Coordination."

I. Indoor Air Quality Management Plan: Comply with requirements specified in Section 01 57 19.11 "Indoor Air Quality Management."

J. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

K. Welding Certificates: Prepare written certification that welding procedures and personnel comply with Contract Documents. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on American Welding Society (AWS) forms. Include names of firms and personnel certified.

L. Installer Certificates: Prepare written statements on manufacturer’s letterhead certifying Installer complies with Contract Documents and, where required, is authorized by manufacturer for this specific Project.

M. Manufacturer Certificates: Prepare written statements on manufacturer’s letterhead certifying manufacturer complies with Contract Documents. Include evidence of manufacturing experience where required.

N. Product Certificates: Prepare written statements on manufacturer's letterhead certifying product complies with Contract Documents.

O. Material Certificates: Prepare written statements on manufacturer's letterhead certifying material complies with Contract Documents.

P. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in Contract Documents.

Q. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.

R. Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.

S. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in Contract Documents.

T. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.

U. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in Contract Documents.

V. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements specified in Section 01 78 23 "Operation and Maintenance Data."

W. Design Data: Prepare written and graphic information, including: performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.

X. Manufacturer's Instructions: Prepare written or published information documenting manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.

Y. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. As applicable, include:

1. Statement on condition of substrates and their acceptability for installation of product.

2. Summary of installation procedures being followed, compliance with requirements and, if not, what corrective action was taken.
3. Results of operational and other tests and a statement of whether observed performance complies with the requirements.

Z. Permit Compliance Products: Prepare required information for compliance with permit provisions. Products include written notification of project startup, suspension, and completion of work; photo documentation of site conditions; reports; and drawings.

PART 3 - EXECUTION

3.1 CONSTRUCTION CONTRACTOR'S REVIEW

A. Review each submittal and check for coordination with other Work of Contract and for compliance with Contract Documents. Note corrections and field dimensions.

3.2 CONTRACTING OFFICER'S ACTION

A. General: Submittals will be disapproved without technical review if identification information is missing, not filled in, or if placed on back of submittal; an incorrect format of submittals is provided; transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of Construction Contractor's approval.

1. Any work done or orders for materials or services placed before approval shall be at Construction Contractor's own risk.

B. Action Submittals: Contracting Officer will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on the Transmittal Form (CM-16). Submittal will be marked as defined below:

1. APPROVED: Acceptable with no corrections.
2. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review is required. Construction Contractor shall address review comments when proceeding with the work.
3. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. Contracting Officer will identify reasons for disapproval. Construction Contractor shall revise and resubmit with changes clearly identified.

C. Informational Submittals: Contracting Officer will review each submittal and will either accept or reject it.

D. Partial submittals are not acceptable, will be considered non-responsive, and will be returned without review.

END OF SECTION 01 33 23

Submittal List with Review Estimate Template

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)								
				INFORMATIONAL				ACTION				
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
013100	1.4.A	Coordination Drawings	X		X				X		
013100	1.4.C	Coordination Digital Data Files	X			X					
013100	1.4.D.1	Letter designating Project Superintendent	X		X						
013100	1.4.D.2	Construction Schedule	X		X						
013100	1.4.D.3	A Comprehensive Schedule of Values	X		X						
013100	1.4.D.4	Accident Prevention Plan	X		X						
013100	1.4.D.5	A List of Subcontractors for this project	X		X						
013100	1.4.D.6	Written statements from Subcontractors certifying compliance with applicable labor standard clauses	X		X						
013100	1.4.D.7	Certificates of Insurance or SF1413 for Contactor and all Subcontractors	X		X						
013100	1.4.D.8	Waste Management Plan	X		X						
013100	1.4.D.9	Quality Control Plan	X		X				X		
013100	1.4.D.10	Temporary Storm Water Pollution Prevention Plan (SWPPP)	X		X						
013100	1.4.D.11	Indoor Air Quality (IAQ) Management Plan	X		X						
013100	1.4.D.12	Construction Contractor Commissioning Plan	X		X				X		
013100	1.4.D.13	Hazardous Materials Management Plan	X		X						
013100	1.4.D.14	Wildlife Exclusion Plan	X		X						
013100	1.4.D.15	Park Data Protection and Access Plan	X		X						
013100	1.4.D.16	List of Requireed Construction Permits	X		X						
013216	1.3.A	Electronic Copies			X						
013216	1.3.B	Schedule of Value	X		X						
013216	1.3.C	Construction Baseline Schedule	X		X						
013216	1.3.D	CPM Reports			X						
013216	1.3.E	Construction Schedule Updates			X						
013216	1.3.F	Construction Schedule Revisions			X						
013233	1.3.A	Construction Images								X	
013233	1.3.B	Closeout Submittals			X						
013513.22	1.3.A	Daily Work Schedule			X						
013523	1.3.A	Accident Prevention Plan	X		X						
013540	1.3	As-builts			X						
013550	1.5.A.1.a	Spill Prevention and Response Plan	X		X						
013550	1.5.A.1.b	MSDS	X							X	
013550	1.5.A.1.c	SWPPP	X		X						
013550	1.5.A.1.d	Hazardous Materials Spill Prevention and Response Plan	X		X						
014000	1.4.A	Quality Control Plan	X		X						
014000	1.4.B	Qualification Data			X						
014000	1.4.C	Contractor Quality Control (CQC) Daily Reports			X						
014000	1.4.D	Test Reports			X						
014000	1.4.E	Accessibility Inspection report			X						
014000	1.4.F	Test and Inspections List / Matrix			X						
014000	1.4.G	Off-Site Inspection Reports		X							
014000	1.4.I	Permits, Licenses, and Certificates		X							
014000	1.4.J	Mockups									X
015719	1.4.B	Product Data				X					
015719	1.4.C	Inspection and Test Reports		X							
015723	1.3.C	Inspection Schedule			X						
015723	1.3.D	Erosion Control Products								X	
016700	1.3.A	Record Submittals			X						
017329	1.2.A	Cutting and Patching Plan			X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
017340	1.2.A	Certificates		X							
017340	1.2.B	Landfill Receipts			X						
017340	1.2.C	Certified Surveys			X						
017340	1.2.D	Quantity Surveys			X						
017419	1.4.A	Waste Management Plan			X						
017419	1.4.B	Progress Documentation			X						
017419	1.4.C	Waste Reduction Calculations			X						
017419	1.4.D	Records of Donations			X						
017419	1.4.E	Records of Sales			X						
017419	1.4.F	Recycling and Processing Facility Records			X						
017419	1.4.G	Landfill and Incinerator Disposa Records			X						
017419	1.4.I	Qualification Data		X							
017419	1.4.K	Progress payment requirements			X						
017419	1.4.L	Closeout Submittals			X						
017700	1.3.A.2	Project Record Documents			X						
017700	1.3.A.3	Environmental Record Docuemnts			X						
017700	1.3.A.4	Operating Instructions				X					
017823	1.2.A	Manual			X						
017900	1.3.A	Instruction Program			X						
017900	1.3.B	Demonstration and Training Video			X						
019114	1.6.A	2-week look ahead			X						
019114	1.6.B	Certificates		X							
019114	1.6.C	Qualifications		X							
019114	1.6.D	Commissioning Plan			X						
019114	1.6.E	Pre-functional checklists					X				
019114	1.6.F	Owner's Proj Requirements					X				
019114	1.6.G	Performance Tests			X						
019114	1.6.H	List of test instrumentation			X						
019114	1.6.I	Deficiency Report			X						
019114	1.6.J	Closout Documentation			X						
024119	1.6.A	Qualifications Statement							X	X	
024119	1.6.B	Schedule of Selective Demolition Activities							X		
024119	1.6.C	Statement of Refrigerant Recovery					X				
028101	1.3.B	Permit for Treatment Storage and Disposal Facility		X							
028101	1.3.C	Bills of Landings and Manifests			X						
028101	1.3.D	Waste Disposal Information			X						
028213	1.7.A.1	Permit for Removal		X							
028213	1.7.A.2	Health and Safety Plan			X						
028213	1.7.A.3	Asbestos Abatement Project Plan			X						
028213	1.7.B	Asbestos-Related Project Submittals			X		X				
028213	1.7.C	Close-Out Report			X						
028333	1.7.A	Health and Safety Plan / Related Submittals			X						
028333	1.7.B	Occupational Exposure Sampling			X			X			
028333	1.7.C	Lead Wast Management and Disposal Plan			X						
028416	1.7.B	Permits		X							
028416	1.7.C	Certifications		X	X						
028416	1.7.D	Certifications		X	X						
028416	1.7.E	Schedule			X						
028416	1.7.F	Waste Handling Procedures			X						
028416	1.7.G	Transport Manifests and Receipts			X						
028416	1.7.H	Shipping Record and Receipt			X						
033000	1.2.A	Product data								X	
033000	1.2.C	Shop drawings, reinforcement							X		
033000	1.2.D	Shop drawings, embedded items							X		
033000	1.2.G	Shop drawings, construction joints							X		
033000	1.2.B	Mix Design		X	X	X					X
033000	1.2.E	Material Certificates		X		X				X	
033000	1.2.F	Material Test Reports			X		X			X	
033000	1.2.H	Corrective Action Plan					X				
040120.63	1.6A	Brick Masonry Repair - Product Data								X	
040120.63	1.6A	Brick Masonry Repair - Qualification Data								X	
040120.64	1.7A	Brick Masonry Repointing - Product Data								X	

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	N170EFMO318539
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
040120.64	1.7B	Brick Masonry Repointing - Shop Drawings							X		
040120.64	1.7C	Brick Masonry Repointing - Qualification Data								X	
042200	1.4.A	Product data								X	
042200	1.4.B	Shop drawings, reinforcement							X		
042200	1.4.C	Samples						X			
042200	1.5.A	Qualification Data								X	
042200	1.5.B	Material Certificates		X		X				X	
042200	1.5.C	Mix Design (Grout, Mortar)		X	X	X					X
042200	1.5.D	Masonry Compressive Strength				X				X	
042200	1.5.E	Procedures					X				
042613	1.2A	Masonry Veneer - Product Data								X	
042613	1.2B	Masonry Veneer - Testing Results								X	
042613	1.2C	Masonry Veneer - Samples						X			
042613	1.2D	Masonry Veneer - Qualifaction Data								X	
044313	1.2A	Adhered Stone Masonry Veneer - Product Data								X	
044313	1.2B	Adhered Stone Masonry Veneer - Testing Results								X	
044313	1.2C	Adhered Stone Masonry Veneer - Samples						X			
051200	1.4.A	Product data								X	
051200	1.4.B	Shop Drawings							X		
051200	1.4.C	Welders's Procedure Specifications			X		X				X
051200	1.4.D	Installer's and Fabricator's Qualification Statement				X	X				
051200	1.4.E	Welders's Certificates		X							X
051200	1.4.F	Paint Compatibility Certificates		X		X				X	
051200	1.4.G	Material Test Reports & SI Reports			X		X			X	
053100	1.3.A	Product data								X	
053100	1.3.B	Shop Drawings							X		
053100	1.3.C	Welders's Certificates		X							X
053100	1.3.D	Welders' Qualification Statement				X	X				
053100	1.3.E	Fabricator's Qualification Statement				X	X				
055000	1.3A	Metal Fabrications - Product Data								X	
055000	1.3B	Metal Fabrications - Shop Drawings							X		
055000	1.3C	Metal Fabrications - Deligated Design									X
055000	1.3D	Metal Fabrications - Designated Design Qualifications								X	
055000	1.4A	Metal Fabrications - Welding Certificates		X							
055000	1.4B	Metal Fabrications - Paint Certificates		X							
055500	1.3A	Shop Drawings							X		
055500	1.3B	Welders's Certificates		X							X
055500	1.3C	Paint Compatibility Certificates		X		X					
055500	1.4.A & B	Welding Qualifications			X		X				X
061000	1.2.A	Product Data		X		X				X	X
061600	1.2.A	Product Data		X		X				X	X
061600	1.2.B	Evaluation Reports			X						
06 18 00	1.2.A	Product Data		X		X				X	X
06 18 00	1.2.B	Shop Drawings							X		
06 18 00	1.2.C	Certificates of Conformance		X							
06 18 00	1.2.D	Material Certificates		X							
062023	1.2A	Interior Finish Carpentry - Product Data								X	
062023	1.2B	Interior Finish Carpentry - Samples						X			
062023	1.2C	Interior Finish Carpentry - Installer Qualifications								X	
064013	1.4A	Exterior Architectural Woodwork - Product Data								X	
064013	1.4B	Exterior Architectural Woodwork - Fire Retardant Treatment								X	
064013	1.4C	Exterior Architectural Woodwork - Shop Drawings							X		
064013	1.4D	Exterior Architectural Woodwork - Samples						X			
064100	1.4A	Architectural Wood Casework - Product Data								X	
064100	1.4B	Architectural Wood Casework - Shop Drawings							X		
064100	1.4C	Architectural Wood Casework - Samples						X			
064100	1.5A	Architectural Wood Casework - Test Reports			X						
064100	1.5B	Architectural Wood Casework - Certification Reports			X						
064100	1.5C	Architectural Wood Casework - Product Certificates		X							
066116	1.4A	Solid Surface Fabrications - Product Data								X	
066116	1.4B	Solid Surface Fabrications - Shop Drawings							X		

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)								
				INFORMATIONAL				ACTION				
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
066116	1.4C	Solid Surface Fabrications - Samples						X			
066116	1.4D	Solid Surface Fabrications - Test Reports								X	
071416	1.5A	Cold Fluid Applied Waterproofing - Product Data								X	
071416	1.5B	Cold Fluid Applied Waterproofing - Qualification Data								X	
071416	1.5C	Cold Fluid Applied Waterproofing - Shop Drawings							X		
071416	1.5D	Cold Fluid Applied Waterproofing - Samples						X			
071416	1.6A	Cold Fluid Applied Waterproofing - Field Quality Control Reports			X						
071800	1.3A	Traffic Coatings - Product Data								X	
071800	1.3B	Traffic Coatings - Sustainability Data								X	
071800	1.3C	Traffic Coatings - Shop Drawings							X		
071800	1.3D	Traffic Coatings - Samples for Selection						X			
071800	1.3E	Traffic Coatings - Samples for Verification						X			
071800	1.3F	Traffic Coatings - Qualification Data								X	
072100	1.2A	Thermal Insulation - Product Data								X	
072100	1.2B	Thermal Insulation - Sustainability Data								X	
072119	1.2A	Foamed In Place Insulation - Product Data								X	
072119	1.2B	Foamed In Place Insulation - Qualification Data								X	
072119	1.3A	Foamed In Place Insulation - Field Reports									X
072726	1.4A	Fluid Applied Membrane - Product Data								X	
072726	1.4B	Fluid Applied Membrane - Shop Drawings							X		
072726	1.4C	Fluid Applied Membrane - Qualification Data								X	
072726	1.4D	Fluid Applied Membrane - Field Reports									X
072726	1.5A	Fluid Applied Membrane - Product Certificates		X							
072726	1.5B	Fluid Applied Membrane - Test Reports			X						
074113	1.4A	Standing Seam Metal Roof - Product Data								X	
074113	1.4B	Standing Seam Metal Roof - Shop Drawings							X		
074113	1.4C	Standing Seam Metal Roof - Samples for Verification						X			
074113	1.4D	Standing Seam Metal Roof - Qualification Data									X
074113	1.5A	Standing Seam Metal Roof - Test Reports			X						
074113	1.5B	Standing Seam Metal Roof - Field Reports			X						
074113	1.5C	Standing Seam Metal Roof - Sample Warranties					X				
074213	1.3A	Formed Metal Wall Panels - Product Data								X	
074213	1.3B	Formed Metal Wall Panels - Shop Drawings							X		
074213	1.3C	Formed Metal Wall Panels - Samples for Selection						X			
074213	1.3D	Formed Metal Wall Panels - Samples for Verification						X			
074213	1.4A	Formed Metal Wall Panels - Qualification Data				X					
074213	1.4B	Formed Metal Wall Panels - Field Reports			X						
074213	1.4C	Formed Metal Wall Panels - Sample Warranties					X				
074213	1.4B	Formed Metal Wall Panels - Field Reports			X						
074213	1.4C	Formed Metal Wall Panels - Sample Warranties					X				
074616	1.3A	Aluminum Cladding - Product Data								X	
074616	1.3B	Aluminum Cladding - Shop Drawings							X		
074616	1.3C	Aluminum Cladding - Samples for Selection						X			
074616	1.3D	Aluminum Cladding - Samples for Verification						X			
074616	1.3E	Aluminum Cladding - Sustainability Data								X	
074616	1.4A	Aluminum Cladding - Qualification Data				X					
074616	1.4B	Aluminum Cladding - Designated Design Qualifications				X					
074616	1.4C	Aluminum Cladding - Sample Warranties					X				
076200	1.3A	Sheet Metal Flashing - Product Data								X	
076200	1.3B	Sheet Metal Flashing - Shop Drawings							X		
076200	1.3C	Sheet Metal Flashing - Samples for Selection						X			
076200	1.3D	Sheet Metal Flashing - Samples for Verification						X			
076200	1.3E	Sheet Metal Flashing - Qualification Data								X	
076200	1.4A	Sheet Metal Flashing - Sample Warranties					X				
077100	1.3A	Roof Specialties - Product Data								X	
077100	1.3B	Roof Specialties - Shop Drawings							X		
077100	1.3C	Roof Specialties - Samples						X			
077100	1.3D	Roof Specialties - Qualification Data								X	
077100	1.4A	Roof Specialties - Sample Warranties					X				
077253	1.2A	Snow Guards - Product Data								X	
077253	1.2B	Snow Guards - Shop Drawings							X		
077253	1.2C	Snow Guards - Samples						X			
077253	1.2D	Snow Guards - Designated Design Calcs									X
077253	1.3A	Snow Guards - Qualification Data				X					
077253	1.3B	Snow Guards - Test Reports			X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)								
				INFORMATIONAL				ACTION				
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
079200	1.2A	Joint Sealants - Product Data								X	
079200	1.2B	Joint Sealants - Samples						X			
079200	1.2C	Joint Sealants - Sealant Schedule									X
081113	1.4A	Hollow Metal Doors and Windows - Product Data								X	
081113	1.4B	Hollow Metal Doors and Windows - Product Details									X
081113	1.4C	Hollow Metal Doors and Windows - Shop Drawings							X		
081113	1.4D	Hollow Metal Doors and Windows - Product Schedule									X
081113	1.5A	Hollow Metal Doors and Windows - Qualification Data				X					
081113	1.5B	Hollow Metal Doors and Windows - Test Reports			X						
081113	1.5C	Hollow Metal Doors and Windows - Field Reports		X							
081416	1.2A	Flush Wood Doors - Product Data								X	
081416	1.2B	Flush Wood Doors - Product Materials									X
081416	1.2C	Flush Wood Doors - Sustainability Data								X	
081416	1.2D	Flush Wood Doors - Shop Drawings							X		
081416	1.2E	Flush Wood Doors - Samples for Selection						X			
081416	1.2F	Flush Wood Doors - Samples for Verification						X			
081416	1.3A	Flush Wood Doors - Qualification Data				X					
081416	1.3B	Flush Wood Doors - Field Reports			X						
081416	1.3C	Flush Wood Doors - Sample Warranties					X				
083113	1.2A	Access Doors and Frames - Product Data								X	
083113	1.2B	Access Doors and Frames - Samples						X			
083113	1.2C	Access Doors and Frames - Project Schedule									X
084113	1.3A	Aluminum Framed Entrances and Storefronts - Product Data								X	
084113	1.3B	Aluminum Framed Entrances and Storefronts - Shop Drawings							X		
084113	1.3C	Aluminum Framed Entrances and Storefronts - Samples for Verification						X			
084113	1.3D	Aluminum Framed Entrances and Storefronts - Designated Design Calcs									X
084113	1.3E	Aluminum Framed Entrances and Storefronts - Sustainability Data								X	
084113	1.4A	Aluminum Framed Entrances and Storefronts - Energy Certs		X							
084113	1.4B	Aluminum Framed Entrances and Storefronts - Test Reports			X						
084113	1.4C	Aluminum Framed Entrances and Storefronts - Field Reports			X						
084113	1.4D	Aluminum Framed Entrances and Storefronts - Deligated Design Qualifications				X					
084113	1.4E	Aluminum Framed Entrances and Storefronts - Sample Warranties					X				
087100	1.3B1	Door Hardware - Product Data								X	
087100	1.3B2	Door Hardware - Riser and Wiring Diagrams									X
087100	1.3B3	Door Hardware - Samples for Verification						X			
087100	1.3B4	Door Hardware - Hardware Schedule									X
087100	1.3B5	Door Hardware - Key Schedule									X
087100	1.3C1	Door Hardware - Qualification Data				X					
088000	1.5A	Glazing - Product Data								X	
088000	1.5B	Glazing - Glass Samples						X			
088000	1.5C	Glazing - Glazing Accessory Samples						X			
088000	1.5D	Glazing - Glazing Schedule									X
088000	1.5E	Glazing - Delegated Design Calcs									X
088000	1.5F	Glazing - Sustainability Data								X	
088000	1.6A	Glazing - Product Certificates		X							
088000	1.6B	Glazing - Test Reports			X						
088000	1.6C	Glazing - Adhesion Test Reports			X						
088000	1.6D	Glazing - Qualification Data				X					
088000	1.6E	Glazing - Sample Warranties					X				
092216	1.2A	Non-Structural Metal Framing - Product Data								X	
092900	1.2A	Gypsum board - Product Data								X	
092900	1.2B	Gypsum board - Test Reports								X	
095123	1.3A	Acoustical Tile Ceilings - Product Data								X	
095123	1.3B	Acoustical Tile Ceilings - Samples						X			
096519	1.3A	Resilient Rubber Flooring - Product Data						X			
096519	1.3B	Resilient Rubber Flooring - Shop Drawings							X		
096519	1.3C	Resilient Rubber Flooring - Samples						X			
096519	1.3D	Resilient Rubber Flooring - Sustainability Data/EPD								X	
096519	1.3E	Resilient Rubber Flooring - Qualifaction Data								X	
096519	1.4A	Resilient Rubber Flooring - Product Test Reports			X						
096723	1.3A	Resinous Epoxy Flooring - Product Data								X	
096723	1.3B	Resinous Epoxy Flooring - Sustainable Data								X	
096723	1.3C	Resinous Epoxy Flooring - Samples						X			
096723	1.3D	Resinous Epoxy Flooring - Qualification Data								X	

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)								
				INFORMATIONAL				ACTION				
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
096813	1.3A	Tile Carpeting - Product Data								X	
096813	1.3B	Tile Carpeting - Shop Drawings							X		
096813	1.3C	Tile Carpeting - Samples for Verification						X			
096813	1.3D	Tile Carpeting - Project Schedule									X
096813	1.3E	Tile Carpeting - Sustainability Data								X	
096813	1.4A	Tile Carpeting - Product Test Reports			X						
099113	1.2A	Exterior Paint - Product Data								X	
099113	1.2B	Exterior Paint - Samples						X			
099113	1.2C	Exterior Paint - Samples for Initial Selection						X			
099113	1.2D	Exterior Paint - Samples for Verification						X			
099113	1.2E	Exterior Paint - Product List									X
099123	1.2A	Interior Paint - Product Data								X	
099123	1.2B	Interior Paint - Samples for Verification						X			
099123	1.2C	Interior Paint - Product List									X
099123	1.2D	Interior Paint - Sustainable Data								X	
099300	1.2A	Finishing - Product Data								X	
099300	1.2B	Finishing - Samples for Initial Selection						X			
099300	1.2C	Finishing - Samples for Verification						X			
099300	1.2D	Finishing - Product List									X
099300	1.2E	Finishing - Sustainable Data									
101423.16	1.2A	Room Signage - Product Data								X	
101423.16	1.2B	Room Signage - Sustainable Data									X
101423.16	1.2C	Room Signage - Shop Drawings							X		
101423.16	1.2D	Room Signage - Samples for Initial Selection						X			
101423.16	1.2E	Room Signage - Samples for Verification						X			
101423.16	1.2F	Room Signage - Product Schedule									X
102113	1.3A	Plastic Toilet Compartments - Product Data								X	
102113	1.3B	Plastic Toilet Compartments - Shop Drawings							X		
102113	1.3C	Plastic Toilet Compartments - Samples for Verification						X			
102113	1.3D	Plastic Toilet Compartments - Project Schedule									X
102239	1.4A	Folding Panel Partitions - Product Data								X	
102239	1.4B	Folding Panel Partitions - Sustainability Data								X	
102239	1.4C	Folding Panel Partitions - Shop Drawings							X		
102239	1.4D	Folding Panel Partitions - Samples for Selection						X			
102239	1.4E	Folding Panel Partitions - Samples for Verification						X			
102239	1.4F	Folding Panel Partitions - Delegated Design Calcs									X
102239	1.5A	Folding Panel Partitions - Coordination Drawings							X		
102239	1.5B	Folding Panel Partitions - Setting Drawings							X		
102239	1.5C	Folding Panel Partitions - Qualification Data									X
102239	1.5D	Folding Panel Partitions - Seismic Certificates		X							
102239	1.5E	Folding Panel Partitions - Product Certificates		X							
102239	1.5F	Folding Panel Partitions - Test Reports			X						
102239	1.5G	Folding Panel Partitions - Field Reports			X						
102239	1.5H	Folding Panel Partitions - Sample Warranties					X				
102800	1.5A	Toilet Accessories - Product Data								X	
102800	1.5B	Toilet Accessories - Samples					X				
102800	1.5C	Toilet Accessories - Product Schedule									X
102800	1.5D	Toilet Accessories - Delegated Design Calcs									X
102800	1.6A	Toilet Accessories - Sample Warranties					X				
107516	1.5A	Flagpole - Product Data								X	
107516	1.5B	Flagpole - Shop Drawings							X		
107516	1.5C	Flagpole - Samples for Verification						X			
107516	1.5D	Flagpole - Delegated-Design Submittal									X
110000	1.2A	Requirements for Misc Equipment - Product Data								X	
122413	1.2A	Roller Window Shades - Product Data								X	
122413	1.2B	Roller Window Shades - Shop Drawings							X		
122413	1.2C	Roller Window Shades - Samples for Verification						X			
122413	1.2D	Roller Window Shades - Product Schedule									X
122413	1.2E	Roller Window Shades - Sustainability Data/VOC								X	
123216	1.2A	Casework - Product Data								X	
123216	1.2B	Casework - Shop Drawings							X		
123216	1.2C	Casework - Samples						X			

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)								
				INFORMATIONAL				ACTION				
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
123661	1.2A	Simulated Stone Countertops - Product Data								X	
123661	1.2B	Simulated Stone Countertops - Shop Drawings							X		
123661	1.2C	Simulated Stone Countertops - Samples for Selection						X			
123661	1.2D	Simulated Stone Countertops - Samples for Verification						X			
140120.71	1.3A	Elevator Rehabilitation - Schedule					X				
140120.71	1.3B	Elevator Rehabilitation - Shop Drawings							X		
140120.71	1.3C	Elevator Rehabilitation - Wiring Diagram									X
210000	1.4 A-P	Submit shop drawings, calculations and product data sheets as one complete stan							X	X	X
211300	1.4 B6	System installer and designer qualifications.					X				
211300	1.4 B1	Hydraulic calculations.			X						
211300	1.4 B2	Sway brace calculations.			X						
211300	1.4 B3	Flexible Sprinkler Assembly details									X
211300	1.4 B4	Trapeze hanger calculations			X						
211319	1.4 B1	Hydraulic calculations.			X						
211319	1.4 B2	Trapeze hanger calculations			X						
212200	1.4 B.1	System installer and designer qualifications.					X				
212200	1.4 B.2-4	Submit shop drawings, calculations and product data sheets as one complete stan							X	X	
212200	1.4 B.5	Sequence of Operations.					X				X
212200	1.4 B.6	Manufacturer's Design Manual.								X	X
220500	1.2	Sleeve-seal Systems								X	
220500	1.2	Grout								X	
220500	1.2	Silicon Sealants								X	
220517	1.2	Product Data								X	
220517	1.3	Field Reports			X						
220518	1.2	Escutcheons								X	
220518	1.2	Floor Plates								X	
220519	1.4	Liquid-in-glass thermometers		X						X	
220519	1.4	Thermowells		X						X	
220519	1.4	Dial-type pressure gages		X						X	
220519	1.4	Gage attachements		X						X	
220519	1.4	Water consumption meter		X						X	
220523	1.2	Brass Ball Valves								X	
220523	1.2	Bronze Swing Check Valves								X	
220529	1.2	Metal Pipe Hangers and Supports								X	
220529	1.2	Trapeze pipe hangers								X	
220529	1.2	Thermal Hanger-Shield Inserts								X	
220529	1.2	Fastner Systems								X	
220529	1.2	Pipe-positioning Systems								X	
220529	1.2	Equipment Supports								X	
220553	1.2	Equipment Labels								X	
220553	1.2	Warning Signs and Labels								X	
220553	1.2	Pipe Labels								X	
220553	1.2	Stencils								X	
220553	1.2	Valve Tags								X	
220719	1.2,1.3	Domestic hot-water piping			X					X	
220719	1.2,1.3	Domestic recirculating hot-water piping			X					X	
220719	1.2,1.3	Supplies and drains for handicap-accessible lavatories and sinks.			X					X	
221116	1.2,1.3	Copper tube and fittings			X					X	
221116	1.2,1.3	Pipe joining materials			X					X	
221116	1.2,1.3	Transition Fittings			X					X	
221116	1.2,1.3	Dielectric fittings			X					X	
221119	1.2	Vaccum breakers								X	
221119	1.2	Backflow preventors								X	
221119	1.2	Balancing valves								X	
221119	1.2	Temperature-actuated, water mixing valves								X	
221119	1.2	Strainers								X	
221119	1.2	Wall hydrants								X	
221119	1.2	Drain Valves								X	
221119	1.2	Water-hammer arresters								X	

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
221123	1.2	Permanently lubriacted, booster pumps								X	
221316	1.2,1.3	Hub-and-spigot, cast-iron soil piping and fittings			X					X	
221316	1.2,1.3	PVC pipe and fittings			X					X	
221316	1.2,1.3	Specialty pipe fittings			X					X	
221319	1.2	Cleanouts								X	
221319	1.2	Cleanouts								X	
221319	1.2	Floor drains								X	
221319	1.2	Roof flashing assemblies								X	
221319	1.2	Miscellaneous sanitary drainage piping specalties								X	
223300	1.2	Commercial, electric, domestic-water heaters			X	X			X	X	
224213.13	1.2	Water closets								X	
224213.13	1.2	Flushometer valves								X	
224213.13	1.2	Toilet seats								X	
224213.16	1.2	Wall-hung urinals								X	
224213.16	1.2	Urinal flushometer valves								X	
224213.16	1.2	Supports								X	
224216.13	1.2	Lavatories								X	
224216.13	1.2	Faucets								X	
224216.16	1.2	Sinks								X	
224216.16	1.2	Utility sinks								X	
224216.16	1.2	Sink faucets								X	
224216.16	1.2	Laminar-flow, faucet-spout outlets								X	
224216.16	1.2	Supply fittings								X	
224216.16	1.2	Waste fittings								X	
224713	1.2	Drinking fountains								X	
230500	1.5	Transitions fittings		X						X	
230500	1.5	Dielectric fittings		X						X	
230500	1.5	Mechanical sleeve seals		X						X	
230500	1.5	Escutcheons		X						X	
230517	1.2	Sleeves without waterstop								X	
230517	1.2	Sleeves with waterstop								X	
230517	1.2	Stack-sleeve fittings								X	
230517	1.2	Sleeve-seal systems								X	
230517	1.2	Grout								X	
230517	1.2	Silicone sealants								X	
230517	1.2	Silicone sealants								X	
230529	1.2	Metal pipe hangers and supports								X	
230529	1.2	Trapeze pipe hangers							X	X	
230529	1.2	Fiberglass pipe hangers								X	
230529	1.2	Metal framing systems							X	X	
230529	1.2	Fiberglass strut systems							X	X	
230529	1.2	Thermal-hanger shield inserts								X	
230529	1.2	Fastener systems								X	
230529	1.2	Pipe stands							X	X	
230529	1.2	Equipment supports							X	X	
230529	1.2	Miscellaneos materials								X	
230553	1.2	Equipment labels								X	
230553	1.2	Warning signs and labels								X	
230553	1.2	Pipe labels								X	
230593	1.3	Strategies and Procedures Plan			X						
230593	1.3	Certified TAB Reports			X						
230713	1.2	Indoor, concealed supply and return air								X	
230713	1.2	Indoor, exposed supply air								X	
230713	1.2	Outdoor, exposed supply and return air								X	
230719	1.2	Insulation		X	X					X	
230719	1.2	Adhesives, Mastics, Sealants		X	X					X	
230719	1.2	Field applied jackets		X	X					X	
230719	1.2	Tapes		X	X					X	
230719	1.2	Securements		X	X					X	
231126	1.2	Pipes, tubes, and fittings		X	X					X	
231126	1.2	Pipes specialties		X	X					X	

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
231126	1.2	Piping and tubing joining materials		X	X					X	
231126	1.2	Valves		X	X					X	
231126	1.2	Pressure regulators		X	X					X	
232300	1.2	Refrigerant pipes and fittings			X					X	
232300	1.2	Refrigerant piping valves and specialites			X					X	
232300	1.3	Refrigerants			X						
233113	1.3	Rectangular duct and fittings								X	
233113	1.3	Round ducts and fittings								X	
233113	1.3	Sheet metal materials								X	
233113	1.3	Sealants and gaskets								X	
233113	1.3	Hangers and supports								X	
233113	1.3	Shop drawings							X		
233300	1.2	Backdraft and pressure relief dampers								X	
233300	1.2	Manual volume dampers								X	
233300	1.2	Control dampers								X	
233300	1.2	Flange connectors								X	
233300	1.2	Flexible connectors								X	
233300	1.2	Duct accessory hardware								X	
233300	1.2	Fire dampers								X	
233300	1.2	Combination fire/smoke dampers								X	
233300	1.2	Turning vanes								X	
233423	1.2	Inline exhaust fan								X	
233713	1.2	Adjustable flade face registers								X	
233713	1.2	Fixed face registers and grilles.								X	
238126	1.2	Indoor units								X	
238126	1.2	Outdoor units								X	
238239	1.2	Propeller unit heaters								X	
260519	1.3	Product data for each type of product indicated								X	
260519	1.3	Field quality-control test reports.			X						
260526	1.2	Product data for each type of product indicated								X	
260526	1.2	As-Built grounding plans			X						
260526	1.2	Field quality-control test reports.			X						
260529	1.5	Product data for each type of product indicated								X	
260529	1.5	Shop Drawings							X	X	
260529	1.5	Welding Certificates		X							
260533	1.4	Product data for each type of product indicated								X	
260544	1.3	Product data for each type of product indicated								X	
260553	1.3	Product data for each type of product indicated								X	
260553	1.3	Identification Schedule			X						
260800	1.4	Construction Checklists			X					X	X
260800	1.5	Close-Out Submittals			X					X	X
260923	1.4	Product data for each type of product indicated								X	
260923	1.4	Shop Drawings							X	X	
260923	1.4	Field quality-control test reports.			X						
260923	1.4	Operation and Maintenance Data				X				X	
260943	1.4	Product data for each type of product indicated								X	
260943	1.4	Shop Drawings							X	X	
260943	1.4	Field quality-control test reports.			X						
260943	1.4	Operation and Maintenance Data				X				X	
260943	1.4	Warranty		X							X
262416	1.4	Product data for each type of product indicated								X	
262416	1.4	Shop Drawings								X	
262416	1.4	Panelboard Schedules							X		
262416	1.4	Field quality-control test reports.			X						
262416	1.5	Close-Out Submittals			X					X	X
262713	1.4	Product data for each type of product indicated								X	
262713	1.4	Field quality-control test reports.			X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
262713	1.4	Operation and Maintenance Data				X				X	
262726	1.4	Product data for each type of product indicated								X	
262726	1.4	Operation and Maintenance Data				X				X	
262816	1.4	Product data for each type of product indicated								X	
262816	1.4	Shop Drawings							X	X	
262816	1.4	Field quality-control test reports.			X						
262816	1.4	Operation and Maintenance Data				X				X	
264113	1.4	Product data for each type of product indicated								X	
264113	1.4	Shop Drawings							X	X	
264113	1.4	Qualification Data		X							
264113	1.4	Field quality-control test reports.			X						
264313	1.4	Product data for each type of product indicated								X	
264313	1.4	Field quality-control test reports.			X						
264313	1.4	Warranty		X							X
265100	1.4	Product data for each type of product indicated								X	
265100	1.4	Shop Drawings							X	X	
265100	1.4	Qualification Data		X							
265100	1.4	Field quality-control test reports.			X						
265100	1.4	Operation and Maintenance Data				X				X	
265100	1.4	Warranty		X							X
270526	1.4	Product data for each type of product indicated								X	
270526	1.4	Shop Drawings							X	X	
270526	1.5	Coordination Drawings							X	X	
270526	1.5	Qualification Data		X							
270526	1.5	Field quality-control test reports.			X						
270526	1.6	Record Drawings							X	X	
270526	1.6	Operation and Maintenance Data				X				X	
270526	1.6	Field quality-control test reports.			X						
270528	1.4	Product data for each type of product indicated								X	
270528	1.4	Shop Drawings							X	X	
270528	1.5	Coordination Drawings							X	X	
270528	1.5	Field quality-control test reports.			X						
270536	1.3	Product data for each type of product indicated								X	
270536	1.3	Shop Drawings							X	X	
270536	1.4	Coordination Drawings							X	X	
270536	1.4	Field quality-control test reports.			X						
270536	1.5	Operation and Maintenance Data				X				X	
270536	1.4	Field quality-control test reports.			X						
270536	1.5	Operation and Maintenance Data				X				X	
270544	1.3	Product data for each type of product indicated								X	
270544	1.4	Coordination Drawings							X	X	
270553	1.3	Product data for each type of product indicated								X	
271116	1.4	Product data for each type of product indicated								X	
271116	1.4	Shop Drawings							X	X	
271116	1.5	Coordination Drawings							X	X	
271116	1.5	Qualification Data		X							
271116	1.5	Field quality-control test reports.			X						
271116	1.6	Record Drawings							X	X	
271116	1.6	Operation and Maintenance Data				X				X	
271116	1.6	Field quality-control test reports.			X						
271323	1.4	Product data for each type of product indicated								X	
271323	1.4	Shop Drawings							X	X	
271323	1.4	Identification Schedule							X	X	
271323	1.5	Qualification Data		X							
271323	1.5	Field quality-control test reports.			X						
271323	1.5	Factory Test Resports									
271323	1.5	Optical Fiber Testing Plan							X	X	
271323	1.6	Operation and Maintenance Data				X				X	
271323	1.6	Warranty		X							X
271323	1.6	Identification Schedule									
271323	1.6	Extra Materials									X
271323	1.6	Field quality-control test reports.			X						
271513	1.4	Product data for each type of product indicated								X	
271513	1.4	Shop Drawings							X	X	

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
271513	1.4	Identification Schedule							X	X	
271513	1.5	Qualification Data		X							
271513	1.5	Field quality-control test reports.			X						
271513	1.5	Factory Test Resports									
271513	1.5	Twisted Pair Cable Testing Plan							X	X	
271513	1.6	Operation and Maintenance Data				X				X	
271513	1.6	Warranty		X							X
271513	1.6	Identification Schedule									
271513	1.6	Extra Materials									X
271513	1.6	Field quality-control test reports.			X						
274100	1.8	Long Lead Items List								X	
274100	1.8	Shop Drawings							X	X	
274100	1.8	Control & DSP Shop Drawings							X	X	
274100	1.8	Written Field Updates including Photo Documentation			X						
274100	1.8	Closeout Documents				X				X	
274100	1.9	Qualification Data		X							
280001	1.4 A-O	Submit shop drawings, calculations and product data sheets as one complete stand-alone package.							X	X	X
280800	1.4B2b	Smoke Detector Sensitivity Report			X						
280800	1.4B2c	Alarm Signal Audibility Levels Reports			X						
281300	1.4	Product data for each type of product indicated								X	
281300	1.5	Coordination Drawings							X	X	
281300	1.5	Qualification Data		X							
281300	1.5	Field quality-control test reports.			X						
281300	1.5	System Testing Plans							X	X	
281300	1.6	Operation and Maintenance Data / Certificates		X		X				X	
281300	1.6	Warranty		X							X
281300	1.6	Software and Firmware Operational Documentation									
281300	1.6	Field quality-control test reports.			X						
281400	1.4	Product data for each type of product indicated								X	
281400	1.4	Shop Drawings							X	X	
281400	1.5	Coordination Drawings							X	X	
281400	1.5	Qualification Data		X							
281400	1.5	Field quality-control test reports.			X						
281400	1.6	Record Documents							X	X	
281400	1.6	Operation and Maintenance Data		X		X				X	
281400	1.6	Warranty		X							X
281400	1.6	Extra Materials									X
281400	1.6	Field quality-control test reports.			X						
281600	1.4	Product data for each type of product indicated								X	
281600	1.4	Shop Drawings							X	X	
281600	1.5	Coordination Drawings							X	X	
281600	1.5	Qualification Data		X							
281600	1.5	Field quality-control test reports.			X						
281600	1.5	System Testing Plans							X	X	
281600	1.6	Record Documents							X	X	
281600	1.6	Operation and Maintenance Data		X		X				X	
281600	1.6	Warranty		X							X
281600	1.6	Software and Firmware Operational Documentation									
281600	1.6	Extra Materials									X
281600	1.6	Field quality-control test reports.			X						
282000	1.4	Product data for each type of product indicated								X	
282000	1.4	Shop Drawings							X	X	
282000	1.5	Coordination Drawings							X	X	
282000	1.5	Qualification Data		X							
282000	1.5	Field quality-control test reports.			X						
282000	1.5	System Testing Plans							X	X	
282000	1.6	Record Documents							X	X	
282000	1.6	Operation and Maintenance Data		X		X				X	
282000	1.6	Warranty		X							X
282000	1.6	Software and Firmware Operational Documentation									
282000	1.6	Field quality-control test reports.			X						
284600	1.4 B	Submit shop drawings, calculations and product data sheets as one complete stand-alone package.							X	X	X
284600	1.4 B2	System installer and designer qualifications.					X				
311000	1.5	Permits for transport of debris			X						
311000	1.5	Demolition procedures				X					

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**N170EFMO318539**
Project Title:	**Rehabilitate Visitor Center**

SUBMITTAL				REQUIREMENTS (Indicate with **X**)							
				INFORMATIONAL				ACTION			
SPECIFICATION SECTIOI	PARAGRAPH NUMBER	DESCRIPTION	PRIOR TO NTP	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
312000	1.4B	Delivery Tickets		X							
312000	1.4C	Product Data								X	
312000	1.4D1	Material Test Reports			X						
312000	1.4D2	Compaction Test Results		X							
312000	1.4E	Preexcavation Photography or Videotape			X						
312316	1.5.A	Sources					X				
312316	1.5.B	Material Test Reports			X						
312316	1.5.C	Sample						X			
312316	1.5.D	Delivery Tickets		X							
312500	1.3	Product Data								X	
321320	1.4.A	Product Data								X	
321320	1.4.B	Design Mixtures			X						
321320	1.4.C	Steel Reinfrocement Shop Drawings							X		
321320	1.4.D	Qualification Data		X							
321320	1.4.E	Material Certificates		X							
321320	1.4.F	Joint Plan							X		
321320	1.4.G	Materials test report			X						
321320	1.4.H	Field quality control reports			X						
323300	1.3.A	Product Data								X	
329113	1.3A	Product Data								X	
329113	1.3B	Sample						X			
329200	1.3A	Product Data								X	
329300	1.5.A	Product Data				X					
329300	1.5.B	Samples for Verificatoin						X			
329300	1.5.C	Qualification Data		X							
329300	1.5.D	Product Certificates		X							
331416	1.4B	Product Data								X	
331416	1.4C	Manufacturer's Certificate		X							
331416	1.4D	Manufacturer's fusion procedure recommendation					X				
331416	1.4F	Disinfection Plan									X
331416	1.4G	Test Reports			X						
334000	1.3.A	Product data								X	
334000	1.3.B	Manufacturer's Certification			X						
334600	1.3.A	Product data								X	
334600	1.3.B	Manufacturer's Certification			X						

Submittal List with Review Estimate Template

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**EFMO/224159**
Project Title:	**Redesign Parking & Replace Curbs, Walks and Pavement**

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
013100	1.4.A	Coordination Drawings		X				X		
013100	1.4.C	Coordination Digital Data Files			X					
013100	1.4.D.1	Letter designating Project Superintendent		X						
013100	1.4.D.2	Construction Schedule		X						
013100	1.4.D.3	A Comprehensive Schedule of Values		X						
013100	1.4.D.4	Accident Prevention Plan		X						
013100	1.4.D.5	A List of Subcontractors for this project		X						
013100	1.4.D.6	Written statements from Subcontractors certifying compliance with applicable labor standard clauses		X						
013100	1.4.D.7	Certificates of Insurance or SF1413 for Contactor and all Subcontractors		X						
013100	1.4.D.8	Waste Management Plan		X						
013100	1.4.D.9	Quality Control Plan		X				X		
013100	1.4.D.10	Traffic Control Plan		X						
013100	1.4.D.11	Under-an-Acre Pollution Prevention Plan (UPPP)		X						
013100	1.4.D.14	List of Required Construction Permits		X						
013216	1.3.A	Electronic Copies		X						
013216	1.3.B	Schedule of Value		X						
013216	1.3.C	Construction Baseline Schedule		X						
013216	1.3.D	CPM Reports		X						
013216	1.3.E	Construction Schedule Updates		X						
013216	1.3.F	Construction Schedule Revisions		X						
013233	1.3.A	Construction Images							X	
013233	1.3.B	Closeout Submittals		X						
013513.22	1.3.A	Daily Work Schedule		X						
013523	1.3.A	Accident Prevention Plan		X						
01 35 40	1.3	As-builts		X						
01 35 50	1.5.A.1.a	Spill Prevention and Response Plan		X						
01 35 50	1.5.A.1.b	MSDS		X					X	
01 35 50	1.5.A.1.c	SWPPP		X						
01 35 50	1.5.A.1.d	Hazardous Materials Spill Prevention and Response Plan		X						
013550	1.5.A.1.a	Spill Prevention and Response Plan		X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**EFMO/224159**
Project Title:	**Redesign Parking & Replace Curbs, Walks and Pavement**

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
013550	1.5.A.1.b	MSDS							X	
013550	1.5.A.1.c	Under-an-Acre Pollution Prevention Plan		X						
013550	1.5.A.1.d	Hazardous Materials Spill Prevention and Response Plan		X						
013591	1.3.A	Historic Preservation Treatment Plan (HPTP)		X						
013591	1.3.B	Alternative Methods and Materials		X						
013591	1.3.C	Photographs or Videotape		X					X	
014000	1.4.A	Quality Control Plan		X						
014000	1.4.B	Qualification Data		X						
014000	1.4.C	Contractor Quality Control (CQC) Daily Reports		X						
014000	1.4.D	Test Reports		X						
014000	1.4.E	Accessibility Inspection report		X						
014000	1.4.F	Off-Site Inspection Reports		X						
014000	1.4.H	Permits, Licenses, and Certificates	X							
015719	1.4.B	Product Data			X					
015719	1.4.C	Inspection and Test Reports	X							
015723	1.3.C	Inspection Schedule		X						
015723	1.3.D	Erosion Control Products							X	
016700	1.3.A	Record Submittals		X						
017329	1.2.A	Cutting and Patching Plan		X						
017340	1.2.A	Certificates	X							
017340	1.2.B	Landfill Receipts		X						
017340	1.2.C	Certified Surveys		X						
017340	1.2.D	Quantity Surveys		X						
017419	1.4.A	Waste Management Plan		X						
017419	1.4.B	Progress Documentation		X						
017419	1.4.C	Waste Reduction Calculations		X						
017419	1.4.D	Records of Donations		X						
017419	1.4.E	Records of Sales		X						
017419	1.4.F	Recycling and Processing Facility Records		X						
017419	1.4.G	Landfill and Incinerator Disposal Records		X						
017419	1.4.I	Qualification Data	X							
017419	1.4.K	Progress payment requirements		X						
017419	1.4.L	Closeout Submittals		X						
017700	1.3.A.2	Project Record Documents		X						
017700	1.3.A.3	Environmental Record Documents		X						
017700	1.3.A.4	Operating Instructions			X					
017823	1.2.A	Manual		X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**EFMO/224159**
Project Title:	**Redesign Parking & Replace Curbs, Walks and Pavement**

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
017900	1.3.A	Instruction Program		X						
017900	1.3.B	Demonstration and Training Video		X						
01 91 14	1.6.A	2-week look ahead		X						
01 91 14	1.6.B	Certificates	X							
01 91 14	1.6.C	Qualifications	X							
01 91 14	1.6.D	Commissioning Plan		X						
01 91 14	1.6.E	Pre-functional checklists				X				
01 91 14	1.6.F	Owner's Project Requirements				X				
01 91 14	1.6.G	Performance Tests		X						
01 91 14	1.6.H	List of test instrumentation		X						
01 91 14	1.6.I	Deficiency Report		X						
01 91 14	1.6.J	Closeout Documentation		X						
024113	1.5.A	Preconstruction Submittals Work Plan		X						
024113	1.5.B	Tank and Piping Removal and Disposal Plan		X						
024113	1.5.C	Test Reports		X						
024113	1.5.E	Permits and Licenses				X				
024113	1.5.F	Waste Manifest Records for Disposal				X				
260519	1.3	Product data for each type of product indicated							X	
260519	1.3	Field quality-control test reports.		X						
260526	1.2	Product data for each type of product indicated							X	
260526	1.2	As-Built grounding plans		X						
260526	1.2	Field quality-control test reports.		X						
260529	1.5	Product data for each type of product indicated							X	
260529	1.5	Shop Drawings						X	X	
260529	1.5	Welding Certificates	X							
260533	1.4	Product data for each type of product indicated							X	
260544	1.3	Product data for each type of product indicated							X	
260533	1.3	Product data for each type of product indicated							X	
260533	1.3	Identification Schedule		X						
265100	1.4	Product data for each type of product indicated							X	
265100	1.4	Shop Drawings						X	X	
265100	1.4	Qualification Data	X							
265100	1.4	Field quality-control test reports.		X						
311000	1.5	Permits for transport of debris		X						
311000	1.5	Demolition procedures			X					
312000	1.4	Delivery Tickets	X							
312000	1.4	Product Data							X	
312000	1.4	Material Test Reports		X						
312000	1.4	Compaction Test Results	X							
312000	1.4	Preexcavation Photography or Videotape		X						

SUBMITTAL LIST

Park Acronym/Project Management Information System (PMIS) Number:	**EFMO/224159**
Project Title:	**Redesign Parking & Replace Curbs, Walks and Pavement**

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
312316	1.5.A	Sources				X				
312316	1.5.B	Material Test Reports		X						
312316	1.5.C	Delivery Tickets	X							
312500	1.3	Product Data							X	
321216	1.3.A	Product Data		X						
321216	1.3.B	Asphalt Mix Design							X	
321216	1.3.C	Test Reports	X							
321320	1.4.A	Product Data							X	
321320	1.4.B	Design Mixtures		X						
321320	1.4.C	Steel Reinforcement Shop Drawings						X		
321320	1.4.D	Qualification Data	X							
321320	1.4.E	Material Certificates	X							
321320	1.4.F	Joint Plan						X		
321320	1.4.G	Materials test report		X						
321320	1.4.H	Field quality control reports		X						
321700	1.2 A	Material Certificates	X					X		
321700	1.2.B	Shop Drawings						X		
321700	1.2 C	Product Data			X					
323113	1.3.A	Product Data			X					
323113	1.3.B	Shop Drawings						X		
323113	1.3.C	Samples for Initial Selection					X			
323113	1.3.D	Samples for Verification					X			
323113	1.3.E	Product Certificates	X							
323113	1.3.F	Field quality control reports		X						
323300	1.3.A	Shop Drawings						X		
334000	1.2.A	Product Data							X	
334000	1.2.B	Manufacturer's Certification							X	
329300	1.5.A	Product Data			X					
329300	1.5.B	Samples for Verification					X			
329300	1.5.C	Qualification Data	X							
329300	1.5.D	Product Certificates	X							
329300	1.5.E	Materials test report		X						
334000	1.3.A	Product data							X	
334000	1.3.B	Manufacturer's Certification							X	

EFMO – N170 318539 / 224159

SECTION 01 35 13.22 – ARCHEOLOGICAL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of protecting archeological resources contained in soil deposits.

B. The Construction Contractor shall consider the entire project site an archeologically sensitive area. An NPS Archeological Monitor is required for all excavation activities, coordinate with the Contracting Officer in advance of all ground disturbing activities to ensure the NPS archeological monitor presence.

1.2 DEFINITIONS

A. Archeological Resources: Archeological resources are physical evidences of past human activity, including evidences of effects of that activity on the environment. Archeological resources represent both prehistoric and historic time periods, found above and below ground and under water.

B. Archeologically Sensitive Areas: Areas having potential to contain significant (National Register eligible) archeological resources. If National Register eligible or listed archeological resources could not be avoided, an appropriate mitigation strategy would be developed in consultation with state historic preservation officer and, if necessary, associated American Indian tribes.

C. Non-sensitive Areas: Areas with little potential of containing significant (National Register eligible) archeological resources.

D. NPS Archeological Monitor: Representative of Government designated to oversee construction activities that could disturb archeological resources.

E. Archeological Resources Protection Act (ARPA) of 1979 (Public Law (P.L.) 96-95; 93 United States Statues at Large (Stat.) 712): defines archeological resources as any material remains of past human life or activities that are of archeological interest and at least 100 years old; Section 4 of the statute describes the requirements that must be met before Federal authorities can issue a permit to excavate or remove any archeological resource on Federal or Indian lands; the curatorial requirements of artifacts, and other materials excavated or removed.

1.3 SUBMITTALS

A. Daily Work Schedule: Detail construction work in archeologically sensitive areas. Submit to Contracting Officer (CO) 30 days before start of ground disturbing site work.

1.4 QUALITY ASSURANCE

A. At least one week before on-site work begins, Contractor shall meet with Contracting Officer and the NPS Archeological Monitor to discuss Daily Work Schedule, equipment, and special methods used in archeologically sensitive areas. Contractor shall ensure approved Daily Work Schedule is followed throughout construction. The Contractor shall identify all ground disturbing work and associated duration within the weekly Three Week Look Ahead report (current week plus 3 week look ahead).

PART 2 - PRODUCTS

2.1 DAILY WORK SCHEDULE

A. Daily Work Schedule is required for work occurring within archeologically sensitive areas. Include all work that is to occur within the area and key the schedule to the drawings to include:

1. Starting and ending dates of ground-disturbing construction.
2. Locations of temporary facilities, such as barriers, field offices, staging areas, sanitary facilities, borrow pits, and haul and access roads.
3. Types of construction, such as clearing, topsoil stripping, structure or trench excavation, landscaping, and post construction clean-up.
4. Methods and equipment used for each type of construction.
5. Plan for relocating work in the event of temporary work stoppages at each archeologically sensitive area

PART 3 - EXECUTION

3.1 BARRICADES

A. Comply with requirements specified in Section 01 50 00 "Temporary Facilities and Controls."

3.2 ARCHEOLOGICAL INVESTIGATION BY NON-NPS PERSONNEL

A. A permit is required for archeological investigations (e.g. excavation, shovel testing, coring, pedestrian survey, underwater archeology, rock art documentation, or other types of reconnaissance including archaeological monitoring of construction) carried out on parklands by non-National Park Service (NPS) personnel, unless carried out under a contract or a cooperative agreement specifically written for archeological investigations. Permits are issued under the Archaeological Resources Protection Act of 1979 (ARPA). The NPS does not issue a permit for archeological investigations carried out by NPS archeologists, or to archeologists working on NPS archeological projects under a contract or cooperative agreement.

B. Applicants should submit a Permit Application (DI Form 1926 (Revision September 2004) Office of Management and Budget (OMB) Number (No.) 1024-0037, approved through 1/31/2008. Permit Application form is available, in PDF (portable document file) format, to the manager of the park in which they propose to work; or to the regional director, with a copy to the park manager.

3.3 OBSERVATION

A. The NPS Archeological Monitor will observe ground-disturbing site work, including construction of temporary facilities, at archeologically sensitive areas, from a safe location mutually agreed on by Contractor and Monitor. As new ground is broken, Monitor will examine excavated materials, using construction layout centerline and perimeter staking as a reference point to record locations of findings.

B. A minimum of 14 calendar days advanced notice to the CO and the NPS archeological monitor is required prior to planned ground disturbance activities in archeologically sensitive areas.

3.4 DISCOVERY OF RESOURCES

A. If the NPS Archeological Monitor discovers resources, immediate relocation of work to a non-excavation area may be required for Monitor to identify and document resources and, if necessary, develop appropriate mitigation plan. While the NPS Archeological Monitor is documenting resources in sensitive areas, Contractor shall relocate work to non-excavation areas where monitoring is not normally required.

B. If resources are discovered while the NPS Archeological Monitor is absent, stop work immediately and report the discovery to the Contracting Officer.

3.5 WORK STOPPAGE

A. Contractor shall plan, schedule, and execute work to prevent stoppages at one area from stopping all work at construction site.

END OF SECTION 01 35 13.22

SECTION 01 35 23 – SAFETY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes establishing an effective accident prevention program and providing a safe working environment for personnel and visitors.

1.2 CONDITIONS PRESENT FOR PROJECT

A. None identified.

1.3 SUBMITTALS

A. Accident Prevention Plan (APP): Submit APP after contract award and before Pre-Construction conference. Contracting Officer (CO) will review proposed APP. If APP requires any revisions or corrections, Contractor shall resubmit Plan within 10 days. No progress payments will be made until the APP is accepted.

1.4 QUALITY ASSURANCE

A. Comply with contract clauses "Accident Prevention" and "Permits and Responsibilities." In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Onsite equipment shall meet 29 CFR 1926 (Code of Federal Regulations) (Occupational Safety and Health Administration (OSHA)) requirements. Failure to comply with requirements of this section and related sections may result in suspension of work.

B. Site Safety Supervisor:

1. Designate authorized onsite representative for preparation and maintenance of the APP.
2. Shall be responsible for:

    a. Implementation and enforcement of the APP
    b. Daily safety inspections
    c. Conducting and documenting weekly and monthly safety meetings
    d. Review of safety requirements at progress meetings
    e. Compilation and maintenance of Safety Data Sheets (SDS) and safety reference materials
    f. Tracking and resolution of safety violations
    g. Site personnel and visitor compliance with site safety and health requirements and APP
    h. Investigation and reporting of accidents and injuries

C. Qualifications of Employees:

1. Physically and able to perform their assigned duties in a safe manner.
2. Do not allow employees whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.
3. Provide operating instructions for equipment. Operators of vehicles, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.

1.5 ACCIDENT REPORTING

A. Reportable Accidents: Defined as: death, occupational disease, and/or traumatic injury to employees or the public; fires; and/or property damage by accident in excess of $100.

1. Notify Contracting Officer immediately in the event of a reportable accident.
2. Fill out and forward an Accident/Property Damage Report Form (CM-22) to Contracting Officer within 7 days of a reportable accident. Obtain form from Contracting Officer.

PART 2 - PRODUCTS

2.1 ACCIDENT PREVENTION PLAN (APP)

A. APP shall be written to comply with OSHA and project requirements (generic plan is not acceptable) including but not limited to:

1. Name and qualifications of supervisor responsible to carry out program.
2. Weekly and monthly safety meetings shall be documented with topics and attendees.
3. First aid and rescue procedures.
4. Job Hazard Analysis (JHA) for each major phase. List of hazards associated and methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Include initial and continuing training.
5. Planning for possible emergency situations, as detailed in Article 1.2. Such planning shall take nature of construction, site conditions, and degree of exposure of persons and property into consideration.
6. Infectious Disease Preparedness:

a. Contractors are responsible for their employees' safety and the safety of job site visitors during the performance of this contract. We encourage Contractors to follow guidance from the Department of Labor (DOL), Occupational Safety and Health Administration (OSHA), the Centers for Disease Control and Prevention (CDC), and all other applicable local, city, and state mandates. We encourage Contractors to develop policies for infection prevention and an Infectious Disease Preparedness and Response Plan.
b. To the extent appropriate, Contractors should include the protective health and safety measures they intend to implement in any accident prevention or safety submittals required under this contract. These plans should contain preventive

measures the Contractor intends to follow while performing work on government property as well as responsive and corrective actions to be taken if an employee exhibits symptoms or tests positive for contagion.

c. Upon contract award, Contractors should communicate with Contracting Officer regarding Contractor decisions and actions to protect the health and safety of workers for the duration of contract performance under which pandemic conditions exist.

2.2 FIRST AID FACILITIES

A. Provide adequate facilities for number of employees and appropriate to construction hazards.

2.3 PERSONNEL PROTECTIVE EQUIPMENT (PPE)

A. Selection shall conform to OSHA 29CFR 1926.95 Subpart E.

PART 3 - EXECUTION

3.1 DAILY SAFETY INSPECTIONS

A. Conduct daily safety inspections and maintain daily safety reports which include:

1. Area/operation inspected
2. Date of inspection
3. Identified hazards
4. Corrective actions taken

3.2 EMERGENCY INSTRUCTIONS

A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.

3.3 FIRE AND LIFE SAFETY

A. Comply with requirements of 2024 International Fire Code.

3.4 HAZARDOUS MATERIALS

A. Hazardous materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.

B. Store hazardous materials in accordance with manufacturer’s and OSHA Subpart D requirements. Maintain Safety Data Sheets (SDS) for each chemical readily available on site.

1. Immediately report spills of hazardous materials to the Park.

2. Maintain a spill emergency response kit.
3. Train employees how to respond to a spill and use emergency response kit.

3.5 PROTECTIVE EQUIPMENT

A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing to another individual.

3.6 SAFETY MEETINGS

A. As a minimum, conduct one weekly 15-minute "toolbox" safety meeting conducted by a foreman or supervisor and attended by construction personnel at worksite. Topics shall coincide with work scheduled for following week. Document and submit meeting minutes to Contracting Officer within one day after meeting.

B. Conduct monthly safety meetings for personnel, contractors, and subcontractors performing work on the site. Notify Contracting Officer of meeting dates and times. Meetings shall be used to: review effectiveness of Contractor's safety effort; resolve current health and safety problems; provide a forum for planning safe construction activities, and for updating Accident Prevention Plan. Contracting Officers Representative will attend meetings and enter results of meetings into the daily log.

3.7 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

A. A hard hat use area shall be designated by Contractor. Hard hat area shall be posted by Contractor in a manner satisfactory to Contracting Officer.

B. It is Contractor's responsibility to require persons working on or visiting site to wear hard hats and PPE in good repair at all times. As a minimum, maintain [six] 6 hard hats and other APP required equipment.

3.8 TRAINING

A. First Aid: Provide training to personnel to ensure prompt and efficient first aid.

B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage.

END OF SECTION 01 35 23

SECTION 01 35 50 – NATURAL & CULTURAL RESOURCES PROTECTION

PART 1 - GENERAL

1.1 DESCRIPTION

A. Effigy Mounds National Monument (EFMO) contains areas of significant natural and cultural sensitivity that must be protected during construction of improvements within the monument boundary. The Contractor shall conform to the requirements set forth in this section to protect identified, suspected, or discovered natural and cultural resources within the project limits. Additional requirements may be described in other specification sections and shall be coordinated with the requirements of this Section.

B. The Contractor shall minimize environmental pollution and damage that may occur as the result of construction operations. The environmental resources, including natural, cultural and physical resources, within the project boundaries and those affected outside the limits of permanent work shall be protected during the entire duration of this contract. The Contractor shall comply with all applicable environmental Federal, State, and local laws and regulations. The Contractor shall be responsible for compliance with this section by its employees and its subcontractor employees. The Contractor shall be responsible for any delays resulting from failure to comply with environmental laws and regulations.

C. This section includes environmental procedures broadly identified as protection of natural, cultural and physical resources. The intention of this section is to ensure those involved with the project act positively to protect the environment, both on site and off site, during all construction operations. All activities and requirements of a general nature are applicable to all areas within the Effigy Mounds National Monument boundary.

D. Work under this Contract is performed pursuant to a Clean Water Act Section 401 Water Quality Certification issued by the Iowa Department of Natural Resources Water Quality Bureau.

E. Work under this Contract is performed pursuant to an Iowa Department of Natural Resources to obtain a National Pollution Discharge Elimination System (NPDES) General Permit for Storm Water Discharges Associated with Construction Activity. The Contractor shall prepare a Storm Water Pollution Prevention Plan (SWPPP) for the project that meets the requirements of the Iowa Department of Natural Resources. Work shall not commence until the Contractor has received approval for the submitted SWPPP. The Contractor shall at all times implement and conform to the requirements of the plan.

F. Related Documents: The Contractor should consult the following sections for related and additional requirements:

1. Submittal Procedures – Section 01 33 23.
2. Temporary Facilities and Controls – Section 01 50 00.
3. Temporary Storm Water Pollution Prevention – Section 01 77 23.
4. Quality Requirements – Section 01 40 00.

1.2 REFERENCES

A. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

1. U.S. National Archives and Records Administration (NARA):

a. 33 CFR 328 Definitions
b. 40 CFR 68 Chemical Accident Prevention Provisions
c. 40 CFR 152 – 186 Pesticide Programs
d. 40 CFR 260 Hazardous Waste Management System: General
e. 40 CFR 261 Identification and Listing of Hazardous Waste
f. 40 CFR 262 Standards Applicable to Generators of Hazardous Waste
g. 40 CFR 279 Standards for the Management of Used Oil
h. 40 CFR 302 Designation, Reportable Quantities, and Notification
i. 40 CFR 355 Emergency Planning and Notification
j. 49 CFR 171 – 178 Hazardous Materials Regulations
k. 42 USC Section 6901
l. National Pollutant Discharge Elimination System (NPDES)

1.3 DEFINITIONS

A. NPS Archeological Monitor: Archaeologist meeting the minimum requirements set forth in the Secretary of the Interior's Standards for Professional Qualification Standards for Archaeology, who is designated to observe ground disturbing activities during project operations to identify, record and collect archaeological deposits exposed during construction activity for compliance with federal historic preservation laws, regulations, and guidelines.

B. Archeologically Sensitive Areas: Areas that have been determined to contain, or have the potential to contain, archaeological deposits eligible for listing on the National Register of Historic Places.

C. Biological Resources: All native and non-native plants and wildlife, the habitats where they are found, and the communities they form.

D. Class III Landfill: Landfill that accepts non-hazardous waste such as household, commercial and industrial waste, including construction, remodeling, repair, and demolition operations.

E. Contractor Generated Hazardous Waste: Contractor generated hazardous waste means materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste materials would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene etc.), waste thinners, excess paints, excess solvents, waste solvents, and excess pesticides, and contaminated pesticide equipment rinse water.

F. Contractor Generated Solid Waste: Solid wastes such as building materials, packaging, rubbish, debris, rubble, food waste, wrappers, spent disposable utensils, leaves, weeds, grass, and other organic materials resulting from construction, remodeling, repair, and demolition operations.

1. Rubbish/Debris: Combustible and non-combustible wastes such as paper, boxes, glass, crockery, tin cans, metal, and lumber scrap.
2. Green Waste: Combustible and non-combustible wastes such as leaves tree trimmings, and other organic materials that result from construction or maintenance and repair work.
3. Sanitary Waste: Refuse and scraps resulting from preparation, cooking, distribution, or consumption of food.

G. Culturally Sensitive Areas: Areas that have been determined to contain, or have potential to contain, Native American traditional resources, or resources to which Native Americans attach cultural and religious significance, and which are eligible for listing on the National Register of Historic Places. Most culturally sensitive areas are delineated on the Construction Drawings; however, the Park Cultural Resource Manager shall be consulted to identify culturally sensitive areas that are not identified on the Drawings.

H. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which could:

1. Adversely affect human health or welfare.
2. Unfavorably alter ecological balances important to human life, affect other species of importance to humanity.
3. Degrade the utility of the environment for aesthetic, cultural, or historical purposes.

I. Environmental Protection: Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

J. Exotic Species: Species of plants or animals that do not naturally occur in a particular area and often interfere with natural biological systems. Also known as alien, introduced, invasive, or non-native species.

K. Hazardous Materials: Any material that can be defined as an explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substance that could lead to death or injury.

L. Hazardous Waste: A hazardous material becomes a hazardous waste if the material has no safe end use in the project.

M. Inert Solids/Inert Waste: Non-liquid solid waste including, but not limited to soil and concrete that does not contain hazardous waste or soluble pollutants at concentrations in excess of water-quality objectives established by local, state, or federal regulatory agencies, and which does not contain significant quantities of decomposable solid waste.

N. Land Application for Discharge Water: The term “Land Application” for discharge water implies that the Contractor shall discharge water at a rate which allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the “waters of the United States” shall occur. Land Application shall be in compliance with all applicable Federal, State, and local laws and regulations.

O. Native American Monitor: Native American meeting tribal qualifications of the tribe most likely associated/affiliated with the project area, and able to identify, record, and protect traditional

cultural resources and resources to which they attach cultural and religious significance eligible for listing on the national Register of Historic Places, and who can act on behalf of the tribe concerning human remains and items protected under the Native American Graves Protection and Repatriation Act that may be inadvertently exposed during construction activity.

P. Natural Resources: All biological and physical resources and their associated processes.

Q. Non-Native: Species of plants or animals that do not naturally occur in a particular area and often interfere with natural biological systems. Also known as alien, introduced, invasive, or exotic species.

R. Nonsensitive Areas: Areas that have a low likelihood of containing significant in-the-ground archeological or cultural resources.

S. Pesticide: Pesticide is defined as any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant.

T. Physical Resources: Soil, water, and air resources. Those naturally occurring elements of the earth, exclusive of plants and wildlife.

U. Cultural Resources: Prehistoric, historic, and recent 20th century artifacts, including charcoal, human bones, ash, fire rocks, and building materials that indicate the presence of past human occupation. Also including historic structures and features such as bridges, trails, buildings and rock alignments, places of spiritual and traditional importance, plant gathering areas, and historic Native American use areas.

V. Root Zones: Typical spread of the roots from the trunk of a tree; normally defined as in a vertical plane with the drip line of the tree branches.

W. Surface Discharge: The term “Surface Discharge” implies that the water is discharged with possible sheeting action and subsequent soil erosion may occur. Waters that are surface discharged may terminate in drainage ditches, storm sewers, creeks, and/or “waters of the United States” and would require a permit to discharge water from the governing agency.

X. Waters of the United States: All waters which are under the jurisdiction of the Clean Water Act, as defined in 33 CFR 328.

Y. Wetlands: Wetlands are areas that are inundated or saturated by surface or ground water at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions. Wetlands generally include swamps, marshes, and bogs.

Z. Wildlife: Includes all fish, mammals, insects, birds, macro and micro-organisms, reptiles, amphibians, etc.

1.4 GENERAL REQUIREMENTS

A. Orientation Training:

1. All Contractor and subcontractor employees shall receive a brief orientation about working in Effigy Mounds National Monument prior to actually performing work. The orientation describes the efforts to be taken by the Contractor and subcontractor employees to protect the natural, cultural and physical resources while working on this and other projects. This orientation also describes mitigation and other environmental protection measures that must be adhered to at all times while in the Park.

1.5 SUBMITTALS

A. The following shall be submitted in accordance with Section 01 33 23, Submittal Procedures.

1. Submittals:

a. Spill Prevention and Response Plan (see Part 3). This plan shall be submitted no less than 20 days prior to the need for approval.
b. MSDSs for all materials proposed for use, all spill response kit contents, reporting structure for spills, and a copy of the work schedule to coordinate with monitors (see Part 3).
c. Temporary Storm Water Pollution Prevention Plan, Section 01 57 23.
d. Hazardous Materials Spill Prevention and Response Plan (see Part 3).

PART 2 - PRODUCTS

2.1 FENCING & BARRICADES

A. Reference Section 01 50 00, “Temporary Facilities and Controls.”

B. Provide temporary barrier for sensitive resource areas as directed by the plans and specifications.

2.2 RECYCLED MATERIALS

A. Construction Safety Fence.

B. When chain-link fence barriers are not required, but a safety fence is prudent, use recycled construction safety barrier fence.

C. Recycled Safety Barrier Fence Suppliers: Jefferson Smurfurt Recycling Company; Environ Products; Hygrade Glove and Safety, or approved equal.

D. Asphalt roads and parking lots and aggregate base rock to be re-used for trench backfill and road base as per plans and specifications.

PART 3 - EXECUTION

3.1 COMPLIANCE WITH FEDERAL AND STATE PERMIT REQUIREMENTS

A. Contractor will comply with all Federal and State permits required for construction-related activities that would include but not be limited to:

1. The Iowa Department of Natural Resource National Pollutant Discharge Elimination System (NPDES) General Permit for Storm Water Discharges Associated with Construction Activities.

3.2 ORIENTATION

A. All contractor and subcontractor employees shall view a government provided orientation video to ensure each is fully aware of the natural and cultural resource protection and mitigation requirements of work. Government staff will provide the initial orientation. Subsequent on-going awareness orientation for new employees and when site conditions change shall be performed by contractor and integrated into construction operation procedures.

B. The Contractor shall maintain a manifest tracking all contractor personnel, when they received their orientation training, and when they started work. Contractor personnel shall be field identifiable as having received their orientation training by means of a readily visible sticker on their hard hat.

3.3 ENVIRONMENTAL PROTECTION MEASURES

A. The Contractor shall comply with these comprehensive waste management measures and shall comply with all applicable Federal, State and local regulations and address all aspects related to the transportation, labeling, storage, handling and disposal of all construction-related hazardous and non-hazardous liquid and solid wastes.

1. Spill Prevention, Control, and Countermeasure (SPCC) Plan:

a. The primary purpose of the Spill Prevention Response Plan (SPRP) is to prevent sewage spills from occurring by proper planning and protection of the project area, and then to respond to any sewage spills that may occur during the course of this project including appropriate notification of staff. The Plan will be general in nature and typical to all phases of the work with site specific plans required for each area involving trenching or any work with the possibility of accessing the existing system.

b. The SPRP is structured in two parts – first a Spill Prevention Plan and then a Spill Response Plan. The Spill Prevention Plan (SPP) includes evaluation of specific conditions, set-up of containment for actual construction work as well as for bypass pumping. Sewer bypasses must be constructed to tie existing lines into the new system and to tie the new system into the existing system. The Spill Response Plan (SRP) includes the initial response to stop and contain a spill, notification of staff, clean-up, and follow-up documentation. The SPP and the SRP together comprise the entire SPRP.

c. All Contractor employees are required to be trained in the Spill Prevention Control in accordance with this SPRP.

2. Non-Hazardous Solid Waste:

a. Waste, trash, and debris shall be controlled at all times and disposed in authorized containers in the Contractor's staging area.
b. All sanitary waste (garbage) must be disposed of in approved, bear-proof disposal bins. Provide lockable, bear-proof dumpsters with lids for waste (garbage) storage. Lids shall be equipped with carabineers/heavy wire lid locks. Verify that dumpster lids are secure at close of work each day.
c. Construction debris (rubbish) may be stored in unlidded dumpsters or construction debris truck/trailers and removed on a regular basis. Do not mingle sanitary or green waste with construction debris.
d. All large, normally open top, waste bins or dumpsters shall be lidded and clearly marked "No Food or Trash."
e. The Contractor shall designate an employee to police the work site daily for waste, wrappers, food packaging and the like. All waste shall be picked up and disposed of in lidded bear-proof dumpsters.
f. Green waste shall be segregated from other non-green waste for processing at disposal site.
g. Burying or burning of trash and debris on site is not permitted. All un-used materials, trash, and debris shall be the property of the Contractor and shall be transported outside of the Effigy Mounds National Monument boundary for disposal in accordance with law.
h. Remove debris from permanently closed spaces prior to enclosing them.

3. Non-Hazardous Liquid Waste:

a. Waste water from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, forms, etc. shall not be allowed to enter water ways or to be discharged prior to being treated to remove pollutants. The Contractor shall dispose of the construction related wastewater off-Government property in accordance with all Federal, State, Regional and Local laws and regulations.
b. Water contaminated with silt, grout, or other construction by-product must be pumped to a holding tank. Location of the holding tank will be proposed by Contractor and approved by Contracting Officer.
c. For discharge of ground water, the Contractor shall obtain a State or Federal permit specific for pumping and discharging ground water prior to surface discharging.

4. Hazardous Materials and Wastes:

a. Identify potentially hazardous substances to be used on the job site.
b. Identify handling procedures to ensure that hazardous substances are not released into the air, water, or ground.
c. Comply with Federal, State, and local laws and regulations for storage, handling, and disposal of these materials.
d. Storage of hazardous or flammable chemicals in the staging area or elsewhere on the site is prohibited except as approved by the Contracting Officer.

e. Hazardous materials shall not be discarded into the jobsite debris or waste-disposal facilities.
f. Empty containers shall be removed from the site and disposed of in a manner prescribed by law.
g. Used lubricants and used oil to be discarded shall be stored in marked corrosion-resistant containers and recycled or disposed in accordance with 40 CFR 279, State, and local laws and regulations.
h. A copy of the Material Safety Data Sheets (MSDS) and the maximum quantity of each hazardous material to be on site at any given time is to be maintained on site and submitted to the Contracting Officer.
i. Before new hazardous materials are brought on site or removed from the site, the MSDS file shall be updated and submitted to the Contracting Officer.

B. Contractor shall provide a Hazardous Materials Spill Prevention, Control and Countermeasure Plan to address spill prevention and response measures for hazardous substances used on site, including fuels. Prior to the start of work, the Contractor shall submit a plan that complies with Effigy Mounds National Monument, Federal and State requirements and allows contractors to properly notify officials in the event of an emergency occurring during construction activities. The Monument requirements include at a minimum:

1. During non-work operations, stationary equipment shall be parked over specially prepared containment pads designed to trap any leaking oil, fuel, or hydraulic fluids.
2. Inspect construction site daily for proper storage of hazardous materials, proper parking of equipment on containment pads, and for hydraulic and oil leaks of equipment, tighten hoses, and ensure they are in good condition.
3. Routine oiling and lubrication shall be conducted in areas with secondary containment using Best Management Practices (BMPs) at all times. Refueling of equipment in wetlands or stream channel areas is not allowed at any time.
4. Contractor shall maintain secondary containment for all equipment operating with fluids (such as drilling) or when direct discharge of leakage, spills, or other source of construction or equipment fluids can flow directly to any streambed, whether flowing with water or dry. Containment shall be designed and installed so as to prevent accidental spills into streambeds in the event of mechanical failure or hose breakage.
5. Contractor shall maintain spill response materials on the project site when using heavy equipment to ensure rapid response to small spills. These materials shall include absorbent pads, booms, or other materials as appropriate to contain oil, hydraulic fluid, solvents, and hazardous material spills. A list of the spill response materials to be kept on site shall be submitted to the Contracting Officer.
6. Contractor shall provide names and phone numbers of appropriate contractor's personnel to be contacted at any time (24 hours per day) regarding accidental release of hazardous substances to air, soil or water. This list shall be submitted to the Contracting Officer and a copy visibly displayed in work areas on site.
7. Contractor shall have the Contracting Officer's and other appropriate Government emergency numbers posted and shall immediately notify the Contracting Officer or other Government representative on any accidental release of hazardous substances to air, soil or water.

C. The Contractor shall comply with these measures to protect cultural and natural resources. In addition, the Contractor must comply with all applicable Federal, State, and local regulations for the protection of these resources.

1. Cultural Resource Protection:

    a. A Government provided Archeological Monitor, and as necessary, will observe all ground-disturbing site work, including construction of temporary facilities at all culturally sensitive areas, from a safe location mutually agreed on by Contractor, Contracting Officer and Monitors. As new ground is broken, Monitors will examine excavated materials, using construction layout centerline and perimeter staking as a reference point to record locations of findings.
    b. Prior to construction, mark with flagging all sensitive cultural resources to be protected within the project area identified per the requirements of the plans and specifications. Proper placement of flagging shall be verified by the Contracting Officer. Upon verification, erect necessary fencing to identify and protect cultural resources from disturbance.
    c. Do not begin ground-penetrating work such as excavation, trenching, drilling, or stump and root removal in culturally sensitive areas without the presence of Archeological Monitor, and if required, Native American Monitor.
    d. If Archeological Monitor discovers resources, immediate relocation of the work to a non-excavation activities area may be required to allow Monitors to take soil samples and record resources. While Monitors are documenting resources in sensitive areas, Contractor shall relocate work to non-sensitive areas.
    e. If resources are discovered while Monitors are absent, stop work immediately and report the discovery to the Contracting Officer.
    f. Stop Work: Cease all activities in the area of discovery and protect the resources discovered. In the event the discovery represents human remains or any objects subject to the Native American Graves Protection and Repatriation Act (NAGPRA), the NPS will follow procedures outlined in the NAGPRA plan of action developed for this project. This will require a stoppage of work in the area of work for a minimum of 30 calendar days. In the event of an inadvertent discovery of Cultural Resources, be prepared to stop work and continue in other areas.

        1) The Contractor shall plan, schedule, and execute the work to prevent stoppages at one area from stopping all work at the construction site.
        2) Adjustments for Work Stoppage. See Federal Acquisition Regulations.

2. Natural Resource Protection:

    a. Plant Condition Inventory: The Contractor and the Contracting Officer or designated representative, shall perform an on-site inventory of trees and other overall vegetation features within or near to the work limits. A print of the contract drawings showing tree locations and a photo record will be used to note condition of trees and vegetation. This annotated drawing will be retained by the Contracting Officer for use during the final walk-through and tree/vegetation assessment. This walk through shall be a part of the project closeout requirements (see Section 01 77 00, “Closeout Procedures”).
    b. On-site inventory shall be scheduled in coordination with the pre-construction conference.
    c. Provide temporary barriers (e.g., orange construction fence) to protect existing trees, plants and critical root zones that are designated to remain, but are: (1) within the construction limits; 2) on or just outside the construction limits; (3) within the clearing limits (i.e., the zone extending 5 feet beyond the staked construction limits);

or (4) on, or just outside the clearing limit line. Barriers shall be in place before construction begins.

d. Trees, shrubs, vines, grasses, and other vegetative features indicated and defined on the Drawings to be preserved shall be clearly identified by marking, fencing, or any other approved techniques. The Contractor shall restore vegetative features damaged or destroyed during construction operations outside the limits of the approved work area.

e. Except in areas indicated on the drawings or specified to be cleared, the Contractor shall not remove, cut, deface, injure, or destroy resources including trees, shrubs, vines, grasses, topsoil, and landforms without approval. No ropes, cables, or guys shall be fastened to or attached to any trees for anchorage unless specifically authorized.

f. Should it be determined during the course of work that additional trees or tree roots require removal, Contractor shall notify the Contracting Officer who will coordinate an inspection and determination by the appropriate authorities whether to remove the tree or not.

g. After tree removal, large roots may remain in the ground. Contractor shall be responsible for carefully removing in-ground tree roots of removed trees to permit excavation, drilling, or other ground penetrating construction activities. During tree root removal, do not use backhoes, chains, or other equipment in a manner that will harm roots of adjacent trees.

h. Minimize disturbance to tree trunks and root zones to prevent damage to trees.

i. Adjust trenches and other excavations to keep them beyond the drip line wherever possible.

    1) Where such adjustments are not practical, maintain the following minimum clearances between the face of trees to be saved and the closest edge of the excavation:

for trees more than 30-inch-in-diameter	10 feet
for trees between 15-inch and 30-inch-in-diameter	8 feet
for trees less than 15-inch-in-diameter	5 feet

    2) Adjust the survey line, as necessary to maintain required clearances.

        a) Notify the Contracting Officer of any proposed trenches or other excavations within the drip line of trees.

j. Steps to Mitigate Damage to Roots Due to Excavation:

    1) Take steps (as called for below) to mitigate damage to tree roots due to excavation, wherever the following circumstances apply:

        a) Wherever excavation must take place within the drip line of oak trees regardless of diameter.
        b) Wherever excavation must take place within the drip line of trees other than oaks, for all trees 12 inches or larger in diameter.

    2) Trees which are anticipated to meet these criteria and therefore require steps to mitigate damage to roots due to excavation are shown on the drawings. Adjustments in trench alignment or other factors may result in variations in

which trees are affected. The Contractor shall accommodate these variations at no additional expense to the Government.

3) Following are the steps which are required to mitigate damage to roots due to excavation:

a) Excavate carefully where tree roots might be encountered. Where roots 2 inches and larger are encountered, hand excavate as required to prevent damage to roots. Tunnel under roots to be saved, hand excavating as necessary.

b) Do not cut roots over 2-inch-in-diameter without approval of Contracting Officer.

c) Cleanly saw-cut roots between 1-inch and 2-inch-in-diameter where they interfere with work; do not cut roots except as necessary. Roots between 1-inch and 2-inch-in-diameter which must be cut shall be cleanly saw-cut near the edge of trench closest to the tree to prevent roots from being dislodged from soil by equipment.

d) Within four hours of excavating, wrap burlap around the ends of cut roots larger than 1-inch-in-diameter and wet the burlap. Keep the burlap moist until the trench is backfilled. During backfilling operations, bring the burlap to within a few inches of the ground surface. Where the excavation is in a paved roadway or path, bring the burlap to within a few inches of the ground surface in a location just off of the edge of the road or path where possible.

e) Thoroughly wet roots and burlap in the excavated area before backfilling. Backfill shall contain as much water as is compatible with compaction.

f) Using a water truck, thoroughly water the excavated area once per week during the months of June, July, August, and September. Take care not to cause disturbances to or runoff into camping areas and other visitor areas during the watering process. Where the trench is in pavement, water the area nearest the trench just off the pavement.

4) Do not drive water trucks where they will disturb vegetation, including meadows and tree root zones. Use long hoses as required.

k. Do not drive over root zones unless work cannot otherwise be reasonably done. Driving over roots compacts the soil and can harm or destroy the tree. If unavoidable, use protective measures, like mats or plywood to prevent soil compaction.

l. Do not cut wheels or make sharp turns with wheeled or tracked equipment in root zones.

m. Do not pile excavated soil against tree trunks.

n. Do not compact soil around roots to a greater degree than surrounding unexcavated soil, except to meet compaction requirements directly under pavement or structures.

o. Preserve tree snags where feasible as potential bat or bird habitat.

3. Plant Appraisal:

a. If the Contractor destroys or injures trees and vegetation designated for protection or outside the work limits, the Contractor will be assessed damages prior to final progress payment.

b. Replacement costs for damaged vegetation will be computed according to the method described in the International Society of Arborculture's 1992 Guide for Plant Appraisal. This method is based on the cost of the largest commonly available tree or shrub, with modifications based on species value, condition, and location. A trained arborist or professional plant appraiser from the Iowa region will be hired by the NPS to make the damage appraisal. The arborist's fees will be included in the damage assessment.

c. This damage appraisal process will be triggered by any of the following types of damage to vegetation outside the work limits or unauthorized disturbance of vegetation within the work limits.

1) Removal of any tree or shrub.
2) Pruning or removal of more than 30 percent of a tree or shrub canopy.
3) Removal or fracture of any limb or trunk that is one of the major structural entities of the damaged plant.
4) Removal or fracture of any limb greater than 12 inches in diameter.
5) Bark damage or removal around more than 30 percent of the trunk circumference.
6) Trenching or soil disturbance within the critical root zone that is deeper than 1-foot unless shown on the Drawings.

d. If the damaged vegetation is protected under the Endangered Species Act or other special legislation, additional penalties may be assessed as per consultation with the U.S. Fish & Wildlife Service.

e. Pruning or removal of vegetation shall be supervised by Contracting Officer. The designated personnel may designate plant species for salvage. When authorized and supervised by the Contracting Officer, the Contractor is exempted from any penalties that might be assessed due to damage to vegetation.

1) Acceptable disturbance to roots is limited to 15 percent of the area under the dripline being either cut or filled. Any tree with more than 50 percent of its roots disturbed should be removed during construction at the direction of the Contracting Officer.
2) Wounds occurring from construction activity may be possible entry sites for disease spores. If a tree is accidentally injured during construction, it may need to be removed at the direction of the Contracting Officer.

f. Trench alignments or other factors may result in variations in which trees are affected. The Contractor shall accommodate these variations at no additional expense to the Government.

g. Minor cuts and damaged areas shall be assessed by the Contracting Officer. Repair to the plant will be at the recommendation of Monument personnel and approval of the Contracting Officer.

4. Protection from Exotic Plant Species:

a. Measures shall be taken to prevent the introduction of exotic species in the project area and staging areas. All earth moving equipment must enter the Park free of dirt, dust, mud, seeds, or other potential contaminant. Equipment exhibiting any dirt or other material attached to frame, tires, wheels, or other parts shall be thoroughly cleaned by the Contractor before entering the Park. All equipment will be directed

for inspection prior to commencing work. Areas inspected shall include, but not be limited to, tracks, track guard/housings, belly pans/under covers, buckets, rippers, and other attachments. Equipment that does not pass inspection will be turned around to the nearest cleaning facility outside the park. The Contractor shall notify the Construction manager at least two work days (not including weekends) prior to bringing any equipment into the Park. Equipment found to have entered the Park with potential contaminants will be removed from the Park at the direction of the Contracting Officer at Contractor's sole expense.

b. Minimize ground disturbance to the greatest extent possible.

c. The contractor shall get approval in writing from the Contracting Officer for fill material that must be used in a way or stored in a location not clearly specified in the contract.

d. Fill materials used within the top 12 inches of finished grade are required to be free of exotic and noxious weed species and shall have the source locations approved by the Contracting Officer. The Contractor shall submit to the Contracting Officer a list of proposed sources for imported fill materials requiring certification 30 calendar days in advance of importing material. The presence of noxious weed species is grounds for rejection of the source. If exotic weed species are found or suspected, the Contractor may be required to strip the top 12 inches of source material and only import sub-surface material and/or sterilize the material, at the Contracting Officer's discretion. If spraying is required, the Contractor shall provide a licensed operator to spray according to applicable state regulations. The Contractor shall not spray any herbicides until approved in writing by the Contracting Officer.

5. Protection from Exotic Animals

a. Drain and flush all pumps, tanks, live wells, buckets and other containers that might carry water contaminated with exotic animals, such as the zebra mussel, prior to bringing equipment into the park. Thoroughly wash all hauling tanks and equipment using a hard spray from a garden hose. If equipment was used in infested waters, use the following steps to clean the equipment:

1) Wash with hot water (140 F or 40 C) or a high-pressure washer (250 pounds per square inch). Remove all aquatic weeds; they can carry zebra mussels.

2) Disinfect equipment. Recent research shows that disinfection of nets and equipment with benzalkonium chloride at typical treatment rates (10 milligrams per liter for 24 hours, 100 milligrams per liter for 3 hours, or 250 milligrams per liter for 15 minutes) will effectively eliminate most exotic animals. Two other commonly used disinfectants, calcium hypochlorite and iodine, are ineffective against zebra mussels.

3) Adult zebra mussels can live more than a week out of water in moist, shaded areas. Dry pumps, nets and other equipment used in infested waters in the sun for two to four days after cleaning. If adult mussels are present, dry equipment for two weeks.

6. Fish and Wildlife Protection:

a. The Contractor and Contractor's employees shall not feed any animals within Effigy Mounds National Monument.

b. The Contractor shall make all reasonable efforts in accordance with the plans and specifications for the protection of threatened or endangered or candidate species

including their habitat in accordance with Federal, State, Regional, and local laws and regulations.

c. Contractor shall maintain routes of escape from excavated pits and trenches for animals that might fall in. During construction activities, Contractor personnel shall maintain vigilance for animals caught in excavations and take appropriate action to free them.

    1) Excavation pits shall have a ramp or incline at either end to allow for human and wildlife escape.
    2) Each morning prior to commencing work activities, Contractor shall inspect construction site for trapped wildlife in excavation pits and carefully remove. If necessary, contact the Contracting Officer for assistance.

d. Bear Precautions: Bears may be present at any location within the Effigy Mounds National Monument boundaries, including at the project site. The Contractor shall incorporate the following precautions in all activities within the Monument boundary.

    1) All food, toiletries, and scented items (i.e., bug spray) shall be placed in bear boxes at the construction site provided by the Contractor. Bear boxes must remain closed and latched at all times, unless items are being retrieved. No food, toiletries, or scented items shall be stored in vehicles or left out.
    2) All food waste and food-related waste shall be disposed of in accordance with Non-Hazardous Solid Wastes requirements described elsewhere within this section.
    3) All vehicles shall be checked daily to ensure that no items that may attract bears remain inside an unattended vehicle. Items that shall not be left in vehicles include canned food, drinks, soap, cosmetics, toiletries, domestic trash, recyclable food containers, ice chests, grocery bags, and unwashed items used for preparing or eating meals.
    4) All windows and doors in recreational vehicles or trailers used for lodging or office space shall be closed and latched when not occupied.
    5) The Contractor shall walk the job site at the end of each day and check for trash, food, and food-related items remaining at the site and dispose of the items in a bear-proof receptacle.
    6) Proper food storage is important to the welfare of the bear population and is required by law. The Contractor shall receive and all Contractor personnel shall read a brochure entitled, The Bears are not to Blame, provided by NPS staff as a courtesy. Contractor staff shall call the Save-a-Bear hotline (209) 372-0322 to report overflowing trash containers, improperly stored food, or bear sightings.

7. Soil:

    a. The Contractor shall confine all activities to areas defined by the drawings and specifications. Stone, soil, or other materials displaced into uncleared areas shall be removed by the Contractor.
    b. Landscape: Land forms and other landscape features indicated and defined on the drawings to be preserved shall be clearly identified by marking, fencing, or other approved techniques. The Contractor shall restore landscape features damaged or

destroyed during construction operations outside the limits of the approved work area.

c. Topsoil shall be salvaged and placed in a separate location from sub-soils and replaced on top of other soils as the trench is backfilled.

8. Air:

a. Measures shall be taken to assure that dust, debris, materials, trash, etc., do not become air borne and travel off the project site.
b. Equipment operations shall be in accordance with all Federal and State air emission and performance laws and standards.
c. Engines shall not be left running for more than 15 minutes while vehicles or other equipment are standing by.
d. Vehicles or equipment with excessive emissions or discharging black smoke will be removed from operation immediately and may not be used until appropriate maintenance and repairs have corrected the emissions problem.
e. Clearing of vegetation shall be minimized to the greatest extent possible to prevent creation of dust sources.
f. Water or other approved compounds will be used to stabilize disturbed soils. Do not apply water when construction caused dust is not present.
g. When hauling dry materials, truck beds will be securely covered to prevent blowing dust or loss of debris.
h. When hauling wet materials, Contractor shall seal or screen truck beds to prevent soil leaks.
i. Contractor shall not exceed a maximum of 15 mph within construction areas in the Park. Slower speeds shall be maintained if necessary to reduce dust formation.
j. At construction zone access points, Contractor shall prevent paved areas from accumulation of mud, soils and other organic materials

9. Water:

a. The Contractor shall adhere at all times to the conditions of U.S. Army Corps of Engineers Nationwide Permit No. 33, Temporary Construction, Access and Dewatering, with the following conditions as a minimum:

1) All work will be subject to the Standard and Technical Conditions of the Certification of the Iowa Department of Natural Resources, a copy which will be provided to the Contractor.

b. Re-grade and restore disturbed areas to preexisting contours to maintain drainage patterns.

10. Noise:

a. Construction noise shall be minimized through use of best available noise control techniques wherever feasible. Sound levels must be kept to a minimum at all times. Equipment and machinery shall not exceed 85 db when measured at 100 linear feet distance. Contractor shall use sound attenuated compressors and generators that comply with the most recent Iowa Department of Transportation standards.
b. Standard noise abatement measures shall include the following elements:

1) A construction schedule that minimizes impacts to adjacent noise-sensitive activities.
2) Use of hydraulically or electrically powered impact tools when feasible.
3) Location of stationary noise sources as far from sensitive activities as possible.
4) All motorized equipment is to have an exhaust muffler in good working order.
5) Engine braking ("jake" brakes) shall not be used in lodging, camping or residential areas. Engine brakes that are used shall be muffled.
6) Continuous noise abatement is required to prevent disturbance and nuisance to Park visitors and workers and to the occupants of adjacent premises and surrounding areas.
7) If the Contracting Officer determines excessive noise is emanating from the construction site, the Contractor may be required to provide sound barriers to deflect noise transmission from visitor areas or other areas impacted by noise.

11. Night Sky:

a. Minimize night lighting during work. If night lighting is necessary, design lighting to be minimal, directed downward, and shielded.

END OF SECTION 01 35 50

SECTION 01 40 00 – QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for quality assurance and quality control.

B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements. Quality of work shall be responsibility of the Contractor.

1. Specified tests, inspections, and related actions do not limit Contractor's other quality assurance and control procedures that facilitate compliance with Contract Document requirements.

C. See Divisions 2 through 49 Sections for specific test and inspection requirements.

1.2 DEFINITIONS

A. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

B. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate actual products incorporated into the work and completed construction comply with requirements.

C. Mockups: Full-size physical assemblies constructed on-site. Mockups are constructed from selections made under Sample submittals; to demonstrate aesthetic effects and, where indicated, qualities of materials and execution; to review coordination, testing, or operation; to show interface between dissimilar materials; and to demonstrate compliance with specified installation tolerances. Mockups are not Samples. Approved mockups establish the standard by which the Work will be judged, unless otherwise indicated. Mockups may be, with the Contracting Officer’s written approval, installed as part of the work in progress, provided that if the mockup is rejected, the work shall be torn out and a new mockup installed, including repair to any damage done to the surrounding work.

D. Preconstruction Testing: Tests and inspections performed specifically for project before products and materials are incorporated into work to verify performance or compliance with specified criteria.

E. Product Testing: Tests and inspections performed by a Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or a testing agency qualified to conduct product testing, to establish product performance and compliance with industry standards.

F. Source Quality Control Testing: Tests and inspections performed at the source, i.e., plant, mill, factory, or shop.

G. Field Quality Control Testing: Tests and inspections performed on-site for installation of work and for completed work.

H. Testing Agency or Laboratory: Entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

I. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.

1. Using a term such as "carpentry" does not imply certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of corresponding generic name.

1.3 CONFLICTING REQUIREMENTS

A. Reference Standards: If compliance with two or more standards is specified and standards establish different or conflicting requirements for minimum quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer before proceeding.

B. Minimum Quality Levels: Quality level shown or specified shall be minimum provided or performed. Actual installation may comply exactly with minimum quality specified, or it may exceed minimum within reasonable limits. To comply with requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Contracting Officer before proceeding.

1.4 SUBMITTALS

A. Quality Control Plan:

1. After contract award and before Pre-Construction conference, submit a written Contractor Quality Control (CQC) plan.
2. If plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
3. Government reserves the right to require changes in plan during contract period as necessary to obtain the quality specified.
4. No change in the approved plan may be made without written concurrence by Contracting Officer.

B. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in form of a recent report on inspection of testing agency by a recognized authority.

C. Contractor Quality Control (CQC) Daily Reports: Submit showing inspections and tests on first workday following date covered by report. Quality Control Supervisor shall utilize DSC Forms.

1. Review Construction Management Representative (CMR) Daily report if applicable and reconcile any differences prior to posting.

D. Test Reports

1. Test reports shall be completed by person performing test.
2. Submit Daily Test Information Sheets with Quality Control Daily Reports.
3. Submit failing test results and proposed remedial actions within four hours of noted deficiency.
4. Submit three copies of complete test results no later than one calendar day after test was performed.

E. Accessibility Inspection Report:

1. Fill out applicable sections of the Accessibility Inspection Report and attach to Contractor Quality Control Daily Report.
2. Utilize attached Accessibility Inspection form to document compliance with Architectural Barriers Act Accessibility Standards (ABAAS).
3. Inspect at various stages of construction as needed to ensure finished product meets standards.
4. Submit report no later than one calendar day after inspection was performed.

F. Test and Inspections List / Matrix: Develop and submit for review a detailed list / matrix for all tests / inspections to be performed that includes the material to be tested / inspected, location(s), relevant codes / criteria, and required certification(s) / license(s) to perform the test / inspection. Matrix shall include coordination with all the Structural Tests and Special Inspections (STSI), mockups, etc. as noted in the drawings and project specifications.

1. A list of testing and inspection requirements has been attached at the end of this specification Section as Appendix D of this Section. The intent is to provide an overall summary of requirements and not a comprehensive testing and inspection list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list.

2. Testing / Inspections Checklist Format. Contractor shall submit a list of testing and inspections in coordination with Appendix D of this Section, in a checklist format approved by Contracting Officer (CO) for use by the Building Code Officer (BCO).

3. NPS BCO and CO shall be notified of inspection requests [two] 2 weeks prior to inspections. NPS Fire Code Official (FCO) and CO shall be notified of fire suppression inspection request thirty (30) days prior to inspections.

G. Off-Site Inspection Reports: Submit prior to shipment.

H. If Contractor Quality Control plan and Quality Control Daily Reports are not submitted as specified, Contracting Officer may retain payments until such time plan(s) is/are accepted and implemented, or may retain payments for work completed on days with no Quality Control Daily Reports.

I. Permits, Licenses, and Certificates: For National Park Service (NPS) records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of work.

J. A list of mockups has been attached at the end of this specification Section. The intent is to provide an overall summary of mockup requirements but not a comprehensive list. Confirm mockup requirements in the technical specification sections of the project manual.

1.5 QUALITY ASSURANCE

A. General: Qualifications paragraphs in this Article establish minimum qualification levels required; individual Specification Sections specify additional requirements.

B. Contractors Quality Control Staff:

1. Contractor's Quality Control Supervisor shall be assigned no other duties.
2. Contractor's designated Quality Control Supervisor shall be on the project site whenever contract work is in progress.

C. Installer Qualifications: Firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent indicated for Project, whose work has resulted in construction with a record of successful in-service performance.

D. Manufacturer Qualifications: Firm experienced in manufacturing products or systems similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

E. Fabricator Qualifications: Firm experienced in producing products similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

F. Professional Engineer Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and is experienced in providing engineering services of kind indicated (including STSI). Engineering services are defined as those performed for installations of system, assembly, or products similar to those indicated for Project in material, design, and extent.

G. Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for activities indicated.

1. Requirement for specialists shall not supersede building codes and regulations governing Work.

2. Substituted workers, if required, shall meet all required qualification requirements indicated.

H. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with experience and capability to conduct testing and inspecting indicated, according to ASTM E 329; and with additional qualifications specified in individual Sections; and where required by Contract, is acceptable to Contracting Officer.

1. Nationally Recognized Testing Laboratory (NRTL): Nationally recognized testing laboratory according to 29 CFR 1910.7 (Code of Federal Regulations).
2. National Voluntary Laboratory Accreditation Program (NVLAP): Testing agency accredited according to National Institute of Standards and Technology's (NIST) National Voluntary Laboratory Accreditation Program.
3. Measuring devices, laboratory equipment, and instruments shall be calibrated at established intervals against certified standards in accordance with NIST requirements. Measuring and testing devices shall be made available for use by Government for verification tests.

I. Factory-Authorized Service Representative Qualifications: Authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products similar in material, design, and extent to those indicated for Project.

J. Mockups: Before installing portions of work requiring mockups, build mockups for each form of construction and finish required to comply with following requirements, using materials indicated for completed work:

1. Build mockups in location and of size indicated; if not indicated, as directed by Contracting Officer.
2. Notify Contracting Officer [seven] 7 days in advance of dates and times when mockups will be constructed.
3. Demonstrate proposed range of aesthetic effects and workmanship.
4. Obtain Contracting Officer's approval of mockups before starting work, fabrication, or construction.
5. Maintain mockups in an undisturbed condition as a standard for judging the completed work.
6. Demolish and remove mockups when directed, unless otherwise indicated.

1.6 QUALITY CONTROL

A. Contractor is responsible for testing and inspections, including STSI, as identified in attached STSI. Inspect and test work as needed to ensure quality of materials, workmanship, construction, finish, and functional performance are in compliance with applicable specifications, drawings, and those required by the Building Code.

1. Engage qualified testing agency to perform quality-control services.
2. Submit appropriate report for each quality-control service.
3. Testing and inspecting requested by Contractor and not required by Contract Documents are Contractor's responsibility.
4. Contracting Officer may designate test locations.

B. Government will engage a third-party inspector to perform routine inspections as identified in the attached Inspection Checklist. Contractor shall coordinate with, cooperate with, and remedy any issues posed by this inspector and confirmed by the Contracting Officer.

C. Manufacturer's Field Services: Where indicated, engage factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.

D. Re-testing/Re-inspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and re-inspecting, for construction of replaced work that failed to comply with Contract Documents.

E. Testing Agency Responsibilities: Cooperate with NPS and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify Contracting Officer and Contractor promptly of irregularities or deficiencies observed in work during performance of services.
2. Determine location from which test samples will be taken and in which in-situ tests are conducted.
3. Conduct and interpret tests and inspections, State in each report whether tested and inspected work complies with or deviates from requirements.
4. Submit 3 copies of certified written report of each test, inspection, and similar quality-control service through Contractor.
5. Do not release, revoke, alter, or increase Contract Document requirements or approve or accept any portion of Work.

F. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide:

1. Access to Work.
2. Incidental labor and facilities necessary to facilitate tests and inspections.
3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
4. Facilities for storage and field curing of test samples.
5. Delivery of samples to testing agencies.
6. Preliminary design mix proposed for material mixes that require control by testing agency.
7. Security and protection for samples and testing and inspecting equipment at Project site.

G. Coordination: Coordinate sequence of activities to accommodate required quality assurance and control services with minimum delay and to avoid removing and replacing construction to accommodate testing and inspecting.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS

2.1 QUALITY CONTROL PLAN

A. Quality Control Plan shall include:

1. List of personnel responsible for quality control and assigned duties. Include each person's qualifications. Include alternate(s) and qualifications.
2. Copy of letter of direction to Contractor's Quality Control Supervisor(s) outlining assigned duties and authorities designated by principal or owner.
3. Names, qualifications / accreditations, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms from laboratories.
4. Methods of performing, documenting, and enforcing quality control of work including Contractor report forms and acknowledgment of NPS forms.
5. Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.
6. Specific discussion regarding mockups, off-site visits, receiving inspections, manufacturers representation, startup requirements, and other aspects of performance specific to Project.
7. Provisions for substantial completion(s) and final inspection(s) per Contract.

PART 3 - EXECUTION

3.1 OFF-SITE CONTROL

A. Items fabricated or assembled off-site shall be inspected for quality control at place of fabrication.

3.2 ON-SITE CONTROL

A. Notification:

1. Notify Contracting Officer at least 48 hours in advance of preparatory phase meeting.
2. Notify Contracting Officer at least 24 hours in advance of initial and follow-up phases.

B. Preparatory Phase: Perform before beginning each feature of work.

1. Review control submittal requirements with personnel directly responsible for quality assurance and quantity control of the work. As a minimum, Contractor's Quality Control Supervisor and foreman responsible for feature of work shall be in attendance.
2. Review applicable specifications sections and drawings related to feature of work.
3. Ensure copies of referenced standards related to sampling, testing, and execution for feature of work are available on site.
4. Ensure provisions have been made for field control testing.
5. Examine work area to ensure preliminary work has been completed.
6. Verify field dimensions and advise Contracting Officer of discrepancies with contract documents.

7. Ensure necessary equipment and materials are at project site and they comply with approved shop drawings and submittals.
8. Document preparatory phase activities and discussions on Contractor's Quality Control Daily Report.

C. Initial Phase:

1. As soon as work begins, inspect and test representative portion of particular feature of work for quality of workmanship.
2. Review control testing procedures to ensure compliance with contract requirements.
3. Document initial phase activities and discussions on Contractor's Quality Control Daily Report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.

D. Follow-Up Phase: Inspect and test as work progresses to ensure compliance with contract requirements until completion of work.

E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases may be required on same feature of work for following reasons:

1. Quality of on-going work is unacceptable.
2. Changes in quality control staff, on-site production supervision, or work crew.
3. Work on particular feature of work is resumed after substantial period of inactivity.

3.3 DOCUMENTATION

A. Maintain Quality Control Daily Reports, Daily Test Report Information Sheets, and Accessibility Inspection Reports of quality control activities and tests. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > Construction Forms section.)

B. Quality Control Daily Reports shall not be substituted for other written reports required under clauses of contract, such as Disputes, Differing Site Conditions, or Changes.

C. Quality Control Daily Reports shall accurately portray all work and materials. The materials shall match the materials as approved from the Construction Documents (CD) requirements. Combustible/flammable materials shall be documented as well.

3.4 ENFORCEMENT

A. Contractor shall stop work on any item or feature pending satisfactory correction of deficiency noted by quality control staff or Contracting Officer.

3.5 REPAIR AND PROTECTION

A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.

1. Provide materials and comply with installation requirements specified in other Specification Sections. Restore patched areas and extend restoration into adjoining areas with durable seams as invisible as possible.
2. Comply with Contract Document requirements for Section 01 73 29 "Cutting and Patching."

B. Protect construction exposed by or for quality-control service activities.

C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 01 40 00

Statement of Structural Tests and Special Inspections

National Park Service (NPS) - Denver Service Center (DSC) | 8-22-24

Park: Effigy Mounds National Monument (EFMO)

Project Management Information System (PMIS) Number: 318539

Project Name: N170 – Rehabilitate Visitor Center

Structural Engineering Firm: Otak, Inc.

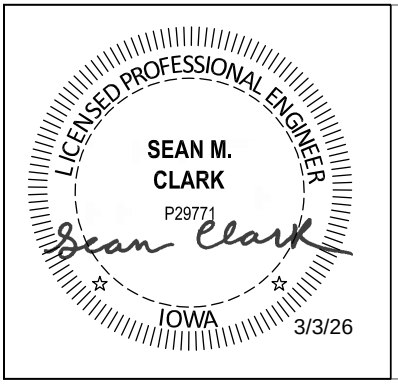
This Statement of Structural Tests and Special Inspections is being submitted as required by Chapter 17 of the **2024 International Building Code** (IBC 2024). It includes the following:

1. Seismic requirements
2. Wind requirements
3. Qualification Requirements for Inspectors and Testing Technicians
4. Listing of Required Structural Tests and Special Inspections

The Construction Contractor's Quality Control Supervisor will provide copies of all special inspection reports and associated documentation to the Contracting Officer (CO). The Construction Contractor will be required to correct all deficiencies discovered in the Special Inspection and Structural Testing program.

Prepared By: Sean Clark
(Type or print name)

Signature: Sean Clark Date: 3/03/2026



CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Sean Clark

Name: Sean Clark

My license renewal date is: 12/31/2026

Pages or sheets covered by this seal: 1-24

Stamped by Professional Engineer (PE) or Structural Engineer (SE)

Seismic and Wind Requirements

Seismic Requirements, IBC 2024 Section 1704.3

Description of seismic-force-resisting system and designated seismic systems subject to special inspections:

Wood outer shell shear walls and plywood diaphragm. CMU (concrete masonry unit) interior shear walls. Rigid concrete floor diaphragm (topping).

Wind Requirements, IBC 2024 Section 1704.3

Description of wind-force-resisting system and designated wind systems subject to special inspections:

Wood outer shell shear walls and plywood diaphragm. CMU (concrete masonry unit) interior shear walls. Rigid concrete floor diaphragm (topping).

Other

I	Inspector Approved by C.O.

Listing of Required Structural Tests and Special Inspections

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Structural Steel (IBC 2024 Section 1705.2; American Institute of Steel Construction: AISC 360-22 Chapter N, AISC 341-22 Chapter J)				
	Prior to Welding (AISC 360-22 Table C-N5.4-1)				
X	1. Welder qualification records and continuity records.	**AWS-CWI**	X		
X	2. Welding procedure specifications (WPS) available.	**AWS-CWI**	X		
X	3. Manufacturer certifications for welding consumables available.	**AWS-CWI**	X		
X	4. Material identification (type/grade).	**AWS-CWI**		X	**20% of welds**
	5. Welder identification system.			X	
	6. Fit-up of groove welds (including joint geometry).				
X	a. Joint preparation.	**AWS-CWI**		X	**20% of welds**
X	b. Dimensions (alignment, root opening, root face, bevel).	**AWS-CWI**		X	**20% of welds**
X	c. Cleanliness (condition of steel surfaces).	**AWS-CWI**		X	**20% of welds**
X	d. Tacking (tack weld quality and location).	**AWS-CWI**		X	**20% of welds**
X	e. Backing type and fit (if applicable).	**AWS-CWI**		X	**20% of welds**
	7. Fit-up of CJP (complete joint penetration) groove welds of HSS (hollow structural sections) T-, Y- and K-joints without backing (including joint geometry).				
X	a. Joint preparation.	**AWS-CWI**		X	**33% of welds**
X	b. Dimensions (alignment, root opening, root face, bevel).	**AWS-CWI**		X	**33% of welds**
X	c. Cleanliness (condition of steel surfaces).	**AWS-CWI**		X	**33% of welds**
X	d. Tacking (tack weld quality and location).	**AWS-CWI**		X	**33% of welds**
X	8. Configuration and finish of access holes.	**AWS-CWI**		X	**33% of welds**
	9. Fit-up of fillet welds.				

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
X	a. Dimensions (alignment, gaps at root).	**AWS-CWI**		X	**20% of welds**
X	b. Cleanliness (condition of steel surfaces).	**AWS-CWI**		X	**20% of welds**
X	c. Tacking (tack weld quality and location).	**AWS-CWI**		X	**20% of welds**
X	10. Check welding equipment.	**AWS-CWI**		X	**20% of welds**
	During Welding (AISC 360-22 Table C-N5.4-2)				
	1. Control and handling of welding consumables				
X	a. Packaging.	**AWS-CWI**		X	**20% of welds**
X	b. Exposure control.	**AWS-CWI**		X	**20% of welds**
X	2. No welding over cracked tack welds.	**AWS-CWI**		X	**20% of welds**
	3. Environmental conditions.				
X	a. Wind speed within limits.	**AWS-CWI**		X	**20% of welds**
X	b. Precipitation and temperature.	**AWS-CWI**		X	**20% of welds**
	4. WPS followed.				
X	a. Settings on welding equipment.	**AWS-CWI**		X	**20% of welds**
X	b. Travel speed.	**AWS-CWI**		X	**20% of welds**
X	c. Selected welding materials.	**AWS-CWI**		X	**20% of welds**
X	d. Shielding gas type/flow rate.	**AWS-CWI**		X	**20% of welds**
X	e. Preheat applied.	**AWS-CWI**		X	**20% of welds**
X	f. Interpass temperature maintained (minimum/maximum).	**AWS-CWI**		X	**20% of welds**
X	g. Proper position (F, V, H, OH).	**AWS-CWI**		X	**20% of welds**
X	h. Intermix of filler metals avoided unless approved (AISC 341-22 Section J7).	**AWS-CWI**		X	**20% of welds**
X	5. Use of qualified welders.	**AWS-CWI**	X		
	6. Welding techniques.				
X	a. Interpass and final cleaning.	**AWS-CWI**		X	**20% of welds**
X	b. Each pass within profile limitations.	**AWS-CWI**		X	**20% of welds**
X	c. Each pass meets quality requirements.	**AWS-CWI**		X	**20% of welds**
	7. Placement and installation of headed stud anchors.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	After Welding (AISC 360-22 Table C-N5.4-3)				
X	1. Welds cleaned.	**AWS-CWI**		X	**20% of welds**
X	2. Size, length and location of welds.	**AWS-CWI**	X		
	3. Welds meet visual acceptance criteria.				
X	a. Crack prohibition.	**AWS-CWI**	X		
X	b. Weld/base-metal fusion.	**AWS-CWI**	X		
X	c. Crater cross section.	**AWS-CWI**	X		
X	d. Weld profiles.	**AWS-CWI**	X		
X	e. Weld size.	**AWS-CWI**	X		
X	f. Undercut.	**AWS-CWI**	X		
X	g. Porosity.	**AWS-CWI**	X		
X	4. Arc strikes.	**AWS-CWI**	X		
X	5. k-area.	**AWS-CWI**	X		
X	6. Weld access holes in rolled heavy shapes and built-up heavy shapes.	**AWS-CWI**	X		
X	7. Backing removed and weld tabs removed (if required).	**AWS-CWI**	X		
X	8. Repair activities.	**AWS-CWI**	X		
X	9. Document acceptance or rejection of welded joint or member.	**AWS-CWI**	X		
X	10. No prohibited welds have been added without the approval of the EOR (Engineer of Record).	**AWS-CWI**	X		
X	11. Placement of reinforcing or contouring fillet welds (if required) (AISC 341-22 Table J7.1).	**AWS-CWI**	X		
	Nondestructive Testing (AISC 360-22 Section N5 and AISC 341-22 Section J7)				
X	1. Risk Category II Structures - Perform Ultrasonic Testing on 10% of CJP groove welds in butt, T- and corner joints subject to transversely applied tension loading, in materials 5/16 inch thick or greater.	**AWS-CWI**		X	**10% of CJP welds**
	2. Risk Category III or IV Structures - Perform Ultrasonic Testing on all CJP groove welds subject to transversely applied tension loading		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	in butt, T- and corner joints, in materials 5/16 inch thick or greater.				
	3. Access Holes - Perform Magnetic Particle Testing or Liquid Penetrant Testing when the flange thickness exceeds 1.5 inches for rolled shapes, or when the web thickness exceeds 1.5 inches for built-up shapes.		X		
	4. Welded Joints Subject to Fatigue - Perform testing per AISC 360-22 Section N5c.		X		
	Nondestructive Testing (AISC 341-22 Section J7)				
	Column Splice and Column to Base Plate PJP Groove Weld (AISC 341-22 Section J7 2b)				
X	1. Perform Ultrasonic Testing on 100% of PJP (partial-joint-penetration) groove welds in column splices and column to base plate welds.	**AWS-CWI**	X		
X	2. CJP Groove weld.	**AWS-CWI**	X		
X	3. Lamellar tearing.	**AWS-CWI**	X		
X	4. Beam cope and access hole.	**AWS-CWI**	X		
X	5. Reduced beam section repair.	**AWS-CWI**	X		
X	6. Weld tab removal.	**AWS-CWI**	X		
	Prior to Bolting (AISC 360-22 Table C-N5.6-1)				
	These inspections are not required for snug-tight joints.				
	1. Manufacturer's certifications available for fastener materials.		X		
	2. Fasteners marked in accordance with ASTM requirements.			X	
	3. Correct fasteners selected for the joint detail (grade, type, bolt length if threads are to be excluded from shear plane).			X	
	4. Correct bolting procedure selected for joint detail.			X	
	5. Connecting elements, including the appropriate faying surface condition and hole preparation, if specified, meet applicable requirements.			X	
	6. Pre-installation verification testing by installation personnel observed and documented for fastener assemblies and methods used.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	7. Protected storage provided for bolts, nuts, washers and other fastener components.			X	
	During Bolting (AISC 360-22 Table C-N5.6-2)				
	These inspections are not required for snug-tight joints.				
	These inspections are not required for pretensioned joints and slip-critical joints, when the installer is using the turn-of-nut method with matchmarking techniques, the direct-tension-indicator method, or the twist-off-type tension control bolt method.				
	1. Fastener assemblies, of suitable condition, placed in all holes and washers (if required) are positioned as required.			X	
	2. Joint brought to the snug-tight condition prior to the pretensioning operation.			X	
	3. Fastener component not turned by the wrench prevented from rotating.			X	
	4. Fasteners are pretensioned in accordance with the RCSC (Resource Council on Structural Connections) Specification, progressing systematically from the most rigid point toward the free edges.			X	
	After Bolting (AISC 360-22 Table C-N5.6-3)				
	Document acceptance or rejection of bolted connections.		X		
	Galvanized Structural Steel Main Members (AISC 360-22 Table C-N5.6-3)				
X	Inspect per AISC 360-22 Section N5.7.	**ICC-SSBSI**		X	**33% of installations**
	Other Inspection Tasks (AISC 360-22 Section N5.8 and AISC 341-22 Section J9)				
X	1. Verify compliance of fabricated steel with the details shown on the approved shop drawings.	**ICC-SSBSI**		X	**33% of installations**
X	2. Verify compliance of the erected steel frame with the details shown on the approved erection drawings, including braces, stiffeners, member locations and joint details.	**ICC-SSBSI**		X	**33% of installations**
	3. Anchor rods and other embedments support structural steel.				

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
X	a. Verify the diameter, grade, type and length of the anchor rod or embedded item.	**ICC-SSBSI**		X	**33% of installations**
X	b. Verify the extent or depth of embedment into the concrete.	**ICC-SSBSI**		X	**33% of installations**
	4. RBS (reduce beam section) requirements, if applicable. (AISC 341-22)				
X	a. Contour and finish.	**AISC 341**		X	**33% of installations**
X	b. Dimensional tolerances.	**AISC 341**		X	**33% of installations**
X	5. Protected zone - no holes or unapproved attachments made by fabricator or erector, as applicable. (AISC 341-22)	**AISC 341**		X	**33% of installations**
	Inspection of H-Piles (AISC 341-22 Section J11)				
	1. Protected zone - no holes or unapproved attachments made by the responsible contractor, as applicable. (AISC 341-22 Table J11.1)			X	
	Structural Stainless Steel (IBC 2024 1705.2.2)				
	1. Special inspections and nondestructive testing in accordance with AISC 370-21.			X	
	Cold-formed Steel Deck (IBC 2024 1705.2.3)				
X	2. Special inspections in accordance with QA/QC-2022 Standard for Quality control (QC) and Quality assurance (QA) for Installation of Steel Deck.	**ICC-SSBSI**		X	**33% of installations**
	Open-Web Steel Joists and Joist Girders (IBC 2024 Table 1705.2.4)				
	1. Installation of open-web steel joists and joist girders.				
	a. End connections - welding or bolted.			X	
	b. Bridging - horizontal or diagonal.				
	1. Standard bridging.			X	
	2. Bridging that differs from the SJI (Steel Joist Institute) specifications listed in Section 2207.1.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Inspection of Composite Structures Prior to Concrete Placement (AISC 341Table J10.1)				
X	1. Material identification of reinforcing steel (Type/Grade).			X	**33% of installations**
X	2. Determination of carbon equivalent for reinforcing steel other than ASTM A706.			X	**33% of installations**
X	3. Proper reinforcing steel size, spacing and orientation.			X	**33% of installations**
X	4. Reinforcing steel has not been rebent in the field.			X	**33% of installations**
X	5. Reinforcing steel has been tied and supported as required.			X	**33% of installations**
X	6. Required reinforcing steel clearances have been provided.			X	**33% of installations**
X	7. Composite member has required size.			X	**33% of installations**
	Inspection of Composite Structures During Concrete Placement (AISC 341-22 Table J10.2)				
X	1. Concrete: Material identification (mix design, compressive strength, maximum large aggregate size, maximum slump).			X	**33% of installations**
X	2. Limits on water added at the truck or pump.			X	**33% of installations**
X	3. Proper placement techniques to limit segregation.			X	**33% of installations**
	Inspection of Composite Structures After Concrete Placement (AISC 341-22 Table J10)				
X	1. Achievement of minimum specified concrete compressive.			X	**33% of installations**
	Cold-formed Steel Trusses Spanning 60-feet or Greater (IBC 2024 Section 1705.2.45)				
	1. Verify temporary installation restraint/bracing installed in accordance with the approved shop drawings.			X	
	2. Verify permanent individual truss member restraint/bracing installed in accordance with the approved shop drawings.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Metal Building Systems (IBC 2024 Section 1705.2.6 - Inspect per Structural Steel requirements above as applicable plus requirements in IBC 2024 Table 1705.2.6)				
	1. Installation of rafter/beam flange braces and column flange braces.			X	
	2. Installation of purlins and girts, including specified lapping.			X	
	3. Purlin and girt restraint/bridging/bracing.			X	
	4. Installation of X-bracing, tightened to remove any sag.			X	
	Concrete Construction (IBC 2024 Section 1705.3)				
X	1. Inspect reinforcing steel, including prestressing tendons, and verify placement.	**ICC-RCSI**		X	**50% of rebar**
	2. Inspection of reinforcing steel welding				
X	a. Verify weldability of reinforcing bars other than ASTM A706.	**PE/SE**	X		
	b. Inspect welding of reinforcement for special moment frames, boundary elements of special structural walls and coupling beams.		X		
X	c. Inspect welded reinforcement splices.	**AWS-CWI**	X		
X	d. Inspect welding of primary tension reinforcement in corbels.	**AWS-CWI**	X		
X	e. Inspect single pass fillet welds, maximum 5/16 inch.	**AWS-CWI**		X	**20% of welds**
X	11. Inspect all other welds.	**AWS-CWI**		X	**20% of welds**
X	3. Inspection of anchors cast in concrete.	**ICC-RCSI**		X	**33% of anchors**
	4. Inspection of anchors post-installed in hardened concrete members.				
X	a. Adhesive anchors installed in horizontally or upwardly inclined orientations to resist sustained tension load.	**ICC-RCSI**	X		
X	b. Mechanical anchors and adhesive anchors not defined in 4a.	**ICC-RCSI**		X	**33% of anchors**
X	5. Verify use of approved design mix.	**ICC-RCSI**		X	**33% of pours**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
X	6. Prior to placement fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete.	**ICC-RCSI**	X		
X	7. Inspect concrete and shotcrete placement for proper application techniques.	**ICC-RCSI**	X		
X	8. Inspect for maintenance of specified curing temperature and techniques.	**ICC-RCSI**		X	**Daily**
	9. Inspection of prestressed concrete:				
	a. Application of prestressing forces.		X		
	b. Grouting of bonded prestressing tendons in the seismic-force-resisting system.		X		
	10. Erection of precast structural members.			X	
	11. For precast concrete diaphragm connections or reinforcement at joints classified as moderate or high deformability elements (MDE or HDE) in structures assigned to Seismic Design Category C, D, E, or F, inspect such connections and reinforcement in the field for:				
	a. Installation of the embedded parts.		X		
	b. Completion of the continuity of reinforcement across joints.		X		
	c. Completion of connections in the field.		X		
	12. Inspect installation tolerances of precast concrete diaphragm connections for compliance with ACI 550.5 (American Concrete Institute).			X	
	13. Verification of in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.			X	
X	14. Inspection formwork for shape, location and dimensions of the concrete member being formed.	**ICC-RCSI**		X	**33% of forms**
	Masonry Construction (IBC 2024 Section 1705.4)				
	1. Inspect masonry construction in accordance with IBC 2024 Section 1705.4 and TMS 402-22 and TMS 602-22 (The Masonry Society).				

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Level 1 Quality Assurance				
	Verification Requirements:				
	1. Prior to construction, verification of compliance of submittals.			X	
	Level 2 Quality Assurance				
	Verification Requirements:				
X	1. Prior to construction, verification of comliance submittals.	**ICC-SMSI**		X	**33% of walls**
X	2. Prior to construction, verification of f'_m and f'_{AAC} except where specifically exempted by the code.	**ICC-SMSI**		X	**33% of walls**
X	3. During construction, verification of slump flow and Visual Stability Index (VSI) when self-consolidating grout is delivered to the project site in accordance with TMS 602-22 Specification Article 1.5 & 1.6.3.	**ICC-SMSI**		X	**33% of walls**
	Inspection Tasks:				
	1. As masonry construction begins, verify the following are in compliance:				
X	a. Proportions of site-prepared mortar.	**ICC-SMSI**		X	**33% of walls**
	b. Grade and size of prestressing tendons and anchorages.			X	
X	c. Grade, type and size or reinforcement, connectors, anchor bolts, and prestressing tendons and anchorages.	**ICC-SMSI**		X	**33% of walls**
	d. Prestressing technique.			X	
	e. Properties of thin-bed mortar for AAC (autoclaved aerated concrete) masonry.		X	X	Continuous inspection is required for the first 5000 square feet of AAC masonry. Periodic inspection is required after the first 5000 square feet of AAC masonry.

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	f. Sample panel construction.			X	
	2. Prior to grouting, verify the following are in compliance:				
X	a. Grout space.	**ICC-SMSI**		X	**33% of walls**
	b. Placement of prestressing tendons and anchorages.			X	
X	c. Placement of reinforcement, connectors, and anchor bolts.	**ICC-SMSI**		X	**33% of anchors**
	d. Proportions of site-prepared grout and prestressing grout for bonded tendons.			X	
	3. Verify the compliance of the following during construction:				
X	a. Materials and procedures with the approved submittals.	**ICC-SMSI**		X	**33% of walls**
X	b. Placement of masonry units and mortar joint construction.	**ICC-SMSI**		X	**33% of walls**
X	c. Size and location of structural members.	**ICC-SMSI**		X	**33% of walls**
X	d. Type, size and location of anchors, including other details of anchorage of masonry to structural members, frames, or other construction.	**ICC-SMSI**		X	**33% of anchors**
	e. Welding of reinforcement.		X		
X	f. Preparation, construction and protection of masonry during cold weather (temperature below 40 degrees Fahrenheit) or hot weather (temperature above 90 degrees Fahrenheit).	**ICC-SMSI**		X	**33% of walls**
	g. Application and measurement of prestressing force.		X		
	h. Placement of grout and prestressing grout for bonded tendons is in compliance.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	i. Placement of AAC masonry units and construction of thin-bed mortar joints.		X	X	Continuous inspection is required for the first 5000 square feet of AAC masonry. Periodic inspection is required after the first 5000 square feet of AAC masonry.
X	4. Observe preparation of grout specimens, mortar specimens and/or prisms.	**ICC-SMSI**		X	**33% of walls**
	Level 3 Quality Assurance				
	Verification Requirements:				
	1. Prior to construction, verification of compliance of submittals.			X	
	2. Prior to construction, verification of f'_m and f'_{AAC} except where specifically exempted by the Code.			X	
	3. During construction, verification of Slump flow and Visual Stability Index (VSI) when self-consolidating grout is delivered to the project site in accordance with TMS 602-22 Specification Article 1.5 & 1.6.3.			X	
	4. During construction, verification of f'_m and f'_{AAC} for every 5000 square feet.			X	
	5. During construction, verification of proportions of materials as delivered to the project site for premixed or preblended mortar, prestressing grout, and grout other than self-consolidating grout.			X	
	Inspection Tasks:				
	1. As masonry construction begins, verify the following are in compliance:				
	a. Proportions of site-prepared mortar.			X	
	b. Grade and size of reinforcement, connectors, anchor bolts, and prestressing tendons and anchorages.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	c. Grade, type and size or reinforcement, connectors, anchor bolts, and prestressing tendons and anchorages.			X	
	d. Prestressing technique.			X	
	e. Properties of thin-bed mortar for AAC (autoclaved aerated concrete) masonry.		X		
	f. Sample panel construction.		X		
	2. Prior to grouting, verify the following are in compliance:				
	a. Grout space		X		
	b. Placement of prestressing tendons and anchorages.			X	
	c. Placement of reinforcement, connectors, and anchor bolts.		X		
	d. Proportions of site-prepared grout and prestressing grout for bonded tendons.			X	
	3. Verify compliance of the following during construction:				
	a. Materials and procedures with the approved submittals.			X	
	b. Placement of masonry units and mortar joint construction.			X	
	c. Size and location of structural members.			X	
	d. Type, size and location of anchors, including other details of anchorage of masonry to structural members, frames, or other construction.		X		
	e. Welding of reinforcement.		X		
	f. Preparation, construction and protection of masonry during cold weather (temperature below 40 degrees Fahrenheit) or hot weather (temperature above 90 degrees Fahrenheit).			X	
	g. Application and measurement of prestressing force.		X		
	h. Placement of grout and prestressing grout for bonded tendons is in compliance.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	i. Placement of AAC masonry units and construction of thin-bed mortar joints.		X		
	4. Observe preparation of grout specimens, mortar specimens and/or prisms.		X		
	Wood Construction (IBC 2024 Section 1705.5)				
X	1. Inspect prefabricated wood structural elements in accordance with Section 1704.2.5.	**I**		X	**33% of elements**
	2. High load diaphragms:				
	a. Verify sheathing grade and thickness.			X	
	b. Verify nominal size of framing members at adjoining panel edges.			X	
	c. Verify nail or staple diameter and length.			X	
	d. Verify number of fastener lines.			X	
	e. Verify spacing between fasteners in each line and at panel edges.			X	
	3. Shearwalls:				
X	a. Verify sheathing grade and thickness.	**I**		X	**33% of elements**
X	b. Verify nominal size of framing members at adjoining panel edges.	**I**		X	**33% of elements**
X	c. Verify nail or staple diameter and length.	**I**		X	**33% of elements**
X	d. Verify number of fastener lines.	**I**		X	**33% of elements**
X	e. Verify spacing between fasteners in each line and at panel edges.	**I**		X	**33% of elements**
X	f. Location and size of holdowns.	**I**		X	**33% of elements**
	4. Verify nailing, bolting, anchoring and fastening of:				
X	a. Drag struts and collectors.	**I**		X	**33% of elements**
X	b. Braces.	**I**		X	**33% of elements**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
X	c. Hold-downs.	**I**		X	**33% of elements**
	5. Metal-plate-connected wood trusses spanning 60 feet or greater:				
	a. Verify temporary installation restraint/bracing installed in accordance with the approved shop drawings.			X	
	b. Verify permanent individual truss member restraint/bracing installed in accordance with the approved shop drawings.			X	
	Mass Timber Construction (IBC 2024 Section 1705.5.3) Mass timber elements in Types IV-A, IV-B and IV-C:				
	1. Inspection of anchorage and connections of mass timber construction to timber deep foundation systems.			X	
	2. Inspect erection of mass timber construction.			X	
	3. Inspection of connections where installation methods are required to meet design loads.			X	
	4. Threaded fasteners:				
	a. Verify use of proper installation equipment.			X	
	b. Verify use of pre-drilled holes where required.			X	
	c. Inspect screws, including diameter, length, head type, spacing, installation angle and depth.			X	
	5. Adhesive anchors installed in horizontal or upwardly inclined orientation to resist sustained tension loads.		X		
	6. Adhesive anchors not defined in preceding cell.			X	
	7. Bolted connections.			X	
	8. Concealed connections.			X	

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Soils (IBC 2024 Section 1705.6)				
X	1. Verify materials below shallow foundations are adequate to achieve the required bearing capacity.	**ICC-SSI or PE/GE**		X	**33% of excavations**
X	2. Verify excavations are extended to proper depth and have reached proper material.	**ICC-SSI or PE/GE**		X	**33% of excavations**
X	3. Perform classification and testing of compacted fill materials.	**ICC-SSI or PE/GE**		X	**33% of excavations**
X	4. During fill placement, verify use of proper materials and procedure in accordance with the provisions of the approved geotechnical report. Verify densities and lift thicknesses during placement and compaction of compacted fill.	**ICC-SSI or PE/GE**	X		
X	5. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly.	**ICC-SSI or PE/GE**		X	**33% of excavations**
	Driven Deep Foundation Elements (IBC 2024 Section 1705.7)				
	1. Verify element materials, sizes and lengths comply with the requirements.		X		
	2. Determine capacities of test elements and conduct additional load tests when required. Refer to project specifications.		X		
	3. Inspect driving operations and maintain complete and accurate records for each element.		X		
	4. Verify element locations and plumbness.		X		
	a. Verify type and size of hammer.		X		
	b. Record number of blows per foot of penetration.		X		
	c. Determine required penetration to achieve specified capacity.		X		
	d. Record pile tip and butt elevations.		X		
	e. Document any damage to any foundation element.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	5. For steel piling, perform additional inspection in accordance with Section 1705.2 and AISC 341-16, Table J10.1.				
	6. For concrete elements and concrete-filled elements, perform additional inspections in accordance with Section 1705.3.				
	7. For specialty elements, perform additional inspections as determined by the registered design professional in responsible charge as required in the project specifications.				
	Cast-in-Place Deep Foundation (IBC 2024 Section 1705.8)				
	1. Inspect drilling operations and maintain complete and accurate records for each element.		X		
	2. Verify element locations and plumbness.		X		
	a. Verify element diameter.		X		
	b. Verify bell diameter (if applicable).		X		
	c. Verify element lengths.		X		
	d. Verify embedment depth into bedrock (if applicable).		X		
	e. Verify adequate end-bearing strata capacity.		X		
	f. Record concrete or grout volumes.		X		
	3. For concrete elements, perform tests and additional inspections in accordance with Section 1705.3.				
	Helical Piles (IBC 2024 Section 1705.10)				
	1. Verify pile locations.		X		
	a. Verify installation equipment used.		X		
	b. Verify pile dimensions.		X		
	c. Verify tip elevations.		X		
	d. Verify final depth.		X		
	e. Verify final installation torque.		X		

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	f. Other data as required by the project specifications.		X		
	Fabricated Items (IBC 2024 Section 1705.11)				
X	1. Inspect fabricated items in accordance with Section 1704.2.5.	**I**	X		
	Wind Resistance (IBC 2024 Section 1705.12)				
X	1. Provide inspections when required by Section 1705.12.	**I**		X	**33% of nailing, bolting, anchoring and other fastening of elements of the main wind force-resisting system, including wood shear walls, wood diaphragms, drag struts, braces and hold-downs. Exception: where the lateral resistance is provided by sheathing of wood structural panels, and the fastener spacing of the sheathing is more than 4 inches (102 mm) on center.**
	Seismic Resistance (IBC 2024 Section 1705.13)				
X	1. Provide inspections when required by Section 1705.13.	**I**		X	**33% of for nailing, bolting, anchoring and other fastening**

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
					of elements of the main seismic force-resisting system, including wood shear walls, wood diaphragms, drag struts, braces and hold-downs. Exception: where the lateral resistance is provided by sheathing of wood structural panels, and the fastener spacing of the sheathing is more than 4 inches (102 mm) on center.
	Testing and Qualification for Seismic Resistance (IBC 2024 Section 1705.14)				
X	1. Test and qualify seismic resistance in accordance with IBC 2024 Section 1705.14 and the project specifications.	**I**		X	**33% of installation of the fire suppression system**
	Sprayed Fire-Resistive Materials (IBC 2024 Section 1705.15)				
	1. Inspect sprayed fire-resistant materials in accordance with IBC 2024 Section 1705.15 and the project specifications.				

Required?	Structural Test or Special Inspection	Required Qualifications	Continuous	Periodic	Frequency of Periodic Test or Inspection
	Intumescent Fire-Resistive Materials (IBC 2024 Section 1705.16)				
	1. Perform inspections in accordance with AWCI 12-B (Association of the Wall and Ceiling Industry) and IBC 2024 Section 1705.16.			X	
	Exterior Insulation and Finish Systems (EIFS) (IBC 2024 Section 1705.17)				
	1. Perform inspections in accordance with project specifications and IBC 2024 Section 1705.17.			X	
	Fire-Resistant Penetrations and Joints (IBC 2024 Section 1705.18)				
	1. Perform inspections in accordance with project specifications and IBC 2024 Section 1705.18.			X	
	Smoke Control (IBC 2024 Section 1705.19)				
	1. Perform testing in accordance with project specifications and IBC 2024 Section 1705.19.				
	Sealing of Mass Timber (IBC 2024 Section 1705.20)				
	1. Perform testing in accordance with project specifications and IBC 2024 Section 1705.20.				

Accessibility Inspection Report - Buildings & Facilities

(Contractor Quality Control (CQC) & Final)

National Park Service (NPS) - Denver Service Center (DSC) | 8-2-24

Park: EFMO

Project Management Information System (PMIS) Number: 224159

Project: Redesign Parking & Replace Curbs, Walks and Pavement

Contract Number:

Report Number:

Date of Inspection:

Final Inspection: ☐Yes ☐No

Notes & Requirements

This report covers the most common accessibility requirements. Expand this form for elements not shown. Inspect for compliance with requirements of the International Building Code (IBC) and Architectural Barriers Act Accessibility Standards (ABAAS), latest available release. Use applicable sections for each inspection. The inspection form is not comprehensive for all IBC or ABAAS requirements or all designs. Refer to IBC Chapter 11 for additional requirements (possibly conflicting, use most restrictive requirements in this case). *(Images shown on this report are screen captures from Access-Board.gov > ABA Accessbility Standards.)*

Measuring Outdoor Improvements

A measuring device shall be capable of measuring to a precision of 1/16 inch. Angular or slope measuring devices shall be capable of measuring to a precision of 0.1 degree. Elevation measuring devices shall be capable of measuring to a precision of 0.01 feet. Measure slope/cross-slope by subtracting elevations and dividing by distance. Measure flatness using a 10-foot metal straightedge with not to exceed 1/4-inch space between hard surface and metal edge. **ABAAS requirements identified refer to post construction measurements which include construction tolerances.**

Upon completion of work, Construction Contractor shall complete a final accessibility inspection. List deficiencies in Section 10 and add to punch list. Correct deficiencies in a timely manner. ABAAS construction non-compliance will be replaced at Construction Contractor's expense unless authorized by Contracting Officer (CO). File this report with CQC Daily Report in official project files.

1. Parking Spaces & Passenger Loading Zones

ABA Accessibility Standards > Chapter 4 Accessible Routes > 406 Curb Ramps & Chapter 5 General Site and Building Elements > 502 Parking Spaces:

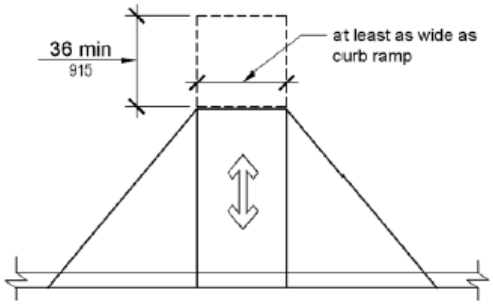


Figure 406.4 Landings at the Top of Curb Ramps

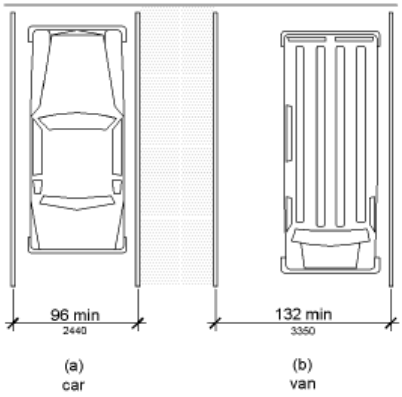


Figure 502.2 Vehicle Parking Spaces

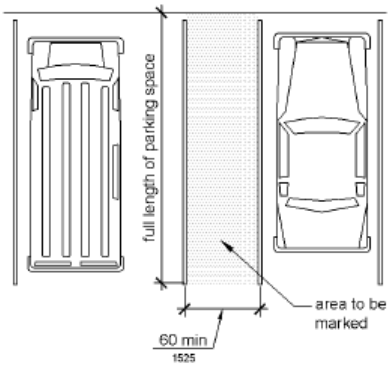


Figure 502.3 Parking Space Access Aisle

Procedure:

Attach copy of grading plan with each accessible parking space and passenger loading zone running and cross slope readings noted. Measure slopes and cross-slopes at 3-foot intervals.

Location/Notes:

_____

_____

_____

A. ☐Yes ☐No ☐Not Applicable

Accessible spaces and access aisles running slope and cross slope is 1:48 (2%) or less. Measure at 3-foot intervals.

_____

B. ☐Yes ☐No ☐Not Applicable
Car spaces are 96 inches wide minimum and van spaces 132 inches wide measured to striping centerline or face of curb.

C. ☐Yes ☐No ☐Not Applicable
Access aisles are 60 inches wide minimum measured to striping centerline and adjoin an accessible route.

D. ☐Yes ☐No ☐Not Applicable
Curb ramp running slope is 1:12 (8.33%) or less and cross slope is 1:48 (2%) or less. Measure at 3-foot intervals.

E. ☐Yes ☐No ☐Not Applicable
Curb ramp has a 36-inch minimum landing length at top and running slope and cross slope are 1:48 (2%) or less.

F. ☐Yes ☐No ☐Not Applicable
Curb ramp flared side slopes are 1:10 (10%) or less.

G. ☐Yes ☐No ☐Not Applicable
Passenger loading zone (drop-off area) running slope and cross slope are 1:48 (2%) or less. Area has flush curb.

Inspection Results:

2. Walking Surfaces & Accessible Routes

ABA Accessibility Standards > *Chapter 3 Building Blocks* > *302 Floor or Ground Surfaces* & *303 Changes in Level*:

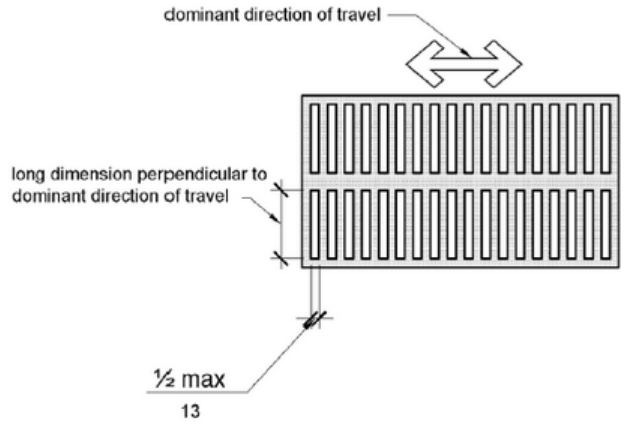


Figure 302.3 Elongated Openings in Floor or Ground Surfaces

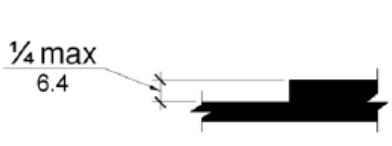


Figure 303.2 Vertical Change in Level

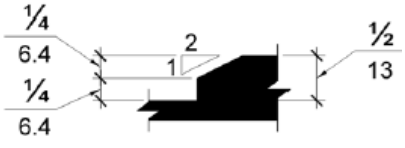


Figure 303.3 Beveled Change in Level

Procedure:

The accessible route is the pedestrian route from accessible parking and passenger loading zones to and between accessible facilities and features. Attach accessible route and/or grading plan with accessible route highlighted and running and cross slope readings noted.

Location/Notes:

A. ☐Yes ☐No ☐Not Applicable
Floor and ground surfaces are stable, firm, and slip resistant.
ABA Accessibility Standards > *Chapter 3 Building Blocks* > *302 Floor or Ground Surfaces*

B. ☐Yes ☐No ☐Not Applicable
Running slope of all walking surfaces on accessible route is 1:20 (5%) or less. Measure at 3-foot intervals.

C. ☐Yes ☐No ☐Not Applicable
Cross slope of walking surfaces is 1:48 (2%) or less. Measure at 3-foot intervals.

D. ☐Yes ☐No ☐Not Applicable
Clear widths of walking surfaces are 36 inches minimum.

E. ☐Yes ☐No ☐Not Applicable
Elongated openings in floor or ground surfaces are 1/2 inch wide or less.

F. ☐Yes ☐No ☐Not Applicable
Curb ramp flared side slopes are 1:10 (10%) or less.

G. ☐Yes ☐No ☐Not Applicable
Elongated openings are perpendicular to direction of travel.

H. ☐Yes ☐No ☐Not Applicable
Accessible route surface changes are 1/2 inch or less with 1/4-inch maximum vertical change.

I. ☐Yes ☐No ☐Not Applicable
Floor drains and grates on accessible route meet all above requirements.

Inspection Results:

3. Ramps

ABA Accessibility Standards > Chapter 4 Accessible Routes > 405 Ramps:

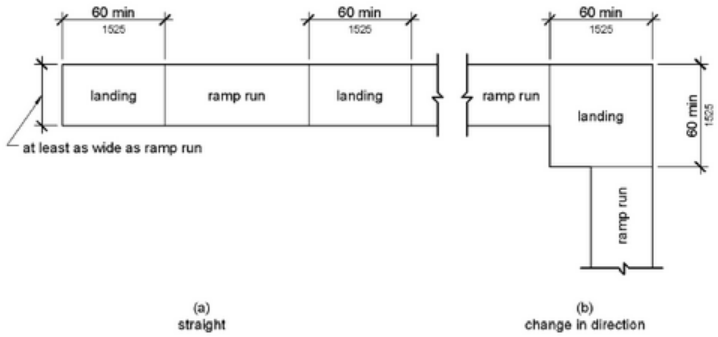


Figure 405.7 Ramp Landings

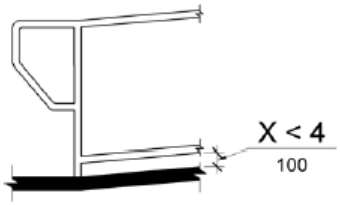


Figure 405.9.2 Curb or Barrier Edge Protection

Procedure:

Ramps are walking surfaces on accessible route steeper than 1:20 (5%) but less than 1:12 (8.33%). Attach grading plan with running and cross slope readings noted.

Location/Notes:

_____

_____

_____

A. ☐Yes ☐No ☐Not Applicable
Running slope is 1:12 (8.33%) or less. Measure at 3-foot intervals.

_____

B. ☐Yes ☐No ☐Not Applicable
Cross slope is 1:48 (2%) or less. Measure at 3-foot intervals.

_____

C. ☐Yes ☐No ☐Not Applicable
Clear width is 36 inches minimum and clear width between handrails is 36 inches minimum.

_____

D. ☐Yes ☐No ☐Not Applicable
Rise for ramp run is 30 inches maximum.

_____

E. ☐Yes ☐No ☐Not Applicable
Ramps have landings at top and bottom of each ramp run.

_____

F. ☐Yes ☐No ☐Not Applicable
Landing clear length is 60 inches minimum and running slope and cross slope of landing is 1:48 (2%) or less.

_____

G. ☐Yes ☐No ☐Not Applicable
Ramps with a rise greater than 6 inches have handrails.

_____

H. ☐Yes ☐No ☐Not Applicable
Ramps and landings have edge protection on both sides preventing passage of 4-inch diameter sphere.

Inspection Results:

4. Handrails

ABA Accessibility Standards > *Chapter 5 General Site and Building Elements* > *505 Handrails*:

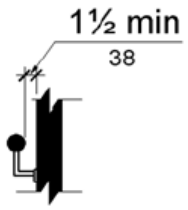


Figure 505.5 Handrail Clearance

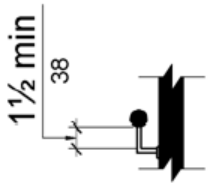


Figure 505.6 Horizontal Projections Below Gripping Surface

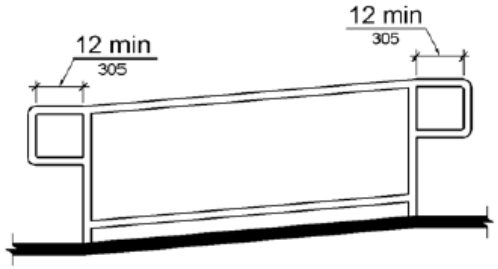


Figure 505.10.1 Top and Bottom Handrail Extension at Ramps

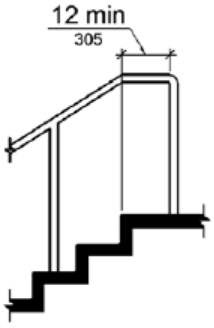


Figure 505.10.2 Top Handrail Extension at Stairs

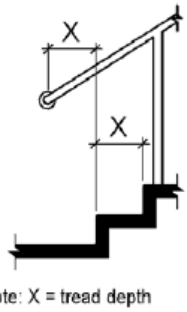


Figure 505.10.3 Bottom Handrail Extension at Stairs

Procedure:

Verify each handrail.

Location/Notes:

A. ☐Yes ☐No ☐Not Applicable
Top of handrail is 34 inches minimum and 38 inches maximum vertically above walking surfaces.

B. ☐Yes ☐No ☐Not Applicable
Handrails extend 12 inches minimum beyond top and bottom of ramp runs.

C. ☐Yes ☐No ☐Not Applicable
Free standing handrails have edge protection preventing passage of 4-inch diameter sphere.

_____

D. ☐Yes ☐No ☐Not Applicable
Handrails extend horizontally 12 inches minimum at top of stairs.

_____

E. ☐Yes ☐No ☐Not Applicable
Handrails extend at slope of stair flight at bottom of stairs minimum one tread depth beyond last riser.

_____

F. ☐Yes ☐No ☐Not Applicable
Handrail clearance is 1-1/2 inches clear minimum to walls and above horizontal attachments.

_____

Inspection Results:

_____

_____

_____

5. Drinking Fountains

ABA Accessibility Standards > Chapter 3 Building Blocks > 307 Protruding Objects:

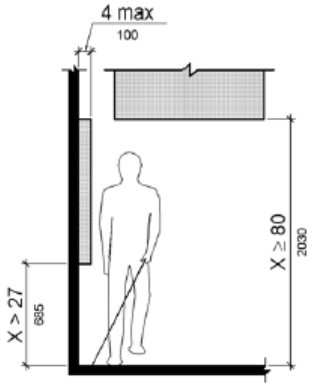


Figure 307.2 Limits of Protruding Objects

Procedure:

Verify drinking fountains do not create a protruding object. (Two spout heights required for each drinking fountain.)

Location/Notes:

_____

_____

_____

A. ☐Yes ☐No ☐Not Applicable
Spout outlet is 36 inches maximum above finish floor or ground for wheelchair accessible spout.

_____

B. ☐Yes ☐No ☐Not Applicable
Spout outlet for standing persons is 38 inches minimum and 43 inches maximum above finish floor.

C. ☐Yes ☐No ☐Not Applicable
Drinking fountain is recessed or has exactly 27 inches maximum space from floor to bottom of fountain. (Fountain must meet ABAAS requirements for knee room and requirements for persons with visual disabilities using a cane.)

Inspection Results:

6. Toilet Compartments

ABA Accessibility Standards > Chapter 6 Plumbing Elements and Facilities > 604 Water Closets and Toilet Compartments, 605 Urinals & 609 Grab Bars:

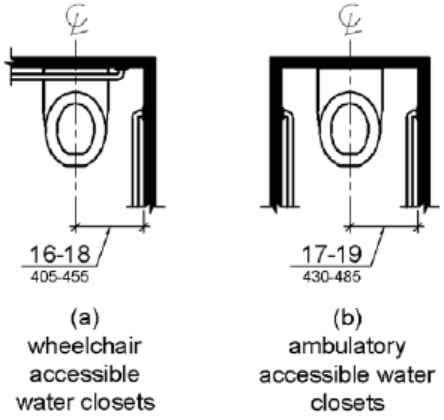


Figure 604.2 Water Closet Location

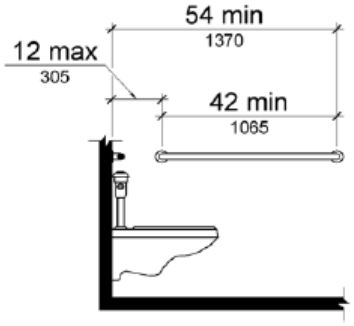


Figure 604.5.1 Side Wall Grab Bar at Water Closets

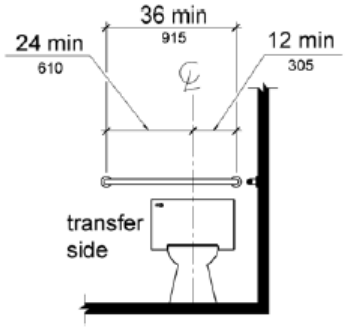


Figure 604.5.2 Rear Wall Grab Bar at Water Closets

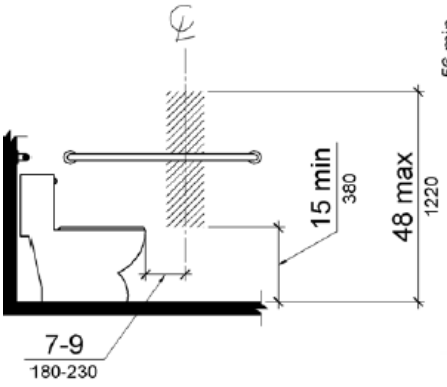


Figure 604.7 Dispenser Outlet Location

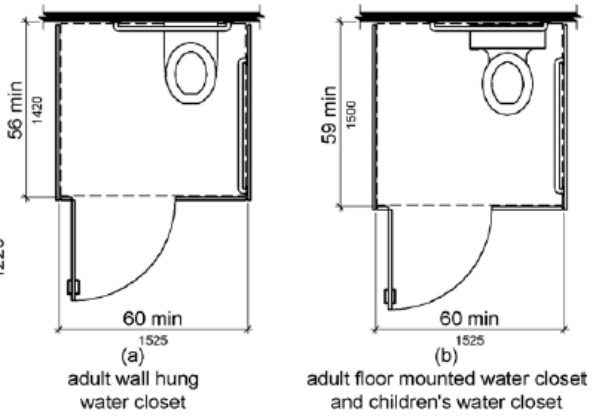


Figure 604.8.1.1 Size of Wheelchair Accessible Toilet Compartment

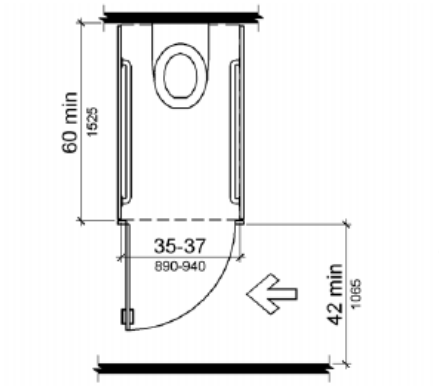


Figure 604.8.2 Ambulatory Accessible Toilet Compartment

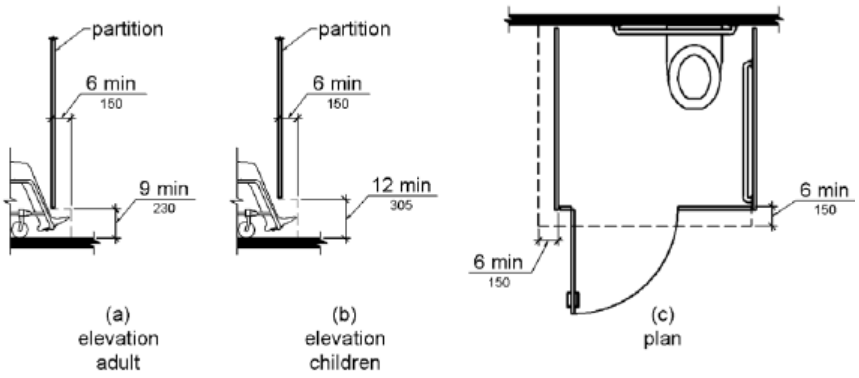
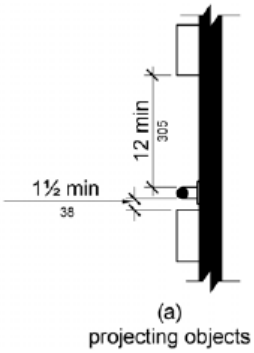


Figure 604.8.1.4 Wheelchair Accessible Toilet Compartment Toe Clearance



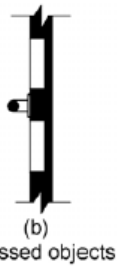


Figure 609.3 Spacing of Grab Bars

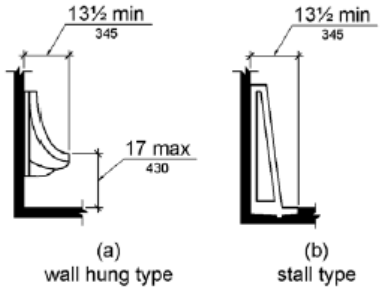


Figure 605.2 Height and Depth of Urinals

Procedure:

For toilet compartments, attach floor plan and note as-built dimensions on diagrams below. (Use additional sheet for each restroom.)

Location/Notes:

_____

_____

_____

A. ☐Yes ☐No ☐Not Applicable
Wheelchair accessible compartment is 60 inches wide minimum and 59 inches deep minimum with wheelchair clear space adjacent to toilet.

_____

B. ☐Yes ☐No ☐Not Applicable
Ambulatory stall is 60 inches deep minimum, width of 35 inches minimum and 37 inches maximum, and toilet is centered in stall.

_____

C. ☐Yes ☐No ☐Not Applicable
Accessible compartment has 32-inch wide door opening minimum with door opening outward, or additional wheelchair clear space for door opening inward.

_____

D. ☐Yes ☐No ☐Not Applicable
Seat height is 17 inches minimum and 19 inches maximum measured to top of seat.

_____

E. ☐Yes ☐No ☐Not Applicable
Urinal rim is 17 inches maximum above finish floor and 13-1/2 inches deep minimum.

F. ☐Yes ☐No ☐Not Applicable
Urinal has a clear floor space of 30 inches wide by 48 inches long minimum for forward approach.

G. ☐Yes ☐No ☐Not Applicable
Side wall grab bar is 42 inches long minimum, located 12 inches maximum from rear wall.

H. ☐Yes ☐No ☐Not Applicable
Rear wall grab bar is 36 inches long minimum, extends from toilet centerline 12 inches minimum one side towards partition or wall and 24 inches minimum other side.

I. ☐Yes ☐No ☐Not Applicable
Grab bar height is 33 inches minimum and 36 inches maximum to top of gripping surface.

J. ☐Yes ☐No ☐Not Applicable
Toilet paper dispensers are 7 inches minimum and 9 inches maximum in front of toilet to centerline of dispenser.

K. ☐Yes ☐No ☐Not Applicable
Toilet paper dispenser outlet is 15 inches minimum and 48 inches maximum above finish floor.

L. ☐Yes ☐No ☐Not Applicable
Centerline of wheelchair stall toilet is 16 inches minimum to 18 inches maximum from side wall.

M. ☐Yes ☐No ☐Not Applicable
Sinks and counters are 34 inches maximum above floor with 27 inches minimum knee space. Drainpipes are insulated or concealed.

N. ☐Yes ☐No ☐Not Applicable
Space between grab bar and wall/partition and projecting objects below is 1-1/2 inches minimum. Space above is 12 inches minimum.

O. ☐Yes ☐No ☐Not Applicable
Wheelchair accessible compartment and ambulatory stall have door pulls on both sides of door near latch.

P. ☐Yes ☐No ☐Not Applicable
Hand dryers, towel and soap dispenser outlets are 15 inches minimum and 48 inches maximum above floor. Bottom edge of reflecting mirror surface above sink or countertops are 40 inches maximum above finished floor.

Inspection Results:

_____

_____

_____

7. Service Counters and Dining and Work Surfaces

ABA Accessibility Standards > Chapter 9 Built-In Elements > 904 Check-Out Aisles and Sales and Service Counters & 902 Dining Surfaces and Work Surfaces:

Procedure:

Service counters and dining and work surfaces are parallel or forward approach. NPS treats service counters as possible employee work areas which are required to be 34 inches maximum.

Location/Notes:

_____

_____

_____

A. ☐Yes ☐No ☐Not Applicable
Parallel Approach. Counter is 36 inches long minimum and 36 inches high maximum.

_____

B. ☐Yes ☐No ☐Not Applicable
Forward Approach. Counter surface is 30 inches long minimum and 36 inches high maximum with knee space under counter. See ABAAS 306 for adequate depth and toe clearance.

_____

C. ☐Yes ☐No ☐Not Applicable
Tops of dining and work surfaces are 28 inches minimum and 34 inches maximum above finished floor with adequate knee and toe clearance.

_____

Inspection Results:

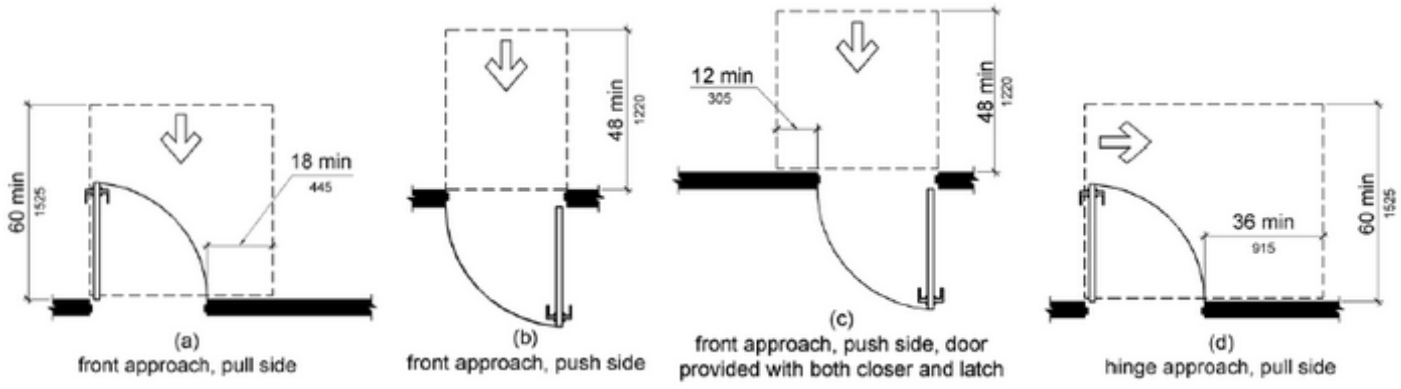
_____

_____

_____

8. Doors

ABA Accessibility Standards > Chapter 4 Accessible Routes > 404 Doors, Doorways, and Gates:



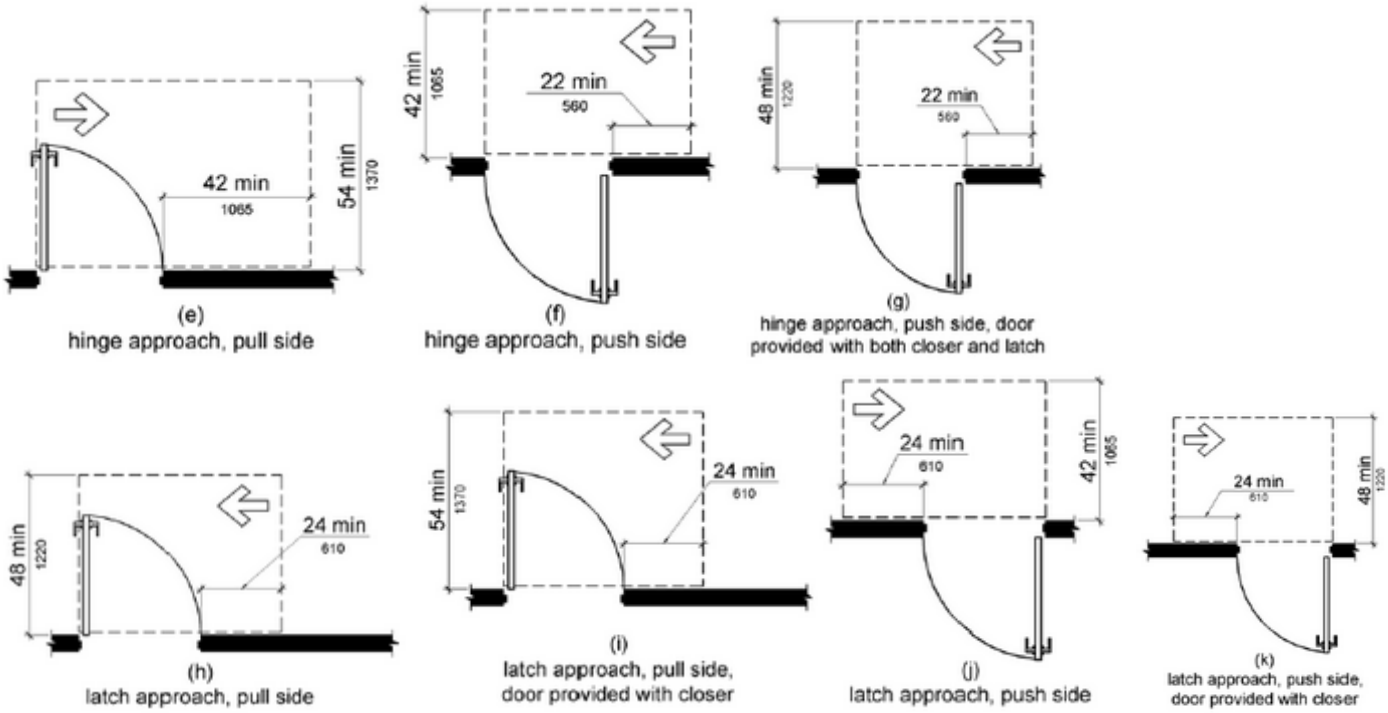


Figure 404.2.4.1 Maneuvering Clearances at Manual Swinging Doors and Gates

Procedure:

Attach accessible route plan with accessible door dimensions and threshold heights.

Location/Notes:

_____

_____

_____

A. ☐Yes ☐No ☐Not Applicable
Door openings on accessible routes provide clear width of 32 inches minimum.

_____

B. ☐Yes ☐No ☐Not Applicable
Door closers move door from open position of 90 degrees to 12 degrees from latch is 5 seconds minimum.

_____

C. ☐Yes ☐No ☐Not Applicable
Spring hinge doors close from open position of 70 degrees to closed position in 1.5 seconds minimum.

_____

D. ☐Yes ☐No ☐Not Applicable
Fire doors have minimum opening force allowable by appropriate administrative authority.

_____

E. ☐Yes ☐No ☐Not Applicable
Interior door and gate hardware located 34 inches minimum and 48 inches maximum above finished floor and operable with hinged doors having opening force of 5 pounds maximum.

_____

F. ☐Yes ☐No ☐Not Applicable
Thresholds at doors on accessible route are beveled 1/2 inch or less with 1/4-inch maximum vertical change.

G. ☐Yes ☐No ☐Not Applicable
Clear floor and wall space provide maneuvering clearances required (all doors on accessible route).

Inspection Results:

9. Signs

ABA Accessibility Standards > Chapter 7 Communication Elements and Features > 703 Signs:

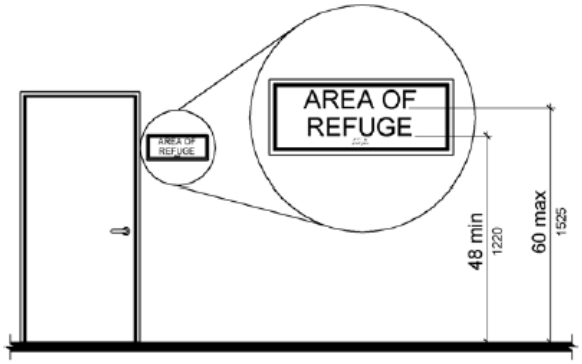


Figure 703.4.1 Height of Tactile Characters Above Finish Floor or Ground

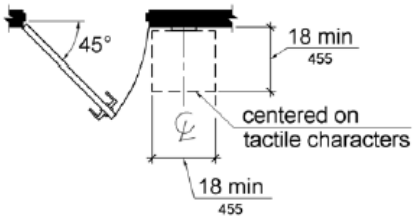


Figure 703.4.2 Location of Tactile Signs at Doors

Location/Notes:

A. ☐Yes ☐No ☐Not Applicable
Parking space signs are 60 inches minimum above finish ground surface measured to bottom of sign.

B. ☐Yes ☐No ☐Not Applicable
Van parking space sign includes designation "van accessible."

C. ☐Yes ☐No ☐Not Applicable
Tactile characters on signs are 48 inches minimum from lowest tactile character and 60 inches maximum above finished floor from baseline of hightest tactile character.

D. ☐Yes ☐No ☐Not Applicable
Visual characters are 40 inches minimum above finished floor.

E. ☐Yes ☐No ☐Not Applicable
Sign is located alongside door at latch side.

_____

F. ☐Yes ☐No ☐Not Applicable
Sign is located with clear floor space of 18 inches by 18 inches minimum.

_____

Inspection Results:

_____

_____

_____

10. Other Accessible Features not Covered by Report

Procedure:
Inspect elevators, lifts, elements for children, etc. Inspect for compliance with ABAAS requirements.

Location/Notes:

_____

_____

_____

Inspection Results:

_____

_____

_____

11. Punch List Items - Summary of Non-Conforming Inspection Results
Add deficiencies. Correct deficiencies in timely manner.

1. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

2. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

3. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

4. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

5. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

6. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

7. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

8. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

9. Deficiency: _____

   Corrective Actions Needed/Taken: _____

   _____

10. Deficiency: _____

    Corrective Actions Needed/Taken: _____

    _____

12. Photographs & Sketches

Attach to report as needed. Identify location of photographs and sketches.

Certification

I certify this inspection report is complete and correct and in compliance with contract documents.

_____

Inspector Name (Print) Inspector Signature Date

Mockup Requirements

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
	04 01 20.63 1.7B	Each Type of Masonry Repair			
	04 01 20.63 1.7B	Brick Replacement			
	04 01 20.64 1.8B	Brick Repointing			
	04 26 13 1.4	Stone Veneer Installation			
	04 26 13 1.4	Brick Replacement/Reinstallation			
	04 43 13 1.4	Adhered Stone Veneer Installation			
	06 40 13 1.6A	Exterior Architectural Woodwork			
Mockups	06 41 00 1.8	Architctural Wood Casework - Lobby Desk			
	06 61 16 1.06B	Solid Surface Fabrications Installation			
	07 18 00 1.5B	Traffic Coating Installation			
	07 42 13.13 1.7	Formed Metal Panel Installation			
	07 46 16 1.8	Aluminum Cladding Installation			
	07 92 00 1.4	Sealant Installations			
	08 41 13 1.7	Aluminum-Framed Entrances and Storefronts Installation			
	08 80 00 1.8	Glazing Installation			
	09 67 23 1.5B	Resinous Epoxy Flooring			
	09 91 13 1.4	Exterior Paint Installation			
	09 91 23 1.4	Interior Paint Installation			
	09 93 00 1.4	Each Type of Stain and Transparent Finish			
	32 13 20 1.4	Site Concrete			

APPENDIX D

Testing / Inspections Requirements

EFMO	Iowa	Effigy Mounds Visitor Center Rehabiliation and Parking Lot Redesign	N170 318539 / 224159

Specification			Test Type	Description	Responsibility	In-Person / Virtual	Completion/ Date	Notes
Number	Name	Paragraph						
01 40 00	Accesibility	1.4.E	Field Quality Assurance	Interior and Exterior	Contractor to coordinate with NPS CO/BCO			Refer to Accessibility Inspection Report.
01 57 23	Temporary Stormwater Pollution Prevention	01	Field Quality Assurance	TESC and SWPPP	Contractor			Contractor shall comply with NPDES requirements per the Iowa DDNR.
01 77 00	Final Inspections	01	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			Contractor shall comply with NPDES requirements per the Iowa DDNR.
02 83 33	Removal and Disposal of Materials Containing Lead	3.2	Laboratory Test	Air Monitoring	Contractor to coordinate with NPS CO/BCO			Submit signed air monitoring test results within three business days.
03 30 00	Concrete		Refer to SSTSI		Contractor			Contractor testing shall include but not limited to, base compaction, footing drains, steel reinforcement, and formwork. Refer to SSTSI. Special inspection required.
04 01 00	Masonry Construction		Refer to SSTSI		Contractor			Refer to SSTSI. Special inspection required.
04 05 19	Post-Installed Anchors	1.4.C	Field & Lab	Anchor Test	Contractor			Refer to SSTSI. Special inspection required.
05 12 00	Structural Steel Framing		Refer to SSTSI		Contractor			Refer to SSTSI. Special inspection required.
05 40 00	Cold Formed Metal Framing		Refer to SSTSI		Contractor			Refer to SSTSI. Special inspection required.
06 10 00 06 16 00	Rough Carpentry		Refer to SSTSI		Contractor			Refer to SSTSI. Special inspection required.
07 14 16	Cold Fluid-Applied Waterproofing	3.7	Field Quality Control	Test and Inspection	Contractor			Engage a qualified testing agency to perform the materials testing. Arrange for waterproofing system manufacturer's technical personnel to inspect condition, prep, application, flashings, protection, and drainage components and to furnish daily reports.
07 18 00	Traffic Coating	3.6	Field Quality Control	Test and Inspection	Contractor			Engage a qualified testing agency to perform the materials testing. Arrange for traffic-coating manufacturer's technical personnel to inspect membrane installation on completion.
07 27 26	Air and Weather Barriers	3.5	Field Quality Control	Quality Assurance	Contractor			Coordinate with ABAA testing agency. Do not cover air barriers until required inspections have been completed.
07 42 13.13	Formed Metal Wall Panels	3.4	Field Quality Control	Quality Assurance	Contractor			Engage a factory-authorized service representative to test and inspect completed metal wall panel installation.
07 42 13.16	Standing-Seam Metal Roof Panels	3.6	Field Quality Control	Quality Assurance	Contractor			Engage a factory-authorized service representative to test and inspect completed metal roof panel installation.
07 62 00	Sheet Metal Flashing and Trim	1.6	Quality Assurance	Surveillance of Work	Contractor to coordinate with NPS CO/BCO			Ascertain compliance with specified requirements
08 11 13	Hollow Metal Doors and Frames	3.3	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			Egress Door Inspections, DSC-CQC Accessibility Inspection Report
08 41 13	Aluminum-Framed Entrances and Storefronts	3.4	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			Water-Spray Test, Water Penetration Test, Egress Door Inspections, DSC-CQC Accessibility Inspection Report

08 14 16	Flush Wood Doors	3.3	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			AWI's Quality Certification Program, Egress Door Inspections, DSC-CQC Accessibility Inspection Report
09 01 90.52	Maintenance Repainting	3.03.A	Field Inspection	Quality Assurance	Contractor to coordinate with NPS CO/BCO			Comply with paint manufacturer's written instructions
10 21 13.19	Plastic Toilet Compartments	3.3	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			DSC-CQC Accessibility Inspection Report
10 22 39	Folding Panel Partitions	3.3	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			NIC Testing
12 32 16	Manufactured Plastic-Laminate-Clad Casework	3.3	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			DSC-CQC Accessibility Inspection Report
14 01 20.71	Elevator Rehabilitation	3.7	Field Quality Control	Quality Assurance	Contractor to coordinate with NPS CO/BCO			
21 05 00	Fire Pump Systems	1.06	Field Inspection	Start Up Reports	Contractor to coordinate with NPS CO/BCO			Veirfy equipment is installed correctly, adjusted and lubricated and operating instructions have been given.
21 13 00	Fire Suppression Testing	3.04	Field Testing	Test Reports	Contractor to coordinate with NPS CO/BCO			Hydrostatic tests for wet systems and air pressure test for dry/preaction systems. Coordinate with fire alarm for inspector's test connection and alarmed isolation valves.
21 22 00	Clean Agent Testing and Commissioning	3.1.E	Field Testing	Test Reports	Contractor to coordinate with NPS CO/BCO			Testing and commissioning of clean agent system to ensure proper installation. Coordinate with fire alarm.
22 10 00	Pipe Testing	3.03	Field Testing	Test Reports	Contractor			Hydrostaic test sanitary waste and vent, cold water and other plumbing piping and submit reports. Air/smoke tests may be used except for plastic piping.
23 05 93	Testing, Adusting and Balancing	1.05	Field Testing	Test Reports	Contractor			Submit reports on AABC, SMACNA and NEBB forms.
26 05 19	Low-Voltage Electrical Power Conductors and Cables	1.3.B	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.5. Submit test reports for review.
26 05 26	Grounding and Bonding for Electrical Systems	1.2.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.4. Submit test reports for review.
26 09 23	Lighting Control Devices	1.4.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.4. Submit test reports for review.
26 09 43	Network Lighting Controls	1.4.D	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.2. Submit test reports for review.
26 24 16	Panelboards	1.4.D	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.4. Submit test reports for review.
26 27 13	Electricity Metering	1.3.B	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.2. Submit test reports for review.
26 28 16	Enclosed Switches and Circuit Breakers	1.4.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.4. Submit test reports for review.
26 41 13	Lightning Protection for Structures	1.4.E	Field Inspection	Field Inspection and Certification	Contractor			See Section 3.3 - system needs to be inspected and certified by LPI or UL.
26 43 13	Transient-Voltage Suppression for Low-Voltage Electrical Power Circuits	1.5.A	Field Quality Control	Test Reports	Contractor			Preform tests and inspections (with assistance from a factory-authorized service representative) per Section 3.2. Submit test reports for review.
27 05 26	Grounding and Bonding for Communication Systems	1.5.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.5. Submit test reports for review.
27 05 36	Cable Trays for Communications Systems	1.4.B	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.5. Submit test reports for review.
27 13 23	Communications Optical Fiber Backbone Cabling	1.5.B	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.6. Submit test reports for review.
27 15 13	Communications Copper Horizontal Cabling	1.5.B	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.9. Submit test reports for review.

27 41 00	Audio Video Systems	1.8.E	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.37. Submit test reports for review.
28 13 00	Access Control Software and Database Management	1.5.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.2. Submit test reports for review.
28 14 00	Access Control System Hardware	1.5.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections per Section 3.2. Submit test reports for review.
28 16 00	Intrusion Detection	1.5.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections (with assistance from a factory-authorized service representative) per Section 3.7. Submit test reports for review.
28 20 00	Video Surveillance	1.5.C	Field Quality Control	Test Reports	Contractor			Preform tests and inspections (with assistance from a factory-authorized service representative) per Section 3.4. Submit test reports for review.
28 16 00	Intrusion Detection	3.70	Field Quality Control	Test Reports	Contractor			System shall be tested to simulate normal use and results are to be documnted under manufactures system engineer.
28 46 00	Fire Alarm and Detection System	3.1R	Field Testing	Standard tests	Contractor			Test and adjust fire alarm system as specified in the spec section.
31 20 00	Earthmoving	3.20 C	Field Quality Control	Subgrade testing	Contractor			Allow testing agency to confirm subgrades and fill comply with requirements
32 13 20	Site Concrete	3.10 B	Concrete Tests	Quality Control	Contractor			Qualified independent testing agency
32 13 20	Site Concrete	3.10 C	Compressive Strength	Quality Control	Contractor			Qualified independent testing agency
32 13 20	Site Concrete	3.11 A	Accessibility Inspection	Quality Control	Contractor to coordinate with NPS CO/BCO			DSC-CQC Accessibility Inspection Report
31 23 16	Topsoil	1.3	Laboratory Test	Soil sample tests	Contractor			Qualified independent testing agency
33 14 16	Site Water Utility Distribution	3.7	Bacterological	Bacterial test	Contractor			
33 14 16	Site Water Utility Distribution	3.8	Hydrostatic	Leakage test	Contractor			
33 46 00	Subdrainage	3.3 A	Field Quality Control	Adequate Slope	Contractor to coordinate with NPS CO/BCO			

SECTION 01 42 00 – REFERENCE STANDARDS

PART 1 - GENERAL

1.1 ENVIRONMENTAL DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114 and as specified herein.

B. Biobased Materials: As defined in the Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) that is composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.

1. Biobased content: Amount of biobased carbon in the material or product as a percentage of weight (mass) of total organic carbon in the material or product.

C. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.

D. Deconstruction: Disassembly of buildings for purpose of recovering materials.

E. DFE (Design for the Environment): A technique that includes elements of resource conservation and pollution prevention as applied in various product sectors. A technique that incorporates approaches which are part of product (or assembly) concept, need and design. Considerations involve material selection, material and energy efficiency, reuse, maintainability and design for disassembly and recyclability. Refer to International Organization for Standardization (ISO) Guide 64 for additional clarification.

F. Environmentally preferable products: Products and services that have a lesser or reduced effect on the environment in comparison to conventional products and services. Refer to EPA’s Final Guidance on Environmentally Preferable Purchasing Program.

G. Non-Renewable Resource: A resource that exists in a fixed amount that cannot be replenished on a human time scale. Non-renewable resources have potential for renewal only by geological, physical, and chemical processes taking place over of millions of years. Examples include iron ore, coal, and oil.

H. Perpetual Resource: A resource that is virtually inexhaustible on a human time scale. Examples include solar energy, tidal energy, and wind energy.

I. Recycled Content Materials: Products that contain pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent Federal Trade Commission (FTC) Guide for Use of Environmental Marketing Claims.

J. Renewable Resource: A resource that is grown, naturally replenished, or cleansed, at a rate which exceeds depletion of the usable supply of that resource. A renewable resource can be exhausted if improperly managed. However, a renewable resource can last indefinitely with proper stewardship. Examples include trees in forests, grasses in grasslands, and fertile soil.

1.2 QUALITY ASSURANCE

A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

B. Publication Dates: Comply with standards in effect as of date of Contract Documents, unless otherwise indicated.

C. Conflicting Requirements: Where compliance with two or more standards is specified, and standards may establish different or conflicting requirements for minimum quantities or quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer (CO) for decision before proceeding.

1.3 INDUSTRY STANDARDS

A. Applicability of Standards: Unless Contract Documents include more stringent requirements, applicable construction industry standards have same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

B. Publication Dates: Comply with standards in effect as of date of Contract Documents unless otherwise indicated.

C. Copies of Standards: Each entity engaged in construction on Project should be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with Contract Documents.

1. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

1.4 ABBREVIATIONS AND ACRONYMS

A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities found in Section 01 42 00 Sources for Reference Publications, Unified Facilities Guide Specifications (UFGS) (accessible via Masters website > Downloads section > click on UFGS Master (WBDG Website).

Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

XX	EXAMPLE Association (The) www.EXAMPLE.org

B. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

DIN	Deutsches Institut fur Normung e.V. www.din.de	49 30 2601-3003
IAPMO	International Association of Plumbing and Mechanical Officials www.iapmo.org	(909) 472-4100
ICC	International Code Council www.iccsafe.org	(888) 422-7233
ICC-ES	ICC Evaluation Service, Inc. icc-es.org	(800) 423-6587 (562) 699-0543

C. Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABA & ABAAS United States Access Board	Architectural Barriers Act (ABA) Architectural Barriers Act Accessibility Standards (ABAAS) www.access-board.gov
CoE	Army Corps of Engineers www.usace.army.mil
CPSC	Consumer Product Safety Commission www.cpsc.gov
DOC	Department of Commerce www.commerce.gov
DOD	Department of Defense www.defense.gov
DOJ	Department of Justice www.justice.gov
DOE	Department of Energy www.energy.gov

EPA	Environmental Protection Agency www.epa.gov
FCC	Federal Communications Commission www.fcc.gov
FDA	Food and Drug Administration www.fda.gov
GSA	General Services Administration www.gsa.gov
LBL	Lawrence Berkeley National Laboratory www.lbl.gov
NCHRP	National Cooperative Highway Research Program (See TRB (Transportation Resource Board))
NIST	National Institute of Standards and Technology www.nist.gov
OSHA	Occupational Safety & Health Administration www.osha.gov
PHS	U.S. Department of Health and Human Services www.hhs.gov
RUS	Rural Utilities Service (See USDA (Department of Agriculture))
SD	State Department www.state.gov
TRB	Transportation Research Board www.nationalacademies.org/trb/transportation-research-board
USDA	Department of Agriculture www.usda.gov
USP	U.S. Pharmacopeia www.usp.org
USPS	Postal Service www.usps.com

D. Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABAAS	Architectural Barriers Act Accessibility Standards www.access-board.gov
CFR	Code of Federal Regulations Available from Government Printing Office www.govinfo.gov/app/collection/cfr
DOD	Department of Defense Military Specifications and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
DSCC	Defense Supply Center Columbus (See FS (Federal Specification))
FED-STD	Federal Standard (See FS (Federal Specification))
FS	Federal Specification Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/ Available from General Services Administration www.gsa.gov Available from National Institute of Building Sciences www.nibs.org
FTMS	Federal Test Method Standard (See FS (Federal Specification))
MIL	(See MILSPEC (Military Specification and Standards))
MIL-STD	(See MILSPEC (Military Specification and Standards))
MILSPEC	Military Specification and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
UFAS	Uniform Federal Accessibility Standards Available from Access Board www.access-board.gov/guidelines-and-standards/buildings-and-sites/about-the-aba-standards/ufas *(UFAS is **only** for housing projects per Fair Housing Act. See also the Fair Housing Act Design Manual, www.huduser.gov/portal/publications/destech/fairhousing)*

1.5 ENVIRONMENTAL REFERENCE STANDARDS

A. American Forest and Paper Association:

1. Sustainable Forestry Initiative

B. American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE):

- ASHRAE 52.2, *Method of Testing General Ventilation Air Cleaning Devices for Removal Efficiency by Particle Size*
- ASHRAE 55, *Thermal Environmental Conditions for Human Occupancy*
- ASHRAE 62.1, *Ventilation for Acceptable Indoor Air Quality*
- ASHRAE 62.2, *Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings*
- ASHRAE/IESNA 90.1, *Energy Standard for Buildings, Except Low-Rise Residential Buildings*
- ASHRAE 90.2, *Energy Efficient Design of Low-Rise Residential Buildings*

C. American Association of State Highway and Transportation Officials (AASHTO):

- M288 Geotextile Specification for Highway Applications
- MP009-06 Standard Specification for Compost for Erosion/Sediment Control (Filter Berms and Filter Socks)
- MP010-03 Standard Specification for Compost for Erosion/Sediment Control (Compost Blankets)

D. American Society for Testing and Materials International (ASTM):

- A478 Standard Specification for Chromium-Nickel Stainless Steel Weaving and Knitting Wire
- A580/A580M Standard Specification for Stainless Steel Wire
- A653/A653M Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
- B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
- C1240 Standard Specification for Silica Fume Used in Cementitious Mixtures
- C128 Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
- C131 Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- C1319 Standard Specification for Concrete Grid Paving Units
- C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings
- C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
- C1371 Standard Test Method for Determination of Emittance of Materials Near Room Temperature Using Portable Emissometers
- C1386 Standard Specification for Precast Autoclaved AERATED Concrete (PAAC) Wall Construction Units
- C1483 Standard Specification for Exterior Solar Radiation Control Coatings on Buildings
- C1549 Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer
- C1601 Standard Test Method for Field Determination of Water Penetration of Masonry Wall Surfaces
- C289 Standard Test Method for Potential Alkali-Silica Reactivity of Aggregates (Chemical Method)

- C311 Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use as a Mineral Admixture in Portland-Cement Concrete
- C33 Standard Specification for Concrete Aggregates
- C593 Standard Specification for Fly Ash and Other Pozzolans for Use With Lime
- C595 Standard Specification for Blended Hydraulic Cements
- C618 Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete
- C67 Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
- C739 Standard Specification for Cellulosic Fiber (Wood-Base) Loose-Fill Thermal Insulation
- C936 Standard Specification for Interlocking Concrete Paver Units
- C989 Standard Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars
- D1435 Standard Practice for Outdoor Weathering of Plastics
- D1557 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 foot pound force per cubic foot (ft-lbf/ft3) (2,700 kilonewton meter per cubic meter (kN-m/m3))
- D1972 Standard Practice for Generic Marking of Plastic Products
- D198 Standard Test Methods of Static Tests of Lumber in Structural Sizes
- D2103 Standard Specification for Polyethylene Film and Sheeting
- D217 Standard Test Methods for Cone Penetration of Lubricating Grease
- D2369 Standard Test Method for Volatile Content of Coatings
- D3273 Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber
- D3786 Standard Test Method for Hydraulic Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
- D3792 Standard Test Method for Water Content of Coatings by Direct Injection Into a Gas Chromatograph
- D3864 Standard Guide for Continual On-Line Monitoring Systems for Water Analysis
- D3960 Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings
- D4017 Standard Test Method for Water in Paints and Paint Materials by Karl Fischer Method
- D4263 Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
- D4444 Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters
- D4491 Standard Test Methods for Water Permeability of Geotextiles by Permittivity
- D4552 Standard Practice for Classifying Hot-Mix Recycling Agents
- D4632 Standard Test Method for Grab Breaking Load and Elongation of Geotextiles
- D4716 Test Method for Determining the (In-plane) Flow Rate per Unit Width and Hydraulic Transmissivity of a Geosynthetic Using a Constant Head
- D4833 Standard Test Method for Index Puncture Resistance of Geotextiles, Geomembranes, and Related Product
- D4840 Standard Guide for Sampling Chain-of-Custody Procedures
- D4887 Standard Test Method for Preparation of Viscosity Blends for Hot Recycled Bituminous Materials
- D5106 Standard Specification for Steel Slag Aggregates for Bituminous Paving Mixtures
- D5116 Standard Guide for Small-Scale Environmental Chamber Determinations of Organic Emissions from Indoor Materials/Products

- D5199 Standard Test Method for Measuring the Nominal Thickness of Geosynthetics
- D5261 Standard Test Method for Measuring Mass per Unit Area of Geotextiles
- D5268 Standard Specification for Topsoil Used for Landscaping Purposes
- D5359 Standard Specification for Glass Cullet Recovered from Waste for Use in Manufacture of Glass Fiber
- D5505 Standard Practice for Classifying Emulsified Recycling Agents
- D5509 Standard Practice for Exposing Plastics to a Simulated Compost Environment
- D5512 Standard Practice for Exposing Plastics to a Simulated Compost Environment Using an Externally Heated Reactor
- D5539 Standard Specification for Seed Starter Mix
- D5957 Standard Guide for Flood Testing Horizontal Waterproofing Installations
- D5603 Standard Classification for Rubber Compounding Materials—Recycled Vulcanizate Particulate Rubber
- D5663 Standard Guide for Validating Recycled Content in Packaging Paper and Paperboard
- D5759 Standard Guide for Characterization of Coal Fly Ash and Clean Coal Combustion Fly Ash for Potential Uses
- D5792 Standard Practice for Generation of Environmental Data Related to Waste Management Activities: Development of Data Quality Objectives
- D5834 Standard Guide for Source Reduction Reuse, Recycling, and Disposal of Solid and Corrugated Fiberboard (Cardboard)
- D5851 Standard Guide for Planning and Implementing a Water Monitoring Program
- D5852 Standard Test Method for Erodibility Determination of Soil in the Field or in the Laboratory by the Jet Index Method
- D6002 Standard Guide for Assessing the Compostability of Environmentally Degradable Plastics
- D6006 Standard Guide for Assessing Biodegradability of Hydraulic Fluid
- D6007 Standard Test Method for Determining Formaldehyde Concentration in Air from Wood Products Using a Small Scale Chamber
- D6046 Standard Classification of Hydraulic Fluids for Environmental Impact
- D6081 Standard Practice for Aquatic Toxicity Testing of Lubricants: Sample Preparation and Results Interpretation
- D6108 Standard Test Method for Compressive Properties of Plastic Lumber and Shapes
- D6109 Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastic Lumber
- D6112 Standard Test Methods for Compressive and Flexural creep and Creep-Rupture of Plastic Lumber and Shapes
- D6117 Standard Test Methods for Mechanical Fasteners In Plastic Lumber and Shapes
- D6155 Standard Specification for Nontraditional Coarse Aggregates for Bituminous Paving Mixtures
- D6245 Standard Guide for Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation
- D6261 Standard Specification for Extruded and Compression Molded Basic Shapes Made from Thermoplastic Polyester (TPES)
- D6262 Standard Specification for Extruded, Compression Molded, and Injection Molded Basic Shapes of Poly(aryl ether ketone) (PAEK)
- D6270 Standard Practice for Use of Scrap Tires in Civil Engineering Applications
- D6329 Standard Guide for Developing Methodology for Evaluating the Ability of Indoor Materials to Support Microbial Growth Using Static Environmental Chambers

- D6330 Standard Practice for Determination of Volatile Organic Compounds (Excluding Formaldehyde) Emissions from Wood-Based Panels Using Small Environmental Chambers Under Defined Test Conditions
- D6345 Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air
- D6400 Standard Specification for Compostable Plastics
- D6435 Standard Test Method for Shear Properties of Plastic Lumber and Plastic Lumber Shapes
- D6629 Standard Guide for Selection of Methods for Estimating Soil Loss by Erosion
- D6662 Standard Specification for Polyolefin-Based Plastic Lumber Decking Boards
- D6712 Standard Specification for Ultra-High-Molecular-Weight Polyethylene (UHMW-PE) Solid Plastic Shapes
- D6886 Standard Test Method for Speciation of the Volatile Organic Compounds (VOCs) in Low VOC Content Waterborne Air-Dry Coatings by Gas Chromatography
- D692 Standard Specification for Coarse Aggregate for Bituminous Paving Mixtures
- D696 Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30°C and 30°C With a Vitreous Silica Dilatometer
- D698 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft3 (600 kN-m/m3))
- D7186 Standard Practice for Quality Assurance Observation of Roof Construction and Repair
- E1021 Standard Test Methods for Measuring Spectral Response of Photovoltaic Cells
- E1038 Standard Test Method for Determining Resistance of Photovoltaic Modules to Hail by Impact with Propelled Ice Balls
- E1039 Standard Test Method for Calibration of Silicon Non-Concentrator Photovoltaic Primary Reference Cells Under Global Irradiation
- E1040 Standard Specification for Physical Characteristics of Nonconcentrator Terrestrial Photovoltaic Reference Cells
- E1105 Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform or Cyclic Static Air Pressure Difference
- E1171 Standard Test Method for Photovoltaic Modules in Cyclic Temperature and Humidity Environments
- E1333 Standard Test Method for Determining Formaldehyde Concentrations in Air and Emission Rates from Wood Products Under Defined Test Conditions Using a Large Chamber
- E1362 Standard Test Method for Calibration of Non-Concentrator Photovoltaic Secondary Reference Cells
- E1433 Standard Guide for Selection of Standards on Environmental Acoustics
- E1462 Standard Test Methods for Insulation Integrity and Ground Path Continuity of Photovoltaic Modules
- E1596 Standard Test Methods for Solar Radiation Weathering of Photovoltaic Modules
- E1597 Standard Test Method for Saltwater Pressure Immersion and Temperature Testing of Photovoltaic Modules for Marine Environments
- E1609 Standard Guide for Development and Implementation of a Pollution Prevention Program
- E1686 Standard Guide for Selection of Environmental Noise Measurements and Criteria
- E1690 Standard Test Method for Determination of Ethanol Extractives in Biomass
- E1721 Standard Test Method for Determination of Acid-Insoluble Residue in Biomass

- E1755 Standard Test Method for Ash in Biomass
- E1758 Standard Test Method for Determination of Carbohydrates in Biomass by High Performance Liquid Chromatography
- E1780 Standard Guide for Measuring Outdoor Sound Received from a Nearby Fixed Source
- E1799 Standard Practice for Visual Inspections of Photovoltaic Modules
- E1802 Standard Test Methods for Wet Insulation Integrity Testing of Photovoltaic Modules
- E1821 Standard Test Method for Determination of Carbohydrates in Biomass by Gas Chromatography
- E1827 Standard Test Methods for Determining Airtightness of Buildings Using an Orifice Blower Door
- E1830 Standard Test Methods for Determining Mechanical Integrity of Photovoltaic Modules
- E1861 Standard Guide for Use of Coal Combustion By-Products in Structural Fills
- E1918 Standard Test Method for Measuring Solar Reflectance of Horizontal and Low-Sloped Surfaces in the Field
- E1971 Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings
- E1980 Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces
- E1991 Standard Guide for Environmental Life Cycle Assessment of Building Materials/Products
- E2047 Standard Test Method for Wet Insulation Integrity Testing of Photovoltaic Arrays
- E2114 Standard Terminology for Sustainability Relative to the Performance of Buildings
- E2128 Standard Guide for Evaluating Water Leakage of Building Walls
- E2129 Standard Practice for Data Collection for Sustainability Assessment of Building Products
- E2397 Standard Practice for Determination of Dead Loads and Live Loads associated with Green Roof Systems
- E2398 Standard Test Method for Water Capture and Media Retention of Geocomposite Drain Layers for Green Roof Systems
- E2399 Standard Test Method for Maximum Media Density for Dead Load Analysis of Green Roof Systems
- E2400 Standard Guide for Selection, Installation, and Maintenance of Plants for Green Roof Systems
- E241 Standard Guide for Limiting Water-Induced Damage to Buildings
- E2432 Standard Guide for General Principles of Sustainability Relative to Buildings
- E408 Standard Test Methods for Total Normal Emittance of Surfaces Using Inspection-Meter Techniques
- E413 Standard Classification for Rating Sound Insulation
- E477 Standard Test Method for Measuring Acoustical and Airflow Performance of Duct Liner Materials and Prefabricated Silencers
- E648 Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source
- E683 Standard Practice for Installation and Service of Solar Space Heating Systems for One- and Two-Family Dwellings
- E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization

- E781 Standard Practice for Evaluating Absorptive Solar Receiver Materials When Exposed to Conditions Simulating Stagnation in Solar Collectors With Cover Plates
- E782 Standard Practice for Exposure of Cover Materials for Solar Collectors to Natural Weathering Under Conditions Simulating Operational Mode
- E823 Standard Practice for Nonoperational Exposure and Inspection of a Solar Collector
- E881 Standard Practice for Exposure of Solar Collector Cover Materials to Natural Weathering Under Conditions Simulating Stagnation Mode
- E90 Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
- E903 Standard Test Method for Solar Absorptance, Reflectance, and Transmittance of Materials Using Integrating Spheres
- E948 Standard Test Method for Electrical Performance of Photovoltaic Cells Using Reference Cells Under Simulated Sunlight
- F1869 Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
- F2034 Standard Specification for Sheet Linoleum Floor Covering
- F2170 Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes

E. Bat Conservation International:

- Bat Approved Bat Houses

F. Carpet and Rug Institute

- Green Label & Green Label Plus Testing Programs, carpet-rug.org/testing/green-label-plus

G. Center for Resource Solutions

- Green-e program

H. Environmental Protection Agency (EPA):

- Comprehensive Procurement Guidelines
- ENERGY STAR
- Environmentally Preferable Purchasing Program Final Guidance
- GreenScapes program
- Heat Island Initiative
- Indoor Air Quality Building Education and Assessment Model (I-BEAM)
- National Environmental Performance Track
- Pollution Prevention (P2)
- Product Stewardship Program
- Significant New Alternatives Policy (SNAP) Program

I. Federal Trade Commission:

- Guide for the Use of Environmental Marketing Claims

J. Forest Stewardship Council:

- Chain-Of-Custody
- Forest Management

K. Green Building Initiative (GBI):

- Green Globes - US

L. Green Seal:

- GC-03 Anti-Corrosive Paints
- GC-12 Occupancy Sensors
- GC-13 Split-Ductless Air-Source Heat Pumps
- GS-05 Compact Fluorescent Lamps
- GS-11 Paints
- GS-13 Windows
- GS-14 Window Films
- GS-31 Electric Chillers
- GS-32 Photovoltaic Modules
- GS-36 Commercial Adhesives
- GS-37 Industrial & Institutional Cleaners

M. International Iron and Steel Institute:

- CO2 Breakthrough Program

N. International Organization of Standardization:

- Guide 64; Guide for Inclusion of Environmental Aspects in Product Standards
- 9660 Information processing -- Volume and file structure of CD-ROM for information interchange
- 14001 Environmental management systems – Specification with guidance for use
- 14004 Environmental Management Systems – General Guidelines on Principles, Systems and Supporting Techniques
- 14020 Environmental labels and declarations – General principles
- 14024 Environmental labels and declarations – Type I environmental labelling - Principles and procedures
- 14040 Environmental management – Life cycle assessment – Principles and framework

O. National Association of Home Builders:

- Advanced Framing Techniques: Optimum Value Engineering

P. National Institute of Building Sciences:

- MOIST program for transfer of heat and moisture
- Whole Building Design Guide

Q. National Institute of Standards and Technology:

- BEES (Building for Environmental and Economic Sustainability) Lifecycle Decision Support Tool

R. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):

- IAQ Guidelines for Occupied Buildings Under Construction

S. Southcoast Air Quality Management District:

- 1168 Adhesive And Sealant Applications

T. US Composting Council:

- Seal of Testing Assurance Program

U. US Department of Agriculture:

- Biobased Products – Definitions and Descriptions

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 42 00

SECTION 01 50 00 – TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 DEFINITIONS

A. Permanent Enclosure: As determined by Contracting Officer (CO), permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and openings are closed with permanent construction or substantial temporary closures.

1.3 USE CHARGES

A. General: Cost or use charges for temporary facilities shall be included in Contract Sum as required.

B. Water Service: Potable water from existing water system is available for use with metering and with payment of use charges. Provide connections and extensions of services as required for construction operations with any additional permit costs.

C. Electric Power Service: Electric power from existing system is available for use with metering and with payment of use charges. Provide connections and extensions of services as required for construction operations with any additional permit costs.

1.4 QUALITY ASSURANCE

A. Electric Service: Comply with National Electrical Contractors Association (NECA), National Electrical Manufacturers Association (NEMA), and Underwriter Laboratories (UL) standards and regulations for temporary electric service. Install service to comply with National Fire Protection Association (NFPA) 70.

B. Environmental Protection: Provide environmental protection as required by agency(ies) with jurisdiction and as indicated in Contract Documents. Coordinate with requirements of the following:

1. Regulatory Requirements
2. Indoor Air Quality (IAQ) Management
3. Noise and Acoustics Management
4. Environmental Management

5. Construction Waste Management

C. Accessible Temporary Egress: Comply with applicable provisions in the United States (U.S.) Architectural & Transportation Barriers Compliance Board's Architectural Barriers Act Accessibility Standard (ABAAS) Accessibility Guidelines.

1.5 PROJECT CONDITIONS

A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before NPS acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Temporary materials may be new or used, but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.

B. Pavement: Comply with Division 32 pavement Sections.

C. Portable Chain-Link Fencing: Minimum 2 inch, 9-gage, galvanized steel, chain-link fabric fencing; minimum 6 feet high with galvanized steel pipe posts; minimum 2-3/8 inch OD line posts and 2-7/8 inch OD corner and pull posts, with 1-5/8 inch OD top and bottom rails. Provide galvanized steel bases for supporting posts.

D. Safety Barrier Fence: Orange plastic fence, minimum height, 4 feet.

E. Barrier Tape: Yellow tape imprinted with "CAUTION: CONSTRUCTION AREA", minimum 2" tall lettering, weather-resistant 3 mil. Low density polyethylene film product.

F. Wood Enclosure Fence: Plywood, 6 feet high, framed with four 2 by 4 inch rails, with preservative-treated wood posts spaced not more than 8 feet apart.

G. Lumber and Plywood: Comply with requirements in Division 06 Section "Miscellaneous Rough Carpentry."

H. Gypsum Board: Minimum 1/2 inch thick by 48 inches wide by maximum available lengths; regular-type panels with tapered edges. Comply with ASTM C 36/C 36M.

I. Insulation: Unfaced mineral-fiber blanket, manufactured from glass, slag wool, or rock wool; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively.

J. Polyethylene Sheet: Reinforced, fire-resistive sheet, 10 mil minimum thickness, with flame-spread rating of 15 or less per ASTM E 84 and passing NFPA 701 Test Method 2.

K. Dust-Control Adhesive-Surface Walk-off Mats: Provide mats minimum 36 by 60 inches.

2.2 TEMPORARY FACILITIES

A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.

B. Contracting Officers Field Office: Field office shall be a separate structure from Contractor's office.

1. Prefabricated, job built, or a mobile unit; excellent condition, structurally sound, non-flammable exterior construction, weather tight, minimum 300 square feet.
2. Operable windows and security screens, adjustable ventilation.
3. Restroom (minimum of lavatory and toilet, with exhaust fan if room is windowless) or accessible Contractor facilities if available.
4. Air conditioner and heater.
5. Interior partition with lockable door to divide office
6. Paneling or freshly painted walls, acoustical tile or painted ceilings, and resilient flooring.
7. Two exterior doors with dead bolts keyed from outside,
8. Minimum 20-square-foot landing and steps at each exterior door.

C. Storage and Fabrication Sheds: Temporary weather tight sheds or other covered facilities for storage of materials subject to weather damage. Number and size of structures shall be subject to Contracting Officer's approval.

D. Toilets: Sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, handicap accessible, sturdy enclosures with privacy locks.

1. Provide separate toilet facilities for men and women.

E. Temporary Visitor Contact Area Restroom Trailer: Sufficiently lighted and ventilated toilet trailer facilities in weatherproof, sight proof, handicap accessible, sturdy enclosures with privacy locks.

1. Utilized for Park visitors at the North Unit temporary access location.

2. Contractor shall provide at the North Unit visitor access location. Contractor shall remove facility at the end of construction.

3. Provide separate toilet facilities for men and women.

4. The Government shall manage daily cleaning and restroom servicing for the restroom facility.

F. Temporary Visitor Contact Station Trailer:

1. Mobile trailer for a Park visitor contact station at the North Unit temporary access location.

2. Contractor shall deliver the trailer to the North Unit visitor access location. Contractor shall remove facility at the end of construction.

3. Mobile trailer shall be approximately 24 feet deep by 64 feet in length.

4. Provide and install the mobile trailer so that it is fully ABA compliant.

G. Truck Wash: A temporary construction vehicle wash system shall be constructed to mitigate the transport of sediment off-site from the project area. Refer to section 01 57 23.

2.3 EQUIPMENT

A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

B. Heating, Ventilation, and Air Conditioning (HVAC) Equipment: Unless Contracting Officer authorizes use of permanent HVAC system, provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.

1. Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.
2. Heating Units: Listed and labeled for type of fuel being consumed, by a testing agency acceptable to agency(ies) with jurisdiction, and marked for intended use.
3. Permanent HVAC System: If Contracting Officer authorizes use of permanent HVAC system for temporary use during construction, provide filter with Minimum Efficiency Reporting Value (MERV) of 8 at each return air grille in system and remove at end of construction.

C. Direct Digital Control (DDC) Internet Connection: Furnish, install and maintain a high-speed connection (Digital Subscriber Line (DSL) or similar) between the project's DDC system and the internet through a Contractor furnished internet service provider. Contractor is responsible for maintenance of this connection and costs associated with internet service provider through warranty period of this contract. Upon completion of warranty period, service shall be transferable to the Government at which time future costs for connection will no longer be the responsibility of Contractor.

D. Contracting Officers Field Office

1. Outlets: Minimum of two, quad outlets with surge protection.
2. Refrigerator: Under counter, 3.2-cubic-foot volume with 0.8-cubic-foot freezer with dedicated power receptacle.
3. Fire Extinguisher: UL listed and FM (Fire Pump Motors) approved, minimum rating of 2-A:10-B:C, dry chemical.
4. First-Aid Kit: General office/light industrial kit.
5. Two desks with five drawers each and two chairs with casters; two drafting tables (minimum 40 inches wide by 5 feet long) and two stools; drawing rack; two 2-drawer and one 4-drawer legal size locking filing cabinets with keys; 8 feet of 12 inch deep shelving; coat rack; two additional guest chairs; two desk lamps; two drafting table lamps; and a maximum/minimum thermometer.
6. Manufactured computer work station, capable of containing CPU (central processing unit), monitor, keyboard, printer; work station chair.
7. Additional tables necessary for FAX machine and copier.
8. Two 5-gallon trash cans and one 30-gallon trash can with lid.
9. Ceiling mounted general lighting fixtures.
10. Contractor must provide internet access for the CO at each project site (2 total locations).

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

1. Locate facilities to limit site disturbance and as directed by the Contracting Officer.

B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

A. General: Install temporary service or connect to existing service.

1. Arrange with utility company, NPS, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services. Acquire necessary permits.

B. Storm Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.

1. Connect temporary sewers to private system indicated as directed by the agency(ies) with jurisdiction.

C. Non-potable water for construction is not available within the park boundaries. Contractor shall furnish non-potable water from a source outside park boundary.

D. Potable water is available on site. Make connections to existing facilities as needed. Facilities must be cleaned and maintained in a condition acceptable to NPS. At Substantial Completion, restore these facilities to condition existing before initial use.

E. Sanitary Facilities: Provide temporary toilets, and wash facilities for use by construction personnel.

1. Place in approved locations secluded from public observation and convenient to work stations. Relocate as work progress requires.
2. Maintain and clean toilet facilities at least weekly.
3. Completely remove sanitary facilities on completion of work.
4. Toilets: Use of existing toilet facilities will be permitted, as long as facilities are cleaned and maintained in a condition acceptable to NPS. At Substantial Completion, restore facilities to condition existing before initial use.

F. Heating and Cooling: Provide temporary heating and cooling required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.

1. Use of permanent heating and cooling system will not be allowed without written authorization from Contracting Officer. When the permanent heating and cooling system

is approved for use as temporary heating and cooling, pay costs until final acceptance. Permanent heating and cooling system shall be sufficiently complete, including controls, to permit safe operation

2. Provide and maintain adequate approved facilities, as required for safety and construction requirements, during the work. Provide ample clearance around stoves, heaters, chimney and vent connections to prevent ignition of combustible material
3. Install and maintain temporary filters when air handing equipment is used for temporary heating and cooling. Install new filters before final acceptance in addition to any extra sets of filters required. Clean coils as determined by Contracting Officer.
4. Warranties for equipment used for temporary heating and cooling shall start on date of Final Acceptance.

G. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient condition required and minimize energy consumption.

H. Electric Power Service: Use of existing electric power service will be permitted, as long as equipment is maintained in a condition acceptable to NPS.

1. When temporary connections are removed, restore existing utility services to original condition.

I. Electric Power Service: Provide electric power service and distribution system of sufficient size, capacity, and power characteristics required for construction operations.

1. Install electric power service overhead unless otherwise indicated.

J. Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.

1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.

K. Telephone Service: No telephone service is available on site for Contractor's use. Make arrangements with Telephone Company and pay costs.

3.3 SUPPORT FACILITIES INSTALLATION

A. General: Comply with the following:

1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 50 feet of building lines. Comply with NFPA 241.
2. Maintain support facilities until near Substantial Completion. Remove structures, equipment, and furnishings, and terminate services after punch list is 100 percent completed or when directed by Contracting Officer. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Contracting Officer.

B. Contracting Officers Field Office:

1. Provide Heat, lights, power, air conditioning, temporary water pressure and sewage holding tanks.
2. Provide office, furnishings, and utility connections no later than 7 days after date of Notice to Proceed. Exact location will be determined by Contracting Officer.
3. Maintain equipment, furnishings, and structures. Provide equipment replacement elements as needed. Provide weekly cleaning services and trash disposal. Maintain and service water and sewer holding tanks as required.

C. Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate for construction operations. Locate temporary roads and paved areas in same location as permanent roads and paved areas. Extend temporary roads and paved areas, within construction limits indicated, as necessary for construction operations.

1. Coordinate elevations of temporary roads and paved areas with permanent roads and paved areas.
2. Prepare subgrade and install subbase and base for temporary roads and paved areas according to Division 31 Section "Earth Moving."
3. Recondition base after temporary use, including removing contaminated material, regrading, proof-rolling, compacting, and testing.
4. Delay installation of final course of permanent hot-mix asphalt pavement until immediately before Substantial Completion. Repair hot-mix asphalt base-course pavement before installation of final course according to Division 32 Section "Asphalt Paving."

D. Traffic Controls: Erect and maintain barricades, lights, danger signals, and warning signs in accordance with Manual on Uniform Traffic Control Devices (MUTCD), Part 6, latest edition.

1. Protect existing site improvements to remain including curbs, pavement, and utilities.
2. Maintain access for fire-fighting equipment and access to fire hydrants.
3. Illuminate barricades and obstructions at night; keep safety lights burning from sunset to sunrise.
4. Adequately barricade and post open cuts in or adjacent to thoroughfares.
5. Protect pedestrian traffic by guardrails or fences.
6. When pedestrian traffic is detoured onto a roadway, provide temporary walkways with protection as required at ends and overhead. For walkways, use lumber running parallel to direction of traffic movement and provide ramps at changes of elevation.
7. Cover pipes, hoses, and power lines crossing sidewalks and walkways with troughs using beveled edge boards.
8. Install Barrier Tape where directed by Contracting Officer. Keep a minimum of two rolls on site.

E. Parking: Use designated areas of existing shown on drawings parking areas for construction personnel.

F. Dewatering Facilities and Drains: Comply with requirements of the agency(ies) with jurisdiction. Maintain Project site, excavations, and construction free of water.

1. Dispose of rainwater in lawful manner that will not result in flooding Project or adjoining properties nor endanger permanent Work or temporary facilities.
2. Remove snow and ice as required to minimize accumulations.

G. Project Identification and Temporary Signs: Provide Project identification and other signs. - Fence, barricade, or otherwise block off the immediate work area to prevent unauthorized entry.

1. Provide temporary, directional signs for construction personnel and visitors.
2. Maintain and touchup signs so they are legible at all times.
3. Erect and maintain sufficient detour signs at road closures and along detour routes.

H. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of agency(ies) with jurisdiction.

I. Existing Elevator Use: Use of existing elevators will be permitted, as long as elevators are cleaned and maintained in a condition acceptable to Contracting Officer. At Substantial Completion, restore elevators to condition existing before initial use, including replacing worn cables, guide shoes, and similar items of limited life.

1. Provide protective coverings, barriers, devices, signs, or other procedures to protect elevator car and entrance doors and frame. If, despite such protection, elevators become damaged, engage elevator Installer to restore damaged work so no evidence remains of correction work. Return items that cannot be refinished in field to shop, make required repairs and refinish entire unit, or provide new units as required.

J. Temporary Stairs: Until permanent stairs are available, provide temporary stairs where ladders are not adequate.

K. Existing Stair Usage: Use of existing stairs will be permitted, as long as stairs are cleaned and maintained in a condition acceptable to Contracting Officer. At Substantial Completion, restore stairs to condition existing before initial use.

1. Provide protective coverings, barriers, devices, signs, or other procedures to protect stairs and to maintain means of egress. If stairs become damaged, restore damaged areas so no evidence remains of correction work.

L. Temporary Use of Permanent Stairs: Cover finished, permanent stairs with protective covering of plywood or similar material so finishes will be undamaged at time of acceptance.

3.4 SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects.

B. Cleaning of Equipment: Contractor shall ensure prior to moving on to Project Area, equipment, is free of soil, seeds, vegetative matter, or other debris that could contain or hold seeds. Ensure equipment has been pressure washed and is free of exotic species. Equipment shall be considered free of soil, seeds, and other debris when visual inspection does not disclose such material. Disassembly of equipment components or specialized inspection tools are not required.

C. Temporary Erosion and Sedimentation Control: Refer to Section 01 57 23 "Temporary Storm Water Pollution Prevention".

D. Tree and Plant Protection: Refer to Section 01 11 00 "Summary of Work".

E. Pest Control: Follow NPS requirements (https://www.nps.gov/orgs/1103/ipm.htm) to minimize attraction and harboring of rodents, roaches, and other pests and perform extermination and control procedures at regular intervals so Project will be free of pests and residues at Substantial Completion. Perform control operations lawfully, using environmentally safe materials. Review the Hazardous Materials Inspection For Effigy Mounds National Monument Visitor's Center for additional protection measure guidance.

F. Site Enclosure Fence: Before construction operations begin, furnish and install chain link fencing to prevent people and animals from easily entering site except by entrance gates.

1. Extent of Fence: As required to enclose entire Project site or portion determined sufficient to accommodate construction operations.
2. Locate vehicular gates to avoid interference with traffic on public thoroughfares.
3. Locate pedestrian entrance gates as required to provide controlled personnel entry.
4. Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide Contracting Officer with two set of keys.

G. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security.

H. Barricades, Warning Signs, and Lights: Comply with requirements of Manual on Uniform Traffic Control Devices (MUTCD), Part 6, latest edition for erecting structurally adequate barricades, including warning signs and lighting.

I. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is not complete, insulate temporary enclosures.

J. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241.

1. Responsible Person: Capable and qualified person shall be placed in charge of fire protection. Responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.
2. Tobacco Use, Smoking, and Vaping: Smoking within buildings or temporary storage sheds is prohibited.
3. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of [NPS] [Agency(ies) with Jurisdiction]. Check with Park; many require "burn permits" for welding.
4. Develop and supervise overall fire-prevention and -protection program for personnel at Project Site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.

5. Provide temporary standpipes and hoses for fire protection. Hang hoses with warning sign stating hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.
6. Hazard Control: Take necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids in existing buildings. Provide adequate ventilation during use of volatile or noxious substances. Ensure cleanup procedures and storage requirements are followed at close of every work session.
7. Spark Arresters: Equip gasoline or diesel powered equipment used during periods of potential fire hazards or in potential forest and grass fire locations with spark arresters approved by United States Department of Agriculture (USDA) Forest Service.

    a. Written determinations of periods and areas of potential fire hazard will be issued by Contracting Officer.

8. Buildings: Furnish a minimum of one extinguisher for each 1,500 square feet of area or major fraction thereof.

    a. Travel distance from any work station to nearest extinguisher shall not exceed 75 ft.

9. Vehicles and Equipment: Provide one extinguisher on each vehicle or piece of equipment.
10. Service and Refueling Areas: Locate areas a minimum of **50** feet from buildings. Shut down equipment before refueling.

3.5 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. Minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Maintenance: Maintain facilities in good operating condition until removal.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on 24-hour basis where required to achieve indicated results and avoid possibility of damage.

C. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.

D. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. NPS reserves right to take possession of Project identification signs.
2. At Substantial Completion, clean and renovate permanent facilities used during construction period.

END OF SECTION 01 50 00

Temporary Road Guide Sign

Sign:
1

Panel Dimensions:
103” x 55”

Corner Radius:
4”

Border:
0.996”

Font:
Clearview Hwy 3-B

Cap Height:
6-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
2

Panel Dimensions:
103” x 47”

Corner Radius:
3.75”

Border:
0.996”

Font:
Clearview Hwy 3-B

Cap Height:
6-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
3

Panel Dimensions:
103” x 47”

Corner Radius:
3.75”

Border:
0.996”

Font:
Clearview Hwy 3-B

Cap Height:
6-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
4

Panel Dimensions:
103” x 47”

Corner Radius:
3.75”

Border:
0.996”

Font:
Clearview Hwy 3-B

Cap Height:
6-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
5

Panel Dimensions:
103” x 55”

Corner Radius:
4”

Border:
0.996”

Font:
Clearview Hwy 3-B

Cap Height:
6-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
6

Panel Dimensions:
124” x 55”

Corner Radius:
4.5”

Border:
0.996”

Font:
Clearview Hwy 3-B

Cap Height:
6-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
7

Panel Dimensions:
67” x 31”

Corner Radius:
2.5”

Border:
0.664”

Font:
Clearview Hwy 3-B

Cap Height:
4-inch

Legend Color:
Black

Background Color:
Orange

Temporary Road Guide Sign

Sign:
8

Panel Dimensions:
67” x 31”

Corner Radius:
2.5”

Border:
0.664”

Font:
Clearview Hwy 3-B

Cap Height:
4-inch

Legend Color:
Black

Background Color:
Orange

SECTION 01 57 19.11 – INDOOR AIR QUALITY MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for Indoor Air Quality (IAQ) management during construction operations.

    a. Control of emissions during construction.
    b. Moisture control during construction.

2. Procedures for testing baseline IAQ. Baseline IAQ requirements, specify maximum indoor pollutant concentrations for acceptance of the facility.

1.2 DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114.

B. Adequate ventilation: Ventilation, including air circulation and air changes, required to cure materials, dissipate humidity, and prevent accumulation of particulates, dust, fumes, vapors, or gases.

C. Hazardous Materials: Any material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have the potential to meet the definition of a Hazardous Waste in accordance with 40 CFR 261. Throughout this specification, hazardous material includes hazardous chemicals.

1. Hazardous materials include pesticides, biocides, and carcinogens as listed by recognized authorities, such as the Environmental Protection Agency (EPA) and the International Agency for Research on Cancer (IARC).

D. Indoor Air Quality (IAQ): Composition and characteristics of air in an enclosed space that affect occupants of that space. Indoor air quality of a space refers to relative quality of air in a building with respect to contaminants and hazards and is determined by the level of indoor air pollution and other characteristics of the air, including impact on thermal comfort such as air temperature, relative humidity and air speed.

E. Interior final finishes: Materials and products exposed to interior occupied spaces; including flooring, wall covering, finish carpentry, and ceilings.

F. Packaged dry products: Materials and products installed in dry form delivered in manufacturer's packaging; including carpets, resilient flooring, ceiling tiles, and insulation.

G. Wet products: Materials and products installed in wet form, including paints, sealants, adhesives, special coatings, and materials which require curing.

1.3 QUALITY ASSURANCE

A. Inspection and Testing Lab Qualifications: Minimum of 5 years of experience in performing types of testing specified herein.

1.4 SUBMITTALS

A. Indoor Air Quality (IAQ) Management Plan: After award and before Pre-construction conference, prepare and submit IAQ Management Plan, including:

1. Procedures for control of emissions during construction.

    a. Identify schedule for application of interior finishes: Identify each interior finish that generates odors, moisture, or vapors or is susceptible to adsorption of odors and vapors. Indicate air handling zone, sequence of application, and curing times.
    b. Identify potential sources of odor and dust.
    c. Identify construction activities likely to produce odor or dust.
    d. Identify areas of project potentially affected, especially occupied areas.
    e. Evaluate potential problems by severity and describe methods of control.
    f. Describe construction ventilation to be provided, including type and duration of ventilation, use of permanent Heating HVAC systems, types of filters and schedule for replacement of filters.
    g. Describe cleaning and dust control procedures.
    h. Describe coordination with commissioning procedures.

2. Procedures for moisture control during construction.

    a. Identify porous materials and absorptive materials.
    b. Identify schedule for inspection of stored and installed porous and absorptive materials.

3. Revise and resubmit Plan as required by Contracting Officer (CO).

    a. Approval of Contractor’s Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations.

B. Product Data:

1. Submit product data for filtration media used during construction and during operation. Include Minimum Efficiency Reporting Value (MERV).
2. Material Safety Data Sheets (MSDS): Submit MSDSs for inclusion in Operation and Maintenance Manual for:

    a. Adhesives

b. Floor and wall patching/leveling materials
c. Caulking and sealants
d. Insulating materials
e. Fireproofing and firestopping
f. Carpet
g. Paint
h. Clear finish for wood surfaces
i. Lubricants
j. Cleaning products

C. Inspection and Test Reports:

1. Moisture control inspections
2. Moisture content testing
3. Moisture penetration testing
4. Microbial Growth testing

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 IAQ MANAGEMENT - EMISSIONS CONTROL

A. During construction operations, follow the recommendations in SMACNA IAQ Guidelines for Occupied Buildings under Construction.

B. HVAC Protection:

1. Seal return registers during construction operations.
2. Provide temporary exhaust during construction operations.
3. To greatest extent possible, isolate and/or shut down return side of HVAC system during construction. When ventilation system must be operational during construction activities, provide temporary filters at air inlets (returns) and at locations for filters prescribed in the design.
4. Contractor shall bear cost of cleaning required due to failure to protect ducts and equipment from construction dust.

C. Source Control: Provide low and zero VOC materials as specified.

D. Pathway Interruption: Isolate areas of work to prevent contamination of clean or occupied spaces. Provide pressure differentials and/or physical barriers to protect clean or occupied spaces.

E. Housekeeping: During construction, maintain project and building products and systems to prevent contamination of building spaces.

F. Temporary Ventilation: For materials/products that generally require ventilation for off gassing, provide an ACH (air changes per hour) of 1.5 or more and as follows:

1. Provide minimum 48-hour pre-ventilation of packaged dry products prior to installation. Remove from packaging and ventilate in a secure, dry, well-ventilated space free from strong contaminant sources and residues. Provide a temperature range of 60 degrees Fahrenheit minimum to 90-degree Fahrenheit maximum continuously during ventilation period. Do not ventilate within limits of Work unless otherwise approved by Contracting Officer.
2. Provide adequate ventilation during and after installation of interior wet products and interior final finishes.
3. Provide filtration media with a Minimum Efficiency Reporting Value (MERV) of 8 as determined by ASHRAE 52.2 during construction and a MERV as described in the construction documents during NPS occupancy. Coordinate with work of Division 23 (15), Heating Ventilating and Air Conditioning (HVAC).

G. Scheduling: Schedule construction operations involving wet products prior to packaged dry products to the greatest extent possible.

H. Flush-Out: After construction ends, prior to occupancy and with all interior finishes installed, perform a building flush-out by supplying a total air volume of 14,000 cubic feet. of outdoor air per square feet of floor area while maintaining an internal temperature of at least 60 degrees Fahrenheit and relative humidity no higher than 60%.

1. Obtain Contracting Officers concurrence that construction is complete enough before beginning flush-out.
2. If additional construction involving materials that produce particulates or any of specified contaminants is conducted during or after flush-out, then flush-out process must be restarted.
3. Install new HVAC filtration media in locations identified to have permanent filtration in contract documents after completion of flush-out and before occupancy or further testing.

3.2 IAQ MANAGEMENT - MOISTURE CONTROL

A. Housekeeping:

1. Keep materials dry. Protect stored on-site and installed absorptive materials from moisture damage.
2. Verify installed materials and products are dry prior to sealing and weatherproofing building envelope.
3. Store interior absorptive materials only after building envelope is sealed and weatherproofed.

B. Inspections: Document and report results of inspections; state whether or not inspections indicate satisfactory conditions.

1. Examine materials for dampness as they arrive. If acceptable to Contracting Officer, dry damp materials completely prior to installation; otherwise, reject materials that arrive damp.
2. Examine materials for mold as they arrive and reject materials that arrive contaminated with mold.

3. Inspect stored and installed absorptive materials regularly for dampness and mold growth. Inspect weekly and after each rain event.

    a. If stored or installed absorptive materials become wet, notify Contracting Officer. Inspect for damage. If acceptable to the Contracting Officer, dry completely prior to closing in assemblies; otherwise, remove (in accordance with the Waste Management Plan) and replace with new materials.

4. Basement: Monitor basement and crawlspace humidity and dehumidify when relative humidity is greater than 70 percent for more than 2 weeks or at first sign of mold growth.
5. Site drainage: Verify final grades of site work and landscaping drain surface water and ground water away from building.
6. Weatherproofing: Inspect moisture control materials as they are being installed. Include:

    a. Air barrier: Verify air barrier is installed without punctures and/or other damage. Verify air barrier is sealed completely.
    b. Flashing: Verify correct shingling of flashing for roof, walls, windows, doors, and other penetrations.
    c. Vapor Barrier: Verify vapor barrier is installed in accordance with Contract documents.
    d. Insulation layer: Verify insulation is installed without voids.
    e. Roofing: In accordance with ASTM D7186 Standard Practice for Quality Assurance Observation of Roof Construction and Repair.

7. Plumbing: Verify satisfactory pressure test of pipes and drains is performed before closing in and insulating lines.
8. HVAC: Inspect HVAC system as specified in Section on Commissioning and, inspect HVAC to verify:

    a. condensate pans are sloped and plumbed correctly;
    b. access panels are installed to allow for inspection and cleaning of coils and ductwork downstream of coils;
    c. ductwork and return plenums are air sealed;
    d. duct insulation is installed and sealed; and
    e. chilled water line and refrigerant line insulation are installed and sealed.]

C. Schedule:

1. Schedule work such that absorptive materials, such as porous insulations, paper-faced gypsum board, ceiling tile, and finish flooring, are not installed until they can be protected from rain and construction-related water.
2. Weather-proof as quickly as possible. Schedule installation of moisture-control materials, including but not limited to air barriers, flashing, exterior sealants and roofing, at earliest possible time.

D. Testing for Moisture Content: Test moisture content of porous materials and absorptive materials to ensure they are dry before sealing them into an assembly. Document and report results of testing. Where tests are not satisfactory, dry materials and retest. If satisfactory results cannot be obtained with retest, remove and replace with new materials.

1. Concrete: [Moisture test prior to finish flooring application as specified in Division 9 (9).] [Moisture test as per one or more of the following; unless otherwise indicated, acceptable upper limits for concrete are < 4% top inch; < 85% headspace relative humidity (RH); <3 pounds per 1000 square feet per day:

    a. ASTM D4263 Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
    b. ASTM F1869 Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
    c. ASTM F2170 Test Method for Determining Relative Humidity in Concrete Floor Slabs Using In Situ Probes]

2. Wood: Moisture test as per ASTM D4444 - Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters; unless otherwise indicated acceptable upper limits for wood products are less than 19% at center of piece; less than 1% at surface.

E. Testing for Moisture Penetration:

1. Windows: Test as per ASTM E1105 Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform or Cyclic Static Air Pressure Difference; unless otherwise indicated, acceptable upper limits are no leakage for 15 minutes.
2. Horizontal Waterproofing (not roofing): Test as per ASTM D5957 Standard Guide for Flood Testing Horizontal Waterproofing Installations; acceptable upper limits are no leakage for 15 minutes.
3. Masonry: Test as per ASTM C1601 Standard Test Method for Field Determination of Water Penetration of Masonry Wall Surfaces; acceptable upper limits are no leakage for 4-hours.
4. Exterior Walls:

    a. Air tightness of the enclosure test: ASTM E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization or ASTM E1827 Standard Test Methods for Determining Air tightness of Buildings Using an Orifice Blower Door; acceptable upper limits are 0.25 CFM/sf or less at 50 Pascal's.
    b. Water Leakage: Review as per ASTM E2128 Standard Guide for Evaluating Water Leakage of Building Walls.

END OF SECTION 01 57 19.11

EFMO – N170 318539 / 224159

SECTION 01 57 19.12 – NOISE AND ACCOUSTICS MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for noise and acoustics management during renovation operations.

1.2 DEFINITIONS

A. Ambient noise level: The total noise associated with a given environment, being usually a composite of normal or existing sounds from all sources near and far, excluding the noise source at issue.

B. Daytime: The hours from 7 A.M. to 9 P.M. on weekdays and 9 A.M. to 9 P.M. on weekends and holidays.

C. Nighttime: All non-daytime hours.

D. Property line: The real or imaginary line along the ground surface and its vertical extension, which separates real property owned or controlled by one person from contiguous real property owned or controlled by another person or from any public right-of-way or from any public space.

E. Receiving noise area: Any real property where people live or work and where noise is heard, excluding the project or source area.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 NOISE MANGEMENT

A. Noise Control: Perform renovation operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer CO).

B. Repetitive and/or intermittent, high-level noise: Permitted only during Daytime.

1. Do not exceed the following dB(A) limitations at 50 feet:

Sound Level in dB(A)	Time Duration of Impact Noise
70	More than 12 minutes in any hour
80	More than 3 minutes in any hour

2. Maximum permissible construction equipment noise levels at 50 feet:

EARTHMOVING	dB(A)	MATERIALS HANDLING	dB(A)
Front Loaders	75	Concrete Mixers	75
Backhoes	75	Concrete Pumps	75
Dozers	75	Cranes	75
Tractors	75	Derricks Impact	75
Scrapers	80	Pile Drivers	95
Graders	75	Jack Hammers	75
Trucks	75	Rock Drills	80
Pavers, Stationary	80	Pneumatic Tools	80
Pumps	75	Saws	75
Generators	75	Vibrators	75
Compressors	75		

C. Ambient Noise:

1. Maximum noise levels (dB (decibel)) for receiving noise area at property line shall be as follows:

   a. Residential receiving area
      Daytime: 65 dB
      Nighttime: 45 dB
   b. Commercial/Industrial receiving area
      Daytime: 67 dB
      Nighttime: 65 dB
   c. In the event the existing local ambient noise level exceeds the maximum allowable receiving noise level (dB), the receiving noise level maximum for construction operations shall be adjusted as follows:
   d. Residential receiving area: Maximum 3 additional dB above the local ambient as measured at property line.
   e. Commercial/Industrial receiving area: Maximum 5 additional dB above the local ambient as measured at the property line.

3.2 FIELD QUALITY CONTROL

A. Assess potential effects of construction noise on adjacent neighbors in accordance with ASTM E1686 and as follows:

1. Ambient noise measurement: Measure at property line at a height of at least four (4) feet above the immediate surrounding surface. Average the ambient noise level over a period of at least 15 minutes.
2. Ambient noise measurement at urban sites: Conduct during morning peak traffic hour between 7 A.M. and 9 A.M. and afternoon peak traffic hour between 4 P.M. and 6 P.M. In

addition, conduct a 24-hour measurement at the proposed project site to document the noise pattern throughout the day. Adjust and weight for seasonal and climatic variations.

B. Monitor noise produced from construction operations in accordance with ASTM E1780.

END OF SECTION 01 57 19.12

EFMO – N170 318539 / 224159

SECTION 01 57 23 – TEMPORARY STORM WATER POLLUTION PREVENTION

PART 1 - GENERAL

1.1 SUMMARY

A. Federal Regulations for controlling discharges of pollutants (including chemicals, erodible material, and trash) from municipal separate storm sewer systems, construction sites, and industrial activities, were brought under the National Pollution Discharge Elimination System (NPDES) permit process by amendments to the Clean Water Act (CWA), and promulgation of federal stormwater regulations issued by the United States Environmental Protection Agency (USEPA). The USEPA uses amount of ground disturbance as a measure of a project potential to generate pollution from erosion. NPDES Phase I regulates discharges from construction sites that disturb 5 acres or more. NPDES Phase II regulations expand existing General Permit requirements under Phase I to include/regulated discharges from construction sites that disturb land equal to or greater than one (1) acre and less than 5 acres, known as Small Construction Activity. Construction disturbances 1 acre and above typically require a formal NPDES permit and a formal Stormwater Pollution Prevention Plan (SWPPP) must be submitted to Agency(ies) with Jurisdiction for review and approval.

B. The work of this section consists of implementing measures to prevent discharges of pollutants, including temporary storm water pollution during construction activities, through compliance with NPDES permit program.

C. Work of this section consists of implementing measures to Temporary Storm Water Pollution during construction activities, through compliance with NPDES permit program.

1.2 DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114.

B. Environmental Pollution and Damage: The presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of the environment for aesthetic, cultural, or historical purposes.

C. National Pollution Discharge Elimination System (NPDES) Phase I: Regulates discharges from construction sites that disturb 5 acres or more.

D. NPDES Phase II: Regulations expand existing General Permit requirements under Phase I to include and regulate discharges from construction sites that disturb land equal to or greater than one (1) acre and less than 5 acres, known as Small Construction Activity.

E. Storm Water Pollution Prevention Plan (SWPPP): Developed and implemented stormwater management measures to protect surface water from pollutants during construction activities

disturbing an acre or more in compliance with federal, state, and local requirements for permit approval under NPDES program.

1.3 SUBMITTALS

A. After contract award and before pre-construction conference, prepare and submit:

1. A SWPPP showing SWPPP satisfies Federal and State NPDES permit requirements.

B. Inspection Schedule: Submit schedule for inspection and monitoring of pollution prevention measures.

C. Inspection Schedule: Submit schedule for inspection and monitoring of storm water pollution prevention measures.

D. Erosion Control Products: Submit manufacturer's product information and installation recommendations for silt fence, filter fabric, erosion control blanket, straw bales, and other materials proposed for use on this project.

1.4 QUALITY ASSURANCE

A. Contractor shall prepare and submit a plan to Contracting Officer (CO) for review and concurrence.

B. Orientation Meeting: Contractor shall arrange and conduct an Erosion and Sediment Control meeting/briefing to inform parties, scheduled to be on-site during project, of measures to be implemented for proper erosion and sediment control (may be included as part of Pre-Construction Meeting).

1. Installation of silt fences, storm drain protection, and other forms of erosion and sediment control shall not begin until after this meeting has occurred.

C. Pollution Prevention and Erosion Control Manager: Contractor shall designate Pollution Prevention and Erosion Control Manager responsible for implementation, inspection, maintenance, and amendments to approved plan.

1. Pollution Prevention and Erosion Control Manager shall be familiar with temporary storm water pollution prevention procedures and Best Management Practices and ensure emergency procedures and plan are updated as needed and available for inspection.

2. When changes in approved plan are required, Pollution Prevention and Erosion Control Manager shall prepare and certify an amendment and submit to Contracting Officer for review and concurrence.

PART 2 - PRODUCTS

2.1 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Provide SWPPP which satisfies Federal and State NPDES permit requirements and includes:

1. Site description.
2. Identification and contract information for Pollution Prevention and Erosion Control Manager.
3. Expected sequencing of operations and construction schedule.
4. Weather monitoring procedure.
5. Descriptions and details Best Management Practices for of pollution prevention and erosion controls, including dust control.
6. Pollution prevention and erosion control plans.
7. Controls for other potential onsite storm water pollutants.
8. Applicable specifications.
9. Maintenance and inspection procedures and forms.
10. Description of potential non-storm water discharges at site.
11. Notice of Intent (NOI) form.
12. Notice of Termination (NOT) form.
13. Contractor and Sub-contractor Certification forms.
14. Other record keeping forms and procedures.
15. Housekeeping Best Management Practices, including vehicle wash-down areas, protection of equipment storage and maintenance areas, and sweeping of roadways related to hauling activities.

PART 3 - EXECUTION

3.1 ENVIRONMENTAL PROTECTION

A. Protection of Natural Resources: Comply with applicable regulations and these specifications. Preserve natural resources within project boundaries and outside limits of work performed under this Contract in their existing condition or restore to an equivalent or improved condition as approved by Contracting Officer.

B. Construction Zone: Arrange construction activities to minimize pollution (i.e., erosion, trash, etc.) to maximum practical extent.

1. Clearing, excavation, and grading shall be limited to those areas of project site necessary for construction. Minimize area exposed and unprotected.
2. Clearly mark and delineate limits of work activities.
3. Equipment shall not be allowed to operate outside limits of work or to disturb existing vegetation.
4. Excavation and grading shall be completed during dry season to maximum extent possible.
5. Material shall be stored away from locations where water is present to greatest extent practicable.

3.2 REGULATORY REQUIREMENTS

A. Permits: Contractor shall obtain required NPDES permits resulting in no impacts to scheduled work. Contractor shall account for possibility of significant lead time in scheduling and executing work.

1. Implement requirements of NPDES permit for erosion control due to storm water runoff during construction.
2. Implement good housekeeping practices, inspections and record keeping.
3. Prior to construction, Contractor and Subcontractors shall sign certifications (included in the plan) that they understand requirements of NPDES permit.
4. Subcontractors shall comply with requirements of NPDES under supervision of Contractor.
5. Accepted plan shall comply with terms and conditions of EPA permit.

B. Notice of Intent (NOI): Contractor shall file a Notice of Intent and formal SWPPP as required to the Agency(ies) with Jurisdiction.

C. Notice of Termination (NOT): After Substantial Completion of Work, file a Notice of Termination (NOT) with the Agency(ies) with Jurisdiction.

D. Contracting Officer Notification: Contractor shall notify Contracting Officer in writing and by telephone of these events:

1. Erosion and sediment control meeting/briefing.
2. Following installation of required sediment control structures.
3. Prior to removal of or modification to sediment control structures.
4. Prior to removal of sediment control structures.

3.3 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Review and Acceptance: Contractor and Contracting Officer will jointly review draft Plan and agree to needed revisions. Contractor shall incorporate revisions, sign, and submit final Plan to Contracting Officer. Final Plan will be the document enforced on the project.

1. Accepted Plan will describe and ensure implementation of practices to be used to reduce pollutants in storm water discharges.
2. Contractor shall maintain current copy of Plan and associated records and forms at jobsite throughout duration of project.
3. Plan shall be available for public inspection and inspection and use of Contracting Officer.
4. Approval of Contractor’s Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.

B. Implementation: Implement Plan as required throughout construction period and maintain erosion control elements in proper working order.

1. Do not perform clearing and grubbing or earthwork until Plan has been implemented.

C. SWPPP (including inspection forms) and data used to complete the NOI shall be provided to Contracting Officer after Substantial Completion of project.

3.4 SITE INSPECTIONS AND PLAN REVISIONS

A. Inspections: Contractor and Contracting Officer will perform a weekly inspection onsite.

1. Inspection shall include disturbed areas not completely stabilized, areas used for storage of materials, locations where vehicles enter or exit site, and other erosion and sediment controls included in the Plan.
2. Inspections shall be documented.
3. Inspection forms shall be retained onsite in Plan notebook throughout construction period.

B. Plan Revisions: It may be necessary to revise Plan during construction to make necessary improvements, revisions, or to respond to unforeseen conditions noted during construction or site inspections.

1. Plan shall specify mechanism whereby revisions may be proposed by Contractor or Contracting Officer.
2. Contractor and Contracting Officer will jointly review each revision to Plan before changes incorporated and implemented. Contractor will then provide revised copy of Plan to Contracting Officer.
3. Accepted modifications will be implemented within 7 calendar days following date of inspection when deficiencies or necessary corrections are first noted.

C. Negligence: Provide additional temporary erosion and pollution controls made necessary by Contractor's errors or negligence at no additional cost to Government.

3.5 HOUSEKEEPING AND SITE MANAGEMENT

A. Store materials onsite in conformance to Federal, state, local, and manufacturer's regulations and specifications. Use Best Management Practices to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.

B. Manage solid waste in conformance to Federal, state, and local regulations. Best Management Practices should be used to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.

C. Include a spill prevention and control plan with provisions placed in SWPPP.

D. Manage hazardous waste (including contaminated soil) in conformance to Federal, state, local and NPS regulations and guidelines.

3.6 EROSION CONTROL MEASURES

A. Erosion control measures shall consist of Best Management Practices for storm water discharges, including silt fencing, barrier protectors, straw bales, temporary soil retention blankets, excelsior drainage filters, sediment traps and berms.

B. Berms and excelsior drainage filters shall be used to form sediment traps and control run-on and run-off into other areas, including creeks, streams, marshes, access roads, well areas, and staging areas.

C. Erosion control measures shall be used to contain only direct precipitation in construction zone. Contained water shall be allowed to percolate into ground or drain slowly through drainage filter sediment traps.

D. Earthen sediment traps or holding ponds shall not be used unless accepted by Contracting Officer.

E. Reduce runoff velocity and direct surface runoff around and away from fuel containment, storage, and borrow areas.

F. Divert surface runoff around and away from cut and fill slopes.

G. Place drainage filters around catch basins to create sediment traps to control run-off from construction area.

H. Excess water used for dust control shall be contained within demolition areas by erosion control measures.

I. Contractor shall prevent deposition of materials onto paved areas. Contractor shall inspect paved areas for deposited materials weekly and remove materials immediately.

J. Furnish, install, maintain, and operate necessary control measures and other equipment necessary to prevent erosion as described in approved SWPPP.

K. Before work begins, sufficient equipment shall be available on site to assure operation and adequacy of erosion control system can be maintained.

3.7 MAINTENANCE OF TEMPORARY FACILITIES

A. Ensure erosion and sediment control structures remain effective throughout excavation and grading operations. Relocate structures as necessary.

B. Inspect control structures after each significant rainfall. Promptly repair breaches which occur.

C. Contractor shall remove entrapped sediment from behind excelsior drainage filter after each storm.

3.8 REPORTING

A. If a discharge occurs or if project receives written notice or order from regulatory agency, Contractor shall immediately notify Contracting Officer and shall file written report to Agency(ies) with Jurisdiction within 7 days of discharge event, notice, or order. Corrective measures shall be implemented immediately following discharge, notice, or order. The report to the Agency(ies) with Jurisdiction shall contain:

1. Date, time, location, nature of operation, and type of discharge, including cause or nature of notice or order.
2. Best Management Practices deployed before discharge event, or prior to receiving notice or order.

3. Date of deployment and type of Best Management Practices deployed after discharge event, or after receiving notice or order, including additional Best Management Practices installed or planned to reduce or prevent re-occurrence.
4. An implementation and maintenance schedule for affected Best Management Practices.

3.9 SEDIMENT DISPOSAL

A. Sediment excavated from temporary sediment control structures shall be disposed on site with general fill, or with topsoil. Sediment shall be allowed to dry out as required before reuse.

B. Contractor shall place sediment removed from traps and other structures where it will not enter a storm drain or watercourse and where it will not immediately reenter the basin.

3.10 REMOVAL OF TEMPORARY STORM WATER POLLUTION CONTROL MEASURES

A. Temporary control measures shall be removed with permission of Contracting Officer within 20 working days after final acceptance of project, and/or once grading is complete and slopes have stabilized.

END OF SECTION 01 57 23

SECTION 01 67 00 – PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and environmental requirements.

1.2 DEFINITIONS

A. Products: Items purchased for incorporating into Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.

1. Named Products: Items identified by manufacturer's product name, including make or model number and other designation shown or listed in manufacturer's published product literature, current as of date of Contract Documents.
2. New Products: Items that have not previously been incorporated into another project or facility, except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.
3. Approved Equals: Product demonstrated and approved through submittal process, or where indicated as a product substitution, to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.

B. Definitions pertaining to sustainable development: As defined in ASTM E2114.

C. Biobased Materials: As defined in Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.

1. Biobased content: Amount of biobased carbon in material or product as a percentage of weight (mass) of total organic carbon in material or product.

D. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.

E. Environmentally preferable products: Products and services with lesser or reduced effect on the environment in comparison to conventional products and services. Refer to Environmental Protection Agency's (EPA) Final Guidance on Environmentally Preferable Purchasing for more information.

F. Stewardship: Responsible use and management of resources in support of sustainability.

G. Sustainability: Maintenance of ecosystem components and functions for future generations.

1. Recycled Content Materials: Products containing pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent with International Organization for Standardization (ISO) 140001 Standard for the Use of Environmental Marketing Claims.
2. Rapidly Renewable Material: Material made from plants typically harvested within a ten-year cycle.
3. Regional Materials: Materials manufactured and extracted, harvested, or recovered within a radius of 500 miles from Project location.

1.3 SUBMITTALS

A. Record Submittals as specified in – Sustainable Design Close-Out Documentation, submit:

1. Affirmative Procurement Reporting Form. Submit on form in Appendix A of this Section, or similar form as approved by Contracting Officer (CO).
2. Submit environmental data in accordance with Table 1 of ASTM E2129 for these products:

a. Masonry
b. Finish Carpentry
c. Plastic Fabrications
d. Building Insulation
e. Roofing
f. Joint Sealers
g. Wood & Plastic Doors
h. Windows
i. Skylights
j. Glazed Curtain Wall
k. Gypsum Board
l. Tile
m. Acoustical Ceilings
n. Resilient Flooring
o. Carpet
p. Wall Coverings
q. Paints & Coatings
r. Toilet Compartments
s. Office Equipment
t. Furnishings & Accessories
u. Renewable Energy Equipment
v. Elevators
w. Plumbing fixtures and equipment.

x. HVAC equipment
y. Lighting equipment

3. Material Safety Data Sheets (MSDS): For each product required by OSHA to have a MSDS, submit an MSDS. MSDS shall be prepared within the previous five years. Include information for MSDS Sections 1 to 16 in accordance with ANSI Z400.1 and as follows:

a. Section 1: Chemical Product and Company Identification
b. Section 2: Composition/Information on Ingredients
c. Section 3: Hazards Identification
d. Section 4: First Aid Measures
e. Section 5: Fire Fighting Measures
f. Section 6: Accidental Release Measures
g. Section 7: Handling and Storage
h. Section 8: Exposure Controls/Person Protection
i. Section 9: Physical and Chemical Properties
j. Section 10: Stability and Reactivity Data
k. Section 11: Toxicological Information. Include data used to determine the hazards cited in Section 3. Identify acute data, carcinogenicity, reproductive effects, and target organ effects. Provide written description of the process used in evaluating chemical hazards relative to preparation of the MSDS.
l. Section 12: Ecological Information. Include data regarding environmental impacts during raw materials acquisition, manufacture, and use. Include data regarding environmental impacts in event of accidental release.
m. Section 13: Disposal Considerations. Include data regarding proper disposal of the chemical. Include information regarding recycling and reuse. Indicate whether or not product is considered to be "hazardous waste" according to United States EPA Hazardous Waste Regulations 40 CFR 261 (Code of Federal Regulations).
n. Section 14: Transportation Information. Identify hazard class for shipping.
o. Section 15: Regulatory Information. Identify federal, state, and local regulations applicable to the material.
p. Section 16: Other Information. Include additional information relative to recycled content, biobased content, and other information regarding environmental and health impacts. Identify the date MSDS was prepared.

4. Chain of Custody: Submit chain-of-custody documentation for sustainable forestry for these products:

a. Rough Carpentry
b. Finish Carpentry
c. Wood Doors
d. Windows
e. Wood Flooring
f. Furnishings & Accessories

1.4 QUALITY ASSURANCE

A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle products using means and methods that prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.

B. Delivery and Handling:

1. Schedule delivery to minimize long-term storage at Project site and prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Project site in undamaged condition; in manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to ensure compliance with Contract Documents. Ensure products are undamaged and properly protected.
5. Obtain materials in biodegradable or recyclable/reusable packaging which uses minimum amount of packaging possible.

C. Storage:

1. Allow for inspection and measurement of quantity or counting of units.
2. Store materials in manner to not endanger Project structure.
3. Store products subject to damage by the elements, under cover in weather tight enclosure above ground, with ventilation adequate to prevent condensation.
4. Store cementitious products and materials on elevated platforms.
5. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
7. Protect stored products from damage and liquids from freezing.
8. Store loose granular materials in well-drained area on solid surfaces to prevent mixing with foreign matter.

1.6 PACKAGING

A. Where Contractor has option to provide one of listed products or equal, preference shall be given to products with minimal packaging and easily recyclable packaging as defined in ASTM D5834.

B. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.

C. Provide minimum 45 percent post-consumer recycled content and minimum 100 percent recovered fiber content of industrial paperboard in accordance with EPA’s Comprehensive Procurement Guidelines and ASTM D5663.

D. Provide minimum 10 percent post-consumer recycled content and minimum 10 percent recovered fiber content of carrier board in accordance with EPA’s Comprehensive Procurement Guidelines and ASTM D5663.

E. Provide minimum 5 percent post-consumer recycled content and minimum 5 percent recovered fiber content of brown papers (e.g., wrapping papers and bags) in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.

1.7 ENVIRONMENTALLY PREFERABLE PRODUCTS

A. Provide environmentally preferable products to greatest extent possible.

1. To greatest extent possible, provide products and materials with a lesser or reduced effect on the environment considering raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and/or disposal of the product.
2. Eliminate use of ozone depleting compounds during and after construction where alternative environmentally preferable products are available, consistent with either Montreal Protocol and Title VI or Clean Air Act Amendments of 1990, or equivalent overall air quality benefits that take into account life cycle impacts.
3. Use products meeting or exceeding EPA's recycled content recommendations for EPA-designated products. Use materials with recycled content such that the sum of post-consumer recycled content plus one-half of the pre-consumer content constitutes at least 10% (based on cost) of total value of the materials in project.

1.8 PRODUCT WARRANTIES

A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.

1. Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for product specifically endorsed by manufacturer to the Government.
2. Special Warranty: Written warranty required by or incorporated into Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for the Government.

B. Special Warranties: Prepare written document containing appropriate terms and identification, ready for execution. Submit draft for approval before final execution.

1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
2. Specified Form: When specified forms are included with Specifications, prepare written document using appropriate form properly executed.
3. Refer to Divisions 2 through 49 Sections for specific content requirements and particular requirements for submitting special warranties.

C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

A. General Product Requirements: Provide products to comply with Contract Documents, undamaged and, unless otherwise indicated, new at time of installation.

1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types produced and used successfully in similar situations on other projects.
3. Government reserves right to limit selection to products with warranties not in conflict with requirements of Contract Documents.
4. Where products are accompanied by term "as selected," Contracting Officer will make selection.
5. Where products are accompanied by term "match sample," sample to be matched is Governments.
6. Descriptive, performance, and reference standard requirements in Specifications establish "salient characteristics" of products.

B. Product Selection Procedures:

1. Product: Where Specifications name single product and manufacturer, provide named product that complies with requirements or approved equal.
2. Manufacturer/Source: Where Specifications name single manufacturer or source, provide product by named manufacturer or source that complies with requirements or approved equal.
3. Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed that complies with requirements or approved equal.
4. Manufacturers: Where Specifications include list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements or approved equal.
5. Available Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.
6. Available Manufacturers: Where Specifications include list of manufacturers, provide a product by one of the manufacturers listed, or an unnamed manufacturer, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.
7. Product Options: Where Specifications indicate sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide specified product, system, or approved equal.
8. Visual Matching Specification: Where Specifications require matching an established Sample, select product that complies with requirements and matches Architect's sample. Contracting Officers decision will be final on whether a proposed product matches.

a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.

9. Visual Selection Specification: Where Specifications include phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.

a. Standard Range: Where Specifications include phrase "standard range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
b. Full Range: Where Specifications include phrase "full range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 APPROVED EQUAL PRODUCTS

A. Conditions: Contracting Officer will consider Contractor's request when the following conditions are satisfied. If following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Evidence proposed product does not require revisions to Contract Documents, that it is consistent with Contract Documents and will produce indicated results and is compatible with other portions of Work.
2. Detailed comparison of significant qualities of proposed product with those named in Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
3. Evidence proposed product provides specified warranty.
4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
5. Samples, if requested.

PART 3 - EXECUTION

3.1 PROTECTION AFTER INSTALLATION

A. Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

END OF SECTION 01 67 00

AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

Project Name: _____ Project Number: _____

Contractor Name: _____ License Number: _____

Contractor Address: _____

Product	Total $ value provided	Total $ value with recycled content Pre-consumer	Total $ value with recycled content Post-consumer	Total $ value with biobased content	Exempted indicate 1,2,3,4	Comments
Hydraulic Mulch (paper based)						
Hydraulic Mulch (wood based)						
Compost						
Parking Stops (Concrete w/ fly ash, slag cement or low cement content)						
Parking Stops (Plastic/Rubber)						
Patio Blocks/Rubber						
Patio Blocks/Plastic						
Playground Surfaces						
Concrete with fly ash						
Concrete with slag cement						
Concrete with low cement content						
Plastic lumber						
Building Insulation						
Rock Wool						
Fiber glass						
Cellulose						
Perlite Comp Board						
Plastic Rigid Foam						
Glass Fiber Reinforced Foam						
Phenolic Rigid Foam						
Ceramic tile						
Resilient flooring						
Floor Tiles/Rubber						
Floor Tiles/Plastic						
Running Tracks						
Carpet (PET)						
Paint						

Product	Total $ value provided	Total $ value with recycled content Pre-consumer	Total $ value with recycled content Post-consumer	Total $ value with biobased content	Exempted indicate 1,2,3,4	Comments
Reprocessed Latex Paint White & Light Colors						
Reprocessed Latex Dark Colors						
Consolidated Latex Paint						
Toilet/Shower partitions (plastic or steel)						
Other						

CERTIFICATION

I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

1. The product does not meet appropriate performance standards.
2. The product is not available within a reasonable time frame.
3. The product is not available competitively (from two or more sources).
4. The product is only available at an unreasonable price (compared with a comparable non-recycled content product.)

Signature: _____ Date: _____

END OF
AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

SECTION 01 73 29 – CUTTING AND PATCHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes procedural requirements for cutting and patching on buildings that do not contain Historic Fabric.

1.2 SUBMITTALS

A. Cutting and Patching Plan: Submit Plan describing procedures at least ten <10> days before cutting and patching will be performed, requesting approval to proceed. Include:

1. Extent: Describe cutting and patching, show how performed, and indicate why they cannot be avoided.
2. Changes to In-Place Construction: Describe anticipated results. Include changes to structural elements and operating components as well as changes in building's appearance and other significant visual elements.
3. Products: List products to be used and firms or entities that will perform Work.
4. Dates: Indicate when cutting and patching will be performed.
5. Utility Services and Mechanical/Electrical Systems: List services/systems that cutting and patching procedures will disturb or affect. List services/systems that will be relocated and those that will be temporarily out of service. Indicate how long services/systems will be disrupted.
6. Structural Elements: Where cutting and patching involve adding reinforcement to structural elements, submit details and engineering calculations showing integration of reinforcement with original structure. Do not cut and patch structural elements in a manner that could change their load carrying capacity or increase deflection.
7. Contracting Officer’s (CO) Approval: Obtain approval of cutting and patching plan before cutting and patching. Approval does not waive right to later require removal and replacement of unsatisfactory work.

1.3 QUALITY ASSURANCE

A. Structural Elements: Do not cut and patch structural elements in a manner that could change their load-carrying capacity or load-deflection ratio.

1. Structural steel (columns, beams, braces, etc.)
2. Concrete beams, walls, footings.
3. Masonry (including CMU) walls, including headers.

B. Operational Elements: Do not cut and patch operating elements and related components in a manner that results in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety.

C. Miscellaneous Elements: Do not cut and patch miscellaneous elements or related components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety.

D. Visual Requirements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch construction exposed on exterior or in occupied spaces in a manner that would, in Contracting Officer's opinion, reduce the building's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

1.4 WARRANTY

A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during cutting and patching operations, by methods and with materials so as not to void existing warranties.

PART 2 - PRODUCTS

2.1 MATERIALS

A. General: Comply with requirements specified in other Sections.

B. In-Place Materials: Use materials identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.

1. If identical materials are unavailable or cannot be used, use materials that will match the visual and functional performance of in-place materials when installed.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine surfaces to be cut and patched and conditions under which cutting and patching are to be performed.

1. Compatibility: Before patching, verify compatibility with and suitability of substrates, including compatibility with in-place finishes or primers.
2. Proceed with installation only after unsafe or unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Temporary Support: Provide temporary support of Work to be cut.

B. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.

C. Adjoining Areas: Avoid interference with use of adjoining areas or interruption of free passage to adjoining areas.

D. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to prevent interruption to occupied areas.

3.3 PERFORMANCE

A. General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at earliest feasible time. Complete without delay.

1. Cut in-place construction to provide for installation of other components or performance of other construction subsequently. Patch as required to restore surfaces to original condition.

B. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer and original Installer's written recommendations.

1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots as small as possible, neatly to size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
2. Finished Surfaces: Cut or drill from exposed or finished side into concealed surfaces.
3. Concrete and masonry: Cut using an abrasive saw or a diamond-core drill.
4. Excavating and Backfilling: Comply with requirements in applicable Division 31 Sections where required by cutting and patching operations.
5. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
6. Proceed with patching after construction operations cutting are complete.

C. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other Work. Patch with durable seams as invisible as possible. Provide materials and comply with installation requirements specified in other Sections.

1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate integrity of installation.
2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will eliminate evidence of patching and refinishing.

3. Floors and Walls: Where walls or partitions that are removed extend one finished area into another. Patch and repair floor and wall surfaces in new space. Provide an even surface of uniform finish, color, texture, and appearance. Remove in-place floor and wall coverings and replace with new materials, if necessary, to achieve uniform color and appearance.
4. Ceilings: Patch, repair, or re-hang in-place ceilings as necessary to provide an even-plane surface of uniform appearance.
5. Exterior Building Enclosure: Patch components in a manner that restores enclosure to a weather tight condition.

D. Cleaning: Clean areas and spaces where cutting and patching are performed. Remove paint, mortar, oils, putty, and similar materials.

END OF SECTION 01 73 29

SECTION 01 73 40 – EXECUTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes general procedural requirements governing execution of Work including:

1. Coordination with utility service providers
2. Construction layout
3. Field engineering and surveying
4. General installation of products
5. Progress cleaning
6. Starting and adjusting
7. Protection of installed construction
8. Correction of the Work

1.2 SUBMITTALS

A. Certificates: Submit certificate signed by land surveyor certifying location and elevation of improvements comply with requirements.

B. Landfill Receipts: Submit copy of receipts issued by landfill facility, licensed to accept hazardous materials, for hazardous waste disposal.

1.3 QUALITY ASSURANCE

A. Land Surveyor Qualifications: Professional land surveyor legally qualified to practice in jurisdiction where Project is located and is experienced in providing land-surveying services of kind indicated.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

A. Existing Conditions: Existence and location of site improvements and other construction indicated as existing are not guaranteed.

1. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Work.
2. Before construction, verify the location and points of connection of utility services.

B. Existing Utilities: Existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify existence and location of underground utilities and other construction affecting Work.

1. Before construction, verify location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; and underground electrical services.
2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.

C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.

1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
2. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
3. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
4. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 COORDINATION WITH UTILITY SERVICE PROVIDERS

A. Coordination with Utility Service Providers: Contact following Utility Service providers, sufficiently in advance to avoid delaying the work, to coordinate Contractor's portion of Work, testing requirements, inspections, etc.

1. Electrical Service Contact: Allamakee-Clayton Electric Cooperative, 229 Hwy. 51, Postville, IA 52162, (563) 864-7611 to coordinate Electrical service requirements.

    a. Construction Contractor Responsibilities: Contractor is responsible for coordination and installation of all electrical work as indicated on the Drawings and per Specifications.

2. Water Service Contact: National Park Service, Jeremy Parker, (563) 873-3491 to coordinate Water service requirements.

    a. Construction Contractor Responsibilities: Contractor is responsible for coordination and installation of all work related to water service as indicated on the Drawings and per Specifications.

3. Wastewater Service Contact: National Park Service, Jeremy Parker, (563) 873-3491 to coordinate Wastewater service requirements.

a. Construction Contractor Responsibilities: Contractor is responsible for coordination and installation of all work related to wastewater as indicated on the Drawings and per Specifications.

4. Telephone Service Contact: Alpine Communications, P.O. Box 1008, 923 Humphrey St., Elkader, IA 52043, (563) 245-4000 to coordinate Telephone service requirements.

a. Construction Contractor Responsibilities: Contractor is responsible for coordination and installation and relocation of all telephone service as indicated on the Drawings and per Specifications.

3.3 PREPARATION

A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of Work are indicated fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying Work.

B. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.

C. Review of Contract Documents and Field Conditions: Immediately on discovery of need for clarification of the Contract Documents caused by differing field conditions outside control of Contractor, submit request for information to Contracting Officer in accordance with Section 01 31 00 "Project Management and Coordination."

3.4 CONSTRUCTION LAYOUT

A. Verification: Verify layout information shown on Drawings, in relation to the existing benchmarks before proceeding to lay out Work. Notify Contracting Officer promptly if discrepancies are discovered.

B. General: Engage a land surveyor to lay out Work using accepted surveying practices.

1. Establish benchmarks and control points to set lines of construction and elsewhere as needed to locate each element of Project.
2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
3. Inform installers of lines and levels to which they must comply.
4. Check location, level and plumb, of every major element as Work progresses.
5. Notify Contracting Officer when deviations from required lines and levels exceed allowable tolerances.
6. Close site surveys with an error of closure equal to or less than the established standard.

C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.

D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical

work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.

E. Record Log: Maintain log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make log available for review by National Park Service (NPS).

3.5 FIELD ENGINEERING

A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning Work. Preserve and protect permanent benchmarks and control points during construction operations. Controls destroyed by Contractor will be replaced by Contractor at their expense.

1. Existing Monuments: All benchmarks, land corners, and triangulation points, established by other surveys, existing within construction area shall be preserved. If existing monuments interfere with Work, secure written permission before removing them.

B. Benchmarks: Establish and maintain a minimum of [two] <2> permanent benchmarks on Project site, referenced to data established by survey control points. Comply with NPS requirements for type and size of benchmark.

1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.

C. Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.

3.6 INSTALLATION

A. General: Locate Work and components of Work accurately in correct alignment and elevation, as indicated.

1. Make vertical work plumb and make horizontal work level.
2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.

B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.

C. Install products at the time and under conditions for best possible results. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.

E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.

F. Templates: Obtain and distribute to parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm adequate provisions are made for locating and installing products to comply with indicated requirements.

G. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of Work.

1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Contracting Officer.
2. Allow for building movement, thermal expansion, and contraction.
3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors to be embedded in concrete or masonry. Deliver to Project site in time for installation.

H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for best visual effect. Fit exposed connections together to form hairline joints.

I. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.

J. Quantity surveys: Shall be conducted, and data derived from these surveys shall be used in computing quantities of work performed and actual construction completed and in place.

1. Contractor shall conduct original and final surveys and surveys for any periods for which progress payments are requested. These surveys shall be conducted under direction of a representative of the Contracting Officer, unless Contracting Officer waives requirement in a specific instance. Government shall make such computations as are necessary to determine quantities of work performed or finally in place. Contractor shall make computations based on surveys for any periods for which progress payments are requested.
2. Promptly upon completing a survey, Contractor shall furnish originals of field notes and other records relating to survey or layout of Work to Contracting Officer. Contractor shall retain copies of all such material furnished to Contracting Officer.

3.7 PROGRESS CLEANING

A. General: Clean Project site, work areas, and common areas daily. Coordinate progress cleaning for joint-use areas where more than one Installer has worked. Enforce requirements strictly. Dispose of materials lawfully.

1. Comply with requirements in National Fire Protection Association (NFPA) 241 for removal of combustible waste materials and debris at the end of every work session.
2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 degrees Fahrenheit.
3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.

B. Site: Maintain Project site free of waste materials and debris.

C. Work Areas: Clean areas where work is in progress to level of cleanliness necessary for proper execution of Work.

1. Remove liquid spills promptly.
2. Where dust would impair proper execution of Work, broom-clean or vacuum entire work area, as appropriate.
3. Contractor shall provide progress cleaning that minimizes sources of food, water, and harborage available to pests.

D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials not hazardous to health or property and will not damage exposed surfaces.

1. Utilize non-toxic cleaning materials and methods.

a. Comply with Green Seal Standard (GS) 37 for general purpose cleaning and bathroom cleaning.
b. Use natural cleaning materials where feasible. Natural cleaning materials include:

1) Abrasive cleaners: substitute 1/2 lemon dipped in borax.
2) Ammonia: substitute vinegar, salt and water mixture, or baking soda and water.
3) Disinfectants: substitute 1/2 cup borax in gallon water.
4) Drain cleaners: substitute 1/4 cup baking soda and 1/4 cup vinegar in boiling water.
5) Upholstery cleaners: substitute dry cornstarch.

E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect from damage and deterioration at time of Substantial Completion.

G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways.

H. Clean and protect construction in progress and adjoining materials already in place during handling and installation. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.

I. Clean and provide maintenance on completed construction as frequently as necessary through remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.

J. Limiting Exposures: Supervise construction operations so that no part of construction completed or in progress, is subject to harmful, dangerous, damaging, or deleterious exposure during construction period.

K. Final Cleaning: At completion of Work, remove remaining waste materials, rubbish, tools, equipment, machinery and surplus materials. Clean exposed surfaces and leave Project clean and ready for occupancy.

1. Provide final cleaning in accordance with ASTM E1971.

3.8 STARTING AND ADJUSTING

A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

B. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.

C. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

D. Manufacturer's Field Service: If factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in Section 01 40 00 "Quality Requirements."

3.9 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

B. Comply with manufacturer's written instructions for temperature and relative humidity.

3.10 CORRECTION OF THE WORK

A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in Section 01 73 29 "Cutting and Patching."

1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

B. Restore permanent facilities used during construction to specified condition.

C. Remove and replace damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.

D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.

E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 01 73 40

SECTION 01 74 19 – CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for:

1. Salvaging nonhazardous demolition and construction waste.
2. Recycling nonhazardous demolition and construction waste.
3. Disposing of nonhazardous demolition and construction waste.

1.2 DEFINITIONS

A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.

B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.

C. Solid Waste: Garbage, debris, sludge, or other discharged material (except hazardous waste) including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations.

D. Debris: Non-hazardous solid waste generated during construction, demolition, or renovation of a structure which exceeds 2.5 inch (60 millimeter) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if mixture is comprised primarily of debris by volume, based on visual inspection.

E. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.

F. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of environment for aesthetic, cultural, or historical purposes.

G. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

H. Hazardous Materials: Material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or

disposal meets or has components which meet or have potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.

I. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.

J. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.

K. Single-use plastic products: Plastic items intended to be disposed of immediately after use, including plastic and polystyrene food and beverage containers, bottles, straws, cups, cutlery, and disposable plastic bags.

1.3 PERFORMANCE REQUIREMENTS

A. General: Project shall minimize creation of construction, deconstruction, and demolition waste to protect and restore natural habitat and resources. Minimize factors contributing to waste such as over packaging, improper storage, ordering error, poor planning, breakage, mishandling, single-use plastic products, and contamination. A Waste Management Plan shall be developed to ensure that existing site and building materials are reused, salvaged, or recycled. Minimize waste disposal in landfills.

B. Salvage /Recycle Requirements: Develop waste management plan resulting in end-of-Project rates for salvage/recycling of 50 percent by weight of total waste generated by the Work. The following waste categories, at a minimum, shall be diverted from a landfill:

1. Land clearing debris (chipped debris can be used on site for mulch or erosion control)
2. Clean dimensional wood, palettes
3. Plywood, OSB (oriented strand board), and particle board
4. Concrete (can be ground and used for fill on site)
5. Asphaltic concrete (can be ground and used for fill on site)
6. Cardboard, paper, packaging, newsprint
7. Metals (from banding, stud trim, piping, rebar, roofing, other trim, steel, iron, galvanized sheet steel, stainless steel, aluminum, copper, zinc, lead, brass, and bronze)
8. Gypsum drywall - unpainted
9. Non-hazardous paint and paint cans
10. Beverage containers: Aluminum, glass, and plastic containers
11. Insulation
12. Ceiling grid and tiles
13. Ductwork
14. Wiring
15. Other mixed construction and demolition waste as appropriate.

C. If waste materials encountered during deconstruction/demolition or construction phase are found to contain lead, asbestos, polychlorinated biphenyls (PCBs), (such as fluorescent lamp ballasts), or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.

D. Existing items and material to be removed during deconstruction/demolition phase shall be reused in construction phase of the Project. Items that cannot be reused shall be recycled. Items

considered for reuse must be in refurbishable condition and must meet quality standards set forth in these specifications. Contractor shall ensure quality of the item(s) in question will meet or exceed accepted industry or trade standards for first quality commercial grade application. During construction, deconstruction, or demolition Contracting Officer (CO) may designate other objects or materials for reuse.

E. Salvage/Recycle Requirements: Government goal is to salvage and recycle as much nonhazardous demolition and construction waste as possible including the following materials. Contractor to verify with CO:

1. Remaining Existing Stone Veneer
2. Remaining Existing Brick
3. Existing WAPs (Wireless Access Points)
4. Existing horizontal wire manager
5. Existing CAT6 patch panel
6. Existing CAT5e patch panel
7. Existing telecom rack in Storage
8. Existing BAS System Switch
9. Existing VC Benches (3)
10. Existing Stanchion (1)
11. Existing Bike Rack (1) (to be reinstalled by contractor)
12. Existing Park Sign (1)
13. Existing Picnic Table (1)
14. Existing Bulletin Cabinets (2)
15. Existing Trash Receptacles (2)
16. Existing Retaining Wall Stone Caps
17. Existing Parking Lot Benches (6)
18. Existing Picnic Tables (4)
19. Existing Trash Receptacles (2)
20. Existing Retaining Wall Stone

1.4 SUBMITTALS

A. Waste Management Plan: After award of contract and prior to scheduled Pre-Construction Conference, Contractor shall submit a draft Waste Management Plan to Contracting Officer for approval. Submit three <3> copies of plan. Revise and resubmit Plan as required by Contracting Officer. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.

B. Progress Documentation: Supplemental to Waste Management Plan, document solid waste disposal, diversion, and cost/revenue analysis and submit completed worksheet on a monthly basis. See Project Waste Management Plan Worksheet Sample, attached to the end of the Division 1 Specifications, and report totals to date for column headings.

C. Waste Reduction Calculations: Before request for Substantial Completion, submit [three] copies of calculated end-of-Project rates for salvage, recycling, and disposal as a percentage of total waste generated by the Work.

D. Records of Donations: Indicate receipt and acceptance of salvageable waste donated to individuals and organizations. Indicate whether organization is tax exempt.

E. Records of Sales: Indicate receipt and acceptance of salvageable waste sold to individuals and organizations. Indicate whether organization is tax exempt.

F. Recycling and Processing Facility Records: Indicate receipt and acceptance of recyclable waste by recycling and processing facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

G. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

H. Qualification Data: For Waste Management Coordinator and refrigerant recovery technician.

I. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating refrigerant that was present was recovered and recovery was performed according to Environmental Protection Agency (EPA) regulations. Include name and address of technician and date refrigerant was recovered.

J. Progress payment requirements:

1. With each Application for payment, submit an updated Project Waste Management Plan worksheet for solid waste disposal and diversion.
2. With each Application for Payment, submit manifests, weight tickets, receipts, and invoices specifically identifying the Project and waste material.

K. Closeout Submittals

1. With Closeout Submittals, submit a summary of a Project Waste Management Plan worksheet for solid waste disposal and diversion.

1.5 QUALITY ASSURANCE

A. Waste Management Coordinator Qualifications: Experienced firm, with record of successful waste management coordination of projects with similar requirements.

B. Refrigerant Recovery Technician Qualifications: Certified by EPA-approved certification program.

C. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.

D. Waste Management Meeting: Conduct separate meeting or cover in Pre-Construction Conference and comply with requirements in Section 01 31 00 "Project Management and Coordination." Review methods and procedures related to waste management including:

1. Review and discuss waste management plan including responsibilities of Waste Management Coordinator.
2. Review requirements for documenting quantities of each type of waste and its disposition.
3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.

4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
5. Review procedures for reduction of single-use plastic products on site.
6. Review waste management requirements for each trade.

PART 2 - PRODUCTS

2.1 WASTE MANAGEMENT PLAN

A. General: Develop plan consisting of waste identification and waste reduction work plan. Include separate sections in plan for demolition and construction waste. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.

B. Waste Identification: Indicate anticipated types and quantities of demolition, and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.

C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.

1. Salvaged Materials for Reuse: For materials salvaged and reused in Project, describe methods for preparing salvaged materials before incorporation into the Work.
2. Salvaged Materials for Sale: For materials sold to individuals and organizations, include list of names, addresses, and telephone numbers.
3. Salvaged Materials for Donation: For materials donated to individuals and organizations, include list of names, addresses, and telephone numbers.
4. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
5. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
6. Handling and Transportation Procedures: Include method used for separating recyclable waste including sizes of containers, container labeling, and designated location on Project site where materials separation will be located.
7. Reduction of single-use plastic products: Include methods for water distribution to include bottle refilling stations if available for personnel.

D. Cost/Revenue Analysis: Indicate total cost of waste disposal as if there was no waste management plan and net additional cost or net savings resulting from implementing waste management plan. Include:

1. Landfill tip fees per ton.
2. If diverted, tip fee savings from landfill diversion.
3. Costs of recycling, salvage, or reuse.
4. Revenue from recycling, salvage, or reuse.
5. Total cost or savings from diversion. (Calculate by using tip fee savings and subtracting costs of recycling or adding revenue from recycling.)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

A. General: Implement waste management plan as approved by Contracting Officer. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during entire duration of Contract.

B. Waste Management Coordinator: Engage waste management coordinator responsible for implementing, monitoring, and reporting status of waste management work plan. Coordinator shall be present at Project site full time for duration of Project.

C. Contractor shall establish contacts with local recycling and reuse companies to set up lines of responsibility. Contractor shall be responsible for coordination in terms of identifying materials, pickup schedules, and standard quality for recycled materials.

D. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.

1. Distribute waste management plan to everyone concerned within three days of submittal return.
2. Distribute waste management plan to entities when they begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.

E. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

F. Separation facilities:

1. Contractor shall designate and Contracting Officer shall approve specific area or areas to facilitate separation of materials for potential reuse, salvage, recycling, and return.
2. Place waste and recycling bins near each other, and close to point of waste generation but out of traffic pattern.
3. Keep recycling and waste bin areas neat, clean, and clearly marked in order to avoid co-mingling of materials.
4. Protect bins during non-working hours from off-site contamination.
5. Check garbage dumpsters periodically for recyclables being thrown away and undocumented materials that could be recycled.

G. Materials handling procedures: Material to be recycled shall be protected from contamination and shall be handled, stored, and transported in a manner that meets requirements set by designated facilities for acceptance. Establish defined area for operations of each trade, especially woodcutting so off-cuts are kept in one area and can be sorted by dimension for future reuse.

3.2 SALVAGING DEMOLITION WASTE

A. Salvaged Items for Reuse in the Work:

1. Clean salvaged items.

2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in secure area until installation.
4. Protect items from damage during transport and storage.
5. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.

B. Salvaged Items for Sale and Donation: Not permitted on Project site.

C. Salvaged Items for Governments Use:

1. Clean salvaged items.
2. Pack or crate items after cleaning. Identify contents of containers.
3. Allow for inspection if necessary.
4. Store items in secure area until delivery to Government.
5. Transport items to storage area off-site designated by Government.
6. Protect items from damage during transport and storage.

3.3 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

A. General: Recycle paper and beverage containers used by on-site workers.

B. Recycling Receivers and Processors: List below is provided for information only; available recycling receivers and processors include:

1. Prairie du Chien Transfer Station
62949 Vineyard Coulee Rd.
Prairie du Chien, WI 53821

C. Recycling Incentives: Revenues, savings, rebates, tax credits, and other incentives received for recycling waste materials shall accrue to Contractor.

D. Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to maximum extent practical.

1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.

a. Inspect containers and bins for contamination and remove contaminated materials if found.

2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
4. Store components off ground and protect from weather.
5. Remove recyclable waste off Government property and transport to recycling receiver or processor.

3.4 RECYCLING DEMOLITION WASTE

A. Asphaltic Concrete Paving: Break up and transport paving to asphalt-recycling facility.

B. Concrete: Remove reinforcement and other metals from concrete and sort with other metals.

1. Pulverize concrete to maximum 1-1/2 inch size.

C. Masonry: Remove metal reinforcement, anchors, and ties from masonry and sort with other metals.

1. Pulverize masonry to maximum 1-1/2 inch size.
2. Clean and stack undamaged, whole masonry units on wood pallets.

D. Wood Materials: Sort and stack members according to size, type, and length. Separate lumber, engineered wood products, panel products, and treated wood materials.

E. Metals: Separate metals by type.

1. Structural Steel: Stack members according to size, type of member, and length.
2. Remove and dispose of bolts, nuts, washers, and other rough hardware.

F. Asphalt Shingle Roofing: Separate organic and glass-fiber asphalt shingles and felts. Remove and dispose of nails, staples, and accessories.

G. Gypsum Board: Stack large clean pieces on wood pallets and store in a dry location. Remove edge trim and sort with other metals. Remove and dispose of fasteners.

H. Acoustical Ceiling Panels and Tile: Stack large clean pieces on wood pallets and store in dry location.

1. Separate suspension system, trim, and other metals from panels and tile and sort with other metals.

I. Carpet and Pad: Roll large pieces tightly after removing debris, trash, adhesive, and tack strips.

1. Store clean, dry carpet and pad in closed container or trailer provided by Carpet Reclamation Agency or carpet recycler.

J. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.

K. Plumbing Fixtures: Separate by type and size.

L. Piping: Reduce piping to straight lengths and store by type and size. Separate supports, hangers, valves, sprinklers, and other components by type and size.

M. Lighting Fixtures: Separate lamps by type and protect from breakage.

N. Electrical Devices: Separate switches, receptacles, switchgear, transformers, meters, panelboards, circuit breakers, and other devices by type.

O. Conduit: Reduce conduit to straight lengths and store by type and size.

P. Electronic Products: Ensure non-usable electronic products are reused, donated, sold, or recycled using environmentally sound management practices at end of life.

3.5 RECYCLING CONSTRUCTION WASTE

A. Packaging:

1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in dry location.
2. Polystyrene Packaging: Separate and bag materials.
3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.

B. Site-Clearing Wastes: Chip brush, branches, and trees at landfill facility.

C. Wood Materials:

1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
2. Clean Sawdust: Bag sawdust not containing painted or treated wood.

D. Gypsum Board: Stack large clean pieces on wood pallets and store in dry location.

1. Clean Gypsum Board: Grind scraps of clean gypsum board using small mobile chipper or hammer mill. Screen out paper after grinding.

3.6 DISPOSAL OF WASTE

A. General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose in landfill or incinerator acceptable to authorities having jurisdiction.

1. Except as otherwise specified, do not allow waste materials to accumulate on-site.
2. Remove and transport debris in manner preventing spillage on adjacent surfaces and areas.

B. Burning: Do not burn waste materials.

C. Burning: Burning of waste materials allowed only at designated areas on Government property, provided required permits are obtained. Provide full-time monitoring for burning materials until fires are extinguished.

D. Disposal: Transport waste materials off Government property and legally dispose of them.

END OF SECTION 01 74 19

Project Waste Management Plan Worksheet Sample

National Park Service (NPS) - Denver Service Center (DSC) | 10-13-21

	A	B	C	D	E	F	G	H	I	J
Material	**Quantity Recycled (in tons)**	**Quantity Salvaged or Reused (in tons)**	**A + B = Total Quantity Diverted from Landfill**	**Quantity to Landfill (in tons)**	**C + D = Total Quantity Generated (in tons)**	**Tip Fee/Ton at Landfill**	**C x F = Tip Fee Savings resulting from Landfill Diversion**	**Cost of Recycling (R), Salvage (S), or Reuse (Re)** (Specify R, S, or Re)	**Revenue from Recycling (R), Salvage (S), or Reuse (Re)** (Specify R, S, or Re)	**G - H + I = Total Cost (-) or Savings (+) from Diversion**
Asphalt/Concrete										
Brick/Masonry/Tile										
Building Materials (doors, windows, fixtures, shingles, lumber, insulation, sheet goods, etc.)										
Carpet										
Carpet Padding, Foam Only										
Cardboard										
Ceiling Tile										
Drywall										
Glass										
Scrap Metal Aluminum										
Copper										
Steel										
Unpainted Wood & Pallets										
Yard Trimmings, Brush, Trees, Stumps, etc.										
Garbage/Trash										
Other										
Column Totals										
	Total Quantity Recycled	Total Quantity Reused or Salvaged	Total Quantity Diverted from Landfill	Total Quantity to Landfill	Total Quantity Generated		Tip Fee Savings from Diversion	Total Cost of Recycling, Salvage, or Reuse	Revenue from Recycling, Salvage, or Reuse	Total Cost (-) or Savings (+) from Diversion

Percentage Diverted = _____ (C divided by E from Column Totals). Should meet 60% diversion goal.

SECTION 01 77 00 – CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for contract closeout, including:

1. Project Record Drawings
2. Closeout Submittals
3. Substantial Completion and Final Inspection
4. Permit Closure and Transfer
5. Final Acceptance of the Work
6. Warranties

1.2 PROJECT RECORD DRAWINGS

A. Maintain one complete full-size set of contract drawings and one full-size set of vendor-supplied drawings. Clearly mark changes, deletions, and additions using National Park Service (NPS) drafting standards to show actual construction conditions. Show additions in red, deletions in green and special instructions in blue.

B. Keep record drawings current. Make record drawings available to Contracting Officer (CO) for inspection at the time of monthly progress payment requests. If project record drawings are not current, Contracting Officer may retain an appropriate amount of progress payment.

C. Submit complete record drawings on completion of total project. Include shop drawings, sketches, and additional drawings to be included in final set, with clear instructions showing the location of these drawings.

1.3 CLOSEOUT SUBMITTALS

A. A list of closeout requirements has been attached at the end of the Division 1 Specifications for your convenience. The intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list. Submit the following before requesting final inspection:

1. Specific warranties, guarantees, workmanship bonds, final certifications, and similar documents.
2. NPS required forms for occupancy, Fire Sprinkler/Alarm acceptance, and other similar forms or certificates.
3. Project Record Documents, operation and maintenance manuals, final completion construction digital images recorded on CD-R (compact disc-recordable) or DVD-R

(digital video disc-recordable) with index and descriptions, and similar final record information.

4. Environmental Record Documents: As specified in Divisions as follows:

    a. IAQ Management Plan: As specified in Section 01 57 19.11 Indoor Air Quality (IAQ) Management.
    b. Product Data for filtration media: As specified in Section 01 57 19.11 Indoor Air Quality (IAQ) Management.
    c. Moisture Control inspections and reports: As specified in Section 01 57 19.11 Indoor Air Quality (IAQ) Management.
    d. Material Safety Data Sheet (MSDS) Data: As specified in Section 01 67 00 Environmental Requirements for Products.
    e. Environmental Product Data: As specified in Section 01 67 00 Environmental Requirements for Products.
    f. Final Summary of Solid Waste Disposal and Diversion: As specified in Section 01 74 19 Construction Waste Management.
    g. Commissioning Report: As specified in Section 01 91 14 Total Building Commissioning.

5. Posted Operating Instructions: As specified in individual sections. Furnish operating instructions attached to or posted adjacent to equipment. Include wiring diagrams, control diagrams, control sequence, start-up, adjustment, operation, lubrication, shut-down, safety precautions, procedures in the event of equipment failure, and other items of instruction recommended by manufacturer.
6. Deliver tools, spare parts, extra materials, and similar items to location designated by Contracting Officer. Label with manufacturer's name and model number where applicable.

    a. Special Tools: One set of special tools required to operate, adjust, dismantle, or repair equipment. Special tools are those not normally found in possession of mechanics or maintenance personnel.

7. Keys and Keying Schedule: Submit keys including duplicates. Wire keys for each lock securely together. Tag and plainly mark with lock number, equipment identification, or panel or switch number, and indicate location, building, and room name or number.
8. Make final changeover of permanent locks and deliver keys to Contracting Officer. Advise Park personnel of changeover in security provisions.
9. Approved pre-functional checklists and functional performance testing reports from commissioning documentation.

    a. Equipment start-up requires coordination with commissioning process. Refer to Section 01 91 14 Total Building Commissioning. Equipment shall not be "temporarily" started for commissioning.

10. Test and balance report.
11. Terminate and remove temporary facilities, mockups, construction tools, and similar elements from Project site, complete final cleaning requirements, including touchup painting.
12. Touch up and repair and restore marred exposed finishes to eliminate visual defects.
13. Instruct NPS personnel in operation, adjustment, and maintenance of products, equipment, and systems. Submit demonstration and training videos.

1.4 FINAL INSPECTION, SUBSTANTIAL COMPLETION AND ACCEPTANCE PROCEDURES

A. Request final inspection in writing when project or designated portion of project is substantially complete. Contracting Officer will proceed with inspection within 10 days of receipt of written request or will advise Contractor of items that prevent project from being substantially complete.

B. If work is determined substantially complete, following final inspection, Contracting Officer will prepare Punch List and issue a Letter of Substantial Completion.

C. If work is not determined substantially complete following final inspection, Contracting Officer will notify Contractor in writing. Contractor shall request new final inspection after completing work. Re-inspection costs may be charged against Contractor in accordance with Inspection of Construction contract clause.

D. Contractor shall complete Punch List within 30 calendar days, documented weather permitting.

1. Prior to requesting final inspection:

a. Complete commissioning requirements of Section 01 91 14 Total Building Commissioning, unless approved in writing by Contracting Officer.

E. If Contractor completes items of work on Punch List and contractually required items, Contracting Officer will issue Letter of final acceptance of work.

F. If Contractor fails to complete work within the time frame, Contracting Officer may correct work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with Inspection of Construction contract clause.

1.5 PERMIT CLOSURE AND TRANSFER

A. When work covered by the permits is complete, create list of tasks required to close or transfer permits to Park. Submit to Contracting Officer for approval.

B. After substantial completion and Punch List completion, permits shall be closed and documented by Agency(ies) with Jurisdiction for the permit.

C. If responsibility for permits is to be transferred to Park, Park shall be informed of permit provisions completed and responsibilities transferring to Park staff.

1.6 WARRANTIES

A. Submittal Time: Submit written warranties on request of Contracting Officer for designated portions of Work where commencement of warranties other than date of Substantial Completion is indicated.

B. Organize warranty documents into an orderly sequence based on table of contents of Project Manual.

1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2 by 11 inch paper.
2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify product or installation. Provide typed description of product or installation, including name of product and name, address, and telephone number of Installer.
3. Identify each binder on front and spine with typed or printed title "WARRANTIES," Project name, and name of Contractor.
4. Warranty Electronic File: Scan warranties and bonds and assemble complete warranty and bond submittal package into a single indexed electronic PDF (portable document format) file with links enabling navigation to each item. Provide bookmarked table of contents at beginning of document.

C. Provide additional copies of each warranty in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 MATERIALS

A. See Division 1 Specification Section "Execution" for information on cleaning agents.

PART 3 - EXECUTION

3.1 FINAL CLEANING

A. General: Conduct final cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:

    a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
    b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
    c. Rake grounds that are neither planted nor paved to smooth, even-textured surface.
    d. Remove tools, construction equipment, machinery, and surplus material from Project site.
    e. Remove snow and ice to provide safe access to building.
    f. Clean exposed exterior and interior hard-surfaced finishes to dirt-free condition, free of stains, films, and foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to original condition.

g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
h. Sweep concrete floors broom clean in unoccupied spaces.
i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo, soil or stains remain.
j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and vision-obscuring materials. Replace chipped or broken glass and transparent materials. Polish mirrors and glass.
k. Remove labels that are not permanent.
l. Touch up, repair, and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

    1) Do not paint over "UL" (Underwriters Laboratories) and similar labels, including mechanical and electrical nameplates.

m. Wipe surfaces of mechanical and electrical equipment[, elevator equipment,] and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
n. Replace parts subject to unusual operating conditions.
o. Clean plumbing fixtures to sanitary condition, free of stains, including stains resulting from water exposure.
p. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
q. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out and noticeably dimmed bulbs, and defective or noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
r. Leave Project clean and ready for occupancy.

C. Pest Control: Engage experienced, licensed exterminator to make a final inspection and rid project of rodents, insects, and other pests. Provide Government with report.

D. Waste Disposal: Comply with requirements of Section 01 74 19 "Construction Waste Management and Disposal."

END OF SECTION 01 77 00

Closeout and Operation & Maintenance (O&M) Requirements **Template**

National Park Service (NPS) - Denver Service Center (DSC) | 1-27-21

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
	024119	Inventory of Removed and Salvaged Items			
	079200	Joint Sealant Schedule			
	087100	Door Hardware - Wiring Diagrams			
	210000	Record Drawings.			
	211300	Record Drawings.			
Project Record Drawings	212200	Record Drawings.			
	260500	1.4 - Record Drawings			
	260526	1.2 - As-Built Grounding Plans			
	265100	1.4 - Shop Drawings			
	270526	1.4 - Shop Drawings			
	270528	1.4 - Shop Drawings			
	270536	1.4 - Shop Drawings			
	271116	1.6 - Record Drawings			
	271513	1.4 - Shop Drawings			
	274100	1.8 - Shop Drawings			
	280001	Record Drawings.			
	281400	1.6 - Record Drawings			
	282000	1.6 - Record Drawings			
	331416	Record Drawings			
	210000	Instruction to Park Staff in proper operation of equipment and systems.			
	233423	Owner Training			
	238126	Owner Training			
System Demonstration and Training	238239	Owner Training			
	238239	Owner Training			
	260800	3.3 - 3.9 - Demonstration Cx Testing for Systems			
	260923	3.6 - Demonstration and Training			
	260943	3.6 - Demonstration and Training			
	270010	3.7 - Demonstration and Training			
	274100	3.37 - Demonstration and Testing			
	280001	Instruction to Park Staff in proper operation of equipment and systems.			
	280010	3.7 - Demonstration and Training			
	281600	3.9 - Demonstration and Training			
	282000	3.7 - Demonstration and Training			
Tools Spare Parts Equipment	066116	Solid Surface Fabrications - Commercial Care and Maintenance Kit			
	211300	Wrenches for each sprinkler type.			
	074616	Aluminum Cladding - Extra Stock Material			
	095123	Acoustical Tile Ceilings - Extra Stock Material			
	096519	Resilient Tile Rubber Flooring - Extra Stock Material			
	096813	Tile Carpeting - Extra Stock Material			
	099113	Exterior Painting - Extra Stock Material			
	099123	Interior Painting - Extra Stock Material			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
Extra Stock	099300	Staining and Transparent Finishing - Extra Stock Material			
	102113.19	Plastic Toilet Compartments - Extra Stock Materials			
	102239	Folding Panel Partitions - Maintenance Materials			
	105116	Wood Lockers - Maintenance Materials			
	211300	Spare Sprinklers.			
	271323	1.6.D.1 - Extra Materials for Patch Panels			
	271323	1.6.D.2 - Extra Materials for Jacks/Connectors			
	271323	1.6.D.1 - Extra Materials for Connecting Blocks			
	271323	1.6.D.2 - Extra Materials for Cover Plates			
	271323	1.6.D.3 - Extra Materials for Jacks			
	271323	1.6.D.4 - Extra Materials for Patch-Panel Units			
	281400	1.6.D.1 - Extra Materials for Credential Readers			
	281400	1.6.D.2 - Extra Materials for Special Tools			
	281600	1.6.E.1 - Extra Materials for Intrusion Detection Devices			
	281600	1.6.E.2 - Extra Materials for Fuses			
	281600	1.6.E.3 - Extra Materials for Tool Kit			
	281600	1.6.E.4 - Extra Materials for Security Fasteners			
Reports	210000	Approved submittal data with review letters.			
	210000	Copies of certificates of code authority inspections, acceptance, code required acceptance tests, letter of conformance, certificates of warranties.			
	210000	Include copy of Clean Agent System acceptance test.			
	210000	Include Commissioning Reports.			
	210000	Water supply flow test used as basis for hydraulic calculations.			
	210000	Hydraulic calculations.			
	210000	Contractor’s Material and Test Certificates for Aboveground Piping/Underground Piping.			
	230500	MFG technical representative/ Factory-authorized service representatives field reports			
	230593	Certified TAB reports			
	260519	1.2 - Test Reports			
	260526	1.2 - Test Reports			
	260923	1.4 - Test Reports			
	260943	1.4 - Test Reports			
	262416	1.4 - Test Reports			
	262713	1.4 - Test Reports			
	262816	1.4 - Test Reports			
	264113	1.4 - Test Reports			
	264313	1.4 - Test Reports			
	265100	1.4 - Test Reports			
	270526	1.5, 1.6 - Test Reports			
	270528	1.5 - Test Reports			
	270536	1.4 - Test Reports			
	271116	1.5, 1.6 - Test Reports			
	271323	1.5, 1.6 - Test Reports			
	271513	1.5, 1.6 - Test Reports			
	274100	1.8 - Written Field Updates including Photo Documentation			
	280001	Copy of approved submittal data with submittal review letters.			
	280001	Copy of start-up and test reports specific to each piece of equipment.			
	280001	Copies of certificates of code authority inspections, acceptance, code required acceptance tests, letter of conformance, certificates of warranties.			
	281300	1.5, 1.6 - Test Reports			
	281400	1.5, 1.6 - Test Reports			
	281600	1.5, 1.6 - Test Reports			
	282000	1.5, 1.6 - Test Reports			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
	140120.71	Copies of all inspection/acceptance certificates and operating permits as required			
	280800	NFPA 72 Record of Completion			
	087100	Copy of Approved Hardware Schedule			
Keys & Keying Schedule	087100	Copy of Approved Keying Schedule with Bitting List			
	087100	Door Hardware Supplier Name, Phone Number and Fax Number			
	102800	Toilet Accessories Maintenance Keys			
	284600	Spare Keys.			
	06 40 13	Exterior Architectural Woodwork - Quality Standard Compliance Certificates			
	06 41 00	Architectural Wood Casework - Quality Standard Compliance Certificates			
	06 61 16	Solid Surface Fabrications - Operational and Maintenance Data			
	06 61 16	Solid Surface Fabrications - Manufacturer's Warranty			
	07 14 16	Cold Fluid Applied Waterproofing - Manufacturer's Warranty			
	07 18 00	Traffic Coatings - Maintenance Data			
	07 18 00	Traffic Coatings - Manufacturer's Warranty			
	07 41 13.16	Standing Seam Metal Roof Panels - Maintenance Data			
	07 41 13.16	Standing Seam Metal Roof Panels - Manufacturer's Special Warranties			
	07 42 13.13	Formed Metal Wall Panels - Maintenance Data			
	07 42 13.13	Formed Metal Wall Panels - Manufacturer's Special Warranties			
	07 46 16	Aluminum Cladding - Maintenance Data			
	07 46 16	Aluminum Cladding - Manufacturer's Special Warranties			
	07 62 00	Sheet Metal Flashing - Manufacturer's Special Warranties			
	07 71 00	Roof Specialties - Maintenance Data			
	07 71 00	Roof Specialties - Manufacturer's Special Warranties			
	07 84 43	Joint Firestopping - Installer Certificates			
	08 14 16	Flush Wood Doors - Quality Standard Compliance Certificates			
	08 14 16	Flush Wood Doors - Manufacturer's Special Warranties			
	08 41 13	Aluminum Framed Entrances and Storefronts - Maintenance Data			
	08 41 13	Aluminum Framed Entrances and Storefronts - Manufacturer's Special Warranties			
	08 71 00	Door Hardware - Operations and Maintenance Data			
	08 71 00	Door Hardware - Manufacturer's Warranties			
	08 80 00	Glazing - Manufacturer's Special Warranties			
	09 65 19	Resilient Tile Rubber Flooring - Maintenance Data			
	09 65 19	Resilient Tile Rubber Flooring - Manufacturer's Special Warranties			
	09 67 23	Resinous Epoxy Flooring - Maintenance Data			
	09 67 23	Resinous Epoxy Flooring - Manufacturer's Warranties			
	09 68 13	Tile Carpeting - Maintenance Data			
	10 21 13.19	Plastic Toilet Compartments - Operations and Maintenance Data			
	10 22 39	Folding Panel Partitions - Operation and Maintenance Data			
	10 22 39	Folding Panel Partitions - Manufacturer's Special Warranties			
	10 28 00	Toilet Accessories - Maintenance Data			
	10 28 00	Toilet Accessories - Manufacturer's Special Warranties			
	10 44 13	Fire Protection Cabinets - Maintenance Data			
	10 44 13	Fire Protection Cabinets - Manufacturer's Warranty			
	10 44 16	Fire Extinguishers - Operation and Maintenance Data			
	10 44 16	Fire Extinguishers - Manufacturer's Warranty			
	10 75 16	Flagpole - Operations and Maintenance Data			
	11 00 00	Misc. Equipment - Maintenance Data			
	11 00 00	Misc. Equipment - Manufacturer's Warranties			
	11 30 13	Residential Appliances - Operation and Maintenance Data			
	12 24 13	Roller Window Shades - Operation and Maintenance Data			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
O&M Data Warranties Guarantees	12 24 13	Roller Window Shades - Manufacturer's Warranties			
	12 32 16	Manufactured Plastic Laminate Clad Casework - Manufacturer's Special Warranties			
	12 36 61	Simulated Stone Countertops - Maintenance Data			
	210000	Copies of certificates of code authority acceptance, code required acceptance tests, test reports and certificates.			
	21 00 00	Cut sheet for each Item of equipment			
	21 00 00	Record Drawings.			
	21 00 00	Product certificates of warranties and guarantees			
	21 00 00	Letter of Conformance			
	21 00 00	Testing and Maintenance Schedule Chart			
	21 00 00	Replacement parts lists and address of distributor.			
	21 00 00	Copy of NFPA 25.			
	21 00 00	Copy of valve charts and whether normally open or normally closed.			
	21 00 00	Copy of drain, auxiliary, and low point drains charts.			
	21 00 00	Copy of approved submittal data along with submittal review letters.			
	21 22 00	O&M Manuals			
	22 05 00	O&M Manuals			
	22 05 19	O&M Manuals			
	22 11 19	O&M Manuals			
	22 11 23	O&M Manuals			
	22 11 23	Warranty			
	22 13 19	O&M Manuals			
	22 33 00	O&M Manuals			
	22 42 13.13	O&M Manuals			
	22 42 13.13	Warranty			
	22 42 13.16	O&M Manuals			
	22 42 16.13	Warranty			
	22 42 16.13	O&M Manuals			
	22 42 16.13	Warranty			
	22 42 16.13	O&M Manuals			
	22 47 13	O&M Manuals			
	22 47 13	Warranty			
	23 11 26	O&M Manuals			
	23 33 00	O&M Manuals			
	23 34 23	O&M Manuals			
	237223.13	O&M Manuals			
	23 81 26	O&M Manuals			
	23 81 26	Warranty			
	23 82 39	O&M Manuals			
	238416.16	O&M Manuals			
	260923	1.4 - Operation and Maintenance Data			
	260943	1.4 - Operation and Maintenance Data			
	262713	1.4 - Operation and Maintenance Data			
	262726	1.4 - Operation and Maintenance Data			
	262816	1.4 - Operation and Maintenance Data			
	265100	1.4 - Operation and Maintenance Data			
	270526	1.6 - Operation and Maintenance Data			
	270536	1.5 - Operation and Maintenance Data			
	271116	1.6 - Operation and Maintenance Data			
	271323	1.6 - Operation and Maintenance Data			
	271513	1.6 - Operation and Maintenance Data			
	28 00 01	Approved submittal data with review letters; indicate installed equipment model numbers.			
	28 00 01	Copy of manufacturer's standard Operations and Maintenance for equipment.			
	28 00 01	Copy of complete parts list for equipment. Include available exploded views of assemblies and sub-assemblies.			
	28 00 01	Warranty.			
	28 00 01	Product certificates of warranties and guarantees.			
	329300	Maintenance Instructions			

Topic	Specification Section	Requirement	Submittal Date	Completed	Received by Park
	329300	Warranty			
	329200	Warranty			
	281300	1.6 - Operation and Maintenance Data			
	281400	1.6 - Operation and Maintenance Data			
	281600	1.6 - Operation and Maintenance Data			
	282000	1.6 - Operation and Maintenance Data			
	284600	Manuals containing the documentation required in NFPA 72; Completion Documentation.			
	284600	One year warranty agreement including parts and labor; begins upon date of substantial completion.			
	284600	Instruction chart.			

SECTION 01 78 23 – OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including:

1. Manuals, general
2. Emergency manuals
3. Operation manuals for systems, subsystems, and equipment
4. Maintenance manuals for care and maintenance of products, materials, finishes, and systems and equipment.

B. See Divisions 2 through 49 Sections for additional operation and maintenance manual requirements for Work in those Sections.

1.2 SUBMITTALS

A. Manual: Submit two copies of each manual in draft form or one electronic copy at least 15 days before final inspection. Contracting Officer (CO) will return copy or edit version with comments within 15 days of receipt.

B. Format: Submit operations and maintenance manuals in following format:

1. PDF (portable document format) electronic file. Assemble each manual into composite electronically indexed file. Submit on digital media acceptable to Contracting Officer.

   a. Name each indexed document file in composite electronic index with applicable item name. Include a complete electronically linked operation and maintenance directory.
   b. Enable inserted reviewer comments on draft submittals.

2. Hard copy manual: In accordance with Part 2 of this Section.

3. Correct or modify each manual to comply with Contracting Officers comments. Submit four [4] copies of each corrected manual within 15 [fifteen] days of receipt of Contracting Officers comments.

1.3 QUALITY ASSURANCE

A. Coordinate with Section 01 91 14 "Total Building Commissioning." Commissioning Agent shall review Operation and Maintenance Manuals for commissioned systems.

PART 2 - PRODUCTS

2.1 MANUALS, GENERAL

A. Organization: Unless otherwise indicated, organize manual into separate sections for each system and subsystem, and separate sections for each piece of equipment not part of a system. Manual shall contain title page, table of contents, and manual contents.

B. Title Page: Enclose title page in transparent plastic sleeve. Include:

1. Project Title
2. Location
3. Park
4. Contract Number
5. Prime Contractors Name and Address
6. Date of Substantial Completion
7. Binder Volume Number

C. Table of Contents: List each product included in manual, identified by product name, indexed to content of the volume, and cross-referenced to Specification Section number in Project Manual.

D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. Assemble instructions for subsystems, equipment, and components of one system into a single binder if needed.

1. Binders: White, commercial quality, hard back, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to hold 8-1/2 by 11 inch paper; with clear plastic window sleeve on front and spine to hold label describing contents and pockets inside covers to hold folded oversize sheets.

a. Cover Sheet: Identify binders on front and spine, with project title, location, park, contract number, prime contractor's name and address, date of substantial completion, and binder volume number. Insert cover sheet into clear plastic view pocket on front of binder. Insert sheet into clear plastic view pocket on spine with title "OPERATION AND MAINTENANCE MANUAL," Project title or name, and subject matter of contents. Indicate volume number for multiple-volume sets.

2. Data: Fill binders to no more than 75 percent of capacity. Punch holes shall not obscure any data. When contents of a single tabbed section cover more than one item, provide colored paper sheets to separate the data for each item.

a. Manufacturers' Data: Provide originals for color or copyrighted data. Black and white data may be originals or clean, good quality reproductions. No copies produced by facsimile transmission and sheets with stamps, such as submittal approval stamps. Include only sheets that apply to items installed; cross out inapplicable data.
b. Vendor Furnished As-Built Drawings: Maximum 24 inch by 36 inch sheets with minimum character or lettering size of 1/8 inch. Reduced-size reproductions may be provided instead of full-size drawings if reproductions are clear and legible. If

reduced-size drawings are used, identify as "REDUCED SIZE" and provide graphic scales, if applicable.

c. Custom Data: Data supplemented by drawings and schematics necessary to describe systems adequately.
d. Equipment Data Sheet: Data, using form at end of this section.
e. Schedules: Schedules reflecting final, as-installed conditions.
f. Poorly reproduced or illegible data will be rejected.

3. Dividers: Divider sheets with Mylar reinforced edges and pre-printed numbered tabs aligned with numbers and title lines on index sheet. Include typed list of products and major components of equipment included in section on each divider, cross-referenced to Specification Section number and title of Project Manual.
4. Protective Plastic Sleeves: Transparent plastic sleeves designed to enclose diagnostic software diskettes for computerized electronic equipment.
5. Drawings: Attach reinforced, punched binder tabs on drawings and bind with text.

a. If oversize drawings are necessary, fold drawings to same size as text pages and use as foldouts.
b. If drawings are too large to be used as foldouts, fold and place drawings in labeled envelopes and bind envelopes in rear of manual. At appropriate locations in manual, insert typewritten pages indicating drawing titles, descriptions of contents, and drawing locations.

2.2 EMERGENCY MANUALS

A. Content: Organize manual into separate section for type of emergency, emergency instructions, and emergency procedures.

B. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component for fire, flood, gas leak, water leak, power failure, water outage, equipment failure, and chemical release or spill.

C. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of National Park Service (NPS) operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.

D. Emergency Procedures: Include instructions on stopping, shutdown instructions for each type of emergency, operating instructions for conditions outside normal operating limits, and required sequences for electric or electronic systems.

2.3 OPERATION AND MAINTENANCE MANUALS

A. Operation Requirements

1. Content: In addition to requirements in Section, include operation data required in individual Specification Sections and equipment descriptions, operating standards, operating procedures, operating logs, wiring and control diagrams, and license requirements.

2. Descriptions: Include:

    a. Product name and model number
    b. Manufacturer's name
    c. Equipment identification with serial number of each component
    d. Equipment function
    e. Operating characteristics
    f. Limiting conditions
    g. Performance curves
    h. Engineering data and tests
    i. Complete nomenclature and number of replacement parts

3. Operating Procedures: Include start-up, break-in, and control procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; and required sequences for electric or electronic systems.
4. Systems and Equipment Controls: Describe sequence of operation, and diagram controls as installed.
5. Piped Systems: Diagram piping as installed and identify color-coding where required for identification.

B. Maintenance Requirements for Systems and Equipment

1. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, and equipment data sheets as described below.
2. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
3. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including maintenance instructions, drawings and diagrams for maintenance, nomenclature of parts and components, and recommended spare parts for each component part or piece of equipment:
4. Maintenance Procedures: Test and inspection instructions, troubleshooting guide, disassembly instructions, and adjusting instructions, and demonstration and training videotape if available, detailing essential maintenance and environmental procedures.
5. Maintenance and Service Schedules: Service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
6. Spare Parts List and Source Information: Lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
7. Warranties and Bonds: Copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.4 PRODUCT MAINTENANCE MANUAL

A. Content: Organize manual into separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.

B. Source Information: List each product included in manual identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.

C. Product Information: Include:

1. Product name and model number
2. Manufacturer's name
3. Color, pattern, and texture
4. Material and chemical composition
5. Reordering information for specially manufactured products

D. Environmental Requirements

1. Identify environmentally preferable products incorporated into Project. Include: product model; manufacturer’s name, address, phone, and website; and local technical representative.

a. Verify plastic products to be incorporated into Project are labeled in accordance with ASTM D1972. Where products are not labeled, provide product data indicating polymeric information in Operation and Maintenance Manual.

1) Type 1: Polyethylene Terephthalate (PET, PETE)
2) Type 2: High Density Polyethylene (HDPE)
3) Type 3: Vinyl (Polyvinyl Chloride or PVC)
4) Type 4: Low Density Polyethylene (LDPE)
5) Type 5: Polypropylene (PP)
6) Type 6: Polystyrene (PS)
7) Type 7: Other. Use of this code indicates that package in question is made with a resin other than the six listed above or is made of more than one resin listed above and used in a multi-layer combination.

b. Describe maintenance procedures associated with environmentally preferable materials and systems. Provide cleaning recommendations in accordance with ASTM E1971 and approved Integrated Pest Management (IPM) plan.

1) Include potential environmental impacts of recommended maintenance procedures and materials.
2) Include potential indoor air quality impacts of recommended maintenance procedures and materials.
3) Where proposed maintenance procedures incorporate composting of plastics, assess potential effect of each type of plastic to be included in composting process in accordance with ASTM D5509 or ASTM D6002

c. Identify take-back programs and appropriate contact information for:

1) Carpet
2) Ceiling Tile
3) Office Equipment

d. Material Safety Data Sheets (MSDS): Include MSDSs as specified.

2. Develop environmental management programs for facility as follows:

a. Waste management program: Develop in accordance with ASTM E1609. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.
b. Indoor Air Quality (IAQ) management program: Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of VOCs (volatile organic compounds) in indoor air in accordance with ASTM D6345.
c. Water management program: Develop water monitoring program for surface and ground water on project site in accordance with ASTM D5851 and consistent with water management program utilized during construction operations.

E. Maintenance Procedures: Include manufacturer's written recommendations and inspection procedures, types of cleaning agents, methods of cleaning, schedule for cleaning and maintenance, and repair instructions.

F. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.

G. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.5 SYSTEMS AND EQUIPMENT MAINTENANCE MANUALS

A. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, as described below.

B. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.

C. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including the following information for each component part or piece of equipment:

1. Standard maintenance instructions and bulletins.
2. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
3. Identification and nomenclature of parts and components.

4. List of items recommended to be stocked as spare parts.

D. Maintenance Procedures: Include the following and items detailing essential maintenance procedures:

1. Test and inspection instructions
2. Troubleshooting guide
3. Precautions against improper maintenance
4. Disassembly; component removal, repair, and replacement; and reassembly instructions
5. Aligning, adjusting, and checking instructions
6. Demonstration and training video recording

E. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.

1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.

F. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.

G. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.

H. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

1. Include procedures and required notifications for warranty claims.

PART 3 - EXECUTION

3.1 GENERAL

A. At start of project, begin accumulating operation and maintenance data and initiate index. Install and index data in binders within 30 days after delivery of items. As custom written data and test results are produced, add to operation and maintenance data file.

B. List of Operation and Maintenance requirements has been attached at end of the Division 1 Specifications for your convenience. Intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list.

C. Keep operation and maintenance data current. Make operation and maintenance binders available to Contracting Officer for inspection at time of monthly progress payment requests. If operation and maintenance binders are not current, Contracting Officer may retain an appropriate amount of the progress payment.

3.2 MANUAL PREPARATION

A. Manual Types

1. Emergency Manual: Assemble complete set of emergency information indicating procedures for use by emergency personnel and by NPS operating personnel for types of emergencies indicated.
2. Product Maintenance Manual: Assemble complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into Work.
3. Operation and Maintenance Manuals: Assemble complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.

B. Manual Contents: Including:

1. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark sheet to identify product or component incorporated into Work. If data include more than one item in a tabular format, identify each item using appropriate references from Contract Documents. Identify data applicable to Work and delete references to information not applicable.
2. Custom Written Data: For data not in manufacturer's standard literature, provide text, drawings, and schematics specifically applicable to installed systems. Include step-by-step descriptions of operating procedures; identification of individual components and their functions; descriptions of how system components relate to one another and operate together to accomplish a common process or function; and sequence of operation for system control circuits. For seasonally operated systems, provide start-up and shutdown instructions.
3. Equipment Data Sheets: For each item of equipment included in operation and maintenance data, provide Equipment Data Sheet using form at the end of this section. For equipment consisting of a driven machine and a driver (for example, a pump and a motor), equipment data shall cover both the driven machine and the driver. For similar type equipment (for example, multiple exhaust fans of the same model and type), provide a single equipment data sheet with an attached schedule listing individual equipment items.
4. Vendor Furnished As-Built Drawings: Provide for each electrical and each mechanical control system.

    a. For each control system, provide control circuit schematic drawings. Identify each wire and terminal block number. Show terminal numbers on control devices. Show control wires and devices remote from control panel.
    b. For each control panel, provide general arrangement drawing showing location of each control component and terminal block on the panel front and interior. Include materials list of panel-mounted control components as well as field-installed control components remote from the panel, identifying components, manufacturer, model number, and initial set points or sensing ranges of devices where applicable.

c. For packaged equipment systems, provide general arrangement drawings showing interrelationships of the various items of equipment and components.
d. In addition to control wiring schematic, provide power wiring schematic drawing showing power flow to each motor. Identify each power conductor. Show over-current protection and motor starting devices.

C. Comply with Section 01 77 00 "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

END OF SECTION 01 78 23

EQUIPMENT DATA SHEET	
Equipment Item: _____ Designation: _____ Function: _____ Location: _____ Project: _____ Model Number: _____ Serial Number: _____	
Manufacturer Address and Phone:	Supplier Address and Phone:
Preventive Maintenance Tasks:	
Nameplate Data:	
Spare Parts Furnished and Other Information:	

SECTION 01 79 00 – DEMONSTRATION AND TRAINING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for instructing National Park Service (NPS) personnel, including:

1. Demonstration of operation of systems, subsystems, and equipment.
2. Training in operation and maintenance of systems, subsystems, and equipment, including environmental considerations.
3. Demonstration and training video.

B. See Divisions 2 through 49 Sections for specific requirements for demonstration and training for products in those Sections.

1.2 GENERAL REQUIRMENTS

A. List of System Demonstration and Training requirements has been attached at end of Division 1 Specifications for your convenience. Intent is to provide overall summary of requirements and not a comprehensive list. Terms and conditions of contract still require satisfaction of requirements of individual specification sections regardless of what is shown on list.

1.3 SUBMITTALS

A. Instruction Program: Submit two copies of outline of instructional program for demonstration and training, including schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.

1. For each training session, Contractor shall submit for approval a proposed outline of subjects to be covered. Training shall not be conducted until outline is approved.

B. Demonstration and Training Video: Submit two copies of each DVD (digital video disc) for training sessions within seven days of end of each training module.

1. Label each DVD with date of demonstration or training, instructor's name, and provide an index of contents. Index shall list start and end time of each subject covered during training session. Sequence of training subjects shall follow sequence listed in approved training outline or as actually conducted

1.4 QUALITY ASSURANCE

A. Facilitator Qualifications: Firm or individual experienced in training or educating maintenance personnel in training program similar in content and extent for this Project, and whose work has resulted in training or education with a record of successful learning performance.

B. Instructor Qualifications: Factory-authorized service representative, complying with requirements in Section 01 40 00 "Quality Requirements," experienced in operation and maintenance procedures and training.

C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by Contracting Officer (CO).

PART 2 - PRODUCTS

2.1 INSTRUCTION PROGRAM

A. Program Structure: Develop instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, and:

1. Fire alarm systems
2. HVAC systems
3. Lighting systems
4. Audio/Visual systems
5. Glazing systems
6. Environmental Topics

a. Overview of environmental issues related to building industry.
b. Overview of environmental issues related to Project.
c. Review of site-specific procedures and management plans implemented during construction:

1) Regulatory Requirements
2) Indoor Air Quality (IAQ) Management
3) Noise and Acoustics Management
4) Environmental Management
5) Construction Waste Management

d. Review of site-specific procedures and management plans to be implemented during operation and maintenance.

1) Include review of environmentally related aspects of Operations and Maintenance Manual.
2) Integrated Pest Management (IPM)

B. Training Modules: Develop a learning objective and teaching outline for each module. Include description of specific skills and knowledge that participant is expected to master. For each module, include instruction for:

1. Basis of System Design, Operational Requirements, and Criteria: Include system and equipment descriptions, operating standards, regulatory requirements, equipment function, operating characteristics, limiting conditions, and performance curves.
2. Documentation: Review emergency, operations, and maintenance manuals; Project Record Documents; identification systems; warranties and bonds; and maintenance service agreements.
3. Emergencies: Instructions on stopping; shutdown instructions; operating instructions for conditions outside normal operating limits; instructions on meaning of warnings, trouble indications, and error messages; and required sequences for electric or electronic systems.
4. Operations: Startup, break-in, control, and safety procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; operating procedures for emergencies and equipment failure; and required sequences for electric or electronic systems.
5. Adjustments: Alignments and checking, noise, vibration, economy, and efficiency adjustments.
6. Troubleshooting: Diagnostic instructions and test and inspection procedures.
7. Maintenance: Inspection procedures, types of cleaning agents, methods of cleaning, procedures for preventive and routine maintenance, and instruction on use of special tools.
8. Repairs: Diagnosis, repair, and disassembly instructions; instructions for identifying parts; and review of spare parts needed for operation and maintenance.

PART 3 - EXECUTION

3.1 INSTRUCTION

A. Facilitator: Engage qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Contracting Officer for number of participants, instruction times, and location.

B. Engage qualified instructors to instruct NPS personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.

C. Scheduling: Provide instruction at mutually agreed on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.

1. Schedule training with NPS through Contracting Officer with at least [seven] days' advance notice.
2. Conduct training sessions after equipment or system has been accepted and turned over to Government. Coordinate with commissioning requirements.
3. Coordinate with Integrated pest management requirements. Refer to specifications section and approved IPM plan.
4. Individual sections specify duration of training required. If no duration is listed, provide training of sufficient duration to adequately cover subjects.

3.2 DEMONSTRATION AND TRAINING VIDEO

A. General: Engage qualified commercial photographer to record demonstration and training video. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice.

1. At beginning of training module, record each chart containing learning objective and lesson outline.

B. Video Format: Digital Video Disc (DVD).

C. Video Recording: Record sessions with high resolution equipment. Instructor's voice shall be clearly audible and understandable on DVD. Utilize supplemental microphone worn by instructor.

D. Narration: Describe scenes on video by audio narration by microphone while video is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

1. DVDs with poor video or audio quality will be rejected and training recorded again.

END OF SECTION 01 79 00

EFMO – N170 318539 / 224159

SECTION 01 81 13 – SUSTAINABLE DESIGN REQUIREMENTS FOR NON-LEED™ PROJECTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes general requirements and procedures for compliance with Federal Sustainability requirements. This project is not seeking Leadership in Energy & Environmental Design (LEED™) certification but shall comply with applicable Federal Sustainability requirements. Requirements include laws (Executive Orders (EO) and regulations), management policies, building codes and standards, Federal directives, and National Park Service (NPS) guidelines.

B. Many Federal requirements can be achieved only through intelligent and integrated design of the project and are beyond control of the Contractor. Certain requirements relate to the products and procedures used for construction, therefore, full cooperation of the Contractor and Subcontractors is essential to successful compliance with Federal requirements.

C. Contractors shall familiarize themselves with relevant requirements and provide necessary information and instruction to subcontractors and installers.

1. Some requirements involve quantifying percentages by weight; these require careful recordkeeping and reporting by Contractor.
2. See Denver Service Center (DSC) Workflows Sustainability Standards for a list of Federal Sustainability requirements. Applicable Federal Sustainability requirements are also summarized on the project's NPS Project Sustainability Checklist. Contractor is responsible for ensuring the elements in the NPS Project Sustainability Checklist, identified by the Architect/Engineer (A/E) team, are incorporated into the construction of the project.

D. Related Sections:

1. See Divisions 1 through 49 Sections for sustainability requirements specific to work of each of these Sections.

1.2 DEFINITIONS

A. Chain-of-Custody Certificates: Certificates signed by manufacturers certifying wood used to make products was obtained from forests certified by a Forest Stewardship Council (FSC)-accredited certification body to comply with FSC Standard STD-01-001, "FSC Principles and Criteria for Forest Stewardship." Certificates shall include evidence manufacturer is certified for chain of custody by an FSC-accredited certification body.

B. LEED™: Leadership in Energy & Environmental Design. Sustainability rating system developed by United States Green Building Council (USGBC).

C. Rapidly Renewable Materials: Materials made from plants typically harvested within 10-year or shorter cycle. Rapidly renewable materials include products made from bamboo, cotton, flax, jute, straw, sunflower seed hulls, vegetable oils, or wool.

D. Recycled Content: Recycled content value of a material assembly shall be determined by weight.

1. "Post-consumer" material: Waste material generated by households or by commercial, industrial, and institutional facilities in their role as end users of product, which can no longer be used for intended purpose.
2. "Pre-consumer" material: Material diverted from waste stream during manufacturing process. Reutilization of materials such as rework, regrind, or scrap generated in a process and capable of being reclaimed within the same process that generated it is excluded.

E. Biopreferred Products: Commercial or industrial products (other than food or feed) composed in whole, or in significant part, of biological products, renewable agricultural materials (including plant, animal, and marine materials), or forestry materials and includes biobased intermediate ingredients or feedstocks.

1.3 FEDERAL SUSTAINABILITY DOCUMENTATION SUBMITTALS

A. Most of Federal sustainability documentation submittals are aggregations of submittals already required in relevant technical specifications. They are mentioned here to ensure they are collected and organized together to efficiently document compliance with sustainability requirements.

B. Provide preliminary submittals to NPS indicating how the following Federal requirements will be met:

1. Recycled Content: List of specified/proposed materials with recycled content. Indicate cost, post-consumer recycled content, and pre-consumer recycled content for each product having recycled content.

2. Certified Wood: Product data and/or chain-of-custody certificates for products containing certified wood. Include statement indicating cost for each certified wood product.

3. Construction Indoor Air Quality (IAQ) Management Plan – During Construction:

a. Construction indoor-air-quality management plan.
b. Product data for temporary filtration media.
c. Product data for filtration media used during occupancy.
d. Construction Documentation: Six photographs at three different times during construction period, along with brief description of the Sheet Metal and Air Conditioning Contractors' National Association (SMACNA) approach employed, documenting implementation of indoor-air-quality management measures, such as protection of ducts and on-site stored or installed absorptive materials.

4. Construction IAQ Management Plan – Before Occupancy:

a. Signed statement describing building air flush-out procedures including dates when flush-out was started and completed and statement filtration media was replaced after flush-out.

b. Product data for filtration media used during flush-out and during occupancy.

5. Low Emitting Materials - Adhesives and Sealants: Product data for adhesives and sealants used inside weatherproofing system indicating Volatile Organic Compound (VOC) content of each product used. Indicate VOC content in g/L (grams per liter) calculated according to 40 CFR 59 Subpart D (Code of Federal Regulations).
6. Low Emitting Materials - Paints and Coatings: Product data for paints and coatings used inside weatherproofing system indicating chemical composition and VOC content of each product used. Indicate VOC content in g/L calculated according to 40 CFR 59 Subpart D.
7. Low Emitting Materials - Flooring: Product data for products containing composite wood or agrifiber products or wood glues indicating they do not contain urea-formaldehyde resin.
8. Biopreferred Products: Provide list of bio-based products used on project.

PART 2 - PRODUCTS

2.1 RECYCLED CONTENT OF MATERIALS

A. Recycled Content: Provide building materials with recycled content such that post-consumer recycled content plus one-half of pre-consumer recycled content constitutes a minimum of [10] percent of cost of materials used for Project.

1. Determine cost of post-consumer recycled content by dividing weight of post-consumer recycled content in item by total weight of item and multiplying by cost of item.
2. Determine cost of pre-consumer recycled content of an item by dividing weight of pre-consumer recycled content in item by total weight of item and multiplying by cost of item.
3. Do not include furniture, mechanical and electrical components, and specialty items such as elevators and equipment in calculation.

2.2 BIOPREFERRED PRODUCTS

A. Use bio-based products found on United States Department of Agriculture (USDA) Biopreferred Products list where applicable on project.

2.3 LOW-EMITTING MATERIALS

A. For applications inside the weatherproofing system, use adhesives and sealants that comply with VOC content limits in Specification Divisions 2 through 49.

B. For field applications inside the weatherproofing system, use paints and coatings that comply with VOC content limits in Specification Divisions 2 through 49.

C. Do not use composite wood or agrifiber products or adhesives that contain urea-formaldehyde resin.

PART 3 - EXECUTION

3.1 REFRIGERANT AND CLEAN-AGENT FIRE-EXTINGUISHING-AGENT REMOVAL

A. Remove Chlorofluorocarbons **(**CFC)-based refrigerants from existing Heating, Ventilation, Air Conditioning, and Refrigeration (HVAC&R) equipment indicated to remain and replace with refrigerants that are not CFC based. Replace or adjust existing equipment to accommodate new refrigerant as described in Division 23 Sections.

B. Remove clean-agent fire-extinguishing agents that contain Hydrochlorofluorocarbons (HCFCs) or halons and replace with agent that does not contain HCFCs or halons. See Division 21 Section “Clean-Agent Fire Extinguishing Systems” for additional requirements.

3.2 MEASUREMENT AND VERIFICATION

A. Coordinate with Divisions 2 through 49 for project requirements regarding installation of building level metering equipment to measure energy, water, and electric usage.

3.3 INDOOR-AIR-QUALITY MANAGEMENT

A. Coordinate with Section 01 57 19.11 "Indoor Air Quality Management" for managing indoor air quality during construction and prior to occupancy.

END OF SECTION 01 81 13

SECTION 01 91 14 – TOTAL BUILDING COMMISSIONING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. General requirements for coordinating and scheduling commissioning.
2. Commissioning meetings
3. Commissioning reports
4. Use of test equipment, instrumentation, and tools for commissioning.
5. Construction checklists, including, but not limited to, installation checks, startup, and performance tests.
6. Commissioning tests
7. Adjusting, verifying, and documenting identified systems and assemblies.

Work included under this section includes a complete and thorough investigation of equipment and systems indicated in Part 3 of section. In order to ensure proper installation and operation of components and systems. Contractor shall perform commissioning as described herein to accomplish the tasks, and goals of commissioning. Systems to be evaluated include but are not limited to:

8. HVAC (Heating, Ventilation, and Air Conditioning) components and equipment.
9. HVAC system: Interaction of cooling, heating, and comfort delivery systems.
10. Building Automation System (BAS): Control hardware and software, sequence of operations, and integration of factory controls with BAS.
11. Lighting Control System and interface with daylighting.
12. Building Envelope (walls, roof, windows, infiltration, etc.)
13. Life Safety Systems (Fire Alarm & Suppression)
14. Electrical and Alternative Energy Systems (propane generator, photovoltaic panels, etc.)
15. Access Control/ Security Systems
16. Lightning Protection

B. Building commissioning activities and documentation are described in the following reference material: United States Green Building Council (USGBC) Leadership in Energy and Environmental Design (LEED™) rating program, American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE) Guideline 0-2005, The Commissioning Process, and National Institute of Building Sciences (NIBS) Guidelines.

C. National Park Service (NPS) personnel, Green Consultant, and Architect/Engineer, are not responsible for construction means, methods, job safety, or management function related to commissioning on job site.

D. Related Sections:

1. 01 31 00 - Project Management and Coordination
2. 01 33 23 - Submittal Procedures
3. 01 40 00 - Quality Requirements
4. 01 57 19.11 - Indoor Air Quality (IAQ) Management
5. 01 57 19.12 - Noise and Acoustics Management
6. 01 77 00 - Closeout Procedures
7. 01 78 23 - Operation and Maintenance Data
8. 01 79 00 - Demonstration and Training
9. 01 81 13 - Sustainable Design Requirements

1.2 DEFINITIONS

A. Acceptance Criteria: Threshold of acceptable work quality or performance specified for a commissioning activity.

B. Basis-of-Design Document: Document prepared by Designer that records concepts, calculations, decisions, and product selections used to comply with Government's Project Requirements and to suit applicable regulatory requirements, standards, and guidelines.

C. Total Building Commissioning (TBC): Quality-focused process for verifying and documenting that facility, systems and assemblies are planned, designed, installed, and tested to comply with Government's Project Requirements. Requirements specified here are limited to construction phase commissioning activities.

D. Construction Checklist: Form used by Contractor to verify appropriate components are on site, ready for installation, correctly installed and functional.

E. Contractor’s Commissioning Representative: (CCxR) Contractor’s designated individual to coordinate, manage, and execute commissioning processes of the contracting organizations.

F. Commissioning Plan (CCxP): Plan that provides structure, schedule and coordination planning for commissioning process proposed specifically for this project. CCxP includes Personnel, activities, and a description of Infrastructure, and list of instruments and logging devices that will be used during Commissioning.

G. Deficiency: Condition in the installation or function of a component, piece of equipment or system not in compliance with Contract Documents, does not perform properly or is not complying with Basis of Design.

H. Functional Performance Test (FPT): Test of dynamic function and operation of equipment and systems using manual (direct observation) or monitoring methods. Functional testing is the dynamic testing of systems (rather than just components) under full operation (e.g., the chiller pump is tested interactively with the chiller functions to see if the pump ramps up and down to maintain the differential pressure setpoint). Systems are tested under various modes, such as during low cooling or heating loads, high loads, component failures, unoccupied, varying outside air temperatures, fire alarm, power failure, etc. The systems are run through all the control system’s sequences of operation and components are verified to be responding as the sequences state. Traditional air or water test and balancing (TAB) is not functional performance testing, in

the commissioning sense of the word. TAB's primary work is setting up the system flows and pressures as specified, while functional testing is verifying that which has already been set up. The CCxR develops the sequentially written functional test procedure forms, and oversees and documents the actual testing, which is performed by the installing contractor or vendor. The CCxR creates worksheets from these forms which include procedures required to accommodate actual equipment, means and methods used in the project. Functional Performance Tests are performed after pre-functional checklists and startup is complete.

I. Manual Test: Using hand-held instruments, control system readouts or direct observation to verify performance (contrasted to analyzing electronically monitored data taken over time to make the "observation").

J. Monitoring: Recording of parameters (flow, current, status, pressure, etc.) of equipment operation using data loggers or the trending capabilities of control systems.

K. Government's Project Requirements: Document originated by Designer that details functional requirements of project and expectations of use and operation, including project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information. Document is updated, with input from Contracting Officer (CO) as required as project is finished.

L. Pre-functional Checklist: List of items to inspect and elementary component tests to conduct to verify proper installation of equipment. Pre-functional checklists are primarily static inspections and procedures to prepare equipment or system for initial operation (e.g., belt tension, oil levels ok, labels affixed, gages in place, sensors calibrated, etc.). However, some pre-functional checklist items entail simple testing of the function of a component, a piece of equipment or system (such as measuring voltage imbalance on a three-phase pump motor of a chiller system). Pre-functional" refers to "before" functional testing. Pre-functional checklists augment and are combined with the equipment manufacturer's start-up checklist.

M. Seasonal Performance Tests: Functional Performance Tests deferred until system(s) will experience seasonal conditions closer to their design conditions.

N. Systems Manual: System focused composite document that includes operational manual, maintenance manual, and additional information of use to Government during Occupancy and Operation Phase.

1.3 COMMISSIONING TEAM

A. Members Appointed by Contractor(s): Individuals, each having authority to act on behalf of the entity he or she represents, explicitly organized to implement the commissioning process through coordinated action.

1. CCxR (Contractor's Commissioning Representative): CCxR shall be approved by Contracting Officer and satisfy as many of the following requirements as possible:

a. Certified in Commissioning by nationally accredited organization (i.e. Associated Air Balance Council (AABC), Association of Energy Engineers (AEE), Building Commissioning Association (BCA), and National Environmental Balancing Bureau (NEBB))

b. Acted as principal Commissioning Authority where total building commissioning approach (including building envelope) was used for at least three projects of comparable size, type, and scope.
c. Technical training in Mechanical, Electrical, and/or fire protection engineering.
d. Past commissioning experience.
e. Knowledge of national codes.
f. Leadership in Energy and Environmental Design (LEED) Accredited Professional.
g. Experience in energy-efficient design and control strategy optimization.
h. Specific experience with specialty systems relative to particular facility type (i.e. Federal blast and progressive collapse requirements, security systems, etc.).

2. Contractor Quality Control (CQC) Supervisor
3. Other Representatives may include Project superintendents, installers, suppliers, and specialists.

B. Members Appointed by Contracting Officer:

1. Representatives of facility user and operation and maintenance personnel.
2. Architect and engineering design professionals.

1.4 CONTRACTOR'S RESPONSIBILITIES

A. Contractor shall assign representatives with expertise and authority to act on its behalf to participate in and perform commissioning process activities including:

1. Perform commissioning tests, as required by technical specifications. Evaluate performance deficiencies identified in test reports and, in collaboration with entity responsible for system and equipment installation, recommend corrective action.
2. Record and resolve commissioning issues.
3. Attend commissioning team meetings held on a weekly basis.
4. Integrate and coordinate commissioning process activities with overall project schedule.
5. Review Construction Checklist attached at end of specification section.
6. Complete electronic construction checklists as contract work is completed and provide to Contracting Officer on a weekly basis.
7. Complete commissioning process test procedures.
8. Provide maintenance orientation and inspection for systems, assemblies, equipment, and components based on contract requirements.
9. Provide Commissioning Plan and documentation for final commissioning documentation.
10. Provide measuring instruments and logging devices to record test data and provide data acquisition equipment to record data for complete range of testing for required test period.

1.5 COMMISSIONING DOCUMENTATION

A. Provide the following information:

1. Review of systems manual, submittals, documents, and other commissioning reports
2. Identification of installed systems, assemblies, equipment, and components including design changes that occurred during the construction phase

3. Commissioning Plan including Process activities and schedule for completing construction checklists and manufacturer's pre-start and startup checklists for systems, assemblies, equipment, and components to be verified and tested
4. Certificate of readiness certifying systems, subsystems, equipment, and associated controls are ready for testing
5. Test and inspection reports and certificates
6. Corrective action documents
7. Testing, adjusting, and balancing reports

1.6 SUBMITTALS

A. Two-week look-ahead schedules: Schedule showing the next two weeks of commissioning related construction activity to include completion dates for each element of commissioning documentation for each major system or subsystem as identified in 1.1.B.

B. Certificates of readiness.

C. Contractor’s Commissioning Representative Qualifications.

D. Commissioning Plan: Submit within 30 calendar days of authorization to proceed.

1. Update as necessary during the work to reflect progress on components and systems.

E. Pre functional checklists.

F. Functional performance test forms: Submit minimum 30 calendar days prior to testing

G. List of test instrumentation, equipment, and monitoring devices. Include:

1. Make, model, serial number, and application for each instrument, equipment, and monitoring device.
2. Brief description of intended use.
3. Calibration record showing:

a. Calibration agency, including name and contact information
b. Last date of calibration
c. Range of values for which calibration is valid
d. Certification of accuracy
e. National Institute of Standards and Technology (NIST) traceability certification for calibration equipment.
f. Due date of the next calibration.

H. Deficiency Report and Resolution Record: Document items of non-compliance in materials, installation or operation. Document results from start-up/pre-functional checklists, functional performance testing, and short-term diagnostic monitoring. Include details of components or systems found to be non-compliant with drawings and specifications. Identify adjustments and alterations required to correct system operation and identify who is responsible for making corrective changes.

1. Update as necessary during work to reflect progress on components and systems. Submit updated versions monthly.

I. Closeout Documentation

1. Closeout documents for commissioned equipment and systems shall be submitted prior to functional performance testing. These include:

   a. Record Documents and Drawings
   b. Start-up certificates for commissioned equipment with start-up requirements
   c. Systems Manual
   d. Include TAB, startup, and Control System check-out reports.

2. Operation and Maintenance (O&M) Submittals (refer to requirements of technical specifications):

   a. Training plan: Include for each training session:

      1) Dates, start and finish times, and locations
      2) Outline of the information to be presented
      3) Names and qualifications of presenters
      4) List of texts and materials required to support training

PART 2 - PRODUCTS

2.1 TEST EQUIPMENT

A. Instrumentation shall:

1. Be of sufficient quality and accuracy to test and measure system performance within tolerances required to determine adequate performance.
2. Be calibrated on manufacturer's recommended intervals calibration tags permanently affixed to instrument being used.
3. Be maintained in good repair and operation condition throughout duration of use on this project.

B. Standard testing equipment required to perform startup and initial checkout and required functional performance testing shall be provided by Contractor for equipment being tested.

C. Required commissioning equipment (sensors, transducers, data loggers, etc.) not integral to the systems or equipment installed shall be provided by Contractors Commissioning Representative and shall not become property of the Government.

2.2 PRE-FUNCTIONAL CHECKLIST:

A. Prepare pre functional checklists for equipment and systems to be commissioned.

B. Pre-functional checklists shall be complementary to Commissioning Plan and Commissioning Schedule.

2.3 FUNCTIONAL TEST PROCEDURE FORMS: Prepare functional test procedure forms for each piece of equipment and each system to be commissioned.

2.4 FUNCTIONAL PERFORMANCE WORKSHEETS:

A. Prepare Functional Performance worksheets, consisting of test procedures and expected results of testing.

2.5 REPORT FORMAT AND ORGANIZATION

A. General Format and Organization:

1. Bind report in three-ring binders.
2. Label front cover and spine of each binder with report title, volume number, project name, Contractor's name, and date of report.
3. Record report on compact disk.
4. Electronic Data: Portable document format (PDF); a single file with outline-organized bookmarks for major and minor tabs and tab contents itemized for specific reports.

B. Commissioning Report:

1. Include table of contents and an index to each test.
2. Include major tabs for each Specification Section.
3. Include minor tabs for each test.
4. Within each minor tab, include:

a. Test specification.
b. Pre-startup reports.
c. Approved test procedures.
d. Test data forms, completed and signed.
e. Commissioning issue reports, showing resolution of issues, and documentation related to resolution of issues pertaining to a single test. Group data forms, commissioning issue reports showing resolution of issues, and documentation related to resolution of issues for each test repetition together within minor tab, in reverse chronological order (most recent on top).

PART 3 - EXECUTION

3.1 COMMISSIONING PROCESS

A. Following activities outline general commissioning tasks (requiring development, execution, etc.) and order in which they occur. Specific Commissioning requirements are found in technical specification Section(s) 21 08 00, 28 08 00, 26 08 00.

1. Commissioning Scoping Meeting
2. Finalize Government’s Project Requirements
3. Commissioning Plan
4. Prepare pre-functional checklists.

5. Prepare functional performance worksheets.
6. Perform Start-Up/Pre-Functional Checks in accordance with manufacturer's recommendations and pre-functional checklists.
7. Functional Performance Testing in accordance with functional performance worksheets
8. Deficiency Report and Resolution Record
9. Operation and Maintenance Documentation
10. Operations and Maintenance Training
11. Deferred Testing

3.2 TOTAL BUILDING COMMISSIONING (TBC) REQUIREMENTS

A. TBC during construction, acceptance, and warranty phases is intended to achieve following specific objectives:

1. Verify that systems and equipment meet Government's Project Requirements.
2. Verify equipment is what was submitted and approved.
3. Verify and document equipment is installed and started per manufacturer's recommendations, industry accepted minimum standards, and Contract Documents.
4. Verify and document equipment and systems receive complete operational checkout by installing contractors.
5. Verify and document equipment capacity and system efficiency.
6. Verify performance of building envelope. Document testing and conformance to Contract Documents.
7. Verify completeness of operations and maintenance materials.
8. Ensure Governments operating personnel are adequately trained on operation and maintenance of building equipment.

3.3 COMMISSIONING SCOPING MEETING

A. Commissioning Scoping Meeting:

1. Schedule, coordinate, and facilitate a scoping meeting.
2. Review each building system to be commissioned, including intended operation, commissioning requirements, and completion and start-up schedules.
3. Establish scope of work, tasks, schedules, deliverables, and responsibilities for implementation of Commissioning Plan.
4. Attendance: Commissioning Team members.

3.4 COMMISSIONING PLAN

A. Commissioning Plan: Develop commissioning plan to identify how commissioning activities will be integrated into general construction and trade activities. Commissioning plan shall identify how commissioning responsibilities are distributed. Intent of plan is to evoke questions, expose issues, and resolve issues with input from entire commissioning team early in construction.

1. Identify who will be responsible for producing various procedures, reports, Contracting Officer notifications and forms.

2. Include commissioning tasks and activities in overall project schedule. Tag individual activities so they can be filtered at later date.
3. List and describe each test/acceptance procedure, including acceptance criteria.

3.5 START-UP/PRE-FUNCTIONAL CHECKLISTS

A. Start-Up/Pre-Functional Checklists: Complete pre-functional checklists prior to start up. Checklist shall help verify that systems are complete and operational, so functional performance testing can be scheduled.

1. Verify equipment installed is what was approved on Submittal.
2. Manufacturer's start-up checklists and other technical documentation guidelines may be used as basis for pre-functional checklists.

3.6 FUNCTIONAL PERFORMANCE TESTING

A. Functional Performance Testing: Test procedures fully describe system configuration and steps required for each test.

1. Test Methods: Functional performance testing and verification may be achieved by direct manipulation of system inputs (i.e. heating or cooling sensors), manipulation of system inputs with building automation system (i.e. software override of sensor inputs), trend logs of system inputs and outputs using building automation system, or short-term monitoring of system inputs and outputs using standalone data loggers. A combination of methods may be required to completely test complete sequence of operations. CCxR shall determine which method or combination of methods is most appropriate.
2. Setup: Each test procedure shall be performed under conditions that simulate normal operating conditions as closely as possible. Where equipment requires integral safety devices to stop/prevent equipment operation unless minimum safety standards or conditions are met, functional performance test procedures shall demonstrate actual performance of safety shutoffs in real or closely simulated conditions of failure.
3. Sampling: Multiple identical pieces of non-life-safety or non-critical equipment may be functionally tested using a sampling strategy. If, after three attempts at testing the specified sample percentage, failures are still present, remaining units shall be tested at Contractors' expense.

B. Prepare functional performance test procedure forms to accommodate actual installed equipment and systems.

C. Coordinate, execute, and record results of functional performance testing.

1. Coordinate retesting as necessary until satisfactory performance is verified.
2. Verify intended operation of individual components and system interactions under various conditions and modes of operation.

3.7 DEFICIENCY REPORT AND RESOLUTION RECORD

A. Deficiency Report and Resolution Record: Document items of non-compliance in materials, installation or operation.

B. Non-Conformance. Non-conformance and deficiencies observed shall be addressed immediately. Notify responsible parties and provide recommended actions to correct deficiencies.

1. Corrections of minor deficiencies identified may be made during tests at discretion of CCxR. In such cases the deficiency and resolution shall be documented on procedure form.
2. For identified deficiencies:

a. If no dispute on deficiency and responsibility to correct it:

1) CCxR documents deficiency and adjustments or alterations required to correct it. Contractor corrects deficiency and notifies CCxR that equipment is ready to be retested.
2) CCxR reschedules test and test is repeated until satisfactory performance is achieved.

b. If there is a dispute about a deficiency or who is responsible:

1) Deficiency is documented CCxR on non-compliance form.
2) Resolutions are made at lowest management level possible. Additional parties are brought into discussions as needed. Contractor shall have responsibility for resolving construction deficiencies. If a design revision is deemed necessary and approved by Contracting Officer, Architect/Engineer (A/E) shall have responsibility for providing design revision. CCxR documents resolution process.
3) Once interpretation and resolution have been decided, appropriate party corrects deficiency and notifies CCxR that equipment is ready to be retested. CCxR reschedules test and test is repeated until satisfactory performance is achieved.

3. Cost of Retesting: Costs for retesting shall be charged to Contractor.

3.8 OPERATIONS AND MAINTENANCE TRAINING

A. Training: Develop Training Plan. Coordinate and execute training programs with CxA.

1. Stress and enhance importance of system interactions, troubleshooting, and long-term preventive maintenance and operation programs.

3.9 DEFERRED TESTING

A. Unforeseen Deferred Tests: If test cannot be completed due to building structure, required occupancy condition, or other deficiency, functional testing may be delayed upon recommendation of CCxR and approval of Contracting Officer. These tests are conducted in same manner as the seasonal tests, as soon as possible.

B. Seasonal Testing

1. Schedule, coordinate, execute, and document additional testing for seasonal variation in operations and control strategies during appropriate season to verify performance of HVAC system and controls. Complete testing during warranty period to fully test sequences of operation.
2. Update O&M manuals and Project Record Drawings as necessary due to testing.

3.10 EQUIPMENT & SYSTEM SCHEDULE

A. Commissioned Equipment and Systems List: Following is a list of systems and equipment to be commissioned organized by system. It includes the percentage of each category that will undergo testing. The intent is to provide an overall summary of commissioned equipment and systems, and not a comprehensive list. Refer to applicable specification sections for more information.

COMMISSIONED EQUIPMENT AND SYSTEMS LIST			
SYSTEM	**EQUIPMENT**	**CC** (Commissioning Coordinator)	**FPT** (Functional Performance Test)

COMMISSIONED EQUIPMENT AND SYSTEMS LIST			
SYSTEM	**EQUIPMENT**	**CC** (Commissioning Coordinator)	**FPT** (Functional Performance Test)

END OF SECTION 01 91 14

EFFIGY MOUNDS NATIONAL MONUMENT

Rehabilitation Visitor Center

N170 EFMO 318539

Division 02 through Division 33

Final Construction Documents

PROJECT SPECIFICATIONS

SECTION 00 01 07 – SEALS PAGE

PART 1 - GENERAL

1.1 SUMMARY

A. Professional seals by Design Professionals and others responsible for preparing Final Construction Documents:

1. Architect

a. Divisions:

1)	Division 07	4)	Division 11
2)	Division 08	5)	Division 12
3)	Division 09	6)	Division 14

b. Sections:

1)	02 41 19	7)	06 20 23	13)	10 22 39
2)	03 30 00	8)	06 40 13	14)	10 28 00
3)	04 01 20.63	9)	06 41 00	15)	10 44 13
4)	04 01 20.64	10)	06 61 16	16)	10 44 16
5)	04 26 13	11)	10 14 23.16	17)	10 51 16
6)	05 50 00	12)	10 21 13.19		

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Architect under the laws of the State of Iowa.

Name: Elizabeth Hallas

My license renewal date is: 06/30/2028

Divisions and sections covered by this seal are as noted above.

2. Civil Engineer

a. Divisions:

1) Division 31
2) Division 33

b. Sections:

1) 02 41 13
2) 10 75 16
3) 32 13 20

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Emily Maggio

Name: Emily Maggio

My license renewal date is: 12/31/2027

Divisions and sections covered by this seal are as noted above.

3. Structural Engineer

a. Divisions:

1) Division 03

b. Sections:

1) 04 22 00
2) 05 12 00
3) 05 31 00
4) 06 10 00
5) 06 16 00
6) 06 18 00

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Name: Sean Clark

My license renewal date is: 12/31/2026

Divisions and sections covered by this seal are as noted above.

4. Fire Protection Engineer

a. Divisions:

1) Division 21

b. Sections:

1) 28 00 01
2) 28 08 00
3) 28 46 00

05/29/26

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Name: Kenton J. Aikens

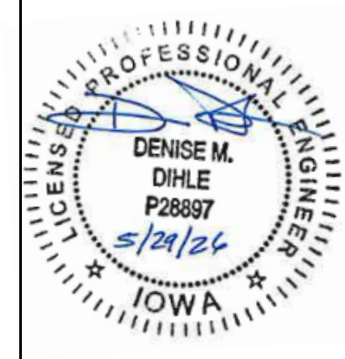
My license renewal date is: 12/31/2026

Divisions and sections covered by this seal are as noted above.

5. Mechanical Engineer

a. Divisions:

1) Division 22
2) Division 23



CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Name: Denise Dihle

My license renewal date is: 12/31/2027

Divisions and sections covered by this seal are as noted above.

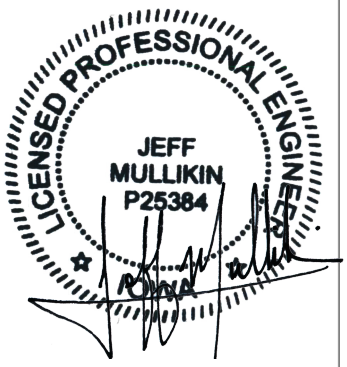
6. Electrical Engineer

a. Divisions:

1) Division 26
2) Division 27

b. Sections:

1) 28 00 10
2) 28 13 00
3) 28 14 00
4) 28 16 00
5) 28 20 00



CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Jeff Mullikin

Name:

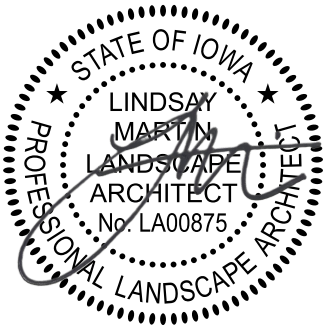
My license renewal date is: 12/31/2026

Divisions and sections covered by this seal are as noted above.

7. Landscape Architect

a. Sections:

1) 04 43 13
2) 31 23 16
3) 32 33 00
4) 32 91 13
5) 32 92 00
6) 32 93 00



5/29/2026

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Landscape Architect under the laws of the State of Iowa.

Name: Lindsay Martin

My license renewal date is: 6/30/2026

Divisions and sections covered by this seal are as noted above.

END OF SECTION 00 01 07

SECTION 02 41 13 – SELECTIVE SITE DEMOLITION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Removal or recycling of existing cement concrete walkways and pads.
2. Removal and return of existing site features to the NPS include:

a. Bench
b. Trash receptacle
c. Retaining wall stone cap
d. Stanchion
e. Park sign
f. Picnic Table

3. Removal and storage of existing site features by Contractor for reinstallation:

a. Bike rack
b. Retaining wall stone veneer

4. Demolition, removal, and disposal of the following items:

a. Retaining wall stone blocks
b. Water line
c. Concrete curb
d. Signage

B. Related Sections include the following:

1. Section 01 57 23 – Temporary Pollution Prevention Plan
2. Section 31 10 00 – Site Clearing
3. Section 31 20 00 – Earth Moving
4. Section 31 25 00 – Erosion and Sediment Controls

1.3 DEFINITIONS

A. Remove: Remove and legally dispose of items except those indicated by CO to be salvaged, reused, reinstalled, or to remain on EFMO property.

B. Remove and Salvage: Remove items indicated; clean, service, and otherwise prepare items for reuse; store and protect against damage. Reinstall items in locations indicated in the plans and specifications or as directed by the CO.

C. Existing to Remain: Protect items identified to remain against damage and soiling during demolition. When permitted by the CO, items may be removed to a suitable, protected storage location during demolition and then cleaned and reinstalled in their original locations.

1.4 MATERIALS OWNERSHIP

A. Except for items or materials indicated to be reused, salvaged, or otherwise indicated to remain on the EFMO property, demolished materials shall become the Contractor's property and shall be removed and legally disposed of at the Contractor's expense.

B. Materials for removal and return to the NPS:

1. The following materials within the limits of disturbance shall be removed and returned to the NPS at the project site:

a. Bench
b. Trash receptacle
c. Retaining wall stone top
d. Stanchion
e. Park sign
f. Picnic Table

C. Materials for removal and storage by Contractor:

1. The following materials within the limits of disturbance shall be removed and stored by the Contractor for reinstallation at the project site:

a. Bike rack
b. Retaining wall stone veneer

D. Materials for removal and disposal:

1. The following materials shall be removed from the site and legally disposed of.

a. Cement concrete walkways and pads
b. Retaining wall stone blocks
c. Water line
d. Concrete curb
e. Signage

1.5 QUALITY ASSURANCE

A. Comply with the local noise ordinances regarding hours of operations and allowable noise levels. Monitor vibrations generated by the demolition activities and received by the existing structures in the immediate vicinity of the project.

1.6 SCHEDULING

A. Notify the Contracting Officer (CO) 48 hours prior to beginning demolition work.

B. Contractor is responsible for verifying location of utilities.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 UTILITY SERVICES

A. Utility requirements: Locate, identify, disconnect, and seal or cap off indicated utility services where indicated.

1. Arrange to shut off indicated utilities with utility companies.

3.2 PREPARATION

A. Install erosion and sediment control measures prior to demolition.

B. Install site protection fencing prior to demolition.

C. Conduct demolition operations and remove debris to ensure minimum interference with roads, driveways, and other adjacent occupied and used facilities.

1. Do not close or obstruct roads, or other adjacent occupied or used facilities without permission from the CO and authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways.

D. Conduct demolition operations to prevent injury to people and damage to facilities to remain. Ensure safe passage of people around demolition area.

1. Erect temporary protection, such as walks, fences, and railings, where required by CO.

2. Protect existing improvements, appurtenances, and landscaping to remain.
3. Erect protective fencing around drip line of individual trees or around perimeter drip line of groups of trees to remain according to requirements specified in Section 311000 “Site Clearing.”
4. A vertical sawcut shall be made between any existing pavement that is to remain and the portion to be removed. Where sawcutting is required, the sawcut shall be a minimum of 3 inches deep. When the existing pavement is more than 3 inches thick, the portion below 3 inches may be broken after the sawcut is made.

3.3 POLLUTION CONTROLS

A. Use water mist, temporary enclosures, and other suitable methods to limit the spread of dust and dirt. Comply with governing environmental protection regulations.

1. Do not create hazardous or objectionable conditions, such as ice, flooding, and pollution, when using water.

2. The Contractor shall be responsible for ensuring that special precautions are undertaken during sawcutting operations. No concrete (asphalt or cement) or concrete by-products are to be discharged into any storm drain or surface water. Sawcutting operations will increase the pH of water, therefore filtering is not acceptable.

B. Remove and transport debris in a manner that will prevent spillage on site improvements or public roads.

3.4 DEMOLITION

A. Remove existing asphalt, concrete and base course material as shown in the plans. Approximate existing asphalt depth, including base course material, is 22 inches. Approximate existing concrete walkway depth, including base course material, is 8 inches.

B. Filling Below-Grade Areas: Completely fill below-grade areas and voids resulting from demolition of existing facilities shown on the plans with soil materials according to requirements specified in Section 31 20 00 – Earth Moving.

C. Damages: Promptly repair damage to adjacent facilities caused by demolition operations.

3.5 DISPOSAL AND RECYCLING OF DEMOLISHED MATERIALS

A. Promptly dispose of or recycle demolished materials: If recycling requires additional time to accomplish, provide for orderly storage of materials. Comply with the waste management plan.

B. Transport demolished materials off EFMO property and legally dispose or recycle material.

END OF SECTION 02 41 13

SECTION 02 41 19 – SELECTIVE BUILDING DEMOLITION

PART 1 - GENERAL

1.1 SUMMARY

A. The Work of this Section Includes:

1. Demolition and removal of selected portions of exterior or interior of building or structure.
2. Removal and salvage of existing items for delivery to Government and removal of existing items for reinstallation.

B. Related Requirements:

1. Section 01 50 00 “Temporary Facilities and Controls” for occupancy requirements.
2. Section 01 73 29 “Cutting and Patching” for incidental cutting and patching not covered by this section.
3. Section 01 74 19 “Construction Waste Management and Disposal” for Government related demolition requirements.
4. Section 02 41 13 “Selective Site Demolition for removal of site furnishings, signage, curb, and retaining walls.
5. Section 31 10 00 “Site Clearing” for site clearing and removal of above- and below-grade improvements not part of selective demolition.
6. Applicable Hazardous Materials Sections are listed in section 1.9 “Field Conditions” below.

1.2 DEFINITIONS

A. Remove: Detach items from existing construction and legally dispose of off-site unless indicated to be removed and salvaged or removed and reinstalled.

B. Salvage and Reinstall: Detach items from existing construction, in a manner to prevent damage; prepare for reuse; and reinstall where indicated.

C. Existing to Remain: Existing items of construction that are not to be removed.

1.3 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition waste becomes property of Contractor.

1.4 COORDINATION

A. Arrange selective demolition schedule so as not to interfere with Government's operations.

1.5 PREINSTALLATION MEETINGS

A. Predemolition Conference: Conduct conference at Project site.

1. Inspect and discuss condition of construction to be selectively demolished.
2. Review structural load limitations of existing structure.
3. Review and finalize selective demolition schedule and verify availability of demolition personnel, equipment, and facilities needed to make progress and avoid delays.
4. Review requirements of work performed by other trades that rely on substrates exposed by selective demolition operations.
5. Review areas where existing construction is to remain and requires protection.
6. Review and finalize protection requirements.
7. Review procedures for noise control.
8. Review storage, protection, and accounting for items to be removed for salvage or reinstallation.

1.6 ACTION SUBMITTALS

A. Qualification Statements: For refrigerant recovery technician.

B. Schedule of Selective Demolition Activities: Indicate the following:

1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity. Ensure Government's on-site operations are uninterrupted.
2. Detailed description of procedures to be used for any element that is to be salvaged for reinstallation or reuse.
3. Temporary interruption of utility services. Indicate how long utility services will be interrupted.
4. Coordination for shutoff, capping, and continuation of utility services.
5. Use of elevator and stairs.
6. Coordination of Government's continuing occupancy of portions of existing building and of Government's partial occupancy of completed Work.

C. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating that all refrigerant that was present was recovered and that recovery was performed in accordance with EPA regulations. Include name and address of technician and date refrigerant was recovered.

1.7 CLOSEOUT SUBMITTALS

A. Inventory: Submit a list of items that have been removed and salvaged. Include annotation of which items were reinstalled and, if applicable, how much of that item remains unused.

1.8 QUALITY ASSURANCE

A. Refrigerant Recovery Technician Qualifications: Type II certified by an EPA-approved certification program.

1.9 FIELD CONDITIONS

A. Government will occupy portions of building immediately adjacent to selective demolition area. Conduct selective demolition so Government's operations will not be disrupted.

1. Occupied portion of building is the room containing the LAN equipment and cabling.

B. Hazardous Materials:

1. Hazardous materials are present in buildings and structures to be selectively demolished. A report on the presence of hazardous materials is on file for review and use. Examine report to become aware of locations where hazardous materials are present.

a. Hazardous material remediation is specified in the following Sections:

1) Section 02 81 01 “Offsite Transportation and Disposal”
2) Section 02 82 13 “Asbestos Abatement”
3) Section 02 83 33 “Removal and Disposal of Material Containing Lead”
4) Section 02 84 16 “Handling of Lamps Containing Mercury”
5) Section 10 81 19 “Rodent and Bird Control” for handling animal bio-waste

b. The Hazardous Materials Report is attached to the specifications as an Appendix.

C. On-site sale of removed items or materials is not permitted.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

B. Standards: Comply with ANSI/ASSP A10.6 and NFPA 241.

C. Sustainable Design Requirements for Building Reuse:

1. Maintain the existing building facade where indicated to remain. Do not demolish such existing construction beyond indicated limits.
2. Maintain the existing building structural systems where indicated to remain. Do not demolish such existing construction beyond indicated limits.
3. Maintain the existing interior partitions and walls where indicated to remain. Do not demolish such existing construction beyond indicated limits.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify that hazardous materials have been remediated before proceeding with building demolition operations.

3.2 PREPARATION

A. Temporary Shoring: Design, provide, and maintain shoring, bracing, and structural supports as required to preserve stability and prevent movement, settlement, or collapse of construction and finishes to remain, and to prevent unexpected or uncontrolled movement or collapse of construction being demolished.

1. Strengthen or add new supports when required during progress of selective demolition.

B. Temporary Protection: Provide temporary barricades and other protection required to prevent injury to people and damage to facilities to remain.

1. Provide protection to ensure safe passage of people around selective demolition area and to and from occupied portions of building.
2. Provide temporary weather protection, during interval between selective demolition of existing construction on exterior surfaces and new construction, to prevent water leakage and damage to structure and interior areas.
3. Protect walls, ceilings, floors, and other existing finish work that are to remain or that are exposed during selective demolition operations.
4. Cover and protect equipment that have not been removed.
5. Comply with requirements for temporary enclosures, dust control, heating, and cooling specified in Section 01 50 00 “Temporary Facilities and Controls.”

C. Existing Items to Remain: Protect construction indicated to remain against damage and soiling during selective demolition. When permitted by Contracting Officer, items may be removed to a suitable, protected storage location and cleaned and reinstalled in their original locations after selective demolition operations are complete.

D. Refrigerant: Before starting demolition, remove refrigerant from mechanical equipment in accordance with 40 CFR 82 and regulations of authorities having jurisdiction.

3.3 UTILITY SERVICES AND BUILDING SYSTEMS

A. Existing Services/Systems to Remain: Maintain utilities and building systems and equipment to remain and protect against damage during selective demolition operations.

B. Existing Services/Systems to Be Removed, Relocated, or Abandoned: Locate, identify, disconnect, and seal or cap off utilities and building systems serving areas to be selectively demolished.

1. Arrange to shut off utilities with utility companies.

2. If disconnection of utilities and building systems will affect adjacent occupied parts of the building, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to those parts of the building.
3. Demolish and remove existing building systems, equipment, and components indicated on Drawings to be removed.

    a. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
    b. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.
    c. Equipment to Be Removed: Disconnect and cap services and remove equipment and components.

3.4 SALVAGE/REINSTALL

A. Salvaged for Reinstallation:

1. Clean and repair items to functional condition adequate for intended reuse.
2. Pack or crate items after cleaning and repairing. Identify contents of containers.
3. Protect items from damage during transport and storage.
4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.
5. For any surplus, items that remain unused by the end of the project, they shall be returned to the Contracting Officer as part of project closeout.

3.5 SELECTIVE DEMOLITION, GENERAL

A. General: Demolish and remove existing construction only to extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:

1. Proceed with selective demolition systematically, from higher to lower level. Complete selective demolition operations above each floor or tier before disturbing supporting members on the next lower level.
2. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping. Temporarily cover openings to remain.
3. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
4. Comply with requirements of NPS hot work permit for any flame-cutting operations.
5. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain portable fire-suppression devices during flame-cutting operations.
6. Maintain adequate ventilation when using cutting torches.

7. Remove decayed, vermin-infested, or otherwise dangerous or unsuitable materials and promptly dispose of off-site. See Section 10 81 19 “Rodent and Bird Control” for more information.
8. Remove structural framing members and lower to ground by method suitable to avoid free fall and to prevent ground impact or dust generation.
9. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.

B. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

1. Do not close or obstruct streets, walks, walkways, or other adjacent occupied or used facilities without permission from Contracting Officer and authorities having jurisdiction. Provide alternate routes around closed or obstructed trafficways if required by authorities having jurisdiction.
2. Use water mist and other suitable methods to limit spread of dust and dirt. Comply with governing environmental-protection regulations. Do not use water when it may damage adjacent construction or create hazardous or objectionable conditions, such as ice, flooding, and pollution.

3.6 SELECTIVE DEMOLITION PROCEDURES FOR SPECIFIC MATERIALS

A. Concrete:

1. Demolish in small sections. Using power-driven saw, cut concrete to a depth of at least 3/4 inch at junctures with construction to remain. Dislodge concrete from reinforcement at perimeter of areas being demolished, cut reinforcement, and then remove remainder of concrete. Neatly trim openings to dimensions indicated.
2. Demolish in sections. Cut concrete full depth at junctures with construction to remain and at regular intervals using power-driven saw, and then remove concrete between saw cuts.

B. Masonry: Demolish in small sections. Cut masonry at junctures with construction to remain, using power-driven saw, and then remove masonry between saw cuts.

1. Where possible, salvage brick masonry for reuse to avoid need for procurement of replacement to match.
2. Salvage all stone masonry for reuse.

C. Concrete Slabs-on-Grade: Saw-cut perimeter of area to be demolished, and then break up and remove.

3.7 CLEANING

A. Clean adjacent areas of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

3.8 SELECTIVE DEMOLITION SCHEDULE

A. Remove: This is not a comprehensive list. Please consult drawings for full demolition scope.

1. Roofing and all roof accessories - trim, gutter, downspouts, and flashing – at gabled roofs.
2. Exterior wood siding and trim
3. Exterior and interior windows and doors, except where noted to remain.
4. Interior ceilings (where not structure) and interior side finish on exterior walls
5. Select interior walls (see drawings)
6. Select electrical fixtures and cabling (see drawings)
7. Plumbing fixtures
8. Insulation in exterior walls, except where noted to remain.
9. Misc. additional per demo drawings

B. Salvaged for Reinstallation: Please confirm this list with the drawings for full salvage scope.

1. Stone Veneer at exterior planters, retaining wall, and building wainscot
2. Brick veneer at new penetrations.
3. Option D: Visitor Center Expanded Deck at NE Corner – Aluminum railing at east side of balcony to be removed for deck expansion has the potential to be salvaged for reinstallation in new configuration. Reuse shall be dependent upon written approval by Contracting Officer after review of salvaged railing to confirm condition suitability for reuse.

C. Existing to Remain: This is not a comprehensive list. Please consult drawings for full demolition scope.

1. Clean agent tank and system
2. Brick veneer at southeast corner
3. Original building roof (inside attic) except where new penetrations are described in the drawings
4. Building structural system and framing, including concrete floor deck and T&G roof deck. Only structural system to be replaced are columns/footings where they are moved/replaced and concrete slab on grade in basement where described in drawings.
5. Elevator equipment, doors, and cab.
6. Aluminum railing around balcony, except where removed for Option D: Visitor Center Expanded Deck at NE Corner
7. Concrete on metal deck bridge to the trail at the north side of the building. Includes metal railing.
8. Select doors in the basement, as described in the drawings.

END OF SECTION 02 41 19

SECTION 02 81 01 – OFF-SITE TRANSPORTATION AND DISPOSAL

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes requirements for the offsite transportation and disposal of wastes generated during the planned historic rehabilitation of the Effigy Mounds National Monument Visitor's Center in Harpers Ferry, Iowa (EFMO #318539), (Project Site).

B. The Contractor will be responsible for sampling, analyzing, and characterizing waste materials to be shipped offsite for final disposal or recycling as required by the Disposal Facility.

C. If building materials coated with Lead-Containing Paint (LCP) and Lead-Based Paint (LBP) are demolished and will be disposed of, it will be the Contractor's sole responsibility to obtain temporary U.S. Environmental Protection Agency (EPA) ID number(s), as and if applicable, on behalf of the National Park Service (NPS) for any hazardous waste materials generated and removed from the Project Site during the execution of this Project. The Contracting Officer will identify appropriate NPS individuals to sign the required paperwork as the generator signatory for the Project.

D. Offsite recycling and disposal facilities will be approved by the Contracting Officer prior to the offsite transportation of materials.

E. Transportation of the materials will comply with the legal load restrictions of the U.S. Department of Transportation (DOT) and the Colorado Department of Transportation (CDOT).

1.2 RELATED SPECIFICATIONS

A. Section 01 74 19: Construction Waste Management and Disposal

B. Section 02 82 13: Asbestos Abatement

C. Section 02 83 13: Removal and Disposal of Material Containing Lead

D. Section 02 84 16: Handling of Lamps Containing Mercury

E. Section 10 81 19: Rodent and Bird Control

1.3 SUBMITTALS

A. The Contractor will include as a component of the Waste Management Plan (described in Section 01 74 19 "*Construction Waste Management and Disposal*") a description of planned means and

methods for transporting and disposing of all the waste materials removed or generated during the Project.

B. Permit for the selected Treatment Storage and/or Disposal Facility.

C. The Contractor will provide an original and/or copies of all Bills of Lading and Manifests for all transported waste material loads from the Project Site, including all required generator, transporter, and waste disposal facility receipt/transfer signatures. These documents will be provided to the Contracting Officer at least weekly and will be accompanied by a detailed electronic waste tracking sheet. The waste tracking sheet will be cumulative and updated weekly. This document will provide the following:

1. Cumulative load numbers for each shipment.
2. A description of the waste material being shipped for disposal.
3. Company transporting the waste material.
4. Estimated waste volumes for each shipment.
5. Certified weight slips for each shipment.
6. The disposal facility of record for each waste stream.
7. The specific Manifest or Bill of Lading specific identifier number for the load being shipped for disposal.

D. At the conclusion of the Project, all waste disposal information collected will be submitted to the Contracting Officer as an electronic deliverable package as part of the required Project closeout record documentation. Payments to the Contractor for waste transportation disposal fees will not be approved until completed disposal tracking documents, as described above, are received by the Contracting Officer.

1.4 WASTE CONTAINERS

A. The Contractor will provide waste containers specific to the individual waste to be handled.

1.5 FIELD VERIFICATION

A. Field verification of all new and existing dimensions affecting the work of this contract by the Contractor is required before ordering products.

1.6 TRANSPORTATION OF WASTES

A. The Contractor will be responsible for transporting all wastes generated during the Project that cannot be salvaged, reused, or recycled. This includes materials generated by final Project Site cleanup activities and the dismantling of the temporary facilities and controls.

B. The Contractor will be responsible for coordinating the number and schedule of vehicles required for the offsite transportation of waste materials generated during the execution of the Project.

C. The Contractor will be responsible for inspecting transportation vehicles before and after loading to ensure compliance with all local, state, and federal regulations for the safe transport of wastes from the Project Site to the receiving facility. The Contractor will provide the necessary labor and materials to ensure all trucks and containers are covered prior to departure.

D. The Contractor will ensure that transporters arriving at the Project Site for loading do not cause undue congestion to local roadways and will stage trucks either within the perimeter of the Project Site or at an offsite staging area approved by the Contracting Officer.

E. The Contractor's transporters will proceed directly from the Project Site to a designated staging area or the designated offsite receiving facility. Temporary staging or storage of material will only be allowed at designated staging areas approved by the Contracting Officer.

F. The Contractor will originate, maintain, and provide the Contracting Officer with a copy of each executed Bill of Lading for all loads shipped offsite. In addition, the Contractor will provide the Contracting Officer with documentation and records verifying the receipt of each truckload by the receiving facility. Such documentation will indicate the actual weight of each load shipped.

G. Transporters will proceed from the Project Site along traffic routes established by the Contractor and approved by the Contracting Officer. Transporters will call back weights after each load and modify the loads accordingly. The Contractor will ensure that trucks leaving the Project Site are within appropriate weight limitations for the local roads along the designated route.

H. The weight measurements for the materials transported to recycling and disposal facilities will be determined by scales certified in accordance with local, state, and federal regulations and requirements.

1.7 DISPOSAL OF WASTES

A. The Contractor will be responsible for the proper disposal of all solid and liquid wastes generated during the execution of this Project in conformance with all local, state, and federal regulations and requirements. Proper disposal requires that the facility accepting the waste be a state-licensed disposal/recycling facility approved for acceptance of the waste based on the results of the characterization testing and analysis.

B. The disposal facilities will be approved by the Contracting Officer prior to transporting waste. The Contractor will not change facilities without prior written consent of the Contracting Officer.

C. If applicable, for wastes determined to have lead toxicity characteristics as a result of LCP or LBP, the Contractor shall obtain temporary EPA ID number(s) on behalf of the NPS, as directed by the Contracting Officer, for hazardous waste generated during the execution of this Project.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 02 81 01

SECTION 02 82 13 – ASBESTOS ABATEMENT

PART 1 - GENERAL

1.1 DESCRIPTION

A. This Section consists of asbestos abatement tasks during the planned historic rehabilitation of the Effigy Mounds National Monument Visitor's Center in Harpers Ferry, Iowa (EFMO #318539), (Project Site).

B. The Contractor will take all necessary reasonable measures to control air and water pollution by any material or equipment used during asbestos abatement.

C. The confirmed Asbestos-Containing Materials (ACM) and presumed ACM present at the Project Site are described below. Estimated quantities of ACM and presumed asbestos to be abated from the planned renovation and rehabilitation areas have also been provided below.

D. The abatement scope of work includes the removal of the following ACM:

ACMs:

1. Carpet Adhesive (VC-CA01): Yellow/tan carpet adhesive beneath green/tan carpet contains 2% chrysotile. There are approximately 8,000 sf of VC-CA01; however, the exact extent could not be determined without removing all flooring finishes. The adhesive is a Category II Non-Friable ACM and can be removed as a Class II Asbestos Work under secondary containment. Potential for disturbance during floor openings for mechanical chases in the main level, removal of casework and countertops in the breakroom, and removal of the ramp in the basement vestibule.
2. Vinyl Baseboard Adhesive (VC-BBA01): Tan adhesive associated with four-inch black vinyl baseboard located in the basement contains 2% chrysotile. (Approximately 100 linear feet [lf]). The adhesive is a Category II Non-Friable ACM and can be removed as a Class II Asbestos Work under secondary containment.
3. Tar Coating on Roof Pipes (VC-Pipe): A tar coating on abandoned roof lead pipes contains 2% chrysotile. Two pipes (approximately 40 lf) were observed in the basement of the Project Site. The coating is a Category I Non-Friable ACM and can be removed as a Class II Asbestos Work under secondary containment. There is a potential for disturbance during the replacement of roofing materials and roof vents.
4. Pipe Wrap (VC-PW02): Pipe wrap (approximately 10 lf), including pliable white wrap with black coating on a loose steel pipe in the mechanical room, contains 5% chrysotile. The pipe wrap is a Friable ACM and can be removed as a Class II Asbestos Work using glovebag methods under a secondary containment. There is a potential for disturbance during the removal of equipment and the new installation in the basement mechanical room.
5. Tar Sealant (VC-TP01): Tar foundation sealant observed in crawlspace UX2 contains 3% chrysotile. The tar foundation sealant is assumed to be located on foundation walls throughout the building. Unknown quantity. If the foundation of the Visitor's Center will be demolished, waste management and prohibitions to recycling concrete with asbestos sealant would be included in the Selective Demolition costs and requirements. The sealant

is a Category I Non-Friable ACM and can be removed as a Class II Asbestos Work under secondary containment. There is a potential for disturbance during the removal of stonework in exterior and wall openings for the installation of mechanical louvers in the crawlspace.

Presumed ACMs:

6. Fire Doors (VC-FD01): A fire-rated door located in the mechanical room is PACM. Door not scheduled for removal during renovation.
7. Metal Panels (VC-CB01): Metal panels were observed on a portion of the 1963-era roof that may contain PACM cement boards. Removal of the metal roof panels (approximately 800 sf) is required to investigate this suspected ACM. Cement boards were identified as Category II Non-Friable PACM. No potential disturbance is scheduled during renovation activities.
8. Chimney Pipe (VC-PI01): A former chimney pipe is disconnected and is lying on the fiberglass attic insulation. The pipe resembles Air Cell, an ACM factory-molded pipe insulation. The pipe is approximately 3 lf and has the potential to have contaminated adjacent fiberglass insulation. Pipe was identified as a Friable PACM and can be removed under a secondary containment, including surrounding contaminated non-ACM fiberglass insulation.

E. ACM floor tile/mastic was not identified via sampling by this inspection, but the potential exists for it to be located beneath portions of carpeted rooms. Landmark inspected beneath carpet corners in multiple locations of each carpeted room of the Project Site.

F. The Contractor will be responsible for the collection and analysis of personnel exposure samples for asbestos in accordance with the federal U.S. Occupational Safety and Health Administration (OSHA) regulations.

G. If any suspect ACM is encountered within the wall or ceiling interiors in unanticipated locations, the unforeseen condition shall be documented and described, including the location, extent, and quantity of suspect ACM and any immediate actions taken for safety or housekeeping purposes, and this information shall then be provided to the National Park Service (NPS).

H. Any ACM of the nature shown or described, but not specifically called out or shown in the plans but which are reasonably implied and necessary for a complete and satisfactory removal job, will be included without additional cost to the government.

I. Appropriate training will be provided by the Contractor prior to personnel performing abatement as required by applicable regulation 29 Code of Federal Regulations (CFR) 1926.1101 (asbestos). All persons conducting asbestos abatement work will receive compliant training per Title 40 of the CFR Protection of the Environment.

J. The Contractor will be responsible for all cleanup and testing of any area contaminated due to the Contractor's own negligence at no additional cost to the NPS.

1.2 REFERENCES AND RELATED SPECIFICATIONS

A. All work under this contract will be done in strict accordance with the applicable federal, state, and local regulations, standards, and codes governing ACM and trace-asbestos removal and other

trade work done in conjunction with this type of asbestos removal work. The most recent edition of any relevant regulation, standard, document, or code will be in effect. Where a conflict among the requirements or within these specifications may exist, the most stringent requirements will be utilized.

B. Section 02 81 01: Offsite Transportation and Disposal.

C. Section 02 83 13: Removal and Disposal of Material Containing Lead.

D. Section 02 84 16: Handling of Lamps Containing Mercury.

E. Section 10 81 19: Rodent and Bird Control.

F. Landmark Environmental, Inc., Hazardous Materials Inspection for Effigy Mounds National Monument Visitors Center, Harpers Ferry, Iowa. Landmark Project No. 24061.001. January 31, 2025.

G. American Society of Safety Engineers (ASSE), ASSE/SAFE Z9.2 Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems, 2012.

H. Underwriters Laboratories (UL), UL Standard for Safety High-Efficiency Particulate, Air Filter Units, UL 586, 2009; reprint 2017.

I. OSHA, 29 CFR 1910.1001, General Industry Standards for Asbestos.

J. OSHA, 29 CFR 1926.1101, Construction Industry Standards for Asbestos.

K. OSHA, 29 CFR 1926.62 Lead.

L. OSHA, 29 CFR 1910.134, Respiratory Protection.

M. OSHA, 29 CFR 1910.145, Accident Prevention.

N. OSHA, 29 CFR 1910.2, Medical Records.

O. OSHA, 29 CFR 1910.38, Emergency/Fire Prevention Programs.

P. OSHA, 29 CFR 1910.1200, Hazard Communication.

Q. U.S. Environmental Protection Agency (EPA), 40 CFR 61 Sub. A, General Provisions.

R. EPA, 40 CFR 61 Sub. B.

S. U.S. Department of Transportation (USDOT) 49 CFR 171 and 172.

T. National Institute for Occupational Safety and Health (NIOSH), Certified Equipment List.

U. NIOSH, Occupational Exposure Sampling Strategy 77-173.

V. NIOSH, Guide to Industrial Respiratory Protection, 87-116.

1.3 DEFINITIONS AND ACRONYMS

A. Asbestos: Includes chrysotile, amosite, crocidolite, tremolite, anthophyllite, actinolite asbestos, and any of these minerals that have been chemically treated and/or altered. Asbestos also includes presumed ACM (PACM), as defined below.

B. Asbestos-Containing Material (ACM): Any material containing more than 1% asbestos.

C. Asbestos-Containing Waste Material (ACWM): Any material that is or is suspected of being contaminated with an ACM and which is to be removed from a work area for disposal.

D. Asbestos Debris: Pieces of ACM that can be identified by color, texture, or composition, or means of dust if the dust is determined by an accredited inspector to be ACM.

E. Authorized Person: Any person authorized by their employer and required by work duties to be present in regulated areas.

F. Class I Asbestos Work: Work activities involving the removal of ACM that is thermal system insulation or a surfacing material.

G. Class II Asbestos Work: Activities involving the removal of ACM, which is not thermal system insulation or a surfacing material. This includes, but is not limited to, the removal of asbestos-containing wallboard systems with ACM drywall joint compound, floor tile and sheeting, roofing and siding shingles, and construction mastics.

H. Critical Barrier: One or more layers of plastic sealed over all openings into a work area or any other similarly placed physical barrier sufficient to prevent airborne asbestos in a work area from migrating into an adjacent area.

I. Competent Person: The Contractor's employee who is capable of identifying existing asbestos in the workplace, who has the authority to take prompt corrective measures to eliminate them, and who is trained and accredited as defined in 29 CFR 1926.1101 and 29 CFR 1926.62. The duties of the Competent Person include but are not limited to the following activities: controlling entry to and exit from the asbestos work area, supervising any employee exposure monitoring required by the standards, ensuring that all employees working in a work area wear the appropriate Personal Protective Equipment (PPE) and are trained in the use of appropriate methods of exposure control and decontamination, and ensuring that engineering controls in use are functioning properly.

J. Disposal Bag: A properly labeled 6-mil leak-tight polyethylene bag used for containerizing and/or transporting asbestos waste to the disposal site.

K. Friable Asbestos Material: Material that contains more than 1% asbestos by weight and can be crumbled, pulverized, or reduced to powder by hand pressure when dry.

L. HEPA Filter: A High-Efficiency Particulate Air (HEPA) filter capable of trapping and retaining 99.97 percent of all mono-dispersed particles (e.g., asbestos fibers) of 0.3 microns in diameter.

M. Industrial Hygiene: An area of specialization in the field of industrial safety and health concerned with predicting, recognizing, assessing, controlling, and preventing environmental stressors in the workplace that can lead to sickness, disease, or other forms of impaired health.

N. Industrial Hygienist: A person having a college or university degree(s) in engineering, chemistry, physics, medicine, or related physical or biological sciences who, by virtue of special studies and training, has acquired competence in industrial hygiene.

O. Intact: ACM that has not crumbled, been pulverized, or otherwise deteriorated so that the asbestos is no longer likely to be bound with its matrix.

P. Negative Initial Exposure Assessment: Assessment by an Industrial Hygienist (IH) that demonstrates that employee exposure during an operation is expected to be consistently below the Permissible Exposure Limit (PEL) for the representative 8-hour Time-Weighted Average (TWA) samples as well as 30-minute Excursion Limit (EL) samples.

Q. Non-Friable Asbestos Material: Material that contains more than 1% asbestos by weight and cannot be crumbled, pulverized, or reduced to a powder by hand pressure when dry.

R. Personal Monitoring: Sampling the asbestos fiber concentrations within the breathing zone of an employee.

S. PACM: Presumed Asbestos-Containing Material means materials that are not wood, masonry, metal, rubber, or fiberglass that should be presumed to contain asbestos until tested.

T. Regulated Area: An area established by the Contractor to demarcate areas where Class I, II, and III asbestos work is conducted; an adjoining area where debris and waste from such asbestos work accumulate; and a work area within which airborne concentrations of asbestos exceed, or there is a reasonable possibility they may exceed, the PEL.

U. Removal: All operations where ACM and/or PACM are taken out or stripped from the Project Site, including demolition operations.

V. Renovation: Modifying any existing structure or portion thereof.

W. Surfactant: A chemical wetting agent added to the water to improve wetting and penetration into a substrate.

X. Time-Weighted Average (TWA): The average concentration of a contaminant in the air during a specific time period.

Y. Visible Emissions: Any emissions containing particulate asbestos material which are visually detectable without the aid of instruments.

Z. Work Area: The area where asbestos-related work or removal operations are performed that is defined and/or isolated to prevent the spread of asbestos dust, fibers, or debris and entry by any unauthorized personnel. A Work Area is a Regulated Area as defined by 29 CFR 1926.1101.

1.4 FIELD VERIFICATION

A. The Contractor shall field-verify all new and existing dimensions affecting work of this contract before ordering products.

1.5 CONSTRUCTION MATERIALS

A. The government will not furnish any materials for this project.

1.6 SALVAGED MATERIALS

A. Excess salvaged materials not reinstalled in the project remain the property of the government.

B. If materials will be reused or reinstalled, stockpile them within the rooms where the materials will be recovered for reuse. If more space is required, coordinate storage location with the NPS building or maintenance manager (whoever the appropriate site representative is).

1.7 PROJECT SUBMITTAL REQUIREMENTS

A. Pre-Project Submittals:

1. Proof that required federal, state, and local permits have been obtained.
2. Health and Safety Related Submittals: The Contractor will submit to the Contracting Officer for review an Asbestos-Specific Health and Safety Program, including, but not limited to, a Respiratory Protection Program, Medical Surveillance Program, and Accident Prevention Program. Additional health and safety-related submittals required are:

    a. The Contractor must submit a resume and copies of professional registrations and certifications for the Contractor's employee who will be the Competent Person for the project. The Competent Person must meet qualifications and have training and certifications defined by OSHA.
    b. Proof that all employees have passed appropriate medical examinations required by OSHA.
    c. Certification that all employees were instructed on the hazards of asbestos exposure, use and fitting of respirators and protective dress, use of showers, work area entry and exit procedures, work methods, and protective measures.
    d. Certification that each employee was properly fit-tested for each type of respirator anticipated to be used on the project.
    e. If rental equipment is to be used in work areas or to transport asbestos-contaminated waste, provide notice to the rental agency stating the intended use of equipment.
    f. Copies of the certificates for each employee for all asbestos training as required by this Section.
    g. Copies of the current physician's written approval to use respiratory protection for all employees subject to medical surveillance requirements defined within OSHA respiratory protection standards.
    h. Copies of qualitative or quantitative fit-tests for each employee wearing a respirator on this project.

3. Asbestos Abatement Project Plan: Develop and submit a written Asbestos Abatement Project Plan to the Contracting Officer for review. This plan shall be prepared by a certified state-licensed Asbestos Project Designer. The plan will be job-specific and list work procedures to be used during the removal of asbestos. This work plan will include, but not be limited to:

a. Construction schedule is in sufficient detail for the Contracting Officer to determine the duration and level of effort required for the project.
b. Description, accompanied by sketches to scale, of the decontamination enclosures and/or planned decontamination procedures.
c. A description of the ACM and trace-asbestos waste handling procedures. Including the name and address of the waste disposal site; the name, address, and telephone number of the entity that will transport ACM waste; and a transport manifest to be used to identify the quantity of waste removed and accepted by the disposal site.

B. Asbestos-Related Project Submittals – Ongoing Requirements:

1. Results of all worker exposure assessment air samples will be made available to applicable personnel per OSHA 29 CFR 1926.1101 within 24 hours of sample collection (if practical or as soon as reasonably possible due to the remote location of the Project Site). Submittals will include sample collection logs and laboratory analysis sheets.
2. After every disposal operation, provide copies of transport manifests, disposal receipts, and Chain-Of-Custody (COC) forms for all asbestos waste materials removed from the site. COC forms will include the date, address of the pickup site, name and address of the Contractor, names of persons responsible for pickup, name and address of the disposal site, the quantity of ACM waste, and the type of containers used. The form will be signed by the Contractor, Contracting Officer, disposal site operator, and hauler if a private hauler is employed.

C. Close-Out Report: Prepare a close-out report with the following items:

1. Description of ACM and trace-asbestos removal activities performed during the project.
2. Pre- and post-work photographs.
3. Description and results of all occupational and environmental sampling.
4. Results of waste characterization sampling.
5. Description and results of the third-party final clearance activities.
6. Copies of all ACWM manifest documents will be completed and signed by the Contracting Officer, the Contractor, the Transporter, and the Disposal facility.

1.8 QUALITY ASSURANCE

A. Qualifications: The workers will be fully qualified and experienced in the techniques of asbestos abatement, the handling and disposal of ACM and trace-asbestos-containing materials, and meet requirements as set forth by OSHA and the EPA.

B. Regulatory Requirements:

1. Comply with the OSHA Safety and Health Standard (29 CFR 1910.1001); OSHA Asbestos Regulation for the Construction Industry (29 CFR 1926.1101); National Emission Standards for Hazardous Air Pollutants (NESHAP) (40 CFR, Part 61, Subpart M); and the Asbestos Hazard Emergency Response Act (AHERA) (40 CFR, Part 763) regulations, and DOT regulations for transporting ACM waste materials.
2. The Environmental Protection Agency (EPA) and the Iowa Department of Natural Resources (IDNR) Air Quality Division require suspect ACM that will be impacted by renovation or demolition activities to be sampled or assumed to contain asbestos by a Certified Asbestos Building Inspector (CABI) before starting those activities.

3. Asbestos workers, contractors, and supervisors must obtain state licenses for all asbestos projects and have completed an EPA-approved training. Asbestos licensing information can be found at https://dial.iowa.gov/licenses/building/asbestos/license. A Contractor engaging in the removal or encapsulation of asbestos shall hold a permit for that purpose. A notification to the Division must be submitted when the total amount of asbestos to be removed or disturbed is above 260 lf, 160 sf, and 35 cubic feet of facility components and above the reporting thresholds as defined in 40 CFR 61.145 as amended on January 16, 1991. For more information about the notification process and form, visit https://dial.iowa.gov/licenses/building/asbestos/asbestos-demolition-renovation-notification.
4. The fees and/or charges for these permits will be included in the Contractor's bid price. Provide copies of all obtained permits to the Contracting Officer. Asbestos permit information can be found at https://dial.iowa.gov/licenses/building/asbestos/permits.
5. Dispose of asbestos waste at an authorized landfill site in accordance with requirements of NESHAP and applicable state and local guidelines and regulations.

IOSHA Regulations for Asbestos:

- Chapter 88B Asbestos Removal and Encapsulation, can be accessed at the following link: https://www.legis.iowa.gov/DOCS/ACO/IC/LINC/Chapter.88b.pdf
- Chapter 155 Asbestos Removal and Encapsulation, is available online at: https://www.legis.iowa.gov/docs/iac/chapter/875.155.pdf

Iowa OSHA is part of the Iowa Division of Labor Services in the Department of Workforce Development. IOSHA handles permitting for asbestos work, and the asbestos program's information can be found at the following web address: https://www.iowaosha.gov/asbestos-abatement

1.9 RECORDKEEPING

A. Daily Project Log: The Contractor's Site Supervisor shall maintain a daily project log to document the following information:

1. Date.
2. Name of Supervisor.
3. Number of workers on site.
4. Name of third-party IH monitoring the Project Site.
5. Equipment utilized.
6. Manometer data (if applicable).
7. Brief description of the daily work activities.
8. List any non-compliance, emergencies, stop-work orders issued by any governmental authorities (with detailed explanation), and descriptions of any other significant events.

1.10 PROJECT CONDITIONS

A. Maintain existing emergency exits and building emergency equipment, such as fire alarms, fire hose equipment, and emergency lighting devices, in operating order.

B. Do not damage the existing building materials not associated with the work or work area. Any damage to building materials not protected within or outside the work area and not associated with the work will be repaired or replaced by the Contractor at no cost to the government.

C. Wall, ceiling, and flooring systems can include multiple layers. Due to the inconsistency of these systems and the potential for overlay materials, all systems will be assumed to be ACMs unless specifically sampled and confirmed to be non-ACM. If the work will impact the wall, ceiling, or flooring systems, any material uncovered that was not previously sampled must be assumed to be asbestos. Samples of the uncovered material will be taken by a competent person, and work will be stopped until the Competent Person allows work to continue. If suspect ACM is encountered in wall or ceiling interiors in unanticipated locations, the unforeseen condition will be documented and described, including the location, extent and quantity of suspect ACM, and any immediate actions taken for safety or housekeeping purposes. The information will be provided to the IH and Contracting Officer.

PART 2 - PRODUCTS

2.1 PERSONAL PROTECTIVE EQUIPMENT

A. Respiratory Protection:

1. Workers who abate asbestos will be provided with personally issued, individually identified respirators that meet the required level of protection. During pre-cleaning, installation of critical barriers, and HEPA-vacuuming activities, the Contractor will utilize (at a minimum) half-face air-purifying respirators with approved HEPA filter cartridges. At a minimum, the Contractor will utilize Powered Air Purifying Respirators (PAPR) for removal and cleanup work unless the Contractor is able to provide an exposure assessment for removal of like materials that indicates the use of half-face negative pressure respirators are sufficient to protect abatement personnel.
2. The Contractor will provide workers with and require the use of respirators approved by NIOSH for asbestos in accordance with OSHA Standard 29 CFR 1926.1101. The minimum respiratory protection allowable will be an approved half-mask air-purifying respirator with HEPA cartridges. Disposable single-use respirators will not be permitted at the Project Site. Initial respiratory protection provided will be chosen prior to commencement of the removal operation based on the data from past exposure assessment monitoring or initial monitoring performed to accurately determine the airborne concentration of asbestos to which employees may be exposed.
3. Workers must perform positive and negative air pressure fit tests each time a respirator is donned, whenever the respirator design so permits. Prior to use, PAPRs will be tested for adequate flow as specified by the manufacturer.
4. Workers will be given a qualitative fit test in accordance with procedures detailed in OSHA 29 CFR 1926.1101 for all respirators to be used. An appropriately administered quantitative fit test may be substituted for the qualitative fit test. Documentation of adequate respirator fit must be provided by the Contractor. Fit tests must be administered annually.

B. Personal Protection Required for Asbestos Abatement:

1. Prior to the commencement of ACM and trace-asbestos removal activities, all personnel who will be required to enter the work areas for handling containerized ACM must have received the required training. Special onsite training on equipment and procedures unique to this job site will be performed as required. The Contractor is responsible for providing

emergency response training, such as Cardiopulmonary Resuscitation (CPR) and evacuation procedures, to all abatement personnel prior to starting the work.

2. All respiratory protection must be provided to the workers through the written respiratory protection program. This program will be posted within the cleanroom area of the worker decontamination enclosure system.
3. Disposable clothing, including head, foot, and full-body protection, will be provided in sufficient quantities and adequate sizes for all workers and authorized visitors. All personnel engaged in asbestos removal work will wear the approved disposable protective clothing constructed of spun-bonded olefin, polypropylene fabrics, or any other material of equivalent resistance to penetration by asbestos. Contractor will provide disposable whole-body clothing, including head covers, gloves, and shoe coverings, for all personnel in the asbestos control areas.
4. Persons with facial hair that may interfere with the respirator's seal shall not enter the work areas.
5. Respirators will be worn at all times when ACM and trace-asbestos-containing material is impacted.
6. Protective eyewear, gloves, rubber boots and/or other necessary footwear will be provided by the Contractor as required for the workers and authorized visitors. Safety shoes may be required for some activities.
7. Eating, drinking, smoking, chewing gum, and tobacco will not be allowed in the work areas.

2.2 HEPA-EQUIPPED FILTRATION EQUIPMENT

A. Vacuum Equipment: HEPA-filtered vacuuming equipment with a filter system capable of collecting and retaining asbestos fibers. Filters are required to be 99.97 percent efficient for retaining fibers 0.3 microns or larger.

B. If full abatement containments are constructed for friable asbestos removal, a sufficient quantity of negative pressure ventilation units will be used, equipped with HEPA filters and operated in accordance with the ASSE/SAFE Z9.2 (local exhaust ventilation requirements), EPA guidance document EPA 560/5- 83-002 (Blue Book), and EPA 560/5-85/024 (Purple Book). The IH will inspect all Negative Air Machines (NAMs) after installation and prior to asbestos removal. This inspection will include visual inspections of HEPA filters for physical/water damage, and the NAM filter logbook will be checked. Filter use will not exceed manufacturers' recommended machine hours (typically 600 to 900 hours). The Contractor will use only those brands manufactured and tested for commercial purposes. No “homemade” units will be permitted. HEPA filters will be approved by UL 586.

C. For friable ACM abatement, provide fully operational negative pressure systems supplying a minimum of one air change every 15 minutes. Determine the volume in cubic feet of the work area by multiplying the floor area by the ceiling height. Determine a total ventilation requirement in cubic feet per minute (cfm) for the work area by dividing this volume by the air change rate.

D. Locate NAMs so that makeup air enters the work area primarily through decontamination facilities and transverses the work area to the negative pressure air machines placed at a maximum distance from the worker entry or the other makeup air sources. Place the end of the unit or its exhaust duct through an opening in the plastic barrier or wall covering. The plastic around the unit or duct will then be sealed with tape. Vent NAM exhaust outside of the building unless authorized in writing by the IH or the Contracting Officer.

2.3 PLASTIC SHEETING

A. All abatement containments and decontamination stations will be constructed of a minimum of 6-mil poly sheeting sized to minimize the frequency of joints.

B. Polyethylene sheeting utilized for the worker decontamination enclosure will be opaque white or black.

2.4 TAPE

A. For use under dry and wet conditions, capable of being cleaned off surfaces without permanent marks or damage to the substratum.

2.5 CONTAINERS

A. Asbestos wastes will be stored in leak-tight, 55-gallon metal or fiberglass drums that have tightly fitting lids and are lined with 6-mil plastic bags. Label the containers in accordance with OSHA 1926.1101.

B. Disposal drums (if applicable) will be metal with locking ring tops and inner 6-mil polyethylene liners.

C. Stick-on warning labels, as per EPA or OSHA requirements, will be used on all disposal drums. All drums must additionally be labeled with a DOT code.

2.6 OTHER MATERIALS

A. Surfactant (wetting agent) will be a 50/50 mixture of polyoxyethylene ether and polyoxyethylene ester, or equivalent, mixed in a proportion as specified by the manufacturer.

B. Encapsulation materials will be either the penetrating or bridging type, pollution-free, non-toxic, with a Class A fire classification as specified herein. The encapsulant shall be flexible when cured and resistant to weathering, oxidation, aging, and abuse. The Contractor will submit product data, use instructions, and recommendations from the manufacturer for products intended for use.

C. Provide all other materials required for temporary construction.

PART 3 - EXECUTION

3.1 HOUSEKEEPING

A. Establish Regulated Areas per OSHA 29 CFR 1926.1101. Demarcate all of the Regulated Areas per OSHA.

B. Keep the project neat, orderly, and in a safe condition at all times. Use wet methods and HEPA-equipped vacuums only to collect all dust, waste, debris, etc., within areas where ACM and trace-

asbestos-containing materials are impacted. Prohibit the use of dry sweeping or vacuuming without using HEPA-equipped vacuums.

C. All powered hand tools and equipment used to impact ACM and trace-asbestos-containing material must be equipped with point-of-contact HEPA-equipped vacuum attachments, and/or the materials must be thoroughly wetted with an effective wetting agent suitable for the material and operation.

D. Access to the work areas must be restricted to "authorized persons" only as defined in OSHA 29 CFR 1926.1101. This may be accomplished using properly demarcated critical barriers, securing adjacent access points, or installing barrier plastic sheeting and signs following the implementation of compliant Negative Exposure Assessment activities and at the direction of the Competent Person.

E. The Contractor will verify the locations of utilities prior to any environmental abatement and interior demolition work. Existing water, sewer, electrical, communications, and gas lines within the boundaries of the environmental abatement work will be located, and non-essential utilities will be properly locked and tagged out at the limits of the environmental abatement or selective demolition work. This work will require a qualified and licensed electrician to verify safety and proper lockout.

F. Unless and until a Negative Exposure Assessment is produced, establish an equipment room or area adjacent to the regulated area for the decontamination of employees and their equipment contaminated with asbestos.

G. Provide enough containers for collecting ACWM and construction debris.

H. Place drop cloths adjacent to areas where impacts to ACM and trace-asbestos-containing material will occur.

I. Use HEPA-equipped vacuum and wet methods to clean up dust, debris, or other waste materials that may contain or be contaminated with ACM and trace-asbestos-containing material.

J. No dry sweeping is permitted at any time during any activities conducted on or around any ACM or trace-asbestos material such that any waste dust, debris, or other items could be contaminated.

K. Thoroughly wet all dry or drying ACM and trace-asbestos-containing material to prevent creating dust.

L. Keep ACWM and trace-asbestos-containing materials adequately wet and in covered containers.

M. Smoking is not permitted.

3.2 FIRE SAFETY

A. The Contractor will take the following precautions against fire:

1. The Contractor will comply fully with requirements of the federal, state, and local government, Owner, and Owner's Representative, including stipulations as outlined below. The Contractor will maintain and enforce all of the regulations imposed and will be

required to secure such protection as may be required. In the event of strikes, these precautions will not be relaxed. The Contractor is responsible for acquainting the local Fire Department with the existing conditions.

2. The Contractor will not impede or void the essential function of emergency fire exits while performing abatement work.
3. At least one qualified person thoroughly familiar with fire protection and prevention will be on duty at all hours while the Contractor's employees are working. This person will patrol the entire work area frequently and will have the authority to take immediate remedial action to eliminate unnecessary fire hazards.
4. Building material storage will be limited to secure areas within the building or, where stored outside, will be kept at least ten feet (10') away from the building. Storage areas will be approved by the Owner's Representative.
5. Fire extinguishers: Maintain and provide approved fire extinguishers throughout accessible areas. Placement, inspection, and maintenance of the fire extinguishers will comply with 29 CFR 1926.130. The Contractor will provide at least one 20 ABC fire extinguisher for every 3,000 sf of the work area, with a maximum distance of 100 feet between fire extinguishers. The Contractor will place (at a minimum) ABC fire extinguishers in the following locations:

    a. Cleanroom of personnel decontamination facility.
    b. Equipment room of personnel decontamination facility.
    c. Within the containment, at least one extinguisher for every 3,000 sf of the work area with a maximum distance of 100 feet between fire extinguishers.

6. Gasoline, oils, and other volatile liquids will be kept outside, to be brought into the building in quantities only as needed. Such storage will be in a well-ventilated location and removed from all open heating or lighting devices. Storage areas will be approved by the NPS.
7. The electrical wiring for construction light and power will be a properly fused Ground Fault Circuit Interrupter (GFCI) installed to conform to basic code requirements and maintained under the supervision of a competent electrician. This also applies to all temporary lines used by the Contractor.
8. During any torch-cutting activities, the Contractor will take necessary steps to comply with 29 CFR 1926 Subpart J (1926.350-1926.353), including provisions addressing transportation and storage of oxygen and acetylene cylinders, fire prevention, and project ventilation. The Contractor will utilize persons trained and knowledgeable in all equipment used onsite.

3.3 FULL ABATEMENT CONTAINMENT PREPARATION

A. Full abatement containments will be constructed of a minimum of two layers of 6-mil poly sheeting on all walls and floors and/or other surfaces not being abated. A single layer of 6-mil poly sheeting will be used to cover all ceilings not being abated in full abatement containments.

B. Isolate the abatement containment area by installing barricade tape and posting caution signs in and around work areas to comply with OSHA 1910.1001(g)(1) and federal, state, and local regulations.

C. Shut down and lock out all heating, cooling, and air-conditioning system components that supply or pass through the work area.

D. Wet wipe or HEPA-vacuum surfaces where critical barriers are to be installed and seal off all openings, such as corridors, doorways, windows, vents, ducts, grilles, diffusers, switch and outlet boxes, and lighting fixtures, with a minimum of one layer of 6-mil poly sheeting sealed with duct tape.

E. Clean all items to be removed from the work area using HEPA vacuum equipment and/or wet cleaning methods. Remove cleaned items to a temporary location as directed by the NPS.

F. Clean items to remain within the work area using HEPA vacuum equipment and/or wet cleaning methods and completely cover with one layer of 6-mil poly sheeting sealed with duct tape. Protect items with temporary barricades, covers, or pads as necessary to prevent damage.

G. Construct a five-stage Personnel Decontamination Unit (PDU) with a functional shower unit and a two-stage Waste Load-Out (WLO) in a location contiguous with the full abatement containment.

H. Install NAMs and establish negative pressure within the work area with only critical barriers, PDU and WLO in place. Install a manometer equipped with a strip-chart recorder to confirm a minimum of -0.02 pressure differential between the interior and exterior of the containment.

I. Pre-clean the entire work area using HEPA-vacuum equipment and/or wet cleaning methods. All waste generated during this activity will be treated as ACWM.

J. Cover floors, then walls, with 6-mil poly sheeting sealed with duct tape.

1. Cover floors with a minimum of two layers of plastic sheeting. Extend plastic at least 24 inches up walls. Cover walls with a minimum of two layers of plastic sheeting to the floor level. Extend the plastic at least 12 inches onto the floor in an overlapping manner with the floor. (e.g., install floor, wall, floor, wall with the wall being the top layer of sheeting.)
2. Install a single layer of 6-mil poly on the ceiling if the ceiling is not being abated, overlapping the ceiling poly and innermost layer of wall poly a minimum of 12 inches.

3.4 SECONDARY ABATEMENT CONTAINMENT PREPARATION

A. Secondary abatement containments will be constructed of a one-layer critical barrier of 6-mil poly sheeting over all windows, doors, vents, and openings or penetrations into the works area and one layer on walls, floors, and ceilings and/or other surfaces not being abated.

B. Isolate the abatement containment area through the use of barricade tape and post caution signs in and around work areas to comply with OSHA 1910.1001(g) (1) and federal, state, and local regulations.

C. Shut down and lock out all heating, cooling, and air-conditioning system components that supply or pass through the work area.

D. Wet wipe or HEPA-vacuum all surfaces where critical barriers are to be installed and seal off all openings, such as corridors, doorways, windows, vents, ducts, grilles, diffusers, switch and outlet boxes, and lighting fixtures, with one layer of 6-mil poly sheeting sealed with duct tape.

E. Clean items to be removed from work areas using HEPA-vacuum equipment and/or wet cleaning methods. Remove cleaned items to a temporary location as directed by the NPS.

F. Clean the items to remain in work areas using HEPA-vacuum equipment and/or wet cleaning methods and completely cover with one layer of 6-mil poly sheeting sealed with duct tape. Protect items with temporary barricades, covers, or pads as necessary to prevent damage.

G. Install a decontamination chamber attached to the secondary containment for personnel decontamination. If a remote shower is used (not directly attached to the secondary containment), a single chamber can be erected for exiting personnel to remove their dirty protective suit and don a new clean suit prior to proceeding to the remote decontamination unit. Install a separate WLO chamber for waste transfer, also attached to the secondary containment.

H. Install NAMs and establish a minimum of 0.02 inches of a water column of negative pressure within the work area prior to erecting secondary containment barriers.

I. Pre-clean the entire work area using HEPA-vacuum equipment and/or wet cleaning methods.

J. Cover porous or non-cleanable surfaces and objects with a minimum of one layer of 6-mil poly sheeting sealed with tape.

K. Install a minimum of one layer of 6-mil poly sheeting on all walls, floors, ceilings, and/or surfaces not being abated. Overlap poly sheeting a minimum of 12 inches at all seams.

3.5 PERSONNEL DECONTAMINATION SYSTEMS – REQUIRED FOR ABATEMENT

A. During asbestos abatement, a worker decontamination system will be provided in at least one pre-designated location for each area where a regulated asbestos area is established. Pre-constructed decontamination trailers are permitted.

B. If decontamination units are constructed onsite, construction plans, including materials and layout, will be submitted as shop drawings, and approved in writing by the Contracting Officer prior to work initiation. Worker decontamination enclosure systems constructed at the worksite will utilize 10-mil opaque black or white polyethylene sheeting or other acceptable materials for privacy.

C. The worker decontamination system will consist of at least a clean room, with a shower room and an equipment room, each separated from the other and from the work area by airlocks.

D. Entry to and exit from all airlocks and the worker decontamination system chambers will be through curtained doorways consisting of three sheets of overlapping polyethylene sheeting. Each sheet will be secured at the top and staggered from left to right for subsequent sheets. The innermost sheets will have weights attached to the bottom to ensure that they hang straight and maintain a seal over the doorway when not in use. Doorway designs, providing equivalent protection and acceptable to the IH, may be utilized.

E. Access between any two rooms within the worker decontamination system will be through an airlock with at least 3 feet separating each curtained doorway, resulting in a five-stage decontamination unit. The pathways into (from clean to contaminated) and out from (contaminated to clean) the work area will be clearly designated.

F. The cleanroom will be sized to adequately accommodate the clothes, equipment, and supplies for the work crew. Clean disposable clothing, replacement filters for respirators, towels, and other necessary items will be provided in adequate supply at the cleanroom. A location for postings will also be provided in this area. Whenever possible, a lockable door will be used to permit access into the cleanroom from outside the work area. Lighting, heat, and electricity will be provided as necessary for comfort. No asbestos-contaminated items may enter this room. The workers will use this area to suit up, store street clothes, don respiratory protection on their way to the work area, and dress in street clothes after showering.

G. The shower room will contain one or more showers as necessary to adequately accommodate the workers. Each showerhead will be supplied with hot and cold water adjustable at the tap. The shower will be constructed with fully sealed walls and an overflow pan base to prohibit leakage of any kind. The Contractor will supply and keep available at all times an adequate supply of soap, shampoo, and towels. Shower water will be drained, collected, and filtered through a system with at least a 5-micron particle size collection capability. All filtered water will be stored for subsequent treatment by the Contractor. The shower should be constructed to ensure against leakage of any kind and will be kept clean of all debris and ACM at all times.

H. The equipment room will be used to store equipment and tools at the end of a shift after they have been decontaminated using a HEPA-filtered vacuum and/or wet cleaning techniques as appropriate. Replacement filters (in sealed containers until used) for HEPA vacuums and negative pressure ventilation equipment, extra tools, containers of surfactant, and any other materials and equipment that may be required during the abatement may also be stored here as needed. A pan filled with water will be located in the work area just outside the equipment room for workers to clean off foot coverings after leaving the work area and prevent excessive contamination of the worker decontamination system. A container lined with a labeled 6-mil polyethylene bag for the collection of disposable clothing will be located in this room. Contaminated footwear (e.g., rubber boots, other reusable footwear) will be stored in this area for reuse the following workday.

I. If full abatement containments are constructed for abatement of friable ACM, the container and equipment decontamination unit will be constructed of a minimum of three chambers, each separated by curtained doorways of three overlapping sheets of poly affixed at the top and staggered at left and right for subsequent sheets. All materials, equipment, and bagged ACM will be decontaminated in this unit. This area will be separate from the personal decontamination unit and will be used for decontamination and temporary storage of bagged waste and equipment. All WLO areas must have a minimum of two separate chambers separated by airlocks. The size of each container/equipment decontamination unit chamber depends on the type of materials to be passed out of the container/equipment decontamination unit. Each separate airlock must be of sufficient size to fully contain, within the individual airlock, any item passed out through the container/equipment decontamination unit. If the regulated area is not a full abatement containment, the waste transfer area will be designated, and all waste will be transferred through that location only.

J. A worker outside the regulated area will hand an unused bag to the worker in the regulated area. A worker in the containment chamber will deposit a sealed bag containing asbestos into this unused bag. The double-bagged material will then be sealed and passed into the waste transfer station, wiped down, and passed outside of the regulated area. If equipment is used to transfer wrapped components, the equipment will pass through the waste transfer, and any visible debris on the wrapped components or the equipment will be wet-wiped or washed with an airless sprayer.

K. The container/equipment decontamination system will be equipped with the necessary facilities to wash and wipe the outside of the bags, drums, or other items prior to loading them into the vehicle for transportation to a landfill. Provisions will be made by the Contractor to prevent any contaminated run-off from leaving the decontamination system. All wastewater from the area will be filtered through a multi-part filter system with a final pore size of 5.0 microns, nominal. The container/equipment decontamination system will be equipped with the necessary facilities to wash and wipe the equipment leaving the containment.

3.6 WORK AREA ENTRY AND EXIT PROCEDURES

A. If a containment is constructed, emergency exits will be established and clearly marked with duct tape arrows or other effective designations to permit easy location from anywhere within the work area. Emergency exit indicator arrow or other effective designations will be placed at a maximum height of three feet to allow for visibility from a crawling position in the event of a fire that fills the work area with smoke. They will be secured to prevent access from uncontaminated areas and still permit emergency exiting. These exits will be properly sealed with polyethylene sheeting, which can be cut to permit egress if needed. These exits may be the worker decontamination system, the waste pass-out airlock, and/or other alternative exits satisfactory to fire officials. The Contractor shall place a razor knife at all emergency exit locations for quick cutting of poly sheeting to allow for quick egress. For uncontained regulated areas, the routes of egress will be clearly identified and communicated to all workers.

1. All workers and authorized personnel will enter the regulated areas through the worker decontamination system and must sign the entry log located in the cleanroom upon entry and exit.
2. All personnel entering any work area will read and be familiar with all posted regulations and personal protection requirements, including workplace entry and exit procedures and emergency procedures. A sign-off sheet will be used to acknowledge that these have been reviewed and understood by all personnel prior to entry.
3. All personnel will proceed first to the cleanroom, remove all street clothes, and put on required respiratory protection, disposable coveralls, head covering, and foot covering. Hard hats, eye protection, and gloves will also be utilized if required. Clean respirators and protective clothing will be provided and utilized by each person for each separate entry into the work area.
4. Personnel wearing designated PPE will proceed from the cleanroom through the shower room and equipment room to the main work area.
5. Before leaving the work area, all personnel will remove gross contamination from the outside of respirators and protective clothing through brushing and/or wet wiping procedures. Each person will clean the bottoms of protective footwear in the walk-off pan just prior to entering the equipment room. Upon entering the equipment room, they will then remove all personal protective equipment except for respirators. Deposit disposable clothing into appropriately labeled containers.
6. Reusable, contaminated footwear will be stored in the equipment room when not in use in work areas. Upon completion of abatement, it will be disposed of as asbestos-contaminated waste. Rubber boots may be decontaminated at the completion of the abatement for reuse.
7. Still wearing respirators, personnel will proceed to the shower area, shower and shampoo to remove residual asbestos contamination, and then clean the outside of the respirator and the exposed face under running water prior to removal of the respirator. Removal of the respirator and cleaning of the face underneath should be the last act prior to exiting the shower. Various types of respirators will require slight modification of these procedures.

An airline respirator with HEPA-filtered disconnect protection may be disconnected in the shower. Care must be taken with a powered air-purifying respirator facepiece since the filter and power pack assembly are not waterproof. A dual-cartridge respirator may be worn into the shower area. Cartridges must be replaced for each new entry into the work area. After showering and drying off, proceed to the cleanroom to don clean disposable clothing if returning to the work area or street clothes if leaving the work area.

B. Cartridges used on half-face or full-face Air Purifying Respirators (APRs) will be disposed of each time a worker leaves the asbestos control area (after showering).

C. If used, PAPR cartridges may be capped after the worker leaves the asbestos control area and has completed initial showering (prior to removing the respirator). When PAPR cartridges become wet or filter loading decreases airflow below 4 cfm, the PAPR cartridges must be disposed of as ACWM.

D. The following provides clarification of decontamination procedures to workers once workers have removed and disposed of disposable clothing (Tyvek suit with hoods and boots and disposable gloves) and have removed and stored all protective footwear, headwear, and eyewear within the equipment room. Workers wearing half-face APRs will proceed to the shower and shower total body (soap and shampoo) with emphasis on running water over the exposed face and the outside of the respirator. Once residual contamination on the head, body, and respirator has been removed, the worker will open a re-sealable bag, remove saturated cartridges, and place cartridges into and seal the re-sealable bag. The worker will complete additional showering and exit to the cleanroom.

E. Replacement (new) respirator cartridges will be stored on shelving in the cleanroom in original packaging. Cartridges will not be stored on the floor or in any other locations where water and/or physical damage may occur.

3.7 ASBESTOS REMOVAL WORK

A. Upon completion of the asbestos removal work, the Contractor will submit a written statement to the NPS attesting that all items containing asbestos have been disposed of in accordance with EPA 40, CFR, Part 61, Subpart M in the approved sanitary landfill(s). Documentation will include proof of regulatory approval and copies of the completed Waste Shipment Records signed by the generator, transporter(s), and disposal site operator and listing the following information:

1. Name, address, and telephone number of the waste generator.
2. Name and address of local, state, or EPA regional agency responsible for administering the asbestos NESHAP program.
3. Quantity of ACWM in cubic meters or yards.
4. Name and telephone number of the disposal site operator.
5. Name and physical location of the disposal site.
6. Date transported.
7. Name, address, and telephone number of the transporter(s).
8. A certification that the contents of the shipment are fully and accurately described by the proper shipping name and classified, packed, marked, and labeled, and are in all respects in proper condition for transport by highway according to all applicable international and government regulations.

3.8 AIR MONITORING

A. Air samples for asbestos may be collected adjacent to the abatement work areas by the IH each day abatement occurs. Asbestos air samples will be analyzed by the IH at the end of each work shift. The purpose of abatement perimeter monitoring is to detect faults in the work area isolation, such as:

1. Contamination of areas/buildings outside of the work area with airborne asbestos fibers.
2. Failure of the filtration system or rupture in the differential pressure system (if used) causing contamination of air outside with airborne asbestos dust.
3. Should any of the above conditions occur, immediately cease abatement activities until the fault is corrected. Do not recommence work until authorized by the IH.

B. If at any time in the abatement process that the outside work area monitoring results indicate that the asbestos concentration is at or above 0.01 fibers per cubic centimeter, CEASE ALL WORK except corrective action. After correcting the cause of unacceptable levels outside the work area, HEPA vacuum all surfaces that could potentially be contaminated; wet wipe, using amended water, all wettable surfaces, and HEPA vacuum a second time. Complete corrective work with no change in the contract sum if high airborne asbestos levels were caused by the Contractor's activities.

C. Aggressive final clearance air samples for asbestos will be collected in each asbestos removal area. The IH will collect clearance air samples for Phase Contrast Microscopy (PCM) analysis. If all of the air samples do not meet the post-abatement standard of 0.010 fibers per cubic centimeter (f/cc), the work area will be re-cleaned until acceptable air levels are achieved at no additional cost to the government. The asbestos abatement action will be considered complete when the clearance air samples result in acceptable air concentrations. Failed clearance costs will be the responsibility of the Contractor.

3.9 ABATEMENT WORK PRACTICES

A. The following sections describe protocols to be used for the various abatement tasks associated with this project. After all PPE and containment controls are implemented, the abatement task sequence will be as follows:

1. Shut down and lock out all heating, ventilation, and air-conditioning (HVAC) units and mechanical or refrigeration equipment within the work area.
2. Initiate pressure differential system using HEPA-filtered ventilation devices. Exhaust must be vented to the outside. Units are to be sized to achieve a minimum of one complete air change within the area every 15 minutes.
3. Construct the worker decontamination system at the entrance to the work area. Construct a waste pass-out decontamination system.
4. Pre-clean areas where critical barriers will be affixed and install critical barriers. Seal off any penetration or entrances into the work area using one layer of 6-mil polyethylene sheeting and tape. Polyurethane sealant foam may be used for small penetrations. Install a manometer equipped with a strip-chart recorder and verify a minimum of -0.020 inches of a water column pressure differential.
5. Pre-clean all mechanical equipment, floors, and walls.
6. Complete containment barrier installation (floors, walls, ceilings) per specification.

7. Approval of worker decontamination system(s) and containment system will be approved in writing prior to initiation of removal or repair activities.
8. Remove and properly dispose of existing friable and non-friable ACM.
9. Upon the completion of removal or repair activities, the Contractor will request a visual inspection of the work area by the IH. All surfaces are to be thoroughly cleaned prior to this request.
10. Upon confirmation from the IH that the scoped ACM has been completely removed and no visible dust or debris remains, the abatement contractor shall remove the wall, floor, and ceiling poly sheeting and package it as ACWM. The abatement contractor supervisor will then request a final visual inspection from the IH.
11. Upon successful completion of the final visual inspection, aggressive sampling of the work area will occur for air clearance with only the PDU, WLO, and critical barriers in place.
12. If air sampling indicates area clearance in accordance with specifications, the area will be declared clean, and all of the remaining polyethylene, tape, debris, etc., will be removed and bagged as asbestos-contaminated waste.
13. An encapsulating agent will be applied to surfaces from which ACM has been stripped if total decontamination or component removal cannot be accomplished. Approval from the IH must be obtained before applying the encapsulant.
14. All asbestos waste must be adequately wet as defined in the EPA/NESHAP Guidance dated December 1990. Per the EPA Guidance, “adequately wet” means to “sufficiently mix or penetrate with liquid to prevent the release of particles.” If visible emissions are observed coming from ACM waste, the material is not adequately wet. The absence of visible emissions is not sufficient evidence of being adequately wet. All asbestos-abated materials, regardless of whether they are first stripped from their substrate, must be adequately wet prior to disposal.

3.10 PROJECT DECONTAMINATION AND FINAL CLEARANCE

A. At the conclusion of the active abatement process, all surfaces in the abatement area should be thoroughly, and completely HEPA vacuumed. The surfaces include but are not limited to ceilings, walls, floors, windows, doors, fixtures of any kind, and appliances. This includes not just abated surfaces but also un-abated surfaces exposed to asbestos dust generated by the abatement process. All areas should be included in this HEPA process, except for spaces that:

1. Were found free of asbestos dust before the abatement process began;
2. Were properly sealed before the abatement process began; and
3. Were never entered during the abatement process.

B. Wet Wiping

1. Use Proper Wet Cleaning Procedures: At the conclusion of the active abatement process and after the first HEPA vacuuming, all surfaces identified as requiring HEPA vacuuming earlier should be thoroughly and completely washed with an amended water solution.
2. Change Cleaning Mixture Regularly: To avoid re-contaminating the area by using overly dirty water, users should carefully follow surface area limits and change cleaning mixtures accordingly. Users must ensure that dirty water does not re-contaminate the environment. This dirty water is potentially hazardous and should be treated accordingly.

C. Final Inspection

1. After the final cleanup is complete, the final inspection will take place. The IH will require 24-hour advanced notice of work areas prepared for the final inspection. The objective of the inspection is to verify abatement completeness and that no visible dust, dirt, or debris is present.
2. Post-abatement visual inspection will be performed to confirm the job completeness and determine whether all surfaces have been abated according to the approved abatement plan. The inspector will present the Contractor with a punch list of items to complete before the inspection process can continue.

3.11 RE-ESTABLISHMENT OF THE WORK AREA AND SYSTEMS

A. Re-establishment of the work area to general access will only occur following the completion of cleanup procedures and/or after clearance air monitoring has been performed and documented to the satisfaction of IH.

B. IH will visually inspect work areas for any remaining visible residues. Evidence of contamination will necessitate additional cleaning and air monitoring. Following satisfactory clearance of the work areas, the remaining polyethylene barriers may be removed and disposed of as asbestos-contaminated waste.

C. Upon successful completion of the final air monitoring, the Contractor will allow the NAMs to operate until all critical barriers are removed and sealed in disposal bags. The barriers will be disposed of as ACM waste. Removal of critical barriers and equipment will be conducted using appropriate PPE.

3.12 DISPOSAL

A. Asbestos waste will be thoroughly wetted and double-bagged in appropriately labeled asbestos disposal bags. Bags will be sealed airtight while the material is still wet. Bag tops will be twisted and sealed with duct tape, then bent over and sealed again with at least three wraps of duct tape. Biological materials may be disposed of in standard construction garbage bags in enclosed roll-off containers.

B. Additional material will not be added to bags, and bags will not be reopened after they have been sealed. Asbestos wastes will be transported by a properly licensed transporter in a covered truck and disposed of at a landfill approved for asbestos. A manifest document will be completed and signed by the Contracting Officer, the Contractor, the Transporter, and the Disposal facility. The Contractor will submit all disposal records to the Contracting Officer.

C. Immediately remove hazardous rubbish from the Project Site. Place other construction debris in refuse containers at least daily. Dispose of refuse at least weekly, in a legal manner, at public or private dumping areas outside the park. Do not burn or bury refuse inside the park.

D. Labels must be affixed to all bags, drums, or other containers containing ACWMs. The following label will be prominently displayed on the waste containers. All warning label print will be readily visible in large bold letter-size print on a contrasting background. Labels will contain the following information:

DANGER
CONTAINS ASBESTOS FIBERS
AVOID CREATING DUST
CANCER AND LUNG DISEASE HAZARD
DO NOT DISTURB WITHOUT PROPER TRAINING AND EQUIPMENT
DO NOT BREATHE ASBESTOS FIBERS

3.13 AIR AND WATER POLLUTION CONTROL

A. Take all necessary reasonable measures to reduce air and water pollution by any material or equipment used during construction.

B. Contractor personnel, building occupants, and visitors to the structures must be protected from exposure to airborne asbestos fibers.

C. Initial exposure assessment sampling will be conducted by the Contractor's Competent Person during activities that impact ACM until a record of potential exposure during each type of activity is established.

D. Samples will be analyzed using PCM according to the NIOSH Counting Method 7400 for Phase Contrast Microscopy.

E. Laboratories performing analysis of the air samples will be rated proficient by the American Industrial Hygiene Association (AIHA) Proficiency Analytical Testing (PAT) program.

F. Results of the air samples will be made available to applicable personnel and will be submitted to the IH within 24 hours of sample collection.

G. Do not allow ACWM or other waste materials to be washed into the lakes or other bodies of water.

3.14 REPORTS

A. Prepare and submit a close-out report at the completion of the work.

END OF SECTION 02 82 13

SECTION 02 83 33 – REMOVAL AND DISPOSAL OF MATERIAL CONTAINING LEAD

PART 1 - GENERAL

1.1 DESCRIPTION

This Section consists of tasks associated with the removal and disposal of material containing lead during the planned historic rehabilitation of the Effigy Mounds National Monument Visitor's Center in Harpers Ferry, Iowa (EFMO #318539) (Project Site). This structure was found to have Lead-Containing Paint (LCP) on interior surfaces. This specification establishes the required environmental controls to be used during impacts to Lead-Based Paint (LBP) or LCP during the planned rehabilitation work.

A. Testing and analysis at the Project Site identified LCP/LBP, including the following painted components:

The following three paints are identified as LBP:

1. PB4: Red paint on metal rafters in the basement contains a lead concentration of 0.81%.
2. PB09: Yellow paint on the concrete floor of the mechanical room contains a lead concentration of 5.7%.
3. PB10: Off-white paint on the concrete walls of the mechanical room contains a lead concentration of 0.53%.

The following three paints are identified as LCP:

1. PB06: The dark blue paint on the doorframe to the mechanical room contains a lead concentration of 0.48%.
2. PB07: The beige/brown paint on the door to the mechanical room contains a lead concentration of 0.39%.
3. PB13: The beige paint over reddish-brown paint on the basement stairwell handrail contains a lead concentration of .069%.

B. The Contractor will be responsible for the collection and analysis of personnel exposure samples for lead in accordance with the U.S. Occupational Safety and Health Administration (OSHA) regulations.

C. Appropriate training will be provided by the Contractor prior to their personnel performing work impacting identified LCP and potentially discovered LBP as required by applicable regulation 29 Code of Federal Regulations (CFR) 1926.62 (lead).

D. Two hours of lead awareness training will be provided to applicable National Park Service (NPS) personnel and their subcontractors working within the areas that contain LBP/LCP who can reasonably be expected to be impacted as part of this work.

E. Work shall be conducted so that no visible emissions are generated by construction that impacts LBP/LCP building components. Manual removal methods that do not generate dust or debris must

be utilized. The Contractor's Competent Person (CP) shall determine if the methods used present a risk of exposure above OSHA's Action Level (AL) of 30 micrograms per cubic meter over an 8-hour Time-Weighted Average (TWA) as required by the OSHA lead construction standard 29 CFR 1926.62.

F. Any employee that may be exposed to lead levels in excess of the OSHA AL will have a minimum of two hours of lead awareness training.

G. The Contractor is responsible for providing an independent Industrial Hygienist (IH). This IH will coordinate with the Contracting Officer to provide waste profile sampling.

H. The Contractor must carefully collect, containerize, characterize for waste disposal, transport, and dispose of any regulated or hazardous wastes generated during this project, including LCP dust and debris and other potentially lead-contaminated wastes. If LCP components or debris are disposed of offsite, a representative waste profile sample shall be collected and analyzed as required by the receiving landfill for waste profiling purposes. The Contractor shall provide temporary staging areas for waste and conduct waste profile sampling; analyses of the sample shall be by Toxicity Characteristic Leaching Procedure (TCLP).

I. The Contractor will be responsible for the cleanup and testing of any area(s) contaminated due to Contractor negligence at no additional cost to the NPS.

1.2 REFERENCES AND RELATED SPECIFICATIONS

A. All work under this contract will be conducted in strict accordance with all applicable federal, state, and local regulations, standards, and codes governing lead remediation and other concurrent trade work. The most recent edition of any relevant regulation, standard, document, or code will be in effect. Where conflict among the requirements or within these specifications exists, the most stringent requirements will be utilized.

B. Section 02 81 01: Offsite Transportation and Disposal.

C. Landmark Environmental, Inc., *Hazardous Materials Inspection for Effigy Mounds National Monument Visitors Center, Harpers Ferry, Iowa. Landmark Project No. 24061.001. January 31, 2025.*

D. American Society of Safety Engineers (ASSE), ASSE/SAFE Z9.2 *Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems*, 2012.

E. Underwriters Laboratories (UL), *UL Standard for Safety High-Efficiency Particulate Air Filter Units, UL 586*, 2009; reprint 2017.

F. Iowa Department of Natural Resources (IDNR) lead resources page at https://www.iowadnr.gov/environmental-protection/land-quality/contaminated-sites/brownfields/lead-based-paint#Proper-Disposal-and-Testing-327.

G. The U.S. Occupational Safety and Health Administration (OSHA), 29 CFR 1910.1025 – General Industry Standards for Lead.

H. OSHA, 29 CFR 1926.62 – Construction Industry Standards for Lead.

I. OSHA, 20 CFR 1910.120 – Hazardous Waste Operations and Emergency Response.

J. OSHA, 29 CFR 1910.134 – Respiratory Protection.

K. OSHA, 29 CFR 1910.145 – Accident Prevention.

L. OSHA, 29 CFR 1910.2 – Medical Records.

M. OSHA, 29 CFR 1910.38 – Emergency/Fire Prevention Programs.

N. OSHA, 29 CFR 1910.1200 – Hazard Communication.

O. The U.S. Environmental Protection Agency (EPA), 40 CFR 260 – Hazardous Waste Management System: General.

P. EPA, 40 CFR 261 – Identification and Listing of Hazardous Waste.

Q. EPA, 40 CFR 262 – Standards Applicable to Generators of Hazardous Waste.

R. EPA, 40 CFR 263 – Standards Applicable to Transporters of Hazardous Waste.

S. The U.S. Department of Transportation (US DOT), 49 CFR 171 and 172 – Hazardous Materials Regulations.

T. National Institute for Occupational Safety and Health (NIOSH), *Certified Equipment List.*

U. NIOSH, *Occupational Exposure Sampling Strategy, 77-173*.

V. NIOSH, *Guide to Industrial Respiratory Protection, 87-116*.

W. ASTM International, *Collection of Settled Dust Samples Using Wipe Sampling Methods for Subsequent Lead Determination E1792*, 2003, reprinted in 2016.

1.3 DEFINITIONS

A. Atomic Absorption Spectrometry (AAS): The laboratory method used to determine lead in bulk and air samples.

B. Authorized Person: Any person(s) authorized by the employer and required by work duties to be present in regulated areas.

C. Critical Barrier: One or more layers of plastic sealed over all openings into a work area(s) or any other similarly placed physical barrier sufficient to prevent airborne dust in a work area(s) from migrating into an adjacent area.

D. Competent Person (CP): The Contractor's employee who identifies lead hazards in the workplace, has the authority to promptly implement corrective measures to eliminate them, and is trained and accredited as defined in 29 CFR 1926.62. The duties of the Competent Person include but are not limited to the following activities: 1) controlling entry to and exit from the lead work area; 2) supervising any employee exposure monitoring required by the standards; 3) ensuring that all employees working within a work area wear the appropriate Personal Protective Equipment (PPE)

and are trained in the use of appropriate methods of exposure control and decontamination, and 4) ensuring that engineering controls in use are functioning properly.

E. Disposal Bag: A properly labeled 6- mil thick, leak-tight polyethylene bag used for containerizing and/or transporting lead waste to the disposal site.

F. Disturbance: Activities that disturb LCP or LBP within the facility.

G. Generator: Any person or site whose processes and actions create hazardous waste per 40 CFR 260.10.

H. HEPA Filter: A High-Efficiency Particulate Air (HEPA) filter capable of trapping and retaining 99.97% of all mono-dispersed particles (e.g., asbestos fibers) of 0.3 microns in diameter.

I. Industrial Hygiene: An area of specialization in the field of industrial safety and health concerned with predicting, recognizing, assessing, controlling, and preventing environmental stressors in the workplace that can lead to sickness, disease, or other forms of impaired health.

J. Industrial Hygienist: A person having a college or university degree(s) in engineering, chemistry, physics, medicine, or related physical or biological sciences who, by virtue of special studies and training, has acquired competence in industrial hygiene.

K. Lead: Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds.

L. Lead-Based Paint (LBP): Paint or other coatings containing at least 0.5% lead by weight or 1.0 milligrams (mg) per square centimeter.

M. Lead-Containing Paint (LCP): Paint or other coatings containing concentrations of lead that are quantifiable through laboratory analysis.

N. Negative Initial Exposure Assessment (IEA): A demonstration by the employer that demonstrates employee exposure during an operation is expected to be consistently below the Permissible Exposure Limit (PEL) for representative 8-hour TWA samples.

O. Personal Monitoring: Sampling of the dust/fiber concentrations within the breathing zone of an employee.

P. Regulated Area: An area established to demarcate the areas where lead-containing dust in the air exceeds, or there is a reasonable possibility it may exceed, the PEL or other applicable industry standards for occupational exposure limits.

Q. Removal: All work operations where lead is taken out of or stripped from structures or substrates, including demolition operations.

R. Renovation: Modifying any existing structure or portion thereof.

S. Time-Weighted Average (TWA): The employee's average airborne exposure in any 8-hour work shift of a 40-hour work week which shall not be exceeded. The 8-hour TWA PEL is the level of exposure established as the highest level of exposure an employee may be exposed to without incurring the risk of adverse health effects.

T. Toxicity: The ability of a substance to cause harmful effects. Toxic waste is harmful or fatal when ingested or absorbed (e.g., containing mercury, lead, etc.). When toxic waste is land-disposed, contaminated liquid may leach from this waste and pollute groundwater.

U. Toxicity Characteristic Leaching Procedure (TCLP): Analytical test designed to determine the mobility of both organic and inorganic analytes present in liquid, solid, and multi-phase wastes. TCLP is used to determine if a waste will be a hazardous waste as defined by 40 CFR Part 261.

V. Transporter: Hazardous waste transporters are individuals or entities that move hazardous waste from one site to another by highway, railway, water, or air per 40 CFR 260.10. This includes transporting hazardous waste from a generator's site to a facility that can recycle, treat, store, or dispose of the waste. It can also include transporting treated hazardous waste to a site for further treatment or disposal.

W. Treatment: Treatment is defined as any method, technique, or process designed to change the physical, chemical, or biological character or composition of any hazardous waste to neutralize such waste, recover energy or material resources from the waste, or render such waste non-hazardous, or less hazardous; safer to transport, store or dispose of; or amenable for recovery, amenable for storage, or reduced in volume.

X. Visible Emissions: Any emissions potentially containing particulate and/or lead-containing dust material that is visually detectable without the aid of instruments.

Y. Work Area: The area(s) where lead-related work or removal operations are performed, which is defined and/or isolated to prevent the spread of lead-containing dust or debris and entry by unauthorized personnel. The work area includes the waste load-out area.

1.4 FIELD VERIFICATION

A. The Contractor shall field-verify all new and existing dimensions affecting the work of this contract before ordering products.

1.5 CONSTRUCTION MATERIALS

A. The Government will not furnish any materials for this project.

1.6 SALVAGED MATERIALS

A. Excess salvaged materials not reinstalled in the project remain the property of the Government.

B. Stockpile material within the rooms where the materials are recovered for reuse. If more space is required, the Contractor shall coordinate storage location with the Contracting Officer.

1.7 PROJECT SUBMITTAL REQUIREMENTS

A. Health and Safety-Related Submittals:

1. The Contractor's IH will prepare submittals on behalf of the Contractor for submission to the Contracting Officer for approval. These submittals will include a Health and Safety Plan that includes a Respiratory Protection Program, Medical Surveillance Program, Accident Prevention Program, and Lead Waste Management and Disposal Plan.
2. The Contractor will submit a resume and copies of professional registrations/certifications for the Contractor's employee, who will act as the CP for the project. The CP must meet qualifications and have the training and certifications defined by OSHA.
3. The Contractor will submit copies of attendance records and certificates for each employee for all lead hazard awareness training required by this Section.
4. The Contractor will submit proof that all employees have passed the appropriate medical examinations required by OSHA for respirator use and work around lead, where the AL may be exceeded.
5. The Contractor will submit copies of current physicians' written medical opinions for all employees subject to the medical surveillance requirements defined in OSHA Respiratory Protection Standards.
6. If the CP determines that respiratory protection is required, the Contractor shall submit certification that each employee was properly fitted with a personally assigned respirator.
7. The Contractor will submit certification that HEPA vacuums, HEPA-equipped negative pressure air machines, and other equipment required to contain airborne lead dust or other particulates meet American National Standards Institute (ANSI) Z9.2 79.
8. For HEPA vacuums used for LBP and LCP debris cleanup or HEPA-equipped negative pressure air machines, the Contractor shall maintain and submit copies of HEPA-filter change logs for each piece of HEPA-equipped equipment at the project site to verify timely filter changes (filters should be changed every 750 hours of use).

B. Occupational Exposure Sampling:

1. After initial exposure monitoring, the Contractor shall submit sampling results to the NPS's Contracting Officer within three working days of sample collection, signed by the testing laboratory employee performing the analysis, the employee who performed the sampling, and the CP.
2. The Contractor shall provide documentation that:

    a. The initial air monitoring is representative of each job classification or previously collected exposure data and represents the worker's job tasks for this project.
    b. The worker exposure data, including a complete description of the job tasks (such as manual demolition and manual scraping, etc.) where lead-containing coatings are present, is submitted.

C. Lead Waste Management and Disposal Plan:

1. The Contractor shall submit a Lead Waste Management and Disposal Plan that complies with applicable requirements of federal, state, and local hazardous waste regulations and shall include:

    a. Identification and classification of wastes associated with the work.
    b. Estimated quantities of wastes to be generated and disposed of.
    c. Names and qualifications of each Contractor who will be transporting, storing, treating, and disposing of the wastes.
    d. Description of the waste containment, removal, and disposal procedures. Proper temporary containment of the waste includes using acceptable leak-tight waste

containers such as 55-gallon drums or other secure and covered storage areas that prevent spillage and exposure to stormwater, as well as proper marking/labeling of the containers. Waste will be cleaned up and containerized daily.

e. Description of any site construction process, such as the use of chemical strippers or caustic agents that may alter or treat waste, rendering hazardous waste non-hazardous.

D. The Contractor is responsible for the waste profile sampling and analysis for disposal of waste streams that include building components with LCP or LBP.

1.8 POST-PROJECT SUBMITTAL REQUIREMENTS

A. After every disposal operation, provide copies of transport manifests and disposal receipts for all lead waste materials removed from the site. The Chain-Of-Custody (COC) form will include the date, address of the pickup site, name and address of the Contractor, names of persons responsible for pickup, name, and address of the disposal site, quantity and type of hazardous waste, and the type of containers used. The form will be signed by the Contractor, Contracting Officer, disposal site operator, and hauler if a private hauler is employed.

B. Closeout Report: Prepare a closeout report with the following items:

1. Description of removal and stabilization activities performed during the project.
2. Description and results of occupational and environmental sampling.
3. Description of cleanup activities.
4. Description and results of confirmatory sampling.
5. Results of waste characterization sampling.
6. COC forms for confirmatory sampling and waste characterization sampling.
7. Description of completed work, including the onsite processing of waste (if any), handling and transportation procedures, and disposal site.
8. Certificates of disposal.
9. Transport manifests.

1.9 QUALITY ASSURANCE

A. Qualifications:

1. Competent Person: The Contractor shall submit the name, address, and telephone number of the CP selected to perform the responsibilities specified in Section 1.3 Definitions, Item D, “Competent Person.” Provide documentation of construction project-related experience with the implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), which shows the ability to assess occupational and environmental exposure to lead, experience with the use of respirators, PPE, and other exposure reduction methods to protect employee health. Submit proper documentation that the CP is trained and certified in accordance with federal, state, and local laws.
2. Workers/Supervisors: The Contractor shall ensure that all workers will be fully qualified and experienced in handling and disposing of lead-painted materials. Submit training certifications and proper documentation for each worker, supervisor, and CP performing LBP/LCP removal activities in accordance with federal, state, and local laws.

3. Analytical Laboratory: The Contractor and IH shall use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) that is accredited by either the American Association for Laboratory Accreditation (AALA) or the American Industrial Hygiene Association (AIHA) and is also a successful participant in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis will be OSHA-approved.
4. The Contractor's Third-Party Industrial Hygiene Consultant: The Contractor shall submit the name, address, and telephone number of a third-party industrial hygiene consultant selected to assist the Contractor with any project submittals, etc. Submit the proper documentation that the consultant is trained and certified as an inspector technician or inspector/risk assessor by the EPA-authorized state (or local) certification and accreditation program.

B. Regulatory Requirements: Develop and implement a written Environmental Control Program covering the employees, visitors, and government personnel who can potentially be exposed to hazardous materials on the project. The program must include, but not be limited to, the following elements: a list of chemicals, labels, and other forms of warning, material Safety Data Sheets (SDS), all employee information and training, procedures, chemical safety and spill response information, and subcontractor notifications.

C. All work under this contract will be performed in strict accordance with all applicable federal, state, and local regulations, standards, and codes governing lead removal and other concurrently performed trade work. The most recent edition of any relevant regulation, standard, document, or code will be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirements will be utilized. Copies of applicable standards, regulations, and codes shall be maintained in the Contractor's project records and be available for access at the worksite during the work.

PART 2 - PRODUCTS

2.1 PERSONAL PROTECTIVE EQUIPMENT

A. Respiratory Protection:

1. If required by the CP, workers shall be provided with personally issued, individually identified respirators that meet the required level of protection. If construction activities generate any airborne dust emissions, then the Contractor will utilize appropriate respirators with HEPA filter cartridges until the initial air sampling exposure assessment demonstrates that the use of the respirators is not required to protect personnel from air concentrations of lead dust. The Contractor will provide workers with and require the use of respirators approved by NIOSH for lead, in accordance with the OSHA Standard 29 CFR 1926.62. Disposable single-use respirators or dust masks are not permitted.
2. Workers must perform positive and negative air pressure fit tests each time a respirator is worn.
3. If respiratory protection is required, workers will be given a qualitative fit test in accordance with procedures detailed in OSHA 29 CFR 1926.62 for all respirators to be used. An appropriately administered quantitative fit test may be substituted for a qualitative fit test. Documentation of adequate respirator fit must be provided to the workers. Fit tests will be administered at the start of the project.

4. Persons having facial hair that may interfere with the seal of a respirator will not be allowed to use respirators.

B. Personnel Protection for Lead-Containing and Lead-Based Paints:

1. If the work generates dust and debris from LBP/LCP, disposable clothing, including head, foot, and full-body protection, shall be provided in sufficient quantities and adequate sizes for all workers and any authorized visitors. All personnel engaged in LCP and LBP demolition work that may potentially exceed the OSHA AL will wear approved disposable protective clothing constructed of spun-bonded olefin or polypropylene fabrics or other material of equivalent resistance to penetration. Disposable whole-body clothing, including head covers, gloves, and shoe coverings, will be provided to and worn by all personnel in the removal work areas. If elastic sleeve closures are not provided, sleeves will be secured with duct tape to the gloves. Washable footwear having a non-skid tracking surface will be provided and used by all personnel within the lead work area.
2. Contaminated clothing will be treated as lead waste and undergo the same disposal procedures utilized for lead waste.
3. Protective eyewear, gloves, rubber boots, and/or other footwear will be provided as required for workers and authorized visitors. Safety shoes may be required for some activities.
4. Eating, drinking, smoking, chewing gum, or tobacco will not be allowed in the LCP/LBP work areas.

2.2 EQUIPMENT

A. HEPA-equipped filtration equipment:

1. Vacuum Equipment: HEPA-filtered vacuuming equipment with a filter system capable of collecting and retaining lead particulates shall be used for housekeeping during work that creates dust and debris from LBP/LCP. The filters must be 99.97% efficient for retaining particles 0.3 microns or larger.
2. Use of HEPA filters will not exceed the manufacturer's recommended machine hours (typically 600 to 900 hours). Verification of proper/timely filter changes for each piece of equipment shall be submitted to NPS in the form of a filter change log. The Contractor will use only those brands manufactured and tested for commercial purposes. No “homemade” units will be permitted. The HEPA filters will be approved by UL Method 586.

B. If utilizing rental equipment that will be impacted by or used to dispose of LCP and LBP or other lead-containing materials, notify the rental agency in writing concerning the intended use of the equipment.

2.3 PLASTIC SHEETING

A. Six mil in thickness and sized to minimize the frequency of joints.

B. Polyethylene sheeting utilized for the worker decontamination enclosure, if required by the CP, will be opaque white or black.

2.4 TAPE

A. For use in dry and wet conditions, and capable of being cleaned off surfaces without permanent marks or damage to the substratum.

2.5 CONTAINERS

A. Air and watertight, 55-gallon metal or fiberglass drums with tightly fitting lids will be used to package the waste. The Contractor will line the drum(s) with 6-mil plastic bags and label containers in accordance with the OSHA requirements and the EPA Resource Conservation and Recovery Act (RCRA) requirements. All containers will carry appropriate labels as required by OSHA, EPA, and U.S. DOT, and the Arizona Department of Transportation (ADOT). No container on the Project Site will be unlabeled as to its contents (empty or otherwise).

PART 3 - EXECUTION

3.1 DECONTAMINATION ENCLOSURE SYSTEMS

A. Worker Decontamination

1. A hygiene facility, as described in OSHA 1926.62, must be utilized during the lead removal activities.

3.2 AIR MONITORING

A. Air samples for lead exposure will be collected by the Contractor for a minimum of the first three days that LCP/LBP work occurs. Lead air samples will be analyzed by an independent accredited laboratory on a rush basis. The purpose of air monitoring is to confirm the protection of workers at the point of operation and to confirm the effectiveness of emission controls established by the Contractor. The Contractor will prevent the contamination of areas/outside of the regulated areas with airborne lead-containing dust. Should any of the above occur, the Contractor shall:

1. Immediately cease removal activities within the regulated areas until the fault is corrected and any contamination outside the regulated area(s) is completely remediated. Do not commence work within the original regulated area until authorized by the Contracting Officer or the Contractor's Industrial Hygienist.
2. Complete the corrective work at no additional cost to the Government if high airborne lead-containing dust levels are identified during the Contractor's activities.

3.3 WORK PRACTICES

The following sections describe protocols for the various LCP and LBP removal tasks associated with this project.

A. Manual or power sanding or grinding of lead-coated surfaces or materials is not permitted unless the tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead is prohibited.

B. Select lead removal processes to minimize contamination of work areas outside the control area with lead-contaminated dust or other debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead.

C. Cleanup

1. Ensure surfaces in the lead control area are free of accumulations of dust and debris.
2. Restrict the spread of dust and debris; keep the waste from being distributed over the work areas.
3. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead operations have been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA-filtered vacuum cleaner and wet mopping and wet wiping the areas.
4. Re-clean areas that show dust or debris.
5. After removing visible dust and debris, wet wipe and HEPA vacuum all surfaces in the controlled area, as applicable.
6. If adjacent areas become contaminated at any time during the work, clean and visually inspect all contaminated areas.
7. The CP will certify in writing that the areas have been cleaned of lead contamination.

3.4 DISPOSAL

A. Additional material will not be added to bags, and bags will not be reopened after they have been sealed. The lead wastes will be transported by a properly licensed transporter and disposed of at a landfill approved for lead.

B. Labels must be affixed by the Contractor to all bags, drums, or other containers containing lead waste. The label will be prominently displayed on the waste containers. All warning label print will be readily visible in large bold letter-size printed on a contrasting background. This applies only to waste that is considered hazardous waste per the waste profile.

3.5 AIR AND WATER POLLUTION CONTROL

A. The Contractor must take all necessary reasonable measures to reduce air and water pollution by any material or equipment used during construction.

B. Contractor personnel and visitors must be protected from exposure to airborne lead-containing dust.

C. Lead air or dust samples will be analyzed by AAS using a laboratory accredited by the Environmental Lead Laboratory Accreditation Program and American Industrial Hygiene Association.

D. The Contractor will not dispose of lead, coating wastes, or oils in storm or sanitary drains.

E. The Contractor will not allow lead or other waste materials to be washed into streams or bodies of water.

3.6 REPORTS

A. Prepare and submit a final closeout report at the completion of the work.

END OF SECTION 02 83 33

SECTION 02 84 16 – HANDLING OF LAMPS CONTAINING MERCURY

PART 1 - GENERAL

1.1 DESCRIPTION

A. The work included in this section consists of the abatement of fluorescent light bulbs that may contain mercury to be performed during the planned historic rehabilitation of the Effigy Mounds National Monument Visitor's Center in Harpers Ferry, Iowa (EFMO #318539), (Project Site). The scope items and requirements for light bulbs are described in Sections B through D below.

B. Other-Regulated Materials (ORMs) are materials that will require special handling or disposal during a renovation project, which impacts these materials. An inventory of Regulated Building Materials (RBMs), including ORMs, is contained in Landmark's *Hazardous Materials Inspection for Effigy Mounds National Monument Visitors Center, Harpers Ferry, Iowa. Landmark Project No. 24061.001. January 31, 2025.*

C. Regulated lighting waste identified at the Project Site includes compact light bulbs. The regulated lights included compact fluorescent bulbs. The U.S. Environmental Protection Agency (EPA) classifies bulbs containing mercury concentrations greater than or equal to 0.2 milligrams per liter (mg/l) as potentially hazardous waste. Where inspected, the ballasts in fluorescent light fixtures were labeled "non-PCB." In most locations where light fixtures were inspected, the fluorescent light tubes had green end caps, indicating low mercury content. Unless otherwise labeled, compact fluorescent light bulbs at the Project Site are presumed to contain mercury in concentrations greater than 0.2 mg/l and should be removed and transported for recycling.

D. The Contractor will be responsible for the cleanup and testing of any area contaminated due to Contractor negligence at no additional cost to the National Park Service (NPS).

1.2 RELATED SPECIFICATIONS

A. Section 02 81 01: Offsite Transportation and Disposal

B. Section 02 82 13: Asbestos Abatement

C. Section 02 83 33: Removal and Disposal of Materials Containing Lead

D. Section 10 81 19: Rodent and Bird Control

1.3 FIELD VERIFICATION

A. Field-verify all new and existing dimensions affecting the work of this contract before ordering products.

1.4 CONSTRUCTION MATERIALS

A. The government will not furnish any materials for this project.

1.5 SALVAGED MATERIALS

A. Excess salvaged materials not reinstalled will remain the property of the government.

B. Stockpile materials within the rooms where materials are recovered for reuse. If more space is required, coordinate storage location with the Contracting Officer.

1.6 DEFINITIONS

A. Authorized Person: Any person authorized by the employer and required by work duties to be present in regulated areas.

B. Competent Person (CP): The Contractor's employee who is capable of identifying existing asbestos and lead hazards in the workplace, who has the authority to take prompt corrective measures to eliminate them, and who is trained/accredited as defined in 29 Code of Federal Regulations (CFR) 1926.1101 and 29 CFR 1926.62. The duties of the Competent Person include, but are not limited to, the following: 1) controlling entry to and exit from the abatement work area; 2) supervising any employee exposure monitoring required by standards; 3) ensuring that all employees working within a work area wear the appropriate personal protective equipment and are trained in the use of appropriate methods of exposure control and decontamination; and 4) ensuring that engineering controls in use are functioning properly.

C. Industrial Hygiene: An area of specialization in the field of industrial safety and health that is concerned with predicting, recognizing, assessing, controlling, and preventing environmental stressors in the workplace that can lead to sickness, disease, or other forms of impaired health.

D. Industrial Hygienist (IH): A person having a college or university degree(s) in engineering, chemistry, physics, medicine, or related physical or biological sciences who, by virtue of special studies and training, has acquired competence in industrial hygiene.

E. Personal Monitoring: Sampling of contaminate concentrations within the breathing zone of an employee.

1.7 SUBMITTALS

A. As specified in Sections 1.7, 1.8, and 1.9.

B. Proof that required federal, state, and local permits have been obtained.

C. Certification that all employees have been instructed on the hazards of asbestos and lead exposure since they will be working in and near areas with known asbestos-containing materials and lead-containing materials.

D. Certification that High-Efficiency Particulate Air (HEPA) vacuums, HEPA-equipped negative pressure air machines, and other equipment required to contain airborne fibers meet American National Standards Institute (ANSI) Z9.2-79.

E. Construction schedule in sufficient detail for the Contracting Officer to determine the duration and level of effort required for the project.

F. Description of waste handling procedures. Include the name and address of the waste disposal site; name, address, and telephone number of the entity that will transport the waste; and a sample of transport manifest to be used to identify the quantity of waste removed and accepted by the disposal site.

G. After every disposal operation, provide copies of transport manifests, disposal receipts, and Chain-Of-Custody (COC) forms for all ORM waste materials removed from the Site. The COC form will include the date, address of the pickup site, name and address of the Contractor, names of persons responsible for pickup, the name and address of the disposal site, quantity of waste, and type of containers used. The form will be signed by the Contractor, the Contracting Officer, the disposal site operator, and the hauler if a private hauler is employed.

H. If compact light bulbs are recycled, provide copies of the shipping record and receipt for recycled materials to the Contracting Officer for review.

1.8 APPLICABLE STANDARDS AND GUIDELINES

A. All work under this contract will be performed in strict accordance with all applicable federal, state, and local regulations, standards, and codes governing universal waste removal. The most recent edition of any relevant regulation, standard, document, or code will be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirements will be utilized.

B. Applicable Publications: The publications, codes, and safety regulations listed below are part of this specification and are included for reference only. It will be the responsibility of the Contractor to observe and follow all applicable guidelines:

29 CFR 1910.145	Accident Prevention
29 CFR 1910.2	Medical Records
29 CFR 1910.38	Emergency/Fire Prevention Programs
29 CFR 1910.1200	Hazard Communication
40 CFR 61 Sub. A	General Provisions
40 CFR 61 Sub. B	EPA
49 CFR 171 And 172	Department of Transportation (DOT)
77-173 NIOSH	Occupational Exposure Sampling Strategy
76-189 NIOSH	Guide to Ind. Respiratory Protection
91-105 NIOSH	Certified Equipment List

1.9 CODES AND REGULATIONS

A. The general applicability of codes, regulations, and standards: Except to the extent that more explicit or more stringent requirements are written directly into the Contract Documents, all

applicable codes, regulations, and standards have the same force and effect (and are made a part of Contract Documents by reference) as if copied directly into the Contract Documents or as if published copies are bound herewith.

B. Contractor Responsibility: The Contractor will assume full responsibility and liability for the compliance with all applicable federal, state, and local regulations pertaining to the work practices, hauling, disposal, and protection of workers, visitors at the Project Site, and persons occupying areas adjacent to the Project Site. The Contractor is responsible for providing all medical examinations and maintaining the medical records of personnel as required by the applicable federal, state, and local regulations. Copies of these records will be kept available for the Contracting Officer's review.

C. OSHA: U.S. Department of Labor, Occupational Safety and Health Administration (OSHA) including but not limited to:

1. Access to Employee Exposure and Medical Records: Title 29, Part 1910, Section 20 of the Code of Federal Regulations;
2. Hazard Communication: Title 29, Part 1910, Section 1200 of the Code of Federal Regulations; and
3. Specifications for Accident Prevention Signs and Tags: Title 29, Part 1910, Section 145 of the Code of Federal Regulations.

D. DOT: U.S. Department of Transportation, including but not limited to:

1. Hazardous Substance: Title 49, Part 171 and 172 of the Code of Federal Regulations.

E. EPA: U.S. Environmental Protection Agency (EPA), including but not limited to:

1. Resource Conservation and Resource Recovery Act (RCRA): Title 40, Part 231-265 of the Code of Federal Regulations.

PART 2 - PRODUCTS

2.1 EQUIPMENT

A. HEPA-equipped filtration equipment.

1. Vacuum Equipment: HEPA-filtered vacuuming equipment equipped with a filter system capable of collecting and retaining dust/fibers. Filters will be 99.97 percent efficient for retaining fibers 0.3 microns or larger.

2.2 CONTAINERS

A. The disposal drums (if applicable) will be of metal with locking ring tops and inner 6-mil polyethylene liners.

B. Stick-on warning labels, as per EPA or OSHA requirements, will be used on all disposal drums. All drums must additionally be labeled with federal and state DOT codes.

PART 3 - EXECUTION

3.1 HOUSEKEEPING

A. Keep the Project Site neat, orderly, and in a safe condition at all times. Use wet methods and HEPA-equipped vacuums only to collect all dust, waste, debris, etc., in areas where asbestos and/or lead dust are encountered. Prohibit the use of dry sweeping or vacuuming without using HEPA-equipped vacuums.

B. Access to the work area must be restricted to “authorized persons” only. This could be accomplished by using properly demarcated critical barriers, securing adjacent access points, or installing barrier plastic sheeting and signs.

C. The Contractor will verify the locations of utilities prior to any environmental abatement and interior demolition work. The existing water, sewer, electrical, communications, and gas lines located within the boundaries of the environmental abatement work will be located, and non-essential utilities will be properly locked and tagged out at the limits of the environmental abatement or the selective planned demolition work. This work will require a qualified and licensed electrician to verify safety and proper lockout.

3.2 FIRE SAFETY

The Contractor will take the following precautions against fire:

A. The Contractor will comply fully with the requirements of federal, state, and local governments, the Owner, and the Owner's Representative, including stipulations as outlined below. The Contractor will maintain and enforce all regulations imposed and will be required to secure such protection as may be required. The Contractor is responsible for acquainting the local fire department with existing conditions.

B. The Contractor will not impede or void the essential function of emergency fire exits while performing abatement work.

C. At least one qualified person thoroughly familiar with fire protection and prevention will be on duty at all hours that the Contractor's employees are working. This person will patrol the entire work area frequently and will have the authority to take immediate remedial action to eliminate unnecessary fire hazards.

D. Building materials storage will be limited to secure areas within the building or, where stored outside, will be kept at least ten feet away from the building. Storage areas will be approved by the Contracting Officer.

E. Fire extinguishers: Maintain and provide approved fire extinguishers throughout all accessible areas. Placement, inspection, and maintenance of fire extinguishers will comply with 29 CFR 1926.130. The Contractor will provide at least one 20 ABC fire extinguisher for every 3,000 square feet of the work area, with a maximum distance of 100 feet between fire extinguishers. The Contractor will place (at a minimum) ABC fire extinguishers in the following locations:

1. Cleanroom of personnel decontamination facility.
2. Equipment room of personnel decontamination facility.
3. Within the containment, at least one extinguisher for every 3,000 square feet of the work area with a maximum distance of 100 feet between fire extinguishers.

F. Gasoline, oils, and other volatile liquids will be kept outside to be brought into the building in quantities only as needed. Such storage will be in a well-ventilated location and removed from all open heating or lighting devices. Storage areas will be approved by the NPS.

G. Electrical wiring for construction light and power will be a properly fused Ground Fault Circuit Interrupter (GFCI) installed to conform to basic code requirements and maintained under the supervision of a competent electrician. This also applies to all temporary lines that are used by the Contractor.

3.3 AIR AND WATER POLLUTION CONTROL

A. Take all necessary reasonable measures to reduce air and water pollution by any material or equipment used during construction.

B. Contractor personnel, building occupants, and visitors to the structures must be protected from exposure during all work activities.

C. Do not allow waste materials to be washed into lakes or other bodies of water.

END OF SECTION 02 84 16

SECTION 03 30 00 – CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes but is not limited to

1. Footings.
2. Foundation walls.
3. Concrete columns/plinths.
4. Architectural concrete for concrete bases of west entry steel columns as shown on the drawings. Includes:

    a. Form-facing materials.
    b. Integral color mixture.

5. Slab-on-grade.
6. Waterstops.
7. Sub-grade vapor retarders.
8. Curing materials.
9. Admixtures.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 05 12 00 “Structural Steel Framing” for coordination of tolerances.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Require representatives of each entity directly concerned with cast-in-place architectural concrete to attend, including the following:

    a. Contractor's superintendent.
    b. Independent testing agency responsible for concrete design mixtures.
    c. Ready-mix concrete manufacturer.
    d. Cast-in-place architectural concrete Subcontractor.

2. Review the following:

    a. Special inspection and testing and inspecting agency procedures for field quality control.
    b. Construction joints, control joints, isolation joints, and joint-filler strips.
    c. Reinforcement accessory installation.
    d. Cold- and hot-weather concreting procedures.

e. Concrete finishes and finishing.
f. Curing procedures.
g. Forms and form-removal limitations.
h. Shoring and reshoring procedures.
i. Concrete repair procedures.
j. Protection of cast-in-place architectural concrete.

1.3 SUBMITTALS

A. Product Data:

1. For each type of product indicated.
2. Architectural Concrete: Forms and Form Ties.
3. Architectural Concrete: Integral Color Admixture, including Pigments.

a. Include brochure of manufacturer's full range of color options for selection of colors for physical samples.

B. Design Mixtures:

1. For each concrete mixture. Submit alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

C. Steel Reinforcement Shop Drawings:

1. Placing drawings that detail fabrication, bending, and placement. Include bar sizes, lengths, material, grade, bar schedules, stirrup spacing, bent bar diagrams, bar arrangement, splices and laps, mechanical connections, tie spacing, hoop spacing, and supports for concrete reinforcement.

D. Embedded Items:

1. Provide drawings, diagrams, and/or templates that detail locations, elevations, tolerances, and installation procedures for embedded items.

E. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.
2. Admixtures.
3. Steel reinforcement and accessories.
4. Curing compounds.
5. Form materials and form-release agents.
6. Vapor retarders.
7. Waterstops.
8. Repair materials.

F. Material Test Reports: For the following, from a qualified testing agency, indicating compliance with requirements:

1. Aggregates. Include service record data indicating absence of deleterious expansion of concrete due to alkali aggregate reactivity.

G. Construction Joint Shop Drawings:

1. Location of construction joints is subject to approval of Contracting Officer.

H. Contractor to submit corrective action plan in case slab Flatness and Levelness requirements are not met.

I. Formwork Shop Drawings:

1. Show formwork construction, including form-liner layout, form-liner termination details, dimensioned locations of form-facing material joints, construction and contraction joints, for joint-sealant details, form-tie locations and patterns, inserts and embedments, cutouts, cleanout panels, and other items that visually affect cast-in-place concrete.

    a. Indicate proposed schedule and sequence of stripping forms, shoring removal, and reshoring installation and removal.

J. Samples: For each of the following materials:

1. Form-liners, 12-by-12-inch Sample, indicating texture.
2. Physical samples of 5-6 options of colored concrete with a grey concrete base. Colors of physical samples shall be selected from the manufacturer's full range.

K. Mockups: Provide a mockup of the column bases, using the same procedures, equipment, ties, materials, finishing procedures, and curing procedures that will be used for producing all architectural concrete, to verify selections made under Sample submittals and to demonstrate typical joints, surface finish, color, texture, tolerances, and standard of workmanship. Build mockups to comply with the following requirements, using materials indicated for the completed Work:

1. Build column base of cast-in-place architectural concrete as shown on Drawings, including any sculptured/formed features.
2. Construct base with the same number of lifts anticipated for all concrete bases.
3. Demonstrate curing, cleaning, and protecting of cast-in-place architectural concrete, finishes, and contraction joints, as applicable.
4. Repair a damaged part of the exposed-face surface to demonstrate materials and techniques proposed for repair to match adjacent undamaged surfaces. Do not purposefully damage the mockup to do this.
5. Also demonstrate materials and techniques proposed for repair of tie holes and surface blemishes to match adjacent undamaged surfaces.
6. Obtain Contracting Officer's approval of mockups before casting remainder of architectural concrete.
7. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

    a. First column base cast in architectural concrete may serve as an in situ mockup, but contractor shall be prepared to remove failed mockups.

8. Contractor shall anticipate that multiple iterations of mockups may be required to meet finish quality expectations.

L. Concrete Repair Informational Submittal: Submit a written, detailed description of materials, methods, equipment, and sequence of operations to be used for repairing architectural concrete, including protection of surrounding materials and Project site.

1. If materials and methods other than those indicated are proposed for any repairs to architectural concrete, add a written description of such materials and methods, including evidence of successful use on comparable projects, and demonstrations to show their effectiveness for this Project and Installer's ability to use such materials and methods properly.

1.4 QUALITY ASSURANCE

A. Installer Qualifications:

1. A qualified installer who employs on Project personnel qualified as ACI-certified Flatwork Technician and Finisher and a supervisor who is an ACI-certified Concrete Flatwork Technician.

B. Manufacturer Qualifications:

1. A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.
2. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."

C. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified according to ASTM C 1077 and ASTM E 329.

1. Personnel conducting field tests shall be qualified as ACI Concrete Field-Testing Technician, Grade 1, according to ACI CP-1 or an equivalent certification.
2. Personnel performing laboratory tests shall be ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician – Grade I. Testing Agency laboratory supervisor shall be an ACI-certified Concrete Laboratory Testing Technician – Grade II.

D. Source Limitations:

1. Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, obtain aggregate from single source, and obtain admixtures from single source from single manufacturer.

E. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:

1. ACI 301, "Specifications for Structural Concrete," Sections 1 through 5.
2. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

F. Comply with AISC “Code of Standard Practice” for concrete tolerances associated with steel construction: including but not limited to anchor rods, anchor bolts, foundation bolts, hold-downs, and embedded items. When a conflict occurs between ACI and AISC tolerance requirements the more restrictive requirements apply.

G. Coordinate locations of hold downs, anchor bolts and shear connections with items being connected and location of structure above. Allow sufficient clearance at connections for required fastener installation.

H. Concrete Testing Service: Engage a qualified independent testing agency to perform material evaluation tests and to design concrete mixtures.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Steel Reinforcement: Deliver, store, and handle steel reinforcement to prevent bending and damage.

B. Waterstops: Store waterstops under cover to protect from moisture, sunlight, dirt, oil, and other contaminants.

PART 2 - PRODUCTS

2.1 CONCRETE MATERIALS

A. Cementitious Material:

1. Portland Cement: ASTM C 150, Type I, gray.
2. Fly Ash: ASTM C618, Class F or C.
3. Slag Cement: ASTM C 989/C 989M, Grade 100 or 120.
4. Use the same type, brand, and source of cement materials throughout Project.

B. Normal-Weight Aggregates:

1. ASTM C 33, Class 3S coarse aggregate or better, graded.
2. Maximum Coarse-Aggregate Size: 3/4 inches nominal.
3. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.
4. Obtain aggregates from a single source with documented service record data of at least 5 years’ satisfactory service in similar applications and service conditions using similar aggregates and cementitious materials.

C. Water: ASTM C 94/C 94M and potable.

D. Chemical Admixtures:

1. Air-Entraining Admixture: ASTM C 260.
2. Water-Reducing Admixture: ASTM C 494, Type A.
3. Retarding Admixture: ASTM C 494/C 494M, Type B.
4. Water-Reducing and Retarding Admixture: ASTM C 494 Type D.
5. High-Range, Water-Reducing Admixture: ASTM C 494, Type F.

6. High-Range, Water-Reducing and Retarding Admixture: ASTM C 494 Type G.
7. Plasticizing and Retarding Admixture: ASTM C 1017, Type II.
8. Provide admixtures certified by manufacturer to be compatible with other admixtures and that will not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride.
9. Burlap jute cloth, weighing 9 oz./sy. when dry, Class 2, AASHTO M 182.
10. Water: Potable.

E. Moisture Retaining Cover:

1. 6 mil, polyethylene film or polyethylene sheet, ASTM C 171

2.2 AUXILIARY PRODUCTS

A. Waterstops:

1. Basis of Design: Coordinate approved product with waterproofing system selected.
2. Self-Expanding Bentonite Strip Waterstops: Manufactured rectangular or trapezoidal strip, butyl rubber with sodium bentonite or other hydrophilic polymers, for adhesive bonding to concrete waterproofing system.

B. Sheet Vapor Retarder:

1. Basis of Design: Stego Industries, LLC; Stego Wrap 15 mil Class A, or approved equal.
2. Salient Characteristics:

a. Reference Standard: Class A, ASTM E 1745.
b. Water Vapor Permeance: Less than 0.01 perms, ASTM F-1249/ASTM E-96.
c. Provide manufacturer's recommended pressure-sensitive tapes, pre-formed boots, and specialized accessories.
d. Warranty: 2 years.

C. Expansion and Isolation Joint-Filler Strips:
1. Asphalt-saturated cellulosic fiber, ASTM D 1751.

D. Bonding Agent:

1. Acrylic emulsion or styrene butadiene, ASTM C 1059, Type II, non-re-dispersible.

E. Repair Underlayment:

1. Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/8 inch and that can be feathered at edges to match adjacent floor elevations.
2. Cement Binder: ASTM C 150, Portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
3. Primer: Product of underlayment manufacturer recommended for substrate, conditions, and application.
4. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch or coarse sand as recommended by underlayment manufacturer.

5. Compressive Strength: Not less than 4100 psi at 28 days when tested according to ASTM C 109/C 109M.

2.3 CONCRETE MIXTURES, GENERAL

A. Mixtures:

1. Prepare design mixtures for each type and strength of concrete, proportioned per laboratory or field test data, or both, according to ACI 301.
2. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.
3. Minimum 28-day compressive strength = 4,000-psi unless noted otherwise on the drawings.

B. Cementitious Materials:

1. Percentages in subparagraphs below repeat ACI 301 limits for concrete exposed to deicing chemicals. Revise to suit Project.
2. Fly Ash or Slag: 20 percent maximum by weight.
3. Other materials, quantities, and combinations of materials: As approved.

C. Water Soluble Chloride-ion Content Limitation:

1. Maximum 0.06 percent by weight of hardened concrete.

D. Admixtures:

1. Use admixtures according to manufacturer's written instructions.
2. Use water-reducing high-range water-reducing or plasticizing admixture in concrete, as required, for placement and workability.
3. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
4. Use water-reducing admixture in pumped concrete, concrete for heavy-use industrial slabs and parking structure slabs, concrete required to be watertight, and concrete with a water-cementitious materials ratio below 0.48.
5. Use corrosion-inhibiting admixture in concrete mixtures where indicated.

E. Color Pigment: ASTM C979/C979M, synthetic mineral-oxide pigments or colored water-reducing admixtures; color stable, free of carbon black, nonfading, and resistant to lime and other alkalis.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Alabama Pigments Company, LLC
b. Brickform; a division of Solomon Colors
c. Butterfield Color, Inc.
d. Dynamic Color Solutions, Inc
e. Euclid Chemical Company (The); a subsidiary of RPM International, Inc.
f. Lambert Corporation

g. Matcrete Inc.
h. Proline Concrete Tools, Inc.
i. Scofield, a Business Unit of Sika Corporation
j. Solomon Colors Inc.
k. Or Approved Equal

2. Source Limitations: Obtain color pigment from single source from single manufacturer.
3. Color: Buff color matching darker shades of veneer stone. Final color to be selected by Contracting Officer from manufacturer's full range during submittal process.

2.4 REINFORCEMENT

A. Steel:

1. Reinforcing Bars: ASTM A 615, Grade 60, deformed.
2. Low-Alloy-Steel Reinforcing bars: ASTM A 706, deformed.
3. Plain-Steel Wire: ASTM A 82.
4. Plane-Steel Welded Wire Reinforcement: ASTM A 185, plain, fabricated from as-drawn steel wire into flat sheets.
5. Recycled Content: Postconsumer recycled content plus one-half of pre-consumer recycled content not less than 60 percent.

2.5 REINFORCEMENT ACCESSORIES

A. Joint Dowel Bars:

1. ASTM A 615/A 615M, Grade 60, plain-steel bars, cut true to length with ends square and free of burrs.

B. Bar Supports:

1. Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice," of greater compressive strength than concrete and as follows:
2. For concrete surfaces exposed to view where legs of wire bar support contact forms, use CRSI Class 1 plastic-protected steel wire or CRSI Class 2 stainless-steel bar supports.

2.6 FABRICATING REINFORCEMENT

A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

2.7 CONCRETE MIXING

A. Ready-Mixed Concrete:

1. Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M and ASTM C 1116/C 1116M, and furnish batch ticket information.

2. When air temperature is between 85- and 90-degrees F, reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 degrees F, reduce mixing and delivery time to 60 minutes.

B. General:

1. Obtain each color, size, type, and variety of concrete mixture from single manufacturer with resources to provide cast-in-place architectural concrete of consistent quality in appearance and physical properties.
2. Prepare design mixtures for each type and strength of cast-in-place architectural concrete proportioned on basis of laboratory trial mixture or field test data, or both, in accordance with ACI 301.
3. Color Pigment: Add color pigment to concrete mixture in accordance with manufacturer's written instructions and to result in hardened concrete color consistent with approved mockup.

2.8 SMOOTH FORMED FINISHED CONCRETE

A. Form-facing panels that will provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.

2.9 CURING MATERIALS

A. For integrally colored concrete, curing materials to be approved by color pigment manufacturer.

B. For concrete indicated to be sealed, curing materials to be compatible with sealer.

2.10 FORM-FACING MATERIALS

A. Source Limitations: Obtain each type of form-facing material from single source from single manufacturer.

B. Form-Facing Panels:

1. Finishes: Architectural concrete.

    a. Steel- and glass-fiber-reinforced plastic, or other approved nonabsorptive panel materials that provide continuous, true, and smooth architectural concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.
    b. Final finish shall be smooth.

2. Finishes: Non-architectural concrete (footings, exterior retaining wall surfaces, locations not visible).

    a. Plywood may be used with proper form-release agent.

C. Form Liners: Units of face design, texture, arrangement, and configuration indicated. Furnish with manufacturer's recommended liquid-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent surface treatments and finishes of

concrete.

D. Chamfer Strips: Metal, rigid plastic, elastomeric rubber, or dressed wood, 1/2 by 1/2 inch, minimum, 3/4 by 3/4 inch maximum; nonstaining; in longest practicable lengths.

E. Form Joint Tape: Compressible foam tape; pressure sensitive; AAMA 800; minimum 1/4 inch thick.

F. Form Joint Sealant: Elastomeric sealant complying with ASTM C920, Type M or Type S, Grade NS, that adheres to form joint substrates, does not stain, does not adversely affect concrete surfaces, and does not impair subsequent treatments and finishes of concrete surfaces.

G. Form-Release Agent: Commercially formulated, colorless form-release agent that does not bond with, stain, or adversely affect architectural concrete surfaces and will not impair subsequent treatments and finishes of architectural concrete surfaces.

1. Formulate form-release agent with rust inhibitor for steel form-facing materials.
2. Form-release agent for form liners to be acceptable to form-liner manufacturer.

H. Surface Retarder: Water-soluble chemical liquid set retarder, for application on form-facing materials, capable of temporarily delaying final hardening of newly placed architectural concrete surface to depth of aggregate exposure specified.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Euclid Chemical Company (The); a subsidiary of RPM International, Inc.
b. Master Builders Solutions
c. W. R. Meadows, Inc
d. Or Approved Equal

2. Source Limitations: Obtain surface retarder from single source from single manufacturer.

I. Form Ties: Factory-fabricated, internally disconnecting ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.

1. For Architectural Concrete: Furnish ties with tapered tie cone spreaders that, when removed, will leave holes no larger than 3/4 inch in diameter on architectural concrete surface.

PART 3 - EXECUTION

3.1 FORMWORK

A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.

B. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical.

1. Install keyways, reglets, recesses, and the like, for easy removal.
2. Do not use rust-stained steel form-facing material.

C. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.

D. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.

E. Chamfer exterior corners and edges of permanently exposed concrete 3/4 inches.

F. Form openings, chases, offsets, sinkages, keyways, reglets, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.

G. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

H. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

I. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

3.2 TOLERANCES

A. Construct formwork so concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117, except for components that are required to coordinate with steel erection.

B. Steel erection coordination. Install anchor rods, accurately located, to elevations required and complying with tolerances in Section 7.5 of AISC's "Code of Standard Practice for Steel Buildings and Bridges."

C. Limit concrete surface irregularities; designated by ACI 347 as abrupt or gradual.

1. Surface Finish-3.0: ACI 117 Class A, 1/8 inch for smooth-formed finished surfaces and all architectural concrete.
2. Class B, 1/4 inch for rough-formed finished surfaces.

D. Construct forms tight enough to prevent loss of concrete mortar.

3.3 EMBEDDED ITEMS

A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

3.4 REMOVING AND REUSING FORMS

A. General: Formwork for sides of beams, walls, columns, and similar parts of the Work that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F for 24 hours after placing concrete. Concrete has to be hard enough to not be damaged by form-removal operations and curing and protection operations need to be maintained.

1. Leave formwork for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved at least 70 percent of its 28-day design compressive strength.
2. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.

B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.

C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Contracting Officer.

3.5 SHORES AND RESHORES

A. Comply with ACI 318 and ACI 301 for design, installation, and removal of shoring and reshoring.

B. Plan sequence of removal of shores and reshore to avoid damage to concrete. Locate and provide adequate reshoring to support construction without excessive stress or deflection.

3.6 VAPOR RETARDERS

A. Sheet Vapor Retarders: Place, protect, and repair sheet vapor retarder according to ASTM E 1643 and manufacturer's written instructions. Place sheets in position with longest dimension parallel with direction of pour.

1. Lap joints and seams 6 inches
2. Seal seams with manufacturer's recommended adhesive or joint tape.
3. Seal all penetrations per manufacturer's recommended method.
4. Repair all damaged areas.
5. Seal vapor retarder at terminating edges to foundation wall per manufacturer's recommended method.

3.7 STEEL REINFORCEMENT

A. General: Comply with CRSI's "Manual of Standard Practice" for placing reinforcement.

1. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.

B. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that would reduce bond to concrete.

C. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover. Do not tack weld crossing reinforcing bars.

1. Weld reinforcing bars according to AWS D1.4/D 1.4M, where indicated.

D. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

E. Install welded wire reinforcement in longest practicable lengths on bar supports spaced to minimize sagging. Lap edges and ends of adjoining sheets at least one mesh spacing. Offset laps of adjoining sheet widths to prevent continuous laps in either direction. Lace overlaps with wire.

3.8 JOINTS

A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.

B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by Contracting Officer.

1. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints unless otherwise indicated. Do not continue reinforcement through sides of strip placements of floors and slabs.
2. Form keyed joints as indicated. Embed keys at least 1-1/2 inches into concrete.
3. Locate joints for beams, slabs, joists, and girders in the middle third of spans. Offset joints in girders a minimum distance of twice the beam width from a beam-girder intersection.
4. Locate horizontal joints in walls and columns at underside of floors, slabs, beams, and girders and at the top of footings or floor slabs.
5. Locate vertical joints beside piers integral with walls, near corners, and in concealed locations where possible.
6. Use a bonding agent at locations where fresh concrete is placed against hardened or partially hardened concrete surfaces.

C. Contraction Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of concrete thickness as follows:

1. Sawed Joints: Form contraction joints with early entry power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inch-wide joints into concrete when cutting action will not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks.

2. Joint Edges: Chamfer edges of saw cuts in exposed locations.

D. Isolation Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.

1. Extend joint-filler strips full width and depth of joint, terminating flush with finished concrete surface unless otherwise indicated.
2. Terminate full-width joint-filler strips not less than 1/2 inch or more than 1 inch below finished concrete surface where joint sealants are indicated.
3. Install joint-filler strips in lengths as long as practicable. Where more than one length is required, lace or clip sections together.

E. Doweled Joints: Install dowel bars and support assemblies at joints where indicated. Lubricate or asphalt coat one-half of dowel length to prevent concrete bonding to one side of joint.

3.9 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.

B. Do not add water to concrete during delivery, at Project site, or during placement.

C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.

1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301. Exterior and interior vibrating may be required to avoid honeycombing.
3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.

D. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.

1. Consolidate concrete during placement operations so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
2. Maintain reinforcement in position on chairs during concrete placement.
3. Screed slab surfaces with a straightedge and strike off to correct elevations.
4. Slope surfaces uniformly to drains where required.
5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.

E. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.

1. When average high and low temperature is expected to fall below 40 deg F for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

F. Hot-Weather Placement: Comply with ACI 301 and as follows:

1. Maintain concrete temperature below 90 deg F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.
2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.10 FINISHING FORMED SURFACES

A. Architectural Concrete Finish: Match Mockup.

1. Comply with ACI 301 Surface Finish-3.0 (SF-3.0).
2. Form-liner Finish: Produce a textured surface free of pockets, streaks, and honeycombs, and of uniform appearance, color, and texture.
3. Maintain uniformity of architectural concrete finishes over construction joints unless otherwise indicated.

B. As-Cast Surface Finishes:

1. ACI 301 Surface Finish SF-1.0: As-cast concrete texture imparted by form-facing material.

    a. Patch voids larger than 1-1/2 inches wide or 1/2 inch deep.
    b. Remove projections larger than 1 inch.
    c. Tie holes do not require patching.
    d. Surface Tolerance: ACI 117 Class D.
    e. Apply to concrete surfaces not exposed to public view.

2. ACI 301Surface Finish SF-2.0: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams.

    a. Not used.

3. ACI 301 Surface Finish SF-3.0:

    a. Patch voids larger than 3/4 inch wide or 1/2 inch deep.
    b. Remove projections larger than 1/8 inch.

c. Patch tie holes.
d. Surface Tolerance: ACI 117 Class A.
e. Locations: Apply to concrete surfaces exposed to public view, to receive a rubbed finish, or to be covered with a coating or covering material applied directly to concrete.

4. Grout-Cleaned Rubbed Finish:

a. Clean concrete surfaces after contiguous surfaces are completed and accessible.
b. Do not clean concrete surfaces as Work progresses.
c. Mix 1 part portland cement to 1-1/2 parts fine sand, complying with ASTM C144 or ASTM C404, by volume, with sufficient water to produce a mixture with the consistency of thick paint. Add white portland cement in amounts determined by trial patches, so color of dry grout matches adjacent surfaces.
d. Wet concrete surfaces.
e. Scrub grout into voids and remove excess grout. When grout whitens, rub surface with clean burlap, and keep surface damp by fog spray for at least 36 hours.
f. Maintain required patterns or variances as shown on Drawings or to match field sample panels or mockups.

C. Related Unformed Surfaces:

1. At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a color and texture matching adjacent formed surfaces.
2. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces unless otherwise indicated.

3.11 FINISHING FLOORS AND SLABS

A. Comply with ACI 302.1R recommendations for screeding, restraightening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.

B. Trowel Finish:

1. After applying float finish, apply first troweling and consolidate concrete by hand or power-driven trowel.
2. Continue troweling passes and restraighten until surface is free of trowel marks and uniform in texture and appearance.
3. Grind smooth any surface defects that would telegraph through applied coatings or floor coverings.
4. Do not add water to concrete surface.
5. Do not apply hard-troweled finish to concrete, which has a total air content greater than 3 percent.
6. Apply a trowel finish to surfaces exposed to view or to be covered with resilient flooring, carpet, ceramic or quarry tile set over a cleavage membrane, paint, or another thin-film-finish coating system.

C. Broom Finish: Apply a broom finish to exterior concrete platforms, steps, ramps, and locations indicated on Drawings.

1. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route.
2. Coordinate required final finish with Contracting Officer before application.

3.12 FINISHING FLOORS AND SLABS

A. General: Comply with ACI 302.1R recommendations for screeding, restraightening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.

B. Slope: Slope exterior surfaces to drain. Slope away from adjacent building components. Maximum cross slope of 1/4 inch per foot. Running slopes as indicated on Drawings.

C. Trowel Finish:

1. After applying float finish, apply first troweling and consolidate concrete by hand or power-driven trowel.
2. Continue troweling passes and restraighten until surface is free of trowel marks and uniform in texture and appearance.
3. Grind smooth any surface defects that would telegraph through applied coatings or floor coverings.
4. Do not add water to concrete surface.
5. Do not apply hard-troweled finish to concrete, which has a total air content greater than 3 percent.
6. Revise locations of trowel finish in first subparagraph below to suit Project, or delete subparagraph and indicate locations on Drawings.
7. Apply a trowel finish to surfaces exposed to view or to be covered with resilient flooring, carpet, ceramic or quarry tile set over a cleavage membrane, paint, or another thin-film-finish coating system.
8. Finish surfaces to the following tolerances, in accordance with ASTM E1155, for a randomly trafficked floor surface:

D. Slabs on Ground – Flatness:

1. Finish and measure surface so gap at any point between concrete surface and an unleveled, freestanding, 10-ft.- long straightedge resting on two high spots and placed anywhere on the surface does not exceed 1/4 inch.

3.13 MISCELLANEOUS CONCRETE ITEMS

A. Filling In: Fill in holes and openings left in concrete structures after work of other trades is in place unless otherwise indicated. Mix, place, and cure concrete, as specified, to blend with in-place construction. Provide other miscellaneous concrete filling indicated or required to complete the Work.

B. Curbs: Provide monolithic finish to interior curbs by stripping forms while concrete is still green and by steel-troweling surfaces to a hard, dense finish with corners, intersections, and terminations slightly rounded.

C. Equipment Bases and Foundations: Provide machine and equipment bases and foundations as shown on Drawings. Set anchor bolts for machines and equipment at correct elevations, complying with diagrams or templates from manufacturer furnishing machines and equipment.

3.14 CONCRETE PROTECTING AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.

D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.

E. Cure concrete according to ACI 308.1, by one or a combination of the following methods:

1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:

    a. Water.
    b. Continuous water-fog spray.
    c. Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces and edges with 12-inch lap over adjacent absorptive covers.

2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.

    a. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
    b. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
    c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer certifies will not interfere with bonding of floor covering used on Project.

3. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall

within three hours after initial application. Maintain continuity of coating and repair damage during curing period.

a. Removal: After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound will not interfere with bonding of floor covering used on Project.

4. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hours later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

F. Protect concrete columns during remaining construction by placing form halves loosely around columns and securing.

3.15 JOINT FILLING

A. Prepare, clean, and install joint filler according to manufacturer's written instructions.

1. Defer joint filling until concrete has aged at least one month(s). Do not fill joints until construction traffic has permanently ceased.

B. Remove dirt, debris, saw cuttings, curing compounds, and sealers from joints; leave contact faces of joint clean and dry.

C. Install semirigid joint filler full depth in saw-cut joints and at least 2 inches deep in formed joints. Overfill joint and trim joint filler flush with top of joint after hardening.

3.16 CONCRETE SURFACE REPAIRS

A. Defective Concrete: Repair and patch defective areas when approved by Contracting Officer. Remove and replace concrete that cannot be repaired and patched to Contracting Officer's approval.

B. Patching Mortar: Mix dry pack patching mortar, consisting of one part portland cement to two and one-half parts fine aggregate passing a No. 16 sieve, using only enough water for handling and placing.

C. Repairing Formed Surfaces: Surface defects include color and texture irregularities, cracks, spalls, air bubbles, honeycombs, rock pockets, fins and other projections on the surface, and stains and other discolorations that cannot be removed by cleaning.

1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 1/2 inch in any dimension to solid concrete. Limit cut depth to 3/4 inch. Make edges of cuts perpendicular to concrete surface. Clean, dampen with water, and brush-coat holes and voids with bonding agent. Fill and compact with patching mortar before bonding agent

has dried. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.

2. Repair defects on surfaces exposed to view by blending white portland cement and standard portland cement so that, when dry, patching mortar will match surrounding color. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place and strike off slightly higher than surrounding surface.
3. Repair defects on concealed formed surfaces that affect concrete's durability and structural performance as determined by Contracting Officer.
4. For formed surfaces exposed to view, fill honeycombs and voids more than 1/4 inch in any dimension with a blend of white Portland cement and standard portland cement. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place.

D. Repairing Unformed Surfaces: Test unformed surfaces, such as floors and slabs, for finish and verify surface tolerances specified for each surface. Correct low and high areas. Test surfaces sloped to drain for trueness of slope and smoothness; use a sloped template.

1. Repair finished surfaces containing defects. Surface defects include spalls, popouts, honeycombs, rock pockets, crazing and cracks in excess of 0.01 inch wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width, and other objectionable conditions.
2. For concrete surfaces scheduled to remain exposed and/or sealed remove all chalk lines, pencil, ink, or other marks.
3. After concrete has cured at least 14 days, correct high areas by grinding.
4. Correct localized low areas during or immediately after completing surface finishing operations by cutting out low areas and replacing with patching mortar. Finish repaired areas to blend into adjacent concrete.
5. Correct other low areas scheduled to receive floor coverings with a repair underlayment. Prepare, mix, and apply repair underlayment and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface. Feather edges to match adjacent floor elevations.
6. Correct other low areas scheduled to remain exposed with a repair topping. Cut out low areas to ensure a minimum repair topping depth of 1/4 inch to match adjacent floor elevations. Prepare, mix, and apply repair topping and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface.
7. Repair defective areas, except random cracks and single holes 1 inch or less in diameter, by cutting out and replacing with fresh concrete. Remove defective areas with clean, square cuts and expose steel reinforcement with at least a 3/4-inch clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding agent. Mix patching concrete of same materials and mixture as original concrete except without coarse aggregate. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.
8. Repair random cracks and single holes 1 inch or less in diameter with patching mortar. Groove top of cracks and cut out holes to sound concrete and clean off dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding agent. Place patching mortar before bonding agent has dried. Compact patching mortar and finish to match adjacent concrete. Keep patched area continuously moist for at least 72 hours.

E. Perform structural repairs of concrete, subject to Contracting Officer's approval, using epoxy adhesive and patching mortar.

F. Repair materials and installation not specified above may be used, subject to Contracting Officer's approval.

G. Repair damaged finished surfaces of cast-in-place architectural concrete when repairing is approved by Contracting Officer.

H. Match repairs to color, texture, and uniformity of surrounding surfaces and to repairs on approved mockups.

I. Remove and replace cast-in-place architectural concrete that cannot be repaired to Contracting Officer's approval.

3.17 FIELD QUALITY CONTROL

A. Testing and Inspecting: Construction contractor shall engage a special inspector and qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.

B. Inspections, unless otherwise noted on Drawings:

1. Steel reinforcement placement.
2. Steel reinforcement welding.
3. Headed bolts and studs.
4. Verification of use of required design mixture.
5. Concrete placement, including conveying and depositing.
6. Concrete slab flatness, on-grade.
7. Curing procedures and maintenance of curing temperature.
8. Verification of concrete strength before removal of shores and forms from beams and slabs.

C. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:

1. Samples for strength tests of each class of concrete placed each day shall be taken not less than once a day, or not less that once for each 150 cubic yards of concrete, or not less than once for each 5,000 square feet of surface area for slabs or walls.
2. Slump: ASTM C 143/C 143M; one test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture. Perform additional tests when concrete consistency appears to change.
3. Air Content: ASTM C 231, pressure method, for normal-weight concrete; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
4. Concrete Temperature: ASTM C 1064/C 1064M; one test hourly when air temperature is 40 deg F and below and when 80 deg F and above, and one test for each composite sample.
5. Unit Weight: ASTM C 567, fresh unit weight of structural lightweight concrete; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
6. Compression Test Specimens: ASTM C 31/C 31M.

    a. Cast and laboratory cure two sets of two standard cylinder specimens for each composite sample.
    b. Cast and field cure two sets of two standard cylinder specimens for each composite sample.

7. Compressive-Strength Tests: ASTM C 39/C 39M; test one set of two laboratory-cured specimens at 7 days and one set of two specimens at 28 days.

    a. Test one set of two field-cured specimens at 7 days and one set of two specimens at 28 days.
    b. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.

8. When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, Contractor shall evaluate operations and provide corrective procedures for protecting and curing in-place concrete.
9. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength, and no compressive-strength test value falls below specified compressive strength by more than 500 psi.
10. Test results shall be reported in writing to Contracting Officer, concrete manufacturer, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7- and 28-day tests.
11. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by Contracting Officer. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C 42/C 42M or by other methods as directed by Contracting Officer.
12. Additional testing and inspecting, at Contractor’s expense, will be performed to determine compliance of replaced or additional work with specified requirements.
13. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

D. Measure floor and slab flatness and levelness within 48 hours of finishing.

3.18 CLEANING

A. Clean cast-in-place architectural concrete surfaces after finish treatment to remove stains, markings, dust, and debris.

B. Wash and rinse surfaces in accordance with concrete finish applicator's written instructions.

1. Protect other Work from staining or damage due to cleaning operations.
2. Do not use cleaning materials or processes that could change the appearance of cast-in-place architectural concrete finishes.

3.19 PROTECTION

A. Protect corners, edges, and surfaces of cast-in-place architectural concrete from damage; use guards and barricades.

B. Protect cast-in-place architectural concrete from staining, laitance, and contamination during remainder of construction period.

3.20 FINAL ACCEPTANCE

A. Final acceptance of completed architectural concrete Work will be determined by Contracting Officer by comparing approved mockups with installed Work, when viewed at a distance of 10 feet.

END OF SECTION 03 30 00

SECTION 04 01 20.63 – BRICK MASONRY REPAIR AND CLEANING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Cleaning brick masonry.
2. Repairing brick masonry.
3. Removing metal fasteners.

B. Related Requirements:

1. Section 04 01 20.64 “Brick Masonry Repointing” for repointing scope.
2. Section 04 26 13 “Masonry Veneer” for replacement bricks and mortar.

1.3 UNIT PRICES

A. Contractor shall state in Proposal, unit prices for patching and lineal feet of masonry crack repair.

B. Unit prices shall include all materials, equipment, labor, overhead and profit for a complete installation.

C. Any increase or decrease in actual square inch of lineal feet of work area to square inch or lineal feet area included in Base Bid shall be paid for by or credited to Government on basis of unit prices, computed total net difference of overage and underage for each type of patching and repointing.

D. See Section 01 27 00 “Definition of Contract Line Items” for unit price definitions.

E. Unit Price No. 2: Masonry Crack Repair

F. Unit Price No. 3: Brick Masonry Replacement

1.4 DEFINITIONS

A. Low-Pressure Spray: maximum 100 psi measured at nozzle at 4 to 6 gpm.

B. Rebuilding (Setting) Mortar: Mortar used to set and anchor masonry in a structure, distinct from pointing mortar installed after masonry is set in place.

C. Saturation Coefficient: Ratio of the weight of water absorbed during immersion in cold water to weight absorbed during immersion in boiling water; used as an indication of resistance of bricks to freezing and thawing.

1.5 SEQUENCING AND SCHEDULING

A. Work Sequence: Perform brick masonry repair work in the following sequence, which includes work specified in this and other Sections:

1. Remove plant growth.
2. Inspect masonry for open mortar joints and point them before cleaning to prevent the intrusion of water and other cleaning materials into the wall.
3. Remove paint.
4. Clean masonry.
5. Rake out mortar from joints surrounding masonry to be replaced and from joints adjacent to masonry repairs along joints.
6. Repair masonry, including replacing existing masonry with new masonry materials.
7. Rake out mortar from joints to be repointed.
8. Point mortar and sealant joints.
9. After repairs and repointing have been completed and cured, perform a final cleaning to remove residues from this work.
10. Where water repellents are to be used on or near masonry work, delay application of these chemicals until after pointing and cleaning.

B. As scaffolding is removed, patch anchor holes used to attach scaffolding. Patch holes in bricks according to “3.6 Brick Masonry Patching.” Patch holes in mortar joints according to Section 04 01 20.64 “Brick Masonry Repointing.”

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include recommendations for product application and use.
2. Include test data substantiating that products comply with requirements.

B. Qualification Data: For brick masonry repair specialist.

1.7 QUALITY ASSURANCE

A. Brick Masonry Repair Specialist Qualifications: Engage an experienced brick masonry repair firm to perform work of this Section. Firm shall have completed work similar in material, design, and extent to that indicated for this Project with a record of successful in-service performance. Experience in only installing masonry is insufficient experience for masonry repair work.

1. Work shall be performed by a firm having not less than five years successful experience in comparable unit masonry restoration projects and employing personnel skilled in unit masonry restoration processes and operations indicated.

2. Field Supervision: Brick masonry repair specialist firm shall maintain experienced full-time supervisors on Project site during times that brick masonry repair work is in progress.
3. Brick Masonry Repair Worker Qualifications: When bricks are being patched, assign at least one worker per crew who is trained and certified by manufacturer of patching compound to apply its products.

B. Mockups: Prepare mockups of brick masonry repair to demonstrate aesthetic effects and to set quality standards for materials and execution and for fabrication and installation.

1. Mockups may be part of the final work product provided they are approved by Contracting Officer. Mockups that have not been approved shall be removed and redone until approved.
2. Masonry Repair: Prepare sample areas for each type of masonry repair work performed. If not otherwise indicated, size each mockup not smaller than two adjacent whole units or approximately 48 inches in least dimension. Construct sample areas in locations in existing walls where directed by Contracting Officer unless otherwise indicated. Demonstrate quality of materials, workmanship, and blending with existing work. Include the following as a minimum:
3. Replacement: At least Four brick units replaced.

    a. Patching: Three small holes at least 1 inch in diameter for each type of brick indicated to be patched.

4. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
5. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver bricks to Project site strapped together in suitable packs or pallets or in heavy-duty cartons and protected against impact and chipping.

B. Deliver packaged materials to Project site in manufacturer's original and unopened containers, labeled with manufacturer's name and type of products.

C. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

D. Store hydrated lime in manufacturer's original and unopened containers. Discard lime if containers have been damaged or have been opened for more than two days.

E. Store sand where grading and other required characteristics can be maintained and contamination avoided.

F. Handle bricks to prevent overstressing, chipping, defacement, and other damage.

1.9 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit brick masonry repair work to be performed according to product manufacturers' written instructions and specified requirements.

B. Temperature Limits: Repair brick masonry only when air temperature is between 40 and 90 deg F and is predicted to remain so for at least seven days after completion of the Work unless otherwise indicated.

C. Cold-Weather Requirements: Comply with the following procedures for masonry repair unless otherwise indicated:

1. When air temperature is below 40 deg F, heat mortar ingredients, masonry repair materials, and existing masonry walls to produce temperatures between 40 and 120 deg F.
2. When mean daily air temperature is below 40 deg F, provide enclosure and heat to maintain temperatures above 32 deg F within the enclosure for seven days after repair.

D. Hot-Weather Requirements: Protect masonry repairs when temperature and humidity conditions produce excessive evaporation of water from mortar and repair materials. Provide artificial shade and wind breaks, and use cooled materials as required to minimize evaporation. Do not apply mortar to substrates with temperatures of 90 deg F and above unless otherwise indicated.

E. For manufactured repair materials, perform work within the environmental limits set by each manufacturer.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Source Limitations: Obtain each type of material for repairing brick masonry (brick, cement, sand, etc.) from single source with resources to provide materials of consistent quality in appearance and physical properties.

2.2 REPAIR MATERIALS

A. Brick Patching Material: Color and Texture to Match Existing Brick

1. Jahn M100 and Jahn M120 restoration mortar.
2. Edison Coatings, Custom System 45 Composite Repair Compound.
3. Or Approved Equal.

B. Epoxies/Injection Grouts for Crack Repair:

1. Jahn M30 and Jahn M40 injection grout.
2. Edison coatings, Flexi-Seal 510 and 510-U Epoxy Crack Filler.
3. Or Approved Equal.

2.3 MASONRY MATERIALS

A. Brick to match existing per Section 04 26 13 “Masonry Veneer.”

2.4 MORTAR MATERIALS

A. Mortar to match existing, per Section 04 26 13 “Masonry Veneer.”

2.5 ACCESSORY MATERIALS

A. Water: Shall be potable and free of injurious amounts of oil, soluble salts, alkali, acids, and other impurities that might strain or otherwise damage masonry.

B. Pins for crack stitching in brick masonry shall be 1808 stainless steel, all-thread rods.

C. Epoxy adhesive for embedding anchors and pins shall be a high modulus epoxy resin conforming to ASTM C 881, Type I, II, and V, Grade 3 epoxy resin adhesive.

D. Cleaning Materials:

1. Water for Cleaning: Clean, potable, free of oils, acids, alkalis, salts and organic matter.
2. Brushes: Fiber bristle only.
3. Detergent: Mild non-ionic detergent soap or masonry detergent. See also Section 04 26 13 “Masonry Veneer.”
4. Do not use cleaners containing hydrochloric acid.
5. Detergent Solutions: Dilute detergent cleaning materials with water to produce solutions of concentration indicated, but not greater than that recommended by detergent manufacturer.

E. Cleaning Equipment:

1. Provide spray equipment for controlled spray application of water and detergent cleaners, if any, at rates indicated for pressure (maximum 100 psi, measured at nozzle to 4 to 6 gpm).
2. For spray application of water and detergents provide fan-shaped spray tip which disperses water in a flat 25 to 30 degree wide spray.
3. For spray application of heated water, provide equipment capable of maintaining temperature, at flow rates indicated, between 140 and 180 degrees F.

F. Cleaners:

1. Cleaner for Removing Light to Moderate Staining from Dense Masonry: Prosoco Light Duty Restoration Cleaner (Basis of Design); clear, gelled liquid.
2. Cleaner to Removing Moderate to Heavy Atmospheric Staining from Dense Masonry: Prosoco Restoration Cleaner/Heavy Duty Restoration Cleaner (Basis of Design); clear liquid, prepared at 1:5 dilution.
3. Cleaner for Removing Asphalt, Tar, Grease, and Oil: Prosoco Asphalt & Tar Remover for Masonry & Concrete (Basis of Design); clear liquid.
4. Cleaner for Removing Mold, Mildew, and Atmosphere Stains: Prosoco Enviro Klean ReVive (Basis of Design); clear, slight amber liquid.

5. Stripper for Removing Lead-based Paint and Other Coatings: Prosoco Enviro Klean SafStrip (Basis of Design); clear gel.
6. Alternative Manufacturers:

    a. Diedrich Technologies, Inc.
    b. EaCo Chem, Inc.
    c. Or Approved Equal

G. Masking Tape: Nonstaining, nonabsorbent material; compatible with mortar, joint primers, sealants, and surfaces adjacent to joints; and that easily comes off entirely, including adhesive.

H. Other Products: Select materials and methods of use based on the following, subject to approval of a mockup:

1. Previous effectiveness in performing the work involved.
2. Minimal possibility of damaging exposed surfaces.
3. Consistency of each application.
4. Uniformity of the resulting overall appearance.
5. Do not use products or tools that could leave residue on surfaces.

PART 3 - EXECUTION

3.1 PROTECTION

A. Prevent mortar from staining face of surrounding masonry and other surfaces.

1. Cover sills, ledges, and other projecting items to protect them from mortar droppings.
2. Keep wall area wet below rebuilding and repair work to discourage mortar from adhering.
3. Immediately remove mortar splatters in contact with exposed masonry and other surfaces.

3.2 MASONRY REPAIR, GENERAL

A. Appearance Standard: Repaired surfaces are to have a uniform appearance as viewed from 20 feet away by Contracting Officer.

3.3 METAL ATTACHMENT REMOVAL

A. Remove metal attachments, abandoned anchors, brackets, wood nailers, and other extraneous items no longer in use unless indicated to remain.

1. Remove items carefully to avoid spalling or cracking masonry.
2. Notify Contracting Officer before proceeding if an item cannot be removed without damaging surrounding masonry. Do the following where directed:

    a. Cut or grind off item approximately 3/4 inch beneath surface and core drill a recess of same depth in surrounding masonry as close around item as practical.

b. Immediately paint exposed end of item with two coats of antirust coating, following coating manufacturer's written instructions and without exceeding manufacturer's recommended dry film thickness per coat. Keep paint off sides of recess.

3. Patch hole where each item was removed unless directed to remove and replace bricks.

3.4 BRICK REMOVAL AND REPLACEMENT

A. At locations indicated, remove bricks that are damaged, spalled, or deteriorated or are to be reused. Carefully remove entire units from joint to joint, without damaging surrounding masonry, in a manner that permits replacement with full-size units.

1. When removing single bricks, remove material from center of brick and work toward outside edges.

B. Support and protect remaining masonry that surrounds removal area.

C. Maintain flashing, reinforcement, lintels, and adjoining construction in an undamaged condition. Coordinate with new flashing, reinforcement, and lintels, which are specified in other Sections.

D. Notify Contracting Officer of unforeseen detrimental conditions including voids, cracks, bulges, and loose units in existing masonry backup, rotted wood, rusted metal, and other deteriorated items.

E. Remove in an undamaged condition as many whole bricks as possible.

1. Remove mortar, loose particles, and soil from brick by cleaning with hand chisels, brushes, and water.
2. Remove sealants by cutting close to brick with utility knife and cleaning with solvents.
3. Store brick for reuse. Store off ground, on skids, and protected from weather.
4. Deliver cleaned brick not required for reuse to Contracting Officer unless otherwise indicated. If declined by the Contracting Officer, salvaged brick becomes the property of the Contractor.

F. Clean masonry surrounding removal areas by removing mortar, dust, and loose particles in preparation for brick replacement.

G. Replace removed damaged brick with other removed brick in good condition, where possible, or with new brick matching existing brick. Do not use broken units unless they can be cut to usable size.

H. Install replacement brick into bonding and coursing pattern of existing brick. If cutting is required, use a motor-driven saw designed to cut masonry with clean, sharp, unchipped edges.

1. Maintain joint width for replacement units to match existing joints.
2. Use setting buttons or shims to set units accurately spaced with uniform joints.

I. Lay replacement brick with rebuilding (setting) mortar and with completely filled bed, head, and collar joints. Butter ends with enough mortar to fill head joints and shove into place. Wet both replacement and surrounding bricks that have ASTM C 67 initial rates of absorption (suction) of more than 30 g/30 sq. in. per min. Use wetting methods that ensure that units are nearly saturated but surface is dry when laid.

1. Tool exposed mortar joints in repaired areas to match joints of surrounding existing brickwork.
2. When mortar is hard enough to support units, remove shims and other devices interfering with pointing of joints.

J. Curing: Cure mortar by maintaining in thoroughly damp condition for at least 72 consecutive hours, including weekends and holidays.

1. Hairline cracking within the mortar or mortar separation at edge of a joint is unacceptable. Completely remove such mortar and repoint.

3.5 BRICK MASONRY PATCHING

A. Patch the following bricks unless another type of repair or replacement is indicated:

1. Bricks with holes.
2. Bricks with chipped edges or corners. Patch chipped edges or corners measuring more than 3/4 inch in least dimension.
3. Bricks with small areas of deep deterioration. Patch deep deteriorations measuring more than 3/4 inch in least dimension and more than 1/4 inch deep.

B. Patching Bricks:

1. Remove loose material from masonry surface. Carefully remove additional material so patch does not have feathered edges but has square or slightly undercut edges on area to be patched and is at least 1/4 inch thick, but not less than recommended in writing by patching compound manufacturer.
2. Mask adjacent mortar joint or rake out for repointing if patch extends to edge of brick.
3. Mix patching compound in individual batches to match each unit being patched. Combine one or more colors of patching compound, as needed, to produce exact match.
4. Rinse surface to be patched and leave damp, but without standing water.
5. Brush-coat surfaces with slurry coat of patching compound according to manufacturer's written instructions.
6. Place patching compound in layers as recommended in writing by patching compound manufacturer, but not less than 1/4 inch or more than 2 inches thick. Roughen surface of each layer to provide a key for next layer.
7. Trowel, scrape, or carve surface of patch to match texture and surrounding surface plane or contour of brick. Shape and finish surface before or after curing, as determined by testing, to best match existing brick.
8. Keep each layer damp for 72 hours or until patching compound has set.
9. Remove and replace patches with hairline cracks or that show separation from brick at edges, and those that do not match adjoining brick in color or texture.

3.6 FINAL CLEANING

A. After mortar has fully hardened, thoroughly clean exposed masonry surfaces of excess mortar and foreign matter; use wood scrapers, stiff-nylon or -fiber brushes, and clean water applied by low-pressure spray.

1. Do not use metal scrapers or brushes.
2. Do not use acidic or alkaline cleaners.

B. Clean adjacent nonmasonry surfaces. Use detergent and soft brushes or cloths.

C. Clean mortar and debris from roof; remove debris from gutters and downspouts. Rinse off roof and flush gutters and downspouts.

D. Remove masking materials, leaving no residues that could trap dirt.

3.7 MASONRY WASTE DISPOSAL

A. Salvageable Materials: If declined by the Contracting Officer, excess masonry materials are Contractor's property.

B. Masonry Waste: Remove masonry waste and legally dispose of off Government property.

END OF SECTION 04 01 20.63

SECTION 04 01 20.64 – BRICK MASONRY REPOINTING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Repointing joints with mortar.
2. Repointing joints with sealant.

B. Related Requirements:

1. Section 04 01 20.63 “Brick Masonry Repair” for related repair and cleaning of existing brick.
2. Section 04 26 13 “Masonry Veneer” for mortar specification, testing, and submittals.
3. Section 07 92 00 “Joint Sealants” for sealant specification and submittals.

1.3 UNIT PRICES

A. Work of this Section is affected by unit prices specified in Section 01 27 00 “Definition of Contract Line Items.”

1. Unit prices apply to authorized work covered by quantity allowances on drawings.
2. Unit prices apply to additions to and deletions from Work as authorized by Change Orders.
3. Unit Price No. 1: Masonry Repointing

B. Unit prices shall include all materials, equipment, labor, overhead and profit for a complete installation.

C. Any increase or decrease in actual square inch of lineal feet of work area included in Base Bid shall be paid for by or credited to Government on basis of unit prices, computed total net difference of overage and underage for repointing.

1.4 DEFINITIONS

A. Low-Pressure Spray: maximum 100 psi measured at nozzle at 4 to 6 gpm.

1.5 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review methods and procedures related to repointing brick masonry including, but not limited to, the following:

    a. Verify brick masonry repointing specialist's personnel, equipment, and facilities needed to make progress and avoid delays.
    b. Materials, material application, sequencing, tolerances, and required clearances.
    c. Quality-control program.
    d. Coordination with building occupants.
    e. Coordination of repointing work with other masonry repair and installation of new or salvaged brick.

1.6 SEQUENCING AND SCHEDULING

A. Order sand and gray portland cement for pointing mortar immediately after approval of Samples and mockups. Take delivery of and store at Project site enough quantity to complete Project.

B. Repointing shall not precede initial masonry cleaning in order to allow accurate color matching.

C. Work Sequence: Perform brick masonry repointing work in the following sequence, which includes work specified in this and other Sections:

1. Remove plant growth.
2. Inspect masonry for open mortar joints and permanently or temporarily point them before cleaning to prevent the intrusion of water and other cleaning materials into the wall.
3. Remove paint.
4. Clean masonry.
5. Rake out mortar from joints surrounding masonry to be replaced and from joints adjacent to masonry repairs along joints.
6. Repair masonry, including replacing existing masonry with new masonry materials.
7. Rake out mortar from joints to be repointed.
8. Point mortar and sealant joints.
9. After repairs and repointing have been completed and cured, perform a final cleaning to remove residues from this work.

1.7 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.
2. Include recommendations for product application and use.
3. Include test data substantiating that products comply with requirements.

B. Shop Drawings:

1. Include plans, elevations, sections, and locations of repointing work on the structure.
2. Show provisions for expansion joints or other sealant joints.

C. Qualification Data: For brick masonry repointing specialist including field supervisors and workers.

1.8 QUALITY ASSURANCE

A. Brick Masonry Repointing Specialist Qualifications: Engage an experienced brick masonry repointing firm to perform work of this Section. Firm shall have completed work similar in material, design, and extent to that indicated for this Project with a record of successful in-service performance. Experience in only installing masonry is insufficient experience for masonry repointing work.

1. Field Supervision: Brick masonry repointing specialist firms shall maintain experienced full-time supervisors on Project site during times that brick masonry repointing work is in progress.

B. Mockups: Prepare mockups of brick masonry repointing to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Repointing: Rake out joints in two separate areas, each approximately 36 inches high by 48 inches wide for each type of repointing required and repoint one of the areas.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Deliver packaged materials to Project site in manufacturer's original and unopened containers, labeled with manufacturer's name and type of products.

B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

C. Store hydrated lime in manufacturer's original and unopened containers. Discard lime if containers have been damaged or have been opened for more than two days.

D. Store sand where grading and other required characteristics can be maintained and contamination avoided.

1.10 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit repointing work to be performed according to product manufacturers' written instructions and specified requirements.

B. Temperature Limits: Repoint mortar joints only when air temperature is between 40 and 90 deg F and is predicted to remain so for at least seven days after completion of the Work unless otherwise indicated.

C. Cold-Weather Requirements: Comply with the following procedures for mortar-joint pointing unless otherwise indicated:

1. When air temperature is below 40 deg F, heat mortar ingredients and existing masonry walls to produce temperatures between 40 and 120 deg F.
2. When mean daily air temperature is below 40 deg F, provide enclosure and heat to maintain temperatures above 32 deg F within the enclosure for seven days after pointing.

D. Hot-Weather Requirements: Protect mortar-joint pointing when temperature and humidity conditions produce excessive evaporation of water from mortar materials. Provide artificial shade and wind breaks and use cooled materials as required to minimize evaporation. Do not apply mortar to substrates with temperatures of 90 deg F and above unless otherwise indicated.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Source Limitations: Obtain each type of material for repointing brick masonry (cement, sand, etc.) from single source with resources to provide materials of consistent quality in appearance and physical properties.

2.2 MORTAR MATERIALS

A. DESIGN INTENT:

1. It is the intent of this project that the mortar shall match the original in color and installation such that the new infill shall be as seamless as possible.
2. See Spec Section 04 26 13 “Masonry Veneer” for mortar testing, specification, and submittals.

2.3 ACCESSORY MATERIALS

A. Masking Tape: Nonstaining, nonabsorbent material; compatible with mortar, joint primers, sealants, and surfaces adjacent to joints; and that easily comes off entirely, including adhesive.

B. Other Products: Select materials and methods of use based on the following, subject to approval of a mockup:

1. Previous effectiveness in performing the work involved.
2. Minimal possibility of damaging exposed surfaces.
3. Consistency of each application.
4. Uniformity of the resulting overall appearance.
5. Do not use products or tools that could leave residue on surfaces.

PART 3 - EXECUTION

3.1 PROTECTION

A. Prevent mortar from staining face of surrounding masonry and other surfaces.

1. Cover sills, ledges, and other projecting items to protect them from mortar droppings.
2. Keep wall area wet below pointing work to discourage mortar from adhering.
3. Immediately remove mortar splatters in contact with exposed masonry and other surfaces.

3.2 MASONRY REPOINTING, GENERAL

A. Appearance Standard: Repointed surfaces are to have a uniform appearance as viewed from 20 feet away by Contracting Officer.

3.3 REPOINTING

A. Rake out and repoint joints to the following extent:

1. All loose or missing joints, up to 25% allowance.
2. Joints indicated as sealant-filled joints.
3. Joints at locations of the following defects:

a. Holes and missing mortar.
b. Cracks that can be penetrated 1/4 inch or more by a knife blade 0.027 inch thick.
c. Cracks 1/16 inch or more in width and of any depth.
d. Hollow-sounding joints when tapped by metal object.
e. Eroded surfaces 1/4 inch or more deep.
f. Deterioration to point that mortar can be easily removed by hand, without tools.
g. Joints filled with substances other than mortar.

B. Do not rake out and repoint joints where not required.

C. Rake out joints as follows, according to procedures demonstrated in approved mockup:

1. Remove mortar from joints to depth of 2 times joint width and not less than that required to expose sound, unweathered mortar. Do not remove unsound mortar more than 2 inches deep; consult Contracting Officer for direction.

2. Remove mortar from brick and other masonry surfaces within raked-out joints to provide reveals with square backs and to expose masonry for contact with pointing mortar. Brush, vacuum, or flush joints to remove dirt and loose debris.
3. Do not spall edges of brick or other masonry units or widen joints. Replace or patch damaged brick or other masonry units as directed by Contracting Officer.

D. Notify Contracting Officer of unforeseen detrimental conditions including voids in mortar joints, cracks, loose masonry units, rotted wood, rusted metal, and other deteriorated items.

E. Pointing with Mortar:

1. Rinse joint surfaces with water to remove dust and mortar particles. Time rinsing application so, at time of pointing, joint surfaces are damp but free of standing water. If rinse water dries, dampen joint surfaces before pointing.
2. Apply pointing mortar first to areas where existing mortar was removed to depths greater than surrounding areas. Apply in layers not greater than 3/8 inch until a uniform depth is formed. Fully compact each layer and allow it to become thumbprint hard before applying next layer.
3. After deep areas have been filled to same depth as remaining joints, point joints by placing mortar in layers not greater than 3/8 inch. Fully compact each layer and allow to become thumbprint hard before applying next layer. Where existing masonry units have worn or rounded edges, slightly recess finished mortar surface below face of masonry to avoid widened joint faces. Take care not to spread mortar beyond joint edges onto exposed masonry surfaces or to featheredge the mortar.
4. When mortar is thumbprint hard, tool joints to match original appearance of joints as demonstrated in approved mockup. Remove excess mortar from edge of joint by brushing.
5. Cure mortar by maintaining in thoroughly damp condition for at least 72 consecutive hours, including weekends and holidays.
6. Hairline cracking within mortar or mortar separation at edge of a joint is unacceptable. Completely remove such mortar and repoint.

F. Pointing with Sealant, to be used at joints where movement can occur, such as below brick ledges: Comply with Section 07 92 00 “Joint Sealants.” and as follows:

1. After raking out, keep joints dry and free of mortar and debris.
2. Clean and prepare joint surfaces. Do not allow primer to spill or migrate onto adjoining surfaces.
3. Fill sealant joints with specified joint sealant.

    a. Install cylindrical sealant backing beneath the sealant. Where space is insufficient for cylindrical sealant backing, install bond-breaker tape.
    b. Install sealant using only proven installation techniques that ensure that sealant is deposited in a uniform, continuous ribbon, without gaps or air pockets, and with complete wetting of the joint bond surfaces equally on both sides. Fill joint flush with surrounding masonry and matching the contour of adjoining mortar joints.
    c. Install sealant as recommended in writing by sealant manufacturer but within the following general limitations, measured at the center (thin) section of the bead:

        1) Fill joints to a depth equal to joint width, but not more than 1/2 inch deep or less than 1/4 inch deep.

d. Tool sealant to form smooth, uniform beads, slightly concave. Remove excess sealant from surfaces adjacent to joint.
e. Sanded Joints: Immediately after first tooling, apply ground-mortar aggregate to sealant, gently pushing aggregate into the surface of sealant. Lightly retool sealant to form smooth, uniform beads, slightly concave. Remove excess sealant and aggregate from surfaces adjacent to joint.
f. Do not allow sealant to overflow or spill onto adjoining surfaces, or to migrate into the voids of adjoining surfaces, particularly rough textures. Remove excess and spillage of sealant promptly as the work progresses. Clean adjoining surfaces by the means necessary to eliminate evidence of spillage, without damage to adjoining surfaces or finishes, as demonstrated in an approved mockup.

G. Repointing work precedes final cleaning of existing masonry, allow mortar to harden at least 30 days before beginning cleaning work.

3.4 FINAL CLEANING

A. After mortar has fully hardened, thoroughly clean exposed masonry surfaces of excess mortar and foreign matter; use wood scrapers, stiff-nylon or -fiber brushes, and clean water applied by low-pressure spray.

1. Do not use metal scrapers or brushes.
2. Do not use acidic or alkaline cleaners.

B. Clean adjacent nonmasonry surfaces. Use detergent and soft brushes or cloths.

C. Clean mortar and debris from roof; remove debris from gutters and downspouts. Rinse off roof and flush gutters and downspouts.

D. Remove masking materials, leaving no residues that could trap dirt.

END OF SECTION 04 01 20.64

SECTION 04 22 00 – CONCRETE UNIT MASONRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Concrete masonry units.
2. Mortar and grout.
3. Steel reinforcing bars.
4. Grout fill.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 03 30 00 “Cast in Place Concrete”

1.2 DEFINITIONS

A. CMU(s): Concrete masonry unit(s).

B. Reinforced Masonry: Masonry containing reinforcing steel in grouted cells.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For the following:

1. Proposed expansion joint spacing, where not shown.

C. Samples for Verification: For each type and color of the following:

1. Nonstandard CMUs.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For testing agency.

B. Material Certificates: For each type and size of the following:

1. Masonry units.

    a. Include material test reports substantiating compliance with requirements.

2. Integral water repellant used in CMUs.
3. Cementitious materials. Include name of manufacturer, brand name, and type.
4. Mortar admixtures.
5. Preblended, dry mortar mixes. Include description of type and proportions of ingredients.
6. Grout mixes. Include description of type and proportions of ingredients.
7. Reinforcing bars.
8. Joint reinforcement.
9. Anchors, ties, and metal accessories.

C. Mix Designs: For each type of mortar and grout. Include description of type and proportions of ingredients.

1. Include test reports for mortar mixes required to comply with property specification. Test according to ASTM C109/C 109M for compressive strength, ASTM C1506 for water retention, and ASTM C91/C 91M for air content.
2. Include test reports, according to ASTM C1019, for grout mixes required to comply with compressive strength requirement.

D. Statement of Compressive Strength of Masonry: For each combination of masonry unit type and mortar type, provide statement of average net-area compressive strength of masonry units, mortar type, and resulting net-area compressive strength of masonry determined according to TMS 602/ACI 530.1/ASCE 6.

E. Cold-Weather and Hot-Weather Procedures: Detailed description of methods, materials, and equipment to be used to comply with requirements.

1.6 QUALITY ASSURANCE

A. Testing Agency Qualifications: Qualified according to ASTM C1093 for testing indicated.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Build mockups for each type of exposed unit masonry construction in sizes approximately 96 inches long by 24 inches high by full thickness, including accessories.

    a. Include a sealant-filled joint at least 16 inches long in each exterior wall mockup, sanded to match mortar.
    b. Include door opening jamb in mockup.

2. Protect accepted mockups from the elements with weather-resistant membrane.
3. Approval of mockups is for color, texture, and blending of masonry units; relationship of mortar and sealant colors to masonry unit colors; tooling of joints; and aesthetic qualities of workmanship.

a. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.

4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Store masonry units on elevated platforms in a dry location. If units are not stored in an enclosed location, cover tops and sides of stacks with waterproof sheeting, securely tied. If units become wet, do not install until they are dry.

B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

C. Store aggregates where grading and other required characteristics can be maintained, and contamination avoided.

D. Deliver preblended, dry mortar mix in moisture-resistant containers. Store preblended, dry mortar mix in delivery containers on elevated platforms in a dry location or in covered weatherproof dispensing silos.

E. Store masonry accessories, including metal items, to prevent corrosion and accumulation of dirt and oil.

1.8 FIELD CONDITIONS

A. Protection of Masonry: During construction, cover tops of walls, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.

1. Extend cover a minimum of 24 inches down both sides of walls, and hold cover securely in place.

B. Do not apply uniform floor or roof loads for at least 12 hours and concentrated loads for at least three days after building masonry walls or columns.

C. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry to be left exposed or painted. Immediately remove grout, mortar, and soil that come in contact with such masonry.

1. Protect base of walls from rain-splashed mud and from mortar splatter by spreading coverings on ground and over wall surface.
2. Protect sills, ledges, and projections from mortar droppings.
3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.

D. Cold-Weather Requirements: Do not use frozen materials or materials mixed or coated with ice or frost. Do not build on frozen substrates. Remove and replace unit masonry damaged by frost or by freezing conditions. Comply with cold-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

1. Cold-Weather Cleaning: Use liquid cleaning methods only when air temperature is 40 deg F and higher and will remain so until masonry has dried for not less than seven days after completing cleaning.

E. Hot-Weather Requirements: Comply with hot-weather construction requirements contained in TMS 602/ACI 530.1/ASCE 6.

PART 2 - PRODUCTS

2.1 MANUFACTURERS, SOURCING

A. Source Limitations for Masonry Units: Obtain exposed masonry units of a uniform texture and color, or a uniform blend within the ranges accepted for these characteristics, from single source from single manufacturer for each product required.

B. Source Limitations for Mortar Materials: Obtain mortar ingredients of a uniform quality, including color for exposed masonry, from single manufacturer for each cementitious component and from single source or producer for each aggregate.

2.2 UNIT MASONRY, GENERAL

A. Masonry Standard: Comply with TMS 602/ACI 530.1/ASCE 6 except as modified by requirements in the Contract Documents.

B. Defective Units: Referenced masonry unit standards may allow a certain percentage of units to contain chips, cracks, or other defects exceeding limits stated. Do not use units where such defects are exposed in the completed Work.

2.3 CONCRETE MASONRY UNITS

A. Shapes: Provide shapes indicated and as follows, with exposed surfaces matching exposed faces of adjacent units unless otherwise indicated.

1. Provide special shapes for lintels, corners, jambs, sashes, movement joints, headers, bonding, and other special conditions.
2. Finish end of units to match face where units are finished.
3. Provide square-edged units for outside corners unless otherwise indicated.

B. CMUs: ASTM C 90.

1. Density Classification: Medium density unless otherwise indicated.
2. Unit Compressive Strength: Provide units with minimum average net-area compressive strength of 2000 psi.

3. Size, General: Manufactured to dimensions 3/8 inch less-than-nominal dimensions indicated.
4. Sizes sand Finishes.

    a. Nominal: by 8 by 16 inches, depth, height by length.

        1) Actual: 7-5/8 by 7-5/8 by 15-5/8 inches.
        2) Finish: Ground faced.
        3) Color: Natural.

2.4 MORTAR AND GROUT MATERIALS

A. Portland Cement: ASTM C150/C150M, Type I or II, except Type III may be used for cold-weather construction. Provide natural color or white cement as required to produce mortar color indicated.

1. Alkali content shall not be more than 0.1 percent when tested according to ASTM C114.

B. Hydrated Lime: ASTM C207, Type S.

C. Portland Cement-Lime Mix: Packaged blend of portland cement and hydrated lime containing no other ingredients.

D. Aggregate for Mortar: ASTM C144.

1. For mortar that is exposed to view, use washed aggregate consisting of natural sand or crushed stone.

E. Aggregate for Grout: ASTM C404.

F. Cold-Weather Admixture: Nonchloride, noncorrosive, accelerating admixture complying with ASTM C494/C494M, Type C, and recommended by manufacturer for use in masonry mortar of composition indicated.

G. Water: Potable.

2.5 MISCELLANEOUS MASONRY ACCESSORIES

A. Compressible Filler: Premolded filler strips complying with ASTM D1056, Grade 2A1; compressible up to 35 percent; of width and thickness indicated; formulated from neoprene.

B. Preformed Control-Joint Gaskets: Made from styrene-butadiene-rubber compound, complying with ASTM D 2000, Designation M2AA-805 and designed to fit standard sash block and to maintain lateral stability in masonry wall; size and configuration as indicated.

2.6 MASONRY CLEANERS

A. Proprietary Acidic Cleaner: Manufacturer's standard-strength cleaner designed for removing mortar/grout stains, efflorescence, and other new construction stains from new masonry without discoloring or damaging masonry surfaces. Use product expressly approved for intended use by cleaner manufacturer and manufacturer of masonry units being cleaned.

2.7 MORTAR AND GROUT MIXES

A. General: Do not use admixtures, including pigments, air-entraining agents, accelerators, retarders, water-repellent agents, antifreeze compounds, or other admixtures unless otherwise indicated.

1. Do not use calcium chloride in mortar or grout.
2. Use portland cement- lime mortar.
3. Add cold-weather admixture (if used) at same rate for all mortar that will be exposed to view, regardless of weather conditions, to ensure that mortar color is consistent.

B. Mortar for Unit Masonry: Comply with ASTM C270, Proportion Specification. Provide the following types of mortar for applications stated unless another type is indicated or needed to provide required compressive strength of masonry.

1. For masonry below grade or in contact with earth, use Type S.
2. For reinforced masonry, use Type S.
3. For partitions, used Type S.

C. Grout for Unit Masonry: Comply with ASTM C476.

1. Use grout of type indicated or, if not otherwise indicated, of type (fine or coarse) that will comply with TMS 602/ACI 530.1/ASCE 6 for dimensions of grout spaces and pour height.
2. Proportion grout in accordance with ASTM C476.
3. Provide grout with a slump of 8 to 11 inches as measured according to ASTM C143/C143M, unless otherwise indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

1. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.
2. Verify that foundations are within tolerances specified.
3. Verify that reinforcing dowels are properly placed.
4. Verify that substrates are free of substances that would impair mortar bond.

B. Before installation, examine rough-in and built-in construction for piping systems to verify actual locations of piping.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

A. Build chases and recesses to accommodate items specified in this and other Sections.

B. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match construction immediately adjacent to opening.

C. Use full-size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor-driven saws; provide clean, sharp, unchipped edges. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed.

3.3 TOLERANCES

A. General: Comply with ACI 530.1/ASCE 6/TMS 602 and as indicated below.

B. Dimensions and Locations of Elements:

1. For dimensions in cross section or elevation, do not vary by more than plus or minus 1/4 inch where abutting other construction. Plus 1/2 and minus 1/4 inch elsewhere.
2. For location of elements in plan, do not vary from that indicated by more than plus or minus 1/2 inch.
3. For location of elements in elevation, do not vary from that indicated by more than plus or minus 1/4 inch in a story height or 1/2 inch total.

C. Lines and Levels:

1. For bed joints and top surfaces of bearing walls, do not vary from level by more than 1/4 inch in 10 feet, or 1/2-inch maximum.
2. For conspicuous horizontal lines, such as lintels, sills, parapets, and reveals, do not vary from level by more than 1/8 inch in 10 feet, 1/4 inch in 20 feet, or 1/2-inch maximum.
3. For vertical lines and surfaces do not vary from plumb by more than 1/4 inch in 10 feet, 3/8 inch in 20 feet, or 1/2-inch maximum.
4. For conspicuous vertical lines, such as external corners, door jambs, reveals, and expansion and control joints, do not vary from plumb by more than 1/8 inch in 10 feet, 1/4 inch in 20 feet, or 1/2-inch maximum.
5. For lines and surfaces, do not vary from straight by more than 1/8 inch in 10 feet, 3/8 inch in 20 feet, or 1/2-inch maximum.
6. For vertical alignment of exposed head joints, do not vary from plumb by more than 1/4 inch in 10 feet, or 1/2-inch maximum.
7. For faces of adjacent exposed masonry units, do not vary from flush alignment by more than 1/16 inch.

D. Joints:

1. For bed joints, do not vary from thickness indicated by more than plus or minus 1/8 inch, with a maximum thickness limited to 1/2 inch.
2. For exposed bed joints, do not vary from bed-joint thickness of adjacent courses by more than 1/16 inch.
3. For exposed head joints, do not vary from thickness indicated by more than plus or minus 1/8 inch.
4. Maintain a 1/4 inch to 3/8 inch space or dimension shown, between masonry face and hollow metal frames.

3.4 LAYING MASONRY WALLS

A. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint thicknesses and for accurate location of openings, movement-type joints, returns, and offsets. Avoid using less-than-half-size units, particularly at corners, jambs, and, where possible, at other locations.

B. Bond Pattern for Exposed Masonry: Unless otherwise indicated, lay exposed masonry in running bond pattern shown.

C. Stopping and Resuming Work: Stop work by stepping back units in each course from those in course below; do not tooth. When resuming work, clean masonry surfaces that are to receive mortar, remove loose masonry units and mortar, and wet masonry if required before laying fresh masonry.

D. Built-in Work: As construction progresses, build in items specified in this and other Sections. Fill in solidly with masonry around built-in items, except where otherwise specified.

E. Retain space between hollow metal frames and masonry for installation of sealant.

F. At all penetrations other than structural penetrations and those shown as solidly filled, at mechanical and plumbing piping and similar, fill space with backer rod and sealant.

G. Where built-in items are to be embedded in cores of hollow masonry units, place a layer of metal lath, wire mesh, or plastic mesh in the joint below, and rod mortar or grout into core.

H. Fill cores in hollow CMUs with grout 24 inches under bearing plates, beams, lintels, posts, and similar items unless otherwise indicated.

3.5 MORTAR BEDDING AND JOINTING

A. Lay hollow CMUs as follows:

1. Bed face shells in mortar and make head joints of depth equal to bed joints.
2. Bed webs in mortar in all courses of piers, columns, and pilasters.
3. Bed webs in mortar in grouted masonry, including starting course on footings.
4. Fully bed entire units, including areas under cells, at starting course on footings where cells are not grouted.

B. Lay solid CMUs with completely filled bed and head joints; butter ends with sufficient mortar to fill head joints and shove into place. Do not deeply furrow bed joints or slush head joints.

C. Tool exposed joints slightly concave when thumbprint hard, using a jointer larger than joint thickness unless otherwise indicated above or on drawings.

D. Cut joints flush for masonry walls to receive, plaster or other direct-applied finishes (other than paint) unless otherwise indicated.

E. Cut joints flush where indicated to receive adhered weather barriers unless otherwise indicated.

3.6 CONTROL AND EXPANSION JOINTS

A. General: Install control- and expansion-joint materials in unit masonry as masonry progresses. Do not allow materials to span control and expansion joints without provision to allow for in-plane wall or partition movement.

B. Form control joints in concrete masonry as indicated on Drawings and using one of the methods where no joints are described on the drawings as acceptable to the Contracting Officer, as follows:

1. Install preformed control-joint gaskets designed to fit standard sash block.
2. Install interlocking units designed for control joints. Install bond-breaker strips at joint. Keep head joints free and clear of mortar, or rake out joint for application of sealant.
3. Install temporary foam-plastic filler in head joints, and remove filler when unit masonry is complete for application of sealant.

3.7 REINFORCED UNIT MASONRY INSTALLATION

A. Temporary Formwork and Shores: Construct formwork and shores as needed to support reinforced masonry elements during construction.

1. Construct formwork to provide shape, line, and dimensions of completed masonry as indicated. Make forms sufficiently tight to prevent leakage of mortar and grout. Brace, tie, and support forms to maintain position and shape during construction and curing of reinforced masonry.
2. Do not remove forms and shores until reinforced masonry members have hardened sufficiently to carry their own weight and other loads that may be placed on them during construction.

B. Placing Reinforcement: Comply with requirements in TMS 602/ACI 530.1/ASCE 6.

C. Grouting: Do not place grout until entire height of masonry to be grouted has attained enough strength to resist grout pressure.

1. Comply with requirements in TMS 602/ACI 530.1/ASCE 6 for cleanouts and for grout placement, including minimum grout space and maximum pour height.
2. Limit height of vertical grout pours to not more than 60 inches or as indicated.

3.8 FIELD QUALITY CONTROL

A. Testing and Inspecting: Construction contractor shall engage special inspectors to perform tests and inspections and prepare reports. Allow inspectors access to scaffolding and work areas as needed to perform tests and inspections. Retesting of materials that fail to comply with specified requirements shall be done at Contractor's expense.

B. Inspections: Special inspections shall be in accordance with the provided statement of special inspections.

3.9 REPAIRING, POINTING, AND CLEANING

A. Remove and replace masonry units that are loose, chipped, broken, stained, or otherwise damaged or that do not match adjoining units. Install new units to match adjoining units; install in fresh mortar, pointed to eliminate evidence of replacement.

B. Pointing: During the tooling of joints, enlarge voids and holes, except weep holes, and completely fill with mortar. Point up joints, including corners, openings, and adjacent construction, to provide a neat, uniform appearance. Prepare joints for sealant application, where indicated.

C. In-Progress Cleaning: Clean unit masonry as work progresses by dry brushing to remove mortar fins and smears before tooling joints.

D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:

1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
2. Test cleaning methods on sample wall panel; leave one-half of panel uncleaned for comparison purposes. Obtain the Contracting Officer’s approval of sample cleaning before proceeding with cleaning of masonry.
3. Protect adjacent stone and nonmasonry surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.
4. Wet wall surfaces with water before applying cleaners; remove cleaners promptly by rinsing surfaces thoroughly with clear water.
5. Clean concrete masonry by applicable cleaning methods indicated in NCMA TEK 8-4A.

3.10 MASONRY WASTE DISPOSAL

A. Salvageable Materials: Unless otherwise indicated, excess masonry materials are Contractor's property. At completion of unit masonry work, remove from Project site.

B. Masonry Waste Recycling: Return broken CMUs not used as fill to manufacturer for recycling.

C. Excess Masonry Waste: Remove excess clean masonry waste and legally dispose of off the Project property.

END OF SECTION 04 22 00

SECTION 04 26 13 – MASONRY VENEER

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Replacement brick.
2. Replacement stone veneer.
3. Mortar materials.
4. Ties and anchors.
5. Embedded flashing.
6. Accessories.
7. Mortar mixes.
8. Installation of salvaged stone veneer.

B. Locations:

1. Replacement bricks at existing brick finish on southeast exterior side of building.
2. Brick at relocated jambs of moved door 005A.
3. Stone veneer at west elevation and at vestibule entry using salvaged stone veneer.
4. Stone veneer at northwest retaining wall and plaza seat wall using salvaged stone veneer.

C. Products Installed but not Furnished under This Section:

1. Steel lintels in masonry veneer.

D. Related Requirements:

1. Section 02 41 13 “Selective Site Demolition” for salvage of existing stone veneer from planters.
2. Section 02 41 19 “Selective Building Demolition” for salvage of existing stone veneer from wall.
3. Section 04 01 20.63 “Brick Masonry Repair” for repair/patching of existing brick.
4. Section 04 01 20.64 “Brick Masonry Repointing” for repointing of existing brick.
5. Section 05 12 00 “Structural Steel Framing” for installing anchor sections of adjustable masonry anchors for connecting to structure.
6. Section 07 62 00 “Sheet Metal Flashing and Trim” for flashing.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Testing Results: Test existing mortar for chemical composition and aggregate ratio. This shall be used to determine the matching mortar mix.

C. Samples:

1. Clay face brick, matching existing
2. Stone veneer, 4”, matching existing
3. Mortar color samples

a. Each type, color, and texture of pointing mortar in the form of sample mortar strips, 6 inches long by width matching existing installation, set in aluminum or plastic channels
b. Include with each Sample a list of ingredients with proportions of each for comparison with mortar test results.
c. Include sample for brick and for stone, both matching existing

D. Qualifications Data:

1. Installers: All masonry flashing installers must complete the International Masonry Institute Flashing Upgrade training course.

1.3 MOCKUPS

A. Sample Panels (Stone Veneer installation): Build sample panels to verify selections made under Sample submittals and to demonstrate aesthetic effects. Comply with requirements in Section 01 40 00 “Quality Requirements” for mockups.

1. Build sample panels for:

a. Stone veneer installation at building wall
b. Stone veneer installation at retaining wall/planter.

2. Build sample panels facing west.
3. Mockup may be part of final work product if approved by Contracting Officer. If not approved, mockup must be removed and work redone until approved.
4. As masonry is to match existing, build panels adjacent and parallel to existing surface.
5. Protect approved sample panels from the elements with weather-resistant membrane.
6. Approval of sample panels is for color, texture, and blending of masonry units; relationship of mortar and sealant colors to masonry unit colors; tooling of joints; aesthetic qualities of workmanship; and other material and construction qualities specifically approved by Contracting Officer in writing.

B. Brick Replacement/Reinstallation: The first installation of the relocated masonry jambs shall function as a mockup to verify selections made under Sample submittals and to demonstrate aesthetic effects. Comply with requirements in Section 01 40 00 “Quality Requirements” for mockups.

1. Mockup may be part of final work product if approved by Contracting Officer. If not approved, mockup must be removed and work redone until approved.
2. Jamb location is an interior location, but if building is not dried in, protect approved sample panel from the elements with weather-resistant membrane.
3. As masonry is to match existing, build panels adjacent and parallel to existing surface.

4. Approval of sample panels is for color, texture, and blending of masonry units; relationship of mortar and sealant colors to masonry unit colors; tooling of joints; aesthetic qualities of workmanship; and other material and construction qualities specifically approved by Contracting Officer in writing.
5. Jamb mockup may serve for mockup of replacement bricks in exterior brick veneer location only if approved by Contracting Officer in writing.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Store masonry units on elevated platforms in a dry location. If units are not stored in an enclosed location, cover tops and sides of stacks with waterproof sheeting, securely tied. If units become wet, do not install until they are dry.

B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

C. Store aggregates where grading and other required characteristics can be maintained and contamination avoided.

D. Deliver preblended, dry mortar mix in moisture-resistant containers. Store preblended, dry mortar mix in delivery containers on elevated platforms in a dry location or in covered weatherproof dispensing silos.

E. Store masonry accessories, including metal items, to prevent corrosion and accumulation of dirt and oil.

1.5 FIELD CONDITIONS

A. Protection of Masonry: During construction, cover tops of veneer, projections, and sills with waterproof sheeting at end of each day's work. Cover partially completed masonry when construction is not in progress.

1. Extend cover a minimum of 24 inches down face of veneer, and hold cover securely in place.

B. Stain Prevention: Prevent grout, mortar, and soil from staining the face of masonry. Immediately remove grout, mortar, and soil that come in contact with masonry.

1. Protect base of walls from rain-splashed mud and from mortar splatter by spreading coverings on ground and over wall surface.
2. Protect sills, ledges, and projections from mortar droppings.
3. Protect surfaces of window and door frames, as well as similar products with painted and integral finishes, from mortar droppings.
4. Turn scaffold boards near the wall on edge at the end of each day to prevent rain from splashing mortar and dirt onto completed masonry.

PART 2 - PRODUCTS

2.1 UNIT MASONRY, SALVAGED STONE

A. DESIGN INTENT, SALVAGED STONE

1. It is the intent of this project to have as much of the stone veneer on this project as possible be sourced from the stone veneer salvaged during the selective demolition portion of this project. (see image previous page of veneer to be salvaged) A priority shall be placed on using salvaged stone on the building. If the use of newer stone veneer to match is needed, the newer stone may be used on the landscape planter.

B. Masonry Standard: Comply with TMS 602, except as modified by requirements in the Contract Documents.

2.2 UNIT MASONRY, BRICK

A. DESIGN INTENT, BRICK

1. It is the intent of this project that the brick under the siding on the south east corner of the building (see top image above) be exposed, repaired if needed, and used as the exterior finish for that corner. That brick is assumed to be the same brick as the brick found in the storage area around the north and north eastern sides of the basement level (see bottom image above). Contractor shall confirm this prior to finding replacement brick. Intent is for replacement brick to match the original in color (and range of variation), texture, and size/proportion such that the replacements shall be as seamless as possible.
2. The brick used for comparison to determine the appropriate color/texture match shall be cleaned to ensure accurate color match and shall represent the range of color variation on the building. It is recommended that the initial cleaning of the brick precedes the selection of matching brick and mortar.

B. Masonry Standard: Comply with TMS 602, except as modified by requirements in the Contract Documents.

2.3 STONE

A. Stone Sources:

1. Acceptable stone sources include salvaged stone from deconstruction of existing stone features and commercial stone sources.
2. Sources and species may be submitted for approval by the CO. New stone shall be from local / regional quarry and stone shall match existing stone in size, proportion, and material.

    a. Stone from local source to match existing shape, color, texture and size.
    b. Specific stone selection to be through visit to materials yard for visual confirmation.

3. Stone shall be sound, hard, well-shaped, clean, durable, and free from structural defects and seams, iron rust, and dirt. Sorting, collection, storage and transportation of stone materials are the responsibility of the Contractor as is coordinating with the sub-contractors. Cutting, splitting, tooling and installing stone as required, is the responsibility

of the contractor and sub-contractors. All chips and waste must be returned hauled off and disposed of legally as per Division 01.

B. Dimensions General:

1. Veneer
2. Minimum depth of face stone in cavity walls and stone veneer is 4 inches as indicated on the Drawings and as approved by CO.

C. Pattern

1. Shaped and tooled stone masonry veneer shall be laid up in a horizontal random pattern, roughly coursed, with a mixture of somewhat rectilinear and somewhat squarer stones. with a variety of sizes spread evenly throughout the wall. Mix and pattern shall match the existing stone veneer installation. See photo under 2.1 this section.

2.4 BRICK

A. General: Provide shapes indicated and as follows, with exposed surfaces matching finish and color of exposed faces of adjacent units.

1. For ends of sills and caps and for similar applications that would otherwise expose unfinished brick surfaces, provide units without cores or frogs and with exposed surfaces finished.

B. Clay Face Brick: Facing brick complying with ASTM C216 , Grade SW , Type FBS or HBS depending on application.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Acme Brick Company.
b. Belden Brick Company (The).
c. Endicott Clay Products Co.
d. General Shale, Inc.
e. Glen-Gery Corporation.
f. Or Approved Equal

2. Initial Rate of Absorption: Less than 30 g/30 sq. in. per minute when tested in accordance with ASTM C67/C67M.
3. Efflorescence: Provide brick that has been tested in accordance with ASTM C67/C67M and is rated “not effloresced.”
4. Surface Coating: Brick with colors or textures produced by application of coatings withstand 50 cycles of freezing and thawing in accordance with ASTM C67/C67M with no observable difference in the applied finish when viewed from 10 ft. or have a history of successful use in Project's area.
5. Application: Use where brick is exposed unless otherwise indicated.
6. Provide clay face brick matching color range, texture, and size of existing adjacent brickwork.

2.5 MORTAR MATERIALS

A. DESIGN INTENT:

1. For stone: It is the intent of this project that the mortar shall match the original in color.
2. For brick: It is the intent of this project that the mortar shall match the original in color and installation such that the new infill shall be as seamless as possible.
3. The mortar used for comparison to determine the appropriate color match shall be clean to ensure accurate color match.
4. Testing shall be conducted to determine the mortar chemical composition and aggregate ratio.

B. Portland Cement: ASTM C150/C150M, Type I or II, except Type III may be used for cold-weather construction. Provide natural color or white cement as required to produce mortar color indicated.

1. Alkali content will not be more than 0.1 percent when tested in accordance with ASTM C114.

C. Hydrated Lime: ASTM C207, Type S.

D. Mortar Pigments: Natural and synthetic iron oxides and chromium oxides, compounded for use in mortar mixes and complying with ASTM C979/C979M. Use only pigments with a record of satisfactory performance in masonry mortar.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Davis Colors.
b. Euclid Chemical Company (The); a subsidiary of RPM International, Inc.
c. Solomon Colors Inc.
d. Or Approved Equal

E. Aggregate for Mortar: ASTM C144.

1. For mortar that is exposed to view, use washed aggregate consisting of natural sand or crushed stone.
2. Colored-Mortar Aggregates: Natural sand or crushed stone of color necessary to produce required mortar color.

F. Water: Potable.

2.6 TIES AND ANCHORS

A. General: Ties and anchors extend at least 1-1/2 inches into veneer but with at least a 5/8-inch cover on outside face.

B. Materials: Provide ties and anchors specified in this article that are made from materials that comply with the following unless otherwise indicated:

1. Hot-Dip Galvanized, Carbon-Steel Wire: ASTM A1064/A1064M, with ASTM A153/A153M, Class B-2 coating.
2. Steel Sheet, Galvanized after Fabrication: ASTM A1008/A1008M, Commercial Steel, with ASTM A153/A153M, Class B coating.

C. Corrugated-Metal Ties: Metal strips not less than 7/8 inch wide with corrugations having a wavelength of 0.3 to 0.5 inch and an amplitude of 0.06 to 0.10 inch made from 0.0635-inch- thick, steel sheet, galvanized after fabrication .

D. Adjustable Masonry-Veneer Anchors:

1. General: Provide anchors that allow vertical adjustment but resist a 100 lbf load in both tension and compression perpendicular to plane of wall without deforming or developing play in excess of 1/16 inch.
2. Seismic Zone Spacing: As this installation is Category B based on seismic design standards, anchors shall be placed per standard practice.
3. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Hohmann & Barnard, Inc.
    b. Wire-Bond.
    c. Heckmann Building Products
    d. Or Approved Equal.

4. Fabricate sheet metal anchor sections and other sheet metal parts from 0.1084-inch- thick steel sheet, galvanized after fabrication .
5. Fabricate wire ties from 0.187-inch- diameter, hot-dip galvanized steel wire unless otherwise indicated.
6. Contractor's Option: Unless otherwise indicated, provide any of the adjustable masonry-veneer anchors specified.
7. Masonry-Veneer Anchors; Vertical Slotted L-Plate: Rib-stiffened, sheet metal anchor section with screw holes at top and bottom, projecting vertical leg with slotted hole for wire tie and washer at face of insulation.
8. Masonry-Veneer Anchors; Double-Pintle Plate: Rib-stiffened, sheet metal anchor section with screw holes at top and bottom, projecting horizontal leg with slots for vertical legs of double pintle wire tie. Provide with seismic tie, clip, and continuous wire in veneer.
9. Masonry-Veneer Anchors; Slotted Plate: Sheet metal anchor section, with screw holes at top and bottom; and raised rib-stiffened strap, stamped into center to provide a slot between strap and base for wire tie. Use self-adhering tape to seal penetration behind anchor plate.
10. Masonry-Veneer Anchors; Slotted Plate with Prongs: Sheet metal anchor section, with screw holes at top and bottom; top and bottom ends bent to form pronged legs of length to match thickness of insulation; and raised rib-stiffened strap, stamped into center to provide a slot between strap and base for wire tie. Use self-adhering tape to seal penetration behind anchor plate.

2.7 EMBEDDED FLASHING

A. Metal Flashing: Provide metal flashing complying with SMACNA's “Architectural Sheet Metal Manual” and as follows:

1. Stainless Steel: ASTM A240/A240M or ASTM A666, Type 316, 0.016 inch thick.
2. Finish: Prefinished/factory finished matched to adjacent materials per detail.
3. Fabricate continuous flashings in sections 96 inches long minimum but not exceeding 12 ft. Provide splice plates at joints of formed, smooth metal flashing.
4. Fabricate through-wall metal flashing embedded in masonry from stainless steel, with ribs at 3-inch intervals along length of flashing to provide an integral mortar bond.
5. Fabricate through-wall flashing with snaplock receiver on exterior face where indicated to receive counterflashing.
6. Fabricate through-wall flashing with drip edge unless otherwise indicated. Fabricate by extending flashing 1/2 inch out from wall, with outer edge bent down 30 degrees.
7. Fabricate through-wall flashing with sealant stop unless otherwise indicated. Fabricate by bending metal back on itself 3/4 inch at exterior face of wall and down into joint 1/4 inch to form a stop for retaining sealant backer rod.
8. Fabricate metal drip edges and sealant stops for sawtooth metal flashing from plain metal flashing of same metal as sawtooth flashing and extending at least 3 inches into wall with hemmed inner edge to receive sawtooth flashing and form a hooked seam. Form hem on upper surface of metal so that completed seam sheds water.
9. Fabricate metal drip edges from stainless steel. Extend at least 3 inches into wall and 1/2 inch out from wall, with outer edge bent down 30 degrees.
10. Fabricate metal sealant stops from stainless steel. Extend at least 3 inches into wall and out to exterior face of wall. At exterior face of wall, bend metal back on itself for 3/4 inch and down into joint 1/4 inch to form a stop for retaining sealant backer rod.
11. Solder metal items at corners.

B. Flexible Flashing: Use one of the following unless otherwise indicated:

1. Butyl Rubber Flashing: Composite, self-adhesive, flashing product consisting of a pliable, butyl rubber compound, bonded to a high-density polyethylene film, aluminum foil, or spunbonded polyolefin to produce an overall thickness of not less than 40 mil.

    a. Accessories: Provide preformed corners, end dams, other special shapes, and seaming materials produced by flashing manufacturer.

2. Elastomeric Thermoplastic Flashing: Composite flashing product consisting of a polyester-reinforced ethylene interpolymer alloy.

    a. Monolithic Sheet: Elastomeric thermoplastic flashing, 40 mil thick.
    b. Self-Adhesive Sheet: Elastomeric thermoplastic flashing, 25 mil thick, with a 15-mil- thick coating of adhesive.

        1) Color: Black.

    c. Accessories: Provide preformed corners, end dams, other special shapes, and seaming materials produced by flashing manufacturer.

3. EPDM Flashing: Sheet flashing product made from ethylene-propylene-diene terpolymer, complying with ASTM D4637/D4637M, 40 mil thick.

C. Adhesives, Primers, and Seam Tapes for Flashings: Flashing manufacturer's standard products or products recommended by flashing manufacturer for bonding flashing sheets to each other and to substrates.

2.8 ACCESSORIES

A. Weep/Vent Products: Use one of the following unless otherwise indicated:

1. Cellular Plastic Weep/Vent: One-piece, flexible extrusion made from UV-resistant polypropylene copolymer, full height and width of head joint and depth 1/8 inch less than depth of outer wythe, in color selected from manufacturer's standard by Contracting Officer.
2. Mesh Weep/Vent: Free-draining mesh; made from polyethylene strands, full height and width of head joint and depth 1/8 inch less than depth of outer wythe; in color selected from manufacturer's standard by Contracting Officer.
3. Vinyl Weep Hole/Vent: Units made from flexible PVC, designed to fit into a head joint and consisting of a louvered vertical leg, flexible wings to seal against ends of masonry units, and a top flap to keep mortar out of the head joint; in color selected by Contracting Officer.

B. Proprietary Acidic Masonry Cleaner: Manufacturer's standard-strength cleaner designed for removing mortar/grout stains, efflorescence, and other new construction stains from new masonry without discoloring or damaging masonry surfaces. Use product expressly approved for intended use by cleaner manufacturer and manufacturer of masonry units being cleaned.

1. Basis of Design: Prosoco SureKlean 600
2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Diedrich Technologies, Inc.; a Hohmann & Barnard company.
    b. EaCo Chem, Inc.
    c. PROSOCO, Inc.
    d. Or Approved Equal.

3. For cleaning existing masonry, see Section 04 01 20.63 “Brick Masonry Repair.”

2.9 MORTAR MIXES

A. General: Do not use admixtures, including pigments, air-entraining agents, accelerators, retarders, water-repellent agents, antifreeze compounds, or other admixtures unless otherwise indicated.

1. Do not use calcium chloride in mortar or grout.
2. Use portland cement-lime mortar unless otherwise indicated.
3. Add cold-weather admixture (if used) at same rate for all mortar that will be exposed to view, regardless of weather conditions, to ensure that mortar color is consistent.

B. Mortar for Unit Masonry: Match existing mortar mix.

C. Pigmented Mortar: Use colored cement product or select and proportion pigments with other ingredients to produce color required. Do not add pigments to colored cement products.

1. Mix to match existing mortar.
2. Application: Use pigmented mortar for exposed mortar joints.

D. Colored-Aggregate Mortar: Produce required mortar color by using colored aggregates and natural color or white cement as necessary to produce required mortar color.

1. Only use if needed to achieve match with existing mortar.
2. Application: Use colored-aggregate mortar for exposed mortar joints.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

1. For the record, prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

A. Leave openings for equipment to be installed before completing masonry. After installing equipment, complete masonry to match the construction immediately adjacent to opening.

B. Use full-size units without cutting if possible. If cutting is required to provide a continuous pattern or to fit adjoining construction, cut units with motor-driven saws; provide clean, sharp, unchipped edges. Allow units to dry before laying unless wetting of units is specified. Install cut units with cut surfaces and, where possible, cut edges concealed.

C. Select and arrange units for exposed unit masonry to produce a uniform blend of colors and textures. Mix units from several pallets or cubes as they are placed.

D. Matching Existing Masonry: Match coursing, bonding, color, and texture of existing masonry.

E. Wetting of Brick: Wet brick before laying if initial rate of absorption exceeds 30 g/30 sq. in. per minute when tested in accordance with ASTM C67/C67M. Allow units to absorb water so they are damp but not wet at time of laying.

3.3 TOLERANCES

A. Dimensions and Locations of Elements:

1. For dimensions in cross section or elevation, do not vary by more than plus 1/2 inch or minus 1/4 inch.
2. For location of elements in plan, do not vary from that indicated by more than plus or minus 1/2 inch.
3. For location of elements in elevation, do not vary from that indicated by more than plus or minus 1/4 inch in a story height or 1/2 inch total.

B. Lines and Levels (for Brick):

1. For bed joints and top surfaces of bearing walls, do not vary from level by more than 1/4 inch in 10 ft., or 1/2-inch maximum.
2. For conspicuous horizontal lines, such as lintels, sills, and caps, do not vary from level by more than 1/8 inch in 10 ft., 1/4 inch in 20 ft., or 1/2-inch maximum.
3. For vertical lines and surfaces, do not vary from plumb by more than 1/4 inch in 10 ft., 3/8 inch in 20 ft., or 1/2-inch maximum.
4. For conspicuous vertical lines, such as external corners, door jambs, reveals, and expansion and control joints, do not vary from plumb by more than 1/8 inch in 10 ft., 1/4 inch in 20 ft., or 1/2-inch maximum.
5. For lines and surfaces, do not vary from straight by more than 1/4 inch in 10 ft., 3/8 inch in 20 ft., or 1/2-inch maximum.
6. For vertical alignment of exposed head joints, do not vary from plumb by more than 1/4 inch in 10 ft., or 1/2-inch maximum.
7. For faces of adjacent exposed masonry units, do not vary from flush alignment by more than 1/16 inch except due to warpage of masonry units within tolerances specified for warpage of units.

C. Joints (for Brick):

1. For all joints, match existing. Do not vary from thickness indicated by more than plus or minus 1/8 inch.

3.4 LAYING MASONRY WALLS

A. Lay out walls in advance for accurate spacing of surface bond patterns with uniform joint thicknesses and for accurate location of openings, movement-type joints, returns, and offsets. Avoid using less-than-half-size units, particularly at corners, jambs, and, where possible, at other locations.

B. Bond Pattern for Exposed Brick Masonry: Match existing adjacent masonry; do not use units with less-than-nominal 4-inch horizontal face dimensions at corners or jambs.

C. Bond Pattern for Exposed Stone Masonry: Unless otherwise indicated, lay exposed masonry in random ashlar pattern; do not use units with less-than-nominal 4-inch horizontal face dimensions at corners or jambs.

D. Stopping and Resuming Work: Stop work by stepping back units in each course from those in course below; do not tooth. When resuming work, clean masonry surfaces that are to receive mortar, remove loose masonry units and mortar, and wet brick if required before laying fresh masonry.

E. Built-in Work: As construction progresses, build in items specified in this and other Sections. Fill in solidly with masonry around built-in items.

F. Fill space between steel frames and masonry solidly with mortar unless otherwise indicated.

3.5 MORTAR BEDDING AND JOINTING

A. Lay solid masonry units with completely filled bed and head joints; butter ends with sufficient mortar to fill head joints and shove into place. Do not deeply furrow bed joints or slush head joints.

B. Tool exposed joints slightly concave when thumbprint hard, using a jointer larger than joint thickness unless otherwise indicated.

1. Joint profile shall match existing mortar installation.

3.6 ANCHORED MASONRY VENEERS

A. Anchor masonry veneers to wall framing with seismic masonry-veneer anchors to comply with the following requirements:

1. Fasten screw-attached and seismic anchors through sheathing to wall framing with metal fasteners of type indicated. Use two fasteners unless anchor design only uses one fastener.
2. Embed tie sections and connector sections and continuous wire in masonry joints.
3. Locate anchor sections to allow maximum vertical differential movement of ties up and down.
4. Space anchors in accordance with seismic and wind loading requirements as described on the Structural General Notes sheet in the drawings. Install additional anchors around openings and perimeter also in accordance with requirements and per the capacity of the selected anchors.

B. At brick veneer: match existing installation airspace width.

C. At stone veneer, re: drawings for airspace width.

D. Keep airspace clean of mortar droppings and other materials during construction. Bevel beds away from airspace, to minimize mortar protrusions into airspace. Do not attempt to trowel or remove mortar fins protruding into airspace.

3.7 ANCHORING MASONRY TO STRUCTURAL STEEL AND CONCRETE

A. Anchor masonry to structural steel and concrete, where masonry abuts or faces structural steel or concrete to comply with the following:

1. Provide an open space not less than existing open space between masonry and structural steel or concrete unless otherwise indicated. Keep open space free of mortar and other rigid materials.
2. Anchor masonry with anchors embedded in masonry joints and attached to structure.
3. Space anchors as indicated, but not more than 24 inches o.c. vertically and 36 inches o.c. horizontally.

3.8 LINTELS

A. Install steel lintels as directed in structural drawings.

3.9 FLASHING, WEEP HOLES, AND VENTS

A. General: Install embedded flashing and weep holes in masonry at shelf angles, lintels, ledges, other obstructions to downward flow of water in wall, and where indicated.

B. Install flashing as follows unless otherwise indicated:

1. Prepare masonry surfaces so they are smooth and free from projections that could puncture flashing. Where flashing is within mortar joint, place through-wall flashing on sloping bed of mortar and cover with mortar. Before covering with mortar, seal penetrations in flashing with adhesive, sealant, or tape as recommended by flashing manufacturer.
2. Extend flashing through veneer, across airspace behind veneer, and up face of sheathing at least 8 inches; with upper edge tucked under water-resistive barrier , lapping at least 4 inches. Fasten upper edge of flexible flashing to sheathing through termination bar.
3. At lintels and shelf angles, extend flashing 6 inches minimum at each end. At heads and sills, extend flashing 6 inches minimum and turn ends up not less than 2 inches to form end dams.
4. Interlock end joints of sawtooth sheet metal flashing by overlapping ribs not less than 1-1/2 inches or as recommended by flashing manufacturer, and seal lap with elastomeric sealant complying with requirements in Section 07 92 00 “Joint Sealants” for application indicated.
5. Install metal drip edges with sawtooth sheet metal flashing by interlocking hemmed edges to form hooked seam. Seal seam with elastomeric sealant complying with requirements in Section 07 92 00 “Joint Sealants” for application indicated.
6. Install metal drip edges beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch back from outside face of wall, and adhere flexible flashing to top of metal drip edge.
7. Install metal flashing termination beneath flexible flashing at exterior face of wall. Stop flexible flashing 1/2 inch back from outside face of wall, and adhere flexible flashing to top of metal flashing termination.
8. Cut flexible flashing off flush with face of wall after masonry wall construction is completed.

C. Install reglets and nailers for flashing and other related construction where they are indicated to be built into masonry.

D. Install weep holes in veneers in head joints of first course of masonry immediately above embedded flashing.

1. Use specified weep/cavity vent products to form weep holes.
2. Space weep holes 24 inches o.c. unless otherwise indicated.

E. Install vents in head joints in exterior wythes at spacing indicated. Use specified weep/cavity vent products to form vents.

1. Close cavities off vertically and horizontally with blocking in manner indicated. Install through-wall flashing and weep holes above horizontal blocking.

3.10 REPAIRING, POINTING, AND CLEANING

A. Remove and replace masonry units that are loose, chipped, broken, stained, or otherwise damaged or that do not match adjoining units. Install new units to match adjoining units; install in fresh mortar, pointed to eliminate evidence of replacement.

B. Pointing: During the tooling of joints, enlarge voids and holes, except weep holes, and completely fill with mortar. Point up joints, including corners, openings, and adjacent construction, to provide a neat, uniform appearance. Prepare joints for sealant application, where indicated.

C. In-Progress Cleaning: Clean unit masonry as work progresses by dry brushing to remove mortar fins and smears before tooling joints.

D. Final Cleaning: After mortar is thoroughly set and cured, clean exposed masonry as follows:

1. Remove large mortar particles by hand with wooden paddles and nonmetallic scrape hoes or chisels.
2. Test cleaning methods on sample wall panel; leave one-half of panel uncleaned for comparison purposes. Obtain Contracting Officer's approval of sample cleaning before proceeding with cleaning of masonry.
3. Protect adjacent stone and nonmasonry surfaces from contact with cleaner by covering them with liquid strippable masking agent or polyethylene film and waterproof masking tape.
4. Wet wall surfaces with water before applying cleaners; remove cleaners promptly by rinsing surfaces thoroughly with clear water.
5. Clean brick by bucket-and-brush hand-cleaning method described in BIA Technical Notes 20.
6. Clean masonry with a proprietary acidic cleaner applied according to manufacturer's written instructions.

3.11 MASONRY WASTE DISPOSAL

A. Salvageable Materials: Unless otherwise indicated, excess masonry materials are Government property. At completion of unit masonry work, return to Contracting Officer.

B. Excess Masonry Waste: Remove excess clean masonry waste that cannot be used as fill, as described above or recycled, and other masonry waste, and legally dispose of off Government's property.

END OF SECTION 04 26 13

SECTION 05 12 00 – STRUCTURAL STEEL FRAMING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Structural steel.
2. Grout.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 03 30 00 “Cast in Place Concrete”
2. Section 05 50 00 “Metal Fabrications”

1.2 DEFINITIONS

A. Structural Steel: Elements of the structural frame indicated on Drawings and as described in AISC 303, “Code of Standard Practice for Steel Buildings and Bridges.”

B. Lateral-Force-Resisting System: Elements of structural-steel frame designated as “LFRS” or along grid lines designated as “LFRS” on Drawings, including columns, beams, and braces and their connections.

C. Heavy Sections: Rolled and built-up sections as follows:

1. Shapes included in ASTM A 6/A 6M with flanges thicker than 1-1/2 inches.
2. Welded built-up members with plates thicker than 2 inches.
3. Column base plates thicker than 2 inches.

D. Protected Zone: Structural members or portions of structural members indicated as “Protected Zone” on Drawings. Connections of structural and nonstructural elements to protected zones are limited.

E. Demand Critical Welds: Those welds, the failure of which would result in significant degradation of the strength and stiffness of the Lateral-Force-Resisting System, and which are indicated as “Demand Critical” on Drawings.

1.3 COORDINATION

A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers’ written recommendations to ensure that shop primers and topcoats are compatible with one another.

B. Coordinate installation of anchorage items to be embedded in or attached to other construction without delaying the Work. Provide setting diagrams, sheet metal templates, instructions, and directions for installation.

1.4 SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: Show fabrication of structural-steel components.

1. Include details of cuts, connections, splices, camber, holes, and other data.
2. Steel layout plan.
3. Include embedment Drawings and tolerance requirements.
4. Indicate welds by standard AWS symbols, distinguishing between shop and field welds, and show size, length, and type of each weld. Show backing bars that are to be removed and supplemental fillet welds where backing bars are to remain.
5. Indicate type, size, and length of bolts, distinguishing between shop and field bolts. Identify pretensioned and slip-critical, high-strength bolted connections.
6. Identify members and connections of the Lateral-Force-Resisting System.
7. Indicate locations and dimensions of protected zones.
8. Identify demand critical welds.

C. Welding Procedure Specifications (WPSs) and Procedure Qualification Records (PQRs): Provide according to AWS D1.1/D1.1M, "Structural Welding Code - Steel," for each welded joint whether prequalified or qualified by testing, including the following:

1. Power source (constant current or constant voltage).
2. Electrode manufacturer and trade name, for demand critical welds.

D. Qualification Data: For Installer and fabricator.

E. Welding certificates.

F. Paint Compatibility Certificates: From manufacturers of topcoats applied over shop primers, certifying that shop primers are compatible with topcoats.

G. Field quality-control and special inspection reports.

1.5 QUALITY ASSURANCE

A. Fabricator Qualifications: A qualified fabricator that fabricates to AISC Quality Standards.

B. Installer Qualifications: A qualified installer who installs in conformance with the AISC Quality Certification Program.

C. Shop-Painting Applicators: Qualified according to AISC's Sophisticated Paint Endorsement P1 or to SSPC-QP 3, "Standard Procedure for Evaluating Qualifications of Shop Painting Applicators."

D. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

1. Welders and welding operators performing work on bottom-flange, demand-critical welds shall pass the supplemental welder qualification testing, as required by AWS D1.8/D1.8M. FCAW-S and FCAW-G shall be considered separate processes for welding personnel qualification.

E. Comply with applicable provisions of the following specifications and documents:

1. AISC 303.
2. AISC 341.
3. AISC 360.
4. RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

1.6 SOURCE QUALITY CONTROL

A. Testing Agency: Contracting Officer may engage a qualified testing agency to perform shop tests and inspections.

1. Provide testing agency with access to places where structural-steel work is being fabricated or produced to perform tests and inspections.

B. Bolted Connections: Inspect and test shop-bolted connections according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

C. Welded Connections: Visually inspect shop-welded connections according to AWS D1.1/D1.1M and the following inspection procedures, at testing agency's option:

1. Liquid Penetrant Inspection: ASTM E 165.
2. Magnetic Particle Inspection: ASTM E 709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration are not accepted.
3. Ultrasonic Inspection: ASTM E 164.
4. Radiographic Inspection: ASTM E 94.

D. In addition to visual inspection, test and inspect shop-welded shear connectors according to requirements in AWS D1.1/D1.1M for stud welding and as follows:

1. Perform bend tests if visual inspections reveal either a less-than-continuous 360-degree flash or welding repairs to any shear connector.
2. Conduct tests according to requirements in AWS D1.1/D1.1M on additional shear connectors if weld fracture occurs on shear connectors already tested.

E. Prepare test and inspection reports.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Store materials to permit easy access for inspection and identification. Keep steel members off ground and spaced by using pallets, dunnage, or other supports and spacers. Protect steel members and packaged materials from corrosion and deterioration.

1. Do not store materials on structure in a manner that might cause distortion, damage, or overload to members or supporting structures. Repair or replace damaged materials or structures as directed.

B. Store fasteners in a protected place in sealed containers with manufacturer's labels intact.

1. Fasteners may be repackaged provided Contracting Officer's testing and inspecting agency observes repackaging and seals containers.
2. Clean and relubricate bolts and nuts that become dry or rusty before use.
3. Comply with manufacturers' written recommendations for cleaning and lubricating ASTM F 1852 fasteners and for retesting fasteners after lubrication.

PART 2 - PRODUCTS

2.1 STRUCTURAL-STEEL MATERIALS

A. W-Shapes: ASTM A 992/A 992M.

B. Channels, Angles, M, S-Shapes: ASTM A 36/A 36M.

C. Plate and Bar: ASTM A 36/A 36M or ASTM A 572/A 572M, Grade 50.

D. Corrosion-Resisting Structural-Steel Shapes, Plates, and Bars: ASTM A 588/A 588M, Grade 50.

E. Cold-Formed Hollow Structural Sections: ASTM A 500/A 500M, Grade C, structural tubing.

F. Corrosion-Resisting, Cold-Formed Hollow Structural Sections: ASTM A 847/A 847M, structural tubing.

G. Steel Pipe: ASTM A 53/A 53M, Type E or Type S, Grade B.

1. Weight Class: as indicated on Drawings Extra strong Double-extra strong.
2. Finish: Black except where indicated to be galvanized.

H. Steel Castings: ASTM A 216/A 216M, Grade WCB with supplementary requirement S11.

I. Steel Forgings: ASTM A 668/A 668M.

J. Welding Electrodes: Comply with AWS requirements.

K. Recycled Content of Steel Products: Post-consumer recycled content plus one-half of pre-consumer recycled content not less than 50 percent.

2.2 BOLTS, CONNECTORS, AND ANCHORS

A. High-Strength Bolts, Nuts, and Washers: ASTM A 325, Type 1, heavy-hex steel structural bolts; ASTM A 563, Grade C, heavy-hex carbon-steel nuts; and ASTM F 436, Type 1, hardened carbon-steel washers; all with plain finish.

B. Zinc-Coated High-Strength Bolts, Nuts, and Washers: ASTM A 325, Type 1, heavy-hex steel structural bolts; ASTM A 563, Grade DH heavy-hex carbon-steel nuts; and ASTM F 436, Type 1, hardened carbon-steel washers.

1. Finish: Hot-dip zinc coating.

C. Machine Bolts, Nuts, and Washers: ASTM A 307, Grade A, steel structural bolts; ASTM A 563A, carbon-steel nuts; and ASTM F 844 steel washers.

D. Shear Connectors: ASTM A 108, Grades 1015 through 1020, headed-stud type, cold-finished carbon steel; AWS D1.1/D1.1M, Type B.

E. Unheaded Anchor Rods: ASTM F 1554, Grade 36.

1. Configuration: as shown.
2. Nuts: ASTM A 563hex carbon steel.
3. Plate Washers: ASTM A 36/A 36M carbon steel.
4. Washers: ASTM F 436, Type 1, hardened carbon steel.
5. Finish: Plain or Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

F. Headed Anchor Rods: ASTM F 1554, Grade 36, straight.

1. Nuts: ASTM A 563hex carbon steel.
2. Plate Washers: ASTM A 36/A 36M carbon steel.
3. Washers: ASTM F 436, Type 1, hardened carbon steel.
4. Finish: Plain or Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

G. Threaded Rods: ASTM A 36/A 36M.

1. Nuts: ASTM A 563hex carbon steel.
2. Washers: ASTM F 436, Type 1, hardened carbon steel.
3. Finish: Plain or Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

H. Eye Bolts and Nuts: Made from cold-finished carbon steel bars, ASTM A 108, Grade 1030.

I. Sleeve Nuts: Made from cold-finished carbon steel bars, ASTM A 108, Grade 1018.

2.3 PRIMER

A. Primer: Fabricator’s standard lead- and chromate-free, nonasphaltic, rust-inhibiting primer complying with MPI#79 and compatible with topcoat.

2.4 GROUT

A. Nonmetallic, Shrinkage-Resistant Grout: ASTM C 1107/C 1107M, factory-packaged, nonmetallic aggregate grout, noncorrosive and non-staining, mixed with water to consistency suitable for application and a 30-minute working time.

2.5 FABRICATION

A. Structural Steel: Fabricate and assemble in shop to greatest extent possible. Fabricate according to AISC 303, "Code of Standard Practice for Steel Buildings and Bridges," and to AISC 360.

1. Camber structural-steel members where indicated.
2. Fabricate beams with rolling camber up.
3. Identify high-strength structural steel according to ASTM A 6/A 6M and maintain markings until structural steel has been erected.
4. Mark and match-mark materials for field assembly.
5. Complete structural-steel assemblies, including welding of units, before starting shop-priming operations.

B. Thermal Cutting: Perform thermal cutting by machine to greatest extent possible.

1. Plane thermally cut edges to be welded to comply with requirements in AWS D1.1/D1.1M.

C. Bolt Holes: Cut, drill, or punch standard bolt holes perpendicular to metal surfaces.

D. Finishing: Accurately finish ends of columns and other members transmitting bearing loads.

E. Cleaning: Clean and prepare steel surfaces that are to remain unpainted according to SSPC-SP 1, "Solvent Cleaning."

F. Shear Connectors: Prepare steel surfaces as recommended by manufacturer of shear connectors. Use automatic end welding of headed-stud shear connectors according to AWS D1.1/D1.1M and manufacturer's written instructions.

G. Holes: Provide holes required for securing other work to structural steel and for other work to pass through steel members.

1. Cut, drill, or punch holes perpendicular to steel surfaces. Do not thermally cut bolt holes or enlarge holes by burning.
2. Baseplate Holes: Cut, drill, or punch holes perpendicular to steel surfaces.
3. Weld threaded nuts to framing and other specialty items indicated to receive other work.

2.6 SHOP CONNECTIONS

A. High-Strength Bolts: Shop install high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts" for type of bolt and type of joint specified.

1. Joint Type: as shown.

B. Weld Connections: Comply with AWS D1.1/D1.1M and AWS D1.8/D1.8M for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.

1. Assemble and weld built-up sections by methods that maintain true alignment of axes without exceeding tolerances in AISC 303 for mill material.

2.7 SHOP PRIMING

A. Shop prime steel surfaces except the following:

1. Surfaces embedded in concrete or mortar. Extend priming of partially embedded members to a depth of 2 inches.
2. Surfaces to be field welded – includes steel designated to receive shear studs or DBA's (developed bar anchors).
3. Surfaces of high-strength bolted, slip-critical connections.
4. Surfaces to receive sprayed fire-resistive materials (applied fireproofing).
5. Galvanized surfaces.

B. Surface Preparation: Clean surfaces to be painted. Remove loose rust and mill scale and spatter, slag, or flux deposits. Prepare surfaces according to the following specifications and standards:

1. SSPC-SP 2, "Hand Tool Cleaning."

C. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils. Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.

2.8 GALVANIZING

A. Hot-Dip Galvanized Finish: Apply zinc coating by the hot-dip process to exposed structural steel according to ASTM A 123/A 123M.

1. Fill vent and drain holes that are exposed in the finished Work unless they function as weep holes, by plugging with zinc solder and filing off smooth.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify, with certified steel erector present, elevations of concrete- and masonry-bearing surfaces and locations of anchor rods, bearing plates, and other embedments for compliance with requirements.

1. Prepare a certified survey of existing conditions. Include bearing surfaces, anchor rods, bearing plates, and other embedments showing dimensions, locations, angles, and elevations.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Provide temporary shores, guys, braces, and other supports during erection to keep structural steel secure, plumb, and in alignment against temporary construction loads and loads equal in intensity

to design loads. Remove temporary supports when permanent structural steel, connections, and bracing are in place unless otherwise indicated.

1. Do not remove temporary shoring supporting composite deck construction until cast-in-place concrete has attained its design compressive strength.

3.3 ERECTION

A. Set structural steel accurately in locations and to elevations indicated and according to AISC 303 and AISC 360.

B. Baseplates Bearing Plates and Leveling Plates: Clean concrete- and masonry-bearing surfaces of bond-reducing materials and roughen surfaces prior to setting plates. Clean bottom surface of plates.

1. Set plates for structural members on wedges, shims, or setting nuts as required.
2. Weld plate washers to top of baseplate.
3. Snug-tighten anchor rods after supported members have been positioned and plumbed. Do not remove wedges or shims but, if protruding, cut off flush with edge of plate before packing with grout.
4. Promptly pack grout solidly between bearing surfaces and plates so no voids remain. Neatly finish exposed surfaces; protect grout and allow to cure. Comply with manufacturer's written installation instructions for shrinkage-resistant grouts.

C. Maintain erection tolerances of structural steel within AISC 303, "Code of Standard Practice for Steel Buildings and Bridges."

D. Align and adjust various members that form part of complete frame or structure before permanently fastening. Before assembly, clean bearing surfaces and other surfaces that are in permanent contact with members. Perform necessary adjustments to compensate for discrepancies in elevations and alignment.

1. Level and plumb individual members of structure.
2. Make allowances for difference between temperature at time of erection and mean temperature when structure is completed and in service.

E. Splice members only where indicated.

F. Do not use thermal cutting during erection.

G. Do not enlarge unfair holes in members by burning or using drift pins. Ream holes that must be enlarged to admit bolts.

H. Shear Connectors: Prepare steel surfaces as recommended by manufacturer of shear connectors. Use automatic end welding of headed-stud shear connectors according to AWS D1.1/D1.1M and manufacturer's written instructions.

3.4 FIELD CONNECTIONS

A. High-Strength Bolts: Install high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts" for type of bolt and type of joint specified.

1. Joint Type: Slip Critical (SC).

B. Weld Connections: Comply with AWS D1.1/D1.1M and AWS D1.8/D1.8M for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.

1. Comply with AISC 303 and AISC 360 for bearing, alignment, adequacy of temporary connections, and removal of paint on surfaces adjacent to field welds.
2. Remove backing bars or runoff tabs where indicated, back gouge, and grind steel smooth.
3. Assemble and weld built-up sections by methods that maintain true alignment of axes without exceeding tolerances in AISC 303, "Code of Standard Practice for Steel Buildings and Bridges," for mill material.

3.5 FIELD QUALITY CONTROL

A. Special Inspections: Construction contractor shall engage a qualified special inspector to perform the following special inspections:

1. Reference Statement of Structural Tests and Special Inspections at the end of Section 01 40 00 – Quality Requirements.

B. Testing Agency: Contractor shall engage a qualified testing agency to perform tests and inspections.

C. Bolted Connections: Inspect and test bolted connections according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

D. Welded Connections: Visually inspect field welds according to AWS D1.1/D1.1M.

1. In addition to visual inspection, test and inspect field welds according to AWS D1.1/D1.1M and the following inspection procedures, at testing agency's option:

a. Liquid Penetrant Inspection: ASTM E 165.
b. Magnetic Particle Inspection: ASTM E 709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration are not accepted.
c. Ultrasonic Inspection: ASTM E 164. Use for all CJP welds.
d. Radiographic Inspection: ASTM E 94.

E. In addition to visual inspection, test and inspect field-welded shear connectors according to requirements in AWS D1.1/D1.1M for stud welding and as follows:

1. Perform bend tests if visual inspections reveal either a less-than-continuous 360-degree flash or welding repairs to any shear connector.
2. Conduct tests according to requirements in AWS D1.1/D1.1M on additional shear connectors if weld fracture occurs on shear connectors already tested.

3.6 REPAIRS AND PROTECTION

A. Touchup Painting: Immediately after erection, clean exposed areas where primer is damaged or missing and paint with the same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

1. Clean and prepare surfaces by SSPC-SP 2 hand-tool cleaning or SSPC-SP 3 power-tool cleaning.

B. Galvanized Surfaces: Clean areas where galvanizing is damaged or missing and repair galvanizing to comply with ASTM A 780/A 780M, A1 Heat Stick Method.

END OF SECTION 05 12 00

SECTION 05 31 00 – STEEL DECKING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Roof deck.
2. Supplementary framing for openings up to and including 18 inches (450 mm).
3. Bearing plates and angles.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 05 12 00 – Structural Steel Framing: Support framing for openings larger than 18 inches (450 mm).

1.2 REFERENCE STANDARDS

A. ASTM A653/A653M – Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2023.

B. AWS D1.3/D1.3M – Structural Welding Code – Sheet Steel; 2018, with Errata (2022).

C. SDI (DM) – Publication No.30, Design Manual for Composite Decks, Form Decks, and Roof Decks; 2007.

1.3 SUBMITTALS

A. Product Data:

1. Provide deck profile characteristics, dimensions, structural properties, and finishes.

B. Shop Drawings:

1. Indicate deck plan, support locations, projections, openings, reinforcement, pertinent details, and accessories.

C. Certificates:

1. Certify that products furnished meet or exceed specified requirements.

D. Welders' Qualification Statement:

1. Welders' certificates in accordance with AWS B2.1/B2.1M and dated no more than 12 months before start of scheduled welding work.

E. Fabricator's Qualification Statement:

1. Provide documentation showing steel fabricator is accredited under IAS AC172.

1.4 QUALITY ASSURANCE

A. Welder Qualifications:

1. Welding processes and welding operators qualified in accordance with AWS D1.1/D1.1M and AWS D1.3/D1.3M and dated no more than 12 months before start of scheduled welding work.

B. Installer Qualifications:

1. Company specializing in performing the work of this Section with minimum 5 years of experience.

PART 2 - PRODUCTS

2.1 STEEL DECK

A. Composite Metal Decking:

1. Galvanized Steel Sheet: ASTM A653/A653M, Structural Steel (SS) Grade 33/230, with G90/Z275 galvanized coating.
2. Primer: Shop coat of manufacturer's standard primer paint over cleaned and phosphatized substrate.
3. Structural Properties:

    a. Section Modulus: As indicated on Drawings.
    b. Span Design: Triple, as able.

4. Minimum Base Metal Thickness: 18 gauge, 0.0478 inch (1.214 mm).
5. Nominal Height: 3 inch (38 mm).
6. Profile: W-Deck.
7. Formed Sheet Width: 36 inch (900 mm).
8. Side Joints: Lapped, mechanically fastened.
9. End Joints: Lapped, mechanically fastened.

2.2 ACCESSORY MATERIALS

A. Bearing Plates and Angles:

1. ASTM A36/A36M steel, galvanized per ASTM A123/A123M.

B. Welding Materials:

1. AWS D1.1/D1.1M.

C. Fasteners:

1. Galvanized hardened steel, self-tapping.

D. Powder Actuated Mechanical Fasteners:

1. Steel; with knurled shank and forged ballistic point. Comply with applicable requirements of ICC-ES AC70.

E. Mechanical Fasteners:

1. Steel; hex washer head, self-drilling, self-tapping.

F. Weld Washers:

1. Mild steel, uncoated, 3/4 inch (19 mm) outside diameter, 1/8 inch (3 mm) thick.

G. Shop and Touch-Up Primer:

1. SSPC-Paint 15, complying with VOC limitations of authorities having jurisdiction.

H. Touch-Up Primer for Galvanized Surfaces:

1. SSPC-Paint 20, complying with VOC limitations of authorities having jurisdiction.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify existing conditions prior to beginning work.

3.2 INSTALLATION

A. Erect metal deck in accordance with SDI Design Manual and manufacturer's instructions. Align and level.

B. On steel supports provide minimum 2 inch (38 mm) bearing.

C. Fasten deck to steel support members at ends and intermediate supports at 12 inches (300 mm) on center maximum, parallel with the deck flute and at each transverse flute using methods specified.

1. Welding: Use fusion welds through weld washers.

D. Clinch lock seam side laps.

E. At mechanically fastened male/female side laps fasten at 12 inches (600 mm) on center maximum.

F. Drive mechanical sidelap connectors completely through adjacent lapped sheets; positively engage adjacent sheets with minimum three-thread penetration.

G. Weld deck in accordance with AWS D1.3/D1.3M.

H. At deck openings from 6 inches (150 mm) to 18 inches (450 mm) in size, provide 2 by 2 by 1/4 inch (50 by 50 by 6 mm) steel angle reinforcement. Place angles perpendicular to flutes; extend minimum two flutes beyond each side of opening and fusion weld to deck at each flute.

I. Immediately after welding deck and other metal components in position, coat welds, burned areas, and damaged surface coating, with touch-up primer.

END OF SECTION 05 31 00

SECTION 05 50 00 – METAL FABRICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Miscellaneous framing and supports.
2. Shelf angles.
3. Pipe handrails at basement ramp
4. Aluminum handrail added to existing aluminum guardrail.
5. Aluminum guardrail to match existing in design and layout (Base scope and Option D)

B. Products furnished, but not installed, under this Section include the following:

1. Loose steel lintels.
2. Steel weld plates and angles for casting into concrete for applications where they are not specified in other Sections.
3. Anchor bolts, steel pipe sleeves, slotted-channel inserts, and wedge-type inserts indicated to be cast into concrete or built into unit masonry.

C. Related Requirements:

1. Section 03 30 00 “Cast in Place Concrete” for balcony floor structure, top.
2. Section 04 22 00 “Concrete Unit Masonry” for installing loose lintels, anchor bolts, and other items built into unit masonry.
3. Section 04 26 13 “Masonry Veneer” for anchors and other items built into unit masonry.
4. Section 05 31 00 “Steel Decking” for balcony floor structure, bottom.
5. Section 05 12 00 “Structural Steel Framing” for steel beams and columns, hollow steel shapes, pipes, and other steel components of the building structure.
6. Section 06 10 00 “Rough Carpentry” for associated blocking and wood structure.
7. Section 09 91 13 “Exterior Painting” for finishing exterior metal fabrications where exposed to view.
8. Section 09 91 23 “Interior Painting” for finishing interior metal fabrications where exposed to view.

1.2 COORDINATION

A. Coordinate selection of shop primers with topcoats to be applied over them. Comply with paint and coating manufacturers' written instructions to ensure that shop primers and topcoats are compatible with one another.

B. Coordinate installation of metal fabrications that are anchored to or that receive other work. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

1.3 ACTION SUBMITTALS

A. Product Data:

1. Fasteners.
2. Shop primers.
3. Shrinkage-resisting grout.

B. Shop Drawings: Show fabrication and installation details. Include plans, elevations, sections, and details of metal fabrications and their connections. Show anchorage and accessory items. Provide Shop Drawings for the following:

1. Miscellaneous framing and supports for applications where framing and supports are not specified in other Sections.
2. Shelf angles.
3. Loose steel lintels.
4. Guardrails to match existing in design and layout.

C. Delegated Design Submittals: For guardrails, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

D. Delegated design engineer qualifications.

1.4 INFORMATIONAL SUBMITTALS

A. Welding certificates.

B. Paint Compatibility Certificates: From manufacturers of topcoats applied over shop primers, certifying that shop primers are compatible with topcoats.

1.5 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel in accordance with the following welding codes:

1. AWS D1.1/D1.1M, “Structural Welding Code – Steel.”
2. AWS D1.2/D1.2M, “Structural Welding Code – Aluminum.”

1.6 FIELD CONDITIONS

A. Field Measurements: Verify actual locations of walls, floor slabs, decks, and other construction contiguous with metal fabrications by field measurements before fabrication.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 “Quality Requirements,” to provide signed shop drawings of guardrails that comply with code requirements.

B. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.2 METALS

A. Metal Surfaces, General: Provide materials with smooth, flat surfaces unless otherwise indicated. For metal fabrications exposed to view in the completed Work, provide materials without seam marks, roller marks, rolled trade names, or blemishes.

B. Steel Plates, Shapes, and Bars: ASTM A 36/A 36M.

C. Steel Tubing: ASTM A 500/A 500M, cold-formed steel tubing.

D. Steel Pipe: ASTM A 53/A 53M, Standard Weight (Schedule 40) unless otherwise indicated.

E. Stainless-Steel Sheet, Strip, and Plate: ASTM A 240/A 240M or ASTM A 666, Type 304.

F. Stainless-Steel Bars and Shapes: ASTM A 276, Type 304.

G. Aluminum Plate and Sheet: ASTM B 209 (ASTM B 209M), Alloy 6061-T6.

H. Aluminum Extrusions: ASTM B 221 (ASTM B 221M), Alloy 6063-T6.

I. Aluminum-Alloy Rolled Tread Plate: ASTM B 632/B 632M, Alloy 6061-T6.

2.3 FASTENERS

A. General: Unless otherwise indicated, provide Type 304 stainless steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B633 or ASTM F1941/F1941M, Class Fe/Zn 5, at exterior walls. Select fasteners for type, grade, and class required.

1. Provide stainless-steel fasteners for fastening stainless steel and aluminum.

B. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A307, Grade A; with hex nuts, ASTM A563; and, where indicated, flat washers.

C. Stainless-Steel Bolts and Nuts: Regular hexagon-head annealed stainless-steel bolts, ASTM F 593; with hex nuts, ASTM F 594; and flat washers; Alloy Group 1.

D. High-Strength Bolts, Nuts, and Washers: ASTM F3125/F3125M, Grade A325, Type 3, heavy-hex steel structural bolts; ASTM A563, Grade DH3, heavy-hex carbon-steel nuts; and where indicated, flat washers.

E. Anchor Bolts: ASTM F1554, Grade 36, of dimensions indicated; with nuts, ASTM A563; and, where indicated, flat washers.

1. Hot-dip galvanize or provide mechanically deposited, zinc coating where item being fastened is indicated to be galvanized.

F. Anchors, General: Capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing in accordance with ASTM E488/E488M, conducted by a qualified independent testing agency.

G. Cast-in-Place Anchors in Concrete: Either threaded or wedge type unless otherwise indicated; galvanized ferrous castings, either ASTM A47/A47M malleable iron or ASTM A27/A27M cast steel. Provide bolts, washers, and shims as needed, all hot-dip galvanized per ASTM F2329/F2329M.

H. Post-Installed Anchors: Torque-controlled expansion anchors or chemical anchors.

1. Material for Interior Locations: Carbon-steel components zinc plated to comply with ASTM B633 or ASTM F1941/F1941M, Class Fe/Zn 5, unless otherwise indicated.
2. Material for Exterior Locations and Where Stainless Steel Is Indicated: Alloy Group 1 stainless steel bolts, ASTM F593, and nuts, ASTM F594.

I. Slotted-Channel Inserts: Cold-formed, hot-dip galvanized-steel box channels (struts) complying with MFMA-4, 1-5/8 by 7/8 inches by length indicated with anchor straps or studs not less than 3 inches long at not more than 8 inches o.c. Provide with temporary filler and tee-head bolts, complete with washers and nuts, all zinc-plated to comply with ASTM B633, Class Fe/Zn 5, as needed for fastening to inserts.

2.4 MISCELLANEOUS MATERIALS

A. Universal Shop Primer: Fast-curing, lead- and chromate-free, universal modified-alkyd primer complying with MPI#79 and compatible with topcoat.

1. Use primer that contains pigments that make it easily distinguishable from zinc-rich primer.

B. Water-Based Primer: Emulsion type, anticorrosive primer for mildly corrosive environments that is resistant to flash rusting when applied to cleaned steel, complying with MPI#107 and compatible with topcoat.

C. Epoxy Zinc-Rich Primer: Complying with MPI#20 and compatible with topcoat.

D. Shrinkage-Resistant Grout: Factory-packaged, nonmetallic, nonstaining, noncorrosive, nongaseous grout complying with ASTM C1107/C1107M. Provide grout specifically recommended by manufacturer for interior and exterior applications.

2.5 FABRICATION, GENERAL

A. Shop Assembly: Preassemble items in the shop to greatest extent possible. Disassemble units only as necessary for shipping and handling limitations. Use connections that maintain structural value of joined pieces. Clearly mark units for reassembly and coordinated installation.

B. Cut, drill, and punch metals cleanly and accurately. Remove burrs and ease edges to a radius of approximately 1/32 inch unless otherwise indicated. Remove sharp or rough areas on exposed surfaces.

C. Form bent-metal corners to smallest radius possible without causing grain separation or otherwise impairing work.

D. Form exposed work with accurate angles and surfaces and straight edges.

E. Weld corners and seams continuously to comply with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.

F. Form exposed connections with hairline joints, flush and smooth, using concealed fasteners or welds where possible. Where exposed fasteners are required, use Phillips flat-head (countersunk) fasteners unless otherwise indicated. Locate joints where least conspicuous.

G. Fabricate seams and other connections that are exposed to weather in a manner to exclude water. Provide weep holes where water may accumulate.

H. Cut, reinforce, drill, and tap metal fabrications as indicated to receive finish hardware, screws, and similar items.

I. Provide for anchorage of type indicated; coordinate with supporting structure. Space anchoring devices to secure metal fabrications rigidly in place and to support indicated loads.

J. Where units are indicated to be cast into concrete or built into masonry, equip with integrally welded steel strap anchors, 1/8 by 1-1/2 inches, with a minimum 6-inch embedment and 2-inch hook, not less than 8 inches from ends and corners of units and 24 inches o.c., unless otherwise indicated.

2.6 MISCELLANEOUS FRAMING AND SUPPORTS

A. General: Provide steel framing and supports not specified in other Sections as needed to complete the Work.

B. Fabricate units from steel shapes, plates, and bars of welded construction unless otherwise indicated. Fabricate to sizes, shapes, and profiles indicated and as necessary to receive adjacent construction.

1. Fabricate units from slotted channel framing where indicated.
2. Furnish inserts for units installed after concrete is placed.

C. Fabricate supports for operable partitions from continuous steel beams of sizes indicated with attached bearing plates, anchors, and braces as recommended by partition manufacturer. Drill or punch bottom flanges of beams to receive partition track hanger rods; locate holes where indicated on operable partition Shop Drawings.

D. Fabricate steel girders for wood frame construction from continuous steel shapes of sizes indicated.

1. Provide bearing plates welded to beams where indicated.
2. Drill or punch girders and plates for field-bolted connections where indicated.
3. Where wood nailers are attached to girders with bolts or lag screws, drill or punch holes at 24 inches o.c.

E. Fabricate steel pipe columns for supporting wood frame construction from steel pipe with steel baseplates and top plates as indicated. Drill or punch baseplates and top plates for anchor and connection bolts and weld to pipe with fillet welds all around. Make welds the same size as pipe wall thickness unless otherwise indicated.

1. Unless otherwise indicated, fabricate from Schedule 40 steel pipe.
2. Unless otherwise indicated, provide 1/2-inch baseplates with four 5/8-inch anchor bolts and 1/4-inch top plates.

F. Galvanize miscellaneous framing and supports where indicated.

G. Prime miscellaneous framing and supports with zinc-rich primer where indicated.

2.7 GENERAL RAILINGS

A. Design, fabricate, and test railing assemblies in accordance with the most stringent requirements of applicable local code.

B. Comply with ASTM E985.

C. Allow for expansion and contraction of members and building movement without damage to connections or members.

D. Dimensions: To match existing, and reference contract drawings.

E. Anchors: To match existing, and reference contract drawings.

F. Provide shop welded fittings to join lengths, seal open ends, and conceal exposed mounting bolts and nuts, including but not limited to elbows, T-shapes, splice connectors, flanges, escutcheons, and wall brackets.

G. Welding Fittings: Factory- or shop-welded from matching pipe, tube, or solid steel shape; seams continuously welded; joints and seams ground smooth.

H. Finishes: Reference paint specification.

I. Fabrication:

1. Accurately form components to suit specific project conditions and for proper connection to building structure.
2. Fit and shop assemble components in largest practical sizes for delivery to site.
3. Fabricate components with joints tightly fitted and secured.

2.8 EXTERIOR RAILINGS

A. Handrail: 1.5 in diameter prefinished aluminum.

B. Handrail Bracket: Prefinished extruded aluminum handrail bracket with anchor plate sized to match post width and provide required support per railing design.

2.9 SHELF ANGLES

A. Fabricate shelf angles from steel angles of sizes indicated and for attachment to concrete framing. Provide horizontally slotted holes to receive 3/4-inch bolts, spaced not more than 6 inches from ends and 24 inches o.c., unless otherwise indicated.

1. Provide mitered and welded units at corners.
2. Provide open joints in shelf angles at expansion and control joints. Make open joint approximately 2 inches larger than expansion or control joint.

B. For cavity walls, provide vertical channel brackets to support angles from backup masonry and concrete.

C. Galvanize and prime shelf angles located in exterior walls.

D. Furnish wedge-type concrete inserts, complete with fasteners, to attach shelf angles to cast-in-place concrete.

2.10 MISCELLANEOUS STEEL TRIM

A. Unless otherwise indicated, fabricate units from steel shapes, plates, and bars of profiles shown with continuously welded joints and smooth exposed edges. Miter corners and use concealed field splices where possible.

B. Provide cutouts, fittings, and anchorages as needed to coordinate assembly and installation with other work.

1. Provide with integrally welded steel strap anchors for embedding in concrete or masonry construction.

C. Galvanize and prime exterior miscellaneous steel trim.

D. Prime exterior miscellaneous steel trim with zinc-rich primer.

2.11 LOOSE STEEL LINTELS

A. Fabricate loose steel lintels from steel angles and shapes of size indicated for openings and recesses in masonry walls and partitions at locations indicated. Fabricate in single lengths for each opening unless otherwise indicated. Weld adjoining members together to form a single unit where indicated.

B. Size loose lintels to provide bearing length at each side of openings equal to one-twelfth of clear span, but not less than 8 inches unless otherwise indicated.

C. Galvanize loose steel lintels located in exterior walls.

2.12 STEEL WELD PLATES AND ANGLES

A. Provide steel weld plates and angles not specified in other Sections, for items supported from concrete construction as needed to complete the Work. Provide each unit with no fewer than two integrally welded steel strap anchors for embedding in concrete.

2.13 GENERAL FINISH REQUIREMENTS

A. Finish metal fabrications after assembly.

B. Finish exposed surfaces to remove tool and die marks and stretch lines, and to blend into surrounding surface.

2.14 STEEL AND IRON FINISHES

A. Galvanizing: Hot-dip galvanize items as indicated to comply with ASTM A153/A153M for steel and iron hardware and with ASTM A123/A123M for other steel and iron products.

1. Do not quench or apply post galvanizing treatments that might interfere with paint adhesion.

B. Preparation for Shop Priming Galvanized Items: After galvanizing, thoroughly clean galvanized surfaces of grease, dirt, oil, flux, and other foreign matter, and treat with metallic phosphate process.

C. Shop prime iron and steel items [not indicated to be galvanized] unless they are to be embedded in concrete, sprayed-on fireproofing, or masonry, or unless otherwise indicated.

1. Shop prime with universal shop primer.

D. Preparation for Shop Priming: Prepare surfaces to comply with requirements indicated below:

1. Exterior Items: SSPC-SP 6/NACE No. 3, “Commercial Blast Cleaning.”
2. Items Indicated to Receive Zinc-Rich Primer: SSPC-SP 6/NACE No. 3, “Commercial Blast Cleaning.”
3. Zinc-Rich Primer: SSPC-SP 6/NACE No. 3, “Commercial Blast Cleaning.”
4. High-Performance Coatings” SSPC-SP 6/NACE No. 3, “Commercial Blast Cleaning.”
5. Powder Coating: SSPC-SP 16, “Brush-off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals.”
6. Other Steel Items: SSPC-SP 3, “Power Tool Cleaning.”
7. Galvanized-Steel Items: SSPC-SP 16, “Brush-off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals.”

E. Shop Priming: Apply shop primer to comply with SSPC-PA 1, “Paint Application Specification No. 1: Shop, Field, and Maintenance Painting of Steel,” for shop painting.

1. Stripe paint corners, crevices, bolts, welds, and sharp edges.

2.15 ALUMINUM FINISHES

A. As-Fabricated Finish: AA-M12.

B. Painted to match existing railing. Cleaning and painting are specified in Section 09 91 13 “Exterior Painting.”

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Cutting, Fitting, and Placement: Perform cutting, drilling, and fitting required for installing metal fabrications. Set metal fabrications accurately in location, alignment, and elevation; with edges and surfaces level, plumb, true, and free of rack; and measured from established lines and levels.

B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections.

C. Field Welding: Comply with the following requirements:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. At exposed connections, finish exposed welds and surfaces smooth and blended so no roughness shows after finishing and contour of welded surface matches that of adjacent surface.

D. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag screws, wood screws, and other connectors.

E. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.

F. Corrosion Protection: Coat concealed surfaces of aluminum that come into contact with grout, concrete, masonry, wood, or dissimilar metals with the following:

1. Cast Aluminum: Heavy coat of bituminous paint.
2. Extruded Aluminum: Two coats of clear lacquer.

3.2 INSTALLATION OF MISCELLANEOUS FRAMING AND SUPPORTS

A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings.

B. Anchor supports for ceiling-braced toilet partitions and operable partitions securely to, and rigidly brace from, building structure.

C. Anchor shelf angles securely to existing construction with expansion anchors, anchor bolts, through bolts, as indicated in the drawings.

D. Support steel girders on solid grouted masonry, concrete, or steel pipe columns. Secure girders with anchor bolts embedded in grouted masonry or concrete or with bolts through top plates of pipe columns.

1. Where grout space under bearing plates is indicated for girders supported on concrete or masonry, install as specified in “Installing Bearing and Leveling Plates” Article.

E. Install pipe columns on concrete footings with grouted baseplates. Position and grout column baseplates as specified in “Installation of Bearing and Leveling Plates” Article.

1. Grout baseplates of columns supporting steel girders after girders are installed and leveled.

3.3 INSTALLATION OF SHELF ANGLES

A. Install shelf angles as required to keep masonry level, at correct elevation, and flush with vertical plane.

3.4 INSTALLATION OF MISCELLANEOUS STEEL TRIM

A. Anchor to concrete construction to comply with manufacturer's written instructions.

3.5 INSTALLATION OF LOOSE BEARING AND LEVELING PLATES

A. Clean concrete and masonry bearing surfaces of bond-reducing materials, and roughen to improve bond to surfaces. Clean bottom surface of plates.

B. Set bearing and leveling plates on wedges, shims, or leveling nuts. After bearing members have been positioned and plumbed, tighten anchor bolts. Do not remove wedges or shims but, if protruding, cut off flush with edge of bearing plate before packing with shrinkage-resistant grout. Pack grout solidly between bearing surfaces and plates to ensure that no voids remain.

3.6 REPAIRS

A. Touchup Painting:

1. Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with same material as used for shop painting to comply with SSPC-PA 1 for touching up shop-painted surfaces.

    a. Apply by brush or spray to provide a minimum 2.0-mil dry film thickness.

2. Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint are specified in Section 09 91 13 “Exterior Painting.” And Section 099123 “Interior Painting.”

B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A780/A780M.

END OF SECTION 05 50 00

SECTION 06 10 00 – ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Framing with dimension lumber.
2. Framing with timber.
3. Framing with engineered wood products.
4. Rooftop equipment bases and support curbs.
5. Wood blocking and nailers.
6. Wood furring.
7. Plywood backing panels.
8. Pressure treating.
9. Framing gaskets.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 05 12 00 “Structural Steel Framing” for coordination with steel framing.
2. Section 06 16 00 “Sheathing”
3. Section 06 18 00 “Glued-Laminated Construction”

1.2 SUBMITTALS

A. Product Data: For each type of process and factory-fabricated product. Indicate component materials and dimensions and include construction and application details.

1. Include data for wood-preservative treatment. Indicate type of preservative used and net amount of preservative retained.
2. For products receiving a waterborne treatment, include statement that moisture content of treated materials was reduced to levels specified before shipment to Project site.
3. Fasteners
4. Framing Anchors and Connectors
5. Metal type and coating recommendations by framing anchor manufacturer at wood preservative treated and fire treated wood.
6. Hold Downs/Ties
7. Sill Sealer Gasket (for sills)
8. Sealer Gasket (for top plates)
9. Water Repellent Preservative

1.3 QUALITY ASSURANCE

A. Testing Agency Qualifications: For testing agency providing classification marking for fire-retardant treated material, an inspection agency acceptable to authorities having jurisdiction that periodically performs inspections to verify that the material bearing the classification marking is representative of the material tested.

B. Source Limitations: Obtain each type of engineered wood product from single source from a single manufacturer.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Stack wood products flat with spacers beneath and between each bundle to provide air circulation. Protect wood products from weather by covering with waterproof sheeting, securely anchored. Provide for air circulation around stacks and under coverings.

PART 2 - PRODUCTS

2.1 LUMBER

A. Lumber: DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Grade lumber by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.

B. Factory mark each piece of lumber with grade stamp of grading agency.

C. For exposed lumber indicated to receive a stained or natural finish, mark grade stamp on end or back of each piece or omit grade stamp and provide certificates of grade compliance issued by grading agency.

D. Dress lumber, S4S, unless otherwise indicated.

E. Maximum Moisture Content of Lumber:

1. At time of fabrication, 19 percent (kiln dried) and dried to a maximum of 15 percent before gypsum wall board installation.
2. Moisture content to be verified by special inspection.

2.2 DIMENSION LUMBER FRAMING

A. Materials:

1. Species and Grade: As indicated on Drawings.

2.3 TONGUE AND GROOVE DECKING

A. Materials:

1. Species and Grade: As indicated on Drawings.

2.4 WOOD-PRESERVATIVE-TREATED LUMBER

A. For lumber, Preservative Treatment by Pressure Process, AWPA U1:

1. Use Category UC2, for interior construction not in contact with ground.
2. Use Category UC3B for exterior construction not in contact with ground.
3. Use Category UC4A for items in contact with ground.

B. For lumber, Preservative Chemicals:

1. Acceptable to jurisdiction and containing no arsenic or chromium.
2. For exposed items indicated to receive a stained or natural finish, chemical formulations shall not require incising, contain colorants, bleed through, or otherwise adversely affect finishes.

C. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Do not use material that is warped or that does not comply with requirements for untreated material.

D. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.

E. For exposed lumber indicated to receive a stained or natural finish, mark end or back of each piece or omit marking and provide certificates of treatment compliance issued by inspection agency.

F. Application: Treat items indicated on Drawings, and the following:

1. All exposed lumber beams, joists, and blocking.
2. Wood sills, sleepers, blocking, furring, stripping, and similar concealed members in contact with masonry or concrete.
3. Wood framing and furring attached directly to the interior of below-grade exterior masonry or concrete walls.
4. Wood framing members that are less than 18 inches above the ground in crawlspaces or unexcavated areas.
5. Wood floor plates that are installed over concrete slabs-on-grade.

2.5 ENGINEERED WOOD PRODUCTS

A. Manufacturers:

1. Coordinate manufacturer’s with proprietary assembly reference requirements.

B. Engineered Wood Products:

1. Products shall contain no urea formaldehyde.
2. Acceptable to authorities having jurisdiction and for which current model code research or evaluation reports exist that show compliance with building code in effect for Project.
3. Allowable design stresses, as published by manufacturer, shall meet or exceed those indicated. Manufacturer's published values shall be determined from empirical data or by rational engineering analysis and demonstrated by comprehensive testing performed by a qualified independent testing agency.

C. Laminated-Veneer Lumber:

1. Structural composite lumber made from wood veneers with grain primarily parallel to member lengths, evaluated and monitored according to ASTM D 5456 and manufactured with an exterior-type adhesive complying with ASTM D 2559.

D. Parallel-Strand Lumber:

1. Structural composite lumber made from wood strand elements with grain primarily parallel to member lengths, evaluated and monitored according to ASTM D 5456 and manufactured with an exterior-type adhesive complying with ASTM D 2559.

E. Rim Boards:

1. Product designed to be used as a load-bearing member and to brace wood I-joists at bearing ends, complying with research or evaluation report for I-joists.
2. Manufacturer: Provide products by same manufacturer as I-joists.
3. Material: All-veneer product.
4. Thickness: 1-3/4 inches.
5. Comply with APA PRR-401, grade. Factory mark rim boards with APA-EWS trademark indicating thickness, grade, and compliance with APA-EWS standard.

2.6 MISCELLANEOUS LUMBER

A. General: Provide miscellaneous lumber indicated and lumber for support or attachment of other construction, including the following:

1. Blocking.
2. Nailers.
3. Rooftop equipment bases and support curbs.
4. Furring.

B. Utility, Stud, or No. 3 grade lumber of any species may be used provided that it is cut and selected to eliminate defects that will interfere with its attachment and purpose.

C. For blocking and nailers used for attachment of other construction, select and cut lumber to eliminate knots and other defects that will interfere with attachment of other work.

D. For furring strips for installing plywood or hardboard paneling, select boards with no knots capable of producing bent-over nails and damage to paneling.

2.7 PLYWOOD BACKING PANELS

A. Equipment Backing Panels: Plywood, DOC PS 1, Exterior, A-C, in thickness indicated or, if not indicated, not less than 3/4-inch nominal thickness.

1. Plywood shall contain no urea formaldehyde.

2.8 FASTENERS

A. General: Fasteners shall be of size and type indicated and shall comply with requirements specified in this article for material and manufacture.

1. All framing nails shall be common nails with full heads, no box nails or clipped heads allowed.
2. Where rough carpentry is exposed to weather, in ground contact, pressure-preservative treated, or in area of high relative humidity, provide fasteners with hot-dip zinc coating complying with ASTM A 153/A 153M and have a minimum coating weight of 2.0 ounces per square foot. or Type 304 stainless steel.

B. Nails: ASTM F 1667.

C. Power-Driven Fasteners: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.

D. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 ICC-ES AC58 ICC-ES AC193 or ICC-ES AC308 as appropriate for the substrate.

1. Material: Carbon-steel components, zinc plated to comply with ASTM B 633, Class Fe/Zn 5.
2. Material: Stainless steel with bolts and nuts complying with ASTM F 593 and ASTM F 594, Alloy Group 1 or 2.
3. GPR (or X-Ray) required for all post-installed concrete anchors. Do not install within 3-inches of any scanned interference.

2.9 METAL FRAMING ANCHORS

A. Manufacturers:

1. Basis of Design: As indicated on Drawings.
2. Approved equal, as indicated on Drawing sheet notes.

B. Product Data:

1. Galvanized-Steel Sheet: Hot-dip, zinc-coated steel sheet complying with ASTM A 653/A 653M, G60 coating designation.

a. Use for interior locations unless otherwise indicated.

2. Hot-Dip or Heavy-Galvanized Steel Sheet: ASTM A 653/A 653M; structural steel (SS), high-strength low-alloy steel Type A (HSLAS Type A), or high-strength low-alloy steel Type B (HSLAS Type B); G185 coating designation; and not less than 0.036 inches thick and 1.85 ounces of zinc per square foot.

    a. Use for wood-preservative-treated lumber, exterior locations and where indicated.

3. Stainless-Steel Sheet: ASTM A 666, Type 304.

    a. Use for wood-preservative-treated lumber when recommended by metal framing anchor manufacturer for treatment type and retention amount and where indicated.

2.10 SILL PLATE GASKET

A. Closed-cell foam, 1/4 inch thick selected from manufacturer's standard widths to suit width of sill members indicated, for air-sealing and capillary break below/under sill plates.

2.11 TOP PLATE GASKET

A. Closed-cell foam, 1/2 inch thick selected from manufacturer's standard widths to suit width of members indicated, for air-sealing both sides of top plates to gypsum board.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Framing Standards:

1. Comply with AF&PA's WCD 1, "Details for Conventional Wood Frame Construction," unless otherwise indicated.

B. Framing with Engineered Wood Products:

1. Install engineered wood products to comply with manufacturer's written instructions.

C. Set rough carpentry to required levels and lines, with members plumb, true to line, cut, and fitted. Fit rough carpentry accurately to other construction. Locate furring, nailers, blocking, and similar supports to comply with requirements for attaching other construction.

D. Install plywood backing panels by fastening to studs; coordinate locations with utilities requiring backing panels. When used install fire-retardant-treated plywood backing panels with classification marking of testing agency exposed to view.

E. Install metal framing anchors to comply with manufacturer's written instructions. Install fasteners through each fastener hole.

F. Do not splice structural members between supports unless otherwise indicated.

G. Provide blocking and framing as indicated and as required to support facing materials, fixtures, specialty items, and trim.

1. Provide metal clips for fastening gypsum board or lath at corners and intersections where framing or blocking does not provide a surface for fastening edges of panels. Space clips not more than 16 inches o.c.

H. Provide fire blocking in furred spaces, stud spaces, and other concealed cavities as indicated and as follows:

1. Fire block furred spaces of walls, at each floor level, at ceiling, and at not more than 96 inches o.c. with solid wood blocking or noncombustible materials accurately fitted to close furred spaces.
2. Fire block concealed spaces of wood-framed walls and partitions at each floor level, at ceiling line of top story, and at not more than 96 inches o.c. Where fire blocking is not inherent in framing system used, provide closely fitted solid wood blocks of same width as framing members and 2-inch nominal thickness.
3. Fire block concealed spaces between floor sleepers with same material as sleepers to limit concealed spaces to not more than 100 sq. ft. and to solidly fill space below partitions.
4. Fire block concealed spaces behind combustible cornices and exterior trim at not more than 20 feet o.c.

I. Sort and select lumber so that natural characteristics do not interfere with installation or with fastening other materials to lumber. Do not use materials with defects that interfere with function of member or pieces that are too small to use with minimum number of joints or optimum joint arrangement.

J. Comply with AWPA M4 for applying field treatment to cut surfaces of preservative-treated lumber.

1. Follow preservative manufacturer's instructions.
2. If allowed, use inorganic boron for items that are continuously protected from liquid water.
3. If allowed, use copper naphthenate for items not continuously protected from liquid water.

K. Securely attach rough carpentry work to substrate by anchoring and fastening as indicated, complying with the following:

1. Table 2304.9.1, "Fastening Schedule," in ICC's International Building Code (IBC).
2. ICC-ES evaluation report for fastener.

L. Use steel common nails unless otherwise indicated. Select fasteners of size that will not fully penetrate members where opposite side will be exposed to view or will receive finish materials. Make tight connections between members. Install fasteners without splitting wood. Drive nails snug but do not countersink nail heads unless otherwise indicated.

M. For exposed work, arrange fasteners in straight rows parallel with edges of members, with fasteners evenly spaced, and with adjacent rows staggered.

3.2 FRAMING GASKETS

A. Sill Plate Gasket:

1. Install sill plate gaskets per manufacturer's instructions.
2. Install at all exterior and interior walls that rest on concrete walls or slabs
3. Install sill plate gaskets at all floor levels at residential corridor walls, at all residential unit separation party walls, and at all residential unit exterior walls. (To create a continuous sill seal at the perimeter of each unit.)

B. Top Plate Gasket:

1. Install at all interior and exterior walls directly below roof structure at top floor. Use full width of gasket, do not cut in half.

3.3 WOOD BLOCKING, AND NAILER INSTALLATION

A. Install where indicated and where required for attaching other work. Form to shapes indicated and cut as required for true line and level of attached work. Coordinate locations with other work involved.

B. Attach items to substrates to support applied loading. Recess bolts and nuts flush with surfaces unless otherwise indicated.

3.4 WOOD FURRING INSTALLATION

A. Install level and plumb with closure strips at edges and openings. Shim with wood as required for tolerance of finish work.

B. Furring to Receive Plywood, Gypsum Board, other panels: Install 1-by-3-inch nominal-size furring vertically at stud spacing.

C. Furring to Align: Shim or trim to establish plumb plane with adjacent finish.

3.5 WALL AND PARTITION FRAMING INSTALLATION

A. General: Provide single bottom plate and double top plates using members of 2-inch nominal thickness whose widths equal that of studs. Fasten plates to supporting construction unless otherwise indicated.

1. Provide continuous horizontal blocking at mid-height of partitions more than 96 inches high, using members of 2-inch nominal thickness and of same width as wall or partitions.

B. Corners and intersections:

1. At fire rated walls, provide studs and blocking as required to maintain fire rating of walls at intersection.

2. Provide number of studs required for structural requirements.
3. Provide sufficient blocking for finish materials if not using metal clips to support finish edges.

C. Frame openings as shown on the Structural Drawings.

3.6 ROOF JOIST FRAMING INSTALLATION

A. General: Install roof joists with crown edge up and support ends of each member with not less than 1-1/2 inches of bearing on wood or metal, or 3 inches on masonry. Attach floor joists as follows:

1. Where supported on wood members, by using metal framing anchors.
2. Where framed into wood supporting members by using metal joist hangers.

B. Frame openings with headers and trimmers supported by metal joist hangers; double headers and trimmers where span of header exceeds 48 inches.

C. Do not notch in middle third of joists; limit notches to one-sixth depth of joist, one-third at ends. Do not bore holes larger than one-third depth of joist; do not locate closer than 2 inches from top or bottom.

D. Provide solid blocking of 2-inch nominal thickness by depth of joist at ends of joists unless nailed to header or band.

E. Lap members framing from opposite sides of beams, girders, or partitions not less than 4 inches or securely tie opposing members together. Provide solid blocking of 2-inch nominal thickness by depth of joist over supports.

3.7 PROTECTION

A. Protect wood that has been treated with inorganic boron (SBX) from weather. If, despite protection, inorganic boron-treated wood becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

END OF SECTION 06 10 00

SECTION 06 16 00 – SHEATHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wall sheathing.
2. Roof sheathing.
3. Underlayment.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 06 10 00 “Rough Carpentry”
2. Section 06 18 00 “Glued-Laminated Construction”

1.2 SUBMITTALS

A. Product Data: For each type of process and factory-fabricated product. Indicate component materials and dimensions and include construction and application details.

1. Sheathing – Wall and Roof: Grade and Type

B. Evaluation Reports: For the following, from ICC-ES:

1. Fire-retardant-treated plywood.

1.3 DELIVERY, STORAGE, AND HANDLING

A. Stack panels flat with spacers beneath and between each bundle to provide air circulation. Protect sheathing from weather by covering with waterproof sheeting, securely anchored. Provide for air circulation around stacks and under coverings.

1.4 PERFORMANCE REQUIREMENTS

A. Fire-Resistance Ratings:

1. As tested according to ASTM E 119, testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
2. Fire-Resistance Ratings: Indicated by design designations from UL’s “Fire Resistance Directory” or from the listings of another qualified testing agency.

PART 2 - PRODUCTS

2.1 WOOD SHEATHING PANELS

A. Exterior Wall, Plywood:

1. DOC PS 1, Exposure 1, sheathing.
2. Span Rating and Thickness: As indicated on Drawings.
3. Edges: Square

B. Interior Wall, Plywood or OSB:

1. DOC PS 2, Exposure 1, sheathing.
2. Span Rating and Thickness: As indicated on Drawings.
3. Edges: Square

C. Exterior Roof, Plywood:

1. DOC PS 1, Exterior, sheathing.
2. Span Rating and Thickness: As indicated on Drawings.
3. Edges: Square.

D. Fire Retardant Plywood: Locations indicated on Drawings.

1. DOC PS 1, Exterior, sheathing.
2. Span Rating and Thickness: As indicated on Drawings.
3. Edges: Square.

2.2 GYPSUM SHEATHING PANELS

A. Exterior Wall, Glass Mat Gypsum:

1. Glass-Mat Gypsum Sheathing, ASTM C 1177/1177M.
2. Type and Thickness: Type X, 5/8 inches thick.
3. Edges: Square.

2.3 ACCESSORIES

A. Fasteners:

1. All exterior sheathing fasteners to have hot-dip zinc coating complying with ASTM A 153/A 153M.
2. Nails, Brads, and Staples: ASTM F 1667.
3. Power-Driven Fasteners: With evaluation report based on ICC-ES AC70.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Do not use materials with defects that impair quality of sheathing or pieces that are too small to use with minimum number of joints or optimum joint arrangement. Arrange joints so that pieces do not span between fewer than three support members.

B. Cut panels at penetrations, edges, and other obstructions of work; fit tightly against abutting construction unless otherwise indicated.

C. Securely attach to substrate by fastening as indicated, complying with the following:

1. Retain one of first two subparagraphs below, as required to comply with requirements of Project and local codes.
2. Table 2304.9.1, “Fastening Schedule,” in the ICC’s International Building Code.
3. Table R602.3(1), “Fastener Schedule for Structural Members,” and Table R602.3(2), “Alternate Attachments,” in the ICC’s International Residential Code for One- and Two-Family Dwellings.
4. ICC-ES evaluation report for fastener.
5. As indicated on Drawings.

D. Use common wire nails unless otherwise indicated. Select fasteners of size that will not fully penetrate members where opposite side will be exposed to view or will receive finish materials. Make tight connections. Install fasteners without splitting wood.

E. Coordinate wall parapet and roof sheathing installation with flashing and joint-sealant installation so these materials are installed in sequence and manner that prevent exterior moisture from passing through completed assembly.

F. Do not bridge building expansion joints; cut and space edges of panels to match spacing of structural support elements.

G. Coordinate sheathing installation with installation of materials installed over sheathing so sheathing is not exposed to precipitation or left exposed at end of the workday when rain is forecast.

3.2 WOOD STRUCTURAL PANEL INSTALLATION

A. General:

1. Comply with applicable recommendations in APA Form No. E30, “Engineered Wood Construction Guide,” for types of structural-use panels and applications indicated and as noted below.

B. Wall Sheathing:

1. Nail to wood framing.
2. Space panels 1/8 inch apart at edges and ends.
3. Install panels with indicated gap between floors when noted on Structural Drawings.

C. Roof Sheathing:

1. Nail to wood framing.
2. Glue to framing if required by assembly.
3. Space panels 1/8 inch apart at edges and ends.

3.3 GYPSUM SHEATHING INSTALLATION

A. Comply with GA-253, with manufacturer's written instructions and as noted below.

1. Fasten gypsum sheathing to wood framing with nails or screws.
2. Install panels with a 1/4-inch gap where non-load-bearing construction abuts structural elements.
3. Install panels with a 1/4-inch gap where they abut masonry or similar materials that might retain moisture, to prevent wicking.
4. Install exterior panels with indicated gap between floors as noted on Drawings.
5. Install panels with face side out. Butt panels together for a light contact at edges and ends with not more than 1/16 inch of open space between panels, except at required gap between floors. Do not force into place.

B. Apply fasteners so heads bear tightly against face of sheathing, but do not cut into facing.

C. Vertical Installation: Install vertical edges centered over studs. Abut ends and edges with those of adjacent panels. Attach at perimeter and within field of panel to each stud.

1. Space fasteners as shown and per assembly references, whichever is more stringent.

END OF SECTION 06 16 00

SECTION 06 18 00 – GLUED-LAMINATED CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes but is not limited to:

1. Framing using structural glued-laminated timber.

B. The Contractor shall be responsible for all inspections and testing.

C. Related Requirements:

1. Section 06 10 00 “Rough Carpentry”
2. Section 06 16 00 “Sheathing”

1.2 SUBMITTALS

A. Product Data: For each type of product.

1. Include data on lumber, adhesives, fabrication, and protection.
2. For preservative-treated wood products. Include chemical treatment manufacturer’s written instructions for handling, storing, installing, and finishing treated material.
3. For connectors. Include installation instructions.
4. Field applied preservative treatment.

B. Shop Drawings:

1. Show layout of glued-laminated construction and dimensions of each member.
2. Indicate species and laminating combination.

C. Certificates of Conformance: Issued by a qualified testing and inspecting agency indicating that structural glued-laminated timber complies with requirements in AITC A190.1.

D. Material Certificates: For preservative-treated wood products, from manufacturer. Indicate type of preservative used and net amount of preservative retained.

1.3 QUALITY ASSURANCE

A. Manufacturer Qualifications: An AITC- or APA-EWS-licensed firm.

1. Factory mark each piece of structural glued laminated timber with AITC Quality Mark or APA-EWS trademark. Place mark on surfaces that will not be exposed in the completed Work.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Comply with provisions in AITC 111.

B. Individually wrap members using plastic-coated paper covering with water-resistant seams.

PART 2 - PRODUCTS

2.1 STRUCTURAL GLUED-LAMINATED TIMBER

A. General:

1. Provide structural glued-laminated timber that complies with AITC A190.1 and AITC 117 or research/evaluation reports acceptable to authorities having jurisdiction (ICC-ES Reports).
2. Factory mark each piece of glued-laminated timber with AITC Quality Mark.
3. Place mark on surfaces that are not exposed in the completed Work.
4. Within individual members use single species.
5. Do not use laminated veneer lumber for individual laminations.
6. Comply with AITC 110.

B. Adhesives:

1. Conform to AITC 405A. 100 percent waterproof.
2. Adhesive shall not contain urea-formaldehyde resins.
3. Use clear adhesive at exposed members.

2.2 APPEARANCE GRADES

A. Concealed Framing:

1. Framing or Industrial

B. Exposed Framing:

1. Architectural

2.3 STRUCTURAL GLUED LAMINATED TIMBER

A. Lumber Grade and Species:

1. Douglas Fir.
2. Comply with structural requirements as indicated on Drawings.
3. Basis of Design: Use Klear Gard 25 wood preservation system by Permapost, or approved equal, for all exposed glulam beams. Salient Characteristics:

a. Full permeation without incising.
b. Wood preservative to UC3B AWPA rating.

2.4 TIMBER CONNECTORS

A. Fabricated Connectors:

1. Basis of Design Products: As indicated on Drawings.
2. Hot-dip galvanize all fabricated connectors.

B. Bolts:

1. Complying with ASTM A 307, Grade A, size as indicated on Drawings.

C. Nuts:

1. Complying with ASTM A 563; with flat washers.

D. Interior Finishes:

1. Finish steel assemblies and fasteners with primer, 2-mil dry film thickness.

E. Exterior Finishes:

1. Hot-dip galvanize steel assemblies and fasteners after fabrication.

2.5 FABRICATION

A. Camber:

1. All beams shall have no camber unless otherwise noted on the drawings.

B. Field Cuts:

1. Use field treatment recommended by wood preservative manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates in areas to receive structural glued-laminated timber, with Installer present, for compliance with requirements, installation tolerances, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. General:

1. Erect structural glued-laminated timber true and plumb and with uniform, close-fitting joints. Provide temporary bracing to maintain lines and levels until permanent supporting members are in place.
2. Handle and temporarily support glued-laminated timber to prevent surface damage, compression, and other effects that might interfere with indicated finish.

B. Cutting:

1. Avoid extra cutting after fabrication. Where field fitting is unavoidable, comply with requirements for shop fabrication.

C. Fabrication, Fitting and Finishing:

1. Predrill for fasteners using timber connectors as templates.
2. Finish exposed surfaces to remove surfacing marks and to provide a finish equivalent to that produced by machine sanding with No. 120 grit sandpaper.
3. Coat crosscuts with end sealer.
4. Use field treatment recommended by preservative manufacturer.
5. Install timber connectors as indicated.
6. Unless otherwise indicated, install bolts with same orientation within each connection and in similar connections.

3.3 ADJUSTING

A. Repair damaged surfaces and finishes after completing erection. Replace damaged structural glued-laminated timber if repairs are not approved by the Contracting Officer.

3.4 PROTECTION

A. Do not remove wrappings on individually wrapped members until they no longer serve a useful purpose, including protection from weather, sunlight, soiling, and damage from work of other trades.

1. Coordinate wrapping removal with finishing work.
2. Retain wrapping where it can serve as a painting shield.
3. Slit underside of wrapping to prevent accumulation of moisture inside the wrapping.

END OF SECTION 06 18 00

SECTION 06 20 23 – INTERIOR FINISH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Interior trim.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry” for furring, blocking, and other carpentry work not exposed to view and for framing exposed to view.
2. Section 06 40 13 “Exterior Architectural Woodwork” for exterior wood trim.
3. Section 06 41 00 “Architectural Wood Casework” for wood portions of lobby transaction desk.
4. Section 07 92 00 “Joint Sealants” for adjacent sealant.
5. Section 09 91 23 “Interior Painting” for adjacent finish.
6. Section 09 93 00 “Staining and Transparent Finishing” for finishing.

1.2 ACTION SUBMITTALS

A. Product Data Submittals: For each type of process and factory-fabricated product. Indicate component materials, dimensions, profiles, textures, and colors and include construction and application details.

1. Include data for wood-preservative treatment from chemical-treatment manufacturer and certification by treating plant that treated materials comply with requirements. Indicate type of preservative used and net amount of preservative retained. Include chemical-treatment manufacturer's written instructions for finishing treated material.
2. For products receiving a waterborne treatment, include statement that moisture content of treated materials was reduced before shipment to Project site to levels specified.

B. Samples: For each exposed product and for each color and texture specified.

C. Installer Qualifications: minimum 5 years documented experience with projects of similar size and scope.

1.3 DELIVERY, STORAGE, AND HANDLING

A. Stack lumber, plywood, and other panels flat with spacers between each bundle to provide air circulation.

1. Protect materials from weather by covering with waterproof sheeting, securely anchored.
2. Provide for air circulation around stacks and under coverings.

B. Deliver interior finish carpentry materials only when environmental conditions comply with requirements specified for installation areas. If interior finish carpentry materials must be stored in other than installation areas, store only where environmental conditions comply with requirements specified for installation areas.

1.4 FIELD CONDITIONS

A. Environmental Limitations: Do not deliver or install interior finish carpentry materials until building is enclosed and weatherproof, wet-work in space is completed and nominally dry, and HVAC system is operating and maintaining temperature and relative humidity at occupancy levels during the remainder of the construction period.

B. Do not install finish carpentry materials that are wet, moisture damaged, or mold damaged.

1. Indications that materials are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
2. Indications that materials are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS

2.1 MATERIALS, GENERAL

A. Certified Wood: The following wood products shall be made from certified wood tracked through a chain-of-custody process. Certified wood documentation shall be provided by sources certified through a forest certification system with principles, criteria, and standards developed using ISO/IEC Guide 59 or the World Trade Organization's “WTO Agreement on Technical Barriers to Trade.”

1. Interior trim.

B. Lumber: DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with applicable rules of any rules-writing agency certified by the American Lumber Standard Committee's (ALSC) Board of Review. Grade lumber by an agency certified by the ALSC's Board of Review to inspect and grade lumber under the rules indicated.

1. Factory mark each piece of lumber with grade stamp of grading agency.
2. For exposed lumber, mark grade stamp on end or back of each piece , or omit grade stamp and provide certificates of grade compliance issued by grading agency.

C. Softwood Plywood: DOC PS 1.

2.2 INTERIOR TRIM

A. Hardwood Lumber Trim for Transparent Finish (Stain or Clear Finish):

1. Species and Grade: White Oak; NHLA Clear.
2. Maximum Moisture Content: 9 percent.

3. Finger Jointing: Not allowed.
4. Gluing for Width: Not allowed.
5. Veneered Material: Not allowed.
6. Face Surface: Surfaced (smooth).
7. Matching: Selected for compatible grain and color.

2.3 MISCELLANEOUS MATERIALS

A. Fasteners for Interior Finish Carpentry: Nails, screws, and other anchoring devices of type, size, material, and finish required for application indicated to provide secure attachment, concealed where possible.

B. Glue: Aliphatic-resin, polyurethane, or resorcinol wood glue recommended by manufacturer for general carpentry use.

C. Multipurpose Construction Adhesive: Formulation, complying with ASTM D3498, that is recommended for indicated use by adhesive manufacturer.

1. Adhesives shall have a VOC content of 70 g/L or less.
2. Adhesive shall comply with the testing and product requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers."

2.4 FABRICATION

A. Back out or kerf backs of the following members, except those with ends exposed in finished work:

1. Interior standing and running trim, except shoe and crown molds.

B. Ease edges of lumber less than 1 inch in nominal thickness to 1/16-inch radius and edges of lumber 1 inch or more in nominal thickness to 1/8-inch radius.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine finish carpentry materials before installation. Reject materials that are wet, moisture damaged, and mold damaged.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean substrates of projections and substances detrimental to application.

B. Before installing interior finish carpentry, condition materials to average prevailing humidity in installation areas for a minimum of 24 hours unless longer conditioning is recommended by manufacturer.

3.3 INSTALLATION, GENERAL

A. Do not use materials that are unsound; warped; improperly treated or finished; inadequately seasoned; too small to fabricate with proper jointing arrangements; or with defective surfaces, sizes, or patterns.

B. Install interior finish carpentry level, plumb, true, and aligned with adjacent materials.

1. Use concealed shims where necessary for alignment.
2. Scribe and cut interior finish carpentry to fit adjoining work. Refinish and seal cuts as recommended by manufacturer.
3. Where face fastening is unavoidable, countersink fasteners, fill surface flush, and sand unless otherwise indicated.
4. Install to tolerance of 1/8 inch in 96 inches for level and plumb. Install adjoining interior finish carpentry with 1/32-inch maximum offset for flush installation and 1/16-inch maximum offset for reveal installation.
5. Coordinate interior finish carpentry with materials and systems in or adjacent to it. Provide cutouts for mechanical and electrical items that penetrate interior finish carpentry.

3.4 INSTALLATION OF INTERIOR TRIM

A. Install trim with minimum number of joints as is practical, using full-length pieces from maximum lengths of lumber available.

1. Do not use pieces less than 24 inches long, except where necessary.
2. Stagger joints in adjacent and related standing and running trim.
3. Miter at returns, miter at outside corners, and cope at inside corners to produce tight-fitting joints with full-surface contact throughout length of joint.
4. Use scarf joints for end-to-end joints.
5. Plane backs of casings to provide uniform thickness across joints where necessary for alignment.
6. Match color and grain pattern of trim for transparent finish (stain or clear finish) across joints.
7. Install trim after gypsum-board joint finishing operations are completed.
8. Install without splitting; drill pilot holes before fastening where necessary to prevent splitting.
9. Fasten to prevent movement or warping.
10. Countersink fastener heads on exposed carpentry work and fill holes.

3.5 ADJUSTING

A. Replace interior finish carpentry that is damaged or does not comply with requirements.

1. Interior finish carpentry may be repaired or refinished if work complies with requirements and shows no evidence of repair or refinishing.

B. Adjust joinery for uniform appearance.

3.6 CLEANING

A. Clean interior finish carpentry on exposed and semi-exposed surfaces.

B. Restore damaged or soiled areas and touch up factory-applied finishes if any.

3.7 PROTECTION

A. Protect installed products from damage from weather and other causes during construction.

B. Remove and replace finish carpentry materials that are wet, moisture damaged, and mold damaged.

1. Indications that materials are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
2. Indications that materials are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

END OF SECTION 06 20 23

SECTION 06 40 13 – EXTERIOR ARCHITECTURAL WOODWORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Exterior roof trim for stained finish.
2. Exterior wood members for benches.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry” for furring, blocking, and other carpentry work not exposed to view and for framing exposed to view.
2. Section 06 40 13 “Interior Finish Carpentry” for interior wood trim.
3. Section 06 41 00 “Architectural Wood Casework” for wood portions of lobby transaction desk.
4. Section 07 92 00 “Joint Sealants” for adjacent sealant.
5. Section 09 93 00 “Staining and Transparent Finishing” for finish as indicated on drawings.

1.2 COORDINATION

A. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of Work specified in other Sections, to ensure that exterior architectural woodwork can be supported and installed as indicated.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings:

1. Include dimensioned plans, elevations, sections, and attachment details.
2. Show large-scale details.
3. Show locations and sizes of furring, blocking, and hanging strips, including blocking and reinforcement concealed by construction and specified in other Sections.
4. Apply AWI Quality Certification Program label to Shop Drawings.

C. Samples for Verification: For the following:

1. Lumber for Exterior Stained Finish: Not less than 5 inches wide by 12 inches long, for each species, with one-half of exposed surface finished.
2. Lumber and Panel Products with Shop-Applied Opaque Finish: 5 inches wide by 12 inches long for lumber and 8 by 10 inches for panels, for each finish system and color.

    a. Finish entire exposed surface.

1.5 CLOSEOUT SUBMITTALS

A. Quality Standard Compliance Certificates: WI Certified Compliance Program certificates.

1.6 QUALITY ASSURANCE

A. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Build mockup (8 foot length minimum) of typical exterior architectural woodwork as shown on Drawings.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless CO specifically approves such deviations by Change Order.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Comply with Architectural Woodwork Standards, Section 2.

B. Store woodwork in installation areas or in areas where environmental conditions comply with requirements specified in “Field Conditions” Article.

1.8 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation of exterior architectural woodwork only when existing and forecasted weather conditions permit work to be performed and at least one coat of specified finish to be applied without exposure to rain, snow, or dampness.

B. Field Measurements: Where exterior architectural woodwork is indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication, and indicate measurements on Shop Drawings.

1. Locate concealed framing, blocking, and reinforcements that support woodwork by field measurements before being concealed by construction, and indicate measurements on Shop Drawings.

C. Established Dimensions: Where exterior architectural woodwork is indicated to fit to other construction, establish dimensions for areas where woodwork is to fit. Provide allowance for trimming at site, and coordinate construction to ensure that actual dimensions correspond to established dimensions.

PART 2 - PRODUCTS

2.1 ARCHITECTURAL WOODWORK, GENERAL

A. Quality Standard: Unless otherwise indicated, comply with the Architectural Woodwork Standards for grades of exterior architectural woodwork indicated for construction, finishes, installation, and other requirements.

1. Provide certificates from AWI certification program indicating that woodwork and installation complies with requirements of grades specified.

2.2 EXTERIOR STANDING AND RUNNING TRIM FOR STAINED FINISH

A. Architectural Woodwork Standards Grade: Premium.

B. Backout or groove backs of flat trim members and kerf backs of other wide, flat members, except for members with ends exposed in finished work.

C. Wood Species: Western red cedar.

1. Do not use plain-sawn softwood lumber with exposed, flat surfaces more than 3 inches wide.

2.3 FASTENERS

A. General: Provide fasteners of size and type indicated, acceptable to authorities having jurisdiction, and that comply with requirements specified in this article for material and manufacture. Provide nails or screws, in sufficient length, to penetrate not less than 1-1/2 inches into wood substrate.

1. For pressure-preservative-treated wood, use stainless steel fasteners.

B. Nails: ASTM F1667.

C. Power-Driven Fasteners: ICC-ES AC70.

D. Wood Screws and Lag Screws: ASME B18.2.1, ASME B18.6.1, or ICC-ES AC233.

2.4 MISCELLANEOUS MATERIALS

A. Blocking, Shims, and Nailers: Softwood or hardwood lumber, kiln-dried to less than 15 percent moisture content.

1. Wood-Preservative Treatment: By pressure process, AWPA U1; Use Category UC3b.

    a. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent.

2.5 FABRICATION

A. Fabricate exterior architectural woodwork to dimensions, profiles, and details indicated.

1. Ease edges to radius indicated for the following:

    a. Edges of Solid-Wood (Lumber) Members: 1/16 inch unless otherwise indicated.

B. Complete fabrication, including assembly, to maximum extent possible before shipment to Project site.

1. Disassemble components only as necessary for shipment and installation.
2. Where necessary for fitting at site, provide allowance for scribing, trimming, and fitting.

2.6 SHOP PRIMING

A. Preparations for Finishing: Comply with the Architectural Woodwork Standards for sanding, filling countersunk fasteners, sealing concealed surfaces, and similar preparations for finishing exterior architectural woodwork, as applicable to each unit of work.

B. Exterior Architectural Woodwork for Transparent Finish:

1. Shop seal surfaces to be concealed
2. Shop seal exposed surfaces with stain (if specified), other required pretreatments, and first coat of finish as specified in Section 099300 “Staining and Transparent Finishing.”

2.7 SHOP FINISHING

A. Finish exterior architectural woodwork with transparent finish at fabrication shop. Defer only final touchup, cleaning, and polishing until after installation.

B. Preparation for Finishing: Comply with the Architectural Woodwork Standards for sanding, filling countersunk fasteners, sealing concealed surfaces, and similar preparations for finishing exterior architectural woodwork, as applicable to each unit of work.

1. Backpriming: Apply one coat of sealer or primer, compatible with finish coats, to concealed surfaces of exterior architectural woodwork. Apply two coats to end-grain surfaces.

C. Transparent Finish: Comply with Section 09 93 00 “Staining and Transparent Finishing.”

PART 3 - EXECUTION

3.1 PREPARATION

A. Before installation, condition exterior architectural woodwork to average prevailing humidity conditions at Project site.

B. Before installing exterior architectural woodwork, examine shop-fabricated work for completion, and complete work as required, including removing packing and backpriming concealed surfaces.

3.2 INSTALLATION

A. Assemble exterior architectural woodwork, and complete fabrication at Project site to the extent that it was not completed during shop fabrication.

B. Install exterior architectural woodwork level, plumb, true in line, and without distortion.

1. Shim as required with concealed shims.
2. Install level and plumb to a tolerance of 1/8 inch in 96 inches.

C. Standing and Running Trim:

1. Install with minimum number of joints possible, using full-length pieces (from maximum length of lumber available) to greatest extent possible.
2. Do not use pieces less than 60 inches long, except where shorter single-length pieces are necessary.
3. Scarf running joints and stagger in adjacent and related members.

D. Scribe and cut exterior architectural woodwork to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts.

E. Anchor exterior architectural woodwork to anchors or blocking built in or directly attached to substrates.

1. Secure with countersunk, concealed fasteners and blind nailing.
2. Use fine finishing nails or finishing screws for exposed fastening, countersunk and filled flush with exterior architectural woodwork.
3. For exposed work, arrange fasteners in straight rows parallel with edges of members, with fasteners evenly spaced and with adjacent rows staggered.
4. For shop-finished items, use filler matching finish of items being installed.

F. Touch up finishing work specified in this Section after installation of exterior architectural woodwork.

1. Fill nail holes with matching filler where exposed.
2. Apply specified finish coats, including stains and paste fillers if any, to exposed surfaces where only sealer/prime coats are shop applied.

G. Field Finishing: See Section 09 93 00 “Staining and Transparent Finishing” for final finishing of installed exterior architectural woodwork.

3.3 REPAIR

A. Repair damaged and defective exterior architectural woodwork, where possible, to eliminate functional and visual defects.

B. Where not possible to repair, replace defective woodwork.

3.4 CLEANING

A. Clean exterior architectural woodwork on exposed and semiexposed surfaces.

END OF SECTION 06 40 13

SECTION 06 41 00 – ARCHITECTURAL WOOD CASEWORK

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wood-veneer-faced casework for stained finish associated with Lobby transaction desk.
2. Substructure and supports for a café height worksurface in the basement vestibule (013).
3. Casework hardware.
4. Shop staining and finishing.

B. Related Requirements:

1. Section 05 50 00 “Metal Fabrications” for misc. metal supports.
2. Section 06 10 00 “Rough Carpentry” for wood furring, blocking, shims, and hanging strips required for installing on pony support wall.
3. Section 06 20 23 “Interior Finish Carpentry” for interior wood trim.
4. Section 06 40 13 “Exterior Architectural Woodwork” for exterior wood trim.
5. Section 12 36 61 “Simulated Stone Countertops.”
6. Section 09 93 00 “Staining and Transparent Finishing” for finishing of other wood.
7. Section 12 32 16 “Manufactured Plastic-Laminate-Clad Casework” for upper and base cabinets in the staff break area (002).

1.2 COORDINATION

A. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of Work specified in other Sections to support loads imposed by installed and fully loaded cabinets.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include panel material description and finish.
2. Include accessories.
3. Include hardware.
4. Include shop finishing.

B. Shop Drawings:

1. Plans, elevations, sections, and attachment details.

a. Show locations and sizes of furring, blocking, and hanging strips, including concealed blocking and reinforcement specified in other Sections.
b. Show locations and sizes of cutouts and holes for items installed in architectural cabinets.
c. Show veneer leaves with dimensions, grain direction, exposed face, and identification numbers indicating the flitch and sequence within the flitch for each leaf.

2. Apply AWI's Quality Certification Program (hereafter AWI QCP) label to Shop Drawings.

C. Samples for Verification: Provide a minimum of three samples showing the full range of color and/or grain to be expected in the finished work, for the following.

1. Lumber for Stained Finish: Not less than 8 inches long by 3/4 inch thick, and as wide as practical, for each species and cut, finished on one side and one edge.
2. Veneered Panel for Stained Finish: 8 by 8 inches by 1/4 inch, representative of and selected from flitches to be used for stained-finished casework.
3. Provide Stain Sample on each type of wood to be visible in transaction desk.
4. Exposed Cabinet Hardware and Accessories: One full-size unit for each type and finish.

1.5 INFORMATIONAL SUBMITTALS

A. Test Reports: Valid AWI “Performance Quality Test Report” indicating compliance with specified ANSI/AWI 0641 Structural Performance Duty Level.

B. Certification Reports:

1. Provide certificates from AWI QCP indicating woodwork complies with requirements of aesthetic grade and duty level specified.

a. Register the Work of this Section with AWI's Quality Certification Program.

C. Product Certificates: For each type of product.

1. Adhesives.

1.6 CLOSEOUT SUBMITTALS

A. Quality Standard Compliance Certificates: AWI QCP Certificate.

1.7 QUALITY ASSURANCE

A. Manufacturer Qualifications: Employs skilled workers who custom fabricate products similar to those required for this Project and whose products have a record of successful in-service performance.

1. Manufacturer's Certification: Licensed participant in AWI's Quality Certification Program.

1.8 MOCKUPS

A. Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. One section of the desk shall be built in its entirety to demonstrate quality of materials and execution for Contracting Officer’s review. Then the remainder of the desk shall be constructed/assembled.
2. If constructed offsite in sections, Contractor shall arrange for the Contracting Officer to visit the shop to view the mockup prior to approvals.
3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Do not deliver cabinets until painting and similar finish operations that might damage architectural cabinets have been completed in installation areas. Store cabinets in installation areas or in areas where environmental conditions comply with requirements specified in “Field Conditions” Article.

1.10 FIELD CONDITIONS

A. Environmental Limitations without Humidity Control: Do not deliver or install cabinets until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature and relative humidity at levels planned for building occupants during the remainder of the construction period.

B. Environmental Limitations with Humidity Control: Do not deliver or install cabinets until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature between 60 and 90 deg F and relative humidity between 25 and 45 percent during the remainder of the construction period.

C. Field Measurements: Where cabinets are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication, and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1. Locate concealed framing, blocking, and reinforcements that support cabinets by field measurements before being enclosed/concealed by construction, and indicate measurements on Shop Drawings.

D. Established Dimensions: Where cabinets are indicated to fit to other construction, establish dimensions for areas where cabinets are to fit. Provide allowance for trimming at site, and coordinate construction to ensure that actual dimensions correspond to established dimensions.

PART 2 - PRODUCTS

2.1 WOOD-VENEER-FACED CASEWORK FOR STAINED FINISH

A. Architectural Woodwork Institute (AWI) Standard:

1. Structural Performance Duty Level: 4.
2. Aesthetic Performance Grade: Premium.
3. Product Data: Verify materials were fabricated within 500 miles of Project site from materials that have been extracted, harvested, or recovered within 500 miles of Project site. If materials are transported by rail or water, multiply the distance by 0.25 to determine the distance to Project site.
4. Certified Wood: Verify wood products are labeled in accordance with the AF&PA's Sustainable Forestry Initiative, certified as “FSC Pure” in accordance with FSC STD-01-001 and FSC STD-40-004, or certified and labeled in accordance with the standards of the Program for Endorsement of Forest Certification.
5. Verify adhesives have a VOC content of 30 g/L or less.

B. Casework Construction Type: Frameless.

C. Door and Drawer-Front:

1. Design: Flush panel.
2. Style: Flush overlay.

    a. Reveal Dimension: As determined by hinge type and clearance.

D. Exposed Exterior Surfaces: As indicated on Drawings.

1. Species: White oak.
2. Cut: Rift cut/rift sawn.
3. Grain Direction: Vertically for drawer fronts, doors, and fixed panels.
4. Veneer Thickness: 1/8” minimum
5. Veneer Matching:

    a. Matching of Veneer Leaves: Book match.
    b. Veneer Matching within Panel Face: Balance match.

6. Edgeband: Solid wood matching exposed surfaces.

E. Exposed Interior Surfaces: As indicated on Drawings.

1. Species: White oak.
2. Cut: Match exposed exterior surfaces.
3. Veneer Thickness: 1/8” minimum
4. Veneer Matching: Match exposed exterior surfaces.
5. Edgeband: Solid wood.

F. Semiexposed Surfaces:

1. Surfaces Other Than Drawer Bodies: Same species wood veneer panel indicated for exposed surfaces.
2. Drawer Subfronts, Backs, and Sides: Solid-hardwood lumber, same species indicated for exposed surfaces.
3. Drawer Bottoms: Hardwood plywood.
4. Shelves: Adjustable.

    a. Loading Capacity: 50 lb/sq. ft.

5. Edgeband: Match exposed surfaces.

2.2 CASEWORK HARDWARE

A. Cabinet Hardware: Provide cabinet hardware and accessory materials associated with architectural cabinets except for items specified in Section 087100 “Door Hardware.”

B. Cabinet Hinges:

1. Frameless Concealed Hinges (European Type): ANSI/BHMA A156.9, B01602, 115-140 degrees of opening, self-closing.
2. Provide two hinges for doors less than 48 inches high, and provide three hinges for doors more than 48 inches high.

C. Swinging Gate Hinges:

1. Double Action Swinging Door Hinges with spring-loaded self-closing, appropriate for installation on “saloon-style” door/gate.
2. 3 inches long
3. Material: Stainless Steel
4. Finish: Brushed
5. Attachments: Product will come with screws. Attachment is planned to be to steel post. Contractor will drill and tap hole to receive screws or provide replacement self-tapping/thread cutting screws of equivalent diameter (to be compatible with holes in hinges) for attachment.
6. Basis of Design: Kolet 2WDH-SS4, or approved equal.

D. Pulls and Knobs:

1. Back-Mounted Pulls: ANSI/BHMA A156.9, B02011, solid metal, 5 inches long, 1/2 inch width minimum, 3-inch centers, brushed aluminum or brushed nickel finish.

    a. Basis of Design: Estmoon Brushed Nickel Cabinet Handles 3-inch hole center drawer pull

E. Shelf Rests and Standards:

1. Adjustable Shelf Standards and Supports: ANSI/BHMA A156.9, B04071; with shelf rests, B04081.
2. Shelf Rests: ANSI/BHMA A156.9, B04013; metal.

F. Drawer Slides: ANSI/BHMA A156.9.

1. Heavy-Duty (Grade 1HD-100 and Grade 1HD-200): Side mount.

    a. Type: Soft close, Full overtravel extension.
    b. Material: Aluminum slides.

G. Locks for Drawer and Hinged-Doors: Mortise type, five-pin tumbler, brass with chrome-plated finish, and complying with ANSI/BHMA A156.11, Grade 1.

1. Provide a minimum of two keys per lock and six master keys.
2. Provide locks where indicated.

H. Door and Drawer Bumpers: Self-adhering, clear silicone rubber.

1. Doors: Provide one bumper at top and bottom of closing edge of each swinging door.
2. Drawers: Provide one bumper on back side of drawer front at each corner.

I. Hardware Finish:

1. Finish: Brushed Nickel or Clear Anodized Aluminum.
2. Concealed Hardware: Manufacturer's standard finish that complies with product class requirements in ANSI/BHMA A156.9.

J. Desk Countertop Support

1. Countertop L-Bracket: Aluminum, 8” deep, industrial strength, load capacity 400 lbs./bracket minimum, color: black.

    a. Basis of Design: The Original Side Wall Free Hanging Shelf Bracket
    b. Or Approved Equal

2. Bridge Bracket: Aluminum, 1” deep, 2.5” wide, industrial strength, load capacity 800 lbs./bracket minimum, color: black.

    a. Basis of Design: The Original Side Wall Free Hanging Shelf Bracket
    b. Or Approved Equal

3. Corner Connector: Aluminum, L-Type, 5 Holes, 90 degree, Plate Angle Bracket

K. Café Countertop Support:

1. Countertop L-Bracket: Front mount, iron, matte black, 14” deep x 10” high minimum.

2.3 FABRICATION

A. Fabricate architectural cabinets to dimensions, profiles, and details indicated. Ease sharp edges and corners.

B. Complete fabrication, including assembly and hardware application, to maximum extent possible before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.

C. Shop-cut openings to maximum extent possible to receive hardware, appliances, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Sand edges of cutouts to remove splinters and burrs.

2.4 SHOP FINISHING

A. Finish architectural cabinets at manufacturer's shop as specified in this Section. Defer only final touchup, cleaning, and polishing until after installation.

B. Preparation for Finishing: Comply with referenced quality standard for sanding, filling countersunk fasteners, sealing concealed surfaces, and similar preparations for finishing architectural cabinets, as applicable to each unit of work.

1. Backpriming: Apply one coat of sealer or primer, compatible with finish coats, to concealed surfaces of cabinets.

C. Stain Finish:

1. Aesthetic Performance Grade: Premium.
2. Wash Coat for Closed-Grain Woods: Apply wash-coat sealer to cabinets made from closed-grain wood before staining and finishing.
3. Staining: Walnut stain, to be confirmed with Contracting Officer during submittal process.
4. Filled Finish for Open-Grain Woods: After staining, apply wash-coat sealer and allow to dry. Apply paste wood filler and wipe off excess. Tint filler to match stained wood.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances, location of reinforcements, and other conditions affecting performance of the Work.

1. Confirm location and adequacy of blocking and supports required for installation.
2. Verify size and locations of cutouts for appliances, special equipment, sinks, plumbing, and other items to be mounted in or adjacent to casework.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Before installation, condition cabinets to humidity conditions in installation areas for not less than 72 hours.

3.3 INSTALLATION OF CASEWORK

A. Install casework to comply with requirements of the structural and aesthetic performance specified.

B. Assemble cabinets and complete fabrication at Project site to extent that it was not completed in the shop.

C. Install cabinets level, plumb, and true in line to a tolerance of 1/8 inch in 96 inches using concealed shims.

1. Scribe and cut cabinets to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts.
2. Install cabinets without distortion so doors and drawers fit openings and are accurately aligned. Adjust hardware to center doors and drawers in openings and to provide unencumbered operation. Complete installation of hardware and accessory items as indicated.
3. Maintain veneer sequence matching of cabinets with transparent finish.

D. Anchor cabinets in accordance with AWI requirements or tested method to anchors or blocking built in or directly attached to framing or substrates.

E. Shop Finishes: Touch up finishing after installation of architectural cabinets. Fill nail holes with matching filler.

1. Apply specified finish coats, including stains and paste fillers if any, to exposed surfaces where only sealer/prime coats are shop applied.

a. Use filler matching finish of items being installed.

3.4 ADJUSTING AND CLEANING

A. Repair damaged and defective cabinets, where possible, to eliminate functional and visual defects. Where impossible to repair, replace architectural cabinets. Adjust joinery for uniform appearance.

B. Clean, lubricate, and adjust hardware and other moving parts to function smoothly.

C. Clean cabinets on exposed and semiexposed surfaces. Touch up finishes to restore damaged or soiled areas.

3.5 FIELD QUALITY CONTROL

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 06 41 00

SECTION 06 61 16 – SOLID SURFACE FABRICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Provide solid surfacing fabrications including but not limited to following:

1. Solid surface wall panels in Restrooms

B. Related Sections: Following description of work is included for reference only and shall not be presumed complete:

1. Provision of indoor air quality requirements: Section 01 81 19, Indoor Air Quality Requirements.
2. Provision of elastomeric joint sealants: Section 07 92 00, Joint Sealants.
3. Provision of access doors and frames in solid surface panels: Section 08 31 13, Access Doors and Frames.
4. Provision of epoxy flooring: Section 09 67 23, Resinous Flooring.
5. Provision of plumbing and plumbing fixtures: Division 22, Plumbing

1.2 REFERENCES

A. Abbreviations and Acronyms:

1. LEED®: Leadership in Energy and Environmental Design.
2. MDF: Medium Density Fiberboard.
3. SCAQMD: South Coast Air Quality Management District.
4. VOC: Volatile Organic Compound.

B. Definitions:

1. Solid Surface: Non-porous, homogeneous material maintaining the same composition throughout the part with a composition of acrylic polymer, aluminum trihydrate filler and pigment.

C. Reference Standards:

1. ANSI/NPA A208.2-09 – Medium Density Fiberboard (MDF) For Interior Applications
2. ASTM C920-14a – Standard Specification for Elastomeric Joint Sealants
3. ASTM D638-10 – Standard Test Method for Tensile Properties of Plastics
4. ASTM D785-08 – Standard Test Method for Rockwell Hardness of Plastics and Electrical Insulating Materials
5. ASTM D790-10 – Standard Test Methods for Flexural Properties of Unreinforced
6. and Reinforced Plastics and Electrical Insulating Materials
7. ASTM D5420-10 – Standard Test Method for Impact Resistance of Flat, Rigid
8. Plastic Specimen by Means of a Striker Impacted by a Falling Weight (Gardner Impact)

9. ASTM E84-14 – Standard Test Method for Surface Burning Characteristics of
10. Building Materials
11. ASTM E228-11 – Standard Test Method for Linear Thermal Expansion of Solid
12. Materials with a Push-Rod Dilatometer
13. ASTM G21-13 – Standard Practice for Determining Resistance of Synthetic
14. Polymeric Materials to Fungi
15. ASTM G22-76(96) – Standard Practice for Determining Resistance of Plastics to
16. Bacteria
17. ASTM G155-13 – Standard Practice for Operating Xenon Arc Light Apparatus for
18. Exposure of Non-Metallic Materials
19. CSA B45.5-11/ IAPMO Z124-2011- Plastic Plumbing Fixtures
20. NFPA 255-06 – Standard Method of Test of Surface Burning Characteristics of
21. Building Materials
22. NSF/ANSI 51-07 – Food Equipment Materials
23. SCAQMD Rule 1168 – Adhesive and Sealant Applications (amended January 2005)
24. UL 723 – Standard for Test for Surface Burning Characteristics of Building Materials
25. UL Environment/- Standard for Chemical Emissions for Building Materials,
26. GREENGUARD – Finishes and Furnishings, Section 7.1 UL 2818
27. UL Environment/ – Gold Standard for Chemical Emissions for Building Materials,
28. GREENGUARD – Finishes and Furnishings, Section 7.1 and 7.2 UL 2818
29. UL 2824 – GREENGUARD Certification Program, Method for Measuring
30. Microbial Resistance from Various Sources Using Static Environmental Chambers

1.3 ADMINISTRATIVE REQUIREMENTS

A. Preinstallation Meetings: Arrange preinstallation meeting 2 weeks prior to commencing work with all parties associated with trade as designated in Contract Documents or as requested by Contracting Officer. Presided over by Contractor, include Contracting Officer who may attend, Subcontractor performing work of this trade, Contracting Officer’s representative, testing company's representative and consultants of applicable discipline. Review Contract Documents for work included under this trade and determine complete understanding of requirements and responsibilities relative to work included, storage and handling of materials, materials to be used, installation of materials, sequence and quality control, Project staffing, restrictions on areas of work and other matters affecting construction, to permit compliance with intent of work of this Section.

1.4 SUBMITTALS

A. Product Data: Indicate Product description including solid surface sheets, sinks, bowls and illustrating full range of standard colors, fabrication information and compliance with specified performance requirements. Submit Product data with resistance to list of chemicals.

B. Shop Drawings: Submit Shop Drawings for work of this Section in accordance with Section

C. 01 33 23. Indicate plans, sections, dimensions, component sizes, edge details, thermosetting requirements, fabrication details, attachment provisions, sizes of furring, blocking, including concealed blocking and coordination requirements with adjacent work. Show locations and sizes of cutouts and holes for plumbing fixtures, faucets, soap dispensers, waste receptacles and other items installed in solid surface.

D. Samples: Submit samples in accordance with Section 01 33 23. Submit minimum 6” x 6” samples. Cut sample and seam together for representation of inconspicuous seam. Indicate full range of color and pattern variation. Approved samples will be retained as standards for work.

E. Test and Evaluation Reports: Submit flammability test reports.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data:

1. Submit manufacturer's care and maintenance data, including repair and cleaning instructions. Include in Project closeout documents.
2. Provide a commercial care and maintenance kit and video. Review maintenance procedures and warranty details with Contracting Officer upon completion.

1.6 QUALITY ASSURANCE

A. Qualifications:

1. Installers: Provide work of this Section executed by competent installers with minimum 5 years experience in the application of Products, systems and assemblies specified and with approval and training of the Product manufacturers.

B. Mock-Ups:

1. Prior to final approval of Shop Drawings, erect 1 full size mock-up of the installation of two adjacent panels at Project site demonstrating quality of materials and execution for Contracting Officer review.
2. Should mock-up not be approved, rework or remake until approval is secured. Remove rejected units from Project site.
3. Approved mock-up will be used as standard for acceptance of subsequent work.
4. Approved mock-ups may remain as part of finished work.

1.7 DELIVERY, STORAGE AND HANDLING

A. Delivery and Acceptance Requirements: Deliver no components to Project site until areas are ready for installation.

B. Storage and Handling Requirements:

1. Store components indoors prior to installation.
2. Handle materials to prevent damage to finished surfaces.

1.8 WARRANTY

A. Manufacturer Warranty: Provide manufacturer's standard warranty for material only for period of 10 years against defects and/or deficiencies in accordance with General Conditions of the Contract. Promptly correct any defects or deficiencies which become apparent within warranty period, to satisfaction of Contracting Officer and at no expense to Government.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturer List: Products of following manufacturers are acceptable subject to conformance to requirements of Drawings, Schedules and Specifications:

1. Corian® by DuPont
2. Samsung Chemical USA
3. Wilsonart Contract

B. Substitution Limitations: Comparable Products from manufacturers listed herein will be accepted provided they meet requirements of this Specification.

2.2 MATERIALS

A. Solid Surface Material: Non-porous, homogeneous material maintaining the same composition throughout the part with a composition of acrylic polymer, aluminum trihydrate filler and pigment; not coated, laminated or of composite construction; meeting following criteria:

1. Basis of Design: Wilsonart Solid Surface, Color: "Beige Travertine" 9236SS.
2. 1/2" thick
3. Flammability: Class 1 and A when tested to UL 723.
4. Matte, with a 60° gloss rating of 5 – 20.
5. Adhesive for Bonding to Other Products: One component silicone to ASTM C920.
6. Sealant: A standard mildew-resistant, FDA/UL® recognized silicone color matched sealant or clear silicone sealants.

2.3 COMPONENTS

A. Wall Panels: 1/2" thick vertical wall panels of solid polymer solid surfacing material, cast to desired profiles and sizes having edge details as indicated on Drawings conforming to CSA B45.5/IAPMO Z124. Ensure sheets are single color, as selected by Contracting Officer.

1. Ensure surfaces have uniform finish.
2. Metal Edge Strips: No metal edging as indicated on Drawings.

B. Top of panel condition: As indicated on Drawings.

C. Bottom of panel condition: As indicated on Drawings.

D. Coordinate with installation of epoxy flooring and integrated epoxy coved base.

E. Fabrication:

1. Fabricate components in shop to greatest extent practical to sizes and shapes indicated, in accordance with approved Shop Drawings and solid polymer manufacturer requirements. Form joints between components using manufacturer's standard joint adhesive without conspicuous joints. Provide factory cutouts for plumbing fittings and bath accessories as indicated on Drawings.
2. Where indicated, thermoform corners and edges or other objects to shapes and sizes indicated on Drawings, prior to seaming and joining. Cut components larger than finished dimensions and sand edges to remove nicks and scratches. Heat entire component uniformly prior to forming.
3. Ensure no blistering, whitening and cracking of components during forming.
4. Fabricate joints between components using manufacturer's standard joint adhesive. Ensure joints are inconspicuous in appearance and without voids. Attach 50 mm (2") wide reinforcing strip of solid polymer material under each joint. Reinforcing strip of solid polymer material is not required when using adhesive.
5. Provide holes and cutouts for plumbing and bath accessories as indicated on Drawings.
6. Rout and finish component edges to a smooth, uniform finish. Rout cutouts, then sand edges smooth. Repair or reject defective or inaccurate work.
7. Finish: Ensure surfaces have uniform finish:
8. Fabrication Tolerances:

a. Variation in Component Size: +/-1/8".
b. Location of Openings: +/-1/8" from indicated location.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:

1. Examine substrates and conditions, with fabricator present for compliance with requirements for installation tolerances and other conditions affecting performance of work. Proceed with installation only after unsatisfactory conditions have been corrected.
2. Verify actual site dimensions and location of adjacent materials prior to commencing work.
3. Notify Contracting Officer in writing of any conditions which would be detrimental to installation.

B. Evaluation and Assessment: Commencement of work implies acceptance of previously completed work.

3.2 INSTALLATION

A. Install components plumb, level, rigid, scribed to adjacent finishes in accordance with reviewed Shop Drawings and Product installation details.

B. Fabricate field joints using manufacturer's recommended adhesive, with joints being inconspicuous in finished work. Exposed joints/seams are not permitted. Keep components and hands clean when making joints. Reinforce field joints as specified herein. Cut and finish component edges with clean, sharp returns.

C. Route radii and contours to template. Anchor securely to base component or other supports. Align adjacent components and form seams to comply with manufacturer's written recommendations using adhesive in color to match work. Carefully dress joints smooth, remove surface scratches and clean entire surface.

D. Seal between wall and components with joint sealant as specified herein and in Section 07 92 00 "Joint Sealants", as applicable.

E. Keep components and hands clean during installation. Remove adhesives, sealants and other stains. Ensure components are clean on date of Substantial Completion of the Work.

F. Coordinate connections of plumbing fixtures with Division 22. Make plumbing connections to fixtures in accordance with Division 22.

3.3 REPAIR

A. Repair minor imperfections and cracked seams and replace areas of severely damaged surfaces in accordance with manufacturer's "Technical Bulletins".

3.4 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

3.5 SITE QUALITY CONTROL

A. Non-Conforming Work: Replace damaged work which cannot be satisfactorily repaired, restored or cleaned, to satisfaction of Contracting Officer at no cost to Government.

3.6 CLEANING

A. Remove excess adhesive and sealant from visible surfaces.

B. Clean surfaces in accordance with manufacturer's "Care and Maintenance Instructions".

3.7 PROTECTION

A. Provide protective coverings to prevent physical damage or staining following installation for duration of Project.

B. Protect surfaces from damage until date of Substantial Completion of the Work.

END OF SECTION 06 61 16

SECTION 07 14 16 – COLD FLUID-APPLIED WATERPROOFING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Polyurethane waterproofing for existing and new foundation walls.
2. Accessory waterproofing system materials.
3. Molded-sheet drainage panels.

B. Related Requirements:

1. Section 03 30 00 “Cast in Place Concrete” for new foundation walls.
2. Section 07 18 00 “Traffic Coatings” for exposed, fluid-applied membrane with an integral wearing surface.
3. Section 07 21 00 “Thermal Insulation” for below-grade foundation insulation.
4. Section 07 92 00 “Joint Sealants” for joint sealants, joint fillers, and joint preparation.

1.2 DEFINITIONS

A. Compatible: Material that will not adversely affect adjacent materials, is chemically compatible with adjacent materials, and where required for bond, achieves adhesive compatibility with adjacent materials.

B. Chemical Compatibility: Material that will not break down, deteriorate, degrade, or prematurely fail when in contact with another material. Material that will not cause chemical breakdown, deterioration, degradation, staining, or premature failure of another material.

C. Adhesive Compatibility: Material that will develop bond strength or provide a suitable surface for another material to develop bond strength complying with requirements when in contact with another material.

1.3 COORDINATION

A. Coordinate Work under this Section with adjacent concrete foundation work, including fill, foundation wall insulation, and , subdrainage systems.

B. Coordinate requirements for concrete formwork to provide suitable substrate for waterproofing and to minimize penetrations in waterproofing.

1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review waterproofing requirements, including, but not limited to, the following:

    a. Minimum concrete curing period.
    b. Forecasted weather conditions.
    c. Special details and sheet flashings.
    d. Repairs.
    e. Field quality control.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, and tested physical and performance properties of waterproofing.
2. Include manufacturer's written instructions for evaluating, preparing, and treating substrate.

B. Qualification Data: For Installer.

C. Shop Drawings:

1. Indicate locations and extent of waterproofing.
2. Include details for substrate joints and cracks, sheet flashings, penetrations, inside and outside corners, expansion-joint conditions, tie-ins with adjoining waterproofing, and other termination conditions.
3. Include setting drawings indicating layout, sizes, sections, profiles, and joint details of pedestal-supported concrete pavers.

D. Samples: For the following materials:

1. Cured sample of waterproofing membrane on suitable rigid substrate, 8 by 8 inches.
2. Drainage panel, 4 by 4 inches.

1.6 INFORMATIONAL SUBMITTALS

A. Field quality control reports.

B. Sample warranties.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: Entity that employs installers and supervisors who are trained and certified by waterproofing manufacturer.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Store materials in their original undamaged containers in a clean, dry, protected location and within the temperature range required by waterproofing manufacturer. Protect stored materials in accordance with manufacturer's written instructions.

B. Remove and replace materials that cannot be applied within their stated shelf life.

1.9 FIELD CONDITIONS

A. Environmental Limitations: Apply waterproofing within the range of ambient and substrate temperatures recommended in writing by waterproofing manufacturer.

1. Do not apply waterproofing to frozen, damp, or wet substrates, when relative humidity exceeds 85 percent, or when temperatures are less than 5 deg F above dew point.
2. Do not apply waterproofing when snow, rain, fog, or mist are present, or when such weather conditions are imminent during application and curing period.

B. Maintain adequate ventilation during application and curing of waterproofing materials.

1.10 WARRANTY

A. Manufacturer's Special Warranty: Manufacturer agrees to repair or remove and replace waterproofing that fails to remain watertight within specified warranty period.

1. Warranty includes leak remediation, including repair, removal, and replacement of drainage panels.
2. Warranty Period: 10 years from date of Substantial Completion.

B. Installer's Special Warranty: Submit warranty signed by Installer, covering the Work of this Section, for warranty period of two years from date of Substantial Completion.

1. Warranty includes leak remediation, including repair, removal, and replacement of drainage panels.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Waterproofing System: Obtain waterproofing materials and molded-sheet drainage panels from manufacturer approved by waterproofing membrane manufacturer.

2.2 POLYURETHANE WATERPROOFING

A. Single-Component, Modified Polyurethane Waterproofing: ASTM C836/C836M and coal-tar free.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Tremco Inc, TREMproof 250GC or comparable product by one of the following:

    a. Carlisle Coatings & Waterproofing Inc
    b. CETCO is a subsidiary of Minerals Technologies Inc.
    c. Henry, a Carlisle Company (formerly Henry Company and Carlisle Coatings & Waterproofing Inc. brands)
    d. MAPEI Corporation
    e. Polycoat Products
    f. Sika Corporation
    g. Tremco Incorporated
    h. Or Approved Equal

2. Salient Characteristics:

    a. Material: Single component, rapid-curing, high solids, modified aliphatic polyurethane, ASTM C 836/C 836M and coal-tar free
    b. Formulated for application to damp and green concrete
    c. Thickness: 120 wet mil
    d. VOC Content: Less than 100 g/L, roller grade
    e. VOC Content: Less than 160 g/L, trowel detailing grade
    f. Hardness, ASTM D 2240: 70-80
    g. Low Temperature Flexibility and Crack Bridging, ASTM C 1305: Pass
    h. Adhesion in Peel, ASTM C 794: 26 lbf/in.

2.3 ACCESSORY WATERPROOFING SYSTEM MATERIALS

A. General: Accessory materials as recommended in writing by waterproofing manufacturer for intended use and compatible with one another and with waterproofing.

B. Primer: Liquid primer as recommended in writing for substrate by waterproofing manufacturer.

C. Elastomeric Detail Sheet: Blended thermoset elastomeric sheet reinforced with polyester woven scrim by waterproofing manufacturer.

D. Surface Conditioner: Liquid, waterborne surface conditioner as recommended in writing for substrate by waterproofing manufacturer.

E. Joint Sealant: Single-component or Multicomponent polyurethane sealant, compatible with waterproofing; ASTM C920, Type M, Class 25 or greater; Grade NS for sloping and vertical applications and Grade P for deck applications; Use NT exposure; and as recommended in writing by waterproofing manufacturer for substrate and joint conditions.

F. Backer Rod: Closed-cell polyethylene foam.

2.4 MOLDED-SHEET DRAINAGE PANELS

A. Nonwoven-Geotextile-Faced, Molded-Sheet Drainage Panels with Polymeric Film: Composite subsurface drainage panel acceptable to waterproofing manufacturer and consisting of a

studded, nonbiodegradable, molded-plastic-sheet drainage core; with a nonwoven, needle-punched geotextile facing with an apparent opening size not exceeding No. 70 sieve laminated to one side of the core and a polymeric film bonded to the other side; and with a vertical flow rate through the core of 9 to 21 gpm per ft..

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

1. Verify that concrete has cured and aged for minimum time period as recommended in writing by waterproofing manufacturer.
2. Verify that substrate is visibly dry and within the moisture limits as recommended in writing by waterproofing manufacturer. Test for capillary moisture by plastic sheet method in accordance with ASTM D4263.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean, prepare, and treat substrates in accordance with waterproofing manufacturer's written instructions. Provide clean, dust-free, and dry substrates for waterproofing application.

B. Mask off adjoining surfaces not receiving waterproofing to prevent spillage and overspray affecting other construction.

C. Remove grease, oil, bitumen, form-release agents, paints, curing compounds, acid residues, and other penetrating contaminants or film-forming coatings from concrete.

D. Remove fins, ridges, and other projections, and fill honeycomb, aggregate pockets, holes, and other voids.

3.3 PREPARATION AT TERMINATIONS, PENETRATIONS, AND CORNERS

A. Prepare surfaces at terminations and penetrations through waterproofing and at expansion joints, drains, sleeves, and corners in accordance with waterproofing manufacturer's written instructions and to recommendations in ASTM C898/C898M.

3.4 TREATMENT OF JOINTS AND CRACKS

A. Prepare, treat, rout, and fill joints and cracks in substrate in accordance with waterproofing manufacturer's written instructions and recommendations in ASTM C898/C898M. Before coating surfaces, remove dust and dirt from joints and cracks in accordance with ASTM D4258.

1. Comply with ASTM C1193 for joint-sealant installation.

2. Apply bond breaker on sealant surface, beneath preparation strip.
3. Prime substrate along each side of joint and apply a single thickness of preparation strip at least 6 inches wide along each side of joint. Apply waterproofing in two separate applications and embed a joint-reinforcing strip in first preparation coat.

B. Install sheet flashing and bond to deck and wall substrates where required in accordance with waterproofing manufacturer's written instructions.

3.5 INSTALLATION OF COLD FLUID-APPLIED WATERPROOFING

A. General: Apply waterproofing in accordance with manufacturer's written instructions and recommendations in ASTM C898/C898M.

B. Start installing waterproofing in presence of manufacturer's technical representative.

C. Apply primer over prepared substrate at manufacturer's recommended rate and allow it to dry.

D. Unreinforced Waterproofing Membrane Applications: Mix materials and apply waterproofing by spray, roller, notched squeegee, trowel, or other application method suitable to slope of substrate.

1. Apply one or more coats of waterproofing to obtain a seamless membrane free of entrapped gases and pinholes, with a minimum dry film thickness of 120 mils.
2. Apply waterproofing to prepared wall terminations and vertical surfaces.
3. Verify manufacturer's recommended wet film thickness of waterproofing every 100 sq. ft.

E. Reinforced Waterproofing Membrane Applications: Mix materials and apply waterproofing by roller, notched squeegee, trowel, or other suitable application method.

1. Apply first coat of waterproofing, embed membrane-reinforcing fabric, and apply second coat of waterproofing to completely saturate reinforcing fabric and to obtain a seamless reinforced membrane free of entrapped gases and pinholes, with an average dry film total thickness of 120 mils.
2. Apply reinforced waterproofing to prepared wall terminations and vertical surfaces.
3. Verify manufacturer's recommended wet film thickness of waterproofing every 100 sq. ft.

F. Cure waterproofing, taking care to prevent contamination and damage to membrane.

3.6 INSTALLATION OF MOLDED-SHEET DRAINAGE PANELS

A. Place and secure molded-sheet drainage panels in accordance with waterproofing manufacturer's written instructions. Use adhesive or another method that does not penetrate waterproofing.

3.7 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections:

1. Testing agency to verify thickness of waterproofing during application for each 600 sq. ft. of installed waterproofing or part thereof.

B. Perform the following tests before overlying construction is placed:

C. Manufacturer's Field Service: Engage a site representative qualified by waterproofing system manufacturer to inspect substrate conditions, surface preparation, membrane application, flashings, protection, and drainage components and to furnish daily reports to Contracting Officer.

1. Final Inspection: Arrange for waterproofing system manufacturer's technical personnel to inspect system installation on completion, in presence of Contracting Officer, and to prepare inspection report.
2. Notify Contracting Officer 48 hours in advance of date and time of inspection.
3. Repair or remove and replace components of waterproofing system where inspections indicate that they do not comply with specified requirements.

D. Additional Tests and Inspections:

1. Additional testing and inspecting, at Contractor's expense, will be performed to determine if replaced or additional work complies with specified requirements.
2. Waterproofing system will be considered defective if it does not pass tests and inspections.

E. Prepare test and inspection reports.

3.8 PROTECTION, REPAIR, AND CLEANING

A. Protect waterproofing system from damage and wear during remainder of construction period.

B. Protect installed drainage panels from damage due to UV light, harmful weather exposures, physical abuse, and other causes. Provide temporary coverings where material is subject to abuse and cannot be concealed and protected by permanent construction immediately after installation.

C. Correct deficiencies in or remove waterproofing system that does not comply with requirements, repair substrates, and repair or reinstall waterproofing system to a condition free of damage and deterioration at time of Substantial Completion and in accordance with warranty requirements.

D. Clean spillage and soiling from adjacent construction using cleaning agents and procedures recommended in writing by manufacturer of affected construction.

END OF SECTION 07 14 16

SECTION 07 18 00 – TRAFFIC COATINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Pedestrian traffic coatings for the following applications:

1. At building vestibule recess.
2. At existing wrap around balcony.
3. Option D: At new extension to wrap around balcony.

B. Related Requirements:

1. Section 03 30 00 “Cast in Place Concrete” for traffic coating substrate.
2. Section 07 62 00 “Sheet Metal Flashing and Trim” for adjacent flashing at wall base.
3. Section 08 41 13 “Aluminum-framed Entrances and Storefronts” for adjacent windows and doors.
4. Section 09 67 23 “Resinous Flooring” for fluid-applied, decorative resinous flooring in restrooms that does not serve as a waterproofing membrane with integral wearing surface.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.3 ACTION SUBMITTALS

A. Product Data Submittals: For each product.

1. Include installation instructions and details, material descriptions, dry- or wet-film thickness requirements, and finish.

B. Sustainable Design Submittals:

1. Product Data: For coatings, indicating VOC content.

C. Shop Drawings: For traffic coatings.

1. Include details for treating substrate joints and cracks, flashings, deck penetrations, and other termination conditions that are not included in manufacturer's product data.

D. Samples for Initial Selection: For each type of exposed finish.

E. Samples for Verification: For each type of exposed finish, prepared on rigid backing.

1. Provide stepped Samples on backing to illustrate buildup of traffic coatings.

F. Qualification Data: For Installer.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance Data: For traffic coatings to include in maintenance manuals.

B. Sample Warranty: For manufacturer's warranty.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

B. Mockups: Build mockups to set quality standards for materials and execution.

1. Build mockup for each traffic coating and substrate to receive traffic coatings.
2. Size: 3 foot x 3 foot (9 sq. ft.) to demonstrate surface preparation, joint and crack treatment, thickness, texture, color, and standard of workmanship.
3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.6 FIELD CONDITIONS

A. Environmental Limitations: Apply traffic coatings within the range of ambient and substrate temperatures recommended in writing by manufacturer. Do not apply traffic coatings to damp or wet substrates, when temperatures are below 40 deg F, when relative humidity exceeds 85 percent, or when temperatures are less than 5 deg F above dew point.

1. Do not apply traffic coatings in snow, rain, fog, or mist, or when such weather conditions are imminent during the application and curing period. Apply only when frost-free conditions occur throughout the depth of substrate.

B. Do not install traffic coating until items that penetrate membrane have been installed.

1.7 WARRANTY

A. Manufacturer's Warranty: Manufacturer agrees to repair or replace traffic coating that fails in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Adhesive or cohesive failures.
b. Abrasion or tearing failures.
c. Surface crazing or spalling.

d. Intrusion of water, oils, gasoline, grease, salt, deicer chemicals, or acids into deck substrate.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain traffic coatings from single source from single manufacturer.

B. Obtain primary traffic-coating materials, including primers, from traffic-coating manufacturer. Obtain accessory materials including aggregates, sheet flashings, joint sealants, and substrate repair materials of types and from sources recommended in writing by primary material manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Material Compatibility: Provide primers; base coat, intermediate coat, and topcoat; and accessory materials that are compatible with one another and with substrate under conditions of service and application, as demonstrated by manufacturer based on testing and field experience.

2.3 TRAFFIC COATINGS

A. Traffic Coating: Traffic-bearing, seamless, high-solids-content, cold liquid-applied, elastomeric, water-resistant membrane system with integral wearing surface for pedestrian traffic service condition; according to ASTM C957/C957M.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

   a. Waterstop/GAF
   b. Sika
   c. Tremco (Basis of design: Vulkem EWS with PUMA Technology)

2. Salient Characteristics:

   a. Material: see coats materials below
   b. Thickness: 100 mil minimum.
   c. Three coat system
   d. Designed for adherence to concrete slab
   e. Slip resistance: minimum wet DCOF (Dynamic Coefficient of Friction) value of 0.55 or greater.
   f. Puncture resistance: greater than 56 lbs, ASTM D5602.
   g. Low Temperature Crack Bridging: Passes ASTM C1305
   h. Elongation: Minimum 30%, ASTM D5147

B. Primer: Liquid primer as recommended in writing for substrate and conditions by traffic-coating manufacturer.

1. Material: Two-Component Modified Epoxy.

C. Preparatory and Base Coats: Polyurethane.

1. Thicknesses: Minimum dry- wet- film thickness as recommended in writing by manufacturer for substrate and service conditions indicated.

D. Intermediate Coat: Polyurethane.

1. Thicknesses: Minimum dry- wet- film thickness as recommended in writing by manufacturer for substrate and service conditions indicated, measured excluding aggregate.
2. Aggregate Content: As recommended in writing by traffic-coating manufacturer for substrate and service conditions indicated.

E. Topcoat: Polyurethane.

1. Thicknesses: Minimum dry- wet- film thickness as recommended in writing by manufacturer for substrate and service conditions indicated, measured excluding aggregate.
2. Aggregate Content: As required to achieve slip-resistant finish.
3. Color: Buff/Beige, color to be confirmed by Contracting Officer from manufacturer's full range.

F. Aggregate: Manufacturer's standard aggregate for each use indicated of particle sizes, shape, and minimum hardness recommended in writing by traffic-coating manufacturer.

G. VOC Content: 100 g/L or less.

H. Low-Emitting Materials: Interior coatings shall comply with the testing and product requirements of the California Department of Public Health's “Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers.”

2.4 ACCESSORY MATERIALS

A. Joint Sealants: As specified in Section 07 92 00 “Joint Sealants.”

B. Adhesive: Contact adhesive recommended in writing by traffic-coating manufacturer.

C. Reinforcing Strip: Fiberglass mesh recommended in writing by traffic-coating manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for maximum moisture content, surface smoothness, and other conditions affecting performance of traffic-coating work.

B. Verify that substrates are visibly dry and free of moisture.

1. Test for moisture according to ASTM D4263.
2. Test for moisture content by measuring with an electronic moisture meter.

C. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of traffic-coating work.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

1. Begin coating application only after substrate construction and penetrating work have been completed.
2. Begin coating application only after minimum concrete-curing and -drying period recommended in writing by traffic-coating manufacturer has passed and after substrates are dry.
3. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

A. Clean and prepare substrates according to ASTM C1127 and manufacturer's written instructions to produce clean, dust-free, dry substrate for traffic-coating application. Remove projections, fill voids, and seal joints if any, as recommended in writing by traffic-coating manufacturer.

B. Priming: Unless manufacturer recommends in writing against priming, prime substrates according to manufacturer's written instructions.

1. Limit priming to areas that will be covered by traffic-coating material on same day. Reprime areas exposed for more time than recommended by manufacturer.

C. Schedule preparation work so dust and other contaminants from process do not fall on wet, newly coated surfaces.

D. Mask adjoining surfaces not receiving traffic coatings to prevent overspray, spillage, leaking, and migration of coatings. Prevent traffic-coating materials from entering deck substrate penetrations and clogging weep holes and drains.

E. Concrete Substrates: Mechanically abrade surface to a uniform profile acceptable to manufacturer, according to ASTM D4259. Do not acid etch.

1. Remove grease, oil, paints, and other penetrating contaminants from concrete.
2. Remove concrete fins, ridges, and other projections.

3. Remove laitance, glaze, efflorescence, curing compounds, concrete hardeners, form-release agents, and other incompatible materials that might affect coating adhesion.
4. Remove remaining loose material to provide a sound surface, and clean surfaces according to ASTM D4258.

3.3 TERMINATIONS AND PENETRATIONS

A. Prepare vertical and horizontal surfaces at terminations and penetrations through traffic coatings and at expansion joints, drains, and sleeves according to ASTM C1127 and manufacturer's written instructions.

B. Provide sealant cants at penetrations and at reinforced and nonreinforced, deck-to-wall butt joints.

C. Terminate edges of deck-to-deck expansion joints with preparatory base-coat strip.

D. Install sheet flashings at deck-to-wall expansion and dynamic joints, and bond to deck and wall substrates according to manufacturer's written recommendations.

3.4 JOINT AND CRACK TREATMENT

A. Prepare, treat, rout, and fill joints and cracks in substrates according to ASTM C1127 and manufacturer's written recommendations. Before coating surfaces, remove dust and dirt from joints and cracks according to ASTM D4258.

1. Comply with recommendations in ASTM C1193 for joint-sealant installation.

B. Apply reinforcing strip in traffic-coating system where recommended in writing by traffic-coating manufacturer.

3.5 INSTALLATION OF TRAFFIC COATINGS

A. Apply traffic coating according to ASTM C1127 and manufacturer's written instructions.

B. Apply coats of specified compositions for each type of traffic coating at locations as indicated on Drawings.

C. Start traffic-coating application in presence of manufacturer's technical representative.

D. Verify that wet-film thickness of each coat complies with requirements every 100 sq. ft..

E. Uniformly broadcast and embed aggregate in each coat indicated to receive aggregate according to manufacturer's written instructions. After coat dries, sweep away excess aggregate.

F. Apply traffic coatings to prepared wall terminations and vertical surfaces to height indicated; omit aggregate on vertical surfaces.

G. Cure traffic coatings. Prevent contamination and damage during coating application and curing.

3.6 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform the following field tests and inspections:

1. Materials Testing:

a. Samples of material delivered to Project site ae to be taken, identified, sealed, and certified in presence of Contractor.
b. Testing agency must perform tests for characteristics specified, using applicable referenced testing procedures.
c. Testing agency must verify thickness of coatings during traffic-coating application for each 600 sq. ft. of installed traffic coating or part thereof.

B. Final Traffic-Coating Inspection: Arrange for traffic-coating manufacturer's technical personnel to inspect membrane installation on completion.

1. Notify Contracting Officer 48 hours in advance of date and time of inspection.

C. Waterproofing will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

3.7 PROTECTING AND CLEANING

A. Protect traffic coatings from damage and wear during remainder of construction period.

B. Clean spillage and soiling from adjacent construction using cleaning agents and procedures recommended by manufacturer of affected construction.

END OF SECTION 07 18 00

SECTION 07 21 00 – THERMAL INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Extruded polystyrene foam-plastic board insulation - at foundation walls.
2. Mineral-wool blanket insulation - in framed exterior walls and in attic.
3. Mineral-wool board insulation, unfaced - on exterior side of exterior wall sheathing.
4. Mineral-wool board insulation, faced – on interior side of masonry exterior walls.

B. Related Requirements:

1. Section 04 26 13 “Masonry Veneer” for adjacent stone wall finish.
2. Section 06 10 00 “Rough Carpentry” for adjacent wood structure.
3. Section 07 14 16 “Cold Fluid-Applied Waterproofing” for foundation waterproofing and drainage panels installed with XPS foundation insulation.
4. Section 07 21 19 “Foamed-in-Place Insulation” for spray-applied polyurethane foam insulation.
5. Section 07 26 00 “Vapor Retarders” for vapor retarders in the wall assemblies.
6. Section 07 27 15 “Nonbituminous Self-adhering Sheet Air Barriers” for vapor permeable air barriers in the wall assemblies.
7. Section 07 27 26 “Fluid-applied Membrane Air Barriers” for fluid applied vapor permeable air barriers in the wall assemblies.
8. Section 07 42 13.13 “Formed Metal Wall Panels” for adjacent decorative wall paneling.
9. Section 07 46 16 “Aluminum Cladding” for adjacent primary wall finish.
10. Section 09 29 00 “Gypsum Board” for sound attenuation blanket used as acoustic insulation.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Environmental Product Declaration: For each product.
2. Third-Party Certifications: For each product.
3. Product Data: For adhesives, indicating VOC content.
4. Laboratory Test Reports: For insulation, indicating compliance with requirements for low-emitting materials.

1.3 DELIVERY, STORAGE, AND HANDLING

A. Protect insulation materials from physical damage and from deterioration due to moisture, soiling, and other sources. Store inside and in a dry location. Comply with manufacturer's written instructions for handling, storing, and protecting during installation.

B. Protect foam-plastic board insulation as follows:

1. Do not expose to sunlight except to necessary extent for period of installation and concealment.
2. Protect against ignition at all times. Do not deliver foam-plastic board materials to Project site until just before installation time.
3. Quickly complete installation and concealment of foam-plastic board insulation in each area of construction.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Surface-Burning Characteristics: Maximum flame-spread and smoke-developed indexes less than Class A, 25 and 450 when tested in accordance with ASTM E84.

B. Fire-Resistance Ratings: Comply with ASTM E119 or UL 263; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Indicate design designations from UL's “Fire Resistance Directory” or from listings of another qualified testing agency.

C. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly.

D. Labeling: Provide identification of mark indicating R-value of each piece of insulation 12 inches and wider in width.

E. Thermal-Resistance Value (R-Value): R-value as indicated below in accordance with ASTM C518.

1. R-Value at Attic: R-49.
2. R-Value at Walls (Cavity) - R-21
3. R-Value at Walls (Continuous) - R-8.4ci
4. R-Value at Walls Below Grade - R7.5ci
5. Under slabs - None

F. Verify insulation complies with requirements of the California Department of Public Health's “Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers.” Verify formaldehyde emissions do not exceed 16.5 mcg/cu. m or 13.5 ppb, whichever is less, except for insulation manufactured without formaldehyde.

G. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 30 percent.

1. Verify adhesives have a VOC content of 70 g/L or less.
2. Verify adhesive complies with testing and product requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers." Verify formaldehyde emissions do not exceed 9 mcg/cu. m or 7 ppb, whichever is less.

2.2 EXTRUDED POLYSTYRENE FOAM-PLASTIC BOARD INSULATION (XPS)

A. Extruded Polystyrene Board Insulation, Type VI: ASTM C578, Type VI, 40 psi minimum compressive strength.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

a. HydroBlok
b. Owens Corning
c. The Dow Chemical Company

2.3 MINERAL-WOOL BLANKET INSULATION

A. Mineral-Wool Blanket Insulation, Unfaced: ASTM C665, Type I (blankets without membrane facing); consisting of fibers; complying with ASTM E136 for combustion characteristics.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Rockwool; Comfortbatt or comparable product by one of the following:

a. Johns Manville; a Berkshire Hathaway company
b. Owens Corning
c. ROCKWOOL
d. Or Approved Equal

2. Salient Characteristics:

a. R-Value: R-21 exterior walls, R-30 attic
b. Density: > 2 lbs/ft^2 (>32 kg/m^3), nominal.
c. Surface Burning Characteristics: Tested in accordance with ASTM E84

1) Unfaced: Flame Spread 0 and Smoke Developed 0

d. Moisture Resistance: Absorption of less than 0.03 percent by volume, when tested in accordance with ASTM C1104.
e. Corrosion Resistance: Non-corrosive/Passed, when tested in accordance with ASTM C665 for Steel & ASTM C795 for Stress Corrosion Cracking Tendency of Austenitic Stainless Steel.
f. Fungi resistance: Zero mold growth to ASTM C1338

g. Recyclability: Material to be capable of being fully recyclable at end of life with the intention of sending zero waste to landfill.
h. Environmental Product Declaration (EPD): Material must be included on a UL Certified EPD in accordance with EN 15804 and ISO 14025.
i. GREENGUARD Gold Certified

2.4 MINERAL-WOOL BOARD INSULATION

A. Mineral-Wool Board Insulation, Unfaced: Type IVB : ASTM C612, Type IVB; unfaced.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Rockwool; Cavityrock or comparable product by one of the following:

a. Johns Manville; a Berkshire Hathaway company
b. Owens Corning
c. ROCKWOOL
d. Or Approved Equal

2. Salient Characteristics:

a. R-value: 4.2 /inch at 75°F.
b. Facing: Unfaced
c. Melting Point – Minimum melting point temperature of 1177°C (2150°F).
d. Surface Burning Characteristics: Tested in accordance with ASTM E84

1) Unfaced: Flame Spread 0 and Smoke Developed 0

e. Monolithic Density (Thickness: 2”): > 4.3 lbs/ft³ (>69 kg/m³), nominal.
f. Moisture Resistance: Absorption of less than 0.03 percent by volume, when tested in accordance with ASTM C1104.
g. Corrosion Resistance: Non-corrosive/Passed, when tested in accordance with ASTM C665 for Steel & ASTM C795 for Stress Corrosion Cracking Tendency of Austenitic Stainless Steel.
h. Fungi resistance: Zero mold growth to ASTM C1338
i. Linear Shrinkage at 1200°F (650°C): Average linear shrinkage in all dimensions not to exceed 0.7% when tested to ASTM C356 at 1200°F (650°C).
j. Tumbling Friability: Material loss not to exceed 15% when tested to ASTM C421 – Standard Test Method for Tumbling Friability of Preformed Block-Type Thermal Insulation.
k. Recyclability: Material to be capable of being fully recyclable at end of life with the intention of sending zero waste to landfill.
l. Environmental Product Declaration (EPD): Material must be included on a UL Certified EPD in accordance with EN 15804 and ISO 14025.
m. DECLARE Certified

B. Mineral-Wool Board Insulation, Faced: Type IVB : ASTM C612, AC528 Humidity-Dependent Vapor Retarder

1. Basis-of-Design Product: Subject to compliance with requirements, provide Rockwool; Smartrock or comparable product by one of the following:

a. Johns Manville; a Berkshire Hathaway company
b. Owens Corning
c. ROCKWOOL
d. Or Approved Equal

2. Salient Characteristics:

a. R-value: 4.2 /inch at 75°F per ASTM C518.
b. Facing: Faced with vapor variable membrane.
c. Surface Burning Characteristics: Tested in accordance with ASTM E84

1) Flame Spread 15 and Smoke Developed 20

d. Dual Density (Thickness: ≥ 2.5"): 6.2 lbs/ft³ (100 kg/m³) outer layer and 3.8 lbs/ft³ (61 kg/m³) inner layer, nominal.
e. Acceptance Criteria for Humidity-Dependent Vapor Retarder with AC528
f. Acceptance Criteria for Factory Bonded Humidity-Dependent Vapor Retarder Membranes to Rigid Insulation Board with AC566
g. Reaction to Moisture (per DIN EN ISO 12572) for Humidity dependent membrane:

1) Test Condition (73 deg, 0/50 % R.H.); Avg Humidity 25% R.H.; Water Vapor Permeance 0.09 perm
2) Test Condition (73 deg, 50/93 % R.H.); Avg Humidity 71.5% R.H.; Water Vapor Permeance 1.19 perm
3) Test Condition (73 deg, 85/95 % R.H.); Avg Humidity 90% R.H.; Water Vapor Permeance 6.03 perm

h. Reaction to Moisture (per ASTM E96) for Mineral Wool Insulation:

1) Water Vapor Transmission (3.75 in. thickness evaluated), Desiccant Method, 27 perm

i. Moisture Resistance: Absorption of 0.16 percent by volume, when tested in accordance with ASTM C1104.
j. Corrosion Resistance: Non-corrosive/Passed, when tested in accordance with ASTM C1617.
k. Fungi resistance: Zero mold growth to ASTM C1338
l. Linear Shrinkage at 1200°F (650°C): Average linear shrinkage in all dimensions not to exceed 0.6% when tested to ASTM C356 at 1200°F (650°C).
m. Recyclability: Material to be capable of being fully recyclable at end of life with the intention of sending zero waste to landfill.
n. DECLARE Certified

2.5 INSULATION FASTENERS

A. Insulation Fastener Accessories: Provide double-pointed weld pins, lagging pins, quilting pins, duct liner pins, insulation hangers, specialty washers, special caps, j-hooks, capacitor discharge annular weld pins, capacitor discharge acoustical lagging pins, and other accessory materials

that are recommended in writing by insulation fastener manufacturer to produce complete insulation supports.

2.6 ACCESSORIES

A. Miscellaneous Application Accessories:

1. Adhesive for Bonding Insulation: Product compatible with insulation and air and water barrier materials, and with demonstrated capability to bond insulation securely to substrates without damaging insulation and substrates.
2. Crack Sealer: Closed-cell insulating foam in aerosol dispenser recommended in writing by insulation manufacturer for filling gaps in board insulation.
3. Clip-and-Pin Components:

    a. Beam/Bar Joist Clips: For beams, bar joists, and Z-type purlins.
    b. Tube Clips: For wood beams and metal tubular framing.
    c. Locking Washers: Aluminum; white to match reflective bubble insulation facing colors.

4. Tape for seams, transition details, fastener heads, and penetrations in faced applications:

    a. Vapor-permeable or highlight vapor semi-permeable.
    b. And compliant with manufacturer recommendations.

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean substrates of substances that are harmful to insulation, including removing projections capable of puncturing insulation or vapor retarders, or those that interfere with insulation attachment.

3.2 INSTALLATION, GENERAL

A. Comply with insulation manufacturer's written instructions applicable to products, applications and applicable codes.

B. Install insulation that is undamaged, dry, and unsoiled and that has not been left exposed to ice, rain, or snow at any time.

C. Install insulation with manufacturer's R-value label exposed after insulation is installed.

D. Extend insulation to envelop entire area to be insulated. Fit tightly around obstructions and fill voids with insulation. Remove projections that interfere with placement.

E. Provide sizes to fit applications and selected from manufacturer's standard thicknesses, widths, and lengths. Apply single layer of insulation units unless multiple layers are otherwise shown or required to make up total thickness or to achieve R-value.

3.3 INSTALLATION OF FOUNDATION WALL INSULATION

A. Butt panels together for tight fit.

B. Anchor Installation: Install board insulation on concrete substrates by adhesively attached, spindle-type insulation anchors as follows:

1. Fasten insulation anchors to concrete substrates with insulation anchor adhesive in accordance with anchor manufacturer's written instructions.
2. Space anchors in accordance with insulation manufacturer's written instructions for insulation type, thickness, and application.
3. Apply insulation standoffs to each spindle to create cavity width indicated on Drawings between concrete substrate and insulation.
4. After adhesive has dried, install board insulation by pressing insulation into position over spindles and securing it tightly in place with insulation-retaining washers, taking care not to compress insulation.
5. Where insulation will not be covered by other building materials, apply capped washers to tips of spindles.

C. Adhesive Installation: Install with adhesive or press into tacky waterproofing or dampproofing in accordance with manufacturer's written instructions.

3.4 INSTALLATION OF CAVITY-WALL INSULATION

A. Mineral-Wool Board Insulation: Install insulation fasteners 4 inches from each corner of board insulation, at center of board, and as recommended by manufacturer.

1. Fit courses of insulation between masonry wall ties and other obstructions, with edges butted tightly in both directions, and with faces flush.
2. Press units firmly against inside substrates.

3.5 INSTALLATION OF INSULATION IN FRAMED CONSTRUCTION

A. Blanket Insulation: Install in cavities formed by framing members in accordance with the following requirements:

1. Use insulation widths and lengths that fill the cavities formed by framing members. If more than one length is required to fill the cavities, provide lengths that will produce a snug fit between ends.
2. Place insulation in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.
3. Maintain 3-inch clearance of insulation around recessed lighting fixtures not rated for or protected from contact with insulation.
4. Attics: Install eave ventilation troughs between roof framing members in insulated attic spaces at vented eaves.
5. For wood-framed construction, install blankets in accordance with ASTM C1320.
6. At attic, with faced blankets having stapling flanges, lap blanket flange over flange of adjacent blanket to maintain continuity of vapor retarder once finish material is installed over it.

B. Miscellaneous Voids: Install insulation in miscellaneous voids and cavity spaces where required to prevent gaps in insulation using the following materials:

1. Detailing Foam Insulation for Voids: Apply in accordance with manufacturer's written instructions.

3.6 INSTALLATION OF BOARD INSULATION

A. Install board insulation in accordance with manufacturer's written instructions per project applications and conditions.

B. At faced board insulation, tape seams and edges in accordance with manufacturer's written instructions with products approved for use with their product.

3.7 PROTECTION

A. Protect installed insulation from damage due to harmful weather exposures, physical abuse, and other causes.

B. Provide temporary coverings or enclosures where insulation is subject to abuse and cannot be concealed and protected by permanent construction immediately after installation.

END OF SECTION 07 21 00

SECTION 07 21 19 – FOAMED-IN-PLACE INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Closed-cell spray polyurethane foam insulation for misc. voids or as noted in drawings.

B. Related Requirements:

1. Section 072100 “Thermal Insulation” for XPS and mineral wool insulation.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Qualification Statements: For Installer.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality control reports.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer.

PART 2 - PRODUCTS

2.1 CLOSED-CELL SPRAY POLYURETHANE FOAM INSULATION

A. Closed-Cell Spray Polyurethane Foam: ASTM C1029.

1. Type and Minimum Compressive Strength:

a. Type II, minimum density of 1.5 lb/cu. ft. and minimum aged R-value at 1-inch thickness of 6.2 deg F x h x sq. ft./Btu at 75 deg F.

2. Surface-Burning Characteristics: Comply with ASTM E84; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

    a. Flame-Spread Index: 25 or less.
    b. Smoke-Developed Index: 450 or less.
    c. Fire-Propagation Characteristics: Passes NFPA 285 and NFPA 276 testing as part of an approved assembly.

PART 3 - EXECUTION

3.1 PREPARATION

A. Verify that substrates are clean, dry, and free of substances that are harmful to insulation.

B. Priming: Prime substrates where recommended by insulation manufacturer. Apply primer to comply with insulation manufacturer's written instructions. Confine primers to areas to be insulated; do not allow spillage or migration onto adjoining surfaces.

3.2 INSTALLATION OF FOAMED-IN-PLACE INSULATION

A. Comply with insulation manufacturer's written instructions applicable to products and applications.

B. Spray insulation to envelop entire area to be insulated and fill voids.

C. Apply in multiple passes to not exceed maximum thicknesses recommended by manufacturer. Do not spray into rising foam.

D. Framed Construction: Install into cavities formed by framing members to achieve thickness indicated on Drawings.

E. Miscellaneous Voids: Apply according to manufacturer's written instructions.

F. Apply barrier coatings in accordance with manufacturer's written instructions and to comply with requirements for listing and labeling for fire-propagation characteristics and surface-burning characteristics specified.

1. Use equipment and techniques best suited for substrate and type of material applied as recommended by coating manufacturer.
2. Apply coatings to prepared surfaces as soon as practical after preparation and before subsequent surface soiling or deterioration.
3. Apply coatings to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Produce sharp lines and color breaks.

3.3 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect spray foam insulation installation, including accessories. Report results in writing.

3.4 PROTECTION

A. Protect installed insulation from damage due to harmful weather exposures, physical abuse, and other causes.

END OF SECTION 07 21 19

SECTION 07 26 00 – VAPOR RETARDERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Polyethylene vapor retarder to be used at interior sides of exterior walls.
2. Reinforced-polyethylene vapor retarder to be used in crawlspace.

B. Related Requirements:

1. Section 033000 “Cast-in-Place Concrete” for under-slab vapor retarders.
2. Section 06 10 00 “Rough Carpentry” for substrate material.
3. Section 072100 “Thermal Insulation” for air barriers integral with insulation products.
4. Section 07 27 26 “Fluid-applied Membrane Air Barrier” for fluid applied vapor-permeable air barriers.
5. Section 09 29 00 “Gypsum Board” for adjacent material.

1.2 ACTION SUBMITTALS

A. Product Data:

1. Polyethylene vapor retarders.
2. Reinforced-polyethylene vapor retarders.

1.3 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For each product, for tests performed by a qualified testing agency.

PART 2 - PRODUCTS

2.1 POLYETHYLENE VAPOR RETARDERS

A. Polyethylene Vapor Retarders: ASTM D4397, minimum 6 mil thickness, with maximum permeance rating of 0.1 perm.

2.2 REINFORCED-POLYETHYLENE VAPOR RETARDERS

A. Reinforced-Polyethylene Vapor Retarders: Sheet with outer layers of polyethylene film laminated to an inner reinforcing layer consisting of either nylon cord or polyester scrim with maximum permeance rating of 0.1 perm.

1. Basis of Design: Reef Industries Griffolyn TX-1600
2. Manufacturers: Subject to compliance with salient characteristics, provide products by one of the following:

    a. ISI Building Products
    b. Reef Industries, Inc.
    c. Viaflex
    d. W. R. Meadows, Inc
    e. Stego
    f. Or Approved Equal.

3. Salient Characteristics:

    a. Class I Vapor-Retarder, with maximum permeance rating of 0.1 perm per ASTM E96
    b. Weight: not less than 50 lb/1000 sqft. per ASTM D751
    c. Not less than three layers, including two layers of polyethelyne and one layer of high strength cord grid.
    d. Cold crack resistance
    e. Class C, ASTM E1745-11 for vapor retarders in contact with soil
    f. 3” Load @ Yield: 110 LBF per ASTM D882
    g. 3” Load @ Break: 92 LBF/2600 PSI per ASTM D882
    h. 3” Elongation @ Break: 475% per ASTM D882
    i. Tongue Tear: 36 LBF per ASTM D2261
    j. Trapezoidal Tear: 36 LBF per ASTM D4533
    k. PPT Resistance: 38 LBF per ASTM D2582
    l. Dart Impact Strength: 1.30 LBS per ASTM D1709
    m. Cold Impact Strength: -40 deg F per ASTM D1790
    n. Color: White

2.3 ACCESSORIES

A. Vapor-Retarder Tape: Pressure-sensitive tape of type recommended by vapor-retarder manufacturer for sealing joints and penetrations in vapor retarder.

B. Adhesive for Vapor Retarders: Product recommended by vapor-retarder manufacturer and has demonstrated capability to bond vapor retarders securely to substrates indicated.

C. Vapor-Retarder Fasteners: Pancake-head, self-tapping steel drill screws; with fender washers.

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean substrates of substances that are harmful to vapor retarders, including removing projections capable of puncturing vapor retarders.

3.2 INSTALLATION OF VAPOR RETARDERS ON FRAMING

A. Place vapor retarders on side of construction indicated on Drawings.

B. Extend vapor retarders to extremities of areas to protect from vapor transmission. Secure vapor retarders in place with adhesives, vapor retarder fasteners, or other anchorage system as recommended by manufacturer. Extend vapor retarders to cover miscellaneous voids in insulated substrates, including those filled with loose-fiber insulation.

C. Seal vertical joints in vapor retarders over framing by lapping no fewer than two studs and sealing with vapor-retarder tape according to vapor-retarder manufacturer's written instructions. Locate all joints over framing members or other solid substrates.

D. Seal joints caused by pipes, conduits, electrical boxes, and similar items penetrating vapor retarders with vapor-retarder tape to create an airtight seal between penetrating objects and vapor retarders.

E. Repair tears or punctures in vapor retarders immediately before concealment by other work. Cover with vapor-retarder tape or another layer of vapor retarders.

3.3 INSTALLATION OF VAPOR RETARDERS IN CRAWL SPACES

A. Install vapor retarders over prepared grade. Lap joints a minimum of 12 inches and seal with manufacturer's recommended tape. Install second layer over pathways to equipment.

B. Extend vapor retarder over footings and seal to foundation wall or grade beam with manufacturer's recommended tape.

1. Extend vapor retarder vertically minimum 24 inches above top of grade in crawlspace.

C. Seal around penetrations such as utilities and columns in order to create a monolithic, airtight membrane at grade surface, perimeter, and all vertical penetrations.

3.4 PROTECTION

A. Protect vapor retarders from damage until concealed by permanent construction.

END OF SECTION 07 26 00

SECTION 07 27 15 – NONBITUMINOUS SELF-ADHERING SHEET AIR BARRIERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Self-adhering air barrier.

1. Vapor-permeable nonbituminous self-adhering sheet air barriers, to be installed at exterior side of building sheathing where indicated in the drawings.
2. Acts as vapor permeable weather barrier.

B. Related Requirements:

1. Section 04 26 13 “Masonry Veneer” for adjacent accent wall finish.
2. Section 06 16 00 “Sheathing” for adjacent substrate.
3. Section 07 21 00 “Thermal Insulation” for adjacent rigid insulation.
4. Section 07 26 00 “Vapor Retarders” for vapor-retarding air barriers
5. Section 07 27 26 “Fluid-applied Membrane Air Barrier” for fluid applied vapor-permeable air barriers.
6. Section 07 42 13.13 “Formed Metal Wall Panels” for adjacent decorative wall paneling.
7. Section 07 46 16 “Aluminum Cladding” for adjacent primary wall finish.
8. Section 08 41 13 “Aluminum-framed Entrances and Storefronts” for adjacent windows.

1.2 DEFINITIONS

A. Air-Barrier Accessory: A transitional component of the air barrier that provides continuity.

B. Air-Barrier Assembly: The collection of air-barrier materials and accessories applied to an opaque wall, including joints and junctions to abutting construction, to control air movement through the wall.

C. Air-Barrier Material: A primary element that provides a continuous barrier to the movement of air.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review air-barrier requirements and installation, special details, mockups, air-leakage and bond testing, air-barrier protection, and work scheduling that covers air barriers.

1.4 ACTION SUBMITTALS

A. Product Data: Self-adhering air barrier. Include manufacturer's written instructions for evaluating, preparing, and treating each substrate; technical data; and tested physical and performance properties of products.

1. Vapor-permeable nonbituminous sheet.

B. Shop Drawings: For air-barrier assemblies.

1. Show locations and extent of air-barrier materials, accessories, and assemblies specific to Project conditions.
2. Include details for substrate joints and cracks, counterflashing strips, penetrations, inside and outside corners, terminations, and tie-ins with adjoining construction.
3. Include details of interfaces with other materials that form part of air barrier.

C. Qualification Data: For Installer. Include list of ABAA-certified installers and supervisors employed by Installer, who work on Project.

1.5 INFORMATIONAL SUBMITTALS

A. Product Certificates: From air-barrier manufacturer, certifying compatibility of air barriers and accessory materials with Project materials that connect to or that come in contact with air barrier.

B. Product Test Reports: For each air-barrier assembly, for tests performed by a qualified testing agency.

C. Field quality-control reports.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1. Installer to be licensed by ABAA in accordance with ABAA's Quality Assurance Program and to employ ABAA-certified installers and supervisors on Project.

B. Mockups: Build mockups to set quality standards for materials and execution.

1. Build integrated mockups of exterior wall assembly 150 sq. ft., incorporating backup wall construction, external cladding, window, storefront, door frame and sill, insulation, ties and other penetrations, and flashing to demonstrate surface preparation, crack and joint treatment, application of air barriers, and sealing of gaps, terminations, and penetrations of air-barrier assembly.

a. Coordinate construction of mockups to permit inspection of air barrier before external insulation and cladding are installed.

b. Include building corner condition, and foundation wall intersection.
c. If Contracting Officer determines mockups do not comply with requirements, reconstruct mockups and apply air barrier until mockups are approved.

2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion if approved in writing by the Contracting Officer.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Remove and replace liquid materials that cannot be applied within their stated shelf life.

B. Protect stored materials from direct sunlight.

1.8 FIELD CONDITIONS

A. Environmental Limitations: Apply air barrier within the range of ambient and substrate temperatures recommended in writing by air-barrier manufacturer.

1. Protect substrates from environmental conditions that affect air-barrier performance.
2. Do not apply air barrier to a damp or wet substrate or during snow, rain, fog, or mist.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain primary air-barrier materials and air-barrier accessories from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Air-Barrier Performance: Air-barrier assembly and seals with adjacent construction to be capable of performing as a continuous air barrier and as a liquid-water drainage plane flashed to discharge to the exterior incidental condensation or water penetration. Air-barrier assemblies to be capable of accommodating substrate movement and of sealing substrate expansion and control joints, construction material changes, penetrations, tie-ins to installed foundation waterproofing, and transitions at perimeter conditions without deterioration and air leakage exceeding specified limits.

B. Air-Barrier Assembly Air Leakage: Maximum 0.04 cfm/sq. ft. of surface area at 1.57 lbf/sq. ft., when tested in accordance with ASTM E2357.

2.3 NONBITUMINOUS SHEET AIR BARRIER

A. Vapor-Permeable Nonbituminous Sheet: Minimum 20-mil- thick, self-adhering sheet consisting of a breathable carrier film or fabric and an adhesive with release liner on adhesive side and if applicable to system, formulated for application with primer that complies with VOC limits.

1. Basis of Design: Tremco Exoair 210AT
2. Manufacturers: Subject to compliance with salient characteristics, provide products by one of the following:

    a. 3M Building and Construction
    b. Carlisle Coatings & Waterproofing Inc
    c. Dorken Systems Inc.
    d. GCP Applied Technologies Inc.
    e. Tremco Commercial Sealants and Waterproofing, part of Tremco CPG
    f. VaproShield LLC
    g. W. R. Meadows, Inc
    h. Or Approved Equal

3. Salient Characteristics:

    a. Thickness: Minimum 20 mil
    b. Color: White or black.
    c. 100% Solids
    d. Application: Self-adhered.
    e. Air Permeance: Maximum 0.004 cfm/sq. ft. of surface area at 1.57-lbf/sq. ft. pressure difference; ASTM E2178.
    f. Facer Basis Weight: 2.95 oz/sy (100 gsm)
    g. Criteria for Water Resistive Barriers: Passes ICC-ES AC38 testing.
    h. Water Resistance: Passes AATCC-127 testing
    i. Maximum VOC: 2 g/L per Method 310
    j. Air Leakage of Material: Minimum 0.003 cfm/sqft
    k. Puncture Resistance: Minimum 40 lbf; ASTM E154/E154M.
    l. Vapor Permeance: Minimum 16.9 perms; ASTM E96/E96M, Desiccant Method, Procedure B.
    m. Adhesion to Substrate: Minimum 20 lbf/sq. in. when tested in accordance with ASTM D4541 as modified by ABAA.
    n. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly.
    o. Low Temperature Flexibility: Passes ASTM D4541 testing
    p. UV Resistance: Can be exposed to sunlight for 150 days in accordance with manufacturer's written instructions.

2.4 ACCESSORY MATERIALS

A. Requirement: Provide primers, transition strips, termination strips, joint sealants, counterflashing strips, flashing sheets and metal termination bars, termination mastic, substrate patching materials, adhesives, tapes, foam sealants, lap sealants, and other accessory materials that are recommended in writing by air-barrier manufacturer to produce a complete air-barrier

assembly and that are compatible with primary air-barrier material and adjacent construction to which they may seal.

B. Primer: Low-VOC liquid primer recommended for substrate by air-barrier material manufacturer.

C. Stainless Steel Sheet: ASTM A240/A240M, Type 304, 0.0187 inch thick, and Series 300 stainless steel fasteners.

D. Preformed Silicone Extrusion: Manufacturer's standard system consisting of cured low-modulus silicone extrusion, sized to fit opening widths, with a single-component, neutral-curing, Class 100/50 (low-modulus) silicone sealant for bonding extrusions to substrates. To be provided where necessary to achieve an IECC code compliant air barrier system.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

1. Verify that substrates are sound and free of oil, grease, dirt, excess mortar, or other contaminants.
2. Verify that substrates have cured and aged for minimum time recommended in writing by air-barrier manufacturer.
3. Verify that substrates are visibly dry and free of moisture.
4. Verify that masonry joints are flush and completely filled with mortar.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 SURFACE PREPARATION

A. Clean, prepare, treat, fill, and seal substrate and joints and cracks in substrate in accordance with manufacturer's written instructions and details. Provide clean, dust-free, and dry substrate for air-barrier application.

B. Mask off adjoining surfaces not covered by air barrier to prevent spillage and overspray affecting other construction.

C. Remove grease, oil, bitumen, form-release agents, paints, curing compounds, and other penetrating contaminants or film-forming coatings from concrete.

D. Remove fins, ridges, mortar, and other projections and fill honeycomb, aggregate pockets, holes, and other voids in concrete with substrate-patching membrane.

E. Remove excess mortar from masonry ties, shelf angles, and other obstructions.

F. At changes in substrate plane, apply sealant or termination mastic beads at sharp corners and edges to form a smooth transition from one plane to another.

G. Cover gaps in substrate plane and form a smooth transition from one substrate plane to another with stainless steel sheet mechanically fastened to structural framing to provide continuous support for air barrier.

H. Bridge isolation joints and discontinuous wall-to-wall, deck-to-wall, and deck-to-deck joints with air-barrier accessory material that accommodates joint movement in accordance with manufacturer's written instructions and details.

3.3 INSTALLATION OF NONBITUMINOUS SHEET AIR BARRIER

A. Install materials in accordance with air-barrier manufacturer's written instructions and details to form a seal with adjacent construction and ensure continuity of air and water barrier.

1. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.

B. Prepare, treat, and seal inside and outside corners and vertical and horizontal surfaces at terminations and penetrations with termination mastic.

C. Apply primer to substrates at required rate and allow it to dry. Limit priming to areas that will be covered by air-barrier sheet on same day. Reprime areas exposed for more than 24 hours.

D. Apply and firmly adhere air-barrier sheets over area to receive air barrier. Accurately align sheets and maintain uniform 2-1/2-inch- minimum lap widths and end laps. Overlap and seal seams, and stagger end laps to ensure airtight installation.

1. Apply sheets in a shingled manner to shed water.
2. Roll sheets firmly to enhance adhesion to substrate.

E. Apply continuous air-barrier sheets over accessory strips bridging substrate cracks, construction, and contraction joints.

F. CMU: Install air-barrier sheet horizontally against the CMU beginning at base of wall. Align top edge of air-barrier sheet immediately below protruding masonry ties or joint reinforcement or ties, and firmly adhere in place.

1. Overlap horizontally adjacent sheets a minimum of 2 inches and roll seams.
2. Apply overlapping sheets with bottom edge slit to fit around masonry reinforcing or ties. Roll firmly into place.
3. Seal around masonry reinforcing or ties and penetrations with termination mastic.
4. Continue the sheet into all openings in the wall, such as doors and windows, and terminate at points to maintain an airtight barrier that is not visible from interior.

G. Seal top of through-wall flashings to air-barrier sheet with an additional 6-inch- wide, transition strip.

H. Seal exposed edges of sheet at seams, cuts, penetrations, and terminations not concealed by metal counterflashings or ending in reglets with termination mastic.

I. Install air-barrier sheet and accessory materials to form a seal with adjacent construction and to maintain a continuous air barrier.

1. Coordinate air-barrier installation with installation of roofing membrane and base flashing to ensure continuity of air barrier with roofing membrane.
2. Install transition strip on roofing membrane or base flashing so that a minimum of 3 inches of coverage is achieved over each substrate.

J. Connect and seal exterior wall air-barrier sheet continuously to roofing-membrane air barrier, concrete below-grade structures, floor-to-floor construction, exterior glazing and window systems, glazed curtain-wall systems, storefront systems, exterior louvers, exterior door framing, and other construction used in exterior wall openings, using accessory materials.

K. At end of each working day, seal top edge of air-barrier material to substrate with termination mastic.

L. Apply joint sealants forming part of air-barrier assembly within manufacturer's recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges.

M. Wall Openings: Prime concealed, perimeter frame surfaces of windows, curtain walls, storefronts, and doors. Apply transition strip or preformed silicone extrusion so that a minimum of 3 inches of coverage is achieved over each substrate. Maintain 3 inches of contact over firm bearing to perimeter frames, with not less than 1 inch of full contact.

1. Transition Strip: Roll firmly to enhance adhesion.
2. Preformed Silicone Extrusion: Set in full bed of silicone sealant applied to walls, frame, and air-barrier material.

N. Fill gaps in perimeter frame surfaces of windows, curtain walls, storefronts, doors, and miscellaneous penetrations of air-barrier material with foam sealant.

O. Repair punctures, voids, and deficient lapped seams in air barrier. Slit and flatten fishmouths and blisters. Patch with air-barrier sheet extending 6 inches beyond repaired areas in all directions.

P. Do not cover air barrier until it has been tested and inspected by testing agency.

Q. Correct deficiencies in or remove air barrier that does not comply with requirements; repair substrates and reapply air-barrier components.

3.4 FIELD QUALITY CONTROL

A. ABAA Quality Assurance Program: Perform examinations, preparation, installation, testing, and inspections under ABAA's Quality Assurance Program.

B. Testing Agency: Engage a qualified testing agency to perform inspections.

C. Inspections: Air-barrier materials, accessories, and installation are subject to inspection for compliance with requirements. Inspections may include the following:

1. Continuity of air-barrier system has been achieved throughout the building envelope with no gaps or holes.
2. Continuous structural support of air-barrier system has been provided.
3. Masonry and concrete surfaces are smooth, clean, and free of cavities, protrusions, and mortar droppings.
4. Site conditions for application temperature and dryness of substrates have been maintained.
5. Maximum exposure time of materials to UV deterioration has not been exceeded.
6. Surfaces have been primed.
7. Laps in sheet materials have complied with the minimum requirements and have been shingled in the correct direction (or mastic applied on exposed edges), with no fishmouths.
8. Termination mastic has been applied on cut edges.
9. Air barrier has been firmly adhered to substrate.
10. Compatible materials have been used.
11. Transitions at changes in direction and structural support at gaps have been provided.
12. Connections between assemblies (air barrier and sealants) have complied with requirements for cleanliness, surface preparation and priming, structural support, integrity, and continuity of seal.
13. All penetrations have been sealed.

D. Repair damage to air barriers caused by testing or during construction; follow manufacturer's written instructions.

E. Prepare test and inspection reports.

3.5 CLEANING AND PROTECTION

A. Protect air-barrier system from damage during application and remainder of construction period, in accordance with manufacturer's written instructions.

1. Protect air barrier from exposure to UV light and harmful weather exposure as recommended in writing by manufacturer. If exposed to these conditions for longer than recommended, remove and replace air barrier or install additional, full-thickness, air-barrier application after repairing and preparing the overexposed materials in accordance with air-barrier manufacturer's written instructions.
2. Protect air barrier from contact with incompatible materials and sealants not approved by air-barrier manufacturer.

B. Clean spills, stains, and soiling from construction that would be exposed in the completed Work, using cleaning agents and procedures recommended in writing by manufacturer of affected construction.

END OF SECTION 07 27 15

SECTION 07 27 26 – FLUID-APPLIED MEMBRANE AIR BARRIERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Medium-build air barrier, water-resistant and vapor permeable, to be installed at existing masonry walls where indicated in the drawings.
2. Acts as vapor permeable weather barrier where installed on exterior side of masonry.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry” for adjacent wood framing.
2. Section 07 21 00 “Thermal Insulation” for adjacent rigid insulation.
3. Section 07 27 15 “Non-bituminous Self-adhering Sheet Air Barriers” for air barrier installed on adjacent framed walls.
4. Section 07 46 16 “Aluminum Cladding” for adjacent primary wall finish.

1.2 DEFINITIONS

A. Air-Barrier Accessory: A transitional component of the air barrier that provides continuity.

B. Air-Barrier Assembly: The collection of air-barrier materials and accessories applied to an opaque wall, including joints and junctions to abutting construction, to control air movement through the wall.

C. Air-Barrier Material: A primary element that provides a continuous barrier to the movement of air.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review air barrier requirements and installation, special details, mockups, air-leakage and bond testing, air barrier protection, and work scheduling that covers air barriers.

1.4 ACTION SUBMITTALS

A. Product Data: Include manufacturer's written instructions for evaluating, preparing, and treating each substrate; technical data; dry film thickness; and tested physical and performance properties of products.

1. Medium-build air barriers, vapor permeable.

B. Shop Drawings: For air barrier assemblies.

1. Show locations and extent of air barrier materials, accessories, and assemblies specific to Project conditions.
2. Include details for substrate joints and cracks, counterflashing strips, penetrations, inside and outside corners, terminations, and tie-ins with adjoining construction.
3. Include details of interfaces with other materials that form part of air barrier.

C. Qualification Data: For Installer.

D. Field quality-control reports.

1.5 INFORMATIONAL SUBMITTALS

A. Product Certificates: From air barrier manufacturer, certifying compatibility of air barriers and accessory materials with Project materials that connect to or that come in contact with the barrier.

B. Product Test Reports: For each air barrier assembly, for tests performed by a qualified testing agency.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Remove and replace liquid materials that cannot be applied within their stated shelf life.

B. Protect stored materials from direct sunlight.

1.8 FIELD CONDITIONS

A. Environmental Limitations: Apply air barrier within the range of ambient and substrate temperatures recommended in writing by air barrier manufacturer.

1. Protect substrates from environmental conditions that affect air barrier performance.
2. Do not apply air barrier to a damp or wet substrate or during snow, rain, fog, or mist.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain primary air barrier materials and air barrier accessories from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Air Barrier Performance: Air barrier assembly and seals with adjacent construction to be capable of performing as a continuous air barrier and as a liquid-water drainage plane flashed to discharge to the exterior incidental condensation or water penetration. Air barrier assemblies to be capable of accommodating substrate movement and of sealing substrate expansion and control joints, construction material changes, penetrations , tie-ins to installed waterproofing, and transitions at perimeter conditions without deterioration and air leakage exceeding specified limits.

B. Air Barrier Assembly Air Leakage: Maximum 0.04 cfm/sq. ft. of surface area at 1.57 lbf/sq. ft., when tested in accordance with ASTM E2357.

C. Air Permeance: Maximum 0.004 cfm/sq. ft. of surface area at 1.57 lbf/sq. ft. pressure difference; ASTM E2178.

D. Ultimate Elongation: Minimum 500 percent; ASTM D412, Die C.

E. Adhesion to Substrate: Minimum 30 lbf/sq. in. when tested in accordance with ASTM D4541.

F. UV Resistance: Can be exposed to sunlight for 360 days in accordance with manufacturer's written instructions.

2.3 MEDIUM-BUILD AIR BARRIERS, VAPOR PERMEABLE

A. Medium-Build, Vapor-Permeable Air Barrier: Synthetic polymer material with an installed dry film thickness, according to manufacturer's written instructions, of 16 to 34 mils over smooth, void-free substrates.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Tremco Exoair 230 or comparable product by one of the following:

    a. 3M Industrial Adhesives and Tapes Division
    b. Carlisle Coatings & Waterproofing Inc
    c. DuPont de Nemours, Inc.
    d. Mar-flex Waterproofing & Building Products
    e. Sto Corp.
    f. Tremco Commercial Sealants and Waterproofing, part of Tremco CPG
    g. W. R. Meadows, Inc
    h. Or Approved Equal

2. Salient Characteristics:

    a. Type: Synthetic Acrylic
    b. Application: Spray, Roller, or Trowel
    c. Vapor Permeance: Minimum 12 perms; ASTM E96/E96M, Method B.
    d. Color: White/off white or black
    e. Solids: 0.53
    f. Thickness (Porous Substrates): Minimum 70 mils wet, 35 mils dry.

g. Application Temperature: Above 40 deg F and rising. If contractor intends to install at lower temperatures, they shall provide manufacturer approved low temperature installation.
h. Crack Bridging: Passes ASTM C1305 testing
i. Water Immersion: Passes ASTM D870 testing
j. Adhesion to Substrate (Concrete): Minimum 38 psi when tested in accordance with ASTM D4541.
k. Air Leakage of Material: Minimum 0.00158 cfm/sqft per ASTM E2178 Free Film Method at 75 Pa; 0.00435 cfm/sqft per ASTM E2178 Free Film Method at 300 Pa.
l. Fire Propagation Characteristics: Passes NFPA 285 testing as part of an approved assembly
m. Flame Spread (E84): 10
n. Smoke Development (E84): 25

2.4 ACCESSORY MATERIALS

A. Provide primers, transition strips, termination strips, joint reinforcing fabric and strips, joint sealants, counterflashing strips, flashing sheets and metal termination bars, termination mastic, substrate patching materials, adhesives, tapes, foam sealants, lap sealants, and other accessory materials that are recommended in writing by air barrier manufacturer to produce a complete air barrier assembly and that are compatible with primary air barrier material and adjacent construction to which they may seal.

B. Primer: Liquid waterborne primer recommended for substrate by air barrier material manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

1. Verify that substrates are sound and free of oil, grease, dirt, excess mortar, or other contaminants.
2. Verify that substrates have cured and aged for minimum time recommended in writing by air barrier manufacturer.
3. Verify that substrates are visibly dry and free of moisture.
4. Verify that masonry joints are flush and completely filled with mortar.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 SURFACE PREPARATION

A. Clean, prepare, treat, fill, and seal substrate and joints and cracks in substrate in accordance with manufacturer's written instructions and details. Provide clean, dust-free, and dry substrate for air barrier application.

B. Mask off adjoining surfaces not covered by air barrier to prevent spillage and overspray affecting other construction.

C. Remove grease, oil, bitumen, form-release agents, paints, curing compounds, and other penetrating contaminants or film-forming coatings from concrete.

D. Remove fins, ridges, mortar, and other projections and fill honeycomb, aggregate pockets, holes, and other voids in concrete with substrate-patching material.

E. Remove excess mortar from masonry ties, shelf angles, and other obstructions.

F. At changes in substrate plane, apply sealant or termination mastic beads at sharp corners and edges to form a smooth transition from one plane to another.

G. Cover gaps in substrate plane and form a smooth transition from one substrate plane to another with stainless steel sheet mechanically fastened to structural framing to provide continuous support for air barrier.

H. Bridge discontinuous wall-to-wall, deck-to-wall, and deck-to-deck joints with air barrier accessory material that accommodates joint movement in accordance with manufacturer's written instructions and details.

3.3 INSTALLATION OF ACCESSORIES

A. Install accessory materials in accordance with air barrier manufacturer's written instructions and details to form a seal with adjacent construction and ensure continuity of air and water barrier.

1. Coordinate the installation of air barrier with installation of roofing membrane and base flashing to ensure continuity of air barrier with roofing membrane.
2. Install transition strip on roofing membrane or base flashing so that a minimum of 3 inches of coverage is achieved over each substrate.
3. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.
4. Apply primer to substrates at required rate and allow it to dry. Limit priming to areas that will be covered by air barrier material on same day. Reprime areas exposed for more than 24 hours.

B. Connect and seal exterior wall air barrier material continuously to roofing-membrane air barrier, concrete below-grade structures, floor-to-floor construction, exterior glazing and window systems, glazed curtain-wall systems, storefront systems, exterior louvers, exterior door framing, and other construction used in exterior wall openings, using accessory materials.

C. At end of each working day, seal top edge of strips and transition strips to substrate with termination mastic.

D. Apply joint sealants forming part of air barrier assembly within manufacturer's recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges.

E. Wall Openings: Prime concealed, perimeter frame surfaces of windows, curtain walls, storefronts, and doors. Apply transition strip so that a minimum of 3 inches of coverage is achieved over each substrate. Maintain 3 inches of full contact over firm bearing to perimeter frames, with not less than 1 inch of full contact.

1. Transition Strip: Roll firmly to enhance adhesion.
2. Preformed Silicone Extrusion: Set in full bed of silicone sealant applied to walls, frame, and air barrier material.

F. Fill gaps in perimeter frame surfaces of windows, storefronts, and doors, and miscellaneous penetrations of air barrier material with foam sealant.

G. Seal strips and transition strips around masonry reinforcing or ties and penetrations with termination mastic.

H. Seal top of through-wall flashings to air barrier with an additional 6-inch- wide, transition strip.

I. Seal exposed edges of strips at seams, cuts, penetrations, and terminations not concealed by metal counterflashings or ending in reglets with termination mastic.

J. Repair punctures, voids, and deficient lapped seams in strips and transition strips. Slit and flatten fishmouths and blisters. Patch with transition strips extending 6 inches beyond repaired areas in strip direction.

3.4 INSTALLATION OF PRIMARY AIR BARRIER MATERIAL

A. Apply air barrier material to form a seal with strips and transition strips and to achieve a continuous air barrier in accordance with air barrier manufacturer's written instructions and details. Apply air barrier material within manufacturer's recommended application temperature ranges.

1. Unless manufacturer recommends in writing against priming, apply primer to substrates at required rate and allow it to dry.
2. Limit priming to areas that will be covered by air barrier material on same day. Reprime areas exposed for more than 24 hours.
3. Where multiple prime coats are needed to achieve required bond, allow adequate drying time between coats.

B. Medium-Build Air Barriers: Apply continuous unbroken air barrier material to substrates according to the following thickness. Apply an increased thickness of air barrier material in full contact around protrusions such as masonry ties.

1. Vapor-Permeable, Medium-Build Air Barrier: Total dry film thickness as recommended in writing by manufacturer to comply with performance requirements, applied in one or more equal coats. Apply additional material as needed to achieve void- and pinhole-free surface, but do not exceed thickness on which required vapor permeability is based.

C. Do not cover air barrier until it has been tested and inspected by testing agency.

D. Correct deficiencies in or remove air barrier that does not comply with requirements; repair substrates and reapply air barrier components.

3.5 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Inspections: Air barrier materials, accessories, and installation are subject to inspection for compliance with requirements. Inspections may include the following:

1. Continuity of air barrier system has been achieved throughout the building envelope with no gaps or holes.
2. Air barrier dry film thickness.
3. Continuous structural support of air barrier system has been provided.
4. Masonry and concrete surfaces are smooth, clean, and free of cavities, protrusions, and mortar droppings.
5. Site conditions for application temperature and dryness of substrates have been maintained.
6. Maximum exposure time of materials to UV deterioration has not been exceeded.
7. Surfaces have been primed, if applicable.
8. Laps in strips and transition strips have complied with minimum requirements and have been shingled in the correct direction (or mastic has been applied on exposed edges), with no fishmouths.
9. Termination mastic has been applied on cut edges.
10. Strips and transition strips have been firmly adhered to substrate.
11. Compatible materials have been used.
12. Transitions at changes in direction and structural support at gaps have been provided.
13. Connections between assemblies (air barrier and sealants) have complied with requirements for cleanliness, surface preparation and priming, structural support, integrity, and continuity of seal.
14. All penetrations have been sealed.

C. Air barriers will be considered defective if they do not pass tests and inspections.

1. Apply additional air barrier material, in accordance with manufacturer's written instructions, where inspection results indicate insufficient thickness.
2. Remove and replace deficient air barrier components for retesting as specified above.

D. Repair damage to air barriers caused by testing; follow manufacturer's written instructions.

E. Prepare test and inspection reports.

3.6 CLEANING AND PROTECTION

A. Protect air barrier system from damage during application and remainder of construction period, in accordance with manufacturer's written instructions.

1. Protect air barrier from exposure to UV light and harmful weather exposure as recommended in writing by manufacturer. If exposed to these conditions for longer than recommended, remove and replace air barrier or install additional, full-thickness, air barrier application after repairing and preparing the overexposed materials in accordance with air barrier manufacturer's written instructions.
2. Protect air barrier from contact with incompatible materials and sealants not approved by air barrier manufacturer.

B. Clean spills, stains, and soiling from construction that would be exposed in the completed work using cleaning agents and procedures recommended in writing by manufacturer of affected construction.

C. Remove masking materials after installation.

END OF SECTION 07 27 26

SECTION 07 41 13.16 – STANDING-SEAM METAL ROOF PANELS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Vertical-rib, seamed-joint, standing-seam metal roof panels.
2. Underlayment.
3. Roof flashing.

B. Related Requirements:

1. Section 07 42 13.13 “Formed Metal Wall Panels” for metal panels used in vertical wall assemblies.
2. Section 07 42 93 “Soffit Panels” for metal panels used in horizontal soffit applications.
3. Section 07 62 00 “Sheet Metal Flashing and Trim” for flashing in assemblies other than the roof.
4. Section 07 71 00 “Roof Specialties” for gutters and downspouts.
5. Section 07 72 53 “Snow Guards” for prefabricated devices designed to hold snow on the roof surface, allowing it to melt and drain off slowly.
6. Section 26 41 13 “Lightning Protection for Structures” for Option B: Lightning Protection.

1.2 COORDINATION

A. Coordinate sizes and locations of roof penetrations with actual equipment provided.

B. Coordinate metal roof panel installation with rain drainage work, flashing, trim, construction of soffits, and other adjoining work to provide a leakproof, secure, and noncorrosive installation.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Meet with Contracting Officer, metal roof panel Installer, metal roof panel manufacturer's representative, structural-support Installer, and installers whose work interfaces with or affects metal roof panels, including installers of roof accessories and roof-mounted equipment.
2. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
3. Review methods and procedures related to metal roof panel installation, including manufacturer's written installation instructions.
4. Examine support conditions for compliance with requirements, including alignment between and attachment to structural members.
5. Review loading limitations of supporting structure during and after roofing.

6. Review flashings, special details, drainage, penetrations, and condition of other construction that affect metal roof panels.
7. Review governing regulations and requirements for insurance, certificates, and tests and inspections if applicable.
8. Review temporary protection requirements for metal roof panel systems during and after installation.
9. Review procedures for repair of metal roof panels damaged after installation.
10. Document proceedings, including corrective measures and actions required, and furnish copy of record to each participant.

1.4 ACTION SUBMITTALS

A. Product Data:

1. For standing-seam metal roof panels. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each type of panel and accessory.

B. Shop Drawings:

1. Include fabrication and installation layouts of metal roof panels; details of edge conditions, joints, panel profiles, corners, anchorages, attachment system, trim, flashings, closures, and accessories; and special details.
2. Accessories: Include details of the flashing, trim, and anchorage systems, at a scale of not less than 1-1/2 inches per 12 inches.

C. Samples for Verification: Actual sample of finished products for each type of exposed finish for metal roof panels and metal panel accessories. Manufacturers' standard size.

D. Qualification Statements: For roof installers.

1.5 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For standing-seam metal roof panels, for tests performed by a qualified testing agency.

B. Field quality-control reports.

C. Sample Warranty: For manufacturer's special warranties.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For metal roof panels.

B. Manufacturer's Warranties: For all products used.

1.7 QUALITY ASSURANCE

A. Roof Installer Qualifications – Entity that employs installers and supervisors who are trained and approved by the manufacturer.

B. Portable Roll-Forming Equipment Certification: UL-certified, portable roll-forming equipment capable of producing metal roof panels warranted by manufacturer to be the same as factory-formed products. Maintain UL certification of portable roll-forming equipment for duration of Work.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver components, metal roof panels, and other manufactured items so as not to be damaged or deformed. Package metal roof panels for protection during transportation and handling.

B. Unload, store, and erect metal roof panels in a manner to prevent bending, warping, twisting, and surface damage.

C. Stack metal roof panels horizontally on platforms or pallets, covered with suitable weathertight and ventilated covering. Store metal roof panels to ensure dryness, with positive slope for drainage of water. Do not store metal roof panels in contact with other materials that might cause staining, denting, or other surface damage.

D. Retain strippable protective covering on metal roof panels during installation.

1.9 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit assembly of metal roof panels to be performed in accordance with manufacturers' written installation instructions and warranty requirements.

1.10 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of metal roof panel systems that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures including rupturing, cracking, or puncturing.
b. Deterioration of metal and other materials beyond normal weathering.

2. Warranty Period: Two years from date of Substantial Completion.

B. Special Warranty on Panel Finishes: Manufacturer agrees to repair finish or replace metal roof panels that show evidence of deterioration of factory-applied finishes within specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:

    a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
    b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
    c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Finish Warranty Period: 20 years from date of Substantial Completion.

C. Special Weathertightness Warranty: Manufacturer agrees to repair or replace standing-seam metal roof panel systems that fail to remain weathertight, including leaks, within specified warranty period.

1. Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide metal roof panel systems capable of withstanding the effects of the following loads when tested in accordance with ASTM E1592:

1. Wind Loads: As indicated on Structural Drawings.
2. Other Design Loads – As noted on the Structural Drawings.
3. Deflection Limits: For wind loads, no greater than 1/180 of the span.
4. Structural Standing-Seam Steel Roof Panel Systems: Unless more stringent requirements are indicated, comply with ASTM E1514.

B. Air Infiltration: Air leakage of not more than 0.06 cfm/sq. ft. when tested in accordance with ASTM E1680 or ASTM E283/E283M at the following test-pressure difference:

1. Test-Pressure Difference: 1.57 lbf/sq. ft..

C. Water Penetration under Static Pressure: No water penetration when tested in accordance with ASTM E1646 at the following test-pressure difference:

1. Test-Pressure Difference: 2.86 lbf/sq. ft..

D. Watertightness: No water penetration when tested in accordance with ASTM E2140 for hydrostatic-head resistance.

E. Wind-Uplift Resistance: Provide metal roof panel assemblies that comply with UL 580 for wind-uplift-resistance class indicated.

1. Uplift Rating: UL 90.

F. FM Approvals Listing: Provide metal roof panels and component materials that comply with requirements in FM Approvals 4471 as part of a metal panel roofing system and that are listed in FM's "Approval Guide" for Class 1 or noncombustible construction, as applicable. Identify materials with FM Approvals markings.

1. Fire/Windstorm Classification: Class 1A- 120.
2. Hail Resistance: MH.

G. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

H. Energy Performance:

1. Provide metal roof panels meeting when tested in accordance with CRRC-1:

a. Three-year, aged Solar Reflectance Index (SRI) of not less than 29 when calculated in accordance with ASTM E1980.

2.2 STANDING-SEAM METAL ROOF PANELS, GENERAL

A. Provide factory-formed metal roof panels designed to be installed by lapping and interconnecting raised side edges of adjacent panels with seamed joint type indicated and mechanically attaching panels to supports using concealed fasteners in side laps. Include all accessories required for weathertight installation.

2.3 VERTICAL-RIB, SEAMED-JOINT, STANDING-SEAM METAL ROOF PANELS

A. Metal Roof Panels: Formed with vertical ribs at panel edges; designed for sequential installation by mechanically attaching panels to supports using concealed clips located under one side of panels, engaging opposite edge of adjacent panels, and mechanically seaming panels together.

1. Structural Support: Over solid deck.
2. Material: Metallic-coated steel.
3. Seam Type: Double folded.
4. Panel Profile: Flat pan.
5. Panel Coverage: 16 inches.
6. Panel Height: 1.5 inches.
7. Clips: Two piece, floating, designed to accommodate thermal movement.

a. Steel Clips: thickness as indicated on approved Shop Drawings, zinc-coated (galvanized) or aluminum-zinc alloy-coated steel sheet.
b. Clip Spacing: As indicated on approved Shop Drawings.

2.4 METAL ROOF PANEL MATERIAL

A. Metallic-Coated Steel Sheet: Zinc-coated (galvanized) steel sheet complying with minimum ASTM A653/A653M, G90 coating designation, or aluminum-zinc alloy-coated steel sheet complying with minimum ASTM A792/A792M, Class AZ50 coating designation; structural quality. Sheet prepainted by the coil-coating process to comply with ASTM A755/A755M.

1. Nominal Thickness: 0.034 inch.
2. Surface: Smooth, flat texture.
3. Exterior Finish: Two-coat fluoropolymer.
4. Color: Surrey Beige (SRI of 40) by Fabral, or alternate as selected by CO from manufacturer's full range.

2.5 UNDERLAYMENT

A. Self-Adhering, High-Temperature Underlayment: Provide self-adhering, cold-applied, sheet underlayment, a minimum of 30 mils thick, consisting of slip-resistant, polyethylene-film top surface laminated to a layer of butyl or SBS-modified asphalt adhesive, with release-paper backing. Provide primer when recommended by underlayment manufacturer.

1. Thermal Stability: Stable after testing at 240 deg F; ASTM D1970/D1970M.
2. Low-Temperature Flexibility: Passes after testing at minus 20 deg F; ASTM D1970/D1970M.

B. Slip Sheet: Manufacturer's recommended slip sheet, of type required for application.

2.6 MISCELLANEOUS MATERIALS

A. Miscellaneous Metal Subframing and Furring: ASTM C645; cold-formed, metallic-coated steel sheet, minimum ASTM A653/A653M, G90 hot-dip galvanized coating designation or ASTM A792/A792M, Class AZ50 coating designation. Provide manufacturer's standard sections as required for support and alignment of metal panel system.

B. Roof Panel Accessories: Provide components required for a complete, weathertight metal roof panel system including trim, copings, fasciae, mullions, sills, corner units, fasteners, flashings, sealants, gaskets, fillers, closure strips, and similar items. Match material and finish of metal roof panels unless otherwise indicated.

1. Closures: Provide closures at eaves and ridges, fabricated of same metal as metal roof panels.
2. Backing Plates: Provide metal backing plates at roof panel end splices, fabricated from material recommended by manufacturer.
3. Closure Strips: Closed-cell, expanded, cellular, rubber or crosslinked, polyolefin-foam or closed-cell laminated polyethylene; minimum 1-inch- thick, flexible closure strips; cut or premolded to match metal roof panel profile. Provide closure strips where indicated or necessary to ensure weathertight construction.

C. Flashing and Trim: Provide flashing and trim formed from same material as metal roof panels as required to seal against weather and to provide finished appearance. Locations include, but are not limited to, eaves, rakes, corners, bases, framed openings, ridges, fasciae, and fillers. Finish flashing and trim with same finish system as adjacent metal roof panels.

D. Roof Panel Fasteners: Self-tapping screws designed to withstand design loads.

E. Roof Panel Sealants: Provide sealant type recommended by manufacturer that are compatible with metal roof panel materials, are nonstaining, and do not damage panel finish.

1. Sealant Tape: Pressure-sensitive, 100 percent solids, gray polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch wide and 1/8 inch thick.
2. Joint Sealant: ASTM C920; elastomeric polyurethane or silicone sealant; of type, grade, class, and use classifications required to seal joints in metal roof panels and remain weathertight; and as recommended in writing by metal roof panel manufacturer.
3. Butyl-Rubber-Based, Solvent-Release Sealant: ASTM C1311.

2.7 FABRICATION

A. Fabricate and finish metal roof panels and accessories at the factory, by manufacturer's standard procedures and processes, as necessary to fulfill indicated performance requirements demonstrated by laboratory testing. Comply with indicated profiles and with dimensional and structural requirements.

B. On-site Fabrication: Subject to compliance with requirements of this Section, metal roof panels may be fabricated on-site using UL-certified, portable roll-forming equipment if panels are of same profile and warranted by manufacturer to be equal to factory-formed panels. Fabricate in accordance with equipment manufacturer's written instructions and to comply with details shown.

C. Provide roof panel profile, including major ribs and intermediate stiffening ribs, if any, for full length of panel.

D. Fabricate metal roof panel joints with factory-installed captive gaskets or separator strips that provide a weathertight seal and prevent metal-to-metal contact, and that minimize noise from movements.

E. Sheet Metal Flashing and Trim: Fabricate flashing and trim to comply with manufacturer's recommendations that apply to design, dimensions, metal, and other characteristics of item indicated.

1. Form exposed sheet metal accessories that are without excessive oil canning, buckling, and tool marks and that are true to line and levels indicated, with exposed edges folded back to form hems.
2. Seams for other than Aluminum: Fabricate nonmoving seams in accessories with flat-lock seams. Tin edges to be seamed, form seams, and solder.
3. Sealed Joints: Form nonexpansion, but movable, joints in metal to accommodate sealant and to comply with manufacturer's recommendations.

4. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not permitted on faces of accessories exposed to view.
5. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal recommended in writing by metal roof panel manufacturer.

    a. Size: As recommended by metal roof panel manufacturer for application, but not less than thickness of metal being secured.

2.8 FINISHES

A. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in same piece are unacceptable. Variations in appearance of other components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

C. Steel Roof Panels and Accessories:

1. Two-Coat Fluoropolymer: Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
2. Siliconized Polyester: Epoxy primer and silicone-modified, polyester-enamel topcoat; with a dry film thickness of not less than 0.2 mil for primer and 0.8 mil for topcoat.
3. Concealed Finish: Apply pretreatment and manufacturer's standard white or light-colored acrylic or polyester backer finish consisting of prime coat and wash coat with a minimum total dry film thickness of 0.5 mil.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal roof panel supports, and other conditions affecting performance of the Work.

1. Examine solid roof sheathing to verify that sheathing joints are supported by framing or blocking and that installation is within flatness tolerances required by metal roof panel manufacturer.

    a. Verify that air- or water-resistive barriers have been installed over sheathing or backing substrate to prevent air infiltration or water penetration.

B. Examine roughing-in for components and systems penetrating metal roof panels to verify actual locations of penetrations relative to seam locations of metal roof panels before installation.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Miscellaneous Supports: Install subframing, furring, and other miscellaneous panel support members and anchorages in accordance with ASTM C754 and metal roof panel manufacturer's written installation instructions.

3.3 INSTALLATION OF UNDERLAYMENT

A. Self-Adhering Sheet Underlayment: Apply primer if required by manufacturer. Comply with temperature restrictions of underlayment manufacturer for installation. Apply at locations indicated on Drawings, wrinkle free, in shingle fashion to shed water, and with end laps of not less than 6 inches staggered 24 inches between courses. Overlap side edges not less than 3-1/2 inches. Roll laps with roller. Cover underlayment within 14 days.

1. Apply over the entire roof surface.

B. Slip Sheet: Apply slip sheet over underlayment before installing metal roof panels.

C. Flashings: Install flashings to cover underlayment to comply with requirements specified in Section 076200 “Sheet Metal Flashing and Trim.”

3.4 INSTALLATION OF STANDING-SEAM METAL ROOF PANELS

A. Install metal roof panels in accordance with manufacturer's written installation instructions and approved Shop Drawings in orientation, sizes, and locations indicated. Anchor metal roof panels and other components of the Work securely in place, with provisions for thermal and structural movement.

1. Shim or otherwise plumb substrates receiving metal roof panels.
2. Flash and seal metal roof panels at perimeter of all openings. Fasten with self-tapping screws. Do not begin installation until air- or water-resistive barriers and flashings that will be concealed by metal roof panels are installed.
3. Install screw fasteners in predrilled holes.
4. Locate and space fastenings in uniform vertical and horizontal alignment.
5. Install flashing and trim as metal roof panel work proceeds.
6. Locate panel splices over, but not attached to, structural supports. Stagger panel splices and end laps to avoid a four-panel lap splice condition.
7. Align bottoms of metal roof panels and fasten with blind rivets, bolts, or self-tapping screws. Fasten flashings and trim around openings and similar elements with self-tapping screws.
8. Provide weathertight escutcheons for pipe- and conduit-penetrating panels.

B. Fasteners:

1. Steel Roof Panels: Use stainless steel fasteners for surfaces exposed to the exterior; use galvanized-steel fasteners for surfaces exposed to the interior.

C. Metal Protection: Where dissimilar metals contact each other or corrosive substrates, protect against galvanic action as recommended in writing by metal roof panel manufacturer.

D. Concealed Clip, Standing-Seam Metal Roof Panel Installation: Fasten metal roof panels to supports with concealed clips at each standing-seam joint at location, spacing, and with fasteners recommended in writing by manufacturer.

1. Install clips to supports with self-tapping fasteners.
2. Install pressure plates at locations indicated in manufacturer's written installation instructions.

E. Roof Panel Joints: Fasten panel joints to substrate in accordance with manufacturer's instructions.

1. Seamed Joint: Crimp standing seams with manufacturer-approved, motorized seamer tool so clip, metal roof panel, and factory-applied sealant are completely engaged.
2. Watertight Installation:

a. Apply a continuous ribbon of sealant or tape to seal joints of metal roof panels, using sealant or tape as recommended in writing by manufacturer as needed to make panels watertight.
b. Provide sealant or tape between panels and protruding equipment, vents, and accessories.
c. At panel splices, nest panels with minimum 6-inch end lap, sealed with sealant and fastened together by interlocking clamping plates.

F. Accessory Installation: Install accessories with positive anchorage to building and weathertight mounting and provide for thermal expansion. Coordinate installation with flashings and other components.

1. Install components required for a complete metal roof panel system including trim, copings, corners, seam covers, flashings, sealants, gaskets, fillers, closure strips, and similar items. Provide types indicated by metal roof panel manufacturers; or, if not indicated, types recommended by metal roof panel manufacturer.

G. Flashing and Trim: Comply with performance requirements and manufacturer's written installation instructions. Provide concealed fasteners where possible and set units true to line and level as indicated. Install work with laps, joints, and seams that will be permanently watertight and weather resistant.

1. Install exposed flashing and trim that is without buckling and tool marks, and that is true to line and levels indicated, with exposed edges folded back to form hems. Install sheet metal flashing and trim to fit substrates and achieve waterproof and weather-resistant performance.

2. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim. Space movement joints at a maximum of 10 ft. with no joints allowed within 24 inches of corner or intersection. Where lapped expansion provisions cannot be used or would not be sufficiently weather resistant and waterproof, form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with mastic sealant (concealed within joints).

H. Pipe and Conduit Penetrations: Fasten and seal to metal roof panels as recommended by manufacturer.

3.5 ERECTION TOLERANCES

A. Installation Tolerances: Shim and align metal roof panels within installed tolerance of 1/4 inch in 20 ft. on slope and location lines as indicated and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.

3.6 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect metal roof panel installation, including accessories. Report results in writing.

B. Remove and replace applications of metal roof panels where tests and inspections indicate that they do not comply with specified requirements.

C. Additional tests and inspections, at Contractor's expense, are performed to determine compliance of replaced or additional work with specified requirements.

D. Prepare test and inspection reports.

3.7 CLEANING AND PROTECTION

A. Remove temporary protective coverings and strippable films, if any, as metal roof panels are installed, unless otherwise indicated in manufacturer's written installation instructions. On completion of metal roof panel installation, clean finished surfaces as recommended by metal roof panel manufacturer. Maintain in a clean condition during construction.

B. Replace metal roof panels that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

END OF SECTION 07 41 13.16

SECTION 07 42 13.13 – FORMED METAL WALL PANELS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Concealed-fastener, aluminum lap-seam metal wall panels.

B. Related Requirements:

1. Section 04 26 13 “Masonry Veneer” for adjacent stone veneer.
2. Section 07 27 26 “Fluid-Applied Membrane Air and Weather Barrier” for adjacent air and weather barrier.
3. Section 07 62 00 “Sheet Metal Flashing and Trim” for adjacent flashing.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Meet with Contracting Officer, metal panel Installer, metal panel manufacturer's representative, structural-support Installer, and installers whose work interfaces with or affects metal panels, including installers of windows.
2. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
3. Review methods and procedures related to metal panel installation, including manufacturer's written instructions.
4. Examine support conditions for compliance with requirements, including alignment between and attachment to structural members.
5. Review flashings, special siding details, wall penetrations, openings, and condition of other construction that affect metal panels.
6. Review governing regulations and requirements for insurance, certificates, and tests and inspections if applicable.
7. Review temporary protection requirements for metal panel assembly during and after installation.
8. Review of procedures for repair of metal panels damaged after installation.
9. Document proceedings, including corrective measures and actions required, and furnish copy of record to each participant.

1.3 ACTION SUBMITTALS

A. Product Data: Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each type of panel and accessory.

1. Concealed-fastener, lap-seam metal wall panels.

B. Shop Drawings:

1. Include fabrication and installation layouts of metal panels; details of edge conditions, joints, panel profiles, corners, anchorages, attachment system, trim, flashings, closures, and accessories; and special details.
2. Accessories: Include details of the flashing, trim, and anchorage systems, at a scale of not less than 1-1/2 inches per 12 inches.

C. Samples for Initial Selection: For each type of metal panel indicated with factory-applied finishes.

1. Include Samples of trim and accessories involving color selection.

D. Samples for Verification: For each type of exposed finish, prepared on Samples of size indicated below:

1. Metal Panels: 12 inches long by actual panel width. Include fasteners, closures, and other metal panel accessories.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Field quality-control reports.

C. Sample Warranties: For special warranties

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For metal panels to include in maintenance manuals.

B. Manufacturer's Warranties: For all products used.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

B. Manufacturer - Company specializing in manufacturing products specified in this section with at least five years of documented experience.

1.7 MOCKUPS

A. First section (complete area between sections of stone veneer) shall be reviewed as a mockup in situ to set quality standards for the remainder of the manufacture and installation of the panels. Subject to approval by the Contracting Officer in writing, mockup shall become part of the completed work.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver components, metal panels, and other manufactured items so as not to be damaged or deformed. Package metal panels for protection during transportation and handling.

B. Unload, store, and erect metal panels in a manner to prevent bending, warping, twisting, and surface damage.

C. Stack metal panels horizontally on platforms or pallets, covered with suitable weathertight and ventilated covering. Store metal panels to ensure dryness, with positive slope for drainage of water. Do not store metal panels in contact with other materials that might cause staining, denting, or other surface damage.

D. Retain strippable protective covering on metal panels during installation.

1.9 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit assembly of metal panels to be performed according to manufacturers' written instructions and warranty requirements.

1.10 COORDINATION

A. Coordinate metal panel installation with masonry veneer, flashing, storefront windows, and other adjoining work to provide a leakproof, secure, and noncorrosive installation.

1.11 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of metal panel systems that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures including rupturing, cracking, or puncturing.
b. Deterioration of metals and other materials beyond normal weathering.

2. Warranty Period: Five years from date of Substantial Completion.

B. Special Warranty on Panel Finishes: Manufacturer's standard form in which manufacturer agrees to repair finish or replace metal panels that show evidence of deterioration of factory-applied finishes within specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Delta E units when tested according to ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested according to ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Finish Warranty Period: 20 years minimum from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Structural Performance: Provide metal panel systems capable of withstanding the effects of the following loads, based on testing according to ASTM E1592:

1. Wind Loads: As indicated on Drawings.
2. Other Design Loads: As indicated on Drawings.
3. Deflection Limits: For wind loads, no greater than 1/180 of the span.

B. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes by preventing buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change (Range): 120 deg F, ambient; 180 deg F, material surfaces.

2.2 CONCEALED-FASTENER, LAP-SEAM METAL WALL PANELS

A. Provide factory-formed metal panels designed to be field assembled by lapping and interconnecting side edges of adjacent panels and mechanically attaching through panel to supports using concealed fasteners and factory-applied sealant inside laps. Include accessories required for weathertight installation.

B. Reveal-Joint, Concealed-Fastener Metal Wall Panels: Formed with vertical panel edges and a flat pan between panel edges; with narrow reveal joint between panels.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Dri-Design - Aluminum Metal Plate Wall Panels System or comparable product by one of the following:

a. Berridge Manufacturing Company
b. CENTRIA, a Nucor Brand
c. Englert, Inc.
d. MBCI; Cornerstone Building Brands
e. Metal Sales & Service, Inc.
f. Morin - A Kingspan Group Company
g. PAC-CLAD; Petersen; a Carlisle company
h. Taylor Metal Products
i. The Bryer Co.

2. Aluminum Sheet: Coil-coated sheet, ASTM B209, alloy as standard with manufacturer, with temper as required to suit forming operations and structural performance required.

   Salient Characteristics:

   a. Material: Aluminum Sheet
   b. Thickness: 12 ga, 0.080 inch, min.
   c. Surface: Smooth, flat finish.
   d. Exterior Finish: Coil coated, Two-coat fluoropolymer.
   e. Basis of Design Color: Dri Design Sienna - to be confirmed by Contracting Officer from manufacturer's full range.

3. Panel Coverage: See drawings for panel layout.
4. Panel Height: 1.25 inches nominal.

2.3 MISCELLANEOUS MATERIALS

A. Miscellaneous Metal Subframing and Furring: G90ASTM A792/A792M, Class AZ50 aluminum-zinc-alloy coating designation unless otherwise indicated. Provide manufacturer's standard sections as required for support and alignment of metal panel system.

B. Panel Accessories: Provide components required for a complete, weathertight panel system including trim, copings, fasciae, mullions, sills, corner units, clips, flashings, sealants, gaskets, fillers, closure strips, and similar items. Match material and finish of metal panels unless otherwise indicated.

   1. Closures: Provide closures at edges, fabricated of same metal as metal panels.
   2. Backing Plates: Provide metal backing plates at panel end splices, fabricated from material recommended by manufacturer.
   3. Closure Strips: Closed-cell, expanded, cellular, rubber or crosslinked, polyolefin-foam or closed-cell laminated polyethylene; minimum 1-inch- thick, flexible closure strips; cut or premolded to match metal panel profile. Provide closure strips where indicated or necessary to ensure weathertight construction.

C. Flashing and Trim: Provide flashing and trim formed from same material as metal panels as required to seal against weather and to provide finished appearance. Locations include, but are not limited to, bases, drips, sills, jambs, corners, endwalls, framed openings, reveals, and fillers. Finish flashing and trim with same finish system as adjacent metal panels.

D. Panel Fasteners: Self-tapping screws designed to withstand design loads. Provide exposed fasteners with heads matching color of metal panels by means of plastic caps or factory-applied coating. Provide EPDM or PVC sealing washers for exposed fasteners.

E. Panel Sealants: Provide sealant type recommended by manufacturer that are compatible with panel materials, are nonstaining, and do not damage panel finish.

   1. Sealant Tape: Pressure-sensitive, 100 percent solids, gray polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch wide and 1/8 inch thick.

2. Joint Sealant: ASTM C920; elastomeric polyurethane or silicone sealant; of type, grade, class, and use classifications required to seal joints in metal panels and remain weathertight; and as recommended in writing by metal panel manufacturer.
3. Butyl-Rubber-Based, Solvent-Release Sealant: ASTM C1311.

2.4 FABRICATION

A. Fabricate and finish metal panels and accessories at the factory, by manufacturer's standard procedures and processes, as necessary to fulfill indicated performance requirements demonstrated by laboratory testing. Comply with indicated profiles and with dimensional and structural requirements.

B. Provide panel profile, including major ribs and intermediate stiffening ribs, if any, for full length of panel.

C. Fabricate metal panel joints with factory-installed captive gaskets or separator strips that provide a weathertight seal and prevent metal-to-metal contact, and that minimize noise from movements.

D. Sheet Metal Flashing and Trim: Fabricate flashing and trim to comply with manufacturer's recommendations and recommendations in SMACNA's “Architectural Sheet Metal Manual” that apply to design, dimensions, metal, and other characteristics of item indicated.

1. Form exposed sheet metal accessories that are without excessive oil canning, buckling, and tool marks and that are true to line and levels indicated, with exposed edges folded back to form hems.
2. Seams for Aluminum: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer. Rivet joints for additional strength.
3. Sealed Joints: Form nonexpansion, but movable, joints in metal to accommodate sealant and to comply with SMACNA standards.
4. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on faces of accessories exposed to view.
5. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal recommended in writing by metal panel manufacturer.

    a. Size: As recommended by SMACNA's “Architectural Sheet Metal Manual” or metal wall panel manufacturer for application but not less than thickness of metal being secured.

2.5 FINISHES

A. Protect mechanical and painted finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in same piece are not acceptable. Variations in appearance of other components are acceptable

if they are within the range of approved Samples and are assembled or installed to minimize contrast.

C. Aluminum Panels and Accessories:

1. Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent polyvinylidene fluoride (PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions .
2. Siliconized Polyester: Epoxy primer and silicone-modified, polyester-enamel topcoat; with a dry film thickness of not less than 0.2 mil for primer and 0.8 mil for topcoat.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal panel supports, and other conditions affecting performance of the Work.

1. Examine wall framing to verify that girts, angles, channels, studs, and other structural panel support members and anchorage have been installed within alignment tolerances required by metal wall panel manufacturer.
2. Examine wall sheathing to verify that sheathing joints are supported by framing or blocking and that installation is within flatness tolerances required by metal wall panel manufacturer.

a. Verify that air- or water-resistive barriers have been installed over sheathing or backing substrate to prevent air infiltration or water penetration.

B. Examine roughing-in for components and systems penetrating metal panels to verify actual locations of penetrations relative to seam locations of metal panels before installation.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Miscellaneous Supports: Install subframing, furring, and other miscellaneous panel support members and anchorages according to ASTM C754 and metal panel manufacturer's written recommendations.

3.3 INSTALLATION OF METAL PANELS

A. Install metal panels according to manufacturer's written instructions in orientation, sizes, and locations indicated. Install panels perpendicular to supports unless otherwise indicated. Anchor metal panels and other components of the Work securely in place, with provisions for thermal and structural movement.

1. Shim or otherwise plumb substrates receiving metal panels.
2. Flash and seal metal panels at perimeter of all openings. Fasten with self-tapping screws. Do not begin installation until air- or water-resistive barriers and flashings that will be concealed by metal panels are installed.
3. Install screw fasteners in predrilled holes.
4. Locate and space fastenings in uniform vertical and horizontal alignment.
5. Install flashing and trim as metal panel work proceeds.
6. Locate panel splices over, but not attached to, structural supports. Stagger panel splices and end laps to avoid a four-panel lap splice condition.
7. Align bottoms of metal panels and fasten with blind rivets, bolts, or self-tapping screws. Fasten flashings and trim around openings and similar elements with self-tapping screws.
8. Provide weathertight escutcheons for pipe- and conduit-penetrating panels.

B. Fasteners:

1. Aluminum Panels: Use aluminum or stainless steel fasteners for surfaces exposed to the exterior; use aluminum or galvanized-steel fasteners for surfaces exposed to the interior.

C. Metal Protection: Where dissimilar metals contact each other or corrosive substrates, protect against galvanic action as recommended in writing by metal panel manufacturer.

D. Lap-Seam Metal Panels: Fasten metal panels to supports with fasteners at each lapped joint at location and spacing recommended by manufacturer.

1. Lap ribbed or fluted sheets one full rib. Apply panels and associated items true to line for neat and weathertight enclosure.
2. Provide metal-backed washers under heads of exposed fasteners bearing on weather side of metal panels.
3. Locate and space exposed fasteners in uniform vertical and horizontal alignment. Use proper tools to obtain controlled uniform compression for positive seal without rupture of washer.
4. Install screw fasteners with power tools having controlled torque adjusted to compress washer tightly without damage to washer, screw threads, or panels. Install screws in predrilled holes.
5. Flash and seal panels with weather closures at perimeter of all openings.

E. Accessory Installation: Install accessories with positive anchorage to building and weathertight mounting, and provide for thermal expansion. Coordinate installation with flashings and other components.

1. Install components required for a complete metal panel system including trim, copings, corners, seam covers, flashings, sealants, gaskets, fillers, closure strips, and similar items. Provide types indicated by metal wall panel manufacturer; or, if not indicated, provide types recommended by metal panel manufacturer.

F. Flashing and Trim: Comply with performance requirements, manufacturer's written installation instructions, and SMACNA's “Architectural Sheet Metal Manual.” Provide concealed fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints, and seams that are permanently watertight.

1. Install exposed flashing and trim that is without buckling and tool marks, and that is true to line and levels indicated, with exposed edges folded back to form hems. Install sheet metal flashing and trim to fit substrates and achieve waterproof performance.
2. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim. Space movement joints at a maximum of 10 feet with no joints allowed within 24 inches of corner or intersection. Where lapped expansion provisions cannot be used or would not be sufficiently waterproof, form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with mastic sealant (concealed within joints).

3.4 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect completed metal wall panel installation, including accessories.

B. Remove and replace metal wall panels where tests and inspections indicate that they do not comply with specified requirements.

C. Additional tests and inspections, at Contractor's expense, are performed to determine compliance of replaced or additional work with specified requirements.

D. Prepare test and inspection reports.

3.5 CLEANING AND PROTECTION

A. Remove temporary protective coverings and strippable films, if any, as metal panels are installed, unless otherwise indicated in manufacturer's written installation instructions. On completion of metal panel installation, clean finished surfaces as recommended by metal panel manufacturer. Maintain in a clean condition during construction.

B. After metal panel installation, clear weep holes and drainage channels of obstructions, dirt, and sealant.

C. Replace metal panels that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

END OF SECTION 07 42 13.13

SECTION 07 46 16 – ALUMINUM CLADDING

PART 1 - GENERAL

1.1 SUMMARY

A.

1. Aluminum cladding outside of the stone veneered area
2. Aluminum cladding soffit at underside of roof, vented and unvented planks.

B. Related Requirements:

1. Section 061000 “Rough Carpentry” for wood furring, grounds, nailers, and blocking.
2. Section 04 26 13 “Masonry Veneer” for adjacent stone veneer finish.
3. Section 07 27 26 “Fluid-Applied Membrane Air and Weather Barrier” for adjacent air and weather barrier.
4. Section 07 42 13.13 “Formed Metal Wall Panels” for decorative wall panels within the stone veneer.
5. Section 07 62 00 “Flashing and Sheet Metal” for flashing that is not part of this wall system.
6. Section 07 92 00 “Joint Sealants” for joint sealants in wall assembly.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings:

1. Include construction details, material descriptions, dimensions of individual components and profiles, assembly description and supports details, and finishes; signed and sealed by the qualified professional engineer responsible for their preparation.

C. Samples for Initial Selection: For aluminum cladding including related accessories.

D. Samples for Verification: Actual sample of finished products for each type, color, texture, and pattern required.

1. Size: Manufacturers' standard size.

E. Sustainable Design Submittals:

1. Product Certificates: For indigenous materials, indicating location of material manufacturer and point of extraction, harvest, or recovery for each raw material. Include distance to Project, means of transportation, and cost for each indigenous material.
2. Environmental Product Declaration (EPD): For each product.

F. Qualification Statements: For Installer.

1.4 INFORMATIONAL SUBMITTALS

A. Delegated Design Engineer Qualifications: For component assemblies.

B. Sample Warranty: For manufacturer's special warranties.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For each type of product, including related accessories.

B. Manufacturer's Warranties: For all products used.

1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials to CO that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Furnish full lengths of aluminum cladding including related accessories, in a quantity equal to 5% of installed cladding.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: Authorized representative who is trained and approved by manufacturer.

B. Delegated Design Engineer Qualifications: A professional engineer who is legally qualified to practice in state where Project is located and who is experienced in providing engineering services of the type indicated.

1.8 MOCKUPS

A. First section (complete area between sections of stone veneer) shall be reviewed as a mockup in situ to set quality standards for the remainder of the manufacture and installation of the panels. Subject to approval by the Contracting Officer in writing, mockup shall become part of the completed work

1.9 DELIVERY, STORAGE, AND HANDLING

A. Deliver and store packaged materials in original containers with labels intact until time of use.

B. Store and protect materials on elevated platforms, under cover, and in a dry location.

1.10 COORDINATION

A. Coordinate siding installation with flashings and other adjoining construction to ensure proper sequencing.

1.11 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace products that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures, including cracking peeling, and deforming.
b. Deterioration of metals, metal finishes, and other materials beyond weathering.

2. Special Warranty shall include warranty against gloss/color retention within the guidelines stated by the American Aluminum Manufacturers Association (AAMA).
3. Warranty Period: 15 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain products, including related accessories, from single source from single manufacturer.

2.2 MATERIALS

A. Extruded Aluminum Cladding

1. Basis of Design – Knotwood Wood Grain Aluminum Cladding
2. Salient Characteristics:

a. Board Size – 4 inches wide
b. Material – 6063 T6 Temper Extruded Aluminum Alloy (complies with ASTM B221-13/B221M-13 for Aluminum Alloy Extrusions)
c. Interlocking installation
d. Horizontal layout
e. 100% recycled aluminum
f. Sublimated Timber Grain finish surface
g. Resistant to scratching
h. ASTM E84 Class A fire rating

B. Extruded Aluminum Soffit Panel – Unvented

1. Basis of Design – Knotwood Soffit Systems, 4” Soffit Panel (ST) KEC100
2. Salient Characteristics:

    a. Material: Extruded 6063 T6 Aluminum
    b. Width: 4 inches nominal
    c. Material thickness: 1/16”
    d. Sublimated Timber Grain finish surface
    e. 100% recycled aluminum
    f. Interlocking installation
    g. Quick install cladding clips

C. Extruded Aluminum Soffit Panel – Vented

1. Basis of Design – Knotwood Soffit Systems, Vented Cladding Board (ST) KECV150
2. Salient Characteristics:

    a. From same manufacturer and compatible with Soffit Panel system
    b. Net Free Area (NFA)

        1) 7.52 sq. in. of Open Slot per Linear Foot
        2) 15.28 sq. in. of Open Slot per Square Foot

    c. Width: 6 inches nominal
    d. Width: 4 inches nominal
    e. Material thickness: 1/16”
    f. Sublimated Timber Grain finish surface
    g. 100% recycled aluminum
    h. Interlocking installation
    i. Quick install cladding clips

D. Accessories and Trim

1. As provided by the manufacturer of the cladding, as part of the cladding system.
2. Finish to match aluminate bonded film finish.

E. Finish

1. Pretreatment – five stage aluminum pretreatment system complying with AAMA 2604 Performance Standard. Waste treatment is a simple pH neutralization.
2. Meets EPA, OSHA, State, and Local environmental requirements and contains no chromates, cyanides, or other heavy metals.
3. Base coat and electrostatic applied architectural powder coating approved to AAMA 2604 Performance Standard.
4. Aluminate finish uses a polyurethane powder coat with ink-based wood grain patterns sublimated into the base powder, effectively tattooing the powder coating.
5. VOC Free Powder Coat
6. Gloss Level: Standard gloss is 30%, plus or minus 5%.
7. Basis of Design Color: Knotwood Kwila, to be confirmed by Contracting Officer from manufacturer's full range.

2.3 FABRICATION

A. Prepare surfaces, pre-treat and coat components in accordance with AAMA 2604 Quality Standards.

B. Wrap and package coated components using methods suitable for transit and covered site storage without damage.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates for compliance with requirements for installation tolerances and other conditions affecting performance of aluminum cladding and related accessories.

B. Verify framing members are ready to receive a cladding system.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean substrates of projections and substances detrimental to application.

B. Prepare surfaces using methods recommended by the manufacturer.

3.3 INSTALLATION OF ALUMINUM CLADDING

A. Install in accordance with manufacturer’s installation instructions.

B. Do not install over dissimilar metals without adequate separation of metals with a barrier.

C. Fasten cladding to structural supports; aligned, level, and plumb.

D. Locate joints over supports.

E. Install expansion control joints where needed.

F. Use concealed fasteners per cladding system.

G. Install cladding and accessories in accordance with best practice, with all joint members plumb and true.

H. Allow for expansion and contraction of materials according to manufacturer’s instructions.

3.4 INSTALLATION OF SOFFIT

A. Install in accordance with manufacturer's installation instructions.

B. Do not install over dissimilar metals without adequate separation of metals with a barrier.

C. Fasten soffit to structural supports; aligned and level.

D. Install vented planks where indicated on the drawings.

E. Confirm vented plank location is not obstructed by building structure, as they are spaced to occur between structural members.

F. Use concealed fasteners per cladding system.

G. Install cladding and accessories in accordance with best practice, with all joint members plumb and true.

H. Allow for expansion and contraction of materials according to manufacturer's instructions.

3.5 ADJUSTING AND CLEANING

A. Remove damaged, improperly installed, or otherwise defective materials and replace with new materials complying with specified requirements.

B. Upon completion of the cladding application, clean the entire area, removing all scrap, packaging and unused materials related to this work.

C. After application of cladding, clean as necessary to remove all fingerprints and soiled areas.

D. Protect installed products until completion of project.

E. Touch-up, repair, or replace damaged products before Substantial Completion.

END OF SECTION 07 46 16

SECTION 07 62 00 – SHEET METAL FLASHING AND TRIM

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Custom flashing and trim fabrications, made from the following:

1. Sheet metal materials.
2. Underlayment.
3. Miscellaneous materials.

B. Related Requirements:

1. Section 061000 “Rough Carpentry” for wood nailers, curbs, and blocking.
2. Section 07 41 13.16 “ Standing-Seam Metal Roof Panels “ for materials and installation of sheet metal flashing and trim integral with roofing.
3. Section 07 42 13.13 “ Formed Metal Wall Panels “ for sheet metal flashing and trim integral with decorative metal wall panels in stone veneer area.
4. Section 07 46 16 “Aluminum Cladding” for metal flashing and trim associated with the aluminum cladding system and soffits.
5. Section 07 7 100 “Roof Specialties” for gutters and downspouts.
6. Section 07 92 00 “Joint Sealants” for adjacent sealants.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review construction schedule. Verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
2. Review special roof details, roof drainage, roof-penetration flashing, equipment curbs, and condition of other construction that affect sheet metal flashing and trim.
3. Review requirements for insurance and certificates if applicable.
4. Review sheet metal flashing observation and repair procedures after flashing installation.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: For sheet metal flashing and trim.

1. Plans, elevations, sections, and attachment details.
2. Fabrication and installation layouts, expansion-joint locations, and keyed details. Distinguish between shop- and field-assembled Work.
3. Identification of material, thickness, weight, and finish for each item and location in Project.
4. Details for forming, including profiles, shapes, seams, and dimensions.

5. Details for joining, supporting, and securing, including layout and spacing of fasteners, cleats, clips, and other attachments. Include pattern of seams.
6. Details of termination points and assemblies.
7. Details of connections to adjoining work.

C. Samples for Initial Selection: Manufacturer's standard color sheets, showing full range of available colors for each type of exposed finish.

D. Samples for Verification: Actual sample of finished products for each type of exposed finish for sheet metal and other metal accessories.

1. Sheet Metal Flashing and Trim: Manufacturers' standard size. Include finished seam with required profile. Include fasteners, cleats, clips, closures, and other attachments.
2. Anodized Aluminum Samples: Samples to show full range to be expected for each color required.

E. Qualification Statements: For fabricator.

1.4 INFORMATIONAL SUBMITTALS

A. Sample Warranty: For manufacturer's special warranties.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For sheet metal flashing and trim, and its accessories.

B. Manufacturer's Warranties: For all products used.

1.6 QUALITY ASSURANCE

A. Fabricator Qualifications: Entity that employs skilled workers who custom fabricate sheet metal flashing and trim similar to that required for this Project and whose products have a record of successful in-service performance.

B. For roof edge flashings and copings that are ANSI/SPRI/FM 4435/ES-1 tested , shop is to be listed as able to fabricate required details as tested and approved.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Do not store sheet metal flashing and trim materials in contact with other materials that might cause staining, denting, or other surface damage.

1. Store sheet metal flashing and trim materials away from uncured concrete and masonry.
2. Protect stored sheet metal flashing and trim from contact with water.

B. Protect strippable protective covering on sheet metal flashing and trim from exposure to sunlight and high humidity, except to extent necessary for period of sheet metal flashing and trim installation.

1.8 COORDINATION

A. Coordinate sheet metal flashing and trim layout and seams with sizes and locations of penetrations to be flashed, and joints and seams in adjacent materials.

B. Coordinate sheet metal flashing and trim installation with adjoining roofing and wall materials, joints, and seams to provide leakproof, secure, and noncorrosive installation.

1.9 WARRANTY

A. Special Warranty on Finishes: Manufacturer agrees to repair finish or replace sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within specified warranty period.

1. Exposed Panel Finish: Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Sheet metal flashing and trim assemblies, including cleats, anchors, and fasteners, are to withstand wind loads, structural movement, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Completed sheet metal flashing and trim are not to rattle, leak, or loosen, and are to remain watertight.

B. Sheet Metal Standard for Flashing and Trim: Comply with NRCA's “The NRCA Roofing Manual: Architectural Metal Flashing, Condensation and Air Leakage Control, and Reroofing” and SMACNA's “Architectural Sheet Metal Manual” requirements for dimensions and profiles shown unless more stringent requirements are indicated.

C. SPRI Wind Design Standard: Manufacture and install roof edge flashings and copings tested in accordance with ANSI/SPRI/FM 4435/ES-1 and capable of resisting the following design pressure:

1. Design Pressure: As indicated on Drawings.

D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.2 SHEET METAL MATERIALS

A. Protect mechanical and other finishes on exposed surfaces from damage by applying strippable, temporary protective film before shipping.

B. Aluminum Sheet: Coil-coated sheet, ASTM B209/B209M, alloy as standard with manufacturer, with temper as required to suit forming operations and structural performance required.

1. Thickness: 0.040 inch.
2. Surface: Smooth, flat.
3. Exposed Coil-Coated Finish:

    a. Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions .

4. Color: Match adjacent materials as selected by Contracting Officer from manufacturer's full range.

    a. At aluminum cladding and soffit: Variegated material, basis of design: dark bronze
    b. At downspouts and gutters: Match medium bronze
    c. At stone: Variegated material, basis of design: light buff
    d. At brick: Variegated material, basis of design: light buff
    e. At storefront windows and doors: Match dark bronze
    f. At hollow metal doors: Match dark bronze

5. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester backer finish, consisting of prime coat and wash coat with minimum total dry film thickness of 0.5 mil.
6. Color Anodic Finish, Coil Coated: AAMA 611, AA-M12C22A42/A44, Class I, 0.018 mm or thicker.

    a. Color: Match where adjacent material is color anodized aluminum.
    b. Color Range: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

2.3 MISCELLANEOUS MATERIALS

A. Provide materials and types of fasteners , protective coatings, sealants, and other miscellaneous items as required for complete sheet metal flashing and trim installation and as recommended by manufacturer of primary sheet metal or manufactured item unless otherwise indicated.

B. Fasteners: Wood screws, annular threaded nails, self-tapping screws, self-locking rivets and bolts, and other suitable fasteners designed to withstand design loads and recommended by manufacturer of primary sheet metal or manufactured item.

1. General: Blind fasteners or self-drilling screws, gasketed, with hex-washer head.
2. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.

C. Sealant Tape: Pressure-sensitive, 100 percent solids, polyisobutylene compound sealant tape with release-paper backing. Provide permanently elastic, nonsag, nontoxic, nonstaining tape 1/2 inch wide and 1/8 inch thick.

2.4 FABRICATION, GENERAL

A. Custom fabricate sheet metal flashing and trim to comply with details indicated and recommendations in cited sheet metal standard that apply to design, dimensions, geometry, metal thickness, and other characteristics of item required.

1. Fabricate sheet metal flashing and trim in shop to greatest extent possible.
2. Fabricate sheet metal flashing and trim in thickness or weight needed to comply with performance requirements, but not less than that specified for each application and metal.
3. Verify shapes and dimensions of surfaces to be covered and obtain field measurements for accurate fit before shop fabrication.
4. Form sheet metal flashing and trim to fit substrates without excessive oil-canning, buckling, and tool marks; true to line, levels, and slopes; and with exposed edges folded back to form hems.
5. Conceal fasteners and expansion provisions where possible. Do not use exposed fasteners on faces exposed to view.

B. Fabrication Tolerances:

1. Fabricate sheet metal flashing and trim that is capable of installation to a tolerance of 1/4 inch in 20 ft. on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.
2. Fabricate sheet metal flashing and trim that is capable of installation to tolerances specified.

C. Expansion Provisions: Form metal for thermal expansion of exposed flashing and trim.

1. Form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with butyl sealant concealed within joints.
2. Use lapped expansion joints only where indicated on Drawings.

D. Sealant Joints: Where movable, nonexpansion-type joints are required, form metal in accordance with cited sheet metal standard to provide for proper installation of elastomeric sealant.

E. Fabricate cleats and attachment devices from same material as accessory being anchored or from compatible, noncorrosive metal.

F. Fabricate cleats and attachment devices of sizes as recommended by cited sheet metal standard for application, but not less than thickness of metal being secured.

G. Seams:

1. Seams for Aluminum: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer.

H. Do not use graphite pencils to mark metal surfaces.

2.5 STEEP-SLOPE ROOF SHEET METAL FABRICATIONS

A. Apron, Step, Cricket, and Backer Flashing: Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

B. Valley Flashing: Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

C. Drip Edges: Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

D. Eave, Rake, Ridge, and Hip Flashing: Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

E. Counterflashing: Shop fabricate interior and exterior corners. Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

F. Flashing Receivers: Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

G. Roof-Penetration Flashing: Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

H. Only use the above specification for roof flashing not covered by Section 07 41 13.16 “Standing-Seam Metal Roof Panels.”

2.6 WALL SHEET METAL FABRICATIONS

A. Through-Wall Flashing: Fabricate continuous flashings in minimum 96-inch- long, but not exceeding 12 ft. long, sections, under copings, and at shelf angles. Fabricate discontinuous lintel, sill, and similar flashings to extend 6 inches beyond each side of wall openings; and form with 2-inch- high, end dams. Fabricate from the following materials:

1. Aluminum 0.032 inch thick.

B. Opening Flashings in Frame Construction: Fabricate head, sill, jamb, and similar flashings to extend distance as shown in drawings beyond wall openings. Form head and sill flashing with 2-inch- high, end dams. Fabricate from the following materials:

1. Aluminum: 0.032 inch thick.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, substrates, and other conditions affecting performance of the Work.

1. Verify compliance with requirements for installation tolerances of substrates.
2. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored.
3. Verify that air- or water-resistant barriers have been installed over substrate to prevent air infiltration or water penetration.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF SHEET METAL FLASHING AND TRIM, GENERAL

A. Install sheet metal flashing and trim to comply with details indicated and recommendations of cited sheet metal standard that apply to installation characteristics required unless otherwise indicated on Drawings.

1. Install fasteners , protective coatings, separators, sealants, and other miscellaneous items as required to complete sheet metal flashing and trim system.
2. Install sheet metal flashing and trim true to line, levels, and slopes. Provide uniform, neat seams with minimum exposure of sealant.
3. Anchor sheet metal flashing and trim and other components of the Work securely in place, with provisions for thermal and structural movement.
4. Install sheet metal flashing and trim to fit substrates and to result in watertight performance.
5. Install continuous cleats with fasteners spaced not more than 12 inches o.c.
6. Space individual cleats not more than 12 inches apart. Attach each cleat with at least two fasteners. Bend tabs over fasteners.
7. Install exposed sheet metal flashing and trim with limited oil-canning, and free of buckling and tool marks.
8. Do not field cut sheet metal flashing and trim by torch.
9. Do not use graphite pencils to mark metal surfaces.

B. Metal Protection: Where dissimilar metals contact each other, or where metal contacts pressure-treated wood or other corrosive substrates, protect against galvanic action or corrosion by painting contact surfaces with bituminous coating or by other permanent separation as recommended by sheet metal manufacturer or cited sheet metal standard.

1. Coat concealed side of stainless steel sheet metal flashing and trim with bituminous coating where flashing and trim contact wood, ferrous metal, or cementitious construction.
2. Underlayment: Where installing sheet metal flashing and trim directly on cementitious or wood substrates, install underlayment and cover with slip sheet.

C. Expansion Provisions: Provide for thermal expansion of exposed flashing and trim.

1. Space movement joints at maximum of 10 ft. with no joints within 24 inches of corner or intersection.
2. Form expansion joints of intermeshing hooked flanges, not less than 1 inch deep, filled with sealant concealed within joints.
3. Use lapped expansion joints only where indicated on Drawings.

D. Fasteners: Use fastener sizes that penetrate wood blocking or sheathing not less than 1-1/4 inches for nails and not less than 3/4 inch for wood screws unless here fastener manufacturer recommendation to achieve maximum pull-out resistance is greater.

E. Conceal fasteners and expansion provisions where possible in exposed work and locate to minimize possibility of leakage. Cover and seal fasteners and anchors as required for a tight installation.

F. Seal joints as required for watertight construction.

1. Use sealant-filled joints unless otherwise indicated.

a. Embed hooked flanges of joint members not less than 1 inch into sealant.
b. Form joints to completely conceal sealant.
c. When ambient temperature at time of installation is between 40 and 70 deg F, set joint members for 50 percent movement each way.
d. Adjust setting proportionately for installation at higher ambient temperatures.

1) Do not install sealant-type joints at temperatures below 40 deg F.

2. Prepare joints and apply sealants to comply with requirements in Section 079200 “Joint Sealants.”

G. Soldered Joints: Clean surfaces to be soldered, removing oils and foreign matter.

1. Pretin edges of sheets with solder to width of 1-1/2 inches; however, reduce pretinning where pretinned surface would show in completed Work.
2. Do not use torches for soldering.
3. Heat surfaces to receive solder, and flow solder into joint.

a. Fill joint completely.
b. Completely remove flux and spatter from exposed surfaces.

4. Stainless Steel Soldering:

a. Tin edges of uncoated sheets, using solder for stainless steel and acid flux.
b. Promptly remove acid-flux residue from metal after tinning and soldering.
c. Comply with solder manufacturer's recommended methods for cleaning and neutralization.

3.3 INSTALLATION OF SLOPED ROOF SHEET METAL FABRICATIONS

A. Install sheet metal flashing and trim to comply with performance requirements , sheet metal manufacturer's written installation instructions, and cited sheet metal standard.

1. Provide concealed fasteners where possible, and set units true to line, levels, and slopes.
2. Install work with laps, joints, and seams that are permanently watertight and weather resistant.

B. Roof Edge Flashing:

1. Install roof edge flashings in accordance with ANSI/SPRI/FM 4435/ES-1.
2. Anchor to resist uplift and outward forces in accordance with recommendations in cited sheet metal standard unless otherwise indicated. Interlock bottom edge of roof edge flashing with continuous cleat anchored to substrate at staggered 3-inch centers.
3. Anchor to resist uplift and outward forces in accordance with recommendations in FM Global Property Loss Prevention Data Sheet 1-49 for FM Approvals' listing for required windstorm classification.

C. Copings:

1. Install copings in accordance with ANSI/SPRI/FM 4435/ES-1.
2. Anchor to resist uplift and outward forces in accordance with recommendations in cited sheet metal standard unless otherwise indicated.

    a. Interlock exterior bottom edge of coping with continuous cleat anchored to substrate at 16-inch centers.
    b. Anchor interior leg of coping with washers and screw fasteners through slotted holes at 24-inch centers.

3. Anchor to resist uplift and outward forces in accordance with recommendations in FM Global Property Loss Prevention Data Sheet 1-49 for specified FM Approvals' listing for required windstorm classification.

D. Pipe or Post Counterflashing: Install counterflashing umbrella with close-fitting collar with top edge flared for elastomeric sealant, extending minimum of 4 inches over base flashing. Install stainless steel draw band and tighten.

E. Counterflashing: Coordinate installation of counterflashing with installation of base flashing.

1. Insert counterflashing in reglets or receivers and fit tightly to base flashing.
2. Extend counterflashing 4 inches over base flashing.
3. Lap counterflashing joints minimum of 4 inches.
4. Secure in waterproof manner by means of interlocking folded seam or blind rivets and sealant unless otherwise indicated.

F. Roof-Penetration Flashing: Coordinate installation of roof-penetration flashing with installation of roofing and other items penetrating roof. Seal with elastomeric sealant and clamp flashing to pipes that penetrate roof.

3.4 INSTALLATION OF WALL SHEET METAL FABRICATIONS

A. Install sheet metal wall flashing to intercept and exclude penetrating moisture in accordance with cited sheet metal standard unless otherwise indicated. Coordinate installation of wall flashing with installation of wall-opening components such as windows, doors, and louvers.

B. Opening Flashings in Frame Construction: Install continuous head, sill, jamb, and similar flashings to extend per drawings beyond wall openings.

3.5 INSTALLATION TOLERANCES

A. Shim and align sheet metal flashing and trim within installed tolerance of 1/4 inch in 20 ft. on slope and location lines indicated on Drawings and within 1/8-inch offset of adjoining faces and of alignment of matching profiles.

3.6 CLEANING

A. Clean and neutralize flux materials. Clean off excess solder.

B. Clean off excess sealants.

3.7 PROTECTION

A. Remove temporary protective coverings and strippable films as sheet metal flashing and trim are installed unless otherwise indicated in manufacturer's written installation instructions.

B. On completion of sheet metal flashing and trim installation, remove unused materials and clean finished surfaces as recommended in writing by sheet metal flashing and trim manufacturer.

C. Maintain sheet metal flashing and trim in clean condition during construction.

D. Replace sheet metal flashing and trim that have been damaged or that have deteriorated beyond successful repair by finish touchup or similar minor repair procedures, as determined by Contracting Officer.

END OF SECTION 07 62 00

SECTION 07 71 00 – ROOF SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Manufactured units for the following applications:

1. Roof-edge drainage systems – gutters and downspouts.

B. Related Requirements:

1. Section 061000 “Rough Carpentry” for wood nailers, curbs, and blocking.
2. Section 07 27 26 “ Fluid-Applied Membrane Air Barriers “ for air barrier and compatibility with and transitions to underlayment for roof specialties.
3. Section 07 41 13.16 “ Standing-Seam Metal Roof Panels “ for roof-edge drainage-system components provided by metal-roof-panel manufacturer.
4. Section 076200 “Sheet Metal Flashing and Trim” for custom- and site-fabricated, sheet metal flashing and trim.
5. Section 077253 “Snow Guards” for prefabricated devices designed to hold snow on the roof surface, allowing it to melt and drain off slowly.
6. Section 079200 “Joint Sealants” for field-applied sealants between roof specialties and adjacent materials.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Meet with Contracting Officer, roofing-system testing and inspecting agency representative, roofing Installer, roofing-system manufacturer's representative, Installer, structural-support Installer, and installers whose work interfaces with or affects roof specialties, including installers of roofing materials and accessories.
2. Examine substrate conditions for compliance with requirements, including flatness and attachment to structural members.
3. Review special roof details, roof drainage, and condition of other construction that will affect roof specialties.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of roof specialty.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

B. Shop Drawings: For roof specialties.

1. Plans, expansion-joint locations, keyed details, and attachments to other work. Distinguish between factory pre manufactured- and field-assembled installation.
2. Details for expansion and contraction; locations of expansion joints, including direction of expansion and contraction.
3. Indicate profile and pattern of seams and layout of fasteners, cleats, clips, and other attachments.
4. Details of termination points and assemblies, including fixed points.
5. Details of special conditions.

C. Samples: For each type of roof specialty and for each color and texture specified.

D. Qualification Statements: For manufacturer.

1.4 INFORMATIONAL SUBMITTALS

A. Sample Warranty: For manufacturer's special warranties.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For roof specialties.

B. Manufacturer's Warranties: For all products used.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Do not store roof specialties in contact with other materials that might cause staining, denting, or other surface damage. Store roof specialties away from uncured concrete and masonry.

B. Protect strippable protective covering on roof specialties from exposure to sunlight and high humidity, except to extent necessary for the period of roof-specialty installation.

1.7 FIELD CONDITIONS

A. Field Measurements: Verify profiles and tolerances of roof-specialty substrates by field measurements before fabrication, and indicate measurements on Shop Drawings.

1.8 COORDINATION

A. Coordinate roof specialties with roofing system, exterior wall system, air barrier, flashing, trim, and roof and wall panels, and other adjoining work to provide a leakproof, weathertight, secure, and noncorrosive installation.

1. Performance Coordination: Coordinate with the Work of roofing and exterior wall Sections to ensure that roof specialties provided under the Work of this Section meet or exceed specified roofing and exterior wall design performance requirements.

B. Confirm and coordinate compatibility of materials and comply with warranty requirements of roofing system manufacturer.

C. Coordinate roof specialties layout and seams with sizes and locations of joints and seams in adjacent materials.

1.9 WARRANTY

A. Special Warranty on Painted Finishes: Manufacturer agrees to repair finishes or replace roof specialties that show evidence of deterioration of factory-applied finishes within specified warranty period.

1. Fluoropolymer Finish: Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

2. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. General Performance: Roof specialties to withstand exposure to weather and resist thermally induced movement without failure, rattling, leaking, or fastener disengagement due to defective manufacture, fabrication, installation, or other defects in construction.

B. SPRI Wind Design Standard: Manufacture and install roof-edge specialties tested in accordance with ANSI/SPRI/FM 4435/ES-1 and capable of resisting the following design pressures:

1. Design Pressure: As indicated on Drawings.

C. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Provide clips that resist rotation and avoid shear stress as a result of thermal movements. Base calculations on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.2 ROOF-EDGE DRAINAGE SYSTEMS

A. Basis of Design Product: SAF Perimeter Systems, Geometric Collection, 6” Model DSGC3-6.

B. Gutters: Manufactured in uniform section lengths not exceeding 30 feet, with matching corner units, ends, outlet tubes, and other accessories. Elevate back edge at least 1/2 inch above front edge. Furnish flat-stock gutter straps, gutter brackets, expansion joints, and expansion-joint covers fabricated from same metal as gutters.

1. Aluminum: 0.040 inch thick.
2. Gutter Profile – D-style gutter (per SMACNA's "Architectural Sheet Metal Manual") with a snap on decorative fascia providing the exterior profile shown in the drawings.
3. Corners: Factory mitered and mechanically pinned and sealed watertight.

C. Decorative Fascia shall continue around the gable ends of the entry porch. The gutter is not intended to wrap along the gable ends. See Drawings.

D. Drainware:

1. Downspout & Elbows: 0.125" thickness. Downspouts shall be manufactured from extruded aluminum, alloy 6063-T5 finished to match gutter fascia mouldings. Downspout elbows shall have heliarc welded joints.

    a. Basis of Design: SAF DS-EX
    b. Size & Locations: As indicated on Drawings.

2. Outlets: At all downspout locations provide aluminum outlets to connect liner to downspout.
3. Wall Brackets: Provide Style 1 Wall brackets at 60" maximum spacing (minimum 2 brackets). Brackets shall be manufactured from 0.125" x 1.00" aluminum, finished to match downspout.

E. Finishes:

1. Aluminum: Two-coat fluoropolymer.

    a. Color: Medium Bronze.
    b. Color to be confirmed through submittal process and approval of Contracting Officer.

2.3 SHEET METAL MATERIALS

A. Aluminum Extrusions and Tubes: ASTM B221, manufacturer's standard alloy and temper for type of use, finished to match assembly where used; otherwise, mill finished.

1. Surface – Smooth, flat.
2. Exterior Finish – Two-coat fluoropolymer.
3. Color – To match roof color. May require custom color.

2.4 MISCELLANEOUS MATERIALS

A. Provide materials and types of fasteners, protective coatings, sealants, and other miscellaneous items required by manufacturer for a complete installation.

B. Fasteners: Roof specialty manufacturer's recommended fasteners, designed to meet performance requirements, suitable for application and metals being fastened. Match finish of exposed fasteners with finish of material being fastened. Provide nonremovable fastener heads to exterior exposed fasteners. Furnish the following unless otherwise indicated:

1. Fasteners for Aluminum Sheet: Aluminum or Series 300 stainless steel.

C. Gaskets: Manufacturer's standard tubular or fingered design of neoprene, EPDM, PVC, or silicone or a flat design of foam rubber, sponge neoprene, or cork.

D. Elastomeric Sealant: ASTM C920, elastomeric polyurethane polymer sealant of type, grade, class, and use classifications required by roofing-specialty manufacturer for each application.

E. Butyl Sealant: ASTM C1311, single-component, solvent-release butyl rubber sealant; polyisobutylene plasticized; heavy bodied for hooked-type joints with limited movement.

F. Bituminous Coating: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

G. Asphalt Roofing Cement: ASTM D4586, asbestos free, of consistency required for application.

2.5 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM/NOMMA AMP 500, “Metal Finishes Manual for Architectural and Metal Products,” for recommendations for applying and designating finishes.

B. Protect mechanical and painted finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, to verify actual locations, dimensions, and other conditions affecting performance of the Work.

B. Examine walls and roof edges for suitable conditions for roof specialties.

C. Verify that substrate is sound, dry, smooth, clean, sloped for drainage where applicable, and securely anchored.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

A. Install roof specialties in accordance with manufacturer's written instructions. Anchor roof specialties securely in place, with provisions for thermal and structural movement. Use fasteners, solder, protective coatings, separators, underlayments, sealants, and other miscellaneous items as required to complete roof-specialty systems.

1. Install roof specialties level, plumb, true to line and elevation; with limited oil-canning and without warping, jogs in alignment, buckling, or tool marks.
2. Provide uniform, neat seams with minimum exposure of sealant.
3. Install roof specialties to fit substrates and to result in weathertight performance. Verify shapes and dimensions of surfaces to be covered before manufacture.
4. Torch cutting of roof specialties is not permitted.
5. Do not use graphite pencils to mark metal surfaces.

B. Metal Protection: Protect metals against galvanic action by separating dissimilar metals from contact with each other or with corrosive substrates by painting contact surfaces with bituminous coating or by other permanent separation as recommended by manufacturer's written installation instructions.

1. Coat concealed side of uncoated aluminum roof specialties with bituminous coating where in contact with wood, ferrous metal, or cementitious construction.
2. Bed flanges in thick coat of asphalt roofing cement where required by manufacturers of roof specialties for waterproof performance.

C. Expansion Provisions: Allow for thermal expansion of exposed roof specialties.

1. Space movement joints at a maximum of 29 feet with no joints within 18 inches of corners or intersections unless otherwise indicated on Drawings.
2. When ambient temperature at time of installation is between 40 and 70 deg F, set joint members for 50 percent movement each way. Adjust setting proportionately for installation at higher ambient temperatures.

D. Fastener Sizes: Use fasteners of sizes that penetrate wood blocking or sheathing not less than 1-1/4 inches for nails and not less than 3/4 inch for wood screws.

E. Seal concealed joints with butyl sealant as required by roof specialty manufacturer.

F. Seal joints as required for weathertight construction. Place sealant to be completely concealed in joint. Do not install sealants at temperatures below 40 deg F.

3.3 INSTALLATION OF ROOF-EDGE DRAINAGE SYSTEMS

A. Install components to produce a complete roof-edge drainage system in accordance with manufacturer's written instructions. Coordinate installation of roof perimeter flashing with installation of roof-edge drainage system.

B. Gutters: Join and seal gutter lengths. Allow for thermal expansion. Attach gutters to firmly anchored gutter supports spaced not more than 24 inches apart. Attach ends with rivets and seal with sealant to make watertight. Slope to downspouts.

1. Install gutter with expansion joints at locations indicated but not exceeding 29 feet apart. Install expansion-joint caps.

C. Downspouts: Join sections with manufacturer's standard joints. Provide hangers with fasteners designed to hold downspouts securely to walls and 1 inch away from walls; locate fasteners at top and bottom and at approximately 60 inches o.c.

1. Connect downspouts to underground drainage system indicated.

3.4 CLEANING AND PROTECTION

A. Clean and neutralize flux materials. Clean off excess solder and sealants.

B. Remove temporary protective coverings and strippable films as roof specialties are installed. On completion of installation, clean finished surfaces, including removing unused fasteners, metal filings, pop rivet stems, and pieces of flashing. Maintain roof specialties in a clean condition during construction.

C. Replace roof specialties that have been damaged or that cannot be successfully repaired by finish touchup or similar minor repair procedures, as determined by Contracting Officer.

END OF SECTION 07 71 00

SECTION 07 72 53 – SNOW GUARDS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Rail-type, seam-mounted snow guards.

B. Related Requirements:

1. Section 07 41 13.16 " Standing-Seam Metal Roof Panels " for standing seam roof on which the snow guard is fastened and roof flashing.
2. Section 07 62 00 "Sheet Metal Fashing and Trim" for flashing and trim.
3. Section 07 71 00 "Roof Specialties" for gutters and downspouts.

1.2 ACTION SUBMITTALS

A. Product Data: Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

1. Rail-type, seam-mounted snow guards.

B. Shop Drawings: Include roof plans showing layouts and attachment details of snow guards.

1. Include details of rail-type snow guards.

C. Samples:

1. Rail-Type Snow Guards: Bracket, 12-inch- long rail, and installation hardware.

a. For units with factory-applied finishes, submit specified color.

D. Delegated Design Submittals: For snow guards, include analysis reports signed and sealed by the qualified professional engineer responsible for their preparation.

1. Include calculation of number and location of snow guards.

1.3 INFORMATIONAL SUBMITTALS

A. Qualification Data: For professional engineer's experience with providing delegated design engineering services of the kind indicated, including documentation that the engineer is licensed in the state in which the Project is located.

B. Product Test Reports: For each type of snow guard, for tests performed by a qualified testing agency, indicating load at failure of attachment to roof system identical to roof system used on this Project.

1.4 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit adhesive-mounted snow guards to be installed, and adhesive cured, according to adhesive manufacturer's written instructions.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design snow guards, including attachment to roofing material and roof deck, as applicable for attachment method, based on the following:

1. Roof snow load.
2. Snow drifting
3. Roof slope.
4. Roof type.
5. Roof dimensions.
6. Roofing substrate type and thickness.
7. Snow guard type.
8. Snow guard fastening method and strength.
9. Snow guard spacing.
10. Coefficient of Friction Between Snow and Roof Surface: 0.
11. Factor of Safety: 2.

B. Performance Requirements: Provide snow guards that withstand exposure to weather and resist thermally induced movement without failure, rattling, or fastener disengagement due to defective manufacture, fabrication, installation, or other defects in construction.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

C. Structural Performance: Snow guards to withstand the effects of gravity loads and the following loads and stresses within limits and under conditions indicated.

1. Snow Loads: As indicated on Drawings.

2.2 RAIL-TYPE SNOW GUARDS

A. Rail-Type, Seam-Mounted Snow Guards:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Alpine Snowguards ASG 4025-AL Double Lock Pipe Style Snow Guard or comparable product by one of the following:

a. Alpine SnowGuards
b. IceBlox Inc.
c. LMCurbs
d. PMC Industries, Inc
e. Rocky Mountain Snow Guards, Inc.
f. TRA Snow and Sun, Inc.

2. Description: Snow guard rails fabricated from metal pipes, bars, or extrusions, anchored to brackets and equipped with two rails.
3. Brackets and Baseplate: ASTM A240/A240M, Type 304 stainless steel; 180-grit polished finish, uniform, directionally textured finish.
4. Bars: ASTM A240/A240M, Type 304 stainless steel; 180-grit polished finish, uniform, directionally textured finish.

    a. Profile: Round with integral track to accept color-matching inserts of material and finish used for metal roof.
    b. Two bars

5. Seam Clamps: ASTM B221 aluminum extrusion or ASTM B85/B85M aluminum casting with stainless steel set screws incorporating round nonpenetrating point; designed for use with applicable roofing system to which clamp is attached.
6. Finish/Color:

    a. Powder Coated Finish to match standing seam roofing color
    b. Compliant to AAMA 2604
    c. Gloss Level – Smooth Matte
    d. Basis of Design TAL Series 138 Tiger Drylac RAL 1019
    e. To be confirmed during submittal process by Contracting Officer

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements for installation tolerances, snow guard attachment, and other conditions affecting performance of the Work.

1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean and prepare substrates for bonding snow guards.

B. Prime substrates according to snow guard manufacturer's written instructions.

3.3 INSTALLATION

A. Install snow guards according to manufacturer's written instructions.

1. Space rows as recommended by manufacturer.

B. Attachment for Standing-Seam Metal Roofing:

1. Do not use fasteners that will penetrate metal roofing or fastening methods that void metal roofing finish warranty.
2. Rail-Type, Seam-Mounted Snow Guards:

a. Install brackets to vertical ribs in straight rows.
b. Secure with stainless steel set screws, incorporating round nonpenetrating point, on same side of standing seam.
c. Torque set screw in accordance with manufacturer's written instructions.
d. Install cross members to brackets.

END OF SECTION 07 72 53

SECTION 07 84 13 – PENETRATION FIRESTOPPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Penetration firestopping systems.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry: for wall framing.
2. Section 07 84 43 “Joint Firestopping” for joints in or between fire-resistance-rated construction.
3. Section 09 29 00 “Gypsum Board” for Type X gyp board and adjacent ceilings.
4. Section 09 91 23 “Interior Painting” for adjacent painted materials.
5. Division 21 for Fire Suppression lines.
6. Division 22 for Plumbing lines.
7. Division 23 for Mechanical ductwork.
8. Division 26 for Electrical conduit and fixtures.
9. Division 27 for Communications conduit and fixtures.
10. Division 28 for Electrical Safety and Security conduit and fixtures.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site .

1.3 ACTION SUBMITTALS

A. Product Data: Penetration firestopping systems.

B. Product Schedule: For each penetration firestopping system. Include location, illustration of firestopping system, and design designation of qualified testing and inspecting agency.

1. Engineering Judgments: Where Project conditions require modification to a qualified testing and inspecting agency's illustration for a particular penetration firestopping system, submit illustration, with modifications marked, approved by penetration firestopping system manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly developed in accordance with current International Firestop Council (IFC) guidelines. Obtain approval of authorities having jurisdiction prior to submittal.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

1.5 PROJECT CONDITIONS

A. Environmental Limitations: Do not install penetration firestopping system when ambient or substrate temperatures are outside limits permitted by penetration firestopping system manufacturers or when substrates are wet because of rain, frost, condensation, or other causes.

B. Install and cure penetration firestopping materials per manufacturer's written instructions using natural means of ventilations or, where this is inadequate, forced-air circulation.

1.6 COORDINATION

A. Coordinate construction of openings and penetrating items to ensure that penetration firestopping systems can be installed according to specified firestopping system design.

B. Coordinate sizing of sleeves, openings, core-drilled holes, or cut openings to accommodate penetration firestopping systems.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics:

1. Perform penetration firestopping system tests by a qualified testing agency acceptable to authorities having jurisdiction.
2. Test in accordance with testing standards referenced in “Penetration Firestopping Systems” Article. Provide rated systems complying with the following requirements:

a. Penetration firestop systems installed with products bearing the classification marking of a qualified product certification agency in accordance with listed system designs published by a qualified testing agency.

1) UL in its online directory “Product iQ.”
2) Intertek Group in its “Directory of Building Products.”
3) FM Approvals in its “Approval Guide.”

2.2 PENETRATION FIRESTOPPING SYSTEMS

A. Penetration Firestopping Systems: Systems that resist spread of fire, passage of smoke and other gases, and maintain original fire-resistance rating of construction penetrated. Penetration firestopping systems are to be compatible with one another, with the substrates forming openings, and with penetrating items if any.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. 3M Building and Construction.
b. Balco; a CSW Industrials Company.

c. Everkem Diversified Products, Inc.
d. Hilti, Inc.
e. Holdrite; a division of Reliance Worldwide Corporation.
f. Specified Technologies Inc.
g. Tremco Incorporated.
h. Or Approved Equal

B. Penetrations in Fire-Resistance-Rated Walls: Penetration firestopping systems with ratings determined in accordance with ASTM E814 or UL 1479 and ASTM E3037.

1. F-Rating: Not less than the fire-resistance rating of the wall penetrated.
2. Membrane Penetrations: Install recessed fixtures such that the required fire resistance will not be reduced.
3. M-Rating: compliant with ASTM 3037.

C. Accessories: Provide components for each penetration firestopping system that are needed to install fill materials and to maintain ratings required. Use only those components specified by penetration firestopping system manufacturer and approved by qualified testing and inspecting agency for conditions indicated.

1. Permanent forming/damming/backing materials.
2. Substrate primers.

2.3 FILL MATERIALS

A. Cast-in-Place Firestop Devices: Factory-assembled devices for use in cast-in-place concrete floors and consisting of an outer sleeve lined with an intumescent strip, a flange attached to one end of the sleeve for fastening to concrete formwork, and a neoprene gasket.

B. Latex Sealants: Single-component latex formulations that do not re-emulsify after cure during exposure to moisture.

C. Firestop Devices: Factory-assembled collars formed from galvanized steel and lined with intumescent material sized to fit specific diameter of penetrant.

D. Intumescent Putties: Nonhardening, water-resistant, intumescent putties containing no solvents or inorganic fibers.

E. Intumescent Wrap Strips: Single-component intumescent elastomeric strips for use around combustible penetrants.

F. Mortars: Prepackaged dry mixes consisting of a blend of inorganic binders, hydraulic cement, fillers and lightweight aggregate formulated for mixing with water at Project site to form a nonshrinking, homogeneous mortar.

G. Pillows/Bags: Compressible, removable, and reusable intumescent pillows encased in fire-retardant polyester or glass-fiber cloth. Where exposed, cover openings with steel-reinforcing wire mesh to protect pillows/bags from being easily removed.

H. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.

I. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants.

J. Wall-Opening Protective Materials: Intumescent, non-curing putty pads or self-adhesive inserts for protection of electrical switch and receptacle boxes.

K. Firestop Plugs: Flexible, re-enterable, intumescent, foam-rubber plug for use in blank round openings and cable sleeves.

L. Fire-Rated Cable Grommet: Molded two-piece grommet made of plenum-grade polymer and foam inner core for sealing small cable penetrations in gypsum walls up to 1/2 inch in diameter.

M. Endothermic Wrap: Flexible, insulating, fire-resistant, endothermic wrap for protecting membrane penetrations of utility boxes, critical electrical circuits, communications lines, and fuel lines.

2.4 MIXING

A. Penetration Firestopping Materials: For those products requiring mixing before application, comply with penetration firestopping system manufacturer's written instructions for accurate proportioning of materials, water (if required), type of mixing equipment, selection of mixer speeds, mixing containers, mixing time, and other items or procedures needed to produce products of uniform quality with optimum performance characteristics for application indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements for opening configurations, penetrating items, substrates, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning: Before installing penetration firestopping systems, clean out openings immediately to comply with manufacturer's written instructions and with the following requirements:

1. Remove from surfaces of opening substrates and from penetrating items foreign materials that could interfere with adhesion of penetration firestopping materials.
2. Clean opening substrates and penetrating items to produce clean, sound surfaces capable of developing optimum bond with penetration firestopping materials. Remove loose particles remaining from cleaning operation.
3. Remove laitance and form-release agents from concrete.

B. Prime substrates where recommended in writing by manufacturer using that manufacturer's recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

3.3 INSTALLATION OF PENETRATION FIRESTOPPING SYSTEMS

A. General: Install penetration firestopping systems to comply with manufacturer's written installation instructions and published drawings for products and applications.

B. Install forming materials and other accessories of types required to support fill materials during their application and in the position needed to produce cross-sectional shapes and depths required to achieve fire ratings.

1. After installing fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not forming permanent components of firestopping.

C. Install fill materials by proven techniques to produce the following results:

1. Fill voids and cavities formed by openings, forming materials, accessories and penetrating items to achieve required fire-resistance ratings.
2. Apply materials so they contact and adhere to substrates formed by openings and penetrating items.
3. For fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.4 IDENTIFICATION

A. Wall Identification: Permanently label walls containing penetration firestopping systems with the words “FIRE AND/OR SMOKE BARRIER – PROTECT ALL OPENINGS,” using lettering not less than 3 inches high and with minimum 0.375-inch strokes.

1. If possible, locate above ceiling at 15 ft. from end of wall and at intervals not exceeding 30 ft.
2. If not possible, locate at top of wall, 6-8” below ceiling at 15 ft. from end of wall and at intervals not exceeding 30 ft.

3.5 CLEANING AND PROTECTION

A. Clean off excess fill materials adjacent to openings as the Work progresses by methods and with cleaning materials that are approved in writing by penetration firestopping system manufacturers and that do not damage materials in which openings occur.

B. Provide final protection and maintain conditions during and after installation that ensure that penetration firestopping systems are without damage or deterioration at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, immediately cut out and remove damaged or deteriorated penetration firestopping material and install new materials to produce systems complying with specified requirements.

END OF SECTION 07 84 13

SECTION 07 84 43 – JOINT FIRESTOPPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Joints in or between fire-resistance-rated construction.

a. Joints at top and bottom of rated walls.
b. Joints where rated walls meet structure.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry: for wall framing.
2. Section 07 84 13 “Penetration Firestopping” for penetrations in fire-resistance-rated walls, horizontal assemblies, and smoke barriers and for wall identification.
3. Section 09 22 16 “Non-structural Metal Framing” for framing of adjacent ceilings.
4. Section 09 29 00 “Gypsum Board” for Type X gyp board and adjacent ceilings.
5. Section 09 65 19 “Resilient Tile Rubber Flooring” for adjacent rubber base.
6. Section 09 91 23 “Interior Painting” for adjacent painted materials.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site .

1.3 ACTION SUBMITTALS

A. Product Data:

1. Joints in or between fire-resistance-rated construction.

B. Product Schedule: For each joint firestopping system. Include location, illustration of firestopping system, and design designation of qualified testing agency.

1. Engineering Judgments: Where Project conditions require modification to a qualified testing agency's illustration for a particular joint firestopping system condition, submit illustration, with modifications marked, approved by joint firestopping system manufacturer's fire-protection engineer as an engineering judgment or equivalent fire-resistance-rated assembly developed in accordance with current International Firestop Council (IFC) guidelines.

1.4 INFORMATIONAL SUBMITTALS

A. Qualification Data: For Installer.

B. Listed System Designs: For each joint firestopping system, for tests performed by a qualified testing agency.

1.5 CLOSEOUT SUBMITTALS

A. Installer Certificates: From Installer indicating that joint firestopping systems have been installed in compliance with requirements and manufacturer's written instructions.

1.6 PROJECT CONDITIONS

A. Environmental Limitations: Do not install joint firestopping systems when ambient or substrate temperatures are outside limits permitted by joint firestopping system manufacturers or when substrates are wet due to rain, frost, condensation, or other causes.

B. Install and cure joint firestopping systems per manufacturer's written instructions using natural means of ventilation or, where this is inadequate, forced-air circulation.

1.7 COORDINATION

A. Coordinate construction of joints to ensure that joint firestopping systems can be installed according to specified firestopping system design.

B. Coordinate sizing of joints to accommodate joint firestopping systems.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain joint firestop systems for each type of joint opening indicated from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics:

1. Perform joint firestopping system tests by a qualified testing agency acceptable to authorities having jurisdiction.
2. Test per testing standards referenced in “Joint Firestopping Systems” Article. Provide rated systems complying with the following requirements:

a. Joint firestop systems installed with products bearing the classification marking of a qualified product certification agency in accordance with Listed System Designs published by a qualified testing agency.

1) UL in its online directory "Product iQ."
2) Intertek Group in its "Directory of Building Products."

2.3 JOINT FIRESTOPPING SYSTEMS

A. Joint Firestopping Systems: Systems that resist spread of fire, passage of smoke and other gases, and maintain original fire-resistance rating of assemblies in or between which joint firestopping systems are installed. Joint firestopping systems must accommodate building movements without impairing their ability to resist the passage of fire and hot gases.

1. Joint firestopping systems that are compatible with one another, with the substrates forming openings, and with penetrating items, if any.
2. Provide products that, upon curing, do not re-emulsify, dissolve, leach, breakdown, or otherwise deteriorate over time from exposure to atmospheric moisture, sweating pipes, ponding water or other forms of moisture.
3. Provide firestop products that do not contain ethylene glycol.

B. Joints in or between Fire-Resistance-Rated Construction: Provide joint firestopping systems with ratings determined per ASTM E1966 or UL 2079.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. 3M Building and Construction.
b. Hilti, Inc.
c. Owens Corning.
d. ROCKWOOL.
e. Tremco Incorporated.
f. Or Approved Equal.

2. Fire-Resistance Rating: Equal to or exceeding the fire-resistance rating of the wall, floor, or roof in or between which it is installed.

C. Exposed Joint Firestopping Systems: Flame-spread and smoke-developed indexes of less than 25 and 450, respectively, as determined per ASTM E84.

1. Verify sealant has a VOC content of 250 g/L or less.

2.4 ACCESSORIES

A. Provide components of joint firestopping systems, including primers and forming materials, that are needed to install elastomeric fill materials and to maintain ratings required. Use only components specified by joint firestopping system manufacturer and approved by the qualified testing agency for conditions indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements for joint configurations, substrates, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning: Before installing joint firestopping systems, clean joints immediately to comply with fire-resistive joint system manufacturer's written instructions and the following requirements:

1. Remove from surfaces of joint substrates foreign materials that could interfere with adhesion of elastomeric fill materials or compromise fire-resistive rating.
2. Clean joint substrates to produce clean, sound surfaces capable of developing optimum bond with elastomeric fill materials. Remove loose particles remaining from cleaning operation.
3. Remove laitance and form-release agents from concrete.

B. Prime substrates where recommended in writing by joint firestopping system manufacturer using that manufacturer's recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

C. Apply a suitable bond-breaker to prevent three-sided adhesion in applications where this condition occurs, such as the intersection of a gypsum wall to floor or roof assembly where the joint is backed by a steel ceiling runner or track.

3.3 INSTALLATION

A. General: Install joint firestopping systems to comply with manufacturer's written installation instructions and published drawings for products and applications indicated.

B. Install forming materials and other accessories of types required to support elastomeric fill materials during their application and in position needed to produce cross-sectional shapes and depths required to achieve fire ratings indicated.

1. After installing elastomeric fill materials and allowing them to fully cure, remove combustible forming materials and other accessories not indicated as permanent components of fire-resistive joint system.

C. Install elastomeric fill materials for joint firestopping systems by proven techniques to produce the following results:

1. Elastomeric fill voids and cavities formed by joints and forming materials as required to achieve fire-resistance ratings indicated.

2. Apply elastomeric fill materials so they contact and adhere to substrates formed by joints.
3. For elastomeric fill materials that will remain exposed after completing the Work, finish to produce smooth, uniform surfaces that are flush with adjoining finishes.

3.4 IDENTIFICATION

A. Wall Identification: Permanently label walls containing firestopping systems with the words "FIRE AND/OR SMOKE BARRIER – PROTECT ALL OPENINGS," using lettering not less than 3 inches high and with minimum 0.375-inch strokes.

1. If possible, locate above ceiling at 15 ft. from end of wall and at intervals not exceeding 30 ft.
2. If not possible, locate at top of wall, 6-8" below ceiling at 15 ft. from end of wall and at intervals not exceeding 30 ft.

3.5 FIELD QUALITY CONTROL

A. Where deficiencies are found or joint firestopping systems are damaged, repair or replace joint firestopping systems so they comply with requirements.

B. Proceed with enclosing joint firestopping systems with other construction only after visual inspection conducted and installations comply with requirements.

3.6 CLEANING AND PROTECTION

A. Clean off excess elastomeric fill materials adjacent to joints as the Work progresses by methods and with cleaning materials that are approved in writing by joint firestopping system manufacturers and that do not damage materials in which joints occur.

B. Provide final protection and maintain conditions during and after installation that ensure joint firestopping systems are without damage or deterioration at time of Substantial Completion. If damage or deterioration occurs despite such protection, cut out and remove damaged or deteriorated joint firestopping systems immediately and install new materials to produce joint firestopping systems complying with specified requirements.

END OF SECTION 07 84 43

SECTION 07 92 00 – JOINT SEALANTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Silicone joint sealants.
2. Nonstaining silicone joint sealants.
3. Urethane joint sealants.
4. Butyl joint sealants.

1.2 ACTION SUBMITTALS

A. Product Data:

1. Joint sealants.
2. Joint sealant backing materials.

B. Samples for Initial Selection: Manufacturer's standard color charts consisting of strips of cured sealants showing the full range of colors available for each product exposed to view.

C. Joint-Sealant Schedule: Include the following information:

1. Joint-sealant application, joint location, and designation.
2. Joint-sealant manufacturer and product name.
3. Joint-sealant formulation.
4. Joint-sealant color.

1.3 CLOSEOUT SUBMITTALS

A. Joint Sealant Schedule, with manufacturers and colors identified for each location.

1.4 MOCKUPS

A. Install sealant in mockups of assemblies specified in other Sections that are indicated to receive joint sealants specified in this Section. Use materials and installation methods specified in this Section.

1. Requires approval of Contracting Officer to proceed.

1.5 FIELD CONDITIONS

A. Do not proceed with installation of joint sealants under the following conditions:

1. When ambient and substrate temperature conditions are outside limits permitted by joint-sealant manufacturer.
2. When joint substrates are wet.
3. Where joint widths are less than those allowed by joint-sealant manufacturer for applications indicated.
4. Where contaminants capable of interfering with adhesion have not yet been removed from joint substrates.

PART 2 - PRODUCTS

2.1 JOINT SEALANTS, GENERAL

A. Compatibility: Provide joint sealants, backings, and other related materials that are compatible with one another and with joint substrates under conditions of service and application, as demonstrated by joint-sealant manufacturer, based on testing and field experience.

B. Sustainability:

1. Confirm the compliance of all joint sealants with the requirements of Section 01 57 19.11 “Indoor Air Quality Management.”

C. Manufacturers:

1. Sika Corporation.
2. Tremco Incorporated.
3. Or Approved Equal.

D. Colors of Exposed Joint Sealants: As selected by Contracting Officer to blend with adjacent materials. .

2.2 NONSTAINING SILICONE JOINT SEALANTS

A. Nonstaining Joint Sealants: No staining of substrates when tested in accordance with ASTM C1248.

B. Silicone, Nonstaining, S, NS, 50, NT: Nonstaining, single-component, nonsag, plus 50 percent and minus 50 percent movement capability, nontraffic-use, neutral-curing silicone joint sealant; ASTM C920, Type S, Grade NS, Class 50, Use NT.

2.3 URETHANE JOINT SEALANTS

A. Urethane, S, NS, 25, NT: Single-component, nonsag, nontraffic-use, plus 25 percent and minus 25 percent movement capability, urethane joint sealant; ASTM C920, Type S, Grade NS, Class 25, Use NT.

B. Urethane, S, P, 25, T, NT: Single-component, pourable, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C920, Type S, Grade P, Class 25, Uses T and NT.

C. Urethane, M, P, 50, T, NT: Multicomponent, pourable, plus 50 percent and minus 50 percent movement capability, traffic- and nontraffic-use, urethane joint sealant; ASTM C920, Type M, Grade P, Class 50, Uses T and NT.

2.4 BUTYL JOINT SEALANTS

A. Butyl-Rubber-Based Joint Sealants: ASTM C1311.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

    a. Bostik; Arkema.
    b. Pecora Corporation.
    c. Or Approved Equal.

2.5 JOINT-SEALANT BACKING

A. Sealant Backing Material, General: Nonstaining; compatible with joint substrates, sealants, primers, and other joint fillers; and approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

    a. Master Builders Solutions.
    b. Or Approved Equal.

B. Cylindrical Sealant Backings: ASTM C1330, Type C (closed-cell material with a surface skin) Type B (bicellular material with a surface skin) , and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.

2.6 MISCELLANEOUS MATERIALS

A. Primer: Material recommended by joint-sealant manufacturer where required for adhesion of sealant to joint substrates indicated, as determined from preconstruction joint-sealant-substrate tests and field tests.

B. Cleaners for Nonporous Surfaces: Chemical cleaners acceptable to manufacturers of sealants and sealant backing materials, free of oily residues or other substances capable of staining or harming joint substrates and adjacent nonporous surfaces in any way, and formulated to promote optimum adhesion of sealants to joint substrates.

C. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine joints indicated to receive joint sealants, with Installer present, for compliance with requirements for joint configuration, installation tolerances, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Surface Cleaning of Joints: Clean out joints immediately before installing joint sealants to comply with joint-sealant manufacturer's written instructions and the following requirements:

1. Remove all foreign material from joint substrates that could interfere with adhesion of joint sealant, including dust, paints (except for permanent, protective coatings tested and approved for sealant adhesion and compatibility by sealant manufacturer), old joint sealants, oil, grease, waterproofing, water repellents, water, surface dirt, and frost.
2. Clean porous joint substrate surfaces by brushing, grinding, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants. Remove loose particles remaining after cleaning operations above by vacuuming or blowing out joints with oil-free compressed air. Porous joint substrates include the following:

    a. Concrete.
    b. Masonry.

3. Remove laitance and form-release agents from concrete.
4. Clean nonporous joint substrate surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants. Nonporous joint substrates include the following:

    a. Metal.
    b. Glass.
    c. Glazed surfaces of ceramic tile.

B. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.

C. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.3 INSTALLATION OF JOINT SEALANTS

A. General: Comply with joint-sealant manufacturer's written installation instructions for products and applications indicated, unless more stringent requirements apply.

B. Sealant Installation Standard: Comply with recommendations in ASTM C1193 for use of joint sealants as applicable to materials, applications, and conditions indicated.

C. Install sealant backings of type indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.

1. Do not leave gaps between ends of sealant backings.
2. Do not stretch, twist, puncture, or tear sealant backings.
3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.

D. Install bond-breaker tape behind sealants where sealant backings are not used between sealants and backs of joints.

E. Install sealants using proven techniques that comply with the following and at the same time backings are installed:

1. Place sealants so they directly contact and fully wet joint substrates.
2. Completely fill recesses in each joint configuration.
3. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.

F. Tooling of Nonsag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants according to requirements specified in subparagraphs below to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint.

1. Remove excess sealant from surfaces adjacent to joints.
2. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.
3. Provide concave joint profile in accordance with Figure 8A in ASTM C1193 unless otherwise indicated.

3.4 CLEANING

A. Clean off excess sealant or sealant smears adjacent to joints as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur.

3.5 PROTECTION

A. Protect joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out, remove, and repair damaged or deteriorated joint sealants immediately so installations with repaired areas are indistinguishable from original work.

3.6 JOINT-SEALANT SCHEDULE

A. Exterior joints in horizontal traffic surfaces:

1. Joint Locations:

    a. Joints between concrete flatwork and building foundation.

2. Joint Sealant: Urethane, M, P, 50, T, NT.
3. Joint-Sealant Color: As selected by Contracting Officer from manufacturer's full range of colors.

B. Exterior joints in vertical surfaces and horizontal nontraffic surfaces:

1. Joint Locations:

    a. Control and expansion joints in unit masonry.
    b. Joints in exterior insulation and finish systems.
    c. Joints between metal panels.
    d. Joints between different materials listed above.
    e. Perimeter joints between materials listed above and frames of doors windows and louvers.

2. Joint Sealant: Silicone, nonstaining, S, NS, 50, NT.
3. Joint-Sealant Color: As selected by Contracting Officer from manufacturer's full range, design intent to match adjacent colors.

C. Interior joints in vertical surfaces and horizontal nontraffic surfaces:

1. Joint Locations:

    a. Vertical joints on exposed surfaces of walls and partitions.

2. Joint Sealant: Urethane, S, NS, 25, NT.
3. Joint-Sealant Color: As selected by Contracting Officer from manufacturer's full range of colors.

D. Mildew-resistant interior joints in vertical surfaces and horizontal nontraffic surfaces:

1. Joint Locations:

a. Joints between plumbing fixtures and adjoining walls, floors, and counters.
b. Joints between solid surface panels.

2. Joint Sealant: Silicone, mildew resistant, acid curing, S, NS, 25, NT.
3. Joint-Sealant Color: As selected by Contracting Officer from manufacturer's full range of colors.

E. Concealed mastics:

1. Joint Locations:

a. Aluminum thresholds.
b. Sill plates.
c. Other joints as indicated on Drawings.

2. Joint Sealant: Butyl-rubber based.
3. Joint-Sealant Color: As selected by Contracting Officer from manufacturer's full range of colors.

END OF SECTION 07 92 00

SECTION 08 11 13 – HOLLOW METAL DOORS AND FRAMES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Interior standard steel doors and frames.
2. Exterior insulated standard steel doors and frames.
3. Interior custom hollow-metal doors and frames.
4. Fire-rated hollow-metal doors and frames.

B. Related Requirements:

1. Section 087100 “Door Hardware” for door hardware for hollow-metal doors.
2. Section 084113 “Aluminum-Framed Entrances and Storefronts” for aluminum-framed entrances and storefronts
3. Section 081416 “Flush Wood Doors” for flush wood doors.
4. Section 088000 ‘Glazing” for glazing in sidelight conditions within hollow metal frames.

1.2 DEFINITIONS

A. Minimum Thickness: Minimum thickness of base metal without coatings in accordance with NAAMM-HMMA 803 or ANSI/SDI A250.8.

B. “Standard-Duty Doors and Frames” described interior-use assemblies fabricated to SDI requirements and containing doors with 0.032-inch (0.8mm) thick, (nominal 20 gage) face sheets and laminated cores. ANSI/SDI A250.8, Level 1; ANSI/SDI A250.4, Level C.

C. “Heavy-Duty Doors and Frames” described exterior-use assemblies fabricated to SDI requirements and containing doors with 0.042-inch (1.0mm) thick, (nominal 18 gage) face sheets and laminated cores. ANSI/SDI A250.8, Level 2; ANSI/SDI A250.4, Level B.

1.3 COORDINATION

A. Coordinate anchorage installation for hollow-metal frames. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors. Deliver such items to Project site in time for installation.

B. Coordinate requirements for installation of door hardware, electrified door hardware, and access control and security systems.

C. Accessibility: Coordinate all door and frame components with related specification sections to ensure compliance with ABAAS accessibility requirements. Including but not limited to door hardware and thresholds.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Interior standard steel doors and frames.
2. Exterior insulated standard steel doors and frames.
3. Fire-rated hollow-metal doors and frames.

B. Product Data Submittals: For each product.

1. Include construction details, material descriptions, core descriptions, fire-resistance ratings, and finishes.

C. Shop Drawings: Include the following:

1. Elevations of each door type.
2. Details of doors, including vertical- and horizontal-edge details and metal thicknesses.
3. Frame details for each frame type, including dimensioned profiles and metal thicknesses.
4. Locations of reinforcement and preparations for hardware.
5. Details of each different wall opening condition.
6. Details of electrical raceway and preparation for electrified hardware, access control systems, and security systems.
7. Details of anchorages, joints, field splices, and connections.
8. Details of accessories.
9. Details of moldings, removable stops, and glazing.

D. Product Schedule: For hollow-metal doors and frames, prepared by or under the supervision of supplier, using same reference numbers for details and openings as those on Drawings. Coordinate with final door hardware schedule.

E. Qualification Data: For door installer.

1.5 INFORMATIONAL SUBMITTALS

A. Fire-rated Door Inspector: Submit documentation of compliance with NFPA 80, Section 5.2.3.1.

B. Egress Door Inspector: Submit documentation of compliance with NFPA 101, Section 7.2.1.15.4.

C. Product Test Reports: For each type of fire-rated hollow-metal door and frame assemblies and thermally rated door assemblies for tests performed by a qualified testing agency indicating compliance with performance requirements.

D. Field quality-control reports.

1.6 QUALITY ASSURANCE

A. Fire-Rated Door Inspector Qualifications: Inspector for field quality-control inspections of fire-rated door assemblies is to meet the qualifications set forth in NFPA 80, Section 5.2.3.1 and the following:

1. Door and Hardware Institute Fire and Egress Door Assembly Inspector (FDAI) certification.

B. Egress Door Inspector Qualifications: Inspector for field quality-control inspections of egress door assemblies is to meet the qualifications set forth in NFPA 101, Section 7.2.1.15.4 and the following:

1. Door and Hardware Institute Fire and Egress Door Assembly Inspector (FDAI) certification

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver hollow-metal doors and frames palletized, packaged, or crated to provide protection during transit and Project-site storage. Do not use nonvented plastic.

1. Provide additional protection to prevent damage to factory-finished units.

B. Deliver welded frames with two removable spreader bars across bottom of frames, tack welded to jambs and mullions.

C. Store hollow-metal doors and frames vertically under cover at Project site with head up. Place on minimum 4-inch- high wood blocking. Provide minimum 1/4-inch space between each stacked door to permit air circulation.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction for fire-protection ratings indicated on Drawings, based on testing at positive pressure in accordance with NFPA 252 or UL 10C.

B. Thermally Rated Door Assemblies at Exterior Doors: Provide door assemblies with U-factor of not more than U-factor of 0.37 when tested in accordance with ASTM C1363 or ASTM E1423.

2.2 INTERIOR HEAVY-DUTY STEEL DOORS AND FRAMES

A. Construct hollow-metal doors and frames to comply with standards indicated for materials, fabrication, hardware locations, hardware reinforcement, tolerances, and clearances, and as specified.

B. Heavy-Duty Doors and Frames: ANSI/SDI A250.8, Level 2; ANSI/SDI A250.4, Level B. At locations indicated in the Door and Frame Schedule on Drawings .

1. Doors:

a. Type: As indicated in the Door and Frame Schedule on Drawings.

b. Thickness: 1-3/4 inches .
c. Face: Metallic-coated steel sheet, minimum thickness of 0.042 inch.
d. Edge Construction: Model 2, Seamless .
e. Edge Bevel: Provide manufacturer's standard beveled or square edges .
f. Core: Kraft-paper honeycomb.
g. Fire-Rated Core: Manufacturer's standard laminated mineral board core for fire-rated doors.

2. Frames:

a. Materials: Metallic-coated steel sheet, minimum thickness of 0.053 inch.
b. Sidelite and Transom Frames: Fabricated from same thickness material as adjacent door frame.
c. Construction: Full profile welded.

3. Exposed Finish: Primed.
4. Glazing:

a. Refer to section 088000 “Glazing” for borrowed lites and sidelight requirements.

2.3 EXTERIOR HEAVY-DUTY STEEL DOORS AND FRAMES

A. Construct hollow-metal doors and frames to comply with standards indicated for materials, fabrication, hardware locations, hardware reinforcement, tolerances, and clearances, and as specified.

B. Heavy-Duty Doors and Frames: ANSI/SDI A250.8, Level 2; ANSI/SDI A250.4, Level B. At locations indicated in the Door and Frame Schedule on Drawings .

1. Doors:

a. Type: As indicated in the Door and Frame Schedule on Drawings.
b. Thickness: 1-3/4 inches.
c. Face: Metallic-coated steel sheet, minimum thickness of 0.042 inch, with minimum A60 coating.
d. Edge Construction: Model 2, Seamless .
e. Edge Bevel: Provide manufacturer's standard beveled or square edges .
f. Top Edge Closures: Close top edges of doors with flush closures of same material as face sheets. Seal joints against water penetration.
g. Bottom Edges: Close bottom edges of doors where required for attachment of weather stripping with end closures or channels of same material as face sheets. Provide weep-hole openings in bottoms of exterior doors to permit moisture to escape.
h. Core: Polyurethane.
i. Maximum U-Factor: 0.37

2. Frames:

   a. Materials: Metallic-coated steel sheet, minimum thickness of 0.053 inch, with minimum A60 coating.
   b. Construction: Full profile welded .

3. Exposed Finish: Primed.

2.4 BORROWED LITES

A. Fabricate of metallic-coated steel sheet, minimum thickness of 0.053 inch.

B. Construction: Full profile welded .

C. Fabricate in one piece except where handling and shipping limitations require multiple sections. Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of metal of same or greater thickness as metal as frames.

D. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated.

2.5 FRAME ANCHORS

A. Jamb Anchors:

1. Type: Anchors of minimum size and type required by applicable door and frame standard, and suitable for performance level indicated.
2. Quantity: Minimum of three anchors per jamb, with one additional anchor for frames with no floor anchor. Provide one additional anchor for each 24 inches of frame height above 7 feet.
3. Postinstalled Expansion Anchor: Minimum 3/8-inch- diameter bolts with expansion shields or inserts, with manufacturer's standard pipe spacer.

B. Floor Anchors: Adjustable-type anchors with extension clips, allowing not less than 2-inch height adjustment. Terminate bottom of frames at top of underlayment.

C. Material: ASTM A879/A879M, Commercial Steel (CS), 04Z coating designation; mill phosphatized.

1. For anchors built into exterior walls, steel sheet complying with ASTM A1008/A1008M or ASTM A1011/A1011M; hot-dip galvanized in accordance with ASTM A153/A153M, Class B.

2.6 MATERIALS

A. Cold-Rolled Steel Sheet: ASTM A1008/A1008M, Commercial Steel (CS), Type B; suitable for exposed applications.

B. Metallic-Coated Steel Sheet: ASTM A653/A653M, Commercial Steel (CS), Type B.

C. Inserts, Bolts, and Fasteners: Hot-dip galvanized in accordance with ASTM A153/A153M.

D. Power-Actuated Fasteners in Concrete: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with clips or other accessory devices for attaching hollow-metal frames of type indicated.

E. Mineral-Fiber Insulation: ASTM C665, Type I (blankets without membrane facing); consisting of fibers manufactured from slag or rock wool; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively; passing ASTM E136 for combustion characteristics.

F. Glazing: Comply with requirements in Section 088000 “Glazing.”

2.7 FABRICATION

A. Door Astragals: Provide overlapping astragal on one leaf of pairs of doors where required by NFPA 80 for fire-performance rating or where indicated. Extend minimum 3/4 inch beyond edge of door on which astragal is mounted or as required to comply with published listing of qualified testing agency.

B. Hollow-Metal Frames: Fabricate in one piece except where handling and shipping limitations require multiple sections. Where frames are fabricated in sections, provide alignment plates or angles at each joint, fabricated of metal of same or greater thickness as frames.

1. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated.
2. Door Silencers: Except on weather-stripped frames, drill stops to receive door silencers as follows. Keep holes clear during construction.

    a. Single-Door Frames: Drill stop in strike jamb to receive three door silencers.
    b. Double-Door Frames: Drill stop in head jamb to receive two door silencers.

C. Hardware Preparation: Factory prepare hollow-metal doors and frames to receive templated mortised hardware, and electrical wiring; include cutouts, reinforcement, mortising, drilling, and tapping in accordance with ANSI/SDI A250.6, the Door Hardware Schedule on Drawings, and templates.

1. Reinforce doors and frames to receive nontemplated, mortised, and surface-mounted door hardware.
2. Comply with BHMA A156.115 for preparing hollow-metal doors and frames for hardware.

D. Glazed Lites: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with mitered hairline joints.

1. Provide stops and moldings flush with face of door, and with beveled stops unless otherwise indicated.

2. Provide fixed frame moldings on outside of exterior and on secure side of interior doors and frames. Provide loose stops and moldings on inside of hollow-metal doors and frames.
3. Coordinate rabbet width between fixed and removable stops with glazing and installation types indicated.
4. Provide stops for installation with countersunk flat- or oval-head machine screws spaced uniformly not more than 9 inches o.c. and not more than 2 inches o.c. from each corner.

2.8 STEEL FINISHES

A. Prime Finish: Clean, pretreat, and apply manufacturer's standard primer.

1. Shop Primer: Manufacturer's standard, fast-curing, lead- and chromate-free primer complying with ANSI/SDI A250.10; recommended by primer manufacturer for substrate; compatible with substrate and field-applied coatings despite prolonged exposure.

PART 3 - EXECUTION

3.1 PREPARATION

A. Remove welded-in shipping spreaders installed at factory. Restore exposed finish by grinding, filling, and dressing, as required to make repaired area smooth, flush, and invisible on exposed faces. Touch up factory-applied finishes where spreaders are removed.

B. Drill and tap doors and frames to receive nontemplated, mortised, and surface-mounted door hardware.

3.2 INSTALLATION

A. Install hollow-metal doors and frames plumb, rigid, properly aligned, and securely fastened in place. Comply with approved Shop Drawings and with manufacturer's written instructions.

B. Hollow-Metal Frames: Comply with ANSI/SDI A250.11 .

1. Set frames accurately in position; plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is complete, remove temporary braces without damage to completed Work.

a. Where frames are fabricated in sections, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces. Touch-up finishes.
b. Install frames with removable stops located on secure side of opening.

2. Fire-Rated Openings: Install frames in accordance with NFPA 80.
3. Floor Anchors: Secure with postinstalled expansion anchors.

a. Floor anchors may be set with power-actuated fasteners instead of postinstalled expansion anchors if so indicated and approved on Shop Drawings.

4. Solidly pack mineral-fiber insulation inside frames.
5. Masonry Walls: Coordinate installation of frames to allow for solidly filling space between frames and masonry with grout or mortar.
6. In-Place Concrete or Masonry Construction: Secure frames in place with postinstalled expansion anchors. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.
7. Installation Tolerances: Adjust hollow-metal frames to the following tolerances:

    a. Squareness: Plus or minus 1/16 inch, measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.
    b. Alignment: Plus or minus 1/16 inch, measured at jambs on a horizontal line parallel to plane of wall.
    c. Twist: Plus or minus 1/16 inch, measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.
    d. Plumbness: Plus or minus 1/16 inch, measured at jambs at floor.

C. Hollow-Metal Doors: Fit and adjust hollow-metal doors accurately in frames, within clearances specified below.

1. Non-Fire-Rated Steel Doors: Comply with ANSI/SDI A250.8.
2. Fire Rated Doors: Install doors with clearances in accordance with NFPA 80.

D. Glazing: Comply with installation requirements in Section 08 80 00 “Glazing” and with hollow-metal manufacturer's written instructions.

3.3 FIELD QUALITY CONTROL

A. Inspection Agency: Engage a qualified inspector to perform inspections and to furnish reports to Contracting Officer.

B. Inspections:

1. Fire-Rated Door Inspections: Inspect each fire-rated door in accordance with NFPA 80, Section 5.2.
2. Egress Door Inspections: Inspect each door equipped with panic hardware, located in an exit enclosure, each electrically controlled egress door, and each door equipped with special locking arrangements in accordance with NFPA 101, Section 7.2.1.15.
3. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

C. Repair or remove and replace installations where inspections indicate that they do not comply with specified requirements.

D. Reinspect repaired or replaced installations to determine if replaced or repaired door assembly installations comply with specified requirements.

E. Prepare and submit separate inspection report for each fire-rated door assembly indicating compliance with each item listen in NFPA 80 and NFPA 101.

3.4 REPAIR

A. Prime-Coat Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat and apply touchup of compatible air-drying, rust-inhibitive primer.

B. Metallic-Coated Surface Touchup: Clean abraded areas and repair with galvanizing repair paint in accordance with manufacturer's written instructions.

C. Touchup Painting: Cleaning and touchup painting of abraded areas of paint are specified in painting Sections.

END OF SECTION 08 11 13

SECTION 08 14 16 – FLUSH WOOD DOORS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Solid-core flush wood veneer-faced doors for stained finish.
2. Fire-rated mineral-core flush wood veneer-faced doors for stained finish.

B. Related Requirements:

1. Section 084113 “Aluminum-Framed Entrances And Storefronts” for conditions where wood doors are to be installed within new aluminum frames.
2. Section 087100 “Door Hardware” for coordination with new door hardware.
3. Section 088000 “Glazing” for glass view panels in flush wood doors and fire rated glazing for fire-rated doors.
4. Section 099300 “Staining and Transparent Finishing” for field finishing doors.

1.2 ACTION SUBMITTALS

A. Product Data:

1. Solid-core flush wood veneer-faced doors for stained finish.
2. Fire-rated mineral core-core flush wood veneer-faced doors for stained finish.

B. Product Data Submittals: For each product, including the following:

1. Door core materials and construction.
2. Door edge construction
3. Door face type and characteristics.
4. Door trim for openings.
5. Door frame construction.
6. Factory-machining criteria.
7. Factory- priming specifications.
8. Fire-resistance ratings, where applicable.

C. Sustainable Design Submittals:

1. Environmental Product Declaration: For each product.
2. Third-Party Certifications: For each product.
3. Product Data: For adhesives, indicating that product contains no urea formaldehyde.
4. Product Data: For composite wood products, indicating that product contains no urea formaldehyde.

D. Shop Drawings: Indicate location, size, and hand of each door; elevation of each type of door; construction details not covered in Product Data; and the following:

1. Door schedule indicating door and frame location, type, size, fire protection rating, and swing.
2. Door elevations, dimension and locations of hardware, lite and louver cutouts, and glazing thicknesses.
3. Details of frame for each frame type, including dimensions and profile.
4. Details of electrical raceway and preparation for electrified hardware, access control systems, and security systems.
5. Dimensions and locations of blocking for hardware attachment.
6. Dimensions and locations of mortises and holes for hardware.
7. Clearances and undercuts.
8. Requirements for veneer matching.
9. Doors to be factory primed and application requirements.
10. Apply AWI Quality Certification WI Certified Compliance Program label to Shop Drawings.
11. Fire-resistance ratings, where applicable.

E. Samples for Initial Selection: For polymer edging.

F. Samples for Verification:

1. Factory finishes applied to actual door face materials, approximately 8 by 10 inches, for each material and finish. For each wood species and stained finish, provide set of three Samples showing typical range of color and grain to be expected in finished Work.
2. Plastic laminate, 6 inches square, for each color, texture, and pattern selected.
3. Polymer edging, in manufacturer's standard colors.
4. Corner sections of doors, approximately 8 by 10 inches, with door faces and edges representing actual materials to be used.
5. Louver blade and frame sections, 6 inches long, for each material and finish specified.
6. Frames for light openings, 6 inches long, for each material, type, and finish required.

G. Qualification Data: For door installer.

1.3 INFORMATIONAL SUBMITTALS

A. Fire-rated Door Inspector: Submit documentation of compliance with NFPA 80, Section 5.2.3.1.

B. Egress Door Inspector: Submit documentation of compliance with NFPA 101, Section 7.2.1.15.4.

C. Product Test Reports: For each type of fire-rated hollow-metal door and frame assemblies and thermally rated door assemblies for tests performed by a qualified testing agency indicating compliance with performance requirements.

D. Field quality-control reports.

E. Sample Warranty: For manufacturer's special warranties.

F. Cleaning Instructions: Submit manufacturer’s cleaning instructions for doors.

1.4 CLOSEOUT SUBMITTALS

A. Quality Standard Compliance Certificates: AWI Quality Certification Program certificates.

B. Manufacturer's Warranties: For all products used.

1.5 QUALITY ASSURANCE

A. Tolerances for Warp, Telegraphing, Squareness, and Prefitting Dimensions: WDMA I.S.1-A.

B. Identifying Label: Each fire-rated door shall bear identifying label indicating:

1. Door manufacturer.
2. Order number.
3. Door number.

C. Fire-Rated Door Inspector Qualifications: Inspector for field quality-control inspections of fire-rated door assemblies is to meet the qualifications set forth in NFPA 80, Section 5.2.3.1 and the following:

1. Door and Hardware Institute Fire and Egress Door Assembly Inspector (FDAI) certification.

D. Egress Door Inspector Qualifications: Inspector for field quality-control inspections of egress door assemblies is to meet the qualifications set forth in NFPA 101, Section 7.2.1.15.4 and the following:

1. Door and Hardware Institute Fire and Egress Door Assembly Inspector (FDAI) certification

E. Manufacturer's Certification: Licensed participant in AWI's Quality Certification Program.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Comply with requirements of referenced standard and manufacturer's written instructions.

B. Package doors individually in plastic bags or cardboard cartons.

C. Mark each door on top and bottom rail with opening number used on Shop Drawings.

1.7 FIELD CONDITIONS

A. Environmental Limitations:

1. Do not deliver or install doors until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, and HVAC system is operating and maintaining temperature and relative humidity at levels designed for building occupants for the remainder of construction period.
2. Do not deliver or install doors until building is enclosed and weathertight, wet work is complete, and HVAC system is operating and maintaining temperature between 60 and 90 deg F and relative humidity between 43 and 70 percent during remainder of construction period.

1.8 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace doors that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Delamination of veneer.
b. Warping (bow, cup, or twist) more than 1/4 inch in a 42-by-84-inch section.
c. Telegraphing of core construction in face veneers exceeding 0.01 inch in a 3-inch span.

2. Warranty also includes installation and finishing that may be required due to repair or replacement of defective doors.
3. Warranty Period for Solid-Core Interior Doors: Life of installation.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Rated Door Assemblies: Assemblies complying with NFPA 80 that are listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction for fire-protection ratings indicated on Drawings, based on testing at positive pressure in accordance with NFPA 252 or UL 10C.

2.2 FLUSH WOOD DOORS AND FRAMES, GENERAL

A. Quality Standard: In addition to requirements specified, comply with AWI's “Architectural Woodwork Standards.”

1. Provide labels certificates from AWI certification program indicating that doors comply with requirements of grades specified.

a. Contractor registers the Work under this Section with the AWI Quality Certification Program at www.awiqcp.org or by calling 855-345-0991.

2. The Contract Documents contain requirements that are more stringent than the referenced quality standard. Comply with the Contract Documents in addition to those of the referenced quality standard.

B. Regional Materials: Wood doors shall be manufactured within 500 miles of Project site from materials that have been extracted, harvested, or recovered, as well as manufactured, within 500 miles of Project site. If materials are transported by rail or water, the distance transported by rail or water shall be multiplied by 0.25 to determine the distance to Project site.

C. Certified Wood: Wood doors shall contain not less than 60 percent certified wood tracked through a chain-of-custody process. Certified wood documentation shall be provided by sources certified through a forest certification system with principles, criteria, and standards developed using ISO/IEC Guide 59 or the World Trade Organization's "Technical Barriers to Trade."

D. Adhesives: Do not use adhesives that contain urea formaldehyde.

E. Composite Wood Products: Products shall be made without urea formaldehyde.

2.3 SOLID-CORE FIVE-PLY FLUSH WOOD VENEER-FACED DOORS FOR STAINED FINISH

A. Product: Subject to compliance with requirement meeting all salient characteristics.

1. Salient characteristics:

a. Compliance: WDMA I.S.1-A.

1) Aesthetic Grade: Custom.
2) Duty Level: Heavy- duty.

b. Door Thickness: 1-3/4 inches.
c. Structural composite lumber (SCL). Factory Sealed.
d. Core Material: Particleboard core, non-rated.
e. Blocking: Provide wood blocking in particleboard-core doors as follows:

1) 5-inch top-rail blocking, in doors indicated to have closers.
2) 5-inch bottom-rail blocking, in exterior doors and doors indicated to have kick, mop, or armor plates.
3) 5-inch midrail blocking, in doors indicated to have exit devices.

f. Veneers:

1) Apply to crossbanded core in hot press using Type I, exterior, water-resistant adhesive.
2) Species: White oak.
3) Cut: Rift cut/rift sawn.
4) Match: Book match.
5) Assembly: Balanced match.
6) Minimum Thickness Before Sanding: 1/42 inch.
7) Grade: NHLA Clear

2.4 FIRE RATED MINERAL-CORE FIVE-PLY FLUSH WOOD VENEER-FACED DOORS FOR STAINED FINISH

A. Product: Subject to compliance with requirement meeting all salient characteristics.

1. Salient characteristics:

a. Fire composite core, 45-minute rated.
b. Compliance: WDMA I.S.1-A.

1) Aesthetic Grade: Custom.
2) Type: FD-5.

c. Door Thickness: 1-3/4 inches.
d. Core material: Fire-Retardant Mineral Core: Does not contain asbestos or added urea formaldehyde.
e. Blocking for Mineral-Core Doors: Provide composite blocking with improved screw-holding capability approved for use in doors of fire-protection ratings indicated on Drawings as follows:

1) 5-inch top-rail blocking.
2) 5-inch bottom-rail blocking, in doors indicated to have protection plates.
3) 5-inch midrail blocking, in doors indicated to have exit devices.

f. Outer Edges: Compatible species as face veneer.
g. Construction: Five plies, hot-pressed bonded (vertical and horizontal edging is bonded to core), with entire unity abrasive planed before veneering.
h. Veneers:

1) Apply to crossbanded core in hot press using Type I, exterior, water-resistant adhesive.
2) Species: White oak.
3) Cut: Rift cut/rift sawn
4) Match: Book match.
5) Assembly: Balanced match.
6) Minimum Thickness Before Sanding: 1/42 inch.
7) Grade: NHLA Clear

B. Fire-rated wood doors where indicated on the Drawings.

C. Coordinate with fire-rated hollow metal frames as indicated on the Drawings.

2.5 FINISHES

A. Doors come from factory prepared for staining or Contractor shall be responsible for preparing doors for staining.

B. See Section 09 93 00 “Staining and Transparent Finishing.”

2.6 LIGHT FRAMES

A. Wood Beads for Light Openings in Wood Doors: Provide manufacturer's standard wood beads unless otherwise indicated.

1. Wood Species: Same species as door faces.
2. Profile: Flush rectangular beads.

B. Wood-Veneered Beads for Light Openings in Fire-Rated Doors: Manufacturer's standard wood-veneered noncombustible beads matching veneer species of door faces and approved for use in doors of fire-protection rating indicated on Drawings. Include concealed metal glazing clips where required for opening size and fire-protection rating indicated.

C. Metal Frames for Light Openings in Fire-Rated Doors: Manufacturer's standard frame formed of 0.048-inch thick cold-rolled steel sheet; factory primed for paint finish; and approved for use in doors of fire-protection rating indicated on Drawings.

2.7 FABRICATION

A. Factory fit doors to suit frame-opening sizes indicated.

1. Comply with clearance requirements of referenced quality standard for fitting unless otherwise indicated.
2. Comply with NFPA 80 requirements for fire-rated doors.

B. Factory machine doors for hardware that is not surface applied.

1. Locate hardware to comply with DHI-WDHS-3.
2. Comply with final hardware schedules, door frame Shop Drawings, ANSI/BHMA-156.115-W, and hardware templates.
3. Coordinate with hardware mortises in metal frames, to verify dimensions and alignment before factory machining.
4. For doors scheduled to receive electrified locksets, provide factory-installed raceway and wiring to accommodate specified hardware.

C. Openings: Factory cut and trim openings through doors.

1. Light Openings: Trim openings with moldings of material and profile indicated.
2. Glazing: Factory install glazing in doors indicated to be factory finished. Comply with applicable requirements in Section 088000 “Glazing.”

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine doors and installed door frames, with Installer present, before hanging doors.

1. Verify that installed frames comply with indicated requirements for type, size, location, and swing characteristics and have been installed with level heads and plumb jambs.

2. Reject doors with defects.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Hardware: For installation, see Section 087100 “Door Hardware.”

B. Install doors to comply with manufacturer's written instructions and referenced quality standard, and as indicated.

C. Job-Fitted Doors:

1. Align and fit doors in frames with uniform clearances and bevels as indicated below.

    a. Do not trim stiles and rails in excess of limits set by manufacturer or permitted for fire-rated doors.

2. Machine doors for hardware.
3. Seal edges of doors, edges of cutouts, and mortises after fitting and machining.
4. Clearances:

    a. Provide 1/8 inch at heads, jambs, and between pairs of doors.
    b. Provide 1/8 inch from bottom of door to top of decorative floor finish or covering unless otherwise indicated on Drawings.
    c. Where threshold is shown or scheduled, provide 1/4 inch from bottom of door to top of threshold unless otherwise indicated.

5. Bevel non-fire-rated doors 1/8 inch in 2 inches at lock and hinge edges.

D. Factory-Fitted Doors: Align in frames for uniform clearance at each edge.

E. Factory-Finished Doors: Restore finish before installation if fitting or machining is required at Project site.

3.3 FIELD QUALITY CONTROL

A. Inspection Agency: Engage a qualified inspector to perform inspections and to furnish reports to Architect.

B. Inspections:

1. Provide inspection of installed Work through AWI's Quality Certification Program , certifying that wood doors and frames, including installation, comply with requirements of AWI's “Architectural Woodwork Standards” for the specified grade.
2. Egress Door Inspections: Inspect each door located in an exit enclosure, each electrically controlled egress door, and each door equipped with special locking arrangements in accordance with NFPA 101, Section 7.2.1.15.

3. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

C. Repair or remove and replace installations where inspections indicate that they do not comply with specified requirements.

D. Reinspect repaired or replaced installations to determine if replaced or repaired door assembly installations comply with specified requirements.

3.4 ADJUSTING

A. Operation: Rehang or replace doors that do not swing or operate freely.

B. Finished Doors: Replace doors that are damaged or that do not comply with requirements. Doors may be repaired or refinished if Work complies with requirements and shows no evidence of repair or refinishing.

END OF SECTION 08 14 16

SECTION 08 31 13 – ACCESS DOORS AND FRAMES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Flush access doors with exposed flanges.

B. Related Requirements:

1. Section 06 61 16 “Solid Surface Fabrications” for wall panels in restrooms.
2. Section 09 29 00 “Gypsum Board” for wall material.
3. Division 22 for Plumbing access requirements.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

PART 2 - PRODUCTS

2.1 ACCESS DOORS AND FRAMES

A. Flush Access Doors with Exposed Flanges:

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

a. Best Access Doors
b. TRUDOOR
c. ACUDOR Products, Inc
d. Or Approved Equal

2. Description: Door face flush with adjacent finish material, with exposed flange and concealed hinge.
3. Locations: Wall.
4. Door Size: 12”x12”
5. Metallic-Coated Steel Sheet for Door: Nominal 0.064 inch, 16 gage, factory finished white.
6. Frame Material: Same material and thickness as door.
7. Edges of flanges: Rounded edges.
8. Latch and Lock: Screwdriver operated cam latch.

2.2 MATERIALS

A. Steel Plates, Shapes, and Bars: ASTM A36/A36M.

B. Steel Sheet: Uncoated or electrolytic zinc coated, ASTM A879/A879M, with cold-rolled steel sheet substrate complying with ASTM A1008/A1008M, Commercial Steel (CS), exposed.

C. Metallic-Coated Steel Sheet: ASTM A653/A653M, Commercial Steel (CS), Type B; with minimum G60 or A60 metallic coating.

D. Aluminum Extrusions: ASTM B221, Alloy 6063.

E. Aluminum Sheet: ASTM B209/B209M, alloy and temper recommended by aluminum producer and finisher for type of use and finish indicated.

F. Frame Anchors: Same material as door face.

G. Inserts, Bolts, and Anchor Fasteners: Hot-dip galvanized steel in accordance with ASTM A153/A153M or ASTM F2329/F2329M.

2.3 FABRICATION

A. General: Provide access door and frame assemblies manufactured as integral units ready for installation.

B. Metal Surfaces: For metal surfaces exposed to view in the completed Work, provide materials with smooth, flat surfaces without blemishes. Do not use materials with exposed pitting, seam marks, roller marks, rolled trade names, or roughness.

C. Doors and Frames: Grind exposed welds smooth and flush with adjacent surfaces. Furnish mounting holes, attachment devices, and fasteners of type required to secure access doors to types of supports indicated.

D. Flush Access Doors: Coordinate access panel locations with plumbing access requirements and solid surface wall panels in restrooms prior to the installation of the solid surface wall panels.

E. Aluminum: After fabrication, apply manufacturer's standard protective coating on aluminum that will come in contact with concrete.

2.4 FINISHES

A. Comply with NAAMM/NOMMA AMP 500 for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

C. Appearance of Finished Work: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

D. Painted Finishes: Comply with coating manufacturer's written instructions for cleaning, conversion coating, and applying and baking finish.

1. Factory Finished: Apply manufacturer's standard baked-enamel or powder-coat finish immediately after cleaning and pretreating, with minimum dry-film thickness of 1 mil for topcoat.

a. Color: White.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF ACCESS DOORS AND FRAMES

A. Comply with manufacturer's written instructions for installing access doors and frames.

3.3 FIELD QUALITY CONTROL

A. Repair or remove and replace installations where inspections indicate that they do not comply with specified requirements.

B. Reinspect repaired or replaced installations to determine if replaced or repaired door assembly installations comply with specified requirements.

3.4 ADJUSTING

A. Adjust doors and hardware, after installation, for proper operation.

END OF SECTION 08 31 13

SECTION 08 41 13 – ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Aluminum-framed entrance and storefront systems.
2. Aluminum-framed windows

B. Related Requirements:

1. Section 081416 “Flush Wood Doors.”
2. Section 087100 “Door Hardware” for door hardware coordination.
3. Section 088000 “Glazing” for all glazing and Low-E Coated & Bird-Friendly Glass coordination.
4. Section 122413 “Roller Window Shades” for shading device coordination.
5. Division 26 for electrical requirements.
6. Division 28 for electronic safety and security requirements.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site .

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Construction details, material descriptions, dimensions of individual components and profiles, and finishes.
2. Operating characteristics, electrical characteristics, and furnished accessories.

B. Shop Drawings:

1. Plans, elevations, sections, full-size details, and attachments to other work.
2. Details of provisions for assembly expansion and contraction and for draining moisture occurring within the assembly to the exterior.
3. Full-size isometric details of each type of vertical-to-horizontal intersection of aluminum-framed entrance and storefront systems, showing the following:

a. Joinery, including concealed welds.
b. Anchorage.
c. Expansion provisions.
d. Glazing.
e. Flashing and drainage.

4. Connection to and continuity with adjacent thermal, weather, air, and vapor barriers.
5. Point-to-point wiring diagrams showing the following:

    a. Power requirements for each electrically operated door hardware.
    b. Location and types of switches, signal device, conduit sizes, and number and size of wires.

6. Signed and sealed by the qualified professional engineer responsible for their preparation.

C. Samples for Verification: Actual sample of finished products for each type of exposed finish.

1. Size: Manufacturers' standard size .

D. Delegated Design Submittals: For aluminum-framed entrances and storefront systems, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

E. Sustainable Design Submittals:

1. Product Data: For sealants, indicating VOC content.

1.4 INFORMATIONAL SUBMITTALS

A. Energy Performance Certificates: For aluminum-framed entrance and storefront systems, accessories, and components, from manufacturer.

1. Basis for Certification: NFRC-certified energy performance values for each aluminum-framed entrance and storefront system.

B. Product Test Reports: For aluminum-framed entrance and storefront systems, for tests performed by manufacturer and witnessed by a qualified testing agency.

C. Field Quality-Control Reports: For aluminum-framed entrance and storefront systems.

D. Delegated Design Engineer Qualifications: For aluminum-framed entrance and storefront systems.

E. Sample Warranty: For manufacturer's special warranties.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For aluminum-framed entrance and storefront systems.

B. Manufacturer's Warranties: For all products used.

1.6 QUALITY ASSURANCE

A. Installer Qualifications:

1. Entity that employs installers and supervisors who are trained and approved by manufacturer.
2. Entity that is certified under the North American Contractor Certification Program (NACC) and that employs installers and supervisors who are trained and approved by manufacturer and who are certified under the Architectural Glass and Metal Technician (AGMT) certification program.

B. Delegated Design Engineer Qualifications: A professional engineer who is legally qualified to practice in state where Project is located and who is experienced in providing engineering services of the type indicated.

C. Egress Door Inspector Qualifications:

1. Inspector for field quality-control inspections of egress door assemblies to comply with qualifications set forth in NFPA 101, Ch. 7 “Means of Egress,” Section “Means of Egress Components,” Article “Inspection of Door Openings.”
2. Inspector for field quality-control inspections of egress door assemblies to be certified under DHI's certification program as a Fire and Egress Door Assembly Inspector (FDAI) or a Certified Fire and Egress Door Assembly Inspector (CFDAI).

D. Product Options: Information on Drawings and in Specifications establishes requirements for aesthetic effects and performance characteristics of assemblies. Aesthetic effects are indicated by dimensions, arrangements, alignment, and profiles of components and assemblies as they relate to sightlines, to one another, and to adjoining construction.

1. Do not change intended aesthetic effects, as judged solely by Contracting Officer, except with Contracting Officer s approval. If changes are proposed, submit comprehensive explanatory data to Contracting Officer for review.

1.7 MOCKUPS

A. Build mockups to set quality standards for fabrication and installation.

1. First installed window or entrance shall be reviewed as a mockup in situ to set quality standards for the remainder of the manufacture and installation of the panels. Subject to approval by the Contracting Officer in writing, mockup shall become part of the completed work.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.8 WARRANTY

A. Special Warranty: Manufacturer and Installer agree to repair or replace components of aluminum-framed entrance and storefront systems that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures, including failed component connections.
b. Faulty operation of door swinging, latching, and thermal qualities where applicable.
c. Deterioration of metals, metal finishes, and other materials beyond normal use.

2. Warranty Period: 10 years from date of Substantial Completion.

B. Special Finish Warranty, Anodized Finishes: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of anodized finishes within specified warranty period.

1. Deterioration includes, but is not limited to, the following:

a. Color fading more than 5 Delta E units when tested in accordance with ASTM D 2244.
b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D 4214.
c. Cracking, peeling, or chipping.

2. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain all components of aluminum-framed entrance and storefront system, including framing and accessories, from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 01 40 00 “Quality Requirements,” to design aluminum-framed entrance and storefront systems.

B. General Performance: Comply with performance requirements specified, as determined by testing of aluminum-framed entrance and storefront systems representing those indicated for this Project without failure due to defective manufacture, fabrication, installation, or other defects in construction.

1. Aluminum-framed entrance and storefront systems to withstand movements of supporting structure, including, but not limited to, twist, column shortening, long-term creep, and deflection from uniformly distributed and concentrated live loads.

2. Failure also includes the following:

    a. Thermal stresses transferring to building structure.
    b. Glass breakage.
    c. Noise or vibration created by wind and thermal and structural movements.
    d. Loosening or weakening of fasteners, attachments, and other components.

C. Structural Loads:

1. Wind Loads: As indicated on Drawings.
2. Other Design Loads: As indicated on Drawings.

D. Deflection of Framing Members Supporting Glass: At design wind load, as follows:

1. Deflection Normal to Wall Plane: Limited to 1/175 of length of span of the framing member for lengths of up to 13 feet 6 inches and to 1/240 of length of span of the framing member plus 1/4 inch for lengths greater than 13 feet 6 inches.
2. Deflection Parallel to Glazing Plane: Limited to amount not exceeding that which reduces glazing bite to less than 75 percent of design dimension and that which reduces edge clearance between framing members and glazing or other fixed components to less than 1/8 inch.

E. Structural: Test in accordance with ASTM E330/E330M as follows:

1. When tested at positive and negative wind-load design pressures, storefront assemblies, including entrance doors, do not evidence deflection exceeding specified limits.
2. When tested at 150 percent of positive and negative wind-load design pressures, storefront assemblies, including entrance doors and anchorage, do not evidence material failures, structural distress, or permanent deformation of main framing members exceeding 0.2 percent of span.
3. Test Durations: As required by design wind velocity, but not less than 10 seconds.

F. Water Penetration under Static Pressure: Test in accordance with ASTM E331 as follows:

1. No evidence of water penetration through fixed glazing and framing areas, including entrance doors, when tested in accordance with a minimum static-air-pressure differential of 20 percent of positive wind-load design pressure, but not less than 10 lbf/sq. ft. .

G. Water Penetration under Dynamic Pressure: Test in accordance with AAMA 501.1 as follows:

1. No evidence of water penetration through fixed glazing and framing areas when tested at dynamic pressure equal to 20 percent of positive wind-load design pressure, but not less than 10 lbf/sq. ft. .
2. Maximum Water Leakage: No uncontrolled water penetrating assemblies or water appearing on assemblies' normally exposed interior surfaces from sources other than condensation. Water leakage does not include water controlled by flashing and gutters, or water that is drained to exterior.

H. Energy Performance: Certified and labeled by manufacturer for energy performance as follows:

1. Thermal Transmittance (U-factor):

a. Fixed Glazing and Framing Areas: U-factor for the system of not more than 0.34 as determined in accordance with NFRC 100.
b. Entrance Doors: U-factor of not more than 0.63 as determined in accordance with NFRC 100.

2. Heat-Gain Coefficient (SHGC):

a. Fixed Glazing and Framing Areas: SHGC for the system of not more than 0.61 for Solar Main Level & 0.38 for Basement Level as determined in accordance with NFRC 200.
b. Entrance Doors: SHGC of not more than 0.53 for Main Level & 0.33 for Basement Level as determined in accordance with NFRC 200.

3. Air Leakage:

a. Fixed Glazing and Framing Areas: Air leakage for the system of not more than 0.06 cfm/sq. ft. at a static-air-pressure differential of 6.24 lbf/sq. ft. when tested in accordance with ASTM E283.
b. Entrance Doors: Air leakage of not more than 1.0 cfm/sq. ft. at a static-air-pressure differential of 1.57 lbf/sq. ft.

4. Condensation Resistance Factor (CRF):

a. Fixed Glazing and Framing Areas: CRF for the system of not less than 70 as determined in accordance with AAMA 1503.
b. Entrance Doors: CRF of not less than 68 as determined in accordance with AAMA 1503.

I. Windborne-Debris Impact Resistance: Passes ASTM E1886 missile-impact and cyclic-pressure tests in accordance with ASTM E1996 for Wind loading as noted on Structural Drawings.

J. Thermal Movements: Allow for thermal movements resulting from ambient and surface temperature changes.

1. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

2.3 ALUMINUM-FRAMED ENTRANCE AND STOREFRONT SYSTEMS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Arcadia Inc.
2. EFCO Corporation
3. Kawneer Company, Inc.; Arconic Corporation

4. Manko Window Systems, Inc.
5. Old Castle Building Envelope (OBE)
6. Or Approved Equal

B. Framing Members: Manufacturer's extruded- or formed-aluminum framing members of thickness required and reinforced as required to support imposed loads.

1. Exterior Framing Construction: Thermally broken.
2. Interior Vestibule Framing Construction: Thermally broken.
3. Glazing System: Retained mechanically with gaskets on four sides.
4. Glazing Plane: Center glazed.
5. Finish: Color: dark bronze finish.
6. Fabrication Method: Field-fabricated stick system.
7. Aluminum: Alloy and temper recommended by manufacturer for type of use and finish indicated.
8. Steel Reinforcement: As required by manufacturer.

C. Backer Plates: Manufacturer's standard, continuous backer plates for framing members, if not integral, where framing abuts adjacent construction.

D. Brackets and Reinforcements: Manufacturer's standard high-strength aluminum with nonstaining, nonferrous shims for aligning system components.

E. Entrance Doors: Manufacturer's standard glazed entrance doors for manual-swing or automatic operation.

1. Door Construction: 1-3/4-inch overall thickness, with minimum 0.125-inch- thick, extruded-aluminum tubular rail and stile members. Mechanically fasten corners with reinforcing brackets that are deeply penetrated and fillet welded or that incorporate concealed tie rods.

    a. Thermal Construction: High-performance plastic connectors separate aluminum members exposed to the exterior from members exposed to the interior.

2. Door Design: Wide stile; 5-inch nominal width.
3. Glazing Stops and Gaskets: Square, snap-on, extruded-aluminum stops and preformed gaskets.

    a. Provide nonremovable glazing stops on outside of door.

4. Finish: Match adjacent storefront framing finish.

2.4 ENTRANCE DOOR HARDWARE

A. Entrance Door Hardware: Hardware not specified in this Section is specified in Section 08 71 00 “Door Hardware.”

B. General: Provide entrance door hardware and entrance door hardware sets indicated in door and frame schedule for each entrance door, to comply with requirements in this Section.

1. Entrance Door Hardware Sets: Provide quantity, item, size, finish or color indicated, and products equivalent in function and comparable in quality to named products.
2. Sequence of Operation: Provide electrified door hardware function, sequence of operation, and interface with other building control systems indicated.
3. Opening-Force Requirements:

    a. Egress Doors: Not more than 15 lbf to release the latch and not more than 30 lbf to set the door in motion.
    b. Accessible Interior Doors: Not more than 5 lbf to fully open door.

C. Cylinders:

1. As specified in Section 087100 “Door Hardware.”

D. Thresholds: BHMA A156.21 raised thresholds beveled with a slope of not more than 1:2, with maximum height of 1/2 inch.

2.5 GLAZING

A. Glazing: Comply with Section 088000 “Glazing.”

B. Glazing Gaskets: Comply with Section 088000 “Glazing.”

C. Glazing Sealants: Comply with Section 088000 “Glazing.”

D. Weatherseal Sealants: ASTM C920 for Type S; Grade NS; Class 25; Uses NT, G, A, and O; chemically curing silicone formulation that is compatible with structural sealant and other system components with which it comes in contact; recommended by structural-sealant, weatherseal-sealant, and structural-sealant-glazed storefront manufacturers for this use.

2.6 MATERIALS

A. Sheet and Plate: ASTM B209.

B. Extruded Bars, Rods, Profiles, and Tubes: ASTM B221.

C. Structural Profiles: ASTM B308/B308M.

2.7 ACCESSORIES

A. Fasteners and Accessories: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding fasteners and accessories compatible with adjacent materials.

1. Use self-locking devices where fasteners are subject to loosening or turning out from thermal and structural movements, wind loads, or vibration.
2. Reinforce members as required to receive fastener threads.
3. Use exposed fasteners with countersunk Phillips screw heads, finished to match framing system.

B. Anchors: Three-way adjustable anchors with minimum adjustment of 1 inch that accommodate fabrication and installation tolerances in material and finish compatible with adjoining materials and recommended by manufacturer.

1. Concrete and Masonry Inserts: Hot-dip galvanized cast-iron, malleable-iron, or steel inserts complying with ASTM A123/A123M or ASTM A153/A153M requirements.

C. Concealed Flashing: Manufacturer's standard corrosion-resistant, nonstaining, nonbleeding flashing compatible with adjacent materials.

D. Bituminous Paint: Cold-applied asphalt-mastic paint containing no asbestos, formulated for 30-mil thickness per coat.

E. Rigid PVC filler.

2.8 FABRICATION

A. Form or extrude aluminum shapes before finishing.

B. Fabricate components that, when assembled, have the following characteristics:

1. Profiles that are sharp, straight, and free of defects or deformations.
2. Accurately fitted joints with ends coped or mitered.
3. Physical and thermal isolation of glazing from framing members.
4. Accommodations for thermal and mechanical movements of glazing and framing to maintain required glazing edge clearances.
5. Fasteners, anchors, and connection devices that are concealed from view to greatest extent possible.

C. Mechanically Glazed Framing Members: Fabricate for flush glazing without projecting stops.

D. Entrance Door Frames: Reinforce as required to support loads imposed by door operation and for installing entrance door hardware.

1. At interior and exterior doors, provide compression weather stripping at fixed stops.

E. Entrance Doors: Reinforce doors as required for installing entrance door hardware.

1. At pairs of exterior doors, provide sliding-type weather stripping retained in adjustable strip and mortised into door edge.
2. At exterior doors, provide weather sweeps applied to door bottoms.

F. Entrance Door Hardware Installation: Factory install entrance door hardware to the greatest extent possible. Cut, drill, and tap for factory-installed entrance door hardware before applying finishes.

G. After fabrication, clearly mark components to identify their locations in Project in accordance with Shop Drawings.

2.9 ALUMINUM FINISHES

A. Color Anodic Finish: AAMA 611, AA-M12C22A42/A44, Class I, 0.018 mm or thicker.

1. Color: Dark bronze.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF ALUMINUM-FRAMED ENTRANCE AND STOREFRONT SYSTEMS

A. Comply with manufacturer's written instructions.

B. Do not install damaged components.

C. Fit joints to produce hairline joints free of burrs and distortion.

D. Rigidly secure nonmovement joints.

E. Install anchors with separators and isolators to prevent metal corrosion and electrolytic deterioration and to prevent impeding movement of moving joints.

F. Seal perimeter and other joints watertight unless otherwise indicated.

G. Metal Protection:

1. Where aluminum is in contact with dissimilar metals, protect against galvanic action by painting contact surfaces with materials recommended by manufacturer for this purpose or by installing nonconductive spacers.
2. Where aluminum is in contact with concrete or masonry, protect against corrosion by painting contact surfaces with bituminous paint.

H. Set continuous sill members and flashing in full sealant bed, as specified in Section 079200 “Joint Sealants,” to produce weathertight installation.

I. Install joint filler behind sealant as recommended by sealant manufacturer.

J. Install components plumb and true in alignment with established lines and grades.

K. Install operable units level and plumb, securely anchored, and without distortion. Adjust weather-stripping contact and hardware movement to produce proper operation.

L. Install entrance doors to produce smooth operation and tight fit at contact points.

1. Exterior Doors: Install to produce weathertight enclosure and tight fit at weather stripping.
2. Field-Installed Entrance Door Hardware: Install surface-mounted entrance door hardware in accordance with entrance door hardware manufacturers' written instructions using concealed fasteners to greatest extent possible.

M. Install glazing as specified in Section 088000 “Glazing.”

3.3 ERECTION TOLERANCES

A. Install aluminum-framed entrance and storefront systems to comply with the following maximum tolerances:

1. Plumb: 1/8 inch in 10 feet; 1/4 inch in 40 feet.
2. Level: 1/8 inch in 20 feet; 1/4 inch in 40 feet.
3. Alignment:

    a. Where surfaces abut in line or are separated by reveal or protruding element up to 1/2 inch wide, limit offset from true alignment to 1/16 inch.
    b. Where surfaces are separated by reveal or protruding element from 1/2 to 1 inch wide, limit offset from true alignment to 1/8 inch.
    c. Where surfaces are separated by reveal or protruding element of 1 inch wide or more, limit offset from true alignment to 1/4 inch.

4. Location: Limit variation from plane to 1/8 inch in 12 feet; 1/2 inch over total length.

3.4 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests.

B. Tests: Perform the following tests on representative areas of aluminum-framed entrance and storefront systems.

1. Water-Spray Test: Before installation of interior finishes has begun, areas designated by Contracting Officer to be tested in accordance with AAMA 501.2 and to not evidence water penetration.

    a. Perform a minimum of two tests in areas as directed by Contracting Officer.
    b. Perform tests in each test area as directed by Contracting Officer. Perform at least three tests, prior to 10, 35, and 70 percent completion.

2. Air Leakage: ASTM E783 at 1.5 times the rate specified for laboratory testing in “Performance Requirements” Article but not more than 0.09 cfm/sq. ft. at a static-air-pressure differential of 1.57 lbf/sq. ft.

    a. Perform a minimum of three tests in areas as directed by Contracting Officer.
    b. Perform tests in each test area as directed by Contracting Officer. Perform at least three tests, prior to 10, 35, and 70 percent completion.

3. Water Penetration: ASTM E1105 at a minimum uniform cyclic static-air-pressure differential of 0.67 times the static-air-pressure differential specified for laboratory testing in “Performance Requirements” Article, but not less than 6.24 lbf/sq. ft., and to not evidence water penetration.

C. Inspection Agency: Engage a qualified inspector to perform inspections.

D. Inspections:

1. Egress Door Inspections: Inspect each aluminum-framed entrance door equipped with panic hardware, located in an exit enclosure, electrically controlled, and equipped with special locking arrangements, in accordance with NFPA 101, Ch. 7 “Means of Egress,” Section “Means of Egress Components,” Article “Inspection of Door Openings.”
2. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

E. Aluminum-framed entrance and storefront systems will be considered defective if they do not pass tests and inspections.

F. Prepare test and inspection reports.

3.5 MAINTENANCE SERVICE

A. Entrance Door Hardware Maintenance:

1. Maintenance Tools and Instructions: Furnish a complete set of specialized tools and maintenance instructions as needed for Government's continued adjustment, maintenance, and removal and replacement of entrance door hardware.
2. Initial Maintenance Service: Beginning at Substantial Completion, provide six months' full maintenance by skilled employees of entrance door hardware Installer. Include quarterly preventive maintenance, repair or replacement of worn or defective components, lubrication, cleaning, and adjusting as required for proper entrance door hardware operation at rated speed and capacity. Use parts and supplies that are the same as those used in the manufacture and installation of original equipment.

END OF SECTION 08 41 13

SECTION 08 71 00 – DOOR HARDWARE

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Mechanical and electrified door hardware
2. Electronic access control system components

B. Section excludes:

1. Windows
2. Cabinets (casework), including locks in cabinets
3. Signage
4. Toilet accessories
5. Overhead doors

C. Related Sections:

1. Division 01 “General Requirements” sections for Allowances, Alternates, Owner Furnished Contractor Installed, Project Management and Coordination.
2. Division 06 Section "Rough Carpentry"
3. Division 06 Section "Finish Carpentry"
4. Division 07 Section "Joint Sealants" for sealant requirements applicable to threshold installation specified in this section.
5. Division 08 Sections:

    a. "Metal Doors and Frames"
    b. "Flush Wood Doors"
    c. "Interior Aluminum Doors and Frames"
    d. "Aluminum-Framed Entrances and Storefronts"
    e. "Steel Doors and Frames"

6. Division 26 "Electrical" sections for connections to electrical power system and for low-voltage wiring.
7. Division 28 "Electronic Safety and Security" sections for coordination with other components of electronic access control system and fire alarm system.

1.2 REFERENCES

A. UL LLC

1. UL 10B - Fire Test of Door Assemblies
2. UL 10C - Positive Pressure Test of Fire Door Assemblies
3. UL 1784 - Air Leakage Tests of Door Assemblies
4. UL 305 - Panic Hardware

B. DHI - Door and Hardware Institute

1. Sequence and Format for the Hardware Schedule
2. Recommended Locations for Builders Hardware
3. Keying Systems and Nomenclature
4. Installation Guide for Doors and Hardware

C. NFPA – National Fire Protection Association

1. NFPA 70 – National Electric Code
2. NFPA 80 – 2016 Edition – Standard for Fire Doors and Other Opening Protectives
3. NFPA 101 – Life Safety Code
4. NFPA 105 – Smoke and Draft Control Door Assemblies
5. NFPA 252 – Fire Tests of Door Assemblies

D. ANSI - American National Standards Institute

1. ANSI A117.1 – 2017 Edition – Accessible and Usable Buildings and Facilities
2. ANSI/BHMA A156.1 - A156.29, and ANSI/BHMA A156.31 - Standards for Hardware and Specialties
3. ANSI/BHMA A156.28 - Recommended Practices for Keying Systems
4. ANSI/WDMA I.S. 1A - Interior Architectural Wood Flush Doors
5. ANSI/SDI A250.8 - Standard Steel Doors and Frames

1.3 SUBMITTALS

A. General:

1. Submit in accordance with Conditions of Contract and Division 01 Submittal Procedures.
2. Prior to forwarding submittal:

a. Review drawings and Sections from related trades to verify compatibility with specified hardware.
b. Highlight, encircle, or otherwise specifically identify on submittals: deviations from Contract Documents, issues of incompatibility or other issues which may detrimentally affect the Work.

B. Action Submittals:

1. Product Data: Submit technical product data for each item of door hardware, installation instructions, maintenance of operating parts and finish, and other information necessary to show compliance with requirements.
2. Riser and Wiring Diagrams: After final approval of hardware schedule, submit details of electrified door hardware, indicating:

a. Wiring Diagrams: For power, signal, and control wiring and including:

1) Details of interface of electrified door hardware and building safety and security systems.
2) Schematic diagram of systems that interface with electrified door hardware.

3) Point-to-point wiring.
4) Risers.

3. Samples for Verification: If requested by Architect, submit production sample of requested door hardware unit in finish indicated and tagged with full description for coordination with schedule.

   a. Samples will be returned to supplier. Units that are acceptable to Architect may, after final check of operations, be incorporated into Work, within limitations of key coordination requirements.

4. Door Hardware Schedule:

   a. Submit concurrent with submissions of Product Data, Samples, and Shop Drawings. Coordinate submission of door hardware schedule with scheduling requirements of other work to facilitate fabrication of other work critical in Project construction schedule.
   b. Submit under direct supervision of a Door Hardware Institute (DHI) certified Architectural Hardware Consultant (AHC) or Door Hardware Consultant (DHC) with hardware sets in vertical format as illustrated by Sequence of Format for the Hardware Schedule published by DHI.
   c. Indicate complete designations of each item required for each opening, include:

      1) Door Index: door number, heading number, and Architect's hardware set number.
      2) Quantity, type, style, function, size, and finish of each hardware item.
      3) Name and manufacturer of each item.
      4) Fastenings and other pertinent information.
      5) Location of each hardware set cross-referenced to indications on Drawings.
      6) Explanation of all abbreviations, symbols, and codes contained in schedule.
      7) Mounting locations for hardware.
      8) Door and frame sizes and materials.
      9) Degree of door swing and handing.
      10) Operational Description of openings with electrified hardware covering egress, ingress (access), and fire/smoke alarm connections.

5. Key Schedule:

   a. After Keying Conference, provide keying schedule that includes levels of keying, explanations of key system's function, key symbols used, and door numbers controlled.
   b. Use ANSI/BHMA A156.28 "Recommended Practices for Keying Systems" as guideline for nomenclature, definitions, and approach for selecting optimal keying system.
   c. Provide 3 copies of keying schedule for review prepared and detailed in accordance with referenced DHI publication. Include schematic keying diagram and index each key to unique door designations.
   d. Index keying schedule by door number, keyset, hardware heading number, cross keying instructions, and special key stamping instructions.

e. Provide one complete bitting list of key cuts and one key system schematic illustrating system usage and expansion. Forward bitting list, key cuts and key system schematic directly to Owner, by means as directed by Owner.
f. Prepare key schedule by or under supervision of supplier, detailing Owner's final keying instructions for locks.

C. Informational Submittals:

1. Provide Qualification Data for Supplier, Installer and Architectural Hardware Consultant.
2. Provide Product Data:

a. Certify that door hardware approved for use on types and sizes of labeled fire-rated doors complies with listed fire-rated door assemblies.
b. Include warranties for specified door hardware.

D. Closeout Submittals:

1. Operations and Maintenance Data: Provide in accordance with Division 01 and include:

a. Complete information on care, maintenance, and adjustment; data on repair and replacement parts, and information on preservation of finishes.
b. Catalog pages for each product.
c. Final approved hardware schedule edited to reflect conditions as installed.
d. Final keying schedule
e. Copy of warranties including appropriate reference numbers for manufacturers to identify project.
f. As-installed wiring diagrams for each opening connected to power, both low voltage and 110 volts.

E. Inspection and Testing:

1. Submit written reports to the Owner and Authority Having Jurisdiction (AHJ) of the results of functional testing and inspection for:

a. Fire door assemblies, in compliance with NFPA 80.
b. Required egress door assemblies, in compliance with NFPA 101.

1.4 QUALITY ASSURANCE

A. Qualifications and Responsibilities:

1. Supplier: Recognized architectural hardware supplier with a minimum of 5 years documented experience supplying both mechanical and electromechanical door hardware similar in quantity, type, and quality to that indicated for this Project. Supplier to be recognized as a factory direct distributor by the manufacturer of the primary materials with a warehousing facility in the Project’s vicinity. Supplier to have on staff, a certified Architectural Hardware Consultant (AHC) or Door Hardware Consultant (DHC) available to Owner, Architect, and Contractor, at reasonable times during the Work for consultation.

2. Installer: Qualified tradesperson skilled in the application of commercial grade hardware with experience installing door hardware similar in quantity, type, and quality as indicated for this Project.
3. Architectural Hardware Consultant: Person who is experienced in providing consulting services for door hardware installations that are comparable in material, design, and extent to that indicated for this Project and meets these requirements:

    a. For door hardware: DHI certified AHC or DHC.
    b. Can provide installation and technical data to Architect and other related subcontractors.
    c. Can inspect and verify components are in working order upon completion of installation.
    d. Capable of producing wiring diagram and coordinating installation of electrified hardware with Architect and electrical engineers.

4. Single Source Responsibility: Obtain each type of door hardware from single manufacturer.

B. Certifications:

1. Fire-Rated Door Openings:

    a. Provide door hardware for fire-rated openings that complies with NFPA 80 and requirements of authorities having jurisdiction.
    b. Provide only items of door hardware that are listed products tested by UL LLC, Intertek Testing Services, or other testing and inspecting organizations acceptable to authorities having jurisdiction for use on types and sizes of doors indicated, based on testing at positive pressure and according to NFPA 252 or UL 10C and in compliance with requirements of fire-rated door and door frame labels.

2. Smoke and Draft Control Door Assemblies:

    a. Provide door hardware that meets requirements of assemblies tested according to UL 1784 and installed in compliance with NFPA 105
    b. Comply with the maximum air leakage of 0.3 cfm/sq. ft. (3 cu. m per minute/sq. m) at tested pressure differential of 0.3-inch wg (75 Pa) of water.

3. Electrified Door Hardware

    a. Listed and labeled as defined in NFPA 70, Article 100, by testing agency acceptable to authorities having jurisdiction.

4. Accessibility Requirements:

    a. Comply with governing accessibility regulations cited in "REFERENCES" article 087100, 1.02.D3 herein for door hardware on doors in an accessible route. This project must comply with all Federal Americans with Disability Act regulations and all Local Accessibility Regulations.

C. Pre-Installation Meetings

1. Keying Conference

a. Incorporate keying conference decisions into final keying schedule after reviewing door hardware keying system including:

1) Function of building, flow of traffic, purpose of each area, degree of security required, and plans for future expansion.
2) Preliminary key system schematic diagram.
3) Requirements for key control system.
4) Requirements for access control.
5) Address for delivery of keys.

2. Pre-installation Conference

a. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
b. Inspect and discuss preparatory work performed by other trades.
c. Inspect and discuss electrical roughing-in for electrified door hardware.
d. Review sequence of operation for each type of electrified door hardware.
e. Review required testing, inspecting, and certifying procedures.
f. Review questions or concerns related to proper installation and adjustment of door hardware.

3. Electrified Hardware Coordination Conference:

a. Prior to ordering electrified hardware, schedule and hold meeting to coordinate door hardware with security, electrical, doors and frames, and other related suppliers.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Inventory door hardware on receipt and provide secure lock-up for hardware delivered to Project site. Promptly replace products damaged during shipping

B. Tag each item or package separately with identification coordinated with final door hardware schedule, and include installation instructions, templates, and necessary fasteners with each item or package. Deliver each article of hardware in manufacturer's original packaging.

C. Maintain manufacturer-recommended environmental conditions throughout storage and installation periods.

D. Provide secure lock-up for door hardware delivered to Project. Control handling and installation of hardware items so that completion of Work will not be delayed by hardware losses both before and after installation.

E. Handle hardware in manner to avoid damage, marring, or scratching. Correct, replace or repair products damaged during Work. Protect products against malfunction due to paint, solvent, cleanser, or any chemical agent.

F. Deliver keys to manufacturer of key control system for subsequent delivery to Owner.

1.6 COORDINATION

A. Coordinate layout and installation of floor-recessed door hardware with floor construction. Cast anchoring inserts into concrete.

B. Installation Templates: Distribute for doors, frames, and other work specified to be factory or shop prepared. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing door hardware to comply with indicated requirements.

C. Security: Coordinate installation of door hardware, keying, and access control with Owner's security consultant.

D. Electrical System Roughing-In: Coordinate layout and installation of electrified door hardware with connections to power supplies and building safety and security systems.

1.7 WARRANTY

A. Manufacturer's standard form in which manufacturer agrees to repair or replace components of door hardware that fail in materials or workmanship within published warranty period.

1. Warranty does not cover damage or faulty operation due to improper installation, improper use or abuse.
2. Warranty Period: Beginning from date of Substantial Completion, for durations indicated in manufacturer's published listings.

a. Mechanical Warranty

1) Locks

a) Schlage L Series: 10 years
b) Schlage ND Series: 10 years

2) Exit Devices

a) Von Duprin: 10 years

3) Closers

a) LCN 4000 Series: 30 years

4) Automatic Operators

a) LCN: 2 years

b. Electrical Warranty

1) Locks

a) Schlage: 3 years

2) Exit Devices

a) Von Duprin: 3 years

1.8 MAINTENANCE

A. Furnish complete set of special tools required for maintenance and adjustment of hardware, including changing of cylinders.

B. Turn over unused materials to Owner for maintenance purposes.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. The Owner requires use of certain products for their unique characteristics and project suitability to ensure continuity of existing and future performance and maintenance standards. After investigating available product offerings, the Awarding Authority has elected to prepare proprietary specifications. These products are specified with the notation: "No Substitute."

1. Where "No Substitute" is noted, submittals and substitution requests for other products will not be considered.

B. Approval of alternate manufacturers and/or products other than those listed as "Scheduled Manufacturer" or "Acceptable Manufacturers" in the individual article for the product category are only to be considered by official substitution request in accordance with section 01 25 00.

C. Approval of products from manufacturers indicated in "Acceptable Manufacturers" is contingent upon those products providing all functions and features and meeting all requirements of scheduled manufacturer's product.

D. Where specified hardware is not adaptable to finished shape or size of members requiring hardware, furnish suitable types having same operation and quality as type specified, subject to Architect's approval.

2.2 MATERIALS

A. Fabrication

1. Provide door hardware manufactured to comply with published templates generally prepared for machine, wood, and sheet metal screws. provide screws according to manufacturer’s recognized installation standards for application intended.
2. Finish exposed screws to match hardware finish, or, if exposed in surfaces of other work, to match finish of this other work including prepared for paint surfaces to receive painted finish.

3. Provide concealed fasteners wherever possible for hardware units exposed when door is closed. Coordinate with “Metal Doors and Frames”, “Flush Wood Doors”, “Stile and Rail Wood Doors” to ensure proper reinforcements. Advise the Architect where visible fasteners, such as thru bolts, are required.

B. Provide screws, bolts, expansion shields, drop plates and other devices necessary for hardware installation.

1. Where fasteners are exposed to view: Finish to match adjacent door hardware material.

C. Cable and Connectors:

1. Where scheduled in the hardware sets, provide each item of electrified hardware and wire harnesses with number and gage of wires enough to accommodate electric function of specified hardware.
2. Provide Molex connectors that plug directly into connectors from harnesses, electric locking and power transfer devices.
3. Provide through-door wire harness for each electrified locking device installed in a door and wire harness for each electrified hinge, electrified continuous hinge, electrified pivot, and electric power transfer for connection to power supplies.

2.3 HINGES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

a. Ives 5PB series

2. Acceptable Manufacturers and Products:

a. Per Architect/Owner's approval

B. Requirements:

1. Provide hinges conforming to ANSI/BHMA A156.1.
2. Provide five knuckle, plain bearing hinges.
3. 1-3/4 inch (44 mm) thick doors, up to and including 36 inches (914 mm) wide:

a. Exterior: Standard weight, bronze or stainless steel, 4-1/2 inches (114 mm) high
b. Interior: Standard weight, steel, 4-1/2 inches (114 mm) high

4. Doors over 1-3/4 inch (44 mm) thick or over 36 inches (914 mm) wide:

a. Exterior: Bronze/stainless steel, 5 inches (127 mm) high
b. Interior: Steel, 5 inches (127 mm) high

5. Adjust hinge width for door, frame, and wall conditions to allow proper degree of opening.
6. Provide three hinges per door leaf for doors 90 inches (2286 mm) or less in height, and one additional hinge for each 30 inches (762 mm) of additional door height.

7. Hinge Pins: Except as otherwise indicated, provide hinge pins as follows:

    a. Steel Hinges: Steel pins
    b. Non-Ferrous Hinges: Stainless steel pins
    c. Out-Swinging Exterior Doors: Non-removable pins
    d. Out-Swinging Interior Lockable Doors: Non-removable pins
    e. Interior Non-lockable Doors: Non-rising pins

2.4 HINGES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

    a. Ives 5BB series

2. Acceptable Manufacturers and Products:

    a. Per Architect/Owner's approval

B. Requirements:

1. Provide hinges conforming to ANSI/BHMA A156.1.
2. Provide five knuckle, ball bearing hinges.
3. 1-3/4 inch (44 mm) thick doors, up to and including 36 inches (914 mm) wide:

    a. Exterior: Standard weight, bronze or stainless steel, 4-1/2 inches (114 mm) high
    b. Interior: Standard weight, steel, 4-1/2 inches (114 mm) high

4. 1-3/4 inch (44 mm) thick doors over 36 inches (914 mm) wide:

    a. Exterior: Heavy weight, bronze/stainless steel, 5 inches (127 mm) high
    b. Interior: Heavy weight, steel, 5 inches (127 mm) high

5. 2 inches or thicker doors:

    a. Exterior: Heavy weight, bronze or stainless steel, 5 inches (127 mm) high
    b. Interior: Heavy weight, steel, 5 inches (127 mm) high

6. Adjust hinge width for door, frame, and wall conditions to allow proper degree of opening.
7. Provide three hinges per door leaf for doors 90 inches (2286 mm) or less in height, and one additional hinge for each 30 inches (762 mm) of additional door height.
8. Hinge Pins: Except as otherwise indicated, provide hinge pins as follows:

    a. Steel Hinges: Steel pins
    b. Non-Ferrous Hinges: Stainless steel pins
    c. Out-Swinging Exterior Doors: Non-removable pins
    d. Out-Swinging Interior Lockable Doors: Non-removable pins
    e. Interior Non-lockable Doors: Non-rising pins

9. Provide hinges with electrified options as scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware. Locate electric hinge at second hinge from bottom or nearest to electrified locking component. Provide mortar guard for each electrified hinge specified.

2.5 CONTINUOUS HINGES

A. Manufacturers:

1. Scheduled Manufacturer:

   a. Ives

2. Acceptable Manufacturers:

   a. Per Architect/Owner's approval

B. Requirements:

1. Provide aluminum geared continuous hinges conforming to ANSI/BHMA A156.26, Grade 1.
2. Provide aluminum geared continuous hinges, where specified in the hardware sets, fabricated from 6063-T6 aluminum.
3. Provide split nylon bearings at each hinge knuckle for quiet, smooth, self-lubricating operation.
4. Provide hinges capable of supporting door weights up to 450 pounds, and successfully tested for 1,500,000 cycles.
5. On fire-rated doors, provide aluminum geared continuous hinges classified for use on rated doors by testing agency acceptable to authority having jurisdiction.
6. Provide aluminum geared continuous hinges with electrified option scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware.
7. Provide hinges 1 inch (25 mm) shorter in length than nominal height of door, unless otherwise noted or door details require shorter length and with symmetrical hole pattern.

2.6 ELECTRIC POWER TRANSFER

A. Manufacturers:

1. Scheduled Manufacturer and Product:

   a. Von Duprin EPT-10

2. Acceptable Manufacturers and Products:

   a. Per Architect/Owner's approval

B. Requirements:

1. Provide power transfer with electrified options as scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware.
2. Locate electric power transfer per manufacturer's template and UL requirements, unless interference with operation of door or other hardware items.

2.7 FLUSH BOLTS

A. Manufacturers:

1. Scheduled Manufacturer:

a. Ives

2. Acceptable Manufacturers:

a. Trimco

B. Requirements:

1. Provide automatic, constant latching, and manual flush bolts with forged bronze or stainless-steel face plates, extruded brass levers, and with wrought brass guides and strikes. Provide 12 inch (305 mm) steel or brass rods at doors up to 90 inches (2286 mm) in height. For doors over 90 inches (2286 mm) in height increase top rods by 6 inches (152 mm) for each additional 6 inches (152 mm) of door height. Provide dust-proof strikes at each bottom flush bolt.

2.8 TWO-POINT LOCK

A. Manufacturer and Product:

1. Scheduled Manufacturer and Product:

a. Schlage LM9200

2. Acceptable Manufacturers and Products:

a. No Substitute, Owner standard

B. Requirements:

1. Provide concealed two-point locking system for use in pair wood door applications manufactured from heavy gauge steel, containing components of steel with a zinc dichromate plating for corrosion resistance. Provide lock case that is multi-function and field reversible for handing without opening case. Cylinders: Refer to "KEYING" article, herein.

2. Concealed Vertical Locking Devices: Vertical latch system in two-point for non-rated or fire rated wood doors up to a 45-minute rating and less bottom latch (LBL) configuration for non-rated or fire rated wood doors up to 20-minute rating.
3. Provide electrified lockset functions as scheduled in the hardware sets.
4. Lever Trim: Solid brass, bronze, or stainless steel, cast or forged in design specified, with wrought roses or escutcheon as scheduled and external lever spring cages. Provide escutcheon trim which does not require the use of a backer plate in wood door applications. Provide thru-bolted levers with 2-piece spindles.

    a. Lever Design: Schlage Rhodes.
    b. Specified lever design is for estimating purpose only. Provide lockset with lever design approved by Architect.

2.9 CYLINDRICAL LOCKS – GRADE 1

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

    a. Schlage ND series

2. Acceptable Manufacturers and Products:

    a. No Substitute, Owner standard

B. Requirements:

1. Provide cylindrical locks conforming to ANSI/BHMA A156.2 Series 4000, Grade 1, and UL Listed for 3-hour fire doors.
2. Indicators: Where specified, provide escutcheon with lock status indicator window on top of lockset rose:

    a. Escutcheon height (including rose) 6.05 inches high by 3.68 inches wide.
    b. Indicator window measuring a minimum 3.52-inch by .60 inch with 1.92 square-inches of front facing viewing area and 180-degree visibility with a total of .236 square-inches of total viewable area.
    c. Provide snap-in serviceable window to prevent tampering. Lock must function if indicator is compromised.
    d. Provide messages color-coded with full text and symbol, as scheduled, for easy visibility.
    e. Unlocked and Unoccupied message will display on white background, and Locked and Occupied message will display on red background.

3. Cylinders: Refer to "KEYING" article, herein.
4. Provide locks with standard 2-3/4 inches (70 mm) backset, unless noted otherwise, with 1/2-inch latch throw. Provide proper latch throw for UL listing at pairs.
5. Provide locksets with separate anti-rotation thru-bolts, and no exposed screws.
6. Provide independently operating levers with two external return spring cassettes mounted under roses to prevent lever sag.
7. Provide standard ASA strikes unless extended lip strikes are necessary to protect trim.

8. Provide electrified options as scheduled in the hardware sets.
9. Lever Trim: Solid cast levers without plastic inserts and wrought roses on both sides.

    a. Lever Design: Schlage Rhodes.
    b. Specified lever design is for estimating purpose only. Provide lockset with lever design approved by Architect.

2.10 DEADBOLTS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

    a. Schlage B600 Series

2. Acceptable Manufacturers and Products:

    a. No Substitute, Owner standard

B. Requirements:

1. Provide grade 1 deadbolt series conforming to ANSI/BHMA A156.
2. Cylinders: Refer to "KEYING" article, herein.
3. Provide deadbolts with standard 2-3/4 inches (70 mm) backset. Provide 2-3/8 inches (60 mm) where noted or if door or frame detail requires. Provide deadbolt with full 1-inch (25 mm) throw, constructed of steel alloy.
4. Provide manufacturer's standard strike.
5. Lock Status Indicator Trim: Where specified, provide escutcheon with lock status indicator widow.

    a. Escutcheon height 4.125 inches, width 2.54 inches. Projection 1.32 inches on thumbturn side and 1.28 inches on cylinder side.
    b. Unlocked and Unoccupied message will display on white background, and Locked and Occupied message will display on red background.
    c. Provide snap-in serviceable window to prevent tampering. Lock must function if indicator is compromised.
    d. Indicator window to provide 180-degree visibility.

2.11 EXIT DEVICES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

    a. Von Duprin 98/35A series

2. Acceptable Manufacturers and Products:

    a. No Substitute, Owner standard

B. Requirements:

1. Provide exit devices tested to ANSI/BHMA A156.3 Grade 1 and UL listed for Panic Exit or Fire Exit Hardware.
2. Cylinders: Refer to "KEYING" article, herein.
3. Provide smooth touchpad type exit devices, fabricated of brass, bronze, stainless steel, or aluminum, plated to standard architectural finishes to match balance of door hardware.
4. Touchpad must extend a minimum of one half of door width. No plastic inserts are allowed in touchpads.
5. Provide exit devices with deadlatching feature for security and for future addition of alarm kits and/or other electrified requirements.
6. Provide exit devices with weather resistant components that can withstand harsh conditions of various climates and corrosive cleaners used in outdoor pool environments.
7. Provide flush end caps for exit devices.
8. Provide exit devices with manufacturer's approved strikes.
9. Provide exit devices cut to door width and height. Install exit devices at height recommended by exit device manufacturer, allowable by governing building codes, and approved by Architect.
10. Mount mechanism case flush on face of doors or provide spacers to fill gaps behind devices. Where glass trim or molding projects off face of door, provide glass bead kits.
11. Provide cylinder or hex-key dogging as specified at non fire-rated openings.
12. Removable Mullions: 2 inches (51 mm) x 3 inches (76 mm) steel tube. Where scheduled as keyed removable mullion, provide type that can be removed by use of a keyed cylinder, which is self-locking when re-installed.
13. Provide factory drilled weep holes for exit devices used in full exterior application, highly corrosive areas, and where noted in hardware sets.
14. Provide electrified options as scheduled.
15. Top latch mounting: double- or single-tab mount for steel doors, face mount for aluminum doors eliminating requirement of tabs, and double tab mount for wood doors.
16. Provide exit devices with optional trim designs to match other lever and pull designs used on the project.

2.12 ELECTRIC STRIKES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

   a. Von Duprin 6000 Series

2. Acceptable Manufacturers and Products:

   a. Per Architect/Owner's approval

B. Requirements:

1. Provide electric strikes designed for use with type of locks shown at each opening.
2. Provide electric strikes UL Listed as burglary resistant that are tested to a minimum endurance test of 1,000,000 cycles.
3. Where required, provide electric strikes UL Listed for fire doors and frames.

4. Provide transformers and rectifiers for each strike as required. Verify voltage with electrical contractor.

2.13 CYLINDERS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

a. Schlage Everest 29 T

2. Acceptable Manufacturers and Products:

a. No Substitute

B. Requirements:

1. Provide cylinders/cores compliant with ANSI/BHMA A156.5; latest revision; cylinder face finished to match lockset; manufacturer's series as indicated. Refer to "KEYING" article, herein.
2. Provide cylinders in the below-listed configuration(s), distributed throughout the Project as indicated.

a. Patented Restricted: cylinder with interchangeable core with patented, restricted keyway.

3. Patent Protection: Cylinders/cores requiring use of restricted, patented keys, patent protected.
4. Nickel silver bottom pins.

2.14 KEYING

A. Scheduled System:

1. New factory registered system:

a. Provide a factory registered keying system, complying with guidelines in ANSI/BHMA A156.28, incorporating decisions made at keying conference.

B. Requirements:

1. Construction Keying:

a. Replaceable Construction Cores.

1) Provide temporary construction cores replaceable by permanent cores, furnished in accordance with the following requirements.

a) 3 construction control keys
b) 12 construction change (day) keys.

2) Owner or Owner's Representative will replace temporary construction cores with permanent cores.

2. Permanent Keying:

a. Provide permanent cylinders/cores keyed by the manufacturer according to the following key system.

1) Master Keying system as directed by the Owner.

b. Forward bitting list and keys separately from cylinders, by means as directed by Owner. Failure to comply with forwarding requirements will be cause for replacement of cylinders/cores involved at no additional cost to Owner.

c. Provide keys with the following features:

1) Material: Nickel silver; minimum thickness of .107-inch (2.3mm)
2) Patent Protection: Keys and blanks protected by one or more utility patent(s).
3) Geographically Exclusive: Where High Security or Security cylinders/cores are indicated, provide nationwide, geographically exclusive key system complying with the following restrictions.

d. Identification:

1) Mark permanent cylinders/cores and keys with applicable blind code for identification. Do not provide blind code marks with actual key cuts.
2) Identification stamping provisions must be approved by the Architect and Owner.
3) Stamp cylinders/cores and keys with Owner's unique key system facility code as established by the manufacturer; key symbol and embossed or stamped with "DO NOT DUPLICATE" along with the "PATENTED" or patent number to enforce the patent protection.
4) Failure to comply with stamping requirements will be cause for replacement of keys involved at no additional cost to Owner.
5) Forward permanent cylinders/cores to Owner, separately from keys, by means as directed by Owner.

e. Quantity: Furnish in the following quantities.

1) Permanent Control Keys: 3.
2) Master Keys: 6.
3) Change (Day) Keys: 3 per cylinder/core that is keyed differently
4) Key Blanks: Quantity as determined in the keying meeting.

2.15 KEY MANAGEMENT SOFTWARE

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product: Overtur Key System Management
2. Acceptable Manufacturers and Products: Keystone Web, Simple K. B.

B. Requirements

1. Provide a cloud based key management system that is SOC2 Type 2 Certified - cyber security compliance framework.
2. Provide system with ability of key system design, including:

    a. Uploading and leveraging current plans in pdf format to determine function and traffic flow.
    b. Uploading and utilizing a current door schedule using the Revit Plugin or in Excel format.
    c. Leveraging a mobile app to survey existing conditions and upload data into project.
    d. Collaborating with stakeholders, make notes, and view real time data.
    e. Conducting key meetings using visual digital tools with capability to filter for keyed openings and applying as many key symbols as necessary to an opening.
    f. Designing key system hierarchy and creating schematic, and visually referencing affected openings.
    g. Creating key schedule with fully configured part numbers.

3. Provide training and onboarding, including:

    a. Providing one-time personalized training by manufacturer via web conference to Owner or Owner identified key system personnel.
    b. Access to software support team and self-service knowledge center.
    c. Setting up a Company Admin to manage all user access and roles
    d. Step by step, white glove assistance to set up your project and key system in Overtur, given the appropriate information is supplied.
    e. Facilitate upload of key holder data and currently assigned keys

2.16 KEY CONTROL SYSTEM

A. Manufacturers:

1. Scheduled Manufacturer:

    a. Telkee

2. Acceptable Manufacturers:

    a. HPC
    b. Lund

B. Requirements:

1. Provide key control system, including envelopes, labels, tags with self-locking key clips, receipt forms, 3-way visible card index, temporary markers, permanent markers, and standard metal cabinet, all as recommended by system manufacturer, with capacity for 150% of number of locks required for Project.

a. Provide complete cross index system set up by hardware supplier, and place keys on markers and hooks in cabinet as determined by final key schedule.
b. Provide hinged-panel type cabinet for wall mounting.

2.17 DOOR CLOSERS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

a. LCN 4040XP series

2. Acceptable Manufacturers and Products:

a. No Substitute, Owner standard

B. Requirements:

1. Provide door closers conforming to ANSI/BHMA A156.4 Grade 1 requirements by BHMA certified independent testing laboratory. ISO 9000 certify closers. Stamp units with date of manufacture code.
2. Provide door closers with fully hydraulic, full rack and pinion action with high strength cast iron cylinder, and full complement bearings at shaft.
3. Cylinder Body: 1-1/2-inch (38 mm) diameter piston with 5/8-inch (16 mm) diameter double heat-treated pinion journal. QR code with a direct link to maintenance instructions.
4. Hydraulic Fluid: Fireproof, passing requirements of UL10C, and requiring no seasonal closer adjustment for temperatures ranging from 120 degrees F to -30 degrees F.
5. Spring Power: Continuously adjustable over full range of closer sizes, and providing reduced opening force as required by accessibility codes and standards. Provide snap-on cover clip, with plastic covers, that secures cover to spring tube.
6. Hydraulic Regulation: By tamper-proof, non-critical valves, with separate adjustment for latch speed, general speed, and backcheck. Provide graphically labelled instructions on the closer body adjacent to each adjustment valve. Provide positive stop on reg valve that prevents reg screw from being backed out.
7. Provide closers with solid forged steel main arms and factory assembled heavy-duty forged forearms for parallel arm closers.
8. Pressure Relief Valve (PRV) Technology: Not permitted.
9. Finish for Closer Cylinders, Arms, Adapter Plates, and Metal Covers: Powder coating finish which has been certified to exceed 100 hours salt spray testing as described in ANSI Standard A156.4 and ASTM B117, or has special rust inhibitor (SRI).
10. Provide special templates, drop plates, mounting brackets, or adapters for arms as required for details, overhead stops, and other door hardware items interfering with closer mounting.
11. Closers shall be capable of being upgraded by adding modular mechanical or electronic components in the field.

2.18 ELECTROMECHANICAL AUTOMATIC OPERATORS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

a. LCN Senior Swing

2. Acceptable Manufacturers and Products:

a. No Substitute, Owner standard

B. Requirements:

1. Provide low energy automatic operator units that are electromechanical design complying with ANSI/BHMA A156.19.

a. Opening: Powered by DC motor working through reduction gears.
b. Closing: Spring force.
c. Manual, hydraulic, or chain drive closers: Not permitted.
d. Operation: Motor is off when door is in closing mode. Door can be manually operated with power on or off without damage to operator. Provide variable adjustments, including opening and closing speed adjustment.
e. Cover: Aluminum.

2. Provide units with manual off/auto/hold-open switch, push and go function to activate power operator, vestibule interface delay, electric lock delay, hold-open delay adjustable from 1 to 32 seconds, and logic terminal to interface with accessories, mats, and sensors.
3. Provide drop plates, brackets, and adapters for arms as required to suit details.
4. Provide motion sensors and/or actuator switches, and receivers for operation as specified. Provide weather-resistant actuators at exterior applications.
5. Provide key switches, with LED's, recommended and approved by manufacturer of automatic operator as required for function as described in operation description of hardware sets. Cylinders: Refer to "KEYING" article, herein.
6. Provide complete assemblies of controls, switches, power supplies, relays, and parts/material recommended and approved by manufacturer of automatic operator for each individual leaf. Actuators control both doors simultaneously at pairs. Sequence operation of exterior and vestibule doors with automatic operators to allow ingress or egress through both sets of openings as directed by Architect. Locate actuators, key switches, and other controls as directed by Architect.

2.19 COMPACT ELECTRO-HYDRAULIC AUTOMATIC OPERATOR

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:

a. LCN 6440XP

2. Acceptable Manufacturers and Products:

a. No Substitute, Owner standard

B. Requirements:

1. Provide units with the following feature sets:

a. Fire alarm input
b. Adjustable spring size
c. Conventional door closer opening and closing forces unless motor is activated
d. Electronic power boost feature
e. Adjustable hold open time
f. On/off switch for manual operation

2. Provide non-handed low energy automatic operator units complying with ANSI/BHMA A156.19, UL 228 & 325, and ANSI 117.1.
3. Provide drop plates, brackets, adapters, and arms as required. Provide complete assemblies of controls, switches and parts/material recommended and approved by manufacturer of automatic operator for each individual opening. Locate actuators and other controls as directed by Architect.
4. Provide units that convert 120VAC via an external power supply or wall plug adapter and receive 24VDC.

2.20 DOOR TRIM

A. Manufacturers:

1. Scheduled Manufacturer:

a. Ives

2. Acceptable Manufacturers:

a. Trimco

B. Requirements:

1. Provide push plates, push bars, pull plates, pulls, and hands-free reversible door pulls with diameter and length as scheduled.

2.21 PROTECTION PLATES

A. Manufacturers:

1. Scheduled Manufacturer:

a. Ives

2. Acceptable Manufacturers:

a. Trimco

B. Requirements:

1. Provide protection plates with a minimum of 0.050 inch (1 mm) thick, beveled four edges as scheduled. Furnish with sheet metal or wood screws, finished to match plates.
2. Sizes plates 2 inches (51 mm) less width of door on single doors, pairs of doors with a mullion, and doors with edge guards. Size plates 1 inch (25 mm) less width of door on pairs without a mullion or edge guards.
3. At fire rated doors, provide protection plates over 16 inches high with UL label.

2.22 OVERHEAD STOPS AND OVERHEAD STOP/HOLDERS

A. Manufacturers:

1. Scheduled Manufacturers:

a. Glynn-Johnson

2. Acceptable Manufacturers:

a. No Substitute, Owner standard

B. Requirements:

1. Provide overhead stop at any door where conditions do not allow for a wall stop or floor stop presents tripping hazard.

2.23 DOOR STOPS AND HOLDERS

A. Manufacturers:

1. Scheduled Manufacturer:

a. Ives

2. Acceptable Manufacturers:

a. Trimco

B. Provide door stops at each door leaf:

1. Provide wall stops wherever possible. Provide concave type where lockset has a push button of thumbturn.
2. Where a wall stop cannot be used, provide universal floor stops.
3. Where wall or floor stop cannot be used, provide overhead stop.
4. Provide roller bumper where doors open into each other and overhead stop cannot be used.

2.24 THRESHOLDS, SEALS, DOOR SWEEPS, AUTOMATIC DOOR BOTTOMS, AND GASKETING

A. Manufacturers:

B. Scheduled Manufacturer:

1. Zero International

C. Acceptable Manufacturers:

1. National Guard
2. Pemko

D. Seals and Gasketing: Provide continuous gasketing on exterior openings, to the head and jambs, forming a continuous seal between the door and the frame. Provide smoke, light, or sound gasketing on interior doors where indicated.

1. Provide self-tapping fasteners for aluminum extruded gasketing being applied to hollow metal frames.

a. Provide non-corrosive fasteners for all exterior applications.
b. Provide security fasteners where indicated.

E. Smoke Labeled Gasketing: At all smoke labeled openings, provide smoke listed perimeter gasketing assemblies complying with NFPA 105 listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction for smoke control ratings indicated based on testing according to UL 1784.

F. Fire Listed Gasketing: Assemblies complying with NFPA 80 that are listed by a testing and inspecting agency acceptable to authorities having jurisdiction for fire ratings indicated based on testing according to UL-10C.

1. Where frame-applied intumescent seals are required by the manufacturer, provide gaskets that comply with UL10C Standard for Positive Pressure Fire Tests of Door Assemblies and UBC 7-2, Fire Tests of Door Assemblies.

G. Meeting-Stile Gasketing: Where indicated, provide meeting-stile gasketing that fastens to the meeting stiles forming a continuous seal when doors are closed.

H. Door Sweeps or Shoes: Where indicated, apply to the bottom of the door to close the gap between the door bottom and finished floor or saddle threshold.

I. Automatic Door Bottoms:

1. Provide closed cell sponge, bulb neoprene. or EPDM type of seal as indicated in hardware sets.

J. Rain Drips:

1. Provide overhead rain drips for out-swinging hollow metal doors that are not covered against 45 degree blowing rain. Aluminum extrusion to be a minimum of .088 inches thick and extend 2.50 inches from the face of the frame, in anodized finish to match door.
2. Door sweeps or shoes with integral rain drip must meet ADA requirements

K. Thresholds: Provide threshold units not less than 4 inches wide, formed to accommodate change in floor elevation where indicated, and fabricated to accommodate door hardware and fit door frames.

1. Provide thresholds conforming to ANSI/BHMA A156.21

2.25 SILENCERS

A. Manufacturers:

1. Scheduled Manufacturer:

a. Ives

2. Acceptable Manufacturers:

a. Trimco

B. Requirements:

1. Provide "push-in" type silencers for hollow metal or wood frames.
2. Provide one silencer per 30 inches (762 mm) of height on each single frame, and two for each pair frame.
3. Omit where gasketing is specified.

2.26 MAGNETIC CATCHES

A. Manufacturers:

1. Scheduled Manufacturer:

a. Ives

2. Acceptable Manufacturers:

a. Trimco

B. Requirements:

1. Provide magnetic catches with self-aligning magnets that can be surface mounted or mortised.

2. Provide magnetic catches in an aluminum case 1-inch wide x 3-1/8-inch-long. Provide dual triple pole (Ives 327), where scheduled, with 14-pound load capacity, and dual double pole catches (Ives 326), where scheduled, with 9-pound load capacity.

2.27 FINISHES

A. FINISH: BHMA 626/652 (US26D); EXCEPT:

1. Hinges at Exterior Doors: BHMA 630 (US32D)
2. Aluminum Geared Continuous Hinges: BHMA 628 (US28)
3. Push Plates, Pulls, and Push Bars: BHMA 630 (US32D)
4. Protection Plates: BHMA 630 (US32D)
5. Overhead Stops and Holders: BHMA 630 (US32D)
6. Door Closers: Powder Coat to Match
7. Wall Stops: BHMA 630 (US32D)
8. Latch Protectors: BHMA 630 (US32D)
9. Weatherstripping: Clear Anodized Aluminum
10. Thresholds: Mill Finish Aluminum

PART 3 - EXECUTION

3.1 EXAMINATION

A. Prior to installation of hardware, examine doors and frames, with Installer present, for compliance with requirements for installation tolerances, labeled fire-rated door assembly construction, wall and floor construction, and other conditions affecting performance. Verify doors, frames, and walls have been properly reinforced for hardware installation.

B. Examine roughing-in for electrical power systems to verify actual locations of wiring connections before electrified door hardware installation.

C. Submit a list of deficiencies in writing and proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Mount door hardware units at heights to comply with the following, unless otherwise indicated or required to comply with governing regulations.

1. Standard Steel Doors and Frames: ANSI/SDI A250.8.
2. Custom Steel Doors and Frames: HMMA 831.
3. Interior Architectural Wood Flush Doors: ANSI/WDMA I.S. 1A
4. Installation Guide for Doors and Hardware: DHI TDH-007-20

B. Install door hardware in accordance with NFPA 80, NFPA 101 and provide post-install inspection, testing as specified in section 1.03.E unless otherwise required to comply with governing regulations.

C. Install each hardware item in compliance with manufacturer's instructions and recommendations, using only fasteners provided by manufacturer.

D. Do not install surface mounted items until finishes have been completed on substrate. Protect all installed hardware during painting.

E. Set units level, plumb and true to line and location. Adjust and reinforce attachment substrate as necessary for proper installation and operation.

F. Drill and countersink units that are not factory prepared for anchorage fasteners. Space fasteners and anchors according to industry standards.

G. Install operating parts so they move freely and smoothly without binding, sticking, or excessive clearance.

H. Hinges: Install types and in quantities indicated in door hardware schedule but not fewer than quantity recommended by manufacturer for application indicated.

I. Lock Cylinders:

1. Install construction cores to secure building and areas during construction period.
2. Replace construction cores with permanent cores as indicated in keying section.
3. Furnish permanent cores to Owner for installation.

J. Wiring: Coordinate with Division 26, ELECTRICAL and Division 28 ELECTRONIC SAFETY AND SECURITY sections for:

1. Conduit, junction boxes and wire pulls.
2. Connections to and from power supplies to electrified hardware.
3. Connections to fire/smoke alarm system and smoke evacuation system.
4. Connection of wire to door position switches and wire runs to central room or area, as directed by Architect.
5. Connections to panel interface modules, controllers, and gateways.
6. Testing and labeling wires with Architect's opening number.

K. Key Control System: Tag keys and place them on markers and hooks in key control system cabinet, as determined by final keying schedule.

L. Continuous Hinges: Re-locate the door and frame fire rating labels where they will remain visible so that the hinge does not cover the label once installed.

M. Door Closers & Auto Operators: Mount closers/operators on room side of corridor doors, inside of exterior doors, and stair side of stairway doors from corridors. Mount closers/operators so they are not visible in corridors, lobbies and other public spaces unless approved by Architect.

N. Overhead Stops/Holders: Mount overhead stops/holders on room side of corridor doors, inside of exterior doors, and stair side of stairway doors.

O. Power Supplies: Locate power supplies as indicated or, if not indicated, above accessible ceilings or in equipment room, or alternate location as directed by Architect.

P. Thresholds:

1. Set thresholds in full bed of sealant complying with requirements specified in Division 07 Section "Joint Sealants."
2. Aluminum thresholds to be cut-in, and scribed around mullions, frame members, and stops. Do not butt to thresholds. Provide a continuous surface across full width of opening from jamb to jamb.
3. Where aluminum panic-type (rabbeted) thresholds with neoprene inserts are specified, undercut doors as required to properly mate with seal in threshold.

Q. Stops: Provide floor stops for doors unless wall or other type stops are indicated in door hardware schedule. Do not mount floor stops where they may impede traffic or present tripping hazard.

R. Perimeter Gasketing:

1. Apply to head and jamb, forming seal between door and frame.
2. Install gasketing in a manner eliminating need to cut any seal to install surface mounted hardware. Install compatible mounting bracket for surface mounted hardware unless minimum 1/4 inch thick solid aluminum seals are provided for mounting of surface applied hardware.

S. Meeting Stile Gasketing: Fasten to meeting stiles, forming seal when doors are closed.

T. Door Bottoms and Sweeps: Apply to bottom of door, forming seal with threshold when door is closed.

3.3 ADJUSTING

A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

1. Spring Hinges: Adjust to achieve positive latching when door can close freely from an open position of 30 degrees.
2. Electric Strikes: Adjust horizontal and vertical alignment of keeper to properly engage lock bolt.
3. Door Closers: Adjust sweep period to comply with accessibility requirements and requirements of authorities having jurisdiction.

B. Occupancy Adjustment: Approximately three to six months after date of Substantial Completion, examine and readjust each item of door hardware, including adjusting operating forces, as necessary to ensure function of doors and door hardware.

3.4 CLEANING AND PROTECTION

A. Clean adjacent surfaces soiled by door hardware installation.

B. Clean operating items per manufacturer's instructions to restore proper function and finish.

C. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of Substantial Completion.

3.5 DOOR HARDWARE SCHEDULE

A. The intent of the hardware specification is to specify the hardware for interior and exterior doors, and to establish a type, continuity, and standard of quality. However, it is the door hardware supplier's responsibility to thoroughly review existing conditions, schedules, specifications, drawings, and other Contract Documents to verify the suitability of the hardware specified.

B. Discrepancies, conflicting hardware, and missing items are to be brought to the attention of the architect with corrections made prior to the bidding process. Omitted items not included in a hardware set should be scheduled with the appropriate additional hardware required for proper application.

C. Hardware items are referenced in the following hardware schedule. Refer to the above specifications for special features, options, cylinders/keying, and other requirements.

D. Hardware Sets:

138931 OPT0450740 Version 3

Legend:
Link to catalog cut sheet
Electrified Opening

Hardware Group No. 01

For use on Door #(s):

016

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER			FINISH	MFR
3	EA	HINGE	5BB1HW SH 4.5 X 4.5 NRP SEC			630	IVE
1	EA	PANIC HARDWARE-EXIT ONLY	98-EO-SNB-SEC			626	VON
1	EA	ELECTRIC STRIKE	6300 FSE 12/24 VAC/VDC			630	VON
1	EA	OH STOP	100S ADJ			630	GLY
1	EA	SURF. AUTO OPERATOR	9542IQ LESS TRACK COV			628	LCN
1	EA	3-WAY KEY SWITCH	8310-806K				LCN
1	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T (MOUNT ON INSIDE)			630	LCN
1	EA	MOUNTING PLATE	9540-18 (AS REQ'D)			ANCLR	LCN
1	EA	RAIN DRIP	142AA (AS REQ'D)			AA	ZER
1	SET	GASKETING	429 @ HEAD & JAMBS			AA	ZER
1	EA	DOOR SWEEP	39A			A	ZER

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	SADDLE THRESHOLD	1/2” HIGH X FULL JAMB DEPTH (OR PER SILL DETAILS)		A	ZER
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

HARDWARE IS FOR EXISTING HOLLOW METAL DOOR/FRAME.
FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

OPERATION:
EXIT ONLY
KEY SWITCH TURNS ON/OFF OPERATOR
ACTUATOR ACTIVATES OPERATOR
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM

Hardware Group No. 02

For use on Door #(s):

108A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	HINGE	5BB1HW SH 4.5 X 4.5 NRP SEC		630	IVE
1	EA	ELECTRIC HINGE	5BB1HW 4.5 X 4.5 CON TW8		630	IVE
1	EA	EU STOREROOM LOCK -W/REX	ND80JDEU RHO TORX RX CON 12V/24V DC		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	LOCK GUARD	LG12		630	IVE
1	EA	SURFACE CLOSER	4040XP SCUSH TBTRX		689	LCN
1	EA	RAIN DRIP	142AA (AS REQ'D)		AA	ZER
1	SET	GASKETING	429 @ HEAD & JAMBS		AA	ZER
1	EA	DOOR SWEEP	39A		A	ZER
1	EA	SADDLE THRESHOLD	1/2” HIGH X FULL JAMB DEPTH (OR PER SILL DETAILS)		A	ZER
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY
PROVIDE LOCK SET WITH LEVER DESIGN MATCHING SELECTION FOR NEW EXTERIOR DOORS

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
PRESENTING VALID CREDENTIAL TEMPORALLY RELEASES LEVER FOR ENTRY
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE LOCK
FREE EGRESS AT ALL TIMES

Hardware Group No. 03

For use on Door #(s):

104A 118A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	HINGE	5BB1HW SH 4.5 X 4.5 NRP SEC		630	IVE
1	EA	ELECTRIC HINGE	5BB1HW 4.5 X 4.5 CON TW8		630	IVE
1	EA	ELEC PANIC HARDWARE	RX-98-EO-ALK-EMERG EXIT RSS-CON-SNB-SEC 9-VOLT BATTERY WITH HARDWIRED OPTION		626	VON
1	EA	FSIC MORTISE CYLINDER HOUSING	26-094 (CAM & RING AS REQ'D)		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	SURFACE CLOSER	4040XP SCUSH TBTRX		689	LCN
1	EA	RAIN DRIP	142AA (AS REQ'D)		AA	ZER
1	EA	GASKETING/WEATHER STRIPPING	8150SBK PSA (HEAD & JAMB)		BK	ZER
1	EA	DOOR SWEEP	39A		A	ZER
1	EA	SADDLE THRESHOLD	1/2” HIGH X FULL JAMB DEPTH (OR PER SILL DETAILS)		A	ZER
1	EA	WIRE HARNESS (FROM DEVICE TO HINGE)	CON-XXX (LENGTH TO SUIT)			SCH
1	EA	POWER SUPPLY WIRE HARNESS	CON-6W			SCH
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

EXIT ONLY WITH ALARM

OPERATION:
DOOR IS FOR EMERGENCY EXIT ONLY
PUSHING EXIT DEVICE SOUNDS ALARM; MANUALLY RESET ALARM
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE DEVICE
FREE EGRESS AT ALL TIMES

Hardware Group No. 04

For use on Door #(s):

007

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	HINGE	5BB1HW SH 5 X 4.5 NRP SEC		652	IVE
1	EA	ELECTRIC HINGE	5BB1HW 5 X 4.5 CON TW8 SEC		652	IVE
1	EA	EU STOREROOM LOCK -W/REX	ND80JDEU RHO TORX RX CON 12V/24V DC		626	SCH
1	EA	LFIC PERMANENT CORE	SEPARATE SYSTEM VERIFY WITH OWNER		626	
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBTRX		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	INTUMESCENT SEALS	488FSBK PSA (@HEAD & JAMBS)		BK	ZER
1	EA	DOOR BOTTOM	364AA-Z49		AA	ZER
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
PRESENTING VALID CREDENTIAL TEMPORALLY RELEASES LEVER FOR ENTRY
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE LOCK
FREE EGRESS AT ALL TIMES

Hardware Group No. 04A

For use on Door #(s):

005C

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	HINGE	5BB1HW SH 5 X 4.5 NRP SEC		652	IVE
1	EA	ELECTRIC HINGE	5BB1HW 5 X 4.5 CON TW8 SEC		652	IVE
1	EA	EU STOREROOM LOCK -W/REX	ND80JDEU RHO TORX RX CON 12V/24V DC		626	SCH
1	EA	LFIC PERMANENT CORE	SEPARATE SYSTEM VERIFY WITH OWNER		626	
1	EA	SURFACE CLOSER	4040XP EDA TBTRX		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	INTUMESCENT SEALS	488FSBK PSA (@HEAD & JAMBS)		BK	ZER
1	EA	DOOR BOTTOM	364AA-Z49		AA	ZER
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
PRESENTING VALID CREDENTIAL TEMPORALLY RELEASES LEVER FOR ENTRY
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE LOCK
FREE EGRESS AT ALL TIMES

Hardware Group No. 04B

For use on Door #(s):

005B

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	HINGE	5BB1HW SH 5 X 4.5 NRP SEC		652	IVE
1	EA	ELECTRIC HINGE	5BB1HW 5 X 4.5 CON TW8 SEC		652	IVE
1	EA	EU STOREROOM LOCK -W/REX	ND80JDEU RHO TORX RX CON 12V/24V DC		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBTRX		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	INTUMESCENT SEALS	488FSBK PSA (@HEAD & JAMBS)		BK	ZER
1	EA	DOOR BOTTOM	364AA-Z49		AA	ZER
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
PRESENTING VALID CREDENTIAL TEMPORALLY RELEASES LEVER FOR ENTRY
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE LOCK
FREE EGRESS AT ALL TIMES

Hardware Group No. 05

For use on Door #(s):

103A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	ENTRANCE LOCK	ND53JD RHO TORX		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

Hardware Group No. 05A

For use on Door #(s):

103B 108B 114

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	ENTRANCE LOCK	ND53JD RHO TORX		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
DOOR MAY BE PROGRAMMED TO REMAIN UNLOCKED ON A SCHEDULE THROUGH ACCESS CONTROL SYSTEM
PRESENTING VALID CREDENTIAL TEMPORARILY RETRACTS LATCHBOLT FOR ENTRY WHEN DOOR IS LOCKED
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
FREE EGRESS AT ALL TIMES

Hardware Group No. 06

For use on Door #(s):

013B 014 015 017A 018A 018B

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	STOREROOM LOCK	ND80JD RHO TORX		626	SCH

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	FSIC CORE	23-030		626	SCH
1	EA	WALL STOP	WS406/407CVX		630	IVE
3	EA	SILENCER	SR64		GRY	IVE

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

Hardware Group No. 07

For use on Door #(s):

001 002B

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	STOREROOM LOCK	ND80JD RHO TORX		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBSRT		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE
3	EA	SILENCER	SR64		GRY	IVE

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

Hardware Group No. 08

For use on Door #(s):

119A 120

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
6	EA	HINGE	5PB1 4.5 X 4		652	IVE
1	EA	MANUAL FLUSH BOLT	FB458 OR FB358 PER DR MATERIAL		626	IVE
1	EA	STOREROOM LOCK	ND80JD RHO TORX		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
2	EA	OH STOP	450S		652	GLY
2	EA	SILENCER	SR64		GRY	IVE

Hardware Group No. 09

For use on Door #(s):

105 106 109

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
6	EA	HINGE	5PB1 4.5 X 4		652	IVE
1	EA	MANUAL FLUSH BOLT	FB458 OR FB358 PER DR MATERIAL		626	IVE
1	EA	ONE WAY DEADBOLT	B661J		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
2	EA	OH STOP	450S		652	GLY
2	EA	SILENCER	SR66		GRY	IVE

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

Hardware Group No. 10

For use on Door #(s):

107

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5PB1 4.5 X 4		652	IVE
1	EA	ONE WAY DEADBOLT	B661J		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	OH STOP	450S		652	GLY
3	EA	SILENCER	SR64		GRY	IVE

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

Hardware Group No. 11

For use on Door #(s):

002A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	PASSAGE SET	ND10S RHO		626	SCH
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBSRT		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

Hardware Group No. 12

For use on Door #(s):

008A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1HW SH 4.5 X 4.5 NRP		652	IVE
1	EA	ENTRANCE LOCK	ND53JD RHO TORX		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	ELECTRIC STRIKE	6400 FSE 12/24 VAC/VDC		630	VON
1	EA	OH STOP	100S ADJ (AS REQ'D)		630	GLY
1	EA	SURF. AUTO OPERATOR	6440XP RW/PA LESS COVER CAWMS (MOUNT AT ROOM SIDE)		AL	LCN
1	EA	ROCKER SWITCH	8310-806R (@ROOM SIDE)			LCN
1	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T (MOUNT AT VESTIBULE SIDE)		630	LCN
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER
1	EA	DOOR BOTTOM	364AA-Z49 (OR 355A PER DR MATERIAL)		AA	ZER

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY

OPERATION:
ROCKER SWITCH TURNS ON OPERATOR DURING BUSINESS HOURS AND TURNS OFF AFTER HOURS
ACTUATOR ACTIVATES OPERATOR WHEN IT IS TURNED ON

Hardware Group No. 13

For use on Door #(s):

004

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	PRIVACY LOCK W/ OUTSIDE INDICATOR	ND40S RHO OS-OCC		626	SCH
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBSRT		689	LCN
1	EA	KICK PLATE	8400 10” X 2” LDW B-CS		630	IVE
1	EA	WALL STOP	WS406/407CCV		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

Hardware Group No. 14

For use on Door #(s):

101

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5		652	IVE
1	EA	PRIVACY LOCK W/ OUTSIDE INDICATOR	ND40S RHO OS-OCC		626	SCH
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBSRT		689	LCN
1	EA	KICK PLATE	8400 10” X 2” LDW B-CS		630	IVE
1	EA	WALL STOP	WS406/407CCV		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

PROVIDE OVERHEAD STOP IN LIEU OF WALL STOP WHERE WALL STOP CAN'T BE INSTALLED

Hardware Group No. 15

For use on Door #(s):

009 010 011

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
6	EA	HINGE	5PB1 4.5 X 4		652	IVE
2	EA	MAGNETIC CATCH	326		673	IVE
2	EA	SINGLE DUMMY TRIM	ND170 RHO		626	SCH
2	EA	OH STOP	450S		652	GLY
2	EA	SILENCER	SR66		GRY	IVE

FIELD VERIFY EXISTING DOOR, FRAME AND HARDWARE CONDITIONS, AND PROVIDE HARDWARE AS NECESSARY
VERIFY HARDWARE FUNCTION

Hardware Group No. 16

For use on Door #(s):

005A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5		652	IVE
1	EA	PUSH PLATE	8200 6” X 16”		630	IVE
1	EA	PULL PLATE	8305 10” 4” X 16”		630	IVE
1	EA	SURF. AUTO OPERATOR	9542IQ LESS TRACK COV		628	LCN

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T		630	LCN
1	EA	MOUNTING PLATE	9540-18 (AS REQ'D)		ANCLR	LCN
3	EA	SILENCER	SR64		GRY	IVE

PUSH PULL SPECIFIED TO WORK WITH OPERATOR.

OPERATION:
ACTUATORS ACTIVATE THE OPERATOR

Hardware Group No. 17

For use on Door #(s):

104B

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
6	EA	HINGE	5BB1HW SH 4.5 X 4.5 NRP		652	IVE
1	EA	FIRE EXIT HARDWARE	9847WDC-L-F-LBR-06-SNB		626	VON
1	EA	FIRE EXIT HARDWARE	9847WDC-L-F-LBRAFL-06-SNB		626	VON
2	EA	FSIC RIM HOUSING	20-079		626	SCH
2	EA	FSIC CORE	23-030		626	SCH
2	EA	SURFACE CLOSER	4040XP CUSH		689	LCN
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

HARDWARE IS FOR FIRE RATED WOOD DOOR IN HM FRAME
COORDINATE WITH WOOD DOOR MANUFACTURER TO PROVIDE FIRE DOOR WITH INTUMESCENT SEAL PER CODE
PROVIDE MAG HOLDER OR ELECTRIC CLOSER/HOLDER AS REQUIRED. CONNECT MAG HOLDER TO FIRE ALARM SYSTEM

UNLOCK BOTH LEVER TRIMS WITH A KEY DURING BUSINESS HOURS AND LOCK THEM AFTER HOURS

Hardware Group No. 18

For use on Door #(s):

112 113

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5		652	IVE
1	EA	PUSH PLATE	8200 6” X 16”		630	IVE
1	EA	PULL PLATE	8305 10” 4” X 16”		630	IVE

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	SURFACE CLOSER	4040XP REG OR PA AS REQ TBSRT		689	LCN
1	EA	KICK PLATE	8400 10” X 2” LDW B-CS		630	IVE
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

Hardware Group No. 20

For use on Door #(s):

006 110 111 115 116 117

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1 4.5 X 4.5		652	IVE
1	EA	ENTRANCE LOCK	ND53JD RHO TORX		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	WALL STOP	WS406/407CCV		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

Hardware Group No. 21

For use on Door #(s):

118B

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP		652	IVE
1	EA	PANIC HARDWARE-CLASSROOM TRIM	LD-98-L-06-SNB-SEC		626	VON
1	EA	FSIC RIM HOUSING	20-079		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	SURFACE CLOSER	4040XP EDA TBTRX		689	LCN
1	EA	WALL STOP	WS406/407CVX		630	IVE
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER

Hardware Group No. A01

For use on Door #(s):

013A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	CONT. HINGE	112XY EPT		628	IVE

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	POWER TRANSFER	EPT10 CON		689	VON
1	EA	ELEC PANIC HARDWARE	LX-RX-QEL-35A-L-NL-360-06-CON-SEC 24 VDC		626	VON
1	EA	FSIC RIM HOUSING	20-079		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
1	EA	OH STOP	100S ADJ		630	GLY
1	EA	SURF. AUTO OPERATOR	9542IQ LESS TRACK COV		628	LCN
1	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T (MOUNT AT INSIDE)		630	LCN
1	EA	MOUNTING PLATE	9540-18 (AS REQ'D)		ANCLR	LCN
1	EA	WEATHERSTRIPPING	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
1	EA	DOOR SWEEP	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
1	EA	THRESHOLD	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
1	EA	WIRE HARNESS (FROM DEVICE TO HINGE)	CON-XXX (LENGTH TO SUIT)			SCH
1	EA	POWER SUPPLY WIRE HARNESS	CON-6W			SCH
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

HARDWARE IS FOR STOREFRONT SYSTEM WITH 1-3/4” THICK DOOR, PROVIDE IVES 027XY AT 2” THICK DOOR
REPLACED EXTERIOR LADDER PULL WITH LEVER TRIM PER 2ND HARDWARE REVIEW

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
DOOR MAY BE PROGRAMMED TO REMAIN UNLOCKED ON A SCHEDULE THROUGH ACCESS CONTROL SYSTEM
PRESENTING VALID CREDENTIAL TEMPORARILY RETRACTS LATCHBOLT AND ACTIVATES OPERATOR FOR ENTRY WHEN DOOR IS LOCKED
INTERIOR ACTUATOR IS ALWAYS ACTIVE
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE DEVICE
FREE EGRESS AT ALL TIMES

Hardware Group No. A02

For use on Door #(s):

100A

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	CONT. HINGE	112XY EPT		628	IVE
2	EA	POWER TRANSFER	EPT10 CON		689	VON
1	EA	ELEC PANIC HARDWARE	LX-RX-QEL-3547A-NL-OP-388-CON-SEC 24 VDC		626	VON
1	EA	ELEC PANIC HARDWARE	RX-QEL-3547A-EO-CON-SEC 24 VDC		626	VON
1	EA	FSIC RIM HOUSING	20-079		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
2	EA	LONG DOOR PULL	9264F 36” 20” O		630	IVE
2	EA	OH STOP	100S ADJ		630	GLY
1	EA	SURFACE CLOSER	4040XP EDAW/62G TBTRX		689	LCN
1	EA	SURF. AUTO OPERATOR	9542IQ LESS TRACK COV (MOUNT ON LEFT-HAND LEAF)		628	LCN
1	EA	PA MOUNTING PLATE	4040XP-18PA		689	LCN
1	EA	CUSH SHOE SUPPORT	4040XP-30 SRT (AS REQ'D)		689	LCN
1	EA	WEATHER RING	8310-801			LCN
1	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T (MOUNT ON EXTERIOR STONE VENEER)		630	LCN
1	EA	VESTIBULE ACTUATOR, TOUCH	8310-855 (MOUNT ON INSIDE)		630	LCN
1	EA	MOUNTING PLATE	9540-18 (AS REQ'D)		ANCLR	LCN
1	EA	WEATHERSTRIPPING	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
1	EA	MEETING STILE	BY ALUMINUM DOOR/FRAME MANUFACTURER			
2	EA	DOOR SWEEP	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
1	EA	THRESHOLD	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
2	EA	WIRE HARNESS (FROM DEVICE TO HINGE)	CON-XXX (LENGTH TO SUIT)			SCH
2	EA	POWER SUPPLY WIRE HARNESS	CON-6W			SCH
1	EA	CREDENTIAL READER	BY DIVISION 28.			
2	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

HARDWARE IS FOR STOREFRONT SYSTEM WITH 1-3/4” THICK DOOR, PROVIDE IVES 027XY AT 2” THICK DOOR

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
DOOR MAY BE PROGRAMMED TO REMAIN UNLOCKED ON A SCHEDULE THROUGH ACCESS CONTROL SYSTEM
PRESENTING VALID CREDENTIAL TEMPORARILY RETRACTS LATCHBOLT FOR ENTRY WHEN DOOR IS LOCKED
EXTERIOR ACTUATOR IS CONTROLLED BY ACCESS CONTROL SYSTEM; INTERIOR ACTUATOR IS ALWAYS ACTIVE
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE DEVICE
FREE EGRESS AT ALL TIMES

Hardware Group No. A03A

For use on Door #(s):

102

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
1	EA	CONT. HINGE	112XY EPT		628	IVE
1	EA	POWER TRANSFER	EPT10 CON		689	VON
1	EA	ELEC PANIC HARDWARE	LX-RX-QEL-35A-NL-OP-388-CON 24 VDC		626	VON
1	EA	FSIC RIM HOUSING	20-079		626	SCH
1	EA	FSIC MORTISE CYLINDER HOUSING	26-094 (CAM & RING AS REQ'D)		626	SCH
2	EA	FSIC CORE	23-030		626	SCH
1	EA	SURF. AUTO OPERATOR	9542IQ LESS TRACK COV		628	LCN
1	EA	3-WAY KEY SWITCH	8310-806K			LCN
2	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T		630	LCN
1	EA	MOUNTING PLATE	9540-18 (AS REQ'D)		ANCLR	LCN
1	EA	GASKETING	488SBK PSA @ HEAD & JAMBS		BK	ZER
1	EA	WIRE HARNESS (FROM DEVICE TO HINGE)	CON-XXX (LENGTH TO SUIT)			SCH
1	EA	POWER SUPPLY WIRE HARNESS	CON-6W			SCH
1	EA	CREDENTIAL READER	BY DIVISION 28.			
1	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)			
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.			

HARDWARE IS FOR STOREFRONT SYSTEM WITH 1-3/4” THICK DOOR, PROVIDE IVES 027XY AT 2” THICK DOOR

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
DOOR MAY BE PROGRAMMED TO REMAIN UNLOCKED ON A SCHEDULE THROUGH ACCESS CONTROL SYSTEM
PRESENTING VALID CREDENTIAL TEMPORARILY RETRACTS LATCHBOLT FOR ENTRY WHEN DOOR IS LOCKED
KEY SWITCH TURNS ON/OFF/HOLD OPEN OPERATOR
EXTERIOR ACTURATOR IS CONTROLLED BY ACCESS CONTROL SYSTEM; INTERIOR ACTUATOR IS ALWAYS ACTIVE
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE DEVICE
FREE EGRESS AT ALL TIMES

Hardware Group No. A05

For use on Door #(s):

100B

Provide each door(s) with the following:

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	CONT. HINGE	112XY EPT		628	IVE
2	EA	POWER TRANSFER	EPT10 CON		689	VON
1	EA	ELEC PANIC HARDWARE	LX-RX-QEL-3547A-NL-OP-LBR-388-CON-SEC 24 VDC		626	VON
1	EA	ELEC PANIC HARDWARE	RX-QEL-3547A-EO-LBR-CON 24 VDC		626	VON
1	EA	FSIC RIM HOUSING	20-079		626	SCH
1	EA	FSIC CORE	23-030		626	SCH
2	EA	LONG DOOR PULL	9264F 36” 20” O		630	IVE
2	EA	OH STOP	100S ADJ		630	GLY
1	EA	SURFACE CLOSER	4040XP EDAW/62G TBTRX		689	LCN
1	EA	SURF. AUTO OPERATOR	9542IQ LESS TRACK COV		628	LCN
1	EA	PA MOUNTING PLATE	4040XP-18PA		689	LCN
1	EA	CUSH SHOE SUPPORT	4040XP-30 SRT (AS REQ'D)		689	LCN
1	EA	4.75” HARDWIRED SQUARE ACTUATOR, TOUCH	8310-853T		630	LCN
1	EA	MOUNTING PLATE	9540-18 (AS REQ'D)		ANCLR	LCN
1	EA	WEATHERSTRIPPING	BY ALUMINUM DOOR/FRAME MANUFACTURER.			
1	EA	MEETING STILE	BY ALUMINUM DOOR/FRAME MANUFACTURER			
1	EA	THRESHOLD	BY ALUMINUM DOOR/FRAME MANUFACTURER.			

QTY		DESCRIPTION	CATALOG NUMBER		FINISH	MFR
2	EA	WIRE HARNESS (FROM DEVICE TO HINGE)	CON-XXX (LENGTH TO SUIT)	↯		SCH
2	EA	POWER SUPPLY WIRE HARNESS	CON-6W	↯		SCH
1	EA	CREDENTIAL READER	BY DIVISION 28.	↯		
2	EA	DOOR POSITION SWITCH	BY DIVISION 28. (AS REQ'D)	↯		
1	EA	LOW VOLTAGE POWER	BY LOW VOLTAGE.	↯		

HARDWARE IS FOR STOREFRONT SYSTEM WITH 1-3/4” THICK DOOR, PROVIDE IVES 027XY AT 2” THICK DOOR
REFERENCE HARDWARE SET FOR EXTERIOR FOR ACTUATOR AT VESTIBULE SIDE

OPERATION:
DOOR IS NORMALLY LATCHED AND SECURED
DOOR MAY BE PROGRAMMED TO REMAIN UNLOCKED ON A SCHEDULE THROUGH ACCESS CONTROL SYSTEM
PRESENTING VALID CREDENTIAL TEMPORALLY RETRACTS LATCHBOLT FOR ENTRY WHEN DOOR IS LOCKED
DOOR IS MONITORED THROUGH ACCESS CONTROL OR SECURITY SYSTEM
DOOR IS SECURED UPON LOSS OF POWER TO THE DEVICE
FREE EGRESS AT ALL TIMES

END OF SECTION 08 71 00

SECTION 08 80 00 – GLAZING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Monolithic glass.
2. Insulating glass.
3. Bird-friendly glass.
4. Fire-protection-rated glazing.
5. Glazing sealants.
6. Glazing tapes.
7. Miscellaneous glazing materials.

B. Related Requirements:

1. Section 084113 "Aluminum-framed Entrances and Storefronts"
2. Section 081113 "Hollow Metal Doors and Frames"
3. Section 081416 "Flush Wood Doors."

1.2 DEFINITIONS

A. Glass Manufacturers: Firms that produce primary glass, fabricated glass, or both, as defined in referenced glazing publications.

B. Glass Thicknesses: Indicated by thickness designations in millimeters in accordance with ASTM C1036.

C. IBC: International Building Code.

D. Interspace: Space between lites of an insulating-glass unit.

E. Fire-Protection-Rated Glazing: Glazing that prevents spread of fire and smoke and complies with requirements for rated openings; incapable of blocking radiant heat.

F. Fire-Resistance-Rated Glazing: Glazing that prevents spread of fire and smoke and radiant heat and complies with requirements for rated walls and rated openings; capable of blocking radiant heat.

1.3 COORDINATION

A. Coordinate glazing channel dimensions to provide necessary bite on glass, minimum edge and face clearances, and adequate sealant thicknesses, with reasonable tolerances to achieve proper safety margins for glazing retention under each design load case, load case combination, and service condition.

1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference on project site.

1. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
2. Review temporary protection requirements for glazing during and after installation.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Glass Samples: For each type of glass product other than clear monolithic vision glass; 12 inches square.

C. Glazing Accessory Samples: For sealants, in 12-inch lengths. Install sealant Samples between two strips of material representative in color of adjoining framing system.

D. Glazing Schedule: List glass types and thicknesses for each size opening and location. Use same designations indicated on Drawings.

E. Delegated Design Submittals: For glass indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by qualified professional engineer responsible for their preparation.

F. Sustainable Design Submittals:

1. Product Data: For sealants, indicating VOC content.

1.6 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of glass and glazing product.

B. Product Test Reports: For fabricated glass and glazing sealants, for tests performed by a qualified testing agency.

1. For glazing sealants, provide test reports based on testing current sealant formulations within previous 36-month period.

C. Preconstruction adhesion and compatibility test report.

D. Qualification Statements: For Installer manufacturers of fabricated glass units, glass testing agency, and sealant testing agency.

E. Sample Warranty: For manufacturer's special warranties.

1.7 CLOSEOUT SUBMITTALS

A. Manufacturer's Warranties: For all products used.

1.8 QUALITY ASSURANCE

A. Fabricated-Glass Manufacturer Qualifications: A qualified manufacturer of fabricated glass units who is approved and certified by primary glass manufacturer.

B. Installer Qualifications: A qualified glazing contractor for this Project who is certified under the North American Contractor Certification Program (NACC) for Architectural Glass & Metal (AG&M) contractors and who employs glazing technicians certified under the Architectural Glass and Metal Technician (AGMT) certification program.

C. Glass Testing Agency Qualifications: A qualified independent testing agency accredited according to the NFRC CAP 1 Certification Agency Program.

D. Sealant Testing Agency Qualifications: An independent testing agency qualified according to ASTM C1021 to conduct the testing indicated.

1.9 MOCKUPS

A. Build mockups to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Install glazing in mockups specified in Section 084113 "Aluminum-Framed Entrances and Storefronts" to match glazing systems required for Project, including glazing methods.
2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless the Contracting Officer specifically approves in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.10 PRECONSTRUCTION TESTING

A. Preconstruction Adhesion and Compatibility Testing: Test each glass product, tape sealant, gasket, glazing accessory, and glass-framing member for adhesion to and compatibility with elastomeric glazing sealants.

1. Testing is not required if data are submitted based on previous testing of current sealant products and glazing materials matching those submitted.
2. Use ASTM C1087 to determine whether priming and other specific joint-preparation techniques are required to obtain rapid, optimum adhesion of glazing sealants to glass, tape sealants, gaskets, and glazing channel substrates.
3. Test no fewer than eight Samples of each type of material, including joint substrates, shims, sealant backings, secondary seals, and miscellaneous materials.
4. Schedule enough time for testing and analyzing results to prevent delaying the Work.
5. For materials failing tests, submit sealant manufacturer's written instructions for corrective measures including use of specially formulated primers.

1.11 DELIVERY, STORAGE, AND HANDLING

A. Protect glazing materials in accordance with manufacturer's written instructions. Prevent damage to glass and glazing materials from condensation, temperature changes, direct exposure to sun, or other causes.

B. Comply with insulating-glass manufacturer's written instructions for venting and sealing units to avoid hermetic seal ruptures due to altitude change.

1.12 FIELD CONDITIONS

A. Environmental Limitations: Do not proceed with glazing when ambient and substrate temperature conditions are outside limits permitted by glazing material manufacturers and when glazing channel substrates are wet from rain, frost, condensation, or other causes.

1. Do not install glazing sealants when ambient and substrate temperature conditions are outside limits permitted by sealant manufacturer or are below 40 deg F.

1.13 WARRANTY

A. Manufacturer's Special Warranty for Coated-Glass Products: Manufacturer agrees to replace coated-glass units that deteriorate within specified warranty period. Deterioration of coated glass is defined as defects developed from normal use that are not attributed to glass breakage or to maintaining and cleaning coated glass contrary to manufacturer's written instructions. Defects include peeling, cracking, and other indications of deterioration in coating.

1. Warranty Period: 10 years from date of Substantial Completion.

B. Manufacturer's Special Warranty for Insulating Glass: Manufacturer agrees to replace insulating-glass units that deteriorate within specified warranty period. Deterioration of insulating glass is defined as failure of hermetic seal under normal use that is not attributed to glass breakage or to maintaining and cleaning insulating glass contrary to manufacturer's written instructions. Evidence of failure is obstruction of vision by dust, moisture, or film on interior surfaces of glass.

1. Warranty Period: 10 years from date of Substantial Completion.

C. Manufacturer's Special Warranty for Fully Tempered Glass: Manufacturer agrees to replace fully tempered glass units that spontaneously break due to nickel sulfide (NiS) inclusions at a rate exceeding 0.3 percent (3/1000) within specified warranty period. Coverage for any other cause is excluded.

1. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Source Limitations for Glass: Obtain coated glass from single source from single manufacturer.

B. Source Limitations for Glazing Accessories: For each product and installation method, obtain from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. General: Installed glazing systems to withstand normal thermal movement and wind and impact loads (where applicable) without failure, including loss or glass breakage attributable to defective manufacture, fabrication, or installation; failure of sealants or gaskets to remain watertight and airtight; deterioration of glazing materials; or other defects in construction.

B. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 "Quality Requirements," to design glazing.

C. Structural Performance: Glazing to withstand the following design loads within limits and under conditions indicated determined in accordance with the IBC and ASTM E1300:

1. Design Wind Pressures: Determine design wind pressures applicable to Project in accordance with ASCE/SEI 7, based on heights above grade indicated on Drawings.

    a. Wind Design Data: As indicated on Drawings.
    b. Basic Wind Speed: 108 mph.
    c. Importance Factor: 1.0.
    d. Exposure Category: C.

2. Design Snow Loads: As indicated on Drawings.
3. Maximum Lateral Deflection: For glass supported on all four edges, limit center-of-glass deflection at design wind pressure to not more than 1/50 times the short-side length or 1 inch, whichever is less.
4. Thermal Loads: Design glazing to resist thermal stress breakage induced by differential temperature conditions and limited air circulation within individual glass lites and insulated glazing units.

D. Safety Glazing: Where safety glazing is indicated, provide glazing that complies with 16 CFR 1201, Category II.

E. Thermal and Optical Performance Properties: Provide glass with performance properties specified, as indicated in manufacturer's published test data, based on procedures indicated below:

1. For monolithic-glass lites, properties are based on units with lites of thickness indicated.
2. For insulating-glass units, properties are based on units of thickness indicated for overall unit and for each lite.

3. U-Factors: Center-of-glazing values, in accordance with NFRC 100 and based on most current non-beta version of LBL's WINDOW computer program, expressed as Btu/sq. ft. x h x deg F.
4. SHGC and Visible Transmittance: Center-of-glazing values, in accordance with NFRC 200 and based on most current non-beta version of LBL's WINDOW computer program.
5. Visible Reflectance: Center-of-glazing values, in accordance with NFRC 300.

2.3 GLASS, GENERAL

A. Glazing Publications: Comply with written instructions of glass product manufacturers and organizations below unless more stringent requirements are indicated. See these publications for glazing terms not otherwise defined in this Section or in referenced standards.

1. NGA Publications: "Laminated Glazing Reference Manual" and "Glazing Manual."
2. AAMA Publications: AAMA GDSG-1 and AAMA TIR A7.
3. IGMA Publication for Sloped Glazing: IGMA TB-3001.
4. IGMA Publication for Insulating Glass: IGMA TM-3000.

B. Safety Glazing Labeling: Where safety glazing is indicated, permanently mark glazing with certification label of the SGCC or another certification agency acceptable to authorities having jurisdiction. Label indicates manufacturer's name, type of glass, thickness, and safety glazing standard with which glass complies.

C. Thickness: Where glass thickness is indicated, it is a minimum. Provide glass that complies with performance requirements and is not less than thickness indicated.

1. Minimum Glass Thickness for Exterior Lites: 1/4 inch.
2. Thickness of Tinted Glass: Provide same thickness for each tint color indicated throughout Project.

D. Strength: Where annealed float glass is indicated, provide annealed float glass, heat-strengthened float glass, or fully tempered float glass as needed to comply with "Performance Requirements" Article. Where heat-strengthened float glass is indicated, provide heat-strengthened float glass or fully tempered float glass as needed to comply with "Performance Requirements" Article. Where fully tempered float glass is indicated, provide fully tempered float glass.

2.4 MONOLITHIC GLASS

A. Clear Annealed Float Glass: ASTM C1036, Type I, Class 1 (clear), Quality-Q3.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. Guardian Glass LLC
b. Vitro Architectural Glass

B. Fully Tempered Float Glass: ASTM C1048, Kind FT (fully tempered), Condition A (uncoated) unless otherwise indicated, Type I, Class 1 (clear) or Class 2 (tinted) as indicated, Quality-Q3.

1. Fabrication Process: By horizontal (roller-hearth) process with roll-wave distortion parallel to bottom edge of glass as installed unless otherwise indicated.

C. Reflective- and Low-E-Coated Vision Glass: ASTM C1376.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Guardian Glass LLC or comparable product by one of the following:

a. Cardinal Glass Industries, Inc.
b. Vitro Architectural Glass
c. Viracon, Inc.

2.5 INSULATING GLASS

A. Insulating Glass: Factory-assembled units consisting of sealed lites of glass separated by a dehydrated interspace, qualified in accordance with ASTM E2190.

1. Sealing System: Dual seal, with manufacturer's standard polyisobutylene and silicone primary and secondary sealants.
2. Desiccant: Molecular sieve or silica gel, or a blend of both.

2.6 BIRD-FRIENDLY GLASS

A. Bird-Friendly Glass: Glass incorporating visual markers or patterns that are visible to birds and have been tested for effectiveness in deterring bird collisions by the American Bird Conservancy.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. Guardian Glass LLC
b. Viracon, Inc.
c. Vitro Architectural Glass

2. Basis-of-Design Product: Subject to compliance with requirements, provide Bird 1st Etch DX22 or comparable product by one of the following:

a. Guardian Glass LLC
b. Viracon, Inc.
c. Vitro Architectural Glass

3. Threat Factor Rating: 15.

2.7 FIRE-PROTECTION-RATED GLAZING

A. General: Listed and labeled by a testing agency acceptable to authorities having jurisdiction, for fire-protection ratings indicated, based on positive-pressure testing in accordance with NFPA 257 or UL 9, including hose-stream test, and shall comply with NFPA 80.

1. Fire-protection-rated glazing required to have a fire-protection rating of 20 minutes shall be exempt from hose-stream test.

B. Fire-Protection-Rated Glazing Labeling: Permanently mark fire-protection-rated glazing with certification label of a testing agency acceptable to authorities having jurisdiction. Label shall indicate manufacturer's name; test standard; whether glazing is permitted to be used in doors or openings; if permitted in openings, whether glazing has passed hose-stream test; whether glazing meets 450 deg F temperature-rise limitation; and fire-resistance rating in minutes.

C. Fire-Protection-Rated Monolithic Glass for Doors and Protected Openings: 19-mm thickness; low-iron fire-protection-rated glass; complying with 16 CFR 1201, Category II. UL listed and tested in accordance with NFPA 252 for fire-rated doors and NFPA 257 for protected openings with hose-stream testing.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. McGrory Glass, Inc
b. SAFTI FIRST Fire Rated Glazing Solutions
c. Schott North America, Inc.

D. Fire-Protection-Rated Monolithic Glass for Doors Only: 19-mm thickness; clear, fire-protection glass; complying with 16 CFR 1201, Category II. UL listed and tested in accordance with NFPA 252 for fire-rated doors with hose-stream testing.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. McGrory Glass, Inc
b. SAFTI FIRST Fire Rated Glazing Solutions

2.8 GLAZING SEALANTS

A. General:

1. Compatibility: Compatible with one another and with other materials they contact, including glass products, seals of insulating-glass units, and glazing channel substrates, under conditions of service and application, as demonstrated by sealant manufacturer based on testing and field experience.
2. Suitability: Comply with sealant and glass manufacturers' written instructions for selecting glazing sealants suitable for applications indicated and for conditions existing at time of installation.

3. Colors of Exposed Glazing Sealants: As selected by Architect from manufacturer's full range of industry colors.

2.9 GLAZING TAPES

A. Back-Bedding Mastic Glazing Tapes: Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with ASTM C1281 and AAMA 800 for products indicated below:

1. AAMA 804.3 tape, where indicated.
2. AAMA 806.3 tape, for glazing applications in which tape is subject to continuous pressure.
3. AAMA 807.3 tape, for glazing applications in which tape is not subject to continuous pressure.

B. Expanded Cellular Glazing Tapes: Closed-cell, PVC or acrylic foam tapes; factory coated with adhesive on both surfaces; and complying with AAMA 800 for the following types:

1. AAMA 810.1, Type 1, for glazing applications in which tape acts as primary sealant.
2. AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

2.10 MISCELLANEOUS GLAZING MATERIALS

A. General: Provide products of material, size, and shape complying with referenced glazing standard, recommended in writing by manufacturers of glass and other glazing materials for application indicated, and with a proven record of compatibility with surfaces contacted in installation.

B. Cleaners, Primers, and Sealers: Types recommended by sealant or gasket manufacturer.

C. Setting Blocks:

1. Type: Silicone with Shore A durometer hardness of 85, plus or minus 5.
2. Type recommended in writing by sealant or glass manufacturer.

D. Spacers:

1. Type: Neoprene blocks or continuous extrusions of hardness required by glass manufacturer to maintain glass lites in place for installation indicated.
2. Type recommended in writing by sealant or glass manufacturer.

E. Edge Blocks:

1. Type: Silicone with Shore A durometer hardness per manufacturer's written instructions.
2. Type recommended in writing by sealant or glass manufacturer.

F. Cylindrical Glazing Sealant Backing: ASTM C1330, Type O (open-cell material), of size and density to control glazing sealant depth and otherwise produce optimum glazing sealant performance.

2.11 FABRICATION OF GLAZING UNITS

A. Fabricate glazing units in sizes required to fit openings indicated for Project, with edge and face clearances, edge and surface conditions, and bite complying with written instructions of product manufacturer and referenced glazing publications, to comply with system performance requirements.

1. Allow for thermal movements from ambient and surface temperature changes acting on glass framing members and glazing components.

a. Temperature Change: 120 deg F, ambient; 180 deg F, material surfaces.

B. Clean-cut or flat-grind vertical edges of butt-glazed monolithic lites to produce square edges with slight chamfers at junctions of edges and faces.

C. Grind smooth and polish exposed glass edges and corners.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine framing, glazing channels, and stops, with Installer present, for compliance with the following:

1. Manufacturing and installation tolerances, including those for size, squareness, and offsets at corners.
2. Presence and functioning of weep systems.
3. Minimum required face and edge clearances.
4. Effective sealing between joints of glass-framing members.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Clean glazing channels and other framing members receiving glass immediately before glazing. Remove coatings not firmly bonded to substrates.

B. Examine glazing units to locate exterior and interior surfaces. Label or mark units as needed so that exterior and interior surfaces are readily identifiable. Do not use materials that leave visible marks in the completed Work.

3.3 GLAZING, GENERAL

A. Comply with combined written instructions of manufacturers of glass, sealants, gaskets, and other glazing materials, unless more stringent requirements are indicated, including those in referenced glazing publications.

B. Protect glass edges from damage during handling and installation. Remove damaged glass from Project site and legally dispose of off Project site. Damaged glass includes glass with edge damage or other imperfections that, when installed, could weaken glass, impair performance, or impair appearance.

C. Apply primers to joint surfaces where required for adhesion of sealants, as determined by preconstruction testing.

D. Install setting blocks in sill rabbets, sized and located to comply with referenced glazing publications, unless otherwise required by glass manufacturer. Set blocks in thin course of compatible sealant suitable for heel bead.

E. Do not exceed edge pressures stipulated by glass manufacturers for installing glass lites.

F. Provide spacers for glass lites where length plus width is larger than 50 inches.

1. Locate spacers directly opposite each other on both inside and outside faces of glass. Install correct size and spacing to preserve required face clearances, unless gaskets and glazing tapes are used that have demonstrated ability to maintain required face clearances and to comply with system performance requirements.
2. Provide 1/8-inch- minimum bite of spacers on glass and use thickness equal to sealant width. With glazing tape, use thickness slightly less than final compressed thickness of tape.

G. Provide edge blocking where indicated or needed to prevent glass lites from moving sideways in glazing channel, as recommended in writing by glass manufacturer and in accordance with requirements in referenced glazing publications.

H. Set glass lites in each series with uniform pattern, draw, bow, and similar characteristics.

I. Set glass lites with proper orientation so that coatings face exterior or interior as specified.

J. Where wedge-shaped gaskets are driven into one side of channel to pressurize sealant or gasket on opposite side, provide adequate anchorage so gasket cannot walk out when installation is subjected to movement.

K. Square cut wedge-shaped gaskets at corners and install gaskets in a manner recommended by gasket manufacturer to prevent corners from pulling away; seal corner joints and butt joints with sealant recommended in writing by gasket manufacturer.

3.4 INSTALLATION OF GLAZING TAPES

A. Position tapes on fixed stops so that, when compressed by glass, their exposed edges are flush with or protrude slightly above sightline of stops.

B. Install tapes continuously, but not necessarily in one continuous length. Do not stretch tapes to make them fit opening.

C. Cover vertical framing joints by applying tapes to heads and sills first, then to jambs. Cover horizontal framing joints by applying tapes to jambs, then to heads and sills.

D. Place joints in tapes at corners of opening with adjoining lengths butted together, not lapped. Seal joints in tapes with compatible sealant approved by tape manufacturer.

E. Do not remove release paper from tape until right before each glazing unit is installed.

F. Apply heel bead of elastomeric sealant.

G. Center glass lites in openings on setting blocks, and press firmly against tape by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings.

H. Apply cap bead of elastomeric sealant over exposed edge of tape.

3.5 INSTALLATION OF GLAZING GASKET (DRY)

A. Cut compression gaskets to lengths recommended by gasket manufacturer to fit openings exactly, with allowance for stretch during installation.

B. Insert soft compression gasket between glass and frame or fixed stop so it is securely in place with joints miter cut and bonded together at corners.

C. Installation with Drive-in Wedge Gaskets: Center glass lites in openings on setting blocks, and press firmly against soft compression gasket by inserting dense compression gaskets formed and installed to lock in place against faces of removable stops. Start gasket applications at corners and work toward centers of openings. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended in writing by gasket manufacturer.

D. Installation with Pressure-Glazing Stops: Center glass lites in openings on setting blocks, and press firmly against soft compression gasket. Install dense compression gaskets and pressure-glazing stops, applying pressure uniformly to compression gaskets. Compress gaskets to produce a weathertight seal without developing bending stresses in glass. Seal gasket joints with sealant recommended in writing by gasket manufacturer.

E. Install gaskets so they protrude past face of glazing stops.

3.6 INSTALLATION OF GLAZING SEALANT

A. Install continuous spacers, or spacers combined with cylindrical sealant backing, between glass lites and glazing stops to maintain glass face clearances and to prevent sealant from extruding into glass channel and blocking weep systems until sealants cure. Secure spacers or spacers and backings in place and in position to control depth of installed sealant relative to edge clearance for optimum sealant performance.

B. Force sealants into glazing channels to eliminate voids and to ensure complete wetting or bond of sealant to glass and channel surfaces.

C. Tool exposed surfaces of sealants to provide a substantial wash away from glass.

3.7 CLEANING

A. Immediately after installation, remove nonpermanent labels and clean surfaces.

B. Wash glass on both exposed surfaces not more than four days before date scheduled for inspections that establish date of Substantial Completion. Wash glass as recommended in writing by glass manufacturer.

3.8 PROTECTION

A. Protect glass from contact with contaminating substances resulting from construction operations. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains.

1. If, despite such protection, contaminating substances do contact with glass, remove substances immediately as recommended in writing by glass manufacturer. Remove and replace glass that cannot be cleaned without damage to coatings.

B. Remove and replace glass that is damaged during construction period.

3.9 MONOLITHIC GLASS SCHEDULE

A. Clear Glass Type: Fully tempered float glass GL-5.

1. Minimum Thickness: 1/4 inch.
2. Safety glazing required.

3.10 INSULATING GLASS SCHEDULE

A. Clear Insulating Glass Type : GL-3, GL-4.

1. Basis-of-Design Product: Guardian Glass LLC.
2. Overall Unit Thickness: 1 inch.
3. Minimum Thickness of Each Glass Lite: 1/4 inch.
4. Outdoor Lite: Annealed Fully tempered float glass.
5. Bird friendly etching on surface #1.
6. GL-3 outdoor lite is annealed and GL-4 outdoor lite is fully tempered.
7. Interspace Content: Air.
8. Indoor Lite: Annealed Fully tempered float glass.
9. GL-3 indoor lite is annealed and GL-4 indoor lite is fully tempered.

10. Winter Nighttime U-Factor: 0.34 maximum.
11. Summer Daytime U-Factor: 0.34 maximum.
12. Safety glazing required.

B. Low-E-Coated, Clear Insulating Glass Type: GL-1 & GL-2.

1. Basis-of-Design Product: SNX 51/23.
2. Overall Unit Thickness: 1 inch.
3. Minimum Thickness of Each Glass Lite: 1/4 inch.
4. Outdoor Lite: Annealed Fully tempered float glass.
5. Bird friendly etching on surface #1.
6. GL-1 outdoor lite is annealed and GL-2 outdoor lite is fully tempered.
7. Interspace Content: Air.
8. Indoor Lite: Annealed Fully tempered float glass.
9. GL-1 indoor lite is annealed and GL-2 indoor lite is fully tempered.
10. Low-E Coating: Sputtered on second surface.
11. Winter Nighttime U-Factor: 0.34 maximum.
12. Summer Daytime U-Factor: 0.34 maximum.
13. Visible Light Transmittance: 51 percent minimum.
14. SHGC: 0.61 maximum.
15. Safety glazing required.

C. Frosted Insulating Glass Type : GL-6.

1. Basis-of-Design Product: Guardian Glass LLC.
2. Overall Unit Thickness: 1 inch.
3. Minimum Thickness of Each Glass Lite: 1/4 inch.
4. Outdoor Lite: Annealed Fully tempered float glass.
5. 50% frosted etching on surface #2.

D. Fire-Rated Clear Glass Type: Fully tempered float glass GL-7.

1. Minimum Thickness: 1/4 inch.
2. Safety glazing required.
3. Fire-rating required: 45-minute.

END OF SECTION 08 80 00

SECTION 09 22 16 – NON-STRUCTURAL METAL FRAMING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Nonstructural steel framing.
2. Grid suspension systems.
3. For drywall ceilings and aluminum soffit if existing soffit framing is not able to be reused.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry” for wood supports and miscellaneous blocking.
2. Section 07 46 16 “Aluminum Cladding” for aluminum soffit materials and accessories.
3. Section 09 29 00 “Gypsum Board” for gypsum board ceiling material.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials in manufacturer's original, unopened, undamaged containers with identification labels intact.

B. Protect materials from corrosion, deformation, and other damage during delivery, storage, and handling in accordance with AISI S202.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Horizontal Deflection: For non-composite wall assemblies, limited to 1/240 of the wall height based on the following horizontal loading:

1. Horizontal Loading: 5 lbf/sq. ft.

2.2 AUXILIARY MATERIALS

A. General: Provide auxiliary materials that comply with referenced installation standards.

1. Fasteners for Steel Framing: Of type, material, size, corrosion resistance, holding power, and other properties required to fasten steel members to substrates.

B. Tie Wire: ASTM A641/A641M, Class 1 zinc coating, soft temper.

1. Suspended-Ceiling Systems:

a. Splicing Carrying Channels and Furring Members: Use double loops of minimum 0.0625-inch- diameter wire.
b. Saddle Tying Main Runners and Cross Furring: Use minimum 0.0625-inch-diameter wire, or double strand of 0.0475-inch- diameter wire.

C. Wire Hangers: ASTM A641/A641M, Class 1 zinc coating, soft temper, 0.162 inch in diameter.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, substrates, and conditions, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Suspended Assemblies: Coordinate installation of suspension systems with installation of overhead structure to ensure that inserts and other provisions for anchorages to building structure have been installed to receive hangers at spacing required to support the Work and that hangers will develop their full strength.

1. Furnish concrete inserts and other devices required to other trades for installation in advance of time needed for coordination and construction.

3.3 INSTALLATION OF NONSTRUCTURAL METAL FRAMING, GENERAL

A. Installation Standard: ASTM C754.

1. Gypsum Board Assemblies: Also comply with requirements in ASTM C840 that apply to framing installation.

B. Install framing and accessories plumb, square, and true to line, with connections securely fastened.

C. Install supplementary framing, and blocking to support fixtures, equipment services, heavy trim, grab bars, toilet accessories, furnishings, or similar construction.

D. Install bracing at terminations in assemblies.

END OF SECTION 09 22 16

SECTION 09 29 00 – GYPSUM BOARD

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Interior gypsum board.
2. Type X gypsum board
3. Trim accessories.
4. Acoustic Insulation in walls.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry” Building framing, sheathing, wood blocking, and execution requirements.
2. Section 07 21 00 “Thermal Insulation” for insulation not related to acoustics.
3. Section 07 21 19 “Foamed-in-place Insulation” for misc. voids or as noted in drawings.
4. Section 07 26 00 “Vapor Retarders” for vapor barrier installed under gypsum board at exterior walls.
5. Section 07 84 13 “Penetration Firestopping” for sealing penetration gaps in fire rated walls.
6. Section 07 84 43 “Joint Firestopping” for sealing transitions fire rated wall construction.
7. Section 07 92 00 “Joint Sealants” for sealing acoustical gaps in construction.
8. Section 09 22 16 "Non-Structural Metal Framing" for nonstructural steel framing and suspension systems that support gypsum board panels.
9. Section 09 91 23 “Interior Painting” for gypsum board finish.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Test Reports: For stud framing products that do not comply with ASTM C645 or ASTM C754, provide independent laboratory reports showing maximum stud heights at required spacings and deflections.

1.3 REFERENCE STANDARDS

A. ASTM A653/A653M – Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2019a.

B. ASTM C645 – Standard Specification for Nonstructural Steel Framing Members; 2018

C. ASTM C840 – Standard Specification for Application and Finishing of Gypsum Board; 2019b.

D. ASTM C1047 – Standard Specification for Accessories For Gypsum Wallboard and Gypsum Veneer Base; 2019.

E. ASTM C1396/C1396M – Standard Specification for Gypsum Board; 2017.

F. ASTM C1629/C1629M – Standard Classification for Abuse-Resistant Nondecorated Interior Gypsum Panel Products and Fiber-Reinforced Cement Panels; 2019.

1.4 DELIVERY, STORAGE, AND HANDLING

A. Store materials inside under cover and keep them dry and protected against weather, condensation, direct sunlight, construction traffic, and other potential causes of damage. Stack panels flat and supported on risers on a flat platform to prevent sagging.

1.5 FIELD CONDITIONS

A. Environmental Limitations: Comply with ASTM C840 requirements or manufacturer's written instructions, whichever are more stringent.

B. Do not install paper-faced gypsum panels until installation areas are enclosed and conditioned.

C. Do not install panels that are wet, moisture damaged, or mold damaged.

1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain each type of gypsum panel and joint finishing material from single source with resources to provide products of consistent quality in appearance and physical properties.

2.2 GYPSUM BOARD, GENERAL

A. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 90 percent.

B. Size: Provide panel products in maximum lengths and widths available that will minimize joints in each area and that correspond with support system specified or indicated on Drawings.

1. Manufacturers – Gypsum-Based Board:

a. American Gypsum Company; www.americangypsum.com
b. CertainTeed Corporation; www.certainteed.com.

c. Continental Building Products; www.continental-bp.com.
d. Georgia-Pacific Gypsum; www.gpgypsum.com.
e. National Gypsum Company; www.nationalgypsum.com.
f. Or Approved Equal.

C. Gypsum Wallboard: Paper-faced gypsum panels as defined in ASTM C1396/C1396M; sizes to minimize joints in place; ends tapered.

1. Thickness:

a. Vertical Surfaces: 5/8 inch.
b. Ceilings: 1/2 inch.

D. Cementitious Backer Board: moisture resistant panels; sizes to minimize joints in place; ends square cut.

1. Thickness: 5/8 inch.

2.3 INTERIOR GYPSUM BOARD

A. Gypsum Wallboard: ASTM C1396/C1396M.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. American Gypsum
b. CertainTeed; SAINT-GOBAIN
c. Georgia-Pacific Gypsum LLC
d. Gold Bond Building Products, LLC provided by National Gypsum Company
e. PABCO Gypsum
f. USG Corporation
g. Or Approved Equal.

2. Thickness: As indicated on Drawings.
3. Long Edges: Tapered.

B. Gypsum Board, Type X: ASTM C1396/C1396M.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. American Gypsum
b. CertainTeed; SAINT-GOBAIN
c. Georgia-Pacific Gypsum LLC
d. Gold Bond Building Products, LLC provided by National Gypsum Company
e. PABCO Gypsum
f. USG Corporation
g. Or Approved Equal.

2. Thickness: As indicated on Drawings.
3. Long Edges: Tapered.

C. The interior wall and ceiling finishes in rooms and corridors shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

2.4 TRIM ACCESSORIES

A. Interior Trim: ASTM C1047.

1. Material: Galvanized-steel sheet or aluminum-coated steel sheet or rolled zinc.
2. Shapes:

    a. Cornerbead.
    b. U-Bead: J-shaped; exposed short flange does not receive joint compound.
    c. Expansion (control) joint.

2.5 JOINT TREATMENT MATERIALS

A. General: Comply with ASTM C475/C475M requirements.

1. Mold-Resistant Joint Compound: Use mold-resistant formulations with mold-resistant panel products.

B. Joint Tape:

1. Interior Gypsum Board: Paper.

C. Joint Compound for Interior Gypsum Board: For each coat, use formulation that is compatible with other compounds applied on previous or for successive coats.

1. Prefilling: At open joints and damaged surface areas, use setting-type taping compound.
2. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use drying-type, all-purpose compound.

    a. Use setting-type compound for installing paper-faced metal trim accessories.

3. Fill Coat: For second coat, use drying-type, all-purpose compound.
4. Finish Coat: For third coat, use drying-type, all-purpose compound.
5. Skim Coat: For final coat of Level 4 finish, use drying-type, all-purpose compound.

2.6 AUXILIARY MATERIALS

A. Provide auxiliary materials that comply with referenced installation standards and manufacturer's written instructions.

B. Fiber-Reinforced-Plastic Panels

1. To be used adjacent to mop sinks to provide 2024 IBC compliant cleanable surfaces.

2. Basis of Design: Crane Composites Glasbord, Smooth Texture, White.
3. Salient Characteristics:

    a. Thickness: 0.075 inches
    b. Formaldehyde Free.
    c. Greenguard Certified for Low Chemical Emissions
    d. Mold and Mildew Free
    e. Meets USDA/FSIS Requirements
    f. Accessories: Manufacturer provided edge trim and adhesive.

C. Laminating Adhesive: Adhesive or joint compound recommended in writing by manufacturer for directly adhering gypsum panels to continuous substrate.

1. Verify adhesives have a VOC content of 50 g/L or less.

D. Steel Drill Screws: ASTM C1002 unless otherwise specified or indicated on Drawings.

E. Sound-Attenuation Blankets: ASTM C665, Type I (blankets without membrane facing) produced by combining thermosetting resins with mineral fibers as follows:

1. Non-Fire-Resistance-Rated Assemblies: slag or rock wool.

F. Thermal Insulation: As specified in Section 07 21 00 "Thermal Insulation."

G. Primer: As recommended in writing by textured finish manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas and substrates including welded hollow-metal frames and support framing, with Installer present, for compliance with requirements and other conditions affecting performance of the Work.

B. Examine panels before installation. Reject panels that are wet, moisture damaged, and mold damaged.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION AND FINISHING OF PANELS, GENERAL

A. Comply with ASTM C840 requirements.

B. Install ceiling panels across framing to minimize the number of abutting end joints and to avoid abutting end joints in central area of each ceiling. Stagger abutting end joints of adjacent panels not less than one framing member.

C. Install panels with face side out. Butt panels together for a light contact at edges and ends with not more than 1/16 inch of open space between panels. Do not force into place.

D. Locate edge and end joints over supports, except in ceiling applications where intermediate supports or gypsum board back-blocking is provided behind end joints. Do not place tapered edges against cut edges or ends. Stagger vertical joints on opposite sides of partitions. Do not make joints other than control joints at corners of framed openings.

E. Form control and expansion joints with space between edges of adjoining gypsum panels.

F. Wood Framing: Install gypsum panels over wood framing, with floating internal corner construction. Do not attach gypsum panels across the flat grain of wide-dimension lumber, including floor joists and headers. Float gypsum panels over these members or provide control joints to counteract wood shrinkage.

3.3 INSTALLATION OF INTERIOR GYPSUM BOARD

A. Install interior gypsum board in the following locations:

1. Gypsum Wallboard: As indicated on Drawings.
2. Gypsum Ceiling Board: As indicated on Drawings.

B. Single-Layer Application:

1. On ceilings, apply gypsum panels before wall/partition board application to greatest extent possible and at right angles to framing unless otherwise indicated on Drawings.
2. On partitions/walls, apply gypsum panels horizontally (perpendicular to framing) unless otherwise specified or indicated on Drawings or required by fire-resistance-rated assembly, and minimize end joints.

    a. Stagger abutting end joints not less than one framing member in alternate courses of panels.
    b. At stairwells and other high walls, install panels horizontally unless otherwise indicated on Drawings or required by fire-resistance-rated assembly.

3. On Z-shaped furring members, apply gypsum panels vertically (parallel to framing) with no end joints. Locate edge joints over furring members.
4. Fastening Methods: Apply gypsum panels to supports with steel drill screws.

C. Multilayer Application:

1. On ceilings, apply gypsum board indicated for base layers before applying base layers on walls/partitions; apply face layers in same sequence. Apply base layers at right angles to framing members and offset face-layer joints one framing member, 16 inches minimum, from parallel base-layer joints, unless otherwise indicated on Drawings or required by fire-resistance-rated assembly.
2. On partitions/walls, apply gypsum board indicated for base layers and face layers vertically (parallel to framing) with joints of base layers located over studs or furring members and face-layer joints offset at least one stud or furring member with base-layer joints unless otherwise indicated on Drawings or required by fire-resistance-rated assembly. Stagger joints on opposite sides of partitions.
3. On Z-shaped furring members, apply base layer vertically (parallel to framing) and face layer either vertically (parallel to framing) or horizontally (perpendicular to framing) with

vertical joints offset at least one furring member. Locate edge joints of base layer over furring members.

4. Fastening Methods: Fasten base layers with screws; fasten face layers with adhesive and supplementary fasteners.

D. Laminating to Substrate: Where gypsum panels are indicated as directly adhered to a substrate (other than studs, joists, furring members, or base layer of gypsum board), comply with gypsum board manufacturer's written instructions and temporarily brace or fasten gypsum panels until fastening adhesive has set.

E. Curved Surfaces:

1. Install panels horizontally (perpendicular to supports) and unbroken, to extent possible, across curved surface plus 12-inch- long straight sections at ends of curves and tangent to them.
2. For double-layer construction, fasten base layer to studs with screws 16 inches o.c. Center gypsum board face layer over joints in base layer, and fasten to studs with screws spaced 12 inches o.c.

3.4 APPLICATION OF JOINT TREATMENT MATERIALS

A. Finishing Panel Products: Treat joints, interior angles, edge trim, control joints, penetrations, fastener heads, surface defects, and elsewhere as required to prepare panel surfaces for decoration. Promptly remove residual joint compound from adjacent surfaces.

B. Prefill open joints and damaged surface areas.

C. Apply joint tape over panel joints, except for trim products specifically indicated as not intended to receive tape.

D. Interior Gypsum Board: Finish panels to levels indicated below and in accordance with ASTM C840:

1. Level 4: At panel surfaces that will be exposed to view unless otherwise indicated.

a. Primer and its application to surfaces are specified in Section 099123 "Interior Painting."

3.5 PROTECTION

A. Protect adjacent surfaces from joint compound and promptly remove from floors and other non-gypsum board surfaces. Repair surfaces stained, marred, or otherwise damaged during gypsum board installation and finishing.

B. Protect installed products from damage from weather, condensation, direct sunlight, construction, and other causes during remainder of the construction period.

C. Remove and replace panels that are wet, moisture damaged, and mold damaged.

1. Indications that panels are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.
2. Indications that panels are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

3.6 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 09 29 00

SECTION 09 51 23 – ACOUSTICAL TILE CEILINGS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Acoustical tiles.
2. Metal suspension system.

B. Related Requirements:

1. Section 09 29 00 “Gypsum Board” for acoustical batt insulation above ceiling at perimeter of offices.
2. Division 21 “Fire Suppression” for sprinklers installed in acoustical tile ceilings.
3. Division 23 “Mechanical” for HVAC fixtures installed in acoustical tile ceilings.
4. Division 26 “Electrical and Lighting” for light fixtures installed in acoustical tile ceilings.
5. Division 28 “Electrical Safety and Security” for security fixtures installed in acoustical tile ceilings.

1.2 REFERENCE STANDARDS

A. ASTM C423 - Standard Test Method for Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method; 2017.Insert your Paragraph.

B. ASTM C635/C635M - Standard Specification for the Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-in Panel Ceilings; 2017.

C. ASTM C636/C636M - Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels; 2013.

D. ASTM E1264 - Standard Classification for Acoustical Ceiling Products; 2014.

1.3 ACTION SUBMITTALS

A. Product Data:

1. For each type of product.

B. Samples: For each exposed product and for each color and texture specified, 4 inches in size.

C. Shop Drawings: Showing ceiling grid proposed layout, including diffusers, lights, etc.

1.4 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Acoustical Ceiling Units: Full-size tiles equal to percent of quantity installed.
2. Suspension-System Components: Quantity of each concealed grid and exposed component equal to 2 percent of quantity installed.

1.5 QUALITY ASSURANCE

A. Suspension System Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum three years documented experience.

B. Acoustical Unit Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum three years documented experience.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver acoustical tiles, suspension-system components, and accessories to Project site and store them in a fully enclosed, conditioned space where they will be protected against damage from moisture, humidity, temperature extremes, direct sunlight, surface contamination, and other causes.

B. Before installing acoustical tiles, permit them to reach room temperature and a stabilized moisture content.

1.7 FIELD CONDITIONS

A. Environmental Limitations: Do not install acoustical tile ceilings until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, work above ceilings is complete, and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.

PART 2 - PRODUCTS

2.1 ACOUSTICAL TILES

A. Basis of Design: Armstrong Cirrus Second Look,

B. Salient Characteristics:

1. Size: 24x48, scored for 2’x2’ and scored for 6”x4’.
2. Edge Profile: Square Tegular.
3. NRC: 0.85 minimum.
4. Recycled Content: 75% minimum.

5. Manufacturers:

    a. Armstrong World Industries, Inc: www.armstrong.com.
    b. CertainTeed Corporation: www.certainteed.com.
    c. Hunter Douglas Architectural: www.hunterdouglasarchitectural.com/#sle.
    d. USG; www.usg.com.
    e. Or Approved Equal.

6. Manufacturer's recommended suspension system.
7. The interior wall and ceiling finishes in rooms and corridors shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

C. Acoustical Units - General: ASTM E1264, Class A.

2.2 ACCESSORIES

A. Attachment Devices: Size for five times the design load indicated in ASTM C635/C635M, Table 1, "Direct Hung," unless otherwise indicated. Comply with seismic design requirements.

B. Support Channels and Hangers: Galvanized steel; size and type to suit application and ceiling system flatness requirement specified.

C. Perimeter Moldings: Same material and finish as grid.

1. At Exposed Grid: Provide L-shaped molding for mounting at same elevation as face of grid.

D. Gasket For Perimeter Moldings: Closed cell rubber sponge tape.

E. Touch-up Paint: Type and color to match acoustical and grid units

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, including structural framing and substrates to which acoustical tile ceilings attach or abut, with Installer present, for compliance with requirements specified in this and other Sections that affect ceiling installation and anchorage and for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine acoustical tiles before installation. Reject acoustical tiles that are wet, moisture damaged, or mold damaged.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Measure each ceiling area and establish layout of acoustical tiles to balance border widths at opposite edges of each ceiling. Avoid using less-than-half-width tiles at borders unless otherwise indicated, and comply with layout shown on reflected ceiling plans.

B. Layout openings for penetrations centered on the penetrating items.

3.3 INSTALLATION OF SUSPENDED ACOUSTICAL TILE CEILINGS

A. Install suspended acoustical tile ceilings in accordance with ASTM C636/C636M and manufacturer's written instructions.

B. Suspend ceiling hangers from building's structural members and as follows:

1. Install hangers plumb and free from contact with insulation or other objects within ceiling plenum that are not part of supporting structure or of ceiling suspension system.
2. Where width of ducts and other construction within ceiling plenum produces hanger spacings that interfere with location of hangers at spacings required to support standard suspension-system members, install supplemental suspension members and hangers in form of trapezes or equivalent devices.
3. Secure flat, angle, channel, and rod hangers to structure, including intermediate framing members, by attaching to inserts, eye screws, or other devices that are secure and appropriate for both the structure to which hangers are attached and the type of hanger involved. Install hangers in a manner that will not cause them to deteriorate or fail due to age, corrosion, or elevated temperatures.
4. Do not support ceilings directly from permanent metal forms or floor deck. Fasten hangers to cast-in-place hanger inserts, post-installed mechanical or adhesive anchors, or power-actuated fasteners that extend through forms into concrete.
5. When steel framing does not permit installation of hanger wires at spacing required, install carrying channels or other supplemental support for attachment of hanger wires.
6. Space hangers not more than 48 inches o.c. along each member supported directly from hangers unless otherwise indicated; provide hangers not more than 6 inches from ends of each member.
7. Size supplemental suspension members and hangers to support ceiling loads within performance limits established by referenced standards.

C. Install edge moldings and trim of type indicated at perimeter of acoustical ceiling area and where necessary to conceal edges of acoustical tiles.

1. Screw attach moldings to substrate at intervals not more than 16 inches o.c. and not more than 3 inches from ends. Miter corners accurately and connect securely.
2. Do not use exposed fasteners, including pop rivets, on molding and trim.

D. Install suspension-system runners so they are square and securely interlocked with one another. Remove and replace dented, bent, or kinked members.

E. Install acoustical tiles in coordination with suspension system and exposed moldings and trim. Place splines or suspension-system flanges into kerfed edges of tiles so tile-to-tile joints are interlocked.

1. Fit adjoining tiles to form flush, tight joints. Scribe and cut tiles for accurate fit at borders and around penetrations through ceiling.
2. Hold tile field in compression by inserting leaf-type, spring-steel spacers between tiles and moldings, spaced 12 inches o.c.

F. Lay out system to a balanced grid design with edge units no less than 50 percent of acoustical unit size.

3.4 ERECTION TOLERANCES

A. Suspended Ceilings: Install main and cross runners level to a tolerance of 1/8 inch in 10 feet, non-cumulative.

B. Moldings and Trim: Install moldings and trim to substrate and level with ceiling suspension system to a tolerance of 1/8 inch in 10 feet, non-cumulative.

C. Maximum Variation from Plumb of Grid Members Caused by Eccentric Loads: 2 degrees.

3.5 ADJUSTING

A. Clean exposed surfaces of acoustical tile ceilings, including trim and edge moldings. Comply with manufacturer's written instructions for cleaning and touchup of minor finish damage.

B. Remove and replace tiles and other ceiling components that cannot be successfully cleaned and repaired to permanently eliminate evidence of damage.

END OF SECTION 09 51 23

SECTION 09 65 19 – RESILIENT TILE RUBBER FLOORING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Rubber floor tiles

a. Textured/molded rubber tiles at main level lobby and work stations area.
b. Smooth rubber tiles at basement level.

2. Rubber base.
3. Rubber stair covering.
4. Accessories:

a. Aluminum transition strip for rubber to sealed concrete transition.
b. Anti slip safety edge at basement ramp landing

B. Related Requirements:

1. Section 06 20 23 “Interior Finish Carpentry” for wood base trim.
2. Section 06 41 00 “Architectural Wood Casework” for built in transaction desk.
3. Section 08 71 00 “Door Hardware” for exterior door thresholds.
4. Section 09 67 23 “Resinous Flooring” for adjacent flooring material.
5. Section 09 68 13 “Tile Carpeting” for adjacent flooring material.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review methods and procedures related to resilient tile and plank flooring installation including, but not limited to, the following:

a. Review delivery, storage, and handling procedures.
b. Review ambient conditions and ventilation procedures.
c. Review subfloor preparation procedures.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include manufacturers' written data on physical characteristics, durability, and fade resistance.
2. Include manufacturers' written installation instructions for each type of substrate.

B. Shop Drawings: For each type of resilient flooring type required.

1. Include flooring layouts, edges, columns, doorways, enclosing partitions, built-in

C. Samples: Full-size units of each color, texture, and pattern of resilient flooring type required.

D. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.
2. Environmental Product Declaration (EPD): For each product.

E. Qualification Data: For Installer.

1.4 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For resilient flooring, for tests performed by a qualified testing agency.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For each type of resilient flooring product type to include:

1. Methods for maintaining resilient flooring, including cleaning and stain-removal products and procedures and manufacturer's recommended maintenance schedule.
2. Precautions for cleaning materials and methods that could be detrimental to resilient flooring.

B. Manufacturer's Special Warranties

1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials , from the same production run, to Contracting Officer that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Resilient Tile and Plank Flooring: Furnish no fewer than 1 box for each 50 boxes or fraction thereof, of each type, color, pattern, and size of resilient flooring product installed.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are competent in techniques required by manufacturer for resilient flooring installation and seaming method indicated.

1. Engage an installer who employs workers for this Project who are trained or certified by resilient flooring manufacturer for installation techniques required.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturers' labels indicating brand name and directions for storing.

B. Store resilient flooring and accessory materials in dry spaces protected from the weather, with ambient temperatures maintained within range recommended by manufacturer, but not less than 50 deg F or more than 90 deg F. Store resilient flooring on flat surfaces.

1.9 FIELD CONDITIONS

A. Environmental Limitations: Do not deliver or install resilient flooring until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, and ambient temperature and humidity conditions are maintained at levels planned for building occupants during the remainder of the construction period.

B. Do not install resilient flooring over concrete slabs until slabs have cured and are sufficiently dry to bond with adhesive and concrete slabs have pH range recommended in writing by resilient flooring manufacturer.

C. Adhesively Applied Products:

1. Maintain temperatures during installation within range recommended in writing by manufacturer, but not less than 70 deg F or more than 95 deg F, in spaces to receive flooring 48 hours before installation, during installation, and 48 hours after installation unless longer period is recommended in writing by manufacturer.
2. After post installation period, maintain temperatures within range recommended in writing by manufacturer, but not less than 55 deg F or more than 95 deg F.
3. Close spaces to traffic during flooring installation.
4. Close spaces to traffic for 48 hours after flooring installation unless manufacturer recommends longer period in writing.

1.10 WARRANTY

A. Special Warranty for Resilient Flooring Products: Manufacturer agrees to repair or replace components of flooring installation that fail in materials or workmanship within specified warranty period.

1. Warranty does not include deterioration or failure of flooring due to unusual traffic, failure of substrate, vandalism, or abuse.
2. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Test-Response Characteristics: For resilient tile flooring, as determined by testing identical products in accordance with ASTM E648 or ASTM E662 by a qualified testing agency.

1. Critical Radiant Flux Classification: Class I, not less than 0.45 W/sq. cm.

B. Verify flooring products comply with requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers." Verify formaldehyde emissions do not exceed 16.5 mcg/cu. m or 13.5 ppb, whichever is less.

C. Verify products comply with requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers." Verify formaldehyde emissions do not exceed 16.5 mcg/cu. m or 13.5 ppb, whichever is less.

D. Verify products comply with requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers."

1. Verify adhesive has a VOC content of 50 g/L or less.

2.2 RESILIENT TILE RUBBER FLOORING

A. Rubber Floor Textured Tiles: Resilient flooring product consisting of modular rubber tiles with smooth square edges and a rough stone-like top surface.

1. Basis-of-Design Product: Subject to compliance with requirements, provide Interface Arago, Calm or comparable product by one of the following:

    a. Interface
    b. American Biltrite
    c. Mannington Mills, Inc
    d. Roppe Corporation; Roppe Holding Company
    e. Tarkett USA
    f. Or Approved Equal

2. Classification: ASTM F1344, Type I-A, Homogeneous Rubber Tile, solid color.
3. Hardness: Manufacturer's standard hardness, when measured using Shore, Type A durometer in accordance with ASTM D2240.
4. Wearing Surface: Molded pattern.

    a. Molded-Pattern Figure: Slate.

5. Thickness: 3mm.
6. Size: 39.5 by 19.8 inches.
7. Colors and Patterns: Basis of Design TBD by park, to be confirmed by Contracting Officer from manufacturer's full range.
8. Recycled Content: Provide manufacturer documentation for recycled content, indicating postconsumer and preconsumer recycled content.
9. The interior wall and ceiling finishes in rooms and corridors shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

B. Rubber Floor Tiles – Smooth: Resilient flooring product consisting of modular vulcanized rubber tiles with smooth square edges and a smooth top surface.

1. Basis-of-Design Product – Subject to Compliance with requirements, provide Interface Sentica, Deep Sea or comparable product by one of the following

    a. Interface
    b. American Biltrite
    c. Mannington Mills, Inc.
    d. Roppe Corporation; Roppe Holding Company
    e. Tarkett USA
    f. Or Approved Equal

2. Classification: ASTM F1344, Type I-B, Homogeneous Rubber Tile.
3. Hardness – Manufacturer's standard hardness, when measured using Shore, Type A durometer in accordance with ASTM D2240.
4. Wearing Surface – Smooth.
5. Thickness – 3 mm
6. Size – 24 by 24 inch
7. Recycled Content: Provide manufacturer documentation for recycled content, indicating postconsumer and preconsumer recycled content.
8. The interior wall and ceiling finishes in rooms and corridors shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

C. Rubber Base – Smooth rubber product with 5/8” toe base with smooth square edges and a smooth top surface.

1. Basis-of-Design Product – Subject to Compliance with requirements, provide Pinnacle Rubber Base by Interface Roppe or comparable product by one of the following:

    a. Interface
    b. Armstrong
    c. Burke by Mannington Commercial
    d. Roppe Corporation; Roppe Holding Company
    e. Tarkett USA
    f. Or Approved Equal

2. Height: 4”
3. Toe: Standard 5/8” toe
4. Accessories: Inside and outside corners
5. Made in the USA
6. Meets Greenguard Gold

D. Rubber Stair Covering

1. By same manufacturer as rubber tile flooring.
2. Basis of design: Subject to compliance with requirements, provide Interface Arago, Calm or comparable product by one of the following:

    a. Interface
    b. American Biltrite

c. Mannington Mills, Inc
d. Roppe Corporation; Roppe Holding Company
e. Tarkett USA
f. Or Approved Equal

3. Salient Characteristics: Shall match either the molded or smooth rubber tile product.
4. Shall cover treads, risers, and nosing.
5. If product is different depth from adjacent rubber floorings, provide aluminum transition strip for sloped transition. Shall be ABAAS Compliant.

2.3 ACCESSORIES

A. Transition strip – Sloped extruded aluminum transition strip at transition of rubber flooring to sealed concrete.

1. Basis of Design Product – Schluter Reno-U
2. Salient Characteristics

a. Material: Extruded aluminum
b. Finish: Mill or Satin Clear Anodized
c. Height: 1/8” (3.5 mm) to 0
d. ABAAS Compliant

B. Anti Slip Safety Edge

1. Basis of Design Product: Subject to compliance with salient characteristics, provide DK Hardware Safestep Anti-Slip Step Edges or comparable product.
2. Salient Characteristics

a. Heavy duty preformed step nosing
b. Anti-slip texture
c. High visibility yellow color at front and top of nosing

2.4 INSTALLATION MATERIALS

A. Trowelable Leveling and Patching Compounds: Latex-modified, Portland-cement-based or blended hydraulic-cement-based formulation provided or approved by flooring manufacturer for applications indicated.

B. Adhesives: Water-resistant type recommended in writing by floor tile and adhesive manufacturers for substrate and conditions indicated.

1. Verify adhesive complies with testing and product requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers." Verify formaldehyde emissions do not exceed 9 mcg/cu. m or 7 ppb, whichever is less.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, moisture content, and other conditions affecting performance of the Work.

1. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of resilient products.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

1. Installation of resilient flooring products indicates acceptance of surfaces and conditions.

3.2 PREPARATION

A. Prepare substrates in accordance with manufacturer's written installation instructions to ensure adhesion of flooring.

B. Concrete Substrates: Prepare in accordance with ASTM F710.

1. Verify that substrates are dry and free of curing compounds, sealers, and hardeners.
2. Remove substrate coatings and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, using mechanical methods recommended by resilient flooring manufacturer. Do not use solvents.
3. Alkalinity Testing: Perform pH testing in accordance with ASTM F710. Proceed with installation only if pH readings are not less than 7.0 and not greater than 8.5.
4. Moisture Testing: Perform tests so that each test area does not exceed 1000 sq. ft. and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.

a. Relative Humidity Test: Using in-situ probes, ASTM F2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.

C. Fill cracks, holes, and depressions in substrates with trowelable leveling and patching compound; remove bumps and ridges to produce a uniform and smooth substrate.

D. Move flooring and installation materials into spaces where they will be installed at least 48 hours in advance of installation unless manufacturer recommends a longer period in writing.

1. Do not install flooring until it is the same temperature as space where it is to be installed.

E. Sweep and vacuum clean substrates to be covered by flooring immediately before installation. After cleaning, examine substrates for moisture, alkaline salts, carbonation, and dust. Proceed with installation only after unsatisfactory conditions have been corrected.

3.3 INSTALLATION OF RESILIENT TILE FLOORING

A. Comply with manufacturer's written installation instructions.

B. Discard broken, cracked, chipped, or deformed tiles.

C. Lay out floor tiles from center marks established with principal walls, discounting minor offsets, so tiles at opposite edges of room are of equal width. Adjust as necessary to avoid using cut widths that equal less than one-half tile at perimeter.

1. Lay tiles square with room axis.

D. Tile Matching: Match floor tiles for color and pattern by selecting tiles from cartons in the same sequence as manufactured and packaged, if so numbered.

E. Scribe, cut, and fit floor tiles to butt neatly and tightly to vertical surfaces and permanent fixtures including built-in furniture, cabinets, pipes, outlets, and door frames.

F. Extend floor tiles into toe spaces, door reveals, closets, and similar openings. Extend floor tiles to center of door openings.

G. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on floor tiles as marked on substrates. Use chalk or other nonpermanent marking device.

H. Adhered Floor Tile: Adhere floor tiles to substrates using a full spread of adhesive applied to substrate to comply with adhesive and flooring manufacturers' written installation instructions, including those for trowel notching, adhesive mixing, and adhesive open and working times, to produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections.

3.4 CLEANING AND PROTECTION

A. Comply with manufacturer's written instructions for cleaning and protecting flooring.

B. Perform the following operations immediately after completing flooring installation:

1. Remove adhesive and other blemishes from surfaces.
2. Sweep and vacuum surfaces thoroughly.
3. Damp-mop surfaces to remove marks and soil.

C. Protect flooring from mars, marks, indentations, and other damage from construction operations and placement of equipment and fixtures during remainder of construction period.

D. Cover resilient flooring until Substantial Completion.

3.5 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 09 65 19

SECTION 09 67 23 – RESINOUS EPOXY FLOORING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Resinous flooring.
2. Integral cove base accessories.

B. Related Sections:

1. Section 09 29 00 "Gypsum Board Assemblies" for wall substrate.
2. Division 22 for plumbing fixtures and fittings.
3. Division 7 for thermal and moisture protection.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review manufacturer's written instructions for substrate preparation and environmental conditions affecting resinous flooring installation.
2. Review details of integral cove bases.
3. Review manufacturer's written instructions for installing resinous flooring systems.
4. Review protection measures for adjacent construction and installed flooring, floor drainage requirements, curbs, base details, and so forth.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include manufacturer's technical data, installation instructions, and recommendations for each resinous flooring component required.

B. Sustainable Design Submittals:

1. Environmental Product Declaration: For each product.
2. Laboratory Test Reports: For flooring products, indicating compliance with requirements for low-emitting materials.
3. Product Data: For adhesives, indicating that product contains no urea formaldehyde.

C. Samples: 6 inches square in size, applied to a rigid backing by Installer for this Project.

D. Qualification Data: For Installer.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance Data: For resinous flooring to include in maintenance manuals.

B. Manufacturer's Special Warranties

1.5 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer.

1. Engage an installer who is certified in writing by resinous flooring manufacturer as qualified to apply resinous flooring systems indicated.
2. Furnish proof that the Contractor and resinous flooring installer foreman have at least 10 years of experience in installation on projects of similar size and character along with three references from persons who were responsible for supervision of these projects. Include name, address, and telephone number of references who will testify to a successful completion of these projects by the Contractor and resinous flooring installer foreman.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Apply full-thickness mockups on 96-inch- square floor area selected by Architect.

    a. Include 96-inch length of integral cove base with inside corner.

2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.
3. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Deliver materials in original packages and containers, with seals unbroken, bearing manufacturer's labels indicating brand name and directions for storage and mixing with other components.

1.7 FIELD CONDITIONS

A. Environmental Limitations: Comply with resinous flooring manufacturer's written instructions for substrate temperature, ambient temperature, moisture, ventilation, and other conditions affecting resinous flooring installation.

B. Lighting: Provide permanent lighting or, if permanent lighting is not in place, simulate permanent lighting conditions during resinous flooring installation.

C. Close spaces to traffic during resinous flooring installation and for 24 hours after installation unless manufacturer recommends a longer period.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Verify flooring products comply with the requirements of Section 01 57 19.11 "Indoor Air Quality Management.

2.2 RESINOUS EPOXY FLOORING

A. Resinous Flooring System: Abrasion-, impact-, and chemical-resistant, aggregate-filled, resin-based monolithic floor surfacing designed to produce a seamless floor and integral cove base.

1. Basis of Design: Everlast
2. Manufacturers:

    a. Duraflex, Inc.
    b. Crown Polymers Corp.
    c. Sika Corporation; Flooring.
    d. Sherwin-Williams High Performance Flooring.
    e. Or Approved EquaL

3. The interior wall and ceiling finishes in rooms and corridors shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

B. System Characteristics:

1. Color and Pattern: Light Chocolate
2. Wearing Surface: Manufacturer's standard wearing surface.
3. Overall System Thickness: 1/8 inch.

C. Patching and Fill Material: Resinous product of or approved by resinous flooring manufacturer and recommended in writing by manufacturer for installation indicated.

D. Primer: Type recommended in writing by resinous flooring manufacturer for substrate and resinous flooring system indicated.

E. System Physical Properties: Provide resinous flooring system with the following minimum physical property requirements when tested in accordance with test methods indicated:

1. Compressive Strength: 10,500 psi minimum in accordance with ASTM C579.
2. Tensile Strength: 3,700 minimum in accordance with ASTM C307.
3. Abrasion Resistance: 4 mg maximum weight loss in accordance with ASTM D4060.
4. Hardness: 80 minimum, Shore D in accordance with ASTM D2240.

2.3 INTEGRAL COVE BASE AND INSTALLATION ACCESSORIES

A. Precast, Integral Cove Base: Impact-resistant, polymer-resin, cove base moldings with a grit profile to promote adhesion of resinous flooring and recommended in writing by resinous flooring manufacturer.

1. Radius Cove: Cove molding with approximately 1-inch radius for adhesive installation at floor-to-wall joint as substrate to receive resinous flooring system to form an integral cove base.
2. Same manufacturer as flooring.
3. Radius Cove Base: 6-inch- high base molding that provides approximately 1-inch radius cove at floor-to-wall joint; for adhesive installation as substrate for resinous flooring system to form an integral cove base.

B. Installation Adhesive: As recommended in writing by accessory manufacturer.

1. Adhesives: Do not use adhesives that contain urea formaldehyde.

C. Transition strip – Sloped extruded aluminum transition strip at transition of epoxy flooring to rubber flooring.

1. Basis of Design Product – Schluter Reno-U
2. Salient Characteristics

a. Material: Extruded aluminum
b. Finish: Mill or Satin Clear Anodized
c. Height: Epoxy Thickness to Rubber Flooring Height
d. ABAAS Compliant

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, with Installer present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

1. Verify that finishes of substrates comply with tolerances and other requirements specified in other Sections and that substrates are free of cracks, ridges, depressions, scale, and foreign deposits that might interfere with adhesion of resinous flooring systems.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Prepare and clean substrates in accordance with resinous flooring manufacturer's written instructions for substrate indicated to ensure adhesion.

B. Concrete Substrates: Provide sound concrete surfaces free of laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants incompatible with resinous flooring.

1. Roughen concrete substrates as follows:

a. Comply with requirements in SSPC-SP 13/NACE No. 6, with a Concrete Surface Profile of 3 or greater in accordance with ICRI Technical Guideline No. 310.2R, unless manufacturer's written instructions are more stringent.

2. Repair damaged and deteriorated concrete in accordance with resinous flooring manufacturer's written instructions.

C. Patching and Filling: Use patching and fill material to fill holes and depressions in substrates in accordance with manufacturer's written instructions.

1. Control Joint Treatment: Treat control joints and other nonmoving substrate cracks to prevent cracks from reflecting through resinous flooring in accordance with manufacturer's written instructions.

D. Resinous Materials: Mix components and prepare materials in accordance with resinous flooring manufacturer's written instructions.

3.3 INSTALLATION

A. Apply components of resinous flooring system in accordance with manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness specified.

1. Coordinate installation of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.
2. Cure resinous flooring components in accordance with manufacturer's written instructions. Prevent contamination during installation and curing processes.
3. Expansion and Isolation Joint Treatment: At substrate expansion and isolation joints, comply with resinous flooring manufacturer's written instructions.

B. Primer: Apply primer over prepared substrate at spreading rate recommended in writing by manufacturer.

C. Waterproofing Membrane: Apply waterproofing membrane over entire substrate surface, in thickness recommended in writing by manufacturer.

1. Apply waterproofing membrane to integral cove base substrates.

D. Integral Cove Base Accessories: Adhesively install precast accessories before applying flooring coats and in accordance with manufacturer's written instructions.

E. Field-Formed Integral Cove Base: Apply cove base mix to wall surfaces before applying flooring coats. Apply in accordance with manufacturer's written instructions and details, including those for taping, mixing, priming, troweling, sanding, and top coating of cove base. Round internal and external corners.

1. Integral Cove Base: 6 inches high.

F. Self-Leveling Body Coats: Apply self-leveling slurry body coats in thickness specified for flooring system.

1. Aggregates: Broadcast aggregates at rate recommended in writing by manufacturer. After resin is cured, remove excess aggregates to provide surface texture indicated.

G. Troweled or Screeded Body Coats: Apply troweled or screeded body coats in thickness specified for flooring system. Hand or power trowel and grout to fill voids. When body coats are cured, remove trowel marks and roughness using method recommended in writing by manufacturer.

H. Grout Coat: Apply grout coat to fill voids in surface of final body coat.

I. Topcoats: Apply topcoats in number indicated for flooring system specified, at spreading rates recommended in writing by manufacturer, and to produce wearing surface specified.

3.4 FIELD QUALITY CONTROL

A. Material Sampling: CO may, at any time and any number of times during resinous flooring installation, require material samples for testing for compliance with requirements.

1. If test results show applied materials do not comply with specified requirements, pay for testing, remove noncomplying materials, prepare surfaces coated with unacceptable materials, and reinstall flooring materials to comply with requirements.

3.5 PROTECTION

A. Protect resinous flooring from damage and wear during the remainder of construction period. Use protective methods and materials, including temporary covering, recommended in writing by resinous flooring manufacturer.

3.6 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 09 67 23

SECTION 09 68 13 – TILE CARPETING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Tile carpeting for auditorium/multi-purpose room, exhibit area, offices, and associated spaces. See architectural plans.
2. Walk off carpet in vestibule.

B. Related Requirements:

1. Section 06 20 23 “Interior Finish Carpentry” for wood base trim.
2. Section 08 71 00 “Door Hardware” for exterior door thresholds.
3. Section 09 65 19 “Resilient Tile Rubber Flooring” for adjacent flooring material and rubber base.

1.2 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review methods and procedures related to carpet tile installation including, but not limited to, the following:

a. Review delivery, storage, and handling procedures.
b. Review ambient conditions and ventilation procedures.
c. Review subfloor preparation procedures.
d. Review layout pattern and locations.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include manufacturer's written data on physical characteristics, durability, and fade resistance.
2. Include manufacturer's written installation recommendations for each type of substrate.

B. Shop Drawings: For carpet tile installation, showing the following:

1. Columns, doorways, enclosing walls or partitions, built-in cabinets, and locations where cutouts are required in carpet tiles.
2. Carpet tile type, color, and dye lot.
3. Type of installation.
4. Pattern of installation.
5. Pattern type, location, and direction.

6. Pile direction.
7. Type, color, and location of insets and borders.
8. Type, color, and location of edge, transition, and other accessory strips.
9. Transition details to other flooring materials.

C. Samples for Verification: Actual sample of finished products for each of the following products and for each color and texture required. Label each Sample with manufacturer's name, material description, color, pattern, and designation indicated on Drawings and in schedules.

1. Carpet Tile: Full-size Sample.
2. Exposed Edge, Transition, and Other Accessory Stripping: 12-inch- long Samples.

D. Product Schedule: For carpet tile. Use same designations indicated on Drawings.

E. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.

1.4 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For carpet tile, for tests performed by a qualified testing agency.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For carpet tiles. Include the following:

1. Methods for maintaining carpet tile, including cleaning and stain-removal products and procedures and manufacturer's recommended maintenance schedule.
2. Precautions for cleaning materials and methods that could be detrimental to carpet tile.

1.6 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials, from the same production run, to CO that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Carpet Tile: Full-size units equal to 5 percent of amount installed for each type indicated, but no fewer than 10 full-size units.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is certified by the International Certified Floorcovering Installers Association at the Commercial II certification level.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Comply with CRI 104.

1.9 FIELD CONDITIONS

A. Comply with CRI 104 for temperature, humidity, and ventilation limitations.

B. Environmental Limitations: Do not deliver or install carpet tiles until spaces are enclosed and weathertight, wet-work in spaces is complete and dry, and ambient temperature and humidity conditions are maintained at levels planned for building occupants during the remainder of the construction period.

C. Do not install carpet tiles over concrete slabs until slabs have cured and are sufficiently dry to bond with adhesive and concrete slabs have pH range recommended in writing by carpet tile manufacturer.

D. Where demountable partitions or other items are indicated for installation on top of carpet tiles, install carpet tiles before installing these items.

1.10 WARRANTY

A. Special Warranty for Carpet Tiles: Manufacturer agrees to repair or replace components of carpet tile installation that fail in materials or workmanship within specified warranty period.

1. Warranty does not include deterioration or failure of carpet tile due to unusual traffic, failure of substrate, vandalism, or abuse.
2. Failures include, but are not limited to, the following:

a. More than 10 percent loss of face fiber, edge raveling, snags, and runs.
b. Loss of tuft-bind strength.
c. Excess static discharge.
d. Delamination.
e. Dimensional instability.

3. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Interface Flooring; www.interface.com.

B. Forbo Flooring; www.forbo.com.

C. Shaw Contract; www.shawcontract.com.

D. Mannington Commercial; www.manningtoncommercial.com.

E. Or Approved Equal.

2.2 PRIMARY TILE CARPET (CPT 1)

A. Basis of Design: Interface Flooring; Light Play; Color: Sahara

B. Salient Characteristics:

1. Construction: Tufted Textured Loop
2. Material: 100% Recycled Content Nylon
3. Dye Method: 100% Solution Dyed
4. Tufted Yarn Weight: 18 oz/sq yd.
5. Installation Method: Non Directional
6. Pile Height: 0.105 in (2.7 mm)
7. Pile Thickness: 0.098 in (2.5 mm)
8. Stitches: 8.3/in
9. Pile Density: 6,612 oz/sq yd.
10. Smoke Density: ASTM E662 maximum 450
11. Flammability: Passes Methenamine Pill Test (DOC-FF1-70)
12. Lightfastness: AATCC 16-E, minimum 4.0 @ 60 AFU's
13. Dimensional Stability: Passes ASTM F2199
14. Traffic Classification: Severe
15. Dimensional Tolerance: Within 1/32 inch of specified size dimensions, as determined by physical measurement.
16. Electrostatic Propensity Less than 3 kV according to AATCC 134.
17. The interior wall and ceiling finishes in rooms and hallways shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

2.3 WALK OFF CARPET (CPT 2)

A. Basis of Design: Interface; Step Repeat Collection; SR999; Color: Midnight

B. Salient Characteristics:

1. Construction: Tufted Textured Loop
2. Material: 100% Recycled Content Nylon
3. Dye Method: 100% Solution Dyed
4. Tufted Yarn Weight: 27 oz/sq yd.
5. Installation Method: Non Directional
6. Pile Height: 0.17 in (4.3 mm)
7. Pile Thickness: 0.127 in (3.2 mm)
8. Stitches: 10/in
9. Pile Density: 7,654 oz/sq yd.
10. Smoke Density: ASTM E662 maximum 450
11. Flammability: Passes Methenamine Pill Test (DOC-FF1-70)
12. Lightfastness: AATCC 16-E, minimum 4.0 @ 60 AFU's
13. Dimensional Stability: AACHEN Din 54318 < 0.10%
14. Traffic Classification: Severe
15. Dimensional Tolerance: Within 1/32 inch of specified size dimensions, as determined by physical measurement.

16. Electrostatic Propensity Less than 3 kV according to AATCC 134.
17. The interior wall and ceiling finishes in rooms and hallways shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

2.4 INSTALLATION ACCESSORIES

A. Trowelable Leveling and Patching Compounds: Latex-modified, hydraulic-cement-based formulation provided or recommended in writing by carpet tile manufacturer.

B. Adhesives: Water-resistant, mildew-resistant, nonstaining, pressure-sensitive types to suit products and subfloor conditions indicated, that comply with flammability requirements for installed carpet tile, and that are recommended in writing by carpet tile manufacturer for releasable installation.

C. Metal Edge/Transition Strips: Extruded aluminum with mill finish of profile and width shown, of height required to protect exposed edge of carpet, and of maximum lengths to minimize running joints.

1. Basis of Design Product – Schluter Schiene
2. Salient Characteristics

    a. Material: Extruded aluminum
    b. Finish: Mill or Satin Clear Anodized
    c. Height: 1/8” (3.5 mm)
    d. ABAAS Compliant

D. Carpet Underlayment: 1/8" Fiber-type pad compatible with carpet per manufacturer.

1. Basis of Design: Interface CushionBacRE
2. Salient Characteristics:

    a. Provides increased sound absorption
    b. 100% Recycled Polyester
    c. Thickness 1/8” (3.2 mm)
    d. Weight: 700 g/sq m.
    e. Anti-Static: AATCC134 <3000V
    f. Warranty: 15 year limited warranty

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for maximum moisture content, alkalinity range, installation tolerances, and other conditions affecting carpet tile performance.

B. Examine carpet tile for type, color, pattern, and potential defects.

C. Concrete Slabs: Verify that finishes comply with requirements specified in Section 033000 "Cast-in-Place Concrete" and that surfaces are free of cracks, ridges, depressions, scale, and foreign deposits.

1. Moisture Testing: Perform tests so that each test area does not exceed 200 sq. ft., and perform no fewer than three tests in each installation area and with test areas evenly spaced in installation areas.

a. Anhydrous Calcium Chloride Test: ASTM F1869. Proceed with installation only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 sq. ft. in 24 hours.
b. Relative Humidity Test: Using in situ probes, ASTM F2170. Proceed with installation only after substrates have a maximum 75 percent relative humidity level measurement.
c. Perform additional moisture tests recommended in writing by adhesive and carpet tile manufacturers. Proceed with installation only after substrates pass testing.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. General: Comply with CRI 104 and with carpet tile manufacturer's written installation instructions for preparing substrates.

B. Use trowelable leveling and patching compounds, in accordance with manufacturer's written instructions, to fill cracks, holes, depressions, and protrusions in substrates. Fill or level cracks, holes and depressions 1/8 inch wide or wider, and protrusions more than 1/32 inch unless more stringent requirements are required by manufacturer's written instructions.

C. Concrete Substrates: Remove coatings, including curing compounds, and other substances that are incompatible with adhesives and that contain soap, wax, oil, or silicone, without using solvents. Use mechanical methods recommended in writing by adhesive and carpet tile manufacturers.

D. Metal Substrates: Clean grease, oil, soil and rust, and prime if recommended in writing by adhesive manufacturer. Rough sand painted metal surfaces and remove loose paint. Sand aluminum surfaces, to remove metal oxides, immediately before applying adhesive.

E. Broom and vacuum clean substrates to be covered immediately before installing carpet tile.

3.3 INSTALLATION

A. General: Comply with CRI 104, Section 10, "Carpet Tile," and with carpet tile manufacturer's written installation instructions.

B. Installation Method: As recommended in writing by carpet tile manufacturer.

C. Maintain dye-lot integrity. Do not mix dye lots in same area.

D. Maintain pile-direction patterns recommended in writing by carpet tile manufacturer.

E. Cut and fit carpet tile to butt tightly to vertical surfaces, permanent fixtures, and built-in furniture including cabinets, pipes, outlets, edgings, thresholds, and nosings. Bind or seal cut edges as recommended in writing by carpet tile manufacturer.

F. Extend carpet tile into toe spaces, door reveals, closets, open-bottomed obstructions, removable flanges, alcoves, and similar openings.

G. Maintain reference markers, holes, and openings that are in place or marked for future cutting by repeating on carpet tile as marked on subfloor. Use nonpermanent, nonstaining marking device.

H. Install pattern parallel to walls and borders.

3.4 CLEANING AND PROTECTION

A. Perform the following operations immediately after installing carpet tile:

1. Remove excess adhesive and other surface blemishes using cleaner recommended in writing by carpet tile manufacturer.
2. Remove yarns that protrude from carpet tile surface.
3. Vacuum carpet tile using commercial machine with face-beater element.

B. Protect carpet tile against damage from construction operations and placement of equipment and fixtures during the remainder of construction period. Use protection methods indicated or recommended in writing by carpet tile manufacturer.

3.5 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 09 68 13

SECTION 09 91 13 – EXTERIOR PAINTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Surface preparation of exterior substrates and application of the following:

1. Primers.
2. Water-based finish coatings.
3. Solvent-based finish coatings.

B. Related Requirements:

1. Section 05 12 00 "Structural Steel Framing" for shop priming structural steel substrates.
2. Section 05 50 00 "Metal Fabrications" for shop priming metal fabrications.
3. Section 09 91 23 "Interior Painting."
4. Section 09 93 00 "Staining and Transparent Finishing" for surface preparation and application of wood stains and transparent finishes on exterior wood substrates.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include preparation requirements and application instructions.
2. Indicate VOC content.

B. Samples: For each type of topcoat product.

C. Samples for Initial Selection: For each type of topcoat product.

D. Samples for Verification: For each type of paint system and each color and gloss of topcoat. Submit actual paint drawdowns as specified below for verification Samples.

1. Submit Samples on rigid backing, 8 inches square.
2. Apply coats on Samples in steps to show each coat required for system.
3. Label each coat of each Sample.
4. Label each Sample for location and application area.

E. Product List: Cross-reference to paint system and locations of application areas. Use same designations indicated on Drawings and in applicable exterior painting schedule articles to cross-reference paint systems specified in this Section. Include color designations.

1.3 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match paint products applied and that are packaged with protective covering for storage and identified with labels describing contents.

1. Paint Products: 5 percent, but not less than 1 gal. of each material and color applied.

1.4 MOCKUPS

A. Apply mockups of each paint system indicated and each color and finish selected to verify selections made under Sample submittals, demonstrate aesthetic effects, and set quality standards for materials and execution.

1. Contracting Officer will select one surface to represent surfaces and conditions for application of each paint system.

a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft.
b. Other Items: Contracting Officer will designate items or areas required.

2. Final approval of color selections will be based on mockups.

a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Contracting Officer at no added cost to Government.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F.

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.6 FIELD CONDITIONS

A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F.

B. Do not apply paints in snow, rain, fog, or mist; when relative humidity exceeds 85 percent; at temperatures less than 5 deg F above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Obtain each paint product from single source from single manufacturer.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Behr Paint Company; Behr Process Corporation.
2. Benjamin Moore & Co.
3. Carboline Company; a subsidiary of RPM International.
4. Conco Paints.
5. Coronado Paint; Benjamin Moore & Co.
6. Diamond Vogel Paint Company.
7. H&C Decorative Concrete Products; a brand of Sherwin-Williams Co.
8. Kelly-Moore Paints.
9. McCormick Paints.
10. PPG Paints; PPG Industries, Inc.
11. Or Approved Equal.

2.2 EXTERIOR PAINTS, GENERAL

A. Exterior Paints: Subject to compliance with requirements, provide one of the products listed in product types below and applicable exterior painting schedule articles for the paint category indicated.

B. Material Compatibility:

1. Materials for use within each paint system must be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer based on testing and field experience.
2. For each coat in a paint system, provide products recommended in writing by topcoat manufacturer for use in paint system and on substrate indicated.

C. Colors: As listed in section 3.6 of this specification, color to confirmed by the Contracting Officer from manufacturer's full range .

1. 100 percent of surface area will be painted with deep tones.

2.3 PRIMERS

A. Exterior Wood Preservative: Solvent-based, zinc or copper napthenate, penetrating antifungal treatment for exterior wood.

B. Zinc-Rich, Inorganic Primer: Corrosion-resistant, inorganic-based, zinc-rich primer formulated for use on prepared ferrous metal substrates subject to severe industrial or marine environments.

C. Alkyd Metal Primer: Corrosion-resistant, solvent-based, alkyd primer formulated for use on prepared ferrous metal substrates subject to industrial and light marine environments.

D. Water-Based, Galvanized-Metal Primer: Corrosion-resistant, pigmented, acrylic primer; formulated for use on cleaned/etched galvanized metal substrates to prepare it for subsequent water-based coatings.

E. Quick-Drying Aluminum Primer: Corrosion-resistant, solvent-based, alkyd or modified-alkyd primer formulated for quick-drying capabilities and for use on prepared aluminum substrates.

2.4 WATER-BASED FINISH COATINGS

A. Exterior Latex Paint: Water-based, pigmented coating; formulated for alkali, mold, microbial, and water resistance and for use on exterior substrates.

1. Gloss and Sheen: Manufacturer's standard gloss.

B. Exterior, Water-Based, Light Industrial Coating: Corrosion-resistant, water-based, pigmented, emulsion coating formulated for resistance to water, alkalis, moderate abrasion, and mild chemical exposure and for use on exterior substrates.

1. Gloss and Sheen Level: Manufacturer's standard gloss.

2.5 SOLVENT-BASED FINISH COATINGS

A. Exterior Alkyd Paint: Solvent-based, pigmented, alkyd coating formulated for mold, microbial, and water resistance and for use on exterior substrates.

1. Gloss and Sheen Level: Manufacturer's standard gloss.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

B. Verify suitability of substrates, including surface conditions and compatibility, with finishes and primers. Proceed with coating application only after unsatisfactory conditions have been corrected.

1. Application of coating indicates acceptance of substrates and conditions.

3.2 PREPARATION

A. Comply with manufacturer's written instructions applicable to substrates and paint systems indicated.

B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection.

C. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and encapsulants.

1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce paint systems specified in this Section.

D. Steel Substrates: Remove loose rust, loose mill scale, loose shop primer, and other loose foreign matter. Clean using methods recommended in writing by paint manufacturer.

E. Shop-Primed Steel Substrates: Clean field welds, bolted connections, and areas where shop paint is abraded. Paint exposed areas with the same material as used for shop priming to comply with SSPC-PA 1 for touching up shop-primed substrates.

F. Galvanized-Metal Substrates: Remove grease and oil residue from galvanized metal by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied paints.

3.3 APPLICATION OF EXTERIOR PAINT PRODUCTS

A. Apply paints in accordance with manufacturer's written instructions.

1. Use applicators and techniques suited for paint and substrate indicated.
2. Paint surfaces behind movable items same as similar exposed surfaces. Before final installation, paint surfaces behind permanently fixed items with prime coat only.
3. Paint both sides and edges of exterior doors and entire exposed surface of exterior door frames.
4. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.
5. Primers specified in applicable exterior painting schedule articles may be omitted on items that are factory primed or factory finished if compatible with intermediate and topcoat coatings and acceptable to intermediate and topcoat paint manufacturers.

B. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, color, and appearance.

C. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

D. Painting, Fire-Suppression, Plumbing, HVAC, Electrical, Communication, and Electronic Safety and Security Work:

1. Paint the following work where exposed to view:

a. Equipment, including panelboards and switch gear.
b. Pipe hangers and supports.
c. Metal conduit.

3.4 CLEANING AND PROTECTION

A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

1. Do not clean equipment with free-draining water and prevent solvents, thinners, cleaners, and other contaminants from entering into waterways, sanitary and storm drain systems, and ground.
2. Dispose of contaminants in accordance with requirements of authorities having jurisdiction.
3. Allow empty paint cans to dry before disposal.
4. Collect waste paint by type and deliver to recycling or collection facility.

B. After completing paint application, clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

C. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Contracting Officer, and leave in an undamaged condition.

D. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.5 EXTERIOR PAINTING SCHEDULE, METAL SUBSTRATES

A. Steel and Iron Substrates:

1. Water-Based, Light Industrial Coating System:

a. Prime Coat: Zinc-rich, inorganic primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Exterior, water-based, light industrial coating, gloss.

2. Alkyd System:

a. Prime Coat: Alkyd metal primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Exterior alkyd paint, gloss.

B. Galvanized Metal Substrates:

1. Latex System:

a. Prime Coat: Water-based, galvanized-metal primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Exterior latex paint, gloss.

2. Water-Based, Light Industrial Coating System:

a. Prime Coat: Water-based, galvanized-metal primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Exterior, water-based, light industrial coating, gloss.

C. Aluminum (Not Anodized or Otherwise Coated) Substrates

1. High Performance Architectural Latex System:

a. Prime Coat: Quick-dry aluminum primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Exterior, high performance architectural latex, semigloss.

3.6 EXTERIOR PAINTING SCHEDULE, COLORS

1. Steel Columns – Medium Bronze. Match prefinished downspouts.
2. Hollow Metal Doors and Door Frames – Dark Bronze / Dark Brown.
3. Metal Louvers – Dark Bronze / Dark Brown.
4. Aluminum Railings – Medium Bronze.
5. Miscellaneous – Review with Contracting Officer.

END OF SECTION 09 91 13

SECTION 09 91 23 – INTERIOR PAINTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes: Surface preparation of interior substrates and application of the following:

1. Primers.
2. Water-based finish coatings.
3. Floor sealers and paints.

B. Related Requirements:

1. Section 05 12 00 "Structural Steel Framing" for shop priming structural steel substrates.
2. Section 05 50 00 "Metal Fabrications" for shop priming metal fabrications.
3. Section 09 29 00 “Gypsum Board” for substrate of painted walls and ceilings.
4. Section 09 91 13 "Exterior Painting."
5. Section 09 93 00 "Staining and Transparent Finishing" for surface preparation and application of wood stains and transparent finishes on interior wood substrates.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include preparation requirements and application instructions.
2. Indicate VOC content.

B. Samples for Verification: For each type of paint system and each color and gloss of topcoat. Submit actual paint drawdowns as specified below for verification Samples.

1. Submit Samples on rigid backing, 8 inches square.
2. Apply coats on Samples in steps to show each coat required for system.
3. Label each coat of each Sample.
4. Label each Sample for location and application area.

C. Product List: Cross-reference to paint system and locations of application areas. Use same designations indicated on Drawings and in applicable interior painting schedule articles to cross-reference paint systems specified in this Section. Include color designations.

D. Sustainable Design Submittals:

1. Product Data: For paints and coatings, indicating VOC content.
2. Environmental Product Declaration (EPD): For each product.
3. Environmental Product Declaration (EPD): For each product.
4. Third-Party Certifications: For each product.
5. Third-Party-Certified Life-Cycle Assessment: For each product.

1.3 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match paint products applied and that are packaged with protective covering for storage and identified with labels describing contents.

1. Paint Products: 5 percent, but not less than 1 gal. of each material and color applied.

1.4 MOCKUPS

A. Apply mockups of each paint system indicated and each color and finish selected to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for materials and execution.

1. Contracting Officer will select one surface to represent surfaces and conditions for application of each paint system.

a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft..
b. Other Items: Contracting Officer will designate items or areas required.

2. Final approval of color selections will be based on mockups.

a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Contracting Officer at no added cost to Government.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.
4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F.

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.6 FIELD CONDITIONS

A. Apply paints only when temperature of surfaces to be painted and ambient air temperatures are between 50 and 95 deg F.

B. Do not apply paints when relative humidity exceeds 85 percent; at temperatures of less than 5 deg F above the dew point; or to damp or wet surfaces.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Obtain each paint product from single source from single manufacturer.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Behr Paint Company; Behr Process Corporation.
2. Benjamin Moore & Co.
3. Carboline Company; a subsidiary of RPM International.
4. Conco Paints.
5. Coronado Paint; Benjamin Moore & Co.
6. Diamond Vogel Paint Company.
7. Dunn-Edwards Corporation (a Nippon Paint Holdings Co. Ltd. company).
8. H&C Decorative Concrete Products; a brand of Sherwin-Williams Co.
9. Hempel (USA), Inc.
10. Insl-X Products; Benjamin Moore & Co.
11. Kelly-Moore Paints.
12. McCormick Paints.
13. PPG Paints; PPG Industries, Inc.
14. Sherwin-Williams Company (The).
15. Valspar; a brand of The Sherwin-Williams Company.
16. Or Approved Equal.

2.2 INTERIOR PAINTS, GENERAL

A. Interior Paints: Subject to compliance with requirements, provide one of the products listed in product types below and applicable interior painting schedule articles for the paint category indicated.

B. Material Compatibility:

1. Materials for use within each paint system must be compatible with one another and substrates indicated, under conditions of service and application as demonstrated by manufacturer based on testing and field experience.
2. For each coat in a paint system, provide products recommended in writing by topcoat manufacturer for use in paint system and on substrate indicated.

C. Material Emissions and Pollutant Control: Verify not less than 85 percent of field-applied paints and coatings that are inside the weatherproofing system comply with either of the following:

1. Low-Emitting Materials: Verify VOC emissions comply with requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers." Verify formaldehyde emissions do not exceed 9 mcg/cu. m or 7 ppb, whichever is less.

2. Verify VOC content does not exceed limits of authorities having jurisdiction and the following:

    a. Flat Coatings: 50 g/L.
    b. Nonflat Coatings: 100 g/L.
    c. Nonflat – High-Gloss Coatings: 150 g/L.
    d. Concrete/Masonry Sealers: 100 g/L.
    e. Floor Coatings: 100 g/L.
    f. Industrial Maintenance Coatings: 250 g/L.
    g. Low-Solids Coatings: 120 g/L.
    h. Mastic Texture Coatings: 100 g/L.
    i. Metallic Pigmented Coatings: 500 g/L.
    j. Pretreatment Wash Primers: 420 g/L.
    k. Primers, Sealers, and Undercoaters: 100 g/L.
    l. Reactive Penetrating Sealers: 350 g/L.
    m. Recycled Coatings: 250 g/L.
    n. Rust-Preventive Coatings: 250 g/L.

D. Colors: As noted in section 3.10 of this specification, to be confirmed by Contracting Officer from manufacturer's full range.

1. 20 percent of surface area will be painted with deep tones.

E. Verify that the permeance rating of the paint on exterior walls is 7 perms, minimum. Perms shall be 7 or higher to prevent trapping moisture.

2.3 PRIMERS

A. Latex Block Filler: Water-based, pigmented, high-solids, emulsion coating formulated to bridge and fill porous surfaces of CMU substrates in preparation for specified subsequent coatings.

B. Alkali-Resistant, Water-Based Primer: Water-based primer formulated for use on alkaline substrates.

C. Interior Latex Primer Sealer: Pigmented, water-based latex sealer; formulated to reduce porosity of substrate for finish coats; for use on new interior plaster, concrete, and gypsum board substrates. Not intended for use on wood or previously painted surfaces.

D. Interior, Institutional Low-Odor/VOC Primer Sealer: Water-based primer sealer with low-odor characteristics and a VOC of less than 10 grams per liter for use on new interior plaster, concrete, and gypsum wallboard surfaces that are subsequently to be painted with latex finish coats.

E. Water-Based, Rust-Inhibitive Primer: Corrosion-resistant, water-based-emulsion primer; formulated for resistance to flash rusting when applied to cleaned, ferrous metal substrates subject to mildly corrosive environments.

F. Water-Based, Galvanized-Metal Primer: Corrosion-resistant, pigmented, acrylic primer; formulated for use on cleaned/etched galvanized metal substrates to prepare it for subsequent water-based coatings.

G. Solvent-Based Bonding Primer: Solvent-based primer formulated to seal substrates and promote adhesion of specified subsequent coatings.

2.4 WATER-BASED FINISH COATINGS

A. Interior Latex Paint: Pigmented, water-based coating for use on interior primed/sealed plaster and gypsum board, and on primed wood and metals.

1. Gloss and Sheen Level: Manufacturer's standard semigloss.

B. Interior, Latex, Institutional Low-Odor/VOC Paint: White or colored latex coating with low-odor characteristics and a VOC of less than 10 grams per liter, for use in areas, such as hospitals and other occupied buildings, where the odor and VOC levels of conventional latex products would preclude their use.

1. Gloss and Sheen Level: Manufacturer's standard flat satin gloss.

C. Interior, Water-Based Light Industrial Coating: Pigmented, water-based emulsion coating for interior primed wood and metal surfaces (e.g., walls, doors, frames, trim, and sash), providing resistance to moderate abrasion and mild chemical exposure and corrosive conditions.

1. Gloss and Sheen Level: Manufacturer's standard semigloss.

2.5 FLOOR SEALERS AND PAINTS

A. Water-Based Concrete Floor Sealer: Clear, water-based, acrylic-copolymer-emulsion sealer formulated for oil, gasoline, alkali, and water resistance and for use on concrete traffic surfaces.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

B. Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:

1. Concrete: 12 percent.
2. Cementitious Composition Board: 12 percent.
3. Masonry (Clay and CMU): 12 percent.
4. Wood: 15 percent.
5. Gypsum Board: 12 percent. Verify that finishing compound is dry and sanded smooth.

C. Verify suitability of substrates, including surface conditions and compatibility, with finishes and primers. Proceed with coating application only after unsatisfactory conditions have been corrected.

1. Application of coating indicates acceptance of surfaces and conditions.

3.2 PREPARATION

A. Comply with manufacturer's written instructions applicable to substrates and paint systems indicated.

B. Remove hardware, covers, plates, and similar items already in place that are removable and are not to be painted. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and painting.

1. After completing painting operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.

C. Clean substrates of substances that could impair bond of paints, including dust, dirt, oil, grease, and incompatible paints and encapsulants.

1. Remove incompatible primers and reprime substrate with compatible primers or apply tie coat as required to produce paint systems indicated.

D. Concrete Substrates: Remove release agents, curing compounds, efflorescence, and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces to be painted exceeds that permitted in manufacturer's written instructions.

E. Masonry Substrates: Remove efflorescence and chalk. Do not paint surfaces if moisture content or alkalinity of surfaces or mortar joints to be painted exceeds that permitted in manufacturer's written instructions.

F. Steel Substrates: Remove loose rust, loose mill scale, loose shop primer, and other loose foreign matter. Clean using methods recommended in writing by paint manufacturer.

G. Shop-Primed Steel Substrates: Clean field welds, bolted connections, and areas where shop paint is abraded. Paint exposed areas with the same material as used for shop priming to comply with SSPC-PA 1 for touching up shop-primed surfaces.

H. Galvanized Metal Substrates: Remove grease and oil residue from galvanized metal by mechanical methods to produce clean, lightly etched surfaces that promote adhesion of subsequently applied paints.

I. Wood Substrates:

1. Scrape and clean knots. Before applying primer, apply coat of knot sealer recommended in writing by topcoat manufacturer for interior use in paint system indicated.
2. Sand surfaces that will be exposed to view and remove sanding dust.
3. Prime edges, ends, faces, undersides, and backsides of wood.
4. After priming, fill holes and imperfections in the finish surfaces with putty or plastic wood filler. Sand smooth when dried.

J. Plastic Trim Fabrication Substrates: Remove dust, dirt, and other foreign material that might impair bond of paints to substrates.

3.3 APPLICATION OF INTERIOR PAINT PRODUCTS

A. Apply paints in accordance with manufacturer's written instructions.

1. Use applicators and techniques suited for paint and substrate indicated.
2. Paint surfaces behind movable equipment and furniture same as similar exposed surfaces. Before final installation, paint surfaces behind permanently fixed equipment or furniture with prime coat only.
3. Paint front and backsides of access panels, removable or hinged covers, and similar hinged items to match exposed surfaces.
4. Do not paint over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.
5. Primers specified in the applicable interior painting schedule articles may be omitted on items that are factory primed or factory finished if compatible with intermediate and topcoat coatings and acceptable to intermediate and topcoat paint manufacturers.

B. If undercoats or other conditions show through topcoat, apply additional coats until cured film has a uniform paint finish, color, and appearance.

C. Apply paints to produce surface films without cloudiness, spotting, holidays, laps, brush marks, roller tracking, runs, sags, ropiness, or other surface imperfections. Cut in sharp lines and color breaks.

D. Painting Fire-Suppression, Plumbing, HVAC, Electrical, Communication, and Electronic Safety and Security Work:

1. Paint the following work where exposed to view in occupied spaces:

a. Equipment, including panelboards.
b. Uninsulated metal piping.
c. Uninsulated plastic piping.
d. Pipe hangers and supports.
e. Metal conduit.
f. Plastic conduit.
g. Duct, equipment, and pipe insulation having cotton or canvas insulation covering or other paintable jacket material.
h. Other items as directed by Contracting Officer.

2. Paint portions of internal surfaces of metal ducts, without liner, behind air inlets and outlets that are visible from occupied spaces.

3.4 CLEANING AND PROTECTION

A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

1. Do not clean equipment with free-draining water and prevent solvents, thinners, cleaners, and other contaminants from entering into waterways, sanitary and storm drain systems, and ground.

2. Dispose of contaminants in accordance with requirements of authorities having jurisdiction.
3. Allow empty paint cans to dry before disposal.
4. Collect waste paint by type and deliver to recycling or collection facility.

B. After completing paint application, clean spattered surfaces. Remove spattered paints by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

C. Protect work of other trades against damage from paint application. Correct damage to work of other trades by cleaning, repairing, replacing, and refinishing, as approved by Contracting Officer, and leave in an undamaged condition.

D. At completion of construction activities of other trades, touch up and restore damaged or defaced painted surfaces.

3.5 INTERIOR PAINTING SCHEDULE, CONCRETE SUBSTRATES

A. Vertical (Nontraffic) Surfaces:

1. Institutional Low-Odor/VOC Latex System:

a. Prime Coat: Interior, institutional low-odor/VOC primer sealer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, latex, institutional low odor/VOC, satin.

2. Water-Based Light Industrial Coating System:

a. Prime Coat: Alkali-resistant, water-based primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, water-based, light industrial coating, semigloss.

B. Horizontal (Traffic) Surfaces:

1. Water-Based Concrete Floor Sealer System:

a. First Coat: Matching topcoat.
b. Topcoat: Water-based concrete floor sealer.

3.6 INTERIOR PAINTING SCHEDULE, MASONRY SUBSTRATES

A. CMU Substrates:

1. Latex System:

a. Block Filler: Interior/exterior latex block filler.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior latex paint, semigloss.

3.7 INTERIOR PAINTING SCHEDULE, METAL SUBSTRATES

A. Steel and Iron Substrates:

1. Institutional Low-Odor/VOC Latex System:

a. Prime Coat: Water-based rust-inhibitive primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, latex, institutional low odor/VOC, satin gloss.

B. Galvanized Metal Substrates:

1. Institutional Low-Odor/VOC Latex System:

a. Prime Coat: Water-based galvanized primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, latex, institutional low odor/VOC, satin.

3.8 INTERIOR PAINTING SCHEDULE, GYPSUM-BASED SUBSTRATES

A. Gypsum Board and Plaster Substrates:

1. Institutional Low-Odor/VOC Latex System:

a. Prime Coat: Interior, institutional low-odor/VOC primer sealer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, latex, institutional low odor/VOC, flat satin gloss.

3.9 INTERIOR PAINTING SCHEDULE, MISCELLANEOUS SUBSTRATES

A. Plastic Trim Fabrication Substrates:

1. Institutional Low-Odor/VOC Latex System:

a. Prime Coat: Water-based bonding primer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, latex, institutional low odor/VOC, satin.

B. Canvas and Cotton Insulation-Covering Substrates: Including pipe and duct coverings.

1. Institutional Low-Odor/VOC Latex System:

a. Prime Coat: Interior latex primer sealer.
b. Intermediate Coat: Matching topcoat.
c. Topcoat: Interior, latex, institutional low odor/VOC, flat satin.

3.10 INTERIOR PAINTING SCHEDULE, COLORS

1. Steel Columns – Medium Bronze. Match exterior columns.
2. Steel Beams Exposed to View – Medium Bronze, match columns.
3. Hollow Metal Doors and Door Frames – Dark Bronze / Dark Brown.
4. Walls, typical – Sherman Williams “City Loft.”
5. Walls, breakroom, stairs, and workroom – Sherman Williams “Taupe Tone.”
6. Walls, restrooms – Sherman Williams “Urbane Bronze.”
7. Ceilings, typical gypsum board – Sherman Williams “Greek Villa” @ 50%.
8. Ceilings, breakroom, stairs, and workroom – Sherman Williams “Taupe Tone” @ 50%.
9. Ceilings, restrooms – Sherman Williams “Urbane Bronze” @ 50%.
10. Miscellaneous – Review with Contracting Officer.

B. Do not paint wood ceiling or acoustic ceiling tile.

END OF SECTION 09 91 23

SECTION 09 93 00 – STAINING AND TRANSPARENT FINISHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wood stains.

a. All wood trim, interior and exterior.
b. Wood header over main entry door.
c. Wood doors.
d. Wood portions of the transaction desk.
e. Wood “ceiling” (underside of original building’s roof deck) in lobby and vestibule.

1) This wood is currently painted and will need paint removal to prepare for staining. If not cleared by the hazardous material report, this shall require testing to confirm paint is not lead-containing.

2. Transparent finishes.

B. Related Requirements:

1. Section 06 20 23 “Interior Finish Carpentry” for interior wood trim.
2. Section 06 40 13 “Exterior Architectural Woodwork” for exterior wood trim.
3. Section 06 41 00 “Architectural Wood Casework” for wood portions of lobby transaction desk.
4. Section 09 91 13 “Exterior Painting” for exterior paint on metal surfaces.
5. Section 09 91 23 "Interior Painting" for interior paint and transparent finishes on concrete floors.

1.2 ACTION SUBMITTALS

A. Product Data:

1. For each type of product.
2. Include preparation requirements and application instructions.
3. Indicate VOC content.

B. Samples for Initial Selection: Manufacturer's standard color sheets, showing full range of available colors for each type of exposed finish.

C. Samples for Verification: Sample for each type of finish system and in each color and gloss of finish required on representative samples of actual wood substrates.

1. Size: 8 inches square.
2. Apply coats on Samples in steps to show each coat required for system.

3. Label each coat of each Sample.
4. Label each Sample for location and application area.

D. Product List: Cross-reference to finish system and locations of application areas. Use same designations indicated on Drawings and in schedules. Include color designations.

E. Sustainable Design Submittals:

1. Product Data: For paints and coatings, indicating VOC content.
2. Laboratory Test Reports: For paints and coatings, indicating compliance with requirements for low-emitting materials.
3. Environmental Product Declaration: For each product.
4. Third-Party Certifications: For each product.
5. Third-Party Certified Life Cycle Assessment: For each product.

1.3 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Material: Furnish extra materials, from the same product run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Stains and Transparent Finishes: 5 percent, but not less than 1 gal. of each material and color applied.

1.4 MOCKUPS

A. Apply mockups of each finish system indicated and each color selected to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Exterior Mockups: For each exterior stained wood element, one section of the exterior wood element shall be finished and installed and reviewed. After the mockup is accepted by the Contracting Officer in writing, the remainder of the exterior wood stain scope for that element may be started.

a. Exterior soffit trim
b. Exterior wood header
c. Exterior bench slats

2. Interior Mockups: For each interior stained wood element, one section or area, as applicable, shall be finished and installed and reviewed. After the mockup is accepted by the Contracting Officer in writing, the remainder of the interior wood stain scope for that element may be started.

a. Interior existing wood deck in lobby ceiling
b. Interior windowsills and aprons
c. Interior base trim

3. Final approval of stain color selections will be based on mockups.

   a. If preliminary stain color selections are not approved, apply additional mockups of additional stain colors selected by Contracting Officer at no added cost to Government.

4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Store materials not in use in tightly covered containers in well-ventilated areas with ambient temperatures continuously maintained at not less than 45 deg F.

1. Maintain containers in clean condition, free of foreign materials and residue.
2. Remove rags and waste from storage areas daily.

1.6 FIELD CONDITIONS

A. Apply finishes only when temperature of surfaces to be finished and ambient air temperatures are between 50 and 95 deg F.

B. Do not apply finishes when relative humidity exceeds 85 percent, at temperatures of less than 5 deg F above the dew point, or to damp or wet surfaces.

C. Do not apply exterior finishes in snow, rain, fog, or mist.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Source Limitations: Obtain each coating product from single source from single manufacturer.

2.2 WOOD STAINS

A. Stain, Interior, Semitransparent, for Interior Wood: Solvent-based, oil or oil/alkyd, semitransparent, pigmented stain for new interior wood surfaces that are to be finished with a clear varnish.

1. Color: Walnut

2.3 TRANSPARENT FINISHES

A. Varnish with UV Inhibitor, Exterior, Semigloss: Solvent-based, alkyd-type, clear semigloss varnish stabilized against UV deterioration for exterior wood surfaces.

1. Gloss Level: Gloss of 35 to 70 units at 60 degrees when tested in accordance with ASTM D523.

B. Varnish, Interior Polyurethane, Moisture Cured, Gloss: Solvent-based, moisture-curing polyurethane clear-coating with a gloss finish for interior wood surfaces,

1. Gloss Level: Gloss of 70 to 85 units at 60 degrees when tested in accordance with ASTM D523.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Applicator present, for compliance with requirements for maximum moisture content and other conditions affecting performance of the Work.

B. Maximum Moisture Content of Exterior Wood Substrates: 15 percent, when measured with an electronic moisture meter.

C. Maximum Moisture Content of Interior Wood Substrates: 13 percent, when measured with an electronic moisture meter.

D. Verify suitability of substrates, including surface conditions and compatibility with existing finishes and primers.

E. Proceed with finish application only after unsatisfactory conditions have been corrected.

1. Beginning finish application constitutes Contractor's acceptance of substrates and conditions.

3.2 PREPARATION, GENERAL

A. Remove hardware, covers, plates, and similar items already in place that are removable. If removal is impractical or impossible because of size or weight of item, provide surface-applied protection before surface preparation and finishing.

1. After completing finishing operations, use workers skilled in the trades involved to reinstall items that were removed. Remove surface-applied protection if any.

B. Clean and prepare surfaces to be finished according to manufacturer's written instructions for each substrate condition and as specified.

1. Remove dust, dirt, oil, and grease by washing with a detergent solution; rinse thoroughly with clean water and allow to dry. Remove grade stamps and pencil marks by sanding lightly. Remove loose wood fibers by brushing.
2. Remove mildew by scrubbing with a commercial wash formulated for mildew removal and as recommended by stain manufacturer.

C. Exterior Wood Substrates:

1. Scrape and clean knots, and apply coat of knot sealer before applying primer.
2. Prime edges, ends, faces, undersides, and backsides of wood.

    a. For solid hide stained wood, stain edges and ends after priming.
    b. For varnish-coated stained wood, stain edges and ends and prime with varnish. Prime undersides and backsides with varnish.

3. Countersink steel nails, if used, and fill with putty or plastic wood filler tinted to final color. Sand smooth when dried.

D. Interior Wood Substrates:

1. Scrape and clean knots, and apply coat of knot sealer before applying primer.
2. Apply wood filler paste to open-grain woods to produce smooth, glasslike finish.
3. Sand surfaces exposed to view and dust off.
4. After priming, fill holes and imperfections in the finish surfaces with putty or plastic wood filler. Sand smooth when dry.

3.3 WOOD CEILING PREPARATION

A. Confirm through testing that the paint being removed does not contain lead.

1. If it does, comply with Section 02 83 33 “Removal and Disposal of Material Containing Lead” for all removal of existing paint on wood ceiling to be exposed and stained.

B. Remove paint to allow natural wood substrate to be stained. Process may use heat or chemical paint removal methods provided that:

1. Removal method is approved by Contracting Officer.
2. Surrounding materials (steel beams, wood framing, and walls to be retained, where applicable) are protected from damage.
3. Adjacent steel beams are primed. If priming is damaged by removal process, Contractor shall re-prime these beams prior to painting them.
4. Follow all manufacturer’s instructions for effective and safe use.
5. Take all manufacturer’s precautions regarding safety and health.

3.4 APPLICATION

A. Apply finishes according to manufacturer's written instructions.

1. Use applicators and techniques suited for finish and substrate indicated.
2. Finish surfaces behind movable equipment and furniture same as similar exposed surfaces.
3. Do not apply finishes over labels of independent testing agencies or equipment name, identification, performance rating, or nomenclature plates.

B. Apply finishes to produce surface films without cloudiness, holidays, lap marks, brush marks, runs, ropiness, or other surface imperfections.

3.5 CLEANING AND PROTECTION

A. At end of each workday, remove rubbish, empty cans, rags, and other discarded materials from Project site.

B. After completing finish application, clean spattered surfaces. Remove spattered materials by washing, scraping, or other methods. Do not scratch or damage adjacent finished surfaces.

C. Protect work of other trades against damage from finish application. Correct damage by cleaning, repairing, replacing, and refinishing, as approved by Contracting Officer, and leave in an undamaged condition.

D. At completion of construction activities of other trades, touch up and restore damaged or defaced finished wood surfaces.

3.6 INTERIOR WOOD-FINISH-SYSTEM SCHEDULE

A. Interior Wood Substrates, Wood Ceiling, Wood Trim, Wood Doors, and Architectural Woodwork:

1. Moisture-Cured Clear Polyurethane over Stain System:

a. Stain Coat: Stain, semitransparent, for interior wood.
b. First Intermediate Coat: Moisture-cured polyurethane matching topcoat.
c. Second Intermediate Coat: Moisture-cured polyurethane matching topcoat.
d. Topcoat: Varnish, polyurethane, moisture cured, gloss.

B. Exterior Wood Substrates, Soffit Trim and Header:

1. Moisture-Cured Clear Polyurethane over Stain System:

a. Stain Coat: Stain, semitransparent, for exterior wood.
b. First Intermediate Coat: Moisture-cured polyurethane matching topcoat.
c. Second Intermediate Coat: Moisture-cured polyurethane matching topcoat.
d. Topcoat: Varnish, polyurethane, moisture cured, gloss.

END OF SECTION 09 93 00

SECTION 10 14 23.16 – ROOM-IDENTIFICATION PANEL SIGNAGE

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes room-identification signs that are directly attached to the building.

1. ABA compliant, code required room signs – interior.
2. Notification signs as identified in the drawings – interior.

1.2 ALLOWANCES

A. See drawings for sign schedule.

1. Include supplemental allowance for 5 additional miscellaneous signs. Assume 12x12 min size for the miscellaneous signs.

B. Confirm the following is included:

1. Code and ABAAS compliant room identification signage at one side of all doors in the building, excepting exterior doors and coiling door.
2. Each sign shall anticipate 4 x 8 inch maximum size and at least 3 colors.
3. Design process with Contracting Officer to develop a sign design and aesthetic for the provided signage. Include at least 3 rounds of design and review prior to finalization. Include at least 3 alternatives in the first round of design. All alternatives shall fit within the budgeted allowance and comply with salient characteristics as noted below.
4. Design process shall include the creation of a room identification schedule that includes door/room numbering and room name determination. Two rounds of review shall be assumed so that Contracting Officer can review with the park and ensure room names and language on notification signs are agreed upon prior to manufacture.

1.3 DEFINITIONS

A. Accessible: In accordance with the accessibility standard.

1.4 COORDINATION

A. Furnish templates for placement of sign-anchorage devices embedded in permanent construction by other installers.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Sustainable Design Submittals:

1. Product Data: For adhesives, indicating VOC content.
2. Laboratory Test Reports: For adhesives, indicating compliance with requirements for low-emitting materials.

C. Shop Drawings: For room-identification signs.

1. Include fabrication and installation details and attachments to other work.
2. Show sign mounting heights, locations of supplementary supports to be provided by other installers, and accessories.
3. Show message list, typestyles, graphic elements, including raised characters and Braille, and layout for each sign at least half size.
4. Location on plan and typical interior elevation relative to door.

D. Samples for Initial Selection: For each type of sign assembly, exposed component, and exposed finish.

1. Include representative Samples of available typestyles and graphic symbols.

E. Samples for Verification: For each type of sign assembly showing all components and with the required finish(es), in manufacturer's standard size unless otherwise indicated and as follows:

1. Room-Identification Signs: Full-size Sample.
2. Variable Component Materials: Full-size Sample of each base material, character (letter, number, and graphic element) in each exposed color and finish not included in Samples above.
3. Exposed Accessories: Full-size Sample of each accessory type.
4. Full-size Samples, if approved, will be returned to Contractor for use in Project.

F. Product Schedule: For room-identification signs. Use same designations indicated on Drawings or specified.

1. Names and numbers of all rooms shall be reviewed and affirmatively approved by the Contracting Officer prior to production of signs.

1.6 FIELD CONDITIONS

A. Field Measurements: Verify locations of anchorage devices and electrical service embedded in permanent construction by other installers by field measurements before fabrication and indicate measurements on Shop Drawings.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Accessibility Standard: Comply with applicable provisions in the ABA standards of the Federal agency having jurisdiction and ICC A117.1.

2.2 ROOM-IDENTIFICATION SIGNS

A. Room-Identification Sign: Sign with smooth, uniform surfaces; with message and characters having uniform faces, sharp corners, and precisely formed lines and profiles; and as follows:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. ACE Sign Systems, Inc.
    b. ASE, Inc.
    c. ASI Sign Systems, Inc.
    d. Best Sign Systems, Inc.
    e. Clarke Systems.
    f. Mohawk Sign Systems.
    g. Poblocki Sign Company, LLC.
    h. Vista System, LLC.
    i. Vomar Products, Inc.

2. Basis of Design: ASI Sign Systems, InTac Eco.
3. Laminated-Sheet Sign: Photopolymer face sheet with raised graphics laminated over subsurface graphics to backing sheet to produce composite sheet.

    a. Composite-Sheet Thickness: 0.125 inch.
    b. Surface-Applied Graphics: low VOC paint and UV digitally printed graphics.
    c. Color(s): To be determined during submittal process.

4. Mounting: Surface mounted to wall with two-face tape.
5. Text and Typeface: Accessible raised characters and Braille, typeface to be determined during deferred design process, variable content as scheduled. Finish raised characters to contrast with background color, and finish Braille to match background color.

    a. Contrast and text size to be compliant with ABA and current ANSI 117.1 standards.
    b. Draft of text for notification signs shall be included in sign schedule in drawings, but final text shall be approved by the Contracting Officer during the submittal process.

2.3 SIGN MATERIALS

A. Acrylic Sheet: ASTM D 4802, category as standard with manufacturer for each sign, Type UVF (UV filtering).

B. Vinyl Film: UV-resistant vinyl film with pressure-sensitive, permanent adhesive; die cut to form characters or images as indicated on Drawings.

C. Paints and Coatings for Sheet Materials: Inks, dyes, and paints that are recommended by manufacturer for optimum adherence to surface and are UV and water resistant for colors and exposure indicated.

2.4 ACCESSORIES

A. Two-Face Tape: Manufacturer's standard high-bond, foam-core tape, 0.045 inch thick, with adhesive on both sides.

2.5 FABRICATION

A. General: Provide manufacturer's standard sign assemblies according to requirements indicated.

1. Preassemble signs and assemblies in the shop to greatest extent possible. Disassemble signs and assemblies only as necessary for shipping and handling limitations. Clearly mark units for reassembly and installation; apply markings in locations concealed from view after final assembly.
2. Mill joints to a tight, hairline fit. Form assemblies and joints exposed to weather to resist water penetration and retention.
3. Conceal connections if possible; otherwise, locate connections where they are inconspicuous.
4. Provide rabbets, lugs, and tabs necessary to assemble components and to attach to existing work. Drill and tap for required fasteners. Use concealed fasteners where possible; use exposed fasteners that match sign finish.

2.6 GENERAL FINISH REQUIREMENTS

A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General: Install signs using mounting methods indicated and according to manufacturer's written instructions.

1. Install signs level, plumb, true to line, and at locations and heights indicated, with sign surfaces free of distortion and other defects in appearance.
2. Install signs so they do not protrude or obstruct according to the accessibility standard.
3. Before installation, verify that sign surfaces are clean and free of materials or debris that would impair installation.

B. Accessibility: Install signs in locations on walls and according to the accessibility standard.

C. Mounting Methods:

1. Two-Face Tape: Clean bond-breaking materials from substrate surface and remove loose debris. Apply tape strips symmetrically to back of sign and of suitable quantity to support weight of sign without slippage. Keep strips away from edges to prevent visibility at sign edges. Place sign in position and push to engage tape adhesive.

3.2 ADJUSTING AND CLEANING

A. Remove and replace damaged or deformed signs and signs that do not comply with specified requirements. Replace signs with damaged or deteriorated finishes or components that cannot be successfully repaired by finish touchup or similar minor repair procedures.

B. Remove temporary protective coverings and strippable films as signs are installed.

C. On completion of installation, clean exposed surfaces of signs according to manufacturer's written instructions, and touch up minor nicks and abrasions in finish. Maintain signs in a clean condition during construction and protect from damage until acceptance by Contracting Officer.

3.3 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 10 14 23.16

SECTION 10 21 13.19 – PLASTIC TOILET COMPARTMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Solid-plastic toilet compartments.

B. Related Requirements:

1. Section 061000 "Rough Carpentry" for blocking.
2. Section 102800 "Toilet Accessories" for accessories mounted on toilet compartments.
3. Section 066116 “Solid Surface Fabrications” for solid surface wall panel coordination.
4. Division 22 for Plumbing fixture coordination.

1.2 COORDINATION

A. Coordinate requirements for blocking, reinforcing, and other supports concealed within wall to ensure that toilet compartments can be supported and installed as indicated.

B. Coordinate requirements for toilet compartment dimensions, floor and fixture clearances, door opening widths and hardware, and knee and toe clearances to ensure that toilet compartments are compliant with ABAAS accessibility requirements.

1.3 ACTION SUBMITTALS

A. Product Data:

1. Solid-plastic toilet compartments:

a. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for toilet compartments.

B. Shop Drawings:

1. Include plans, elevations, sections, details, and attachment details.
2. Show locations of cutouts for compartment-mounted toilet accessories.
3. Show locations of centerlines of toilet fixtures.
4. Show locations of floor drains.

C. Samples for Verification: Actual sample of finished products for each type of toilet compartment, hardware, and accessory.

1. Size: Manufacturer's standard size.

D. Product Schedule: For toilet compartments, prepared by or under the supervision of supplier, detailing location and selected colors for toilet compartment material.

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For toilet compartments.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Extra Stock Materials: Furnish extra materials to CO that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Door Hinges: Two hinge(s) with associated fasteners.
2. Latch and Keeper: Two latch(es) and keeper(s) with associated fasteners.
3. Door Bumper: Two bumper(s) with associated fasteners.
4. Door Pull: Two door pull(s) with associated fasteners.
5. Fasteners: 10 fasteners of each size and type.

1.6 FIELD CONDITIONS

A. Field Measurements: Verify actual locations of toilet fixtures, walls, columns, ceilings, and other construction contiguous with toilet compartments by field measurements, and coordinate before fabrication.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain plastic toilet compartments from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 50 percent.

B. Fire Performance: Tested in accordance with, and pass the acceptance criteria of, NFPA 286.

C. Structural Performance: Where grab bars are mounted on toilet compartments, design panels to comply with the following requirements:

1. Panels are able to withstand a concentrated load on grab bar of at least 250 lbf applied at any direction and at any point, without deformation of panel.

D. Regulatory Requirements: Comply with applicable provisions in ICC A117.1 for toilet compartments designated as accessible.

2.3 SOLID-PLASTIC TOILET COMPARTMENTS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. All American Metal Corp
2. American Sanitary Partition Corporation
3. ASI Global Partitions
4. General Partitions Mfg. Corp
5. Hadrian Inc.; Zurn Industries, LLC
6. Scranton Products

B. Toilet-Enclosure Style: Floor-and-ceiling anchored.

C. Door, Panel, and Pilaster Construction: Solid, high-density polyethylene (HDPE) material, not less than 1 inch thick, seamless, with eased edges, and with homogenous color throughout thickness of material. Provide with no-sightline system consisting of door and pilaster lapped edges on strike side of door and door and pilaster lapped edges on hinge side of door (unless continuous hinge is used).

1. Basis of Design Color: Hadrian, Grey 239, to be confirmed by CO during sample submittal. Another color may be selected by CO from manufacturer's full range.

D. Pilaster Shoes: Manufacturer's standard design; stlainless steel.

E. Pilaster Sleeves (Caps): Manufacturer's standard design; stainless steel.

F. Brackets (Fittings):

1. Stirrup Type: Ear or U-brackets, stainless steel.
2. Full-Height (Continuous) Type: Manufacturer's standard design; stainless steel.

2.4 HARDWARE AND ACCESSORIES

A. Door Hardware and Accessories, Heavy Duty: Manufacturer's heavy-duty institutional operating hardware and accessories. Coordinate to ensure that toilet compartments, hardware and accessories are compliant with ABAAS accessibility requirements.

B.

1. Hinges: Manufacturer's minimum 0.062-inch- thick, stainless steel continuous, cam type that swings to a closed or partially open position, allowing emergency access by lifting door. Mount with through bolts.
2. Latch and Keeper: Manufacturer's heavy-duty, surface-mounted, cast stainless steel latch unit, designed to resist damage due to slamming, with combination rubber-faced door strike and keeper, and with provision for emergency access. Provide units that comply with regulatory requirements for accessibility at toilet enclosures designated as accessible. Mount with through bolts.

3. Coat Hook: Manufacturer's heavy-duty, combination cast stainless steel hook and rubber-tipped bumper, sized to prevent inswinging door from hitting compartment-mounted accessories. Mount with through bolts.
4. Door Bumper: Manufacturer's heavy-duty, rubber-tipped, cast stainless steel bumper at outswinging doors. Mount with through bolts.
5. Door Pull: Manufacturer's heavy-duty, cast stainless steel pull at outswinging doors that complies with regulatory requirements for accessibility. Provide units on both sides of doors at toilet enclosures designated as accessible. Mount with through bolts.

C. Anchorages and Fasteners: Manufacturer's standard exposed fasteners of stainless steel, finished to match items they are securing, with theft-resistant-type heads. Provide sex-type bolts for through-bolt applications. For concealed anchors, use stainless steel, hot-dip galvanized steel, or other rust-resistant, protective-coated steel compatible with related materials.

2.5 MATERIALS

A. Stainless Steel Sheet: ASTM A240/A240M or ASTM A666, Type 304, stretcher-leveled standard of flatness.

B. Stainless Steel Castings: ASTM A743/A743M.

2.6 FABRICATION

A. Fabricate toilet compartment components to sizes indicated. Coordinate requirements and provide cutouts for through-partition toilet accessories where required for attachment of toilet accessories.

B. Floor-and-Ceiling-Anchored Units: Manufacturer's standard corrosion-resistant anchoring assemblies at pilasters and walls, with leveling adjustment nuts at tops and bottoms of pilasters. Provide shoes and sleeves (caps) at pilasters to conceal anchorage.

C. Door Size and Swings: Unless otherwise indicated, provide 24-inch- wide, inswinging doors for standard toilet enclosures and 36-inch- wide, outswinging doors with a minimum 32-inch- wide, clear opening for toilet enclosures designated as accessible.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas and conditions, with Installer present, for compliance with requirements for fastening, support, alignment, operating clearances, and other conditions affecting performance of the Work.

1. Confirm location and adequacy of blocking and supports required for installation.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. General: Comply with manufacturer's written installation instructions. Install units rigid, straight, level, and plumb. Secure units in position with manufacturer's recommended anchoring devices.

1. Maximum Clearances:

a. Pilasters and Panels or Screens: 1/2 inch.
b. Panels or Screens and Walls: 1 inch.

2. Full-Height (Continuous) Brackets: Secure panels or screens to walls and to pilasters with full-height brackets.

a. Locate bracket fasteners, so holes for wall anchors occur in masonry or tile joints.
b. Align brackets at pilasters with brackets at walls.

B. Floor-and-Ceiling-Anchored Units: Secure pilasters to supporting construction and level, plumb, and tighten. Hang doors and adjust, so doors are level and aligned with panels, when doors are in closed position.

3.3 FIELD QUALITY CONTROL

A. Inspection Agency: Engage a qualified inspector to perform inspections and to furnish reports to Contracting Officer.

B. Inspections:

1. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

3.4 ADJUSTING

A. Hardware Adjustment: Adjust and lubricate hardware in accordance with hardware manufacturer's written instructions for proper operation. Set hinges on inswinging doors to hold doors open approximately 30 degrees from closed position when unlatched. Set hinges on outswinging doors to return doors to fully closed position.

END OF SECTION 10 21 13.19

SECTION 10 22 39 – FOLDING PANEL PARTITIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Operable acoustical panel partitions.

B. Related Requirements:

1. Section 055000 "Metal Fabrications" for supports that attach supporting tracks to overhead structural system.
2. Section 051200 “Structural Steel Framing” for steel beam and column necessary to support the operable partition.
3. Electrical and communications Sections for electrical service and connections for motor operators, controls, and limit switches and for system disconnect switches.

1.2 DEFINITIONS

A. NIC: Noise Isolation Class.

B. NRC: Noise Reduction Coefficient.

C. STC: Sound Transmission Class.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Operable acoustical panel partitions.

B. Sustainable Design Submittals:

1. Chain-of-Custody Certificates: For certified wood products.
2. Product Data: For composite wood products, indicating that product contains no urea formaldehyde.

C. Shop Drawings: For operable panel partitions.

1. Include plans, elevations, sections, attachment details, and numbered panel installation sequence.
2. Indicate stacking and operating clearances. Indicate location and installation requirements for hardware and track, blocking, and direction of travel.

D. Samples for Initial Selection: For each type of exposed material, finish, covering, or facing.

1. Fabric sample for Auditorium side.
2. Vinyl sample for Lobby side.
3. Include samples of accessories involving color selection.

E. Samples for Verification: For each type of exposed material, finish, covering, or facing, prepared on Samples of size indicated below:

1. Textile Facing Material: Full width by not less than 36-inch- long section of fabric from dye lot to be used for the Work, with specified treatments applied. Show complete pattern repeat.
2. Panel Facing Material: Manufacturer's standard-size unit, not less than 6 inches square.
3. Panel Edge Material: Not less than 3 inches long.
4. Chair Rail: Manufacturer's standard-size unit, 6 inches long.
5. Hardware: One of each exposed door-operating device.

F. Delegated Design Submittals: For operable panel partitions.

1. Include design calculations for seismic restraints that brace tracks to structure above.

G. Coordination Drawings: Reflected ceiling plans, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:

1. Partition track, track supports and bracing, switches, turning space, and storage layout.
2. Suspended ceiling components.
3. Structural members to which suspension systems will be attached.
4. Size and location of initial access modules for acoustical tile.
5. Items penetrating finished ceiling including the following:

    a. Lighting fixtures.
    b. HVAC ductwork, outlets, and inlets.
    c. Speakers.
    d. Sprinklers.
    e. Smoke detectors.
    f. Access panels.

6. Plenum acoustical barriers.

H. Setting Drawings: For embedded items and cutouts required in other work, including support-beam, mounting-hole template.

I. Qualification Data: For Installer.

1.5 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Certificates: For operable panel partitions, tracks, accessories, and components, from manufacturer. Include seismic capacity of partition assemblies to remain in vertical position during a seismic event and the following:

1. Basis for Certification: Indicate whether certification is based on analysis, testing, or experience data, in accordance with ASCE/SEI 7.
2. Detailed description of partition anchorage devices on which the certification is based and their installation requirements.

B. Product Certificates: For each type of operable panel partition.

1. Include approval letter signed by manufacturer acknowledging government-furnished panel facing material complies with requirements.

C. Product Test Reports: For each operable panel partition, for tests performed by a qualified testing agency.

D. Field quality-control reports.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For operable panel partitions to include in maintenance manuals.

1. In addition to items specified in Section 017823 "Operation and Maintenance Data," include the following:

a. Panel finish facings and finishes for exposed trim and accessories. Include precautions for cleaning materials and methods that could be detrimental to finishes and performance.
b. Seals, hardware, track, track switches, carriers, and other operating components.

B. Sample Warranty: For manufacturer's special warranty.

1.7 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials, from the same production run, that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Panel Finish-Facing Material: Furnish full width in quantity to cover both sides of two panels when installed for each type of finish, fabric and vinyl.

1.8 QUALITY ASSURANCE

A. Installer Qualifications: An entity that employs installers and supervisors who are trained and approved by manufacturer.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Protectively package and sequence panels in order for installation. Clearly mark packages and panels with numbering system used on Shop Drawings. Do not use permanent markings on panels.

1.10 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of operable panel partitions that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Faulty operation of operable panel partitions.
b. Deterioration of metals, metal finishes, and other materials beyond normal use.

2. Warranty Period: 10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 01 40 00 "Quality Requirements," to design seismic bracing of tracks to structure above.

B. Seismic Performance: Operable panel partitions are to withstand the effects of earthquake motions determined in accordance with ASCE/SEI 7.

1. The term "withstand" means "the partition panels will remain in place without separation of any parts when subjected to the seismic forces specified."

C. Acoustical Performance: Provide operable panel partitions tested by a qualified testing agency for the following acoustical properties in accordance with test methods indicated:

1. Sound-Transmission Requirements: Operable panel partition assembly tested for laboratory sound-transmission loss performance in accordance with ASTM E90, determined by ASTM E413, and rated for not less than the STC indicated.
2. Noise-Reduction Requirements: Operable panel partition assembly, identical to partition tested for STC, tested for sound-absorption performance in accordance with ASTM C423, and rated for not less than the NRC indicated.
3. Noise-Isolation Requirements: Installed operable panel partition assembly, identical to partition tested for STC, tested for NIC in accordance with ASTM E336, determined by ASTM E413, and rated for 10 dB less than STC value indicated.

2.2 OPERABLE ACOUSTICAL PANEL PARTITIONS

A. Operable Acoustical Panel Partitions: Partition system, including panels, seals, finish facing, suspension system, operators, and accessories.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. Modernfold, Inc, Acousti-Seal Encore
b. KWIK-WALL Company
c. Moderco Inc

B. Panel Operation: Manually operated, individual panels.

C. Panel Construction: As required to support panel from suspension components and with reinforcement for hardware attachment. Fabricate panels with tight hairline joints and concealed fasteners. Fabricate panels so finished in-place partition is rigid; level; plumb; aligned, with tight joints and uniform appearance; and free of bow, warp, twist, deformation, and surface and finish irregularities.

D. Dimensions: Fabricate operable acoustical panel partitions to form an assembled system of dimensions indicated and verified by field measurements.

1. Panel Width: Equal widths.

E. STC: Not less than 54.

F. Panel Weight: 10 lb./sq. ft. maximum.

G. Panel Thickness: Nominal dimension of 4 inches.

H. Panel finishes: Refer to section 2.4 below.

I. Panel Materials:

1. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.
2. Adhesives: Do not use adhesives that contain urea formaldehyde.
3. Adhesives: Use adhesives that comply with Section 01 57 19.11 “Indoor Air Quality Management.”
4. Composite Wood Products: Products shall be made without urea formaldehyde.
5. Composite Wood Products: Products shall comply with with Section 01 57 19.11 “Indoor Air Quality Management.”
6. Composite Wood Products: Products shall be made using ultra-low-emitting formaldehyde resins as defined in the California Air Resources Board's "Airborne Toxic Control Measure to Reduce Formaldehyde Emissions from Composite Wood Products" or shall be made with no added formaldehyde.

7. Composite Wood Products: Formaldehyde emission rates shall not be greater than the following when tested according to ASTM D6007 or ASTM E1333:

    a. Hardwood Plywood: 0.05 ppm.
    b. Particleboard: 0.09 ppm.
    c. MDF More Than 5/16 Inch (8 mm) Thick: 0.11 ppm.
    d. MDF 5/16 Inch (8 mm) or Less in Thickness: 0.13 ppm.

8. Composite Wood Products: Products shall be made without urea formaldehyde.
9. Steel Frame: Steel sheet, 0.0508-inch (16 gauge) nominal minimum thickness for uncoated steel.
10. Steel Face/Liner Sheets: Tension-leveled steel sheet, 0.0299-inch (21 gauge) minimum nominal thickness for uncoated steel.
11. Aluminum: Alloy and temper recommended by aluminum producer and finisher for type of use, corrosion resistance, and finish indicated; ASTM B221 for extrusions; manufacturer's standard strengths and thicknesses for type of use.

    a. Frame Reinforcement: Manufacturer's standard steel.

12. Gypsum Board: ASTM C1396/C1396M.
13. Cement Board: ASTM C1288.
14. Particleboard: ANSI A208.1.
15. Medium-Density Fiberboard: ANSI A208.2.
16. Plywood: DOC PS 1.
17. The interior wall and ceiling finishes in rooms and corridors shall be tested by an approved testing agency and shall comply with IBC Section 803 and Table 803.13.

J. Panel Closure: Manufacturer's standard unless otherwise indicated.

1. Initial Closure: Fixed jamb.
2. Final Closure: Constant-force, lever-operated mechanical closure expanding from panel edge to create a constant-pressure acoustical seal.

K. Hardware: Manufacturer's standard as required to operate operable panel partition and accessories; with decorative, protective finish.

1. Hinges: Full leaf butt hinges, attached directly to the panel frame with welded hinge anchor plates within panel to further support hinge mounting to frame. Lifetime warranty on hinges. Hinges mounted into panel edge or vertical astragal are not acceptable.

2.3 SEALS

A. Vertical Interlocking Sound Seals between panels: Roll-formed steel astragals, with reversible tongue and groove configuration in each panel edge for universal panel operation. Rigid plastic or aluminum astragals or astragals in only one panel edge are not acceptable.

B. Horizontal Top Seals: Continuous contact extruded vinyl bulb shape with pairs of non-contacting vinyl fingers to prevent distortion without the need for mechanically operated parts.

C. Horizontal Bottom Seals

1. SM2 – Manually activated seals providing nominal 2-inch (51 mm) operating clearance with an operating range of +1/2-inch (13 mm) to –1-1/2-inch (38 mm). Seal shall be operable from panel edge or face. Extended seal shall exert nominal 120 pounds (265 kg) downward force to the floor throughout operating range.

2.4 PANEL FINISH FACINGS

A. Description: Finish facings for panels that comply with indicated fire-test-response characteristics and that are factory applied to operable panel partitions with appropriate backing, using mildew-resistant nonstaining adhesive as recommended by facing manufacturer's written instructions.

1. Apply one-piece, seamless facings free of air bubbles, wrinkles, blisters, and other defects, with edges tightly butted with no gaps or overlaps. Horizontal seams are not permitted. Tightly secure and conceal raw and selvage edges of facing for finished appearance.
2. Where facings with directional or repeating patterns, directional weave, directional grain, or matching grain are indicated, mark facing top and attach facing in same direction.
3. Match facing pattern 72 inches above finished floor.

B. Vinyl on Lobby side: Manufacturer's standard, mildew-resistant, washable, vinyl wall covering; complying with WA-101, Type III-Heavy Duty; Class A.

1. Reinforced heavy-duty vinyl with woven backing weighing not less than 30 ounces (850 g) per lineal yard.
2. Antimicrobial Treatment: Additives capable of inhibiting growth of bacteria, fungi, and yeasts.
3. Basis of Design: Modernfold Vinyl Feather Grass

C. Fabric Wall Covering on Auditorium side: Manufacturer's standard fabric from same dye lot, treated to resist stains.

1. Basis of Design: Modernfold, Tek-Wall Ulster 009 Handprint

D. Cap-Trimmed Edges: Protective perimeter-edge trim with tight hairline joints concealing edges of panel and finish facing, finished as follows:

1. Steel, Painted: Finished with manufacturer's color as selected by Contracting Officer from manufacturer's full range.
2. Aluminum: Finished with manufacturer's standard dark bronze anodic finish.

E. Trimless Edges: Fabricate exposed panel edges so finish facing wraps uninterrupted around panel, covering edge and resulting in an installed partition with facing visible on vertical panel edges, without trim, for minimal sightlines at panel-to-panel joints.

2.5 SUSPENSION SYSTEMS

A. Suspension Tracks:

1. #17 Suspension System

a. Minimum 11-gage, 0.12-inch (3 mm) roll-formed steel track, suitable for either direct mounting to a wood header or supported by adjustable steel hanger brackets, supporting the load-bearing surface of the track, connected to structural support by pairs of 3/8-inch (9.5 mm) diameter threaded rods. Aluminum track is not acceptable.
b. Exposed track soffit: Steel, integral to track, and pre-painted off-white.

2. Carriers:

a. Right Angle Turn: Two carriers of low-friction polymer, reinforced with steel, that permit panels to traverse L, T, and X intersections without mechanical switching, on all panels except hinged closure panels.
b. “Smart Track”: Two all-steel trolleys with steel-tired ball bearing wheels. Non-steel tires are not acceptable. Suspension system shall provide automatic indexing of panels into stack area using preprogrammed switches and trolleys without electrical, pneumatic, or mechanical activation.

3. Warranty period: Five (5) years.

B. Track Intersections, Switches, and Accessories: As required for operation, storage, track configuration, and layout indicated for operable panel partitions, and compatible with partition assembly specified. Fabricate track intersections and switches from steel or aluminum.

1. Curve-and-Diverter Switches: Allow radius turns to divert panels to an auxiliary track.
2. L Intersections: Allow panels to change 90 degrees in direction of travel.
3. T Intersections: Allow panels to pass through or change 90 degrees to another direction of travel.
4. X Intersections: Allow panels to pass through or change travel direction full circle in 90-degree increments, and allow one partition to cross track of another.
5. Multidirectional Switches: Adjustable switch configuring track into L, T, or X intersections and allowing panels to be moved in all pass-through, 90-degree change, and cross-over travel direction combinations.
6. Center carrier stop.

C. Steel Finish: Manufacturer's standard, factory-applied, corrosion-resistant, protective coating unless otherwise indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine flooring, floor levelness, structural support, and opening, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of operable panel partitions.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF OPERABLE PANEL PARTITIONS

A. Install operable panel partitions and accessories after other finishing operations, including painting, have been completed in area of partition installation.

B. Install panels in numbered sequence indicated on Shop Drawings.

C. Broken, cracked, chipped, deformed, or unmatched panels are not acceptable.

D. Broken, cracked, deformed, or unmatched gasketing or gasketing with gaps at butted ends is not acceptable.

E. Light-Leakage Test: Illuminate one side of partition installation and observe vertical joints and top and bottom seals for voids. Adjust partitions for alignment and full closure of vertical joints and full closure along top and bottom seals. Perform test and make adjustments before NIC testing.

3.3 FIELD QUALITY CONTROL

A. NIC Testing: Contracting Officer a qualified testing agency to perform tests and inspections.

1. Testing Extent: Testing agency is to randomly select [one] operable panel partition installation(s) for testing.
2. Testing Methodology: Perform testing of installed operable panel partition for noise isolation in accordance with ASTM E336, determined by ASTM E413, and rated for not less than NIC indicated. Adjust and fit partitions to comply with NIC test method requirements.

B. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

C. An operable panel partition installation will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

3.4 ADJUSTING

A. Adjust operable panel partitions, hardware, and other moving parts to function smoothly, and lubricate as recommended by manufacturer.

B. Adjust to operate smoothly and easily, without binding or warping.

C. Verify that safety devices are properly functioning.

3.5 MAINTENANCE SERVICE

A. Maintenance Service: Beginning at Substantial Completion, maintenance service is to include 12 months' full maintenance by manufacturer's authorized service representative. Include quarterly preventive maintenance, repair or replacement of worn or defective components, lubrication, cleaning, and adjusting as required for proper operable-partition operation. Parts and supplies are to be manufacturer's authorized replacement parts and supplies.

3.6 DEMONSTRATION

A. Engage a factory-authorized service representative to train government maintenance personnel to adjust, operate, and maintain operable panel partitions.

END OF SECTION 10 22 39

SECTION 10 28 00 – TOILET ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Related Requirements:

1. Section 061000 "Rough Carpentry" for blocking.
2. Section 066116 “Solid Surface Fabrications” for solid surface wall panel coordination.
3. Section 10 21 13.19 “Plastic Toilet Compartments” for toilet partition coordination.
4. Division 22 for Plumbing fixture coordination.
5. Division 26 for Electrical fixture coordination.

1.2 COORDINATION

A. Coordinate accessory locations with other work to prevent interference with clearances required for access by people with disabilities, and for proper installation, adjustment, operation, cleaning, and servicing of accessories.

B. Coordinate mounting location and height requirements for accessories to ensure that toilet compartments are compliant with ABAAS accessibility requirements.

1.3 ACTION SUBMITTALS

A. Product Data: For each product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.
2. Include anchoring and mounting requirements, including requirements for cutouts in other work and substrate preparation.
3. Include electrical characteristics.

B. Samples: For each exposed product and for each finish specified, full size.

1. Approved full-size Samples will be returned and may be used in the Work.

C. Product Schedule: Indicating types, quantities, sizes, and installation locations by room of each accessory required.

1. Identify locations using room designations indicated.
2. Identify accessories using designations indicated.

1.4 INFORMATIONAL SUBMITTALS

A. Sample Warranty: For manufacturer's special warranties.

1.5 CLOSEOUT SUBMITTALS

A. Maintenance Data: For accessories to include in maintenance manuals.

B. Manufacturer's Warranties: For all products used.

1.6 WARRANTY

A. Manufacturer's Special Warranty for Mirrors: Manufacturer agrees to repair or replace mirrors that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, visible silver spoilage defects.
2. Warranty Period: 15 years from date of Substantial Completion.

B. Manufacturer's Special Warranty for Hand Dryers: Manufacturer agrees to repair or replace hand dryers that fail in materials or workmanship within specified warranty period.

1. Warranty Period:10 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Structural Performance: Design accessories and fasteners to comply with the following requirements:

1. Grab Bars: Installed units are able to resist 250 lb. concentrated load applied in any direction and at any point.

2.2 WASHROOM ACCESSORIES

A. Source Limitations: Obtain each type of public-use washroom accessory from single source from single manufacturer.

B. Toilet Tissue (Jumbo-Roll) Dispenser:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Bobrick Washroom Equipment, Inc. B-2892 or comparable product by one of the following:

a. ASI-American Specialties, Inc.
b. Bradley Corporation
c. Or Approved Equal.

2. Description: Two-roll unit with sliding panel to expose other roll.
3. Mounting: Surface mounted.

4. Capacity: 10-inch diameter rolls.
5. Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (stain).
6. Lockset: Tumbler type.
7. Refill Indicator: Pierced slots at front.

C. High-Speed Air Hand Dryer:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Dyson Airblade V or comparable product by one of the following:

    a. ASI-American Specialties, Inc.
    b. Bradley Corporation
    c. Or Approved Equal.

2. Description: High-speed, warm -air hand dryer for rapid hand drying.
3. Mounting is Wall mounted
4. Operation: Infrared-sensor activated with timed power cutoff switch.

    a. Automatic Shutoff: At 60 seconds.

5. Cover Material and Finish: Stainless steel material.
6. Electrical Requirements: 115 V, 13 A, 1000 W

D. Automatic Soap Dispenser:

1. Basis-of-Design Product: Subject to compliance with requirements, provide GOJO Foaming Soap Push Dispenser or comparable product by one of the following:

    a. ASI-American Specialties, Inc.
    b. Bradley Corporation
    c. Or Approved Equal.

2. Mounting: Wall mounted.
3. Capacity: 40-fl oz.
4. Materials: Stainless steel material.
5. Accessories: Wall mounted drip tray.

E. Paper Towel Dispenser:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Tork Matic Hand Towel Dispenser or comparable product by one of the following:

    a. Bobrick
    b. ASI-American Specialties, Inc.
    c. Bradley Corporation
    d. Or Approved Equal.

2. Mounting: Wall mounted
3. Materials: Stainless steel material.
4. Compatible towels: Hand towel rolls.

F. Grab Bar:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Bobrick Washroom Equipment, Inc or comparable product by one of the following:

    a. ASI-American Specialties, Inc.
    b. Bradley Corporation
    c. Or Approved Equal.

2. Mounting: Flanges with concealed fasteners.
3. Material: Stainless steel, 0.05 inch thick.

    a. Finish: Smooth, ASTM A480/A480M No. 4 finish (satin) on ends and slip-resistant texture in grip area.

4. OD: 1-1/2 inches.
5. Configuration and Length: Side, rear, and vertical grab bars as indicated on drawings.

G. Sanitary-Napkin Disposal Unit:

1. Basis-of-Design Product: Subject to compliance with requirements, Bobrick B-270 comparable product by one of the following:

    a. ASI-American Specialties, Inc.
    b. Bradley Corporation
    c. Or Approved Equal.

2. Mounting: Surface mounted.
3. Door or Cover: Self-closing, disposal-opening cover.
4. Receptacle: Removable.
5. Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (stain).

H. Hook:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Bobrick Washroom Equipment, Inc B-9542 Coat hook or comparable product by one of the following:

    a. ASI-American Specialties, Inc.
    b. Bradley Corporation
    c. Or Approved Equal.

2. Description: Combination door bumper and coat hook.
3. Mounting: Concealed.
4. Material and Finish: Stainless steel, ASTM A480/A480M No. 4 finish (stain).

I. Mirror:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Bobrick B-290 series or comparable product by one of the following:

    a. ASI-American Specialties, Inc.

b. Bradley Corporation
c. Or Approved Equal.

2. Mounting: Wall mounted
3. Materials: Stainless steel angle frame material.
4. Size: As selected by Contracting Officer.

J. Plumbing Below Sink P-Trap and Pipe Cover:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Trap Gear or comparable product by one of the following:

a. ASI-American Specialties, Inc.
b. Bradley Corporation
c. Or Approved Equal.

2. Materials: Soft ADA Trap Cover Slim-Fit Design, Molded PVC Finish
3. Fusion-bonded full length Velcro.

2.3 CHILDCARE ACCESSORIES

A. Source Limitations: Obtain each type of childcare accessory from single source from single manufacturer.

B. Diaper-Changing Station:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Koala Kare Products KB-310-SSWM or comparable product by one of the following:

a. ASI-American Specialties, Inc.
b. Diaper Deck & Co.
c. Or Approved Equal.

2. Description: Horizontal unit that opens by folding down from stored position and with child-protection strap.

a. Engineered to support minimum of 250 lb. static load when opened.

3. Mounting: Surface mounted, with unit projecting not more than 4 inches from wall when closed.
4. Operation: By pneumatic shock-absorbing mechanism.
5. Material and Finish: HDPE with plastic-laminate insert in color selected by Contracting Officer.
6. Liner Dispenser: Provide built-in dispenser for disposable sanitary liners.
7. Color: White Granite or as selected by Contracting Officer.

C. Source Limitations: Obtain custodial accessories from single source from single manufacturer.

2.4 ADULT CARE ACCESSORIES

A. Source Limitations: Obtain each type of accessory from single source from single manufacturer.

B. Adult-Changing Station:

1. Basis-of-Design Product: Subject to compliance with requirements, provide Foundations Model 100SSE-SM or comparable product by one of the following:

    a. ASI-American Specialties, Inc.
    b. Diaper Deck & Co.
    c. Or Approved Equal.

2. Description: Horizontal unit that opens by folding down from stored position and with smooth nylon belt.

    a. Engineered to support minimum of 400 lb. static load when opened.

3. Mounting: Surface mounted, with unit projecting not more than 4 inches from wall when closed .
4. Operation: By pneumatic shock-absorbing mechanism.
5. Material and Finish: Full body 16 gauge, 304 brushed stainless steel material, seamless welds.
6. Liner Dispenser: Provide built-in dispenser for disposable sanitary liners.

C. Source Limitations: Obtain custodial accessories from single source from single manufacturer.

2.5 MATERIALS

A. Stainless Steel: ASTM A240/A240M or ASTM A666, Type 304, 0.031-inch- minimum nominal thickness unless otherwise indicated.

B. Brass: ASTM B19, flat products; ASTM B16/B16M, rods, shapes, forgings, and flat products with finished edges; or ASTM B30, castings.

C. Steel Sheet: ASTM A1008/A1008M, Designation CS (cold rolled, commercial steel), 0.036-inch- minimum nominal thickness.

D. Galvanized-Steel Sheet: ASTM A653/A653M, with G60 hot-dip zinc coating.

E. Galvanized-Steel Mounting Devices: ASTM A153/A153M, hot-dip galvanized after fabrication.

F. Fasteners: Screws, bolts, and other devices of same material as accessory unit, unless otherwise recommended by manufacturer or specified in this Section, and tamper and theft resistant where exposed, and of stainless or galvanized steel where concealed.

G. Chrome Plating: ASTM B456, Service Condition Number SC 2 (moderate service).

H. Mirrors: ASTM C1503, Mirror Glazing Quality, clear-glass mirrors, nominal 6.0 mm thick.

2.6 FABRICATION

A. General: Fabricate units with tight seams and joints, and exposed edges rolled. Hang doors and access panels with full-length, continuous hinges. Equip units for concealed anchorage and with corrosion-resistant backing plates.

B. Keys: Provide universal keys for internal access to accessories for servicing and resupplying. Provide minimum of six keys to government representative.

PART 3 - EXECUTION

3.1 INSTALLATION OF TOILET ACCESSORIES

A. Install accessories in accordance with manufacturers' written instructions, using fasteners appropriate to substrate indicated and recommended by unit manufacturer. Install units level, plumb, and firmly anchored in locations and at heights indicated.

1. Remove temporary labels and protective coatings.

B. Grab Bars: Install to comply with specified structural-performance requirements.

3.2 ADJUSTING AND CLEANING

A. Adjust accessories for unencumbered, smooth operation. Replace damaged or defective items.

B. Clean and polish exposed surfaces in accordance with manufacturer's written instructions.

3.3 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 10 28 00

SECTION 10 44 13 – FIRE PROTECTION CABINETS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Fire-protection cabinet.

B. Related Requirements:

1. Section 06 10 00 “Rough Carpentry” for framing of recessed openings and blocking for attachment.
2. Section 09 29 00 “Gypsum Board” for adjacent wall finish.
3. Section 10 44 16 "Fire Extinguishers" for portable, hand-carried fire extinguishers accommodated by fire-protection cabinets.
4. Division 21 for Fire Suppression Equipment.
5. Division 28 for Electrical Safety and Security, including fire alarms.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of specified product.

1. Show door hardware, cabinet type, trim style, and panel style. Include roughing-in dimensions and details showing semirecessed- or surface-mounting method and relationships of box and trim to surrounding construction.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: For fire-protection cabinets.

1.4 COORDINATION

A. Coordinate size of fire-protection cabinets to ensure that type and capacity of fire extinguishers indicated are accommodated.

B. Coordinate sizes and locations of fire-protection cabinets with wall depths.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain fire-protection cabinets, accessories, and fire extinguishers from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Fire-Rated Fire-Protection Cabinets: Listed and labeled to comply with requirements in ASTM E814 for fire-resistance rating of walls where they are installed.

2.3 FIRE-PROTECTION CABINETS

A. Fire-Protection Cabinet:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Babcock-Davis
    b. Croker; a Division of Morris Group International
    c. Guardian Fire Equipment, Inc
    d. JL Industries; Activar Construction Products Group, Inc.
    e. Larsen's Manufacturing Company
    f. Nystrom, Inc.
    g. Potter Roemer LLC; a Division of Morris Group International
    h. Or Approved Equal.

2. Fire-Protection Cabinet Type: Suitable for fire extinguisher.
3. Cabinet Construction: Nonrated and One-hour fire rated.

    a. Fire-Rated Cabinets: Construct fire-rated cabinets with double walls fabricated from 0.043-inch- thick cold-rolled steel sheet lined with minimum 5/8-inch- thick fire-barrier material. Provide factory-drilled mounting holes.
    b. Fire-Rated Cabinets to be installed in 1-hour rated walls. Non-rated cabinets to be installed in non-rated walls.

4. Cabinet Material: Cold-rolled steel sheet.

    a. Shelf: Same metal and finish as cabinet.

5. Semirecessed Cabinet: One-piece combination trim and perimeter door frame overlapping surrounding wall surface, with exposed trim face and wall return at outer edge (backbend).

    a. Rolled-Edge Trim: 2-1/2-inch backbend depth.

6. Surface-Mounted Cabinet: Cabinet box fully exposed and mounted directly on wall with no trim.

7. Cabinet Trim Material: Steel sheet.
8. Door Material: Steel sheet.
9. Door Style: Fully glazed panel with frame Solid opaque panel with frame.
10. Door Glazing: Clear transparent acrylic sheet.
11. Door Hardware: Manufacturer's standard door-operating hardware of proper type for cabinet type, trim style, and door material and style indicated.

    a. Provide manufacturer's standard.
    b. Provide manufacturer's standard hinge, permitting door to open 180 degrees.

12. Accessories:

    a. Mounting Bracket: Manufacturer's standard steel, designed to secure fire extinguisher to fire-protection cabinet, of sizes required for types and capacities of fire extinguishers indicated, with plated or baked-enamel finish.
    b. Identification: Lettering complying with authorities having jurisdiction for letter style, size, spacing, and location.

        1) Identify fire extinguisher in fire-protection cabinet with the words "FIRE EXTINGUISHER."

            a) Location: Applied to cabinet door.
            b) Application Process: Application Process: Silk-screened, Decals, or Pressure-sensitive vinyl letters.
            c) Lettering Color: Red.
            d) Orientation: Vertical.

13. Materials:

    a. Cold-Rolled Steel: ASTM A1008/A1008M, Commercial Steel (CS), Type B.

        1) Finish: Baked enamel, TGIC polyester powder coat, HAA polyester powder coat, epoxy powder coat, or polyester/epoxy hybrid powder coat, complying with AAMA 2603.
        2) Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.
        3) Color: White.

    b. Tempered Float Glass: ASTM C1048, Kind FT, Condition A, Type I, Quality q3, 3 mm thick, Class 1 (clear).
    c. Transparent Acrylic Sheet: ASTM D4802, Category A-1 (cell-cast sheet), 6 mm thick, with Finish 1 (smooth or polished).

2.4 FABRICATION

A. Fire-Protection Cabinets: Provide manufacturer's standard box (tub) with trim, frame, door, and hardware to suit cabinet type, trim style, and door style indicated.

1. Weld joints and grind smooth.
2. Miter corners and grind smooth.

3. Provide factory-drilled mounting holes.
4. Prepare doors and frames to receive locks.
5. Install door locks at factory.

B. Cabinet Doors: Fabricate doors according to manufacturer's standards, from materials indicated and coordinated with cabinet types and trim styles.

1. Fabricate door frames with tubular stiles and rails and hollow-metal design, minimum 1/2 inch thick.
2. Fabricate door frames of one-piece construction with edges flanged.
3. Miter and weld perimeter door frames and grind smooth.

C. Cabinet Trim: Fabricate cabinet trim in one piece with corners mitered, welded, and ground smooth.

2.5 GENERAL FINISH REQUIREMENTS

A. Comply with NAAMM AMP 500for recommendations for applying and designating finishes.

B. Protect mechanical finishes on exposed surfaces of fire-protection cabinets from damage by applying a strippable, temporary protective covering before shipping.

C. Finish fire-protection cabinets after assembly.

D. Appearance of Finished Work: Noticeable variations in same piece are unacceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine roughing-in for cabinets to verify actual locations of adjacent conduit/piping before cabinet installation.

B. Examine walls and partitions for suitable framing depth and blocking where semirecessed cabinets will be installed.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Prepare recesses for semirecessed fire-protection cabinets as required by type and size of cabinet and trim style.

3.3 INSTALLATION OF FIRE-PROTECTION CABINETS

A. General: Install fire-protection cabinets in locations indicated.

1. Fire-Protection Cabinet Mounting Height: 42 inches above finished floor to top of fire extinguisher.

B. Fire-Protection Cabinets: Fasten cabinets to structure, square and plumb.

1. Provide inside latch and lock for break-glass panels.
2. Fasten mounting brackets to inside surface of fire-protection cabinets, square and plumb.
3. Fire-Rated Cabinets:
   a. Seal through penetrations with firestopping sealant as specified in Section 07 84 13 "Penetration Firestopping."

C. Identification:

1. Apply decals or vinyl lettering at locations indicated.

3.4 ADJUSTING

A. Remove temporary protective coverings and strippable films, if any, as fire-protection cabinets are installed unless otherwise indicated in manufacturer's written installation instructions.

B. Adjust fire-protection cabinet doors to operate easily without binding. Verify that integral locking devices operate properly.

3.5 CLEANING

A. On completion of fire-protection cabinet installation, clean interior and exterior surfaces as recommended by manufacturer.

B. Touch up marred finishes, or replace fire-protection cabinets that cannot be restored to factory-finished appearance. Use only materials and procedures recommended or furnished by fire-protection cabinet and mounting bracket manufacturers.

C. Replace fire-protection cabinets that have been damaged or have deteriorated beyond successful repair by finish touchup or similar minor repair procedures.

3.6 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 10 44 13

SECTION 10 44 16 – FIRE EXTINGUISHERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes portable, hand-carried fire extinguishers.

B. Related Requirements:

1. Section 10 44 13 "Fire Protection Cabinets."
2. Division 21 for Fire Suppression Equipment.
3. Division 28 for Electrical Safety and Security, including fire alarms.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product. Include rating and classification, material descriptions, dimensions of individual components and profiles, and finishes for fire extinguisher.

B. Submittals shall be reviewed by Authority Having Jurisdiction in addition to the Contracting Officer, to confirm compliance.

1.3 INFORMATIONAL SUBMITTALS

A. Warranty: Sample of special warranty.

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For fire extinguishers to include in maintenance manuals.

1.5 COORDINATION

A. Coordinate type and capacity of fire extinguishers with fire-protection cabinets to ensure fit and function.

1.6 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace fire extinguishers that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

    a. Failure of hydrostatic test according to NFPA 10 when testing interval required by NFPA 10 is within the warranty period.
    b. Faulty operation of valves or release levers.

2. Warranty Period: Six years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. NFPA Compliance: Fabricate and label fire extinguishers to comply with NFPA 10, "Portable Fire Extinguishers."

B. Fire Extinguishers: Listed and labeled for type, rating, and classification by an independent testing agency acceptable to authorities having jurisdiction.

1. Provide fire extinguishers approved, listed, and labeled by FM Global.

2.2 PORTABLE, HAND-CARRIED FIRE EXTINGUISHERS

A. Fire Extinguishers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Amerex Corporation
    b. Ansul; brand of Johnson Controls International plc, Building Solutions North America
    c. Babcock-Davis
    d. Badger Fire Protection; a Carrier company
    e. Croker; a Division of Morris Group International
    f. Guardian Fire Equipment, Inc
    g. JL Industries; Activar Construction Products Group, Inc.
    h. Larsen's Manufacturing Company
    i. Nystrom, Inc.
    j. Potter Roemer LLC; a Division of Morris Group International
    k. Pyro-Chem; brand of Johnson Controls International plc, Building Solutions North America
    l. Or Approved Equal.

2. Source Limitations: Obtain fire extinguishers, fire-protection cabinets, and accessories, from single source from single manufacturer.
3. Valves: Manufacturer's standard.
4. Handles and Levers: Manufacturer's standard.
5. Instruction Labels: Include pictorial marking system complying with NFPA 10, Appendix B.

B. Multipurpose (ABC) Dry-Chemical Type: UL-rated 10 lb. nominal capacity, with monoammonium phosphate-based dry chemical in manufacturer's standard enameled container.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine fire extinguishers for proper charging and tagging.

1. Remove and replace damaged, defective, or undercharged fire extinguishers.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. General: Install fire extinguishers in locations indicated and in compliance with requirements of authorities having jurisdiction.

END OF SECTION 10 44 16

SECTION 10 51 16 – WOOD LOCKERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Laminate-faced wood lockers.

B. Related Sections:

1. Section 06 10 00 “Rough Carpentry” for miscellaneous wood blocking and supports.
2. Section 06 20 23 “Interior Finish Carpentry” for adjacent wood base.
3. Section 09 29 00 “Gypsum Board” for adjacent wall material.
4. Section 09 68 13 “Tile Carpeting” for adjacent floor material.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for each type of wood locker.

a. Identify accessibility compliant features clearly.

B. Sustainable Design Submittals:

1. Environmental Product Declaration: For each product.
2. Product Data: For composite wood products, indicating that product contains no urea formaldehyde.
3. Product Data: For adhesives, indicating VOC content.
4. Laboratory Test Reports: For adhesives, indicating compliance with requirements for low-emitting materials.

C. Shop Drawings: For wood lockers.

1. Include plans, elevations, sections, and attachment details.
2. Show details full size.
3. Show locations and sizes of furring, blocking, and hanging strips, including concealed blocking and reinforcement specified in other Sections.
4. Show locations and sizes of cutouts and holes for items installed in lockers.
5. Show locker fillers, trim, base, sloping tops, and accessories.
6. Show locker identification system and numbering sequence.

D. Samples: For each type of the following:

1. Thermally fused laminate overlay panels

a. Manufacturer’s full range of colors
b. Physical samples to allow for accurate color/texture representation.

2. Exposed cabinet hardware and accessories, one unit for each type and finish.

E. Qualification Data: For Installer.

1.3 INFORMATIONAL SUBMITTALS

A. Sample Warranty: For special warranty.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance Data: For adjusting, repairing, and replacing locker doors and latching mechanisms to include in maintenance manuals.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Units of the following locker hardware items equal to 10 percent of amount installed for each type and finish installed, but no fewer than three (3) units:

a. Hinges.
b. Pulls.
c. Shelf rests.
d. Blank identification plates and holders.
e. Hooks.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Do not deliver lockers until painting and similar operations that could damage lockers have been completed in installation areas. If lockers must be stored in other-than-installation areas, store only in areas where environmental conditions are the same as those in final installation location, and comply with requirements specified in "Field Conditions" Article.

1.7 FIELD CONDITIONS

A. Environmental Limitations: Do not deliver or install wood lockers until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature between 60 and 90 deg F and relative humidity between 43 and 70 percent during the remainder of the construction period.

B. Field Measurements: Where lockers are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication, and indicate measurements on Shop Drawings.

1. Locate concealed framing, blocking, and reinforcements that support lockers by field measurements before being enclosed, and indicate measurements on Shop Drawings.

C. Established Dimensions: Where wood lockers are indicated to fit to other construction, establish dimensions for areas where lockers are to fit. Provide allowance for trimming at site, and coordinate construction to ensure that actual dimensions correspond to established dimensions.

1.8 COORDINATION

A. Coordinate sizes and locations of concealed wood support bases.

1. Requirements are specified in Section 06 10 00 "Rough Carpentry."

B. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of work specified in other Sections to ensure that wood lockers can be supported and installed as indicated.

1.9 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of lockers that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Structural failures.
b. Faulty operation of locks and other hardware.
c. Deterioration of wood, wood finishes, and other materials beyond normal use.

2. Warranty Period: Three years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Accessibility Standard: For lockers indicated to be accessible, comply with applicable provisions in the ABA standards of the Federal agency having jurisdiction and ICC A117.1.

2.2 LAMINATE-FACED WOOD LOCKERS

A. Basis of Design: Salsbury 15” Wide Premier Wood Locker

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Salsbury Industries
2. Club Resource Group
3. Famous Lockers

4. Hollman, Inc
5. Ideal Products, Inc
6. List Industries Inc.
7. Treeforms
8. Or Approved Equal.

C. Construction Style: Manufacturer's standard.

D. Final Assembly: Manufacturer's standard factory or knocked-down assembly.

E. Locker Size:

1. Width – 6 lockers across @ 15” per locker.
2. Height – 2 lockers high @ 36” per locker.
3. Depth – 1 locker deep @ 18” per locker.

F. Locker Body: Fabricated from industrial grade particleboard-core panels covered on both sides with Thermally fused laminate overlay.

1. Side Panels: Manufacturer's standard 3/4 inch thick.
2. Back Panel: Manufacturer's standard 1/2 inch thick.
3. Top Panel: Manufacturer's standard 3/4 inch thick.
4. Bottom Panel: Manufacturer's standard 3/4 inch thick.
5. Exposed Panel Edges: 1-mm-thick PVC.

G. Flush, Laminate-Faced Doors: Thermoset laminate veneer applied over 3/4-inch- thick, industrial grade particleboard core.

1. Thermally Fused Laminate Veneer
2. PVC Edges: 1 mm thick.

H. End Panels: Match style, material, construction, and finish of laminate-faced wood doors.

I. Shelves: Fabricated from industrial grade particleboard-core panels covered on both sides with thermally fused laminate overlay; adjustable.

1. Thickness: 3/4 inch.
2. Exposed Edges: 1-mm-thick PVC.

J. Corners and Filler Panels: 3/4-inch- thick panel. Match style, material, construction, and finish of wood-faced wood doors.

K. Continuous Finish Base:

1. Minimum 4” high.
2. Laminate-faced, 3/4-inch- thick panel that matches door faces.
3. Fabricated in lengths as long as practical to enclose base and base ends of lockers.

L. Factory finish wood lockers as specified in this Section. Defer only final touchup, cleaning, and polishing until after installation.

2.3 MATERIALS

A. Solid Wood: Clear hardwood lumber, selected for compatible grain and color.

B. Composite Wood: Provide materials that comply with requirements of referenced quality standard for each type of woodwork and quality grade specified unless otherwise indicated.

1. Thermally fused laminate Panels: Particleboard or medium-density fiberboard finished with thermally fused, melamine-impregnated decorative paper and complying with requirements of NEMA LD 3, Grade VGL, for Test Methods 3.3, 3.4, 3.6, 3.8, and 3.10.
2. Particleboard: ANSI A208.1, Grade M-2.
3. Softwood Plywood: DOC PS 1**,** medium-density overlay.
4. Veneer-Faced Panel Products (Hardwood Plywood): HPVA HP-1.

C. Adhesives: Do not use adhesives that contain urea formaldehyde.

D. Furring, Blocking, Shims, and Hanging Strips: Softwood or hardwood lumber, kiln dried to less than 15 percent moisture content.

E. Anchors: Material, type, size, and finish as required for each substrate for secure anchorage. Provide metal expansion sleeves or expansion bolts for post-installed anchors. Use nonferrous-metal or hot-dip galvanized anchors and inserts at inside face of exterior walls and at floors.

F. Wood Support Base: 2-by-6-inch nominal-size lumber treated with manufacturer's standard preservative-treatment, non-pressure process.

2.4 HARDWARE

A. Cam Padlock Hasp: Surface mounted, steel; finished to match other locker hardware.

B. Frameless Hinges (European Type): Fully concealed, self-closing, soft close hinge, nickel-plated steel, with not less than 125 degrees of opening.

1. Provide two hinges for doors 36 inches high and less.
2. Provide three hinges for doors more than 36 inches high.

C. Knobs: Metal; back mounted; 1-inch- diameter, flat round shape.

D. Accessible Handle: Metal, fixed, graspable lever handle and rose trim; surface mounted.

E. Shelf Rests: BHMA A156.9, B04013.

F. Hooks: Manufacturer's standard, ball-pointed aluminum or steel; finished to match other locker hardware. Attach hooks with at least two fasteners.

1. Provide one double-prong ceiling hook and two single-prong wall hooks for each compartment of double-tier lockers.

G. Exposed Hardware Finish: Satin nickel unless otherwise indicated.

2.5 ACCESSORIES

A. Provide adjustable shelf in bottom row of lockers to provide ABAAS accessibility compliance requirements.

1. Shelf shall be same construction as other shelves in the product.

2.6 FABRICATION

A. Fabricate each locker with shelves, an individual door and frame, an individual top, a bottom, and a back, and with common intermediate uprights separating compartments.

1. Fabricate lockers to dimensions, profiles, and details indicated.
2. Ease edges of corners of solid-wood members to 1/16-inch radius.

B. Fabricate lockers square, rigid, without warp, and with finished faces flat and free of dents, scratches, and chips. Accurately factory machine components for attachments. Make joints tight and true.

1. Fabricate lockers using manufacturer's standard construction, with joints made with dowels, dados, or rabbets. Dado side panels to receive shelving except where indicated to be adjustable.

C. Accessible Lockers: Fabricate as follows:

1. Locate bottom shelf no lower than 15 inches above the floor.
2. Where hooks, coat rods, or additional shelves are provided, locate no higher than 48 inches above the floor.

D. Number Identification Plates: Inlay number plates flush in each locker door, near top, centered.

E. Complete fabrication, including assembly, finishing, and hardware application, to maximum extent possible, before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.

1. Trial fit assemblies at fabrication shop that cannot be shipped completely assembled. Install dowels, screws, bolted connectors, and other fastening devices that can be removed after trial fitting. Verify that the parts fit as intended, and check measurements of assemblies against field measurements indicated on Shop Drawings before disassembling for shipment.
2. Use only manufacturer's nuts, bolts, screws, and other devices for assembly.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine walls and floors or support bases, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Verify that furring is attached to concrete and masonry walls that are to receive lockers.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Condition lockers to average prevailing humidity conditions in installation areas before installation.

B. Before installing lockers, examine factory-fabricated work for completeness and complete work as required, including removal of packing.

3.3 INSTALLATION

A. Install wood support base with 1/2-inch- thick, plywood top.

B. Knocked-Down Lockers: Assemble with manufacturer's standard fasteners, with no exposed fasteners on face frames.

C. Install lockers level, plumb, and true; use concealed shims.

D. Connect groups of lockers together with manufacturer's standard fasteners, through predrilled holes, with no exposed fasteners on face frames. Fit lockers accurately together to form flush, tight, hairline joints.

E. Install lockers without distortion so doors and drawers fit openings properly and are accurately aligned. Adjust hardware to center doors and drawers in openings, providing unencumbered operation. Complete installation of hardware and accessory items as indicated.

1. Installation Tolerance: No more than 1/8 inch in 96-inch sag, bow, or other variation from a straight line. Shim as required with concealed shims.
2. Maintain veneer sequence matching of wood-faced wood lockers.

F. Locker Anchorage:

1. Fasten lockers through wood locker base, at ends, and not more than 36 inches o.c. with No. 8 flush-head wood screws sized for 1-inch penetration into wood base.
2. Fasten lockers through back, near top and bottom, at ends with No. 8 flush-head wood screws sized for 1-inch penetration into wood framing, blocking, or furring and spaced not more than 16 inches o.c.

G. Scribe and cut corner and filler panels to fit adjoining work using fasteners concealed where practical. Repair damaged finish at cuts.

H. Install number identification plates after lockers are in place.

1. Attach number identification plate on each locker door, near top, centered, with at least two screws with finish matching the plate.

3.4 ADJUSTING

A. Clean, lubricate, and adjust hardware. Adjust doors to operate easily without binding.

3.5 PROTECTION

A. Protect lockers from damage, abuse, dust, dirt, stain, or paint. Do not permit use during construction.

B. Touch up marred finishes, or replace lockers that cannot be restored to factory-finished appearance. Use only materials and procedures recommended or furnished by locker manufacturer.

3.6 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 10 51 16

SECTION 10 75 16 – GROUND-SET FLAGPOLES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes ground-set flagpoles made from aluminum.

1. Delegated design.

B. Government-Furnished Material: Flags.

1.3 SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, operating characteristics, fittings, accessories, and finishes for flagpoles.

B. Shop Drawings: For flagpoles.

1. Include plans, elevations, and attachment details. Show general arrangement, jointing, fittings, accessories, grounding, anchoring, and support.
2. Include section, and details of foundation system.

C. Samples for Verification: For each type of exposed finish, in manufacturer's standard sizes.

D. Delegated-Design Submittal: For flagpoles.

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For flagpoles to include in operation and maintenance manuals.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Spiral wrap flagpoles with heavy paper and enclose in a hard fiber tube or other protective container.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain flagpoles as complete units, including fittings, accessories, bases, and anchorage devices, from single source from single manufacturer.

2.2 PERFORMANCE REQUIREMENTS

A. Delegated Design: Engage a qualified professional engineer, as defined in Section 01 40 00 "Quality Requirements," to design flagpole assemblies.

B. Seismic Performance: Flagpole assemblies shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

C. Structural Performance: Flagpole assemblies, including anchorages and supports, shall withstand design loads indicated within limits and under conditions indicated.

1. Wind Loads: Determine according to NAAMM FP 1001. Basic wind speed for Project location is indicated on Structural Drawings.
2. Base flagpole design on flags of maximum standard size suitable for use with flagpole or flag size indicated, whichever is more stringent.

2.3 ALUMINUM FLAGPOLES

A. Basis of Design: Flagpole Warehouse: Architectural Series Internal Halyard Flagpole 35’ Satin – EC35IH https://www.flagpolewarehouse.com/ or Model / Manufacturer provided from Park.

B. Aluminum Flagpoles: Cone-tapered flagpoles fabricated from seamless extruded tubing complying with ASTM B241/B241M, Alloy 6063, with a minimum wall thickness of 3/16 inch.

C. Exposed Height: 35 feet

D. Construct flagpoles in one piece if possible. If more than one piece is necessary, comply with the following:

1. Fabricate shop and field joints without using fasteners, screw collars, or lead calking.
2. Provide flush hairline joints using self-aligning, snug-fitting, internal sleeves.

E. Foundation Support: Provide one of the following.

1. Metal Foundation Tube: Manufacturer's standard corrugated-steel foundation tube, 0.060-inch wall thickness with 3/16-inch steel bottom plate and support plate; 3/4-inch-diameter, steel ground spike; and steel centering wedges welded together. Galvanize foundation tube after assembly. Furnish loose hardwood wedges at top of foundation tube for plumbing pole.

2. Sleeve for Aluminum Flagpole: Fiberglass or PVC pipe foundation sleeve, made to fit flagpole, for casting into concrete foundation.

F. Flashing Collar: Same material and finish as flagpole.

2.4 FITTINGS

A. Finial Ball: Flush-seam ball, sized as indicated or, if not indicated, to match flagpole-butt diameter.

1. 0.063-inch spun aluminum, finished to match flagpole.
2. Spun stainless steel, finished to match flagpole.

B. Internal Halyard, Cam Cleat System: 5/16-inch-diameter, braided polypropylene halyard; cam cleat; and concealed revolving truck assembly with plastic-coated counterweight and sling. Furnish flush access door secured with cylinder lock. Finish truck assembly to match flagpole.

1. Halyard Flag Snaps: Stainless-steel swivel snap hooks with neoprene or vinyl covers. Furnish two per halyard.
2. Plastic Halyard Flag Clips: Made from injection-molded, UV-stabilized, acetal resin (Delrin). Clips attach to flag and have two eyes for inserting both runs of halyards. Furnish two per halyard.

2.5 MISCELLANEOUS MATERIALS

A. Nonshrink, Nonmetallic Grout: Factory-packaged, nonstaining, noncorrosive, nongaseous grout complying with ASTM C1107/C1107M.

B. Drainage Material: Crushed stone, or crushed or uncrushed gravel; coarse aggregate.

C. Sand: ASTM C33/C33M, fine aggregate.

D. Elastomeric Joint Sealant: Multicomponent nonsag urethane joint sealant complying with requirements in Section 07 92 00 "Joint Sealants."

E. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D1187/D1187M.

2.6 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, AA-M12C22A31.

2.7 STAINLESS-STEEL FINISHES

A. Surface Preparation: Remove tool and die marks and stretch lines, or blend into finish.

B. Polished Finishes: Grind and polish surfaces to produce uniform finish, free of cross scratches.

1. When polishing is completed, passivate and rinse surfaces. Remove embedded foreign matter and leave surfaces chemically clean.
2. Directional Satin Finish: No. 4.

PART 3 - EXECUTION

3.1 PREPARATION

A. Prepare uncoated metal flagpoles that are set in foundation tubes by painting below-grade portions with a heavy coat of bituminous paint.

B. Foundation Excavation: Excavate to neat clean lines in undisturbed soil. Remove loose soil and foreign matter from excavation and moisten earth before placing concrete. Place and compact drainage material at excavation bottom.

C. Provide forms where required due to unstable soil conditions and for perimeter of flagpole base at grade. Secure and brace forms to prevent displacement during concreting.

D. Foundation Tube: Place foundation tube, center, and brace to prevent displacement during concreting. Place concrete. Plumb and level foundation tube and allow concrete to cure.

E. Sleeves: Locate and secure sleeves in forms by bracing to reinforcement and forms.

F. Anchor Bolts: Locate and secure anchor bolts in forms with templates and by tying to reinforcement.

G. Place concrete, as specified in Section 03 30 00 "Cast-in-Place Concrete." Compact concrete in place by using vibrators. Moist-cure exposed concrete for no fewer than seven days or use nonstaining curing compound.

H. Trowel exposed concrete surfaces to a smooth, dense finish, free of trowel marks, and uniform in texture and appearance. Provide positive slope for water runoff to perimeter of concrete base.

3.2 FLAGPOLE INSTALLATION

A. General: Install flagpoles where indicated and according to Shop Drawings and manufacturer's written instructions.

B. Foundation Tube: Place flagpole in tube, seated on bottom plate between steel centering wedges, and install hardwood wedges to secure flagpole in place. Place and compact sand in foundation tube and remove hardwood wedges. Seal top of foundation tube with a 2-inch layer of elastomeric joint sealant and cover with flashing collar.

C. Baseplate: Cast anchor bolts in concrete foundation. Install baseplate on washers placed over leveling nuts on anchor bolts and adjust until flagpole is plumb. After flagpole is plumb, tighten retaining nuts and fill space under baseplate solidly with nonshrink, nonmetallic grout. Finish exposed grout surfaces smooth and slope 45 degrees away from edges of baseplate.

END OF SECTION 10 75 16

SECTION 10 81 19 – RODENT AND BIRD CONTROL

PART 1 - GENERAL

1.1 DESCRIPTION

A. This Section consists of abating health hazards associated with rodents and birds and furnishing, installing, and maintaining barriers to protect from rodent/bird entry or intrusions during planned historic rehabilitation of the Effigy Mounds National Monument Visitor's Center in Harpers Ferry, Iowa (EFMO #318539), (Project Site).

B. The Project Site had evidence of rodent and insect instruction, including rodent feces and insect nests, per Landmark's January 31, 2025 Hazardous Material Survey Report for Effigy Mounds National Monument Visitor's Center. Any rodent or bird intrusion removal and exclusion work should be performed by a licensed and insured pest control contractor.

C. All removal of rodent feces, urine, and nesting materials must be conducted in accordance with the National Park Service's (NPS) *Hantavirus Risk Reduction Worker Protection* guidelines and requirements.

D. All removal of bird feces, urine, and nesting materials must be conducted in accordance with the *Histoplasmosis Protecting Workers from Risk* guidance document produced by National Institute for Occupational Safety and Health (NIOSH).

E. Locations on the exterior building envelope that are open and allow rodent and bird species to enter the interior of the Project Site will require treatment to exclude these animals. This project will also require the disinfection of surfaces contaminated with rodent feces and bird nesting areas prior to the start of work that will disturb these areas.

F. Items to be protected from animal entry include but are not limited to, interior ceilings, base trim, window and door trim, walls, floors, and other materials installed or replaced according to project requirements.

G. Neoprene seals, spray-in-place foam, and similar products commonly used to close openings are not rodent-proof. Even in new buildings, utility pipes, electrical conduit (often at meters or circuit breaker panels), water and gas lines, and communication cables may have large openings that permit the entry of rodents and birds. Once animals have entered walls, they generally have ready access to much of a building via holes for utility pipes and wires in the framing, overhead ceilings, beneath raised floors, or other types of construction adjacent to utility enclosures.

H. The Contractor will maintain a clean and debris-free Project Site. If the areas are known to have serious rodent infiltration problems, the Contractor will use exterior surface materials that have a hard, smooth surface on at least the lower levels that will be subject to contact with the ground surface, ornamental plantings, fences, and other potential areas of rodent harborage.

I. The Contractor will tightly rodent and bird-proof any electrical penetrations associated with solar panels that may be installed for this building and openings associated with other potential planned utility alterations.

J. During construction, the Contractor will implement measures to keep the job site free of rubbish, foodstuff, and other items as necessary to avoid attracting animals and will trap/bait rodents as necessary to control rodent intrusion.

1.2 REFERENCE DOCUMENTATION

A. All work will also comply with the following:

1. National Park Service, *Rodent Exclusion Manual,* Published by the U.S. Department of the Interior, National Park Service, Natural Resource Stewardship and Science: Fort Collins, CO, September 2014.
2. National Park Service, *Hantavirus Risk Reduction Worker Protection* – Revised Edition, December 2013.
3. National Park Service, *Rodent Exclusion Manual*, September 2014.
4. National Institute for Occupational Safety and Health, Histoplasmosis: *Protecting Workers at Risk* – Revised Edition, December 2004.

1.3 FIELD VERIFICATION

A. Field-verify all new and existing dimensions affecting the work of this contract before ordering products.

1.4 CONSTRUCTION MATERIALS

A. The Government will not furnish any materials for this project.

1.5 PROJECT CONDITIONS

A. Do not damage any existing building materials not associated with the work or work area. Any damage to building materials not protected in or outside the work areas and not associated with the work will be repaired or replaced by the Contractor at no cost to the Government.

B. The Government will supply electricity and water service to the Project Site. The Contractor will provide the necessary temporary utility connections and the water source needed to complete the scope of work. If any generators are provided by the Contractor for temporary electric service, a written Spill Prevention, Control, and Countermeasure (SPCC) plan and equipment will also be required. The Contractor will be required to operate generators at noise levels that do not exceed applicable regulatory limits.

PART 2 - PRODUCTS

2.1 PERSONAL PROTECTIVE EQUIPMENT

A. Respiratory Protection:

1. Workers who abate rodent droppings and/or bird nesting material wastes will be provided with personally issued and individually identified respirators that meet the required level of protection. During the pre-cleaning, installation of critical barriers, and High-Efficiency Particulate Air (HEPA) vacuuming activities, the Contractor will utilize (at a minimum) half-face air-purifying respirators with approved HEPA filter cartridges. At a minimum, the Contractor will also utilize the half-face air-purifying respirators with approved HEPA-filter cartridges for the removal and cleanup work. The respiratory cartridges used will be, at a minimum, N-100 or P-100 filters.
2. The Contractor will provide workers with, and require the use of, respirators approved by the NIOSH in accordance with the OSHA Standard 29 Code of Federal Regulations (CFR) 1926.1101. Minimum respiratory protection will be an approved half-mask air-purifying respirator with HEPA cartridges. Disposable single-use respirators will not be permitted. The initial respiratory protection provided will be chosen prior to the commencement of removal operations based on data from past exposure assessment monitoring or the initial monitoring performed to accurately determine the airborne concentration of dust/fibers to which employees may be exposed.
3. Workers must perform positive and negative air pressure fit tests each time a respirator is donned and whenever the respirator design permits. Powered air-purifying respirators will be tested for adequate flow as specified by the manufacturer.
4. Workers will be given a qualitative fit test in accordance with procedures detailed in the OSHA 29 CFR 1926.1101 requirements for all respirators to be used. An appropriately administered quantitative fit test may be substituted for a qualitative fit test. Documentation of adequate respirator fit must be provided to the Contractor. Fit tests will be administered annually.

B. Personal Protection Required for Rodent Removal/Rodent-Contaminated Material Handling:

1. Prior to the commencement of cleaning of animal-infested areas, all personnel who will be required to enter the work areas for handling animal-infested and contaminated materials will have hazard awareness training. Special onsite training on equipment and procedures unique to this job site will be performed as required. Training in emergency response and evacuation procedures will also be provided.
2. All respiratory protection will be provided to the workers through a written respiratory protection program. This program will be available at the Project Site.
3. Disposable clothing, including head, foot, and full body protection, will be provided in sufficient quantities and in adequate sizes for all workers and authorized visitors. All personnel engaged in the animal removal work will wear approved disposable protective clothing constructed of spun-bonded olefin, polypropylene fabrics, or other material of equivalent resistance to penetration. A full-body suit is recommended instead of a separate

set of coveralls, head covers, and shoe covers. Disposable whole-body clothing, including head covers, gloves, and shoe coverings, will be provided to and worn by all personnel in the abatement work area. If elastic sleeve closures are not provided, sleeves will be secured with duct tape to the gloves. Washable footwear having a non-skid tracking surface will be provided and used by all personnel within the abatement work area.

4. Persons with facial hair that may interfere with the respirator's seal will not be allowed to enter the work area.
5. Respirators will be worn at all times if animal-contaminated materials are impacted to the extent that airborne emissions are created.
6. Protective eyewear, gloves, rubber boots, and/or other footwear will be provided as required for workers and authorized visitors. Safety shoes may be required for some activities.
7. Eating, drinking, smoking, and chewing gum or tobacco will not be allowed in the work areas.

2.2 EQUIPMENT

A. The materials utilized may be new or used but will be suitable for the intended purpose. Rodent exclusion barriers will be structurally adequate, neat in appearance, and consistent with historical preservation or the historical fabric on or within the structure.

2.3 SCREENS AND BARRIERS

A. Use only metal window screening materials where windows or doors are accessible to animals. Avoid unnecessary ledges outside of the windows. When necessary, screen ventilation openings and windows with a woven/welded galvanized hardware cloth. Such screening work is critical in a building where high animal pressures are found. For large openings or where the screen may be subject to abuse, add crossbars to support the hardware cloth. If the opening is an access route, install the screen on a hinged frame.

2.4 DISINFECTING SOLUTIONS

A. Two types of disinfecting solutions are recommended for the cleanup of animal-contaminated materials:

1. General-Purpose Household Disinfectant: Prepare according to the label if not prediluted. Almost any agent commercially available in the U.S. is sufficient as long as the label states that it is a disinfectant product. Effective agents include those based on phenols, quaternary ammonium compounds, and hypochlorite.
2. Hypochlorite Solution: A chlorine solution that is freshly prepared by mixing 1½ cups of household bleach in one gallon of water (or a 1:10 solution) can be used in place of a commercial disinfectant product. When using a chlorine solution, avoid spilling the mixture on clothing or other items that might be damaged by bleach. Wear rubber, latex,

vinyl, or nitrile gloves when preparing and using chlorine solutions. Chlorine solutions should be prepared fresh daily.

PART 3 - EXECUTION

3.1 PROTECTION OF STRUCTURES

A. Rodents and birds seek shelter behind, under, or in appliances, sinks, cabinets, drawers, stored goods, wall voids, false ceilings, and other undisturbed areas.

B. Rodents can climb the outside of vertical pipes and conduits up to 3 inches (7.6 cm) in diameter and the outsides of larger pipes attached to buildings by bracing themselves between the wall and pipe.

C. Rodents can also climb the inside of vertical pipes, wall voids, or structural/expansion seams and joints between 1.5 and 4 inches (3.8 and 10.2 cm) in diameter. All building installations must include systems that prevent these intrusions.

D. Junctures, where utilities (i.e., pipes, cables, etc.) enter the structure, require special consideration in preventing rodent entry.

E. Exclusion must include no exterior openings higher or wider than one-quarter inch.

F. Close openings around pipes and electric cables where they enter structures with Portland cement mortar, Concrete Patch®, masonry, or metal collars.

3.2 RODENT REMOVAL

A. Completely remove rodents when approved by the Contracting Officer. Do not use poisons due to the potential for negative environmental or health effects.

B. The Federal Migratory Bird Act from the U.S. Fish & Wildlife Service forbids Contractors from moving or destroying a nest that has eggs, chicks, or young birds in it. If live birds and eggs are present at the Project Site, the Contracting Officer needs to be informed, and the Contractor must seek approval from the NPS to cautiously work around any live animals and their offspring/eggs.

3.3 CLEANUP OF ANIMAL URINE, DROPPINGS, AND CONTAMINATED SURFACES

A. Prior to cleaning areas such as sheds, outbuildings, or attics, ventilate the area by opening doors or windows to the outdoors for at least 30 minutes. Provide cross ventilation where possible.

B. Utilize appropriate Personal Protective Equipment (PPE).

C. Spray all animal urine and the droppings with a disinfectant or chlorine solution until thoroughly soaked.

D. To avoid generating potentially infectious aerosols, do not vacuum or sweep the animal urine, droppings, or contaminated surfaces until they have been disinfected.

E. Use a paper towel to absorb the animal urine and pick up the droppings. Place the paper towel in the garbage.

F. After animal droppings and urine have been removed, disinfect items that might have been contaminated by animals or their urine and droppings.

1. Mop floors with a disinfectant or chlorine solution.
2. Disinfect countertops, cabinets, drawers, and other durable surfaces with a disinfectant or chlorine solution.
3. Disinfect carpets with a disinfectant or commercial-grade steam cleaner or shampoo.
4. Steam-clean or shampoo rugs and upholstered furniture.
5. Leave books, papers, and other items that cannot be thrown away or cleaned with a liquid disinfectant outdoors in the sunlight for several hours or in an indoor area free of rodents for approximately one week before cleanup. After that time, the virus should no longer be infectious. However, to further reduce risk, wear rubber, latex, vinyl, or nitrile gloves and wipe the items with a cloth moistened with disinfectant.

3.4 CLEAN UP OF DEAD ANIMALS AND NESTS

A. Utilize appropriate PPE.

B. Use insect repellent (containing DEET) on clothing, socks, and arms to reduce the risk of fleabites that might transmit the plague.

C. Spray dead animals and nests with a disinfectant or a chlorine solution, soaking them thoroughly. Wait 10 minutes before disturbing to ensure the inactivation of the virus.

D. Place the dead animal or nest in a plastic bag or remove the dead rodent from the trap and place it within a plastic bag. When the cleanup is complete (or when the bag is full), seal the bag, place it into a second plastic bag, and seal the second bag. Dispose of the material in the double bag by burning it or discarding it in a covered trash can that is regularly emptied.

E. Clean up the surrounding area(s) as described in Section 3.3 – Cleanup of Animal Urine and Droppings and Contaminated Surfaces above.

3.5 VERIFICATION OF RODENT EXCLUSION ACTIVITIES

A. The Contractor will procure the services of a third-party rodent and bird exclusion specialist to verify the animal exclusion activities.

B. The selected third-party specialist will document the measures taken to prevent animals from entering the Project Site.

END OF SECTION 10 81 19

SECTION 11 00 00 – REQUIREMENTS FOR MISCELLANEOUS EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Art Hangers: To be installed inside and outside of the building, see drawings for locations.
2. Option D: Dual viewers on extended balcony.

B. Related Requirements:

1. Section 03 30 00 “Cast in Place Concrete” for balcony floor structure, top.
2. Section 05 31 00 “Steel Decking” for balcony floor structure, bottom.
3. Section 07 46 16 “Aluminum Cladding” for art hanger substrate at exterior installation.
4. Section 09 29 00 “Gypsum Board” for art hanger substrate at interior installation.

1.2 SUBMITTALS

A. Product Data for all equipment.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: For all equipment.

B. Manufacturer’s Warranty: For all equipment.

PART 2 - PRODUCTS

2.1 ART HANGERS

A. Basis of Design: Gallery System, GalleryOne art hanging system

B. Salient Characteristics:

1. Stainless steel track, approximately 3/8” wide, 3/4" tall, with opening for track at bottom edge. Grey plastic end caps.
2. Stainless steel cable hangers, 6’-6” long, with track heads designed to attach to stainless steel track.

2.2 DUAL VIEWER

A. Basis of Design: Hi Spy Viewing, Dual Viewer

B. Salient Characteristics:

1. Long distance viewer with 20x binocular function
2. Style: Dual
3. Two swiveling heads, one mounted at ABA compliant height (3 feet 9 inches AFF), one mounted at standing height (5 feet AFF)
4. Construction: Aluminum
5. Finish: Powder coated
6. Color: Green, to be confirmed by Contracting Officer from manufacturer's full color selection.
7. Coin Acceptance: Non-coin
8. Waterproof Optics
9. Support (Main Tube) Diameter: 6 inches minimum, 0.1875 inch wall thickness minimum
10. Height of ABA compliant arm: 29 inches
11. Orientation of Arm: From perspective of the person standing using the upright viewer, ABA head is to their right.
12. Horizontal Movement: Tapered roller bearing system, 360 degrees.
13. Vertical Movement: Bronze bearing on aluminum trunnion, 30 degrees up, 30 degrees down.
14. Viewing Head Security: Pop lock over security bolt.

C. Warranty: 5 years minimum on all parts.

PART 3 - EXECUTION

3.1 PRE-INSTALLATION

A. Confirm slab structure and traffic coating installation are complete prior to installation to avoid finishing around equipment.

B. Confirm condition of mounting location and that structure to be attached to are acceptable to installer prior to installation. Installation of equipment indicates acceptance of substrate.

3.2 INSTALLATION

A. Per manufacturer's written instructions.

3.3 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 11 00 00

SECTION 11 30 13 – RESIDENTIAL APPLIANCES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Refrigerator

B. Related Requirements:

1. Div. 22 Plumbing for related water hookup.
2. Div. 26 Electrical and Lighting for related power hookup.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include installation details, material descriptions, dimensions of individual components, and finishes for each appliance.
2. Include rated capacities, operating characteristics, electrical characteristics, and furnished accessories.

B. Sustainable Design Submittals:

1. Product Data: For indicated products, indicating compliance with requirements for ENERGY STAR product labeling.

1.3 INFORMATIONAL SUBMITTALS

A. Qualification Data: For manufacturer.

B. Product Certificates: For each type of appliance.

C. Field quality-control reports.

D. Sample Warranties: For manufacturers' special warranties.

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For each residential appliance to include in operation and maintenance manuals.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Maintains, within 50 miles of Project site, a service center capable of providing training, parts, and emergency maintenance repairs.

1.6 WARRANTY

A. Refrigerator/Freezer, Freezer, and Icemaker, Sealed System: Full warranty, including parts and labor for on-site service on the product.

1. Warranty Period for Sealed Refrigeration System: Five years from date of Substantial Completion.
2. Warranty Period for Other Components: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Electrical Appliances: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Accessibility: Comply with applicable provisions in the ABA standards of the Federal agency having jurisdiction and ICC A117.1.

2.2 REFRIGERATION APPLIANCES

A. Refrigerator/Freezer: Two-door refrigerator/freezer with freezer on bottom and complying with AHAM HRF-1.

1. Basis of Design: GE Energy Star 17.5 Cu. Ft. Counter Depth French Door Refrigerator
2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Amana; Whirlpool Corporation
    b. Bosch US; BSH Home Appliances Corporation
    c. Frigidaire
    d. GE Appliances; Haier Group
    e. Jennair; Whirlpool Corporation
    f. Kenmore; Transformco SR Brands LLC
    g. KitchenAid; Whirlpool Corporation
    h. LG Electronics USA, Inc.; LG Electronics Inc.
    i. Maytag; Whirlpool Corporation
    j. Miele
    k. Samsung Electronics America, Inc. (SEA); Samsung Electronics Co., Ltd. (SEC)
    l. Thermador; BSH Home Appliances Corporation
    m. Viking Range, LLC; Middleby Corporation
    n. Whirlpool Corporation
    o. Or Approved Equal

3. Type: Freestanding.
4. Dimensions:

    a. Width: 33 inches maximum.
    b. Depth: 31 inches.
    c. Height: 70 inches maximum.

5. Storage Capacity:

    a. Refrigeration Compartment Volume: 12.5 cu. ft.
    b. Freezer Volume: 5.01 cu. ft. maximum.
    c. Shelf Area: Five adjustable glass shelves minimum.

6. General Features:

    a. ABAAS/ADA Compliant
    b. Dispenser in door for ice.
    c. Built-in water-filtration system.
    d. Dual refrigeration systems.
    e. Separate temperature controls for each compartment.
    f. Temperature controls shall be within ABAAS compliant reach range.
    g. Handles shall protrude from face of door/drawer. No recessed handles.

7. Refrigerator Features:

    a. Interior light in refrigeration compartment.
    b. Compartment Storage: vegetable crisper and meat compartment.
    c. Door Storage: Clear compartments and 1 gal. milk-container storage.
    d. Temperature-controlled meat/deli bin.
    e. Shelves and drawers shall be clear.
    f. Layout shall be adjustable.

8. Freezer Features: One freezer compartment configured as pull-out drawer.

    a. Automatic defrost.
    b. Interior light in freezer compartment.
    c. Automatic icemaker and storage bin.
    d. Interior drawers shall divide storage such that upper level of drawers shall be within ABAAS compliant reach range.
    e. Interior drawers shall be clear or coated wire.

9. Energy Star: Provide appliances that qualify for the EPA/DOE Energy Star product-labeling program.
10. Color/Finish: Manufacturer's standard – Fingerprint resistant stainless steel or white.

2.3 GENERAL FINISH REQUIREMENTS

A. Protect mechanical finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, power connections, and other conditions affecting installation and performance of residential appliances.

B. Examine roughing-in for piping systems to verify actual locations of piping connections before appliance installation.

C. Examine walls, ceilings, and roofs for suitable conditions where refrigerator will be installed.

D. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

E. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF RESIDENTIAL APPLIANCES

A. Install appliances according to manufacturer's written instructions.

B. Place units in final locations after finishes have been completed in each area. Verify that clearances are adequate to properly operate equipment.

3.3 FIELD QUALITY CONTROL

A. Tests and Inspections: Perform the following tests and inspections with the assistance of a factory-authorized service representative:

1. Perform visual, mechanical, and electrical inspection and testing for each appliance according to manufacturers' written recommendations. Certify compliance with each manufacturer's appliance-performance parameters.
2. Leak Test: After installation, test for leaks. Repair leaks and retest until no leaks exist.
3. Operational Test: After installation, start units to confirm proper operation.
4. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and components.

B. An appliance will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

3.4 DEMONSTRATION

A. Engage a factory-authorized service representative to train Government's maintenance personnel to adjust, operate, and maintain residential appliances.

END OF SECTION 11 30 13

SECTION 12 24 13 – ROLLER WINDOW SHADES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Manually operated, single-roller shades as indicated on Drawings.
2. Manually operated, blackout double-roller shades at the Auditorium space.

B. Related Requirements:

1. Section 061000 "Rough Carpentry" for wood blocking and grounds for mounting roller shades and accessories.
2. Section 079200 "Joint Sealants" for sealing the perimeters of installation accessories for light-blocking shades with a sealant.
3. Section 084113 “Aluminum-Framed Entrances and Storefronts” for coordination with windows.
4. Section 092900 “Gypsum Board” for coordination with adjacent gypsum board walls, ceilings, and accessories.

1.2 ACTION SUBMITTALS

A. Product Data:

1. For each type of product.

a. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

B. Shop Drawings:

1. Show fabrication and installation details for roller shades, including shadeband materials, their orientation to rollers, and their seam and batten locations.

C. Samples for Verification: Actual sample of finished products for each type of roller shade including actual shadeband material and installation accessories.

1. Size: Manufacturers' standard size.

D. Product Schedule: For fascia mounted roller shades.

E. Sustainable Design Submittals:

1. Provide GREENGUARD certification with documentation for no-VOC roller shades are eco-friendly window coverings made from materials with low or no Volatile Organic Compounds (VOCs).

1.3 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For roller shades.

B. Manufacturer's Warranty: For all products used.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: Fabricator of products.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver roller shades in factory packages, marked with manufacturer, product name, and location of installation using same designations indicated on Drawings.

1.6 FIELD CONDITIONS

A. Environmental Limitations: Do not install roller shades until construction and finish work in spaces, including painting, is complete and dry and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.

B. Field Measurements: Where roller shades are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Allow clearances for operating hardware of operable glazed units through entire operating range. Notify Architect of installation conditions that vary from Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

PART 2 - PRODUCTS

2.1 SOURCE LIMITATIONS

A. Obtain roller shades from single source from single manufacturer.

2.2 MANUALLY OPERATED ROLLER WINDOW SHADES

A. Manually Operated, Single-Roller Shades: For interior use in rectangular openings.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Draper, Inc.
b. Hunter Douglas Architectural Window Coverings
c. Lafayette Interior Fashions
d. Legrand Shading Systems; Legrand North America, LLC
e. Levolor Inc.

f. Lutron Electronics Co., Inc
g. MechoShade Systems, LLC
h. Or Approved Equal

2. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

   a. Roller Drive-End Location: Right side of interior face of shade.
   b. Direction of Shadeband Roll: Regular, from back (exterior face) of roller.
   c. Shadeband-to-Roller Attachment: Removable spline fitting into integral channel in tube.

3. Roller-Coupling Assemblies: Coordinated with operating mechanism and designed to join up to six inline rollers into a multiband shade that is operated by one roller drive-end assembly.
4. Mounting Hardware: Brackets or endcaps, with endcap covers where exposed, corrosion resistant and compatible with roller assembly, operating mechanism, installation accessories, and installation location and conditions indicated.
5. Installation Accessories:

   a. Front Fascia: Aluminum extrusion that conceals front and underside of roller and shadeband assembly and attaches to roller endcaps without exposed fasteners.

      1) Shape: L-shaped.
      2) Height: Manufacturer's standard height required to conceal roller and shadeband assembly when shade is fully open, but not less than 4 inches.

   b. Exposed Headbox: Extruded-aluminum enclosure including front fascia, L-shaped top/back cover, and removable bottom closure, that fully conceals roller and shadeband assembly, and attaches to roller endcaps without exposed fasteners.

      1) Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than 4 inches.

   c. Installation Accessories Color and Finish: Anodized Aluminum, Dark Bronze.

B. Manually Operated, Double-Roller Shades: For interior use in rectangular openings.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

   a. Draper, Inc.
   b. Hunter Douglas Architectural Window Coverings
   c. Lafayette Interior Fashions
   d. Legrand Shading Systems; Legrand North America, LLC
   e. Levolor Inc.
   f. Lutron Electronics Co., Inc

g. MechoShade Systems, LLC
h. Or Approved Equal

2. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

   a. Double-Roller Mounting Configuration: Offset, outside roller over and inside roller under.
   b. Inside Roller:

      1) Drive-End Location: Right side of interior face of shade.
      2) Direction of Shadeband Roll: Regular, from back (exterior face) of roller.

   c. Outside Roller:

      1) Drive-End Location: Right side of interior face of shade.
      2) Direction of Shadeband Roll: Regular, from back (exterior face) of roller.

   d. Shadeband-to-Roller Attachment: Manufacturer's standard method.

3. Roller-Coupling Assemblies: Coordinated with operating mechanism and designed to join up to six inline rollers into a multiband shade that is operated by one roller drive-end assembly.
4. Mounting Hardware: Double roller brackets or endcaps, with endcap covers where exposed, corrosion resistant and compatible with roller mounting configuration, roller assemblies, operating mechanisms, installation accessories, and installation location and conditions indicated.
5. Installation Accessories:

   a. Front Fascia: Aluminum extrusion that conceals front and underside of roller and shadeband assembly and attaches to roller endcaps without exposed fasteners.

      1) Shape: L-shaped.
      2) Height: Manufacturer's standard height required to conceal roller and shadeband assembly when shade is fully open, but not less than 4 inches.

   b. Exposed Headbox: Extruded-aluminum enclosure including front fascia, L-shaped top/back cover, and removable bottom closure, that fully conceals roller and shadeband assembly, and attaches to roller endcaps without exposed fasteners.

      1) Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than 4 inches.

   c. Side Channels: Extruded aluminum "'U" channel with light seals, designed to eliminate light gaps at sides of shades as shades are drawn down. Provide side channels with shadeband guides or other means of aligning shadebands with channels at tops.

d. Bottom (Sill) Channel or Angle: Extruded aluminum "U" channel or "L" angle with light seals, designed to eliminate light gaps at bottoms of shades when shades are closed.
e. Installation Accessories Color and Finish: Anodized Aluminum, Dark Bronze.

2.3 MANUAL OPERATION

A. Manual Corded Operation:

1. Chain-and-Clutch Operating Mechanisms: With continuous-loop bead chain and clutch that stops shade movement when bead chain is released; permanently adjusted and lubricated.

a. Bead Chains: Stainless steel bead chain with WCMA A100.1 compliant tension device installed on roller window shade by manufacturer and mounted on jamb.

1) Loop Length: Full length of roller shade.
2) Limit Stops: Provide upper and lower ball stops.

2.4 SHADEBAND MATERIALS

A. Light-Filtering Fabric: Woven fabric, stain and fade resistant.

1. Source: Roller shade manufacturer.
2. Type: 100% Glass Yarn.
3. Weight: 7.67 oz./sq. yd..
4. Basis of Design: Ecoglass 2800 Series
5. Openness Factor: 1 percent.
6. Color: Pewter.

B. Light-Blocking Fabric: Opaque fabric, stain and fade resistant.

1. Source: Roller shade manufacturer.
2. Type: 100% Glass Yarn, Acrylic coated + White Flock.
3. Weight: 14.75 oz./sq. yd..
4. Basis of Design: Ecoglass Blackout 2820 Series
5. Color: Pewter.

2.5 FABRICATION OF ROLLER WINDOW SHADES

A. Unit Sizes: Fabricate units in sizes to fill window and other openings as follows, measured at 74 deg F:

1. Between (Inside) Jamb Installation: Width equal to jamb-to-jamb dimension of opening in which shade is installed less 1/4 inch per side or 1/2-inch total, plus or minus 1/8 inch. Length equal to head-to-sill or -floor dimension of opening in which shade is installed less 1/4 inch, plus or minus 1/8 inch.

B. Shadeband Fabrication: Where width-to-length ratio of shadeband is equal to or greater than 1:4, provide battens and seams at uniform spacings along shadeband length to ensure shadeband tracking and alignment through its full range of movement without distortion of the material.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, operational clearances, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF ROLLER WINDOW SHADES

A. Install roller shades level, plumb, aligned and centered on openings, and aligned with adjacent units in accordance with manufacturer's written instructions.

B. Roller Shade Locations: As indicated on Drawings.

3.3 ADJUSTING

A. Adjust and balance roller window shades to operate smoothly, easily, safely, and free from binding or malfunction through full operational range.

3.4 CLEANING AND PROTECTION

A. Clean roller window shade surfaces, after installation, in accordance with manufacturer's written instructions.

B. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and Installer, that ensure that roller window shades are without damage or deterioration at time of Substantial Completion.

C. Replace damaged roller shades that cannot be repaired, in a manner approved by Contracting Officer, before time of Substantial Completion.

END OF SECTION 12 24 13

SECTION 12 32 16 – MANUFACTURED THERMALLY FUSED LAMINATE-CLAD CASEWORK

PART 1 - GENERAL

1.1 SUMMARY

1. Thermally fused laminate-clad casework in break room.
2. Hardware and accessories.

B. Related Requirements:

1. Section 061000 "Rough Carpentry" for wood blocking for anchoring casework.
2. Section 096519 "Resilient Tile Rubber Flooring" for resilient base applied to thermally fused-laminate-clad casework.
3. Section 12 36 61 "Simulated Stone Countertops" for countertop applied to thermally fused-laminate casework.
4. Division 22 for plumbing fixtures and fittings.

1.2 DEFINITIONS

A. Definitions in the AWI "Architectural Woodwork Standards" apply to the Work of this Section.

1.3 COORDINATION

A. Coordinate sizes and locations of framing, blocking, furring, reinforcements, and other related units of Work specified in other Sections to ensure that casework can be supported and installed as indicated.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Thermally fused-laminate-clad casework.
2. Hardware and accessories.

B. Shop Drawings: For thermally fused-laminate-clad casework.

1. Include plans, elevations, sections, and attachments to other work including blocking and reinforcements required for installation.
2. Indicate types and sizes of casework.
3. Show fabrication details, including types and locations of hardware.
4. Indicate locations of and clearances from adjacent walls, doors, windows, other building components, and equipment.
5. Apply WI's Certified Compliance Program label to Shop Drawings.

C. Samples: For casework and hardware finishes.

1.5 CLOSEOUT SUBMITTALS

A. Manufacturer’s Warranty: For all products used.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: An authorized representative who is trained and approved by manufacturer.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Protect finished surfaces during handling and installation with protective covering of polyethylene film or other suitable material.

1.8 FIELD CONDITIONS

A. Environmental Limitations: Do not deliver or install casework until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature and relative humidity at levels planned for building occupants during remainder of construction period. Maintain temperature and relative humidity during remainder of construction period in range recommended for Project location by the AWI "Architectural Woodwork Standards."

B. Field Measurements: Where casework is indicated to fit to existing construction, verify dimensions of existing construction by field measurements before fabrication and indicate measurements on Shop Drawings. Provide fillers and scribes to allow for trimming and fitting.

1.9 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of casework that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Delamination of components or other failures of glue bond.
b. Warping of components.
c. Failure of operating hardware.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR CASEWORK

A. Quality Standard: Unless otherwise indicated, comply with the AWI "Architectural Woodwork Standards" for grades of casework indicated for construction, finishes, installation, and other requirements.

1. Grade: Custom.
2. Provide labels and certificates from WI certification program indicating that casework complies with requirements of grades specified.

    a. Contractor to register the Work under this Section with AWI's Quality Certification Program at www.awiqcp.org or by calling 855-345-0991.

2.2 THERMALLY FUSED-LAMINATE CLAD CASEWORK

A. Source Limitations: Obtain from single source from single manufacturer.

B. Basis of Design: Premium Laminate: Panolam Surface Systems, Cressida color.

C. Design: Frameless cabinet construction with the following door and drawer-front style: Flush overlay

D. Thermally fused Laminate:

1. Door and drawer fronts: Vertical with continuous vertical matching.
2. Face Frame Members: Lengthwise.
3. End Panels: Vertical.
4. Bottoms and Tops of Units: Side to side.
5. Knee Space Panels: Vertical.
6. Aprons: Horizontal.

E. Exposed Materials:

1. Thermally fused-Laminate Grade: HGS VGS.
2. Edgebanding: Thermally fused laminate matching adjacent surfaces.

F. Semiexposed Materials:

1. Thermally fused Laminate: Grade VGS unless otherwise indicated. Provide thermally fused laminate for semi-exposed surfaces unless otherwise indicated.

    a. Colors and Patterns: As indicated by manufacturer's designations as selected from manufacturer's full range.
    b. Provide thermally fused laminate of same grade as exposed surfaces for interior faces of doors and drawer fronts and other locations where opposite side of component is exposed.

G. Concealed Materials:

1. Plywood: Hardwood plywood.

2.3 HARDWARE AND ACCESSORIES

A. Hardware: Unless otherwise indicated, provide manufacturer's standard satin-finish, commercial-quality, heavy-duty hardware.

1. Use threaded metal or plastic inserts with machine screws for fastening to particleboard except where hardware is through-bolted from back side.

B. Frameless Concealed Hinges (European Type): ANSI/BHMA A156.9, Type B01602 , self-closing.

1. Provide two hinges for doors less than 48 inches high, and provide three hinges for doors more than 48 inches high.
2. Degrees of Opening: 135 degrees.

C. Back-Mounted Pulls: ANSI/BHMA A156.9, B02011, solid metal, 5 inches long, 1/2 inch width minimum, 3-inch centers, brushed aluminum or brushed nickel finish.

1. Basis of Design: Estmoon Brushed Nickel Cabinet Handles 3-inch hole center drawer pull

D. Shelf Rests and Standards:

1. Adjustable Shelf Standards and Supports: ANSI/BHMA A156.9, B04071; with shelf rests, B04081.
2. Shelf Rests: ANSI/BHMA A156.9, B04013; metal.

E. Door and Drawer Bumpers: Self-adhering, clear silicone rubber.

1. Doors: Provide one bumper at top and bottom of closing edge of each swinging door.
2. Drawers: Provide one bumper on back side of drawer front at each corner.

F. Drawer Slides: ANSI/BHMA A156.9.

1. Heavy-Duty (Grade 1HD-100 and Grade 1HD-200): Side mount.

a. Type: Soft close, Full overtravel extension.
b. Material: Aluminum slides.

2.4 MATERIALS

1. Hardwood Plywood: 0.05 ppm.
2. Particleboard: 0.09 ppm.
3. MDF More Than 5/16 Inch Thick: 0.11 ppm.
4. MDF 5/16 Inch or Less in Thickness: 0.13 ppm.

B. Composite Wood Products: Products shall be made without urea formaldehyde.

C. Maximum Moisture Content for Lumber: 7 percent for hardwood and 12 percent for softwood.

D. Hardwood Plywood: HPVA HP-1, particleboard core except where veneer core is indicated.

E. Softwood Plywood: DOC PS 1.

F. Particleboard: ANSI A208.1, Grade M-2.

1. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 15 percent.

G. MDF: Medium-density fiberboard, ANSI A208.2, Grade 130.

1. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 15 percent.

H. Hardboard: ANSI A135.4, Class 1 tempered.

1. Recycled Content: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 15 percent.

I. PVC Edgebanding for Thermally fused Laminate: Rigid PVC extrusions, through color with satin finish, 3.0 mm thick at doors and drawer fronts, 1.0 mm thick elsewhere.

J. Thermally Fused Laminate Panels: Particleboard or MDF finished with thermally fused, melamine-impregnated decorative paper and complying with requirements of ISO 4586.

1. Edgebanding for Thermally Fused Laminate (TFL) Panels: PVC or polyester edgebanding matching thermally fused laminate panels.

K. Adhesives: Do not use adhesives that contain urea formaldehyde.

2.5 FABRICATION

A. Thermally fused-Laminate-Clad Cabinet Construction: As required by referenced quality standard, but not less than the following:

1. Bottoms and Ends of Cabinets, and Tops of Wall Cabinets and Tall Cabinets: 3/4-inch particleboard.
2. Shelves: 3/4-inch- thick plywood
3. Backs of Casework: 1/4-inch- thick, veneer-core hardwood plywood dadoed into sides, bottoms, and tops where not exposed.
4. Drawer Fronts: 3/4-inch particleboard.
5. Drawer Sides and Backs: 1/2-inch- thick solid-wood or veneer-core hardwood plywood with glued dovetail or multiple-dowel joints.
6. Drawer Bottoms: 1/4-inch- thick hardwood plywood glued and dadoed into front, back, and sides of drawers. Use 1/2-inch material for drawers more than 24 inches wide.
7. Drawer Bodies: Steel drawer pans formed from 0.0359-inch- thick metal, metallic phosphate treated, and finished with manufacturer's standard two-coat, baked-enamel finish consisting of prime coat and thermosetting topcoat with a minimum dry film thickness of 1 mil for topcoat and 2 mils for system.

8. Cabinet Doors:

    a. 48 Inches (1220 mm) High or Less: 3/4 inch thick, with particleboard or MDF cores and solid-wood stiles and rails.
    b. 48 Inches (1220 mm) or More in Height: 1-1/8 inches thick, with particleboard cores.

B. Filler Strips: Provide as needed to close spaces between casework and walls, ceilings, and equipment. Fabricate from same material and with same finish as casework.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas, with Installer present, for compliance with requirements for installation tolerances, location of framing and reinforcements, and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Grade: Install casework to comply with same quality standard grade as item to be installed.

B. Install casework level, plumb, and true in line; shim as required using concealed shims. Where casework abuts other finished work, apply filler strips and scribe for accurate fit, with fasteners concealed where practical.

C. Base Cabinets: Set cabinets straight, level, and plumb. Adjust subtops within 1/16 inch of a single plane. Align similar adjoining doors and drawers to a tolerance of 1/16 inch. Bolt adjacent cabinets together with joints flush, tight, and uniform.

D. Wall Cabinets: Hang cabinets straight, level, and plumb. Adjust fronts and bottoms within 1/16 inch of a single plane. Fasten cabinets to hanging strips, masonry, framing, wood blocking, or reinforcements in walls and partitions. Align similar adjoining doors to a tolerance of 1/16 inch.

E. Fasten casework to adjacent units and to masonry, framing, wood blocking, or reinforcements in walls and partitions to comply with the AWI "Architectural Woodwork Standards."

F. Install hardware uniformly and precisely. Set hinges snug and flat in mortises unless otherwise indicated. Adjust and align hardware so moving parts operate freely and contact points meet accurately. Allow for final adjustment after installation.

G. Adjust operating hardware so doors and drawers operate smoothly without warp or bind. Lubricate operating hardware as recommended by manufacturer.

3.3 FIELD QUALITY CONTROL

A. Inspections: Provide inspection of installed Work through AWI's Quality Certification Program certifying that woodwork, including installation, complies with requirements of the Architectural Woodwork Standards for the specified grade.

1. Inspection entity to prepare and submit report of inspection.
2. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

3.4 CLEANING

A. Repair or remove and replace defective work as directed on completion of installation.

B. Clean finished surfaces, touch up as required, and remove or refinish damaged or soiled areas to match original factory finish, as approved by Contracting Officer.

END OF SECTION 12 32 16

SECTION 12 36 61 – SIMULATED STONE COUNTERTOPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Quartz agglomerate countertops.

a. Transaction Desk
b. Breakroom Counters
c. Basement Vestibule Café Counter

2. Quartz agglomerate backsplashes.
3. Accessories.

B. Related Requirements:

1. Section 055000 "Metal Fabrications" for cantilever supports.
2. Division 22 for plumbing fixtures and fittings.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of countertop material .

B. Shop Drawings:

1. Plans, sections, details, edge and backsplash profiles, and attachment to other work.
2. Locations and details of joints.
3. Locations, quantity, and type of supports/brackets.
4. Locations and sizes of cutouts and holes for items installed in countertop.

C. Samples for Initial Selection: For each type of material exposed to view.

D. Samples for Verification:

1. Countertop material, 4 inches square with edge detail.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: For solid surface material countertops to include in maintenance manuals. Include product data for care products used or recommended by Installer and names, addresses, and telephone numbers of local sources for products.

1.4 QUALITY ASSURANCE

A. Fabricator Qualifications: Shop that employs skilled workers who custom-fabricate countertops similar to that required for this Project, and whose products have a record of successful in-service performance.

B. Installer Qualifications: Fabricator of countertops.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver countertops only after casework and supports on which they will be installed have been completed in installation areas.

B. Store countertops in areas where environmental conditions comply with requirements specified in "Field Conditions" Article.

C. Keep surfaces of countertops covered with protective covering during handling and installation.

1.6 FIELD CONDITIONS

A. Field Measurements: Where countertops are indicated to fit to other construction, verify dimensions of countertops by field measurements after base cabinets are installed but before countertop fabrication is complete and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Quality Standard: Unless otherwise indicated, comply with AWI 1236 for grades of simulated stone countertops indicated for construction, finishes, installation, and other requirements.

1. Provide inspections of fabrication and installation together with labels and certificates from AWI certification program indicating that countertops comply with requirements of grade specified.
2. The Contract Documents contain more stringent requirements than that of the referenced quality standard. Comply with requirements of the Contract Documents in addition to those of referenced quality standard.

2.2 QUARTZ AGGLOMERATE COUNTERTOPS

A. Grade: Premium

B. Manufacturers:

1. Caesarstone: www.caesarstoneus.com.
2. Corian Quartz: www.corian.com

3. Wilsonart Quartz: www.wilsonart.com/quartz
4. Or Approved Equal.

C. Basis of Design: Wilsonart Quartz, 2cm thick, quartz material.

1. Eased Edge.
2. Color: Tumbled Stone

D. Backsplashes and Transaction Desk Waterfall Edge: 3cm- thick quartz to match countertop.

2.3 ACCESSORIES

A. Support Brackets:

1. Countertop Recessed L-style bracket:

a. Type: Hidden.
b. Style Basis of Design: Original Granite Bracket; originalgranitebracket.com. L Bracket Countertop Support
c. Material: Steel.
d. Color: Black powder coat.
e. Install minimum 36" on center, and within 3" of each unsupported end, or as directed by bracket and countertop manufacturer.

2.4 FABRICATION

A. Fabricate countertops in sizes and shapes required to comply with requirements indicated.

B. Fabricate tops with shop-applied edges [and backsplashes] unless otherwise indicated. Comply with solid surface material manufacturer's written instructions for adhesives, sealers, fabrication, and finishing.

C. Joints:

1. Fabricate countertops in sections for joining in field.

a. Joint Locations: Not within 18 inches of a sink or cooktop and not where a countertop section less than 36 inches long would result, unless unavoidable.

D. Cutouts and Holes:

1. Undercounter Plumbing Fixtures: Make cutouts for fixtures in shop using template or pattern furnished by fixture manufacturer. Form cutouts to smooth, even curves.

a. Provide vertical edges, slightly eased at juncture of cutout edges with top and bottom surfaces of countertop and projecting 3/16 inch into fixture opening.

2. Fittings: Drill countertops in shop for grommets, plumbing fittings, undercounter soap dispensers, and similar items.

2.5 INSTALLATION MATERIALS

A. Plywood: Exterior softwood plywood complying with DOC PS 1, Grade C-C Plugged, touch sanded.

B. Adhesive: Product recommended by manufacturer.

C. Sealant for Countertops: Comply with applicable requirements in Section 079200 "Joint Sealants."

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates to receive countertops and conditions under which countertops will be installed, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of countertops.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION OF SIMULATED STONE COUNTERTOPS

A. Assemble countertops and complete fabrication at Project site to the extent that it was not completed in the shop.

B. Countertop Installation:

1. Scribe and cut countertops to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts.
2. Anchor securely by screwing through corner blocks of base cabinets or other supports into underside of countertop.
3. Anchor wall cleating necessary for proper setting for countertops not supported by casework.
4. Install countertops level to a tolerance of 1/8 inch in 8 ft., 1/4 inch maximum. Do not exceed 1/64-inch difference between planes of adjacent units.
5. Fasten subtops to cabinets by screwing through subtops into cornerblocks of base cabinets. Shim as needed to align subtops in a level plane.
6. Secure countertops to subtops with adhesive according to manufacturer's written instructions. Align adjacent surfaces and, using adhesive in color to match countertop, form seams to comply with quartz agglomerate manufacturer's written instructions. Carefully dress joints smooth, remove surface scratches, and clean entire surface.
7. Bond joints with adhesive and draw tight as countertops are set. Mask areas of countertops adjacent to joints to prevent adhesive smears.

    a. Install metal splines in kerfs in countertop edges at joints. Fill kerfs with adhesive before inserting splines and remove excess immediately after adjoining units are drawn into position.

b. Clamp units to temporary bracing, supports, or each other to ensure that countertops are properly aligned and joints are of specified width.

8. Install backsplashes and end splashes by adhering to wall and countertops with adhesive. Mask areas of countertops and splashes adjacent to joints to prevent adhesive smears.
9. Install aprons to backing and countertops with adhesive. Mask areas of countertops and splashes adjacent to joints to prevent adhesive smears. Fasten by screwing through backing. Predrill holes for screws as recommended by manufacturer.
10. Seal joints between countertop and backsplash, if any, and joints where countertop and backsplash abut walls. Comply with Section 079200 "Joint Sealants."

3.3 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 12 36 61

SECTION 14 01 20.71 – ELEVATOR REHABILITATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Passenger elevator upgrade:

a. Field verify elevator conditions prior to bid.
b. Intent is to reuse existing elevator equipment in situ. Floor levels are not expected to change but stops will need to be checked at the conclusion of the finish work. Controls will remain where they are. Finish upgrades to include replacement wall panels, new light fixture, and new flooring.

B. Related Sections:

1. Section 09 51 23: “Acoustical Tile Ceiling”
2. Section 09 65 19: “Resilient Tile Rubber Flooring.”
3. Section 09 90 00: “Painting and Coating.”
4. Section 28 31 11: “Digital, Addressable Fire-Alarm System.”
5. Division 21 for fire suppression.
6. Division 26 for electrical connections.
7. Appendix D: TKE Elevator Modernization Work Coordination Handbook

1.2 SYSTEM DESCRIPTION

A. Existing elevator is Dover Elevator Co., Model Fleetwood-21 Oildraulic Passenger Elevator.

1.3 SUBMITTALS

A. Schedule of required work needed for elevator rehabilitation.

B. Shop Drawings as required for scope:

1. Include dimensioned drawings showing plans, elevations, sections and large-scale details indicating revised service at each landing.

C. Wiring Diagram:

1. Diagram detailing wiring for revisions to power, signal and control systems differentiating clearly between existing-installed wiring and revised wiring.
2. Indicate maximum and average power demands.

D. Closeout Documentation:

1. Provide Contracting Officer with copies of all inspection/acceptance certificates and operating permits as required by governing authorities to allow normal, unrestricted use of elevators.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: Engage elevator manufacturer or an installer approved by elevator manufacturer and who has completed elevator revisions similar in material, design, and extent to that indicated for Project which have resulted in installations with a record of successful in-service performance.

B. Regulatory Requirements: In addition to local governing regulations, comply with applicable requirements of ASME/ANSI A17.1, Safety Code for Elevators and Escalators (hereafter referred to as the "Code").

1.5 WARRANTY

A. Guarantee that materials and workmanship of apparatus and finishes installed under these specifications will be first-class in every respect, and make good any defects not due to ordinary wear and tear, or improper use or care which may develop within one year from date of completion of Work.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Reuse existing elevator equipment and cab.

B. Intent is to reuse as much of the existing equipment and controls as possible.

2.2 WORK BY ELEVATOR COMPANY

A. Elevator company shall submit work plan required for cab, controls, switches, interface with new mechanical, electrical, fire suppression, and alarm systems to assure a fully operational, and complete elevator assembly that will meet approvals of State Inspector.

B. Alarm response and fire department recall functions shall be updated to conform with requirements of fire department that responds to this site.

2.3 FINISHES

A. Entrance Frames and Door Finishes: Clean and repair existing doors and frames. Refinish or replace stainless steel paneling. See images of existing panels next page. Red arrows indicate areas where finish is damaged.

Inspection
Reports
Located in
Administration

B. Car Door and Front Finishes: Clean and repair existing doors and frames. Refinish or replace stainless steel paneling. See images of existing panels previous page. Red arrows indicate areas where finish is damaged.

C. Cab Wall Finish: Replace existing plastic laminate wall panels. Color to be selected from manufacturer's full selection.

D. Cab Ceiling: Remove existing cab ceiling panels and replace with new to match.

E. Cab Accessories: Salvage and reinstall existing aluminum handrails and wall padding hooks.

F. Floor Finish: Remove the existing carpeting and replace with rubber flooring to match adjacent lobby level floor. Color/Style: per Section 09 65 19: "Resilient Tile Rubber Flooring," to be confirmed by Contracting Officer in submittal process.

2.4 WORK BY OTHERS

A. Smoke Detectors:

1. Smoke detectors shall be furnished and installed in the room that accesses the elevator at each floor and the elevator machine room according to NFPA No. 72E, Automatic Fire Detectors, Chapter 4.
2. Activating a smoke detector in any elevator lobby, or associated elevator machine rooms other than designated floor, shall cause car to return non-stop to designated floor.
3. If smoke detector at designated floor is activated, car shall return to an alternate floor approved by enforcing authority unless Phase I key operated switch is in "ON" position.
4. When Phase I service is in "BYPASS" position, normal elevator service shall be restored independent of smoke detectors. (Dover will provide terminals in machine room to receive sensor signals.)
5. Smoke detectors and/or smoke detector systems shall not be self-resetting.

B. Electrical Power: Provide electrical power for light, tools, hoists, etc., during installation as well as electric current for starting, testing and adjusting elevator.

C. Any modification or installation of lights and/or electrical outlets in machine room and/or pit.

D. Addition of a sump pump in the elevator pit.

E. Provide a safe and dry on-site storage area for elevator material.

PART 3 - EXAMINATION

A. Prior to commencing elevator installation, examine hoistways, hoistway openings, pits, and machine rooms, as constructed.

B. Notify Contractor in writing of any dimensional discrepancies or other conditions detrimental to proper installation or performance of elevator work.

C. Do not proceed with elevator revision until unsatisfactory conditions have been corrected in a manner acceptable to Installer.

3.2 INSTALLATION

A. Comply with manufacturer's instructions and recommendations for work required during installation.

B. Provide welded connections for installation of elevator work where bolted connections are not required for subsequent removal or for normal operation, adjustment, inspection, maintenance, and replacement of worn parts.

C. Comply with AWS standards for workmanship and for qualifications of welding operators.

D. Coordinate elevator work with work of other trades for proper time and sequence to avoid construction delays.

E. Use benchmarks, lines, and levels designated by Contractor to ensure dimensional coordination of the work.

F. Mount rotating and vibrating elevator equipment and components on vibration-isolation mounts, designed to effectively prevent transmission of vibrations to structure and thereby to reduce structure-borne noise from elevator system to a noise criterion (NC) of 25 or less in occupied spaces.

G. Lubricate operating parts of systems, including ropes, if any, as recommended by manufacturers.

H. Reduce clearances to minimum, safe, workable dimension at each landing.

I. Leveling Tolerance: 1/2 inch, up or down, regardless of load and direction of travel.

J. Coordinate with other trades to facilitate and ensure proper grouting of sills.

3.3 FIELD QUALITY CONTROL

A. Upon nominal completion of each elevator installation and before permitting use of elevator (either temporary or permanent), perform acceptance tests as required and recommended by Code and by governing regulations or agencies.

B. Load each elevator to its rated capacity and operate continuously for thirty minutes over its full travel distance, stopping at each level and proceeding immediately to the next.

C. Record failures of elevator to perform as required.

D. Advise Contractor and Contracting Officer, and inspection department of governing agencies in advance of dates and times tests are to be performed on elevators.

3.4 PROTECTION

A. At time of Substantial Completion of elevator work (or portion thereof), provide suitable protective coverings, barriers, devices, signs, or such other methods or procedures to protect elevator work from damage or deterioration.

B. Maintain protective measures throughout remainder of construction period.

C. Provide similar protective measures for elevator units that will be placed in temporary service, including inspection and maintenance service during period of temporary service.

3.5 DEMONSTRATION

A. Instruct Government personnel in proper use, operations, and daily maintenance of elevators.

B. Review emergency provisions, including emergency access and procedures to be followed at time of failure in operation and other building emergencies.

C. Train Government personnel in normal procedures to be followed in checking for sources of operational failures or malfunctions.

D. Confer with Contracting Officer on requirements for a complete elevator maintenance program.

E. Make a final check of each elevator operation with Contracting Officer present and just prior to date of Substantial Completion.

F. Determine that control systems and operating devices are functioning properly.

3.6 MAINTENANCE

A. Provide one year of maintenance for elevator.

3.7 INSPECTION

A. Contractor will commence inspection immediately after finishing work. Observations, measurements, and other notes will be recorded using DSC-CQC Accessibility Inspection Report as basis. Report can include digital images, product literature, drawings and sketches, etc. to communicate conditions, especially items not in compliance with ABAAS. Submit report to CO within 24 hours of commencing inspection.

END OF SECTION 14 01 20.71

SECTION 21 00 00 – FIRE SUPPRESSION BASIC REQUIREMENTS

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Work included in this Section applies to Division 21, Fire Suppression work to provide materials, labor, tools, permits, incidentals, and other services to provide and make ready for Park's use of fire protection systems for proposed project.

B. Contract Documents include, but are not limited to, Specifications including Division 00, Procurement and Contracting Requirements and Division 01, General Requirements, Drawings, Addenda, and Contracting Officer/Contractor Agreement. Confirm requirements before commencement of work.

C. Definitions:

1. Provide: To furnish and install, complete and ready for intended use.
2. Furnish: Supply and deliver to project site, ready for unpacking, assembly and installation.
3. Install: Includes unloading, unpacking, assembling, erecting, installation, applying, finishing, protecting, cleaning and similar operations at project site as required to complete Item of work furnished.
4. Approved or Approved Equivalent: To possess the same performance qualities and characteristics and fulfill the utilitarian function without any decrease in quality, durability or longevity. For equipment/products defined by the Contractor as "equivalent," substitution requests must be submitted to Contracting Officer for consideration, in accordance with Division 01, General Requirements, and approved by the Contracting Officer.
5. Authority Having Jurisdiction (AHJ): Indicates reviewing authorities, including local fire marshal, insurance underwriter, Contracting Officer, and other reviewing entities whose approval is required to obtain systems acceptance.

1.2 RELATED SECTIONS

A. Content of Section applies to Division 21, Fire Suppression Contract Documents.

B. Related Work:

1. Additional conditions apply to this Division including, but not limited to:

a. Specifications including Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.
b. Drawings
c. Addenda
d. Contracting Officer/Contractor Agreement
e. Codes, Standards, Public Ordinances and Permits

1.3 REFERENCES AND STANDARDS

A. References and Standards per Division 00, Procurement and Contracting Requirements and Division 01, General Requirements, individual Division 21, Fire Suppression Sections and those listed in this Section.

B. Codes to include latest adopted editions, including current amendments, supplements and local jurisdiction requirements in effect as of the date of the Contract Documents, of/from:

1. International Code Council:

a. International Building Code (IBC)
b. International Fire Code (IFC)
c. International Existing Building Code (IEBC)

C. Reference standards and guidelines include but are not limited to the latest adopted editions from:

1. ABA – Architectural Barriers Act
2. AHRI – Air-Conditioning Heating & Refrigeration Institute
3. ANSI – American National Standards Institute
4. ASCE – American Society of Civil Engineers

a. ASCE-7 Minimum Design Loads for Buildings and Other Structures

5. ASHRAE – American Society of Heating, Refrigerating and Air-Conditioning Engineers

a. ASHRAE Guideline 0, the Commissioning Process

6. ASME – American Society of Mechanical Engineers
7. ASPE – American Society of Plumbing Engineers
8. ASSE – American Society of Sanitary Engineering
9. ASTM – ASTM International
10. AWWA – American Water Works Association
11. CFR – Code of Federal Regulations
12. EPA – Environmental Protection Agency
13. ETL – Electrical Testing Laboratories
14. FCC – Federal Communications Commission
15. FM – FM Global

a. FM Global – FM Global Approval Guide

16. ICC – International Code Council
17. IEC – International Electrotechnical Commission
18. ICC-ESR – International Code Council Evaluation Service Reports
19. HI – Hydraulic Institute Standards
20. ISO – International Organization for Standardization
21. MSS – Manufacturers Standardization Society
22. NEC – National Electric Code
23. NEMA – National Electrical Manufacturers Association
24. NFPA – National Fire Protection Association:

a. NFPA 13 – Standard for the Installation of Sprinkler Systems
b. NFPA 24 – Standard for Installation of Private Fire Service Mains and Their Appurtenances
c. NFPA 25 – Standard for Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems
d. NFPA 70 – National Electrical Code
e. NFPA 72 – National Fire Alarm and Signaling Code
f. NFPA 2001 – Clean Agent Fire Extinguishing Systems

25. OSHA – Occupational Safety and Health Administration
26. UL – Underwriters Laboratories Inc.

D. See Division 21, Fire Suppression individual Sections for additional references.

1.4 SUBMITTALS

A. See Division 01, General Requirements for Submittal Procedures as well as specific individual Division 21, Fire Suppression sections.

B. Provide drawings in format and software release equal to the design documents. Drawings to be the same sheet size and scale as the Contract Documents.

C. "No Exception Taken" constitutes that review is for general conformance with the design concept expressed in the Contract Documents for the limited purpose of checking for conformance with information given. Any action is subject to the requirements of the Contract Documents. Contractor is responsible for the dimensions and quantity and will confirm and correlate at the job site, fabrication processes and techniques of construction, coordination of the work with that of all other trades, and the satisfactory performance of the work.

D. Provide product submittals and shop drawings in electronic format only. Electronic format must be submitted via zip file via e-mail. For electronic format, provide one file per division containing one bookmarked PDF file with each bookmark corresponding to each Specification Section. Arrange bookmarks in ascending order of Specification Section number. Individual submittals sent piecemeal in a per Specification Section method will be returned without review or comment. Copy Contracting Officer on all transmissions/submissions.

E. Submit shop drawings, calculations and product data sheets as one complete stand-alone package to the Contracting Officer.

F. Product Data: Provide Manufacturer's descriptive literature for products specified in Division 21, Fire Suppression Sections.

G. Identify/mark each submittal in detail. Note what differences, if any, exist between the submitted item and the specified item. Failure to identify the differences will be considered cause for disapproval. If differences are not identified and/or not discovered during the submittal review process, Contractor remains responsible for providing equipment and materials that meet the Specifications and Drawings.

1. Label submittal to match numbering/references as shown in Contract Documents. Highlight and label applicable information to individual equipment or cross out/remove

extraneous data not applicable to submitted model. Clearly note options and accessories to be provided, including field installed Item. Highlight connections by/to other trades.

2. Include technical data, installation instructions and dimensioned drawings for products, equipment and devices installed, furnished or provided. Reference Division 21, Fire Suppression specification Sections for specific Item required in product data submittal outside of these requirements.
3. See Division 21, Fire Suppression Sections for additional submittal requirements outside of these requirements.

H. Maximum of two reviews provided of complete submittal package. Arrange for additional reviews and/or early review of long-lead items. Incomplete submittal documentation will be returned to without review. Contractor must arrange for, and bear costs of, the reviewing entity's additional reviews.

I. Resubmission Requirements: Make corrections or changes in submittals as required, and in consideration of Contracting Officer's comments. Identify Contracting Officer's comments and provide an individual response to each of the Contracting Officer's comments. Cloud changes in the submittals and further identify changes which are in response to Contracting Officer's comments.

J. Structural/Seismic: Provide weights, dimensions, mounting requirements and like information required for mounting, seismic bracing, and support. Indicate manufacturer's installation and support requirements to meet ASCE 7-16 requirements for non-structural components. Provide engineered seismic drawings and equipment seismic certification. Equipment Importance Factor as specified in Division 01 and in Structural documents.

K. Trade Coordination: Include physical characteristics, electrical characteristics, device layout plans, wiring diagrams, and connections as required per Division 21, Fire Suppression coordination documents. For equipment with electrical connections, furnish copy of approved submittal for inclusion in Division 26, Electrical and Division 28, Electronic Safety submittals.

L. Make provisions for openings in building for admittance of equipment prior to start of construction or ordering of equipment.

M. Substitutions and Variation from Basis of Design:

1. The Basis of Design designated product where provided in the specifications or on the drawings, establishes the parameters used for space planning coordination with other disciplines.
2. If substitutions from the basis of design and/or equivalent equipment/products are being proposed, it is the responsibility of parties concerned, involved in, and furnishing the substitute and/or equivalent equipment to verify and compare the characteristics and requirements of that furnished to that specified and/or shown. If greater capacity and/or more materials and/or more labor is required for the rough-in, circuitry or connections than for the item specified and provided for, then provide compensation for additional charges required for the proper rough-in, circuitry and connections for the equipment being furnished. Coordinate with the requirements of "Submittals". For any product marked "or approved equivalent", a substitution request must be submitted to Contracting Officer for approval prior to purchase, delivery or installation.

N. Shop Drawings:

1. Provide coordinated Shop Drawings which include physical characteristics of all systems, equipment and piping layout, pipe layout, hanger layout, drains, location of drain discharge, risers, valves, details, water test information, physical device layout plans, and control wiring diagrams. Reference individual Division 21, Fire Suppression Sections for additional requirements for shop drawings outside of these requirements.
2. Shop Drawings, hydraulic calculations, trapeze hanger calculations, and the like, to be prepared under the direct supervision and control of a Professional competent to do such work and licensed in the state of Iowa.
3. Provide Shop Drawings which indicate information required by NFPA 13 and NFPA 2001. Include room names and fire sprinkler occupancy hazard classifications.
4. Provide Shop Drawings illustrating information for Hydraulic Information Sign for each hydraulic remote area calculated.
5. Utilizing the Reflected Ceiling backgrounds, provide Shop Drawings illustrating locations of fire sprinklers and piping.
6. Provide Shop Drawings illustrating each type of hanger, including fasteners to structure.
7. Provide details for any hanger or attachment to be attached to any I-joist, structural insulated panels (SIPs), cross laminated timber, and similar engineered structural products according to the specifications of the engineered product manufacturer.
8. Provide Shop Drawings illustrating information for Sprinkler System General Information Sign.
9. Shop Drawings to include a cross-sectional view that shows the sprinkler heads and piping in relation to the building's architectural and structural information. View to be chosen based on a location that will display the most information.
10. When required, provide Coordination Drawings.
11. Provide Shop Drawings indicating access panel locations, size and elevation for approval prior to installation.
12. Provide details of hanger attachments to structure and to piping. Include details on the size and load capacities of fasteners.
13. Clearly indicate the elevation of the highest sprinkler in relation to the elevation of the flow test pressure gauge monitor hydrant.
14. Provide details of flexible sprinkler hose fitting per manufacturer's schedule of equivalent feet used in hydraulic calculations, showing device length, maximum number of 90-degree bends and expected radius of bends.
15. Provide a schedule of signage to be installed at each flexible sprinkler hose fitting.
16. On the drawings, provide a list of number, model, temperature, sprinkler Identification number, manufacturer, orifice, deflector type, thermal sensitivity and pressure rating, quantity of each type to be contained in the spare sprinkler cabinet and the issue date or revision date of the list.
17. Spare sprinkler head cabinet size indicating the number of spare sprinkler head to be contained therein.

O. Samples: Provide samples when requested by individual Sections.

P. Resubmission Requirements:

1. Make any corrections or change in submittals when required. Provide submittals as specified. The Contracting Officer will not be required to edit and/or interpret the Contractor's submittals. Indicate changes for the resubmittal in a cover letter with reference

to page(s) changed and reference response to comment. Clearly indicate changes on Drawings and cloud changes in the submittals.

2. Resubmit for review until review indicates no exceptions taken or make "corrections as noted".

Q. Operation and Maintenance (O&M) Manuals:

1. Submit, at one time, electronic files (PDF format) of manufacturer's operation and maintenance instruction manuals and parts lists for equipment or Item requiring servicing. Include valve charts. Submit data when work is substantially complete and in same order format as submittals. Include name and location of source parts and service for each piece of equipment.

    a. Include copies of certificates of code authority acceptance, code-required acceptance tests, test reports and certificates.
    b. Include Warranty per Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Sections.
    c. Catalog description of each Item of equipment actually installed on job.
    d. Instructions for operation and maintenance of fire suppression systems composed of operating instructions, maintenance instructions and manufacturer's literature as follows:

        1) Testing and Maintenance Schedule Chart: Provide an 8-1/2- by 11-inch typewritten list of each item of installed equipment requiring testing inspection, lubrication or service, describing and scheduling performance of maintenance.
        2) Manufacturer's Literature: Provide copies of manufacturer's instructions for operation and maintenance of fire suppression equipment, including replacement parts list with name and address of nearest distributor. Mark each copy with equipment identification label as listed in equipment schedule, i.e., F-5 etc.

    e. Include product certificates of warranties and guarantees.
    f. Include Record Drawings.
    g. Include copy of water supply flow test used as basis for hydraulic calculations.
    h. Include hydraulic calculations.
    i. Include Contractor’s Material and Test Certificates for Aboveground Piping/Underground Piping.
    j. Include a copy of NFPA 25.
    k. Include a copy of valve charts and whether normally open or normally closed.
    l. Include a copy of drain, auxiliary, and low point drains charts.
    m. Include a copy of the list to be included in the spare sprinkler head box.
    n. Include copy of Clean Agent System Acceptance Test and Room Pressure Test.
    o. Include Commissioning Reports.
    p. Include copy of approved submittal data along with submittal review letters received from Contracting Officer. Data to clearly indicate installed equipment model numbers. Delete or cross out data pertaining to other equipment not specific to this project.
    q. Include copy of manufacturer's standard Operations and Maintenance for equipment. At front of each tab, provide routine maintenance documentation for scheduled equipment. Include manufacturer's recommended maintenance schedule

and highlight maintenance required to maintain warranty. Furnish list of routine maintenance parts, including part numbers, sizes, and quantities relevant to each piece of equipment: i.e., belts, motors, lubricants, and filters.

r. Include copy of complete parts list for equipment. Include available exploded views of assemblies and sub-assemblies.

s. Include copy of startup and test reports specific to each piece of equipment.

t. Incomplete submittal documentation will be returned to without review. Contractor must arrange for, and bear costs of, the reviewing entity's additional reviews.

2. Thoroughly instruct Park Staff as directed by the Contracting Officer in proper operation of equipment and systems. Where noted in individual Sections, training will include classroom instruction with applicable training aids and systems demonstrations. Field instruction per Section 21 00 00, Fire Suppression Basic Requirements, Article titled "Demonstration".

3. Copies of certificates of code authority inspections, acceptance, code required acceptance tests, letter of conformance and other special guarantees, certificates of warranties, specified elsewhere or indicated on Drawings.

R. Record Drawings:

1. Maintain at site at least one set of Drawings for recording "As-constructed" conditions. Indicate on Drawings changes to original documents by referencing revision documents, include buried elements, location of cleanouts, and location of concealed mechanical Item. Include items changed by field orders, supplemental instructions, and constructed conditions.

2. Record Drawings are to include equipment and fixture/connection schedules that accurately reflect "as constructed or installed" for project.

3. At completion of project, input changes to original project on CAD Drawings and make one set of black-line drawings created from CAD Files in version/release equal to contract drawings. Submit CAD Files and drawings upon substantial completion.

4. Invert elevations and dimensioned locations for water services and drainage piping below grade extending to 5-feet outside building line.

5. Record Drawings to include site information or reference site information for complete understanding of the fire protection system between the building and the point of connection to the water supply and location of flow test pressure hydrants.

6. See Division 21, Fire Suppression individual Sections for additional items to include in Record Drawings.

S. Calculations: Submit hydraulic, clean agent, and the like calculations.

1. Hydraulic Calculations:

a. Include friction losses between the hydraulically most remote design area and the hydrant flow test pressure hydrant.

b. Hydraulic calculations to be performed on a nationally recognized fire sprinkler hydraulic calculation computer program, with cover sheets in the format required by the latest edition of NFPA 13. Hydraulic calculations performed "by hand" or not on a nationally recognized fire sprinkler hydraulic calculations computer program will be returned without review by Contracting Officer.

c. Provide one or more hydraulic calculations for each hydraulically most remote area.

d. Where it is not obvious which area is most hydraulically remote, perform and submit for review additional hydraulic calculations proving the hydraulically most remote area.
e. For grid systems, either provide “peaked” hydraulic calculations, or provide two additional sets of hydraulic calculations for each hydraulically most remote area.
f. Include pressure losses between the highest sprinkler and the elevation of the pressure gauge monitor hydrant of the flow test.
g. Include friction loss for flexible branch line connectors per manufacturer's schedule of equivalent feet for device length, maximum number of bends and expected radius of bends.
h. When flexible sprinkler hose fittings are added to an existing system, provide hydraulic calculations verifying the design flow rate will be achieved."
i. For Future Tenant Improvement Spaces: Include in hydraulic calculations friction loss allowances for future installation of flexible sprinkler head connectors so that flexible connectors may be installed in the future without revisions to the overhead system.

1.5 QUALITY ASSURANCE

A. Regulatory Requirements: Work and materials installed to conform with all local, State and Federal codes, and other applicable laws and regulations. Where code requirements are at variance with Contract Documents, meet code requirements as a minimum requirement and include costs necessary to meet these in Contract. Machinery and equipment are to comply with OSHA requirements, as currently revised and interpreted for equipment manufacturer requirements. Install equipment provided per manufacturer recommendations.

B. Whenever this Specification calls for material, workmanship, arrangement or construction of higher quality and/or capacity than that required by governing codes, higher quality and/or capacity take precedence.

C. Drawings are intended to be diagrammatic and reflect the Basis of Design manufacturer's equipment. They are not intended to show every Item in its exact dimensions, or details of equipment or proposed systems layout. Verify actual dimensions of systems (i.e., piping) and equipment proposed to assure that systems and equipment will fit in available space. Contractor is responsible for design and construction costs incurred for equipment other than Basis of Design, including, but not limited to, architectural, structural, electrical, HVAC, fire sprinkler, and plumbing systems.

D. Manufacturer's Instructions: Follow manufacturer's written instructions. If in conflict with Contract Documents, obtain clarification. Notify Contracting Officer in writing before starting work.

E. Items shown on Drawings are not necessarily included in Specifications or vice versa. Confirm requirements in all Contract Documents.

F. Provide products that are UL listed.

G. Piping Insulation products to contain less than 0.1 percent by weight PBDE in all insulating materials.

1.6 WARRANTY

A. Provide written warranty covering the work for a period of one year from date of Substantial Completion in accordance with Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Sections under this Division can require additional and/or extended warranties that apply beyond basic warranty under Division 01, General Requirements and the General Conditions. Confirm requirements in all Contract Documents.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Articles, fixtures, and equipment of a kind to be standard product of one manufacturer, including but not limited to sprinkler heads, pipe, fittings, hangers and bracing materials.

2.2 STANDARDS OF MATERIALS AND WORKMANSHIP

A. Base contract upon furnishing materials as specified. Materials, equipment, and fixtures used for construction are to be new, latest products listed in manufacturer's printed catalog data and are to be UL, ETL, FM, and ICC-ES listed and labeled for their intended fire protection function or be approved by State, County, and City authorities prior to procurement and installation.

B. Names and manufacturer's names denote character and quality of equipment desired and are not to be construed as limiting competition.

C. Hazardous Materials:

1. Comply with local, State of Iowa, and Federal regulations relating to hazardous materials.
2. Comply with Division 00, Procurement and Contracting Requirements and Division 01, General Requirements for this project relating to hazardous materials.
3. Do not use any materials containing a hazardous substance. If hazardous materials are encountered, do not disturb; immediately notify Contracting Officer.

PART 3 - EXECUTION

3.1 ACCESSIBILITY AND INSTALLATION

A. Confirm Accessibility and Installation requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Install equipment requiring access (i.e., drains, control operators, valves, motors, engines, pumps, controllers, air compressors, gauges, fill cups, tanks, cleanouts and the like) so that they may be serviced, reset, replaced or recalibrated by service people with normal service tools and

equipment. Do not install equipment in obvious passageways, doorways, scuttles or crawlspaces which would impede, or block intended usage.

C. Install equipment and products complete as directed by manufacturer's installation instructions. Obtain installation instructions from manufacturer prior to rough-in of equipment and examine instructions thoroughly. When requirements of installation instructions conflict with Contract Documents, request clarification from Contracting Officer prior to proceeding with installation. This includes proper installation methods, sequencing, and coordination with other trades and disciplines.

D. Earthwork:

1. Confirm Earthwork requirements in Contract Documents.
2. In absence of specific requirements, comply with the following:

    a. Perform excavation, dewatering, shoring, bedding, and backfill required for installation of work in this Division in accordance with the provisions specified. Contact utilities and locate existing utilities prior to excavation. Repair any work damaged during excavation or backfilling.
    b. Excavation: Do not excavate under footings, foundation bases, or retaining walls.
    c. Provide protection of underground systems. Review the project Geotechnical Report for references to corrosive or deleterious soils which will reduce the performance or service life of underground systems materials.

E. Firestopping:

1. Confirm Firestopping requirements in Division 07, Thermal and Moisture Protection.
2. In absence of specific requirements, comply with individual Division 21, Fire Suppression Sections and coordinate location and protection level of fire and/or smoke rated walls, ceilings, and floors. When these assemblies are penetrated, seal around piping, ductwork and equipment with approved firestopping material. Install firestopping material complete as directed by manufacturer's installation instructions. Meet requirements of ASTM International E814, Standard Test Method for Fire Tests of Through-Penetration Fire Stops.

F. Pipe Installation:

1. Provide installation of piping systems coordinated to account for expansion and contraction of piping materials and building as well as anticipated settlement or shrinkage of building. Install work to prevent damage to piping, equipment, and building and its contents. Provide piping offsets, loops, expansion joints, sleeves, anchors or other means to control pipe movement and minimize forces on piping. Verify anticipated settlement and/or shrinkage of building with Project Structural Engineer. Verify construction phasing, type of building construction products and rating coordinating installation of piping systems.
2. Include provisions for servicing and removal of equipment without dismantling piping.

G. Plenums: Provide plenum rated materials that meet the requirements to be installed in plenums. Immediately notify Contracting Officer of discrepancy.

3.2 REVIEW AND OBSERVATION

A. Confirm Review and Observation requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Notify Contracting Officer in writing at following stages of construction so that they may, at their option, visit site for review and construction observation:

1. Underground piping installation prior to backfilling.
2. Prior to covering walls.
3. Prior to ceiling cover/installation.
4. When main systems, or portions of, are being tested and ready for inspection by AHJ.
5. When mains or branchlines are to be permanently concealed by construction or insulation systems.
6. When fire suppression systems, or portions of, are being tested and ready for inspection by AHJ.

C. Bear responsibility and cost to make piping accessible, to expose concealed lines, or to demonstrate acceptability of the system. If Contractor fails to notify Contracting Officer at times prescribed above, costs incurred by removal of such work are the responsibility of the Contractor.

D. Final Punch: Costs incurred by additional trips required due to incomplete systems will be the responsibility of the Contractor.

3.3 CUTTING AND PATCHING

A. Confirm Cutting and Patching requirements in Division 01, General Requirements. In absence of specific requirements, comply with individual Division 21, Fire Suppression Sections and the following:

B. In absence of specific requirements, comply with individual Division 21, Fire Suppression Sections and the following:

1. Cutting and patching performed under Division 21, Fire Suppression includes, but is not limited to:

    a. Cutting and patching of plaster or partitions.
    b. Cutting and patching of finished ceilings.

2. Perform cutting and patching by skilled craftsmen in trade of work to be performed. Fill holes which are cut oversized for completed work. Match refinished areas with existing adjacent finish in a manner acceptable to Contracting Officer.
3. When masonry to concrete construction must be penetrated, provide a steel pipe sleeve in opening and grout in place in a neat manner. Leave grout surface to match existing finish. Provide escutcheons. If sleeves are not provided, core drill penetrations.
4. Locate concealed utilities to eliminate possible service interruption or damage.
5. Additional work required by lack of proper coordination will be provided at no additional cost adjustment to the Contract.

6. Proposed floor cutting/core drilling/sleeve locations to be approved by Project Structural Engineer. Submit proposed locations to Contracting Officer/Project Structural Engineer. Where slabs are of post tension construction, perform x-ray scan of proposed penetration locations and submit scan results including proposed penetration locations to Project Structural Engineer/Contracting Officer for approval. Where slabs are of waffle type construction, show column cap extent and cell locations relative to proposed penetration(s).
7. Cutting, patching and repairing for work specified in this Division including plastering, masonry work, concrete work, carpentry work, and painting included under this Section will be performed by skilled craftspeople of each respective trade in conformance with appropriate Division of Work.
8. Additional openings required in building construction to be made by drilling or cutting. Use of jack hammer is specifically prohibited. Patch openings in and through concrete and masonry with grout.
9. Restore new or existing work that is cut and/or damaged to original condition. Patch and repair specifically where existing items have been removed. This includes repairing and painting walls, ceilings, etc. where existing conduit and devices are removed as part of this project. Where alterations disturb lawns, landscaping, paving, and walks, surfaces to be repaired, refinished and left in condition matching existing prior to commencement of work.
10. Repair mutilation of building around pipes, equipment, hangers, and braces.

3.4 EQUIPMENT SELECTION AND SERVICEABILITY

A. Replace or reposition equipment which is too large or located incorrectly to permit servicing at no additional cost adjustment to the Contract.

3.5 DELIVERY, STORAGE AND HANDLING

A. Confirm requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.

B. In absence of specific requirements, comply with individual Division 21, Fire Suppression Sections and the following:

1. Handle materials delivered to project site with care to avoid damage and deterioration. Store materials in original containers which identify manufacturer, name, brand and model numbers on site inside building or protected from weather, sun, dirt and construction dust. Insulation and lining that becomes wet from improper storage and handling to be replaced before installation. Products and/or materials that become damaged due to water, dirt and/or dust as a result of improper storage to be replaced before installation.
2. Protect equipment and pipe to avoid damage. Close pipe openings with caps or plugs. Keep motors and bearings in watertight and dustproof covers during entire course of installation.
3. Protect bright finished shafts, bearing housings and similar items until in service.

3.6 DEMONSTRATION

A. Confirm Demonstration requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Upon completion of work and adjustment of equipment and test systems, demonstrate to Contracting Officer that equipment furnished and installed or connected under provisions of these Specifications functions in manner required. Provide field instruction to Maintenance Staff as specified in Division 01, General Requirements and individual Division 21, Fire Suppression Sections.

C. Manufacturer's Field Services: Furnish services of a qualified person at time approved by Contracting Officer to instruct maintenance personnel, correct defects or deficiencies, and demonstrate to satisfaction of Contracting Officer that entire system is operating in satisfactory manner and complies with requirements of other trades that may be required to complete work. Complete instruction and demonstration prior to final job site observations.

D. Prior to acceptance of work and during time designated by Contracting Officer, provide necessary qualified personnel to operate system for a period of two hours.

E. Instruct the Park Staff as directed by the Contracting Officer in the operation of the sprinkler system, including main valve position (open or closed) recognition, system drainage, system testing, dry pipe valve reset and the relation to the fire alarm system.

F. Upon completion of work and adjustment of equipment, test systems to demonstrate to Contracting Officer that equipment is furnished and installed or connected under provisions of these Specifications.

3.7 CLEANING

A. Confirm Cleaning requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Upon completion of installation, except for sprinklers, thoroughly clean exposed portions of equipment, removing temporary labels and traces of foreign substances. Throughout work, remove construction debris and surplus materials accumulated during work.

C. Sprinklers may not be cleaned except for vacuuming in a manner in which no part of the sprinkler is touched by the vacuuming equipment. Replace sprinklers which bear traces of foreign substances with sprinklers of same model, temperature, K-factor, orifice, finish, style, orientation, and the like.

3.8 INSTALLATION

A. Confirm Installation requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Install equipment in accordance with manufacturer's installation instructions, plumb and level and firmly anchored to vibration isolators. Maintain manufacturer's recommended clearances.

C. Start-up equipment, in accordance with manufacturer's start-up instructions, in the presence of manufacturer's representative. Test controls and demonstrate compliance with requirements. Replace damaged or malfunctioning controls and equipment. Provide pump impellers to obtain Basis of Design design capacities.

D. Provide miscellaneous supports/metals required for installation of equipment and piping.

3.9 PAINTING

A. Confirm requirements in Division 01, General Requirements and Division 09, Finishes.

3.10 DEMOLITION

A. Confirm Demolition requirements in Division 01, General Requirements and Division 02, Existing Conditions.

B. In absence of specific requirements, comply with individual Sections in Division 21, Fire Suppression and the following:

1. Scope:

    a. It is the intent of these documents to provide necessary information and adjustments to fire protection system required to meet code, and accommodate installation of new work.
    b. Coordinate with Contracting Officer so that work can be scheduled not to interrupt operations, normal activities, building access or access to different areas.
    c. Existing Conditions: Determine exact location of existing utilities and equipment before commencing work, pay for damages caused by failure to locate and preserve underground utilities. Replace damaged Item with new material to match existing. Promptly notify Contracting Officer if utilities are found which are not shown on Drawings.

2. Equipment and Piping: Unless otherwise directed, equipment, piping, or fittings being removed as part of demolition process are the Park's property. Remove other item not scheduled to be reused or relocated from job site as directed by Contracting Officer.
3. Unless specifically indicated on Drawings, remove exposed, unused piping to behind finished surfaces (floor, walls, ceilings, etc.). Cap piping and patch surfaces to match surrounding finish.
4. Unless specifically indicated on Drawings, remove unused equipment, fittings, rough-ins, and connectors. Removal is to be to a point behind finished surfaces (floors, walls, and ceilings).
5. Coordinate demolition of existing fire suppression systems with Contractor. Where applicable or possible, portions of fire suppression demolition work may be performed by Contractor. Verify with local AHJ as to limitations of demolition by others and not fire suppression trades. Coordinate extent of demolition of fire suppression work to be done by others and supervise this work. No extra costs will be approved by replacement of systems due to improper or excessive demolition.

3.11 ACCEPTANCE

A. Confirm requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.

B. In absence of specific requirements, comply with individual Sections in Division 21, Fire Suppression. System cannot be considered for acceptance until work is completed and demonstrated to Contracting Officer that installation is in strict compliance with Specifications, Drawings and manufacturer's installation instructions, particularly in reference to following:

1. Testing reports including Contractor's Material and Test Certificate for Underground Piping, Contractor's Material and Test Certificate for Aboveground Piping, Contractor's Material and Test Certificate for Private Fire Service Mains, Fire pump acceptance test data report, and the like.
2. Cleaning
3. Operation and Maintenance Manuals
4. Training of Operating Personnel
5. Record Drawings
6. Warranty and Guaranty Certificates
7. Start-up/Test Document and Commissioning Reports
8. Letter of Conformance

3.12 FIELD QUALITY CONTROL

A. Confirm Field Quality Control requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 21, Fire Suppression Sections.

B. Upon completion of installation of equipment, sprinklers, hose valves and piping and after units are water pressurized, test system to demonstrate capability and compliance with requirements. When possible, correct malfunctioning items at site, then retest to demonstrate compliance; otherwise remove and replace with new Item and proceed with retesting.

C. Inspect each installed Item for damage to finish. If feasible, restore and match finish to original, except fire sprinklers, at site; otherwise, remove Item and replace with new Item. Feasibility and match to be judged by Contracting Officer. Remove cracked or dented Item and replace with new Item.

D. Fire sprinklers may not be reused, or cleaned, except for dusting. Replace damaged, field painted, oversprayed, overcoated or field coated sprinklers with new sprinklers of same manufacturer, model, finish, K-factor and performance characteristics. Where identical replacement sprinklers are not available, provide sprinklers of similar finish, style, K-factor and performance characteristics.

3.13 LETTER OF CONFORMANCE

A. Provide Letter of Conformance and copies of manufacturers' warranties and extended warranties with a statement that fire suppression items were installed in accordance with manufacturer's recommendations, UL listings and FM Global approvals.

B. Include Letter of Conformance, copies of manufacturers' warranties and extended warranties in Operation and Maintenance Manuals.

3.14 ELECTRICAL INTERLOCKS

A. Where equipment motors are to be electrically interlocked with other equipment for simultaneous operation, utilize fire protection equipment wiring diagrams to coordinate with electrical systems so that proper wiring of equipment involved is affected.

3.15 CONNECTIONS TO EXISTING

A. Prior to connection of piping to existing piping or utilities, field verify existing conditions and exact sizes and locations of existing piping.

B. Provide additional offsets, transitions, joints, cut-ins, and replace portions of existing as required to facilitate connections of new.

END OF SECTION 21 00 00

SECTION 21 05 00 – COMMON WORK RESULTS FOR FIRE SUPPRESSION

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included:

1. Aboveground Black Steel Pipe and Fittings
2. Wall and Floor Penetrations and Sleeves
3. Switches, Valve Supervisory
4. Switches, Water Detector
5. Hangers and Supports
6. Anchors and Attachments
7. Pipe Stands
8. Gauges
9. Alarm Bells
10. Valves
11. Backflow Prevention Devices
12. Pipe, Valve, and Fire Protection Equipment Identification
13. Signs
14. Drains

1.2 RELATED SECTIONS

A. Contents of Division 21, Fire Suppression and Division 01, General Requirements apply to this Section.

B. In addition, reference the following:

1. Division 22, Plumbing
2. Division 23, Heating, Ventilating and Air Conditioning
3. Division 26, Electrical
4. Division 28, Electronic Safety
5. Division 31, Earthwork
6. Section 21 00 00, Fire Suppression Basic Requirements
7. Section 21 08 00, Commissioning of Fire Suppression Systems
8. Section 21 13 00, Fire Suppression Sprinkler Systems
9. Section 21 13 19, Fire Suppression Preaction Sprinkler Systems
10. Section 21 22 00, Clean-Agent Fire-Extinguishing Systems

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. Meet requirements of ASCE 7, Minimum Design Loads for Buildings and Other Structures, by American Society of Civil Engineers, latest adopted edition.

1.4 SUBMITTALS

A. Submittals as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, meet the following:

1. Material and Equipment: Listed for its intended fire protection use in current UL Fire Protection Equipment Directory, or UL Online Certifications Directory for Fire Protection, FM Global Approval Guide, and International Code Council Evaluation Service Reports. All material and equipment to be new and from a current manufacturer.
2. Provide per AHJ requirements.
3. References to product Specifications for materials are listed according to accepted ANSI, ASTM, ASME, AWWA and other base standards. Materials to meet latest approved versions of these standards.
4. Fire Suppression Screw-Thread Connections: Comply with local fire department/fire marshal regulations for sizes, threading and arrangement of connections for fire department equipment to fire department connections.
5. Manufacturers: Unless an item is marked "No substitutions", submit substitution request for materials of other than named manufacturers.
6. Noise and Vibration:

    a. Install vibration isolators and measures required to prevent noise and vibration from being transmitted to occupied areas. Select equipment to operate within noise coefficient (NC) design level for particular type of installation in relation to its location.
    b. After installation, make proper adjustments to reduce noise and vibration to acceptable levels as defined by Contracting Officer.
    c. In acoustically sensitive areas, design system in a manner that minimizes the number of wall penetrations.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.7 SYSTEM IMPAIRMENT

A. When returning a water-based fire protection system to service after impairment or control valve closure, verify the system is in working order by performing a main drain test per NFPA 25.

PART 2 - PRODUCTS

2.1 GENERAL

A. Salient Characteristics as outlined in Articles below:

1. Physical
2. Functional
3. Performance

2.2 ABOVEGROUND BLACK STEEL PIPE AND FITTINGS

A. Pipe Sizes up to and Including 4-inches:

1. Physical: Carbon steel.
2. Functional: None.
3. Performance:

a. ASTM A53, ASTM A135, or ASTM A795.
b. Schedule 40 only.

B. Joints:

1. Physical:

a. Threaded, flanged or bevel welded.
b. Piping installed in plenums or shafts to have welded joints.

2. Functional: None.
3. Performance: None.

C. Threaded Fittings:

1. Physical:

a. Malleable iron.
b. Cast iron.

2. Functional: Fittings.

3. Performance:

    a. Malleable Iron: Class 150 and Class 300, ANSI B16.3.
    b. Cast Iron: Class 125 and 250, ANSI B16.3.

D. Flanged Fittings:

1. Physical:

    a. Cast iron.
    b. Raised ground face, bolt holes spot faced.

2. Functional: Fittings.
3. Performance: Class 125 and 250, ASME B16.1.

E. Welded Fittings:

1. Physical: Steel.
2. Functional: Fittings.
3. Performance:

    a. Carbon Steel: Long radius, standard weight or extra strong.
    b. Factory Wrought Steel Buttweld Fittings: ASME B16.9.
    c. Buttwelding Ends for Pipe, Valves, Flanges and Fittings: ASME B16.25.
    d. Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and Elevated Temperatures: ASTM A234.
    e. Steel Pipe Flanges and Flanged Fittings: ASME B16.5.
    f. Forged Steel Fittings, Socket Welded and Threaded: ASME B16.11.

F. Grooved Couplings:

1. Physical: Iron.
2. Functional: Fittings.
3. Performance:

    a. UL 213, AWWA C606, ASTM A536 ductile iron or ASTM A47 malleable iron, with enamel finish and grooves or shoulders designed to accept grooved couplings.
    b. Synthetic-rubber gasket with central-cavity, pressure-responsive design and ASTM A183 carbon-steel bolts and nuts.
    c. FM Global approved.

G. Anti-Microbial Coating:

1. Physical: None.
2. Functional: Inhibits corrosion from microbiological organisms.
3. Performance: Factory-applied coating to inhibit corrosion from microbiological organisms.

2.3 WALL AND FLOOR PENETRATIONS AND SLEEVES

A. Physical: None.

B. Functional: None.

C. Performance: Below grade and high water table areas to utilize waterproof elastomeric compound.

2.4 SWITCHES, VALVE SUPERVISORY

A. Physical: None.

B. Functional: None.

C. Performance: Provide to mount on applicable, compatible valve (OS&Y gate, or PIV), with SPDT switches to match requirements of fire alarm system.

2.5 SWITCHES, WATER DETECTOR

A. Water Flow Switches:

1. Physical: None.
2. Functional: None.
3. Performance:

    a. Vane-type; SPDT switches; electronic retard, adjustable time delay (0 to 75 seconds).
    b. Wet Sprinkler Systems: 450 PSI, 18-feet per second, 4-10 gpm.

B. Pressure Operated Alarm Switches:

1. Physical: None.
2. Functional: None.
3. Performance: Pressure actuated with SPDT electrical switches and adjustable time delay (0 to 75 seconds).

2.6 HANGERS AND SUPPORTS

A. Physical:

1. Ferrous.
2. Hanger rods to be zinc electroplated carbon steel.
3. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finish.

B. Functional: None.

C. Performance: Use carbon steel pipe hangers and supports, metal trapeze pipe hangers and attachments for general service applications.

2.7 ANCHORS AND ATTACHMENTS

A. General Anchors and Attachments:

1. Physical: Ferrous.
2. Functional: None.
3. Performance:

    a. Anchor supports to masonry, concrete and block walls per anchoring system manufacturer's recommendations, or as modified by project Structural Engineer.
    b. ITW Buildex Sammys with FM Approval only are not allowed in certain seismic zones.

B. Attachments in Concrete:

1. Physical: None.
2. Functional: None.
3. Performance: For cast-in-place anchors for hangers, verify listing is for hangers, braces, or both.

2.8 PIPE STANDS

A. Adjustable Pipe Saddle Support with Yoke:

1. Physical: Steel pipe with steel saddle.
2. Functional: Designed to support horizontal pipe from floor stanchion.
3. Performance:

    a. U-bolt and hex nuts to hold pipe securely to saddle or pipe clamp type.
    b. ANSI/MSS SP-69; SP-58. Type 37.

B. Base Stand:

1. Physical: Steel pipe welded to steel base plate.
2. Functional: None.
3. Performance: Base stand.

2.9 PRESSURE GAUGES

A. Physical:

1. Stainless steel case.
2. Bronze bourdon tube or spring type.
3. 3-1/2-inch, dial type.

B. Functional: Pressure gauge.

C. Performance: 0 to 300 PSI.

2.10 ALARM BELLS

A. Physical: Weatherproof.

B. Functional: Alarm bell.

C. Performance:

1. Minimum weatherproof backbox, typical 90 dBA at 10-feet.
2. Provide sign that reads, "When Bell Rings - Call 911".

2.11 VALVES

A. OS&Y Gate Valves:

1. Physical:

a. 2-1/2-inches and Larger: Iron body.
b. 2-inches and Smaller: Bronze body.

2. Functional: Valve.
3. Performance: None.

B. NRS Gate Valves:

1. Physical:

a. Iron body.
b. Non-rising stem with indicator post.

2. Functional: Valve.
3. Performance: None.

C. Underground Butterfly Valves:

1. Physical: Telescopic barrel type.
2. Functional: Valve.
3. Performance: None.

D. Swing Check Valves:

1. Physical:

a. Iron body.
b. Rubber and bronze faced checks.

2. Functional: Valve.
3. Performance: None.

E. Wafer Check Valves:

1. Physical:

a. Iron body.
b. Rubber seat.

2. Functional: Valve.
3. Performance: Iron body, rubber spring actuated.

F. Butterfly Valves:

1. Physical:

a. Ductile iron body.
b. Manufactured with factory-installed tamper switches.

2. Functional: Valve.
3. Performance: None.

G. Pressure Relief Valves:

1. Physical: Bronze body, stainless steel spring.
2. Functional: Valve.
3. Performance: None.

H. Automatic Ball Drip Drain Valves:

1. Physical: Bronze.
2. Functional: Valve.
3. Performance: Spring-type.

I. Three-Way Gauge Valves:

1. Physical: Brass.
2. Functional:

a. Valve.
b. Rated to 300 psi.

3. Performance: None.

J. Automatic Air-Release Valves for Wet Systems:

1. Physical: None.
2. Functional:

a. Valve.
b. Rated to 175 psi.

3. Performance:

    a. Automatic float-type with shutoff mounted in a water retention pan.
    b. Single set 24VAC@2A for electronic supervision.
    c. Ball valve switch with cover tamper.

K. Ball Valves:

1. Physical:

    a. Brass body.
    b. Brass stem.
    c. Forged brass ball disc.

2. Functional: Valve.
3. Performance: None.

2.12 BACKFLOW PREVENTION DEVICES

A. Double Check Valve Assembly:

1. Physical: None.
2. Functional: None.
3. Performance:

    a. Two check valves in series with OS&Y gate or butterfly valves at each end.
    b. Provide detector if required by local utility.
    c. UL listed or FM Global Approved for fire suppression service as an assembly.
    d. Approved by local and state authorities, including project's State Department of Health for the position in which it is installed.

2.13 PIPE, VALVE, AND FIRE PROTECTION EQUIPMENT IDENTIFICATION

A. Physical:

1. Plastic.
2. Corrosion-resistant metal.

B. Functional: None.

C. Performance:

1. Engraved plastic laminate or corrosion resistant metal sign or plastic equipment marker.
2. Corrosion-resistant chain or permanent adhesive.

2.14 SIGNS

A. Physical:

1. Plastic.
2. Corrosion-resistant metal.

B. Functional: Signage.

C. Performance:

1. Engraved plastic laminate or corrosion resistant metal sign or plastic equipment marker.
2. Corrosion-resistant chain or permanent adhesive.

2.15 DRAINS

A. Physical: Reference Aboveground Black Steel Pipe and Fittings.

B. Functional: Drainage.

C. Performance:

1. Route drain to exterior.
2. Provide with 45-degree turn down elbow.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. Install in conformance with UL Listing, FM Approval or ICC-ES requirements and restrictions.

B. Install per manufacturer's instructions and recommendations.

3.2 ABOVEGROUND BLACK STEEL PIPE AND FITTINGS

A. Piping Routing:

1. Orient horizontal routes parallel with walls and beam lines.
2. Install piping as shown or described by diagrams, details and notations on Drawings or, if not indicated, install piping to provide the shortest route which does not obstruct usable space or block access for servicing the building and its equipment.
3. Install piping in concealed spaces above finished ceilings. Prior to design and installation. obtain pre-approval by Contracting Officer for exposed piping.
4. In open-to-structure areas which are open to public view, route exposed piping to minimize visual impact. Obtain Contracting Officer's approval of exposed piping installation.
5. Coordinate installation with other trades. Route piping as required to avoid building structure, equipment, plumbing piping, HVAC piping, ductwork, lighting fixtures, electrical conduits and bus ducts and similar work. Final location of lighting will have

priority over final sprinkler locations. Provide drains to trapped sections of system which result from such routing. Other trades take precedence for installation space.

6. Support piping adjacent to walls, overhead construction, columns and other structural and permanent enclosure elements of the building. Limit clearance to 2-inches wherever furring is indicated for concealment of piping. Allow for insulation thickness. Locate insulated piping to provide minimum 1-inch clearance outside insulation.
7. Wherever possible in finished and occupied spaces, conceal piping from view by locating within column or beam enclosures, hollow wall construction, or above suspended ceilings. Do not encase horizontal routes in solid partitions, except where approved.
8. General Electrical Equipment Clearances: Do not route piping through electrical rooms, transformer vaults, elevator equipment rooms and other electrical or electronic equipment spaces and enclosures. Do not route piping above electric power or lighting panel, switchgear, low voltage panel, or similar electric device.
9. Rooms Protected by Alternative Systems: Route water filled and dry system piping around rooms protected by pre-action systems, clean agent systems, gaseous suppression systems and other alternative fire suppression systems.
10. Install pipe runs to minimize obstruction to other work.
11. Pitch all dry and pre-action system piping 1/4-inch per 10-feet for mains and 1/2-inch per 10-feet for branch lines, including pipe passing through both warm and cold areas.

B. Couplings:

1. Install where indicated on Drawings and on each side of pieces of equipment to permit easy removal of equipment.
2. Deburr cut edges.

C. Pipe Penetrations: Wire pipe cutout coupon at point of pipe penetration.

D. Pipe and Pipe Fittings:

1. Expansion and Flexibility: Install work with due regard for expansion and contraction to prevent damage to the piping, equipment, building and its contents. Provide piping offsets, loops, approved type expansion joints, sway bracing, wire restraints, vertical restraints, flexible couplings or other means to control pipe movement and to minimize pipe forces.
2. Coordinate support of pipe 4-inches and larger with Structural Engineer.
3. Provide clearances around piping per NFPA 13.
4. Install dry and pre-action welded pipe with welds facing vertically up, or where this is not possible, as close as possible to vertical between 46 degrees and 234 degrees. Intent is to minimize corrosion caused by moisture in the bottom of pipes.

3.3 WALL AND FLOOR PENETRATIONS AND SLEEVES

A. Escutcheons: Install on exposed pipes passing through walls or floors.

1. Pipe Sleeves: Lay out work in advance of pouring concrete and furnish and set sleeves necessary to complete work.
2. Floor Sleeves: Provide sleeves on pipes passing through concrete or masonry construction. Extend sleeve 1-inch above finished floor. Caulk pipes passing through floor with nonshrinking fire and water resistant grout or approved equivalent caulking compound.

Caulk/seal piping penetrations through fire rated building walls and floors with listed, fire-rated assemblies. Provide fire-rated assemblies per local AHJ requirements.

3. Wall Sleeves: Provide sleeves on pipes passing through concrete or masonry construction. Provide sleeve flush with finished face of wall. Caulk pipes passing through walls with non-shrinking caulking compound. Caulk/seal piping penetrations through fire rated building walls and floors with listed, fire-rated assemblies. Provide fire-rated assemblies per local AHJ requirements.
4. Beam Sleeves:

    a. Coordinate with trades for locations of pipe sleeves in reinforced concrete and steel beams. Penetrations must be indicated on structural shop drawings. See Drawings and Specifications for specific sleeve location limitations. Pipe sleeve locations must be indicated on reinforced concrete and steel beam shop drawings.
    b. Field cutting of beams not allowed without written approval of Structural Engineer. No extra costs allowed for failure to coordinate beam penetrations prior to reinforced concrete and steel beam shop drawing submittal.

5. Penetrations in Fire-Rated Wall/Floor Assemblies:

    a. Reference Division 07, Thermal and Moisture Protection.
    b. Coordinate with Drawings location of fire rated walls, ceilings and floors. When these assemblies are penetrated, seal around piping and equipment with approved firestopping material.
    c. Provide proper sizing when providing sleeves or core-drilled holes to accommodate the penetration. Firestop voids between sleeve or core-drilled hole and pipe passing through to meet the requirements of ASTM E814 and NFPA.
    d. Install firestopping material complete as directed by manufacturer's installation instructions. Meet requirements of ASTM E814.

3.4 SWITCHES, VALVE SUPERVISORY

A. Coordinate with Division 28, Electronic Safety

3.5 SWITCHES, WATER DETECTOR

A. Wire pipe cutout coupon at point of connection of switch to pipe.

B. Flow switches: Connect to system side of valves and drain connections.

C. Coordinate with Division 28, Electronic Safety.

3.6 HANGERS AND SUPPORTS

A. Installation of pipe hangers, inserts and supports to conform to NFPA 13. Provide adjustable hangers, inserts, brackets, clamps, supplementary steel and other accessory materials required for proper support of pipe lines and equipment. Provide supplementary materials for proper support and attachment of hangers.

B. Select size of hangers and supports to exactly fit pipe size for bare piping.

C. Select size of hangers and supports to exactly fit around piping insulation with saddle or shield for insulated piping.

3.7 ANCHORS AND ATTACHMENTS

A. In post-tension construction, determine location of post-tension cables and install anchors to avoid contact or interference with post-tension cables. Coordinate with Structural.

B. Do not use powder-driven attachments.

C. Building Attachments and Inserts: Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves and flanges, for sizes NPS 2-1/2 and larger. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.

D. Hanger and Support Attachments:

1. Concrete:

    a. Before Pouring: Prior to installation, coordinate locations of cast in place concrete inserts with other trades. Install in accordance with manufacturer's instructions.
    b. After Pouring:

        1) Where supports in slabs are required after concrete has been poured, provide drilled-in threaded inserts (mechanical-expansion anchors), installed in accordance with manufacturer's recommendations.
        2) Install mechanical-expansion anchors after concrete is completely cured and in accordance with manufacturer's installation instructions.
        3) Where anchors are to be installed in post-tension construction, determine and avoid locations of post-tension cables prior to drilling.

2. Metal Floor Deck: Support hangers per UL Listing or FM Approval for selected concrete insert before pouring of concrete topping, or from beam clamps fastened to structural steel.
3. Steel Joists: Support hangers from beam clamps fastened to bar joists or to auxiliary steel between bar joists as required.
4. C-Clamp Hangers: Do not attach to one side of double-angle bottom members.
5. Locate and install hangers, supports and attachments connecting to I-joists, structural insulated panels (SIPs), cross laminated timber and similar engineered structural products according to the structural product manufacturer specifications.

E. Make available to the Contracting Officer information required to verify the anchorage, sway bracing and restraint of fire protection systems.

3.8 PIPE STANDS

A. Secure to floor.

B. Install to maintain pipe level and plumb.

C. Securely attach to supported pipe by u-bolt.

3.9 GAUGES

A. Install gauges conveniently and accessibly located with reference to finished building for repairs, removal and service.

B. Install with dial positioned for maximum visibility.

3.10 ALARM BELLS

A. Locate exterior alarm bells at 8-feet above finished grade. Coordinate with Contracting Officer.

B. Coordinate with Division 26, Electrical and Division 28, Electronic Safety.

3.11 VALVES

A. General:

1. Provide post indicator on buried control valves.
2. Inspect valves for leaks. Adjust or replace packing to stop leaks. Replace valve if leak persists.
3. Install valves where required for proper operation, testing and drainage. Locate valves so as to be accessible and so that separate support can be provided when necessary. Install conveniently and accessibly located with reference to finished building for repairs, removal and service.

B. Swing Check Valves: Install in horizontal position with hinge pin horizontally perpendicular to centerline of pipe. Install for proper direction of flow.

C. Wafer Check Valves: Install between two flanges in horizontal or vertical position, position for proper direction of flow.

D. Pressure Relief Valves: Provide piping to permanent drain.

3.12 BACKFLOW PREVENTION DEVICES

A. Install conveniently and accessibly located with reference to finished building for repairs, removal and service.

B. Provide listed backflow assembly at sprinkler system water source connection. Coordinate with local utility; conform to their installation requirements.

C. Provide method of forward flow testing at full system demand without dismantling any part of the system. Indicate location, method of testing and location of test drain discharge on submittal and As-Built Drawings. Provide signage as required by NFPA 13. Locate drainage for forward testing where damage will not occur, including damage to landscaping.

D. Chain and padlock in "open" position. Provide two sets of keys.

E. Provide control valve supervisory switches connected to the fire alarm system.

F. Provide Water Bureau approved Bypass-Meter and compatible Touch-Pad. Touch-Pad unit must be accessible from right-of-way.

G. Reduced Pressure Backflow Preventer:

1. Locate within 5-feet of finished floor near drain shown on Plumbing Drawings or an existing drain of sufficient size which can accept full discharge of relief valve without doing damage or arrange and pay for installation of a suitable size drain.
2. Provide drain piping to sanitary sewer. Coordinate with Division 22, Plumbing.

3.13 PIPE, VALVE, AND FIRE PROTECTION EQUIPMENT IDENTIFICATION

A. Install engraved plastic laminate or corrosion resistant metal sign or plastic equipment marker, secured with corrosion-resistant chain or permanent adhesive on or near each Item of fire suppression equipment and each operational device, as specified in this specification if not otherwise specified for each Item or device.

B. Provide signs for the following general categories of equipment and operational devices: Valves, drains, pumps, standpipes, tanks and similar equipment.

C. Each new piece of equipment to bear a permanently attached identification plate, listing manufacturer's name, capacities, sizes and characteristics.

D. Piping to bear the manufacturer's name, schedule of thickness, size and ASTM identification number

E. Provide valve tag on every valve, control device, main drain, auxiliary drain, and drum drip in each system. Exclude check valves and valves within factory fabricated equipment units. List each tagged valve in valve schedule for each piping system.

F. List each tagged item and its location in valve schedule; identify on fire suppression drawings.

G. Install framed, glass or rigid transparent plastic covered, mounted valve schedule and valve location drawing in main riser or fire pump room.

H. Provide identification sign on ceiling tile below valve location.

I. Provide permanent identification sign at pressure regulating valves stating required setting of pressure regulator.

J. Adjusting: Relocate fire suppression identification device which has become visually blocked.

K. Cleaning: Clean face of identification devices and glass frames of valve charts.

3.14 SIGNS

A. General Information Signs: Provide a general information sign used to determine system design basis and information relevant to the inspection, testing and maintenance requirements required by NFPA 25, Standard for the Inspection, Testing and Maintenance of Water-Based Fire Protection Systems. Such general information is to be provided with a permanently marked weatherproof metal or rigid plastic sign, secured with corrosion-resistant wire, chain, or other acceptable means. Such signs are to be placed at each system control rise loop and auxiliary system control valve. The sign is to include the following information:

1. Name and Location of the Facility Protected
2. Presence of High-Piled and/or Rack Storage
3. Maximum Height of Storage Planned
4. Flow Test Data
5. Location of Auxiliary Drains and Low Point Drains
6. Original Results of Main Drain Flow Test
7. Name of Installing Contractor or Designer
8. Indication of presence and location of other auxiliary systems.

B. Dry Signs: At system riser supplying dry systems, provide the following information: volume in gallons contained in each system.

3.15 DRAINS

A. Locate drain connections within 7-feet of floor. Provide piping capable of being fully drained.

B. Provide a drain vent at top of vertical drains. Coordinate with Division 22, Plumbing.

C. Coordinate location of auxiliary drains with Contracting Officer. Contracting Officer to approve location before drain is installed.

D. Protect drains from tampering and accidental operation.

E. Protect drain discharge at the exterior with a turned-down 45 degree elbow.

END OF SECTION 21 05 00

SECTION 21 08 00 – COMMISSIONING OF FIRE SUPPRESSION SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included: Test Equipment

B. Summary:

1. Fire suppression commissioning requirements are included in this Section.
2. Commissioning is the process of reviewing system installation and operation to confirm the systems are installed and operating per the design intent and contract documents. It does not replace the official testing and acceptance of the Authority Having Jurisdiction (AHJ).

C. Definitions:

1. Reference Division 21, Fire Suppression and Division 01, General Requirements.
2. Commissioning Authority: Commissioning Agent, representing the Contracting Officer and directing commissioning activities.

1.2 RELATED SECTIONS

A. Contents of Division 21, Fire Suppression and Division 01, General Requirements apply to this Section.

B. In addition, reference Section 01 91 14, Total Building Commissioning.

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Division 21, Fire Suppression and Division 01, General Requirements.

B. In addition, meet the following:

1. ASHRAE Guideline 0 – The Commissioning Process
2. NFPA 3 – Recommended Practice for Commissioning and Integrated Testing of Fire Protection and Life Safety Systems

1.4 SUBMITTALS

A. Submittals as required by Division 21, Fire Suppression and Division 01, General Requirements.

B. In addition, provide:

1. Reference Section 01 91 14, Total Building Commissioning, for specific Commissioning submittal requirements.
2. In addition, submit NFPA 13 Record of Completion.

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Division 21, Fire Suppression and Division 01, General Requirements.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by Division 21, Fire Suppression and Division 01, General Requirements.

B. In addition, provide:

1. Manufacturer's Warranty.
2. Commissioning, inspecting, and testing not to modify terms or time periods of fire suppression equipment, systems, and controls warranties including related equipment and systems, and adjacent work.
3. Fire Suppression system warranties to start from date of Commissioning Agent acceptance.

1.7 COORDINATION

A. Coordination as required by Division 21, Fire Suppression and Division 01, General Requirements.

B. Reference Section 01 91 14, Total Building Commissioning, for requirements pertaining to coordination during the commissioning process.

PART 2 - PRODUCTS

2.1 TEST EQUIPMENT

A. Provide testing equipment required to perform startup, initial checkout and functional performance testing for the equipment being tested under Division 21, Fire Suppression. Furnish two-way radios for each testing participant.

B. Testing equipment to be of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the Specifications.

PART 3 - EXECUTION

3.1 GENERAL DOCUMENTATION REQUIREMENTS

A. With assistance from the installing contractors, the Commissioning Authority will prepare Pre-Functional Checklists for commissioned systems.

B. Red-Lined Drawings:

1. Verify equipment, devices and wiring are shown correctly on red-lined drawings.
2. Record the red-lined drawing changes, as a result of Functional Testing and incorporate into the final As-Built Drawings.

C. Operation and Maintenance Data:

1. Submit a copy of O&M literature within 45 days of each submittal acceptance for use during the commissioning process for commissioned equipment and systems.
2. The Commissioning Authority will review the O&M literature once for conformance to project requirements.
3. The Commissioning Authority will receive a copy of the final approved O&M literature once corrections have been made by the Contractor.

D. Demonstration and Training:

1. Provide demonstration and training as required by the specifications.
2. Submit complete training plan and schedule to the Commissioning Authority four weeks prior to training.
3. Submit training agenda for each training session to the Commissioning Authority one week prior the training session.
4. Notify the Commissioning Authority at least 72 hours in advance of scheduled tests so that testing may be observed by the Commissioning Authority and Contracting Officer. Submit copies of the test record to the Commissioning Authority and Contracting Officer.
5. Engage a Factory-authorized service representative to train Contracting Officer's maintenance personnel to operate and maintain specific equipment.
6. Train Contracting Officer's maintenance personnel on procedures and schedules for trouble shooting, servicing, and maintaining equipment.
7. Review data in O&M Manuals.

3.2 CONTRACTOR'S RESPONSIBILITIES

A. Provide submittals to the Commissioning Authority for review.

B. Provide information requested by the Commissioning Authority for final commissioning documentation.

C. Prepare preliminary schedule for system orientation and inspections, operation and maintenance manual submissions, training sessions and functional performance testing for Contracting Officer. Distribute preliminary schedule to commissioning team members.

D. With input from the Commissioning Authority, update the construction schedule as required with detailed commissioning tasks throughout the construction period.

E. During the startup and initial checkout process, execute the related portions of the prefunctional checklists for commissioned equipment.

F. Provide ladders, radios, and test equipment to facilitate the commissioning process.

G. Gather operation and maintenance literature on equipment, and assemble in binders as required by the specifications. Submit to Commissioning Authority 45 days after submittal acceptance.

H. Provide any proprietary equipment required to verify the system functionality.

I. Pretest all system components and controls and correct all deficiencies prior to scheduling the Commissioning Authority for functional performance testing.

J. Coordinate with the Commissioning Authority to provide 48-hour advance notice so that the witnessing of equipment and system testing can begin.

K. Perform commissioning tests at the direction of the Commissioning Authority.

L. Participate in, and schedule vendors and contractors to participate in the training sessions.

3.3 TESTING PREPARATION

A. Provide written notification to the Commissioning Authority that work has been completed in accordance with the contract documents, and that the equipment, systems, and sub-systems are operating as required, and that testing has been completed.

B. Detection devices are installed, powered, programmed, labeled and pre-tested.

C. Notification appliances are installed, powered, synchronized, and pre-tested.

D. Smoke/fire dampers are powered and pre-tested and provided with access panel for visual inspection.

E. Smoke barriers and smoke control fans are started up and air flow balanced.

F. Fire suppression associated supply and/or exhaust fan shutdown are programmed and pre-tested.

G. Fire suppression control panel programmed per fire suppression sequence of operation.

H. Communication between the FACP and Central Station verified.

I. Check safety cutouts, alarms, and interlocks with life-safety systems during each mode of operation.

J. Prior to scheduling the Commissioning Authority for commissioning tests, certify in writing that testing procedures have been completed and that testing reports have been submitted, discrepancies corrected, and corrective work approved.

3.4 GENERAL TESTING REQUIREMENTS

A. Functional performance testing of the system, as observed by the Commissioning Authority, may not begin until the Fire Suppression contractor has completed all required pre-testing checklists and documentation.

B. The Commissioning Authority along with the fire suppression contractor to prepare detailed testing plans, procedures, and checklists for fire suppression systems, subsystems, and devices.

C. Provide technicians, instrumentation, and tools to perform commissioning test at the direction of the Commissioning Authority.

D. Scope of testing includes the entire fire suppression installation.

E. Test operating modes, interlocks, control responses, and responses to abnormal or trouble conditions, and verify proper response of the system control equipment.

F. If tests cannot be completed because of a deficiency outside the scope of the system, document the deficiency and report it to the Contracting Officer. After deficiencies are resolved, reschedule tests.

3.5 DEFICIENCIES/NON-CONFORMANCE, COST OF RETESTING, FAILURE DUE TO MANUFACTURER DEFECT

A. Reference Section 01 91 14 Total Building Commissioning, for requirements pertaining to deficiencies/non-conformance, cost of retesting, or failure due to manufacturer defect.

3.6 OPERATION AND MAINTENANCE MANUALS

A. The Operation and Maintenance Manuals to conform to Contract Documents requirements as stated in Division 21, Fire Suppression and Division 01, General Requirements.

3.7 TRAINING OF CONTRACTING OFFICER PERSONNEL

A. Contractor Training Responsibilities:

1. Provide the Commissioning Authority with a training plan four weeks before the planned training.
2. Provide designated Contracting Officer personnel with comprehensive training in the understanding of the systems and the operation and maintenance of each major piece of commissioned equipment or system.
3. Training starts with classroom sessions, if necessary, followed by hands on training on each piece of equipment, which illustrates the various modes of operation, including startup, shutdown, power failure, etc.
4. During any demonstration, should the system fail to perform in accordance with the requirements of the O&M manual or sequence of operations, the system will be repaired or adjusted as necessary and the demonstration repeated.

5. The appropriate trade or manufacturer's representative provides the instructions on each major piece of equipment. This person may be the start-up technician for the piece of equipment, the installing contractor or manufacturer's representative. Practical building operating expertise as well as in-depth knowledge of modes of operation of the specific piece of equipment are required. More than one party may be required to execute the training.
6. The training sessions follows the outline in the Table of Contents of the operation and maintenance manual and illustrate whenever possible the use of the O&M manuals and record drawings for reference.
7. Training Includes:

    a. Use the printed installation, operation and maintenance instruction material included in the O&M manuals.
    b. Include a review of the written O&M instructions emphasizing safe and proper operating requirements, preventative maintenance, special tools needed and spare parts inventory suggestions.
    c. Discuss warranties and guarantees.
    d. Cover common troubleshooting problems and solutions.
    e. Explain information included in the O&M manuals and the location of plans and manuals in the facility.
    f. Discuss any peculiarities of equipment installation or operation.

8. Fully explain and demonstrate the operation, function and overrides of any local packaged controls, not controlled by the central control system.

B. Schedule training after functional testing is complete, unless approved otherwise by the Contracting Officer.

END OF SECTION 21 08 00

SECTION 21 13 00 – FIRE SUPPRESSION SPRINKLER SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included:

1. Sprinklers
2. Flexible Sprinkler Hose Fitting Assembly - For fire sprinklers in suspended ceilings which are supplied by a wet pipe system.
3. Riser Manifold
4. Inspector's Test Connection
5. Dry-Pipe Valve
6. Dry System Drum Drip Drains
7. Nitrogen System
8. Wet System Air Vent
9. Spare Sprinkler Cabinet
10. Sprinkler Guards

B. Scope:

1. Wet-Pipe Sprinkler System.
2. Dry-pipe sprinkler system and/or dry barrel sprinklers for areas subject to 40 degrees F or less.
3. Preaction sprinkler system in collections/work room.

C. Coordinate location and type of tamper, flow and pressure switches and fire alarm system.

D. Provide electrical connections and wiring as required for a complete and operable system. Includes but is not limited to bells, air compressors, sump pumps, fire pumps, jockey pumps and pump controllers.

1.2 RELATED SECTIONS

A. Contents of Division 21, Fire Suppression and Division 01, General Requirements apply to this Section.

B. In addition, reference the following:

1. Division 22, Plumbing
2. Division 23, Heating, Ventilating and Air-Conditioning
3. Division 26, Electrical
4. Division 28, Electronic Safety
5. Section 21 00 00, Fire Suppression Basic Requirements
6. Section 21 05 00, Common Work Results for Fire Suppression

7. Section 21 08 00, Commissioning of Fire Suppression Systems
8. Section 21 13 19, Fire Suppression Preaction Sprinkler Systems
9. Section 21 22 00, Clean-Agent Fire-Extinguishing Systems

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.4 SUBMITTALS

A. Submittals as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, provide:

1. Hydraulic calculations.
2. Sway brace calculations.
3. Details of flexible sprinkler hose fitting assembly, including number and radius of bends, corresponding to equivalent feet used in hydraulic calculations. Provide details of sign to be installed at each flexible sprinkler hose fitting assembly.
4. Trapeze hanger details and calculations, including size, length and material. Additionally, provide size, weight and number of pipes to be carried on the trapeze.
5. On submittal and As-Built drawings, provide text of sprinkler list to be installed in the spare sprinkler cabinet.
6. Designer and installer proof of qualifications.

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, meet the following requirements:

1. Fire sprinkler system to be designed by someone meeting or exceeding one or more of the following requirements:

    a. A fire protection engineer.
    b. A NICET III technician for the type of system designed.
    c. As approved by the AHJ.

2. Fire sprinkler system to be installed by someone meeting or exceeding one or more of the following requirements:

    a. State or municipal certified/licensed sprinkler contractor.
    b. NICET Level II for the type of system being installed.
    c. As approved by the AHJ.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.7 SYSTEM DESCRIPTION

A. Provide coverage for entire building. Field verify field conditions prior to submittal of offer. Adjust offer to provide protection features in accordance with applicable codes and interpretations by AHJ. Provide design and installation based on more stringent requirements if this specification and AHJ requirements differ from Code.

B. Design Parameters:

1. Building Areas: Corridors, Restrooms, Museum Space, Conference, Offices.

    a. Occupancy Classification: Light.
    b. Outside Hose Allowance: 100 GPM.

2. Building Areas: Mechanical Rooms, Electrical Rooms, Storage.

    a. Occupancy Classification: Ordinary Group 1.
    b. Outside Hose Allowance: 250 GPM.

3. Design parameters above are NFPA 13 minimums. Provide increased design densities, design areas and hose allowances to meet requirements of AHJ.

C. Sprinkler system design to include a 10 percent pressure and flow cushion between system demand point and available water supplies.

D. Extend hydraulic calculations from hydraulically most remote design area back to location of pressure hydrant of flow test or effective point of water supply where characteristics of water supply are known.

1.8 EXTRA STOCK

A. Provide extra sprinklers per code.

B. Provide suitable wrenches for each sprinkler type and metal storage cabinet in riser room.

1.9 CONTROL VALVES

A. Sprinkler system control valves to be OS&Y or butterfly valves located inside building in a room with outside door.

PART 2 - PRODUCTS

2.1 GENERAL

A. Salient Characteristics as outlined in Articles below:

1. Physical
2. Functional
3. Performance

2.2 SPRINKLERS

A. Finished Areas:

1. Physical:

a. Type: Glass-bulb.
b. Style: Concealed.
c. Finish: White polyester.
d. Escutcheon: White polyester.
e. Coverplate for Concealed Sprinklers: Flat plate, white.

2. Functional: Fire suppression applications.
3. Performance: Quick response.

B. Nonfinished Areas:

1. Physical:

a. Type: Glass-bulb.
b. Finish: Brass.

2. Functional: Fire suppression applications.
3. Performance: Quick response.

C. Dry Sprinklers:

1. Physical:

a. Type: Glass-bulb.
b. Style: Concealed, recessed.
c. Finish: Chrome, white polyester.
d. Escutcheon: Chrome, white polyester.
e. Coverplate for Concealed Sprinklers: Flat plate; chrome, white.
f. Dry Sprinkler Boot: Manufactured for use with the dry sprinkler it protects.

2. Functional: Fire suppression applications.
3. Performance: Quick response.

2.3 FLEXIBLE SPRINKLER HOSE FITTING ASSEMBLY

A. Physical:

1. Fully welded non-mechanical fittings, stainless steel, braided, with minimum 1-inch true-bore internal corrugated hose diameter.
2. Ceiling Bracket: Galvanized steel, direct attachment type, with integrated snap-on clip ends and removable flexible hose attachment with set screw.

B. Functional: None.

C. Performance:

1. Leak-tested.
2. 175 PSI.
3. FM1637, UL 2443.

2.4 RISER MANIFOLD

A. Physical: Provide cover tamper switch when required by AHJ.

B. Functional: None.

C. Performance:

1. Water-flow alarm, gauge, integral pressure relief valve connected to drain, pressure gauge with 3-way gauge control valve and drain valve.
2. Integral pressure relief valve connected to drain, sight glass, smooth bore orifice union of same size as smallest orifice sprinkler installed.

2.5 INSPECTOR'S TEST CONNECTION

A. Dry System Inspector’s Test Connection:

1. Physical:

a. Bronze body.
b. Brass stem.
c. Steel handle.
d. Chrome-plated bronze ball.
e. Teflon valve seat.

2. Functional: Inspector's test connection for dry system.
3. Performance:

a. Tamper and corrosion resistant orifice equivalent to smallest sprinkler orifice.
b. Sight flow connection.

2.6 DRY-PIPE VALVE

A. Physical: None.

B. Functional: Valve.

C. Performance:

1. Differential or low pressure actuator type.
2. Trim as recommended by manufacturer for variable pressure service, including air maintenance device, electric low pressure alarm switch, priming valves and test, main drain and pressure gauges.
3. External reset.

2.7 DRY SYSTEM DRUM DRIP DRAINS

A. Physical: Steel.

B. Functional: Drainage.

C. Performance:

1. Normally open upper ball valve with lever handle.
2. Normally closed lower ball valve with lever handle.

2.8 NITROGEN SYSTEM

A. Physical: Steel cabinet.

B. Functional: None.

C. Performance:

1. Complete system package from one manufacturer, consisting of the following:

    a. Nitrogen Generation System.
    b. Oilless Air Compressor.
    c. Building Monitoring System.
    d. Alarm Integration.
    e. Auto Purge System.
    f. Auto Purge Remote Annunciator.
    g. Supervisory Gas Monitoring.
    h. Air Maintenance Device.

2. All components to be compatible with each other.
3. Install with corrosion-inhibiting system from same manufacturer.

2.9 WET SYSTEM AIR VENT

A. Physical: Brass.

B. Functional: Venting.

C. Performance: UL 2573 with ball valve supervisory switch.

2.10 SPARE SPRINKLER CABINET

A. Physical:

1. Welded steel with hinged steel cover.
2. Red enamel or polyester coated finish inside and out.
3. Enamel coated finish.

B. Functional: Storage of spare sprinklers and associated tools, etc.

C. Performance: Sized to accommodate a minimum of two spare sprinklers of each Sprinkler Identification Number (SIN), manufacturer, model, orifice, deflector type, temperature and thermal sensitivity, or a minimum of six sprinklers for facilities having under 300 sprinklers, or a minimum of 12 sprinklers for facilities having 300 to 1000 sprinklers, or a minimum of 24 sprinklers for facilities having over 1000 sprinklers, whichever is more.

2.11 SPRINKLER GUARDS

A. Physical: Metal.

B. Functional: None.

C. Performance: Listed for use with sprinkler model to which it is attached.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. Install per manufacturer's requirements and recommendations.

3.2 SPRINKLERS

A. Center sprinklers in center or quarter points of suspended ceiling tile.

B. Align sprinklers with architectural column lines, lighting, diffusers and other ceiling features. In unfinished ceilings, route piping to minimize visual impact. Sprinklers and piping not so aligned are to be removed and replaced at no additional cost adjustment to the Contract.

C. Install dry sprinklers in a manner which does not trap water.

D. Affix permanent sign, label or decal at each flexible sprinkler hose fitting assembly anchoring component limiting the relocation of the sprinkler.

3.3 FLEXIBLE SPRINKLER HOSE FITTING ASSEMBLY

A. Install flexible sprinkler hose fitting assemblies where pendent sprinkler heads are located in acoustic ceiling tiles.

B. Install with no more bends than are included in equivalent footage used in hydraulic calculations.

C. Maintain manufacturer's recommended bending radius as included in equivalent footage used in hydraulic calculations.

D. Affix permanent sign, label or decal at each flexible sprinkler hose fitting assembly anchoring component limiting the relocation of the sprinkler.

3.4 RISER MANIFOLD

A. Install so valves and gauges are conveniently and accessibly located with reference to finished building for repairs, removal and service.

B. Provide connection to drain.

C. Pipe pressure relief valve to drain.

D. Install with supervised control valve(s) and check valve.

3.5 INSPECTOR'S TEST CONNECTION

A. Locate where full flow discharge or pressure relief valve discharge will not do damage, including damage to landscaping and will not cause dangerous conditions to walking surfaces or discoloration to building surfaces.

B. Locate within 5-feet of finished floor.

3.6 DRY-PIPE VALVE

A. Install with sufficient access for ease of reset.

3.7 DRY SYSTEM DRUM DRIP DRAINS

A. Locate within 7-feet of finished floor.

B. Coordinate drain locations with Contracting Officer prior to design and installation of dry sprinkler system.

3.8 NITROGEN SYSTEM

A. Install per manufacturer's recommendations.

B. Provide a nitrogen purge valve at farthest point from nitrogen generator.

3.9 WET SYSTEM AIR VENT

A. Locate at a point in the system that will vent the most air.

B. Connect at top of pipe.

C. Locate so as not to interfere with sprinkler spray pattern.

D. Locate where it can be easily accessed for inspection and cleaning.

E. Pipe output of air vent to drain with an indirect connector or to exterior where it will not cause damage.

3.10 SPARE SPRINKLER CABINET

A. Attach to wall at the main sprinkler system riser.

B. Locate so cover is easy to open and readily accessible.

C. Locate in an area with a temperature between 40 and 100 degrees Fahrenheit.

D. Locate sprinkler wrenches inside cabinet.

E. Inside the cabinet, provide a list of sprinklers installed in the property, including sprinkler identification number, manufacturer, model, orifice, deflector type, thermal sensitivity and pressure rating, quantity of each type to be contained in the cabinet and issue or revision date of the list.

3.11 SPRINKLER GUARDS

A. Install per manufacturer's instructions and recommendations.

END OF SECTION 21 13 00

SECTION 21 13 19 – FIRE SUPPRESSION PREACTION SPRINKLER SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included:

1. Preaction Deluge Valves
2. Inspector's Test Connection
3. Sprinklers
4. Nitrogen System
5. Preaction Control Panel
6. High/Low Pressure Alarm Switch
7. Alarm Pressure Switches
8. Valve Tamper Switches and Supervisory Pressure Switches
9. Circuit Conductors
10. Locks and Keys
11. Instruction Charts
12. Caution and Advisory Signs and Placards
13. Drum Drip Drains

B. Scope: Preaction sprinkler system.

C. Coordinate location and type of tamper, flow and pressure switches.

D. Provide electrical connections and wiring as required for a complete and operable system. Includes but is not limited to Nitrogen systems, compressors, bells, control equipment and power supplies.

1.2 RELATED SECTIONS

A. Contents of Division 21, Fire Suppression and Division 01, General Requirements apply to this Section.

B. In addition, reference the following:

1. Division 22, Plumbing
2. Division 23, Heating, Ventilating and Air-Conditioning
3. Division 26, Electrical
4. Division 28, Electronic Safety
5. Section 21 00 00, Fire Suppression Basic Requirements
6. Section 21 05 00, Common Work Results for Fire Suppression
7. Section 21 08 00, Commissioning of Fire Suppression Systems
8. Section 21 13 00, Fire Suppression Sprinkler Systems
9. Section 21 22 00, Clean-Agent Fire-Extinguishing Systems

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.4 SUBMITTALS

A. Submittals as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, provide:

1. Hydraulic calculations.
2. Trapeze hanger details and calculations, including size, length and material. Additionally, provide size, weight and number of pipes to be carried on the trapeze.

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

1.7 SYSTEM DESCRIPTION

A. Provide a complete and operable double-interlocked preaction sprinkler system for Collections Storage/Work Room.

B. Field verify field conditions prior to submittal to offer. Adjust offer to provide protection features in accordance with applicable codes and interpretations by AHJ. Provide design and installation based on more stringent requirements if this specification or AHJ requirements differ from Code.

C. Design Parameters:

1. Building Area: Collections Storage.

    a. Occupancy Classification: Ordinary Group 1.
    b. Outside Hose Allowance: 250 GPM.

2. Design parameters above are NFPA 13 minimums. Provide increased design densities, design areas and hose allowances to meet requirements of AHJ.

D. Sprinkler system design to include a 10 percent pressure and flow cushion between system demand point and available water supplies.

E. Extend hydraulic calculations from hydraulically most remote design area back to location of pressure hydrant of flow test or effective point of water supply where characteristics of water supply are known.

F. Develop cost-effective designs that may include use of extended coverage sprinklers and design area reductions as allowed by NFPA 13.

1.8 EXTRA STOCK

A. Provide extra sprinklers per code.

B. Provide suitable wrenches for each sprinkler type and metal storage cabinet in riser room.

C. Inside the cabinet, provide a list of sprinklers installed in the property, including sprinkler identification number, manufacturer, model, orifice, deflector type, thermal sensitivity and pressure rating, quantity of each type to be contained in the cabinet an issue or revision date of the list.

PART 2 - PRODUCTS

2.1 GENERAL

A. Substitutions:

1. Reference Specification Section 01 67 00, Product Requirements.
2. Submit product data showing features and certification by Contractor that the design will comply with contract documents.

B. Equipment to be supplied by a certified manufacturer representative.

C. Salient Characteristics as outlined in Articles below:

1. Physical
2. Functional
3. Performance

2.2 PREACTION DELUGE VALVES

A. Physical:

1. Electric/electric release.
2. Internal check diaphragm.

B. Functional: Valve.

C. Performance: External reset.

2.3 INSPECTOR'S TEST CONNECTIONS

A. Physical:

1. Bronze body.
2. Brass stem.
3. Steel handle.
4. Chrome-plated bronze ball.
5. Teflon valve seat.

B. Functional: Inspector's test connection for system.

C. Performance:

1. Tamper and corrosion resistant orifice equivalent to smallest sprinkler orifice.
2. Sight flow connection.

2.4 SPRINKLERS

A. Physical:

1. Type: Glass-bulb, solder link.
2. Style: Upright.
3. Finish: Brass.

B. Functional: Fire suppression applications.

C. Performance: Quick-response.

2.5 NITROGEN SYSTEM

A. Physical: Steel cabinet.

B. Functional: None.

C. Performance:

1. Complete system package from one manufacturer, consisting of the following:

a. Nitrogen Generation System.
b. Oilless Air Compressor.
c. Building Monitoring System.
d. Alarm Integration.
e. Auto Purge System.
f. Auto Purge Remote Annunciator.
g. Supervisory Gas Monitoring.
h. Air Maintenance Device.

2. All components to be compatible with each other.
3. Install with corrosion-inhibiting system from same manufacturer.

2.6 PREACTION CONTROL PANEL

A. Physical: See Division 28, Electronic Safety.

B. Functional:

1. Utilize the building fire alarm system as the preaction sprinkler system control panel.
2. Programmed control point activation includes shutdown of HVAC systems and other required fire safety functions. See Division 28, Electronic Safety.

C. Performance: See Division 28, Electronic Safety.

2.7 AIR-SAMPLING SMOKE DETECTORS

A. Physical:

1. Aspirating, photo-electric or laser type, high-sensitivity smoke detector.
2. Provide sampling pipe network of CPVC or pipe as required by the manufacturer and Code.

B. Functional: Smoke detection.

C. Performance:

1. Minimum programmable sensitivity range of 0.5 to 4 percent / foot obscuration.
2. Minimum two alarm outputs and one trouble output to be monitored by the preaction control panel.

2.8 HIGH/LOW PRESSURE ALARM SWITCH

A. Physical: None.

B. Functional: Alarm switch.

C. Performance:

1. Pressure actuated to detect 10 PSI increase or decrease.
2. SPDT switches.

2.9 ALARM PRESSURE SWITCHES

A. Physical: None.

B. Functional: Alarm switch.

C. Performance: Monitor each alarm pressure switch with preaction control panel.

2.10 VALVE TAMPER SWITCHES AND SUPERVISORY PRESSURE SWITCHES

A. Physical: None.

B. Functional: Alarm switch.

C. Performance: Monitor each alarm pressure switch with preaction control panel.

2.11 CIRCUIT CONDUCTORS

A. Physical:

1. Copper or optical fiber.
2. Color code and label.
3. Type FPL, FPLR and FPLP as required by NEC.
4. Minimum signaling line circuit and initiating device circuit wire size: AWG18.
5. Minimum notification appliance circuit wire size: AWG14, or as approved by Contracting Officer.

B. Functional: Circuit conductors.

C. Performance:

1. Provide conductors in compliance with the requirements of the manufacturer.
2. Color code and label.

2.12 LOCKS AND KEYS

A. Physical:

1. Provide same standard lock and key for each key operated switch and lockable panel and cabinet.
2. Provide five keys of each type.

B. Functional: Lock and key system to secure Items.

C. Performance: None.

2.13 INSTRUCTION CHARTS

A. Physical:

1. Frame to be stainless steel or aluminum with polycarbonate or glass cover.
2. Confirm make and model with Contracting Officer prior to ordering.

B. Functional:

1. Printed, framed instruction chart for operators, showing steps to be taken when signal is received (normal, alarm, supervisory and trouble).

2. Easily readable from normal operator's station.

C. Performance: Suitable for mounting.

2.14 CAUTION AND ADVISORY SIGNS AND PLACARDS

A. Physical: None.

B. Functional: Cautionary and advisory signage.

C. Performance:

1. Provide manual release station operation placards.
2. Provide signs and placards compliant with NFPA 13 and manufacturer requirements.

2.15 DRUM DRIP DRAINS

A. Salient Characteristics:

1. Normally open upper ball valve.
2. Normally closed lower ball valve.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. Install per manufacturer's instructions and recommendations.

B. Coordinate with Division 28, Electronic Safety.

C. Obtain approval of system design from AHJ prior to installation. Do not begin installation without approval from AHJ and submittal review comments from Contracting Officer.

D. Install in accordance with applicable codes, NFPA 13, NFPA 72, NFPA 70 and the Contract Documents.

E. Provide pipe, fittings, detection and control system, shutdowns, auxiliary system interfaces, signs, wiring, conduit and outlet boxes, piping, hangers, etc., required for the erection of a complete system as described in these specifications, as shown on Drawings and as required by AHJ.

F. Provide wiring to meet the requirements of national, state and local electrical codes. Provide color coded wiring as recommended and specified by the preaction detection system manufacturer.

G. Conceal wiring, conduit, boxes and supports .

H. Provide complete conduit system for wiring.

I. At junction boxes and termination points, provide identification tags on wires and cables.

J. Route wiring and piping to avoid blocking access to equipment requiring service, access, or adjustment.

K. Provide machine printed address label on addressable devices visible from the floor without magnification.

L. Obtain Contracting Officer's approval of locations of devices before installation.

M. Install instruction placards in or adjacent to control panel.

N. Install warning and advisory placards adjacent to manual release stations and notification appliances at the entrance to the protected space and in the protected space.

O. Coordinate location of auxiliary drains with Contracting Officer. Contracting Officer to approve location before drain is installed. Protect valves from tampering or accidental operation. Coordinate with Division 22, Plumbing.

P. Double Interlock System Alarm Sequence of Operation:

1. Alarm signal from clean agent system control panel causes system to enter “alarm” mode including following operations:

    a. Local annunciation of alarm zone and condition and audible and visual alarm signal at control panel.
    b. Manual "acknowledge" function at control panel to silence audible alarm signal. Visual signal remains displayed until initiating alarm is cleared.
    c. Operate HVAC fan shutdown and damper closure in the protected space.

2. Operation of sprinkler system fusible element in conjunction with an alarm condition from the clean agent system control panel opens preaction valve filling the fire sprinkler pipes with water and discharging water from the operated sprinkler.
3. Supervisory Sequence of Operation: See Division 28, Electronic Safety for supervisory condition sequences.
4. Trouble Sequence of Operation: See Division 28, Electronic Safety for trouble condition sequences.

Q. Fire Safety Systems Interfaces and Functions:

1. Provide power and control conduit, wiring, boxes and terminations to power devices and interface devices to preaction system.
2. Provide alarm, supervisory and trouble relay output connections to the building fire alarm system. Coordinate with Division 28, Electronic Safety, for providing input modules to monitor the preaction sprinkler system
3. HVAC Control: Provide control wiring from relay contacts to activate fan shutdown and damper closure for the protected space.

R. Inspection and Testing for Completion:

1. System inspection, testing and commissioning to be performed by a certified manufacturer's representative.

2. Perform testing and inspection in accordance with NFPA 13. In addition, test the fire control and detection system in accordance with NFPA 72. Document each inspection and test.
3. Provide the services of the installer's supervisor or person with equivalent qualifications to supervise inspection and testing, correction and adjustments.
4. Provide tools, software and supplies required to accomplish inspection and testing.
5. Prepare for testing by ensuring work is complete and correct; perform preliminary tests as required to test system.
6. Correct defective work, adjust for proper operation and retest until entire system complies with Contract Documents.
7. Notify Contracting Officer seven days prior to beginning completion inspections and tests.
8. Notify authorities having jurisdiction and comply with their requirements for scheduling inspections and tests and for observation by their personnel.
9. Diagnostic Period: After successful completion of inspections and tests, operate system in normal mode for at least 14 days without any system or equipment malfunctions.

    a. Record system operations and malfunctions.
    b. If a malfunction occurs, start diagnostic period over after correction of malfunction.
    c. Contracting Officer will provide attendant operator personnel during diagnostic period; schedule training to allow Contracting Officer personnel to perform normal duties.
    d. At end of successful diagnostic period, submit completed test documentation.

S. Contracting Officer Personnel Instruction:

1. Comply with all requirements of Division 01, General Requirements and Section 21 00 00, Fire Suppression Basic Requirements.
2. Provide the Following Instruction to Designated Contracting Officer Personnel:

    a. Hands-On Instruction: On-site, using operational system.
    b. Classroom Instruction: Contracting Officer furnished classroom, on-site or at other local facility.
    c. Factory Instruction: At control unit manufacturer's training facility.

3. Basic Operation: One-hour sessions for attendant personnel, security officers and engineering staff, combination of classroom and hands-on.

    a. Initial Training: One session pre-closeout.
    b. Refresher Training: One session post-occupancy.

4. Detailed Operation: Two-hour sessions for staff, combination of classroom and hands-on:

    a. Initial Training: One session pre-closeout.
    b. Refresher Training: One session post-occupancy.

5. Furnish the services of instructors and teaching aids; have copies of operation and maintenance data and record drawings available during instruction.
6. Provide means of evaluation of trainees suitable to type of training given; report results to Contracting Officer.

T. Closeout:

1. Comply with all requirements of Division 01, General Requirements and Section 21 00 00, Fire Suppression Basic Requirements.
2. Closeout Demonstration: Demonstrate proper operation of functions to Contracting Officer.

    a. Be prepared to conduct any of the required tests.
    b. Have at least one copy of operation and maintenance data, preliminary copy of project record drawings, input/output matrix and operator instruction chart(s) available during demonstration.
    c. Have authorized technical representative of control unit manufacturer present during demonstration.
    d. Demonstration may be combined with inspection and testing required by AHJ. Notify AHJ in time to schedule demonstration.
    e. Repeat demonstration until successful.

3. Substantial Completion of the project cannot be achieved until inspection and testing is successfully completed.
4. Diagnostic Period:

    a. Specified diagnostic period without malfunction has been completed.
    b. Approved operating and maintenance data has been delivered.
    c. Aspects of operation have been demonstrated to Contracting Officer.
    d. Final acceptance of the fire alarm system has been given by authorities having jurisdiction.
    e. Occupancy permit has been granted.
    f. Specified pre-closeout instruction is complete.
    g. Perform post-occupancy instruction within three months after date of occupancy.

U. Inspection, Testing and Maintenance, Post-Completion:

1. Reference Division 01, General Requirements and Section 21 00 00, Fire Suppression Basic Requirements, for additional requirements relating to maintenance service.
2. Perform routine inspection, testing and preventive maintenance required by NFPA 72 and NFPA 13, including:

    a. Maintenance of fire safety interface connected to fire suppression system.
    b. Repairs required, unless due to improper use, accidents, or negligence beyond the control of the maintenance contractor.

3. Record keeping required by NFPA 13, NFPA 72 and authorities having jurisdiction.
4. Provide system test and inspection during the sixth and twelfth months after substantial completion. Correct deficiencies and replace defective components at no cost to the Contract.
5. Provide trouble call-back service upon notification by Contracting Officer:
6. Provide on-site response within two hours of notification.
7. Maintain a log at each fire alarm control unit, listing the date and time of each inspection and call-back visit, the condition of the system, nature of the trouble, correction performed, and parts replaced.
8. Submit duplicate of each log entry to Contracting Officer upon completion of site visit.
9. Comply with Contracting Officer's requirements for access to facility and security.

3.2 PREACTION DELUGE VALVES

A. Install with sufficient access for ease of reset.

3.3 INSPECTOR'S TEST CONNECTIONS

A. Route water supply flow test connections to a location which can accept the flow under wide-open flow and pressure for a sufficient time to assure a proper test and which will not cause damage, including to landscaping.

B. Install conveniently and accessibly located with reference to finished building for repairs, removal and service.

3.4 SPRINKLERS

A. Center sprinklers in center or quarter points of suspended ceiling tile.

B. Align sprinklers with architectural column lines, lighting, diffusers and other ceiling features. In unfinished ceilings, route piping to minimize visual impact. Sprinklers and piping not so aligned are to be removed and replaced at no additional cost to the Contract.

C. Space sprinkler heads to meet requirements of ceiling baffles installed to create hot / cold aisles where baffles have been installed.

D. Comply with sprinkler layouts as shown in Construction Documents to meet architectural constraints. These may be more conservative than Code maximums. Notify Contracting Officer if layout does not meet Code requirements.

3.5 NITROGEN SYSTEM

A. Coordinate fire alarm system monitoring of Nitrogen system.

B. Install per manufacturer's recommendations with corrosion-inhibiting system.

3.6 PREACTION CONTROL PANEL

A. Circuits:

1. Signaling Line Circuits (SLC): Class B.
2. Notification Appliance Circuits (NAC): Class B.

B. Power Sources:

1. Primary: Dedicated branch circuits of facility power distribution system.
2. Secondary: Storage batteries.
3. Capacity: Sufficient to operate fire alarm system under normal supervisory condition for 24 hours and operate alarm signals for five minutes at end of standby period.

3.7 AIR-SAMPLING SMOKE DETECTORS

A. Reference 3.01, General Installation Requirements.

3.8 MANUAL RELEASE STATIONS

A. Reference 3.01, General Installation Requirements.

3.9 HIGH/LOW PRESSURE ALARM SWITCH

A. Reference 3.01, General Installation Requirements.

3.10 ALARM PRESSURE SWITCHES

A. Reference 3.01, General Installation Requirements.

3.11 VALVE TAMPER SWITCHES AND SUPERVISORY PRESSURE SWITCHES

A. Reference 3.01, General Installation Requirements.

3.12 CIRCUIT CONDUCTORS

A. Provide wiring to meet the requirements of national, state and local electrical codes. Provide color coded wiring as recommended and specified by the fire alarm and detection system manufacturer.

B. Provide Type FPLR cable when in a riser application or FPLP cable when installed in plenums.

3.13 LOCKS AND KEYS

A. Deliver keys to Contracting Officer.

3.14 INSTRUCTION CHARTS

A. Locate charts as to be easily readable from operator's station.

B. Obtain approval of Contracting Officer prior to mounting. Mount in location acceptable to Contracting Officer.

C. Provide extra copy of Operation and Maintenance data submittal.

3.15 CAUTION AND ADVISORY SIGNS AND PLACARDS

A. Mount signs and placards at manual release stations and as required by NFPA 13, NFPA 72 and manufacturer's recommendations.

3.16 DOCUMENT STORAGE CABINET

A. Obtain approval of Contracting Officer prior to mounting.

B. Mount in location acceptable to Contracting Officer.

3.17 DRUM DRIP DRAINS

A. Locate within 7-feet of finished floor.

B. Coordinate drain locations with Contracting Officer prior to design and installation of dry sprinkler system.

END OF SECTION 21 13 19

SECTION 21 22 00 – CLEAN-AGENT FIRE-EXTINGUISHING SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included:

1. Novec™ 1230 Fire Protection Fluid Fire Extinguishing Clean Agent System for the Collections Storage Area.
2. Clean Agent System Control Unit
3. Manual Release Stations
4. Photoelectric Type Analog Detectors
5. Air-Sampling Smoke Detectors
6. Abort Station
7. Keyed Maintenance Switch
8. Combination Horn/Strobes
9. Strobes
10. Surface Backboxes
11. Agent
12. Agent Storage
13. Control Heads
14. Alarm Pressure Switch
15. Agent Distribution
16. Discharge Nozzles
17. Circuit Conductors
18. Surge Protection
19. Locks and Keys
20. Instruction Charts
21. Caution and Advisory Signs and Placards
22. Document Storage Cabinet
23. Batteries

B. Provide design for the modification/extension of the existing Novec™ 1230 clean agent suppression system as required in these contract documents.

1.2 RELATED SECTIONS

A. Contents of Division 21, Fire Suppression and Division 01, General Requirements apply to this Section.

B. In addition, reference the following:

1. Division 22, Plumbing
2. Division 23, Heating, Ventilating and Air Conditioning
3. Division 26, Electrical
4. Division 28, Electronic Safety

5. Section 21 00 00, Fire Suppression Basic Requirements
6. Section 21 05 00, Common Work Result for Fire Suppression
7. Section 21 08 00, Commissioning of Fire Suppression Systems
8. Section 21 13 00, Fire Suppression Sprinkler Systems
9. Section 21 13 19, Fire Suppression Preaction Sprinkler Systems

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, meet the following:

1. NFPA 2001 - Standard on Clean Agent Fire Extinguishing Systems, latest edition.
2. National Electrical Manufacturers Association (NEMA) Publication: Enclosures for Industrial Controls and Systems.
3. U.S. Environmental Protection Agency, Protection of Stratospheric Ozone 59 FR 13044 (SNAP).
4. Department of Transportation (DOT).
5. Title 49 Code of Federal Regulations Parts 100 to 199.

1.4 SUBMITTALS

A. Submittals as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, provide:

1. Manufacturer's Certificate of Acceptance of qualifications of contractor to install, test, maintain, and recharge manufacturer's equipment.
2. Material and equipment information to include manufacturer's catalog cut sheet and technical data for each component or device used in system.
3. Provide Shop Drawings in accordance with NFPA 2001 and the following:

    a. Field installation layout drawings having a scale of not less than 1/8-inch = 1-foot-0-inches or 1:100 detailing the location of agent storage tanks, nozzles, pipe runs, including pipe sizes and lengths, control panel, detectors, manual pull stations, abort stations, audible and visual alarms, etc.
    b. Indicate locations, installation details, and operation details of equipment associated with the Clean Agent fire extinguishing system. Provide floor plans showing equipment locations, piping, conduit runs and other details. Draw to a large scale elevations, cross section and other details.
    c. Provide sequences of operation, symbol legend.
    d. Illustrate the agent distribution piping system and location of nozzles and initiating devices.
    e. Auxiliary details and information such as maintenance, door holders, special sealing requirements, and equipment shutdown.
    f. Separate layouts, or drawings, provided for each level, (i.e., room, subfloor, above ceiling) and for mechanical and electrical work.

g. Drawings showing the location of devices include point-to-point conduit runs and a description of the method(s) used for detector mounting.
h. Provide an internal control panel wiring diagram which includes power supply requirements and field wiring termination points.

4. Calculations: Provide computer input and output calculations, with ancillary manual calculations as required to verify agent amounts, pipe sizes, nozzle selections, end pressures, and other derived system parameters. Provide calculations for the battery stand-by power supply, taking into consideration the power requirements of alarms, initiating devices, and auxiliary components under full load conditions.
5. A complete sequence of operations, shutdown functions, remote signaling, damper operation, time delay, and agent discharge for each zone or system; electrical schematics and connection diagrams to completely describe operation of controls.
6. System Design Documentation: Provide a copy of manufacturer's design manual, indicating allowable tee flow mass flow rate splits, vertical and horizontal tee installations, system pressure and mass flow rate balance, and parameters which are restricted in their use by UL listed calculation procedure for system.
7. Operation and Maintenance Manuals: Provide complete operation and maintenance instruction manuals (three copies for each system) to the Contracting Officer. Provide information outlining operation and maintenance procedures that will be required of Contracting Officer. Information to explain any special knowledge or tools Contracting Officer will be required to employ and all spare parts that should be readily available. Detail aspects of system operation and maintenance, including electrical schematics of circuits, a written description of the system design, drawing illustrating control logic and equipment used in the system, checklists and procedures for emergency situations, troubleshooting techniques, and maintenance operation.

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Section 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

B. In addition, meet the following refire extinguishing system to be designed by someone meeting or exceeding one or more of the following requirements:

1. Fire extinguishing system to be designed by someone meeting or exceeding one or more of the following requirements:

a. A fire protection engineer.
b. A NICET III technician for the type of system designed.
c. As approved by the AHJ.

2. Fire extinguishing system to be installed by someone meeting or exceeding one or more of the following requirements:

a. State or municipal certified/licensed sprinkler contractor.
b. NICET Level II for the type of system being installed.
c. As approved by the AHJ.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by 21 00 00, Fire Suppression Basic Requirements and Division 01, General Requirements.

PART 2 - PRODUCTS

2.1 GENERAL

A. Salient Characteristics as outlined in Articles below:

1. Physical
2. Functional
3. Performance

2.2 NOVEC™ 1230 FIRE PROTECTION FLUID FIRE EXTINGUISHING CLEAN AGENT SYSTEM

A. Physical: Existing 3M (NOVEC 1230™).

B. Functional: N/A, existing.

C. Performance: N/A, existing.

2.3 CLEAN AGENT SYSTEM CONTROL UNIT

A. Physical: Existing Autopulse A-10.

B. Functional: N/A, existing.

C. Performance: N/A, existing.

2.4 MANUAL RELEASE STATIONS

A. Physical:

1. Electric.
2. Red finish.
3. Semi-flush mounted.
4. Marked “Agent Release.”

B. Functional: Double action.

C. Performance:

1. Stations do not allow closure without keyed reset.
2. Provide each release station with instruction placard.

2.5 PHOTOELECTRIC TYPE ANALOG DETECTORS

A. Physical:

1. LED source, multiple cell.
2. Visual latching operation indicator.
3. Insect screen.
4. Functional test switch.
5. Two-wire operation.
6. Vandal-resistant locking feature.

B. Functional: Smoke detection.

C. Performance: 360-degree smoke entry.

2.6 AIR-SAMPLING SMOKE DETECTORS

A. Physical:

1. Flush mounted units where installed in finished areas.
2. Surface mounted units are acceptable in unfinished areas, unless otherwise noted.

B. Functional:

1. Aspirating, photoelectric or laser type, high-sensitivity smoke detector.
2. Minimum programmable sensitivity range of 0.0015 to 3 percent/foot obscuration.
3. Minimum two alarm outputs (alert and fire) and three trouble outputs (maintenance, fault and isolate) to be monitored by fire control panel.
4. Provide sampling pipe network of CPVC or pipe as required by the manufacturer and code.

C. Performance: None.

2.7 ABORT STATION

A. Physical: "Dead Man" type, non-latching requiring sustained pressure to operate abort function.

B. Functional: Abort station.

C. Performance: Provide each abort station with instruction placard.

2.8 KEYED MAINTENANCE SWITCH

A. Physical: None.

B. Functional: Maintenance switch,

C. Performance: Capable of bypassing releasing functions.

2.9 COMBINATION HORN/STROBES

A. Physical:

1. Flush wall mount.
2. Red finish.
3. Insect-proof.

B. Functional: Multi-candela fire alarm horn/strobes.

C. Performance:

1. Must be compatible with fire alarm control equipment and notification appliance circuit panels.
2. Meet the latest requirements of NFPA 72, ANSI 117.1 and UL 1971.
3. Candela rating as required by NFPA 72 for the space where used.

2.10 STROBES

A. Physical:

1. Flush wall mount.
2. Red finish.
3. Insect-proof.

B. Functional: Multi-candela fire alarm strobes.

C. Performance:

1. Must be compatible with fire alarm control equipment and notification appliance circuit panels.
2. Meet the latest requirements of NFPA 72, ANSI 117.1 and UL 1971.
3. Candela rating as required by NFPA 72 for the space where used.

2.11 SURFACE BACKBOXES

A. Physical: None.

B. Functional: Provide manufacturer surface backboxes where devices cannot be installed recessed.

C. Performance: Must be compatible with control equipment and notification appliance circuit panels.

2.12 AGENT

A. Physical:

1. Novec™ 1230 Fire Protection Fluid manufactured by 3M Company.
2. Super pressurized to 360 PSI or as required by system design per manufacturer requirements.

B. Functional: Fire suppression.

C. Performance: None.

2.13 AGENT STORAGE

A. Physical: Existing agent tank.

B. Functional: N/A, existing.

C. Performance: N/A, existing.

2.14 CONTROL HEADS

A. Physical: N/A, existing.

B. Functional: N/A, existing.

C. Performance: N/A, existing.

2.15 ALARM PRESSURE SWITCH

A. Physical: N/A, existing.

B. Functional: N/A, existing.

C. Performance: N/A, existing.

2.16 AGENT DISTRIBUTION

A. Physical: Schedule 40 piping, minimum.

B. Functional: Distribution of Agent gas.

C. Performance: Piping and fittings in accordance with current NFPA and manufacturer's requirements.

2.17 DISCHARGE NOZZLES

A. Physical: None.

B. Functional: Agent discharge nozzles.

C. Performance:

1. Manufacturer nozzles designed for distribution of Agent.
2. 180 or 360 degree discharge pattern; orifice sized per engineered calculations.

2.18 CIRCUIT CONDUCTORS

A. Physical:

1. Copper.
2. Color code and label.

B. Functional: Circuit conduction.

C. Performance:

1. Type FPL, FPLR and FPLP as required by NEC.
2. Provide conductors in compliance with the requirements of the manufacturer.
3. Minimum circuit wire size: AWG18.

2.19 SURGE PROTECTION

A. Physical: None.

B. Functional: Surge protection.

C. Performance:

1. In accordance with IEEE C62.41 B3 combination waveform and NFPA 70; except for optical fiber conductors.
2. Provide for alternating current circuits powering fire alarm equipment.
3. Initiating Device Circuits, Notification Appliance Circuits and Communications Circuits: Rated to protect applicable equipment; for 24V(dc) maximum dc clamping voltage of 36 V(dc), line-to-ground and 72V(dc), line-to-line.

2.20 LOCKS AND KEYS

A. Physical:

1. Provide same standard lock and key for each key operated switch and lockable panel and cabinet.
2. Provide five keys of each type.

B. Functional: Keyed locks.

C. Performance: None.

2.21 INSTRUCTION CHARTS

A. Physical:

1. Frame to be stainless steel or aluminum with polycarbonate or glass cover.
2. Confirm make and model with Contracting Officer prior to ordering.

B. Functional:

1. Printed instruction chart for operators, showing steps to be taken when signal is received (normal, alarm, supervisory, and trouble).
2. Easily readable from normal operator's station.

C. Performance:

1. Provide one for each control unit where operations are to be performed.
2. Mount in location acceptable to Contracting Officer.
3. Provide extra copy with operation and maintenance data submittal.

2.22 CAUTION AND ADVISORY SIGNS AND PLACARDS

A. Physical: Signs and placards to be compliant with current NFPA and manufacturer requirements.

B. Functional: Cautionary and advisory signage.

C. Performance:

1. Provide entrance warning signs, area warning signs.
2. Provide manual release station operation and abort station operation placards.

2.23 DOCUMENT STORAGE CABINET

A. Physical:

1. Suitable to store as-built drawings, operation and maintenance manual, system data file disk, spare parts, and tools.
2. Constructed from steel with baked enamel finish.
3. Size adequate for storage of full-size as-built drawings, operation and maintenance manual, and system data file disk.

B. Functional: Storage of as-built drawings, operation and maintenance manual, and system data file disk.

C. Performance: Suitable for mounting.

2.24 BATTERIES

A. Physical: Sealed gel cell type.

B. Functional: Provides battery power.

C. Performance: Provide additional cabinet, if required due to space limitations in control panels.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. Install per manufacturer's instructions and recommendations.

B. Design Criteria

1. Design Fire Extinguishing Clean Agent System in compliance with code as interpreted by the AHJ, current NFPA and the manufacturer's requirements.
2. Shop drawings and calculations to be prepared under the direct supervision and control of a Professional competent to do such work and licensed in the state of Iowa. Drawings and calculations to bear the seal and wet signature of Professional. Comply with all requirements of Division 01, General Requirements and Section 21 00 00, Fire Suppression Basic Requirements.
3. Design system for the Collections Storage Area and Work Room.
4. Separately calculate and distribute agent amounts to each of above-mentioned spaces. Provide agent from a single container with piping to each of these spaces, or provide agent to each space from a separate storage container. Actuate these spaces together from a single integrated detection and actuation system.
5. Agent Design Concentration Novec ™ 1230 Fire Protection Fluid: Minimum 4.2 percent by volume for Class A hazards. Minimum 5.85 percent by volume for Class B hazards. Determine concentration at maximum temperature anticipated in the protected space.
6. Provide design with quantity of agent required to obtain the minimum uniform concentration for 10 minutes. Take into consideration such factors as unclosable openings, run-down time of fans, time required for dampers to close, and any other feature of the facility that could affect establishing and maintaining uniform concentration. Examine the architectural and mechanical features of the applicable spaces required to hold the agent for the required period.
7. In addition to Code requirements, provide the following:

    a. Manual release stations at exits from the protected space within 5-feet of the exit door operator.
    b. Provide air sampling smoke detection for the protected space. Provide sampling points on a maximum of 250 SF area per sampling point or as required by NFPA 72, whichever is more restrictive.
    c. Provide discharge pressure switch on discharge piping.

8. Circuits:

    a. Signaling Line Circuits (SLC): Class B.
    b. Notification Appliance Circuits (NAC): Class B.

9. Power Sources:

    a. Primary: Dedicated branch circuits of facility power distribution system.
    b. Secondary: Storage batteries.
    c. Capacity: Sufficient to operate fire alarm system under normal supervisory condition for 24 hours and operate alarm signals for five minutes at end of standby period.

10. System Operation:

    a. Alarm Sequence of Operation:

        1) Activation of one automatic spot smoke detector or activation of the air sampling smoke detection system on fire alert detector causes system to enter "alarm" mode including following operations:

            a) Local annunciation of alarm zone and condition, and audible and visual alarm signal at control panel.
            b) Provide manual "acknowledge" function at control panel to silence audible alarm signal. Visual signal remains displayed until initiating alarm is cleared.
            c) Transmit “alarm” signal to building fire alarm control panel.
            d) Operate the alarm notification appliances using Temporal 3 in the protected space.
            e) Transmit signal to close fire/smoke dampers for the protected space.
            f) Transmit signal to initiate shutdown of air handling equipment for the protected space.

        2) Activation of spot smoke detector and activation of the air sampling smoke detection system on fire alert causes system to enter "pre-discharge" mode including following operations:

            a) Local annunciation of alarm zone and condition, and audible and visual alarm signal at control panel indicating pre-discharge condition.
            b) Provide manual "acknowledge" function at control panel to silence audible alarm signal. Visual signal remains displayed until initiating alarm is cleared.
            c) Change the alarm notification appliances output to 120 pulses per minute in the protected space.
            d) Time delay to allow for egress from the space and equipment rundown and closure.
            e) Actuate the preaction sprinkler releasing solenoid valve.

        3) After expiration of the time delay the system to enter the "discharge" mode including the following operation:

a) Local annunciation of alarm zone and condition, and audible and visual alarm signal at control panel indicating discharge condition.
b) Provide manual "acknowledge" function at control panel to silence audible alarm signal. Visual signal remains displayed until initiating alarm is cleared.
c) Discharge of the agent to the protected space.
d) Initiate the discharge strobe located at the entrances to the protected space.
e) Change the alarm notification appliances output to steady in the protected space.

4) Activation of the electric manual release, mechanical manual release causes the system to enter perform the sequences listed for alarm, pre-discharge and discharge functions except the abort and time delay functions are bypassed. If the mechanical manual release is operated, the alarm pressure switch on the discharge piping will initiate shutdown/closure operations.

b. Supervisory Sequence of Operation: Low pressure switch activation causes system to enter "supervisory" mode including the following operations:

1) Local annunciation of condition and audible and visual supervisory signal at control panel.
2) Provide manual "acknowledge" function at control panel to silence audible supervisory signal. Visual signal remains displayed until initiating supervisory is cleared.
3) Transmit "supervisory" signal to building fire alarm control panel.

c. Trouble Sequence of Operation: System trouble, including single ground or open of supervised circuit, or power or system failure, causes system to enter "trouble" mode including the following operations:

1) Local annunciation of condition and audible and visual trouble signal at control panel.
2) Provide manual "acknowledge" function at control panel to silence audible trouble signal. Visual signal remains displayed until initiating trouble is cleared.
3) Transmit "trouble" signal to building fire alarm control panel.

C. Installation

1. Installer:

a. Trained by supplier to design, install, test, and maintain fire suppression systems.
b. Experienced firm regularly engaged in the installation of automatic Clean Agent, or similar, fire suppression systems, in strict accordance with applicable codes and standards.
c. Minimum of five years' experience in design, installation, and testing, of Clean Agent, or similar fire suppression systems. Provide on request a list of systems of a similar nature and scope.

d. Maintain, or have access to, a Clean Agent recharging station. Provide proof of ability to recharge the largest Clean Agent system within 12 hours after a discharge. Include amount of bulk agent storage available.
e. Authorized stocking distributor of manufacturer for the equipment included in system so that immediate replacement parts can be made from inventory. Provide available telephone communications on a 24-hour, 7-days-a-week basis, service personnel available for emergency service at all times. Respond to emergency situation within 2 hours of receiving an emergency trouble call.

2. Obtain approval of system design from AHJ prior to installation. Do not begin installation without approval from AHJ and submittal review comments from Contracting Officer.
3. Install in accordance with applicable codes, NFPA 2001, NFPA 72, NFPA 70, and the Contract Documents.
4. In accordance with manufacturer's instructions, provide storage containers, extinguishing agent, nozzles, pipe, fittings, manual release stations, abort stations, detection and control system, notification appliances, shutdowns, auxiliary system interfaces, signs, wiring, conduit and outlet boxes, piping, bracing, hangers, etc., required for the erection of a complete system as described in these specifications, as shown on Drawings, and as required by AHJ.
5. Provide wiring to meet the requirements of Federal, State and local Electrical Codes. Provide color coded wiring as recommended and specified by the Clean Agent detection system manufacturer.
6. Provide complete conduit system for wiring.
7. At junction boxes and termination points, provide identification tags on wires and cables.
8. Route wiring and piping to avoid blocking access to equipment requiring service, access, or adjustment.
9. Provide machine printed zone label on initiating devices visible from the floor without magnification.
10. Obtain Contracting Officer's approval of locations of devices before installation.
11. Install instruction placards in or adjacent to control panel.
12. Install warning and advisory placards adjacent to manual release stations, notification appliances, abort stations, at the entrance to the protected space and in the protected space.
13. Provide control panel and remote power supplies with 120VAC dedicated circuit per NFPA requirements.
14. Do not install cabinets or equipment below the battery cabinet. Do not locate battery and charging system cabinets in ceiling space.
15. Provide alarm pressure switch to initiate fire safety functions if system is released via mechanical, manual release.
16. Provide pre-discharge notification appliances.
17. Provide discharge notification appliances.

D. Fire Safety Systems Interfaces

1. Fire Safety Interfaces and Functions:

a. Division 26, Electrical provides power and control conduit, wiring, boxes and terminations to interface devices to Clean Agent system.
b. Fire Alarm System: Provide alarm, supervisory and trouble relay output connections to the building fire alarm system.
c. Doors: Provide control relays and connection to release magnetic hold open doors. Verify requirements and quantities prior to offer.

d. HVAC Systems:

1) Fire/Smoke Dampers and Smoke Dampers:

a) Verify requirements, locations and quantities with Division 23, HVAC prior to offering.
b) Provide required smoke detectors, relays, wiring, and the like.
c) Connect control and power wiring to dampers per manufacturer's instructions.
d) Verify quantities, location and requirements of dampers with Division 23, HVAC Drawings and Specifications, and mechanical system installer.
e) Provide control wiring, transformers and power connections for an operable damper system.

2) Air Moving Systems:

a) Verify requirements, locations and quantities with Division 23, HVAC prior to offering.
b) Provide control wiring from fire suppression system relay contacts to air handling equipment controllers for the protected space.

E. Inspection and Testing for Completion:

1. Certified manufacturer representative to perform system inspection, testing and commissioning.
2. Perform testing and inspection in accordance with NFPA 2001 for approval of a new system except agent discharge test. In addition, test the fire control and detection system in accordance with NFPA 72. Document each inspection and test.
3. Provide the services of the installer's supervisor or person with equivalent qualifications to supervise inspection and testing, correction, and adjustments.
4. Provide tools, software, and supplies required to accomplish inspection and testing.
5. Prepare for testing by ensuring that work is complete and correct; perform preliminary tests as required to test system.
6. Correct defective work, adjust for proper operation, and retest until entire system complies with Contract Documents.
7. Conduct room pressurization tests for each protected space to determine the presence of openings, which would affect the agent concentration levels. Conducted test using the Retro-Tec Corp. Door Fan system, or equivalent, with integrated computer program. Test in accordance with current NFPA Standards.

a. If room pressurization testing indicates openings exist which would result in leaks or loss of the extinguishing agent, adequately seal the protected space against agent loss or leakage. Inspect work to ascertain that the protected spaces have been adequately and properly sealed. The suppression system installing contractor is responsible for the success of the room pressurization tests. If the first room pressurization test is not successful, in accordance with these Specifications, determine and correct the cause of the test failure.
b. Conduct additional room pressurization tests, at no additional cost, until a successful test is obtained.

8. Notify Contracting Officer seven days prior to beginning completion inspections and tests. Notify AHJ and comply with their requirements for scheduling inspections and tests and for observation by their personnel.
9. Diagnostic Period: After successful completion of inspections and tests, Operate system in normal mode for at least 14 days without any system or equipment malfunctions.

    a. Record system operations and malfunctions.
    b. If a malfunction occurs, start diagnostic period over after correction of malfunction.
    c. Contracting Officer will provide attendant operator personnel during diagnostic period; schedule training to allow Contracting Officer's personnel to perform normal duties.
    d. At end of successful diagnostic period, submit completed test documentation.

F. Contracting Officer's Personnel Instruction

1. Provide the Following Instruction to designated personnel:

    a. Hands-On Instruction: On-site, using operational system.
    b. Classroom Instruction: Contracting Officer furnished classroom, on-site or at other local facility.
    c. Factory Instruction: At control unit manufacturer's training facility.

2. Basic Operation: One hour sessions for attendant personnel, security officers, and engineering staff; combination of classroom and hands-on:

    a. Initial Training: One session pre-closeout or post-occupancy.
    b. Refresher Training: One session post-occupancy.

3. Detailed Operation: Two-hour sessions for engineering and maintenance staff; combination of classroom and hands-on:

    a. Initial Training: One session pre-closeout or post-occupancy.
    b. Refresher Training: One session post-occupancy.

4. Furnish the services of instructors and teaching aids; have copies of operation and maintenance data and record drawings available during instruction.
5. Provide means of evaluation of trainees suitable to type of training given; report results to Contracting Officer.

G. Closeout:

1. Comply with all requirements of Division 01, General Requirements and Section 21 00 00, Fire Suppression Basic Requirements.
2. Closeout Demonstration: Demonstrate proper operation of functions to Contracting Officer.

    a. Be prepared to conduct any of the required tests.
    b. Have at least one copy of operation and maintenance data, preliminary copy of project record drawings, input/output matrix, and operator instruction chart(s) available during demonstration.

c. Have authorized technical representative of control unit manufacturer present during demonstration.
d. Demonstration may be combined with inspection and testing required by AHJ; notify AHJ in time to schedule demonstration.
e. Repeat demonstration until successful.

3. Substantial Completion of the project cannot be achieved until inspection and testing is successful and:

a. specified diagnostic period without malfunction has been completed.
b. approved operating and maintenance data has been delivered.
c. aspects of operation have been demonstrated to Contracting Officer.
d. final acceptance of the fire alarm system has been given by authorities having jurisdiction.
e. occupancy permit has been granted.
f. specified pre-closeout instruction is complete.

4. Perform post-occupancy instruction within three months after Substantial Completion.

H. Inspection, Testing and Maintenance Post Completion

1. See Division 01, General Requirements and Section 21 00 00, Fire Suppression Basic Requirements, for additional requirements relating to maintenance service.
2. Provide to Contracting Officer, at no extra cost, a written maintenance contract for entire manufacturer's warranty period, to include the work described below.
3. Provide to Contracting Officer, a proposal for later acceptance, for a maintenance contract for entire warranty period, to include the work described below; include the total cost of contract, proposal to be valid at least until 30 days after date of Substantial Completion.
4. Perform routine inspection, testing, and preventive maintenance required by NFPA 72, NFPA 2001, and the manufacturer, including:

a. Maintenance of fire safety interface connected to fire suppression system.
b. Repairs required, unless due to improper use, accidents, or negligence beyond the control of the maintenance contractor.
c. Conduct room pressurization test for each protected space to verify the space is adequately sealed against leakage. Conduct test per NFPA 2001. If the protected test fails, identify and correct the cause of failure. conduct additional tests and corrections, as needed, until the room is adequately sealed and passes the pressurization test.

5. Record keeping required by NFPA 2001, NFPA 72 and authorities having jurisdiction.
6. Provide system test and inspection during the sixth and eleventh month after substantial completion. Correct deficiencies and replace defective components at no cost to the Contracting Officer.
7. Provide trouble call-back service upon notification by Contracting Officer:
8. Provide on-site response within 24 hours of notification.
9. Maintain a log at each fire alarm control unit, listing the date and time of each inspection and call-back visit, the condition of the system, nature of the trouble, correction performed, and parts replaced.
10. Submit duplicate of each log entry to Contracting Officer upon completion of site visit.
11. Comply with Contracting Officer's requirements for access to facility and security.

3.2 NOVEC™ 1230 FIRE PROTECTION FLUID FIRE EXTINGUISHING CLEAN AGENT SYSTEM:

A. Reference 3.01, General Installation Requirements.

3.3 CLEAN AGENT SYSTEM CONTROL UNIT

A. Reference 3.01, General Installation Requirements.

3.4 MANUAL RELEASE STATIONS

A. Provide machine printed zone labels on devices.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.5 PHOTOELECTRIC TYPE ANALOG DETECTORS

A. Provide machine printed zone labels on devices. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.6 AIR-SAMPLING SMOKE DETECTORS

A. Provide machine printed zone labels on devices. Labels to be visible from the floor without magnification.

3.7 ABORT STATION

A. Reference 3.01, General Installation Requirements.

3.8 KEYED MAINTENANCE SWITCH

A. Reference 3.01, General Installation Requirements.

3.9 COMBINATION HORN/STROBES

A. Provide machine printed labels on notification appliances with appliance circuit number and sequence. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.10 STROBES

A. Provide machine printed labels on notification appliances with appliance circuit number and sequence. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.11 SURFACE BACKBOXES

A. Provide manufacturer's weatherproof backbox listed for use in areas where the device or appliance is subject to humidity in excess of listed rating.

B. Provide manufacturer surface backboxes where devices cannot be installed recessed.

3.12 AGENT

A. Adjust quantity of agent stored in cylinders as needed for new space configuration.

3.13 AGENT STORAGE

A. Existing.

3.14 CONTROL HEADS

A. Reference 3.01, General Installation Requirements.

3.15 ALARM PRESSURE SWITCH

A. Reference 3.01, General Installation Requirements.

3.16 AGENT DISTRIBUTION

A. Reference 3.01, General Installation Requirements.

3.17 DISCHARGE NOZZLES

A. Install in center point of ceilings tiles where installed in suspended ceilings.

3.18 CIRCUIT CONDUCTORS

A. Provide wiring to meet the requirements of national, state and local electrical codes. Provide color coded wiring as recommended and specified by the fire alarm and detection system manufacturer.

B. Provide Type FPLR cable when in a riser application or FPLP cable when installed in plenums.

3.19 SURGE PROTECTION

A. Reference 3.01, General Installation Requirements.

3.20 LOCKS AND KEYS

A. Deliver to Contracting Officer.

3.21 INSTRUCTION CHARTS

A. Install chart adjacent to control panel.

3.22 CAUTION AND ADVISORY SIGNS AND PLACARDS

A. Reference 3.01, General Installation Requirements.

3.23 DOCUMENT STORAGE CABINET

A. Provide document storage cabinet adjacent to fire alarm control panel.

3.24 BATTERIES

A. Provide machine printed label with installation date.

END OF SECTION 21 22 00

SECTION 22 05 00 – COMMON WORK RESULTS FOR PLUMBING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes:

1. Concrete equipment base construction requirements.
2. Equipment nameplate data requirements.
3. Nonshrink grout for equipment installations.
4. Field-fabricated metal and wood equipment supports.
5. Installation requirements common to equipment specification sections.
6. Cutting and patching.
7. Touch-up painting and finishing.

B. The plans and specifications are complimentary and shall be used together in order to fully describe the Work. In the case of a conflict between the plans and specifications, the plans take precedence.

C. The CONTRACTING OFFICER Technical Representative has based the drawings and design on non-certified information furnished by various equipment manufacturers. It is incumbent on the part of the CONTRACTOR to include in the bid all material and labor needed to install the actual equipment furnished.

D. Related Sections:

1. The following is work of Division 26 sections of the Specifications:

a. Power supply wiring from power source to power connection on equipment. Include starters, disconnects, and required electrical devices, except where specified as furnished, or factory-installed, by manufacturer.
b. Interlock wiring between field-installed equipment, except where specified as factory installed. Interlock wiring, as used in this specification, is defined as that wiring between electrically-interlocked equipment for the purpose of controlling one piece or pieces of equipment by the operation (on, off, etc.) of another piece or pieces of associated equipment.

2. Pipe and pipe fitting materials as specified in piping system sections.

1.3 DEFINITIONS

A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe chases, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawlspaces, and tunnels.

B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.

C. Exposed, Exterior Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.

D. Concealed, Interior Installations: Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and in chases.

E. Concealed, Exterior Installations: Concealed from view and protected from weather conditions and physical contact by building occupants but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.

F. The following are industry abbreviations for plastic materials:

1. ABS: Acrylonitrile-butadiene-styrene plastic.
2. CPVC: Chlorinated polyvinyl chloride plastic.
3. PE: Polyethylene plastic.
4. PVC: Polyvinyl chloride plastic.

G. The following are industry abbreviations for rubber materials:

1. EPDM: Ethylene-propylene-diene terpolymer rubber.
2. NBR: Acrylonitrile-butadiene rubber.

1.4 REFERENCES

A. Applicable Standards:

1. American Society for Testing and Materials (ASTM):

a. A47 – Ferritic Malleable Iron Castings.
b. A53 – Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless.
c. A126 – Gray Iron castings for Valves, Flanges, and Pipe Fittings.
d. A536 – Ductile Iron Castings.
e. B32 – Solder Metal.
f. C1107 – Packaged Dry, Hydraulic-Cement Grout (Nonshrink).
g. D709 – Laminated Thermosetting Materials.

2. American Society of Mechanical Engineers (ASME):

a. Boiler and Pressure Vessel Code.
b. A13.1 – Scheme for the Identification of Piping Systems.
c. B1.20.1 – Pipe Threads, General Purpose (Inch).
d. B16.20 – Ring-Joint Gaskets and Grooves for Steel Pipe Flanges.

e. B16.21 – Nonmetallic Flat Gaskets for Pipe Flanges.
f. B18.2.1 – Square and Hex Bolts and Screws-Inch Series.
g. B31 Series – Code for Pressure Piping.

3. American Welding Society (AWS):

a. Soldering Manual, latest.
b. Brazing Manual, latest.
c. A5.8 – Filler Metals for Brazing.
d. D1.1 – Structural Welding Code for Steel.
e. D10.12 – Recommended Practices and Procedures for Welding Low Carbon Steel Pipe.

1.5 SUBMITTALS

A. General, all Division 22 sections of the Specifications: Follow the procedures specified in Division 01. Prepare maintenance manuals in accordance with Division 01.

B. Welder certificates signed by Contractor certifying that welders comply with requirements specified under the "Quality Assurance" article of this section.

1.6 QUALITY ASSURANCE

A. Qualify welding processes and operators for structural steel according to AWS D1.1.

B. All welding on pressure piping shall conform with the requirements of the American National Standard Code for Pressure Piping, ANSI B31.3, "Chemical Plant and Petroleum Refinery Piping." All welds on piping having working pressures of 300 psig or greater shall be subjected to a full X-ray examination and will not be accepted until all welds meet the requirements of ANSI B31.1, "Power Piping." Faulty welds shall be removed at no additional cost to CONTRACTING OFFICER. X-ray testing shall be performed by others at no additional cost to the Contractor.

1. Certify that each welder has passed AWS qualification tests for the welding processes involved and that certification is current.

C. Pressure Vessels: Prior to installation and acceptance, any power boiler, low-pressure heating boiler, or unfire pressure vessel operated at pressures of 15 pounds per square inch or greater, furnished under this contract will be stamped with ASME Boiler and Pressure Vessel Code Symbol and a National Board of Boiler and Pressure Vessel Inspector's number, thus certifying that the vessel has been fabricated and tested per the provisions of the ASME Boiler and Pressure Vessel Code. Manufacturers' data reports (unless exempted by the ASME Code) will be filed with the National Board in Columbus, Ohio. Two copies of these data reports shall be submitted to CONTRACTING OFFICER. Testing, certification, and registration will be at the expense of the Contractor.

D. Electrical Characteristics for Plumbing Equipment: Equipment of higher electrical characteristics may be furnished provided such proposed equipment is approved in writing and connecting electrical services, circuit breakers, and conduit sizes are appropriately modified. If

minimum energy ratings or efficiencies are specified, equipment shall comply with requirements.

1.7 DELIVERY, STORAGE, AND PROTECTION

A. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe-end damage and prevent entrance of dirt, debris, and moisture.

B. Protect stored pipes and tubes from moisture and dirt. Elevate above grade. When stored inside, do not exceed structural capacity of the floor.

C. Protect flanges, fittings, and piping specialties from moisture and dirt.

D. Store plastic pipes protected from direct sunlight. Support to prevent sagging and bending.

E. Materials and equipment furnished by others.

1. When equipment or materials are indicated to be furnished by others (F.B.O.) or by government furnished equipment (GFE) to the CONTRACTOR for installation and connection, the CONTRACTOR shall make a complete check of all materials and furnish a receipt to the CONTRACTING OFFICER detailing the products received and the condition of the products delivered to him. After executing the receipt and acceptance by the CONTRACTOR, the CONTRACTOR shall assume full responsibility for the safe keeping, handling, and installation of the materials and equipment furnished by others or furnished by government, until completed installation and final approval by the CONTRACTING OFFICER Technical Representative.
2. If the CONTRACTOR fails to issue said receipt it shall be assumed that all equipment and materials were then delivered to the CONTRACTOR in the proper quantities and in perfect condition.

1.8 PROJECT SITE CONDITIONS

A. Altitude Ratings: Unless otherwise noted, all specified equipment capacities, air quantities, etc., are for an altitude at sea level.

1.9 SEQUENCING AND SCHEDULING

A. Coordinate plumbing equipment installation with other building components.

B. Arrange for chases, slots, and openings in building structure during progress of construction, to allow for plumbing installations.

C. Coordinate the installation of required supporting devices and set sleeves in poured-in-place concrete and other structural components, as they are constructed.

D. Sequence, coordinate, and integrate installations of plumbing materials and equipment for efficient flow of the work. Coordinate installation of large equipment requiring positioning prior to closing in the building.

E. Interruption of Plumbing Utilities:

1. The Contractor shall not interrupt any main interior or exterior plumbing utility without written request for an outage and a subsequent approval of CONTRACTING OFFICER Construction Manager nor shall he interrupt any branch line to an outlet or item of equipment without approval from the CONTRACTING OFFICER Construction Manager.
2. Written request for outages shall be submitted seven calendar days in advance of the outage date. This request will delineate the particular utility or service in question, the time the service will be interrupted and the approximate hours the utility shall be off.
3. Unless otherwise noted on the drawings, or directed, any tie-ins or connections to existing utilities or equipment that necessitate interruptions of service shall be performed on a during non-standard hours
4. The work to be performed during the interruption, will be preceded by all possible preparation, and will be carefully coordinated to minimize the duration of the interruption and work will proceed continuously until the system is restored to normal.
5. Unless otherwise directed, the manipulation of existing main valves to isolate piping, the shutdown of equipment will be done by CONTRACTING OFFICER maintenance personnel.

F. Coordinate installation of identifying devices after completion of covering and painting, where devices are applied to surfaces. Install identifying devices prior to installation of acoustical ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 ELECTRIC MOTORS

A. General: All motors (except as noted) shall conform to the following specifications:

1. Physical: None
2. Functional:

    a. Comply with requirements in this Section except when stricter requirements are specified in fire suppression equipment schedules or sections.
    b. 1/2 hp and Small: Single-phase.
    c. Larger than 1/2 hp shall be three-phase, except where specifically noted otherwise
    d. Comply with NEMA MG 1 unless otherwise indicated.

3. Performance: None

2.2 POWER FACTOR CORRECTION

A. Capacitors shall be installed on all motors 25 hp and larger that have an uncorrected power factor of less than 90% at rated load. KVAR size shall be as required to correct motor power factor to 95% lagging.

B. Capacitors shall have:

1. Individual unit cells
2. All welded steel housings
3. Non-flammable synthetic liquid impregnant
4. Craft tissue insulation
5. Aluminum foil electrodes

C. Each capacitor shall be internally fused.

2.3 IDENTIFICATION

PART 3 - EXECUTION

3.1 ERECTION INSTALLATION APPLICATION

A. Piping Systems – Common Requirements:

1. General: Install piping as described below, except where system sections specify otherwise. Individual piping system specification sections in Division 22 specify piping installation requirements unique to the piping system.
2. General Locations and Arrangements: Drawings (plans, schematics, and diagrams) indicate general location and arrangement of piping systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated, except where deviations to layout are approved on coordination drawings.
3. Install piping at indicated slope.
4. Install components having pressure rating equal to or greater than system operating pressure.
5. Install piping in concealed interior and exterior locations, except in equipment rooms and service areas.
6. Install piping free of sags and bends.
7. Install exposed interior and exterior piping at right angles or parallel to building walls. Diagonal runs are prohibited, except where indicated.
8. Install piping tight to slabs, beams, joists, columns, walls, and other building elements. Allow sufficient space above removable ceiling panels to allow for ceiling panel removal.
9. Install piping to allow application of insulation plus 1-inch clearance around insulation.
10. Locate groups of pipes parallel to each other, spaced to permit valve servicing.
11. Install fittings for changes in direction and branch connections.
12. Escutcheons: Where uncovered exposed pipes pass through floors, finished walls, or finished ceilings, they shall be fitted with chromium-plated cast-brass plates on chromium-plated pipe, or with cast-iron or steel plates on ferrous pipe. Plates shall be large enough to completely close the holes around the pipes and shall be square, octangular, or round, with the least dimension not less than 1-1/2 inches or more than 2-1/2 inches larger than the diameter of the pipe. Plates shall be secured in an approved manner.
13. Verify final equipment locations for roughing in.

14. Refer to equipment specifications in other sections of these specifications for roughing-in requirements.
15. Piping Joint Construction: Join pipe and fittings as follows and as specifically required in individual piping system specification sections.

    a. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
    b. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.
    c. Soldered Joints: Construct joints according to AWS "Soldering Manual."
    d. Brazed Joints: Construct joints according to AWS "Brazing Manual."
    e. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1.
    f. Flanged Joints: Align flange surfaces parallel. Select appropriate gasket material, size, type, and thickness for service application. Install gasket concentrically positioned. Assemble joints by sequencing bolt tightening to make initial contact of flanges and gaskets as flat and parallel as possible. Use suitable lubricants on bolt threads. Tighten bolts gradually and uniformly using torque wrench.

16. Piping Connections: Except as otherwise indicated, make piping connections as specified below.

    a. Install unions, in piping 2 inches and smaller, adjacent to each valve and at final connection to each piece of equipment having 2-inches or smaller threaded pipe connection.
    b. Install flanges, in piping 2-1/2 inches and larger, adjacent to flanged valves and at final connection to each piece of equipment having flanged pipe connection.
    c. Wet Piping Systems (Water): Install dielectric coupling and nipple fittings to connect piping materials of dissimilar metals.

B. Equipment Installation – Common Requirements:

1. Install equipment to provide the maximum possible head room, where mounting heights are not indicated.
2. Install equipment according to approved submittal data. Portions of the Work are shown only in diagrammatic form. Refer conflicts to the CONTRACTING OFFICER.
3. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, except where otherwise indicated.
4. Install plumbing equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. Connect equipment for ease of disconnecting, with minimum of interference with other installations. Extend grease fittings to an accessible location.
5. Install equipment giving right-of-way to piping systems installed at a required slope.
6. Shaft Alignment: All motors and pumps (or drives) connected by a shaft coupling, whether factory or field assembled, shall be aligned during installation using a dial indicator applied to both ends of both shafts for a full 360 degrees prior to operation. Alignment of the shafts shall be less than the maximum allowable tolerances as recommended by the coupling or equipment manufacturer. Alignment of shafts shall be rechecked after several hours of operation and equipment has reached operating temperature.

C. Painting and Finishing:

1. Field painting requirements are specified in Division 09.
2. Damage and Touch-Up: Repair marred and damaged factory-painted finishes with materials and procedures to match original factory finish.

D. Concrete Bases: Construct concrete equipment bases of dimensions indicated, but not less than 4 inches larger in both directions than supported unit. Follow supported equipment manufacturer's setting templates for anchor bolt and tie locations. Use 3000 psi, 28-day compressive strength concrete and reinforcement as specified in Division 3. Housekeeping pads under pumps, etc., shall be 3-1/2A inches thick with #4 reinforcing bars 12 inches on center each way unless otherwise noted.

E. Erection of Metal Supports and Anchorage:

1. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor mechanical materials and equipment.
2. Field Welding: Comply with AWS D1.1 “Structural Welding Code – Steel.”

F. Cutting and Patching:

1. Cut, channel, chase, and drill floors, walls, partitions, ceilings, and other surfaces necessary for plumbing installations. Perform cutting by skilled mechanics of the trades involved.
2. Repair cut surfaces to match adjacent surfaces.

END OF SECTION 22 05 00

SECTION 22 05 17 – SLEEVES AND SLEEVE SEALS FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Sleeves without waterstop.
2. Sleeves with waterstop.
3. Stack-sleeve fittings.
4. Sleeve-seal systems.
5. Grout.
6. Silicone sealants.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

PART 2 - PRODUCTS

2.1 SLEEVES WITHOUT WATERSTOP

A. Physical Requirements

1. Cast-Iron Pipe Sleeves: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends.
2. Steel Pipe Sleeves: ASTM A53/A53M, Type E, Grade B, Schedule 40, hot-dip galvanized, with plain ends.
3. Steel Sheet Sleeves: ASTM A653/A653M, 0.0239-inch minimum thickness; hot-dip galvanized, round tube closed with welded longitudinal joint.

2.2 SLEEVES WITH WATERSTOP

A. Physical Requirements

1. Description: Manufactured PVC/HDPE, sleeve-type, waterstop assembly, made for imbedding in concrete slab or wall.

2.3 STACK-SLEEVE FITTINGS

A. Physical Requirements

1. Description: Manufactured, galvanized cast-iron sleeve with integral cast flashing flange for use in waterproof floors and roofs. Include clamping ring, bolts, and nuts for membrane flashing.

    a. Underdeck Clamp: Clamping ring with setscrews.

2.4 SLEEVE-SEAL SYSTEMS

A. Physical Requirements

1. Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.

    a. Designed to form a hydrostatic seal of 20 psig.
    b. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size.
    c. Pressure Plates: Carbon steel
    d. Connecting Bolts and Nuts: Carbon steel, with ASTM B633 coating of length required to secure pressure plates to sealing elements.

2.5 GROUT

A. Physical Requirements

1. Description: Nonshrink, for interior and exterior sealing openings in non-fire-rated walls or floors.
2. Standard: ASTM C1107/C1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

B. Functional Requirements

1. Design Mix: 5000 psi, 28-day compressive strength.
2. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 INSTALLATION OF SLEEVES – GENERAL

A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide1-inch annular clear space between piping and concrete slabs and walls.

1. Sleeves are not required for core-drilled holes.

C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.

1. Permanent sleeves are not required for holes in slabs formed by molded-PE or -PP sleeves.
2. Cut sleeves to length for mounting flush with both surfaces.

    a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.

3. Using grout, seal space outside of sleeves in slabs and walls without sleeve-seal system.

D. Install sleeves for pipes passing through interior partitions.

1. Cut sleeves to length for mounting flush with both surfaces.
2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
3. Seal annular space between sleeve and piping or piping insulation; use sealants appropriate for size, depth, and location of joint.

E. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke-Barrier Penetrations: Maintain indicated fire or smoke rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping and fill materials specified in Section 017329 “Cutting and Patching.”

3.2 INSTALLATION OF SLEEVES WITH WATERSTOP

A. Install sleeve with waterstop as new walls and slabs are constructed.

B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.

C. Secure nailing flanges to concrete forms.

D. Using grout or silicone sealant, seal space around outside of sleeves.

3.3 INSTALLATION OF STACK-SLEEVE FITTINGS

A. Install stack-sleeve fittings in new slabs as slabs are constructed.

1. Install fittings that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
2. Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing. Comply with requirements for flashing specified in Section 076200 “Sheet Metal Flashing and Trim.”
3. Install section of cast-iron soil pipe to extend sleeve to 3 inches above finished floor level.

4. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
5. Using waterproof silicone sealant, seal space between top hub of stack-sleeve fitting and pipe.

B. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke Barrier Penetrations: Maintain indicated fire or smoke rating of floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping specified in Section 017329 “Cutting and Patching.”

3.4 INSTALLATION OF SLEEVE-SEAL SYSTEMS

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building, and passing through exterior walls.

B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal-system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.5 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Leak Test: After allowing for a full cure, test sleeves and sleeve seals for leaks. Repair leaks and retest until no leaks exist.
2. Sleeves and sleeve seals will be considered defective if they do not pass tests and inspections.

B. Prepare test and inspection reports.

3.6 SLEEVE SCHEDULE

A. Use sleeves and sleeve seals for the following piping-penetration applications:

1. Exterior Concrete Walls above and below Grade:

a. Sleeves with waterstops.

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

2. Concrete Slabs-on-Grade:

a. Sleeves with waterstops.

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

3. Concrete Slabs above Grade:

    a. Sleeves with waterstops or stack-sleeve fittings.

4. Interior Walls and Partitions:

    a. Sleeves without waterstops.

END OF SECTION 23 05 17

SECTION 22 05 18 – ESCUTCHEONS FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Escutcheons.
2. Floor plates.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 ESCUTCHEONS

A. Physical: None

B. Functional:

1. One-Piece, Steel Type: With polished, chrome-plated finish and setscrew fastener.
2. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped steel with polished, chrome-plated finish and spring-clip fasteners.
3. One-Piece, Stamped-Steel Type: With polished, chrome-plated finish and spring-clip fasteners.
4. Split-Plate, Stamped-Steel Type: With polished, chrome-plated finish; concealed hinge; and spring-clip fasteners.

C. Performance: None

2.2 FLOOR PLATES

A. Physical: None

B. Functional:

1. Split Floor Plates: Cast brass with concealed hinge.

C. Performance: None

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.

B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of insulated piping and with OD that completely covers opening.

1. Escutcheons for New Piping:

a. Piping with Fitting or Sleeve Protruding from Wall: One-piece, deep pattern.
b. Chrome-Plated Piping: One-piece steel with polished, chrome-plated finish.
c. Insulated Piping: One-piece steel with polished, chrome-plated finish.
d. Insulated Piping: One-piece stamped steel or split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.

2. Escutcheons for Existing Piping to Remain:

a. Chrome-Plated Piping: Split-casting, stamped steel with concealed hinge with polished, chrome-plated finish.
b. Insulated Piping: Split-plate, stamped steel with concealed hinge with polished, chrome-plated finish
c. Bare Piping at Wall and Floor Penetrations in Finished Spaces: Split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.
d. Bare Piping at Ceiling Penetrations in Finished Spaces: Split-plate, stamped steel with concealed hinge with polished, chrome-plated finish.

C. Install floor plates for piping penetrations of equipment-room floors.

D. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.

1. New Piping: Split floor plate.
2. Existing Piping: Split floor plate.

3.2 FIELD QUALITY CONTROL

A. Using new materials, replace broken and damaged escutcheons and floor plates.

END OF SECTION 22 05 18

SECTION 22 05 19 – METERS AND GAGES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Liquid-in-glass thermometers.
2. Thermowells.
3. Dial-type pressure gages.
4. Gage attachments.
5. Water Consumption Meter

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of meter and gage.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For meters and gages to include in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 LIQUID-IN-GLASS THERMOMETERS

A. Metal-Case, Compact-Style, Liquid-in-Glass Thermometers:

1. Standard: ASME B40.200.
2. Case: Cast aluminum; 6-inch nominal size.
3. Case Form: Back angle unless otherwise indicated.
4. Tube: Glass with magnifying lens and blue or red organic liquid.

5. Tube Background: Nonreflective aluminum with permanently etched scale markings graduated in deg F.
6. Window: Glass or plastic.
7. Stem: Aluminum or brass and of length to suit installation.

    a. Design for Thermowell Installation: Bare stem.

8. Connector: 3/4 inch, with ASME B1.1 screw threads.
9. Accuracy: Plus or minus 1 percent of scale range or one scale division, to a maximum of 1.5 percent of scale range.

B. Plastic-Case, Compact-Style, Liquid-in-Glass Thermometers:

1. Standard: ASME B40.200.
2. Case: Plastic; 6-inch nominal size.
3. Case Form: Back angle unless otherwise indicated.
4. Tube: Glass with magnifying lens and blue or red organic liquid.
5. Tube Background: Nonreflective with permanently etched scale markings graduated in deg F.
6. Window: Glass or plastic.
7. Stem: Aluminum or brass and of length to suit installation.

    a. Design for Thermowell Installation: Bare stem.

8. Connector: 3/4 inch, with ASME B1.1 screw threads.
9. Accuracy: Plus or minus 1 percent of scale range or one scale division, to a maximum of 1.5 percent of scale range.

2.2 THERMOWELLS

A. Physical: None

B. Functional:

a. Standard: ASME B40.200.
b. Description: Pressure-tight, socket-type fitting made for insertion into piping tee fitting.
c. Material for Use with Copper Tubing: CNR or CUNI.
d. Type: Stepped shank unless straight or tapered shank is indicated.
e. External Threads: NPS 1/2, NPS 3/4, or NPS 1, ASME B1.20.1 pipe threads.
f. Internal Threads: 1/2, 3/4, and 1 inch, with ASME B1.1 screw threads.
g. Bore: Diameter required to match thermometer bulb or stem.
h. Insertion Length: Length required to match thermometer bulb or stem.
i. Lagging Extension: Include on thermowells for insulated piping and tubing.
j. Bushings: For converting size of thermowell's internal screw thread to size of thermometer connection.

C. Performance: None

D. Heat-Transfer Medium: Mixture of graphite and glycerin.

2.3 PRESSURE GAGES

A. Direct-Mounted, Metal-Case, Dial-Type Pressure Gages:

1. Physical: None
2. Functional:

a. Standard: ASME B40.100.
b. Case: Sealed type; cast aluminum or drawn steel; 4-1/2-inch nominal diameter.
c. Pressure-Element Assembly: Bourdon tube unless otherwise indicated.
d. Match pressure connection size in first subparagraph below with gage attachment size.
e. Pressure Connection: Brass, with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads and bottom-outlet type unless back-outlet type is indicated.
f. Movement: Mechanical, with link to pressure element and connection to pointer.
g. Dial: Nonreflective aluminum with permanently etched scale markings graduated in psi.
h. Pointer: Dark-colored metal.
i. Window: Glass or plastic.
j. Ring: Metal.
k. Accuracy: Grade A, plus or minus 1 percent of middle half of scale range.

B. Performance: None

1. Direct-Mounted, Plastic-Case, Dial-Type Pressure Gages:

a. Physical: None
b. Functional:
c. Standard: ASME B40.100.
d. Case: Sealed type; plastic; 4-1/2-inch nominal diameter.
e. Pressure-Element Assembly: Bourdon tube unless otherwise indicated.
f. Match pressure connection size in first subparagraph below with gage attachment size.
g. Pressure Connection: Brass, with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads and bottom-outlet type unless back-outlet type is indicated.
h. Movement: Mechanical, with link to pressure element and connection to pointer.
i. Dial: Nonreflective aluminum with permanently etched scale markings graduated in psi.
j. Pointer: Dark-colored metal.
k. Window: Glass or plastic.
l. Accuracy: Grade A, plus or minus 1 percent of middle half of scale range.

C. Performance: None

2.4 GAGE ATTACHMENTS

A. Physical: None

B. Functional:

1. Snubbers: ASME B40.100, brass; with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads and piston-type surge-dampening device. Include extension for use on insulated piping.
2. Valves: Brass ball, with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads.

C. Performance: None

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install thermowells with socket extending one-third of pipe diameter and in vertical position in piping tees.

B. Install thermowells of sizes required to match thermometer connectors. Include bushings if required to match sizes.

C. Install thermowells with extension on insulated piping.

D. Fill thermowells with heat-transfer medium.

E. Install direct-mounted thermometers in thermowells and adjust vertical and tilted positions.

F. Install direct-mounted pressure gages in piping tees with pressure gage located on pipe at the most readable position.

G. Install valve and snubber in piping for each pressure gage for fluids.

H. Install thermometers in the following locations:

1. Inlet and outlet of each water heater.

I. Install pressure gages in the following locations:

1. Building water service entrance into building.
2. Inlet and outlet of each pressure-reducing valve.
3. Suction and discharge of each domestic water pump.

J. Install meters and gages adjacent to machines and equipment to allow service and maintenance of meters, gages, machines, and equipment.

K. Adjust faces of meters and gages to proper angle for best visibility.

3.2 THERMOMETER SCHEDULE

A. Thermometers at inlet and outlet of each domestic water heater shall be one of the following:

1. Metal or Plastic case, compact-style, liquid-in-glass type.

B. Retain option in first paragraph below to allow Contractor to select thermometer materials from those retained.

C. Retain option in first paragraph below to allow Contractor to select thermometer materials from those retained.

D. Insert additional paragraphs for thermometer applications.

E. Thermometer stems shall be of length to match thermowell insertion length.

3.3 THERMOMETER SCALE-RANGE SCHEDULE

A. Scale Range for Domestic Cold-Water Piping: 0 to 100 deg F.

B. Scale Range for Domestic Hot-Water Piping: 20 to 240 deg F.

3.4 PRESSURE-GAGE SCHEDULE

A. Pressure gages at discharge of each water service into building shall be one of the following:

1. Sealed, direct-mounted, metal case.
2. Sealed, direct-mounted, plastic case.

B. Pressure gages at inlet and outlet of each water pressure-reducing valve shall be one of the following:

1. Sealed, direct-mounted, metal case.
2. Sealed, direct-mounted, plastic case.
3. Test plug with EPDM self-sealing rubber inserts.

C. Pressure gages at suction and discharge of each domestic water pump shall be one of the following:

1. Sealed, direct-mounted, metal case.
2. Sealed, direct-mounted, plastic case.
3. Test plug with EPDM self-sealing rubber inserts.

3.5 PRESSURE-GAGE SCALE-RANGE SCHEDULE

A. Scale Range for Water Service Piping: 0 to 160 psi.

B. Scale Range for Domestic Water Piping: 0 to 100 psi.

END OF SECTION 22 05 19

SECTION 22 05 23 – GENERAL-DUTY VALVES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Brass ball valves.
2. Bronze swing check valves.

B. Related Sections:

1. Section 220553 “Identification for Plumbing Piping and Equipment” for valve tags and schedules.
2. Section 221116 “Domestic Water Piping” for valves applicable only to this piping.
3. Section 221319 “Sanitary Waste Piping Specialties” for valves applicable only to this piping.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of valve indicated.

1.3 QUALITY ASSURANCE

A. ASME Compliance: ASME B16.10 and ASME B16.34 for ferrous valve dimensions and design criteria.

B. NSF Compliance: NSF 61 for valve materials for potable-water service.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR VALVES

A. Physical: None

B. Functional:

1. Plumbing valve applications specified in this Section are limited to NPS 12 (DN 300).
2. Refer to valve schedule articles for applications of valves.
3. Caution: Revise pressure ratings and insert temperature ratings in valve articles if valves with higher ratings are required.
4. Valve Pressure and Temperature Ratings: Not less than indicated and as required for system pressures and temperatures.
5. Valve Sizes: Same as upstream piping unless otherwise indicated.

6. Valve Actuator Types:

    a. Gear Actuator: For quarter-turn valves NPS 8 and larger.
    b. Handwheel: For valves other than quarter-turn types.
    c. Handlever: For quarter-turn valves NPS 6 and smaller.

7. Valves in Insulated Piping: With 2-inch stem extensions and the following features:

    a. Ball Valves: With extended operating handle of non-thermal-conductive material, and protective sleeve that allows operation of valve without breaking the vapor seal or disturbing insulation.

8. Valve-End Connections:

    a. Flanged: With flanges according to ASME B16.1 for iron valves.
    b. Solder Joint: With sockets according to ASME B16.18.
    c. Threaded: With threads according to ASME B1.20.1.

C. Performance: None

2.2 BRASS BALL VALVES

A. Physical: None

B. Functional:

1. Two-Piece, Full-Port, Brass Ball Valves with Brass Trim:

    a. See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers. See Section 016000 “Product Requirements.”
    b. Description:

        1) Standard: MSS SP-110.
        2) SWP Rating: 150 psig.
        3) CWP Rating: 600 psig.
        4) Body Design: Two piece.
        5) Body Material: Forged brass.
        6) Ends: Threaded.
        7) Seats: PTFE or TFE.
        8) Stem: Brass.
        9) Ball: Chrome-plated brass.
        10) Port: Full.

C. Performance: None

2.3 BRONZE SWING CHECK VALVES

A. Physical: None

B. Functional:

1. Class 125, Bronze Swing Check Valves with Bronze Disc:

a. See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers. See Section 016000 “Product Requirements.”
b. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
c. Description:

1) Standard: MSS SP-80, Type 3.
2) CWP Rating: 200 psig.
3) Body Design: Horizontal flow.
4) Body Material: ASTM B 62, bronze.
5) Ends: Threaded.
6) Disc: Bronze.

2. Class 125, Bronze Swing Check Valves with Nonmetallic Disc:

a. See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers. See Section 016000 “Product Requirements.”
b. Description:

1) Standard: MSS SP-80, Type 4.
2) CWP Rating: 200 psig.
3) Body Design: Horizontal flow.
4) Body Material: ASTM B 62, bronze.
5) Ends: Threaded.
6) Disc: PTFE or TFE.

C. Performance: None

PART 3 - EXECUTION

3.1 VALVE INSTALLATION

A. Install valves with unions or flanges at each piece of equipment arranged to allow service, maintenance, and equipment removal without system shutdown.

B. Locate valves for easy access and provide separate support where necessary.

C. Install valves in horizontal piping with stem at or above center of pipe.

D. Install valves in position to allow full stem movement.

E. Install swing check valves for proper direction of flow and in horizontal position with hinge pin level.

3.2 ADJUSTING

A. Adjust or replace valve packing after piping systems have been tested and put into service but before final adjusting and balancing. Replace valves if persistent leaking occurs.

3.3 GENERAL REQUIREMENTS FOR VALVE APPLICATIONS

A. If valve applications are not indicated, use the following:

1. Shutoff Service: Ball valves.
2. Throttling Service Ball valves.
3. Pump-Discharge Check Valves:

    a. Sanitary Waste: Iron swing check valves with lever and weight or spring.

B. If valves with specified SWP classes or CWP ratings are not available, the same types of valves with higher SWP class or CWP ratings may be substituted.

C. Select valves, except wafer types, with the following end connections:

1. For Copper Tubing, NPS 2 and Smaller: Threaded ends except where solder-joint valve-end option is indicated in valve schedules below.
2. For Steel Piping, NPS 2 and Smaller: Threaded ends.

3.4 DOMESTIC, HOT- AND COLD-WATER VALVE SCHEDULE

A. Pipe NPS 2 and Smaller:

1. Bronze and Brass Valves: May be provided with solder-joint ends instead of threaded ends.
2. Bronze Angle Valves: Class 125, bronze.
3. Ball Valves: Two-piece, full port, brass or bronze.
4. Bronze Swing Check Valves: Class 125, bronze nonmetallic disc.

END OF SECTION 22 05 23

SECTION 22 05 29 – HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal pipe hangers and supports.
2. Trapeze pipe hangers.
3. Thermal hanger-shield inserts.
4. Fastener systems.
5. Pipe-positioning systems.
6. Equipment supports.

B. Related Requirements:

1. Section 220548 “Vibration and Seismic Controls for Plumbing Piping and Equipment” for vibration isolation devices.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Delegated-Design Submittal: For trapeze hangers indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.3 INFORMATIONAL SUBMITTALS

A. Welding certificates.

1.4 QUALITY ASSURANCE

A. Structural-Steel Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M.

B. Pipe Welding Qualifications: Qualify procedures and operators according to “2015 ASME Boiler and Pressure Vessel Code, Section IX.”

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Physical: None

B. Functional:

1. Delegated Design: Engage a qualified professional engineer, as defined in Section 014000 “Quality Requirements,” to design trapeze pipe hangers and equipment supports.
2. Structural Performance: Hangers and supports for plumbing piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.
3. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.
4. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.
5. Design seismic-restraint hangers and supports for piping and equipment and obtain approval from authorities having jurisdiction.

C. Performance: None

2.2 METAL PIPE HANGERS AND SUPPORTS

A. Physical: None

B. Functional:

1. Carbon-Steel Pipe Hangers and Supports:

a. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
b. Galvanized Metallic Coatings: Pregalvanized, hot-dip galvanized, or electro-galvanized.
c. Nonmetallic Coatings: Plastic coated or epoxy powder coated.
d. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
e. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel or stainless steel.

2. Stainless-Steel Pipe Hangers and Supports:

a. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
b. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
c. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.

3. Copper Pipe and Tube Hangers:

   a. Description: MSS SP-58, Types 1 through 58, copper-coated-steel, factory-fabricated components.
   b. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel.

C. Performance: None

2.3 TRAPEZE PIPE HANGERS

A. Physical: None

B. Functional:

1. Description: MSS SP-58, Type 59, shop- or field-fabricated pipe-support assembly, made from structural-carbon-steel shapes, with MSS SP-58 carbon-steel hanger rods, nuts, saddles, and U-bolts.

C. Performance: None

2.4 THERMAL HANGER-SHIELD INSERTS

A. Physical: None

B. Functional:

1. Insulation-Insert Material for Cold Piping: ASTM C552, Type II cellular glass with 100-psig or ASTM C591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength and vapor barrier.
2. Insulation-Insert Material for Hot Piping: ASTM C552, Type II cellular glass with 100-psig or ASTM C591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength.
3. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.
4. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.
5. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

C. Performance: None

2.5 FASTENER SYSTEMS

A. Physical: None

B. Functional:

1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

2. Mechanical-Expansion Anchors: Insert-wedge-type anchors, for use in hardened portland cement concrete, with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.
3. Indoor Applications: Zinc-coated or stainless steel.
4. Outdoor Applications: Stainless steel.

C. Performance: None

2.6 PIPE-POSITIONING SYSTEMS

A. Physical: None

B. Functional:

1. Description: IAPMO PS 42 positioning system composed of metal brackets, clips, and straps for positioning piping in pipe spaces; for plumbing fixtures in commercial applications.

C. Performance: None

2.7 EQUIPMENT SUPPORTS

A. Physical: None

B. Functional:

1. Description: Welded, shop- or field-fabricated equipment support made from structural-carbon-steel shapes.

C. Performance: None

2.8 MATERIALS

A. Physical: None

B. Functional:

1. Aluminum: ASTM B221.
2. Carbon Steel: ASTM A1011/A1011M.
3. Structural Steel: ASTM A36/A36M carbon-steel plates, shapes, and bars; black and galvanized.
4. Stainless Steel: ASTM A240/A240M.
5. Grout: ASTM C1107/C1107M, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.

    a. Properties: Nonstaining, noncorrosive, and nongaseous.
    b. Design Mix: 5000-psi, 28-day compressive strength.

C. Performance: None

PART 3 - EXECUTION

3.1 APPLICATION

A. Comply with requirements in Section 078413 “Penetration Firestopping” for firestopping materials and installation, for penetrations through fire-rated walls, ceilings, and assemblies.

B. Strength of Support Assemblies: Where not indicated, select sizes of components, so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb.

3.2 HANGER AND SUPPORT INSTALLATION

A. Metal Pipe-Hanger Installation: Comply with MSS SP-58. Install hangers, supports, clamps, and attachments as required to properly support piping from building structure.

B. Metal Trapeze Pipe-Hanger Installation: Comply with MSS SP-58. Arrange for grouping of parallel runs of horizontal piping, and support together on field-fabricated trapeze pipe hangers.

1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size, or install intermediate supports for smaller-diameter pipes as specified for individual pipe hangers.
2. Field fabricate from ASTM A36/A36M carbon-steel shapes selected for loads being supported. Weld steel according to AWS D1.1/D1.1M.

C. Thermal Hanger-Shield Installation: Install in pipe hanger or shield for insulated piping.

D. Fastener System Installation:

1. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches thick in concrete, after concrete is placed and completely cured. Use operators that are licensed by powder-actuated tool manufacturer. Install fasteners according to powder-actuated tool manufacturer's operating manual.
2. Install mechanical-expansion anchors in concrete, after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

E. Pipe-Positioning-System Installation: Install support devices to make rigid supply and waste piping connections to each plumbing fixture.

F. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.

G. Equipment Support Installation: Fabricate from welded-structural-steel shapes.

H. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.

I. Install lateral bracing with pipe hangers and supports to prevent swaying.

J. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms, and install reinforcing bars through openings at top of inserts.

K. Load Distribution: Install hangers and supports, so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment.

L. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.

M. Insulated Piping:

1. Attach clamps and spacers to piping.

    a. Piping Operating Above Ambient Air Temperature: Clamp may project through insulation.
    b. Piping Operating Below Ambient Air Temperature: Use thermal hanger-shield insert with clamp sized to match OD of insert.
    c. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.

2. Install MSS SP-58, Type 39 protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.

    a. Option: Thermal hanger-shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

3. Install MSS SP-58, Type 40 protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.

    a. Option: Thermal hanger-shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

4. Shield Dimensions for Pipe: Not less than the following:

    a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.

5. Thermal Hanger Shields: Install with insulation of same thickness as piping insulation.

3.3 EQUIPMENT SUPPORTS

A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor.

B. Grouting: Place grout under supports for equipment, and make bearing surface smooth.

C. Provide lateral bracing, to prevent swaying, for equipment supports.

3.4 METAL FABRICATIONS

A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers and equipment supports.

B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.

C. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work.

3.5 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.6 PAINTING

A. Touchup: Clean field welds and abraded, shop-painted areas. Paint exposed areas immediately after erecting hangers and supports. Use same materials as those used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.

B. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas, and apply galvanizing-repair paint to comply with ASTM A780/A780M.

3.7 HANGER AND SUPPORT SCHEDULE

A. Specific hanger and support requirements are in Sections specifying piping systems and equipment.

B. Comply with MSS SP-58 for pipe-hanger selections and applications that are not specified in piping system Sections.

C. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finishes.

D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.

E. Use carbon-steel pipe hangers and supports and metal trapeze pipe hangers and attachments for general service applications.

F. Use stainless-steel pipe hangers and stainless-steel or corrosion-resistant attachments for hostile environment applications.

G. Use copper-plated pipe hangers and copper or stainless-steel attachments for copper piping and tubing.

H. Use padded hangers for piping that is subject to scratching.

I. Use thermal hanger-shield inserts for insulated piping and tubing.

J. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of noninsulated or insulated, stationary pipes NPS 1/2 to NPS 30.
2. Yoke-Type Pipe Clamps (MSS Type 2): For suspension of up to 1050 deg F pipes NPS 4 to NPS 24, requiring up to 4 inches of insulation.
3. Carbon- or Alloy-Steel, Double-Bolt Pipe Clamps (MSS Type 3): For suspension of pipes NPS 3/4 to NPS 36, requiring clamp flexibility and up to 4 inches of insulation.
4. Steel Pipe Clamps (MSS Type 4): For suspension of cold and hot pipes NPS 1/2 to NPS 24 if little or no insulation is required.
5. Pipe Hangers (MSS Type 5): For suspension of pipes NPS 1/2 to NPS 4, to allow off-center closure for hanger installation before pipe erection.
6. Adjustable, Swivel Split- or Solid-Ring Hangers (MSS Type 6): For suspension of noninsulated, stationary pipes NPS 3/4 to NPS 8 .Adjustable, Steel Band Hangers (MSS Type 7): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
7. Adjustable Band Hangers (MSS Type 9): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
8. Adjustable, Swivel-Ring Band Hangers (MSS Type 10): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
9. Split Pipe Ring with or without Turnbuckle Hangers (MSS Type 11): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 8.
10. Extension Hinged or Two-Bolt Split Pipe Clamps (MSS Type 12): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 3.
11. U-Bolts (MSS Type 24): For support of heavy pipes NPS 1/2 to NPS 30.
12. Clips (MSS Type 26): For support of insulated pipes not subject to expansion or contraction.
13. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30, from two rods if longitudinal movement caused by expansion and contraction occurs.
14. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes NPS 2-1/2 to NPS 24, from single rod if horizontal movement caused by expansion and contraction occurs.
15. Complete Pipe Rolls (MSS Type 44): For support of pipes NPS 2 to NPS 42 if longitudinal movement caused by expansion and contraction occurs but vertical adjustment is unnecessary.
16. Pipe Roll and Plate Units (MSS Type 45): For support of pipes NPS 2 to NPS 24 if small horizontal movement caused by expansion and contraction occurs and vertical adjustment is unnecessary.
17. Adjustable Pipe Roll and Base Units (MSS Type 46): For support of pipes NPS 2 to NPS 30 if vertical and lateral adjustment during installation, in addition to expansion and contraction, is required.

K. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24.
2. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps.

L. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel Turnbuckles (MSS Type 13): For adjustment of up to 6 inches for heavy loads.
2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11 split pipe rings.
4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.

M. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel or Malleable-Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
2. Top-Beam C-Clamps (MSS Type 19): For use under roof installations with bar-joist construction, to attach to top flange of structural shape.
3. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
4. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
5. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
6. C-Clamps (MSS Type 23): For structural shapes.
7. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
8. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
9. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
10. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.
11. Malleable-Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
12. Welded-Steel Brackets: For support of pipes from below or for suspending from above by using clip and rod. Use one of the following for indicated loads:

    a. Light (MSS Type 31): 750 lb.
    b. Medium (MSS Type 32): 1500 lb.
    c. Heavy (MSS Type 33): 3000 lb.

13. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.
14. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
15. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.

N. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel-Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
3. Thermal Hanger-Shield Inserts: For supporting insulated pipe.

O. Spring Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Restraint-Control Devices (MSS Type 47): Where indicated to control piping movement.
2. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1-1/4 inches .
3. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41 roll hanger with springs.
4. Spring Sway Braces (MSS Type 50): To retard sway, shock, vibration, or thermal expansion in piping systems.
5. Variable-Spring Hangers (MSS Type 51): Preset to indicated load, and limit variability factor to 25 percent to allow expansion and contraction of piping system from hanger.
6. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load, and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.
7. Variable-Spring Trapeze Hangers (MSS Type 53): Preset to indicated load, and limit variability factor to 25 percent to allow expansion and contraction of piping system from trapeze support.
8. Constant Supports: For critical piping stress and if necessary to avoid transfer of stress from one support to another support, critical terminal, or connected equipment. Include auxiliary stops for erection, hydrostatic test, and load-adjustment capability. These supports include the following types:

    a. Horizontal (MSS Type 54): Mounted horizontally.
    b. Vertical (MSS Type 55): Mounted vertically.
    c. Trapeze (MSS Type 56): Two vertical-type supports and one trapeze member.

P. Comply with MSS SP-58 for trapeze pipe-hanger selections and applications that are not specified in piping system Sections.

Q. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.

R. Use pipe-positioning systems in pipe spaces behind plumbing fixtures to support supply and waste piping for plumbing fixtures.

END OF SECTION 22 05 29

SECTION 22 05 53 – IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Equipment labels.
2. Warning signs and labels.
3. Pipe labels.
4. Stencils.
5. Valve Tags.

1.2 ACTION SUBMITTAL

A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS

2.1 EQUIPMENT LABELS

A. Physical: None

B. Functional:

1. Metal Labels for Equipment:

a. Retain and punctuate multiple material options in first subparagraph below to allow Contractor to select most cost-effective material.
b. Material and Thickness: Brass, 0.032-inch or Stainless steel, 0.025-inch minimum thickness, and having predrilled or stamped holes for attachment hardware.
c. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
d. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
e. Fasteners: Stainless-steel rivets or self-tapping screws.
f. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

2. Plastic Labels for Equipment:

a. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.

b. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
c. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
d. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
e. Fasteners: Stainless-steel rivets or self-tapping screws.
f. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

1) In first paragraph below, describe required label content. The objective of labeling equipment is to coordinate it with Drawings, including plans, details, and schedules. This will allow other information, such as capacities and operating characteristics, to be obtained.
2) Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified.
3) Equipment Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules), plus the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data.

C. Performance: None

2.2 WARNING SIGNS AND LABELS

A. Physical: None

B. Functional:

1. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.
2. Maximum Temperature: Able to withstand temperatures up to 160 deg F.
3. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
4. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
5. Fasteners: Stainless-steel rivets or self-tapping screws.
6. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

7. Label Content: Include caution and warning information, plus emergency notification instructions.

C. Performance: None

2.3 PIPE LABELS

A. Physical: None

B. Functional:

1. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction.
2. Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to cover full circumference of pipe and to attach to pipe without fasteners or adhesive.
3. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
4. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.

    a. Flow-Direction Arrows: Integral with piping system service lettering to accommodate both directions, or as separate unit on each pipe label to indicate flow direction.
    b. Lettering Size: At least 1-1/2 inches high.

C. Performance: None

2.4 Stencils:

A. Physical: None

B. Functional:

1. Stencils: Prepare with letter sizes according to ASME A 13.1 for piping; minimum letter height of 2 inches for equipment labels, and similar operational instructions.
2. Stencil Material: Fiberboard or metal.
3. Stencil Paint: Exterior, gloss enamel black unless otherwise indicated. Paint may be in pressurized spray-can form.
4. Identification Paint: Exterior enamel in colors according to ASME A13.1 unless otherwise indicated.

C. Performance: None

2.5 Valve Tags:

A. Physical: None

B. Functional:

1. Valve Tags: Stamped or engraved with 1/4-inch letters for piping system abbreviation and 1/2-inch numbers.
2. Tag Materials: Brass, 0.032-inch minimum thickness, and having pre-drilled or stamped holes for attachment hardware.

3. Valve Schedules: For each piping system, on 8-/2-by-11-inch bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open or closed), and variations for identification. Mark valves for emergency shutoff and similar special uses.
4. Valve tag schedule shall be included in operation and maintenance data.

C. Performance: None

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean piping and equipment surfaces of substances that could impair bond of identification devices, including dirt, oil, grease, release agents, and incompatible primers, paints, and encapsulants.

3.2 EQUIPMENT LABEL INSTALLATION

A. Install or permanently fasten labels on each major item of mechanical equipment.

B. Locate equipment labels where accessible and visible.

3.3 PIPE LABEL INSTALLATION

A. Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Near each valve and control device.
2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.
3. Near penetrations through walls, floors, ceilings, and inaccessible enclosures.
4. At access doors, manholes, and similar access points that permit view of concealed piping.
5. Near major equipment items and other points of origination and termination.
6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.
7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.

B. Pipe Label Color Schedule backgrounds and text color shall comply with ASME/ANSI A13.1 standard for commercial buildings.

END OF SECTION 22 05 53

SECTION 22 07 19 – PLUMBING PIPING INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes insulating the following plumbing piping services:

1. Domestic hot-water piping.
2. Domestic recirculating hot-water piping.
3. Supplies and drains for handicap-accessible lavatories and sinks.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

1.4 QUALITY ASSURANCE

A. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E 84 by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.
2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

B. Comply with the following applicable standards and other requirements specified for miscellaneous components:

1. Supply and Drain Protective Shielding Guards: ICC A117.1.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Physical: None

B. Functional:

1. If retaining more than one type of insulation in this article, indicate where each type applies in insulation system schedules.
2. Comply with requirements in “Piping Insulation Schedule, General,” “Indoor Piping Insulation Schedule,” “Outdoor, Aboveground Piping Insulation Schedule,” and “Outdoor, Underground Piping Insulation Schedule” articles for where insulating materials shall be applied.
3. See "Product Characteristics" Article in Evaluations for comparisons and temperature ranges for insulation material properties.
4. Products shall not contain asbestos, lead, mercury, or mercury compounds.
5. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.
6. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.
7. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.
8. Cellular Glass: Inorganic, incombustible, foamed or cellulated glass with annealed, rigid, hermetically sealed cells. Factory-applied jacket requirements are specified in “Factory-Applied Jackets” Article.

    a. Special-Shaped Insulation: ASTM C 552, Type III.
    b. If retaining both types of insulation in first two subparagraphs below, indicate where each type applies in insulation system schedules.
    c. Preformed Pipe Insulation without Jacket: Comply with ASTM C 552, Type II, Class 1.
    d. Preformed Pipe Insulation with Factory-Applied ASJ or ASJ-SSL: Comply with ASTM C 552, Type II, Class 2.
    e. Factory fabricate shapes according to ASTM C 450 and ASTM C 585.

9. Mineral-Fiber, Preformed Pipe Insulation:

    a. Type I, 850 Deg F Materials: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 547, Type I, Grade A, without factory-applied jacket or with factory-applied ASJ or with factory-applied ASJ-SSL. Factory-applied jacket requirements are specified in “Factory-Applied Jackets” Article.

C. Performance: None

2.2 INSULATING CEMENTS

A. Physical: None

B. Functional:

1. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement: Comply with ASTM C 449.

C. Performance: None

2.3 ADHESIVES

A. Physical: None

B. Functional:

1. Military Specification in this article was the only standard available when this Section was written. MIL-A-3316C was last updated in October 1987.
2. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated, unless otherwise indicated.
3. Product attributes in first paragraph below are based on Foster Brand products; there are variations among manufacturers.
4. Cellular-Glass Adhesive: Two-component, thermosetting urethane adhesive containing no flammable solvents, with a service temperature range of minus 100 to plus 200 deg F.

    a. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
    b. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

5. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
6. ASJ Adhesive, and FSK Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.
7. PVC Jacket Adhesive: Compatible with PVC jacket.

C. Performance: None

2.4 MASTICS

A. Physical: None

B. Functional:

1. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.

2. Vapor-Barrier Mastic: Water based; suitable for indoor use on below-ambient services.

    a. Water-Vapor Permeance: ASTM E 96/E 96M, Procedure B, 0.013 perm at 43-mil dry film thickness.
    b. Service Temperature Range: Minus 20 to plus 180 deg F.
    c. Solids Content: ASTM D 1644, 58 percent by volume and 70 percent by weight.
    d. Color: White.

3. Breather Mastic: Water based; suitable for indoor and outdoor use on above-ambient services.

a. Water-Vapor Permeance: ASTM F 1249, 1.8 perms at 0.0625-inch dry film thickness.
b. Service Temperature Range: Minus 20 to plus 180 deg F.
c. Solids Content: 60 percent by volume and 66 percent by weight.
d. Color: White.

C. Performance: None

2.5 SEALANTS

A. Physical: None

B. Functional:

1. Joint Sealants for Cellular-Glass Products:

a. Materials shall be compatible with insulation materials, jackets, and substrates.
b. Permanently flexible, elastomeric sealant.
c. Service Temperature Range: Minus 100 to plus 300 deg F.
d. Color: White or gray.

2. FSK and Metal Jacket Flashing Sealants:

a. Materials shall be compatible with insulation materials, jackets, and substrates.
b. Fire- and water-resistant, flexible, elastomeric sealant.
c. Service Temperature Range: Minus 40 to plus 250 deg F.
d. Color: Aluminum.

3. ASJ Flashing Sealants, and Vinyl, PVDC, and PVC Jacket Flashing Sealants:

a. Materials shall be compatible with insulation materials, jackets, and substrates.
b. Fire- and water-resistant, flexible, elastomeric sealant.
c. Service Temperature Range: Minus 40 to plus 250 deg F.
d. Color: White.

C. Performance: None

2.6 FACTORY-APPLIED JACKETS

A. Physical: None

B. Functional:

1. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

a. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C 1136, Type I.
b. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C 1136, Type I.

c. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C 1136, Type II.

C. Performance: None

2.7 FIELD-APPLIED JACKETS

A. Physical: None

B. Functional:

1. Field-applied jackets shall comply with ASTM C 921, Type I, unless otherwise indicated.

2. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D 1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.

a. Adhesive: As recommended by jacket material manufacturer.

1) Color: White or Color-code jackets based on system. Color as selected by Architect.
2) Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.

b. Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

3. Underground Direct-Buried Jacket: 125-mil- thick vapor barrier and waterproofing membrane consisting of a rubberized bituminous resin reinforced with a woven-glass fiber or polyester scrim and laminated aluminum foil.

C. Performance: None

2.8 TAPES

A. Physical: None

B. Functional:

1. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136.

a. Width: 3 inches.
b. Thickness: 11.5 mils.
c. Adhesion: 90 ounces force/inch in width.
d. Elongation: 2 percent.
e. Tensile Strength: 40 lbf/inch in width.
f. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

2. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

    a. Width: 2 inches.
    b. Thickness: 6 mils.
    c. Adhesion: 64 ounces force/inch in width.
    d. Elongation: 500 percent.
    e. Tensile Strength: 18 lbf/inch in width.

C. Performance: None

2.9 SECUREMENTS

A. Physical: None

B. Functional

1. Aluminum Bands: ASTM B 209, Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, 1/2 inch or 3/4 inch wide with wing seal or closed seal.
2. Staples: Outward-clinching insulation staples, nominal 3/4-inch- wide, stainless steel or Monel.

3. Wire: 0.080-inch nickel-copper alloy or 0.062-inch soft-annealed, stainless steel or 0.062-inch soft-annealed, galvanized steel.

C. Performance: None

2.10 PROTECTIVE SHIELDING GUARDS

A. Physical: None

B. Functional:

1. Protective Shielding Pipe Covers:

    a. Description: Manufactured plastic wraps for covering plumbing fixture hot-water supply or hot- and cold-water supplies and trap and drain piping. Comply with Americans with Disabilities Act (ADA) requirements.

2. Protective Shielding Piping Enclosures:

    a. Description: Manufactured plastic enclosure for covering plumbing fixture hot- and cold-water supplies and trap and drain piping. Comply with ADA requirements.

C. Performance: None

PART 3 - EXECUTION

3.1 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

B. Coordinate insulation installation with the trade installing heat tracing. Comply with requirements for heat tracing that apply to insulation.

C. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless-steel surfaces, use demineralized water.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of piping including fittings, valves, and specialties.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item of pipe system as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.

G. Keep insulation materials dry during application and finishing.

H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

I. Install insulation with least number of joints practical.

J. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.
2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.
4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.

K. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

L. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth.
2. Cover circumferential joints with 3-inch- wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.
3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 2 inches o.c.

    a. For below-ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.
5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to pipe flanges and fittings.

M. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

N. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

O. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

P. For above-ambient services, do not install insulation to the following:

1. Vibration-control devices.
2. Testing agency labels and stamps.
3. Nameplates and data plates.
4. Cleanouts.

3.3 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.
4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

C. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions.

1. Comply with requirements in Section 078413 “Penetration Firestopping” for firestopping and fire-resistive joint sealers.

D. Insulation Installation at Floor Penetrations:

1. Pipe: Install insulation continuously through floor penetrations.
2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 078413 “Penetration Firestopping.”

3.4 GENERAL PIPE INSULATION INSTALLATION

A. Requirements in this article generally apply to all insulation materials except where more specific requirements are specified in various pipe insulation material installation articles.

B. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:

1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity unless otherwise indicated.
2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below-ambient services, provide a design that maintains vapor barrier.
6. Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.

7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below-ambient services and a breather mastic for above-ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
8. For services not specified to receive a field-applied jacket except for flexible elastomeric and polyolefin, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
9. Stencil or label the outside insulation jacket of each union with the word "union." Match size and color of pipe labels.

C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.

D. Install removable insulation covers at locations indicated. Installation shall conform to the following:

1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel or aluminum bands. Select band material compatible with insulation and jacket.
3. Construct removable valve insulation covers in same manner as for flanges, except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.5 INSTALLATION OF CELLULAR-GLASS INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above-ambient services, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below-ambient services, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of cellular-glass block insulation of same thickness as pipe insulation.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available. Secure according to manufacturer's written instructions.
2. When preformed sections of insulation are not available, install mitered sections of cellular-glass insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of cellular-glass insulation to valve body.
2. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.

3.6 INSTALLATION OF MINERAL-FIBER PREFORMED PIPE INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with factory-applied jackets on above-ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.
4. For insulation with factory-applied jackets on below-ambient surfaces, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with mineral-fiber blanket insulation.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed sections are not available, install mitered sections of pipe insulation to valve body.
3. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
4. Install insulation to flanges as specified for flange insulation application.

3.7 FIELD-APPLIED JACKET INSTALLATION

A. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints. Seal with manufacturer's recommended adhesive.

1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

3.8 FINISHES

A. Insulation with ASJ, Glass-Cloth, or Other Paintable Jacket Material: Paint jacket with paint system identified below and as specified in Section 099113 “Exterior Painting” and Section 099123 “Interior Painting.”

1. Flat Acrylic Finish: Two finish coats over a primer that is compatible with jacket material and finish coat paint. Add fungicidal agent to render fabric mildew proof.

a. Finish Coat Material: Interior, flat, latex-emulsion size.

B. Color: Final color as selected by Architect. Vary first and second coats to allow visual inspection of the completed Work.

C. Do not field paint aluminum or stainless-steel jackets.

3.9 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Tests and Inspections:

1. Inspect pipe, fittings, strainers, and valves, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to three locations of straight pipe, three locations of threaded fittings, three locations of welded fittings, two locations of threaded strainers, two locations of welded strainers, three locations of threaded valves, and three locations of flanged valves for each pipe service defined in the "Piping Insulation Schedule, General" Article.

C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.10 PIPING INSULATION SCHEDULE, GENERAL

A. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.

B. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:

1. Underground piping.
2. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.11 INDOOR PIPING INSULATION SCHEDULE

A. Domestic Cold Water: Insulation shall be one of the following:

1. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
2. Cellular Glass: 1-1/2 inch thick,

B. Domestic Hot and Recirculated Hot Water: Insulation shall be one of the following:

1. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
2. Cellular Glass: 2 inch thick,

C. Exposed Sanitary Drains, Domestic Cold Water, and Stops for Plumbing Fixtures for People with Disabilities: Insulation shall be one of the following:

1. Flexible Elastomeric: 3/4 inch or 1 inch thick.
2. Mineral-Fiber, Preformed Pipe Insulation, Type I: 1/2 inch or 1 inch thick.

3.12 INDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Piping, Concealed:

1. PVC, Color-Coded by System: 20 mils thick.

D. Piping, Exposed:

1. PVC: 20 mils thick.
2. Aluminum, Smooth Embossed: 0.020 inch thick.

3.13 UNDERGROUND, FIELD-INSTALLED INSULATION JACKET

A. For underground direct-buried piping applications, install underground direct-buried jacket over insulation material.

END OF SECTION 22 07 19

SECTION 22 11 16 – DOMESTIC WATER PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Copper tube and fittings.
2. Piping joining materials.
3. Transition fittings.
4. Dielectric fittings.

1.2 ACTION SUBMITTALS

A. Product Data: For transition fittings and dielectric fittings.

1.3 INFORMATIONAL SUBMITTALS

A. System purging and disinfecting activities report.

B. Field quality-control reports.

PART 2 - PRODUCTS

2.1 PIPING MATERIALS

A. Physical: None

B. Functional:

1. Comply with requirements in “Piping Schedule” Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.
2. Potable-water piping and components shall comply with NSF 14 and NSF 61 Annex G. Plastic piping components shall be marked with “NSF-pw.”
3. Comply with NSF 372 for low lead.

C. Performance: None

2.2 COPPER TUBE AND FITTINGS

A. Physical: None

B. Functional:

1. Hard Copper Tube: ASTM B 88, Type L water tube, drawn temper.
2. Soft Copper Tube: ASTM B 88, Type L water tube, annealed temper.
3. Cast-Copper, Solder-Joint Fittings: ASME B16.18, pressure fittings.
4. Bronze Flanges: ASME B16.24, Class 150, with solder-joint ends.
5. Copper Unions:

    a. MSS SP-123.7
    b. Cast-copper-alloy, hexagonal-stock body.
    c. Ball-and-socket, metal-to-metal seating surfaces.
    d. Solder-joint or threaded ends.

C. Performance: None

2.3 PIPING JOINING MATERIALS

A. Physical: None

B. Functional:

1. Pipe-Flange Gasket Materials:

    a. AWWA C110/A21.10, rubber, flat face, 1/8 inch thick or ASME B16.21, nonmetallic and asbestos free unless otherwise indicated.
    b. Full-face or ring type unless otherwise indicated.

2. Metal, Pipe-Flange Bolts and Nuts: ASME B18.2.1, carbon steel unless otherwise indicated.
3. Solder Filler Metals: ASTM B 32, lead-free alloys.
4. Flux: ASTM B 813, water flushable.
5. Brazing Filler Metals: AWS A5.8M/A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing unless otherwise indicated.
6. Solvent Cements for Joining CPVC Piping and Tubing: ASTM F 493.
7. Solvent Cements for Joining PVC Piping: ASTM D 2564. Include primer according to ASTM F 656.
8. Plastic, Pipe-Flange Gaskets, Bolts, and Nuts: Type and material recommended by piping system manufacturer unless otherwise indicated.

C. Performance: None

2.4 TRANSITION FITTINGS

A. Physical: None

B. Functional:

1. General Requirements:

    a. Same size as pipes to be joined.

b. Pressure rating at least equal to pipes to be joined.
c. End connections compatible with pipes to be joined.

2. Fitting-Type Transition Couplings: Manufactured piping coupling or specified piping system fitting.

3. Plastic-to-Metal Transition Fittings:

a. Description:

1) CPVC or PVC one-piece fitting with manufacturer's Schedule 80 equivalent dimensions.
2) One end with threaded brass insert and one solvent-cement-socket or threaded end.

4. Plastic-to-Metal Transition Unions:

a. Description:

1) CPVC or PVC four-part union.
2) Brass or stainless-steel threaded end.
3) Solvent-cement-joint or threaded plastic end.
4) Rubber O-ring.
5) Union nut.

C. Performance: None

2.5 DIELECTRIC FITTINGS

A. Physical: None

B. Functional:

1. General Requirements: Assembly of copper alloy and ferrous materials with separating nonconductive insulating material. Include end connections compatible with pipes to be joined.
2. Dielectric Unions:

a. Standard: ASSE 1079.
b. Pressure Rating: 125 psig minimum at 180 deg F or 150 psig or 250 psig.
c. End Connections: Solder-joint copper alloy and threaded ferrous.

3. Dielectric Nipples:

a. Standard: IAPMO PS 66.
b. Electroplated steel nipple complying with ASTM F 1545.
c. Pressure Rating and Temperature: 300 psig at 225 deg F.
d. End Connections: Male threaded or grooved.
e. Lining: Inert and noncorrosive, propylene.

C. Performance: None

PART 3 - EXECUTION

3.1 EARTHWORK

A. Comply with requirements in Section 312000 “Earth Moving” for excavating, trenching, and backfilling.

3.2 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of domestic water piping. Indicated locations and arrangements are used to size pipe and calculate friction loss, expansion, and other design considerations. Install piping as indicated unless deviations to layout are approved on coordination drawings.

B. Install copper tubing under building slab according to CDA’s “Copper Tube Handbook.”

C. Install shutoff valve, hose-end drain valve, strainer, pressure gage, and test tee with valve inside the building at each domestic water-service entrance. Comply with requirements for pressure gages in Section 220519 “Meters and Gages for Plumbing Piping” and with requirements for drain valves and strainers in Section 221119 “Domestic Water Piping Specialties.”

D. Install shutoff valve immediately upstream of each dielectric fitting.

E. Install water-pressure-reducing valves downstream from shutoff valves. Comply with requirements for pressure-reducing valves in Section 221119 “Domestic Water Piping Specialties.”

F. Install domestic water piping level with 0.25 percent slope downward toward drain or without pitch and plumb.

G. Rough-in domestic water piping for water-meter installation according to utility company’s requirements.

H. Install seismic restraints on piping. Comply with requirements for seismic-restraint devices in Section 220548 “Vibration and Seismic Controls for Plumbing Piping and Equipment.”

I. Install piping concealed from view and protected from physical contact by building occupants unless otherwise indicated and except in equipment rooms and service areas.

J. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

K. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal, and coordinate with other services occupying that space.

L. Install piping to permit valve servicing.

M. Install nipples, unions, special fittings, and valves with pressure ratings the same as or higher than the system pressure rating used in applications below unless otherwise indicated.

N. Install piping free of sags and bends.

O. Install fittings for changes in direction and branch connections.

P. Install unions in copper tubing at final connection to each piece of equipment, machine, and specialty.

Q. Install pressure gages on suction and discharge piping for each plumbing pump and packaged booster pump. Comply with requirements for pressure gages in Section 220519 “Meters and Gages for Plumbing Piping.”

R. Install thermostats in hot-water circulation piping. Comply with requirements for thermostats in Section 221123 “Domestic Water Pumps.”

S. Install thermometers on inlet and outlet piping from each water heater. Comply with requirements for thermometers in Section 220519 “Meters and Gages for Plumbing Piping.”

T. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 220517 “Sleeves and Sleeve Seals for Plumbing Piping.”

U. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 220517 “Sleeves and Sleeve Seals for Plumbing Piping.”

3.3 JOINT CONSTRUCTION

A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.

B. Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.

C. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:

1. Apply appropriate tape or thread compound to external pipe threads.
2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.

D. Brazed Joints for Copper Tubing: Comply with CDA’s “Copper Tube Handbook,” “Brazed Joints” chapter.

E. Soldered Joints for Copper Tubing: Apply ASTM B 813, water-flushable flux to end of tube. Join copper tube and fittings according to ASTM B 828 or CDA’s “Copper Tube Handbook.”

F. Flanged Joints: Select appropriate asbestos-free, nonmetallic gasket material in size, type, and thickness suitable for domestic water service. Join flanges with gasket and bolts according to ASME B31.9.

G. Joints for Dissimilar-Material Piping: Make joints using adapters compatible with materials of both piping systems.

3.4 TRANSITION FITTING INSTALLATION

A. Install transition couplings at joints of dissimilar piping.

B. Transition Fittings in Underground Domestic Water Piping:

1. Fittings for NPS 1-1/2 and Smaller: Fitting-type coupling.
2. Fittings for NPS 2 and Larger: Sleeve-type coupling.

C. Transition Fittings in Aboveground Domestic Water Piping NPS 2 and Smaller: Plastic-to-metal transition fittings or unions.

3.5 DIELECTRIC FITTING INSTALLATION

A. Install dielectric fittings in piping at connections of dissimilar metal piping and tubing.

B. Dielectric Fittings for NPS 2 and Smaller: Use dielectric couplings or couplings or nipples or nipples or unions.

3.6 HANGER AND SUPPORT INSTALLATION

A. Comply with requirements for seismic-restraint devices in Section 220548 “Vibration and Seismic Controls for Plumbing Piping and Equipment.”

B. Comply with requirements for pipe hanger, support products, and installation in Section 220529 “Hangers and Supports for Plumbing Piping and Equipment.”

1. Vertical Piping: MSS Type 8 or 42, clamps.
2. Individual, Straight, Horizontal Piping Runs:

    a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.
    b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.
    c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.

3. Multiple, Straight, Horizontal Piping Runs 100 Feet or Longer: MSS Type 44, pipe rolls. Support pipe rolls on trapeze.
4. Base of Vertical Piping: MSS Type 52, spring hangers.

C. Support vertical piping and tubing at base and at each floor.

D. Rod diameter may be reduced one size for double-rod hangers, to a minimum of 3/8 inch.

E. Install hangers for copper tubing with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 3/4 and Smaller: 60 inches with 3/8-inch rod.
2. NPS 1 and NPS 1-1/4: 72 inches with 3/8-inch rod.
3. NPS 1-1/2 and NPS 2: 96 inches with 3/8-inch rod.
4. NPS 2-1/2: 108 inches with 1/2-inch rod.

F. Install supports for vertical copper tubing every 10 feet.

G. Install hangers for steel piping with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/4 and Smaller: 84 inches with 3/8-inch rod.
2. NPS 1-1/2: 108 inches with 3/8-inch rod.
3. NPS 2: 10 feet with 3/8-inch rod.
4. NPS 2-1/2: 11 feet with 1/2-inch rod.

H. Install supports for vertical steel piping every 15 feet.

I. Support piping and tubing not listed in this article according to MSS SP-58 and manufacturer's written instructions.

3.7 CONNECTIONS

A. Drawings indicate general arrangement of piping, fittings, and specialties.

B. When installing piping adjacent to equipment and machines, allow space for service and maintenance.

C. Connect domestic water piping to exterior water-service piping. Use transition fitting to join dissimilar piping materials.

D. Connect domestic water piping to water-service piping with shutoff valve; extend and connect to the following:

1. Water Heaters: Cold-water inlet and hot-water outlet piping in sizes indicated, but not smaller than sizes of water heater connections.
2. Plumbing Fixtures: Cold- and hot-water-supply piping in sizes indicated, but not smaller than that required by plumbing code.
3. Equipment: Cold- and hot-water-supply piping as indicated, but not smaller than equipment connections. Provide shutoff valve and union for each connection. Use flanges instead of unions for NPS 2-1/2 and larger.

3.8 IDENTIFICATION

A. Identify system components. Comply with requirements for identification materials and installation in Section 220553 "Identification for Plumbing Piping and Equipment."

B. Label pressure piping with system operating pressure.

3.9 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Piping Inspections:

a. Do not enclose, cover, or put piping into operation until it has been inspected and approved by authorities having jurisdiction.
b. During installation, notify authorities having jurisdiction at least one day before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction:

1) Roughing-in Inspection: Arrange for inspection of piping before concealing or closing in after roughing in and before setting fixtures.
2) Final Inspection: Arrange for authorities having jurisdiction to observe tests specified in "Piping Tests" Subparagraph below and to ensure compliance with requirements.

c. Reinspection: If authorities having jurisdiction find that piping will not pass tests or inspections, make required corrections and arrange for reinspection.
d. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.

2. Piping Tests:

a. Fill domestic water piping. Check components to determine that they are not air bound and that piping is full of water.
b. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired. If testing is performed in segments, submit a separate report for each test, complete with diagram of portion of piping tested.
c. Leave new, altered, extended, or replaced domestic water piping uncovered and unconcealed until it has been tested and approved. Expose work that was covered or concealed before it was tested.
d. Cap and subject piping to static water pressure of 50 psig above operating pressure, without exceeding pressure rating of piping system materials. Isolate test source and allow it to stand for four hours. Leaks and loss in test pressure constitute defects that must be repaired.
e. Repair leaks and defects with new materials, and retest piping or portion thereof until satisfactory results are obtained.
f. Prepare reports for tests and for corrective action required.

B. Domestic water piping will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

3.10 ADJUSTING

A. Perform the following adjustments before operation:

1. Close drain valves, hydrants, and hose bibbs.
2. Open shutoff valves to fully open position.
3. Open throttling valves to proper setting.

4. Adjust balancing valves in hot-water-circulation return piping to provide adequate flow.

    a. Manually adjust ball-type balancing valves in hot-water-circulation return piping to provide hot-water flow in each branch.
    b. Adjust calibrated balancing valves to flows indicated.

5. Remove plugs used during testing of piping and for temporary sealing of piping during installation.
6. Remove and clean strainer screens. Close drain valves and replace drain plugs.
7. Remove filter cartridges from housings and verify that cartridges are as specified for application where used and are clean and ready for use.
8. Check plumbing specialties and verify proper settings, adjustments, and operation.

3.11 CLEANING

A. Clean and disinfect potable domestic water piping as follows:

1. Purge new piping and parts of existing piping that have been altered, extended, or repaired before using.
2. Use purging and disinfecting procedures prescribed by authorities having jurisdiction; if methods are not prescribed, use procedures described in either AWWA C651 or AWWA C652 or follow procedures described below:

    a. Flush piping system with clean, potable water until dirty water does not appear at outlets.
    b. Fill and isolate system according to either of the following:

        1) Fill system or part thereof with water/chlorine solution with at least 50 ppm of chlorine. Isolate with valves and allow to stand for 24 hours.
        2) Fill system or part thereof with water/chlorine solution with at least 200 ppm of chlorine. Isolate and allow to stand for three hours.

    c. Flush system with clean, potable water until no chlorine is in water coming from system after the standing time.
    d. Repeat procedures if biological examination shows contamination.
    e. Submit water samples in sterile bottles to authorities having jurisdiction.

B. Prepare and submit reports of purging and disinfecting activities. Include copies of water-sample approvals from authorities having jurisdiction.

C. Clean interior of domestic water piping system. Remove dirt and debris as work progresses.

3.12 PIPING SCHEDULE

A. Transition and special fittings with pressure ratings at least equal to piping rating may be used in applications below unless otherwise indicated.

B. Flanges and unions may be used for aboveground piping joints unless otherwise indicated.

C. Fitting Option: Extruded-tee connections and brazed joints may be used on aboveground copper tubing.

D. Aboveground domestic water piping, NPS 2 and smaller, shall be one of the following:

1. Hard copper tube, ASTM B 88, Type L; cast- or wrought-copper, solder-joint fittings; and brazed or soldered joints.

E. Aboveground domestic water piping, NPS 2-1/2 to NPS 4, shall be one of the following:

1. Hard copper tube, ASTM B 88, Type L; cast- or wrought-copper, solder-joint fittings; and brazed or soldered joints.

END OF SECTION 22 11 16

SECTION 22 11 19 – DOMESTIC WATER PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Vacuum breakers.
2. Backflow preventers.
3. Balancing valves.
4. Temperature-actuated, water mixing valves.
5. Strainers.
6. Wall hydrants.
7. Drain valves.
8. Water-hammer arresters.

B. Related Requirements:

1. Section 220519 “Meters and Gages for Plumbing Piping” for thermometers, pressure gages, and flow meters in domestic water piping.
2. Section 221116 “Domestic Water Piping” for water meters.
3. Section 224713 “Drinking Fountains” for water filters for water coolers.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

1.4 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR PIPING SPECIALTIES

A. Physical: None

B. Functional:

1. Potable-water piping and components shall comply with NSF 61 and NSF 14.

C. Performance: None

2.2 PERFORMANCE REQUIREMENTS

A. Physical: None

B. Functional:

1. Minimum Working Pressure for Domestic Water Piping Specialties: 125 psig unless otherwise indicated.

C. Performance: None

2.3 VACUUM BREAKERS

A. Physical: None

B. Functional:

1. Pipe-Applied, Atmospheric-Type Vacuum Breakers:

a. Standard: ASSE 1001.
b. Size: NPS 1/4 to NPS 3, as required to match connected piping.
c. Body: Bronze.
d. Inlet and Outlet Connections: Threaded.
e. Finish: Chrome plated.

2. Hose-Connection Vacuum Breakers:

a. Standard: ASSE 1011.
b. Body: Bronze, nonremovable, with manual drain.
c. Outlet Connection: Garden-hose threaded complying with ASME B1.20.7.
d. Finish: Chrome or nickel plated.

C. Performance: None

2.4 BACKFLOW PREVENTERS

A. Physical: None

B. Functional:

1. Intermediate Atmospheric-Vent Backflow Preventers :

a. Standard: ASSE 1012.
b. Operation: Continuous-pressure applications.
c. Body: Bronze.

d. End Connections: Union, solder joint.
e. Finish: Chrome plated.

2. Reduced-Pressure-Principle Backflow:

a. Standard: ASSE 1013.
b. Operation: Continuous-pressure applications.
c. Pressure Loss: 12 psig maximum, through middle third of flow range.
d. Body: Bronze for NPS 2 and smaller; cast iron with interior lining that complies with AWWA C550 or that is FDA approved or steel with interior lining that complies with AWWA C550 or that is FDA approved or stainless steel for NPS 2-1/2 and larger.
e. End Connections: Threaded for NPS 2 and smaller; flanged for NPS 2-1/2 and larger.
f. Configuration: Designed for horizontal, straight-through flow.
g. Accessories:

1) Valves NPS 2 and Smaller: Ball type with threaded ends on inlet and outlet.
2) Valves NPS 2-1/2 and Larger: Outside-screw and yoke-gate type with flanged ends on inlet and outlet.
3) Air-Gap Fitting: ASME A112.1.2, matching backflow-preventer connection.

C. Performance: None

2.5 BALANCING VALVES

A. Physical: None

B. Functional:

1. Memory-Stop Balancing Valves

a. Standard: MSS SP-110 for two-piece, copper-alloy ball valves.
b. Pressure Rating: 400-psig minimum CWP.
c. Size: NPS 2 or smaller.
d. Body: Copper alloy.
e. Port: Standard or full port.
f. Ball: Chrome-plated brass.
g. Seats and Seals: Replaceable.
h. End Connections: Solder joint or threaded.
i. Handle: Vinyl-covered steel with memory-setting device.

C. Performance: None

2.6 TEMPERATURE-ACTUATED, WATER MIXING VALVES

A. Physical: None

B. Functional:

1. Water-Temperature Limiting Devices

    a. Standard: ASSE 1017.
    b. Pressure Rating: 125 psig
    c. Type: Thermostatically controlled, water mixing valve.
    d. Material: Bronze body with corrosion-resistant interior components.
    e. Connections: Threaded inlets and outlet.
    f. Accessories: Check stops on hot- and cold-water supplies, and adjustable, temperature-control handle.
    g. Valve Finish: Chrome plated.

2. Primary, Thermostatic, Water Mixing Valves:

    a. Standard: ASSE 1017.
    b. Pressure Rating: 125 psig minimum unless otherwise indicated.
    c. Type: Exposed-mounted , thermostatically controlled, water mixing valve.
    d. Material: Bronze body with corrosion-resistant interior components.
    e. Connections: Threaded inlets and outlet.
    f. Accessories: Manual temperature control, check stops on hot- and cold-water supplies, and adjustable, temperature-control handle.
    g. Valve Finish: Polished, chrome plated.

C. Performance: None

2.7 STRAINERS FOR DOMESTIC WATER PIPING

A. Physical: None

B. Functional:

1. Y-Pattern Strainers:

    a. Pressure Rating: 125 psig minimum unless otherwise indicated.
    b. Body: Bronze for NPS 2 and smaller; cast iron with interior lining that complies with AWWA C550 or that is FDA approved, epoxy coated and for NPS 2-1/2 and larger.
    c. End Connections: Threaded for NPS 2 and smaller; flanged for NPS 2-1/2 and larger.
    d. Screen: Stainless steel with round perforations unless otherwise indicated.

    e. Perforation Size:

        1) Strainers NPS 2 and Smaller: 0.020 inch.
        2) Strainers NPS 2-1/2 to NPS 4: 0.045 inch.

    f. Drain: Factory-installed, hose-end drain valve.

C. Performance: None

2.8 WALL HYDRANTS

A. Physical: None

B. Functional:

1. Standard: ASME A112.21.3M for concealed-outlet, self-draining wall hydrants.
2. Pressure Rating: 125 psig.
3. Operation: Loose key.
4. Casing and Operating Rod: Of length required to match wall thickness. Include wall clamp.
5. Inlet: NPS 3/4.
6. Outlet, Concealed: With integral vacuum breaker and garden-hose thread complying with ASME B1.20.7.
7. Box: Deep, flush mounted with cover.
8. Box and Cover Finish: Polished nickel bronze.
9. Operating Keys(s): One with each wall hydrant.

C. Performance: None

2.9 ICE MAKER BOX

A. Physical: None

B. Functional:

1. Ice Maker Outlet Box with Hammer Arrestor:

a. Standard: ASME A112.18.1
b. Materials:

1) Valve Body: No lead brass
2) Outlet Box: ABS
3) Arrestor Body: Stainless steel
4) Arrestor Piston: GFPP with EPDM o-rings
5) Bracket: Galvanized Steel.

c. Pressure Rating: 150 psig.
d. Temperature Rating: 200F
e. Integral hammer arrestor

C. Performance: None

2.10 DRAIN VALVES

A. Physical: None

B. Functional:

1. Ball-Valve-Type, Hose-End Drain Valves <Insert drawing designation if any>:

a. Standard: MSS SP-110 for standard-port, two-piece ball valves.
b. Pressure Rating: 400-psig minimum CWP.
c. Size: NPS 3/4.
d. Body: Copper alloy.
e. Ball: Chrome-plated brass.
f. Seats and Seals: Replaceable.
g. Handle: Vinyl-covered steel.
h. Inlet: Threaded or solder joint.
i. Outlet: Threaded, short nipple with garden-hose thread complying with ASME B1.20.7 and cap with brass chain.

C. Performance: None

2.11 WATER-HAMMER ARRESTERS

A. Physical: None

B. Functional:

1. Water-Hammer Arresters :

a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

b. Basis-of-Design Product: Subject to compliance with requirements, provide or comparable product by one of the following:

1) AMTROL, Inc.
2) Josam Company.
3) MIFAB, Inc.
4) Precision Plumbing Products, Inc.
5) Sioux Chief Manufacturing Company, Inc.
6) Smith, Jay R. Mfg. Co.; Division of Smith Industries, Inc.
7) Tyler Pipe; Wade Div.
8) Watts Drainage Products.
9) Zurn Industries, LLC; Plumbing Products Group; Specification Drainage Products.

a) Standard: ASSE 1010 or PDI-WH 201.
b) Type: Metal bellows or Copper tube with piston.
c) Size: ASSE 1010, Sizes AA and A through F, or PDI-WH 201, Sizes A through F.

C. Performance: None

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install backflow preventers in each water supply to mechanical equipment and systems and to other equipment and water systems that may be sources of contamination. Comply with authorities having jurisdiction.

1. Locate backflow preventers in same room as connected equipment or system.
2. Install drain for backflow preventers with atmospheric-vent drain connection with air-gap fitting, fixed air-gap fitting, or equivalent positive pipe separation of at least two pipe diameters in drain piping and pipe-to-floor drain. Locate air-gap device attached to or under backflow preventer. Simple air breaks are unacceptable for this application.
3. Do not install bypass piping around backflow preventers.

B. Install water regulators with inlet and outlet shutoff valves and bypass with memory-stop balancing valve. Install pressure gages on inlet and outlet.

C. Install balancing valves in locations where they can easily be adjusted.

D. Install temperature-actuated, water mixing valves with check stops or shutoff valves on inlets and with shutoff valve on outlet.

1. Install cabinet-type units recessed in or surface mounted on wall as specified.

E. Install Y-pattern strainers for water on supply side of each control valve, water pressure-reducing valve, solenoid valve, and pump.

F. Install water-hammer arresters in water piping according to PDI-WH 201.

3.2 CONNECTIONS

A. Comply with requirements for ground equipment in Section 260526 “Grounding and Bonding for Electrical Systems.”

B. Fire-retardant-treated-wood blocking is specified in Section 260519 “Low-Voltage Electrical Power Conductors and Cables” for electrical connections.

3.3 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Test each pressure vacuum breaker, reduced-pressure-principle backflow preventer, double-check, backflow-prevention assembly, and double-check, detector-assembly backflow preventer according to authorities having jurisdiction and the device’s reference standard.

B. Domestic water piping specialties will be considered defective if they do not pass tests and inspections.

C. Prepare test and inspection reports.

3.4 ADJUSTING

A. Set field-adjustable pressure set points of water pressure-reducing valves.

B. Set field-adjustable flow set points of balancing valves.

C. Set field-adjustable temperature set points of temperature-actuated, water mixing valves.

END OF SECTION 22 11 19

SECTION 22 11 23 – DOMESTIC WATER PUMPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Permanently lubricated, booster pumps.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1.3 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

B. 1 year Warranty

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. UL Compliance: Comply with UL 778 for motor-operated water pumps.

PART 2 - PRODUCTS

2.1 PERMANENTLY LUBRICATED, PUMPS

A. Physical: None

B. Functional:

1. Basis-of-Design Product: Subject to compliance with requirements, provide or comparable product by one of the following:

a. Armstrong Pumps Inc.
b. Bell & Gossett Domestic Pump; ITT Corporation.
c. Grundfos Pumps Corp.
d. TACO Incorporated.
e. WILO USA LLC – WILO Canada Inc.

2. Description: Factory-assembled and -tested, in-line, close-coupled, pump.
3. Pump Construction:

    a. Body: Bronze, lead-free with threaded or companion-flange connections.
    b. Impeller: Plastic.
    c. Motor: Single speed, unless otherwise indicated.

C. Performance: Per plumbing schedule in construction drawings.

2.2 MOTORS

A. Physical: None

B. Functional:

1. Comply with NEMA designation, temperature rating, service factor, enclosure type.

    a. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

C. Performance: None

2.3 CONTROLS

A. Physical: None

B. Functional:

1. Thermostats: Electric; adjustable for control of hot-water circulation pump.

    a. Type: Water-immersion temperature sensor, for installation in piping.
    b. Range: 50 to 125 deg F.
    c. Enclosure: NEMA 250.
    d. Operation of Pump: On or off.
    e. Transformer: Provide if required.
    f. Power Requirement: 24 V, ac.
    g. Settings: Start pump at 105 deg F and stop pump at 120 deg F.

2. Timers: Electric, for control of hot-water circulation pump.

    a. Type: Programmable, seven-day clock with manual override on-off switch.
    b. Enclosure: NEMA 250 suitable for wall mounting.
    c. Operation of Pump: On or off.
    d. Transformer: Provide if required.
    e. Power Requirement: 24-V ac.
    f. Programmable Sequence of Operation: Up to two on-off cycles each day for seven days.

C. Performance: None

PART 3 - EXECUTION

3.1 PUMP INSTALLATION

A. Comply with HI 1.4.

B. Install in-line, seal-less centrifugal pumps with shaft horizontal unless otherwise indicated.

C. Install horizontally mounted, in-line, close-coupled centrifugal pumps with shaft horizontal.

D. Install continuous-thread hanger rods and [spring hangers] [spring hangers with vertical-limit stop] of size required to support pump weight.

1. Comply with requirements for vibration isolation devices specified in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment." Fabricate brackets or supports as required.
2. Comply with requirements for hangers and supports specified in Section 220529 "Hangers and Supports for Plumbing Piping and Equipment."

E. Install thermostats in hot-water return piping.

F. Install timers on wall in mechanical room.

3.2 CONNECTIONS

A. Comply with requirements for piping specified in Section 221116 "Domestic Water Piping." Drawings indicate general arrangement of piping, fittings, and specialties.

B. Install piping adjacent to pumps to allow service and maintenance.

C. Connect domestic water piping to pumps. Install suction and discharge piping equal to or greater than size of pump nozzles.

1. Install flexible connectors adjacent to pumps in suction and discharge piping of the following pumps:

a. Horizontally mounted, in-line, close-coupled centrifugal pumps.
b. Comply with requirements for flexible connectors specified in Section 221116 "Domestic Water Piping."

2. Install shutoff valve and strainer on suction side of each pump, and check, shutoff, and throttling valves on discharge side of each pump. Install valves same size as connected piping. Comply with requirements for valves specified in Section 220523 "General-Duty Valves for Plumbing Piping" and comply with requirements for strainers specified in Section 221119 "Domestic Water Piping Specialties."
3. Install pressure gage and snubber at suction of each pump and pressure gage and snubber at discharge of each pump. Install at integral pressure-gage tappings where provided or install pressure-gage connectors in suction and discharge piping around pumps. Comply with requirements for pressure gages and snubbers specified in Section 220519 "Meters and Gages for Plumbing Piping."

D. Connect thermostats and timers to pumps that they control.

3.3 ADJUSTING

A. Adjust domestic water pumps to function smoothly and lubricate as recommended by manufacturer.

B. Adjust initial temperature set points.

C. Set field-adjustable switches and circuit-breaker trip ranges as indicated.

END OF SECTION 22 11 23

SECTION 22 13 16 – SANITARY WASTE AND VENT PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Hub-and-spigot, cast-iron soil pipe and fittings.
2. PVC pipe and fittings.
3. Specialty pipe fittings.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Certificates: For waste and vent piping, accessories, and components, from manufacturer.

B. Field quality-control reports.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Physical: None

B. Functional:

1. Components and installation shall be capable of withstanding the following minimum working pressure unless otherwise indicated:

a. Soil, Waste, and Vent Piping: 10-foot head of water.

2. Seismic Performance: Soil, waste, and vent piping and support and installation shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

C. Performance: None

2.2 PIPING MATERIALS

A. Physical: None

B. Functional:

1. Piping materials shall bear label, stamp, or other markings of specified testing agency.
2. Comply with requirements in “Piping Schedule” Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.

C. Performance: None

2.3 HUBLESS, CAST-IRON SOIL PIPE AND FITTINGS

A. Physical: None

B. Functional:

1. Pipe and Fittings: ASTM A 888, Service class.
2. Couplings: ASTM C1277, 36 gage type 301/304 stainless steel.
3. Gaskets: ASTM C 564, neoprene.
4. Calking Materials: ASTM B 29, pure lead and oakum or hemp fiber.

C. Performance: None

2.4 PVC PIPE AND FITTINGS

A. Physical: None

B. Functional:

1. Comply with NSF 14, “Plastics Piping Systems Components and Related Materials,” for plastic piping components. Include marking with “NSF-dwv” for plastic drain, waste, and vent piping and “NSF-sewer” for plastic sewer piping.
2. Cellular-Core PVC Pipe: ASTM F 891, Schedule 40.
3. PVC Socket Fittings: ASTM D 2665, made to ASTM D 3311, drain, waste, and vent patterns and to fit Schedule 40 pipe.
4. Adhesive Primer: ASTM F 656.
5. Solvent Cement: ASTM D 2564.

C. Performance: None

2.5 SPECIALTY PIPE FITTINGS

A. Physical: None

B. Functional:

1. Transition Couplings:

    a. Fitting-Type Transition Couplings: Manufactured piping coupling or specified piping system fitting.
    b. Unshielded, Nonpressure Transition Couplings:

        1) Standard: ASTM C 1173.
        2) Description: Elastomeric, sleeve-type, reducing or transition pattern. Include shear ring and corrosion-resistant-metal tension band and tightening mechanism on each end.
        3) End Connections: Same size as and compatible with pipes to be joined.
        4) Sleeve Materials:

            a) For Cast-Iron Soil Pipes: ASTM C 564, neoprene.
            b) For Plastic Pipes: ASTM F 477, elastomeric seal or ASTM D 5926, PVC.
            c) For Dissimilar Pipes: ASTM D 5926, PVC or other material compatible with pipe materials being joined.

    c. Shielded, Nonpressure Transition Couplings:

        1) Standard: ASTM C 1460.
        2) Description: Elastomeric or rubber sleeve with full-length, corrosion-resistant outer shield and corrosion-resistant-metal tension band and tightening mechanism on each end.
        3) End Connections: Same size as and compatible with pipes to be joined.

C. Performance: None

PART 3 - EXECUTION

3.1 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems.

1. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations.
2. Install piping as indicated unless deviations to layout are approved on coordination drawings.

B. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.

C. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

D. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.

E. Install piping to permit valve servicing.

F. Install piping at indicated slopes.

G. Install piping free of sags and bends.

H. Install fittings for changes in direction and branch connections.

I. Install piping to allow application of insulation.

J. Install seismic restraints on piping. Comply with requirements for seismic-restraint devices specified in Section 220548 “Vibration and Seismic Controls for Plumbing Piping and Equipment.”

K. Make changes in direction for soil and waste drainage and vent piping using appropriate branches, bends, and long-sweep bends.

1. Sanitary tees and short-sweep 1/4 bends may be used on vertical stacks if change in direction of flow is from horizontal to vertical.
2. Use long-turn, double Y-branch and 1/8-bend fittings if two fixtures are installed back to back or side by side with common drain pipe.

    a. Straight tees, elbows, and crosses may be used on vent lines.

3. Do not change direction of flow more than 90 degrees.
4. Use proper size of standard increasers and reducers if pipes of different sizes are connected.

    a. Reducing size of waste piping in direction of flow is prohibited.

L. Lay buried building waste piping beginning at low point of each system.

1. Install true to grades and alignment indicated, with unbroken continuity of invert. Place hub ends of piping upstream.
2. Install required gaskets according to manufacturer’s written instructions for use of lubricants, cements, and other installation requirements.
3. Maintain swab in piping and pull past each joint as completed.

M. Install soil and waste and vent piping at the following minimum slopes unless otherwise indicated:

1. Building Sanitary Waste: 2 percent downward in direction of flow for piping NPS 3 and smaller; 1 2 percent downward in direction of flow for piping NPS 4 and larger.
2. Horizontal Sanitary Waste Piping: 2 percent downward in direction of flow.
3. Vent Piping: 1 percent down toward vertical fixture vent or toward vent stack.

N. Install cast-iron soil piping according to CISPI’s “Cast Iron Soil Pipe and Fittings Handbook,” Chapter IV, “Installation of Cast Iron Soil Pipe and Fittings.”

O. Install aboveground PVC piping according to ASTM D 2665.

P. Install underground PVC piping according to ASTM D 2321.

Q. Plumbing Specialties:

1. Install backwater valves in sanitary waster gravity-flow piping.

a. Comply with requirements for backwater valves specified in Section 221319 “Sanitary Waste Piping Specialties.”

2. Install cleanouts at grade and extend to where building sanitary drains connect to building sanitary sewers in sanitary waste gravity-flow piping.

a. Comply with requirements for cleanouts specified in Section 221319 “Sanitary Waste Piping Specialties.”

3. Install drains in sanitary waste gravity-flow piping.

a. Comply with requirements for drains specified in Section 221319 “Sanitary Waste Piping Specialties.”

R. Do not enclose, cover, or put piping into operation until it is inspected and approved by authorities having jurisdiction.

S. Install sleeves for piping penetrations of walls, ceilings, and floors.

1. Comply with requirements for sleeves specified in Section 220517 “Sleeves and Sleeve Seals for Plumbing Piping.”

T. Install sleeve seals for piping penetrations of concrete walls and slabs.

1. Comply with requirements for sleeve seals specified in Section 220517 “Sleeves and Sleeve Seals for Plumbing Piping.”

U. Install escutcheons for piping penetrations of walls, ceilings, and floors.

1. Comply with requirements for escutcheons specified in Section 220518 “Escutcheons for Plumbing Piping.”

3.2 JOINT CONSTRUCTION

A. Grooved Joints: Cut groove ends of pipe according to AWWA C606. Lubricate and install gasket over ends of pipes or pipe and fitting. Install coupling housing sections, over gasket, with keys seated in piping grooves. Install and tighten housing bolts.

B. Plastic, Nonpressure-Piping, Solvent-Cement Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:

1. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.
2. PVC Piping: Join according to ASTM D 2855 and ASTM D 2665 appendixes.

3.3 SPECIALTY PIPE FITTING INSTALLATION

A. Transition Couplings:

1. Install transition couplings at joints of piping with small differences in ODs.
2. In Waste Drainage Piping: Unshielded Shielded, nonpressure transition couplings.

3.4 HANGER AND SUPPORT INSTALLATION

A. Comply with requirements for seismic-restraint devices specified in Section 220548 “Vibration and Seismic Controls for Plumbing Piping and Equipment.”

B. Comply with requirements for pipe hanger and support devices and installation specified in Section 220529 “Hangers and Supports for Plumbing Piping and Equipment.” Section 220548 “Vibration Controls for Plumbing Piping and Equipment.”

1. Install carbon-steel pipe hangers for horizontal piping in noncorrosive environments.
2. Install stainless-steel fiberglass pipe hangers for horizontal piping in corrosive environments.
3. Install carbon-steel pipe support clamps for vertical piping in noncorrosive environments.
4. Install stainless-steel pipe support clamps for vertical piping in corrosive environments.
5. Vertical Piping: MSS Type 8 or Type 42, clamps.
6. Install individual, straight, horizontal piping runs:

    a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.
    b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.
    c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.

7. Multiple, Straight, Horizontal Piping Runs 100 Feet or Longer: MSS Type 44, pipe rolls. Support pipe rolls on trapeze.
8. Base of Vertical Piping: MSS Type 52, spring hangers.

C. Support horizontal piping and tubing within 12 inches of each fitting, valve, and coupling.

D. Support vertical piping and tubing at base and at each floor.

E. Rod diameter may be reduced one size for double-rod hangers, with 3/8-inch minimum rods.

F. Install hangers for cast-iron soil piping with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/2 and NPS 2: 60 inches with 3/8-inch rod.
2. NPS 3: 60 inches with 1/2-inch rod.
3. NPS 4 and NPS 5: 60 inches with 5/8-inch rod.

G. Install supports for vertical cast-iron soil piping every 15 feet.

H. Install hangers for PVC piping with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/2 and NPS 2: 48 inches with 3/8-inch rod.

2. NPS 3: 48 inches with 1/2-inch rod.
3. NPS 4 and NPS 5: 48 inches with 5/8-inch rod.

I. Install supports for vertical PVC piping every 48 inches.

J. Support piping and tubing not listed above according to MSS SP-58 and manufacturer's written instructions.

3.5 CONNECTIONS

A. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Connect soil and waste piping to exterior sanitary sewerage piping. Use transition fitting to join dissimilar piping materials.

C. Connect waste and vent piping to the following:

1. Plumbing Fixtures: Connect waste piping in sizes indicated, but not smaller than required by plumbing code.
2. Plumbing Fixtures and Equipment: Connect atmospheric vent piping in sizes indicated, but not smaller than required by authorities having jurisdiction.
3. Plumbing Specialties: Connect waste and vent piping in sizes indicated, but not smaller than required by plumbing code.
4. Install test tees (wall cleanouts) in conductors near floor and floor cleanouts with cover flush with floor.
5. Install horizontal backwater valves with cleanout cover flush with floor in pit with pit cover flush with floor.
6. Comply with requirements for backwater valves cleanouts and drains specified in Section 221319 "Sanitary Waste Piping Specialties."
7. Equipment: Connect waste piping as indicated.
    a. Provide shutoff valve if indicated and union for each connection.
    b. Use flanges instead of unions for connections NPS 2-1/2 and larger.

D. Where installing piping adjacent to equipment, allow space for service and maintenance of equipment.

E. Make connections according to the following unless otherwise indicated:

1. Install unions, in piping NPS 2 and smaller, adjacent to each valve and at final connection to each piece of equipment.
2. Install flanges, in piping NPS 2-1/2 and larger, adjacent to flanged valves and at final connection to each piece of equipment.

3.6 IDENTIFICATION

A. Identify exposed sanitary waste and vent piping.

B. Comply with requirements for identification specified in Section 220553 “Identification for Plumbing Piping and Equipment.”

3.7 FIELD QUALITY CONTROL

A. During installation, notify authorities having jurisdiction at least 24 hours before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction.

1. Roughing-in Inspection: Arrange for inspection of piping before concealing or closing-in after roughing-in and before setting fixtures.
2. Final Inspection: Arrange for final inspection by authorities having jurisdiction to observe tests specified below and to ensure compliance with requirements.

B. Reinspection: If authorities having jurisdiction find that piping will not pass test or inspection, make required corrections and arrange for reinspection.

C. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.

D. Test sanitary waste and vent piping according to procedures of authorities having jurisdiction or, in absence of published procedures, as follows:

1. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired.

    a. If testing is performed in segments, submit separate report for each test, complete with diagram of portion of piping tested.

2. Leave uncovered and unconcealed new, altered, extended, or replaced waste and vent piping until it has been tested and approved.

    a. Expose work that was covered or concealed before it was tested.

3. Roughing-in Plumbing Test Procedure: Test waste and vent piping except outside leaders on completion of roughing-in.

    a. Close openings in piping system and fill with water to point of overflow, but not less than 10-foot head of water.
    b. From 15 minutes before inspection starts to completion of inspection, water level must not drop.
    c. Inspect joints for leaks.

4. Finished Plumbing Test Procedure: After plumbing fixtures have been set and traps filled with water, test connections and prove they are gastight and watertight.

    a. Plug vent-stack openings on roof and building drains where they leave building. Introduce air into piping system equal to pressure of 1-inch wg.
    b. Use U-tube or manometer inserted in trap of water closet to measure this pressure.

c. Air pressure must remain constant without introducing additional air throughout period of inspection.
d. Inspect plumbing fixture connections for gas and water leaks.

5. Repair leaks and defects with new materials and retest piping, or portion thereof, until satisfactory results are obtained.
6. Prepare reports for tests and required corrective action.

3.8 CLEANING AND PROTECTION

A. Clean interior of piping. Remove dirt and debris as work progresses.

B. Protect sanitary waste and vent piping during remainder of construction period to avoid clogging with dirt and debris and to prevent damage from traffic and construction work.

C. Place plugs in ends of uncompleted piping at end of day and when work stops.

D. Exposed PVC Piping: Protect plumbing vents exposed to sunlight with two coats of water-based latex paint.

E. Repair damage to adjacent materials caused by waste and vent piping installation.

3.9 PIPING SCHEDULE

A. Flanges and unions may be used on aboveground pressure piping unless otherwise indicated.

B. Aboveground, soil and waste piping NPS 4 and smaller shall be any of the following:

1. Service class, cast-iron soil pipe and fittings; gaskets; and gasketed joints.

C. Aboveground, soil and waste piping NPS 5 and larger shall be any of the following:

1. Service class, cast-iron soil pipe and fittings; gaskets; and gasketed joints.

D. Aboveground, vent piping NPS 4 and smaller shall be any of the following:

1. Service class, cast-iron soil pipe and fittings; gaskets; and gasketed joints.

E. Aboveground, vent piping NPS 5 and larger shall be any of the following:

1. Service class, cast-iron soil pipe and fittings; gaskets; and gasketed joints.

F. Underground, soil, waste, and vent piping NPS 4 and smaller shall be any of the following:

1. Service class, cast-iron soil piping; gaskets; and gasketed joints.
2. Solid wall Cellular-core PVC pipe, PVC socket fittings, and solvent-cemented joints.

G. Underground, soil and waste piping NPS 5 and larger shall be any of the following:

1. Service class, cast-iron soil piping; gaskets; and gasketed joints.
2. Solid-wall Cellular-core PVC pipe; PVC socket fittings; and solvent-cemented joints.

END OF SECTION 22 13 16

SECTION 22 13 19 – SANITARY WASTE PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes the following sanitary drainage piping specialties:

1. Cleanouts.
2. Floor drains.
3. Roof flashing assemblies.
4. Miscellaneous sanitary drainage piping specialties.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated. Include rated capacities, operating characteristics, and accessories for grease interceptors.

1.3 QUALITY ASSURANCE

A. Drainage piping specialties shall bear label, stamp, or other markings of specified testing agency.

PART 2 - PRODUCTS

2.1 CLEANOUTS

A. Physical: None

B. Functional:

1. Exposed Cast-Iron Cleanouts:

a. Standard: ASME A112.36.2M for cast iron for cleanout test tee.
b. Size: Same as connected drainage piping
c. Body Material: Hubless, cast-iron soil pipe test tee as required to match connected piping.
d. Closure: Countersunk.
e. Closure Plug Size: Same as or not more than one size smaller than cleanout size.

2. Cast-Iron Floor Cleanouts:

a. Standard: ASME A112.36.2M for threaded, adjustable housing cleanout.
b. Size: Same as connected branch.
c. Type: Threaded, adjustable housing.

d. Body or Ferrule: Cast iron.
e. Clamping Device: Required.
f. Outlet Connection: Spigot.
g. Closure: Brass plug with straight threads and gasket or Cast-iron plug.
h. Adjustable Housing Material: Cast iron with threads.
i. Frame and Cover Material and Finish: Nickel-bronze, copper alloy.
j. Frame and Cover Shape: Round.
k. Top Loading Classification: Light Duty.
l. Riser: ASTM A 74, Service class, cast-iron drainage pipe fitting and riser to cleanout.

3. Cast-Iron Wall Cleanouts:

a. Standard: ASME A112.36.2M. Include wall access.
b. Size: Same as connected drainage piping.
c. Body: Hubless, cast-iron soil pipe test tee as required to match connected piping.
d. Closure: Countersunk plug.
e. Closure Plug Size: Same as or not more than one size smaller than cleanout size.
f. Wall Access: Round, flat, chrome-plated brass or stainless-steel cover plate with screw.
g. Wall Access: Round, nickel-bronze, copper-alloy, or stainless-steel wall-installation frame and cover.

C. Performance: None

2.2 FLOOR DRAINS

A. Physical: None

B. Functional:

1. Cast-Iron Floor Drains:

a. Standard: ASME A112.6.3.
b. Body Material: Gray iron.
c. Clamping Device: Required.
d. Outlet: Bottom.
e. Backwater Valve: Not required.
f. Top or Strainer Material: Nickel bronze.
g. Top Shape: Round.
h. Top Loading Classification: Light Duty.

C. Performance: None

2.3 MISCELLANEOUS SANITARY DRAINAGE PIPING SPECIALTIES

A. Physical: None

B. Functional:

1. Deep-Seal Traps:

a. Description: Cast-iron or bronze casting, with inlet and outlet matching connected piping and cleanout trap-seal primer valve connection.
b. Size: Same as connected waste piping.

1) NPS 2: 4-inch- minimum water seal.
2) NPS 2-1/2 and Larger: 5-inch- minimum water seal.

2. Air-Gap Fittings:

a. Standard: ASME A112.1.2, for fitting designed to ensure fixed, positive air gap between installed inlet and outlet piping.
b. Body: Bronze or cast iron.
c. Inlet: Opening in top of body.
d. Outlet: Larger than inlet.
e. Size: Same as connected waste piping and with inlet large enough for associated indirect waste piping.

3. Sleeve Flashing Device:

a. Description: Manufactured, cast-iron fitting, with clamping device, that forms sleeve for pipe floor penetrations of floor membrane. Include galvanized-steel pipe extension in top of fitting that will extend 1 inch above finished floor and galvanized-steel pipe extension in bottom of fitting that will extend through floor slab.
b. Size: As required for close fit to riser or stack piping.

4. Vent Caps:

a. Description: Cast-iron body with threaded or hub inlet and vandal-proof design. Include vented hood and setscrews to secure to vent pipe.
b. Size: Same as connected stack vent or vent stack.

C. Performance: None

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install cleanouts in aboveground piping and building drain piping according to the following, unless otherwise indicated:

1. Size same as drainage piping up to NPS 4. Use NPS 4 for larger drainage piping unless larger cleanout is indicated.
2. Locate at each change in direction of piping greater than 45 degrees.
3. Locate at minimum intervals of 50 feet for piping NPS 4 and smaller and 100 feet for larger piping.
4. Locate at base of each vertical soil and waste stack.

B. For floor cleanouts for piping below floors, install cleanout deck plates with top flush with finished floor.

C. For cleanouts located in concealed piping, install cleanout wall access covers, of types indicated, with frame and cover flush with finished wall.

D. Install floor drains at low points of surface areas to be drained. Set grates of drains flush with finished floor, unless otherwise indicated.

1. Position floor drains for easy access and maintenance.
2. Set floor drains below elevation of surrounding finished floor to allow floor drainage. Set with grates depressed according to the following drainage area radii:

    a. Radius, 30 Inches or Less: Equivalent to 1 percent slope, but not less than 1/4-inch total depression.
    b. Radius, 30 to 60 Inches: Equivalent to 1 percent slope.
    c. Radius, 60 Inches or Larger: Equivalent to 1 percent slope, but not greater than 1-inch total depression.

3. Install floor-drain flashing collar or flange so no leakage occurs between drain and adjoining flooring. Maintain integrity of waterproof membranes where penetrated.
4. Install individual traps for floor drains connected to sanitary building drain, unless otherwise indicated.

E. Install roof flashing assemblies on sanitary stack vents and vent stacks that extend through roof.

F. Install flashing fittings on sanitary stack vents and vent stacks that extend through roof.

G. Assemble open drain fittings and install with top of hub 1 inch above floor.

H. Install deep-seal traps on floor drains and other waste outlets, if indicated.

I. Install air-gap fittings on draining-type backflow preventers and on indirect-waste piping discharge into sanitary drainage system.

J. Install sleeve flashing device with each riser and stack passing through floors with waterproof membrane.

K. Install vent caps on each vent pipe passing through roof.

L. Install traps on plumbing specialty drain outlets. Omit traps on indirect wastes unless trap is indicated.

3.2 CONNECTIONS

A. Comply with requirements in Section 221316 “Sanitary Waste and Vent Piping” for piping installation requirements. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Install piping adjacent to equipment to allow service and maintenance.

C. Grease Interceptors: Connect inlet and outlet to unit and connect flow-control fitting and vent to unit inlet piping. Install valve on outlet of automatic drawoff-type unit.

3.3 FLASHING INSTALLATION

A. Fabricate flashing from single piece unless large pans, sumps, or other drainage shapes are required. Join flashing according to the following if required:

1. Lead Sheets: Burn joints of lead sheets 6.0-lb/sq. ft., 0.0938-inch thickness or thicker. Solder joints of lead sheets 4.0-lb/sq. ft., 0.0625-inch thickness or thinner.

B. Install sheet flashing on pipes, sleeves, and specialties passing through or embedded in floors and roofs with waterproof membrane.

1. Pipe Flashing: Sleeve type, matching pipe size, with minimum length of 10 inches, and skirt or flange extending at least 8 inches around pipe.
2. Sleeve Flashing: Flat sheet, with skirt or flange extending at least 8 inches around sleeve.
3. Embedded Specialty Flashing: Flat sheet, with skirt or flange extending at least 8 inches around specialty.

C. Set flashing on floors and roofs in solid coating of bituminous cement.

D. Secure flashing into sleeve and specialty clamping ring or device.

E. Install flashing for piping passing through roofs with counterflashing or commercially made flashing fittings, according to Section 076200 “Sheet Metal Flashing and Trim.”

F. Extend flashing up vent pipe passing through roofs and turn down into pipe, or secure flashing into cast-iron sleeve having calking recess.

3.4 LABELING AND IDENTIFYING

A. Equipment Nameplates and Signs: Install engraved plastic-laminate equipment nameplate or sign on or near each grease interceptor.

B. Distinguish among multiple units, inform operator of operational requirements, indicate safety and emergency precautions, and warn of hazards and improper operations, in addition to identifying unit. Nameplates and signs are specified in Section 220553 “Identification for Plumbing Piping and Equipment.”

3.5 PROTECTION

A. Protect drains during remainder of construction period to avoid clogging with dirt or debris and to prevent damage from traffic or construction work.

B. Place plugs in ends of uncompleted piping at end of each day or when work stops.

END OF SECTION 22 13 19

SECTION 22 33 00 – ELECTRIC, DOMESTIC-WATER HEATERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Commercial, electric, domestic-water heaters.

1.2 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Commercial domestic-water heaters shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term “withstand” means “the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified.”

1.3 ACTION SUBMITTALS

A. Product Data: For each type and size of domestic-water heater indicated.

1.4 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Certificates: For commercial domestic-water heaters, accessories, and components, from manufacturer.

B. Domestic-Water Heater Labeling: Certified and labeled by testing agency acceptable to authorities having jurisdiction.

C. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

B. Warranty: Sample of special warranty.

1.6 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. ASHRAE/IESNA Compliance: Applicable requirements in ASHRAE/IESNA 90.1.

C. ASME Compliance: Where ASME-code construction is indicated, fabricate and label commercial, domestic-water heater storage tanks to comply with ASME Boiler and Pressure Vessel Code: Section VIII, Division 1.

D. NSF Compliance: Fabricate and label equipment components that will be in contact with potable water to comply with NSF 61, "Drinking Water System Components – Health Effects."

1.7 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of electric, domestic-water heaters that fail in materials or workmanship within specified warranty period.

1. Warranty Periods: From date of Substantial Completion.

a. Commercial, Electric, Domestic-Water Heaters:

1) Tank: Five years.
2) Controls and Other Components: One year.

PART 2 - PRODUCTS

A. Physical:

1. Storage-Tank Construction: ASME-code, steel.

a. Interior Finish: Comply with NSF 61 and NSF 372 barrier materials for potable-water tank linings, including extending lining material into tappings.

B. Functional:

1. Factory-Installed, Storage-Tank Appurtenances:

a. Anode Rod: Replaceable magnesium.
b. Drain Valve: Corrosion-resistant metal with hose-end connection.
c. Insulation: Comply with ASHRAE/IES 90.1.
d. Jacket: Steel with enameled finish or high-impact composite material.
e. Heating Elements: Electric, screw-in or bolt-on immersion type arranged in multiples of three.
f. Temperature Control: Adjustable thermostat.
g. Safety Controls: High-temperature-limit and low-water cutoff devices or systems.
h. Relief Valves: ASME rated and stamped for combination temperature-and-pressure relief valves. Include one or more relief valves with total relieving capacity at least as great as heat input, and include pressure setting less than working-pressure rating of domestic-water heater. Select one relief valve with sensing element that extends into storage tank.

C. Performance: Per plumbing schedule within construction documents.

2.2 DOMESTIC-WATER HEATER ACCESSORIES

A. Domestic-Water Expansion Tanks:

1. Physical:

a. Steel pressure-rated tank constructed with welded joints and factory-installed, butyl-rubber diaphragm. Include air precharge to minimum system-operating pressure at tank.

b. Construction:

1) Tappings: Factory-fabricated steel, welded to tank before testing and labeling. Include ASME B1.20.1 pipe thread.

2) Interior Finish: Comply with NSF 61 and NSF 372 barrier materials for potable-water tank linings, including extending finish into and through tank fittings and outlets.

3) Air-Charging Valve: Factory installed.

c. Drain Pans: Corrosion-resistant metal with raised edge. Include dimensions not less than base of domestic-water heater, and include drain outlet not less than 1-1/4″ with ASME B1.20.1 pipe threads.

2. Functional:

a. Piping-Type Heat Traps: Field-fabricated piping arrangement in accordance with ASHRAE/IES 90.1 or ASHRAE 90.2.

b. Combination Temperature-and-Pressure Relief Valves: ASME rated and stamped. Include relieving capacity at least as great as heat input, and include pressure setting less than working-pressure rating of domestic-water heater. Select relief valves with sensing element that extends into storage tank.

c. Vacuum Relief Valves: ANSI Z21.22/CSA 4.4.

3. Performance: Per plumbing schedule within construction documents.

2.3 SOURCE QUALITY CONTROL

A. Factory Tests: Test and inspect domestic-water heaters specified to be ASME-code construction, in accordance with ASME Boiler and Pressure Vessel Code.

B. Hydrostatically test **commercial** domestic-water heaters to minimum of one and one-half times pressure rating before shipment.

C. Electric, domestic-water heaters will be considered defective if they do not pass tests and inspections.

D. Prepare test and inspection reports.

PART 3 - EXECUTION

3.1 DOMESTIC-WATER HEATER INSTALLATION

A. Commercial, Electric, Domestic-Water Heater Mounting: Install commercial, electric, domestic-water heaters in location indicated on plans and per manufacturer requirements.

1. Maintain manufacturer's recommended clearances.
2. Arrange units so controls and devices that require servicing are accessible.
3. Install anchor bolts to elevations required for proper attachment to supported equipment.

B. Install electric, domestic-water heaters level and plumb, according to layout drawings, original design, and referenced standards. Maintain manufacturer's recommended clearances. Arrange units so controls and devices needing service are accessible.

1. Install shutoff valves on domestic-water-supply piping to domestic-water heaters and on domestic-hot-water outlet piping. Comply with requirements for shutoff valves specified in Section 220523 "General-Duty Valves for Plumbing Piping."

C. Install commercial, electric, domestic-water heaters with seismic-restraint devices. Comply with requirements for seismic-restraint devices specified in Section 220548 "Vibration and Seismic Controls for Plumbing Piping and Equipment."

D. Install thermometers on outlet piping of electric, domestic-water heaters. Comply with requirements for thermometers specified in Section 220519 "Meters and Gages for Plumbing Piping."

3.2 CONNECTIONS

A. Comply with requirements for piping specified in Section 221116 "Domestic Water Piping." Drawings indicate general arrangement of piping, fittings, and specialties.

B. Where installing piping adjacent to electric, domestic-water heaters, allow space for service and maintenance of water heaters. Arrange piping for easy removal of domestic-water heaters.

3.3 IDENTIFICATION

A. Identify system components. Comply with requirements for identification specified in Section 220553 "Identification for Plumbing Piping and Equipment."

3.4 FIELD QUALITY CONTROL

A. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

2. Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.
3. Operational Test: After electrical circuitry has been energized, start units to confirm proper operation.
4. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

B. Electric, domestic-water heaters will be considered defective if they do not pass tests and inspections. Comply with requirements in Section 014000 “Quality Requirements” for retesting and reinspecting requirements and Section 017300 “Execution” for requirements for correcting the Work.

C. Prepare test and inspection reports.

END OF SECTION 22 33 00

SECTION 22 42 13.13 – COMMERCIAL WATER CLOSETS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Water closets.
2. Flushometer valves.
3. Toilet seats.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For flushometer valves and electronic sensors to include in operation and maintenance manuals.

B. Manufacturer warranty.

PART 2 - PRODUCTS

2.1 WALL-MOUNTED WATER CLOSETS

A. Water Closets: Wall mounted, top spud, accessible.

1. Physical: None
2. Functional:

a. Bowl:

1) Standards: ASME A112.19.2/CSA B45.1 and ASME A112.19.5.
2) Material: Vitreous china.
3) Type: Siphon jet.
4) Style: Flushometer valve.
5) Height: Handicapped/elderly, complying with ICC/ANSI A117.1..
6) Rim Contour: Elongated.
7) Water Consumption: 1.28 gal. per flush.
8) Spud Size and Location: NPS 1-1/2; top.

b. Toilet Seat:

1) Compatible with the water closet.

c. Support:

1) Standard: ASME A112.6.1M.
2) Description: Waste-fitting assembly, as required to match drainage piping material and arrangement with faceplates, couplings gaskets, and feet; bolts and hardware matching fixture.
3) Water-Closet Mounting Height: Handicapped/elderly according to ICC/ANSI A117.1.

3. Performance: None

2.2 FLUSHOMETER VALVES

A. Physical: None

B. Functional:

1. Exposed Sensor Flushometer:

a. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on drawings or comparable product.
b. Standard: ASSE 1037.
c. Minimum Pressure Rating: 125 psig.
d. Features: Include integral check stop and backflow-prevention device.
e. Material: Brass body with corrosion-resistant components.
f. Exposed Flushometer-Valve Finish: Chrome plated.
g. Panel Finish: Chrome plated or stainless steel.
h. Style: Exposed.
i. Consumption: 1.28 gal. per flush.
j. Minimum Inlet: NPS 1.
k. Minimum Outlet: NPS 1-1/4.

C. Performance: None

PART 3 - EXECUTION

3.1 INSTALLATION

A. Water-Closet Installation:

1. Install level and plumb according to roughing-in drawings.
2. Install floor-mounted water closets on bowl-to-drain connecting fitting attachments to piping or building substrate.

3. Install accessible, wall-mounted water closets at mounting height for handicapped/elderly, according to ICC/ANSI A117.1.

B. Support Installation:

1. Install supports, affixed to building substrate, for floor-mounted, back-outlet water closets.
2. Use carrier supports with waste-fitting assembly and seal.
3. Install wall-mounted, back-outlet water-closet supports with waste-fitting assembly and waste-fitting seals; and affix to building substrate.

C. Flushometer-Valve Installation:

1. Install flushometer-valve, water-supply fitting on each supply to each water closet.
2. Attach supply piping to supports or substrate within pipe spaces behind fixtures.
3. Install lever-handle flushometer valves for accessible water closets with handle mounted on open side of water closet.
4. Install actuators in locations that are easy for people with disabilities to reach.

D. Install toilet seats on water closets.

E. Wall Flange and Escutcheon Installation:

1. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations and within cabinets and millwork.
2. Install deep-pattern escutcheons if required to conceal protruding fittings.
3. Comply with escutcheon requirements specified in Section 220518 “Escutcheons for Plumbing Piping.”

F. Joint Sealing:

1. Seal joints between water closets and walls and floors using sanitary-type, one-part, mildew-resistant silicone sealant.
2. Match sealant color to water-closet color.
3. Comply with sealant requirements specified in Section 079200 “Joint Sealants.”

3.2 CONNECTIONS

A. Connect water closets with water supplies and soil, waste, and vent piping. Use size fittings required to match water closets.

B. Comply with water piping requirements specified in Section 221116 “Domestic Water Piping.”

C. Comply with soil and waste piping requirements specified in Section 221316 “Sanitary Waste and Vent Piping.”

D. Where installing piping adjacent to water closets, allow space for service and maintenance.

3.3 ADJUSTING

A. Operate and adjust water closets and controls. Replace damaged and malfunctioning water closets, fittings, and controls.

B. Adjust water pressure at flushometer valves to produce proper flow.

3.4 CLEANING AND PROTECTION

A. Clean water closets and fittings with manufacturers' recommended cleaning methods and materials.

B. Install protective covering for installed water closets and fittings.

C. Do not allow use of water closets for temporary facilities unless approved in writing by Contracting Officer.

END OF SECTION 22 42 13.13

SECTION 22 42 13.16 – COMMERCIAL URINALS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Wall-hung urinals.
2. Urinal flushometer valves.
3. Supports.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes for urinals.
2. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For flushometer valves and electronic sensors to include in operation and maintenance manuals.

1.5 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that are packaged with protective covering for storage and identified with labels describing contents.

1. Flushometer-Valve Repair Kits: Equal to 10 percent of amount of each type installed, but no fewer than one of each type.

PART 2 - PRODUCTS

2.1 WALL-HUNG URINALS

A. Physical: None

B. Functional:

1. Urinals – Wall Hung, Bottom Outlet, Washout:

a. Fixture:

1) Standards: ASME A112.19.2/CSA B45.1 and ASME A112.19.5/CSA B45.15.
2) Material: Vitreous china.
3) Drain: Separate removable chrome-plated dome strainer with chrome-plated, NPS 1-1/2 tailpiece.
4) Strainer or Trapway: Manufacturer's standard strainer and NPS 1-1/2 tailpiece.
5) Inlet Spud Size and Location: NPS 3/4; top.
6) Outlet Size and Location: NPS 1-1/2; bottom.
7) Color: White.
8) Water Consumption: 0.125 gpf.

b. Waste Fitting:

1) Standard: ASME A112.18.2/CSA B125.2.
2) Trap:
3) Size: NPS 1-1/2.
4) Material, Chrome Plated: Two-piece, cast-brass trap and swivel elbow with 0.032-inch- thick brass tube to wall and chrome-plated brass or steel wall flange.
5) Material, Stainless Steel: Two-piece trap and swivel elbow with 0.012-inch-thick, stainless steel tube to wall; and stainless steel wall flange.

c. Support: Type II urinal carrier with hanger and bearing plates. Include rectangular, steel uprights.
d. Urinal Mounting Height: Handicapped/elderly according to ICC A117.1.

C. Performance: None

2.2 URINAL FLUSHOMETER VALVES

A. Physical: None

B. Functional:

1. Battery-Powered, Solenoid-Actuator, Piston Flushometer Valves:

a. Standard: ASSE 1037/ASME 112.1037/CSA B125.37.

b. Minimum Pressure Rating: 125 psig.
c. Features: Include integral check stop and backflow-prevention device.
d. Material: Brass body with corrosion-resistant components.
e. Exposed Flushometer-Valve Finish: Chrome plated.
f. Panel Finish: Chrome plated or stainless steel.
g. Style: Exposed.
h. Actuator: Solenoid complying with UL 1951; listed and labeled as defined in NFPA 70, by a qualified testing agency; and marked for intended location and application.
i. Trip Mechanism: Battery-powered electronic sensor complying with UL 1951; listed and labeled as defined in NFPA 70, by a qualified testing agency; and marked for intended location and application.
j. Consumption: 0.5 gal. per flush.
k. Minimum Inlet: NPS 1
l. Minimum Outlet: NPS 1-1/4

C. Performance: None

2.3 SUPPORTS

A. Physical: None

B. Functional:

1. Type II Urinal Carrier:

a. Standard: ASME A112.6.1M.

2. Type I Sink Carrier:

a. Standard: ASME A112.6.1M.

C. Performance: None

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine roughing-in of water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before urinal installation.

B. Examine walls and floors for suitable conditions where urinals will be installed.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Urinal Installation:

1. Install urinals level and plumb according to rough-in drawings.
2. Install wall-hung, back-outlet urinals onto waste fitting seals and attached to supports.
3. Install wall-hung, bottom-outlet urinals with tubular waste piping attached to supports.
4. Install accessible, wall-mounted urinals at mounting height for the handicapped/elderly, according to ICC A117.1.
5. Install trap-seal liquid in waterless urinals.

B. Support Installation:

1. Install supports, affixed to building substrate, for wall-hung urinals.
2. Use off-floor carriers with waste fitting and seal for back-outlet urinals.
3. Use carriers without waste fitting for urinals with tubular waste piping.
4. Use chair-type carrier supports with rectangular steel uprights for accessible urinals.

C. Flushometer-Valve Installation:

1. Install flushometer-valve water-supply fitting on each supply to each urinal.
2. Attach supply piping to supports or substrate within pipe spaces behind fixtures.
3. Install lever-handle flushometer valves for accessible urinals with handle mounted on open side of compartment.
4. Install fresh batteries in battery-powered, electronic-sensor mechanisms.

D. Wall Flange and Escutcheon Installation:

1. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations.
2. Install deep-pattern escutcheons if required to conceal protruding fittings.
3. Comply with escutcheon requirements specified in Section 220518 “Escutcheons for Plumbing Piping.”

E. Joint Sealing:

1. Seal joints between urinals and walls and floors using sanitary-type, one-part, mildew-resistant silicone sealant.
2. Match sealant color to urinal color.
3. Comply with sealant requirements specified in Section 079200 “Joint Sealants.”

3.3 PIPING CONNECTIONS

A. Connect urinals with water supplies and soil, waste, and vent piping. Use size fittings required to match urinals.

B. Comply with water piping requirements specified in Section 221116 “Domestic Water Piping.”

C. Comply with soil and waste piping requirements specified in Section 221316 “Sanitary Waste and Vent Piping.”

D. Where installing piping adjacent to urinals, allow space for service and maintenance.

3.4 ADJUSTING

A. Operate and adjust urinals and controls. Replace damaged and malfunctioning urinals, fittings, and controls.

B. Adjust water pressure at flushometer valves to produce proper flow.

C. Install fresh batteries in battery-powered, electronic-sensor mechanisms.

3.5 CLEANING AND PROTECTION

A. Clean urinals and fittings with manufacturers' recommended cleaning methods and materials.

B. Install protective covering for installed urinals and fittings.

C. Do not allow use of urinals for temporary facilities unless approved in writing by Owner.

END OF SECTION 22 42 13.16

SECTION 22 42 16.13 – COMMERCIAL LAVATORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Lavatories.
2. Faucets.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

1.4 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For lavatories and faucets to include in operation and maintenance manuals.

B. Manufacturer warranty.

PART 2 - PRODUCTS

2.1 VITREOUS-CHINA, WALL-MOUNTED LAVATORIES

A. Physical: None

B. Functional:

1. Copy “Lavatory” Paragraph below and re-edit for each type of vitreous-china, wall-mounted lavatory with back required.
2. Insert number to complete drawing designation. Use these designations on Drawings to identify each lavatory.
3. Lavatory: Vitreous china, wall mounted, with back.

a. Retain “Manufacturers” Subparagraph and list of manufacturers below to require products from manufacturers listed or a comparable product from other manufacturers.
b. Retain “Basis-of-Design Product” Subparagraph and list of manufacturers below to identify a specific product or a comparable product from manufacturers listed.

Retain option and delete insert note if manufacturer's name and model number are indicated on Drawings.

c. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings or comparable product:

d. Fixture:

1) Standard: ASME A112.19.2/CSA B45.1.
2) Type: For wall hanging.
3) Faucet-Hole Punching: One hole.
4) Faucet-Hole Location: Top.
5) Color: White.
6) Mounting Material: Chair carrier.

e. Sensor Faucet:

1) Flow Rate: 0.5 gpm
2) Spray Type: Multi-Laminar
3) Sensor Type: Infrared
4) Mounting Type: Single Hole.
5) Finish: Polished Chrome

f. Support: ASME A112.6.1M, Type II, concealed-arm lavatory carrier.

C. Performance: None

2.2 SUPPLY FITTINGS

A. Physical: None

B. Functional:

1. NSF Standard: Comply with NSF/ANSI 61, "Drinking Water System Components – Health Effects," for supply-fitting materials that will be in contact with potable water.
2. Standard: ASME A112.18.1/CSA B125.1.
3. Supply Piping: Chrome-plated-brass pipe or chrome-plated copper tube matching water-supply piping size. Include chrome-plated-brass or stainless-steel wall flange.
4. Supply Stops: Chrome-plated-brass, one-quarter-turn, ball-type or compression valve with inlet connection matching supply piping.
5. Operation: Loose key.
6. Risers:

a. NPS 3/8.
b. ASME A112.18.6, braided-stainless-steel, flexible hose riser.

C. Performance: None

2.3 WASTE FITTINGS

A. Physical: None

B. Functional:

1. Standard: ASME A112.18.2/CSA B125.2.
2. Drain: Grid type with NPS 1-1/4 offset and straight tailpiece.
3. Trap:

a. Size: NPS 1-1/2 by NPS 1-1/4 or NPS 1-1/4.
b. Material: Stainless-steel, two-piece trap and swivel elbow with 0.012-inch-thick stainless-steel tube to wall; and stainless-steel wall flange.

C. Performance: None

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine roughing-in of water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before lavatory installation.

B. Examine counters and walls for suitable conditions where lavatories will be installed.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install lavatories level and plumb according to roughing-in drawings.

B. Install supports, affixed to building substrate, for wall-mounted lavatories.

C. Install accessible wall-mounted lavatories at handicapped/elderly mounting height for people with disabilities or the elderly, according to ICC/ANSI A117.1.

D. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons if required to conceal protruding fittings. Comply with escutcheon requirements specified in Section 220518 “Escutcheons for Plumbing Piping.”

E. Seal joints between lavatories and counters and walls using sanitary-type, one-part, mildew-resistant silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 079200 “Joint Sealants.”

F. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible lavatories. Comply with requirements in Section 220719 “Plumbing Piping Insulation.”

3.3 CONNECTIONS

A. Connect fixtures with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.

B. Comply with water piping requirements specified in Section 221116 “Domestic Water Piping.”

C. Comply with soil and waste piping requirements specified in Section 221316 “Sanitary Waste and Vent Piping.”

3.4 ADJUSTING

A. Operate and adjust lavatories and controls. Replace damaged and malfunctioning lavatories, fittings, and controls.

B. Adjust water pressure at faucets to produce proper flow.

C. Install fresh batteries in battery-powered, electronic-sensor mechanisms.

3.5 CLEANING AND PROTECTION

A. After completing installation of lavatories, inspect and repair damaged finishes.

B. Clean lavatories, faucets, and other fittings with manufacturers’ recommended cleaning methods and materials.

C. Provide protective covering for installed lavatories and fittings.

D. Do not allow use of lavatories for temporary facilities unless approved in writing by Contracting Officer.

END OF SECTION 22 42 16.13

SECTION 22 42 16.16 – COMMERCIAL SINKS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Sinks.
2. Utility sinks.
3. Sink faucets.
4. Laminar-flow, faucet-spout outlets.
5. Supply fittings.
6. Waste fittings.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. CLOSEOUT SUBMITTALS

1.3 Maintenance data.

PART 2 - PRODUCTS

2.1 SINKS

A. Physical: None

B. Functional:

1. Sinks.

a. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings or comparable product.
b. Fixture:

1) Standard: ASME A112.19.3/CSA B45.2.
2) Type: Drop-in.
3) Nominal Size: Refer to plans.
4) Material: Stainless-Steel, 18 gauge.
5) Rim Guard: Self-rimming top mount.

C. Performance: None

2.2 UTILITY SINKS

A. Physical: None

B. Functional:

1. Mop Service Basin

a. Basis-of-Design Product: Subject to compliance with requirements, provide product indicated on Drawings or comparable product.
b. Fixture:

1) Molded Stone.
2) Number of Compartments: One.
3) Overall Dimensions: 24 x 24″ basin with a minimum of 10″ high walls no less than 1″ wide.
4) Floor mounted.
5) Accessories:
6) Hose and Hose bracket
7) Mop Hanger
8) Stainless Steel Wall guard
9) Pail Hook

C. Performance: None

2.3 SINK FAUCETS

A. Physical: None

B. Functional:

1. NSF Standard: Comply with NSF/ANSI 61, “Drinking Water System Components – Health Effects,” for faucet-spout materials that will be in contact with potable water.
2. Single Handle Pull-Down Faucets:

a. Standard: ASME A112.18.1/CSA B125.1.
b. General: Include hot- and cold-water indicators; coordinate faucet inlets with supplies and fixture hole punchings; coordinate outlet with spout and sink receptor.
c. Body Type: Single hole.
d. Finish: Chrome plated.
e. Maximum Flow Rate: 1.8 gpm.
f. Handle(s): Lever.
g. Mounting Type: Deck, exposed.
h. Spout Type: Rigid, swivel. Spout rotates 360 degrees.
i. Vacuum Breaker: Not required for hose outlet.
j. Spout Outlet: Aerator.

3. Manual Sink Faucets:

a. Standard: ASME A112.18.1/CSA B125.1.

b. General: Include hot- and cold-water indicators; coordinate faucet inlets with supplies and fixture hole punchings; coordinate outlet with spout and sink receptor.
c. Body: 3-3/8″ body, adjustable arms 3″ 8-3/8″ centers
d. Finish: Chrome plated.
e. Maximum Flow Rate: 1.8 gpm.
f. Handle(s): Vandal Proof, 2-3/8″ Lever Handle.
g. Mounting Type: Wall-mounted.
h. Spout Type: Rigid.
i. Vacuum Breaker: Required, Integral.

C. Performance: None

2.4 SUPPLY FITTINGS

A. Physical: None

B. Functional:

1. NSF Standard: Comply with NSF/ANSI 61, "Drinking Water System Components – Health Effects," for supply-fitting materials that will be in contact with potable water.
2. Standard: ASME A112.18.1/CSA B125.1.
3. Supply Piping: Chrome-plated brass pipe or chrome-plated copper tube matching water-supply piping size. Include chrome-plated brass or stainless-steel wall flange.
4. Supply Stops: Chrome-plated brass, one-quarter-turn, ball-type or compression valve with inlet connection matching supply piping.
5. Operation: Loose key.
6. Risers:

    a. NPS 3/8
    b. ASME A112.18.6, braided stainless-steel flexible hose.

C. Performance: None

2.5 WASTE FITTINGS

A. Physical: None

B. Functional:

1. Standard: ASME A112.18.2/CSA B125.2.
2. Drain: Grid type with NPS 1-1/2 offset and straight tailpiece.
3. Trap:

    a. Size: NPS 1-1/2.
    b. Material: Stainless-steel, two-piece trap and swivel elbow with 0.012-inch- thick stainless-steel tube to wall; and stainless-steel wall flange.

C. Performance: None

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine roughing-in of water supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before sink installation.

B. Examine walls, floors, and counters for suitable conditions where sinks will be installed.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install sinks level and plumb according to roughing-in drawings.

B. Install supports, affixed to building substrate, for wall-hung sinks.

C. Install accessible wall-mounted sinks at handicapped/elderly mounting height according to ICC/ANSI A117.1.

D. Set floor-mounted sinks in leveling bed of cement grout.

E. Install water-supply piping with stop on each supply to each sink faucet.

1. Exception: Use ball, gate, or globe valves if supply stops are not specified with sink. Comply with valve requirements specified in Section 220523 “General-Duty Valves for Plumbing Piping.”
2. Install stops in locations where they can be easily reached for operation.

F. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons if required to conceal protruding fittings. Comply with escutcheon requirements specified in Section 220518 “Escutcheons for Plumbing Piping.”

G. Seal joints between sinks and counters, floors, and walls using sanitary-type, one-part, mildew-resistant silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 079200 “Joint Sealants.”

H. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible sinks. Comply with requirements in Section 220719 “Plumbing Piping Insulation.”

3.3 CONNECTIONS

A. Connect sinks with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.

B. Comply with water piping requirements specified in Section 221116 “Domestic Water Piping.”

C. Comply with soil and waste piping requirements specified in Section 221316 “Sanitary Waste and Vent Piping.”

3.4 ADJUSTING

A. Operate and adjust sinks and controls. Replace damaged and malfunctioning sinks, fittings, and controls.

B. Adjust water pressure at faucets to produce proper flow.

3.5 CLEANING AND PROTECTION

A. After completing installation of sinks, inspect and repair damaged finishes.

B. Clean sinks, faucets, and other fittings with manufacturers' recommended cleaning methods and materials.

C. Provide protective covering for installed sinks and fittings.

D. Do not allow use of sinks for temporary facilities unless approved in writing by Contracting Officer.

END OF SECTION 22 42 16.16

SECTION 22 47 13 – DRINKING FOUNTAINS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes drinking fountains and related components.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of drinking fountains.

1.3 CLOSEOUT SUBMITTALS

A. Maintenance Data: For drinking fountains to include in maintenance manuals.

B. Manufacturer Warranty.

PART 2 - PRODUCTS

2.1 DRINKING FOUNTAINS

A. Physical: None

B. Functional:

1. Drinking Fountains: Stainless steel, wall mounted.

a. Standards:

1) Comply with ASME A112.19.3/CSA B45.4.
2) Comply with NSF 61.

b. Receptor Shape: Rectangular.
c. Back Panel: Stainless-steel wall plate behind drinking fountain with integral bottle filler.
d. Bubblers: Two, bi-level, located on deck.
e. Control: Push bar.
f. Drain: Grid type with NPS 1-1/4 tailpiece.
g. Supply Piping: NPS 3/8 with shutoff valve.
h. Drain Piping: ASME A112.18.2/CSA B125.2, NPS 1-1/4 chrome-plated brass P-trap and waste.
i. Support: ASME A112.6.1M, Type III lavatory carrier.

C. Performance: None

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine roughing-in for water-supply and sanitary drainage and vent piping systems to verify actual locations of piping connections before fixture installation.

B. Examine walls and floors for suitable conditions where fixtures will be installed.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install fixtures level and plumb according to roughing-in drawings. For fixtures indicated for children, install at height required by authorities having jurisdiction.

B. Install off-the-floor carrier supports, affixed to building substrate, for wall-mounted fixtures.

C. Install water-supply piping with shutoff valve on supply to each fixture to be connected to domestic-water distribution piping. Use ball, gate, or globe valve. Install valves in locations where they can be easily reached for operation. Valves are specified in Section 220523 “General-Duty Valves for Plumbing Piping.”

D. Install trap and waste piping on drain outlet of each fixture to be connected to sanitary drainage system.

E. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons where required to conceal protruding fittings. Comply with escutcheon requirements specified in Section 220518 “Escutcheons for Plumbing Piping.”

F. Seal joints between fixtures and walls using sanitary-type, one-part, mildew-resistant, silicone sealant. Match sealant color to fixture color. Comply with sealant requirements specified in Section 079200 “Joint Sealants.”

G. Adjust fixture flow regulators for proper flow and stream height.

3.3 CONNECTIONS

A. Connect fixtures with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.

B. Comply with water piping requirements specified in Section 221116 “Domestic Water Piping.”

C. Install ball, gate, or globe shutoff valve on water supply to each fixture. Comply with valve requirements specified in Section 220523 “General-Duty Valves for Plumbing Piping.”

D. Comply with soil and waste piping requirements specified in Section 221316 “Sanitary Waste and Vent Piping.”

3.4 CLEANING

A. After installation, inspect unit. Remove paint splatters and other spots, dirt, and debris. Repair damaged finish to match original finish.

B. Clean fixtures, on completion of installation, according to manufacturer's written instructions.

C. Provide protective covering for installed fixtures.

D. Do not allow use of fixtures for temporary facilities unless approved in writing by Contracting Officer.

END OF SECTION 22 47 13

SECTION 23 05 00 – COMMON WORK RESULTS FOR HVAC

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes:

1. Piping materials and installation instructions common to most piping systems.
2. Equipment nameplate data requirements.
3. Nonshrink grout for equipment installations.
4. Field-fabricated metal and wood equipment supports.
5. Installation requirements common to equipment specification sections.
6. Mechanical demolition.
7. Cutting and patching.

B. The plans and specifications are complimentary and shall be used together in order to fully describe the Work. In the case of a conflict between the plans and specifications, the plans take precedence.

C. The engineer has based the drawings and design on non-certified information furnished by various equipment manufacturers. It is incumbent on the part of the CONTRACTOR to include in the bid all material and labor needed to install the actual equipment furnished.

D. Related Sections:

1. The following is work of Electrical Contractor:

a. Power supply wiring from power source to power connection on equipment. Include starters, disconnects, and required electrical devices, except where specified as furnished, or factory-installed, by manufacturer.

E. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with the Contract Document requirements.

1. Specific quality-assurance and -control requirements for individual construction activities are specified in the Sections that specify those activities. Requirements in those Sections may also cover production of standard products.

2. Specified tests, inspections, and related actions do not limit Contractor's other quality-assurance and -control procedures that facilitate compliance with the Contract Document requirements.
3. Requirements for Contractor to provide quality-assurance and -control services required by Contracting Officer**,** or authorities having jurisdiction are not limited by provisions of this Section.
4. Specific test and inspection requirements are not specified in this Section.

1.3 DEFINITIONS

A. Finished Spaces: Spaces other than mechanical and electrical equipment rooms, furred spaces, pipe chases, unheated spaces immediately below roof, spaces above ceilings, unexcavated spaces, crawlspaces, and tunnels.

B. Exposed, Interior Installations: Exposed to view indoors. Examples include finished occupied spaces and mechanical equipment rooms.

C. Exposed, Exterior Installations: Exposed to view outdoors or subject to outdoor ambient temperatures and weather conditions. Examples include rooftop locations.

D. Concealed, Interior Installations: Concealed from view and protected from physical contact by building occupants. Examples include above ceilings and in chases.

E. Concealed, Exterior Installations: Concealed from view and protected from weather conditions and physical contact by building occupants but subject to outdoor ambient temperatures. Examples include installations within unheated shelters.

1.4 REFERENCES

A. Applicable Standards:

1. American Society for Testing and Materials (ASTM):

a. A47 – Ferritic Malleable Iron Castings.
b. A53 – Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless.
c. B32 – Solder Metal.
d. C1107 – Packaged Dry, Hydraulic-Cement Grout (Nonshrink).
e. D709 – Laminated Thermosetting Materials.

2. American Society of Mechanical Engineers (ASME):

a. Boiler and Pressure Vessel Code.
b. A13.1 – Scheme for the Identification of Piping Systems.
c. B1.20.1 – Pipe Threads, General Purpose (Inch).
d. B16.20 – Ring-Joint Gaskets and Grooves for Steel Pipe Flanges.
e. B16.21 – Nonmetallic Flat Gaskets for Pipe Flanges.
f. B18.2.1 – Square and Hex Bolts and Screws-Inch Series.
g. B31 Series – Code for Pressure Piping.

3. American Welding Society (AWS):

    a. Soldering Manual, latest.
    b. Brazing Manual, latest.
    c. A5.8 – Filler Metals for Brazing.
    d. D1.1 – Structural Welding Code for Steel.
    e. D10.12 – Recommended Practices and Procedures for Welding Low Carbon Steel Pipe.

1.5 SUBMITTALS

A. Submit manufacturer's data sheets on all system components, including the following:

1. Transition fittings
2. Dielectric fittings
3. Mechanical sleeve seals
4. Escutcheons

B. General, all Division 23 sections of the Specifications: Follow the procedures specified in Division 1. Prepare maintenance manuals in accordance with Division 1 sections of the Specifications.

C. Welder certificates signed by Contractor certifying that welders comply with requirements specified under the "Quality Assurance" article of this section.

1.6 QUALITY ASSURANCE

A. Qualify welding processes and operators for structural steel according to AWS D1.1.

B. All welding on pressure piping shall conform with the requirements of the American National Standard Code for Pressure Piping, ANSI B31.3, "Chemical Plant and Petroleum Refinery Piping." All welds on piping having working pressures of 300 psig or greater shall be subjected to a full X-ray examination and will not be accepted until all welds meet the requirements of ANSI B31.1, "Power Piping." Faulty welds shall be removed at no additional cost to the client. X-ray testing shall be performed by others at no additional cost to the Contractor.

1. Certify that each welder has passed AWS qualification tests for the welding processes involved and that certification is current.

C. Electrical Characteristics for Mechanical Equipment: Equipment of higher electrical characteristics may be furnished provided such proposed equipment is approved in writing and connecting electrical services, circuit breakers, and conduit sizes are appropriately modified. If minimum energy ratings or efficiencies are specified, equipment shall comply with requirements.

D. Manufacturer's Technical Representative's Field Reports: Prepare written information documenting manufacturer's technical representative's tests and inspections specified in other Sections. Include the following:

1. Name, address, and telephone number of technical representative making report.
2. Statement on condition of substrates and their acceptability for installation of product.
3. Statement that products at Project site comply with requirements.
4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
6. Statement whether conditions, products, and installation will affect warranty.
7. Other required items indicated in individual Specification Sections.

E. Factory-Authorized Service Representative's Reports: Prepare written information documenting manufacturer's factory-authorized service representative's tests and inspections specified in other Sections. Include the following:

1. Name, address, and telephone number of factory-authorized service representative making report.
2. Statement that equipment complies with requirements.
3. Results of operational and other tests and a statement of whether observed performance complies with requirements.
4. Statement whether conditions, products, and installation will affect warranty.
5. Other required items indicated in individual Specification Sections.

F. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

G. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

H. Installer Qualifications: A firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful in-service performance.

I. Manufacturer's Technical Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to observe and inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

J. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

1.7 DELIVERY, STORAGE, AND PROTECTION

A. Deliver pipes and tubes with factory-applied end caps. Maintain end caps through shipping, storage, and handling to prevent pipe-end damage and prevent entrance of dirt, debris, and moisture.

B. Protect stored pipes ducts and tubes from moisture and dirt. Elevate above grade. When stored inside, do not exceed structural capacity of the floor.

C. Protect flanges, fittings, and ducting/piping specialties from moisture and dirt.

D. Materials and equipment furnished by others.

1. When equipment or materials are indicated to be furnished by others (F.B.O.) to the CONTRACTOR for installation and connection, the CONTRACTOR shall make a complete check of all materials and furnish a receipt to the owner detailing the products received and the condition of the products delivered to him. After executing the receipt and acceptance by the CONTRACTOR, the CONTRACTOR shall assume full responsibility for the safe keeping, handling, and installation of the materials and equipment furnished by others or furnished by government, until completed installation and final approval by the contractor officer.
2. If the CONTRACTOR fails to issue said receipt it shall be assumed that all equipment and materials were then delivered to the CONTRACTOR in the proper quantities and in perfect condition.

1.8 PROJECT SITE CONDITIONS

A. Altitude Ratings: Unless otherwise noted, all specified equipment capacities, air quantities, etc., are for an altitude of 1,000 feet above sea level. Adjustments to manufacturers' ratings must be made accordingly.

1.9 SEQUENCING AND SCHEDULING

A. Coordinate mechanical piping, ducting, and equipment installation with other building components.

B. Arrange for chases, slots, and openings in building structure during progress of construction, to allow for mechanical installations.

C. Coordinate the installation of required supporting devices and set sleeves in poured-in-place concrete and other structural components, as they are constructed.

D. Sequence, coordinate, and integrate installations of mechanical materials and equipment for efficient flow of the work. Coordinate installation of large equipment requiring positioning prior to closing in the building.

E. Interruption of Mechanical Utilities:

1. The Contractor shall not interrupt any main interior or exterior mechanical utility without written request for an outage and a subsequent approval of owner nor shall he interrupt any branch line to an outlet or item of equipment without approval from the owner.
2. Written request for outages shall be submitted no less than two business days in advance of the outage date. This request will delineate the particular utility or service in question, the time the service will be interrupted and the approximate hours the utility shall be off. Shutdowns shall not exceed 3 hours.

3. Unless otherwise noted on the drawings, or directed, any tie-ins or connections to existing utilities or equipment that necessitate interruptions of service shall be performed during approved hours and shall not exceed 3 hours.
4. The work to be performed during the interruption, will be preceded by all possible preparation, and will be carefully coordinated to minimize the duration of the interruption and work will proceed continuously until the system is restored to normal.

F. Coordinate installation of identifying devices after completion of covering and painting, where devices are applied to surfaces. Install identifying devices prior to installation of acoustical ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Grout: Nonshrink, Nonmetallic Grout: ASTM C1107, Grade B.

1. Physical: None
2. Functional:

a. Characteristics: Post-hardening, volume-adjusting, dry, hydraulic-cement grout, nonstaining, noncorrosive, nongaseous, and recommended for interior and exterior applications.
b. Design Mix: 5000 psi, 28-day compressive strength.

3. Performance: None

PART 3 - EXECUTION

3.1 HVAC DEMOLITION

A. Disconnect, demolish, and remove HVAC systems, equipment, and components indicated to be removed.

1. Piping to be removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
2. Equipment to be removed: Disconnect and cap services and remove equipment.
3. Equipment to be removed and reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.

B. If pipe, insulation, or equipment to remain is damaged in appearance or is unserviceable, remove damaged or unserviceable portions and replace with new products of equal capacity and quality.

3.2 ERECTION INSTALLATION APPLICATION

A. Mechanical Systems – Common Requirements:

1. General: Install piping and ducting as described below, except where system sections specify otherwise. Individual piping system specification sections in Division 23 specify piping installation requirements unique to the piping system.
2. General Locations and Arrangements: Drawings (plans, schematics, and diagrams) indicate general location and arrangement of mechanical systems. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install as indicated, except where deviations to layout are approved on coordination drawings.
3. Install piping at indicated slope.
4. Install components having pressure rating equal to or greater than system operating pressure.
5. Install piping in concealed interior and exterior locations, except in equipment rooms and service areas.
6. Install ducting and piping free of sags and bends.
7. Install exposed interior and exterior ducting and piping at right angles or parallel to building walls. Diagonal runs are prohibited, except where indicated.
8. Install ducting and piping tight to slabs, beams, joists, columns, walls, and other building elements. Allow sufficient space above removable ceiling panels to allow for ceiling panel removal.
9. Install ducting and piping to allow application of insulation plus 1-inch clearance around insulation for all exterior insulated ducts and pipes.
10. Locate groups of pipes parallel to each other, spaced to permit valve servicing.
11. Install fittings for changes in direction and branch connections.
12. Escutcheons: Where uncovered exposed pipes and ducts pass through floors, finished walls, or finished ceilings, they shall be fitted with chromium-plated cast-brass plates on chromium-plated pipe, or with cast-iron or steel plates on ferrous pipe. Plates shall be large enough to completely close the openings around the ducts and pipes and shall be square, octangular, or round, with the least dimension not less than 1-1/2 inches or more than 2-1/2 inches larger than the diameter of the duct/pipe. Plates shall be secured in an approved manner.
13. Install sleeves for pipes passing through concrete and masonry walls, concrete floor and roof slabs, and where indicated.
14. Verify final equipment locations for roughing in.
15. See equipment specifications in other sections of these specifications for roughing-in requirements.
16. Piping Joint Construction: Join pipe and fittings as follows and as specifically required in individual piping system specification sections.

    a. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.
    b. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.
    c. Soldered Joints: Construct joints according to AWS “Soldering Manual.”
    d. Brazed Joints: Construct joints according to AWS “Brazing Manual.”
    e. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1.
    f. Flanged Joints: Align flange surfaces parallel. Select appropriate gasket material, size, type, and thickness for service application. Install gasket concentrically

positioned. Assemble joints by sequencing bolt tightening to make initial contact of flanges and gaskets as flat and parallel as possible. Use suitable lubricants on bolt threads. Tighten bolts gradually and uniformly using torque wrench.

17. Piping Connections: Except as otherwise indicated, make piping connections as specified below.

    a. Install unions, in piping 2 inches and smaller, adjacent to each valve and at final connection to each piece of equipment having 2-inches or smaller threaded pipe connection.
    b. Install flanges, in piping 2-1/2 inches and larger, adjacent to flanged valves and at final connection to each piece of equipment having flanged pipe connection.
    c. Wet Piping Systems (WaterInstall dielectric coupling and nipple fittings to connect piping materials of dissimilar metals.

B. Equipment Installation – Common Requirements:

1. Install equipment to provide the maximum possible head room, where mounting heights are not indicated.
2. Install equipment according to approved submittal data. Portions of the Work are shown only in diagrammatic form. Refer conflicts to the engineer.
3. Install equipment level and plumb, parallel and perpendicular to other building systems and components in exposed interior spaces, except where otherwise indicated.
4. Install mechanical equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. Connect equipment for ease of disconnecting, with minimum of interference with other installations. Extend grease fittings to an accessible location.
5. Install equipment giving right-of-way to piping systems installed at a required slope.
6. Shaft Alignment: All motors and pumps (or drives) connected by a shaft coupling, whether factory or field assembled, shall be aligned during installation using a dial indicator applied to both ends of both shafts for a full 360 degrees prior to operation. Alignment of the shafts shall be less than the maximum allowable tolerances as recommended by the coupling or equipment manufacturer. Alignment of shafts shall be rechecked after several hours of operation and equipment has reached operating temperature.

C. Identification Tags and Labels:

1. Materials:

    a. Pipe Labels: Pipe labels shall be self-adhesive labels, all temperature Perma-Code pipe markers No. B-500, manufactured by the W. H. Brady Company. The background color code for all markers shall conform to the American National Standard ANSI A-13.1 “Scheme for the Identification of Piping Systems.” This standard establishes four basic backgrounds as follows: Yellow for dangerous materials, bright blue for protective materials, red for fire protection equipment, and green for safe materials.
    b. Tags: Tags shall be aluminum, brass or laminated plastic 2" x 1" minimum with edges ground smooth or rolled. Each tag shall be punched to receive tie wires or chain. Letters and Numbers shall be evenly spaced and stamped or engraved into the surface.

2. Installation:

a. Identification of Piping:

1) Identify all piping according to the following procedures:

a) Bare pipes to be marked shall first be wiped clean of dirt, dust, grease, and moisture. Markers to be installed on painted piping shall be applied only after completion of final coat of paint. Insulated pipes shall first be painted to a smooth, hard surface in the area the label is to be applied. Labels shall be applied, using pressure, so that it lies smooth and flat. After application on insulated pipes, the label shall be stapled securely to the insulation. The labels shall be applied to the pipe so that the lettering is in the most legible position. For overhead piping apply markers on the lower half of the pipe where view is unobstructed, so that markers can be read at a glance from floor level. The wording on the labels shall correspond directly to the wording in the mechanical symbol lists, regardless of whether or not it is standard wording for the designated manufacturer.

b) Use an arrow marker with each pipe content marker. The arrow shall always point away from the pipe marker and in the direction of flow, with background color and height the same as content marker. If flow can be in both directions, use two arrow markers.

c) Apply pipe marker and arrow marker at each valve, at every point of pipe entry or exit through wall or ceiling, on each riser and branch of tee, and every 20 feet on long continuous lines or at every bay or aisle to show proper identification of pipe content and direction of flow.

b. Valves: All main service valves, including fire protection, located inside the building shall be tagged and identified as to the type of service. All valves controlling branch mains or risers to various portions of the building shall be tagged and identified as to the areas served.

c. Controls: All automatic controls, control panels, zone valves, pressure electric, electric pressure switches, relays and starters shall be clearly tagged and identified. Wording shall be identical to that on the control diagram in the contract drawings.

d. Pumps: All pumps shall be identified as to service with aluminum or brass tags secured by tie wires.

D. Erection of Metal Supports and Anchorage:

1. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor mechanical materials and equipment.
2. Field Welding: Comply with AWS D1.1 “Structural Welding Code – Steel.”

E. Cutting and Patching:

1. Cut, channel, chase, and drill floors, walls, partitions, ceilings, and other surfaces necessary for mechanical installations. Perform cutting by skilled mechanics of the trades involved.
2. Repair cut surfaces to match adjacent surfaces.

F. Grouting:

1. Install nonmetallic, nonshrink grout for mechanical equipment base bearing surfaces, pump and other equipment base plates and anchors. Mix grout according to manufacturer's printed instructions.
2. Clean surfaces that will come into contact with grout.
3. Provide forms for placement of grout, as required.
4. Avoid air entrapment when placing grout.
5. Place grout, completely filling equipment bases.
6. Place grout on concrete bases to provide a smooth bearing surface for equipment.
7. Place grout around anchors.
8. Cure placed grout according to manufacturer's printed instructions.

END OF SECTION 23 05 00

SECTION 23 05 17 – SLEEVES AND SLEEVE SEALS FOR HVAC PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Sleeves without waterstop.
2. Sleeves with waterstop.
3. Stack-sleeve fittings.
4. Sleeve-seal systems.
5. Grout.
6. Silicone sealants.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

PART 2 - PRODUCTS

2.1 SLEEVES WITHOUT WATERSTOP

A. Physical Requirements

1. Cast-Iron Pipe Sleeves: Cast or fabricated of cast or ductile iron and equivalent to ductile-iron pressure pipe, with plain ends.
2. Steel Pipe Sleeves: ASTM A53/A53M, Type E, Grade B, Schedule 40, hot-dip galvanized, with plain ends.
3. Steel Sheet Sleeves: ASTM A653/A653M, 0.0239-inch minimum thickness; hot-dip galvanized, round tube closed with welded longitudinal joint.

2.2 SLEEVES WITH WATERSTOP

A. Physical Requirements

1. Description: Manufactured PVC/HDPE, sleeve-type, waterstop assembly, made for imbedding in concrete slab or wall.

2.3 STACK-SLEEVE FITTINGS

A. Physical Requirements

1. Description: Manufactured, galvanized cast-iron sleeve with integral cast flashing flange for use in waterproof floors and roofs. Include clamping ring, bolts, and nuts for membrane flashing.

    a. Underdeck Clamp: Clamping ring with setscrews.

2.4 SLEEVE-SEAL SYSTEMS

A. Physical Requirements

1. Description: Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.

    a. Designed to form a hydrostatic seal of 20 psig.
    b. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size.
    c. Pressure Plates: Carbon steel
    d. Connecting Bolts and Nuts: Carbon steel, with ASTM B633 coating of length required to secure pressure plates to sealing elements.

2.5 GROUT

A. Physical Requirements

1. Description: Nonshrink, for interior and exterior sealing openings in non-fire-rated walls or floors.
2. Standard: ASTM C1107/C1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

B. Functional Requirements

1. Design Mix: 5000 psi, 28-day compressive strength.
2. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 INSTALLATION OF SLEEVES – GENERAL

A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide1-inch annular clear space between piping and concrete slabs and walls.

1. Sleeves are not required for core-drilled holes.

C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.

1. Permanent sleeves are not required for holes in slabs formed by molded-PE or -PP sleeves.
2. Cut sleeves to length for mounting flush with both surfaces.

a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.

3. Using grout, seal space outside of sleeves in slabs and walls without sleeve-seal system.

D. Install sleeves for pipes passing through interior partitions.

1. Cut sleeves to length for mounting flush with both surfaces.
2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
3. Seal annular space between sleeve and piping or piping insulation; use sealants appropriate for size, depth, and location of joint.

E. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke-Barrier Penetrations: Maintain indicated fire or smoke rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping and fill materials specified in Section 017329 “Cutting and Patching.”

3.2 INSTALLATION OF SLEEVES WITH WATERSTOP

A. Install sleeve with waterstop as new walls and slabs are constructed.

B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.

C. Secure nailing flanges to concrete forms.

D. Using grout or silicone sealant, seal space around outside of sleeves.

3.3 INSTALLATION OF STACK-SLEEVE FITTINGS

A. Install stack-sleeve fittings in new slabs as slabs are constructed.

1. Install fittings that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation.
2. Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing. Comply with requirements for flashing specified in Section 076200 “Sheet Metal Flashing and Trim.”
3. Install section of cast-iron soil pipe to extend sleeve to 3 inches above finished floor level.

4. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
5. Using waterproof silicone sealant, seal space between top hub of stack-sleeve fitting and pipe.

B. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke Barrier Penetrations: Maintain indicated fire or smoke rating of floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping specified in Section 017329 “Cutting and Patching.”

3.4 INSTALLATION OF SLEEVE-SEAL SYSTEMS

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at service piping entries into building, and passing through exterior walls.

B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal-system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.5 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Leak Test: After allowing for a full cure, test sleeves and sleeve seals for leaks. Repair leaks and retest until no leaks exist.
2. Sleeves and sleeve seals will be considered defective if they do not pass tests and inspections.

B. Prepare test and inspection reports.

3.6 SLEEVE SCHEDULE

A. Use sleeves and sleeve seals for the following piping-penetration applications:

1. Exterior Concrete Walls above and below Grade:

a. Sleeves with waterstops.

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

2. Concrete Slabs-on-Grade:

a. Sleeves with waterstops.

1) Select sleeve size to allow for 1-inch annular clear space between piping and sleeve for installing sleeve-seal system.

3. Concrete Slabs above Grade:

    a. Sleeves with waterstops or stack-sleeve fittings.

4. Interior Walls and Partitions:

    a. Sleeves without waterstops.

END OF SECTION 23 05 17

SECTION 23 05 29 – HANGERS AND SUPPORTS FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Metal pipe hangers and supports.
2. Trapeze pipe hangers.
3. Fiberglass pipe hangers.
4. Metal framing systems.
5. Fiberglass strut systems.
6. Thermal-hanger shield inserts.
7. Fastener systems.
8. Pipe stands.
9. Equipment supports.
10. Miscellaneous Materials.

1.3 DEFINITIONS

A. MSS: Manufacturers Standardization Society of The Valve and Fittings Industry Inc.

1.4 PERFORMANCE REQUIREMENTS

A. Delegated Design: Design trapeze pipe hangers and equipment supports, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

B. Structural Performance: Hangers and supports for HVAC piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.

1. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.
2. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.
3. Design seismic-restraint hangers and supports for piping and equipment and obtain approval from authorities having jurisdiction.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Shop Drawings: Signed and sealed by a qualified professional engineer. Show fabrication and installation details and include calculations for the following; include Product Data for components:

1. Trapeze pipe hangers.
2. Metal framing systems.
3. Fiberglass strut systems.
4. Pipe stands.
5. Equipment supports.

C. Delegated-Design Submittal: For trapeze hangers indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1. Detail fabrication and assembly of trapeze hangers.
2. Design Calculations: Calculate requirements for designing trapeze hangers.

1.6 INFORMATIONAL SUBMITTALS

A. Welding certificates.

1.7 QUALITY ASSURANCE

A. Structural Steel Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, “Structural Welding Code – Steel.”

B. Pipe Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

PART 2 - PRODUCTS

2.1 METAL PIPE HANGERS AND SUPPORTS

A. Carbon-Steel Pipe Hangers and Supports:

1. Physical:

a. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
b. Galvanized Metallic Coatings: Pregalvanized or hot dipped.
c. Nonmetallic Coatings: Plastic coating, jacket, or liner.
d. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
e. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel stainless steel.

2. Function: None
3. Performance: None

B. Stainless-Steel Pipe Hangers and Supports:

1. Physical:

a. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.
b. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.
c. Hanger Rods: Continuous-thread rod, nuts, and washer made of stainless steel .

2. Function: None
3. Performance: None

C. Copper Pipe Hangers:

1. Physical:

a. Description: MSS SP-58, Types 1 through 58, copper-coated-steel, factory-fabricated components.
b. Hanger Rods: Continuous-thread rod, nuts, and washer made of copper-coated steel stainless steel.

2. Function: None
3. Performance: None

2.2 TRAPEZE PIPE HANGERS

1. Physical: MSS SP-69, Type 59, shop- or field-fabricated pipe-support assembly made from structural carbon-steel shapes with MSS SP-58 carbon-steel hanger rods, nuts, saddles, and U-bolts.

a. Function: None
b. Performance: None

2.3 METAL FRAMING SYSTEMS

A. MFMA Manufacturer Metal Framing Systems:

1. Physical:

a. Description: Shop- or field-fabricated pipe-support assembly for supporting multiple parallel pipes.
b. Standard: MFMA-4.
c. Channels: Continuous slotted steel channel with inturned lips.
d. Channel Nuts: Formed or stamped steel nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.
e. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel stainless steel.

f. Retain one of four subparagraphs below for coating.
g. Metallic Coating: Electroplated zinc Hot-dipped galvanized Mill galvanized In-line, hot galvanized Mechanically-deposited zinc.
h. Paint Coating: Vinyl Vinyl alkyd Epoxy Polyester Acrylic Amine Alkyd
i. Plastic Coating: PVC Polyurethane Epoxy Polyester.

2. Function: None
3. Performance: None

B. Non-MFMA Manufacturer Metal Framing Systems:

1. Physical:

a. Description: Shop- or field-fabricated pipe-support assembly made of steel channels, accessories, fittings, and other components for supporting multiple parallel pipes.
b. Standard: Comply with MFMA-4.
c. Channels: Continuous slotted steel channel with inturned lips.
d. Channel Nuts: Formed or stamped steel nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.
e. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel stainless steel.
f. Coating: Zinc Paint PVC.

2. Function: None
3. Performance: None

2.4 FIBERGLASS STRUT SYSTEMS

A. Physical:

a. Description: Shop- or field-fabricated pipe-support assembly similar to MFMA-4 for supporting multiple parallel pipes.
b. Channels: Continuous slotted fiberglass or other plastic channel with inturned lips.
c. Channel Nuts: Fiberglass nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.
d. Hanger Rods: Continuous-thread rod, nuts, and washer made of fiberglass stainless steel.

B. Function: None

C. Performance: None

2.5 THERMAL-HANGER SHIELD INSERTS

A. Physical:

1. Insulation-Insert Material for Cold Piping: ASTM C 552, Type II cellular glass with 100-psig or ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength and vapor barrier.
2. Insulation-Insert Material for Hot Piping: Water-repellent treated, ASTM C 533, Type I calcium silicate with 100-psig ASTM C 552, Type II cellular glass with 100-psig or ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength.
3. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

B. Function:

1. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.
2. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.

2.6 FASTENER SYSTEMS

A. Physical:

1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.
2. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated stainless- steel anchors, for use in hardened portland cement concrete; with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

B. Function: None

C. Performance: None

2.7 PIPE STANDS

A. Physical:

1. General Requirements for Pipe Stands: Shop- or field-fabricated assemblies made of manufactured corrosion-resistant components to support roof-mounted piping.
2. Compact Pipe Stand: One-piece plastic unit with integral-rod roller, pipe clamps, or V-shaped cradle to support pipe, for roof installation without membrane penetration.
3. Low-Type, Single-Pipe Stand: One-piece plastic stainless-steel base unit with plastic roller, for roof installation without membrane penetration.
4. High-Type, Single-Pipe Stand:

    a. Description: Assembly of base, vertical and horizontal members, and pipe support, for roof installation without membrane penetration.

b. Base: Plastic Stainless steel.
c. Vertical Members: Two or more cadmium-plated-steel or stainless-steel, continuous-thread rods.
d. Horizontal Member: Cadmium-plated-steel or stainless-steel rod with plastic or stainless-steel, roller-type pipe support.

5. High-Type, Multiple-Pipe Stand:

a. Description: Assembly of bases, vertical and horizontal members, and pipe supports, for roof installation without membrane penetration.
b. Bases: One or more; plastic.
c. Vertical Members: Two or more protective-coated-steel channels.
d. Horizontal Member: Protective-coated-steel channel.
e. Pipe Supports: Galvanized-steel, clevis-type pipe hangers.

6. Curb-Mounted-Type Pipe Stands: Shop- or field-fabricated pipe supports made from structural-steel shapes, continuous-thread rods, and rollers, for mounting on permanent stationary roof curb.

B. Function: None

C. Performance: None

2.8 EQUIPMENT SUPPORTS

A. Physical:

1. Description: Welded, shop- or field-fabricated equipment support made from structural carbon-steel shapes.

B. Function: None

C. Performance: None

2.9 MISCELLANEOUS MATERIALS

A. Physical:

1. Structural Steel: ASTM A 36/A 36M, carbon-steel plates, shapes, and bars; black and galvanized.
2. Grout: ASTM C 1107, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.

a. Properties: Nonstaining, noncorrosive, and nongaseous.
b. Design Mix: 5000-psi, 28-day compressive strength.

B. Function: None

C. Performance: None

PART 3 - EXECUTION

3.1 HANGER AND SUPPORT INSTALLATION

A. Metal Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Install hangers, supports, clamps, and attachments as required to properly support piping from the building structure.

B. Metal Trapeze Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Arrange for grouping of parallel runs of horizontal piping, and support together on field-fabricated trapeze pipe hangers.

1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size or install intermediate supports for smaller diameter pipes as specified for individual pipe hangers.
2. Field fabricate from ASTM A 36/A 36M, carbon-steel shapes selected for loads being supported. Weld steel according to AWS D1.1/D1.1M.

C. Metal Framing System Installation: Arrange for grouping of parallel runs of piping, and support together on field-assembled metal framing systems.

D. Fiberglass Strut System Installation: Arrange for grouping of parallel runs of piping, and support together on field-assembled fiberglass struts.

E. Fastener System Installation:

1. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches thick in concrete after concrete is placed and completely cured. Use operators that are licensed by powder-actuated tool manufacturer. Install fasteners according to powder-actuated tool manufacturer’s operating manual.
2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer’s written instructions.

F. Pipe Stand Installation:

1. Pipe Stand Types except Curb-Mounted Type: Assemble components and mount on smooth roof surface. Do not penetrate roof membrane.
2. Curb-Mounted-Type Pipe Stands: Assemble components or fabricate pipe stand and mount on permanent, stationary roof curb. See Section 077200 “Roof Accessories” for curbs.

G. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.

H. Equipment Support Installation: Fabricate from welded-structural-steel shapes.

I. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.

J. Install lateral bracing with pipe hangers and supports to prevent swaying.

K. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.

L. Load Distribution: Install hangers and supports so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment.

M. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.

N. Insulated Piping:

1. Attach clamps and spacers to piping.

    a. Piping Operating above Ambient Air Temperature: Clamp may project through insulation.
    b. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.
    c. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.

2. Install MSS SP-58, Type 39, protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.

    a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

3. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.

    a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

4. Shield Dimensions for Pipe: Not less than the following:

    a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.
    b. NPS 4: 12 inches long and 0.06 inch thick.
    c. NPS 5 and NPS 6: 18 inches long and 0.06 inch thick.
    d. NPS 8 to NPS 14: 24 inches long and 0.075 inch thick.
    e. NPS 16 to NPS 24: 24 inches long and 0.105 inch thick.

5. Pipes NPS 8 and Larger: Include wood or reinforced calcium-silicate-insulation inserts of length at least as long as protective shield.
6. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.2 EQUIPMENT SUPPORTS

A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor.

B. Grouting: Place grout under supports for equipment and make bearing surface smooth.

C. Provide lateral bracing, to prevent swaying, for equipment supports.

3.3 METAL FABRICATIONS

A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers and equipment supports.

B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.

C. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work; and with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
2. Obtain fusion without undercut or overlap.
3. Remove welding flux immediately.
4. Finish welds at exposed connections so no roughness shows after finishing and so contours of welded surfaces match adjacent contours.

3.4 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

C. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.

D. Touchup: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal are specified in Section 099113 “Exterior Painting” Section 099123 “Interior Painting” and Section 099600 “High Performance Coatings.”

E. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

3.5 HANGER AND SUPPORT SCHEDULE

A. Specific hanger and support requirements are in Sections specifying piping systems and equipment.

B. Comply with MSS SP-69 for pipe-hanger selections and applications that are not specified in piping system Sections.

C. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finish.

D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.

E. Use carbon-steel pipe hangers and supports metal trapeze pipe hangers and metal framing systems and attachments for general service applications.

F. Use stainless-steel pipe hangers and fiberglass pipe hangers and fiberglass strut systems and stainless-steel or corrosion-resistant attachments for hostile environment applications.

G. Use copper-plated pipe hangers and copper or stainless-steel attachments for copper piping and tubing.

H. Use padded hangers for piping that is subject to scratching.

I. Use thermal-hanger shield inserts for insulated piping and tubing.

J. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of noninsulated or insulated, stationary pipes NPS 1/2 to NPS 30.
2. Yoke-Type Pipe Clamps (MSS Type 2): For suspension of up to 1050 deg F, pipes NPS 4 to NPS 24, requiring up to 4 inches of insulation.
3. Carbon- or Alloy-Steel, Double-Bolt Pipe Clamps (MSS Type 3): For suspension of pipes NPS 3/4 to NPS 36, requiring clamp flexibility and up to 4 inches of insulation.
4. Steel Pipe Clamps (MSS Type 4): For suspension of cold and hot pipes NPS 1/2 to NPS 24 if little or no insulation is required.
5. Pipe Hangers (MSS Type 5): For suspension of pipes NPS 1/2 to NPS 4, to allow off-center closure for hanger installation before pipe erection.
6. Adjustable, Swivel Split- or Solid-Ring Hangers (MSS Type 6): For suspension of noninsulated, stationary pipes NPS 3/4 to NPS 8.
7. Adjustable, Steel Band Hangers (MSS Type 7): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
8. Adjustable Band Hangers (MSS Type 9): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
9. Adjustable, Swivel-Ring Band Hangers (MSS Type 10): For suspension of noninsulated, stationary pipes NPS 1/2 to NPS 8.
10. Split Pipe Ring with or without Turnbuckle Hangers (MSS Type 11): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 8.

11. Extension Hinged or Two-Bolt Split Pipe Clamps (MSS Type 12): For suspension of noninsulated, stationary pipes NPS 3/8 to NPS 3.
12. U-Bolts (MSS Type 24): For support of heavy pipes NPS 1/2 to NPS 30.
13. Clips (MSS Type 26): For support of insulated pipes not subject to expansion or contraction.
14. Pipe Saddle Supports (MSS Type 36): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate.
15. Pipe Stanchion Saddles (MSS Type 37): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate, and with U-bolt to retain pipe.
16. Adjustable Pipe Saddle Supports (MSS Type 38): For stanchion-type support for pipes NPS 2-1/2 to NPS 36 if vertical adjustment is required, with steel-pipe base stanchion support and cast-iron floor flange.
17. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30, from two rods if longitudinal movement caused by expansion and contraction might occur.
18. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes NPS 2-1/2 to NPS 24, from single rod if horizontal movement caused by expansion and contraction might occur.
19. Complete Pipe Rolls (MSS Type 44): For support of pipes NPS 2 to NPS 42 if longitudinal movement caused by expansion and contraction might occur but vertical adjustment is not necessary.
20. Pipe Roll and Plate Units (MSS Type 45): For support of pipes NPS 2 to NPS 24 if small horizontal movement caused by expansion and contraction might occur and vertical adjustment is not necessary.
21. Adjustable Pipe Roll and Base Units (MSS Type 46): For support of pipes NPS 2 to NPS 30 if vertical and lateral adjustment during installation might be required in addition to expansion and contraction.

K. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24.
2. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps.

L. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches for heavy loads.
2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
3. Swivel Turnbuckles (MSS Type 15): For use with MSS Type 11, split pipe rings.
4. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
5. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.

M. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.

2. Top-Beam C-Clamps (MSS Type 19): For use under roof installations with bar-joist construction, to attach to top flange of structural shape.
3. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.
4. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.
5. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
6. C-Clamps (MSS Type 23): For structural shapes.
7. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
8. Side-Beam Clamps (MSS Type 27): For bottom of steel I-beams.
9. Steel-Beam Clamps with Eye Nuts (MSS Type 28): For attaching to bottom of steel I-beams for heavy loads.
10. Linked-Steel Clamps with Eye Nuts (MSS Type 29): For attaching to bottom of steel I-beams for heavy loads, with link extensions.
11. Malleable-Beam Clamps with Extension Pieces (MSS Type 30): For attaching to structural steel.
12. Welded-Steel Brackets: For support of pipes from below or for suspending from above by using clip and rod. Use one of the following for indicated loads:

    a. Light (MSS Type 31): 750 lb.
    b. Medium (MSS Type 32): 1500 lb.
    c. Heavy (MSS Type 33): 3000 lb.

13. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.
14. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.
15. Horizontal Travelers (MSS Type 58): For supporting piping systems subject to linear horizontal movement where headroom is limited.

N. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel-Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
3. Thermal-Hanger Shield Inserts: For supporting insulated pipe.

O. Spring Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Restraint-Control Devices (MSS Type 47): Where indicated to control piping movement.
2. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1-1/4 inches.
3. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41, roll hanger with springs.
4. Spring Sway Braces (MSS Type 50): To retard sway, shock, vibration, or thermal expansion in piping systems.
5. Variable-Spring Hangers (MSS Type 51): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from hanger.

6. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.
7. Variable-Spring Trapeze Hangers (MSS Type 53): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from trapeze support.
8. Constant Supports: For critical piping stress and if necessary to avoid transfer of stress from one support to another support, critical terminal, or connected equipment. Include auxiliary stops for erection, hydrostatic test, and load-adjustment capability. These supports include the following types:

    a. Horizontal (MSS Type 54): Mounted horizontally.
    b. Vertical (MSS Type 55): Mounted vertically.
    c. Trapeze (MSS Type 56): Two vertical-type supports and one trapeze member.

P. Comply with MSS SP-69 for trapeze pipe-hanger selections and applications that are not specified in piping system Sections.

Q. Comply with MFMA-103 for metal framing system selections and applications that are not specified in piping system Sections.

R. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.

END OF SECTION 23 05 29

SECTION 23 05 53 – IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Equipment labels.
2. Warning signs and labels.
3. Pipe labels.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Equipment-Label Schedule: Include a listing of all equipment to be labeled with the proposed content for each label.

PART 2 - PRODUCTS

2.1 EQUIPMENT LABELS

A. Metal Labels for Equipment:http://www.specagent.com/LookUp/?ulid=9462&mf=&src=wd

1. Material and Thickness: Brass, 0.032-inch or aluminum, 0.032-inch minimum thickness, with predrilled or stamped holes for attachment hardware.
2. Letter and Background Color: As indicated for specific application under Part 3.
3. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.
4. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances of up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.
5. Fasteners: Stainless steel rivets or self-tapping screws.
6. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

2.2 WARNING SIGNS AND LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, with predrilled holes for attachment hardware.

B. Letter and Background Color: As indicated for specific application under Part 3.

C. Maximum Temperature: Able to withstand temperatures of up to 160 deg F.

D. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

E. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances of up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.

F. Fasteners: Stainless steel rivets or self-taping screws.

G. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

H. Arc-Flash Warning Signs: Provide arc-flash warning signs in locations and with content in accordance with requirements of OSHA and NFPA70E.

I. Label Content: Include caution and warning information plus emergency notification instructions.

2.3 PIPE LABELS

A. Physical:

1. General Requirements for Manufactured Pipe Labels: Preprinted, color coded, with lettering indicating service and showing flow direction in accordance with ASME A13.1.
2. Letter and Background Color: As indicated for specific application under Part 3.
3. Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to cover full circumference of pipe and to attach to pipe without fasteners or adhesive.
4. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.

B. Function:

1. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings. Also include:

    a. Pipe size.
    b. Flow-Direction Arrows: Include flow-direction arrows on main distribution piping. Arrows may be either integral with label or applied separately.
    c. Lettering Size: Size letters in accordance with ASME A13.1 for piping

C. Performance: None

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean piping and equipment surfaces of incompatible primers, paints, and encapsulants, as well as dirt, oil, grease, release agents, and other substances that could impair bond of identification devices.

3.2 INSTALLATION, GENERAL REQUIREMENTS

A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

B. Coordinate installation of identifying devices with locations of access panels and doors.

C. Install identifying devices before installing acoustical ceilings and similar concealment.

D. Locate identifying devices so that they are readily visible from the point of normal approach.

3.3 INSTALLATION OF EQUIPMENT LABELS, WARNING SIGNS, AND LABELS

A. Permanently fasten labels on each item of mechanical equipment.

B. Sign and Label Colors:

1. White letters on an ANSI Z535.1 safety-blue background

C. Locate equipment labels where accessible and visible.

D. Arc-Flash Warning Signs: Provide arc-flash warning signs on electrical disconnects and other equipment where arc-flash hazard exists, as indicated on Drawings, and in accordance with requirements of OSHA and NFPA 70E, and other applicable codes and standards.

3.4 INSTALLATION OF PIPE LABELS

A. Install pipe labels showing service and flow direction with permanent adhesive on pipes.

B. Pipe-Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Within 3 ft. of each valve and control device.
2. At access doors, manholes, and similar access points that permit view of concealed piping.
3. Within 3 ft. of equipment items and other points of origination and termination.
4. Spaced at maximum intervals of 25 ft. along each run. Reduce intervals to 10 ft in areas of congested piping, ductwork, and equipment.

C. Do not apply plastic pipe labels or plastic tapes directly to bare pipes conveying fluids at temperatures of 125 deg F or higher. Where these pipes are to remain uninsulated, use a short section of insulation or use stenciled labels.

D. Flow-Direction Arrows: Use arrows to indicate direction of flow in pipes, including pipes where flow is allowed in both directions.

END OF SECTION 23 05 53

SECTION 23 05 93 – TESTING, ADJUSTING, AND BALANCING FOR HVAC

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Balancing Air Systems:

a. Constant-volume air systems.

2. Testing, adjusting, and balancing of equipment

1.2 DEFINITIONS

A. AABC: Associated Air Balance Council.

B. NEBB: National Environmental Balancing Bureau.

C. TAB: Testing, adjusting, and balancing.

D. TABB: Testing, Adjusting, and Balancing Bureau.

E. TAB Specialist: An independent entity meeting qualifications to perform TAB work.

F. TDH: Total dynamic head.

1.3 INFORMATIONAL SUBMITTALS

A. Strategies and Procedures Plan: Within 30 days of Contractor's Notice to Proceed, submit TAB strategies and step-by-step procedures as specified in "Preparation" Article.

B. Certified TAB reports: Documentation indicating that Work complies with ASHRAE/IES 90.1, Section 6.7.2.3 – "System Balancing."

1.4 QUALITY ASSURANCE

A. TAB Specialists Qualifications: Certified by AABC.

1. TAB Field Supervisor: Employee of the TAB specialist and certified by AABC.
2. TAB Technician: Employee of the TAB specialist and certified by AABC as a TAB technician.

B. TAB Specialists Qualifications: Certified by NEBB or TABB.

1. TAB Field Supervisor: Employee of the TAB specialist and certified by NEBB or TABB.
2. TAB Technician: Employee of the TAB specialist and certified by NEBB or TABB as a TAB technician.

C. Instrumentation Type, Quantity, Accuracy, and Calibration: Comply with requirements in ASHRAE 111, Section 4, "Instrumentation."

D. ASHRAE/IES 90.1 Compliance: Applicable requirements in ASHRAE/IES 90.1, Section 6.7.2.3 – "System Balancing."

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine the Contract Documents to become familiar with Project requirements and to discover conditions in systems designs that may preclude proper TAB of systems and equipment.

B. Examine installed systems for balancing devices, such as test ports, gage cocks, thermometer wells, flow-control devices, balancing valves and fittings, and manual volume dampers. Verify that locations of these balancing devices are applicable for intended purpose and are accessible.

C. Examine the approved submittals for HVAC systems and equipment.

D. Examine design data including HVAC system descriptions, statements of design assumptions for environmental conditions and systems output, and statements of philosophies and assumptions about HVAC system and equipment controls.

E. Examine ceiling plenums and underfloor air plenums used for supply, return, or relief air to verify that they are properly separated from adjacent areas. Verify that penetrations in plenum walls are sealed and fire-stopped if required.

F. Examine equipment performance data including fan and pump curves.

1. Relate performance data to Project conditions and requirements, including system effects that can create undesired or unpredicted conditions that cause reduced capacities in all or part of a system.
2. Calculate system-effect factors to reduce performance ratings of HVAC equipment when installed under conditions different from the conditions used to rate equipment performance. To calculate system effects for air systems, use tables and charts found in AMCA 201, "Fans and Systems," or in SMACNA's "HVAC Systems – Duct Design." Compare results with the design data and installed conditions.

G. Examine system and equipment installations and verify that field quality-control testing, cleaning, and adjusting specified in individual Sections have been performed.

H. Examine test reports specified in individual system and equipment Sections.

I. Examine HVAC equipment and verify that bearings are greased, belts are aligned and tight, filters are clean, and equipment with functioning controls is ready for operation.

J. Examine terminal units, such as variable-air-volume boxes, and verify that they are accessible and their controls are connected and functioning.

K. Examine strainers. Verify that startup screens have been replaced by permanent screens with indicated perforations.

L. Examine control valves for proper installation for their intended function of throttling, diverting, or mixing fluid flows.

M. Examine heat-transfer coils for correct piping connections and for clean and straight fins.

N. Examine system pumps to ensure absence of entrained air in the suction piping.

O. Examine operating safety interlocks and controls on HVAC equipment.

P. Report deficiencies discovered before and during performance of TAB procedures. Observe and record system reactions to changes in conditions. Record default set points if different from indicated values.

3.2 PREPARATION

A. Prepare a TAB plan that includes strategies and step-by-step procedures for balancing the systems.

B. Perform system-readiness checks of HVAC systems and equipment to verify system readiness for TAB work. Include, at a minimum, the following:

1. Airside:

    a. Duct systems are complete with terminals installed.
    b. Volume are open and functional.
    c. Clean filters are installed.
    d. Fans are operating, free of vibration, and rotating in correct direction.
    e. Automatic temperature-control systems are operational.
    f. Windows and doors are installed.
    g. Suitable access to balancing devices and equipment is provided.

2. Hydronics:

    a. Verify leakage and pressure tests on water distribution systems have been satisfactorily completed.
    b. Piping is complete with terminals installed.
    c. Water treatment is complete.
    d. Systems are flushed, filled, and air purged.
    e. Strainers are pulled and cleaned.
    f. Control valves are functioning in accordance with the sequence of operation.
    g. Shutoff and balance valves have been verified to be 100 percent open.

h. Pumps are started and proper rotation is verified.
i. Pump gauge connections are installed directly at pump inlet and outlet flanges or in discharge and suction pipe prior to valves or strainers.
j. Variable-frequency controllers' startup is complete and safeties are verified.
k. Suitable access to balancing devices and equipment is provided.

3.3 GENERAL PROCEDURES FOR TESTING AND BALANCING

A. Perform testing and balancing procedures on each system according to the procedures contained in NEBB's "Procedural Standards for Testing, Adjusting, and Balancing of Environmental Systems" or SMACNA's "HVAC Systems – Testing, Adjusting, and Balancing" and in this Section.

B. Cut insulation, ducts, pipes, and equipment cabinets for installation of test probes to the minimum extent necessary for TAB procedures.

1. After testing and balancing, patch probe holes in ducts with same material and thickness as used to construct ducts.
2. After testing and balancing, install test ports and duct access doors that comply with requirements in Section 233300 "Air Duct Accessories."
3. Install and join new insulation that matches removed materials. Restore insulation, coverings, vapor barrier, and finish according to Section 230713 "Duct Insulation," and Section 230716 "HVAC Equipment Insulation."

C. Mark equipment and balancing devices, including damper-control positions, valve position indicators, fan-speed-control levers, and similar controls and devices, with paint or other suitable, permanent identification material to show final settings.

D. Take and report testing and balancing measurements in inch-pound (IP) units.

3.4 GENERAL PROCEDURES FOR BALANCING AIR SYSTEMS

A. Prepare test reports for both fans and outlets. Obtain manufacturer's outlet factors and recommended testing procedures. Cross-check the summation of required outlet volumes with required fan volumes.

B. Prepare schematic diagrams of systems' "as-built" duct layouts.

C. For variable-air-volume systems, develop a plan to simulate diversity.

D. Determine the best locations in main and branch ducts for accurate duct-airflow measurements.

E. Check airflow patterns from the outdoor-air louvers and dampers and the return- and exhaust-air dampers through the supply-fan discharge and mixing dampers.

F. Locate start-stop and disconnect switches, electrical interlocks, and motor starters.

G. Verify that motor starters are equipped with properly sized thermal protection.

H. Check dampers for proper position to achieve desired airflow path.

I. Check for airflow blockages.

J. Check condensate drains for proper connections and functioning.

K. Check for proper sealing of air-handling-unit components.

L. Verify that air duct system is sealed as specified in Section 233113 “Metal Ducts.”

3.5 TESTING, ADJUSTING, AND BALANCING OF HVAC EQUIPMENT

A. Test, adjust, and balance HVAC equipment indicated on Drawings, including, but not limited to, the following:

1. Motors.
2. Fans and ventilators.
3. Terminal units.
4. Air-handling units.
5. Heating and ventilating units.
6. Split-system air conditioners.
7. Coils.

3.6 PROCEDURES FOR CONSTANT-VOLUME AIR SYSTEMS

A. Adjust fans to deliver total indicated airflows within the maximum allowable fan speed listed by fan manufacturer.

1. Measure total airflow.

    a. Set outside-air, return-air, and relief-air dampers for proper position that simulates minimum outdoor-air conditions.
    b. Where duct conditions allow, measure airflow by main Pitot-tube traverse. If necessary, perform multiple Pitot-tube traverses close to the fan and prior to any outlets, to obtain total airflow.
    c. Where duct conditions are unsuitable for Pitot-tube traverse measurements, a coil traverse may be acceptable.

2. Measure fan static pressures as follows:

    a. Measure static pressure directly at the fan outlet or through the flexible connection.
    b. Measure static pressure directly at the fan inlet or through the flexible connection.
    c. Measure static pressure across each component that makes up the air-handling system.
    d. Report artificial loading of filters at the time static pressures are measured.

3. Review Contractor-prepared shop drawings and Record drawings to determine variations in design static pressures versus actual static pressures. Calculate actual system-effect factors. Recommend adjustments to accommodate actual conditions.

4. Obtain approval from Contracting Officer for adjustment of fan speed higher or lower than indicated speed. Comply with requirements in HVAC Sections for air-handling units for adjustment of fans, belts, and pulley sizes to achieve indicated air-handling-unit performance.
5. Do not make fan-speed adjustments that result in motor overload. Consult equipment manufacturers about fan-speed safety factors. Modulate dampers and measure fan-motor amperage to ensure that no overload occurs. Measure amperage in full-cooling, full-heating, economizer, and any other operating mode to determine the maximum required brake horsepower.

B. Adjust volume dampers for main duct, submain ducts, and major branch ducts to indicated airflows.

1. Measure airflow of submain and branch ducts.
2. Adjust submain and branch duct volume dampers for specified airflow.
3. Re-measure each submain and branch duct after all have been adjusted.

C. Adjust air inlets and outlets for each space to indicated airflows.

1. Set airflow patterns of adjustable outlets for proper distribution without drafts.
2. Measure inlets and outlets airflow.
3. Adjust each inlet and outlet for specified airflow.
4. Re-measure each inlet and outlet after they have been adjusted.

D. Verify final system conditions.

1. Re-measure and confirm that minimum outdoor, return, and relief airflows are within design. Readjust to design if necessary.
2. Re-measure and confirm that total airflow is within design.
3. Re-measure all final fan operating data, speed, volts, amps, and static profile.
4. Mark all final settings.
5. Test system in economizer mode. Verify proper operation and adjust if necessary.
6. Measure and record all operating data.
7. Record final fan-performance data.

3.7 TOLERANCES

A. Set HVAC system’s airflow rates and water flow rates within the following tolerances:

1. Supply Fans and Equipment with Fans: Plus or minus 10 percent.
2. Air Outlets and Inlets: Plus or minus 10 percent.

B. Maintaining pressure relationships as designed shall have priority over the tolerances specified above.

3.8 FINAL REPORT

A. General: Prepare a certified written report; tabulate and divide the report into separate sections for tested systems and balanced systems.

1. Include a certification sheet at the front of the report's binder, signed and sealed by the certified testing and balancing engineer.
2. Include a list of instruments used for procedures, along with proof of calibration.
3. Certify validity and accuracy of field data.

B. Final Report Contents: In addition to certified field-report data, include the following:

1. Fan curves.
2. Manufacturers' test data.
3. Field test reports prepared by system and equipment installers.
4. Other information relative to equipment performance; do not include Shop Drawings and Product Data.

C. General Report Data: In addition to form titles and entries, include the following data:

1. Title page.
2. Name and address of the TAB specialist.
3. Project name.
4. Project location.
5. Contracting Officer's name and address.
6. Engineer's name and address.
7. Contractor's name and address.
8. Report date.
9. Signature of TAB supervisor who certifies the report.
10. Table of Contents with the total number of pages defined for each section of the report. Number each page in the report.
11. Summary of contents including the following:

    a. Indicated versus final performance.
    b. Notable characteristics of systems.
    c. Description of system operation sequence if it varies from the Contract Documents.

12. Nomenclature sheets for each item of equipment.
13. Data for terminal units, including manufacturer's name, type, size, and fittings.
14. Notes to explain why certain final data in the body of reports vary from indicated values.
15. Test conditions for fans and pump performance forms including the following:

    a. Settings for outdoor-, return-, and exhaust-air dampers.
    b. Conditions of filters.
    c. Cooling coil, wet- and dry-bulb conditions.
    d. Face and bypass damper settings at coils.
    e. Fan drive settings including settings and percentage of maximum pitch diameter.

f. Inlet vane settings for variable-air-volume systems.
g. Settings for supply-air, static-pressure controller.
h. Other system operating conditions that affect performance.

D. System Diagrams: Include schematic layouts of air and hydronic distribution systems. Present each system with single-line diagram and include the following:

1. Quantities of outdoor, supply, return, and exhaust airflows.
2. Water and steam flow rates.
3. Duct, outlet, and inlet sizes.
4. Pipe and valve sizes and locations.
5. Terminal units.
6. Balancing stations.
7. Position of balancing devices.

E. Air-Handling-Unit Test Reports: For air-handling units with coils, include the following:

1. Unit Data:

a. Unit identification.
b. Location.
c. Make and type.
d. Model number and unit size.
e. Manufacturer's serial number.
f. Unit arrangement and class.
g. Discharge arrangement.
h. Sheave make, size in inches, and bore.
i. Center-to-center dimensions of sheave and amount of adjustments in inches.
j. Number, make, and size of belts.
k. Number, type, and size of filters.

2. Motor Data:

a. Motor make, and frame type and size.
b. Horsepower and rpm.
c. Volts, phase, and hertz.
d. Full-load amperage and service factor.
e. Sheave make, size in inches, and bore.
f. Center-to-center dimensions of sheave and amount of adjustments in inches.

3. Test Data (Indicated and Actual Values):

a. Total airflow rate in cfm.
b. Total system static pressure in inches wg.
c. Fan rpm.
d. Discharge static pressure in inches wg.
e. Filter static-pressure differential in inches wg.
f. Cooling-coil static-pressure differential in inches wg.
g. Outdoor airflow in cfm.
h. Outdoor-air damper position.
i. Vortex damper position.

F. Fan Test Reports: For supply, return, and exhaust fans, include the following:

1. Fan Data:

    a. System identification.
    b. Location.
    c. Make and type.
    d. Model number and size.
    e. Manufacturer's serial number.
    f. Arrangement and class.
    g. Sheave make, size in inches, and bore.
    h. Center-to-center dimensions of sheave and amount of adjustments in inches.

2. Motor Data:

    a. Motor make, and frame type and size.
    b. Horsepower and rpm.
    c. Volts, phase, and hertz.
    d. Full-load amperage and service factor.
    e. Sheave make, size in inches, and bore.
    f. Center-to-center dimensions of sheave, and amount of adjustments in inches.
    g. Number, make, and size of belts.

3. Test Data (Indicated and Actual Values):

    a. Total airflow rate in cfm.
    b. Total system static pressure in inches wg.
    c. Fan rpm.
    d. Discharge static pressure in inches wg.
    e. Suction static pressure in inches wg.

G. Air-Terminal-Device Reports:

1. Unit Data:

    a. System and air-handling unit identification.
    b. Location and zone.
    c. Apparatus used for test.
    d. Area served.
    e. Make.
    f. Number from system diagram.
    g. Type and model number.
    h. Size.
    i. Effective area in sq. ft.

2. Test Data (Indicated and Actual Values):

    a. Airflow rate in cfm.
    b. Air velocity in fpm.
    c. Preliminary airflow rate as needed in cfm.

d. Preliminary velocity as needed in fpm.
e. Final airflow rate in cfm.
f. Final velocity in fpm.
g. Space temperature in deg F.

H. Instrument Calibration Reports:

1. Report Data:

a. Instrument type and make.
b. Serial number.
c. Application.
d. Dates of use.
e. Dates of calibration.

3.9 VERIFICATION OF TAB REPORT

A. The TAB specialist's test and balance engineer shall conduct the inspection in the presence of NPS, Construction Manager, and commissioning authority.

B. Commissioning authority shall randomly select measurements, documented in the final report, to be rechecked. Rechecking shall be limited to either 10 percent of the total measurements recorded or the extent of measurements that can be accomplished in a normal 8-hour business day.

C. If rechecks yield measurements that differ from the measurements documented in the final report by more than the tolerances allowed, the measurements shall be noted as "FAILED."

D. If the number of "FAILED" measurements is greater than 10 percent of the total measurements checked during the final inspection, the testing and balancing shall be considered incomplete and shall be rejected.

E. If TAB work fails, proceed as follows:

1. TAB specialists shall recheck all measurements and make adjustments. Revise the final report and balancing device settings to include all changes; resubmit the final report and request a second final inspection.
2. If the second final inspection also fails, Contracting Officer may contract the services of another TAB specialist to complete TAB work according to the Contract Documents and deduct the cost of the services from the original TAB specialist's final payment.

3. If the second verification also fails, NPS, design professional, or Contracting Officer may contact AABC Headquarters regarding the AABC National Performance Guaranty.

F. Prepare test and inspection reports.

3.10 ADDITIONAL TESTS

A. Within 90 days of completing TAB, perform additional TAB to verify that balanced conditions are being maintained throughout and to correct unusual conditions.

B. Seasonal Periods: If initial TAB procedures were not performed during near-peak summer and winter conditions, perform additional TAB during near-peak summer and winter conditions.

END OF SECTION 23 05 93

SECTION 23 07 13 – DUCT INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes insulating the following duct services:

1. Indoor, concealed suppl and return air.
2. Indoor, exposed supply air.
3. Outdoor, exposed supply,and return air.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1.3 QUALITY ASSURANCE

A. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E 84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Physical:

1. Products shall not contain asbestos, lead, mercury, or mercury compounds.
2. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.
3. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.
4. Mineral-Fiber Blanket Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 553, Type II and ASTM C 1290, Type III with factory-applied FSK jacket. Factory-applied jacket requirements are specified in “Factory-Applied Jackets” Article.

B. Function: None

C. Performance: None

2.2 ADHESIVES

A. Physical:

1. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.
2. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
3. FSK Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.

B. Function: None

C. Performance: None

2.3 MASTICS

A. Physical:

1. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.
2. Breather Mastic: Water based; suitable for indoor and outdoor use on above ambient services.

    a. Water-Vapor Permeance: ASTM F 1249, 1.8 perms at 0.0625-inch dry film thickness.
    b. Service Temperature Range: Minus 20 to plus 180 deg F.
    c. Solids Content: 60 percent by volume and 66 percent by weight.
    d. Color: White.

B. Function: None

C. Performance: None

2.4 SEALANTS

A. Physical:

1. FSK/FRK Sealants:
2. Materials shall be compatible with insulation materials, jackets, and substrates.
3. Fire- and water-resistant, flexible, elastomeric sealant.
4. Service Temperature Range: Minus 40 to plus 250 deg F.
5. Color: Aluminum.

B. Function: None

C. Performance: None

2.5 FACTORY-APPLIED JACKETS

A. Physical:

1. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

    a. FSK/FRK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C 1136, Type II.

B. Function: None

C. Performance: None

2.6 FIELD-APPLIED JACKETS

A. Physical:

1. Field-applied jackets shall comply with ASTM C 921, Type I, unless otherwise indicated.
2. FSK Jacket: Aluminum-foil-face, fiberglass-reinforced scrim with kraft-paper backing.
3. Self-Adhesive Outdoor Jacket: 60-mil- thick, laminated vapor barrier and waterproofing membrane for installation over insulation located aboveground outdoors; consisting of a rubberized bituminous resin on a crosslaminated polyethylene film covered with stucco-embossed aluminum-foil facing.

B. Function: None

C. Performance: None

2.7 TAPES

A. Physical:

1. FSK/FRK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C 1136.

    a. Width: 3 inches.
    b. Thickness: 6.5 mils.

2. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.http://www.specagent.com/LookUp/?ulid=3099&mf=04&src=wd

    a. Width: 2 inches.
    b. Thickness: 3.7 mils.

B. Function: None.

C. Performance:

1. FSK/FRK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C 1136.

    a. Adhesion: 90 ounces force/inch in width.
    b. Elongation: 2 percent.
    c. Tensile Strength: 40 lbf/inch in width.

2. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.http://www.specagent.com/LookUp/?ulid=3099&mf=04&src=wd

    a. Adhesion: 100 ounces force/inch in width.
    b. Elongation: 5 percent.
    c. Tensile Strength: 34 lbf/inch in width.

2.8 SECUREMENTS

A. Physical:

1. Aluminum Bands: ASTM B 209, Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, 3/4 inch wide with wing seal or closed seal.
2. Insulation Pins and Hangers:

    a. Metal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following requirements:

        1) Baseplate: Perforated, galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
        2) Spindle: Copper- or zinc-coated, low-carbon steel or Aluminum or Stainless steel, fully annealed, 0.106-inch- diameter shank, length to suit depth of insulation indicated.
        3) Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.

    b. Nonmetal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate fastened to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following requirements:

        1) Baseplate: Perforated, nylon sheet, 0.030 inch thick by 1-1/2 inches in diameter.
        2) Spindle: Nylon, 0.106-inch- diameter shank, length to suit depth of insulation indicated, up to 2-1/2 inches.
        3) Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.

c. Self-Sticking-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place. Comply with the following requirements:

1) Baseplate: Galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
2) Spindle: Copper- or zinc-coated, low-carbon steel or Aluminum or Stainless steel, fully annealed, 0.106-inch- diameter shank, length to suit depth of insulation indicated.
3) Adhesive-backed base with a peel-off protective cover.

d. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch-thick, galvanized-steel or aluminum or stainless-steel sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.

1) Protect ends with capped self-locking washers incorporating a spring steel insert to ensure permanent retention of cap in exposed locations.

e. Nonmetal Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch- thick nylon sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.

3. Staples: Outward-clinching insulation staples, nominal 3/4-inch- wide, stainless steel or Monel.

B. Function: None

C. Performance: None

PART 3 - EXECUTION

3.1 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of ducts and fittings.

B. Install insulation materials, vapor barriers or retarders, jackets, and thicknesses required for each item of duct system as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Keep insulation materials dry during application and finishing.

G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

H. Install insulation with least number of joints practical.

I. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.
2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

J. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

1. apply vapor-barrier mastic on seams and joints and at ends adjacent to duct flanges and fittings.

K. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

L. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

M. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

3.3 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.

4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

C. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Terminate insulation at fire damper sleeves for fire-rated wall and partition penetrations. Externally insulate damper sleeves to match adjacent insulation and overlap duct insulation at least 2 inches.

1. Comply with requirements in Section 078413 “Penetration Firestopping” for firestopping and fire-resistive joint sealers.

D. Insulation Installation at Floor Penetrations:

1. Duct: For penetrations through fire-rated assemblies, terminate insulation at fire damper sleeves and externally insulate damper sleeve beyond floor to match adjacent duct insulation. Overlap damper sleeve and duct insulation at least 2 inches.
2. Seal penetrations through fire-rated assemblies.

3.4 INSTALLATION OF MINERAL-FIBER INSULATION

A. Blanket Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.

2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.
3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:

a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.
b. On duct sides with dimensions larger than 18 inches, place pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.
c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.
d. Do not overcompress insulation during installation.
e. Impale insulation over pins and attach speed washers.
f. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.

4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from one edge and one end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch

o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.

a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.

1) Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to two times the insulation thickness, but not less than 3 inches.

5. Overlap unfaced blankets a minimum of 2 inches on longitudinal seams and end joints. At end joints, secure with steel bands spaced a maximum of 18 inches o.c.
6. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.
7. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch- wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.

3.5 INDOOR, FIELD-APPLIED JACKET INSTALLATION

A. Where FSK jackets are indicated, install as follows:

1. Draw jacket material smooth and tight.
2. Install lap or joint strips with same material as jacket.
3. Secure jacket to insulation with manufacturer's recommended adhesive.
4. Install jacket with 1-1/2-inch laps at longitudinal seams and 3-inch- wide joint strips at end joints.
5. Seal openings, punctures, and breaks in vapor-retarder jackets and exposed insulation with vapor-barrier mastic.

B. Where metal jackets are indicated, install with 2-inch overlap at longitudinal seams and end joints. Overlap longitudinal seams arranged to shed water. Seal end joints with weatherproof sealant recommended by insulation manufacturer. Secure jacket with stainless-steel bands 12 inches o.c. and at end joints.

3.6 FINISHES

A. Do not field paint aluminum or stainless-steel jackets.

3.7 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Tests and Inspections:

1. Inspect ductwork, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location(s) for each duct system defined in the "Duct Insulation Schedule, General" Article.

C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.8 DUCT INSULATION SCHEDULE, GENERAL

A. Plenums and Ducts Requiring Insulation:

1. Indoor, rectangular duct concealed supply and return air.
2. Indoor, round duct concealed supply and return air.
3. Indoor, exposed supply and return air.
4. Outdoor, exposed supply, and return.

B. Items Not Insulated:

1. Fibrous-glass ducts.
2. Exhaust ducts
3. Factory-insulated flexible ducts.
4. Factory-insulated plenums and casings.
5. Vibration-control devices.
6. Factory-insulated access panels and doors.

3.9 INDOOR DUCT AND PLENUM INSULATION SCHEDULE

A. Concealed, Rectangular Supply-Air Duct and Plenum Insulation: Mineral fiber blanket, 1.5″ exterior wrap.

B. Concealed, Rectangular Return-Air Duct and Plenum Insulation: Mineral fiber blanket, 1.5″ exterior wrap.

C. Concealed, Round Supply-Air Duct and Plenum Insulation: Mineral fiber blanket, 1.5″ exterior wrap.

D. Concealed, Round Return-Air Duct and Plenum Insulation: Mineral fiber blanket, 1.5″ exterior wrap.

E. Ductwork shall be uninsulated where noted specifically on plans.

END OF SECTION 23 07 13

SECTION 23 07 19 – HVAC PIPING INSULATIONhttp://www.productmasterspec.com/construction_specifications?mf04=230719&update=ms2021_4&utm_source=MasterSpec&utm_medium=Word

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes insulation for HVAC piping systems.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Material test reports.

C. Field quality-control reports.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or craft training program, certified by the Department of Labor, Bureau of Apprenticeship and Training.

1.5 COORDINATION

A. Coordinate clearance requirements with piping Installer for piping insulation application. Before preparing piping Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

B. Coordinate installation and testing of heat tracing.

1.6 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products in accordance with ASTM E84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation, jacket materials, adhesive, mastic, tapes, and cement material containers with appropriate markings of applicable testing agency.

1. All Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.

2.2 INSULATION MATERIALS

A. Physical:

1. Comply with requirements in “Piping Insulation Schedule, General,” “Indoor Piping Insulation Schedule,” articles for where insulating materials shall be applied.
2. Products shall not contain asbestos, lead, mercury, or mercury compounds.
3. Products that come into contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested in accordance with ASTM C871.
4. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable in accordance with ASTM C795.
5. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.
6. Cellular Glass: Inorganic, incombustible, foamed or cellulated glass with annealed, rigid, hermetically sealed cells. Comply with ASTM C552.

    a. Preformed Pipe Insulation without Jacket: Type II, Class 1, unfaced.
    b. Preformed Pipe Insulation with Jacket: Type II, Class 2, with factory-applied ASJ jacket.
    c. Fabricated shapes in accordance with ASTM C450, ASTM C585, and ASTM C1639.
    d. Factory-applied jacket requirements are specified in “Factory-Applied Jackets” Article.

7. Glass-Fiber, Preformed Pipe: Glass fibers bonded with a thermosetting resin; suitable for maximum use temperature up to 850 deg F in accordance with ASTM C411. Comply with ASTM C547.

    a. Preformed Pipe Insulation: Type I, Grade A with factory-applied ASJ.
    b. Fabricated shapes in accordance with ASTM C450 and ASTM C585.
    c. Factory-applied jacket requirements are specified in “Factory-Applied Jackets” Article.

8. Mineral Wool, Preformed Pipe: Mandrel-wound mineral wool fibers bonded with a thermosetting resin, unfaced; suitable for maximum use temperature up to 1200 deg F in accordance with ASTM C447. Comply with ASTM C547.

a. Preformed Pipe Insulation: Type II, Grade A with factory-applied ASJ.
b. Fabricated shapes in accordance with ASTM C450 and ASTM C585.

B. Function: None.

C. Performance: None.http://www.specagent.com/LookUp/?ulid=3204&mf=&src=wd

2.3 ADHESIVES

A. Physical:

1. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.
2. Cellular-Glass Adhesive: Two-component, thermosetting urethane adhesive containing no flammable solvents, with a service temperature range of minus 100 to plus 200 deg F
3. Flexible Elastomeric and Polyolefin Adhesive: Solvent-based adhesive.

    a. Flame-spread index shall be 25 or less and smoke-developed index shall be 50 or less as tested in accordance with ASTM E84.
    b. Wet Flash Point: Below 0 deg F
    c. Service Temperature Range: 40 to 200 deg F
    d. Color: Black

4. Glass-Fiber and Mineral Wool Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
5. ASJ Adhesive and FSK and PVDC Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A, for bonding insulation jacket lap seams and joints.
6. PVC Jacket Adhesive: Compatible with PVC jacket.

B. Function: None

C. Performance: None

2.4 MASTICS AND COATINGS

A. Physical:

1. Materials shall be compatible with insulation materials, jackets, and substrates.
2. Vapor-Retarder Mastic, Water Based: Suitable for indoor use on below-ambient services.

    a. Water-Vapor Permeance: Comply with ASTM E96/E96M or ASTM F1249.
    b. Service Temperature Range: 0 to plus 180 deg
    c. Comply with MIL-PRF-19565C, Type II, for permeance

3. Breather Mastic: Water based; suitable for indoor and outdoor use on above-ambient services.

    a. Water-Vapor Permeance: ASTM E96/E96M, greater than 1.0 perm at manufacturer's recommended dry film thickness.
    b. Service Temperature Range: 0 to plus 180 deg F
    c. Color: White

B. Function: None

C. Performance: None

2.5 SEALANTS

A. Physical:

1. Materials shall be as recommended by the insulation manufacturer and shall be compatible with insulation materials, jackets, and substrates.
2. Joint Sealants:

    a. Permanently flexible, elastomeric sealant.

        1) Service Temperature Range: Minus 150 to plus 250 deg F
        2) Color: White or gray.

3. FSK and Metal Jacket Flashing Sealants:

    a. Fire- and water-resistant, flexible, elastomeric sealant.
    b. Service Temperature Range: Minus 40 to plus 250 deg F
    c. Color: Aluminum.

4. ASJ Flashing Sealants and PVDC and PVC Jacket Flashing Sealants:

    a. Fire- and water-resistant, flexible, elastomeric sealant.
    b. Service Temperature Range: Minus 40 to plus 250 deg F.
    c. Color: White.

B. Function: None

C. Performance: None

2.6 FACTORY-APPLIED JACKETS

A. Physical:

1. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

    a. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C1136, Type I.
    b. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C1136, Type I.
    c. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C1136, Type II.
    d. ASJ+: Aluminum foil reinforced with glass scrim bonded to a kraft paper interleaving with an outer film leaving no paper exposed; complying with ASTM C1136, Types I, II, III, IV, and VII.

e. PSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with polyethylene backing; complying with ASTM C1136, Type II.

B. Function: None

C. Performance: None

2.7 FIELD-APPLIED JACKETS

A. Physical:

1. Field-applied jackets shall comply with ASTM C1136, Type I, unless otherwise indicated.
2. FSK Jacket: Aluminum-foil-face, fiberglass-reinforced scrim with kraft-paper backing.
3. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.

    a. Adhesive: As recommended by jacket material manufacturer.
    b. Color: White.
    c. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.

        1) Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

4. Self-Adhesive Indoor/Outdoor Jacket (Non-Asphaltic): Vapor barrier and waterproofing jacket for installation over insulation located aboveground outdoors or indoors. Specialized jacket has five layers of laminated aluminum and polyester film with low-temperature acrylic pressure-sensitive adhesive. Outer aluminum surface is coated with UV-resistant coating for protection from environmental contaminants.http://www.specagent.com/LookUp/?ulid=14583&mf=&src=wd

    a. Permeance: 0.00 perm as tested in accordance with ASTM F1249.
    b. Flamespread/Smoke Developed: 25/50 as tested in accordance with ASTM E84.
    c. Aluminum Finish: Smooth.

5. PVDC-SSL Jacket: PVDC jacket with a self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip.http://www.specagent.com/LookUp/?ulid=3238&mf=&src=wd

2.8 TAPES

A. Physical:

B. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C1136.

1. Width: 3 inches.

2. Thickness: 11.5 mils

C. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C1136.

1. Width: 3 inches.
2. Thickness: 6.5 mils.

D. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

1. Width: 2 inches.
2. Thickness: 6 mils

E. Function: None

F. Performance:

1. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C1136.

a. Adhesion: 90 ounces force/inch in width.
b. Tensile Strength: 40 lbf/inch in width.
c. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

2. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C1136.

a. Adhesion: 90 ounces force/inch in width.
b. Tensile Strength: 40 lbf/inch in width.
c. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

3. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

a. Adhesion: 64 ounces force/inch in width.
b. Tensile Strength: 18 lbf/inch in width.

2.9 SECUREMENTS

A. Physical:

1. Bands:

a. Stainless Steel: ASTM A240/A240M, Type 304; 0.015 inch thick, wide with wing seal closed seal.
b. Aluminum: ASTM B209, Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, wide with wing seal or closed seal.
c. Springs: Twin spring set constructed of stainless steel, with ends flat and slotted to accept metal bands. Spring size is determined by manufacturer for application.

2. Staples: Outward-clinching insulation staples, nominal 3/4 inch wide, stainless steel or Monel.
3. Wire: 0.080-inch nickel-copper alloy.http://www.specagent.com/LookUp/?ulid=3247&mf=&src=wd

PART 3 - EXECUTION

3.1 PREPARATION

A. Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

B. Clean and prepare surfaces to be insulated. Before insulating, apply a corrosion coating to insulated surfaces as follows:

1. Stainless Steel: Coat 300 series stainless steel with an epoxy primer 5 mils thick and an epoxy finish 5 mils thick if operating in a temperature range between 140 and 300 deg F. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.
2. Carbon Steel: Coat carbon steel operating at a service temperature of between 32 and 300 deg F with an epoxy coating. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.

C. Coordinate insulation installation with the tradesman installing heat tracing. Comply with requirements for heat tracing that apply to insulation.

D. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless steel surfaces, use demineralized water.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of piping, including fittings, valves, and specialties.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and of thicknesses required for each item of pipe system, as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, compress, or otherwise damage insulation or jacket.

D. Install insulation with longitudinal seams at top and bottom (12 o'clock and 6 o'clock positions) of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.

G. Keep insulation materials dry during storage, application, and finishing. Replace insulation materials that get wet during storage or in the installation process before being properly covered and sealed in accordance with the Contract Documents.

H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

I. Install insulation with least number of joints practical.

J. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.
2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends attached to structure with vapor-barrier mastic.
3. Install Insert materials and insulation to tightly join the Insert. Seal insulation to insulation Inserts with adhesive or sealing compound recommended by insulation material manufacturer.
4. Cover Inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.

K. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

L. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth, but not to the extent of creating wrinkles or areas of compression in the insulation.
2. Cover circumferential joints with 3-inch- wide strips, of same material as insulation jacket. Secure strips with adhesive and outward-clinching staples along both edges of strip, spaced 4 inches o.c.
3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward-clinching staples along edge at 2 inches o.c.
4. For below-ambient services, apply vapor-barrier mastic over staples.
5. Cover joints and seams with tape, in accordance with insulation material manufacturer's written instructions, to maintain vapor seal.
6. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to pipe flanges and fittings.

M. Cut insulation in a manner to avoid compressing insulation.

N. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

O. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches in similar fashion to butt joints.

P. For above-ambient services, do not install insulation to the following:

1. Vibration-control devices.
2. Testing agency labels and stamps.
3. Nameplates and data plates.

3.3 PENETRATIONS

A. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

B. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions.

1. Comply with requirements in Section 078413 “Penetration Firestopping” for firestopping and fire-resistive joint sealers.

C. Insulation Installation at Floor Penetrations:

1. Pipe: Install insulation continuously through floor penetrations.
2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 078413 “Penetration Firestopping.”

3.4 GENERAL PIPE INSULATION INSTALLATION

A. Requirements in this article generally apply to all insulation materials, except where more specific requirements are specified in various pipe insulation material installation articles below.

B. Insulation Installation on Fittings, Valves, Strainers, Flanges, Mechanical Couplings, and Unions:

1. Install insulation over fittings, valves, strainers, flanges, mechanical couplings, unions, and other specialties with continuous thermal and vapor-retarder integrity unless otherwise indicated.
2. Insulate pipe elbows using prefabricated fitting insulation made from same material and density as that of adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
3. Insulate tee fittings with prefabricated fitting insulation of same material and thickness as that used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
4. Insulate valves using prefabricated fitting insulation sectional pipe insulation of same material, density, and thickness as that used for adjacent pipe. Overlap adjoining pipe insulation by not less than 2 times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
5. Insulate strainers using prefabricated fitting insulation or sectional pipe insulation of same material, density, and thickness as that used for adjacent pipe. Overlap adjoining pipe insulation by not less than 2 times the thickness of pipe insulation, or one pipe

diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers, so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below-ambient services, provide a design that maintains vapor barrier.

6. Insulate flanges, mechanical couplings, and unions using a section of oversized preformed pipe insulation to fit. Overlap adjoining pipe insulation by not less than 2 times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Stencil or label the outside insulation jacket of each union with the word “union” matching size and color of pipe labels.
7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below-ambient services and a breather mastic for above-ambient services. Reinforce the mastic with reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.

C. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.

D. Install removable insulation covers. Installation shall conform to the following:

1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as that of adjoining pipe insulation.
2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union at least 2 times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless steel or aluminum bands. Select band material compatible with insulation and jacket.
3. Construct removable valve insulation covers in same manner as for flanges, except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.5 INSTALLATION OF CELLULAR-GLASS INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of insulation to pipe with wire or bands, and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with jackets on above-ambient services, secure laps with outward-clinched staples at 6 inches o.c.

4. For insulation with jackets on below-ambient services, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive, as recommended by insulation material manufacturer, and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install prefabricated pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of cellular-glass block insulation of same thickness as that of pipe insulation. Where voids are difficult to fill with block insulation, fill the voids with a fibrous insulation material suitable for the specific operating temperature.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install prefabricated sections of same material as that of straight segments of pipe insulation when available. Secure according to manufacturer's written instructions.
2. When preformed sections of insulation are not available, install mitered or routed sections of cellular-glass insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install prefabricated sections of cellular-glass insulation to valve body.
2. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
3. Install insulation to flanges as specified for flange insulation application.

3.6 INSTALLATION OF GLASS-FIBER AND MINERAL WOOL INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands, and tighten bands without deforming insulation materials.
2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
3. For insulation with jackets on above-ambient surfaces, secure laps with outward-clinched staples at 6 inches o.c.
4. For insulation with jackets on below-ambient surfaces, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive, as recommended by insulation material manufacturer, and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install prefabricated pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.

3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with glass-fiber or mineral-wool blanket insulation.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install prefabricated sections of same material as that of straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:

1. Install prefabricated sections of same material as that of straight segments of pipe insulation when available.
2. When prefabricated sections are not available, install fabricated sections of pipe insulation to valve body.
3. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
4. Install insulation to flanges as specified for flange insulation application.

3.7 FINISHES

A. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

B. Color: White, or directed otherwise by the Contracting Officer. Vary first and second coats to allow visual inspection of the completed Work.

C. Do not field paint aluminum or stainless steel jackets.

3.8 FIELD QUALITY CONTROL

A. Owner will engage a qualified testing agency to perform tests and inspections.

B. Engage a qualified testing agency to perform tests and inspections.

C. All insulation applications will be considered defective if they do not pass tests and inspections.

D. Prepare test and inspection reports.

3.9 PIPING INSULATION SCHEDULE, GENERAL

A. Insulation conductivity and thickness per pipe size shall comply with schedules in this Section or with requirements of authorities having jurisdiction, whichever is more stringent.

B. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.

C. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:

1. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.10 INDOOR PIPING INSULATION SCHEDULE

A. Refrigerant Suction and Hot-Gas Piping:

1. All Pipe Sizes: Insulation is one of the following:

a. Cellular Glass: 1-1/2 inches thick.
b. Glass-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
c. Mineral Wool, Preformed Pipe Insulation, Type II: 1 inch thick.

B. Refrigerant Liquid Piping:

1. All Pipe Sizes: Insulation is one of the following:

a. Cellular Glass: 1-1/2 inches thick.
b. Glass-Fiber, Preformed Pipe Insulation, Type I: 1 inch thick.
c. Mineral Wool, Preformed Pipe Insulation, Type II: 1 inch thick.

3.11 INDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

END OF SECTION 23 07 19

SECTION 23 11 26 – FACILITY LIQUEFIED-PETROLEUM GAS PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Pipes, tubes, and fittings.
2. Piping specialties.
3. Piping and tubing joining materials.
4. Valves.
5. Pressure regulators.

1.2 PERFORMANCE REQUIREMENTS

A. Minimum Operating-Pressure Ratings:

1. For Piping Containing Only Vapor:

a. Piping and Valves: 125 psig unless otherwise indicated.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Data: Submit certification that vaporizer, storage container supports, accessories, and components will withstand seismic forces defined in Section 230548 "Vibration and Seismic Controls for HVAC." Include the following:

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

B. Welding certificates.

C. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

1.6 QUALITY ASSURANCE

A. Steel Support Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, “Structural Welding Code – Steel.”

B. Pipe Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

PART 2 - PRODUCTS

2.1 PIPES, TUBES, AND FITTINGS

A. Steel Pipe: ASTM A 53/A 53M, black steel, Schedules 40 and 80, Type E or S, Grade B.

1. Malleable-Iron Threaded Fittings: ASME B16.3, Class 150, standard pattern.
2. Wrought-Steel Welding Fittings: ASTM A 234/A 234M for butt welding and socket welding.
3. Unions: ASME B16.39, Class 150, malleable iron with brass-to-iron seat, ground joint, and threaded ends.

4. Protective Coating for Underground Piping: Factory-applied, three-layer coating of epoxy, adhesive, and PE.

    a. Joint Cover Kits: Epoxy paint, adhesive, and heat-shrink PE sleeves.

B. PE Pipe: ASTM D 2513, SDR 11.

1. PE Fittings: ASTM D 2683, socket-fusion type or ASTM D 3261, butt-fusion type with dimensions matching PE pipe.
2. PE Transition Fittings: Factory-fabricated fittings with PE pipe complying with ASTM D 2513, SDR 11; and steel pipe complying with ASTM A 53/A 53M, black steel, Schedule 40, Type E or S, Grade B.

3. Anodeless Service-Line Risers: Factory fabricated and leak tested.

    a. Underground Portion: PE pipe complying with ASTM D 2513, SDR 11 inlet.
    b. Casing: Steel pipe complying with ASTM A 53/A 53M, Schedule 40, black steel, Type E or S, Grade B with corrosion-protective coating covering.
    c. Aboveground Portion: PE transition fitting.
    d. Outlet shall be threaded suitable for welded connection.
    e. Tracer wire connection.

f. Ultraviolet shield.
g. Stake supports with factory finish to match steel pipe casing or carrier pipe.

4. Transition Service-Line Risers: Factory fabricated and leak tested.

a. Underground Portion: PE pipe complying with ASTM D 2513, SDR 11 inlet connected to steel pipe complying with ASTM A 53/A 53M, Schedule 40, Type E or S, Grade B, with corrosion-protective coating for aboveground outlet.
b. Outlet shall be threaded suitable for welded connection.
c. Bridging sleeve over mechanical coupling.
d. Factory-connected anode.
e. Tracer wire connection.
f. Ultraviolet shield.
g. Stake supports with factory finish to match steel pipe casing or carrier pipe.

2.2 PIPING SPECIALTIES

A. Flexible Piping Joints:

1. Approved for LPG service.
2. Stainless-steel bellows with woven, flexible, bronze, wire-reinforcing protective jacket.
3. Minimum working pressure of 250 psig and 250 deg F operating temperature.
4. Maximum 36-inch length for liquid LPG lines.

B. Y-Pattern Strainers:

1. Body: ASTM A 126, Class B, cast iron with bolted cover and bottom drain connection.
2. End Connections: Threaded ends for NPS 2 (DN 50) and smaller.
3. Strainer Screen: [40] [60]-mesh startup strainer and perforated stainless-steel basket with 50 percent free area.
4. CWP Rating: 125 psig (862 kPa).

2.3 JOINING MATERIALS

A. Welding Filler Metals: Comply with AWS D10.12/D10.12M for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

2.4 MANUAL GAS SHUTOFF VALVES

A. Two-Piece, Full-Port, Bronze Ball Valves with Bronze Trim: MSS SP-110.

1. Body: Bronze, complying with ASTM B 584.
2. Ball: Chrome-plated bronze.
3. Stem: Bronze; blowout proof.
4. Seats: Reinforced TFE; blowout proof.
5. Packing: Threaded-body packnut design with adjustable-stem packing.
6. Ends: Threaded, flared, or socket as indicated in “Underground Manual Gas Shutoff Valve Schedule” and “Aboveground Manual Gas Shutoff Valve Schedule” Articles.

7. CWP Rating: 600 psig
8. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
9. Service: Suitable for LPG service with "WOG" indicated on valve body.

B. Bronze Plug Valves: MSS SP-78.

1. Body: Bronze, complying with ASTM B 584.
2. Plug: Bronze.
3. Ends: Threaded or socket as indicated in "Underground Manual Gas Shutoff Valve Schedule" and "Aboveground Manual Gas Shutoff Valve Schedule" Articles.
4. Operator: Square head or lug type with tamperproof feature where indicated.
5. Pressure Class: 125 psig.
6. Listing: Valves NPS 1 and smaller shall be listed and labeled by an NRTL acceptable to authorities having jurisdiction.
7. Service: Suitable for LPG service with "WOG" indicated on valve body.

C. PE Ball Valves: Comply with ASME B16.40.

1. Body: PE.
2. Ball: PE.
3. Stem: Acetal.
4. Seats and Seals: Nitrile.
5. Ends: Plain or fusible to match piping.
6. CWP Rating: 80 psig
7. Operating Temperature: Minus 20 to plus 140 deg F
8. Operator: Nut or flat head for key operation.
9. Include plastic valve extension.
10. Include tamperproof locking feature for valves where indicated on Drawings.

D. Valve Boxes:

1. Cast-iron, two-section box.
2. Top section with cover with "GAS" lettering.
3. Bottom section with base to fit over valve and barrel a minimum of 5 inchesin diameter.
4. Adjustable cast-iron extensions of length required for depth of bury.
5. Include tee-handle, steel operating wrench with socket end fitting valve nut or flat head and with stem of length required to operate valve.

2.5 EARTHQUAKE VALVES

A. Earthquake Valves: Comply with ASCE 25.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. Pacific Seismic Products, Inc.
b. Or Equal.

2. Listing: Listed and labeled by an NRTL acceptable to authorities having jurisdiction.
3. Maximum Operating Pressure: 60 psig.
4. Cast-aluminum body with stainless-steel internal parts.
5. Nitrile-rubber, reset-stem o-ring seal.
6. Valve position, open or closed, indicator.
7. Composition valve seat with clapper held by spring or magnet locking mechanism.
8. Level indicator.
9. End Connections: Threaded for valves NPS 4 and smaller.

2.6 EARTHQUAKE VALVES

A. Earthquake Valves, Maximum Operating Pressure of 60 psig (414 kPa): Comply with ASCE 25.

1. Listing: Listed and labeled by an NRTL acceptable to authorities having jurisdiction.
2. Maximum Operating Pressure: [0.5 psig (3.45 kPa)] [7 psig (48 kPa)] [60 psig (414 kPa)].
3. Cast-aluminum body with stainless-steel internal parts.
4. Nitrile-rubber, reset-stem o-ring seal.
5. Valve position, open or closed, indicator.
6. Composition valve seat with clapper held by spring or magnet locking mechanism.
7. Level indicator.
8. End Connections: Threaded for valves NPS 2 (DN 50) and smaller.

2.7 PRESSURE REGULATORS

A. General Requirements:

1. Single stage and suitable for LPG.
2. Steel jacket and corrosion-resistant components.
3. Elevation compensator.
4. End Connections: Threaded for regulators NPS 2 (DN 50) and smaller.

B. Line Pressure Regulators: Comply with ANSI Z21.80.

1. Body and Diaphragm Case: Cast iron or die-cast aluminum.
2. Springs: Zinc-plated steel; interchangeable.
3. Diaphragm Plate: Zinc-plated steel.
4. Seat Disc: Nitrile rubber resistant to gas impurities, abrasion, and deformation at the valve port.
5. Orifice: Aluminum; interchangeable.
6. Seal Plug: Ultraviolet-stabilized, mineral-filled nylon.
7. Single-port, self-contained regulator with orifice no larger than required at maximum pressure inlet and no pressure sensing piping external to the regulator.
8. Pressure regulator shall maintain discharge pressure setting downstream and not exceed 150 percent of design discharge pressure at shutoff.
9. Overpressure Protection Device: Factory mounted on pressure regulator.

10. Atmospheric Vent: Factory- or field-installed, stainless-steel screen in opening if not connected to vent piping.
11. Maximum Inlet Pressure: [2 psig (13.8 kPa)] [5 psig (34.5 kPa)] [10 psig (69 kPa)].

2.8 LABELING AND IDENTIFYING

A. Detectable Warning Tape: Acid- and alkali-resistant PE film warning tape manufactured for marking and identifying underground utilities, a minimum of 6 incheS wide and 4 mils thick, continuously inscribed with a description of utility, with metallic core encased in a protective jacket for corrosion protection, detectable by metal detector when tape is buried up to 30 inches deep; colored yellow.

PART 3 - EXECUTION

3.1 EARTHWORK

A. Comply with requirements in Section 312000 “Earth Moving” for excavating, trenching, and backfilling.

3.2 OUTDOOR PIPING INSTALLATION

A. Comply with NFPA 58 and the International Fuel Gas Code requirements for installation and purging of LPG piping.

B. NFPA 54 requires a minimum of 18 inches of cover over buried LPG piping, or 12 inches with shielding. Pipe with less than 12 inches of cover must be installed in a containment conduit.

C. Install underground, LPG piping buried at least 36 inches below finished grade. Comply with requirements in Section 312000 “Earth Moving” for excavating, trenching, and backfilling.

1. If LPG piping is installed less than 36 inches below finished grade, install it in containment conduit.

D. Install underground, PE, LPG piping according to ASTM D 2774.

E. Steel Piping with Protective Coating:

1. Apply joint cover kits to pipe after joining to cover, seal, and protect joints.
2. Repair damage to PE coating on pipe as recommended in writing by protective coating manufacturer.
3. Replace pipe having damaged PE coating with new pipe.

F. CSA B149.1 requires protective coating for Type G and Type L (Type B) copper pipe and tube installed underground.

G. Install fittings for changes in direction and branch connections.

H. Install pressure gage upstream and downstream from each service regulator.

3.3 3.03 VALVE INSTALLATION

A. Install manual gas shutoff valve for each gas appliance.

B. Install underground valves with valve boxes.

C. Install regulators and overpressure protection devices with maintenance access space adequate for servicing and testing.

D. Install earthquake valves aboveground outside buildings according to listing.

E. Install anode for metallic valves in underground PE piping.

3.4 3.05 PIPING JOINT CONSTRUCTION

A. Ream ends of pipes and tubes and remove burrs.

B. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

C. Threaded Joints:

1. Thread pipe with tapered pipe threads complying with ASME B1.20.1.
2. Cut threads full and clean using sharp dies.
3. Ream threaded pipe ends to remove burrs and restore full ID of pipe.
4. Apply appropriate tape or thread compound to external pipe threads unless dryseal threading is specified.
5. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.

D. Welded Joints:

1. Construct joints according to AWS D10.12/D10.12M, using qualified processes and welding operators.
2. Bevel plain ends of steel pipe.
3. Patch factory-applied protective coating as recommended by manufacturer at field welds and where damage to coating occurs during construction.

E. PE Piping Heat-Fusion Joints: Clean and dry joining surfaces by wiping with clean cloth or paper towels. Join according to ASTM D 2657.

1. Plain-End Pipe and Fittings: Use butt fusion.
2. Plain-End Pipe and Socket Fittings: Use socket fusion.

3.5 3.06 HANGER AND SUPPORT INSTALLATION

A. Install seismic restraints on piping.

B. Comply with requirements for pipe hangers and supports.

C. Install hangers for horizontal steel piping with the following maximum spacing and minimum rod sizes:

1. NPS 1 and Smaller: Maximum span, 96 inches; minimum rod size, 3/8 inch.
2. NPS 1-1/4: Maximum span, 108 inches; minimum rod size, 3/8 inch.
3. NPS 1-1/2 and NPS 2: Maximum span, 108 inches; minimum rod size, 3/8 inch.

3.6 LABELING AND IDENTIFYING

A. Install detectable warning tape directly above gas piping, 12 inches below finished grade, except 6 inches below subgrade under pavements and slabs.

3.7 3.13 FIELD QUALITY CONTROL

A. Test, inspect, and purge LPG according to NFPA 58 and the International Fuel Gas Code and requirements of authorities having jurisdiction.

B. LPG piping will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

3.8 OUTDOOR PIPING SCHEDULE

A. Aboveground LPG liquid piping shall be the following:

1. NPS 2 and Smaller: Schedule 40 steel pipe, malleable-iron threaded fittings and threaded and seal welded joints. Coat pipe and fittings with protective coating for steel piping. Paint piping, color per architectural plans.

B. Underground LPG vapor piping shall be one of the following:

1. PE pipe and fittings joined by heat-fusion; service-line risers with tracer wire terminated in an accessible location.

3.9 INDOOR PIPING SCHEDULE FOR SYSTEM PRESSURES MORE THAN 0.5 PSIG AND LESS THAN 5 PSIG

A. Aboveground LPG liquid piping shall be the following:

1. NPS 2 and Smaller: Schedule 40 steel pipe, malleable-iron threaded fittings and threaded and seal welded joints. Coat pipe and fittings with protective coating for steel piping.

B. Underground, below building, LPG vapor piping shall be one of the following:

1. PE pipe and fittings joined by heat-fusion; service-line risers with tracer wire terminated in an accessible location.

3.10 ABOVEGROUND MANUAL GAS SHUTOFF VALVE SCHEDULE

A. Aboveground Liquid Piping:

1. Two-piece, full-port, bronze ball valves with bronze trim.

B. Distribution piping valves for pipe NPS 2 and smaller shall be[one of] the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

C. Valves in branch piping for single appliance shall be[one of] the following:

1. Two-piece, full-port, bronze ball valves with bronze trim.
2. Bronze plug valve.

END OF SECTION 23 11 26

SECTION 23 23 00 – REFRIGERANT PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Refrigerant pipes and fittings.
2. Refrigerant piping valves and specialties.
3. Refrigerants.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of valve, refrigerant piping, and refrigerant piping specialty.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

1.4 QUALITY ASSURANCE

A. Comply with ASHRAE 15, “Safety Code for Refrigeration Systems.”

B. Comply with ASME B31.5, “Refrigeration Piping and Heat Transfer Components.”

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Line Test Pressure for Refrigerant R-454B:

1. Suction Tubing for Refrigeration and Air-Conditioning Applications Other than Heat Pumps: 300 psig.
2. Suction Tubing for Heat-Pump Applications: 535 psig.
3. Hot-Gas and Tubing Lines: 535 psig.

2.2 PHYSICAL REQUIREMENTS

A. COPPER TUBE AND FITTINGS

B. Copper Tube: ASTM B 88, or L.

C. Wrought-Copper Fittings: ASME B16.22.

D. Wrought-Copper Unions: ASME B16.22.

E. Solder Filler Metals: ASTM B 32. Use 95-5 tin antimony or alloy HB solder to join copper socket fittings on copper pipe.

F. Brazing Filler Metals: AWS A5.8/A5.8M.

G. Flexible Connectors:

1. Body: Tin-bronze bellows with woven, flexible, tinned-bronze-wire-reinforced protective jacket.
2. End Connections: Socket ends.
3. Offset Performance: Capable of minimum 3/4-inch misalignment in minimum 7-inch-long assembly.
4. Working Pressure Rating: Factory test at minimum 500 psig.
5. Maximum Operating Temperature: 250 deg F.

H. Copper Pressure-Seal Fitting for Refrigerant Piping:

1. Standard: UL 207; certified by UL for field installation. Certification as a UL-recognized component alone is unacceptable.
2. Housing: Copper.
3. O-Rings: HNBR or compatible with specific refrigerant.
4. Tools: Manufacturer’s approved special tools.
5. Minimum Rated Pressure: 700 psig.

I. VALVES AND SPECIALTIES

1. Diaphragm Packless Valves:

a. Body and Bonnet: Forged brass or cast bronze; globe design with straight-through or angle pattern.
b. Diaphragm: Phosphor bronze and stainless steel with stainless-steel spring.
c. Operator: Rising stem and hand wheel.
d. Seat: Nylon.
e. End Connections: Socket, union, or flanged.
f. Working Pressure Rating: 500 psig.
g. Maximum Operating Temperature: 275 deg F.

2. Packed-Angle Valves:

a. Body and Bonnet: Forged brass or cast bronze.
b. Packing: Molded stem, back seating, and replaceable under pressure.
c. Operator: Rising stem.
d. Seat: Nonrotating, self-aligning polytetrafluoroethylene.
e. Seal Cap: Forged-brass or valox hex cap.
f. End Connections: Socket, union, threaded, or flanged.
g. Working Pressure Rating: 500 psig.
h. Maximum Operating Temperature: 275 deg F.

3. Check Valves:

    a. Body: Ductile iron, forged brass, or cast bronze; globe pattern.
    b. Bonnet: Bolted ductile iron, forged brass, or cast bronze; or brass hex plug.
    c. Piston: Removable polytetrafluoroethylene seat.
    d. Closing Spring: Stainless steel.
    e. Retain "Manual Opening Stem" Subparagraph below for optional manual opening feature.
    f. Manual Opening Stem: Seal cap, plated-steel stem, and graphite seal.
    g. End Connections: Socket, union, threaded, or flanged.
    h. Maximum Opening Pressure: 0.50 psig.
    i. Working Pressure Rating: 500 psig.
    j. Maximum Operating Temperature: 275 deg F.

4. Service Valves:

    a. Body: Forged brass with brass cap including key end to remove core.
    b. Core: Removable ball-type check valve with stainless-steel spring.
    c. Seat: Polytetrafluoroethylene.
    d. End Connections: Copper spring.
    e. Working Pressure Rating: 500 psig.

5. Solenoid Valves: Comply with AHRI 760 and UL 429; listed and labeled by a National Recognized Testing Laboratory (NRTL).

    a. Body and Bonnet: Plated steel.
    b. Solenoid Tube, Plunger, Closing Spring, and Seat Orifice: Stainless steel.
    c. Seat: Polytetrafluoroethylene.
    d. End Connections: Threaded.
    e. Electrical: Molded, watertight coil in NEMA 250 enclosure of type required by location with 1/2-inch conduit adapter, and 24-V ac coil.
    f. Working Pressure Rating: 400 psig.
    g. Maximum Operating Temperature: 240 deg F.

6. Safety Relief Valves: Comply with 2010 ASME Boiler and Pressure Vessel Code; listed and labeled by an NRTL.

    a. Body and Bonnet: Ductile iron and steel, with neoprene O-ring seal.
    b. Piston, Closing Spring, and Seat Insert: Stainless steel.
    c. Seat: Polytetrafluoroethylene.
    d. End Connections: Threaded.
    e. Working Pressure Rating: 400 psig.
    f. Maximum Operating Temperature: 240 deg F.

7. Straight-Type Strainers:

    a. Body: Welded steel with corrosion-resistant coating.
    b. Screen: 100-mesh stainless steel.
    c. End Connections: Socket or flare.
    d. Working Pressure Rating: 500 psig.
    e. Maximum Operating Temperature: 275 deg F.

8. Angle-Type Strainers:

   a. Body: Forged brass or cast bronze.
   b. Drain Plug: Brass hex plug.
   c. Screen: 100-mesh monel.
   d. End Connections: Socket or flare.
   e. Working Pressure Rating: 500 psig.
   f. Maximum Operating Temperature: 275 deg F.

9. Moisture/Liquid Indicators:

   a. Body: Forged brass.
   b. Window: Replaceable, clear, fused glass window with indicating element protected by filter screen.
   c. Indicator: Color coded to show moisture content in parts per million (ppm).
   d. Minimum Moisture Indicator Sensitivity: Indicate moisture above 60 ppm.
   e. End Connections: Socket or flare.
   f. Working Pressure Rating: 500 psig.
   g. Maximum Operating Temperature: 240 deg F.

2.3 FUNCTIONAL REQUIREMENTS:

A. None.

2.4 REFRIGERANTS

A. ASHRAE 34, R-454

PART 3 - EXECUTION

3.1 PIPING APPLICATIONS FOR REFRIGERANT R-454B

A. Suction Lines: Copper, Type ACR, annealed-temper tubing and wrought-copper fittings with brazed or soldered joints.

B. Hot-Gas and Liquid Lines: Copper, Type ACR, annealed- or drawn-temper tubing and wrought-copper fittings with brazed or soldered joints.

C. Safety-Relief-Valve Discharge Piping: Copper, Type ACR, annealed- or drawn-temper tubing and wrought-copper fittings with brazed or soldered joints.

3.2 VALVE AND SPECIALTY APPLICATIONS

A. Install service valves for gage taps at inlet and outlet of hot-gas bypass valves and strainers if they are not an integral part of valves and strainers.

B. Install a check valve at the compressor discharge and a liquid accumulator at the compressor suction connection.

C. Install safety relief valves where required by 2010 ASME Boiler and Pressure Vessel Code. Pipe safety-relief-valve discharge line to outside according to ASHRAE 15.

D. Install moisture/liquid indicators in liquid line at the inlet of the thermostatic expansion valve or at the inlet of the evaporator coil capillary tube.

E. Install strainers upstream from and adjacent to the following unless they are furnished as an integral assembly for the device being protected:

1. Compressor.

F. Install filter dryers in liquid line between compressor and thermostatic expansion valve.

G. Install receivers sized to accommodate pump-down charge.

H. Install flexible connectors at compressors.

3.3 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems; indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations. Install piping as indicated unless deviations to layout are approved on Shop Drawings.

B. Install refrigerant piping according to ASHRAE 15.

C. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.

D. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

E. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.

F. Install piping adjacent to machines to allow service and maintenance.

G. Install piping free of sags and bends.

H. Install fittings for changes in direction and branch connections.

I. Select system components with pressure rating equal to or greater than system operating pressure.

J. Install piping as short and direct as possible, with a minimum number of joints, elbows, and fittings.

K. Install refrigerant piping in protective conduit where installed belowground.

L. Install refrigerant piping in rigid or flexible conduit in locations where exposed to mechanical injury.

M. Slope refrigerant piping as follows:

1. Install horizontal hot-gas discharge piping with a uniform slope downward away from compressor.
2. Install horizontal suction lines with a uniform slope downward to compressor.
3. Install traps and double risers to entrain oil in vertical runs.
4. Liquid lines may be installed level.

N. When brazing or soldering, remove solenoid-valve coils and sight glasses; also remove valve stems, seats, and packing, and accessible internal parts of refrigerant specialties. Do not apply heat near expansion-valve bulb.

O. Install piping with adequate clearance between pipe and adjacent walls and hangers or between pipes for insulation installation.

P. Identify refrigerant piping and valves according to Section 230553 “Identification for HVAC Piping and Equipment.”

Q. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 230517 “Sleeves and Sleeve Seals for HVAC Piping.”

R. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 230517 “Sleeves and Sleeve Seals for HVAC Piping.”

S. Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Section 230518 “Escutcheons for HVAC Piping.”

3.4 PIPE JOINT CONSTRUCTION

A. Ream ends of pipes and tubes and remove burrs.

B. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

C. Soldered Joints: Construct joints according to ASTM B 828 or CDA’s “Copper Tube Handbook.”

D. Brazed Joints: Construct joints according to AWS’s “Brazing Handbook,” Chapter “Pipe and Tube.”

1. Use Type BCuP (copper-phosphorus) alloy for joining copper socket fittings with copper pipe.
2. Use Type BAg (cadmium-free silver) alloy for joining copper with bronze or steel.

3.5 INSTALLATION OF HANGERS AND SUPPORTS

A. Comply with requirements for seismic restraints in Section 230548 “Vibration and Seismic Controls for HVAC.”

B. Comply with Section 230529 “Hangers and Supports for HVAC Piping and Equipment” for hangers, supports, and anchor devices.

C. Install the following pipe attachments:

1. Adjustable steel clevis hangers for individual horizontal runs less than 20 feet long.
2. Roller hangers and spring hangers for individual horizontal runs 20 feet or longer.
3. Pipe Roller: MSS SP-58, Type 44 for multiple horizontal piping 20 feet or longer, supported on a trapeze.
4. Spring hangers to support vertical runs.
5. Copper-clad hangers and supports for hangers and supports in direct contact with copper pipe.

D. Install hangers for copper tubing, with maximum horizontal spacing and minimum rod diameters, to comply with MSS-58, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

E. Support horizontal piping within 12 inches of each fitting.

F. Support vertical runs of copper tubing to comply with MSS-58, locally enforced codes, and authorities having jurisdiction requirements, whichever are most stringent.

3.6 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Comply with ASME B31.5, Chapter VI.
2. Test refrigerant piping, specialties, and receivers. Isolate compressor, condenser, evaporator, and safety devices from test pressure if they are not rated above the test pressure.
3. Test high- and low-pressure side piping of each system separately at not less than the pressures indicated in “Performance Requirements” Article.

    a. Fill system with nitrogen to the required test pressure.
    b. System shall maintain test pressure at the manifold gage throughout duration of test.
    c. Test joints and fittings with electronic leak detector or by brushing a small amount of soap and glycerin solution over joints.
    d. Remake leaking joints using new materials, and retest until satisfactory results are achieved.

B. Prepare test and inspection reports.

3.7 SYSTEM CHARGING

A. Charge system using the following procedures:

1. Install core in filter dryers after leak test but before evacuation.
2. Evacuate entire refrigerant system with a vacuum pump to 500 micrometers. If vacuum holds for 12 hours, system is ready for charging.
3. Break vacuum with refrigerant gas, allowing pressure to build up to 2 psig.
4. Charge system with a new filter-dryer core in charging line.

3.8 ADJUSTING

A. Adjust thermostatic expansion valve to obtain proper evaporator superheat.

B. Adjust high- and low-pressure switch settings to avoid short cycling in response to fluctuating suction pressure.

C. Adjust set-point temperature of air-conditioning or chilled-water controllers to the system design temperature.

D. Perform the following adjustments before operating the refrigeration system, according to manufacturer's written instructions:

1. Open shutoff valves in condenser water circuit.
2. Verify that compressor oil level is correct.
3. Open compressor suction and discharge valves.
4. Open refrigerant valves except bypass valves that are used for other purposes.
5. Check open compressor-motor alignment and verify lubrication for motors and bearings.

E. Replace core of replaceable filter dryer after system has been adjusted and after design flow rates and pressures are established.

END OF SECTION 23 23 00

SECTION 23 31 13 – METAL DUCTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Rectangular ducts and fittings.
2. Round ducts and fittings.
3. Sheet metal materials.
4. Sealants and gaskets.
5. Hangers and supports.

B. Related Sections:

1. Section 230593 “Testing, Adjusting, and Balancing for HVAC” for testing, adjusting, and balancing requirements for metal ducts.
2. Section 233300 “Air Duct Accessories” for dampers, sound-control devices, duct-mounting access doors and panels, turning vanes, and flexible ducts.

1.2 PERFORMANCE REQUIREMENTS

A. Structural Performance: Duct hangers and supports shall withstand the effects of gravity loads and stresses within limits and under conditions described in SMACNA’s “HVAC Duct Construction Standards – Metal and Flexible” or ASCE/SEI 7.

B. Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ANSI/ASHRAE 62.1.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Shop Drawings: For first and second floor mechanical rooms

1. Fabrication, assembly, and installation, including plans, elevations, sections, components, and attachments to other work.
2. Factory- and shop-fabricated ducts and fittings.
3. Duct layout indicating sizes, configuration, and static-pressure classes.
4. Elevation of top of ducts.
5. Dimensions of main duct runs from building grid lines.
6. Fittings.
7. Reinforcement and spacing.
8. Seam and joint construction.
9. Penetrations through fire-rated and other partitions.
10. Equipment installation based on equipment being used on Project.

11. Locations for duct accessories, including dampers, turning vanes, and access doors and panels.
12. Hangers and supports, including methods for duct and building attachment and vibration isolation.

1.4 INFORMATIONAL SUBMITTALS

A. Welding certificates.

1.5 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel according to the following:

1. AWS D1.1/D1.1M, "Structural Welding Code – Steel," for hangers and supports.
2. AWS D1.2/D1.2M, "Structural Welding Code – Aluminum," for aluminum supports.
3. AWS D9.1M/D9.1, "Sheet Metal Welding Code," for duct joint and seam welding.

B. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1, Section 5 – "Systems and Equipment" and Section 7 – "Construction and System Start-up."

C. ASHRAE/IES Compliance: Applicable requirements in ASHRAE/IES 90.1, Section 6.4.4 – "HVAC System Construction and Insulation."

PART 2 - PRODUCTS

2.1 PHYSICAL REQUIREMENTS

A. RECTANGULAR DUCTS AND FITTINGS

1. General Fabrication Requirements: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible" based on indicated static-pressure class unless otherwise indicated.
2. Transverse Joints: Select joint types and fabricate according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 2-1, "Rectangular Duct/Transverse Joints," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."
3. Longitudinal Seams: Select seam types and fabricate according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 2-2, "Rectangular Duct/Longitudinal Seams," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."
4. Elbows, Transitions, Offsets, Branch Connections, and Other Duct Construction: Select types and fabricate according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Chapter 4, "Fittings and Other Construction," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

B. ROUND DUCTS AND FITTINGS

1. General Fabrication Requirements: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Chapter 3, "Round, Oval, and Flexible Duct," based on indicated static-pressure class unless otherwise indicated.
2. Transverse Joints: Select joint types and fabricate according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 3-1, "Round Duct Transverse Joints," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

    a. Transverse Joints in Ducts Larger Than 60 Inches in Diameter: Flanged.

3. Longitudinal Seams: Select seam types and fabricate according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 3-2, "Round Duct Longitudinal Seams," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

    a. Fabricate round ducts larger Than 90 inches in diameter with butt-welded longitudinal seams.

4. Tees and Laterals: Select types and fabricate according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 3-5, "90 Degree Tees and Laterals," and Figure 3-6, "Conical Tees," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

C. SHEET METAL MATERIALS

1. General Material Requirements: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.
2. Galvanized Sheet Steel: Comply with ASTM A 653/A 653M.

    a. Galvanized Coating Designation: G90.
    b. Finishes for Surfaces Exposed to View: Mill phosphatized.

3. Reinforcement Shapes and Plates: ASTM A 36/A 36M, steel plates, shapes, and bars; black and galvanized.

    a. Where black- and galvanized-steel shapes and plates are used to reinforce aluminum ducts, isolate the different metals with butyl rubber, neoprene, or EPDM gasket materials.

4. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.

D. SEALANT AND GASKETS

1. General Sealant and Gasket Requirements: Surface-burning characteristics for sealants and gaskets shall be a maximum flame-spread index of 25 and a maximum smoke-developed index of 50 when tested according to UL 723; certified by an NRTL.
2. Two-Part Tape Sealing System:

    a. Tape: Woven cotton fiber impregnated with mineral gypsum and modified acrylic/silicone activator to react exothermically with tape to form hard, durable, airtight seal.
    b. Tape Width: 4 inches.
    c. Sealant: Modified styrene acrylic.
    d. Water resistant.
    e. Mold and mildew resistant.
    f. Maximum Static-Pressure Class: 10-inch wg, positive and negative.
    g. Service: Indoor and outdoor.
    h. Service Temperature: Minus 40 to plus 200 deg F.
    i. Substrate: Compatible with galvanized sheet steel (both PVC coated and bare), stainless steel, or aluminum.
    j. For indoor applications, sealant shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
    k. Sealant shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

3. Water-Based Joint and Seam Sealant:

    a. Application Method: Brush on.
    b. Solids Content: Minimum 65 percent.
    c. Shore A Hardness: Minimum 20.
    d. Water resistant.
    e. Mold and mildew resistant.
    f. VOC: Maximum 75 g/L (less water).
    g. Maximum Static-Pressure Class: 10-inch wg, positive and negative.
    h. Service: Indoor or outdoor.
    i. Substrate: Compatible with galvanized sheet steel (both PVC coated and bare), stainless steel, or aluminum sheets.

4. Flanged Joint Sealant: Comply with ASTM C 920.

    a. General: Single-component, acid-curing, silicone, elastomeric.
    b. Type: S.
    c. Grade: NS.
    d. Class: 25.
    e. Use: O.
    f. For indoor applications, sealant shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

g. Sealant shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

5. Flange Gaskets: Butyl rubber, neoprene, or EPDM polymer with polyisobutylene plasticizer.
6. Round Duct Joint O-Ring Seals:

   a. Seal shall provide maximum leakage class of 3 cfm/100 sq. ft. at 1-inch wg and shall be rated for10-inch wg static-pressure class, positive or negative.
   b. EPDM O-ring to seal in concave bead in coupling or fitting spigot.
   c. Double-lipped, EPDM O-ring seal, mechanically fastened to factory-fabricated couplings and fitting spigots.

E. HANGERS AND SUPPORTS

1. Hanger Rods for Noncorrosive Environments: Cadmium-plated steel rods and nuts.
2. Hanger Rods for Corrosive Environments: Electrogalvanized, all-thread rods or galvanized rods with threads painted with zinc-chromate primer after installation.
3. Strap and Rod Sizes: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Table 5-1, "Rectangular Duct Hangers Minimum Size," and Table 5-2, "Minimum Hanger Sizes for Round Duct."
4. Steel Cables for Galvanized-Steel Ducts: Galvanized steel complying with ASTM A 603.
5. Steel Cables for Stainless-Steel Ducts: Stainless steel complying with ASTM A 492.
6. Steel Cable End Connections: Cadmium-plated steel assemblies with brackets, swivel, and bolts designed for duct hanger service; with an automatic-locking and clamping device.
7. Duct Attachments: Sheet metal screws, blind rivets, or self-tapping metal screws; compatible with duct materials.
8. Trapeze and Riser Supports:

   a. Supports for Galvanized-Steel Ducts: Galvanized-steel shapes and plates.
   b. Supports for Stainless-Steel Ducts: Stainless-steel shapes and plates.
   c. Supports for Aluminum Ducts: Aluminum or galvanized steel coated with zinc chromate.

2.2 FUNCTIONAL REQUIREMENTS:

1. None.

2.3 PERFORMANCE REQUIREMENTS:

1. None.

PART 3 - EXECUTION

3.1 DUCT INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of duct system. Indicated duct locations, configurations, and arrangements were used to size ducts and calculate friction loss for air-handling equipment sizing and for other design considerations. Install duct systems as indicated unless deviations to layout are approved on Shop Drawings and Coordination Drawings.

B. Install ducts according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible" unless otherwise indicated.

C. Install ducts in maximum practical lengths.

D. Install ducts with fewest possible joints.

E. Install factory- or shop-fabricated fittings for changes in direction, size, and shape and for branch connections.

F. Unless otherwise indicated, install ducts vertically and horizontally, and parallel and perpendicular to building lines.

G. Install ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building.

H. Install ducts with a clearance of 1 inch, plus allowance for insulation thickness.

I. Route ducts to avoid passing through transformer vaults and electrical equipment rooms and enclosures.

J. Where ducts pass through non-fire-rated interior partitions and exterior walls and are exposed to view, cover the opening between the partition and duct or duct insulation with sheet metal flanges of same metal thickness as the duct. Overlap openings on four sides by at least 1-1/2 inches.

K. Where ducts pass through fire-rated interior partitions and exterior walls, install fire dampers. Comply with requirements in Section 233300 "Air Duct Accessories" for fire and smoke dampers.

L. Protect duct interiors from moisture, construction debris and dust, and other foreign materials. Comply with SMACNA's "IAQ Guidelines for Occupied Buildings Under Construction," Appendix G, "Duct Cleanliness for New Construction Guidelines."

3.2 INSTALLATION OF EXPOSED DUCTWORK

A. Protect ducts exposed in finished spaces from being dented, scratched, or damaged.

B. Trim duct sealants flush with metal. Create a smooth and uniform exposed bead. Do not use two-part tape sealing system.

C. Grind welds to provide smooth surface free of burrs, sharp edges, and weld splatter. When welding stainless steel with a No. 3 or 4 finish, grind the welds flush, polish the exposed welds, and treat the welds to remove discoloration caused by welding.

D. Maintain consistency, symmetry, and uniformity in the arrangement and fabrication of fittings, hangers and supports, duct accessories, and air outlets.

E. Repair or replace damaged sections and finished work that does not comply with these requirements.

3.3 DUCT SEALING

A. Seal ducts for duct static-pressure, seal classes, and leakage classes specified in "Duct Schedule" Article according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."

B. Seal ducts at a minimum to the following seal classes according to SMACNA's "HVAC Duct Construction Standards – Metal and Flexible":

1. Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible."
2. Outdoor, Supply-Air Ducts: Seal Class A.
3. Outdoor, Exhaust Ducts: Seal Class C.
4. Outdoor, Return-Air Ducts: Seal Class C.
5. Unconditioned Space, Supply-Air Ducts in Pressure Classes 2-Inch wg and Lower: Seal Class B.
6. Unconditioned Space, Supply-Air Ducts in Pressure Classes Higher Than 2-Inch wg: Seal Class A.
7. Unconditioned Space, Exhaust Ducts: Seal Class C.
8. Unconditioned Space, Return-Air Ducts: Seal Class B.
9. Conditioned Space, Supply-Air Ducts in Pressure Classes 2-Inch wg and Lower: Seal Class C.
10. Conditioned Space, Supply-Air Ducts in Pressure Classes Higher Than 2-Inch wg: Seal Class B.
11. Conditioned Space, Exhaust Ducts: Seal Class B.
12. Conditioned Space, Return-Air Ducts: Seal Class C.

3.4 HANGER AND SUPPORT INSTALLATION

A. Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Chapter 5, "Hangers and Supports."

B. Building Attachments: Concrete inserts, powder-actuated fasteners, or structural-steel fasteners appropriate for construction materials to which hangers are being attached.

1. Where practical, install concrete inserts before placing concrete.
2. Install powder-actuated concrete fasteners after concrete is placed and completely cured.
3. Use powder-actuated concrete fasteners for standard-weight aggregate concretes or for slabs more than 4 inches thick.
4. Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.
5. Do not use powder-actuated concrete fasteners for seismic restraints.

C. Hanger Spacing: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Table 5-1, "Rectangular Duct Hangers Minimum Size," and Table 5-2, "Minimum Hanger Sizes for Round Duct," for maximum hanger spacing; install hangers and supports within 24 inches of each elbow and within 48 inches of each branch intersection.

D. Hangers Exposed to View: Threaded rod and angle or channel supports.

E. Support vertical ducts with steel angles or channel secured to the sides of the duct with welds, bolts, sheet metal screws, or blind rivets; support at each floor and at a maximum intervals of 16 feet.

F. Install upper attachments to structures. Select and size upper attachments with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

3.5 CONNECTIONS

A. Make connections to equipment with flexible connectors complying with Section 233300 "Air Duct Accessories."

B. Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible" for branch, outlet and inlet, and terminal unit connections.

3.6 DUCT CLEANING

A. Clean new duct system(s) before testing, adjusting, and balancing.

B. Use service openings for entry and inspection.

1. Create new openings and install access panels appropriate for duct static-pressure class if required for cleaning access. Provide insulated panels for insulated or lined duct. Patch insulation and liner as recommended by duct liner manufacturer. Comply with Section 233300 "Air Duct Accessories" for access panels and doors.
2. Disconnect and reconnect flexible ducts as needed for cleaning and inspection.
3. Remove and reinstall ceiling to gain access during the cleaning process.

C. Particulate Collection and Odor Control:

1. When venting vacuuming system inside the building, use HEPA filtration with 99.97 percent collection efficiency for 0.3-micron-size (or larger) particles.
2. When venting vacuuming system to outdoors, use filter to collect debris removed from HVAC system, and locate exhaust downwind and away from air intakes and other points of entry into building.

D. Clean the following components by removing surface contaminants and deposits:

1. Air outlets and inlets (registers, grilles, and diffusers).
2. Supply, return, and exhaust fans including fan housings, plenums (except ceiling supply and return plenums), scrolls, blades or vanes, shafts, baffles, dampers, and drive assemblies.

3. Air-handling unit internal surfaces and components including mixing box, coil section, air wash systems, spray eliminators, condensate drain pans, humidifiers and dehumidifiers, filters and filter sections, and condensate collectors and drains.
4. Coils and related components.
5. Return-air ducts, dampers, actuators, and turning vanes except in ceiling plenums and mechanical equipment rooms.
6. Supply-air ducts, dampers, actuators, and turning vanes.
7. Dedicated exhaust and ventilation components and makeup air systems.

E. Mechanical Cleaning Methodology:

1. Clean metal duct systems using mechanical cleaning methods that extract contaminants from within duct systems and remove contaminants from building.
2. Use vacuum-collection devices that are operated continuously during cleaning. Connect vacuum device to downstream end of duct sections so areas being cleaned are under negative pressure.
3. Use mechanical agitation to dislodge debris adhered to interior duct surfaces without damaging integrity of metal ducts, duct liner, or duct accessories.
4. Clean fibrous-glass duct liner with HEPA vacuuming equipment; do not permit duct liner to get wet. Replace fibrous-glass duct liner that is damaged, deteriorated, or delaminated or that has friable material, mold, or fungus growth.
5. Clean coils and coil drain pans according to NADCA 1992. Keep drain pan operational. Rinse coils with clean water to remove latent residues and cleaning materials; comb and straighten fins.
6. Provide drainage and cleanup for wash-down procedures.
7. Antimicrobial Agents and Coatings: Apply EPA-registered antimicrobial agents if fungus is present. Apply antimicrobial agents according to manufacturer’s written instructions after removal of surface deposits and debris.

3.7 START UP

A. Air Balance: Comply with requirements in Section 230593 “Testing, Adjusting, and Balancing for HVAC.”

3.8 DUCT SCHEDULE

A. Fabricate ducts with galvanized sheet steel except as otherwise indicated and as follows:

B. Supply Ducts:

1. Ducts Connected to Air Handling Units and VAV boxes:

a. Pressure Class: Positive 3-inch wg.
b. Minimum SMACNA Seal Class: A.
c. SMACNA Leakage Class for Rectangular: 12.
d. SMACNA Leakage Class for Round and Flat Oval: 12.

2. Ducts Connected to Equipment Not Listed Above:

a. Pressure Class: Positive 3-inch wg.
b. Minimum SMACNA Seal Class: A.

c. SMACNA Leakage Class for Rectangular: 12.
d. SMACNA Leakage Class for Round and Flat Oval: 12.

C. Return Ducts:

1. Ducts Connected to Air Handling Units and VAV Boxes:

a. Pressure Class: Positive or negative 2-inch wg.
b. Minimum SMACNA Seal Class: A.
c. SMACNA Leakage Class for Rectangular: 12.
d. SMACNA Leakage Class for Round and Flat Oval: 12.

2. Ducts Connected to Equipment Not Listed Above:

a. Pressure Class: Positive or negative 2-inch wg.
b. Minimum SMACNA Seal Class: A.
c. SMACNA Leakage Class for Rectangular: 12.
d. SMACNA Leakage Class for Round and Flat Oval: 12.

D. Exhaust Ducts:

1. Ducts Connected to Fans Exhausting (ASHRAE 62.1, Class 1 and 2) Air:

a. Pressure Class: Negative 2-inch wg.
b. Minimum SMACNA Seal Class: A if negative pressure, and A if positive pressure.
c. SMACNA Leakage Class for Rectangular: 12.
d. SMACNA Leakage Class for Round and Flat Oval: 12.

2. Ducts Connected to Equipment Not Listed Above:

a. Pressure Class: Positive or negative 2-inch wg.
b. Minimum SMACNA Seal Class: A if negative pressure, and A if positive pressure.
c. SMACNA Leakage Class for Rectangular: 12.
d. SMACNA Leakage Class for Round and Flat Oval: 12.

E. Intermediate Reinforcement:

1. Galvanized-Steel Ducts: Galvanized steel or carbon steel coated with zinc-chromate primer.

F. Elbow Configuration:

1. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 4-2, "Rectangular Elbows."

a. Velocity 1000 fpm or Lower:

1) Radius Type RE 1 with minimum 0.5 radius-to-diameter ratio.
2) Mitered Type RE 4 without vanes.

b. Velocity 1000 to 1500 fpm:

1) Radius Type RE 1 with minimum 1.0 radius-to-diameter ratio.
2) Radius Type RE 3 with minimum 0.5 radius-to-diameter ratio and two vanes.
3) Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

c. Velocity 1500 fpm or Higher:

1) Radius Type RE 1 with minimum 1.5 radius-to-diameter ratio.
2) Radius Type RE 3 with minimum 1.0 radius-to-diameter ratio and two vanes.
3) Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

2. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 4-2, "Rectangular Elbows."

a. Radius Type RE 1 with minimum 1.5 radius-to-diameter ratio.
b. Radius Type RE 3 with minimum 1.0 radius-to-diameter ratio and two vanes.
c. Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

3. Round Duct: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 3-4, "Round Duct Elbows."

a. Minimum Radius-to-Diameter Ratio and Elbow Segments: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Table 3-1, "Mitered Elbows." Elbows with less than 90-degree change of direction have proportionately fewer segments.

1) Velocity 1000 fpm or Lower: 0.5 radius-to-diameter ratio and three segments for 90-degree elbow.
2) Velocity 1000 to 1500 fpm: 1.0 radius-to-diameter ratio and four segments for 90-degree elbow.
3) Velocity 1500 fpm or Higher: 1.5 radius-to-diameter ratio and five segments for 90-degree elbow.
4) Radius-to Diameter Ratio: 1.5.

b. Round Elbows, 12 Inches and Smaller in Diameter: Stamped or pleated.
c. Round Elbows, 14 Inches and Larger in Diameter: Standing seam or Welded.

G. Branch Configuration:

1. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 4-6, "Branch Connection."

    a. Rectangular Main to Rectangular Branch: 45-degree entry.
    b. Rectangular Main to Round Branch: Spin in.

2. Round: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible," Figure 3-5, "90 Degree Tees and Laterals," and Figure 3-6, "Conical Tees." Saddle taps are permitted in existing duct.

    a. Velocity 1000 fpm or Lower: 90-degree tap.
    b. Velocity 1000 to 1500 fpm: Conical tap.
    c. Velocity 1500 fpm or Higher: 45-degree lateral.

END OF SECTION 23 31 13

SECTION 23 33 00 – AIR DUCT ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Backdraft and pressure relief dampers.
2. Manual volume dampers.
3. Control dampers.
4. Flange connectors.
5. Flexible connectors.
6. Duct accessory hardware.
7. Fire Dampers
8. Combination Fire/Smoke Dampers
9. Turning vanes.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

PART 2 - PRODUCTS

2.1 ASSEMBLY DESCRIPTION

A. Comply with NFPA 90A, “Installation of Air Conditioning and Ventilating Systems,” and with NFPA 90B, “Installation of Warm Air Heating and Air Conditioning Systems.”

B. Comply with SMACNA’s “HVAC Duct Construction Standards – Metal and Flexible” for acceptable materials, material thicknesses, and duct construction methods unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.

2.2 MATERIALS

A. Galvanized Sheet Steel: Comply with ASTM A653/A653M.

1. Galvanized Coating Designation: G60.
2. Exposed-Surface Finish: Mill phosphatized.

B. Reinforcement Shapes and Plates: Galvanized-steel reinforcement where installed on galvanized sheet metal ducts; compatible materials for aluminum and stainless-steel ducts.

C. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.

2.3 BACKDRAFT AND PRESSURE RELIEF DAMPERS

A. Physical:

1. Description: Gravity balanced.
2. Frame: Hat-shaped, 0.05-inch-thick, galvanized sheet steel.
3. Blades: Multiple single-piece blades, end pivoted, maximum 6-inch width, 0.025-inch-thick, roll-formed aluminum with sealed edges.
4. Blade Action: Parallel.
5. Blade Seals: Neoprene, mechanically locked.
6. Blade Axles:

    a. Material: Galvanized steel or Aluminum.
    b. Diameter: 0.20 inch.

7. Tie Bars and Brackets: Aluminum or Galvanized steel.
8. Return Spring: Adjustable tension.
9. Bearings: Steel ball.

B. Function:

1. Accessories:

    a. Adjustment device to permit setting for varying differential static pressure.
    b. Screen Mounting: Rear mounted.
    c. Screen Material: Galvanized steel or Aluminum.
    d. Screen Type: Insect.
    e. 90-degree stops.

C. Performance:

1. Maximum Air Velocity: 1000 fpm.
2. Maximum System Pressure: 0.5-inch wg.

2.4 MANUAL VOLUME DAMPERS

A. Physical:

1. Standard, Steel, Manual Volume Dampers:

    a. Standard leakage rating, with linkage outside airstream.
    b. Suitable for horizontal or vertical applications.

c. Frames:

1) Frame: Hat-shaped, 0.094-inch-thick, galvanized sheet steel.
2) Mitered and welded corners.
3) Flanges for attaching to walls and flangeless frames for installing in ducts.

d. Blades:

1) Multiple or single blade.
2) Parallel- or opposed-blade design.
3) Stiffen damper blades for stability.
4) Galvanized-steel, 0.064 inch thick.

e. Blade Axles: Galvanized steel.
f. Bearings:

1) Molded synthetic or Stainless-steel sleeve.
2) Dampers in ducts with pressure classes of 3-inch wg or less shall have axles full length of damper blades and bearings at both ends of operating shaft.

g. Tie Bars and Brackets: Galvanized steel.

2. Jackshaft:

a. Size: 1-inch diameter.
b. Material: Galvanized-steel pipe rotating within pipe-bearing assembly mounted on supports at each mullion and at each end of multiple-damper assemblies.
c. Length and Number of Mountings: As required to connect linkage of each damper in multiple-damper assembly.

3. Damper Hardware:

a. Zinc-plated, die-cast core with dial and handle made of 3/32-inch- thick zinc-plated steel, and a 3/4-inch hexagon locking nut.
b. Include center hole to suit damper operating-rod size.
c. Include elevated platform for insulated duct mounting.

2.5 FLEXIBLE CONNECTORS

A. Physical:

1. Materials: Flame-retardant or noncombustible fabrics.
2. Coatings and Adhesives: Comply with UL 181, Class 1.
3. Instrument Test Holes: Cast iron or cast aluminum to suit duct material, including screw cap and gasket. Size to allow insertion of pitot tube and other testing instruments and of length to suit duct-insulation thickness.
4. Adhesives: High strength, quick setting, neoprene based, waterproof, and resistant to gasoline and grease.

B. Function:

1. None.

C. Performance:

1. None

2.6 CONTROL DAMPERS

A. General Requirements:

1. Unless otherwise indicated, use parallel-blade configuration for two-position control, equipment isolation service, and when mixing two airstreams. For other applications, use opposed-blade configuration.
2. Factory or field assemble multiple damper sections to provide a single damper assembly of size required by the application.

B. Performance:

1. Leakage:

a. Class IA: Leakage shall not exceed 3 cfm/sq. ft. against 1-inch wg differential static pressure.
b. Class I: Leakage shall not exceed 4 cfm/sq. ft. against 1-inch wg differential static pressure.
c. Class II: Leakage shall not exceed 10 cfm/sq. ft. against 1-inch wg differential static pressure.
d. Class III: Leakage shall not exceed 40 cfm/sq. ft. against 1-inch wg differential static pressure.

2. Pressure Drop: 0.05 inch wg at 1500 fpm across a 24-by-24-inch damper when tested in accordance with AMCA 500-D, Figure 5.3.
3. Velocity: Up to 3000 fpm.
4. Temperature: Minus 25 to plus 180 deg F.
5. Pressure Rating: Damper close-off pressure equal to fan shutoff pressure with a maximum blade deflection of 1/200 of blade length.

C. Physical Construction:

1. Linkage out of airstream.
2. Suitable for horizontal or vertical airflow applications.
3. Frames:

a. Hat, U, or angle shaped.
b. 16-gauge- thick, galvanized sheet steel.
c. Mitered and welded corners.
d. Flanges for attaching to walls and flangeless frames for installing in ducts.

4. Blades:

    a. Multiple blade with maximum blade width of 8 inches.
    b. Parallel blade design.
    c. Galvanized steel.
    d. 16-gauge- thick single skin thick air foil dual skin].

5. Blade Edging Seals:

    a. Replaceable Closed-cell neoprene.
    b. Inflatable seal blade edging, or replaceable rubber seals.

6. Blade Jamb Seal: Flexible stainless steel, compression type.
7. Blade Axles: 1/2-inch diameter; galvanized steel.
8. Blade-Linkage Hardware: Zinc-plated steel and brass; ends sealed against blade bearings. Linkage mounted out of air stream.
9. Bearings:

    a. Oil-impregnated bronze
    b. Dampers mounted with vertical blades to have thrust bearings at each end of every blade.

D. Function Requirements:

1. Damper Actuator – Electric:

    a. Electric – 120 V ac.
    b. UL 873, plenum rated.
    c. Two position with fail-safe spring return.

        1) Sufficient motor torque and spring torque to drive damper fully open and fully closed with adequate force to achieve required damper seal.
        2) Minimum 90-degree drive rotation.

    d. Clockwise or counterclockwise drive rotation as required for application.
    e. Environmental Operating Range:

        1) Temperature: Minus 40 to plus 130 deg F
        2) Humidity: 5 to 95 percent relative humidity noncondensing.

    f. Environmental enclosure: NEMA 2.
    g. Actuator to be factory mounted and provided with a single-point wiring connection.

2. Controllers, Electrical Devices, and Wiring:

    a. Comply with requirements for electrical devices and connections specified in Section 230923 “Direct Digital Control (DDC) System for HVAC.”
    b. Electrical Connection: 115 V, single phase, 60 Hz.

3. FIRE DAMPERS

a. Type: Static; rated and labeled in accordance with UL 555 by an NRTL.
b. Closing rating in ducts up to 4-inch wg static pressure class and minimum 2000 fpm velocity.
c. Fire Rating: 1-1/2 and 3 hours.
d. Frame: Curtain type with blades inside airstream; fabricated with roll-formed galvanized steel; with mitered and interlocking corners; gauge in accordance with UL listing.
e. Mounting Sleeve: Factory- or field-installed, galvanized sheet steel; gauge in accordance with UL listing.
f. Mounting Orientation: Vertical or horizontal as indicated.
g. Blades: Roll-formed galvanized sheet steel, interlocking. Material gauge is to be in accordance with UL listing.
h. Horizontal Dampers: Include blade lock and stainless steel closure spring.
i. Heat-Responsive Device:

1) Replaceable, 165 deg F rated, fusible links.

2.7 COMBINATION FIRE AND SMOKE DAMPERS

A. General Requirements:

1. Label to indicate conformance to UL 555 and UL 555S by an NRTL.
2. Label to indicate conformance to NFPA 80 and NFPA 90A by an NRTL.
3. Unless otherwise indicated, use parallel-blade configuration.

B. Closing rating in ducts up to 4-inch wg static pressure class and minimum 2000 fpm velocity.

C. Fire Rating: 1-1/2 and 3 hours.

D. Performance:

1. AMCA Certification: Test and rate in accordance with AMCE Publication 511.
2. Leakage:

a. Class IA: Leakage shall not exceed 3 cfm/sq. ft. against 1-inch wg differential static pressure.
b. Class I: Leakage shall not exceed 4 cfm/sq. ft. against 1-inch wg differential static pressure.
c. Class II: Leakage shall not exceed 10 cfm/sq. ft. against 1-inch wg differential static pressure.

3. Pressure Drop: 0.05 in. wg at 1500 fpm across a 24-by-24-inch damper when tested in accordance with AMCA 500-D, Figure 5.3.
4. Velocity: Up to 3000 fpm.
5. Temperature: Minus 25 to plus 180 deg F.
6. Pressure Rating: Damper close-off pressure equal to fan shutoff pressure with a maximum blade deflection of 1/200 of blade length.

E. Physical Construction:

1. Suitable or horizontal or vertical airflow applications.
2. Linkage out of airstream.
3. Frame:

    a. Hat shaped.
    b. Galvanized sheet steel with welded corners and mounting flange.
    c. Gauge is to be in accordance with UL listing.

4. Blades:

    a. Roll-formed, horizontal, v-groove, galvanized sheet steel.
    b. Maximum width and gauge in accordance with UL listing.

5. Blade Edging Seals:

    a. Silicone rubber.

6. Blade Jamb Seal: Flexible stainless steel, compression type.
7. Blade Axles: 1/2-inch- diameter; galvanized steel; blade-linkage hardware of zinc-plated steel and brass; ends sealed against blade bearings. Linkage mounted out of airstream.
8. Bearings:

    a. Oil-impregnated bronze
    b. Factory installed, galvanized sheet steel.
    c. Length to suit wall or floor application with factory-furnished silicone caulking.
    d. Gauge in accordance with UL listing.

9. Heat-Responsive Device:

    a. Resettable, 165 deg F rated, fusible links.

10. Master control panel for use in dynamic smoke-management systems.

F. Functional Requirements:

1. Damper Actuator – Electric:

    a. Electric – 120 V ac.
    b. UL 873, plenum rated.
    c. Designed to operate in smoke-control systems complying with UL 555S requirements.
    d. Two position with fail-safe spring return.

        1) Sufficient motor torque and spring torque to drive damper fully open and fully closed with adequate force to achieve required damper seal.
        2) Maximum 15-second full-stroke closure.
        3) Minimum 90-degree drive rotation.

    e. Clockwise or counterclockwise drive rotation as required for application.

f. Environmental Operating Range:

1) Temperature: Minus 40 to plus 130 deg F.
2) Humidity: 5 to 95 percent relative humidity noncondensing.

g. Environmental Enclosure: NEMA 2.
h. Actuator to be factory mounted and provided with single-point wiring connection.

2. Controllers, Electrical Devices, and Wiring:

a. Comply with requirements for electrical devices and connections specified in Section 230923 "Direct Digital Control (DDC) System for HVAC."
b. Electrical Connection: 115 V, single phase, 60 Hz.

2.8 TURNING VANES

A. Physical Requirements:

1. Manufactured Turning Vanes for Nonmetal Ducts: Fabricate curved blades of resin-bonded fiberglass with acrylic polymer coating; support with bars perpendicular to blades set; set into vane runners suitable for duct mounting.
2. General Requirements: Comply with SMACNA's "HVAC Duct Construction Standards – Metal and Flexible"; Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."
3. Vane Construction:

a. Double wall.
b. Single wall for ducts up to 48 inches wide and double wall for larger dimensions.

B. Function Requirements:

1. None.

C. Performance Requirements:

1. None.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install duct accessories according to applicable details in SMACNA's "HVAC Duct Construction Standards – Metal and Flexible" for metal ducts and in NAIMA AH116, "Fibrous Glass Duct Construction Standards," for fibrous-glass ducts.

B. Install duct accessories of materials suited to duct materials; use galvanized-steel accessories in galvanized-steel and fibrous-glass ducts, stainless-steel accessories in stainless-steel ducts, and aluminum accessories in aluminum ducts.

C. Install backdraft dampers at inlet of exhaust fans or exhaust ducts as close as possible to exhaust fan unless otherwise indicated.

D. Set dampers to fully open position before testing, adjusting, and balancing.

E. Install test holes at fan inlets and outlets and elsewhere as indicated.

F. Install access doors with swing against duct static pressure.

G. Label access doors according to Section 230553 “Identification for HVAC Piping and Equipment” to indicate the purpose of access door.

H. Install flexible connectors to connect ducts to equipment.

I. Install duct test holes where required for testing and balancing purposes.

3.2 FIELD QUALITY CONTROL

A. Tests and Inspections:

1. Operate dampers to verify full range of movement.
2. Inspect locations of access doors and verify that purpose of access door can be performed.
3. Operate fire and smoke dampers to verify full range of movement and verify that proper heat-response device is installed.
4. Inspect turning vanes for proper and secure installation.

END OF SECTION 23 33 00

SECTION 23 34 23 – HVAC POWER VENTILATORS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Inline Exhaust Fan

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Data: For fans, accessories, and components, from manufacturer.

B. Product Certificates: Submit certificates that specified equipment will withstand required wind forces, from manufacturer.

C. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.

B. NFPA Compliance: Comply with NFPA 90A for design, fabrication, and installation of unit components.

C. ASHRAE 62.1 Compliance: Applicable requirements in ASHRAE 62.1, Section 5 – "Systems and Equipment" and Section 7 – "Construction and Startup."

D. ASHRAE/IES 90.1 Compliance: Applicable requirements in ASHRAE/IES 90.1, Section 6 – "Heating, Ventilating, and Air-Conditioning."

E. Seismic Performance: HVAC power ventilators shall withstand the effects of earthquake motions determined according to ASCE/SEI 7. See Section 230548 "Vibration and Seismic Controls for HVAC."

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."
2. Component Importance Factor: 1.5 1.0.

F. Wind Performance: Air-handling units shall withstand the effects of wind determined in accordance with to ASCE/SEI 7. See Section 230548 "Vibration and Seismic Controls for HVAC." Section 230548.13 "Vibration Controls for HVAC."

2.2 INLINE EXHAUST FANS

A. Physical:

1. Housing: Steel, lined with acoustical insulation.
2. Fan Wheel: Centrifugal wheels directly mounted on motor shaft. Fan shrouds, motor, and fan wheel removable for service.
3. Back-draft damper: Integral.
4. Grille: Stainless steel Aluminum Painted aluminum, louvered grille with flange on intake and thumbscrew or spring retainer attachment to fan housing.
5. Electrical Requirements: Junction box for electrical connection on housing and receptacle for motor plug-in.

B. Functional Accessories:

1. Variable-Frequency Motor Controller: Solid-state control to reduce speed from 100 to less than 50 percent.
2. Manual Starter Switch: Single-pole rocker switch assembly with cover and pilot light.
3. Time-Delay Switch: Assembly with single-pole rocker switch, timer, and cover plate.
4. Motion Sensor: Motion detector with adjustable shutoff timer.
5. Ceiling Radiation Damper: Fire-rated assembly with ceramic blanket, stainless steel springs, and fusible link.
6. Filter: Washable aluminum to fit between fan and grille.
7. Isolation: Rubber-in-shear vibration isolators.
8. Manufacturer's standard roof jack or wall cap, and transition fittings.

2.3 MOTORS

A. Comply with NEMA designation, temperature rating, service factor, and efficiency requirements for motors specified in Section 230513 “Common Motor Requirements for HVAC Equipment.”

1. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

2.4 SOURCE QUALITY CONTROL

A. AMCA Certification for Fan Sound Performance Rating: Test, rate, and label in accordance with AMCA 311.

B. AMCA Certification for Fan Energy Index (FEI): Test, rate, and label in accordance with AMCA 211

C. UL Standards: Power ventilators shall comply with UL 705. Power ventilators for use for restaurant kitchen exhaust shall also comply with UL 762.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Install power ventilators level and plumb.

B. Equipment Mounting:

1. Secure roof-mounted fans to roof curbs with zinc-plated hardware. See Section 077200 “Roof Accessories” for installation of roof curbs.
2. Ceiling Units: Suspend units from structure; use steel wire or metal straps.
3. Comply with requirements for vibration isolation and seismic-control devices specified in Section 230548 “Vibration and Seismic Controls for HVAC.”

C. Install units with clearances for service and maintenance.

D. Label units according to requirements specified in Section 230553 “Identification for HVAC Piping and Equipment.”

3.2 DUCTWORK CONNECTIONS

A. Drawings indicate general arrangement of ducts and duct accessories. Make final duct connections with flexible connectors. Flexible connectors are specified in Section 233300 “Air Duct Accessories.”

3.3 ELECTRICAL CONNECTIONS

A. Connect wiring according to Section 260519 “Low-Voltage Electrical Power Conductors and Cables.”

B. Ground equipment according to Section 260526 “Grounding and Bonding for Electrical Systems.”

C. Install electrical devices furnished by manufacturer, but not factory mounted, according to NFPA 70 and NECA 1.

1. Nameplate shall be laminated acrylic or melamine plastic signs, as specified in Section 260553 “Identification for Electrical Systems.”
2. Nameplate shall be laminated acrylic or melamine plastic signs with a black background and engraved white letters at least 1/2 inch high.

3.4 CONTROL CONNECTIONS

A. Install control and electrical power wiring to field-mounted control devices.

B. Connect control wiring according to Section 260523 “Control-Voltage Electrical Power Cables.”

3.5 STARTUP SERVICE:

A. Perform startup service.

1. Complete installation and startup checks in accordance with manufacturer’s written instructions.
2. Verify that shipping, blocking, and bracing are removed.
3. Verify that unit is secure on mountings and supporting devices and that connections to ducts and electrical components are complete. Verify that proper thermal-overload protection is installed in motors, starters, and disconnect switches.
4. Verify that cleaning and adjusting are complete.
5. For direct-drive fans, verify proper motor rotation direction and verify fan wheel free rotation and smooth bearing operation.
6. For belt-drive fans, disconnect fan drive from motor, verify proper motor rotation direction, and verify fan wheel free rotation and smooth bearing operation. Reconnect fan drive system, align and adjust belts, and install belt guards.
7. Adjust damper linkages for proper damper operation.
8. Verify lubrication for bearings and other moving parts.
9. Verify that manual and automatic volume control and fire and smoke dampers in connected ductwork systems are in fully open position.
10. Disable automatic temperature-control operators, energize motor and confirm proper motor rotation and unit operation, adjust fan to indicated rpm, and measure and record motor voltage and amperage.
11. Shut unit down and reconnect automatic temperature-control operators.
12. Remove and replace malfunctioning units and retest as specified above.

3.6 ADJUSTING

A. Adjust damper linkages for proper damper operation.

B. Lubricate bearings.

C. Comply with requirements in Section 230593 “Testing, Adjusting, and Balancing for HVAC.”

3.7 CLEANING

A. After completing system installation and testing, adjusting, and balancing and after completing startup service, clean fans internally to remove foreign material and construction dirt and dust.

3.8 FIELD QUALITY CONTROL

A. Testing Agency: Contracting Officer Contractor will engage a qualified testing agency to perform tests and inspections.

B. Perform tests and inspections with the assistance of a factory-authorized service representative.

1. Fan Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
2. Test and adjust controls and safeties.
3. Fans and components will be considered defective if they do not pass tests and inspections.
4. Prepare test and inspection reports.

3.9 DEMONSTRATION

A. Train Owner’s maintenance personnel to adjust, operate, and maintain centrifugal fans.

END OF SECTION 23 34 23

SECTION 23 37 13 – DIFFUSERS, REGISTERS, AND GRILLES

PART 1 - GENERAL

1.1 ACTION SUBMITTALS

A. Product Data: For each type of product indicated, include the following:

1. Data Sheet: Indicate materials of construction, finish, and mounting details; and performance data including throw and drop, static pressure drop, and noise ratings.
2. Diffuser, Register, and Grille Schedule: Indicate drawing designation, room location, quantity, model number, size, and accessories furnished.

PART 2 - PRODUCTS

2.1 SOURCE QUALITY CONTROL

A. Verification of Performance: Rate diffusers, registers, and grilles according to ASHRAE 70, "Method of Testing for Rating the Performance of Air Outlets and Inlets."

B. Physical:

1. Refer to mechanical schedule on drawings.

C. Functional Requirements:

1. Refer to mechanical schedule on drawings.

D. Performance Requirements:

1. Refer to mechanical schedule on drawings.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install diffusers, registers, and grilles level and plumb.

B. Ceiling-Mounted Outlets and Inlets: Drawings indicate general arrangement of ducts, fittings, and accessories. Air outlet and inlet locations have been indicated to achieve design requirements for air volume, noise criteria, airflow pattern, throw, and pressure drop. Make final locations where indicated, as much as practical. For units installed in lay-in ceiling panels, locate units in the center of panel. Where architectural features or other items conflict with installation, notify Architect for a determination of final location.

C. Install diffusers, registers, and grilles with airtight connections to ducts and to allow service and maintenance of dampers, air extractors, and fire dampers.

3.2 ADJUSTING

A. After installation, adjust diffusers, registers, and grilles to air patterns indicated, or as directed, before starting air balancing.

END OF SECTION 23 37 13

SECTION 23 72 23.13 – PACKAGED INDOOR HEAT WHEEL ENERGY RECOVERY UNITS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Packaged, indoor, heat wheel energy-recovery units.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include packaged, indoor, heat wheel, energy-recovery-unit rated capacities, operating characteristics, furnished specialties, and accessories.
2. Fans:

a. Certified fan-performance curves with system operating conditions indicated.
b. Certified fan-sound power ratings.
c. Fan construction and accessories.
d. Motor ratings, electrical characteristics, and motor accessories.

B. Shop Drawings: For air-to-air energy-recovery equipment.

1. Include plans, elevations, sections, details, and mounting attachment details.
2. Include details of equipment assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Include diagrams for power, signal, and control wiring.

1.3 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For air-to-air energy-recovery equipment to include in maintenance manuals.

1.4 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of packaged, indoor, heat wheel energy-recovery units that fail in materials or workmanship within specified warranty period.

1. Warranty Period for Packaged Energy-Recovery Units: Two years.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. ASHRAE Compliance:

1. Applicable requirements in ASHRAE 62.1.
2. Capacity ratings for air-to-air energy-recovery equipment shall comply with ASHRAE 84.

B. ASHRAE/IES 90.1 Compliance: Applicable requirements in ASHRAE/IES 90.1.

2.2 PACKAGED, INDOOR, HEAT WHEEL ENERGY-RECOVERY UNITS

A. Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1.

B. Housing: Manufacturer's standard construction with corrosion-protection coating and exterior finish, hinged access doors with neoprene gaskets for inspection and access to internal parts, minimum 1- inch- thick, R-4 thermal insulation, knockouts for electrical connections, exterior drain connection, and lifting lugs.

C. Heat Wheel:

1. Casing:

a. Manufacturer's standard construction with standard factory finish.
b. Slide-in, slide-out cassette style for easy access.

2. Rotor:

a. Aluminum or polymer segmented wheel, strengthened with radial spokes, with nontoxic, noncorrosive, silica-gel coating.

3. Drive: Fractional horsepower motor and gear reducer and self-adjusting multilink belt around outside of rotor.

a. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

D. Supply and Exhaust Fans: Forward-curved, centrifugal fan. Assembly shall be mounted on heavy gauge galvanized rails and further mounted on 1.125-inch-thick neoprene vibration isolators.

1. Motor and Drive: Belt driven with adjustable sheaves; motor mounted on adjustable base.
2. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

E. Filters:

1. Description: MERV 13

F. Controls:

1. Control Panel: Solid-state, programmable, microprocessor-based control unit for wall mounting within mechanical room. Coordinate occupancy schedule to match AHU's and dehumidifiers.

PART 3 - EXECUTION

A. Install packaged, indoor, heat wheel energy-recovery units, so supply and exhaust airstreams flow in opposite directions, and rotation is away from exhaust side to purge section to supply side.

1. Install access doors in both supply and exhaust ducts, both upstream and downstream, for access to wheel surfaces, drive motor, and seals.
2. Install removable panels or access doors between supply and exhaust ducts on building side for bypass during startup.
3. Access doors and panels are specified in Section 233300 "Air Duct Accessories."

B. Equipment Mounting:

C. Suspended Units: Suspend unit from structural-steel support frame, using threaded steel rods and spring hangers.

D. Install units with clearances for service and maintenance.

E. Do not operate fan system until filters (temporary or permanent) are in place. Replace temporary filters used during construction and testing with new, clean filters.

3.2 FIELD QUALITY CONTROL

A. Perform tests and inspections with the assistance of a factory-authorized service representative.

B. Tests and Inspections:

1. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

C. Packaged, indoor, heat wheel energy-recovery equipment will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

3.3 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain air-to-air energy-recovery units.

END OF SECTION 23 72 23.13

SECTION 23 81 26 – SPLIT-SYSTEM AIR-CONDITIONERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes split-system air-conditioning and heat-pump units consisting of separate evaporator-fan and compressor-condenser components.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1.3 INFORMATIONAL SUBMITTALS

A. Certificates: For certification required in “Quality Assurance” Article.

B. Installation instructions.

C. Source quality-control reports.

D. Startup service reports.

E. Sample warranty.

1.4 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

1.5 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. ASHRAE Compliance:

1. Fabricate and label refrigeration system to comply with ASHRAE 15, “Safety Standard for Refrigeration Systems.”
2. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1, Section 4 – “Outdoor Air Quality,” Section 5 – “Systems and Equipment,” Section 6 – “ Procedures,” and Section 7 – “Construction and System Start-up.”

C. ASHRAE/IES Compliance: Applicable requirements in ASHRAE/IES 90.1.

1.6 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of split-system air-conditioning units that fail in materials or workmanship within specified warranty period.

1. Warranty Period: 2 years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 INDOOR UNITS

A. Physical:

1. Floor-Mounted, Evaporator-Fan Components:
2. Cabinet: Enameled steel with removable panels on front and ends in color selected by Contracting Officer.

a. Insulation: Faced, glass-fiber duct liner.
b. Drain Pans: Galvanized steel, with connection for drain; insulated.

3. Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and thermal-expansion valve. Comply with ARI 206/110.
4. Fan: Direct drive, centrifugal.
5. Fan Motors:

a. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements specified in Section 230513 "Common Motor Requirements for HVAC Equipment."
b. Multitapped, multispeed with internal thermal protection and permanent lubrication.

6. Air Filtration Section:

a. General Requirements for Air Filtration Section:

1) Comply with NFPA 90A.
2) Minimum MERV according to ASHRAE 52.2.
3) Filter-Holding Frames: Arranged for flat or angular orientation, with access doors on both sides of unit. Filters shall be removable from one side or lifted out from access plenum.

b. Disposable Panel Filters:

1) Factory-fabricated, viscous-coated, flat-panel type.
2) Thickness: 2 inches.
3) MERV according to ASHRAE 52.2: 13.
4) Media: Interlaced glass fibers sprayed with nonflammable adhesive.
5) Frame: Galvanized steel, with metal grid on outlet side, steel rod grid on inlet side, and hinged; with pull and retaining handles.

7. Wall-Mounted, Evaporator-Fan Components:

    a. Cabinet: Enameled steel with removable panels on front and ends in color selected by Architect, and discharge drain pans with drain connection.
    b. Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and thermal-expansion valve. Comply with ARI 206/110.
    c. Fan: Direct drive, centrifugal.
    d. Fan Motors:

        1) Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements specified in Section 230513 "Common Motor Requirements for HVAC Equipment."
        2) Multitapped, multispeed with internal thermal protection and permanent lubrication.
        3) NEMA Premium (TM) efficient motors as defined in NEMA MG 1.
        4) Controllers, Electrical Devices, and Wiring: Comply with requirements for electrical devices and connections specified in electrical Sections.
        5) Mount unit-mounted disconnect switches on exterior of unit.

8. Condensate Drain Pans:

    a. Fabricated with one percent slope in at least two planes to collect condensate from cooling coils (including coil piping connections, coil headers, and return bends) and humidifiers, and to direct water toward drain connection.

        1) Depth: A minimum of 1 inch.

    b. Single-wall, stainless-steel sheet.
    c. Drain Connection: Located at lowest point of pan and sized to prevent overflow. Terminate with threaded nipple on one end of pan.

9. Air Filtration Section:

    a. General Requirements for Air Filtration Section:

        1) Comply with NFPA 90A.
        2) Minimum MERV according to ASHRAE 52.2.
        3) Filter-Holding Frames: Arranged for flat or angular orientation, with access doors on both sides of unit. Filters shall be removable from one side or lifted out from access plenum.

B. Performance:

1. Refer to mechanical schedules on drawings.

2.2 OUTDOOR UNITS

A. Physical:

1. Air-Cooled, Compressor-Condenser Components:

a. Casing: Steel, finished with baked enamel in color selected by Architect, with removable panels for access to controls, weep holes for water drainage, and mounting holes in base. Provide brass service valves, fittings, and gage ports on exterior of casing.

b. Compressor: Hermetically sealed with crankcase heater and mounted on vibration isolation device. Compressor motor shall have thermal- and current-sensitive overload devices, start capacitor, relay, and contactor.

1) Compressor Type: Scroll.
2) Two-speed compressor motor with manual-reset high-pressure switch and automatic-reset low-pressure switch.
3) Refrigerant: R-410A.
4) Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins and liquid subcooler. Comply with ARI 206/110.

c. Heat-Pump Components: Reversing valve and low-temperature-air cutoff thermostat.

d. Fan: Aluminum-propeller type, directly connected to motor.

e. Motor: Permanently lubricated, with integral thermal-overload protection.

f. Low Ambient Kit: Permits operation down to -40 deg F.

g. Mounting Base: Polyethylene.

B. Performance:

1. Refer to mechanical schedules on drawings.

2.3 Functional Requirements:

A. Control equipment and sequence of operation are specified in Section 230923 “Direct Digital Control (DDC) System for HVAC” and Section 230993.11 “Sequence of Operations for HVAC DDC.”

B. Thermostat: Low voltage with subbase to control compressor and evaporator fan.

C. Automatic-reset timer to prevent rapid cycling of compressor.

D. Refrigerant Line Kits: Soft-annealed copper suction and liquid lines factory cleaned, dried, pressurized, and sealed; factory-insulated suction line with flared fittings at both ends.

E. Drain Condensate: Copper.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install units level and plumb.

B. Install evaporator-fan components using manufacturer's standard mounting devices securely fastened to building structure.

C. Install roof-mounted, compressor-condenser components on equipment supports specified in Section 077200 "Roof Accessories." Anchor units to supports with removable, cadmium-plated fasteners.

D. Equipment Mounting:

1. Comply with requirements for vibration isolation and seismic control devices specified in Section 230548 "Vibration and Seismic Controls for HVAC."

E. Install and connect precharged refrigerant tubing to component's quick-connect fittings. Install tubing to allow access to unit.

3.2 CONNECTIONS

A. Piping installation requirements are specified in other Sections. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Where piping is installed adjacent to unit, allow space for service and maintenance of unit.

C. Duct Connections: Duct installation requirements are specified in Section 233113 "Metal Ducts." Drawings indicate the general arrangement of ducts. Connect supply and return ducts to split-system air-conditioning units with flexible duct connectors. Flexible duct connectors are specified in Section 233300 "Air Duct Accessories."

3.3 FIELD QUALITY CONTROL

A. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

B. Tests and Inspections:

1. Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.
2. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
3. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

C. Remove and replace malfunctioning units and retest as specified above.

D. Prepare test and inspection reports.

END OF SECTION 23 81 26

SECTION 23 82 39.16 – PROPELLER UNIT HEATERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes propeller unit heaters with electric heating coils.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include rated capacities, operating characteristics, furnished specialties, and accessories.

1.3 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

PART 2 - PRODUCTS

2.1 MANUFACTURERS:

A. See schedule on plan for Basis of Design information.

B. Or Approved Equal.

C. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 PHYSICAL DESCRIPTION

A. Assembly including casing, coil, fan, and motor in horizontal discharge configuration with adjustable discharge louvers.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.3 PERFORMANCE REQUIREMENTS

A. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1, Section 5 – "Systems and Equipment" and Section 7 – "Construction and Startup."

B. ASHRAE/IESNA 90.1 Compliance: Applicable requirements in ASHRAE/IESNA 90.1, Section 6 – "Heating, Ventilating, and Air-Conditioning."

C. Per Mechanical Schedules.

2.4 HOUSINGS

A. Finish: Manufacturer's standard baked enamel applied to factory-assembled and -tested propeller unit heaters before shipping.

B. Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1.

C. Discharge Louver: Adjustable fin diffuser for horizontal units and conical diffuser for vertical units.

2.5 COILS

A. General Coil Requirements: Test and rate electric propeller unit-heater coils according to ASHRAE 33.

2.6 FAN AND MOTOR

A. Fan: Propeller type with aluminum wheel directly mounted on motor shaft in the fan venturi.

B. Motor: Permanently lubricated.

2.7 CONTROLS

A. Control Devices:

1. Unit-mounted thermostat.

2.8 FUNCTION

1. Garage/Basement Freeze Protection.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive unit heaters for compliance with requirements for installation tolerances and other conditions affecting performance of the work.

B. Examine roughing-in for electrical connections to verify actual locations before unit heater installation.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install propeller unit heaters to comply with NFPA 90A.

B. Install propeller unit heaters level and plumb.

C. Install wall-mounted thermostats and switch controls in electrical outlet boxes at heights to match lighting controls. Verify location of thermostats and other exposed control sensors with drawings and room details before installation.

3.3 CONNECTIONS

A. Comply with safety requirements in UL 1995.

B. Ground equipment according to Section 260526 “Grounding and Bonding for Electrical Systems.”

C. Connect wiring according to Section 260519 “Low-Voltage Electrical Power Conductors and Cables.”

3.4 FIELD QUALITY CONTROL

A. Contractor to furnish field inspection and testing with the assistance of a factory-authorized service representative:

1. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.
2. Test and adjust controls and safety devices. Replace damaged and malfunctioning controls and equipment.

B. Units will be considered defective if they do not pass tests and inspections.

C. Prepare test and inspection reports.

END OF SECTION 23 82 39.16

SECTION 23 84 16.16 – INDOOR, MECHANICAL DEHUMIDIFICATION UNITS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Indoor, mechanical dehumidification units.

1.2 ACTION SUBMITTALS

A. Product Data:

1. For each type of dehumidifier product.

a. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.

B. Sustainable Design Submittals:

1. Product Data: For energy performance.

1.3 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For dehumidification units to include in emergency, operation, and maintenance manuals.

1.4 WARRANTY

A. Special Warranty: Manufacturer agrees to repair or replace components of indoor, mechanical dehumidification units that fail in materials or workmanship within specified warranty period.

1. Warranty Period for Compressors: Manufacturer’s standard, but not less than two years from date of Substantial Completion.
2. Warranty Period for Refrigerant Coils: Manufacturer’s standard, but not less than five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. ASHRAE 15 Compliance: “Safety Standard for Refrigeration Systems.”

2.2 INDOOR, MECHANICAL DEHUMIDIFICATION UNITS

A. Casings:

1. Casing: Single -wall construction with interior corrosion-resistant coating and exterior baked-enamel steel or aluminum, stainless steel fasteners, knockouts for electrical and piping connections, condensate drain connection, and lifting lugs.

a. Access: Removable panels with neoprene compression gaskets and cam latches.
b. Insulation: Minimum 1-inch- thick, glass-fiber-insulation fill or closed cell foam, with thermal breaks.

B. Refrigeration System:

1. Refrigerant Coil with R-454b:
2. Compressors: Hermetic, scroll compressors with integral vibration isolators and crankcase heaters that de-energize during compressor operation; with thermal-expansion valves, filter-dryers, sight glasses, compressor service valves, and liquid- and suction-line service valves.

C. Controls:

1. Operating Control, Factory-Installed Microprocessor Controller:

a. Factory-installed operator panel with backlit display, capable of being remotely mounted, allows menu-driven display for navigation and control of unit.
b. Integral clock.
c. Personal computer interface.
d. Factory programmed.
e. Field-Mounted Sensors:

1) Relative space humidity. Humidistat to control unit operation, occupancy schedule to match associated AHU. Humidistat to control to 50% space relative humidity (adjustable).

f. Integral diagnostics.

D. Verification of Performance: Factory test and rate dehumidification units according to AHRI 910.

PART 3 - EXECUTION

3.1 INSTALLATION OF INDOOR, MECHANICAL DEHUMIDIFICATION UNITS

A. Equipment Mounting:

1. Install dehumidification units on unistrut equipment base(s) with waffle pad vibration isolators.

3.2 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner’s maintenance personnel to adjust, operate, and maintain dehumidification units.

END OF SECTION 23 84 16.16

SECTION 26 05 00 – COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section

B. Applicable codes include the following:

1. International Building Code 2024
2. ASHRAE 90.1-2016
3. International Fire Code 2024
4. National Electric Code 2023
5. NPS Building Standards

1.2 SUMMARY

A. Section Includes:

1. Electrical equipment coordination and installation.
2. Common electrical installation requirements.

1.3 COORDINATION

A. Coordinate arrangement, mounting, and support of electrical equipment:

1. To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.
2. To provide for ease of disconnecting the equipment with minimum interference to other installations.
3. To allow right of way for piping and conduit installed at required slope.
4. To connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 COMMON REQUIREMENTS FOR ELECTRICAL INSTALLATION

A. Comply with NECA 1.

B. Measure indicated mounting heights to bottom of unit for suspended items and to center of unit for wall-mounting items, except as otherwise indicated in drawings.

C. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide maximum possible headroom consistent with these requirements.

D. Equipment: Install to facilitate service, maintenance, and repair or replacement of components of both electrical equipment and other nearby installations. Connect in such a way as to facilitate future disconnecting with minimum interference with other items in the vicinity.

E. Right of Way: Give to piping systems installed at a required slope.

F. Exposed raceways/conduit: Where conduit or raceway is routed in an exposed location, the conduit and raceway shall be painted to match adjacent finishes. Coordinate required finishes with Contracting Officer.

END OF SECTION 26 05 00

SECTION 26 05 19 – LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

B. Adjust list below to suit Project.

1. Building wires and cables rated 600 V and less.
2. Connectors, splices, and terminations rated 600 V and less.

1.3 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Field quality-control test reports.

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 CONDUCTORS AND CABLES

A. Salient Characteristics

1. Physical:

a. Conductors shall be new and unused. Wire and cable shall be copper single conductor type with 600 V insulation, unless otherwise noted. Conductor shall be soft annealed Class B, per ASTM B-3 for solid wire and ASTM B-8 for stranded wire. Conductors shall be minimum 98% conductive.

2. Functional:

a. Insulation shall be flame retardant, heat resistant polyvinyl chloride (PVC), ethylene propylene (EP) or polyethylene (PE) with minimum insulation thicknesses per table 310-13 of the NEC. The insulation shall conform to the requirements of UL 83 ICEA S-68-516 for EP, ICEA S-61-402 for PVC and PE.
b. Type THWN or THHN wire and cable shall have a outer nylon jacket conforming to UL-83. Cables shall be manufactured to meet the standards of Insulated Cable Engineer's Association (ICEA).
c. Copper Conductors: Comply with NEMA WC 70.
d. Conductor Insulation: Comply with NEMA WC 70 for Types THWN-2 or XHHW-2.

3. Performance:

a. None.

2.2 CONNECTORS AND SPLICES

A. Salient Characteristics:

1. Physical:

a. Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 CONDUCTOR MATERIAL APPLICATIONS

A. Feeders: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

B. Branch Circuits: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

2.4 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

A. Exposed Feeders: Type THHN-THWN-2, single conductors in raceway.

B. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspaces: Type THHN-THWN-2, single conductors in raceway.

C. Feeders Concealed in Concrete, below Slabs-on-Grade, and Underground: Type XHHW-2, single conductors in raceway.

D. Exposed Branch Circuits, Including in Crawlspaces: Type THWN-2, single conductors in raceway.

E. Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN-THWN-2, single conductors in raceway. Metal-clad cable, Type MC is not allowed.

F. Branch Circuits Concealed in Concrete, below Slabs-on-Grade, and Underground: Type XHHW-2, single conductors in raceway.

G. Cord Drops and Portable Appliance Connections: Type SO, hard service cord with stainless-steel, wire-mesh, strain relief device at terminations to suit application.

H. Class 1 Control Circuits: Type THHN-THWN-2, in raceway.

I. Class 2 Control Circuits: Type THHN-THWN-2, in raceway.

2.5 INSTALLATION OF CONDUCTORS AND CABLES

A. Conceal cables in finished walls, ceilings, and floors, unless otherwise indicated.

B. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.

C. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.

D. Support cables according to Division 26 Section “Hangers and Supports for Electrical Systems.”

E. Identify and color-code conductors and cables according to Division 26 Section “Identification for Electrical Systems.”

2.6 CONNECTIONS

A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.

B. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.

C. Wiring at Outlets: Install conductor at each outlet, with at least 12 inches (300 mm) of slack.

D. Provide crimped ferrules for stranded wires at any screw terminal locations.

2.7 FIELD QUALITY CONTROL

A. Perform tests and inspections and prepare test reports.

B. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.

C. Test Reports: Prepare a written report to record the following:

1. Test procedures used.
2. Test results that comply with requirements.
3. Test results that do not comply with requirements and corrective action taken to achieve compliance with requirements.

D. Remove and replace malfunctioning units and retest as specified above.

END OF SECTION 26 05 19

SECTION 26 05 26 – GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Other Informational Submittals: Plans showing dimensioned as-built locations of grounding features specified in Part 3 “Field Quality Control” Article, including the following:

1. Ground rods.
2. Grounding arrangements and connections for separately derived systems.

C. Field quality-control test reports.

1.3 SUMMARY

A. This Section includes methods and materials for grounding systems and equipment.

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with UL 467 for grounding and bonding materials and equipment.

PART 2 - PRODUCTS

2.1 CONDUCTORS

A. Salient Characteristics:

1. Physical:

a. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.

b. Bare Copper Conductors:

1) Solid Conductors: ASTM B 3.
2) Stranded Conductors: ASTM B 8.
3) Tinned Conductors: ASTM B 33.
4) Sizes and types of conductors in four subparagraphs below are typical examples. 28-kcmil bonding cable in first subparagraph below is slightly larger than No. 6 AWG.
5) Bonding Cable: 28 kcmil, 14 strands of No. 17 AWG conductor, 1/4 inch (6 mm) in diameter.
6) Bonding Conductor: No. 4 or No. 6 AWG, stranded conductor.
7) Bonding Jumper: Copper tape, braided conductors, terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.
8) Tinned Bonding Jumper: Tinned-copper tape, braided conductors, terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.

2. Functional:

a. None.

3. Performance:

a. None.

2.2 CONNECTORS

A. Salient Characteristics:

1. Physical:

a. Listed and labeled by a nationally recognized testing laboratory acceptable to authorities having jurisdiction for applications in which used, and for specific types, sizes, and combinations of conductors and other items connected.

2. Functional:

a. Bolted Connectors for Conductors and Pipes: Copper or copper alloy, bolted pressure-type, with at least two bolts.
b. Pipe Connectors: Clamp type, sized for pipe.
c. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

3. Performance:

a. None.

2.3 GROUNDING ELECTRODES

A. Salient Characteristics:

1. Physical:

a. Ground Rods: Copper-clad; 5/8” in diameter by 96 inches in length.

2. Functional:

a. None.

3. Performance:

a. None

2.4 GROUNDING BUSBARS

A. Salient Characteristics:

1. Physical:

a. Grounding Bus: Predrilled rectangular bars of annealed copper, ¼ by 4 inches in cross section, with 9/32-inch holes spaced 1-1/8 inches apart. Stand-off insulators for mounting shall comply with UL 891 for use in switchboards, 600 V and shall be Lexan or PVC, impulse tested at 5000V.

2. Functional:

a. None.

3. Performance:

a. None.

PART 3 - EXECUTION

3.1 APPLICATIONS

A. Conductors: Install solid conductor for No. 8 AWG and smaller, and stranded conductors for No. 6 AWG and larger, unless otherwise indicated.

3.2 EQUIPMENT GROUNDING

A. Install insulated equipment grounding conductors with all feeders and branch circuits.

B. Install insulated equipment grounding conductors with the following items, in addition to those required by NFPA 70:

1. Feeders and branch circuits.
2. Lighting circuits.
3. Receptacle circuits.
4. Single-phase motor and appliance branch circuits.
5. Three-phase motor and appliance branch circuits.
6. Flexible raceway runs.

C. Air-Duct Equipment Circuits: Install insulated equipment grounding conductor to duct-mounted electrical devices operating at 120 V and more, including air cleaners, heaters, dampers, humidifiers, and other duct electrical equipment. Bond conductor to each unit and to air duct and connected metallic piping.

D. Water Heater, Heat-Tracing, and Antifrost Heating Cables: Install a separate insulated equipment grounding conductor to each electric water heater and heat-tracing cable. Bond conductor to heater units, piping, connected equipment, and components.

3.3 INSTALLATION

A. Grounding Conductors: Route along shortest and straightest paths possible, unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

B. Common Ground Bonding with Lightning Protection System: Comply with NFPA 780 and UL 96 when interconnecting with lightning protection system. Bond electrical power system ground directly to lightning protection system grounding conductor at closest point to electrical service grounding electrode. Use bonding conductor sized same as system grounding electrode conductor, and install in conduit.

C. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance, except where routed through short lengths of conduit.

1. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install so vibration is not transmitted to rigidly mounted equipment.
2. Use exothermic-welded connectors for outdoor locations, but if a disconnect-type connection is required, use a bolted clamp.

D. Grounding and Bonding for Piping:

1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes, using a bolted clamp connector or by bolting a lug-type connector to a pipe flange, using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.
2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
3. Bond each aboveground portion of gas piping system downstream from equipment shutoff valve.

E. Bonding Interior Metal Ducts: Bond metal air ducts to equipment grounding conductors of associated fans, blowers, electric heaters, and air cleaners. Install tinned bonding jumper to bond across flexible duct connections to achieve continuity.

F. Ground Rod: Re-use existing otherwise, drive rods until tops are 2 inches (50 mm) below finished floor or final grade, unless otherwise indicated.

G. Concrete-Encased Grounding Electrode (Ufer Ground): Re-use existing

3.4 FIELD QUALITY CONTROL

A. Perform the following tests and inspections and prepare test reports:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements listed below.
2. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal, and at individual ground rod. Make tests at ground rods before any conductors are connected.

    a. Measure ground resistance not less than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.
    b. Perform tests by fall-of-potential method according to IEEE 81.

B. Report measured ground resistances that exceed the following values:

1. Power and Lighting Equipment or System with Capacity 500 kVA and Less; 10 ohms.

END OF SECTION 26 05 26

SECTION 26 05 29 – HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Hangers and supports for electrical equipment and systems.
2. Construction requirements for concrete bases.

1.3 DEFINITIONS

A. EMT: Electrical metallic tubing.

B. IMC: Intermediate metal conduit.

C. RMC: Rigid metal conduit.

1.4 PERFORMANCE REQUIREMENTS

A. Design supports for multiple raceways, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

B. Design supports for multiple raceways capable of supporting combined weight of supported systems and its contents.

C. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.

D. Rated Strength: Adequate in tension, shear, and pullout force to resist maximum loads calculated or imposed for this Project, with a minimum structural safety factor of five times the applied force.

1.5 SUBMITTALS

A. Product Data: For the following:

1. Steel slotted support systems.
2. Nonmetallic slotted support systems.

B. Shop Drawings: Show fabrication and installation details and include calculations for the following:

1. Adjust list below to suit Project.
2. Steel slotted channel systems. Include Product Data for components.
3. Nonmetallic slotted channel systems. Include Product Data for components.
4. Equipment supports.

C. Welding certificates.

1.6 QUALITY ASSURANCE

A. Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

B. Comply with NFPA 70.

1.7 COORDINATION

A. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.

B. Coordinate installation of roof curbs, equipment supports, and roof penetrations. These items are specified in Division 07 Section.

PART 2 - PRODUCTS

2.1 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

A. Steel Slotted Support Systems:

1. Salient Characteristics:

a. Physical:

1) Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
2) Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4.
3) Painted Coatings: Manufacturer's standard painted coating applied according to MFMA-4.
4) Channel Dimensions: Selected for applicable load criteria.

b. Functional:

1) None.

c. Performance:

1) Comply with MFMA-4, factory-fabricated components for field assembly.

B. Nonmetallic Slotted Support Systems:

1. Salient Characteristics:

a. Physical:

1) Structural-grade, factory-formed, glass-fiber-resin channels and angles with 9/16-inch- (14-mm-) diameter holes at a maximum of 8 inches (200 mm) o.c., in at least 1 surface.

b. Functional:

1) Fittings and Accessories: Products of channel and angle manufacturer and designed for use with those items.
2) Fitting and Accessory Materials: Same as channels and angles.

c. Performance:

1) Rated Strength: Selected to suit applicable load criteria.

C. Raceway and Cable Supports:

1. Salient Characteristics:

a. Physical, Functional and Performance:

1) As described in NECA 1 and NECA 101.

D. Conduit and Cable Support Devices:

1. Salient Characteristics:

a. Physical:

1) Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.

b. Functional:

1) None.

c. Performance:

1) None.

E. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:

1. Powder-Actuated Fasteners:

a. Salient Characteristics:

1) Physical:

a) Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.

2) Functional:

a) None.

3) Performance:

a) None.

2. Mechanical-Expansion Anchors:

a. Salient Characteristics:

1) Physical:

a) Insert-wedge-type, zinc-coated steel, for use in hardened portland cement concrete with tension, shear, and pullout capacities appropriate for supported loads and building materials in which used.

2) Functional:

a) None.

3) Performance:

a) None.

3. Concrete Inserts:

a. Salient Characteristics:

1) Physical:

a) Steel or malleable-iron, slotted support system units similar to MSS Type 18; complying with MFMA-4 or MSS SP-58.

2) Functional:

a) None.

3) Performance:

a) None.

4. Clamps for Attachment to Steel Structural Elements:

a. Salient Characteristics:

1) Physical:

a) MSS SP-58, type suitable for attached structural element.

2) Functional:

a) None.

3) Performance:

a) None.

5. Through Bolts:

a. Salient Characteristics:

1) Physical:

a) Structural type, hex head, and high strength. Comply with ASTM A 325.

2) Functional:

a) None.

3) Performance:

a) None.

6. Toggle Bolts:

a. Salient Characteristics:

1) Physical:

a) All-steel springhead type.

2) Functional:

a) None.

3) Performance:

a) None.

7. Hanger Rods:

a. Salient Characteristics:

1) Physical:

a) Threaded steel.

2) Functional:

a) None.

3) Performance:

a) None.

2.2 FABRICATED METAL EQUIPMENT SUPPORT ASSEMBLIES

A. Description: Welded or bolted, structural-steel shapes, shop or field fabricated to fit dimensions of supported equipment.

B. Materials: Comply with requirements in Division 05 Section “Metal Fabrications” for steel shapes and plates.

PART 3 - EXECUTION

3.1 APPLICATION

A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems except if requirements in this Section are stricter.

B. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMT, IMC, and RMC as required by NFPA 70. Minimum rod size shall be 1/4 inch (6 mm) in diameter.

C. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1-1/2-inch (38-mm) and smaller raceways serving branch circuits and communication systems above suspended ceilings and for fastening raceways to trapeze supports.

3.2 SUPPORT INSTALLATION

A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this Article.

B. Raceway Support Methods: In addition to methods described in NECA 1, EMT may be supported by openings through structure members, as permitted in NFPA 70.

C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb (90 kg).

D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:

1. To Wood: Fasten with lag screws or through bolts.
2. To New Concrete: Bolt to concrete inserts.
3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
4. To Existing Concrete: Expansion anchor fasteners.
5. Retain first subparagraph below if powder-actuated devices are allowed. Consider deleting if Project contains both lightweight and standard-weight concrete or more than one thickness of concrete slab.
6. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inches (100 mm) thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inches (100 mm) thick.
7. To Steel: Welded threaded studs complying with AWS D1.1/D1.1M, with lock washers and nuts.
8. To Light Steel: Sheet metal screws.
9. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate.

E. Drill holes for expansion anchors in concrete at locations and to depths that avoid reinforcing bars.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

A. Comply with installation requirements in Division 05 Section “Metal Fabrications” for site-fabricated metal supports.

B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.

C. Field Welding: Comply with AWS D1.1/D1.1M.

3.4 PAINTING

A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils (0.05 mm).

B. Touchup: Comply with requirements in Division 09 painting Sections for cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.

C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION 26 05 29

SECTION 26 05 33 – RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes raceways, fittings, boxes, enclosures, and cabinets for electrical wiring.

1.3 DEFINITIONS

A. EMT: Electrical metallic tubing.

B. FMC: Flexible metal conduit.

C. IMC: Intermediate metal conduit.

D. LFMC: Liquidtight flexible metal conduit.

E. RNC: Rigid nonmetallic conduit.

F. RSC: Rigid steel conduit

1.4 SUBMITTALS

A. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.

1.5 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 METAL CONDUIT AND TUBING

A. Salient Characteristics:

1. Physical:

a. Fittings for Conduit (Including all Types and Flexible and Liquidtight), EMT, and Cable: NEMA FB 1; listed for type and size raceway with which used, and for application and environment in which installed.

1) Fittings for EMT: Steel, set-screw or compression type.
2) Coating for Fittings for PVC-Coated Conduit: Minimum thickness, 0.040 inch (1 mm), with overlapping sleeves protecting threaded joints.

b. Joint Compound for Rigid Steel Conduit or IMC: Listed for use in cable connector assemblies, and compounded for use to lubricate and protect threaded raceway joints from corrosion and enhance their conductivity.

2. Functional:

a. None.

3. Performance:

a. Rigid Steel Conduit: ANSI C80.1.
b. Aluminum Rigid Conduit: ANSI C80.5.
c. IMC: ANSI C80.6.
d. PVC-Coated Steel Conduit: PVC-coated Rigid Steel Conduit.

1) Comply with NEMA RN 1.
2) Coating Thickness: 0.040 inch (1 mm), minimum.

e. EMT: ANSI C80.3.
f. LFMC: Flexible steel conduit with PVC jacket.

2.2 NONMETALLIC CONDUIT AND TUBING

A. Salient Characteristics:

1. Physical:

a. RNC: NEMA TC 2, Type EPC-40-PVC, unless otherwise indicated. Type EPC-80-PVC for areas that will be subjected to vehicular travel.
b. Fittings for RNC: NEMA TC 3; match to conduit or tubing type and material.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 SURFACE RACEWAYS

A. Salient Characteristics:

1. Physical:

a. Surface Metal Raceways: Galvanized steel with snap-on covers. Manufacturer's standard enamel finish in color selected by the Contracting Officer.

2. Functional:

a. None.

3. Performance:

a. None.

2.4 BOXES, ENCLOSURES, AND CABINETS

A. Salient Characteristics:

1. Physical:

a. Metal Floor Boxes: Cast or sheet metal, semi-adjustable, rectangular.
b. Nonmetallic Floor Boxes: Nonadjustable, round.
c. Cast-Metal Access, Pull, and Junction Boxes: Cast aluminum with gasketed cover.
d. Cast-Metal Outlet and Device Boxes: Ferrous alloy, Type FD, with gasketed cover.
e. Hinged-Cover Enclosures: Type 1, with continuous-hinge cover with flush latch, unless otherwise indicated.

1) Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
2) Nonmetallic Enclosures: Plastic.

f. Cabinets: Type 1, galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.

1) Hinged door in front cover with flush latch and concealed hinge.
2) Key latch to match panelboards.
3) Metal barriers to separate wiring of different systems and voltage.
4) Accessory feet where required for freestanding equipment.

2. Functional:

a. None.

3. Performance:

a. Sheet Metal Outlet and Device Boxes: NEMA OS 1.
b. Cast-Metal Outlet and Device Boxes: NEMA FB 1.
c. Nonmetallic Outlet and Device Boxes: NEMA OS 2.
d. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
e. Cast-Metal Access, Pull, and Junction Boxes: NEMA FB 1
f. Hinged-Cover Enclosures: NEMA 250.
g. Cabinets: NEMA 250.

2.5 HANDHOLES AND BOXES FOR EXTERIOR UNDERGROUND WIRING

A. Salient Characteristics:

1. Physical:

a. Color of Frame and Cover: Gray.
b. Configuration: Units shall be designed for flush burial and have open bottom, unless otherwise indicated.
c. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure.
d. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
e. Cover Legend: Molded lettering, “ELECTRIC.” or “TELEPHONE.” depending upon the cabling the box is enclosing.
f. Conduit Entrance Provisions: Conduit-terminating fittings shall mate with entering ducts for secure, fixed installation in enclosure wall.
g. Handholes and boxes are to be fiberglass with Polymer-Concrete Fram and Cover. Sheet-molded fiberglass-reinforced polyester-resin enclosure joined to polymer-concrete top ring or frame.

2. Functional:

a. None.

3. Performance:

a. Comply with SCTE 77, TIER 15.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

A. Outdoors: Apply raceway products as specified below, unless otherwise indicated:

1. Exposed Conduit: Rigid steel conduit.
2. Concealed Conduit, Aboveground: EMT.
3. Retain subparagraph below if Project includes small amounts of exterior underground wiring, 600 V and less, and Division 26 Section “Underground Ducts and Raceways for Electrical Systems” is not being used.

4. Underground Conduit: For horizontal underground conduit, utilize RNC, Type EPC-80-PVC, direct buried. Underground bends are to utilize PVC Coated Rigid Steel prefabricated sections. Vertical underground conduit (risers) shall be PVC Coated Rigid Steel.
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.
6. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.

B. Comply with the following indoor applications, unless otherwise indicated:

1. Exposed, Not Subject to Physical Damage: EMT.
2. Exposed, Not Subject to Severe Physical Damage: EMT.
3. Exposed, Subject to Severe Physical Damage: RSC.
4. Concealed in Ceilings and Interior Walls and Partitions: EMT.
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.
6. Damp or Wet Locations: IMC.
7. Raceways for Optical Fiber or Communications Cable in Spaces Used for Environmental Air: EMT.
8. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 3R, stainless steel in damp or wet locations.

C. Minimum Raceway Size: 3/4-inch (21-mm) trade size.

D. Raceway Fittings: Compatible with raceways and suitable for use and location.

1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings, unless otherwise indicated.
2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with that material. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer.

E. Install nonferrous conduit or tubing for circuits operating above 60 Hz. Where aluminum raceways are installed for such circuits and pass through concrete, install in nonmetallic sleeve.

F. Do not install aluminum conduits in contact with concrete.

3.2 INSTALLATION

A. Comply with NECA 1 for installation requirements applicable to products specified in Part 2 except where requirements on Drawings or in this Article are stricter.

B. Keep raceways at least 6 inches (150 mm) away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.

C. Complete raceway installation before starting conductor installation.

D. Support raceways as specified in Division 26 Section “Hangers and Supports for Electrical Systems.”

E. Install no more than the equivalent of three 90-degree bends in any conduit run except for communications conduits, for which fewer bends are allowed.

F. Conceal conduit and EMT within finished walls, ceilings, and floors, unless otherwise indicated.

G. Raceways Embedded in Slabs:

1. Run conduit larger than 1-inch (27-mm) trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support.
2. RNC, Type EPC-40-PVC, rigid steel conduit, or IMC before rising above the floor.

H. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.

I. Raceway Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors, including conductors smaller than No. 4 AWG.

J. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb (90-kg) tensile strength. Leave at least 12 inches (300 mm) of slack at each end of pull wire.

K. Install raceway sealing fittings at suitable, approved, and accessible locations and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install raceway sealing fittings at the following points:

1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
2. Where otherwise required by NFPA 70.

L. Expansion-Joint Fittings for RNC: Install in each run of aboveground conduit that is located where environmental temperature change may exceed 30 deg F (17 deg C), and that has straight-run length that exceeds 25 feet (7.6 m).

1. Install expansion-joint fittings for each of the following locations, and provide type and quantity of fittings that accommodate temperature change listed for location:

a. Outdoor Locations Not Exposed to Direct Sunlight: 125 deg F (70 deg C) temperature change.
b. Outdoor Locations Exposed to Direct Sunlight: 155 deg F (86 deg C) temperature change.
c. Indoor Spaces: Connected with the Outdoors without Physical Separation: 125 deg F (70 deg C) temperature change.
d. Attics: 135 deg F (75 deg C) temperature change.

2. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per deg F (0.06 mm per meter of length of straight run per deg C) of temperature change.

3. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at the time of installation.

M. Flexible Conduit Connections: Use maximum of 72 inches (1830 mm) of flexible conduit for recessed and semirecessed lighting fixtures, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.

1. Use LFMC in damp or wet locations subject to severe physical damage.
2. Use LFMC in damp or wet locations not subject to severe physical damage.

N. Set metal floor boxes level and flush with finished floor surface.

O. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

P. Install plastic bushings at conduit terminations prior to pulling wire.

3.3 INSTALLATION OF UNDERGROUND CONDUIT

A. Direct-Buried Conduit:

1. Excavate trench bottom to provide firm and uniform support for conduit. Prepare trench bottom as indicated in drawings for pipe less than 6 inches (150 mm) in nominal diameter.
2. Install backfill as indicated in drawings.
3. After installing conduit, backfill and compact. Start at tie-in point, and work toward end of conduit run, leaving conduit at end of run free to move with expansion and contraction as temperature changes during this process. Firmly hand tamp backfill around conduit to provide maximum supporting strength. After placing controlled backfill to within 12 inches (300 mm) of finished grade, make final conduit connection at end of run and complete backfilling with normal compaction as indicated in drawings.
4. Install manufactured duct elbows for stub-ups at poles and equipment and at building entrances through floor unless otherwise indicated. Encase elbows for stub-up ducts throughout length of elbow.
5. Install manufactured PVC coated rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through floor.

    a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches (75 mm) of concrete for a minimum of 12 inches (300 mm) on each side of the coupling.
    b. For stub-ups at equipment mounted on outdoor concrete bases and where conduits penetrate building foundations, extend steel conduit horizontally a minimum of 60 inches (1500 mm) from edge of foundation or equipment base. Install insulated grounding bushings on terminations at equipment.

6. Underground Warning Tape: Comply with requirements in Section 260553 “Identification for Electrical Systems.”

3.4 INSTALLATION OF UNDERGROUND HANDHOLES AND BOXES

A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.

B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch (12.5-mm) sieve to No. 4 (4.75-mm) sieve and compacted to same density as adjacent undisturbed earth.

C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 1 inch (25 mm) above finished grade.

D. Install handholes and boxes with bottom below the frost line, 30” below grade.

E. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables, but short enough to preserve adequate working clearances in the enclosure.

F. Field-cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.5 PROTECTION

A. Provide final protection and maintain conditions that ensure coatings, finishes, and cabinets are without damage or deterioration at time of Substantial Completion.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 26 05 33

SECTION 26 05 44 – SLEEVES AND SLEEVE SEALS FOR ELECTRICAL RACEWAYS AND CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Sleeves for raceway and cable penetration of non-fire-rated construction walls and floors.
2. Sleeve-seal systems.
3. Sleeve-seal fittings.
4. Grout.
5. Silicone sealants.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 SLEEVES

A. Salient Characteristics:

1. Physical:

a. Wall Sleeves:

1) Steel Pipe Sleeves: Type E, Grade B, Schedule 40, zinc coated, plain ends.
2) Cast-Iron Pipe Sleeves: Cast or fabricated “wall pipe,” equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop unless otherwise indicated.

b. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies: Galvanized-steel sheet; 0.0239-inch (0.6-mm) minimum thickness; round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.
c. PVC-Pipe Sleeves: Schedule 40
d. Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.

e. Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.
f. Sleeves for Rectangular Openings:

1) Material: Galvanized sheet steel.
2) Minimum Metal Thickness:

a) For sleeve cross-section rectangle perimeter less than 50 inches (1270 mm) and with no side larger than 16 inches (400 mm), thickness shall be 0.052 inch (1.3 mm).
b) For sleeve cross-section rectangle perimeter 50 inches (1270 mm) or more and one or more sides larger than 16 inches (400 mm), thickness shall be 0.138 inch (3.5 mm).

2. Functional:

a. None.

3. Performance:

a. Wall Sleeves:

1) Steel Pipe Sleeves: ASTM A 53/A 53M

b. PVC-Pipe Sleeves: ASTM D 1785

2.2 SLEEVE-SEAL SYSTEMS

A. Salient Characteristics:

1. Physical:

a. Modular sealing device, designed for field assembly, to fill annular space between sleeve and raceway or cable.
b. Sealing Elements: EPDM rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
c. Pressure Plates: Carbon steel
d. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 SLEEVE-SEAL FITTINGS

A. Salient Characteristics:

1. Physical:

a. Manufactured plastic, sleeve-type, waterstop assembly made for embedding in concrete slab or wall. Unit shall have plastic or rubber waterstop collar with center opening to match piping OD..

2. Functional:

a. None.

3. Performance:

a. None.

2.4 GROUT

A. Salient Characteristics:

1. Physical:

a. Nonshrink; recommended for interior and exterior sealing openings in non-fire-rated walls or floors.

2. Functional:

a. Premixed and factory packaged.

3. Performance:

a. ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
b. Design Mix: 5000-psi (34.5-MPa), 28-day compressive strength.

2.5 SILICONE SEALANTS

A. Salient Characteristics:

1. Physical:

a. Silicone Sealants:

1) Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below.

b. Silicone Foams:

1) Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.

2. Functional:

a. Silicone Sealants:

1) Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces that are not fire rated.

3. Performance:

a. Silicone Sealants:

1) Sealant shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION FOR NON-FIRE-RATED ELECTRICAL PENETRATIONS

A. Comply with NECA 1.

B. Comply with NEMA VE 2 for cable tray and cable penetrations.

C. Sleeves for Conduits Penetrating Above-Grade Non-Fire-Rated Concrete and Masonry-Unit Floors and Walls:

1. Interior Penetrations of Non-Fire-Rated Walls and Floors:

a. Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint. Comply with requirements in Section 079200 "Joint Sealants."
b. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall so no voids remain. Tool exposed surfaces smooth; protect material while curing.

2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
3. Size pipe sleeves to provide 1/4-inch (6.4-mm) annular clear space between sleeve and raceway or cable unless sleeve seal is to be installed or unless seismic criteria require different clearance.

4. Install sleeves for wall penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.

5. Install sleeves for floor penetrations. Extend sleeves installed in floors [2 inches (50 mm)] above finished floor level. Install sleeves during erection of floors.

D. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies:

1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.
2. Seal space outside of sleeves with approved joint compound for gypsum board assemblies.

E. Roof-Penetration Sleeves: Seal penetration of individual raceways and cables with flexible boot-type flashing units applied in coordination with roofing work.

F. Aboveground, Exterior-Wall Penetrations: Seal penetrations using steel pipe sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch (25-mm) annular clear space between pipe and sleeve for installing mechanical sleeve seals.

G. Underground, Exterior-Wall and Floor Penetrations: Install cast-iron pipe sleeves. Size sleeves to allow for 1-inch (25-mm) annular clear space between raceway or cable and sleeve for installing sleeve-seal system.

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at raceway entries into building.

B. Install type and number of sealing elements recommended by manufacturer for raceway or cable material and size. Position raceway or cable in center of sleeve. Assemble mechanical sleeve seals and install in annular space between raceway or cable and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

3.3 SLEEVE-SEAL-FITTING INSTALLATION

A. Install sleeve-seal fittings in new walls and slabs as they are constructed.

B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.

C. Secure nailing flanges to concrete forms.

D. Using grout, seal the space around outside of sleeve-seal fittings.

END OF SECTION 26 05 44

SECTION 26 05 53 – IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Adjust list below to suit Project.
2. Identification for raceway and metal-clad cable.
3. Identification for conductors and communication and control cable.
4. Underground-line warning tape.
5. Warning labels and signs.
6. Instruction signs.
7. Equipment identification labels.
8. Miscellaneous identification products.

1.3 SUBMITTALS

A. Product Data: For each electrical identification product indicated.

B. Identification Schedule: An index of nomenclature of electrical equipment and system components used in identification signs and labels.

1.4 QUALITY ASSURANCE

A. Comply with ANSI A13.1 and ANSI C2.

B. Comply with NFPA 70.

C. Comply with 29 CFR 1910.145.

1.5 COORDINATION

A. Coordinate identification names, abbreviations, colors, and other features with requirements in the Contract Documents, Shop Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual, and with those required by codes, standards, and 29 CFR 1910.145. Use consistent designations throughout Project.

B. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

C. Coordinate installation of identifying devices with location of access panels and doors.

D. Install identifying devices before installing acoustical ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 RACEWAY AND METAL-CLAD CABLE IDENTIFICATION MATERIALS

A. Salient Characteristics:

1. Physical:

a. Color for Printed Legend:

1) Power Circuits: Black letters on an orange field.
2) Legend: Indicate system or service and voltage, if applicable.

b. Self-Adhesive Vinyl Labels: Preprinted, flexible label laminated with a clear, weather- and chemical-resistant coating and matching wraparound adhesive tape for securing ends of legend label.

c. Snap-Around Labels: Slit, pretensioned, flexible, preprinted, color-coded acrylic sleeves, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action.

d. Snap-Around, Color-Coding Bands: Slit, pretensioned, flexible, solid-colored acrylic sleeves, 2 inches (50 mm) long, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action.

e. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; 2 inches (50 mm) wide; compounded for outdoor use.

2. Functional:

a. None.

3. Performance:

a. Comply with ANSI A13.1 for minimum size of letters for legend and for minimum length of color field for each raceway and cable size.

2.2 CONDUCTOR AND COMMUNICATION- AND CONTROL-CABLE IDENTIFICATION MATERIALS

A. Salient Characteristics:

1. Physical:

a. Color-Coding Conductor Tape: Colored, self-adhesive vinyl tape not less than 3 mils (0.08 mm) thick by 1 to 2 inches (25 to 50 mm) wide.
b. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.
c. Aluminum Wraparound Marker Labels: Cut from 0.014-inch- (0.35-mm-) thick aluminum sheet, with stamped, embossed, or scribed legend, and fitted with tabs and matching slots for permanently securing around wire or cable jacket or around groups of conductors.
d. Metal Tags: Brass or aluminum, 2 by 2 by 0.05 inch (50 by 50 by 1.3 mm), with stamped legend, punched for use with self-locking nylon tie fastener.
e. Write-On Tags: Polyester tag, 0.010 inch (0.25 mm) thick, with corrosion-resistant grommet and polyester or nylon tie for attachment to conductor or cable.

1) Marker for Tags: Permanent, waterproof, black ink marker recommended by tag manufacturer.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 UNDERGROUND-LINE WARNING TAPE

A. Salient Characteristics:

1. Physical:

a. Tape Characteristics:

1) Detectable three-layer laminate, consisting of a printed pigmented polyolefin film, a solid aluminum-foil core, and a clear protective film that allows inspection of the continuity of the conductive core, bright-colored, compounded for direct-burial service.
2) Overall Thickness: 5 mils (0.125 mm).
3) Foil Core Thickness: 0.35 mil (0.00889 mm).
4) Weight: 28 lb/1000 sq. ft. (13.7 kg/100 sq. m).
5) 3-Inch (75-mm) Tensile According to ASTM D 882: 70 lbf (311.3 N), and 4600 psi (31.7 MPa).

b. Color and Printing:

1) Inscriptions for Red-Colored Tapes: ELECTRIC LINE, HIGH VOLTAGE.
2) Inscriptions for Orange-Colored Tapes: TELEPHONE CABLE, CATV CABLE, COMMUNICATIONS CABLE, OPTICAL FIBER CABLE.
3) Inscriptions for Blue-Colored Tapes: CONTROL CIRCUIT CABLE.

2. Functional:

a. Tape:

1) Recommended by manufacturer for the method of installation and suitable to identify and locate underground electrical utility lines.
2) Printing on tape shall be permanent and shall not be damaged by burial operations.
3) Tape material and ink shall be chemically inert, and not subject to degrading when exposed to acids, alkalis, and other destructive substances commonly found in soils.

3. Performance:

a. Color and Printing:

1) Comply with ANSI Z535.1 through ANSI Z535.5.

2.4 WARNING LABELS AND SIGNS

A. Salient Characteristics:

1. Physical:

a. Self-Adhesive Warning Labels: Factory printed, multicolor, pressure-sensitive adhesive labels, configured for display on front cover, door, or other access to equipment, unless otherwise indicated.
b. Baked-Enamel Warning Signs: Preprinted aluminum signs, punched or drilled for fasteners, with colors, legend, and size required for application. 1/4-inch (6.4-mm) grommets in corners for mounting. Nominal size, 7 by 10 inches (180 by 250 mm).
c. Metal-Backed, Butyrate Warning Signs: Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs with 0.0396-inch (1-mm) galvanized-steel backing; and with colors, legend, and size required for application. 1/4-inch (6.4-mm) grommets in corners for mounting. Nominal size, 10 by 14 inches (250 by 360 mm).

2. Functional:

a. Warning label and sign shall include, but are not limited to, the following legends:

1) Workspace Clearance Warning: “WARNING – OSHA REGULATION – AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES (915 MM).”

3. Performance:

a. Comply with NFPA 70 and 29 CFR 1910.145.

2.5 INSTRUCTION SIGNS

A. Salient Characteristics:

1. Physical:

a. Engraved, laminated acrylic or melamine plastic, minimum 1/16 inch (1.6 mm) thick for signs up to 20 sq. in. (129 sq. cm) and 1/8 inch (3.2 mm) thick for larger sizes.

1) Engraved legend with black letters on white face.
2) Punched or drilled for mechanical fasteners.
3) Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2. Functional:

a. None.

3. Performance:

a. None.

2.6 EQUIPMENT IDENTIFICATION LABELS

A. Salient Characteristics:

1. Physical:

a. Adhesive Film Label: Machine printed, in black, by thermal transfer or equivalent process. Minimum letter height shall be 3/8 inch (10 mm).
b. Self-Adhesive, Engraved, Laminated Acrylic or Melamine Label: Adhesive backed, with white letters on a dark-gray background. Minimum letter height shall be 3/8 inch (10 mm).
c. Engraved, Laminated Acrylic or Melamine Label: Punched or drilled for screw mounting. White letters on a dark-gray background. Minimum letter height shall be 3/8 inch (10 mm).
d. Stenciled Legend: In nonfading, waterproof, black ink or paint. Minimum letter height shall be 1 inch (25 mm).

2. Functional:

a. None.

3. Performance:

a. None.

2.7 MISCELLANEOUS IDENTIFICATION PRODUCTS

A. Salient Characteristics:

1. Physical:

a. Cable Ties: Fungus-inert, self-extinguishing, 1-piece, self-locking, Type 6/6 nylon cable ties.

1) Minimum Width: 3/16 inch (5 mm).
2) Tensile Strength: 50 lb (22.6 kg), minimum.
3) Temperature Range: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
4) Color: Black, except where used for color-coding.

b. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

2. Functional:

a. None.

3. Performance:

a. Paint: Paint materials and application requirements are specified in Division 09 painting Sections.

PART 3 - EXECUTION

3.1 APPLICATION

A. Accessible Raceways and Cables of Auxiliary Systems: Identify the following systems with color-coded, self-adhesive vinyl tape applied in bands:

1. Fire Alarm System: Red.
2. Fire-Suppression Supervisory and Control System: Red and yellow.
3. Combined Fire Alarm and Security System: Red and blue.
4. Security System: Blue and yellow.
5. Mechanical and Electrical Supervisory System: Green and blue.
6. Telecommunication System: Green and yellow.
7. Control Wiring: Green and red.

B. Power-Circuit Conductor Identification: For secondary conductors No. 1/0 AWG and larger in vaults, pull and junction boxes, manholes, and handholes use color-coding conductor tape. Identify source and circuit number of each set of conductors. For single conductor cables, identify phase in addition to the above.

C. Branch-Circuit Conductor Identification: Where there are conductors for more than three branch circuits in same junction or pull box, use color-coding conductor tape. Identify each ungrounded conductor according to source and circuit number.

D. Conductors to Be Extended in the Future: Attach write-on tags to conductors and list source and circuit number.

E. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, signal, sound, intercommunications, voice, and data connections.

1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.
2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
3. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and Operation and Maintenance Manual.

F. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Comply with 29 CFR 1910.145 and apply self-adhesive warning labels. Identify system voltage with black letters on an orange background. Apply to exterior of door, cover, or other access.

1. Equipment Requiring Workspace Clearance According to NFPA 70: Unless otherwise indicated, apply to door or cover of equipment but not on flush panelboards and similar equipment in finished spaces.

G. Equipment Identification Labels: On each unit of equipment, install unique designation label that is consistent with wiring diagrams, schedules, and Operation and Maintenance Manual. Apply labels to disconnect switches and protection equipment, central or master units, control panels, control stations, terminal cabinets, and racks of each system. Systems include power, lighting, control, communication, signal, monitoring, and alarm systems unless equipment is provided with its own identification.

1. Labeling Instructions:

a. Indoor Equipment: Adhesive film label. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on 1-1/2-inch- (38-mm-) high label; where 2 lines of text are required, use labels 2 inches (50 mm) high.
b. Outdoor Equipment: Engraved, laminated acrylic or melamine label.
c. Elevated Components: Increase sizes of labels and letters to those appropriate for viewing from the floor.

2. Equipment to Be Labeled:

a. Panelboards, electrical cabinets, and enclosures.
b. Disconnect switches.
c. Enclosed circuit breakers.

H. Arch Flash Hazard warning label: Refer to Specification 26 05 73 section 3.5(B) for labeling requirements associated with the Arc Flash Hazard Analysis study.

3.2 INSTALLATION

A. Verify identity of each item before installing identification products.

B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.

C. Apply identification devices to surfaces that require finish after completing finish work.

D. Self-Adhesive Identification Products: Clean surfaces before application, using materials and methods recommended by manufacturer of identification device.

E. Attach nonadhesive signs and plastic labels with screws and auxiliary hardware appropriate to the location and substrate.

F. System Identification Color Banding for Raceways and Cables: Each color band shall completely encircle cable or conduit. Place adjacent bands of two-color markings in contact, side by side. Locate bands at changes in direction, at penetrations of walls and floors, at 50-foot (15-m) maximum intervals in straight runs, and at 25-foot (7.6-m) maximum intervals in congested areas.

G. Color-Coding for Phase Identification, 600 V and Less: Use the colors listed below for ungrounded branch-circuit conductors.

1. Color shall be factory applied.
2. Colors for 240/120-V Circuits:

    a. Phase A: Black.
    b. Phase B: Red.

3. Field-Applied, Color-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 6 inches (150 mm) from terminal points and in boxes where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding. Locate bands to avoid obscuring factory cable markings.

H. Aluminum Wraparound Marker Labels and Metal Tags: Secure tight to surface of conductor or cable at a location with high visibility and accessibility.

I. Painted Identification: Prepare surface and apply paint according to Division 09 painting Sections.

END OF SECTION 26 05 53

SECTION 26 05 73 – ELECTRICAL SYSTEM STUDIES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplemental General Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes computer-based, fault-current, overcurrent protective device coordination, and arc flash hazard analysis studies. Protective devices shall be set based on results of the protective device coordination study.

1.3 SUBMITTALS

A. Product Data: For computer software program to be used for studies.

B. Product Certificates: For coordination-study and fault-current-study computer software programs, certifying compliance with IEEE 399.

C. Qualification Data: For coordination-study specialist.

D. Other Action Submittals: The following submittals shall be made after the approval process for system protective devices has been completed. Submittals shall be in digital form.

1. Coordination-study input data, including completed computer program input data sheets.
2. Study and Equipment Evaluation Reports.
3. Coordination-Study Report.

1.4 QUALITY ASSURANCE

A. Studies shall use computer programs that are distributed nationally and are in wide use. Software algorithms shall comply with requirements of standards and guides specified in this Section. Manual calculations are not acceptable.

B. Coordination-Study Specialist Qualifications: An entity experienced in the application of computer software used for studies, having performed successful studies of similar magnitude on electrical distribution systems using similar devices.

1. Professional engineer, licensed in the state where Project is located, shall be responsible for the study. All elements of the study shall be performed under the direct supervision and control of engineer and Contracting Officer.

C. Comply with IEEE 242 for short-circuit currents and coordination time intervals.

D. Comply with IEEE 399 for general study procedures.

PART 2 - PRODUCTS

2.1 COMPUTER SOFTWARE PROGRAM REQUIREMENTS

A. Salient Characteristics:

1. Physical:

a. None

2. Functional:

a. Computer software program shall be capable of plotting and diagramming time-current-characteristic curves as part of its output. Computer software program shall report device settings and ratings of all overcurrent protective devices and shall demonstrate selective coordination by computer-generated, time-current coordination plots.

b. Analytical features of fault-current-study computer software program shall include "mandatory," "very desirable," and "desirable" features as listed in IEEE 399.

3. Performance:

a. Comply with IEEE 399.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine Project overcurrent protective device submittals for compliance with electrical distribution system coordination requirements and other conditions affecting performance. Devices to be coordinated are indicated on Drawings.

1. Proceed with coordination study only after relevant equipment submittals have been assembled. Overcurrent protective devices that have not been submitted and approved prior to coordination study may not be used in study.

3.2 POWER SYSTEM DATA

A. Gather and tabulate the following input data to support coordination study:

1. Product Data for overcurrent protective devices specified in other Division 26 Sections and involved in overcurrent protective device coordination studies. Use equipment

designation tags that are consistent with electrical distribution system diagrams, overcurrent protective device submittals, input and output data, and recommended device settings.

2. Impedance of utility service entrance.
3. Electrical Distribution System Diagram: In hard-copy and electronic-copy formats, showing the following:

    a. Circuit-breaker and fuse-current ratings and types.
    b. Relays and associated power and current transformer ratings and ratios.
    c. Utility transformer kilovolt amperes, primary and secondary voltages, connection type, impedance, and X/R ratios.
    d. Cables: Indicate conduit material, sizes of conductors, conductor material, insulation, and length.
    e. Busway ampacity and impedance.
    f. Motor horsepower and code letter designation according to NEMA MG 1.

4. Data sheets to supplement electrical distribution system diagram, cross-referenced with tag numbers on diagram, showing the following:

    a. Special load considerations, including starting inrush currents and frequent starting and stopping.
    b. Motor full-load current, locked rotor current, service factor, starting time, type of start, and thermal-damage curve.
    c. Ratings, types, and settings of utility company's overcurrent protective devices.
    d. Special overcurrent protective device settings or types stipulated by utility company.
    e. Time-current-characteristic curves of devices indicated to be coordinated.
    f. Manufacturer, frame size, interrupting rating in amperes rms symmetrical, ampere or current sensor rating, long-time adjustment range, short-time adjustment range, and instantaneous adjustment range for circuit breakers.
    g. Panelboards, switchboards, motor-control center ampacity, and interrupting rating in amperes rms symmetrical.

3.3 FAULT-CURRENT STUDY

A. Calculate the maximum available short-circuit current in amperes rms symmetrical at circuit-breaker positions of the electrical power distribution system. The calculation shall be for a current immediately after initiation and for a three-phase bolted short circuit at each of the following:

    1. Service disconnects.
    2. Branch circuit panelboard.

B. Study electrical distribution system from normal and alternate power sources throughout electrical distribution system for Project. Include studies of system-switching configurations and alternate operations that could result in maximum fault conditions.

C. Calculate momentary and interrupting duties on the basis of maximum available fault current.

D. Calculations to verify interrupting ratings of overcurrent protective devices shall comply with IEEE 141 and IEEE 242.

1. Low-Voltage Circuit Breakers: IEEE 1015 and IEEE C37.20.1.
2. Low-Voltage Fuses: IEEE C37.46.

E. Study Report:

1. Show calculated X/R ratios and equipment interrupting rating (1/2-cycle) fault currents on electrical distribution system diagram.

F. Equipment Evaluation Report:

1. For 600-V overcurrent protective devices, ensure that interrupting ratings are equal to or higher than calculated 1/2-cycle symmetrical fault current.
2. For devices and equipment rated for asymmetrical fault current, apply multiplication factors listed in the standards to 1/2-cycle symmetrical fault current.
3. Verify adequacy of phase conductors at maximum three-phase bolted fault currents; verify adequacy of equipment grounding conductors and grounding electrode conductors at maximum ground-fault currents. Ensure that short-circuit withstand ratings are equal to or higher than calculated 1/2-cycle symmetrical fault current.

3.4 COORDINATION STUDY

A. Perform coordination study using approved computer software program. Prepare a written report using results of fault-current study. Comply with IEEE 399.

1. Calculate the maximum and minimum 1/2-cycle short-circuit currents.
2. Calculate the maximum and minimum ground-fault currents.

B. Comply with IEEE 141 and IEEE 242 recommendations for fault currents and time intervals.

C. Conductor Protection: Protect cables against damage from fault currents according to ICEA P-32-382, ICEA P-45-482, and conductor melting curves in IEEE 242. Demonstrate that equipment withstands the maximum short-circuit current for a time equivalent to the tripping time of the primary relay protection or total clearing time of the fuse. To determine temperatures that damage insulation, use curves from cable manufacturers or from listed standards indicating conductor size and short-circuit current.

D. Coordination-Study Report: Prepare a written report indicating the following results of coordination study:

1. Tabular Format of Settings Selected for Overcurrent Protective Devices:

a. Device tag.
b. Relay-current transformer ratios; and tap, time-dial, and instantaneous-pickup values.
c. Circuit-breaker sensor rating; and long-time, short-time, and instantaneous settings.
d. Fuse-current rating and type.

2. Coordination Curves: Prepared to determine settings of overcurrent protective devices to achieve selective coordination. Graphically illustrate that adequate time separation exists between devices installed in series, including power utility company's upstream devices. Prepare separate sets of curves for the switching schemes and for emergency periods where the power source is local generation. Show the following information:

    a. Device tag.
    b. Voltage and current ratio for curves.
    c. No damage, melting, and clearing curves for fuses.
    d. Cable damage curves.
    e. Maximum fault-current cutoff point.

E. Completed data sheets for setting of overcurrent protective devices.

3.5 ARC FLASH HAZARD ANALYSIS STUDY

A. Furnish an Arc Flash Hazard Analysis Study for the entire electrical system. Analysis shall be inclusive of individual disconnecting devices (not light switches or receptacles) per the requirements set forth in NFPA 70E – Standard for Electrical Safety in the workplace. The arc flash hazard analysis shall be performed according to the IEEE 1584 equations that are presented in NFPA 70E.

1. The flash protection boundary and the incident energy shall be calculated at all significant locations in the electrical distribution system (switchboards, switchgear, motor-control centers, panelboards, busway and splitters) where work could be performed on energized parts.
2. The Arc-Flash Hazard Analysis shall include all significant locations in 240 volt system where work could be performed on energized parts.
3. Safe working distances shall be based upon the calculated arc flash boundary considering an incident energy of 1.2 cal/cm2.
4. When appropriate, the short circuit calculations and the clearing times of the phase overcurrent devices will be retrieved from the short-circuit and coordination study model.
5. The short-circuit calculations and the corresponding incident energy calculations for multiple system scenarios must be compared and the greatest incident energy must be uniquely reported for each equipment location. Calculations must be performed to represent the maximum and minimum contributions of fault current magnitude for all normal and emergency operating conditions. The minimum calculation will assume that the utility contribution is at a minimum and will assume a minimum motor contribution (all motors off). Conversely, the maximum calculation will assume a maximum contribution from the utility and will assume the maximum amount of motors to be operating.
6. The incident energy calculations must consider the accumulation of energy over time when performing arc flash calculations on buses with multiple sources. Iterative calculations must take into account the changing current contributions, as the sources are interrupted or decremented with time. Fault contribution from motors should be decremented as follows:

    a. Fault contribution from induction motors should not be considered beyond 3-5 cycles.

b. Fault contribution from synchronous motors should be decayed to match the actual decrement of each as closely as possible.

7. For each equipment location with a separately enclosed main device (where there is adequate separation between the line side terminals of the main protective device and the work location), calculations for incident energy and flash protection boundary shall include both the line and load side of the main breaker.
8. When performing incident energy calculations on the line side of a main breaker (as required per above), the line side and load side contributions must be included in the fault calculation.
9. Mis-coordination should be checked amongst all devices within the branch containing the immediate protective device upstream of the calculation location and the calculation should utilize the fastest device to compute the incident energy for the corresponding location.
10. Arc Flash calculations shall be based on actual overcurrent protective device clearing time. Maximum clearing time will be capped at 2 seconds based on IEEE 1584-2002 section B.1.2. Where it is not physically possible to move outside of the flash protection boundary in less than 2 seconds during an arc flash event, a maximum clearing time based on the specific location shall be utilized.

B. Based on the results of the Arc Flash Hazard Analysis study, produce and install a warning label (orange $\leq$ 40 cal/cm^2) or danger label (red > 40 cal/cm^2) per NFPA 70 & 70E for each piece of equipment as specified above in accordance with ANSI Z535. The label must be readable in both indoor and outdoor environments for at least 3 years and contain the following information:

1. Arc hazard boundary (inches).
2. Working distance (inches)
3. Arc flash incident energy at the working distance (calories/cm2).
4. PPE category and description including the glove rating.
5. Voltage rating of the equipment.
6. Limited approach distance (inches).
7. Restricted approach distance (inches).
8. Prohibited approach distance (inches).
9. Equipment/bus name.
10. Date prepared.
11. Supplier name and address.

END OF SECTION 26 05 73

SECTION 26 08 00 – COMMISSIONING OF ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes Cx process requirements for the following electrical components, systems, assemblies, and equipment:

1. Electrical equipment connected to Normal power systems, including the following:

a. Distribution and branch-circuit panelboards.
b. Luminaires
c. Lighting Controls
d. Fire Alarm System
e. Communication System
f. Grounding System

B. Related Requirements:

1. Section 019114 “Total Building Commissioning” for general Cx process requirements and CxA responsibilities.

1.3 DEFINITIONS

A. BoD: Basis-of-Design Document, as defined in Section 019114 “Total Building Commissioning”

B. Cx: Commissioning, as defined in Section 019114 “Total Building Commissioning”

C. CxA: Commissioning Authority, as defined in Section 019114 “Total Building Commissioning”

D. Low Voltage: 600 V and below.

E. Normal Power Systems: A power system that provides primary power to a facility.

F. OPR: Owner's Project Requirements, as defined in Section 019114 “Total Building Commissioning”

G. “Systems,” “Assemblies,” “Subsystems,” “Equipment,” and “Components”: Where these terms are used together or separately, they shall mean “as-built” systems, assemblies, subsystems, equipment, and components.

1.4 INFORMATIONAL SUBMITTALS

A. Construction Checklists: Draft construction checklists will be created by CxA for Contractor review.

B. Construction Checklists: Include the following and comply with requirements in Section 019114 “Total Building Commissioning” for construction checklists:

1. Panelboards.
2. Luminaires.
3. Lighting Controls
4. Fire Alarm
5. Communications

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For electrical systems and components to include in operation and maintenance manuals.

1.6 QUALITY ASSURANCE

A. Electrical Testing Technician Qualifications: Technicians to perform electrical Construction Checklist verification tests, Construction Checklist verification test demonstrations, Cx tests, and Cx test demonstrations shall have the following minimum qualifications:

1. Journey level or equivalent skill level. Vocational school four-year-program graduate or an Associate's degree in electrical systems, or similar field. Degree may be offset by three years' experience as an apprentice or a journey-level electrician. Generally, required knowledge includes electrical and HVAC&R concepts, building operations, and application and use of tools and instrumentation to measure performance of electrical equipment, assemblies, and systems.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 CONSTRUCTION CHECKLISTS

A. Provided by CxA, completed by contractor.

3.2 CONSTRUCTION CHECKLIST REVIEW

A. Review and provide written comments on draft construction checklists. CxA will create required draft construction checklists and provide them to Contractor.

B. Return draft Construction Checklist review comments within [10] days of receipt.

C. When review comments have been resolved, CxA will provide final construction checklists, marked “Approved for Use, (date).”

D. Use only construction checklists, marked “Approved for Use, (date).”

3.3 GENERAL TESTING REQUIREMENTS

A. Certify that electrical systems, subsystems, and equipment have been installed, calibrated, and started and that they are operating according to the Contract Documents and approved Shop Drawings and submittals.

B. If tests cannot be completed because of a deficiency outside the scope of the electrical system, document the deficiency and report it to Contracting Officer. After deficiencies are resolved, reschedule tests.

3.4 CX TESTS FOR ELECTRICAL SYSTEMS

A. Verification of Power System Operation:

1. Prerequisites: Acceptance of results for construction checklists for Division 26 electrical components associated with Normal power system.
2. Equipment and Systems to Be Tested: Division 26 electrical equipment.
3. Test Purpose: Verify operation of power system.
4. Test Conditions: Energize components of Normal power system, one at a time. Equipment and Systems to Be Tested: Division 26 electrical equipment.

B. Acceptance Criteria: Operation of equipment according to OPR.

3.5 CX TESTS FOR LUMINAIRES.

A. Verification of Luminaires:

1. Prerequisites: Acceptance of results for construction checklists for Division 26 Luminaires.
2. Equipment and Systems to Be Tested: Division 26 Luminaires.
3. Test Purpose: Verify operation of Luminaires.
4. Test Conditions: Energize components of Luminaires.

B. Acceptance Criteria: Operation of equipment according to OPR.

3.6 CX TESTS FOR LIGHTING CONTROLS.

A. Verification of Lighting Controls:

1. Prerequisites: Acceptance of results for construction checklists for Division 26 Lighting Controls.
2. Equipment and Systems to Be Tested: Division 26 Lighting Controls.

3. Test Purpose: Verify operation of Lighting Controls.
4. Test Conditions: Energize components of Lighting Controls.

B. Acceptance Criteria: Operation of equipment according to OPR.

3.7 CX TESTS FOR FIRE ALARM SYSTEM.

A. Verification of Fire Alarm System:

1. Prerequisites: Acceptance of results for construction checklists for Division 26 Digital Addressable Fire-Alarm System.
2. Equipment and Systems to Be Tested: Division 28 Fire Alarm System.
3. Purpose: Verify operation of Fire Alarm System.
4. Test Conditions: Energize components of Fire Alarm System.

B. Acceptance Criteria: Operation of equipment according to OPR.

3.8 CX TESTS FOR COMMUNICATIONS.

A. Verification of Communications:

1. Prerequisites: Acceptance of results for construction checklists for Division 27 Communications System.
2. Equipment and Systems to Be Tested: Division 27 Communications System.
3. Purpose: Verify operation of Communications to comply with cabling warranty requirements.
4. Test Conditions: Energize components of Communications System.

B. Acceptance Criteria: Operation of equipment according to OPR.

3.9 CX TESTS FOR GROUNDING SYSTEM

A. Verification of Grounding and Bonding:

1. Prerequisites: Acceptance of results for construction checklists for Division 26 Grounding and Bonding.
2. Equipment and Systems to Be Tested: Division 26 Grounding and Bonding.
3. Purpose: Verify operation of grounding systems to comply with grounding warranty requirements.
4. Test Conditions: Energize components grounding system.

B. Acceptance Criteria: Operation of equipment according to OPR.

END OF SECTION 26 08 00

SECTION 26 09 23 – LIGHTING CONTROL DEVICES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following lighting control devices:

1. Indoor occupancy sensors.

B. Related Sections include the following:

1. Division 26 Section “Wiring Devices” for manual light switches.

1.3 DEFINITIONS

A. LED: Light-emitting diode.

B. PIR: Passive infrared.

1.4 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Shop Drawings: Show installation details for occupancy and light-level sensors.

1. Interconnection diagrams showing field-installed wiring.

C. Field quality-control test reports.

D. Operation and Maintenance Data: For each type of product to include in emergency, operation, and maintenance manuals.

1.5 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.6 COORDINATION

A. Coordinate layout and installation of ceiling-mounted devices with other construction that penetrates ceilings or is supported by them, including light fixtures, HVAC equipment, smoke detectors, and partition assemblies.

PART 2 - PRODUCTS

2.1 INDOOR OCCUPANCY SENSORS

A. Salient Characteristics:

1. Physical:

a. General Description: Wall mounted, 120v units which do not require a separate low voltage power pack.
b. Sensor Output: Contacts rated to operate the connected relay. Sensor shall be powered from the relay unit.
c. Mounting:

1) Sensor: Suitable for mounting in any position on a standard outlet box.
2) Time-Delay and Sensitivity Adjustments: Recessed and concealed behind hinged door.

d. Indicator: LED, to show when motion is being detected during testing and normal operation of the sensor.
e. Bypass Switch: Override the on function in case of sensor failure.
f. Automatic Light-Level Sensor: Adjustable from 2 to 200 fc (21.5 to 2152 lx); keep lighting off when selected lighting level is present.
g. Auxiliary Contacts: Where mechanical equipment is indicated for occupancy sensor control, auxiliary contact shall have a minimum horsepower rating suitable for the connected equipment, to be verified with final mechanical shop drawings.
h. Dual-Technology Type: Wall/ceiling mounted; detect occupancy by using a combination of PIR and ultrasonic detection methods in area of coverage. Particular technology or combination of technologies that controls on-off functions shall be selectable in the field by operating controls on unit.

1) Sensitivity Adjustment: Separate for each sensing technology.
2) Detector Sensitivity: Detect occurrences of 6-inch- (150-mm-) minimum movement of any portion of a human body that presents a target of not less than 36 sq. in. (232 sq. cm), and detect a person of average size and weight moving not less than 12 inches (305 mm) in either a horizontal or a vertical manner at an approximate speed of 12 inches/s (305 mm/s).
3) Detection Coverage (Standard Room): Detect occupancy anywhere within a circular area of 1000 sq. ft. (93 sq. m) when mounted on a 96-inch- (2440-mm-) high ceiling.

2. Functional:

a. Operation: Unless otherwise indicated, turn lights on when covered area is occupied and off when unoccupied; with a time delay for turning lights off, adjustable over a minimum range of 1 to 60 minutes.

3. Performance:

a. Sensor Output: Comply with UL 773A.

PART 3 - EXECUTION

3.1 SENSOR INSTALLATION

A. Install and aim sensors in locations to achieve not less than 90 percent coverage of areas indicated. Do not exceed coverage limits specified in manufacturer's written instructions.

3.2 WIRING INSTALLATION

A. Wiring Method: Install wiring in raceways. Comply with Division 26 Section “Low-Voltage Electrical Power Conductors and Cables.” Minimum conduit size shall be 3/4 inch

B. Wiring within Enclosures: Comply with NECA 1. Separate power-limited and nonpower-limited conductors according to conductor manufacturer's written instructions.

C. Size conductors according to lighting control device manufacturer's written instructions, unless otherwise indicated.

D. Splices, Taps, and Terminations: Make connections only on numbered terminal strips in junction, pull, and outlet boxes; terminal cabinets; and equipment enclosures.

3.3 IDENTIFICATION

A. Identify components and power and control wiring according to Division 26 Section “Identification for Electrical Systems.”

3.4 FIELD QUALITY CONTROL

A. Perform the following field tests and inspections and prepare test reports:

1. After installing time switches and sensors, and after electrical circuitry has been energized, adjust and test for compliance with requirements.
2. Operational Test: Verify operation of each lighting control device, and adjust time delays.

B. Lighting control devices that fail tests and inspections are defective work.

3.5 ADJUSTING

A. Occupancy Adjustments: When requested within 12 months of date of Substantial Completion, provide on-site assistance in adjusting sensors to suit occupied conditions. Provide up to one visit to Project during other-than-normal occupancy hours for this purpose.

3.6 DEMONSTRATION

A. Engage a factory-authorized service representative to train Park's maintenance personnel to adjust, operate, and maintain lighting control devices.

END OF SECTION 26 09 23

SECTION 26 09 43 – NETWORK LIGHTING CONTROLS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes manually operated, digital lighting controls with external signal source relays and control module.

B. Related Sections include the following:

1. Division 26 Section “Lighting Control Devices” for occupancy sensors.

1.3 DEFINITIONS

A. Low Voltage: As defined in NFPA 70 for circuits and equipment operating at less than 50 V or for remote-control, signaling and power-limited circuits.

B. Monitoring: Acquisition, processing, communication, and display of equipment status data, metered electrical parameter values, power quality evaluation data, event and alarm signals, tabulated reports, and event logs.

C. Power Line Carrier: Use of radio-frequency energy to transmit information over transmission lines whose primary purpose is the transmission of power.

D. RS-485: A serial network protocol, similar to RS-232, complying with TIA/EIA-485-A.

1.4 SUBMITTALS

A. Product Data: For control modules, power distribution components, manual switches and plates, and conductors and cables.

B. Shop Drawings: Detail assemblies of standard components, custom assembled for specific application on this Project.

1. Outline Drawings: Indicate dimensions, weights, arrangement of components, and clearance and access requirements.
2. Block Diagram: Show interconnections between components specified in this Section and devices furnished with power distribution system components. Indicate data communication paths and identify networks, data buses, data gateways, concentrators,

and other devices to be used. Describe characteristics of network and other data communication lines.

3. Wiring Diagrams: Power, signal, and control wiring. Coordinate nomenclature and presentation with a block diagram.

C. Coordination Drawings: Submit evidence that lighting controls are compatible with connected monitoring and control devices and systems specified in other Sections.

1. Show interconnecting signal and control wiring and interfacing devices that prove compatibility of inputs and outputs.
2. For networked controls, list network protocols and provide statements from manufacturers that input and output devices meet interoperability requirements of the network protocol.

D. Field Quality-Control Reports:

1. Test procedures used.
2. Test results that comply with requirements.
3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

E. Software and Firmware Operational Documentation:

1. Software operating and upgrade manuals.
2. Program Software Backup: On a magnetic media or compact disc, complete with data files.
3. Device address list.
4. Printout of software application and graphic screens.

F. Operation and Maintenance Data: For lighting controls to include in emergency, operation, and maintenance manuals.

G. Warranty: Special warranty specified in this Section.

1.5 QUALITY ASSURANCE

A. Source Limitations: Obtain lighting control module and power distribution components through one source from a single manufacturer.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

C. Comply with 47 CFR, Subparts A and B, for Class A digital devices.

D. Comply with NFPA 70.

1.6 COORDINATION

A. Coordinate lighting control components to form an integrated interconnection of compatible components.

1. Match components and interconnections for optimum performance of lighting control functions.

2. Coordinate lighting controls with that in Sections specifying distribution components that are monitored or controlled by power monitoring and control equipment.

1.7 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of lighting controls that fail in materials or workmanship or from transient voltage surges within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Failure of software input/output to execute switching or dimming commands.
b. Failure of modular relays to operate under manual or software commands.
c. Damage of electronic components due to transient voltage surges.

2. Warranty Period: Two years from date of Substantial Completion.
3. Extended Warranty Period for Electrically Held Relays: 10 years from date of Substantial Completion.

1.8 EXTRA MATERIALS

A. Furnish extra materials described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Electrically Held Relays: Equal to 25 percent of amount installed for each size indicated, but no fewer than 2 relays.

1.9 SOFTWARE SERVICE AGREEMENT

A. Technical Support: Beginning with Substantial Completion, provide software support for two years.

B. Upgrade Service: Update software to latest version at Project completion. Install and program software upgrades that become available within two years from date of Substantial Completion. Upgrading software shall include operating system. Upgrade shall include new or revise licenses for use of the software.

1. Provide 30-day notice to NPS to allow scheduling and access to system and to allow NPS to upgrade computer equipment, if necessary.

PART 2 - PRODUCTS

2.1 SYSTEM REQUIREMENTS

A. Expandability: System shall be capable of increasing the number of control functions in the future by 25 percent of current capacity; to include equipment ratings, housing capacities, spare relays, terminals, number of conductors in control cables, and control software.

B. Performance Requirements: Manual switches, an internal timing and control unit, and external sensors or other control signal sources send a signal to a PC-based programmable-system control module that processes the signal according to its programming and routes an open or close command to one or more relays in the power-supply circuits, or routes variable commands to one or more dimmers, for groups of lighting fixtures or other loads.

2.2 CONTROL MODULE

A. Control Module Description: Comply with UL 916 (CSA C22.2, No. 205); microprocessor-based, solid-state, 365-day timing and control unit. Output circuits shall be switched on or off by internally programmed time signals or by program-controlled analog or digital signals from external sources. Output circuits shall be pilot-duty relays compatible with power switching devices. An integral keypad shall provide local programming and control capability. A key-locked cover and a programmed security access code shall protect keypad use. An integral alphanumeric LCD or LED shall display menu-assisted programming and control.

2.3 POWER DISTRIBUTION COMPONENTS

A. Modular Relay Panel: Comply with UL 508 (CSA C22.2, No. 14) and UL 916 (CSA C22.2, No. 205); factory assembled with modular single-pole relays, power supplies, and accessory components required for specified performance.

1. Cabinet: Steel with hinged, locking door.

a. Barriers separate low-voltage and line-voltage components.
b. Directory: Mounted on back of door. Identifies each relay as to load groups controlled and each programmed pilot device if any.
c. Control Power Supply: Transformer and full-wave rectifier with filtered dc output.

2. Single-Pole Relays: Mechanically held unless otherwise indicated; split-coil, momentary-pulsed type.

a. Low-Voltage Leads: Plug connector to the connector strip in cabinet and pilot light power where indicated.
b. Rated Capacity (Mounted in Relay Panel): 20 A, 125-V ac for tungsten filaments; 20 A, 277-V ac for ballasts.
c. Endurance: 50,000 cycles at rated capacity.
d. Mounting: Provision for easy removal and installation in relay cabinet.

2.4 MANUAL SWITCHES AND PLATES

A. Push-Button Switches: Modular, momentary-contact, low-voltage type.

1. Match color specified in Division 26 Section “Wiring Devices.”

2. Integral green LED pilot light to indicate when circuit is on.
3. Internal white LED locator light to illuminate when circuit is off.

B. Manual, Maintained Contact, Full- or Low-Voltage Switch: Comply with Division 26 Section “Wiring Devices.”

C. Wall-Box Dimmers: Comply with Division 26 Section “Wiring Devices.”

D. Wall Plates: Single and multigang plates as specified in Division 26 Section “Wiring Devices.”

E. Legend: Engraved or permanently silk-screened on wall plate where indicated. Use designations indicated on Drawings.

2.5 CONDUCTORS AND CABLES

A. Power Wiring to Supply Side of Class 2 Power Source: Not smaller than No. 12 AWG, complying with Division 26 Section “Low-Voltage Electrical Power Conductors and Cables.”

B. Classes 2 and 3 Control Cables: Multiconductor cable with copper conductors not smaller than No. 18 AWG, complying with Division 26 Section “Low-Voltage Electrical Power Conductors and Cables.”

C. Class 1 Control Cables: Multiconductor cable with copper conductors not smaller than No. 12 AWG, complying with Division 26 Section “Low-Voltage Electrical Power Conductors and Cables.”

D. Digital and Multiplexed Signal Cables: Unshielded, twisted-pair cable with copper conductors, complying with TIA/EIA-568-B.2, Category 6a for horizontal copper cable and with Division 27 Section “Communications Horizontal Cabling.”

PART 3 - EXECUTION

3.1 WIRING INSTALLATION

A. Comply with NECA 1.

B. Wiring Method: Install wiring in raceways. Comply with Division 26 Section “Low-Voltage Electrical Power Conductors and Cables.” Minimum conduit size shall be 3/4 inch.

C. Wiring within Enclosures: Bundle, lace, and train conductors to terminal points. Separate power-limited and non-power-limited conductors according to conductor manufacturer's written instructions.

D. Size conductors according to lighting control device manufacturer's written instructions, unless otherwise indicated.

E. Splices, Taps, and Terminations: Make connections only on numbered terminal strips in terminal cabinets, equipment enclosures, and in junction, pull, and outlet boxes.

F. Identify components and power and control wiring according to Division 26 Section "Identification for Electrical Systems."

3.2 FIELD QUALITY CONTROL

A. Perform the following field tests and inspections and prepare test reports:

1. Test for circuit continuity.
2. Verify that the control module features are operational.
3. Check operation of local override controls.
4. Test system diagnostics by simulating improper operation of several components selected by Contracting Officer.

3.3 SOFTWARE INSTALLATION

A. Install and program software with initial settings of adjustable values. Make backup copies of software and user-supplied values. Provide current licenses for software.

3.4 STARTUP SERVICE

A. Perform startup service.

1. Complete installation and startup checks according to manufacturer's written instructions.
2. Activate luminaires and verify that all lamps are operating at 100 percent.
3. Confirm correct communications wiring and program the lighting control system according to approved configuration schedules, time-of-day schedules, and input override assignments.

3.5 ADJUSTING

A. Occupancy Adjustments: When requested within 12 months of date of Substantial Completion, provide on-site assistance in adjusting sensors and to assist NPS personnel in making program changes to suit actual occupied conditions. Provide up to two visits to Project during other than normal occupancy hours for this purpose.

3.6 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain lighting controls. Refer to Division 01 Section “Demonstration and Training.”

END OF SECTION 26 09 43

SECTION 26 24 16 – PANELBOARDS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Lighting and appliance branch-circuit panelboards.

1.3 DEFINITIONS

A. SVR: Suppressed voltage rating.

B. TVSS: Transient voltage surge suppressor.

1.4 SUBMITTALS

A. Product Data: For each type of panelboard, switching and overcurrent protective device, transient voltage suppression device, accessory, and component indicated. Include dimensions and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes.

B. Shop Drawings: For each panelboard and related equipment.

1. Include dimensioned plans, elevations, sections, and details. Show tabulations of installed devices, equipment features, and ratings.
2. Detail enclosure types and details for types other than NEMA 250, Type 1.
3. Detail bus configuration, current, and voltage ratings.
4. Short-circuit current rating of panelboards and overcurrent protective devices.

5. Detail features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.

6. Include time-current coordination curves for each type and rating of overcurrent protective device included in panelboards. Submit on translucent log-log graft paper; include selectable ranges for each type of overcurrent protective device.

C. Panelboard Schedules: For installation in panelboards. Submit final versions after load balancing.

D. Field Quality-Control Reports:

1. Test procedures used.
2. Test results that comply with requirements.
3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For panelboards and components to include in emergency, operation, and maintenance manuals. In addition to items specified in Section 017823 "Operation and Maintenance Data," include the following:

1. Manufacturer's written instructions for testing and adjusting overcurrent protective devices.
2. Time-current curves, including selectable ranges for each type of overcurrent protective device that allows adjustments.

1.6 QUALITY ASSURANCE

A. Source Limitations: Obtain panelboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.

B. Product Selection for Restricted Space: Drawings indicate maximum dimensions for panelboards including clearances between panelboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

D. Comply with NEMA PB 1.

E. Comply with NFPA 70.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Handle and prepare panelboards for installation according to NECA 407.

1.8 PROJECT CONDITIONS

A. Environmental Limitations:

1. Do not deliver or install panelboards until spaces are enclosed and weathertight, wet work in spaces is complete and dry, work above panelboards is complete, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.

2. Rate equipment for continuous operation under the following conditions unless otherwise indicated:

    a. In first subparagraph below, select first option for panelboards with fused switches; select second option for panelboards with circuit breakers.
    b. Ambient Temperature: 10 deg F to plus 140 deg F.
    c. Altitude: 3,400 feet.

B. Service Conditions: NEMA PB 1, usual service conditions, as follows:

1. Ambient temperatures within limits specified.
2. Altitude: 3,400 feet.

1.9 COORDINATION

A. Coordinate layout and installation of panelboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, encumbrances to workspace clearance requirements, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Coordinate sizes and locations of concrete bases with actual equipment provided. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.

1.10 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace transient voltage suppression devices that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Five years from date of Substantial Completion.

1.11 EXTRA MATERIALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Keys: Two spares for each type of panelboard cabinet lock.
2. Circuit Breakers Including GFCI and Ground Fault Equipment Protection (GFEP) Types: Quantity as indicated on electrical drawings.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR PANELBOARDS

A. Salient Characteristics:

1. Physical:

a. Enclosures: Flush- and surface-mounted cabinets.

1) Rated for environmental conditions at installed location

a) All Locations: NEMA 250, Type 4X.

b. Front: Secured to box with concealed trim clamps. For surface-mounted fronts, match box dimensions; for flush-mounted fronts, overlap box.
c. Hinged Front Cover: Entire front trim hinged to box and with standard door within hinged trim cover.
d. Skirt for Surface-Mounted Panelboards: Same gage and finish as panelboard front with flanges for attachment to panelboard, wall, and ceiling or floor.
e. Gutter Extension and Barrier: Same gage and finish as panelboard enclosure; integral with enclosure body. Arrange to isolate individual panel sections.
f. Finishes:

1) Panels and Trim: Steel, factory finished immediately after cleaning and pretreating with manufacturer's standard two-coat, baked-on finish consisting of prime coat and thermosetting topcoat.
2) Back Boxes: Galvanized steel.
3) Fungus Proofing: Permanent fungicidal treatment for overcurrent protective devices and other components.

g. Phase, Neutral, and Ground Buses:

1) Material: Hard-drawn copper, 98 percent conductivity.
2) Equipment Ground Bus: Full capacity separate ground bus for feeder and branch-circuit equipment grounding conductors; bonded to box.
3) Neutral Bus: Neutral bus rated 100 percent of phase bus and UL listed as suitable for nonlinear loads.

h. Conductor Connectors: Suitable for use with conductor material and sizes.

1) Material: Hard-drawn copper, 98 percent conductivity.
2) Main and Neutral Lugs: Mechanical type.
3) Ground Lugs and Bus-Configured Terminators: Mechanical type.
4) Feed-Through Lugs: Mechanical type, suitable for use with conductor material. Locate at opposite end of bus from incoming lugs or main device.

2. Functional:

a. Incoming Mains Location: Top and bottom.
b. Service Equipment Label: NRTL labeled for use as service equipment for panelboards or load centers with one or more main service disconnecting and overcurrent protective devices.
c. Future Devices: Mounting brackets, bus connections, filler plates, and necessary appurtenances required for future installation of devices.
d. Panelboard Short-Circuit Current Rating: Fully rated to interrupt symmetrical short-circuit current available at terminals.

3. Performance:

a. Fabricate and test panelboards according to IEEE 344.

2.2 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS

A. Salient Characteristics:

1. Physical:

a. Column-Type Panelboards: Narrow gutter extension, with cover, to overhead junction box equipped with ground and neutral terminal buses.

2. Functional:

a. Mains: Circuit breaker or lugs only.
b. Branch Overcurrent Protective Devices: Bolt-on circuit breakers, replaceable without disturbing adjacent units.
c. Doors: Concealed hinges; secured with flush latch with tumbler lock; keyed alike.

3. Performance:

a. Panelboards: NEMA PB 1, lighting and appliance branch-circuit type.

2.3 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

A. Salient Characteristics:

1. Physical:

a. Thermal-Magnetic Circuit Breakers: Inverse time-current element for low-level overloads, and instantaneous magnetic trip element for short circuits. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger. Used for all breakers unless otherwise noted.
b. Ground-Fault Equipment Protection (GFEP) Circuit Breakers: Class B ground-fault protection (30-mA trip). Refer to plans for GFI breaker locations.
c. Molded-Case Circuit-Breaker (MCCB) Features and Accessories:

1) Standard frame sizes, trip ratings, and number of poles.

2) Lugs: Compression style, suitable for number, size, trip ratings, and conductor materials.
3) Application Listing: Appropriate for application; Type SWD for switching fluorescent lighting loads; Type HID for feeding fluorescent and high-intensity discharge (HID) lighting circuits.
4) Ground-Fault Protection: Integrally mounted relay and trip unit with adjustable pickup and time-delay settings, push-to-test feature, and ground-fault indicator.
5) Auxiliary Contacts: One SPDT switch with "a" and "b" contacts; "a" contacts mimic circuit-breaker contacts and "b" contacts operate in reverse of circuit-breaker contacts.
6) Handle Padlocking Device: Fixed attachment, for locking circuit-breaker handle in off position.
7) Handle Clamp: Loose attachment, for holding circuit-breaker handle in on position.

2. Functional:

a. None.

3. Performance:

a. Molded-Case Circuit Breaker (MCCB): Comply with UL 489, with interrupting capacity to meet available fault currents.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Receive, inspect, handle, and store panelboards according to NECA 407.

B. Examine panelboards before installation. Reject panelboards that are damaged or rusted or have been subjected to water saturation.

C. Examine elements and surfaces to receive panelboards for compliance with installation tolerances and other conditions affecting performance of the Work.

D. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install panelboards and accessories according to NECA 407.

B. Equipment Mounting: Install panelboards on concrete bases, 4-inch (100-mm) nominal thickness. Comply with requirements for concrete base specified in Division 03 Section "Cast-in-Place Concrete."

1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch (450-mm) centers around full perimeter of base.

2. For panelboards, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
4. Install anchor bolts to elevations required for proper attachment to panelboards.
5. Attach panelboard to the vertical finished or structural surface behind the panelboard.

C. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from panelboards.

D. Mount top of trim [90 inches (2286 mm)] above finished floor unless otherwise indicated.

E. Mount panelboard cabinet plumb and rigid without distortion of box. Mount recessed panelboards with fronts uniformly flush with wall finish and mating with back box.

F. Install overcurrent protective devices and controllers not already factory installed.

1. Set field-adjustable, circuit-breaker trip ranges.

G. Install filler plates in unused spaces.

H. Stub four 1-inch (27-GRC) empty conduits from panelboard into accessible ceiling space or space designated to be ceiling space in the future. Stub four 1-inch (27-GRC) empty conduits into raised floor space or below slab not on grade.

I. Arrange conductors in gutters into groups and bundle and wrap with wire ties after completing load balancing.

J. Comply with NECA 1.

3.3 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs complying with Division 26 Section “Identification for Electrical Systems.”

B. Create a directory to indicate installed circuit loads after balancing panelboard loads; incorporate final room designations. Obtain approval before installing. Use a computer or typewriter to create directory; handwritten directories are not acceptable.

C. Panelboard Nameplates: Label each panelboard with a nameplate complying with requirements for identification specified in Division 26 Section “Identification for Electrical Systems.”

D. Device Nameplates: Label each branch circuit device in distribution panelboards with a nameplate complying with requirements for identification specified in Division 26 Section “Identification for Electrical Systems.”

3.4 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Acceptance Testing Preparation:

1. Test insulation resistance for each panelboard bus, component, connecting supply, feeder, and control circuit.
2. Test continuity of each circuit.

C. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

D. Panelboards will be considered defective if they do not pass tests and inspections.

E. Prepare test and inspection reports, including a certified report that identifies panelboards included and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.5 ADJUSTING

A. Adjust moving parts and operable component to function smoothly, and lubricate as recommended by manufacturer.

B. Set field-adjustable circuit-breaker trip ranges as indicated.

C. Load Balancing: After Substantial Completion, but not more than 60 days after Final Acceptance, measure load balancing and make circuit changes.

1. Measure as directed during period of normal system loading.
2. Perform load-balancing circuit changes outside normal occupancy/working schedule of the facility and at time directed. Avoid disrupting critical 24-hour services such as fax machines and on-line data processing, computing, transmitting, and receiving equipment.
3. After circuit changes, recheck loads during normal load period. Record all load readings before and after changes and submit test records.
4. Tolerance: Difference exceeding 20 percent between phase loads, within a panelboard, is not acceptable. Rebalance and recheck as necessary to meet this minimum requirement.

3.6 PROTECTION

A. Temporary Heating: Apply temporary heat to maintain temperature according to manufacturer's written instructions.

END OF SECTION 26 24 16

SECTION 26 27 13 – ELECTRICITY METERING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes equipment for electricity metering by National Park Service, Demand Metering.

1.3 SUBMITTALS

A. Product Data: Include construction details, material descriptions, dimensions of individual components and profiles, and finishes. Describe electrical characteristics, features, and operating sequences, both automatic and manual. Include the following:

1. Electricity-metering equipment.

B. Field quality-control test reports.

C. Operation and Maintenance Data: For electricity-metering equipment to include in emergency, operation, and maintenance manuals.

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Receive, store, and handle modular meter center as specified in NECA 400.

PART 2 - PRODUCTS

2.1 EQUIPMENT FOR ELECTRICITY METERING BY UTILITY COMPANY

A. Salient Characteristics:

1. Physical:

a. Current-Transformer Cabinets: Comply with requirements of electrical power utility company.
b. Meter Sockets: Comply with requirements of electrical power utility company.

2. Functional:

a. None.

3. Performance:

a. None.

2.2 EQUIPMENT FOR ELECTRICITY DEMAND METERING

A. Salient Characteristics:

1. Physical:

a. Kilowatt-Hour/Demand Meter: Electronic single- and three-phase meters, measuring electricity use and demand.

2. Functional:

a. Basis of design product, E-MON Green Class, or approved equal.
b. Voltage and Phase Configuration: Meter shall be designed for use on circuits with voltage rating and phase configuration indicated for its application.
c. Display: Digital liquid crystal, indicating accumulative kilowatt hours, current time and date, current demand, historic peak demand, and time and date of historic peak demand.
d. Programmable Contact Module: Unit shall have push-button switches and a display for setting the demand level at which an integral set of Form C contacts shall be operated to initiate indicated action.
e. Enclosure: NEMA 250, Type 1 minimum, with hasp for padlocking or sealing.
f. Identification: Comply with Division 26 Section “Identification for Electrical Systems.”
g. Memory Backup: Self-contained to maintain memory throughout power outages of 72 hours, minimum.
h. Sensors: Current-sensing type, with current or voltage output, selected for optimum range and accuracy for ratings of circuits indicated for this application.
i. Type: Solid core.

3. Performance:

   a. Meter Accuracy: Nationally recognized testing laboratory certified to comply with ANSI C12.1.
   b. Meet the requirements of EPACT 2005.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with equipment installation requirements in NECA 1.

B. Install equipment for utility company metering. Install raceways and equipment according to utility company's written requirements. Provide empty conduits for metering leads and extend grounding connections as required by utility company.

C. Install modular meter center according to NECA 400 switchboard installation requirements.

3.2 FIELD QUALITY CONTROL

A. Test electricity-metering installation for proper operation, accuracy, and usability of output data.

1. Connect a load of known kilowatt rating, 1.5 kW minimum, to a circuit supplied by metered feeder.
2. Turn off circuits supplied by metered feeder and secure them in off condition.
3. Run test load continuously for eight hours, minimum, or longer to obtain a measurable meter indication. Use test load placement and setting that ensures continuous, safe operation.
4. Check and record meter reading at end of test period and compare with actual electricity used based on test load rating, duration of test, and sample measurements of supply voltage at test load connection. Record test results.
5. Repair or replace deficient or malfunctioning metering equipment, or correct test setup; then retest. Repeat for each meter in installation until proper operation of entire system is verified.

END OF SECTION 26 27 13

SECTION 26 27 26 – WIRING DEVICES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Receptacles, receptacles with integral GFCI, and associated device plates.
2. Toggle switches.

B. Related Sections include the following:

1. Division 26 Section “Lighting Control Devices” for wall-switch occupancy sensors.

1.3 DEFINITIONS

A. GFCI: Ground-fault circuit interrupter.

B. Pigtail: Short lead used to connect a device to a branch-circuit conductor.

1.4 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Operation and Maintenance Data: For wiring devices to include in all manufacturers' packing label warnings and instruction manuals that include labeling conditions.

1.5 QUALITY ASSURANCE

A. Source Limitations: Obtain each type of wiring device and associated wall plate through one source from a single manufacturer. Insofar as they are available, obtain all wiring devices and associated wall plates from a single manufacturer and one source.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

C. Comply with NFPA 70.

1.6 COORDINATION

A. Receptacles for Government-Furnished Equipment: Match plug configurations.

1. Cord and Plug Sets: Match equipment requirements.

PART 2 - PRODUCTS

2.1 GENERAL WIRING-DEVICE REQUIREMENTS

A. Salient Characteristics:

1. Physical:

a. Wiring Devices, Components, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2. Functional:

a. None.

3. Performance:

a. Comply with NFPA 70.
b. Devices that are manufactured for use with modular plug-in connectors may be substituted under the following conditions:

1) Connectors shall comply with UL 2459 and shall be made with stranding building wire.
2) Devices shall comply with the requirements in this Section.

2.2 STRAIGHT BLADE RECEPTACLES

A. Salient Characteristics:

1. Physical:

a. Convenience Receptacles, 125 V, 20 A.

2. Functional:

a. Configuration 5-20R

3. Performance:

a. Comply with NEMA WD 1, NEMA WD 6, and UL 498.

2.3 GFCI RECEPTACLES

A. Salient Characteristics:

1. Physical:

a. Straight blade, non-feed-through type.
b. Duplex GFCI Convenience Receptacles, 125 V, 20 A.

2. Functional:

a. Include indicator light that is lighted when device is tripped.
b. Exterior Locations: Provide with in-use cover. Device shall be corrosion resistant, suitable for salt air conditions.

3. Performance:

a. Comply with NEMA WD 1, NEMA WD 6, UL 498, and UL 943, Class A

2.4 SPECIAL EQUIPMENT RECEPTACLES

A. Salient Characteristics:

1. Physical:

a. Voltage and NEMA configuration as indicated on drawings

2. Functional:

a. None.

3. Performance:

a. Comply with NEMA WD 1, NEMA WD 6, and UL 498.

2.5 TOGGLE SWITCHES

A. Salient Characteristics:

1. Physical:

a. Switches, 120/277 V, 20 A

2. Functional:

a. None.

3. Performance:

a. Comply with NEMA WD 1, UL 20, and FS W-S

2.6 WALL PLATES

A. Salient Characteristics:

1. Physical:

a. Single and combination types to match corresponding wiring devices.
b. Exhibit Light Switch Cover: Type 1, with continuous-hinge cover with surface mount latch, no window, unless otherwise indicated.

1) Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
2) Nonmetallic Enclosures: Plastic.

2. Functional:

a. Plate-Securing Screws: Metal with head color to match plate finish.
b. Material for Office, Admin, Exhibit, and Auditorium Spaces: Raw Brass.
c. Material for all other Finished Spaces: Thermoplastic.
d. Material for Unfinished Spaces: Galvanized steel.
e. Material for Damp Locations: Thermoplastic with spring-loaded lift cover, and listed and labeled for use in “wet locations.”

3. Performance:

a. Wet-Location, Weatherproof Cover Plates: NEMA 250, complying with type 3R weather-resistant, die-cast aluminum with lockable cover.

2.7 FINISHES

A. Color: Wiring Device outlet colors are to be white for locations with steel or thermoplastic cover plates, and to be almond for all raw brass locations. Confirm with Contracting Officer prior to providing submittals.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with NECA 1, including the mounting heights listed in that standard, unless otherwise noted.

B. Coordination with Other Trades:

1. Take steps to ensure that devices and their boxes are protected. Do not place wall finish materials over device boxes and do not cut holes for boxes with routers that are guided by riding against outside of the boxes.

2. Keep outlet boxes free of plaster, drywall joint compound, mortar, cement, concrete, dust, paint, and other material that may contaminate the raceway system, conductors, and cables.
3. Install wiring devices after all wall preparation, including painting, is complete.

C. Conductors:

1. Do not strip insulation from conductors until just before they are spliced or terminated on devices.
2. Strip insulation evenly around the conductor using tools designed for the purpose. Avoid scoring or nicking of solid wire or cutting strands from stranded wire.
3. The length of free conductors at outlets for devices shall meet provisions of NFPA 70, Article 300, without pigtails.
4. Existing Conductors:

    a. Replace all existing conductors.

D. Device Installation:

1. Replace all devices that have been in temporary use during construction or that show signs that they were installed before building finishing operations were complete.
2. Keep each wiring device in its package or otherwise protected until it is time to connect conductors.
3. Do not remove surface protection, such as plastic film and smudge covers, until the last possible moment.
4. Connect devices to branch circuits using pigtails that are not less than 6 inches (152 mm) in length.
5. When there is a choice, use side wiring with binding-head screw terminals. Wrap solid conductor tightly clockwise, 2/3 to 3/4 of the way around terminal screw. Pressure plate connections are also acceptable. Stab in connections are not allowed.
6. Use a torque screwdriver when a torque is recommended or required by the manufacturer.
7. When conductors larger than No. 12 AWG are installed on 15- or 20-A circuits, splice No. 12 AWG pigtails for device connections.
8. Tighten unused terminal screws on the device.
9. When mounting into metal boxes, remove the fiber or plastic washers used to hold device mounting screws in yokes, allowing metal-to-metal contact.

E. Receptacle Orientation: Install ground pin of vertically mounted receptacles down, and on horizontally mounted receptacles to the right.

F. Device Plates: Do not use oversized or extra-deep plates. Repair wall finishes and remount outlet boxes when standard device plates do not fit flush or do not cover rough wall opening.

G. Arrangement of Devices: Unless otherwise indicated, mount flush, with long dimension vertical and with grounding terminal of receptacles on top. Group adjacent switches under single, multigang wall plates.

3.2 INSPECTION

A. Contractor will perform inspection of sub-contractor work immediately after work in this section is completed.

B. Contractor will record observations for conformance with ABAAS and SDC Accessibility Standards using the DSC, CQC Accessibility Inspection Report.

1. Report may be supplemented by sketches, notation, digital images, product literature, red-line contract documents, etc. to illustrate and explain conditions.

C. Coordinate inspections according to updated construction schedules.

D. Submit report to Contracting Officer within 24 hours of commencing inspection.

END OF SECTION 26 27 26

SECTION 26 28 16 – ENCLOSED SWITCHES AND CIRCUIT BREAKERS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following individually mounted, enclosed switches and circuit breakers:

1. Fusible switches.
2. Nonfusible switches.
3. Molded-case circuit breakers.
4. Molded-case switches.
5. Enclosures.

1.3 DEFINITIONS

A. GD: General duty.

B. GFCI: Ground-fault circuit interrupter.

C. HD: Heavy duty.

D. RMS: Root mean square.

E. SPDT: Single pole, double throw.

1.4 SUBMITTALS

A. Product Data: For each type of enclosed switch, circuit breaker, accessory, and component indicated. Include dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, and finishes.

1. Enclosure types and details for types other than NEMA 250, Type 1.
2. Current and voltage ratings.
3. Short-circuit current rating.
4. Features, characteristics, ratings, and factory settings of individual overcurrent protective devices and auxiliary components.

B. Shop Drawings: Diagram power, signal, and control wiring.

C. Field quality-control test reports including the following:

1. Test procedures used.
2. Test results that comply with requirements.
3. Results of failed tests and corrective action taken to achieve test results that comply with requirements.

D. Operation and Maintenance Data: For enclosed switches and circuit breakers to include in emergency, operation, and maintenance manuals. Include the following items at a minimum:

1. Manufacturer's written instructions for testing and adjusting enclosed switches and circuit breakers.
2. Time-current curves, including selectable ranges for each type of circuit breaker.

1.5 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with NFPA 70.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for enclosed switches and circuit breakers, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.

1.6 PROJECT CONDITIONS

A. Environmental Limitations: Rate equipment for continuous operation under the following conditions, unless otherwise indicated:

1. Ambient Temperature: Not less than minus 22 deg F (minus 30 deg C) and not exceeding 104 deg F (40 deg C).
2. Altitude: 8,200 Feet

1.7 COORDINATION

A. Coordinate layout and installation of switches, circuit breakers, and components with other construction, including conduit, piping, equipment, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

PART 2 - PRODUCTS

2.1 FUSIBLE AND NONFUSIBLE SWITCHES

A. Salient Characteristics:

1. Physical:

a. Fusible Switch, 600A and Smaller: NEMA KS 1, Type HD, with clips or bolt pads to accommodate specified fuses, lockable handle with capability to accept two padlocks, and interlocked with cover in closed position.
b. Nonfusible Switch, 600A and Smaller: NEMA KS 1, Type HD, lockable handle with capability to accept two padlocks, and interlocked with cover in closed position.

2. Functional:

a. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.
b. Neutral Kit: Internally mounted; insulated, capable of being grounded, and bonded; and labeled for copper and aluminum neutral conductors.
c. Auxiliary Contact Kit: Auxiliary set of contacts arranged to open before switch blades open.

3. Performance:

a. None.

2.2 MOLDED-CASE CIRCUIT BREAKERS AND SWITCHES

A. Salient Characteristics:

1. Physical:

a. Molded-Case Circuit Breaker:

1) Thermal-Magnetic Circuit Breakers: Inverse time-current element for low-level overloads and instantaneous magnetic trip element for short circuits. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.
2) GFCI Circuit Breakers: Single- and two-pole configurations with 5-mA trip sensitivity.

b. Molded-Case Switches:

1) Molded-case circuit breaker with fixed, high-set instantaneous trip only, and short-circuit withstand rating equal to equivalent breaker frame size interrupting rating.

2. Functional:

a. Molded-Case Circuit-Breaker:

1) Standard frame sizes, trip ratings, and number of poles.
2) Lugs: Mechanical style suitable for number, size, trip ratings, and conductor material.
3) Application Listing: Type SWD for switching fluorescent lighting loads; Type HACR for heating, air-conditioning, and refrigerating equipment.
4) Ground-Fault Protection: Integrally mounted relay and trip unit with adjustable pickup and time-delay settings, push-to-test feature, and ground-fault indicator.

b. Molded-Case Switch:

1) Lugs: Mechanical style suitable for number, size, trip ratings, and material of conductors.
2) Application Listing: Type HACR for heating, air-conditioning, and refrigerating equipment.
3) Shunt Trip: 120-V trip coil energized from separate circuit, set to trip at 55 percent of rated voltage. Provide "dummy" trip unit where required for proper operation

3. Performance:

a. Molded-Case Circuit Breaker and Switches:

1) NEMA AB 1, with interrupting capacity to meet available fault currents.

2.3 ENCLOSURES

A. Salient Characteristics:

1. Physical:

a. None.

2. Functional:

a. None.

3. Performance:

a. NEMA AB 1 and NEMA KS 1 to meet environmental conditions of installed location.

1) Outdoor Locations: NEMA 250, Type 3R.
2) Other Wet or Damp Indoor Locations: NEMA 250, Type 3R.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine elements and surfaces to receive enclosed switches and circuit breakers for compliance with installation tolerances and other conditions affecting performance.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Comply with applicable portions of NECA 1, NEMA PB 1.1, and NEMA PB 2.1 for installation of enclosed switches and circuit breakers.

B. Mount individual wall-mounting switches and circuit breakers with tops at uniform height, unless otherwise indicated. Anchor floor-mounting switches to concrete base.

C. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.

3.3 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs as specified in Division 26 Section “Identification for Electrical Systems.”

B. Enclosure Nameplates: Label each enclosure with engraved metal or laminated-plastic nameplate as specified in Division 26 Section “Identification for Electrical Systems.”

3.4 FIELD QUALITY CONTROL

A. Perform the following field tests and inspections and prepare test reports:

1. Perform each electrical test and visual and mechanical inspection stated in NETA ATS, Section 7.5 for switches and Section 7.6 for molded-case circuit breakers. Certify compliance with test parameters.
2. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.

3.5 CLEANING

A. On completion of installation, vacuum dirt and debris from interiors; do not use compressed air to assist in cleaning.

B. Inspect exposed surfaces and repair damaged finishes.

END OF SECTION 26 28 16

SECTION 26 41 13 – LIGHTNING PROTECTION FOR STRUCTURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes lightning protection for buildings.

1.3 DEFINITIONS

A. LPI: Lightning Protection Institute.

B. NRTL: National recognized testing laboratory.

1.4 SUBMITTALS

A. Product Data: For air terminals and mounting accessories.

B. Shop Drawings: Detail lightning protection system, including air-terminal locations, conductor routing and connections, and bonding and grounding provisions. Include indications for use of raceway, data on how concealment requirements will be met, and calculations required by NFPA 780 for bonding of grounded and isolated metal bodies.

C. Qualification data for firms and persons specified in “Quality Assurance” Article to demonstrate their capabilities and experience. Include data on listing or certification by an NRTL or LPI.

D. Certification, signed by Contractor, that roof adhesive for air terminals is approved by manufacturers of both the terminal assembly and the single-ply membrane roofing material.

E. Field inspection reports indicating compliance with specified requirements.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Engage an experienced installer who is an NRTL or who is certified by LPI as a Master Installer/Designer.

B. Listing and Labeling: As defined in NFPA 780, “Definitions” Article.

1.6 COORDINATION

A. Coordinate installation of lightning protection with installation of other building systems and components, including electrical wiring, supporting structures and building materials, metal bodies requiring bonding to lightning protection components, and building finishes.

B. Coordinate installation of air terminals attached to roof systems with roofing manufacturer and Installer.

PART 2 - PRODUCTS

2.1 LIGHTNING PROTECTION SYSTEM COMPONENTS

A. Salient Characteristics:

1. Physical:

a. Roof-Mounting Air Terminals: NFPA Class I, copper, 12” length, 1/2” diameter unless otherwise indicated.

1) Single-Membrane, Roof-Mounting Air Terminals: Designed for single-membrane roof materials.

b. Stack-Mounting Air Terminals: Solid copper.
c. Ground Rods and Ground Loop Conductors: Comply with Division 26 Section “Grounding and Bonding for Electrical Systems” and with standards referenced in this Section.

2. Functional:

a. None.

3. Performance:

a. Comply with UL 96.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install lightning protection components and systems according to UL 96A and NFPA 780.

B. Install conductors with direct paths from air terminals to ground connections. Avoid sharp bends and narrow loops.

C. Conceal the following conductors, except as otherwise noted.

1. System conductors.
2. Down conductors.
3. Interior conductors.
4. Conductors within normal view from exterior locations at grade within 200 feet (60 m) of building.
5. Notify Contracting Officer at least 48 hours in advance of inspection before concealing lightning protection components.

D. Cable Connections: Use approved exothermic-welded connections for all conductor splices and connections between conductors and other components, except those above single-ply membrane roofing.

E. Air Terminals on Single-Ply Membrane Roofing: Comply with adhesive manufacturer's written instructions.

F. Bond extremities of vertical metal bodies exceeding 60 feet (18 m) in length to lightning protection components.

G. A counterpoise installation based on requirements in Division 26 Section “Grounding and Bonding for Electrical Systems” may be used as a ground loop required by NFPA 780, provided counterpoise conductor meets or exceeds minimum requirements in NFPA 780.

1. Bond ground terminals to counterpoise conductor.
2. Bond grounded metal bodies on building within 12 feet (3.6 m) of ground to counterpoise conductor.
3. Bond grounded metal bodies on building within 12 feet (3.6 m) of roof to counterpoise conductor.

H. Bond lightning protection components with intermediate-level interconnection loop conductors to grounded metal bodies of building at 60-foot (18-m) intervals.

3.2 CORROSION PROTECTION

A. Do not combine materials that can form an electrolytic couple that will accelerate corrosion in the presence of moisture unless moisture is permanently excluded from junction of such materials.

B. Use conductors with protective coatings where conditions would cause deterioration or corrosion of conductors.

3.3 FIELD QUALITY CONTROL

A. UL Inspection: Provide inspections as required to obtain a UL Master Label for system.

B. Provide an inspection by an inspector certified by LPI to obtain an LPI certification.

END OF SECTION 26 41 13

SECTION 26 43 13 – TRANSIENT VOLTAGE SUPPRESSION (SURGE PROTECTION) FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes field-mounted SPDs for low-voltage (120 to 600 V) power distribution and control equipment.

1.3 DEFINITIONS

A. Inominal: Nominal discharge current.

B. MCOV: Maximum continuous operating voltage.

C. Mode(s), also Modes of Protection: The pair of electrical connections where the VPR applies.

D. MOV: Metal-oxide varistor; an electronic component with a significant non-ohmic current-voltage characteristic.

E. OCPD: Overcurrent protective device.

F. SCCR: Short-circuit current rating.

G. SPD: Surge protective device.

H. VPR: Voltage protection rating.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
2. Copy of UL Category Code VZCA certification, as a minimum, listing the tested values for VPRs, Inominal ratings, MCOVs, type designations, OCPD requirements, model numbers, system voltages, and modes of protection.

1.5 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

B. Sample Warranty: For manufacturer's special warranty.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: For SPDs to include in maintenance manuals.

1.7 WARRANTY

A. Manufacturer's Warranty: Manufacturer agrees to replace or replace SPDs that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 GENERAL SPD REQUIREMENTS

A. Salient Characteristics:

1. Physical:

a. MCOV of the SPD shall be the nominal system voltage.

2. Functional:

a. None.

3. Performance:

a. SPD with Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
b. Comply with NFPA 70.
c. Comply with UL 1449.

2.2 SERVICE ENTRANCE SUPPRESSOR

A. Salient Characteristics:

1. Physical:

a. SPDs with the following features and accessories:

1) Internal thermal protection that disconnects the SPD before damaging internal suppressor components.
2) Indicator light display for protection status.
3) Surge counter.

2. Functional:

a. Peak Surge Current Rating: The minimum single-pulse surge current withstand rating per phase shall not be less than 200 kA. The peak surge current rating shall be the arithmetic sum of the ratings of the individual MOVs in a given mode.
b. Protection modes and UL 1449 VPR for grounded wye circuits with 240/120 V, single-phase, three-wire circuits shall not exceed the following:

1) Line to Neutral: 700 V for 240/120 V.
2) Line to Ground: 1200 V for 240/120 V.
3) Line to Line: 1000 V for 240/120 V.

c. SCCR: Equal or exceed 100 kA.
d. Inominal Rating: 20 kA.

3. Performance:

a. SPDs: Comply with UL 1449, Type 1, listed as a Type 1 device, suitable for use in a Type 1 or 2 installation.
b. Comply with UL 1283.

2.3 ENCLOSURES

A. Outdoor Enclosure: NEMA 4.

2.4 CONDUCTORS AND CABLES

A. Power Wiring: Same size as SPD leads, complying with Section 260519 “Low-Voltage Electrical Power Conductors and Cables.”

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with NECA 1.

B. Install an OCPD or disconnect as required to comply with the UL listing of the SPD.

C. Install SPDs with conductors between suppressor and points of attachment as short and straight as possible, and adjust circuit-breaker positions to achieve shortest and straightest leads. Do not splice and extend SPD leads unless specifically permitted by manufacturer. Do not exceed manufacturer's recommended lead length. Do not bond neutral and ground.

D. Use crimped connectors and splices only. Wire nuts are unacceptable.

E. Wiring:

1. Power Wiring: Comply with wiring methods in Section 260519 “Low-Voltage Electrical Power Conductors and Cables.”

3.2 FIELD QUALITY CONTROL

A. Perform the following tests and inspections with the assistance of a factory-authorized service representative.

1. Compare equipment nameplate data for compliance with Drawings and Specifications.
2. Inspect anchorage, alignment, grounding, and clearances.
3. Verify that electrical wiring installation complies with manufacturer's written installation requirements.

B. An SPD will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

3.3 STARTUP SERVICE

A. Complete startup checks according to manufacturer's written instructions.

B. Do not perform insulation-resistance tests of the distribution wiring equipment with SPDs installed. Disconnect SPDs before conducting insulation-resistance tests, and reconnect them immediately after the testing is over.

C. Energize SPDs after power system has been energized, stabilized, and tested.

END OF SECTION 26 43 13

SECTION 26 51 00 – LIGHTING FIXTURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Adjust list below to suit Project.
2. Lighting fixtures, lamps, and ballasts.
3. Emergency lighting units.
4. Exit signs.
5. Lighting fixture supports.

B. Related Sections include the following:

1. Division 26 Section “Lighting Control Devices” for automatic control of lighting, including time switches, photoelectric relays, occupancy sensors, and multipole lighting relays and contactors.

1.3 DEFINITIONS

A. CRI: Color-rendering index.

B. LER: Luminaire efficacy rating.

C. Luminaire: Complete lighting fixture, including ballast housing if provided.

1.4 SUBMITTALS

A. Product Data: For each type of lighting fixture, arranged in order of fixture designation. Include data on features, accessories, finishes, and the following:

1. Physical description of lighting fixture including dimensions.
2. Emergency lighting units including battery and charger.
3. Driver.
4. Energy-efficiency data.
5. Life, output, and energy-efficiency data for lamps.
6. Photometric data, in IESNA format, based on laboratory tests of each lighting fixture type, outfitted with lamps, ballasts, and accessories identical to those indicated for the lighting fixture as applied in this Project.

a. For indicated fixtures, photometric data shall be certified by a qualified independent testing agency. Photometric data for remaining fixtures shall be certified by the manufacturer.
b. Photometric data shall be certified by a manufacturer's laboratory with a current accreditation under the National Voluntary Laboratory Accreditation Program (NVLAP) for Energy Efficient Lighting Products.

B. Shop Drawings: Show details of nonstandard or custom lighting fixtures. Indicate dimensions, weights, methods of field assembly, components, features, and accessories.

1. Wiring Diagrams: Power wiring.

C. Qualification Data: For agencies providing photometric data for lighting fixtures.

D. Operation and Maintenance Data: For lighting equipment and fixtures to include in emergency, operation, and maintenance manuals.

E. Warranties: Special warranties specified in this Section.

1.5 QUALITY ASSURANCE

A. Luminaire Photometric Data Testing Laboratory Qualifications: Provided by an independent agency, with the experience and capability to conduct the testing indicated, that is an NRTL as defined by OSHA in 29 CFR 1910.7.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

C. Comply with NFPA 70.

1.6 COORDINATION

A. Coordinate layout and installation of lighting fixtures and suspension system with other construction that penetrates ceilings or is supported by them, including HVAC equipment, fire-suppression system, and partition assemblies.

PART 2 - PRODUCTS

2.1 LIGHTING FIXTURES AND COMPONENTS, GENERAL REQUIREMENTS

A. Salient Characteristics:

1. Physical:

a. Metal Parts: Free of burrs and sharp corners and edges.
b. Sheet Metal Components: Steel, unless otherwise indicated. Form and support to prevent warping and sagging.

c. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position.

d. Reflecting surfaces shall have minimum reflectance as follows, unless otherwise indicated:

1) White Surfaces: 85 percent.
2) Specular Surfaces: 83 percent.
3) Diffusing Specular Surfaces: 75 percent.
4) Laminated Silver Metallized Film: 90 percent.

e. Plastic Diffusers and Covers:

1) Acrylic Lighting Diffusers: 100 percent virgin acrylic plastic. High resistance to yellowing and other changes due to aging, exposure to heat, and UV radiation.
2) Lens Thickness: At least 0.125 inch (3.175 mm) minimum unless different thickness is indicated.
3) UV stabilized.

f. Glass: Annealed crystal glass, unless otherwise indicated.

2. Functional:

a. Refer to drawing sheets.

3. Performance:

a. Refer to drawing sheets.

2.2 EXIT SIGNS

A. Salient Characteristics:

1. Physical:

a. Internally Lighted Signs:

1) Lamps for AC Operation: LEDs, 70,000 hours minimum rated lamp life.
2) Self-Powered Exit Signs (Battery Type): Integral automatic charger in a self-contained power pack.

a) Revise first subparagraph below to specify different battery. See Evaluations for discussion of battery types. Verify that battery selection is available as UL-listed unit and coordinate with Part 1 “Warranty” Article.
b) Battery: Sealed, maintenance-free, nickel-cadmium type.
c) Charger: Fully automatic, solid-state type with sealed transfer relay.

d) Operation: Relay automatically energizes lamp from battery when circuit voltage drops to 80 percent of nominal voltage or below. When normal voltage is restored, relay disconnects lamps from battery, and battery is automatically recharged and floated on charger.
e) Test Push Button: Push-to-test type, in unit housing, simulates loss of normal power and demonstrates unit operability.
f) LED Indicator Light: Indicates normal power on. Normal glow indicates trickle charge; bright glow indicates charging at end of discharge cycle.
g) Integral Self-Test: Factory-installed electronic device automatically initiates code-required test of unit emergency operation at required intervals. Test failure is annunciated by an integral audible alarm and flashing red LED.

2. Functional:

a. Refer to drawing sheets.

3. Performance:

a. Comply with UL 924; for sign colors, visibility, luminance, and lettering size, comply with authorities having jurisdiction.
b. Refer to drawing sheets.

2.3 EMERGENCY LIGHTING UNITS

A. Salient Characteristics:

1. Physical:

a. Battery: Sealed, maintenance-free, lead-acid type.
b. Charger: Fully automatic, solid-state type with sealed transfer relay.
c. Operation: Relay automatically turns lamp on when power supply circuit voltage drops to 80 percent of nominal voltage or below. Lamp automatically disconnects from battery when voltage approaches deep-discharge level. When normal voltage is restored, relay disconnects lamps from battery, and battery is automatically recharged and floated on charger.
d. Test Push Button: Push-to-test type, in unit housing, simulates loss of normal power and demonstrates unit operability.
e. LED Indicator Light: Indicates normal power on. Normal glow indicates trickle charge; bright glow indicates charging at end of discharge cycle.
f. Integral Time-Delay Relay: Holds unit on for fixed interval of 15 minutes when power is restored after an outage.
g. Integral Self-Test: Factory-installed electronic device automatically initiates code-required test of unit emergency operation at required intervals. Test failure is annunciated by an integral audible alarm and flashing red LED.

2. Functional:

a. Refer to drawing sheets.

3. Performance:

a. Self-contained units complying with UL 924.
b. Refer to drawing sheets.

2.4 SOLID STATE LIGHTING FIXTURES

A. Housing, where applicable:

1. Steel bonderized or equal rust projected, or aluminum, rigid construction. Minimum gauge thickness shall be as follows:

a. Interior locations: No. 20 gauge steel, no. 16 gauge aluminum.

B. Finish:

1. Baked enamel finish (except when otherwise specified).
2. Concealed interior surfaces (this applies to interior hardware, circuit boards, etc.) matte black.
3. Concealed exterior surfaces: matte black.
4. Visible surfaces: color and texture as specified below for each fixture type or as selected.
5. Exterior fixture finish: refer to Lighting Fixture Schedule.

C. Light Emitting Diode (LED) requirements:

1. Correlated color temperature (CCT) for phosphor-coated white LEDs must have one of the following designated CCT's and fall within the following binning standards:

a. Outdoor:

1) 2700K defined as 2725 +/- 145K
2) Amber

b. Indoor:

1) 3500K defined as 3465 +/- 245K
2) 4000K defined as 3985 +/- 275K

2. Color spatial uniformity shall be limited to variations in chromaticity for different directions within 0.004 from the weighted average point on the CIE 1976 (u',v') diagram.
3. Color maintenance shall be limited to a maximum change in chromaticity of 0.007 on the CIE 1976 (u',v') diagram over the lifetime of the product.
4. Color rendering index.

a. Color rendering index to be determined using ANSI C78.377-2008 and applicable IESNA standards.
b. Laboratory tests must be produced using specific module(s)/array(s) and power supply combination that will be used in production.
c. Manufacturers must provide a test report from a laboratory accredited by NVLAP or one of its MRA signatories.

5. Lumen depreciation.

    a. Lumen depreciation to be measured using IESNA LM-80-08 standard for IES approved method of measuring lumen maintenance of LED light sources.
    b. Phosphor coated white LED module(s)/array(s) shall deliver at least 70% of initial lumens for a minimum of 35,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.
    c. Colored LED module(s)/array(s) shall deliver at least 50% of initial lumens for a minimum of 35,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.

6. Acceptable LED manufacturers:

    a. CREE

        1) Nichia
        2) Osram Opto Semiconductors
        3) Philips Lumileds

D. Luminaire Efficacy:

1. Luminaire efficiency shall be measured using IESNA LM-79-08 standard for electrical and photometric measurements of solid state lighting products.
2. Manufacturer shall provide published luminaire efficacy, which is defined as luminaire light output divided by luminaire input power measured in a 25 degree Celsius environment. Efficacy shall include power supply, thermal, optical, and fixture losses. Minimum efficacy of 80 lumens per watt.

E. Thermal Management:

1. Solid state light fixture shall not exceed LED manufacturer's maximum junction temperature requirements when operated in-situ at fixture manufacturer's maximum ambient operating temperature and 100% light output.
2. Solid state light fixtures shall be thermally protected using one of more of the following thermal management techniques:

    a. Metal core board
    b. Gap pad
    c. Internal monitoring firmware

3. Solid state lighting fixture housing shall be designed to transfer heat from the LED board to the outside environment.

F. Power Supplies/Drivers:

1. Power supply shall have a power factor of 0.90 or greater for primary application.
2. Power supply input current shall have Total Harmonic Distortion (THD) of less than 20%.
3. Power supply shall have a minimum operating temperature of minus 20 degrees Celsius of below when used in luminaires intended for outdoor applications.
4. Power supply output operating frequency to be equal to or greater than 120Hz.

5. Power supply shall operate with sustained input variations of +/- 10% (voltage and frequency) with no damage to the driver.
6. Power supply shall tolerate sustained open circuit and short circuit output conditions without damage and without need for external fuses or trip devices.
7. Power supply output shall be regulated to +/- 5% across published load range.
8. Power supply shall have a Class A sound rating.
9. Power supply outputs shall have current limiting protection.
10. Power supply shall operate LEDs at constant and regulated current levels. LEDs shall not be overdriven beyond the diode manufacturer's specified nominal voltage and current.

G. Solid State Lighting Controls:

1. Control interface to dimmable power supplies shall consist of one of the following:

   a. Line Voltage Dimming. Controls to be rated for magnetic or electronic low voltage transformer operation.
   b. Low voltage (0-10V) control. Controls to be compatible with either current sink or current source operation.

2. Dimmable LED power supplies shall use pulse width modulation (PWM) to regulate power to LEDs

   a. Dimmable power supplies shall have 12-bit or greater resolution to obtain flicker-free operation throughout the dimming range.

H. System Installation:

1. All hardwired connections to solid state lighting fixtures shall be reverse polarity protected and provide high voltage protection in the event connections are reversed or shorted during the installation process.
2. All solid state lighting fixtures (100% of each lot) shall undergo a minimum eight-hour burn-in test during manufacturing.
3. Solid state lighting installations shall be UL listed as a low-voltage lighting system including, but not limited to, luminaire, power supply, controller, keypad, and wiring.

I. Warranty: Luminaires, drivers, and controllers for solid state lighting systems shall be covered by a two-year warranty against defects in workmanship or material. Warranty shall include in-warranty service program providing for payment of authorized labor charges incurred in replacement of inoperative in-warranty equipment.

2.5 LIGHTING FIXTURE SUPPORT COMPONENTS

A. Comply with Division 26 Section “Hangers and Supports for Electrical Systems” for channel- and angle-iron supports and nonmetallic channel and angle supports.

B. Single-Stem Hangers: 1/2-inch (13-mm) steel tubing with swivel ball fittings and ceiling canopy. Finish same as fixture.

C. Twin-Stem Hangers: Two, 1/2-inch (13-mm) steel tubes with single canopy designed to mount a single fixture. Finish same as fixture.

D. Wires: ASTM A 641/A 641M, Class 3, soft temper, zinc-coated steel, 12 gage (2.68 mm).

E. Wires for Humid Spaces: ASTM A 580/A 580M, Composition 302 or 304, annealed stainless steel, 12 gage (2.68 mm).

F. Rod Hangers: 3/16-inch (5-mm) minimum diameter, cadmium-plated, threaded steel rod.

G. Hook Hangers: Integrated assembly matched to fixture and line voltage and equipped with threaded attachment, cord, and locking-type plug.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Lighting fixtures: Set level, plumb, and square with ceilings and walls. Install lamps in each fixture.

B. Suspended Lighting Fixture Support:

1. Pendants and Rods: Where longer than 48 inches (1200 mm), brace to limit swinging.
2. Stem-Mounted, Single-Unit Fixtures: Suspend with twin-stem hangers.
3. Continuous Rows: Use tubing or stem for wiring at one point and tubing or rod for suspension for each unit length of fixture chassis, including one at each end.

C. Adjust aimable lighting fixtures to provide required light intensities.

D. Retrofit power, lighting and controls will impact interior surfaces in public spaces that are highly character defining. If new installations are not concealed it potentially changes the historic appearance and surface anchorage may result in loss of fabric that degrades the quality of the resource. New equipment and materials choices must be concealed or compatible with the historic surface to avoid obtrusiveness. Major alterations should be relegated to least important locations. Careful choices for materials, finishes, fixtures, and service runs can help preserve the character of the property.

END OF SECTION 26 51 00

SECTION 27 00 10 – SUPPLEMENTAL REQUIREMENTS FOR COMMUNICATIONS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Supplemental requirements generally applicable to the Work specified in Division 27.

a. Work specified in Division 27 may be applicable to, or in coordination with, trades other than division 27. It is the contractor's responsibility to coordinate with other trades to define and complete the scope of work.
b. Information herein shall be considered supplementary, and in addition to, system specific specifications.

B. Related Requirements:

1. Refer to the division 26 specifications for further information regarding:

a. Abbreviations, acronyms, definitions, and additional coordination drawing submittal requirements.

2. Section 280010 "Supplemental Requirements for Electronic Safety and Security" for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.
3. Refer to the specifications below:

a. 270526 – Grounding And Bonding For Communication Systems
b. 270528 – Pathways For Communication Systems
c. 270529 – Hangers And Supports For Communication Systems
d. 270536 – Cable Trays For Communications Systems
e. 270544 – Sleeves And Sleeve Seals For Communication Pathways And Cabling
f. 270553 – Identification For Communication Systems
g. 271116 – Communications Racks, Frames, And Enclosures
h. 271323 – Communications Optical Fiber Backbone Cabling
i. 271513 – Communications Copper Horizontal Cabling
j. 274100 – Audio-Visual Systems

1.2 REFERENCES

A. Abbreviations and Acronyms for Communications:

1. LAN: Local area network.
2. PoE: Power over Ethernet.
3. POTS: Plain old telephone service. See "public switched telephone network."
4. TCP/IP: Transmission control protocol/Internet protocol.
5. WAN: Wide area network.

B. Definitions for Communications:

1. Calling Party Control (CPC): A momentary break in phone line loop current, which is used to signal voicemail and other automated telephone company services that distant party has hung up.
2. Private Branch Exchange (PBX): Analog telephone switch that routes calls internal to a business or organization so a direct external line for each phone is unnecessary.
3. Public Switched Telephone Network (PSTN): Analog telephone technology that uses twisted-pair cables from a telephone-provider central office for the transmission medium. PSTN refers to the telephone network; POTS refers to the individual subscriber line.
4. Remote Office Phone System (ROPS): VoIP system that allows phones for a business or organization located anywhere in the world with internet connectivity to behave similar to phones connected to a PBX.
5. Ringer Equivalence Number (REN): The loading effect of a single traditional telephone ringing circuit. TIA-968 defines REN 1 as an impedance of 7000 Ω at 20 Hz (Type A ringer) or 8000 Ω from 15 Hz to 68 Hz (Type B ringer). The sum of the RENs for all devices on a subscriber line circuit may not exceed the maximum permitted REN for the subscriber line.
6. Voice over Internet Protocol (VoIP): Digital telephone packet technology that uses the internet for its transmission medium.

1.3 COORDINATION

A. The Technology drawings and specifications provide the scope and requirements of construction applicable to multiple trades and systems. These systems are inclusive of, but not limited to, structured cabling, low voltage, audio-visual, and other such systems. Each system requires the coordination of multiple contracting trades and disciplines. It is the responsibility of each contractor to review all technology scope and to coordinate with other disciplines.

B. Contractor shall coordinate with all other trades prior to construction, at minimum, contractor shall coordinate with the general, electrical, security, and other low voltage contractors. It is the contractor's responsibility to ensure all other contractor trades are aware and knowledgeable of the system requirements and associated responsibilities as defined in these construction documents.

C. Contractor shall always maintain a working set of the construction documents in field (These may be printed or virtual). These documents shall always represent the latest information as well as capture all changes to final constructed solution(s). Any cost associated with an unaccepted installation, and subsequent reinstallation, due to the wrong documentation reference is the sole responsibility of the contractor and shall not be a cost transferred to the contracting officer. At a minimum, the working set of construction documents shall include:

1. The latest version of construction documents
2. All changes associates with change orders, RFIs, Addenda, email, and/or other such documentation
3. All field coordinated changes and/or changes to be captured in as-built drawings.

D. For any and all discrepancies, contractor shall provide written notification to contracting officer and general contractor prior to commencement of work and procurement of material.

E. Interruption of Existing Telephone, Internet, CATV, Cellular, or other such Service: Do not interrupt service to facilities occupied by end user or others unless permitted under the following conditions:

1. Notify Contracting Officer no fewer than 20 business days in advance of proposed interruption of service.
2. Do not proceed with interruption of service without Contracting Officer's written permission.

F. Multi-disciplinary coordination requirements: Ensure that all trades and integrators who are impacted or dependent on the Division 27 scope of work (either via scheduling, constructability, support, or other such means) are contacted and informed of coordination items relevant to their scope of work at least 10 days prior to commencement of work. Furthermore, ensure all trades and integrators whose work has a similar effect on the Division 27 scope of work are contacted for information regarding items relevant to the scope of work at least 10 days prior to work commencement.

1.4 PREINSTALLATION MEETINGS

A. Communications Preconstruction Conference: Schedule conference with General Contractor and the Contracting Officer not later than 10 business days after notice to proceed. Agenda topics include, but are not limited to, the following:

1. Installation schedule for communications systems.
2. Installation schedule for all systems effecting, or effected by, the communications systems scope of work.
3. Value analysis proposals and requests for substitution of communications equipment.
4. Utility service coordination, outages, inspections, and/or activation.
5. Sustainability activities.
6. Submission of action submittals specified in Division 27.
7. Submission of information submittals specified in Division 27.
8. Submission of closeout submittals specified in Division 27.
9. Orders placed for major equipment.
10. Arrival of major equipment on-site.
11. Preinstallation meetings specified in Division 27.
12. System startup, testing, and commissioning activities for communications equipment.
13. System startup, testing, and commissioning activities for Work specified in other divisions that depends on Work specified in Division 27.
14. Requests for special inspections.
15. Requests for inspections by authorities having jurisdiction.

1.5 ACTION SUBMITTALS

A. Submission of action submittals shall be in accordance with Division 01 specifications. Refer to each specification section for submittal requirements specific to each scope of work.

B. The items listed below outline the multiple categories of action submittals, and the associated items comprising each category, that may be included in a system's specification. Refer to each division 27 specification for the system specific submittal requirements.

1. Product Data

    a. Manufacturer and Part number

        1) Shall be highlighted to indicate version of equipment to be utilized when multiple models/variations of product are present on documentation.

2. Shop Drawings

    a. Contractor shall provide Site Plans, Overall plans, Floor plans, Ceiling plans, One-line Diagrams, Wiring Diagrams, and/or Elevations views as needed to indicate system specific fabrication, installation, layout, connectivity, and coordination.
    b. Cabling Plan for backbone cabling indicating the origination point, termination point, and general path of cable including the pathways utilized along path.

3. Identification Schedule

    a. An electronic copy of the system specific labeling schedule including, at minimum, an itemized list of the component(s) to be labeled, the associated label per component, and component information such as manufacturer, part number, and installed location.
    b. Scaled details drawings indicating components to be labeled and location of label on components.

1.6 INFORMATIONAL SUBMITTALS

A. Submission of information submittals shall be in accordance with Division 01 specifications. Refer to each specification section for submittal requirements specific to each scope of work.

B. Information submittals shall be submitted to General Contractor at monthly intervals unless otherwise noted/directed by General Contractor.

C. At each issuance, a combined submittal shall be submitted that includes the information submittals from all the division 27 specifications referenced herein.

1. Where multiple specification sections indicate the same or similar recorded item, it is acceptable for the contractor to submit a combined submittal to address multiple specification sections.

    a. For example, Field Quality Control Reports can be combined to encompass multiple specification sections. It must be noted on report which sections are included in report.

D. The items listed below outline the multiple categories of information submittals, and the associated items comprising each category, that may be included in a system's specification. Refer to each division 27 specification for the system specific submittal requirements.

1. Coordination Drawings

    a. Drawings indicating specific coordination items between the division 27 scope of work components with elements from architectural, structural, and other trades.

2. Installer Qualification Data

    a. Certification information of each title as specified.

3. Field Quality Control Reports

    a. Reports that include the following:

        1) Adherence and/or adjustments to installation schedule
        2) Adherence and/or adjustments to procurement schedule
        3) System Start-up, Testing, Commissioning and Turn-Over schedules
        4) System outages and downtime
        5) Major Equipment order placing and arrival schedules
        6) Waste-management and work site cleanliness adherence
        7) Testing and Verification Results

4. Factory Test Reports

    a. Manufacturer generated test reports validating the products viability prior to being shipped.

5. System Testing Plans

    a. A scheduled plan to execute the testing requirements of a specific product in accordance with manufacturer requirements, industry recommendations, and specification requirements.
    b. Plan shall include testing procedures, testing equipment, pass/fail thresholds per test, testing parameters of equipment and test, and the proposed schedule for testing plan.

6. Certificates

    a. Seismic-Load Performance Certificates: Provide special certification as indicated in Paragraph 13.2.2 “Special Certification Requirements for Designated Seismic Systems” of ASCE/SEI 7-16 for designated seismic-load systems identified on Drawings or in the Specifications.

        1) Include the following information:

            a) Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
            b) Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
            c) Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.
            d) Detailed description of conduit support devices and interconnections on which the certification is based and their installation requirements.
            e) Provide equipment manufacturer's written certification for each designated active electrical seismic device and system, stating that it will remain operable following the design earthquake. Certification must be based on requirements of ASCE/SEI 7, including shake table

testing per ICC-ES AC156 or a similar nationally recognized testing standard procedure acceptable to authorities having jurisdiction.

f) Submit evidence demonstrating compliance with these requirements for approval to authorities having jurisdiction after review and acceptance by qualified structural professional engineer.

b. Wind-Load Performance Certificates: Provide special certification for designated wind-load systems and components identified on Drawings or in the Specifications.

1) Include the following information:

a) Provide equipment manufacturer's written certification for each designated system and component, stating that it will remain in place and operable following the design wind event and comply with requirements of authorities having jurisdiction.

b) Certification must be based on ICC-ES or similar nationally recognized testing standard procedures acceptable to authorities having jurisdiction.

1.7 CLOSEOUT SUBMITTALS

A. Submission of closeout submittals shall be in accordance with Division 01 specifications. Refer to each specification section for submittal requirements specific to each scope of work.

B. A combined submittal shall be submitted to the Contracting Officer that includes the closeout submittals from all the division 27 specifications referenced herein.

1. Where multiple specification sections indicate the same or similar recorded item, it is acceptable for the contractor to submit a combined submittal to address multiple specification sections.

a. For example, Record Documentation can be combined to encompass multiple specification sections. It must be noted on document which sections are included in document.

C. The items listed below outline the multiple categories of closeout submittals, and the associated items comprising each category, that may be included in a system’s specification. Refer to each division 27 specification for the system specific submittal requirements.

1. Record Documentation:

a. Including the installed location of equipment and components.
b. Including the major routes of backbone, utility, and grounding cables.

2. Operation and Maintenance Data:

a. Provide emergency operation, normal operation, and preventive maintenance manuals for each system, equipment, and device listed below:
b. Manufacturer's operating specifications.

c. User's guides.
d. Schedule of maintenance material items recommended to be stored at Project site.
e. Detailed instructions covering operation under both normal and abnormal conditions.
f. Instructions for setting field-adjustable components.
g. Manufacturer's instructions for testing, adjusting, and maintaining component/equipment.

3. Warranty Information

a. Manufacturer’s and Contractor’s Warranty Documents
b. Maintenance Contracts and Service Agreements

4. Software and Firmware Operational Documentation: Provide software and firmware operational documentation, including the following:

a. Software operating and upgrade manuals.
b. Names, versions, and website addresses for locations of installed software.
c. Device address list.
d. Printout of software application and graphic screens.
e. Testing and adjusting of panic and emergency power features.
f. Provide the program Software Backup username and password for approved online or cloud solution.
g. Provide to Contracting Officer the upgrades and unrestricted licenses for installed and backup software, including operating systems and programming tools required for operation and maintenance.

5. Identification Schedule

a. The finalized version of the identification schedule updated to include label changes from in-field/coordinated changes.
b. A single identification schedule encompassing all associated specification sections shall be submitted.

6. Extra Materials

a. Furnish to Contracting Officer extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

7. Field Test and Inspection Reports

a. Preform testing per system specification and submit associated report.
b. All test reports shall be combined and submitted as a single file and combined report.

1.8 QUALIFICATIONS

A. Communications Design Professional: Design professional possessing active qualifications specified in Section 014000 “Quality Requirements” and the following:

1. Expertise in design of communications infrastructure and distribution equipment.
2. BICSI Registered Communications Distribution Designer (RCDD) certification.

B. Communications Cable Installer: Entity possessing active qualifications specified in Section 014000 “Quality Requirements” and the following:

1. Training and manufacturer certification to install, splice, and terminate communications cabling.
2. Installation Supervisor: BICSI Technician (TECH) certification.
3. Copper Installers: 30 percent of on-site employees possess BICSI Copper Installer 2 (INSTC) certification. Remaining employees possess BICSI Installer 1 (INST1) certification.
4. Fiber Installers: 30 percent of on-site employees possess BICSI Optical Fiber Installer 2 (INSTF) certification. Remaining employees possess BICSI Installer 1 (INST1) certification.

C. Communications Testing Agency: Entity possessing active credentials from a qualified electrical testing laboratory recognized by authorities having jurisdiction.

1. On-site communications testing supervisor must have BICSI Technician (TECH)] certification and documented training, and be experienced with testing communications equipment in accordance with BICSI and ANSI/TIA testing standards.

1.9 FIELD CONDITIONS

A. Modeling, analysis, product selection, installation, and quality control for work specified in Division 27 must comply with requirements specified in Section 260011 “Facility Performance Requirements for Electrical.”

PART 2 - PRODUCTS

2.1 COMMUNICATIONS EQUIPMENT

A. Products shown within the associated division 27 specifications, and/or within associated drawings/schedules, shall be considered as basis of design, unless specifically noted.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:

1. Immediately upon initial (1st day) arrival to site, contractor shall coordinate with General Contractor for all site and project specific information. Contractor shall adhere to all General Contractor requirements throughout duration of project.
2. Prior to commencement of work, contractor shall reference and/or inspect the following conditions:

a. All safety and security protocols specific to site construction.
b. The most current version of construction documents as specified in section 1.3.C herein.
c. The most updated/recent version of project and installation schedule(s).
d. All coordinated work previously completed and coordinated work that is uncompleted but scheduled for prior completion.
e. Current construction conditions affecting scope of work.

3. For any and all questions, concerns, coordination, or other project specific information, especially those affecting schedule, quality of work, or system acceptance, contractor shall inform General Contractor in as timely a manner as possible.

B. Preinstallation Testing:

1. It is the contractor's responsibility to ensure that all material and equipment is tested for acceptance pursuant to the manufacturer's specifications and parameters prior to installation.
2. All material and equipment shall be tested for acceptance immediately upon being received by contractor.

    a. If material and/or equipment is received off-site, material and/or equipment shall be retested and confirmed to be acceptable upon arrival on-site.

3. Any material and/or equipment that fails testing procedure and is unacceptable for installation shall be noted and communicated to manufacturer. Dependent upon scope and breadth of failures, contractor shall coordinate required replacement material and maintain installation schedule. If failures affect the installation schedule and/or project schedule, contractor shall inform the General Contractor.

3.2 PREPARATION

A. Protection of In-Place Conditions:

1. Contractor shall always maintain a neat and orderly workspace while on site. The workspace shall not interfere with, or in any other form, disallow other trades to complete their scope of work or maintain proper schedule of work.
2. All tools, material, equipment, trash, etc., shall be maintained to be always neat and orderly while on site.
3. All material and equipment that has been installed shall be protected in place. It is the contractor's responsibility to ensure all installed equipment is protected in place through means of protective coverings, enclosed/restricted spaces, and coordination with other trades. The items of note include, but are not limited to:

    a. Capping/Taping the open ends of unused conduit and open backboxes while protecting pull string.
    b. Covering of all active equipment and termination components to ensure any openings, ports, or other such component orifices are protected from elements such as water, dust, and construction debris and/or material.

c. Isolation of telecommunication rooms and equipment from general construction whenever possible to minimize water, dust, and construction debris and/or material.
d. Protection of active equipment from tampering, especially pre-configured or recently configured.
e. Proper bonding and grounding of all equipment immediately upon installation.

3.3 INSTALLATION OF COMMUNICATIONS WORK

A. Unless more stringent requirements are specified in the Contract Documents or manufacturers' instructions, contractors shall complete scope of work in a neat and workmanlike manner and comply with NFPA 70, NECA NEIS 1, and BICSI N1 for installation of work specified in Division 27. Consult Contracting Officer for resolution of conflicting requirements.

B. For all work specified in Division 27, contractor shall provide a complete and functional (turnkey) system. Coordination with General Contractor, Contracting Officer, and other trades may be required. Reference the Responsibility Matrix on Technology sheets for information per each system.

3.4 SYSTEM STARTUP

A. Commissioning Activities:

1. Contractor shall reference and adhere to manufacturer’s startup/activation requirements for commissioning.
2. Commissioning effort shall ensure that system is complete and operational prior to final testing, system turnover and substantial completion.

3.5 FIELD QUALITY CONTROL

A. Administrant for Communications Equipment, Systems, and Material Tests and Inspections:

1. Contractor shall administer and perform tests and inspections.
2. Contractor may utilize assistance of factory-authorized service representative, or a third-party certified testing authority, at no cost to Contracting Officer.

3.6 CLEANING

A. Waste Management:

1. Proper disposal and on-site management of all waste material, packaging, and other such extra items not scheduled for install is required. It is the contractor’s responsibility to ensure all waste management procedures and protocols are observed and practiced with regards to any and all waste accumulated.
2. Prior to initial (1st day) arrival on site, contractor shall coordinate with General Contractor for current site conditions, rules, regulations, and requirements regarding the accumulation and disposal of waste.

3. The requirements below are in addition to General Contractor defined site requirements, not in lieu of, and shall not supersede any site requirements. Any discrepancy between the below and the site requirements shall be communicated to General Contractor for further clarification.

    a. All trash or waste accumulated shall be properly disposed of at the end of each day.
    b. Any material deemed hazardous must be disposed of immediately. Disposal of material shall be communicated to, and coordinated with, General Contractor.

3.7 CLOSEOUT ACTIVITIES

A. Demonstration:

1. Contractor shall coordinate with Contracting Officer and General Contractor to schedule time and attendance for a contractor provided demonstration of the operability of the following systems and equipment:

    a. Telecommunication Equipment Room
    b. Structured Cabling System (Horizontal and Backbone) and Infrastructure
    c. AV system

2. It is acceptable for contractor to engage sub-contractors, manufacturer representatives, and/or third parties, to conduct demonstration. Any engagement of such shall be at no cost to Contracting Officer.
3. At Contracting Officer's request, Contractor shall record demonstration(s) or allow Contracting Officer to record demonstration(s). Any contractor provided recording shall be at no extra cost to Contracting Officer.

B. Training:

1. Train end-user's maintenance personnel on any system and/or equipment defined herein. Training shall be administered by contractor(s) and shall include at least 1 hour of training per section on how to adjust, operate, update, and maintain the system(s) below:

    a. Section 271116 "Communications Rack, Frames, and Enclosures"
    b. Section 271323 "Communications Optical Fiber Backbone Cabling"
    c. Section 271513 "Communications Copper Horizontal Cabling"
    d. Section 274100 "Audio-Visual Systems"

    At Contracting Officer's request, Contractor shall record training session(s) or allow Contracting Officer to record session(s). Any contractor provided recording shall be at no extra cost to Contracting Officer.

END OF SECTION 27 00 10

SECTION 27 05 26 – GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 "Supplemental Requirements for Communications" for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

1. In addition to information listed herein, contractor shall reference the below standards:

a. ANSI/TIA-607-B – Generic Telecommunications Bonding and Grounding (Earthing) For Customer Premises.
b. ANSI/NECA/BICSI 607 – BICSI Standard for Telecommunication Grounding and Bonding, Planning and Installation Methods for Commercial Building
c. NFPA 70 (NEC)

1.2 SUMMARY

A. Section Includes:

1. Selection and installation of communications busbars.
2. Selection and installation of communications bonding conductors.

1.3 DEFINITIONS

A. BBC: Backbone bonding conductor, for connecting multiple SBBs serving the same floor.

B. PBB: Primary bonding busbar, located in main distribution frame room, ideally near electrical service entrance.

C. RBB: Rack bonding busbar, located in equipment cabinets and racks.

D. SBB: Secondary bonding busbar, located in intermediate distribution frame rooms.

E. TBB: Telecommunications bonding backbone, for connecting SBBs to PBB.

F. TBC: Telecommunications bonding conductor, for connecting PBB to intersystem bonding termination device at electrical service entrance.

G. TEBC: Telecommunications equipment bonding conductor, for connecting RBBs to SBBs or PBB.

H. UBC: Unit bonding conductor, for connecting individual communications equipment to RBBs or SBBs.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. The following products shall be included:

a. Grounding Busbars
b. Conductors
c. Connectors

B. Shop Drawings

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

B. Installer Qualification Data

1. Refer to Division 26 specifications and requirements for electrical contractor installation requirements and qualification requirements.
2. Qualifications per specification 270010 “Supplemental Requirements for Communications”.
3. RCDD certification for Installer, installation supervisor, and field inspector.
4. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.

a. Manufacturer’s Published Instructions: Record copy of official installation and testing instructions issued to Installer by manufacturer for the following:

1) Installing wire connector on conductor.
2) Recommended torque values.

C. Field Quality Control Reports

1.6 CLOSEOUT SUBMITTALS

A. Record Documentation: Project record documents in accordance with Section 017839 "Project Record Documents" must include locations of PBB and SBBs, and routing of TBC, TBBs, and BBCs.

B. Operation and Maintenance Data.

C. Field Test and Inspection Report.

PART 2 - PRODUCTS

2.1 GROUNDING BUSBARS

A. Unless otherwise indicated in this Section or on Drawings, provide products specified in Section 260526 "Grounding and Bonding for Electrical Systems."

B. Must be UL listed and have a hole pattern configuration in line with recommendations of ANSI/BICSI and ANSI/TIA 607 standards.

C. Must be made of copper, or copper alloy having a minimum of 95% conductivity as specified by the International Annealed Copper Standard (IACS).

D. PBB:

1. Dimensions: 1/4 inch thick by 4 inch high by 20 inch wide (6.3 mm thick by 100 mm high by 500 mm wide)
2. Stand-Off Distance: 4 inch (100 mm)

2.2 CONDUCTORS

A. Comply with UL 486A-486B.

B. Insulated Conductors: Stranded copper wire, green or green with yellow stripe insulation, insulated for 600 V, and complying with UL 83.

1. Ground wire for custom-length equipment ground jumpers shall be No. 6 AWG, 19-strand, UL-listed, Type THHN wire.

C. Bare Copper Conductors:

1. Solid Conductors: ASTM B 3.
2. Stranded Conductors: ASTM B 8.
3. Tinned Conductors: ASTM B 33.

2.3 CONNECTORS

A. Irreversible connectors listed for the purpose. Listed by an NRTL as complying with NFPA 70 for specific types, sizes, and combinations of conductors and other items connected. Comply with UL 486A-486B.

B. Compression Wire Connectors: Crimp-and-compress connectors that bond to the conductor when the connector is compressed around the conductor. Comply with UL 467.

1. Electroplated tinned copper, C and H shaped.

C. Busbar Connectors: Cast silicon bronze, solderless compression-type, mechanical connector; with a long barrel and two holes spaced on 5/8- or 1-inch (15.8- or 25.4-mm) centers for a two-bolt connection to the busbar.

2.4 LABELING

A. Comply with TIA/EIA-606-A and UL 969 for a system of labeling materials, including label stocks, laminating adhesives, and inks used by label printers.

B. Adhesive Film Label with Clear Protective Overlay: Machine printed, in black, by thermal transfer or equivalent process. Minimum letter height shall be 3/8 inch (10 mm). Overlay shall provide a weatherproof and UV-resistant seal for label.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine facility's grounding electrode system and equipment grounding for compliance with requirements for maximum ground-resistance level and other conditions affecting performance of grounding and bonding of electrical system.

B. Inspect test results of grounding system measured at point of TBC connection.

C. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

D. Proceed with connection of TBC only after unsatisfactory conditions have been corrected.

3.2 SELECTION OF COMMUNICATIONS BONDING CONDUCTORS

A. Unless otherwise indicated in this Section or on Drawings, provide products specified in Section 260526 "Grounding and Bonding for Electrical Systems."

B. Communications Busbar Connections:

1. Unless otherwise specified, contractor shall refer to information below for sizing of grounding conductor cables:

a. For any application(s) where the conductor length does not exceed 100 feet (30.5m), a minimum of 6AWG conductor is acceptable.
b. For any application(s) where the conductor length is greater than 100 feet (30.5m):

1) Determine the size of the grounding electrode conductor (GEC).
2) Increase the size of the bonding conductor by one trade size for every 100 feet (30.5m) of length.

C. Cable Tray Connections:

1. Cable Tray Equipment Grounding Conductor: 6 AWG.
2. Cable Tray Bonding Jumper: If not supplied by cable manufacturer, provide bonding jumper not smaller than 6 AWG and not longer than 12 inch (300 mm). If jumper is wire, it must be terminated with lug having at least one hole and a standard-length barrel for

single crimp. If jumper is flexible braid, it must be terminated with two-hole ferrule. Attach with bonding screw or connector provided by cable tray manufacturer.

D. Underground Connections: Not smaller than 2 AWG. Provide welded connectors, except bolted connectors may be used in handholes or manholes and as otherwise indicated on Drawings.

3.3 INSTALLATION OF BONDING FOR COMMUNICATIONS

A. Comply with manufacturer's published instructions.

B. Reference Standards:

1. Bonding of Communications: Unless more stringent requirements are specified in Contract Documents or manufacturers' published instructions, comply with BICSI N3.
2. Consult Contracting Officer for resolution of conflicting requirements.

C. Special Techniques:

1. Busbars:

a. Indicate locations of grounding busbars on Drawings. Install busbars horizontally, on insulated spacers 18 inch (450 mm) above finished floor unless otherwise indicated.
b. Where indicated on both sides of doorways, route bus up to top of door frame, across top of doorway, and down; connect to horizontal bus.

2. Conductors:

a. Stacking of conductors under a single bolt is not permitted when connecting to busbars.
b. Assemble wire connector to conductor, complying with manufacturer's published instructions and as follows:

1) Use crimping tool and die specific to connector.
2) Pretwist conductor.
3) Apply antioxidant compound to bolted and compression connections.

c. Install in straightest and shortest route between origination and termination point, and no longer than required. Bend radius must not be smaller than 10 times diameter of conductor. No single bend may exceed 90 degrees.
d. Install without splices.
e. Support conductors at not more than 36 inch (900 mm) intervals.
f. Outside telecommunications rooms, install conductors in metric designator 21 (trade size 3/4) PVC-80 conduit until conduit enters telecommunications room. Install bonding conductors in EMT-A or EMT-SS when routed through plenum. Do not install bonding conductors in EMT-S unless otherwise indicated on Drawings.

1) If bonding conductor must be installed in EMT-S or other ferrous metallic raceway, bond conductor to raceway using grounding bushing that complies with Section 270528 "Pathways for Communications Systems," and bond both ends of raceway to SBB.

3. Provide TBC and terminate ends to PBB and intersystem bonding termination device at electrical service entrance in accordance with Section 250.94, "Bonding for Communication Systems," of NFPA 70.
4. Busbar Interconnections: Bond SBBs to PBB with TBBs. If more than one TBB is installed, bond TBBs together. Install BBCs where required by TIA-607.
5. Structural Steel: Where structural steel of steel frame building is readily accessible within room or space, bond each SBB and PBB to vertical steel of building frame.
6. Communications Enclosures: Bond metallic enclosures of telecommunications equipment with UBCs to nearest SBB or PBB.
7. Equipment Racks: Bond metallic components of enclosures to RBB using UBCs. Provide top-mounted RBB if not provided by enclosure or rack manufacturer. Bond RBB to SBB with TEBC. Power connection must comply with NFPA 70; equipment grounding conductor in power cord of cord- and plug-connected equipment must be considered supplemental to bonding requirements in this Section.
8. Shielded Cable: Bond shield of shielded cable to SBB in communications rooms and spaces. Comply with TIA-568.1 and TIA-568.2 when grounding shielded balanced twisted-pair cables.
9. Primary Protector: Bond to PBB with insulated bonding conductor.
10. Electrical Power Panelboards: Where electrical panelboards for communications equipment are located in same room or space, bond each ground bar of panelboard to SBB.
11. Cable Trays: Provide continuous electrical path by installing bonding clips and jumpers. Bond each end to nearest SBB.
12. Conduits/Sleeves: Bond all conduit and/or sleeve pathways to nearest SBB or PBB.
13. Ladder Racks/Runways: Provide continuous electrical path by installing bonding clips and jumpers. Bond each end to nearest SBB.
14. Access Floors: Bond metal parts of access floors to SBB.

3.4 IDENTIFICATION

A. Comply with Section 270553 "Identification for Communications Systems."

B. Labels must be preprinted or computer-printed type.

1. Label PBB(s) with "ts-PBB," where "ts" is telecommunications space identifier for location of PBB.
2. Label SBB(s) with "ts-SBB," where "ts" is telecommunications space identifier for location of SBB.
3. Label TBC, TBBs, and BBCs at attachment points with legend: "WARNING! COMMUNICATIONS BONDING CONDUCTOR. DO NOT REMOVE OR DISCONNECT!"

3.5 FIELD QUALITY CONTROL

A. Acceptance Testing Preparation:

1. Prior to acceptance testing, contractor shall preform initial testing and inspection checks to ensure system is in proper order. All insufficiencies or failures in system should be discovered and mitigated prior to acceptance testing procedure.
2. Coordinate with Contracting Officer and the authority having jurisdiction (AHJ) prior to system acceptance testing for any witness requirements.

B. Tests and Inspections:

1. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with calibrated torque wrench according to manufacturer's published instructions.
2. Test bonding connections of system using AC earth ground-resistance tester, taking two-point bonding measurements in each telecommunications equipment room containing PBB or SBB, using process recommended by BICSI N1. Conduct tests with facility in operation.

    a. Measure resistance between PBB and electrical service intersystem termination point. Maximum acceptable value is 100 mΩ.

        1) If measured resistance from electrical service equipment to ground exceeds 1 Ω, notify Contracting Officer and include recommendations to reduce resistance to ground.

    b. Measure resistance between SBBs and PBB. Maximum acceptable value is 100 mΩ.

3. Test for ground loop currents using digital clamp-on ammeter, with full scale not more than 10 A, displaying current in increments of 0.01 A at accuracy of plus or minus 2.0 percent.

    a. With grounding infrastructure completed and communications system electronics operating, measure current in bonding conductors connected to PBB and to SBBs. Maximum acceptable AC current level is 1.00 A.

C. Nonconforming Work:

1. Communications bonding will be considered defective if it does not pass tests and inspections.
2. Remove and replace defective units and retest.

D. Collect, assemble, and submit test and inspection reports.

3.6 PROTECTION

A. After installation, protect busbars and conductors from construction activities. Remove and replace items that are contaminated, defaced, damaged, or otherwise caused to be unfit for use prior to acceptance by Contracting Officer.

END OF SECTION 27 05 26

SECTION 27 05 28 – PATHWAYS FOR COMMUNICATIONS SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

C. Section 260533 “Raceways and Boxes for Electrical Systems” for conduits, wireways, surface raceways, boxes, enclosures, cabinets, handholes, and faceplate adapters serving electrical systems.

1.2 SUMMARY

A. Section Includes:

1. Metal conduits and fittings.
2. Nonmetallic conduits and fittings.
3. Innerduct pathways
4. Surface Metal Raceways
5. Surface Nonmetallic Raceways
6. Boxes, enclosures, and cabinets.
7. Polymer-concrete handholes and boxes for exterior underground cabling.

1.3 DEFINITIONS

A. ARC: Aluminum rigid conduit.

B. GRC: Galvanized rigid conduit.

C. IMC: Intermediate metal conduit.

D. RTRC: Reinforced thermosetting resin conduit.

E. EMT: Electrical Metallic Tubing.

F. FMC: Flexible Metallic Conduit

1.4 ACTION SUBMITTALS

A. Product data:

1. The following products shall be included:

a. Surface pathways
b. Boxes, enclosures, and cabinets.
c. Underground handholes and boxes.

B. Shop Drawings:

1. Pathway layout, showing route and components to scale, with relationship between the pathway and adjacent structural, electrical, and mechanical elements. Include the following:

a. Coordination with structure, other architectural elements, and other trade elements such as lighting, plumbing, and HVAC elements.
b. Structural connection points and connection components including clamps, brackets, hanger rods, splice-plate connectors, expansion-joint assemblies, straight lengths, and fittings.
c. Vertical and horizontal offsets and transitions.
d. Vertical elevation of pathway above the floor or bottom of ceiling structure.
e. Coordination with access panels.
f. Underground ducts, piping, and structures in location of underground enclosures and handholes.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Pathways, fittings, and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term “withstand” means “the equipment will remain in place without separation of any parts when subjected to the seismic forces specified.”

2.2 METAL CONDUITS AND FITTINGS

A. Description: Metal raceway of circular cross section with manufacturer-fabricated fittings.

B. General Requirements for Metal Conduits and Fittings:

1. Listed and labeled as defined in NFPA 70, by a nationally recognized testing laboratory, and marked for intended location and application.
2. Comply with TIA-569-D.

C. GRC: Comply with ANSI C80.1 and UL 6.

D. ARC: Comply with ANSI C80.5 and UL 6A.

E. IMC: Comply with ANSI C80.6 and UL 1242.

F. PVC-Coated Steel Conduit: PVC-coated GRC.

1. Comply with NEMA RN 1.
2. Coating Thickness: 0.040 inch (1 mm), minimum.

G. EMT: Comply with ANSI C80.3 and UL 797.

H. FMC: Aluminum

I. Fittings for Metal Conduit: Comply with NEMA FB 1 and UL 514B.

1. Conduit Fittings for Hazardous (Classified) Locations: Comply with UL 1203 and NFPA 70.
2. Fittings for EMT:

a. Material: Steel
b. Type: compression

3. Expansion Fittings: PVC or steel to match conduit type, complying with UL-467, rated for environmental conditions where installed, and including flexible external bonding jumper.
4. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch (1 mm), with overlapping sleeves protecting threaded joints.

J. Joint Compound for IMC, GRC, or ARC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.3 NONMETALLIC CONDUITS AND FITTINGS

A. Description: Nonmetallic raceway of circular section with manufacturer-fabricated fittings.

B. General Requirements for Nonmetallic Conduits and Fittings:

1. Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
2. Comply with TIA-569-D.

C. RNC: Type EPC-40-PVC & Type EPC-80-PVC, complying with NEMA TC 2 and UL 651 unless otherwise indicated.

D. Rigid HDPE: Comply with UL 651A.

E. Continuous HDPE: Comply with UL 651A.

F. RTRC: Comply with UL 2515A and NEMA TC 14.

G. Fittings: Comply with NEMA TC 3; match to conduit or tubing type and material.

H. Solvents and Adhesives: As recommended by, and in respect to, each manufacturer listed above.

2.4 INNERDUCT PATHWAYS

A. Description: Comply with UL 2024; flexible-type pathway, able to be installed into existing conduit or routed independently in plenum environment, and approved for plenum and general-use (dependent on specific product) installation unless otherwise indicated.

B. Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.

C. Comply with TIA-569-D.

2.5 SURFACE METAL RACEWAYS:

A. Galvanized steel with snap-on covers. Prime coating (ready for field painting) and painted to match surrounding color. Coordinate with contracting officer prior to color application.

2.6 SURFACE NONMETALLIC RACEWAYS:

A. Two-piece construction, manufactured of rigid PVC with texture and color selected by Contracting Officer.

2.7 BOXES, ENCLOSURES, AND CABINETS

A. Description: Enclosures for communications.

B. General Requirements for Boxes, Enclosures, and Cabinets:

1. Comply with TIA-569-D.
2. Boxes, enclosures, and cabinets installed in wet locations shall be listed and labeled as defined in NFPA 70, by an NRTL, and marked for use in wet locations.
3. Box extensions used to accommodate new building finishes shall be of same material as recessed box.

4. Device Box Dimensions: 4 inches square by 2-1/8 inches deep (100 mm square by 60 mm deep) at minimum

    a. Refer to drawing notes, details, and other such information for the use of any other size/type of device box (commonly referred to as backbox also)

5. Gangable boxes are prohibited.

C. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.

D. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, aluminum, Type FD, with gasketed cover.

E. Metal Floor Boxes:

1. Material: sheet metal
2. Type: Semi-adjustable
3. Shape: Rectangular.
4. Metal floor boxes shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

F. Nonmetallic Floor Boxes:

1. Any use of nonmetallic floor boxes is prohibited without prior approval by Contracting Officer, unless otherwise noted.

G. Conduit Bodies: A pull box that is manufactured to facilitate a change in the direction of the pathway at 90-degree increments, while also adhering to ANSI/TIA 568 and BICSI standards for cabling and pathway bend radius requirements. It is acceptable to use this device to optimize pathway routing near or at a penetration only.

1. For all locations where a conduit body may be installed, contractor shall submit the location and specification of device to be used to the Contracting Officer prior to procurment for approval.
2. Device shall have lid and gasket to protect cabling from exposure to elements and other such debris.

H. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.

I. Nonmetallic Outlet and Device Boxes: Comply with NEMA OS 2 and UL 514C.

J. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250 (Type indicated below) with continuous-hinge cover with flush latch unless otherwise indicated. Shall be manufactured of steel, finished inside and out with manufacturer's standard enamel.

1. The following NEMA 250 enclosure types shall be utilized for each scenario at a minimum, it is the contractor’s responsibility to increase the enclosure type rating as required for each installation.

    a. Interior to Telecommunication Room – Type 2 at minimum
    b. Interior to building, outside of Telecommunication Room – Type 12 at minimum

c. Exterior Environment – Type 4 at minimum
d. Any environment where corrosion, moisture, chemical, or other such hazardous or deteriorating substances may come in direct contact – Type 4x at minimum

K. Cabinets:

1. NEMA 250 (Type indicated below) galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.

a. The following NEMA 250 cabinet types shall be utilized for each scenario at a minimum, it is the contractor's responsibility to increase the cabinet type rating as required for each installation.

1) Interior to Telecommunication Room – Type 2 at minimum
2) Interior to building, outside of Telecommunication Room – Type 12 at minimum
3) Exterior Environment – Type 4 at minimum
4) Any environment where corrosion, moisture, chemical, or other such hazardous or deteriorating substances may come in direct contact – Type 4x at minimum

2. Hinged door in front cover with flush latch and concealed hinge.
3. Key latch to match panelboards.
4. Metal barriers to separate wiring of different systems and voltage.
5. Accessory feet where required for freestanding equipment.
6. Nonmetallic cabinets shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.8 POLYMER-CONCRETE HANDHOLES

A. Description: Molded of sand and aggregate; bound together with polymer resin; and reinforced with steel, fiberglass, or a combination of the two.

B. General Requirements for Polymer Concrete Handholes:

1. Boxes and handholes for use in underground systems shall be listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
2. Boxes installed in wet areas shall be listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
3. Comply with TIA-569-D.

C. Configuration: Designed for flush burial with open bottom unless otherwise indicated.

D. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure and handhole location.

1. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
2. Cover Legend: Molded lettering, ["COMMUNICATIONS"].

a. Coordinate with Contracting Officer prior to procurement of cover to ensure labeling/lettering is acceptable. Contractor shall provide alternate labeling/lettering per Contracting Officer's request.

3. Cover Rating: Tier 15 rating unless otherwise noted

E. Handholes

1. Refer to drawings and schedules for more information.
2. Installed handhole size and installation parameters shall be in accordance with BICSI standards, existing and new conditions, and future growth.

F. Source quality control for underground enclosures:

1. Handhole and Pull-Box Prototype Test: Test prototypes of handholes and boxes for compliance with SCTE 77. Strength tests shall be for specified tier ratings of products supplied.

a. Strength tests of complete boxes and covers shall be by either an independent testing agency or manufacturer. A qualified registered professional engineer shall certify tests by manufacturer.

PART 3 - EXECUTION

3.1 PATHWAY APPLICATION

A. All visible pathways, such as exposed conduit, exterior conduit, or any other such pathway that is visible, shall be painted to match the surrounding color/material. Color chosen shall be coordinated with the contracting officer for approval prior to procurement and application.

B. All pathway routing that is exposed/visible shall be routed in such a way as to minimize its visibility and visible disruption to the building/environment.

C. All surface mounted/visible/exposed pathway routing shall be coordinated with the contracting officer prior to installation.

D. Outdoors: Apply pathway products as specified below unless otherwise indicated:

1. Exposed Conduit: GRC
2. Concealed Conduit, Aboveground: GRC
3. Underground Conduit, direct buried in non-traffic area: Type EPC-40-PVC
4. Underground Conduit, direct buried in traffic area: Type EPC-80-PVC
5. Underground Conduit, concrete encased in traffic area: Type EPC-80-PVC
6. Boxes and Enclosures, Aboveground: NEMA 250, Type 4 and/or Type 4X as previously indicated.

E. Indoors: Apply pathway products as specified below unless otherwise indicated:

1. Exposed, Not Subject to Physical Damage: EMT
2. Exposed, Not Subject to Severe Physical Damage: EMT
3. Exposed and Subject to Severe Physical Damage: GRC. Pathway locations include the following:

    a. Loading dock
    b. Corridors used for traffic of mechanized carts, forklifts, and pallet-handling units

4. Concealed in Ceilings and Interior Walls and Partitions: EMT
5. Damp, Wet, and/or Caustic Locations (Natatorium, high humidity areas, chemical storage, etc.): GRC

    a. This does not include submerged areas and/or pathways.

6. Pathways for Optical-Fiber or Communications Cable in Spaces Used for Environmental Air: EMT
7. Pathways for Optical-Fiber or Communications-Cable Risers in Vertical Shafts: EMT
8. Pathways for Concealed General-Purpose Distribution of Optical-Fiber or Communications Cable: EMT
9. Boxes and Enclosures: NEMA 250, Type 2 and/or Type 12.

    a. Type 4X, stainless steel units in institutional and commercial kitchens and damp or wet locations.

F. Minimum Pathway Size: 1-inch (25-mm) trade size for copper and aluminum cables, and 1 inch (25 mm) for optical-fiber cables.

G. Pathway Fittings: Compatible with pathways and suitable for use and location.

1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with this type of conduit. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer and apply in thickness and number of coats recommended by manufacturer.
3. EMT: Use compression, steel fittings. Comply with NEMA FB 2.10.
4. All conduit fittings that facilitate a change in pathway direction (such as a 90-degree bend) shall be:

    a. For conduits with an internal diameter of 2 inches or less, a sweeping bend with a bend radius greater than 6 times the pathway diameter, or
    b. For conduits with an internal diameter greater than 2 inches, a sweeping bend with a bend radius greater than 10 times the pathway diameter, or

H. Do not install aluminum conduits, boxes, or fittings in contact with concrete or earth.

I. Install surface pathways only where indicated on Drawings.

J. Do not install nonmetallic conduit where ambient temperature exceeds 120 deg F (49 deg C).

K. Pathways indicated on the T-sheets shall be dedicated for use by the associated systems only. No other trade may route through, utilize, or otherwise be attached to these pathways. In addition, refer to specification 271313 and/or 271513 for the separation requirements of pathway when an EMI source is present.

3.2 INSTALLATION

A. Comply with the following standards for installation requirements except where requirements on Drawings or in this Section are stricter:

1. NECA 1.
2. NECA/BICSI 568.
3. TIA-569-D.
4. NECA 101
5. NECA 102.
6. NECA 105.
7. NECA 111.

B. Comply with NFPA 70 limitations for types of pathways allowed in specific occupancies and number of floors.

C. Comply with requirements in Section 078413 “Penetration Firestopping” for firestopping materials and installation for penetrations through fire-rated walls, ceilings, and assemblies.

D. Comply with requirements in Section 270529 “Hangers and Supports for Communications Systems” for hangers and supports.

E. Comply with requirements in Section 270544 “Sleeves and Sleeve Seals for Communications Pathways and Cabling” for sleeves and sleeve seals for communications.

F. Keep pathways at least 6 inches (150 mm) away from parallel runs of flues and steam or hot-water pipes. Install horizontal pathway runs above water and steam piping.

G. Complete pathway installation before starting conductor installation.

H. Arrange stub-ups so curved portions of bends are not visible above finished slab.

I. Install no more than the equivalent of two 90-degree bends in any pathway run. Support within 12 inches (300 mm) of changes in direction. Utilize long radius ells for all optical-fiber cables.

J. Conceal rigid conduit within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.

K. Support conduit within 12 inches (300 mm) of enclosures to which attached.

L. Pathways Embedded in Slabs:

1. Run conduit larger than 1-inch (27-mm) trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support. Secure pathways to reinforcement at maximum 10-foot (3-m) intervals.

2. Arrange pathways to cross building expansion joints at right angles with expansion fittings. Comply with requirements for expansion joints specified in this article.
3. Arrange pathways to keep a minimum of 1 inch (25 mm) of concrete cover in all directions.
4. Do not embed threadless fittings in concrete unless specifically approved by Contracting Officer for each specific location.

M. Stub-ups to Above Recessed Ceilings:

1. Use EMT for pathways, unless otherwise noted
2. Use a conduit bushing or insulated fitting to terminate stub-ups not terminated in hubs or in an enclosure.

N. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of pathway and fittings before making up joints. Follow compound manufacturer's written instructions.

O. Coat field-cut threads on PVC-coated pathway with a corrosion-preventing conductive compound prior to assembly.

P. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install insulated bushings on conduits terminated with locknuts.

Q. Install pathways square to the enclosure and terminate at enclosures with locknuts. Install locknuts hand tight plus one additional quarter-turn.

R. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure, to assure a continuous ground path.

S. Cut conduit perpendicular to the length. For conduits of 2-inch (50-mm) trade size and larger, use roll cutter or a guide to ensure cut is straight and perpendicular to the length.

T. Install pull wires in empty pathways. Use polypropylene or monofilament plastic line with not less than 200-lb (90-kg) tensile strength. Leave at least 12 inches (300 mm) of slack at each end of pull wire. Secure pull wire, so it cannot fall into conduit. Cap pathways designated as spare alongside pathways in use.

U. Surface Pathways:

1. Install surface pathway for surface telecommunications outlet boxes only where indicated on Drawings.
2. Install surface pathway with a minimum 2-inch (50-mm) radius control at bend points.
3. Secure surface pathway with screws or other anchor-type devices at intervals not exceeding 48 inches (1200 mm) and with no less than two supports per straight pathway section. Support surface pathway according to manufacturer's written instructions. Tape and glue are not acceptable support methods.

V. Pathways for Optical-Fiber and Communications Cable: Install pathways, metal and nonmetallic, rigid and flexible, as follows:

1. 1-Inch (25-mm) Trade Size and Larger: Install pathways in maximum lengths of 75 feet (23 m).
2. Install with a maximum of two 90-degree bends or equivalent for each length of pathway unless Drawings show stricter requirements. Separate lengths with pull or junction boxes or terminations at distribution frames or cabinets where necessary to comply with these requirements.

W. Install pathway-sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed pathways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install pathway-sealing fittings according to NFPA 70.

X. Install devices to seal pathway interiors at accessible locations. Locate seals, so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all pathways at the following points:

1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
2. Where an underground service pathway enters a building or structure.
3. Where otherwise required by NFPA 70.

Y. Comply with manufacturer's written instructions for solvent welding PVC conduit and fittings.

Z. Expansion-Joint Fittings:

1. Install in each run of aboveground RNC that is located where environmental temperature change may exceed 30 deg F (17 deg C), and that has straight-run length that exceeds 25 feet (7.6 m). Install in each run of aboveground RMC and EMT that is located where environmental temperature change may exceed 100 deg F (55 deg C), and that has straight-run length that exceeds 100 feet (30 m).
2. Install type and quantity of fittings that accommodate temperature change listed for each of the following locations:

    a. Outdoor Locations: 125 deg F (70 deg C) temperature change.
    b. Indoor Spaces Connected with Outdoors without Physical Separation: 125 deg F (70 deg C) temperature change.
    c. Attics: 135 deg F (75 deg C) temperature change.

3. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per deg F (0.06 mm per meter of length of straight run per deg C) of temperature change for PVC conduits. Install fitting(s) that provide expansion and contraction for at least 0.000078 inch per foot of length of straight run per deg F (0.0115 mm per meter of length of straight run per deg C) of temperature change for metal conduits.
4. Install expansion fittings at all locations where conduits cross building or structure expansion joints.
5. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at time of installation. Install conduit supports to allow for expansion movement.

AA. Hooks:

1. Size to allow a minimum of 25 percent future capacity without exceeding design capacity limits.
2. Shall be supported by dedicated support wires. Do not use ceiling grid support wire or support rods.
3. Hook spacing shall allow no more than 6 inches (150 mm) of slack.
4. The lowest point of the cables shall be no less than 6 inches (150 mm) adjacent to ceilings, mechanical ductwork and fittings, luminaires, power conduits, power and telecommunications outlets, and other electrical and communications equipment.
5. Space hooks no more than 5 feet (1.5 m) and no less than 2 feet (0.6 m) o.c., with distances between hooks alternating so not to allow electro-magnetic induction formation from consistent spacing.
6. Provide a hook at each change in direction.

BB. Mount boxes at heights indicated on Drawings. If mounting heights of boxes are not individually indicated, give priority to ADA requirements. Install boxes with height measured to center of box unless otherwise indicated.

CC. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block and install box flush with surface of wall. Prepare block surface to provide a flat surface for a raintight connection between box and cover plate or supported equipment and box.

DD. Horizontally separate boxes mounted on opposite sides of walls, so they are not in the same vertical channel.

EE. Support boxes of three gangs or more from more than one side by spanning two framing members or mounting on brackets specifically designed for the purpose.

FF. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.

GG. Set metal floor boxes level and flush with finished floor surface.

HH. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

3.3 INSTALLATION OF UNDERGROUND CONDUIT

A. Direct-Buried Conduit:

1. Excavate trench bottom to provide firm and uniform support for conduit. Prepare trench bottom as specified in Section 312000 “Earth Moving” for pipe of less than 6 inches (150 mm) in nominal diameter.
2. Install backfill as specified in Section 312000 “Earth Moving.”
3. After installing conduit, backfill and compact. Start at tie-in point, and work toward end of conduit run, leaving conduit at end of run free to move with expansion and contraction as temperature changes during this process. Firmly hand tamp backfill around conduit to provide maximum supporting strength. After placing controlled backfill to within 12 inches (300 mm) of finished grade, make final conduit connection at end of run and

complete backfilling with normal compaction as specified in Section 312000 “Earth Moving.”

4. Install manufactured elbows for stub-ups at poles and equipment and at building entrances through floor unless otherwise indicated. Encase elbows for stub-up throughout length of elbow.

    a. Material of elbow shall be consistent with the pathway installed.

5. Underground Warning Tape: Comply with requirements in Section 270553 “Identification for Communications Systems.”
6. Coordinate with civil, structural, and any other trades with underground routing; coordinate with Contracting Officer for trenching locations and project specific requirements.

3.4 INSTALLATION OF UNDERGROUND HANDHOLES AND BOXES

A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.

B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch (12.5-mm) sieve to No. 4 (4.75-mm) sieve and compacted to same density as adjacent undisturbed earth.

C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 1 inch (25 mm) above finished grade.

D. Install handholes with bottom below frost line, at least 48 inches below grade.

E. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables, but short enough to preserve adequate working clearances in enclosure.

F. Field cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.5 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR COMMUNICATIONS PENETRATIONS

A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 270544 “Sleeves and Sleeve Seals for Communications Pathways and Cabling.”

3.6 FIRESTOPPING

A. Install firestopping at penetrations of fire-rated floor and wall assemblies. Comply with requirements in Section 078413 “Penetration Firestopping.”

3.7 PROTECTION

A. Protect coatings, finishes, and cabinets from damage or deterioration.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 27 05 28

SECTION 27 05 36 – CABLE TRAYS FOR COMMUNICATIONS SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

C. Section 260536 “Cable Trays for Electrical Systems” for cable trays and accessories serving electrical systems.

1.2 SUMMARY

A. Section Includes:

1. Ladder Runway and Ladder Rack Cable Tray.
2. Cable tray accessories.
3. Warning signs.

1.3 ACTION SUBMITTALS

A. Product Data:

1. The following products shall be included:

a. Ladder Runway and Ladder Rack Cable Tray
b. Cable tray accessories

B. Shop Drawings:

1. Cable tray layout, showing cable tray route to scale, with relationship between the tray and adjacent structural, electrical, and mechanical elements. Include the following:

a. Coordination with structure, other architectural elements, and other trade elements such as lighting, plumbing, and HVAC elements.
b. Structural connection points and connection components including clamps, brackets, hanger rods, splice-plate connectors, expansion-joint assemblies, straight lengths, and fittings.
c. Vertical and horizontal offsets and transitions.
d. Clearances for access above and to sides of cable trays.

e. Vertical elevation of cable trays above the floor or bottom of ceiling structure.
f. Load calculations to show dead and live loads as not exceeding manufacturer's rating for tray and its support elements.

1.4 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

B. Field Quality Control Reports.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Cable trays and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term “withstand” means “the cable trays will remain in place without separation of any parts when subjected to the seismic forces specified.”

B. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes in cable tray installed outdoors.

1. Temperature Change: 120 deg F (67 deg C) ambient temperature.

2.2 GENERAL REQUIREMENTS FOR CABLE TRAYS

A. Cable Trays and Accessories: Identified as defined in NFPA 70 and marked for intended location, application, and grounding.

1. Source Limitations: Obtain cable trays and components from single manufacturer.

B. Sizes and Configurations: See the equipment schedule on drawings for specific requirements for types, materials, sizes, and configurations.

C. Structural Performance: See articles for individual cable tray types for specific values for the following parameters:

1. Uniform Load Distribution: Capable of supporting a uniformly distributed load on the indicated support span when supported as a simple span and tested according to NEMA VE 1.
2. Concentrated Load: A load applied at midpoint of span and centerline of tray.
3. Load and Safety Factors: Applicable to both side rails and rung capacities.

2.3 LADDER RUNWAY AND LADDER RACK CABLE TRAY

A. Description:

1. Configuration: Two longitudinal side rails with transverse rungs swaged or welded to side rails, complying with NEMA VE 1.
2. Width: 12 inches (300 mm), unless otherwise indicated on Drawings.
3. Straight Section Lengths: 10 feet (3.0 m), except where shorter lengths are required to facilitate tray assembly.
4. Rung Spacing: 9 inches (225 mm) on center.
5. Minimum Cable-Bearing Surface for Rungs: 7/8-inch (22-mm) width with radius edges.
6. No portion of the rungs shall protrude below the bottom plane of side rails.
7. Structural Performance of Each Rung: Capable of supporting a maximum cable load, with a safety factor of 1.5, plus a 200-lb (90-kg) concentrated load, when tested according to NEMA VE 1.
8. Class Designation: Comply with NEMA VE 1, Class 12A.
9. Splicing Assemblies: Bolted type using serrated flange locknuts.
10. Splice-Plate Capacity: Splices located within support span shall not diminish rated loading capacity of cable tray.

B. Materials and Finishes:

1. Aluminum:

   a. Materials: Alloy 6063-T6 according to ANSI H35.1/H 35.1M for extruded components, and Alloy 5052-H32 according to ANSI H35.1/H 35.1M for fabricated parts.
   b. Hardware: Chromium-zinc-plated steel, ASTM F1136

2. Finish shall be powder coated in black, unless otherwise noted.

2.4 CABLE TRAY ACCESSORIES

A. Fittings: Tees, crosses, risers, elbows, and other fittings as indicated, of same materials and finishes as cable tray.

B. Barrier Strips: Same materials and finishes as for cable tray.

C. Cable tray supports and connectors, including bonding jumpers, as recommended by cable tray manufacturer.

2.5 WARNING SIGNS

A. Comply with requirements for identification in Section 270553 “Identification for Communications Systems.”

B. Lettering: 1-1/2-inch- (40-mm-) high, black letters on yellow background with legend “Warning! Not To Be Used as Walkway, Ladder, or Support for Ladders or Personnel.”

2.6 SOURCE QUALITY CONTROL

A. Testing: Test and inspect cable trays according to NEMA VE 1.

PART 3 - EXECUTION

3.1 CABLE TRAY INSTALLATION

A. Install cable trays according to NEMA VE 2.

B. Install cable trays as a complete system, including fasteners, hold-down clips, support systems, barrier strips, adjustable horizontal and vertical splice plates, elbows, reducers, tees, crosses, cable dropouts, adapters, covers, and bonding.

C. Install cable trays so that the tray is accessible for cable installation and all splices are accessible for inspection and adjustment.

D. Remove burrs and sharp edges from cable trays.

E. Join aluminum cable tray with splice plates; use four square neck-carriage bolts and locknuts.

F. Fasten cable tray supports to building structure and install seismic restraints.

G. Design fasteners and supports to carry cable tray, the cables, and a concentrated load of 200 lb (90 kg). Comply with requirements in Section 270529 “Hangers and Supports for Communications Systems.

H. Place supports so that spans do not exceed maximum spans on schedules and provide clearances shown on Drawings. Install intermediate supports when cable weight exceeds the load-carrying capacity of the tray rungs.

I. Construct supports from channel members, threaded rods, and other appurtenances furnished by cable tray manufacturer. Arrange supports in trapeze or wall-bracket form as required by application.

J. Locate and install supports according to NEMA VE 2. Do not install more than one cable tray splice between supports.

K. For Ladder Runway connected to equipment, ladder runway shall be supported independent of connections to equipment. Do not carry weight of ladder runway on equipment enclosure.

L. Install expansion connectors where cable trays cross building expansion joints and in cable tray runs that exceed dimensions recommended in NEMA VE 2. Space connectors and set gaps according to applicable standard.

M. Make changes in direction and elevation using manufacturer's recommended fittings. For any transitions that are vertical to a new pathway or cable tray, a cable tray dropout or other similar fitting type shall be utilized.

N. Make cable tray connections using manufacturer's recommended fittings.

O. Seal penetrations through fire and smoke barriers. Comply with requirements in Section 078413 “Penetration Firestopping.”

P. Install cable trays with enough workspace to permit access for installing cables.

Q. Install warning signs in visible locations on or near cable trays after cable tray installation.

3.2 CABLE TRAY GROUNDING

A. Ground cable trays according to NFPA 70 unless additional grounding is specified. Comply with requirements in Section 270526 “Grounding and Bonding for Communications Systems.”

B. Cable trays shall be bonded together with splice plates listed for grounding purposes or with listed bonding jumpers.

C. Cable trays with single-conductor power conductors shall be bonded together with a grounding conductor run in the tray along with the power conductors and bonded to the tray at 72-inch (1800-mm) intervals. The grounding conductor shall be sized according to NFPA 70, Article 250.122, “Size of Equipment Grounding Conductors,” and Article 392, “Cable Trays.”

D. When using powder-coat painted cable tray as a grounding conductor, completely remove coating at all splice contact points or ground connector attachment. After completing splice-to-grounding bolt attachment, repair the coated surfaces with coating materials recommended by cable tray manufacturer.

E. Bond cable trays to power source for cables contained within with bonding conductors sized according to NFPA 70, Article 250.122, “Size of Equipment Grounding Conductors.”

3.3 CABLE INSTALLATION

A. No cable other than network/voice cabling for designated division 27 systems shall utilize the cable tray for communication systems, unless specifically noted. Coordinate with other trades to ensure separate routing methods are utilized for other system cabling.

1. All cabling routed on, through, or via cable tray shall be coordinated with division 27 trade partner to minimize cable burns, kinks, and cable tray capacity/loading.
2. It is acceptable to provide and install bridal rings attached to cable tray for dedicated use by a division 27 system. Installation of bridal rings shall not conflict with cable tray clearances, with architectural elements, or with other trade’s devices/equipment.

B. Install cables only when each cable tray run has been completed and inspected.

C. Fasten cables on horizontal runs with cable clamps or cable ties according to NEMA VE 2. Tighten clamps only enough to secure the cable, without indenting the cable jacket. Install cable ties with a tool that includes an automatic pressure-limiting device.

D. Fasten cables on vertical runs to cable trays every 18 inches (450 mm).

E. Fasten and support cables that pass from one cable tray to another or drop from cable trays to equipment enclosures. Fasten cables to the cable tray at the point of exit and support cables independent of the enclosure. The cable length between cable trays or between cable tray and enclosure shall be no more than 72 inches (1800 mm).

F. Tie MI cables down every 36 inches (900 mm) where required to provide a 2-hour fire rating and every 72 inches (1800 mm) elsewhere.

G. In existing construction, remove inactive or dead cables from cable trays.

3.4 CONNECTIONS

A. Remove paint from all connection points before making connections. Repair paint after the connections are completed.

B. Connect pathways to cable trays according to requirements in NEMA VE 2 and NEMA FG 1.

3.5 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. After installing cable trays and after electrical circuitry has been energized, survey for compliance with requirements.
2. Visually inspect cable insulation for damage. Correct sharp corners, protuberances in cable trays, vibrations, and thermal expansion and contraction conditions, which may cause or have caused damage.
3. Verify that the number, size, and voltage of cables in cable trays do not exceed that permitted by NFPA 70. Verify that communications or data-processing circuits are separated from power circuits by barriers or are installed in separate cable trays.
4. Verify that there are no intruding items such as pipes, hangers, or other equipment in the cable tray.
5. Remove dust deposits, industrial process materials, trash of any description, and any blockage of tray ventilation.
6. Visually inspect each cable tray joint and each ground connection for mechanical continuity. Check bolted connections between sections for corrosion. Clean and retorque in suspect areas.
7. Check for improperly sized or installed bonding jumpers.
8. Check for missing, incorrect, or damaged bolts, bolt heads, or nuts. When found, replace with specified hardware.
9. Perform visual and mechanical checks for adequacy of cable tray grounding; verify that all takeoff raceways are bonded to cable trays. Test entire cable tray system for continuity. Maximum allowable resistance is 1 ohm.

B. Prepare test and inspection reports.

3.6 PROTECTION

A. Protect installed cable trays and cables.

1. Install temporary protection for cables in open trays to safeguard exposed cables against falling objects or debris during construction. Temporary protection for cables and cable tray can be constructed of wood or metal materials and shall remain in place until the risk of damage is over.
2. Repair damage to galvanized finishes with zinc-rich paint recommended by cable tray manufacturer.
3. Repair damage to paint finishes with matching touchup coating recommended by cable tray manufacturer.

END OF SECTION 27 05 36

SECTION 27 05 44 – SLEEVES AND SLEEVE SEALS FOR COMMUNICATIONS PATHWAYS AND CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Manufactured Sleeve Seal Systems.
2. Conduit or Field Fabricated sleeves.
3. Sealants and Devices

1.3 ACTION SUBMITTALS

A. Product data:

1. The following products shall be included:

a. Manufactured Sleeve Seal Systems
b. Conduit or Field Fabricated sleeves
c. Sealant and Devices

1.4 INFORMATION SUBMITTALS

A. Coordination Drawings

1. Indicating all penetration locations and penetrating component.
2. Details indicating each penetration type, associated components, and associated structural coordination.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data

PART 2 - PRODUCTS

2.1 MANUFACTURED SLEEVE SEAL SYSTEM

A. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and cable. Shall be an engineered, re-enterable cable pathway that is specifically manufactured for facilitating a floor, ceiling, or wall penetration. The device shall match or exceed the fire-rating of the penetrated material, at minimum.

1. Sealing Elements: Self-sealing intumescent material

B. Pathway shall be able to be retrofitted to existing cable and penetration and shall be able to be removed from existing cabling, without cutting or splicing existing cables.

C. It is acceptable to utilize grommet type sleeves (typically allowing a single cable or small cable bundle through penetration), round type sleeves, and square type sleeves. Number of cables per rated penetration shall be in accordance with manufacturer.

2.2 CONDUIT OR FIELD FABRICATED SLEEVES

A. Pipe Sleeves, PVC:

1. Description: ASTM D1785, Schedule 40

B. Sheet Metal Sleeves, Galvanized Steel, Round:

1. Description: Galvanized-steel sheet; thickness not less than 0.0239-inch (0.6-mm); round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.

2.3 SEALANTS AND DEVICES

A. Description: Products and material that have been tested for specific fire resistance, smoke, or acoustical rated construction, conforming to penetrating item type, filling annular space between cable and pathway, and filling the space between pathway and penetration.

1. Sealants – Single component latex formulations that, upon cure, do not re-emulsify during exposure to moisture.
2. Putty – Intumescent, non-hardening, water-resistant putties containing no solvents, inorganic fibers or silicone compounds.
3. Pillows – Re-enterable, non-curing, mineral fiber core encapsulated on six sides with intumescent coating contained in a flame retardant poly bag.

PART 3 - EXECUTION

3.1 INSTALLATION OF SLEEVES, SEALANTS, AND DEVICES THROUGH NON-FIRE-RATED BARRIERS

A. Comply with NECA 1.

B. Sleeves for Conduits Penetrating Above-Grade, Non-Fire-Rated, Concrete and Masonry-Unit Floors and Walls:

1. Interior Penetrations of Non-Fire-Rated Walls and Floors:

a. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall or floor so no voids remain. Tool exposed surfaces smooth; protect material while curing.
b. Seal annular space between sleeve and pathway or cable, using joint sealant appropriate for size, depth, and location of joint.

2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
3. Size pipe sleeves to provide 1/4-inch (6.4-mm) annular clear space between sleeve and pathway or cable, unless sleeve seal system is to be installed or seismic criteria require different clearance.
4. Install sleeves for wall penetrations. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.
5. Install sleeves for floor penetrations. Extend sleeves installed in floors 6 inches (50 mm) above finished floor level. Install sleeves during erection of floors.

C. Sleeves for Conduits Penetrating Non-Fire-Rated Wall Assemblies:

1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.
2. Seal space outside of sleeves with approved joint compound for wall assemblies.

D. Roof-Penetration Sleeves: Seal penetration of individual pathways and cables with flexible boot-type flashing units applied in coordination with roofing work.

E. Aboveground, Exterior-Wall Penetrations: Seal penetrations using steel pipe sleeves and mechanical sleeve seal systems. Size sleeves to allow for 1-inch (25-mm) annular clear space between pipe and sleeve for installing mechanical sleeve seals.

F. Underground, Exterior-Wall and Floor Penetrations:

1. For new construction, install steel pipe sleeves with integral waterstops. Size sleeves to allow for 1-inch (25-mm) annular clear space between pathway or cable and sleeve for installing sleeve seal system. Install sleeve during construction of floor or wall.
2. For existing wall, install steel pipe sleeves through core drill penetration. Size sleeves to allow for 1-inch (25-mm) annular clear space between pathway or cable and sleeve for installing sleeve seal system. Grout sleeve into wall or floor opening.

3.2 INSTALLATION OF SLEEVES, SEALANTS, AND DEVICES THROUGH RATED BARRIERS

A. Install sleeves in existing walls without compromising structural integrity of walls. Do not cut structural elements without reinforcing the wall to maintain the designed weight bearing and wall stiffness.

B. Install conduits and cable with no crossings within the sleeve.

C. Fill opening around conduits and cables with sealant without leaving voids.

D. Install type and number of sealing elements recommended by manufacturer for pathway or cable material and size. Position pathway or cable in center of sleeve. Assemble mechanical sleeve seals and install in annular space between pathway or cable and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

END OF SECTION 27 05 44

SECTION 27 05 53 – IDENTIFICATION FOR COMMUNICATIONS SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Labels
2. Underground-line warning tape.
3. Cable straps
4. Miscellaneous identification products.

1.3 ACTION SUBMITTALS

A. Product data:

1. The following products shall be included:

a. Labels
b. Underground-line warning Tape

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Comply with NFPA 70 and TIA 606-B.

B. Comply with ANSI Z535.4 for safety signs and labels.

C. Adhesive-attached labeling materials, including label stocks, laminating adhesives, and inks used by label printers, shall comply with UL 969.

D. Thermal Movements: Allow for thermal movements from ambient and surface temperature changes.

1. Temperature Change: 120 deg F (67 deg C), ambient.

2.2 COLOR AND LEGEND REQUIREMENTS

A. Equipment Identification Labels:

1. Black letters on a white field.

2.3 LABELS

A. Vinyl Wraparound Labels: Preprinted, flexible labels laminated with a clear, weather- and chemical-resistant coating and matching wraparound clear adhesive tape for securing label ends.

B. Self-Adhesive Labels: Vinyl, thermal, transfer-printed, 3-mil- (0.08-mm-) thick, multicolor, weather- and UV-resistant, pressure-sensitive adhesive labels, configured for intended use and location.

1. Minimum Nominal Size:

a. 1-1/2 by 6 inches (37 by 150 mm) for raceway and conductors.
b. 3-1/2 by 5 inches (76 by 127 mm) for equipment.
c. As required by authorities having jurisdiction.

2.4 UNDERGROUND-LINE WARNING TAPE

A. Tape:

1. Recommended by manufacturer for the method of installation and suitable to identify and locate underground communications utility lines.
2. Printing on tape shall be permanent and shall not be damaged by burial operations.
3. Tape material and ink shall be chemically inert and not subject to degradation when exposed to acids, alkalis, and other destructive substances commonly found in soils.

B. Color and Printing:

1. Comply with ANSI Z535.1, ANSI Z535.2, ANSI Z535.3, and ANSI Z535.4.
2. Inscriptions for Orange-Colored Tapes: “TELEPHONE CABLE, CATV CABLE, COMMUNICATIONS CABLE, OPTICAL-FIBER CABLE”

C. Tag, Detectable:

1. Detectable three-layer laminate, consisting of a printed pigmented polyolefin film, a solid aluminum-foil core, and a clear protective film that allows inspection of the continuity of the conductive core; bright colored, continuous-printed on one side with the inscription of the utility, compounded for direct-burial service.

2. Width: 3 inches (75 mm).
3. Overall Thickness: 5 mils (0.125 mm).
4. Foil Core Thickness: 0.35 mil (0.00889 mm).
5. Weight: 28 lb/1000 sq. ft. (13.7 kg/100 sq. m).
6. Tensile according to ASTM D882: 70 lbf (311.3 N) and 4600 psi (31.7 MPa).

2.5 CABLE STRAP

A. Plenum-Rated Velcro Ties/Straps

1. Hook & Loop Strap, Cinch, and Tape style are acceptable.
2. Minimum Width: 3/4 inch (19.1 mm).
3. RoHS Compliant
4. Temperature Range: 0 to plus 220 deg F (Minus 18 to plus 104 deg C).
5. Color: Black.

2.6 MISCELLANEOUS IDENTIFICATION PRODUCTS

A. Paint: Comply with requirements in painting Sections for paint materials and application requirements. Retain paint system applicable for surface material and location (exterior or interior).

B. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 PREPARATION

A. Self-Adhesive Identification Products: Before applying communications identification products, clean substrates of substances that could impair bond, using materials and methods recommended by manufacturer of identification product.

3.2 INSTALLATION

A. Verify and coordinate identification names, abbreviations, colors, and other features with requirements in other Sections requiring identification applications, Drawings, Shop Drawings, manufacturer's wiring diagrams, and operation and maintenance manual. Use consistent designations throughout Project.

B. Install identifying devices before installing acoustical ceilings and similar concealment.

C. Verify identity of each item before installing identification products.

D. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and operation and maintenance manual.

E. Apply identification devices to surfaces that require finish after completing finish work.

F. Install signs with approved legend to facilitate proper identification, operation, and maintenance of communications systems and connected items.

G. Elevated Components: Increase sizes of labels, signs, and letters to those appropriate for viewing from the floor.

H. Vinyl Wraparound Labels:

1. Secure tight to surface of raceway or cable at a location with high visibility and accessibility.
2. Attach labels that are not self-adhesive type with clear vinyl tape, with adhesive appropriate to the location and substrate.
3. Provide label 6 inches (150 mm) from cable end.

I. Self-Adhesive Labels:

1. On each item, install unique designation label that is consistent with wiring diagrams, schedules, and operation and maintenance manual.
2. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on 1-1/2-inch- (38-mm-) high label; where two lines of text are required, use labels 2 inches (50 mm) high.

J. Underground-Line Warning Tape:

1. During backfilling of trenches, install continuous underground-line warning tape directly above cable or raceway at 6 to 8 inches (150 to 200 mm) below finished grade. Use multiple tapes where width of multiple lines installed in a common trench or concrete envelope exceeds 16 inches (400 mm) overall.
2. Install underground-line warning tape for direct-buried cables and cables in raceways.

K. Cable Straps: General purpose, except as listed below:

1. In Spaces Handling Environmental Air: Plenum rated.

3.3 IDENTIFICATION SCHEDULE

A. Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment. Install access doors or panels to provide view of identifying devices.

B. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, pull points, and locations with high visibility. Identify by system and circuit designation.

C. Accessible Fittings for Raceways and Cables within Buildings: Identify covers of each junction and pull box with self-adhesive labels containing wiring system legend.

1. System legends shall be as follows:

a. Network/Voice

b. Audiovisual
c. Security

1) For Surveillance, Access Control, and/or Intrusion

D. Faceplates/Surface Mount Box:

1. Label individual faceplate with self-adhesive labels. Place label at top of faceplate. Each faceplate shall be labeled with its individual, sequential designation, numbered clockwise when entering room from primary egress, composed of the following, in the format shown:

a. [Cable Origination Room #] : [Outlet #]

2. Label individual ports with self-adhesive labels. Place label above each port. Each port label shall be composed of the following, in the format shown:

a. [Outlet Port #]

E. Equipment Room Labeling:

1. Racks, Frames, and Enclosures: Identify front and rear of each with self-adhesive labels. Each shall be labeled sequentially, beginning with the device closest to the wall.

a. [Rack #]

2. Patch Panels: Label each individual patch panel in room sequentially, starting at the top of rack and working down, with self-adhesive labels. Patch panel numbering shall restart at one at each rack. Each patch panel label shall be composed of the following, in the format shown:

a. [Patch Panel #]

3. Patch Panel Port: Label each port with a self-adhesive label, place label directly above port in designated area. Each port shall be composed of the following, in the format shown:

a. [Cable Termination Room #] : [Outlet #] : [Outlet Port #]

F. Backbone Cables:

1. Label each cable within 6 inches of strand/conductor breakout with a vinyl-wraparound label indicating the location of the far or other end of the backbone cable. Each label shall be composed of the following, in the format shown:

a. [Room Number of other end of cable] : [Patching Device] : [Patching Device #]
b. Patching Device Legend:

1) Fiber Patch Panel
2) Copper Patch Panel
3) 110-Block
4) OTHER

G. Horizontal Cables: Label each cable within 6 inches of termination with a vinyl-wraparound label. Label shall match the port label as referenced above.

1. For cables terminated at patch panel:

a. Refer to label specified at Patch Panel Port

2. For cables terminated at faceplate/surface mount box:

a. Refer to label specified at faceplate/surface mount box

H. Locations of Underground Lines: Underground-line warning tape for copper, coaxial, hybrid copper/fiber, and optical-fiber cable.

I. Instructional Signs: Self-adhesive labels.

J. Warning Labels for Indoor Cabinets, Boxes, and Enclosures: Laminated-acrylic or melamine-plastic sign

1. Apply to exterior of door, cover, or other access.

END OF SECTION 27 05 53

SECTION 27 11 16 – COMMUNICATIONS RACKS, FRAMES, AND ENCLOSURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

C. Section 270528 “Pathways for Communications Systems” for materials and installation requirements for underground pathways.

1.2 SUMMARY

A. Section Includes:

1. Backboards.
2. Equipment Racks.
3. Power Distribution Unit (PDU)
4. Grounding.
5. Labeling.

1.3 DEFINITIONS

A. Access Provider: An operator that provides a circuit path or facility between the service provider and user. An access provider can also be a service provider.

B. BICSI: Building Industry Consulting Service International.

C. LAN: Local area network.

D. RCDD: Registered communications distribution designer.

E. Service Provider: The operator of a telecommunications transmission service delivered through access provider facilities.

F. TGB: Telecommunications grounding bus bar.

G. TMGB: Telecommunications main grounding bus bar.

1.4 ACTION SUBMITTALS

A. Product data:

1. The following products shall be included:

a. Equipment Racks.
b. Power Distribution Unit (PDU)
c. Grounding.

B. Shop Drawings:

1. Telecommunications Room/Space layout.
2. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Equipment Racks and Cabinets: Include workspace requirements and access for cable connections.
4. Grounding: Indicate location of PBB/SBB and its mounting detail showing standoff insulators and wall-mounting brackets.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

B. Installer Qualification Data:

1. Qualifications per specification 270010 “Supplemental Requirements for Communications”.
2. RCDD certification for Installer, installation supervisor, and field inspector.
3. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data.

B. Warranty Documentation.

C. Identification Schedule.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: Cabling installer must have personnel certified by BICSI on staff.

1. Layout Responsibility: Preparation of Shop Drawings shall be under direct supervision of RCDD.

2. Installation Supervision: Installation shall be under direct supervision of Technician, who shall be present at all times when Work of this Section is performed at Project site.
3. Field Inspector: Currently registered by BICSI as RCDD to perform on-site inspection.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Equipment shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term “withstand” means “the unit will remain in place without separation of any parts when subjected to the seismic forces specified.”

B. UL listed.

C. RoHS compliant.

D. Compliant with requirements of the Payment Card Industry Data Security Standard.

2.2 BACKBOARDS

A. Backboards: Plywood, fire-retardant treated, 3/4 by 48 by 96 inches (19 by 1220 by 2440 mm).

1. Plywood that is not fire-retardant treated may be utilized under the following condition:

a. Non-treated plywood is painted with a minimum of 2-coats of intumescent paint to ensure a 2-hour fire-rating of painted plywood.

B. Backboard Paint: Plywood shall be painted white, unless otherwise noted. Contractor shall coordinate with Contracting Officer prior to painting for final color and paint type determination.

1. For fire-retardant treated plywood, ensure that the fire-retardant label remain visible after all paint is applied and plywood is installed.

C. Comply with requirements for plywood backing panels specified in Section 061000 “Rough Carpentry.”

2.3 EQUIPMENT RACKS

A. Description: Two- and four- post racks with threaded rails designed for mounting telecommunications equipment. Width is compatible with EIA standards. Modular units designed for telecommunications terminal support and coordinated with dimensions of units to be supported.

1. For equipment details and specific information, refer to equipment schedules on drawings.

2. The below information is applicable at a minimum, unless otherwise noted:

    a. Height: As indicated on Drawings
    b. Width: Compatible with 19 inch EIA standard
    c. Depth: A minimum of 2 foot (24 inches) in usable depth
    d. Material: Extruded Aluminum
    e. Finish: Baked -Polyester Powder Coated
    f. Color: Refer to schedules.

B. Floor-Mounted Racks:

1. A floor mounted rack able to be secured to floor without required mounting to adjacent wall.

    a. Base shall have a minimum of four mounting holes for permanent attachment to floor.

2. Shall have provisions for Vertical and horizontal cable management channels, top and bottom cable troughs, grounding lug, and grounding attachment.

    a. Refer to schedules on drawings for more information

3. Top shall have provisions for attaching to cable tray or ceiling.
4. Self-leveling.

C. Cable Management:

1. Metal, with integral wire retaining fingers.
2. Baked-polyester powder coat finish.
3. Vertical cable management panels shall have front and rear channels, with covers.
4. Provide horizontal crossover cable manager at the top of each relay rack, with a minimum height of two rack units each, unless otherwise noted.

2.4 POWER DISTRIBUTION UNIT (PDU)

A. Power Strips: Comply with UL 1363.

B. Description: Rack mounted, vertical and/or horizontal power distribution unit(s) (PDU) dedicated to the supply and routing of power from local uninterruptable power supply (UPS) and/or power outlet. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

1. A minimum of twelve (12) 20-A, 120-V ac, NEMA WD 6, Configuration 5-20R receptacles. A minimum of six (6) receptacles shall be mounted on front and on back. Refer to schedules for basis of design.
2. LED indicator lights for power and protection status.
3. LED indicator lights for reverse polarity and open outlet ground.
4. Circuit Breaker and Thermal Fusing: Unit continues to supply power if protection is lost.
5. Cord connected with 15-foot (4.5-m) line cord.
6. Rocker-type on-off switch, illuminated when in on position.

2.5 GROUNDING

A. Comply with requirements in Section 270526 “Grounding and Bonding for Communications Systems” for grounding conductors and connectors.

B. Telecommunication Grounding Bars: Rectangular bars of hard-drawn solid copper, accepting conductors ranging from No. 14 to No. 2/0 AWG, NRTL listed as complying with UL 467, and complying with TIA-606-B. Predrilling shall be with holes for use with lugs specified in this Section.

1. Wall Mount Telecommunications Primary Busbar (PBB): Refer to specification 270526 and drawing schedules.
2. Wall Mount Telecommunications Secondary Busbar (SBB): Refer to specification 270526 drawing schedules.
3. Cabinet-Mounted TGB: Terminal block, with stainless-steel or copper-plated hardware for attachment to cabinet.
4. Rack-Mounted Horizontal TGB: Designed for mounting in 19 inch (482.6 mm) equipment racks. Include a copper splice bar for transitioning to an adjoining rack, and stainless-steel or copper-plated hardware for attachment to the rack.
5. Rack-Mounted Vertical TGB: 36 inches (914.4 mm) long, with stainless-steel or copper-plated hardware for attachment to rack.

2.6 LABELING

A. Comply with TIA-606-B and UL 969 for a system of labeling materials, including label stocks, laminating adhesives, and inks used by label printers.

PART 3 - EXECUTION

3.1 ENTRANCE FACILITIES / EQUIPMENT ROOMS

A. Contact telecommunications service provider and arrange for installation of demarcation point, protected entrance terminals, and a housing when so directed by service provider.

1. Coordinate with Contracting Officer as required.
2. If direction for service provider differs from construction drawings, contact the Contracting Officer for coordination.

B. Coordinate all underground and/or wall penetrations with other trades. Refer to drawings, schedules, matrices, and related specifications for pathway and cabling requirements.

C. Comply with requirements in Section 270528 “Pathways for Communications Systems” for materials and installation requirements for underground and/or buried pathways.

3.2 INSTALLATION

A. Comply with NECA 1.

B. Comply with BICSI TDMM for layout of communications equipment spaces.

C. Comply with BICSI ITSIMM for installation of communications equipment spaces.

D. Bundle, lace, and train conductors and cables to terminal points without exceeding manufacturer's limitations on bending radii. Install lacing bars and distribution spools.

E. For vertical routed cabling, secure cabling at intervals required to be in accordance with manufactures limitations on tension and compression.

F. Coordinate layout and installation of communications equipment in racks and room. Coordinate service entrance configuration with service provider.

1. Meet jointly with system providers, equipment suppliers, and Contracting Officer to exchange information and agree on details of equipment configurations and installation interfaces.
2. Record agreements reached in meetings and distribute them to other participants.
3. Adjust configurations and locations of distribution frames, cross-connects, and patch panels in equipment spaces to accommodate and optimize configuration and space requirements of telecommunications equipment.
4. Adjust configurations and locations of equipment with distribution frames, cross-connects, and patch panels of cabling systems of other communications, electronic safety and security, and related systems that share space in equipment room.

G. Coordinate location of power raceways and receptacles with locations of communications equipment requiring electrical power to operate.

H. Coordinate layout of ladder runway and ladder rack with HVAC components to ensure air flow is not restricted by pathway and/or cable utilizing the pathway.

I. Coordinate layout of ladder runway and ladder rack with penetrations to ensure installation heights and cable bend radii are acceptable.

J. Backboards:

1. Install from 6 inches (150 mm) to 8 feet 6 inches (2588 mm) above finished floor. If plywood is fire rated, ensure that fire-rating stamp is visible after installation.
2. Paint all sides of backboard with two coats of paint leaving fire rating stamp visible.
3. Comply with requirements for backboard installation in BICSI's “Information Technology Systems Installation Methods Manual” and TIA-569-D.

3.3 GROUNDING

A. Comply with NECA/BICSI 607.

B. Install grounding according to BICSI ITSIMM, “Bonding, Grounding (Earthing) and Electrical Protection” Ch.

C. Locate PBB/SBB to minimize length of bonding conductors. Fasten to wall, allowing proper clearance behind PBB/SBB for standoff. Connect with a minimum No. 4 AWG grounding electrode conductor from PBB/SBB to suitable electrical building ground. Connect rack TGB to near PBB/SBB.

1. Where shielded or armored cabling are utilized, bond the shielding and/or armor to patch panel. Ensure patch panel is properly bonded to rack or rack TGB and in turn, the in-room PBB/SBB.

D. All conduit, cable tray, ladder rack, ladder runway, sleeves and other such metallic routing equipment shall be grounded properly. Pathway within the telecommunication room shall be bonded directly to local PBB/SBB.

3.4 SLEEVE AND SLEEVE SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 270544 “Sleeves and Sleeve Seals for Communications Pathways and Cabling.”

3.5 FIRESTOPPING

A. Comply with requirements in Section 078413 “Penetration Firestopping.”

B. Comply with TIA-569-D, Annex A, “Firestopping.”

C. Comply with BICSI's “Information Technology Systems Installation Methods Manual,” “Firestopping Practices” Ch.

3.6 IDENTIFICATION

A. Coordinate system components, wiring, and cabling complying with TIA-606-B. Comply with requirements in Section 270553 “Identification for Communications Systems.”

B. Comply with requirements in Section 099123 “Interior Painting” for painting backboards. For fire-resistant plywood, do not paint over manufacturer's label.

C. Labels shall be machine printed. Type shall be 3/16 inch (5 mm).

END OF SECTION 27 11 16

SECTION 27 13 23 – COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Type OFNR optical fiber cable.
2. Type OFCR optical fiber cable.
3. Type OFNP optical fiber cable.
4. Type OFCP optical fiber cable.
5. Types OFN and OFNG optical fiber cable.
6. Types OFC and OFCG optical fiber cable.
7. Optical fiber cable connecting hardware, patch panels, and cross-connects.

1.3 DEFINITIONS

A. Conductive Cable: Cable containing non-current-carrying electrically-conductive members such as metallic strength members and metallic vapor barriers.

B. Cross-Connect: A facility enabling termination of cable elements and their interconnection or cross-connection.

C. Type OFCP: Conductive cable for use in plenums, ducts, and other spaces used for environmental air.

D. Type OFCR: Conductive cable for use as riser in vertical shafts or from floor to floor.

E. Type OFNP: Nonconductive cable for use in plenums, ducts, and other spaces used for environmental air.

F. Type OFNR: Nonconductive cable for use as riser in vertical shafts or from floor to floor.

G. Types OFC and OFCG: Conductive cable for general purpose use.

H. Types OFN and OFNG: Nonconductive cable for general purpose use.

1.4 ACTION SUBMITTALS

A. Product data:

1. The following products shall be included:

a. Optical Fiber Cable
b. Optical Fiber Cable connecting hardware, patch panels, and cross-connects

B. Shop Drawings:

1. Cabling Plan

C. Identification Schedule

1.5 INFORMATIONAL SUBMITTALS

A. Installer Qualification Data:

1. Qualifications per specification 270010 “Supplemental Requirements for Communications”.
2. RCDD certification for Installer, installation supervisor, and field inspector.
3. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.

B. Field Quality Control Reports

C. Factory Test Reports.

D. Optical fiber cable testing plan.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data.

B. Warranty Documentation.

C. Identification Schedule.

D. Extra Material:

1. Patch-Panel Units: One of each type.
2. Jacks/Connectors: 10 of each type.

E. Field test and inspection reports.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Environmental Limitations: Do not deliver or install cables and connecting materials until wet-work in spaces is complete and dry, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during remainder of construction period.

B. Test cables upon receipt at Project site.

1. Test optical fiber cable while on reels. Use optical time domain reflectometer to verify cable length and locate cable defects, splices, and connector, including loss value of each. Retain test data and include record in maintenance data.

PART 2 - PRODUCTS

2.1 OPTICAL FIBER CABLE

A. Cable Type

1. OFNR: This category covers jacketed optical fiber cable for use as risers in vertical runs in shaft or between floors within buildings in accordance with Article 770 of NFPA 70 containing no electrically conductive materials.
2. OFCR: This category covers jacketed optical fiber cable for use as risers in vertical runs in shaft or between floors within buildings in accordance with Article 770 of NFPA 70 containing noncurrent-carrying electrically conductive materials.
3. OFNP: This category covers jacketed optical fiber cable for use in vertical runs in plenums, ducts, or other spaces used for environmental air within buildings in accordance with Article 770 of NFPA 70 containing no electrically conductive materials.
4. OFCP: This category covers jacketed optical fiber cable for use in vertical runs in plenums, ducts, or other spaces used for environmental air within buildings in accordance with Article 770 of NFPA 70 containing noncurrent-carrying electrically conductive materials.
5. OFN and OFNG: This category covers jacketed optical fiber cable for general use within buildings in accordance with Article 770 of NFPA 70 containing no electrically conductive materials.
6. OFC and OFCG: This category covers jacketed optical fiber cable for general use within buildings in accordance with Article 770 of NFPA 70 containing noncurrent-carrying electrically conductive materials.

B. Performance Criteria:

1. Regulatory Requirements: Listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.
2. Listing Criteria: UL CCN QAYK; including UL 1651.
3. General Characteristics:

a. Performance: TIA-568.3.
b. Inside Plant Mechanical Properties: ICEA S-83-596.

c. Inside-Outside Plant Mechanical Properties: ICEA S-104-696.
d. Jacket:

1) Cable cordage jacket, fiber, unit, and group color in accordance with TIA-598.
2) Imprinted with fiber count, fiber type, and aggregate length at regular intervals not to exceed 40 inch (1 m).

C. OS2 Single-Mode Optical Fiber Cable:

1. Source Limitations: Obtain products from single manufacturer.
2. Options:

a. Configuration: ISP/OSP, Loose tube, optical fiber cable; refer to schedule on drawing for strand count and other information.
b. Maximum Attenuation: 0.5 dB/km at 1310 nm wavelength; 0.5 dB/km at 1550 nm wavelength.
c. Jacket Color: Refer to schedule on drawings.

3. Additional Characteristics for OS2 Single-mode Fiber Optic Cabling:

a. Construction: TIA-492CAAA; 9 µm core diameter, 125 µm cladding diameter with low water peak.
b. Minimum Overfilled Modal Bandwidth-Length Product: 500 MHz-km at 850 nm wavelength; 500 MHz-km at 1300 nm wavelength.

2.2 OPTICAL FIBER CABLE HARDWARE

A. Performance Criteria:

1. Fiber Optic Connector Intermateability Standard (FOCIS) specifications of TIA-604 series.
2. TIA-568.3.

B. Cross-Connects and Patch Panels: Modular panels housing multiple-numbered, duplex cable connectors.

1. Number of Connectors per Field: One for each fiber of cable or cables assigned to field, plus spares and blank positions adequate to suit specified expansion criteria.

C. Patch Cords: Factory-made, dual-fiber cables in lengths and colors as specified in schedules on drawings.

D. Connector Type: Type LC complying with TIA-604-10 connectors.

E. Jacks and Jack Assemblies:

1. Female; quick-connect, simplex and/or duplex; fixed telecommunications connector designed for termination of single optical fiber cable.
2. Insertion loss not more than 0.75 dB.

3. Marked to indicate transmission performance.
4. Designed to snap-in to patch panel or faceplate.

2.3 GROUNDING

A. Comply with requirements in Section 270526 “Grounding and Bonding for Communications Systems” for grounding conductors and connectors.

B. Comply with TIA-607-B.

2.4 IDENTIFICATION PRODUCTS

A. Comply with TIA-606-B and UL 969 for a system of labeling materials, including label stocks, laminating adhesives, and inks used by label printers.

PART 3 - EXECUTION

3.1 PREPARATION

A. Coordinate backbone cabling with protectors and demarcation point provided by communications service provider.

3.2 INSTALLATION OF OPTICAL FIBER BACKBONE CABLES

A. Optical fiber backbone cabling system must provide interconnections between communications equipment rooms, main terminal space, and entrance facilities in telecommunications cabling system structure. Cabling system consists of backbone cables, intermediate and main cross-connects, mechanical terminations, and patch cords or jumpers used for backbone-to-backbone cross-connection.

B. Backbone cabling cross-connects may be located in communications equipment rooms or at entrance facilities. Bridged taps and splitters may not be used as part of backbone cabling.

C. Comply with BICSI N1, NECA NEIS 1, and NECA NEIS 301.

D. Backbone cabling system must comply with transmission standards in TIA-568.1.

E. Telecommunications Pathways and Spaces: Comply with TIA-569.

F. Wiring Methods:

1. In Raceway: Install cables in raceways and cable trays for the entire route of cable outside of equipment rooms, telecommunication rooms, or other such rooms housing networking equipment and terminations, unless otherwise noted.

    a. Comply with requirements for pathways specified in Section 270528 “Pathways for Communications Systems.”

2. In Enclosures: Bundle, lace, and train cables within enclosures. Connect to terminal points with no excess and without exceeding manufacturer's limitations on bending radii. Provide and use lacing bars and distribution spools.

G. Optical Fiber Cabling Installation:

1. Comply with TIA-568.1 and TIA-568.3.
2. Comply with BICSI ITSIMM, Ch. 6, “Cable Termination Practices.”
3. Terminate all cables; no cable may contain unterminated elements. Make terminations only at indicated outlets, terminals, cross-connects, and patch panels.
4. Cables may not be spliced. Secure and support cables at intervals not exceeding 30 inch (760 mm) and not more than 6 inch (150 mm) from cabinets, boxes, fittings, outlets, racks, frames, and terminals.
5. Install lacing bars to restrain cables, to prevent straining connections, and to prevent bending cables to smaller radii than minimums recommended by manufacturer.
6. Bundle, lace, and train cable to terminal points without exceeding manufacturer's limitations on bending radii, but not less than radii specified in BICSI ITSIMM, “Cabling Termination Practices” Chapter. Use lacing bars and distribution spools.
7. Do not install bruised, kinked, scored, deformed, or abraded cable. Do not splice cable between termination, tap, or junction points. Remove and discard cable if damaged during installation and replace it with new cable.
8. Cold-Weather Installation: Bring cable to room temperature before dereeling. Heat lamps may not be used for heating.
9. In communications equipment room, provide 10 ft (3 m) long service loop on each end of cable.
10. Pulling Cable: Comply with BICSI ITSIMM, Ch. 4, “Pulling Cable.” Monitor cable pull tensions.
11. Cable may be terminated on connecting hardware that is rack or cabinet mounted.

    a. No fiber cable may be left unterminated (or dark). Ensure all fiber installed are terminated properly, it is the contractors responsibility to review construction documents and ensure all cabling termination points are made available.

H. Group connecting hardware for cables into separate logical fields.

I. Splicing

1. Splicing existing or new fiber optic cabling is not acceptable except for splicing onto pigtail type cable assemblies at fiber termination points. New cable shall be provided and installed to ensure no splicing along fiber path is utilized. For instances where fiber termination onto a pigtail cable assembly is accepted:

    a. All such connections shall be fusion spliced.
    b. All fusion splices shall have a maximum loss of .3 dB.

3.3 FIRESTOPPING

A. Comply with requirements in Section 078413 “Penetration Firestopping.”

B. Comply with requirements in Section 270544 “Sleeves and Sleeve Seals for Communications Pathways and Cabling.”

C. Comply with TIA-569, Annex A, “Firestopping.”

D. Comply with BICSI ITSIMM, “Firestopping” Chapter.

3.4 GROUNDING

A. Install grounding in accordance with BICSI ITSIMM, “Grounding (Earthing), Bonding, and Electrical Protection” Chapter.

B. Comply with TIA-607 and NECA/BICSI-607.

C. Refer to 270526 “Grounding and Bonding for Communications Systems.”

3.5 IDENTIFICATION

A. Identify system components, wiring, and cabling complying with TIA-606. Comply with requirements for identification specified in Section 270553 “Identification for Communications Systems.”

B. Cable Schedule: Install in prominent location in each equipment room and wiring closet. List incoming and outgoing cables and their designations, origins, and destinations. Protect with rigid frame and clear plastic cover. Furnish electronic copy of final comprehensive schedules for Project.

C. Cabling Administration Drawings: Show building floor plans with cabling administration-point labeling. Identify labeling convention and show labels for telecommunications closets, backbone pathways and cables, entrance pathways and cables, terminal hardware and positions, horizontal cables, work areas and workstation terminal positions, grounding buses and pathways, and equipment grounding conductors.

D. Cable and Wire Identification:

1. Label each cable within 4 inch (100 mm) of each termination and tap, where it is accessible in cabinet or junction or outlet box, and elsewhere as indicated.
2. Each wire connected to building-mounted devices is not required to be numbered at device if color of wire is consistent with associated wire connected and numbered within panel or cabinet.
3. Exposed Cables and Cables in Cable Trays and Wire Troughs: Label each cable at intervals not exceeding 15 ft (4.5 m).
4. Label each unit and field within distribution racks and frames.

5. Identification within Connector Fields in Equipment Rooms and Wiring Closets: Label each connector and each discrete unit of cable-terminating and connecting hardware. Where similar jacks and plugs are used for both voice and data communication cabling, use different color for jacks and plugs of each service.

3.6 FIELD QUALITY CONTROL

A. Field tests and inspections must be witnessed by the Contracting Officer or the Contracting Officer's chosen representative. The authority having jurisdiction (AHJ) shall be included at their discretion, coordinate with Contracting Officer for scheduling and attendees requirements.

B. Tests and Inspections:

1. Perform manufacturer's recommended test and inspections.
2. Visually inspect optical fiber jacket materials for qualified electrical testing laboratory certification markings. Inspect cabling terminations in communications equipment rooms for compliance with color-coding for pin assignments, and inspect cabling connections for compliance with TIA-568.1.
3. Visually inspect cable placement, cable termination, grounding and bonding, equipment and patch cords, and labeling of all components.
4. For each fiber installed, perform a link-loss budget calculation that will be compared with measured results.
5. Optical Fiber Cable Tests:

    a. All test performed on Optical Fiber cabling that uses a laser or light-emitting diode (LED) in a test set are to be carried out with a safety precaution in accordance with ANSI Z136.2.
    b. Test instruments must meet or exceed applicable requirements in TIA-568.1. Use only test cords and adapters that are qualified by test equipment manufacturer for channel or link test configuration.
    c. Both Tier 1 and Tier 2 testing shall be applied to each fiber strand as defined in TIA-568.3-D.

        1) Encircled Flux (EF) testing shall be utilized when testing multimode fiber optic cable.

    d. Link End-to-End Attenuation Tests:

        1) Multimode fiber optic cable link measurements: Test at 850 or 1300 nm in one direction in accordance with TIA-526-14, Method B, One Reference Jumper.
        2) Single-mode fiber optic cable link measurements: Test at 1310 or 1550 nm
        3) Attenuation test results must be less than those calculated in accordance with equation in TIA-568.1.

    e. All test reports shall be combined into a single document and provided to the Contracting Officer at completion of testing.

C. Nonconforming Work:

1. Cables will be considered defective if they do not pass tests and inspections.
2. Remove and replace defective cables and retest.

D. Collect, assemble, and submit test and inspection reports.

1. Data for each measurement must be documented.
2. Data for field quality-control report submittals must be printed in summary report that is formatted similar to Table 10.1 in BICSI TDMM, or transferred from instrument to computer, saved as text files, and printed and submitted.

END OF SECTION 27 13 23

SECTION 27 15 13 – COMMUNICATIONS COPPER HORIZONTAL CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 270010 “Supplemental Requirements for Communications” for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Category 6a twisted pair cable.
2. Twisted pair cable hardware, including panels and jacks.

1.3 DEFINITIONS

A. Cross-Connect: A facility enabling the termination of cable elements and their interconnection or cross-connection.

B. EMI: Electromagnetic interference.

C. FTP: Shielded twisted pair.

D. F/FTP: Overall foil screened cable with foil screened twisted pair.

E. F/UTP: Overall foil screened cable with unscreened twisted pair.

F. IDC: Insulation displacement connector.

G. LAN: Local area network.

H. Jack: Also commonly called an “outlet,” it is the fixed, female connector.

I. Plug: Also commonly called a “connector,” it is the removable, male telecommunications connector.

J. RCDD: Registered Communications Distribution Designer.

K. Screen: A metallic layer, either a foil or braid, placed around a pair or group of conductors.

L. Shield: A metallic layer, either a foil or braid, placed around a pair or group of conductors.

M. S/FTP: Overall braid screened cable with foil screened twisted pair.

N. S/UTP: Overall braid screened cable with unscreened twisted pairs.

O. UTP: Unscreened (unshielded) twisted pair.

1.4 ACTION SUBMITTALS

A. Product data:

1. The following products shall be included:

a. Each type of Category Cabling
b. Twisted pair cabling hardware

B. Shop Drawings:

1. Reviewed and stamped by RCDD.
2. Wiring diagrams and installation details of telecommunications equipment, to show location and layout of telecommunications equipment, including the following:

a. Telecommunications cable termination locations.
b. Typical telecommunications details.
c. Patch Panel Configuration with racks and cable labels.
d. Telecommunication room locations and cable origination locations.

C. Identification Schedule

1.5 INFORMATIONAL SUBMITTALS

A. Installer Qualification Data:

1. Qualifications per specification 270010 “Supplemental Requirements for Communications”.
2. RCDD certification for Installer, installation supervisor, and field inspector.
3. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.

B. Field Quality Control Reports

C. Factory Test Reports.

D. Twisted Pair Cable Testing Plan.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data

B. Warranty Documentation

C. Identification Schedule

D. Extra Material

1. Connecting Blocks: One of each type.
2. Cover Plates: Five of each type.
3. Jacks: Ten of each type.
4. Patch-Panel Units: One of each type.

E. Field test and inspection reports.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Environmental Limitations: Do not deliver or install cables and connecting materials until wet-work in spaces is complete and dry, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during remainder of construction period.

B. Test cables upon receipt at Project site.

1. Test each pair of twisted pair cable for open and short circuits.

1.8 QUALITY ASSURANCE

A. Installer Qualifications: Cabling Installer must have personnel certified by BICSI on staff.

1. Layout Responsibility: Preparation of Shop Drawings, cabling administration Drawings, and field-testing program development by an RCDD.
2. Installation Supervision: Installation shall be under the direct supervision of Technician, who shall be present at all times when Work of this Section is performed at Project site.
3. Testing Supervisor: Currently certified by BICSI as an RCDD to supervise on-site testing.

B. Testing Agency Qualifications: An NRTL.

1. Testing Agency's Field Supervisor: Currently certified by BICSI as an RCDD to supervise on-site testing.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

D. Telecommunications Pathways and Spaces: Comply with TIA-569-D.

E. Grounding: Comply with TIA-607-B.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. General Performance: Horizontal cabling system shall comply with transmission standards in TIA-568-C.1, when tested according to test procedures of this standard.

2.2 GENERAL CABLE CHARACTERISTICS

A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction as complying with the applicable standard and NFPA 70 for the following types:

1. Communications, Plenum Rated:

a. Type CMP complying with UL 1685.

2. Communications, Non-Plenum Rated:

a. Type CMR complying with UL 1666.

B. RoHS compliant.

C. It is a requirement that all equipment and components within a channel (patch cable to patch cable), inclusive of, but not limited to, cabling, jacks, plugs, faceplates, patch panels, patch cables, etc., shall be of a single manufacturer. All installation, testing, and warranty requirements shall be in-line with the chosen manufacturer's requirements.

2.3 CATEGORY 6a TWISTED PAIR CABLE

A. Description: Four-pair, balanced-twisted pair cable, with internal spline, certified to meet transmission characteristics of Category 6a cable at frequencies up to 500 MHz.

B. Standard: Comply with TIA-568-C.2 for Category 6a cables.

C. Conductors: 100-ohm, 23 AWG solid copper.

D. Shielding/Screening: Unshielded twisted pairs (UTP).

E. Cable Rating: Plenum.

F. Jacket: Refer to schedules.

2.4 TWISTED PAIR CABLE HARDWARE

A. Description: Hardware designed to connect, splice, and terminate twisted pair copper communications cable.

B. General Requirements for Twisted Pair Cable Hardware:

1. Comply with the performance requirements of Category 6a.
2. Comply with TIA-568-C.2, IDC type, with modules designed for punch-down caps or tools.
3. Cables shall be terminated with connecting hardware of same category or higher.

C. Manufacturer Limitations: Obtain twisted pair cable hardware from single manufacturer that is from same manufacturer of structured cabling. The structured cabling solution shall be an end-to-end solution from a single manufacturer, pursuant to, and in line with, the specific manufacturers product solution warranty.

D. Patch Panel: Modular panels housing numbered jack units with IDC-type connectors at each jack location for permanent termination of pair groups of installed cables.

1. Features:

a. Universal T568A and T568B wiring labels.
b. Labeling areas adjacent to conductors.
c. Replaceable connectors.
d. 24 or 48 ports.

2. Construction: 16-gauge steel and mountable on 19-inch (483 mm) equipment racks.
3. Number of Jacks per Field: One for each four-pair cable indicated at minimum. Refer to rack elevation and schedules.
4. A minimum of one patch panel shall be dedicated for analog voice communication applications. It shall be connected to the wall mounted 110-block via a 25-pair CAT5E cable. Horizontal cabling scheduled for analog voice use shall be patched from horizontal patch panel to the voice dedicated patch panel.

E. Patch Cords: Factory-made, four-pair cables in lengths as specified in drawing schedules; terminated with an eight-position modular plug at each end.

1. Patch cords shall have bend-relief-compliant boots and color-coded icons to ensure performance. Patch cords shall have latch guards to protect against snagging.

F. Jacks and Jack Assemblies:

1. Female; eight position; modular; fixed telecommunications connector designed for termination of a single four-pair, 100-ohm, unshielded or shielded twisted pair cable.
2. Designed to snap-in to a patch panel or cover plate.
3. Standard: Comply with TIA-568-C.2.
4. Marked to indicate transmission performance.

G. Cover Plate:

1. Two, Four, and/or Six port, vertical single gang cover plates designed to mount to single gang wall boxes.
2. Plastic Cover Plate: High-impact plastic. Coordinate color with Section 260533 “Raceway and Boxes for Electrical Systems.”

3. Metal Cover Plate: Stainless steel, complying with requirements in Section 260533 "Raceway and Boxes for Electrical Systems."
4. For use with snap-in jacks accommodating any combination of twisted pair, optical fiber, and coaxial work area cords.

    a. Flush mounting jacks, positioning the cord at a 45-degree angle.

H. Legend:

1. Machine printed, in the field, using adhesive-tape label.
2. Snap-in, clear-label covers and machine-printed paper inserts.

2.5 IDENTIFICATION PRODUCTS

A. Comply with TIA-606-B and UL 969 for a system of labeling materials, including label stocks, laminating adhesives, and inks used by label printers.

PART 3 - EXECUTION

3.1 COPPER HORIZONTAL CABLING DESCRIPTION

A. Horizontal cable and its connecting hardware provide the means of transporting signals between the telecommunications outlet/connector and the horizontal cross-connect located in the communications equipment room. This cabling and its connecting hardware are called "permanent link," a term that is used in the testing protocols.

1. Horizontal cabling shall contain no more that one transition point or consolidation point between the horizontal cross-connect and the telecommunications outlet/connector.
2. Bridged taps and splices shall not be installed in the horizontal cabling.
3. Splitters shall not be installed as part of the optical fiber cabling.

B. The maximum allowable horizontal cable length is 295 feet. This maximum allowable length does not include an allowance for the length of 16 feet to the workstation equipment. The maximum allowable length does not include an allowance for the length of 16 feet in the horizontal cross-connect, when cabling leaves the communications equipment room.

3.2 SELECTION OF COMMUNICATIONS COPPER HORIZONTAL CABLING

A. All cable selection shall be as specified in section 2 and on the construction drawings. Final cable selection shall be approved through submittal process.

B. CMP (Plenum rated) Cabling: Plenum cabling shall be provided and installed in all Inside Plant (ISP) applications.

1. It is acceptable to utilize CMR (Riser rated) cabling interior to the telecommunications space when specified and/or with Contracting Officer's approval.
2. It is at the contractor's risk to utilize any non-CMP cabling for ISP applications.
3. It is acceptable to use ISP/OSP plenum rated cable within the building.

C. Where cabling leaves the footprint of the building and utilized underground pathway, exterior pathway, and/or is exposed to the exterior environment, OSP or ISP/OSP rated cabling shall be utilized.

3.3 WIRING METHODS

A. Routing:

1. Unless otherwise and specifically indicated, cables shall be installed into, and route through raceways and/or cable trays as indicated below. Cables shall route via:

a. Conduit(s) in all cases from telecom room to outlet/device. Cable shall not be routed freely or exposed in any way outside of the telecom room, unless specifically noted.

B. Wiring within Enclosures, cabinets, consoles and other such non-architectural equipment.

1. Cable shall route freely without raceway unless equipment provides raceway. Bundle, lace, and train cables within enclosures. Connect to terminal points with no excess and without exceeding manufacturer's limitations on bending radii. Provide and use lacing bars and distribution spools.

3.4 INSTALLATION OF PATHWAYS

A. Comply with requirements for demarcation point, cabinets, and racks specified in Section 271116 “Communications Racks, Frames, and Enclosures.”

B. Comply with Section 270528 “Pathways for Communications Systems.”

C. Comply with Section 270529 “Hangers and Supports for Communications Systems.”

D. Comply with Section 270536 “Cable Trays for Communications Systems.”

E. Drawings indicate general arrangement of pathways and fittings.

3.5 INSTALLATION OF TWISTED-PAIR HORIZONTAL CABLES

A. Comply with NECA 1 and NECA/BICSI 568.

B. General Requirements for Cabling:

1. Comply with TIA-568-C.0, TIA-568-C.1, and TIA-568-C.2.
2. Comply with BICSI's “Information Transport Systems Installation Methods Manual (ITSIMM), Ch. 5, “Copper Structured Cabling Systems,” “Cable Termination Practices” Section.
3. Install 110-style IDC termination hardware unless otherwise indicated.
4. Do not untwist twisted pair cables more than 1/2 inch (12 mm) from the point of termination to maintain cable geometry.

5. Terminate all conductors; no cable shall contain unterminated elements. Make terminations only at indicated outlets, terminals, cross-connects, and patch panels.
6. MUTOA shall not be used as a cross-connect point.
7. Consolidation points may be used only for making a direct connection to equipment outlets:

    a. Do not use consolidation point as a cross-connect point, as a patch connection, or for direct connection to workstation equipment.
    b. Locate consolidation points for twisted-pair cables at least 49 feet (15 m) from communications equipment room.

8. Cables may not be spliced. Secure and support cables at intervals not exceeding 30 inches (760 mm) and not more than 6 inches (150 mm) from cabinets, boxes, fittings, outlets, racks, frames, and terminals.
9. Install lacing bars to restrain cables, prevent straining connections, and prevent bending cables to smaller radii than minimums recommended by manufacturer.
10. Bundle, lace, and train conductors to terminal points without exceeding manufacturer's limitations on bending radii, but not less than radii specified in BICSI Information Transport Systems Installation Methods Manual, Ch. 5, “Copper Structured Cabling Systems,” “Cable Termination Practices” Section. Use lacing bars and distribution spools.
11. Do not install bruised, kinked, scored, deformed, or abraded cable. Do not splice cable between termination, tap, or junction points. Remove and discard cable if damaged during installation, and replace it with new cable.
12. Cold-Weather Installation: Bring cable to room temperature before dereeling. Heat lamps shall not be used for heating.
13. Install a 10-foot- (3-m-) long service loop on each end of cable. One loop shall be located in communication room and one shall be located in accessible ceiling near cable termination. Refer to drawing details for more information.
14. Pulling Cable: Comply with BICSI Information Transport Systems Installation Methods Manual, Ch. 5, “Copper Structured Cabling Systems,” “Pulling and Installing Cable” Section. Monitor cable pull tensions.

C. Open-Cable Installation:

1. Install cabling with horizontal and vertical cable guides in telecommunications spaces with terminating hardware and interconnection equipment.

D. Group connecting hardware for cables into separate logical fields.

E. Separation from EMI Sources:

1. Comply with recommendations from BICSI's “Telecommunications Distribution Methods Manual” and TIA-569-D for separating unshielded copper communication cable from potential EMI sources, including electrical power lines and equipment.
2. Separation between open communications cables or cables in nonmetallic raceways and unshielded power conductors and electrical equipment shall be as follows:

    a. Electrical Equipment Rating Less Than 2 kVA: A minimum of 5 inches (127 mm).
    b. Electrical Equipment Rating between 2 and 5 kVA: A minimum of 12 inches (300 mm).

c. Electrical Equipment Rating More Than 5 kVA: A minimum of 24 inches (600 mm).

3. Separation between communications cables in grounded metallic raceways and unshielded power lines or electrical equipment shall be as follows:

a. Electrical Equipment Rating Less Than 2 kVA: A minimum of 2-1/2 inches (64 mm).
b. Electrical Equipment Rating between 2 and 5 kVA: A minimum of 6 inches (150 mm).
c. Electrical Equipment Rating More Than 5 kVA: A minimum of 12 inches (300 mm).

4. Separation between communications cables in grounded metallic raceways, power lines, and electrical equipment located in grounded metallic conduits or enclosures shall be as follows:

a. Electrical Equipment Rating Less Than 2 kVA: No requirement.
b. Electrical Equipment Rating between 2 and 5 kVA: A minimum of 3 inches (76 mm).
c. Electrical Equipment Rating More Than 5 kVA: A minimum of 6 inches (150 mm).

5. Separation between Communications Cables and Electrical Motors and Transformers, 5 kVA or HP and Larger: A minimum of 48 inches (1200 mm).
6. Separation between Communications Cables and Fluorescent Fixtures: A minimum of 5 inches (127 mm).

3.6 FIRESTOPPING

A. Comply with requirements in Section 078413 “Penetration Firestopping.”

B. Comply with requirements in Section 270544 “Sleeves and Sleeve Seals for Communications Pathways and Cabling”.

C. Comply with TIA-569-D, Annex A, “Firestopping.”

D. Comply with “Firestopping Systems” Article in BISCI's “Telecommunications Distribution Methods Manual.”

3.7 GROUNDING

A. Comply with requirements in Section 270526 “Grounding and Bonding for Communications Systems” for grounding conductors and connectors.

B. Install grounding according to the “Grounding, Bonding, and Electrical Protection” chapter in BICSI's “Telecommunications Distribution Methods Manual.”

C. Comply with TIA-607-B and NECA/BICSI-607.

D. Bond metallic equipment to the grounding bus bar, using not smaller than a No. 6 AWG equipment grounding conductor.

3.8 IDENTIFICATION

A. Identify system components, wiring, and cabling complying with TIA-606-B. Comply with requirements for identification specified in Section 270553 “Identification for Communications Systems.”

1. Color-code cross-connect fields and apply colors to voice and data service backboards, connections, covers, and labels per indication on drawing schedules and rack elevations.

B. Cable Schedule: Install in a prominent location in each equipment room and wiring closet. List incoming and outgoing cables and their designations, origins, and destinations. Protect with rigid frame and clear plastic cover. Furnish an electronic copy of final comprehensive schedules for Project.

C. Cabling Administration Drawings: Show building floor plans with cabling administration-point labeling. Identify labeling convention and show labels for telecommunications closets, terminal hardware and positions, horizontal cables, work areas and workstation terminal positions, grounding buses and pathways, and equipment grounding conductors.

3.9 FIELD QUALITY CONTROL

A. Field tests and inspections must be witnessed by the Contracting Officer, coordinate with the Contracting Officer for scheduling and attendees requirements.

B. Tests and Inspections:

1. Perform manufacturer’s recommended test and inspections.
2. Factory test cables on reels according to TIA-568-C.1.
3. Factory test twisted pair cables according to TIA-568-C.2.
4. Visually inspect jacket materials for NRTL certification markings. Inspect cabling terminations in communications equipment rooms for compliance with color-coding for pin assignments, and inspect cabling connections for compliance with TIA-568-C.1.
5. Visually inspect cable placement, cable termination, grounding and bonding, equipment and patch cords, and labeling of all components.
6. Test twisted pair cabling for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors. Test operation of shorting bars in connection blocks. Test cables after termination but not cross-connection.

    a. Test instruments shall meet or exceed applicable requirements in TIA-568-C.2. Perform tests with a tester that complies with performance requirements in “Test Instruments (Normative)” Annex, complying with measurement accuracy specified in “Measurement Accuracy (Informative)” Annex. Use only test cords and adapters that are qualified by test equipment manufacturer for channel or link test configuration.

b. The following testing procedures shall be applied:

1) Continuity
2) Insertion Loss
3) NEXT
4) Return Loss

c. All test reports shall be combined into a single document and provided to the Contracting Officer at completion of testing.

C. Data for each measurement shall be documented. Data for submittals shall be printed in a summary report that is formatted similarly to Table 10.1 in BICSI's "Telecommunications Distribution Methods Manual," or shall be transferred from the instrument to the computer, saved as text files, printed, and submitted.

D. Nonconforming Work:

1. End-to-end cabling will be considered defective if it does not pass tests and inspections.
2. Remove and replace cabling where test results indicate that they do not comply with specified requirements.

E. Collect, assemble, and submit test and inspection reports.

END OF SECTION 27 15 13

SECTION 27 41 00 – AUDIO VIDEO SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. The term “provide” used throughout this specification and drawings shall mean “furnish, install, test, and certify”.

B. Extent of Audio Video Systems infrastructure work is indicated by DIVISION 26 electrical and other specification sections, drawings and schedules and is hereby defined to include, but not by way of limitation, the provisions of:

1. Raceway systems including but not limited to conduits, cable trays, sleeves, surface raceways, pull-boxes, junction boxes, back-boxes, etc. as required and specified in this section and/or select DIVISION 26 sections.
2. Power distribution within equipment racks including programable power sequencing of AC circuits.

C. Extent of Audio Video Systems Infrastructure as indicated by DIVISION 27 “Communications” and those associated specification sections that address Ethernet and IT requirements shall include but not be limited to:

1. Patch cords, jumper cables, and cross-connect cables to interconnect wiring terminals as well as electronic equipment.
2. Network connectivity: a channel is defined as the connection from one active device to another, including any patch cords and equipment cords and shall not exceed 100 meters total length.
3. Testing of all IT category type cable infrastructure and grounding systems as noted by specification, drawings, and applicable industry standards.

D. General requirements also include but are not limited to:

1. AV Head-end Room hardware and component fit-out including rack enclosures, cable tray, backboard, and raceways for terminating cable and installation of electronic equipment.
2. Grounding and bonding of all metallic hardware components back to the nearest grounding bus, i.e. AV grounding bar including (AVGB) or similar, but not limited to equipment racks, cabinets, cable trays, ladder rack, metallic cable sheaths, wall mounted wiring terminals, conduits, sleeves, metallic ductwork, and frames..
3. The use of Velcro type cable ties is required; the use of plastic zip ties shall not be permitted. No exceptions.
4. Fire stopping as required. Contractor shall provide fire stopping for all low-voltage openings (including empty low voltage raceway) once cable installation is complete or to use products such as EZ-Path or similar.
5. Seismic bracing of all equipment racks, ladder-type or wire basket cable trays, ceiling recessed equipment such as loudspeakers, plenum boxes etcetera as required by code and by local governing jurisdiction.

6. Preparation and submission of product data, shop drawings, testing reports, as-built drawings, and cabling documentation as required in this specification.
7. Construction and Installation warranties.
8. Manufacturer components, channel and solutions warranties.
9. Installation and testing of all system and components shall be documented.
10. Onsite administrative and user training.

1.2 REFERENCES

A. Audiovisual and Integrated Experience Association (AVIXA):

1. ANSI / INFOCOMM 2M – Standard Guide for Audiovisual Design and Coordination Processes
2. ANSI / INFOCOMM 10 – Audiovisual Systems Performance Verification
3. ANSI / INFOCOMM A102.01 – Audio Coverage Uniformity in Listener Areas
4. AVIXA F502.01 – Rack Building for Audiovisual Systems
5. AVIXA F502.02 – Rack Design for Audiovisual Systems
6. INFOCOMM F501.01 – Cable Labeling for Audiovisual Systems

B. Building Industry Consulting Service International (BISCI):

1. ANSI / BICSI 001 – Information and Communication Technology Systems Design and Implementation Best Practices for Educational Institutions and Facilities
2. ANSI / BICSI N1 – Installation Practices for Telecommunications and ICT Cabling and Related Cabling Infrastructure
3. ANSI / BICSI N3 – Planning and Installation Methods for the Bonding and Grounding of Telecommunication and ICT Systems and Infrastructure

C. Code of Federal Regulations (CFR):

1. 29 CFR 1910.268 – Occupation Safety and Health Standards (OSHA) – Telecommunications
2. 36 CFR 1191 – Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

D. Electronic Components Industry Association (ECIA):

1. ECIA EIA/ECA 310-E – Cabinets, Racks, Panels, and Associated Equipment

E. National Fire Protection Association (NFPA):

1. NFPA 70 – National Electrical Code
2. NFPA 72 – National Fire Alarm and Signaling Code

    a. (muting of sound systems during a life safety event)

3. NFPA 90A – Standard for the Installation of Air-Conditioning and Ventilating Systems

F. Telecommunications Industry Association (TIA):

1. TIA-568.0 – Generic Telecommunications Cabling for Customer Premises
2. TIA-568.1 – Commercial Building Telecommunications Infrastructure Standard
3. TIA-568.2 – Balanced Twisted-Pair Telecommunications Cabling and Components Standards
4. TIA-568.3 – Optical Fiber Cabling Components Standard
5. TIA-568.4 – Broadband Coaxial Cabling and Components Standard
6. TIA-569 – Telecommunications Pathways and Spaces
7. TIA-606 – Administration Standard for Telecommunications Infrastructure
8. TIA-607 – Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
9. TIA-862 – Structured Cabling Infrastructure Standard for Intelligent Building Systems
10. TIA-1152 – Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling

G. Underwriters Laboratories (UL):

1. UL 263 – Fire Tests of Building Construction and Materials
2. UL 1480 – Speakers for Fire Alarm and Signaling Systems, Including Accessories
3. UL 2043 – Fire Test for Heat and Visible Smoke Release for Discrete Products and Their Accessories Installed in Air-Handing Spaces
4. UL 62368-1 – Audio/Video, Information and Communication Technology Equipment – Part 1: Safety Requirements

H. International Code Council (ICC):

1. IBC – International Building Code

I. National Electrical Manufacturer's Association (NEMA)

J. Society of Motion Picture and Television Engineers (SMPTE)

1.3 DEFINITIONS

A. The terms "AV Contractor", "AVC", and "Contractor" shall refer to the Audio-Video Systems Contractor who has been awarded the contract for this scope and who has responsibility for the performance of the work specified in this specification.

B. The term "EC" shall refer to the electrical contractor.

C. The term "GC" shall refer to the general contractor.

D. The term "SCC" shall refer to the structured cabling contractor.

E. The term "specified elsewhere" shall refer to the material and work which is related to this contract and for which the Contractor is not responsible except as otherwise detailed herein. Some of all of these items may be included in the overall electrical contract.

F. The terms “not in contract” and “NIC” shall refer to work or equipment that is not in the contractor covered in this specification.

G. The terms “Contracting Officer furnished equipment” and “OFE” shall refer to equipment which shall be provided by the Contracting Officer or user to the Contractor. The Contractor shall be responsible for installing the equipment at the project site in good functional order as detailed herein. Where the equipment is existing or is used, the AV Contractor shall verify the operation and functionality of these devices prior to installation. Any deficiencies shall be brought to the Contracting Officer’s attention immediately via written documentation.

H. The terms “provided by others” and “PBO” shall refer to work or equipment that is provided by another trade, party, or team not directly responsible for the Audio-Video Systems but that interfaces with this scope of work.

I. The term “complete” shall refer to equipment and systems that shall be provided with all necessary accessories to complete a turn-key solution (i.e., “equipment racks” shall include doors, locks, side panels, covers, blank and vented panels, thermal regulation system, cooling fans, cable management, AC power distribution, UPS and power sequencing, and et cetera.). If the Contractor is unclear about a particular piece of equipment, system accessories, or design intent, then it is the sole responsibility of the Contractor to immediately contact the Contracting Officer via the RFI process for clarification prior to the bid submittal.

J. The term “furnish” shall indicate the responsibility to ship or deliver the item to the job site for receipt, staging and installation by others.

K. The term “install” or “installation” shall indicate the responsibility of receiving the item at the job site, providing adequate storage, unpacking or uncrating the item, physically securing the item or otherwise making ready the item for its intended use by following the instruction and approved methods of the manufacturer and those contained herein.

L. The term “provide” shall refer to the responsibility to both furnish, install, test and certify equipment.

M. The term “shall” is mandatory; the term “will” is informative; the term “should” is advisory.

1.4 SCOPE OF WORK

A. The work as described herein shall be furnished and installed by one (1) firm alone, hereafter referred to as the AV Contractor or AVC, for undivided responsibility. The AV Contractor shall provide all equipment, cabling and audio / video / control signals for the AV Systems described herein.

B. The work shall be provided complete, and the systems shall be fully operational as shown on the Audio-Video System drawings. It is expected that the AVC shall be accountable for a functional system Turn-key solution regardless of the packages selected by the Contracting Officer and that any deficiencies or added expenses created by the Contracting Officers’ selections be brought to the Contracting Officers attention before acceptance of the contract.

C. The AV Contractor shall provide equipment and materials, whether specifically mentioned herein or not, to ensure a complete and operating system. Equipment and materials designated as Provided by Others (PBO), Not in Contract (NIC), and/or Contracting Officer Furnished Equipment (OFE) are specifically exempted from this requirement.

D. The AVC shall provide all software programming required to provide fully operational systems.

E. The specifications and drawings shall be read and used together. System features which are mentioned in one document are not necessarily shown in the other. In case of conflict between the drawings and specifications, request clarification via the pre-bid RFI process. These pre-bid RFI's are due one (1) week before the bids are due. All pre-bid RFI responses shall be distributed to each Bidder.

F. The AVC shall coordinate with the Contracting Officer and/or their network staff et cetera to fully understand their network protocols and the security feature set they utilize so that the AV Network shall be installed and remain secure to outside cyberthreats at all times.

1.5 SYSTEM DESCRIPTIONS AND FUNCTIONS

A. General: Provide audio video systems that promote an enjoyable and educational experience for the guests and staff during events and while visiting facility. The AV systems must be easy to use, set up and maintained by the facility's technical support staff.

B. AV Systems are anticipated for the following areas:

1. Auditorium / Multipurpose Room 118
2. Exhibit & Retail Lobby 102
3. Public Address and background Music in public areas

C. Auditorium general audio video system requirements:

1. Audio Systems:

a. Provide full frequency quality Dolby 7.1 surround sound reinforcement with locations as shown on the drawings.
b. Full frequency loudspeakers

1) Front and rear speakers are recessed ceiling mount units

c. Subwoofer & Surround Loudspeakers are ceiling recessed units
d. A networked digital signal distribution format called Dante shall be used.
e. Wireless microphone System with headset and handheld microphones
f. Ceiling microphone array for presentations that do not require a wireless microphone
g. Provide a digital signal processor (DSP) system that shall utilize the Dante signal protocol as a signal distribution solution.
h. Provide a Dolby surround sound processor
i. Bluetooth receiver for bring your own devices (BYOD)
j. Assistive Listening System

1) This system utilizes AURA / Bluetooth technologies for guest ease of use allowing them to stream audio to their personal smart phone which in turn can connect to hearing aids via Bluetooth technology.
2) Belt pack receivers are still required by code.
3) It is required per the federal ADA code

2. Video System

a. Provide large 10,000 lumen professional grade WUXGA (UHD capable) resolution video projection system.
b. Video flat panel monitors shall be sized based on room size and occupancy.
c. Provide local active HDMI & USB-C wall plates connectivity rated at 4k resolution.
d. Pan Tilt Zoom (PTZ) camera system.
e. Video / audio network internet streaming
f. Wireless video connectivity for OFE BYOD laptops
g. Professional grade 4k Blu-ray player

3. Control System:

a. The AV systems will be operated by the low voltage control system using local wall mounted fixed touch panels. The control system shall operate:

1) AV systems on / off. These can be scheduled events.
2) System presets including PTZ camera presets and adjustments
3) Audio and video source selection
4) Audio and video zone distribution selection
5) CATV channel selection where required
6) System audio volume up / down
7) Audio and or video mute
8) Automatic sound systems mute during a fire alarm / life safety “event”. The FA contractor to provide a dry contact closure & relays or similar signals to the control system CPU located in head-end rack. Cabling etcetera for this functionality shall be provided by the FA Contractor.
9) Each touch panel can be password protection enabled if desired by the Contracting Officer.

b. Note: this is not a complete list of all the functionality the project requires. AVC to coordinate with the team during the shop drawing process. AVC shall coordinate a meeting with the Contracting Officer to discuss touch panel control requirements, GUI layout etcetera.
c. Note: the control system can utilize global scheduling presets such as auto system of/ off based on time of date etcetera. AVC to coordinate the system requirements with the Contracting Officer during the shop drawing process.

4. Ethernet Network Requirements (dedicated): AVC shall closely coordinate with the Contracting Officers IT staff with all AV systems network requirements including but not limited to:

a. Gbps Ethernet backbone including active managed PoE (and or PoE+) switches which are Contracting Officer provided.
b. VLAN / dedicated AV Subnet etcetera

c. Patch-bay and patch cabling
d. Cable plant certification (if applicable)
e. Ethernet switch / PoE (+) requirements
f. Gbps typical for most applications, i.e. control and audio applications
g. Gbps for 4k & 8k video applications

5. Equipment rack

a. Provide sequenced power distribution with the racks utilizing vertical power management.
b. Automatic thermal protection system
c. AVC shall provide blank plates to fill in any open areas in the front of the rack. This is to encourage convection airflow inside the rack.
d. Provide a UPS, rack mounted for use with the control CPU and digital signal processors (DSP) etcetera, but not anticipated for audio power amplifiers.
e. For pricing purposes, include a new rack at the facility.

1) Upon site investigation and review of existing rack, if the existing rack can be reused and adhere to the requirements herein, the existing rack shall be reused and a credit shall be provided to the contracting officer.

D. Conference Room general system requirements:

1. Audio Systems: Provide a Video Sound Bar (VSB) that will be mounted below the professional grade flat panel monitor
2. Microphone array will be internal to the VSB.
3. PTZ camera is integral to the VSB
4. HDMI and USB-C laptop connectivity will be provided.
5. Control System a 10.0” touch panel will be provided for system ease of use.

1.6 AV CONTRACTOR’S GENERAL RESPONSIBILITIES

A. The Contractor shall be responsible for delivering a complete and fully functional turnkey system to the Contracting Officer. The Contractor shall provide equipment and materials, whether specifically mentioned herein or not, to ensure a complete and operating system. The “Not in Contract” (NIC), “Provided by Others” (PBO) and “Contracting Officer Furnished Equipment” (OFE) designated equipment and materials are specifically exempted from this requirement.

B. AV systems, infrastructure, raceway and equipment are sensitive to environmental conditions including but not limited to temperature, dirt, dust, and water. The AV contractor shall ensure the storage and installation of all AV Systems components are sequenced and scheduled accordingly to prevent any damage, loss of performance, and warranty void of such systems. All miss-installed components shall be replaced with new parts and re-installed at the AV Contractor’s expense.

C. It is expected that the Contractor shall be accountable for a functional system regardless of the packages selected by the Contracting Officer and that any deficiencies or added expenses created by the Contracting Officers’ selections be brought to the Contracting Officers attention before acceptance of the contract.

D. AV Ethernet Network Security: The AV Contractor shall extensively coordinate with the Contracting Officer and /or their network staff etcetera to fully understand their network protocols and the security feature set they utilize so that the AV Network shall be installed and remain secure to outside cyberthreats at all times.

1.7 RELATED WORK BY OTHER TRADE CONTRACTORS

A. General Contractor (Contractor): It is the responsibility of the GC to furnish and install all steel not shown on the Structural drawings but required for the Audio-Video equipment rigging and support. Additionally, the GC shall provide plywood backing to support video flat panel monitors, video walls, etcetera.

B. Electrical: Is it the responsibility of the EC to furnish and install the work described in the Electrical documentation as well as all electrical infrastructure for AV systems. All high-voltage work shall be done by the EC only.

1. It is anticipated that the AV equipment rack grounding bar shall be provided and installed by the EC / Tele-Data Contractors. See drawing detail sheets for additional information.

C. Tele-data: It is the responsibility of the Tele-Data / IT Contractor to furnish and install network data drops (including those designated as Dante signal types) and wireless LAN access point equipment that shall be utilized by the AV Systems and one-line diagrams. AVC shall closely coordinate their requirements with the Tele-Data Contractor.

D. Fire Alarm / Life Safety: The AVC shall coordinate the requirement of a contact closure & relay devices from the Fire Alarm Panel to the AV Systems low voltage control CPU. It is required so that the AV System shall mute during an “event”. All cabling etcetera from the FA panel to the AV head-end is by the FA Contractor.

1.8 SUBMITTALS

A. Long Lead-Time Items: Prior to submittal of Shop Documentation, and within one (1) month of contract award, the AVC shall submit a list of anticipated long lead-time items (i.e., items with a lead time of eight (8) weeks or longer) for early approval. It is expected that all items listed in this submittal be furnished immediately upon approval.

B. Shop Documentation: The Shop drawing and data sheet submittals shall be submitted together, as one (1) package. The Shop Documentation submittal shall include the following items to be submitted simultaneously, no exceptions.

1. Bill of Materials: A complete list with quantities of equipment to be provided for the systems, including that required for items which are to be fabricated by the AV Contractor (i.e., a bill of materials). Products shall be listed in the same order as found herein. No exceptions.
2. Product Data: A complete set of manufacturer’s equipment specifications with detailed technical information describing and illustrating all components and materials to be used in the installation of the AV systems. This would include those items listed by the manufacturer and model number herein and any other components needed to provide

complete functional systems. Highlight in YELLOW what devices and accessories are to be installed, No exceptions.

3. Cabling Schedule: A complete materials list containing cable type, cable marker identifier, connector type, and origin and destination location for each cable.
4. Wireless Microphone Frequency Coordination:

    a. Provide on-site scan of UHF radio spectrum, from 470-608 MHz, using a calibrated spectrum analyzer (i.e., RF Venue). Show the noise floor in dBm with a horizontal line.
    b. Using the scanned spectrum as well as the digital television (DTV) station database information available from the microphone system manufacturer's website, determine the specific frequency bands best suited for the project, to minimize intermodulation distortion and channel conflicts, and maximize channel count. Include the specific frequency band for each component on the shop drawings (below).
    c. Submit proposed frequencies for each channel of wireless microphones within the proposed frequency band, taking into account second- and third-order harmonics to prevent intermodulation distortion, and avoiding DTV stations. Indicate and include any additional sources of RF in the space, such as in-ear monitors (IEMs), wireless intercom headsets, etcetera. Provide at least 1 spare frequency for every 4 channels of wireless microphones.
    d. After installation, during the testing phase, perform "real-world" frequency coordination, using "war gaming" techniques. Verify in sequence that there is no RF signal on each microphone receiver when its transmitter is turned off.

5. Shop Drawings: The shop drawings shall be submitted after the award of contract and be of a scale suitable for use for fabrication. AV Contractor shall make the following shop drawings submittals:

    a. A one-line signal flow block diagram showing exactly the manner in which the AV Contractor proposes to install the system. Show all switches, modifications to equipment, relay and control circuits. Show all equipment/apparatus items which are required for realization of the functions described herein.
    b. AC Power sequencing one-line that will be used in larger applications. Assume each power amplifier shall have its own circuit. This must be coordinated with the EC.
    c. Detail diagrams of items which are to be fabricated by the AV Contractor and/or which the AVC intends to fabricate or has fabricated, including, but not limited to, the custom panels and receptacle plates. They shall show materials, finishes, panel/control markings, and complete dimensions when applicable.
    d. Detailed diagrams showing equipment rack layouts.
    e. Detailed diagrams showing loudspeaker mounting details with all hardware identified by manufacturer and model number.
    f. A signal loss diagram for the coaxial distribution of video signals with calculations to verify that the signal strength at each endpoint is within industry specifications.
    g. Any wireless microphone or monitor system shall indicate its proposed frequency band (such as G50 or J50A) on the one-line signal flow interconnect diagram.
    h. Important Note: Submit sight-line calculations regarding FDA 3640 with elevations drawings of each location where a laser projector is to be installed and how its light path does not provide a hazard to human sight. AVC shall coordinate with projector manufacturer regarding this important issue.

i. AVC shall maintain on-site a hardcopy of their shop drawings detailing the latest as-built conditions.

C. Control & DSP System Shop Submittal:

1. GUI layouts that demonstrate page flips and system operation.
2. Domain Manager GUI layout and routing maps

D. Written field updates, including photos etcetera (once the site installation begins) shall be required every two (2) weeks.

E. Closeout Documents: Prior to Final Acceptance, AV Contractor shall submit the following:

1. Test reports, as specified herein. These documents are required twenty-one (21) calendar days before the anticipated final acceptance testing, else the appointment shall be rescheduled.
2. Updated control system and DSP source code and executable files, with all tunings and adjustments incorporated.

    a. It Shall be noted that all programming code, executables etcetera shall be provided to (and owned) by the Contracting Officership in perpetuity.

3. Updated frequency coordination, including specific “real world” frequency selection, showing no intermodulation distortion or channel conflicts.
4. Complete and accurate Final As-Built Drawings prior to final acceptance testing.
5. O & M manuals / documentation, programmable software files and executables as specified herein.
6. Written warranty, as specified herein, to take effect after the Final Acceptance Testing has occurred and any punch list-items have been completed.

1.9 QUALITY ASSURANCE

A. Manufacturer’s Qualifications: All electronic products shall be designed and marketed by the manufacturer for application in professional sound amplifying, reproduction and recording systems.

B. AV Contractor’s Qualifications: AVC shall meet all the requirements of the AV Contractor Qualification Statement, as described herein.

1. The AV Contractor shall have on their full-time payroll at least on staff engineer having five (5) years minimum experience as an audio design engineer. In place of a qualified staff engineer, the AV Contractor may retain a consulting engineer to direct the project. Prior to bid acceptance, the qualifications of the consulting engineer shall be submitted for the approval of the Contracting Officer.

    a. The staff or consulting engineer shall:
    b. Provide all technical liaison between the AV Contractor and the Contracting Officer.
    c. Represent the AV Contractor at meetings and conferences and be present at the job site for final inspection.

d. Be responsible for the supervision of all technical and engineering work required to execute the contract and approve and sign the shop drawings.

2. The senior AVC project staff presented in the AV Contractor Qualification Statement shall remain on the project (at a minimum) in a supervisory role until the completion of all AV-related work.

C. Minimum Performance Requirements: To meet the minimum performance requirements, the AV Contractor shall be responsible for:

1. Installing Assistive Listening System components in conformance with ADAAG (including 216.10 and 706) code requirements, as required by law for indoor / outdoor spaces.
2. Each component's conformance with the manufacturer's published specifications and other requirements as stated herein.
3. Detailed checking of each item of equipment provided, each portion of the installation, and of the complete installation to find and remedy any defects therein.
4. Setting the system gain structure, including initial adjustment of digital signal processing, and amplifier gain controls, and, under the direction of the Contracting Officer, making the final adjustment of these items and, if so directed, re-aiming loudspeakers, resetting loudspeaker levels and providing additional equalization and DSP programming, as necessary.

D. Final Acceptance: Final Acceptance will be contingent upon issuance by the Contracting Officer of a letter of acceptance stating that the work has been completed and is in accordance with the contract documents. This is when the warranty period shall begin and not at "Substantial Completion".

1.10 DELIVERY, STORAGE, AND HANDLING

A. The AV Contractor shall ensure the storage and installation of all AV Systems components are sequenced and scheduled accordingly to prevent any damage, loss of performance, and warranty void of such systems.

B. All incorrectly installed components shall be replaced with new parts and re-installed at the AV Contractor's expense.

1.11 WARRANTY

A. Provide a written warranty, signed by the AV Contractor, due when the AVC issues the As-Built Documentation. Warranty period shall equal the requirements described the DIVISION 00 sections.

B. Include the following provisions:

1. Warranty all equipment and the installation to be free of faulty workmanship.
2. Warranty all components, including solid state devices, to be free of defects.

3. Paint and exterior finishes, fuses and lamps are excluded from above warranty, except when damage or failure results from defective materials or workmanship covered by the warranty.

C. If, within the warranty period of the installation or within a longer period of time as may be prescribed by law or by the terms of any applicable special warranty required by the Contract Documents or provided by a manufacturer, any of the work or equipment is found to be defective or not in accordance with the Contract Documents, the AV Contractor shall correct it promptly including all parts and labor after receipt of notice from the Contracting Officer to do so unless the Contracting Officer has previously given the AV Contractor a written acceptance of such condition. This obligation shall survive termination of the contract. The Contracting Officer shall give such notice promptly after discovery of the condition. Such notice shall be provided by Contracting Officer representatives, to be identified, either verbally or in writing.

D. Nothing contained in the Contract Documents shall be construed to establish a shorter period of limitation with respect to any other obligation which the AV Contractor might have under the Contract Documents or any manufacturer's warranty. The establishment of the time period noted above, after the date of Final Acceptance or such longer period of time as may be prescribed by law or by the terms of any warranty required by the Contract Documents, relates only to the specific obligation of the Contractor to correct the work or equipment, and has no relationship to the time within which their obligation to comply with the Contract Documents may be sought to be enforces, nor to the time within which proceedings may be commenced to establish the Contractor's liability with respect to their obligations other than specifically to correct the work or equipment.

E. The Contracting Officer reserves the right to expand or add to the system during the warranty period using firm(s) other than the AV Contractor for such expansion without affecting the Contractor's responsibilities, provided that the expansion is done by a firm which is an authorized dealer or agent for the equipment of system being expanded.

F. Structured Cable:

1. The manufacturer shall provide a warranty with a minimum term of twenty-five (25) years for structured cabling and all communications cable infrastructure components. This warranty shall cover all components including cables, jacks, patch panels, and wiring panels, etc. to maintain the specified performance, physical criteria, and applications assurance.
2. Any such components, link, or channel shall be replaced by the manufacturer at no cost to Contracting Officer during this period.
3. The Contractor and Manufacturer shall submit all information and documentation on Warranty

G. Service Warranty:

1. Replace defective materials and repair faulty workmanship within 72 hours of discovery at no cost to the Contracting Officer during the period of the Warranty.
2. If during the warranty period, system operation is not fully restored, or a working temporary solution is not deemed acceptable by the Contracting Officer, within four (4) business days, the Contracting Officer reserves the right to require the Contractor to provide on-site manufacturer's service technicians at no additional cost.

3. At no additional charge, provide maintenance service for a period of one (1) year after the Contracting Officer's Final Acceptance of installation. This service shall consist of at least one (1) visit six (6) months and one (1) visit eleven (11) months after Final Acceptance for regular maintenance responsibilities, checking equipment and systems operations / functionality, and rigging safety. The AVC shall make all necessary adjustments in a timely manner.
4. If, during the warranty period, any component is out of service for more than four (4) days due to unavailability of parts or service, supply and install an identical new component. If an identical component is not available, temporarily substitute equivalent equipment, but only with the written approval of the Contracting Officer.

PART 2 - PRODUCTS

2.1 DEVICE QUANTITIES

A. Every attempt has been made to provide an accurate and detailed equipment schedule. However, when a discrepancy is found, the quantities found on the one-lines shall take precedence over the equipment schedule specified herein. AVC shall issue a pre-bid RFI requesting clarification no later than five (5) working days prior to the bid due date. Responses shall be provided to all bidders prior to the bid due date.

2.2 EQUIPMENT PERFORMANCE SPECIFICATIONS

A. For all items which are identified by part number and manufacturer, performance specifications which are published in the most recent manufacturer's data sheets available at the time of bidding this contract shall be applicable to the present work as though fully written out herein.

B. For those items which are not identified by part number and manufacturer, the AV Contractor may select items which conform to the functional and/or technical specifications. For all such items, the AV Contractor shall submit for approval all technical data which is available from the supplier or manufacturer.

C. All terminated and bulk cable types shall be submitted in the shop documentation prior to purchase and installation.

2.3 EQUIPMENT SCHEDULE:

A. The following schedule provides devices that meet the functional criteria for the "Basis of Design" for this project.

B. Products and devices submitted for consideration must meet the "equal or better" criteria to be considered viable for this project.

Tag	Description	Manufacturer	Model Number	Notes
	Audio Equipment			
S1	Ceiling loudspeaker, large format w/ tile bridge and enclosure, complete	Atlas (or approved equal)	FAP63T (or approved equal)	70.7 volt / Tap @ 32 W
S2	Directional recessed ceiling loudspeaker	Bose (or approved equal)	EdgeMax EM180 (or approved equal)	Left. Center, & Right and rear surround sound loudspeakers
S3	Ceiling subwoofer, large format, w/ tile bridge and enclosure, complete	Sonance (or approved equal)	PS-C83RWT (or approved equal)	Color coordinated with contracting officer
	Digital Signal Processor, w/ modules & Dante, complete	QSC (or approved equal)	CORE 24F (or approved equal)	Quantities as required; all licenses included
	Surround sound digital audio processor	Extron (or approved equal)	SSP 200 (or approved equal)	
	Amplifier, 300W x 8 channels, w/ Dante	Crown (or approved equal)	**DCi 8 | 300DA (or approved equal)**	
MC1	Microphone array, complete	Shure (or approved equal)	MXA920 (or approved equal)	Ceiling Mounted
	Wireless microphone system w/ Dante UHF system, batteries, Battery charger. Etcetera, complete	Shure (or approved equal)	ULXD4D-GV (or approved equal)	Combination system w/ hand held and head set microphones, submit broadcast frequency ranges in Shop Document Submittal
WM1	Wireless Antenna, remote	Shure (or approved equal)	UA864US (or approved equal)	
	RF bandpass filter	RF Venue (or approved equal)	BPF custom (or approved equal)	NOTE: AVC to select the best band-pass filter for RF microphone frequency to be installed
BT1	Bluetooth receiver	Symetrix (or approved equal)	**xIO** Bluetooth (or approved equal)	Color coordinated with contracting officer.
ALS	ALS system, Wi-Fi based, with Dante, w/ earbuds, neck loops, rechargeable batteries, chargers, placard etcetera, complete	Listen Tech (or approved equal)	ListenAURA (or approved equal)	All quantities shall be based on ADA code requirements. Receiver quantities based on occupancy
	Analog to AES67 / Dante Converter	Crestron (or approved equal)	DM-NAX-AUD-USB (or approved equal)	

Tag	Description	Manufacturer	Model Number	Notes
M1	Push to talk microphone & base	Sennheiser (or approved equal)	MEG 14-40-L II & MAT 153-S (or approved equal)	
	Boundary wireless microphone system, complete	**Shure (or approved equal)**	**ULXD/6C (or approved equal)**	**Requires wireless receiver**
	Video Equipment			
AV1	HDMI / USB-C to HDBT Transmitter	Crestron (or approved equal)	DM-NVX-E20-2G-x-T (or approved equal)	
	HDBT Encoder	Crestron (or approved equal)	DM-NVX-E760 (or approved equal)	
AV2	2 XLR Microphone Input Plate, Dante	Symetrix	X|O –XLR (or approved equal)	
	HDMI Decoder, w. Dante	Crestron (or approved equal)	DM-NVX-363 (or approved equal)	
~~FP98~~	~~Flat panel Monitor, 98"D~~	~~Sharp (or approved equal)~~	~~PN-ME982 (or approved equal)~~	
~~FP86~~	~~Flat panel Monitor, 86"D~~	~~Sharp (or approved equal)~~	~~PN-ME862 (or approved equal)~~	
FP75	Flat panel Monitor, 75"D	Sharp (or approved equal)	PN-ME752 (or approved equal)	
FP32	**Flat panel Monitor, 32" D**	**Sharp (or approved equal)**	**PN-M322 (or approved equal)**	
	Portable video flat panel monitor cart	**B Tech (or approved equal)**	**BT8506 (or approved equal)**	
	Digital signage player w/ 64G micro SD card	Brightsign (or approved equal)	**XT1144 (or approved equal)**	Contracting Officer provided
PS1	Recessed motorized tab-tensioned projection screen wl LVC, wall switch, etcetera, complete	Draper (or approved equal)	Access V (or approved equal)	**16:9 Aspect ratio 73.5" x 130.6" (Matte White)**
PJ1	Video projector, WUXGA (/ UHD), 10,000 lumens, w/ lens etcetera, complete	Sharp NEC (or approved equal)	NP-PA-1004UL-W-41 (or approved equal)	
PTZ1	HD PTZ Camera	Vaddio (or approved equal)	**RoboSHOT 30E HDBT (or approved equal)**	
VSB	Conference AV system w/ touch panel control, complete	Cisco (or approved equal)	Room Bar (or approved equal)	Conference Room 120

Tag	Description	Manufacturer	Model Number	Notes
	4k Blu-ray player w/ RS-232 Control	Tascam (or approved equal)	BD-MP4K (or approved equal)	
	Wireless (WIFI) video display receiver	ScreenBeam (or approved equal)	**1100 Plus** (or approved equal)	
	Network & Control System			
	Control System CPU, complete	Crestron (or approved equal)	CP4N (or approved equal)	
TP10	Touch Panel, 10.0", fixed, complete	Crestron (or approved equal)	TSW 1070-B-S (or approved equal)	
	Ethernet PoE+ switch, AV Dedicated	Netgear (or approved equal)	4250 series (or approved equal)	AV Network, Quantities as required
USB	USB-C Extender	C2G (or approved equal)	Superbooster® 1-Port USB (or approved equal)	Data only
	Hardware / Miscellaneous			
-	~~Flat panel wall mount, tilting, large~~	~~Chief (or approved equal)~~	~~LTM1U (or approved equal)~~	-
	~~Large Flat Panel Mobile AV Cart~~	~~Legrand (or approved equal)~~	~~PFCUB (or approved equal)~~	
-	~~Flat panel wall mount, tilting, large~~	~~Chief (or approved equal)~~	~~XTM1U (or approved equal)~~	~~98.0" UNIT~~
BB1	Flat panel back box	Chief (or approved equal)	PAC 526 (or approved equal)	AVC provided EC installed
PJ1	Projector mount	Chief (or approved equal)	RPA Series (or approved equal)	
PB1	Ceiling plenum box	Chief (or approved equal)	CMA473 (or approved equal)	
	Ceiling plenum box mount	Chief (or approved equal)	CMA440 (or approved equal)	
	Equipment Rack, wall mount w/ locking doors & panels, fans, thermal sensor system, blank panels etcetera, complete	Middle Atlantic (or approved equal)	BGR-45SA-32 (or approved equal)	
	AV equipment rack AC power sequencer & power distribution and management	Lyntec (or approved equal)	NPAC-120-x (or approved equal)	AVC shall submit power sequencing one-line in Shop Drawings

Tag	Description	Manufacturer	Model Number	Notes
	UPS, rack mounted	APC (or approved equal)	SMT1500RM2UNC (or approved equal)	
	Cabling, bulk & pre-terminated	Custom (or approved equal)	Custom (or approved equal)	See Part 2.05 below for additional info on approved cabling
	CAT6A / RJ45 Patch Bay	Belden (or approved equal)	Custom (or approved equal)	Contractor provided and installed, CAT6A shielded or better
	END			

2.4 WIRING AND CONDUCTORS

A. General Notes:

1. Any specific cable AWG gauge detailed in the drawing package supersedes this document. These cable types are cited to illustrate the type and quality of cable required. Plenum cable must be utilized as required. Unless otherwise noted, cables from other reputable manufacturers will be considered acceptable only if data sheets are submitted and approved by the Contracting Officer prior to installation. The AV Contractor must verify cable lengths and confirm the suitability of the cables listed above but may change based on field conditions. Where signal loss is beyond anticipated norms, the AVC shall coordinate with manufacturers and the Contracting Officer to select a cable that will meet or exceed the requirements. No exceptions.
2. Cables running in plenum areas without conduit shall be plenum rated cable and match the specified cable below. It is the responsibility of the bidder to inspect the electrical drawings and verify in what spaces plenum cable shall be used. No claims for additional monies, based on the use of plenum cable, will be allowed.
3. All cables (except video and pulse cables which must be cut to an electrical length) shall be cut to the length dictated by the run. No splices shall be permitted in any pull boxes without prior permission of the Contracting Officer. For equipment mounted in drawers or on slides, the interconnecting cables shall be provided with a service loop of appropriate length based on the bend radius of the cable etcetera.

2.5 FABRICATED MATERIALS

A. Designation / Engraving

1. All equipment controls, receptacles and all indicators shall have, unless otherwise noted, Lamacoid / Phenolic, permanently engraved, or silk-screened fully descriptive identification labels. The resolution of silk-screened labels shall not be less than 90,000 dots/sq. in.
2. The word “engraved” shall mean that the identification shall be engraved into the indicated panel, plate or control surface.

3. All engraving shall be with 5/32-inch high characters, unless otherwise noted, and shall be filled with engraver's enamel of a contrasting color.

B. Receptacle Plates:

1. Receptacles shall be screwed, bolted or flush-riveted to the plate.
2. Plates shall be marked to indicate function and circuit of the receptacle, for example: MICROPHONE, USB-C VIDEO, USB-C DATA etcetera. Each receptacle shall have a discrete and unique identifier, for example: MIC 1, MIC 2, etc. All identifiers shown on the Contract drawings are for reference only.
3. All wall plates shall be powder coated rather than having an adonized finish.
4. Color and finish of blank panels and custom assembly panels shall match adjacent equipment panels to the extent possible. The finish of wall mounted receptacle panels shall be coordinated with the Contracting Officer.

C. Terminals and Terminations:

1. Microphone cables shall not be spliced or otherwise interrupted from termination to termination. All cables shall have visible adhesive identifying markers on each end.
2. Wiring in terminal cabinets shall be tied and clamped neatly to backboards or cable-forms.
3. Terminals shall, unless otherwise specified, be one of the following types:

    a. Wire-wrap type terminal block.
    b. Barrier strips with screw terminals.

4. Each terminal strip shall have a legibly marked identification strip.

D. Structural Assemblies:

1. Provide holes required for securing other components or assemblies to structural steel framing and for passage of other components through steel framing members as shown on final shop drawings. Cut, drill or punch holes perpendicular to metal surfaces. Do not flame cut holes or enlarge holes by burning.
2. In fabricating mounting brackets and other steel components requiring bends, the radii of the bends shall not be less than three times the thickness of the steel being bent.
3. All bolts and nuts used in the fabrication of mounting hardware shall be Grade 5 or better, i.e. rated five (5) times the anticipated load.

    a. Surface Mount Equipment Color:
    b. Surface mounted AV Equipment, Wherever possible AVC shall coordinate the color of that device, i.e. touch panels, loudspeakers, projectors etcetera with the Contracting Officer prior to purchase.

2.6 TESTING EQUIPMENT FOR SYSTEMS PERFORMANCE VERIFICATION

A. The AV Contractor shall furnish and utilize the following test equipment for their own use during their initial tests and adjustments and, if necessary, during Final Acceptance testing of the system by the Contracting Officer. Equipment listed in this article is not intended to be provided to the Contracting Officer or Project site.

1. Data Cable Analyzer: Hardware and software requirements include but are not limited to:

    a. Level III accuracy in accordance with ANSI/TIA-1152
    b. Permanent Link Adapters: RJ45 Adapter must meet requirements for NEXT, FEXT and Return Loss in accordance with ANSI/TIA-568-C.2 Annex C. Twisted pair Category 6A, 7 or 7A cords are not permitted as their performance degrades with use and can cause false Return Loss failures.
    c. Results Storage: Must be capable of storing and reporting results

2. 250 MHz Dual Trace Oscilloscope: With external video trigger capabilities
3. Audio Analyzer: Hardware and software requirements include but are not limited to:

    a. Class 1 device which shall be calibrated in the field. Shall meet ANSI S1.4 specifications for Type I sound level meters.
    b. 1/12-octave real time FFT spectrum analyzer, with a minimum of six (6) curve integration averaging
    c. 1/12-octave band filter set meeting ANSI S1.11 specifications
    d. Audio signal test measurement set
    e. SPL functionality including dBA and dBC weighted scale functionality
    f. STI-PA software or discrete device where intelligibility is required as part of the project scope

4. Audio Signal Generator: Continuously variable from 20.0 Hz to 20.0 kHz within +1.00 dB with less than 0.50 percent THD. Include the ability to produce Pink and White noise, sine wave (variable frequency) and sweep generator, with variable output: -60.0 dB-V to +4.00 dBm
5. Distortion Analyzer: Tunable 1/12-octave band wave analyzer allowing readings down to 1.00 percent of the fundamental for measuring THD, hum amplitude and noise spectrum shape

B. All field test instruments shall be within the calibration period recommended by the manufacturer.

PART 3 - EXECUTION

3.1 INSTALLATION PRACTICES

A. General:

1. The AV Contractor shall coordinate with EC on conduit/junction box locations for audio equipment and routing of audio, control, and power cables/conduits from terminals, poke-thru /floor and pull boxes, wall plates, stub-ups etcetera to system equipment racks. This shall be done prior to the installation of drywall, ceilings etcetera.
2. Installation shall include the delivery, unloading, setting in place, fastening to walls, floors, ceilings, counters, or other structures where required, interconnecting wiring of the system components, equipment alignment and adjustment, and all other work whether or not expressly required herein which is necessary to result in complete operational systems.
3. The contractor shall construct AV equipment racks and sub-assemblies, including all equipment to be installed therein off-site. All wiring work, labeling of wiring, cable dressing, hardware supports, and connection panels, cable wiring documentation, socket

installation, ventilation and power supply regulation and filtering component installation shall be performed in AV Contractor's own premises.

4. The Contractor shall not deliver equipment in original packaging to the site for installation. All such equipment will be unpacked and checked thoroughly in AV Contractor's premises. The contractor shall test such equipment as it is received to ensure that it conforms to the manufacturer's specifications. On no account shall the Contracting Officer be liable for any delays of completion of the installed system due to defective equipment being received by Contractor.
5. Keep the job adequately staffed at all times. Designate a field supervisor to be present on the job site and in responsible charge during all phases of installation and check out. Maintain the same supervisor throughout the execution of the work unless circumstances beyond the control of the AV Contractor intervene. Install the system in cooperation with other trades in order to achieve coordinated progress and satisfactory final results. Watch for conflicts with work of other trades on the job. Execute without claim for extra payment, moderate moves or changes as are necessary to accommodate other equipment or preserve symmetry and pleasing appearance.
6. In the installation of equipment and cable, consideration shall be given not only to operational efficiency, but also to overall aesthetic factors. Any issue regarding a visual aesthetic shall be brought to the attention of the GC and Contracting Officer immediately so that all parties can coordinate and provide an adequate solution in a timely manner.
7. For the purposes of coordination with Contracting Officer s specified furniture, the AV Contractor will ensure that such equipment or mounting hardware is compatible with and suitable for installation in that furniture. It shall be the AV Contractor's responsibility to ensure they coordinate with the furniture and / or millwork contractor and that parties shall exchange and follow their Shop Drawings to ensure that dimensions and structural supports are adequate for the AV equipment installation. It is the AV Contractor's responsibility that the request and delivery of such critical coordination information is satisfactorily executed and in as much as the Contractor has control over the delivery of such information, the AV Contractor shall deliver it as requested by the Contracting Officer.
8. The equipment specified herein must be capable of operation in environments of normal humidity, dust and temperature. AV Contractor should avoid installing equipment where extreme environmental conditions can occur and shall be responsible for protection of equipment and related wiring from such conditions.
9. The AV Contractor shall take precautions to prevent electromagnetic and electrostatic hum. Install the equipment to provide safe operation. Provide ventilation as required to maintain equipment within the manufacturer's specified temperature limits.
10. The AV Contractor shall do all cutting and patching necessary for proper installation of the system and shall repair any damage done by themselves or their workmen. Any required penetration of slabs or CMU walls shall be made by a cutting method rather than by a concussive method.
11. All installation practices shall be in accordance with, but not limited to, the general design and construction requirements of the Contracting Officer, and these specifications and drawings. Installation shall be performed in accordance with the applicable standards, requirements and recommendations of the Contracting Officer. Before commencing work, the Contractor shall familiarize all project team members with all of these requirements.
12. If, in the opinion of the AV Contractor, an installation practice is desired or required, which is contrary to these specifications or drawings, a written request for modification shall be made to the Contracting Officer. Modifications shall not commence without written approval.

13. During the installation, and up to the date of issuance of the Final Acceptance Memorandum, the AV Contractor shall be under obligation to protect their finished and unfinished work against damage and loss. In the event of such damage or loss, they shall replace or repair such work at no cost to the Contracting Officer.

B. Security:

1. Provide a project security program to:

a. Protect Work, stored products, and construction equipment from theft and vandalism.
b. Protect Contracting Officer's operations at site from theft, vandalism or damage from AV Contractor's work or employees.
c. Maintain security program throughout construction period, until Contracting Officer occupancy or Contracting Officer acceptance precludes the need for AV Contractor security.

C. Coordination:

1. All work shall be reviewed and coordinated with the GC prior to commencing work.
2. The AV Contractor shall coordinate all AV equipment connections, panels and control locations with the Contracting Officer prior to installation.
3. Coordinate installation with Structural, Electrical, Mechanical, Plumbing, Fire Protection, and other trades to eliminate disruption and/or conflict with other systems.
4. Sequence installation of the AV Systems and infrastructure with other work to minimize possibility of damage and soiling during remainder of construction.

3.2 GENERAL INSTALLATION PROCEDURES

A. Locate all apparatuses requiring adjustments, cleaning or similar attention so that they will be accessible.

B. All boxes, equipment, wall plates, loudspeakers, antennae, cabling etcetera shall be secured plumb and square.

C. All supporting structures and enclosures supplied by the AV Contractor, not having a standard factory paint finish, shall be painted. Paint specifications and color shall be supplied by the Contracting Officer.

D. Clearly, logically and permanently mark switches, connectors, jacks, relays, receptacles, cables and cable terminations.

E. Etch, engrave, silk screen or apply in a similar permanent manner all legends and markings on all custom panels and receptacle plates. Embossed tape, press type, etc., are not acceptable.

F. Provide all cables necessary for interconnection of permanently mounted equipment. Use terminations required to achieve full function of equipment as specified herein.

G. Exercise care in wiring, to avoid damage to the cables and to the equipment. Make all joints and connections with rosin-core solder or with mechanical connectors approved for Class I wiring. Execute all wiring in strict adherence to standard broadcast and BICSI procedures. AVC shall adhere to the cable manufacturers' recommended cable bend radius, no exceptions.

H. Run lines in separate metallic conduits for line level circuits (up to +30.0 dBm), loudspeaker circuits (above +30.0 dBm) and power circuits. Non-metallic or PVC conduits for AV system wiring are not acceptable. Confirm with EC that all metallic raceways have been properly bonded. Use only cables which are insulated from the conduit and from each other for the entire conduit length.

I. All shielded audio cable shall be bonded at both ends. Do not ground audio cable shields only at the source end. Preserve continuity of microphone shields at both connecting points. Connect all audio grounds in this system to a common point.

J. All equipment shall be firmly secured in place unless the equipment has been documented to be portable in nature, either within this Specification or as shown in the Construction Documents.

K. All fastenings and supports shall be adequate to support their loads with a safety factor of five (5) times the load weight or as required by ESTA , whichever is greater. Wind shear must be taken into account when installing all outdoor pole mounted devices.

L. All loudspeakers shall be installed according to the manufacturer's instructions. All loudspeakers shall be installed with a secondary seismic safety harness assembly in addition to the specified loudspeaker mount. No exceptions.

M. It is the responsibility of the AVC to test all loudspeakers and/or loudspeaker line arrays for sound pressure levels to verify that the full frequency spectrum coverage area is uniform throughout the listening area. AVC shall reposition and or adjust aiming angles as necessary until uniform full frequency coverage is reached per the specification. This work shall be done as part of this contract and no additional funds shall be provided to the AVC to accomplish this requirement. No exceptions.

3.3 BONDING

A. The Contractor shall be responsible for correcting any signal grounding problems within the AV System (excluding Division 26 grounds) including but not limited to electromagnetic / electrostatic hums, ground loops anomalies, and distortions.

B. A bonding buss bar shall be installed at each equipment rack location. The copper buss bar shall be sized to accommodate all connections plus future expansion. See drawing details for additional information.

1. In locations where multiple AV equipment racks exist, they shall be grounded to each other as well as grounding to the AV Grounding Bar (AVGB).

C. Ground all components according to the following methods:

1. Equipment having a power cord without a grounding conductor connected to chassis: Furnish and install 14 AWG grounding conductor from the component's metallic chassis to grounding buss bar within rack.
2. Equipment having a power cord with a grounding conductor connected to chassis: Do not install an additional grounding conductor.

D. Shielded audio signal cables shall have the shields terminated at both ends. No exceptions.

E. All video and data shielded cables shall have the shielded conductors terminated on both ends.

3.4 LAN & ETHERNET NETWORK REQUIREMENTS FOR AV SYSTEMS

A. The Contractor shall coordinate with the Contracting Officer regarding the Ethernet switch requirements and system topography necessary for the AV Systems:

1. 1 G Ethernet switches for audio systems
2. 10 Gbps Ethernet switches for media servers (including audio and video recording media servers).
3. IP Address quantity for AV System
4. Wireless access points for dedicated AV Systems.

3.5 PATCH PANEL ASSIGNMENTS

A. The Contractor shall provide patch panels that shall be the same category as the cable being passed through them and match the more stringent category rating if multiple cable types are terminated there. The AVC shall provide shielded terminations that also match the category of the most stringent cable type.

B. All patch panels shall be wired so that the signal "sources" (outputs from) appear on the upper row of a row pair; and all "loads" (inputs to) appear on the lower row of a row pair.

C. All audio and video patch panel designation strips shall utilize alphanumeric identifications and descriptive information. The jack position in each horizontal row shall be numbered sequentially from left to right. The horizontal jack rows shall be lettered sequentially from top to bottom. The alphanumeric identification of each jack shall be included on the shop documentation functional block drawings, as well as on reproductions of these drawings, which shall be mounted in an appropriate location near the patch bays.

3.6 EQUIPMENT RACK INSTALLATION

A. Supply completely assembled and fully shop-tested racks with all internal wiring completed as required to provide a complete system, to the extent that such items are not provided by others.

B. All power cables, low voltage control cables, and high level cables shall be run on the right side of an equipment rack as viewed from the rear. All other cables shall be run on the left side of an equipment as viewed from the rear.

C. When dressing the equipment rack, cable ties shall be placed at appropriate intervals of approximately six (6) inches for vertical bundles, four (4) inches for horizontal bundles. No cable ties shall cause the cable to exceed the manufacturer's recommended bend radius or deform the cable.

D. All vertical cable bundles internal to AV system equipment racks shall be attached to the rack frame utilizing a cable management system.

E. Use only Velcro-type cable straps to mount and bundle all cables. The use of plastic wire ties is prohibited. No exceptions.

F. All equipment racks shall by fully dressed out and cabled shall not protrude from the back of the rack.

G. All cable entry shall be through the tops of equipment racks or through entrance holes in the base of the rack. No cable shall enter racks through front, rear, or side panel openings.

H. Millwork installed or pivoting racks shall be provided with a sufficient amount of dressed cabling or harness to allow free and untethered travel of the rack from the permanent locking position to the extended "service" position.

3.7 CABLE INSTALLATION

A. All cables for this project must conform to the latest version of NFPA 70 as well as local code requirements.

B. Where existing cabling is to be abandoned, it shall be completely removed as directed in the NFPA 70 / NFPA 72. No exceptions.

C. All cables, regardless of length, shall be marked with permanent wrap-around number letter cable markers at both ends. There shall be no unmarked cables at any place in the system. Wire labels done by hand in the field must be replaced with computer-generated labels for legibility. Marking codes used on cables shall match the codes / labels shown on the AVC shop drawings and on run sheets.

D. CAT6A / 4k cable runs shall not exceed the most current BICSI telecommunications cabling standards, i.e. 100 meters inclusive of all cross-connects, patches etcetera. When the distances are longer, the AVC must coordinate with the cable manufacturer to verify performance is still within the acceptable range of performance for this scope of work. No exceptions.

E. Internal and inter-rack cabling shall be neatly strapped, dressed, labeled, and adequately supported without pinching the cable bundle.

F. Use only Velcro-type cable straps to mount and bundle all cables. The use of plastic wire ties is prohibited. No exceptions.

G. Terminal blocks, boards, strips, or connectors shall be furnished for all cables that interface with racks, cabinets, consoles, or equipment modules. The use of "wire nuts" to terminate or connect cabling is strictly prohibited.

H. All cables and bundles shall be grouped according to the signals being carried. In order to reduce signal intermodulation distortion, separate groups shall be formed for the following cables:

1. Power loudspeaker cables
2. Low voltage system cables
3. Video cables / Category cables
4. Audio cables carrying signal levels less than -20.0 dBm
5. Audio cables carrying signal levels between -20.0 dBm and +20.0 dBm
6. Audio cables carrying signal levels greater than +20.0 dBm

I. All cables shall be continuous lengths without splices. All system wire, after being cut and stripped, shall have the wire strands twisted back to their original lay and be terminated by approved soldered or mechanical means. No bare wire terminations shall be allowed, unless specified on the drawings. No exceptions. Heat-shrink tubing shall be used to insulate the ground or drain wire.

J. All solder connections shall be made with rosin-core solder using temperature-controlled solder stations. No cold or cracked solder joints are acceptable. Any connections, which do not appear to be clean and shiny, or which show signs of cracking, shall be re-soldered by the AV Contractor before Final Acceptance testing of the system.

K. Mechanical connections using insulations, crimp-type connectors shall be bonded to the connector by soldering the wire to the metal part of the connector.

L. Connections made with screw actuated pressure type terminal strips shall be made by stripping approximately 1/4-inch of insulation from the stranded connector. The un-tinned wire shall be inserted into the terminal and tightened.

M. All wire bundles are to be neat and combed free of cable crossovers.

N. No cable shall be installed with a bend radius less than that recommended by the cable manufacturer.

O. All wire markers / labels shall face a common direction.

P. All cables shall have proper connector housing.

Q. All cables located in the ceiling areas, excluding corridors, where conduit as not been provided for are to be J-hooked and kept separate from data cables with crossovers at ninety (90) degrees. Large conduit sleeves shall be required for all inaccessible ceiling areas or hard lid ceilings.

R. All cable passing through penetrations in fire rated walls must be fire blocked after cable installation by the AV Contractor.

3.8 CLEAN UP AND REPAIR

A. During the installation, the Contractor shall be expected to maintain a clean and safe working environment.

B. Upon completion of the work the Contractor shall remove all their refuse and rubbish from and about the premises and shall leave the relevant areas and equipment clean and in an operational state. The Contractor shall be responsible for repairing any damage caused to the premises by the AV Contractor's installation activities, at no cost to the Contracting Officer.

3.9 PROTECTION

A. Protect installed products until completion of the project.

B. Touch-up, repair or replace damaged products before substantial completion.

3.10 AV CONTRACTOR SYSTEM CHECKOUT CHECKLIST

A. General:

1. Before Final Acceptance tests are scheduled, the Contractor shall perform a complete system checkout. The Contractor shall furnish all required test equipment and shall perform all work necessary to determine and/or modify performance of the system to meet the requirements of this specification.
2. For any audio or video system tests performed which yield results indicating failure to meet the performance requirements of this specification, the Contractor shall be held responsible for all costs related to adjustments and aiming of components required to rectify the failed conditions.
3. Maintain onsite documentation of all field related changes / as-builts etcetera as well as performance tests for reference by the Contracting Officer during the site surveys and Final Acceptance testing.
4. Adjust, balance, and align all equipment for optimum quality and to meet the manufacturer’s published specifications. Establish and note normal settings for all device controls and record these settings in the Operation and Maintenance (O & M) Manual.
5. Check all low voltage control system functions and presets, from all controlling devices to all controlled devices, for proper operation.

B. General Audio System Tests:

1. General Inspection and Adjustment: Measure and subsequentially document that all individual components are performing in accordance with each manufacturer’s published specifications. Specifically, examine frequency response, audible harmonic distortion, and signal-to-noise (S/N) ratio. Replace any components found to be defective.
2. Freedom from Parasitic Oscillation and Radiofrequency (RF) Pickup: Check to ensure that the system is free from spurious oscillation and radio-frequency pickup, both in the absence of any audio input signal and also when the system is driven to full output from 20.0 Hz to 20.0 kHz. Utilize a dual trace oscilloscope as specified.
3. Loudspeaker Phasing: Perform phasing checks of loudspeaker lines by means of a DC source at one end of each line and a voltmeter at the other end. Phase all loudspeaker lines identically with respect to color coding.
4. Freedom from Buzzes, Rattles and Objectionable Distortion: Apply a slow sine wave sweep from 20.0 Hz to 20.0 kHz at a level 6.00 dB below rated power amplifier output voltage. Listen carefully for buzzes, rattles, and objectionable distortion. Correct any causes of these defects, unless the cause is clearly outside the sound amplification system

equipment and installation, in which case, the cause shall be brough to the attention of the Contracting Officer.

5. Gain Control Settings: Establish tentative normal settings for all gain controls. All gain controls on rack-mounted equipment shall be adjusted for optimum S/N ratio and signal balance. Settings may require further adjustment by the AV Contractor as a result of testing by the Contracting Officer. After final gain and attenuator settings have been established, mark all control settings with an adhesive indicator dot or arrow on the amplifiers.
6. Freedom from Switching Transient Noise: Operate all AC power sequencing equipment while listening for clicks and pops in the system outputs. Eliminate any found.
7. Initial Listening Test: Listen to normal program material to be sure that there are no remaining defects.

C. Loudspeaker Equalization Tests – 70.7 Volt Distributed Systems

1. Perform all tests and adjustments required to obtain specified performance. Tests and adjustments shall be performed when all ceiling tiles, acoustical treatment panels, sound screens, furnishings, etcetera are in their final position and ready for personnel. HVAC and lighting shall be operating at normal conditions.
2. Measure the ambient sound level in 1/12-octave bands with all HVAC systems and lights on. Record data as 1/3-octave band readings. Also measure and record the ambient sound level as a single wideband C-weighted (dBC) reading.
3. Individually test each loudspeaker circuit or zone extensively for uniform sound pressure levels (SPL) and adjust the system as necessary.
4. With the pink noise (or sweep generator) off and power amplifier on, and with all amplifier attenuation controls set at an 80.0% rotation (or +80.0 dB where applicable) in all areas, the system hum and noise must be inaudible. Any mechanical rattles or audio distortion must be corrected at this time.
5. Apply the pink noise or sweep generator signal to system and adjust for a working level for the equalization of the system of at least +20.0 dB above the ambient sound level. Staff must use ear protection.
6. Check for uniformity of coverage utilizing a 32-band, 1/12-octave Class 1 spectrum analyzer. The level in all areas must not deviate more than +/- 2.00 dBC. Correct any out of tolerance deviations before proceeding to the next zone / loudspeaker circuit.
7. Document all findings. Provide the Contracting Officer “Before” and “After” spreadsheet and resultant frequency response curves and graphical test results for a representative quantity of each speaker type or zone. Note microphone test locations on floor-plan drawings.

D. Loudspeaker Equalization Tests – Low Impedance Loudspeaker Systems

1. Perform all tests and adjustments required to obtain specified performance. Tests and adjustments shall be performed when all ceiling tiles, acoustical treatment panels, sound screens, furnishings, etcetera are in their final position and ready for personnel. HVAC and lighting shall be operating at normal conditions.
2. Measure the ambient sound level in 1/12-octave bands with all HVAC systems and lights on. Record data as 1/3-octave band readings. Also measure and record the ambient sound level as a single wideband C-weighted (dB-C) reading.
3. The DSP shall utilize the loudspeaker manufacturer’s Finite Impulse Response (FIR) files as an initial starting point for equalization.

4. Using an AC impedance meter or bridge, measure all loudspeaker circuit's impedance at 1.00 kHz while disconnected from the amplifier output and note the data. This will assure that the speaker circuits are free of faults and will not overload the amplifier.
5. With the pink noise (or sweep generator) off and power amplifier on, and with all amplifier attenuation controls set at an 80.0% rotation (or +80.0 dB where applicable) in all areas, the system hum and noise must be inaudible. Any mechanical rattles or audio distortion must be corrected at this time.
6. Perform time domain alignment of full frequency loudspeaker and separated subwoofer units. It is recommended that the AVC coordinate with the loudspeaker manufacturer regarding this complex procedure.

    a. Utilize and record an impulse response sweep for both the full frequency loudspeakers and the subwoofers separately.
    b. Time align the full frequency loudspeaker's 1st peak with that on the subwoofer's 1st peak.
    c. Verify time alignment by testing various locations in the space and document microphone locations on floor-plan drawings that correspond to testing results.
    d. Consider using AFMG SysTime V1.3.7 (or newer) as an alignment software tool, or similar.
    e. It is understood that not all locations can be perfectly aligned but the AVC shall adjust the system for the best overall performance with the space under test.

7. Apply the pink noise or sweep generator signal to system and adjust for a working level for the equalization of the system of at least +20.0 dB above the ambient sound level. Hearing protection must be provided for all individuals present.
8. Check for uniformity of coverage utilizing a 32-band, 1/12-octave Class 1 spectrum analyzer. The level in all areas must not deviate more than +/- 2.00 dBC. Correct any out of tolerance deviations before proceeding.
9. System Ring-Out (Feedback Reduction): Once the sound system has been equalized to obtain a flat response, each microphone input (hardwired & wireless) shall ring out using the following procedure or similar. Hearing protection must be provided for all individuals present:

    a. Attach oscilloscope to the output of the sound system.
    b. Turn on one (1) microphone.
    c. Raise system volume until feedback begins.
    d. Use the oscilloscope to find the largest peak.
    e. Add a very narrow parametric filter to "notch" out that frequency on the corresponding microphone channel.
    f. Repeat feedback procedure to eliminate several feedback frequencies. When using wireless microphones, be sure to walk about the space to excite other potential feedback frequencies.
    g. Once feedback has been essentially eliminated from the microphone, (in software) add a compression module or block with moderate compression ratio, attack and release settings. When the microphone is used the effect shall not be noticeable.
    h. Verify that the compression algorithm has not altered the feedback suppression in place.
    i. Test remaining microphone input(s) using the same general procedure.

10. Document all findings. Provide the Contracting Officer “Before” and “After” spreadsheet and resultant frequency response curves and graphical test results for all low impedance full frequency range loudspeakers, sub-woofers and loudspeaker clusters, array devices etcetera.

E. Acoustical Tests

1. Make all necessary adjustments to ensure proper operation of the system. Tests and adjustments shall be performed when all ceiling tiles, acoustical treatment panels, sound screens, furnishings, etcetera are in their final position and ready for personnel. HVAC and lighting shall be operating at normal conditions.
2. The overall space average acoustical frequency response within each area served shall be within the limits given by the Contracting Officer within this specification. Test signals shall be broadband “pink” noise applied to any system input. Measurements shall be made using 1/12-octave band filters centered on ANSI preferred frequencies. Sweep tests can also be utilized.
3. Throughout the coverage area of the system, amplified sound levels shall not vary more than +/- 3.00 dB as measured using an octave band of “pink” noise centered at 4.00 kHz as the test signal.
4. Measurements of system performance shall be made using a calibrated ANSI S1.4, Type I sound level meter set for “slow” meter damping and flat response. The microphone shall be positioned 4.00 feet above the floor within the area served by the system. The system gain shall be set to provide octave band levels at least 10.0 dB above background noise levels in any octave band at the measuring locations for these tests.
5. Low Impedance Systems: Measurements shall be taken in a grid fashion at twenty (20) foot intervals in all directions. Provide a floor-plan map to add clarity that may be used during Final Acceptance Testing by the Contracting Officer.
6. Volt Systems: Measurements shall be taken on axis as well as at the point of overlapping speaker coverage.
7. Upon completion of above tests and any necessary adjustments, submit two (2) copies of a written report presenting test results in graphical format with units clearly marked, including numerical values in a table where necessary, for review by the Contracting Officer prior to Final Acceptance Testing, final tuning and demonstration. With this report, submit written certification that the installation conforms to the requirements stated herein, is complete in all respects, and is ready for inspection, testing and final tuning.

F. Balanced Twisted Pair Cable Testing (CAT6A or newer):

1. All outlets, cables, patch panels, and associated components shall be fully assembled and labeled prior to testing. Any testing performed on incomplete systems shall be redone upon completion of work.
2. Testing shall be performed on each cabling segment. Sampling is not acceptable
3. Testing of horizontal cabling shall not be performed on test equipment with marginal pass/fail notification disabled. If the tester is capable of indicating tests that pass with a measured value closer to the limit than the guaranteed accuracy of the tester, the test result shall be marked (typically as “Pass” or “Fail”).
4. Marginal pass results will not be accepted. The contractor shall correct issues and retest at no expense to the Contracting Officer.

G. General Video Systems Tests:

1. Check to determine that all projectors, projector bases, carts, tables, and mirrors are rigid and vibration-less in operation.
2. Test and record all optical projection images for average light level, light fall-off, image alignment, and size to comply with the manufacturer's performance standards and specification drawings.

H. HDBT Distributed Video System Tests:

1. AVC shall test every possible routing of sources, matrix switching cross-points and end point sinks etcetera and document the results for future reference. This procedure shall take place after the initial category cable point-to-point testing.
2. The Contractor shall provide to the Contracting Officer a copy of the following test results in electronic format in order to verify the AV video equipment has been installed and configured correctly:

    a. The number of HDCP KDVs Keys supported by each source.
    b. The video timing, HDCP use and audio format of each source when operating.
    c. The video timings and supported audio formats for each connected sink.
    d. The video timings and supported audio formats presented in the EDID to each source. The preferred video timing shall be indicated.
    e. The length of cable used on all shielded twisted pair cable used for AV distribution.
    f. The data rate supported by each twisted pair cable used for AV distribution per table below:

3.11	Video Format	3.12	Data Frame Rate (Gbps)
3.13	4320p (8k / 60 fps) + HDR1	3.14	80.19
3.15	4320p (8k / 60 fps)	3.16	64.15
3.17	2160p (4k / 60 fps) + HDR1	3.18	20.05
3.19	2160p (4k / 60 fps)	3.20	17.82
3.21	UHD 3840 x 2160 (60 fps)	3.22	11.9
3.23	1080p + HDR1	3.24	5.57
3.25	1080p (60 fps)	3.26	4.46

3.27	1600 x 1200	3.28	4.86
3.29	1920 x 1200	3.30	4.62
3.31	1080p	3.32	4.44
3.33	1360 x 768	3.34	2.54
3.35	720p / 1080i	3.36	2.22

a. Notes:

1) HDR is defined as 10-bit color (Dolby Vision is optimized at 12-bit color. Not used here.)
2) Chromas Subsampling used for this table was 4:4:4

2. Check that the image is correctly displayed on the video monitors and/or the video projector. Image must completely fill the imaging area and be "square" without the use of any type of "key-stoning" on video projection systems. No exceptions.
3. Test each input using crosshatch signal, checkerboard signal, Moiré and "H" pattern signals, color balance and greyscale / contrast etcetera at the scan rates indicated in items above.
4. Repeat tests at each connection location.

B. Blu-ray / Universal Format Player Tests:

1. Insert the pre-recorded 4k disc into the deck and check operation of the control panel's transport controls as well as picture image quality. Also check that the audio signal is head from the left and right speakers, where applicable.
2. Check that the transport controls are logically presented on the touch screen GUI
3. Check that the audio and video signals are recorded and can be played back on the picture monitor(s) and/or the video projector and the left, right and surround loudspeakers etcetera, where applicable.

C. Network Distributed Video System Tests:

1. AVC shall test every possible routing of sources and end point sinks etcetera and document the results for future preference. This procedure shall take place after the initial category cable point-to-point testing as detailed above.

D. Video Coaxial Distributed Digital Cable System Tests:

1. AVC shall test and verify all gain losses as documented in the Shop Drawing submittals. Any endpoint that requires additional gain or attenuation shall be in compliance with the signal strength the sink device requires.

E. Bluetooth Device Tests:

1. Document which Bluetooth Standard Profiles are being used in the project: A2DP, HFP, HSP etcetera.
2. Document the release version of Bluetooth being used, i.e. 5.2 or newer.
3. Provide a discrete name for all Bluetooth receivers via manufacturer's software.
4. Verify each receiver's ability to perform pairing.
5. Document connectivity range to verify it is within manufacturer's specifications.
6. Document names, test results, and software package used.

F. Assistive Listening System (ALS) Tests:

1. Verify that the entire scope area to be covered by the ALS provides clear audible sound with no discernable distortion.

3.37 DEMONSTRATION AND FINAL ACCEPTANCE TESTING

A. General:

1. Final Acceptance Testing shall not be performed until the AV Contractor's system checkout has been completed, documented, and results submitted to the Contracting Officer for approval.
2. Upon approval of the AVC test reports by the Contracting Officer, and at a time established by the Contracting Officer, demonstrate the operation of each major component of the system and the completed installation. Typically ten (10) working days after the AVC test documentation / As-Built Drawings have been submitted for approval.

B. Site Testing: To be conducted in the presence of the Contracting Officer and shall consist of the following:

1. Final As-Built Drawings, run sheets, O & M Manuals, submitted test results, and other required documents, as detailed in the Construction Documents shall be on hand. One (1) complete set of these documents shall be delivered to the Contracting Officer at this time (one complete set shall have been delivered to the Contracting Officer prior to the scheduling of Final Acceptance Testing in PDF format).
2. A physical inventory will be taken of all equipment on site and will be compared to equipment lists in the contract documents.
3. The AVC shall demonstrate the operation of all system equipment installed.
4. Both subjective and objective tests shall be required by the Contracting Officer to determine compliance with the specifications. The AVC shall be responsible for providing test equipment for this exercise. Therefore, be prepared to repeat and verify test results previously submitted to the Contracting Officer. It is anticipated that final loudspeaker tuning / equalization adjustments shall be made at the direction of the Contracting Officer.

C. After demonstration, assist as required in the following acceptance tests:

1. Listening tests: These tests may include speech intelligibility survey and subjective aural evaluations by observers at various positions under various operating conditions, using live speech and/or recorded music material.

2. Equipment tests: Any measurements of frequency response, distortion, noise or other characteristics and any operational tests deemed necessary may be performed on any item or group of items to determine conformity with these requirements.

D. Control System Operation: AVC shall coordinate with the Contracting Officer to review the operation of the AV system prior to and during installation. During the second training session, the Contracting Officer can request additional changes and updates to the GUI and system functionality if they so choose without additional cost to the project.

E. If the need for adjustment or modification becomes fully evident during demonstration and testing, continue working until the installation operates fully in accordance with the requirement of this specification.

F. In the event significant adjustments are required during the Final Acceptance Testing phase, or defective equipment must be repaired or replaced, tests may be suspended or continued at the option of the Contracting Officer.

3.38 DOCUMENTATION

A. Project Record: The AV Contractor shall supply one (1) clean full-size print of the system drawings which show all changes which were made during fabrication and installation to the Contracting Officer.

B. Maintenance Information:

1. The AV Contractor shall provide technical information for all electronic apparatuses, including but not limited to schematic diagrams and parts lists, manufacturer's installation instructions, operating instructions, and technical specifications.
2. The AV Contractor submittal shall include all shop drawings prepared and used by the AVC, as well as those which were not required to be submitted for approval. This shall include, but not be limited to, wiring diagrams, schedules for identification of building wiring and installation details useful to a maintenance technician.

C. Operation and Maintenance (O & M) Manuals:

1. Provide one (1) hard copy of an instruction manual to the Contracting Officer containing the following:

    a. Table of Contents
    b. Bill of Materials
    c. Instructions for operating the system in all modes of operation and for fulfilling all functional requirements.
    d. List of settings and adjustments for semi-fixes controls.
    e. Manufacturer's sheets of specifications, operating instructions, and service information arranged alphabetically by manufacturer and then by model number.
    f. Detailed wiring diagrams and the simplified one-line diagram.
    g. All system and device software files, executable, GUI etcetera.

D. Quick Reference Guides:

1. AVC to provide the Contracting Officer with two (2) laminated “Quick-Reference Guides” detailing the operation of each standalone AV system.
2. This shall include both touch-panel layouts and general system “how-to” quick-referenced.
3. This guide should be no longer than two (2) pages where both sides of a standard letter sized sheet are used.

3.39 TRAINING AND INSTRUCTION

A. A minimum of sixteen (16) hours of training shall be provided by the AV Contractor, at a time mutually agreed upon between the Contracting Officer and AV Contractor.

1. Eight (8) hours prior to Final Acceptance Testing.
2. Eight (8) hours one (1) month following the initial training. This will allow the Contracting Officer to acquaint themselves with the system so that they can provide follow up questions as to the systems operation.

B. During the training, the participants shall be given opportunities for “hands-on” experience with operating the controls. There shall be visual and audible demonstrations using the systems themselves as aids. Personnel for instruction and training for the AV systems shall be designated by the Contracting Officer.

C. In the event the AVC does not have qualified instructors on staff for projects with sophisticated or complex equipment, the Contractor will provide a manufacturer’s representative for such instruction at no additional cost to the Contracting Officer.

D. Training will be video recorded by the AVC on a DVD / Blu-ray disc or flash drive as designated by the Contracting Officer. AVC will supply the equipment necessary to record the training sessions. The AV Contractor shall provide two (2) electronic copies to the Contracting Officer and one (1) full-size set of documentation.

END OF SECTION 27 41 00

SECTION 28 00 01 – ELECTRONIC SAFETY BASIC REQUIREMENTS

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Work included in this Section applies to Division 28, Electronic Safety work to provide materials, labor, tools, permits, incidentals, and other services to provide and make ready for Contracting Officer's use of electronic safety systems for proposed project.

B. Contract Documents include, but are not limited to, Specifications including Division 00, Procurement and Contracting Requirements and Division 01, General Requirements, Drawings, Addenda, and Contracting Officer/Contractor Agreement. Confirm requirements before commencement of work.

C. Definitions:

1. Provide: To furnish and install, complete and ready for intended use.
2. Furnish: Supply and deliver to project site, ready for unpacking, assembly and installation.
3. Install: Includes unloading, unpacking, assembling, erecting, installing, applying, finishing, protecting, cleaning and similar operations at project site as required to complete items of work furnished.
4. Approved or Approved Equivalent: To possess the same performance qualities and characteristics and fulfill the utilitarian function without any decrease in quality, durability or longevity. For equipment/products defined by the Contractor as “equivalent,” substitution requests must be submitted to Contracting Officer for consideration, in accordance with Division 01, General Requirements, and approved by the Contracting Officer.
5. Authority Having Jurisdiction (AHJ): Indicates reviewing authorities having jurisdiction, including local fire marshal, Fire Code Official, Contracting Officer, and other reviewing entity whose approval is required to obtain systems acceptance.

1.2 RELATED SECTIONS

A. Contents of Section apply to Division 28, Electronic Safety Contract Documents.

B. Related Work:

1. Additional conditions apply to this Division including, but not limited to:

a. Specifications including Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.
b. Drawings
c. Addenda
d. Contracting Officer/Contractor Agreement
e. Codes, Standards, Public Ordinances and Permits

C. Contents of Division 26, Electrical apply to this Section.

1.3 REFERENCES AND STANDARDS

A. References and Standards per Division 00, Procurement and Contracting Requirements and Division 01, General Requirements, individual Division 28, Electronic Safety Sections and those listed in this Section.

B. Codes to include latest adopted editions, including current amendments, supplements and local jurisdiction requirements in effect as of the date of the Contract Documents, of/from:

1. International Code Council:

    a. International Building Code (IBC)
    b. International Fire Code (IFC)
    c. International Existing Building Code (IEBC)

C. Reference standards and guidelines include but are not limited to the latest adopted editions from:

1. ABA – Architectural Barriers Act
2. ADA – Americans with Disabilities Act
3. ANSI – American National Standards Institute
4. ASCE – American Society of Civil Engineers
5. ASHRAE – American Society of Heating, Refrigerating and Air-Conditioning Engineers
6. ASHRAE Guideline 0, the Commissioning Process
7. ASME – American Society of Mechanical Engineers
8. ASTM – ASTM International
9. CFR – Code of Federal Regulations
10. ETL – Electrical Testing Laboratories
11. FM – FM Global
12. ISO – International Organization for Standardization
13. NEC – National Electric Code
14. NEMA – National Electrical Manufacturers Association
15. NFPA – National Fire Protection Association
16. OSHA – Occupational Safety and Health Administration
17. SMACNA – Sheet Metal and Air Conditioning Contractors' National Association
18. UL – Underwriters Laboratories Inc.

D. See Division 28, Electronic Safety individual Sections for additional references.

1.4 SUBMITTALS

A. See Division 01, General Requirements for Submittal Procedures.

B. Provide drawings in format and software release equal to the design documents. Drawings to be the same sheet size and scale as the Contract Documents.

C. “No Exception Taken” constitutes that review is for general conformance with the design concept expressed in the Contract Documents for the limited purpose of checking for conformance with information given. Any action is subject to the requirements of the Contract Documents. Contractor is responsible for the dimensions and quantity and will confirm and correlate at the

job site, fabrication processes and techniques of construction, coordination of the work with that of all other trades, and the satisfactory performance of the work.

D. Provide product submittals and shop drawings in electronic format only, submitted via zip file via e-mail or posted to ftp site. Provide one file per division containing one bookmarked PDF file with each bookmark corresponding to each Specification Section. Arrange bookmarks in ascending order of Specification Section number. Individual submittals sent piecemeal in a per Specification Section method will be returned without review or comment. Copy Contracting Officer on all transmissions/submissions.

E. Product Data: Provide manufacturer's descriptive literature for products specified in Division 28, Electronic Safety Sections.

F. Identify/mark each submittal in detail. Note what difference, if any, exist between the submitted item and the specified item. Failure to identify the differences will be considered cause for disapproval. If differences are not identified and/or not discovered during the submittal review process, Contractor remains responsible for providing equipment and materials that meet the specifications and drawings.

1. Label submittal to match numbering/references as shown in Contract Documents. Highlight and label applicable information to individual equipment or cross out/remove extraneous data not applicable to submitted model. Clearly note options and accessories to be provided, including field installed items. Highlight connections by/to other trades.
2. Include technical data, installation instructions and dimensioned drawings for products, equipment and devices installed, furnished or provided. Reference individual Division 28, Electronic Safety specification Sections for specific items required in product data submittal outside of these requirements.
3. See Division 28, Electronic Safety individual Sections for additional submittal requirements outside of these requirements.

G. Provide complete submittal package meeting all Division 28, Electronic Safety requirements. Incomplete submittal documentation will be returned to without review.

H. Resubmission Requirements: Make corrections or changes in submittals as required, and in consideration of Contracting Officer’s comments. Identify Contracting Officer’s comments and provide an individual response to each of the Contracting Officer’s comments. Cloud changes in the submittals and further identify changes which are in response to Contracting Officer’s comments.

I. Structural/Seismic: Provide weights, dimensions, mounting requirements and like information required for mounting, seismic bracing, and support. Indicate manufacturer's installation and support requirements to meet ASCE 7-16 requirements for non-structural components. Provide engineered seismic drawings and equipment seismic certification. Equipment Importance Factor as specified in Division 01 and in Structural documents.

J. Trade Coordination: Include physical characteristics, electrical characteristics, device layout plans, wiring diagrams, and connections as required per Division 28, Electronic Safety Coordination Documents. For equipment with electrical connections, furnish copy of approved submittal for inclusion in Division 26, Electrical and Division 28, Electronic Safety submittals.

K. Make provisions for openings in building for admittance of equipment prior to start of construction or ordering of equipment.

L. Substitutions and Variation from Basis of Design:

1. The Basis of Design designated product where provided in the specifications or on the drawings, establishes the parameters used for space planning coordination with other disciplines.
2. If substitutions from the basis of design and/or equivalent equipment/products are being proposed, it is the responsibility of parties concerned, involved in, and furnishing the substitute and/or equivalent equipment to verify and compare the characteristics and requirements of that furnished to that specified and/or shown. If greater capacity and/or more materials and/or more labor are required for the rough-in, circuitry or connections than for the item specified and provided for, then provide compensation for additional charges required for the proper rough-in, circuitry and connections for the equipment being furnished. Coordinate with the requirements of "Submittals." For any product marked "or approved equivalent," a substitution request must be submitted to Contracting Officer for approval prior to purchase, delivery or installation.
3. Where manufacturer equipment or model numbers are indicated with no exceptions, substitutions will be rejected.

M. Shop Drawings:

1. Provide coordinated shop drawings which include physical characteristics of all systems, device layout plans, and control wiring diagrams. Reference individual Division 28, Electronic Safety specification Sections for additional requirements for shop drawings outside of these requirements.
2. Provide Shop Drawings indicating access panel locations, size and elevation for approval prior to installation.

N. Samples: Provide samples when requested by individual Sections.

O. Resubmission Requirements:

1. Make any corrections or change in submittals when required by Contracting Officer's review comments. Provide submittals as specified. The Contracting Officer will not be required to edit and/or interpret the Contractor's submittals. Indicate changes for the resubmittal in a cover letter with reference to page(s) changed and reference response to comment. Cloud changes in the submittals.
2. Resubmit for review until review indicates no exception taken or "make corrections noted."
3. When submitting drawings for Contracting Officer's re-review, clearly indicate changes on drawings and "cloud" any revisions. Submit a list describing each change.

P. Operation and Maintenance (O&M) Manuals:

1. Reference individual Division 28, Electronic Safety Specification Sections for additional requirements for operations and maintenance manuals.
2. Submit, at one time, electronic files (PDF format) of manufacturer's operation and maintenance instruction manuals and parts lists for equipment or items requiring servicing. Submit data when work is substantially complete and in same order format as submittals. Include name and location of source parts and service for each piece of equipment.

a. Include copy of approved submittal data along with submittal review letters received from Contracting Officer. Data to clearly indicate installed equipment model numbers. Delete or cross out data pertaining to other equipment not specific to this project.
b. Include copy of manufacturer's standard Operations and Maintenance for equipment. At front of each tab, provide routine maintenance documentation for scheduled equipment. Include manufacturer's recommended maintenance schedule and highlight maintenance required to maintain warranty. Furnish list of routine maintenance parts, including part numbers, sizes and quantities relevant to each piece of equipment.
c. Include copy of complete parts list for equipment. Include available exploded views of assemblies and sub-assemblies.
d. Include Warranty per Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 28, Electronic Safety Sections.
e. Include product certificates of warranties and guarantees.
f. Include copy of start-up and test reports specific to each piece of equipment.
g. Include commissioning reports.
h. Incomplete submittal documentation will be returned to without review. Contractor must arrange for, and bear costs of, the reviewing entity’s additional reviews.

3. Thoroughly instruct the Park Staff as directed by the Contracting Officer in proper operation of equipment and systems. Where noted in individual Sections, training will include classroom instruction with applicable training aids and systems demonstrations. Field instruction per Section 28 00 01, Electronic Safety Basic Requirements Article titled “Demonstration.”
4. Copies of certificates of code authority inspections, acceptance, code required acceptance tests, letter of conformance and other special guarantees, certificates of warranties, specified elsewhere or indicated on Drawings.

Q. Record Drawings:

1. Maintain at site at least one set of drawings for recording “as-constructed” conditions. Indicate on drawings changes to original documents by referencing revision document, and include buried elements and location of concealed items. Include items changed by addenda, field orders, supplemental instructions, and constructed conditions.
2. Record Drawings are to include equipment locations, calculations, and schedules that accurately reflect “as constructed or installed” for project.
3. At completion of project, input changes to original project on CAD Drawings and make one set of black-line drawings created from CAD Files in version/release equal to contract drawings. Submit CAD Files and drawings upon substantial completion.
4. See Division 28, Electronic Safety individual Sections for additional items to include in Record Drawings.

1.5 QUALITY ASSURANCE

A. Regulatory Requirements: Work and materials installed to conform with all local, State and Federal codes, and other applicable laws and regulations. Where code requirements are at variance with Contract Documents, meet code requirements as a minimum requirement and include costs necessary to meet these in Contract. Machinery and equipment are to comply with

OSHA requirements, as currently revised and interpreted for equipment manufacturer requirements. Install equipment provided per manufacturer recommendations.

B. Whenever this Specification calls for material, workmanship, arrangement or construction of higher quality and/or capacity than that required by governing codes, higher quality and/or capacity take precedence.

C. Drawings are intended to be diagrammatic and reflect the Basis of Design manufacturer's equipment. They are not intended to show every item in its exact dimensions, or details of equipment or proposed systems layout. Verify actual dimensions of systems (e.g., cable tray, panels, etc.) and equipment proposed to assure that systems and equipment will fit in available space. Contractor is responsible for design and construction costs incurred for equipment other than Basis of Design, including, but not limited to, architectural, structural, electrical, HVAC, fire sprinkler, and plumbing systems.

D. Manufacturer's Instructions: Follow manufacturer's written instructions. If in conflict with Contract Documents, obtain clarification. Notify Contracting Officer in writing before starting work.

E. Items shown on Drawings are not necessarily included in Specifications or vice versa. Confirm requirements in all Contract Documents.

F. Provide products that are UL listed.

1.6 WARRANTY

A. Provide written warranty covering the work for a period of one year from date of Substantial Completion in accordance with Division 00, Procurement and Contracting Requirements, and individual Division 28, Electronic Safety Sections.

B. Sections under this Division can require additional and/or extended warranties that apply beyond basic warranty under Division 01, General Requirements and the General Conditions. Confirm requirements in all Contract Documents.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Articles, fixtures, and equipment of a kind to be standard product of one manufacture, including but not limited to panels, devices and equipment unless otherwise specified in individual Division 28, Electronic Safety Sections.

2.2 STANDARDS OF MATERIALS AND WORKMANSHIP

A. Base contract upon furnishing materials as specified. Materials, equipment, and fixtures used for construction are to be new, latest products listed in manufacturer's printed catalog data and are to be UL, ETL, or FM listed and labeled or be approved by State, County, and City authorities prior to procurement and installation.

B. Names and manufacturer's names denote character and quality of equipment desired and are not to be construed as limiting competition.

C. Hazardous Materials:

1. Comply with local, State of Iowa, and Federal regulations relating to hazardous materials.
2. Comply with Division 00, Procurement and Contracting Requirements and Division 01, General Requirements for this project relating to hazardous materials.
3. Do not use any materials containing a hazardous substance. If hazardous materials are encountered, do not disturb; immediately notify Contracting Officer.

PART 3 - EXECUTION

3.1 ACCESSIBILITY AND INSTALLATION

A. Confirm Accessibility and Installation requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 28, Electronic Safety Sections.

B. Install equipment having components requiring access (i.e., devices, equipment, electrical boxes, panels, etc.) so that they may be serviced, reset, replaced or recalibrated by service people with normal service tools and equipment. Do not install equipment in obvious passageways, doorways, scuttles or crawlspaces which would impede, or block intended usage.

C. Install equipment and products complete as directed by manufacturer's installation instructions. Obtain installation instructions from manufacturer prior to rough-in of equipment and examine instructions thoroughly. When requirements of installation instructions conflict with Contract Documents, request clarification from Contracting Officer prior to proceeding with installation. This includes proper installation methods, sequencing and coordination with other trades and disciplines.

D. Earthwork:

1. Confirm Earthwork requirements in Contract Documents.
2. In absence of specific requirements, comply with individual Division 28, Electronic Safety Sections and the following:

    a. Perform excavation, dewatering, shoring, bedding, and backfill required for installation of work in this Division in accordance with related earthwork divisions. Contact utilities and locate existing utilities prior to excavation. Repair any work damaged during excavation or backfilling.
    b. Excavation: Do not excavate under footings, foundation bases, or retaining walls.
    c. Provide protection of underground systems. Review the project Geotechnical Report for references to corrosive or deleterious soils which will reduce the performance or service life of underground systems materials.

E. Firestopping:

1. Confirm Firestopping requirements in Division 07, Thermal and Moisture Protection.

2. In absence of specific requirements, comply with individual Division 28, Electronic Safety Sections and coordinate location and protection level of fire and/or smoke rated walls, ceilings, and floors. When these assemblies are penetrated, seal around conduit, raceway and equipment with approved firestopping material. Install firestopping material complete as directed by manufacturer's installation instructions. Meet requirements of ASTM E814, Standard Test Method for Fire Tests of Through-Penetration Fire Stops.

F. Plenums: In plenums, provide plenum rated materials that meet the requirements to be installed in plenums.

3.2 SEISMIC CONTROL

A. Confirm Seismic Control requirements in Division 01, General Requirements, Structural documents, and individual Division 28 Electronic Safety Sections.

B. Earthquake resistant designs for Electronic Safety (Division 28) systems and equipment to conform to regulations of jurisdiction having authority.

C. Restraints which are used to prevent disruption of function of piece of equipment because of application of horizontal force to be such that forces are carried to frame of structure in such a way that frame will not be deflected when apparatus is attached to a mounting base and equipment pad, or to structure in normal way, utilizing attachments provided. Secure equipment and distribution systems to withstand a force in direction equal to value defined by jurisdiction having authority.

D. Provide means to prohibit excessive motion of safety equipment during earthquake.

3.3 REVIEW AND OBSERVATION

A. Confirm Review and Observation requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 28, Electronic Safety Sections.

B. Notify Contracting Officer in writing at following stages of construction so that they may, at their option, visit site for review and construction observation:

1. Underground conduit and wire installation prior to backfilling.
2. Prior to covering walls when electronic safety systems installation is started.
3. Prior to ceiling cover/installation.
4. When main systems, or portions of, are being tested and ready for inspection by AHJ.

C. Final Punch: Costs incurred by additional trips required due to incomplete systems will be the responsibility of the Contractor.

3.4 CUTTING AND PATCHING

A. Confirm Cutting and Patching Requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.

B. In absence of specific requirements, comply with individual Division 28, Electronic Safety Sections and the following:

1. Proposed floor cutting/core drilling/sleeve locations to be approved by Project Structural Engineer. Submit proposed locations to Contracting Officer/Project Structural Engineer. Where slabs are of post tension construction, perform x-ray scan of proposed penetration locations and submit scan results including proposed penetration locations to Project Structural Engineer/Contracting Officer for approval. Where slabs are of waffle type construction, show column cap extent and cell locations relative to proposed penetration(s).
2. Cutting, patching and repairing for work specified in this Division including plastering, masonry work, concrete work, carpentry work, and painting included under this Section will be performed by skilled craftsmen of each respective trade in conformance with appropriate Division of Work.
3. Additional openings required in building construction to be made by drilling or cutting. Use of jack hammer is specifically prohibited. Patch openings in and through concrete and masonry with grout.
4. Restore new or existing work that is cut and/or damaged to original condition. Patch and repair specifically where existing items have been removed. This includes repairing and painting walls, ceilings, etc. where existing conduit and devices are removed as part of this project. Where alterations disturb lawns, paving, and walks, repair, refinish and leave in condition matching existing prior to commencement of work.
5. Additional work required by lack of proper coordination will be provided at no additional cost adjustment to the Contract.

3.5 EQUIPMENT SELECTION AND SERVICEABILITY

A. Replace or reposition equipment which is too large or located incorrectly to permit servicing, at no additional cost adjustment to the Contract.

3.6 DELIVERY, STORAGE AND HANDLING

A. Confirm requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.

B. In absence of specific requirements, comply with the individual Division 28, Electronic Safety Sections and the following:

1. Handle materials delivered to project site with care to avoid damage. Store materials on site inside building or protected from weather, dirt and construction dust.
2. Protect equipment and pipe to avoid damage. Close conduit openings with caps or plugs. Keep motors and bearings in watertight and dustproof covers during entire course of installation.
3. Protect devices, panels and similar items until in service.
4. Products and/or materials that become damaged due to water, dirt and/or dust as a result of improper storage to be replaced before installation.

3.7 DEMONSTRATION

A. Confirm Demonstration requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements, and individual Division 28, Electronic Safety Sections.

B. Upon completion of work and adjustment of equipment, test systems, demonstrate to Contracting Officer that equipment furnished and installed or connected under provisions of these Specifications functions in manner required. Provide field instruction to Contracting Officer's Staff as specified in Division 01, General Requirements, and individual Division 28, Electronic Safety Sections.

C. Manufacturer's Field Services: Furnish services of a qualified factory certified instructor at time approved by Contracting Officer, to instruct maintenance personnel, correct defects or deficiencies, and demonstrate to satisfaction of Contracting Officer that entire system is operating in satisfactory manner and complies with requirements of other trades that may be required to complete work. Complete instruction and demonstration prior to final job site observations.

3.8 CLEANING

A. Confirm cleaning requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 28 Sections.

B. Upon completion of installation, thoroughly clean exposed portions of equipment, removing temporary labels and traces of foreign substances. Throughout work, remove construction debris and surplus materials accumulated during work.

3.9 INSTALLATION

A. Confirm Installation requirements in Division 00, Procurement and Contracting Requirements, Division 01, General Requirements, and individual Division 28, Electronic Safety Sections.

B. Install equipment in accordance with manufacturer's installation instructions, plumb and level and firmly anchored to building structure. Maintain manufacturer's recommended clearances.

C. Startup equipment, in accordance with manufacturer's start-up instructions, and in presence of manufacturer's representative. Test controls and demonstrate compliance with requirements. Replace damaged or malfunctioning controls and equipment.

D. Provide miscellaneous supports required for installation of equipment, conduit and wiring.

3.10 PAINTING

A. Confirm Painting requirements in Division 01, General Requirements and Division 09, Finishes.

3.11 DEMOLITION

A. Confirm requirements in Division 01, General Requirements and Division 02, Existing Conditions. In the absence of specific requirements, comply with individual Division 28, Electronic Safety Sections and the following:

1. Scope:

   a. It is the intent of these documents to provide necessary information and adjustments to electronic safety system required to meet code, and accommodate installation of new work.
   b. Existing Conditions: Determine exact location of existing utilities and equipment before commencing work, pay for damages caused by failure to locate and preserve underground utilities. Replace damaged items with new material to match existing. Promptly notify Contracting Officer if utilities are found which are not shown on Drawings.
   c. Coordinate with Contracting Officer so that work can be scheduled not to interrupt operations, normal activities, building access, access to different areas. Contracting Officer will cooperate to best of their ability to assist in coordinated schedule, but will remain final authority as to time of work permitted.

2. Examination:

   a. Determine exact location of existing utilities and equipment before commencing work, pay for damages caused by failure to locate and preserve utilities. Replace damaged items with new material to match existing.
   b. Verify that abandoned wiring and equipment serve only abandoned facilities.
   c. Demolition drawings are based on casual field observation and existing record documents.

      1) Verify accuracy of information shown and provide such labor and material as is necessary to accomplish work.
      2) Verify location and number of electronic safety system devices, panels, etc. in field.

   d. Report discrepancies to Contracting Officer before disturbing existing installation.

3. Promptly notify Contracting Officer if systems are found which are not shown on Drawings.
4. Execution:

   a. Remove existing electronic safety equipment, devices and associated wiring from walls, ceilings, floors, and other surfaces scheduled for remodeling, relocation, or demolition unless shown as retained or relocated on Drawings.
   b. Provide temporary wiring and connections to maintain electrical continuity of existing systems during construction. Remove or relocate electrical boxes, conduit, wiring and equipment as encountered in removed or remodeled areas in existing construction affected by this work.
   c. Remove and restore wiring which serves usable existing outlets clear of construction or demolition.

d. If existing junction boxes will be made inaccessible, or if abandoned outlets serve as feed through boxes for other existing electrical equipment which is being retained, provide new conduit and wire to bypass abandoned outlets.
e. If existing conduits pass through partitions or ceiling which are being removed or remodeled, provide new conduit and wire to reroute clear of construction or demolition and maintain service to existing load.
f. Extend circuiting and devices in existing walls to be furred out.
g. Remove abandoned wiring to source of supply.
h. Remove exposed abandoned conduit, including abandoned conduit above accessible ceiling finishes. Cut conduit flush with walls and floors, and patch surfaces.
i. Disconnect and remove electrical devices and equipment serving utilization equipment that has been removed.
j. Maintain access to existing electrical installations which remain active. Modify installation or provide access panel as appropriate.
k. Remove abandoned wiring to leave site clean.
l. If existing electrical equipment contains PCBs (Polychlorinated Biphenyl), replace with new non-PCB equipment. Dispose of material containing PCBs as required by federal and local regulations.
m. Repair adjacent construction and finishes damaged during demolition work.
n. Maintain access to existing electrical installations which remain active. Modify installation or provide access panel as appropriate.

5. Existing Fire Alarm System: Maintain existing system in service during construction. Disable system only to make switchovers and connections.

a. Notify Contracting Officer before partially or completely disabling system.
b. Notify local fire service.
c. Make notifications at least five working days in advance.
d. Make temporary connections to maintain service in areas adjacent to work area.

3.12 ACCEPTANCE

A. Confirm requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.

B. In absence of specific requirements, comply with individual Division 28, Electronic Safety Sections. System cannot be considered for acceptance until work is completed and demonstrated to Contracting Officer that installation is in strict compliance with Specifications, Drawings and manufacturer's installation instructions, particularly in reference to following:

1. Cleaning
2. Operation and Maintenance Manuals
3. Training of Operating Personnel
4. Record Drawings
5. Warranty and Guaranty Certificates
6. Start-up/Test Documents and Commissioning Reports

3.13 FIELD QUALITY CONTROL

A. Confirm requirements in Division 00, Procurement and Contracting Requirements and Division 01, General Requirements.

B. In absence of specific requirements, comply with individual Division 28, Electronic Safety Sections and the following tests:

1. Conduct tests of equipment and systems to demonstrate compliance with requirements specified. Reference individual Specification Sections for required tests. Document tests and include in Closeout Documents.
2. During site evaluations by Contracting Officer, provide appropriate personnel with tools to remove and replace trims, covers, and devices so that proper evaluation of installation can be performed.

3.14 LETTER OF CONFORMANCE

A. Provide Letter of Conformance, copies of manufacturers' warranties and extended warranties with a statement in letter that electronic safety systems were installed in accordance with manufacturer's recommendations, UL listings and FM Global approvals.

B. Include Letter of Conformance, copies of manufacturers' warranties and extended warranties in operating and maintenance manuals.

END OF SECTION 28 00 01

SECTION 28 00 10 – SUPPLEMENTAL REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Supplemental requirements generally applicable to the Work specified in Division 28.

a. Work specified in Division 28 may be applicable to, or in coordination with, trades other than division 28. It is the contractor's responsibility to coordinate with other trades to define and complete the scope of work.

B. Related Requirements:

1. Refer to the division 26 specifications for further information regarding:

a. Abbreviations, acronyms, definitions, and additional coordination drawing submittal requirements.

2. Section 270010 "Supplemental Requirements for Communications" for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.
3. Refer to the specifications below:

a. 281300 – Access Control Software and Database Management
b. 281400 – Access Control System Hardware
c. 281600 – Intrusion Detection
d. 282000 – Video Surveillance

1.2 REFERENCES

A. Abbreviations and Acronyms for Electronic Safety and Security:

1. ACU: (mass notification) Autonomous control unit.
2. BMS: Building management system.
3. FACU: Fire-alarm control unit.
4. HPSA: (mass notification) High power speaker array.
5. LOC: (mass notification) Local operator console.
6. PoE: Power over Ethernet.
7. POTS: Plain old telephone service. See "public switched telephone network."
8. SCBA: Self-contained breathing apparatus.
9. TCP/IP: Transmission control protocol/Internet protocol.

B. Definitions for Electronic Safety and Security:

1. Calling Party Control (CPC): A momentary break in phone line loop current, which is used to signal voicemail and other automated telephone company services that distant party has hung up.
2. Private Branch Exchange (PBX): Analog telephone switch that routes calls internal to a business or organization so a direct external line for each phone is unnecessary.
3. Public Switched Telephone Network (PSTN): Analog telephone technology that uses twisted-pair cables from a telephone-provider central office for the transmission medium. PSTN refers to the telephone network; POTS refers to the individual subscriber line.
4. Remote Office Phone System (ROPS): VoIP system that allows phones for a business or organization located anywhere in the world with internet connectivity to behave similar to phones connected to a PBX.
5. Ringer Equivalence Number (REN): The loading effect of a single traditional telephone ringing circuit. TIA-968 defines REN 1 as an impedance of 7000 Ω at 20 Hz (Type A ringer) or 8000 Ω from 15 Hz to 68 Hz (Type B ringer). The sum of the RENs for all devices on a subscriber line circuit may not exceed the maximum permitted REN for the subscriber line.
6. Voice over Internet Protocol (VoIP): Digital telephone packet technology that uses the internet for its transmission medium.

1.3 COORDINATION

A. The Technology drawings and specifications provide the scope and requirements of construction applicable to multiple trades and systems. These systems are inclusive of, but not limited to, video surveillance, access control, intrusion detection, and other such communication systems. Each system requires the coordination of multiple contracting trades and disciplines. It is the responsibility of each contractor to review all technology scope and to coordinate with other disciplines.

B. Contractor shall coordinate with all other trades prior to construction, at minimum, contractor shall coordinate with the general, electrical, and other low voltage contractors. It is the contractor’s responsibility to ensure all other contractor trades are aware and knowledgeable of the system requirements and associated responsibilities as defined in these construction documents.

C. Contractor shall always maintain a working set of the construction documents in field (These may be printed or virtual). These documents shall always represent the latest information as well as capture all changes to final constructed solution(s). Any cost associated with an unaccepted installation, and subsequent reinstallation, due to the wrong documentation reference is the sole responsibility of the contractor and shall not be a cost transferred to the contracting officer. At a minimum, the working set of construction documents shall include:

1. The latest version of construction documents
2. All changes associates with change orders, RFIs, Addenda, email, and/or other such documentation
3. All field coordinated changes and/or changes to be captured in as-built drawings.

D. For any and all discrepancies, contractor shall provide written notification to contracting officer and general contractor prior to commencement of work and procurement of material.

E. Interruption of Existing Security System: Do not interrupt security system to facilities occupied by Contracting Officer or others unless permitted under the following conditions:

1. Notify Contracting Officer no fewer than 20 business days in advance of proposed interruption of security system.
2. Do not proceed with interruption of security system without Contracting Officer's written permission.

F. Multi-disciplinary coordination requirements: Ensure that all trades and integrators who are impacted or dependent on the Division 28 scope of work (either via scheduling, constructability, support, or other such means) are contacted and informed of coordination items relevant to their scope of work at least 10 days prior to commencement of work. Furthermore, ensure all trades and integrators whose work has a similar effect on the Division 28 scope of work are contacted for information regarding items relevant to the scope of work at least 10 days prior to work commencement.

1.4 PREINSTALLATION MEETINGS

A. Electronic Safety and Security Preconstruction Conference: Schedule conference with General Contractor and Contracting Officer not later than 10 days after notice to proceed. Agenda topics include, but are not limited to, the following:

1. Installation schedule for security systems.
2. Installation schedule for all systems effecting, or effected by, the security scope of work.
3. Value analysis proposals and requests for substitution of communications equipment.
4. Utility service coordination, outages, inspections, and/or activation.
5. Sustainability activities.
6. Submission of action submittals specified in Division 28.
7. Submission of information submittals specified in Division 28.
8. Submission of closeout submittals specified in Division 28.
9. Orders placed for major equipment.
10. Arrival of major equipment on-site.
11. Preinstallation meetings specified in Division 28.
12. System startup, testing, and commissioning activities for communications equipment.
13. System startup, testing, and commissioning activities for Work specified in other divisions that depends on Work specified in Division 28.
14. Requests for special inspections.
15. Requests for inspections by authorities having jurisdiction.

1.5 ACTION SUBMITTALS

A. Submittals shall be provided in accordance with Division 01 specifications.

B. The items listed below outline the multiple categories of action submittals, and the associated items comprising each category, that may be included in a system's specification. Refer to each division 28 specification for the system specific submittal requirements.

1. Product Data

    a. Manufacturer and Part number

        1) Shall be highlighted to indicate version of equipment to be utilized when multiple models/variations of product are present on documentation.

2. Shop Drawings

    a. Contractor shall provide Site Plans, Overall plans, Floor plans, Ceiling plans, One-line Diagrams, Wiring Diagrams, and/or Elevations views as needed to indicate system specific fabrication, installation, layout, connectivity, and coordination.

1.6 INFORMATIONAL SUBMITTALS

A. Submittals shall be provided in accordance with Division 01 specifications.

B. Information submittals shall be submitted to General Contractor at monthly intervals unless otherwise noted/directed by General Contractor.

C. At each issuance, a combined submittal shall be submitted that includes the information submittals from all the division 28 specifications referenced herein.

1. Where multiple specification sections indicate the same or similar recorded item, it is acceptable for the contractor to submit a combined submittal to address multiple specification sections.

    a. For example, Field Quality Control Reports can be combined to encompass multiple specification sections. It must be noted on report which sections are included in report.

D. The items listed below outline the multiple categories of information submittals, and the associated items comprising each category, that may be included in a system's specification. Refer to each division 28 specification for the system specific submittal requirements.

1. Coordination Drawings

    a. Drawings indicating specific coordination items between the division 28 scope of work components with elements from architectural, structural, and other trades.

2. Installer Qualification Data

    a. Certification information of each title as specified.

3. Field Quality Control Reports

    a. Reports that include the following:

        1) Adherence and/or adjustments to installation schedule
        2) Adherence and/or adjustments to procurement schedule
        3) System Start-up, Testing, Commissioning and Turn-Over schedules

4) System outages and downtime
5) Major Equipment order placing and arrival schedules
6) Waste-management and work site cleanliness adherence
7) Testing and Verification Results

4. Factory Test Reports

a. Manufacturer generated test reports validating the products viability prior to being shipped.

5. System Testing Plans

a. A scheduled plan to execute the testing requirements of a specific product in accordance with manufacturer requirements, industry recommendations, and specification requirements.
b. Plan shall include testing procedures, testing equipment, pass/fail thresholds per test, testing parameters of equipment and test, and the proposed schedule for testing plan.

6. Certificates

a. Seismic-Load Performance Certificates: Provide special certification as indicated in Paragraph 13.2.2 “Special Certification Requirements for Designated Seismic Systems” of ASCE/SEI 7-16 for designated seismic-load systems identified on Drawings or in the Specifications.

1) Include the following information:

a) Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
b) Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
c) Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.
d) Detailed description of conduit support devices and interconnections on which the certification is based and their installation requirements.
e) Provide equipment manufacturer's written certification for each designated active electrical seismic device and system, stating that it will remain operable following the design earthquake. Certification must be based on requirements of ASCE/SEI 7, including shake table testing per ICC-ES AC156 or a similar nationally recognized testing standard procedure acceptable to authorities having jurisdiction.
f) Submit evidence demonstrating compliance with these requirements for approval to authorities having jurisdiction after review and acceptance by qualified structural professional engineer.

b. Wind-Load Performance Certificates: Provide special certification for designated wind-load systems and components identified on Drawings or in the Specifications.

1) Include the following information:

a) Provide equipment manufacturer's written certification for each designated system and component, stating that it will remain in place and operable following the design wind event and comply with requirements of authorities having jurisdiction.
b) Certification must be based on ICC-ES or similar nationally recognized testing standard procedures acceptable to authorities having jurisdiction.

1.7 CLOSEOUT SUBMITTALS

A. Submittals shall be provided in accordance with Division 01 specifications.

B. A combined submittal shall be submitted to the Contracting Officer that includes the closeout submittals from all the division 28 specifications referenced herein.

1. Where multiple specification sections indicate the same or similar recorded item, it is acceptable for the contractor to submit a combined submittal to address multiple specification sections.

a. For example, Record Documentation can be combined to encompass multiple specification sections. It must be noted on document which sections are included in document.

C. The items listed below outline the multiple categories of closeout submittals, and the associated items comprising each category, that may be included in a system's specification. Refer to each division 28 specification for the system specific submittal requirements.

1. Record Documentation:

a. Including the installed location of equipment and components.

2. Operation and Maintenance Data:

a. Provide emergency operation, normal operation, and preventive maintenance manuals for each system, equipment, and device listed below:
b. Manufacturer's operating specifications.
c. User's guides.
d. Schedule of maintenance material items recommended to be stored at Project site.
e. Detailed instructions covering operation under both normal and abnormal conditions.
f. Instructions for setting field-adjustable components.
g. Manufacturer's instructions for testing, adjusting, and maintaining component/equipment.

3. Warranty Information

a. Manufacturer's and Contractor's Warranty Documents
b. Maintenance Contracts and Service Agreements

4. Software and Firmware Operational Documentation: Provide software and firmware operational documentation, including the following:

    a. Software operating and upgrade manuals.
    b. Names, versions, and website addresses for locations of installed software.
    c. Device address list.
    d. Printout of software application and graphic screens.
    e. Testing and adjusting of panic and emergency power features.
    f. Provide the program Software Backup username and password for approved online or cloud solution.
    g. Provide to Contracting Officer the upgrades and unrestricted licenses for installed and backup software, including operating systems and programming tools required for operation and maintenance.

5. Extra Materials

    a. Furnish to Contracting Officer extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

6. Field Test and Inspection Reports

    a. Preform testing per system specification and submit associated report.
    b. All test reports shall be combined and submitted as a single file and combined report.

1.8 QUALIFICATIONS

A. Qualified Regional Manufacturer: Manufacturer, possessing qualifications specified in Section 014000 “Quality Requirements,” that maintains a service center capable of providing training, parts, and emergency on-site repairs to Project site with response time less than forty-eight hours.

B. Security Design Professional: Design professional with active Physical Security Professional (PSP) certification issued by American Society for Industrial Security (ASIS) and with documented experience managing installation of security systems in similar size and complexity.

C. Security Testing Agency: Entity possessing active credentials from a qualified electrical testing laboratory recognized by authorities having jurisdiction.

1. On-site security testing supervisor must have an ASIS certification and documented training and experience with testing security equipment in accordance with ASIS testing standards.

PART 2 - PRODUCTS

2.1 SECURITY EQUIPMENT

A. Products shown within the associated division 28 specifications, and/or within associated drawings/schedules, shall be considered as basis of design, unless specifically noted.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verification of Conditions:

1. Immediately upon initial (1st day) arrival to site, contractor shall coordinate with General Contractor for all site and project specific information. Contractor shall adhere to all General Contractor requirements throughout duration of project.
2. Prior to commencement of work, contractor shall reference and/or inspect the following conditions:

    a. All safety and security protocols specific to site construction.
    b. The most current version of construction documents as specified in section 1.3.C herein.
    c. The most updated/recent version of project and installation schedule(s).
    d. All coordinated work previously completed and coordinated work that is uncompleted but scheduled for prior completion.
    e. Current construction conditions affecting scope of work.

3. For any and all questions, concerns, coordination, or other project specific information, especially those affecting schedule, quality of work, or system acceptance, contractor shall inform General Contractor in as timely a manner as possible.

B. Preinstallation Testing:

1. It is the contractor's responsibility to ensure that all material and equipment is tested for acceptance pursuant to the manufacturer's specifications and parameters prior to installation.
2. All material and equipment shall be tested for acceptance immediately upon being received by contractor.

    a. If material and/or equipment is received off-site, material and/or equipment shall be retested and confirmed to be acceptable upon arrival on-site.

3. Any material and/or equipment that fails testing procedure and is unacceptable for installation shall be noted and communicated to manufacturer. Dependent upon scope and breadth of failures, contractor shall coordinate required replacement material and maintain installation schedule. If failures affect the installation schedule and/or project schedule, contractor shall inform the General Contractor.

3.2 PREPARATION

A. Protection of In-Place Conditions:

1. Contractor shall always maintain a neat and orderly workspace while on site. The workspace shall not interfere with, or in any other form, disallow other trades to complete their scope of work or maintain proper schedule of work.
2. All tools, material, equipment, trash, etc., shall be maintained to be always neat and orderly while on site.
3. All material and equipment that has been installed shall be protected in place. It is the contractor's responsibility to ensure all installed equipment is protected in place through means of protective coverings, enclosed/restricted spaces, and coordination with other trades. The items of note include, but are not limited to:

    a. Capping/Taping the open ends of unused conduit and open backboxes while protecting pull string.
    b. Covering of all active equipment and termination components to ensure any openings, ports, or other such component orifices are protected from elements such as water, dust, and construction debris and/or material.
    c. Isolation of telecommunication rooms and equipment from general construction whenever possible to minimize water, dust, and construction debris and/or material.
    d. Protection of active equipment from tampering, especially pre-configured or recently configured.
    e. Proper bonding and grounding of all equipment immediately upon installation.

3.3 INSTALLATION OF ELECTRONIC SAFETY AND SECURITY WORK

A. Unless more stringent requirements are specified in the Contract Documents or manufacturers' instructions, comply with NFPA 70, NFPA 72, NFPA 731, and NECA NEIS 1 for installation of Work specified in Division 28. Consult Contracting Officer for resolution of conflicting requirements.

3.4 SYSTEM STARTUP

A. Commissioning Activities:

1. Contractor shall reference and adhere to manufacturer's startup/activation requirements for commissioning.
2. Commissioning effort shall ensure that system is complete and operational prior to final testing, system turnover and substantial completion.

3.5 FIELD QUALITY CONTROL

A. Administrant for Security Tests and Inspections:

1. Contractor shall administer and perform tests and inspections.

2. Contractor may utilize assistance of factory-authorized service representative, or a third-party certified testing authority, at no cost to Contracting Officer.

3.6 CLEANING

A. Waste Management:

1. Proper disposal and on-site management of all waste material, packaging, and other such extra items not scheduled for install is required. It is the contractor's responsibility to ensure all waste management procedures and protocols are observed and practiced with regards to any and all waste accumulated.
2. Prior to initial (1st day) arrival on site, contractor shall coordinate with General Contractor for current site conditions, rules, regulations, and requirements regarding the accumulation and disposal of waste.
3. The requirements below are in addition to General Contractor defined site requirements, not in lieu of, and shall not supersede any site requirements. Any discrepancy between the below and the site requirements shall be communicated to General Contractor for further clarification.

    a. All trash or waste accumulated shall be properly disposed of at the end of each day.
    b. Any material deemed hazardous must be disposed of immediately. Disposal of material shall be communicated to, and coordinated with, General Contractor.

3.7 CLOSEOUT ACTIVITIES

A. Demonstration:

1. Contractor shall coordinate with Contracting Officer and general contractor to schedule time and attendance for a contractor provided demonstration of the operability of the following systems and equipment:

    a. Access Control System (ACS)
    b. Video Surveillance System (VSS)
    c. Intrusion Detection System (IDS)

2. It is acceptable for contractor to engage sub-contractors, manufacturer representatives, and/or third parties, to conduct demonstration. Any engagement of such shall be at no cost to Contracting Officer.
3. At Contracting Officer's request, Contractor shall record demonstration(s) or allow Contracting Officer to record demonstration(s). Any contractor provided recording shall be at no extra cost to Contracting Officer.

B. Training:

1. Train Contracting Officer's maintenance personnel on any system and/or equipment defined herein. Training shall be administered by contractor(s) over a duration as specified below. This shall include the training of how to adjust, operate, update, and maintain the system(s) as specified in:

a. Section 281300 “Access Control Software and Database Management”

1) 1 hour at minimum

b. Section 281400 “Access Control System Hardware”

1) 1 hour at minimum

c. Section 281600 “Intrusion Detection”

1) 3 hours at minimum

d. Section 282000 “Video Surveillance”

1) 3 hours at minimum

2. At Contracting Officer’s request, Contractor shall record training session(s) or allow Contracting Officer to record session(s). Any contractor provided recording shall be at no extra cost to Contracting Officer.

END OF SECTION 28 00 10

SECTION 28 08 00 – COMMISSIONING OF FIRE ALARM

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included: Test Equipment

B. Summary:

1. Fire alarm commissioning requirements are included in this Section.
2. Commissioning is the process of reviewing system installation and operation to confirm the systems are installed and operating per the design intent and contract documents. It does not replace the official testing and acceptance of the authority having jurisdiction.

C. Definitions:

1. Reference Division 28, Electronic Safety and Division 01, General Requirements.
2. Commissioning Authority: Commissioning Agent, representing the Contracting Officer and directing commissioning activities.

1.2 RELATED SECTIONS

A. Contents of Division 28, Electronic Safety and Division 01, General Requirements apply to this Section.

B. In addition, reference Section 01 91 14, Total Building Commissioning.

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Division 28, Electronic Safety and Division 01, General Requirements.

B. In addition, meet the following:

1. ASHRAE Guideline 0 – The Commissioning Process
2. NFPA 3 – Recommended Practice for Commissioning and Integrated Testing of Fire Protection and Life Safety Systems

1.4 SUBMITTALS

A. Submittals as required by Division 28, Electronic Safety and Division 01, General Requirements.

B. In addition, provide:

1. Reference Section 01 91 14, Total Building Commissioning, for specific Commissioning submittal requirements.
2. In addition, submit the following:

a. NFPA 72 Record of Completion
b. Smoke Detector Sensitivity Report
c. Alarm Signal Audibility (dB) Levels Report

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Division 28, Electronic Safety and Division 01, General Requirements.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by Division 28, Electronic Safety and Division 01, General Requirements.

B. In addition, provide:

1. Manufacturer's Warranty.
2. Commissioning, inspecting, and testing not to modify terms or time periods of fire alarm equipment, systems, and controls warranties including related equipment and systems, and adjacent work.

1.7 COORDINATION

A. Coordination as required by Division 28, Electronic Safety and Division 01, General Requirements. Reference Section 01 91 14, Total Building Commissioning, for requirements pertaining to coordination during the commissioning process.

B. In addition, reference Section 01 91 14, Total Building Commissioning, for requirements pertaining to coordination during the commissioning process.

PART 2 - PRODUCTS

2.1 TEST EQUIPMENT

A. Provide testing equipment required to perform startup, initial checkout and functional performance testing for the equipment being tested under Division 28, Electronic Safety. Furnish two-way radios for each testing participant.

B. Testing equipment to be of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the Specifications.

PART 3 - EXECUTION

3.1 GENERAL DOCUMENTATION REQUIREMENTS

A. With assistance from the installing contractors, the Commissioning Authority will prepare Pre-Functional Checklists for commissioned systems.

B. Red-Lined Drawings:

1. Verify equipment, devices and wiring are shown correctly on red-lined drawings.
2. Record the red-lined drawing changes, as a result of Functional Testing and incorporate into the final As-Built Drawings.

C. Operation and Maintenance Data:

1. Submit a copy of O&M literature within 45 days of each submittal acceptance for use during the commissioning process for commissioned equipment and systems.
2. The Commissioning Authority will review the O&M literature once for conformance to project requirements.
3. The Commissioning Authority will receive a copy of the final approved O&M literature once corrections have been made by the Contractor.

D. Demonstration and Training:

1. Provide demonstration and training as required by the specifications.
2. Submit complete training plan and schedule to the Commissioning Authority four weeks prior to training.
3. Submit training agenda for each training session to the Commissioning Authority one month prior the training session.
4. Notify the Commissioning Authority at least one week in advance of scheduled tests so that testing may be observed by the Commissioning Authority and Contracting Officer. Submit copies of the test record to the Commissioning Authority and Contracting Officer.
5. Engage a Factory-authorized service representative to train Contracting Officer's maintenance personnel to operate and maintain specific equipment.
6. Train Contracting Officer's maintenance personnel on procedures and schedules for trouble shooting, servicing, and maintaining equipment.
7. Review data in O&M Manuals.

3.2 CONTRACTOR'S RESPONSIBILITIES

A. Provide submittals to the Commissioning Authority for review.

B. Provide information requested by the Commissioning Authority for final commissioning documentation.

C. Prepare preliminary schedule for system orientation and inspections, operation and maintenance manual submissions, training sessions and functional performance testing for Contracting Officer. Distribute preliminary schedule to commissioning team members.

D. With input from the Commissioning Authority, update the construction schedule as required with detailed commissioning tasks throughout the construction period.

E. During the startup and initial checkout process, execute the related portions of the prefunctional checklists for commissioned equipment.

F. Provide ladders, radios, and test equipment to facilitate the commissioning process.

G. Gather operation and maintenance literature on equipment, and assemble in binders as required by the specifications. Submit to Commissioning Authority 45 days after submittal acceptance.

H. Provide any proprietary equipment required to verify the system functionality.

I. Pretest all system components and controls and correct all deficiencies prior to scheduling the Commissioning Authority for functional performance testing.

J. Coordinate with the Commissioning Authority to provide one week's advance notice so that the witnessing of equipment and system testing can begin.

K. Perform commissioning tests at the direction of the Commissioning Authority.

L. Participate in, and schedule vendors and contractors to participate in the training sessions.

3.3 TESTING PREPARATION

A. Provide written notification to the Commissioning Authority that the work has been completed in accordance with the contract documents, the equipment, systems, and sub-system are operating as required, and that successful pre-testing has been completed.

1. Detection devices are installed, powered, programmed, labeled and pre-tested.
2. Notification appliances are installed, powered, synchronized, and pre-tested.
3. Fire alarm associated supply and/or exhaust fan shutdown are programmed and pre-tested.
4. Elevator recall programmed and pre-tested.
5. Fire alarm control panel programmed per fire alarm sequence of operation.
6. Communication between the FACP and Central Station verified.
7. Check safety cutouts, alarms, and interlocks with life-safety systems during each mode of operation.
8. Prior to scheduling the Commissioning Authority for commissioning tests, certify in writing that testing procedures have been completed and that testing reports have been submitted, discrepancies corrected, and corrective work approved.

3.4 GENERAL TESTING REQUIREMENTS

A. Functional performance testing of the system, as observed by the Commissioning Authority, may not begin until the Fire Alarm contractor has completed all required pre-testing checklists and documentation.

B. The Commissioning Authority along with the fire alarm contractor to prepare detailed testing plans, procedures, and checklists for fire alarm systems, subsystems, and devices.

C. Provide technicians, instrumentation, and tools to perform commissioning test at the direction of the Commissioning Authority.

D. Scope of testing includes the entire system installation.

E. Test operating modes, interlocks, control responses, and responses to abnormal or trouble conditions, and verify proper response of the system control equipment.

F. If tests cannot be completed because of a deficiency outside the scope of the system, document the deficiency and report it to the Contracting Officer. After deficiencies are resolved, reschedule tests.

3.5 FIRE ALARM SYSTEMS, SUBSYSTEMS AND EQUIPMENT TESTING PROCEDURES

A. Functional Performance Testing Procedures to include the following components and parties involved:

1. Fire Alarm Control Panel: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all initiating devices, alarming devices and programming associated with the fire alarm control panel operate per the fire alarm sequence of operation and contract documents.
2. Fire Alarm Remote Annunciator: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all initiating devices, alarming devices and programming associated with the fire alarm control panel operate per the fire alarm sequence of operation and contract documents.
3. Initiation Devices: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all initiating devices are located, labeled, annunciate correctly at the FACP and function per the fire alarm sequence of operation and contract documents.
4. Manual Pull Stations: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all manual pull stations are located, labeled, annunciate correctly at the FACP and function per the fire alarm sequence of operation and contract documents.
5. Smoke Detectors: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all smoke detection devices are located, labeled, annunciate at the FACP and function per the fire alarm sequence of operation and contract documents. This may be conducted using either a listed canned aerosol smoke approved by the manufacturer or other method approved.
6. Duct Mounted Smoke Detectors: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all duct smoke detection devices are located, labeled, annunciate at the FACP and function per the fire alarm sequence of operation and contract documents. Verify air flow through detector by using a manometer or other manufacturer approved test.
7. Heat Detectors: The Commissioning Authority will witness the Fire Alarm contractor perform field tests using a heat gun as described in the Functional Performance Tests to ensure all heat detection devices are located, labeled, annunciate at the FACP and function per the fire alarm sequence of operation and contract documents. Do not use application of heat to test non-restorable fixed temperature detectors.

8. Fire Alarm Notification Appliances: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all notification devices are located, labeled, and operates in sync with each other per the contract documents.
9. Fire Suppression Flow and Tamper Switches: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all flow and tamper detection switches are located, labeled, annunciate at the FACP and function per the fire alarm sequence of operation and contract documents.
10. Elevator Interface: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure the elevators recall to the appropriate floor upon fire alarm activation per the contract documents.
11. Air Sampling Type Smoke Detector: The Commissioning Authority will witness the Fire Alarm contractor perform field tests as described in the Functional Performance Tests to ensure all detection devices are located, labeled, annunciate at the FACP and function per the fire alarm sequence of operation and contract documents. This may be conducted using canned aerosol smoke approved by the manufacturer, pressure testing and testing of relay devices.

3.6 DEFICIENCIES/NON-CONFORMANCE, COST OF RETESTING, FAILURE DUE TO MANUFACTURER DEFECT

A. Reference Section 01 91 14, Total Building Commissioning, for requirements pertaining to deficiencies/non-conformance, cost of retesting, or failure due to manufacturer defect.

3.7 OPERATION AND MAINTENANCE (O&M) MANUALS

A. O&M Manuals to conform to Contract Documents requirements as stated in Division 28, Electronic Safety and Division 01, General Requirements.

3.8 TRAINING OF CONTRACTING OFFICER PERSONNEL

A. Contractor Training Responsibilities:

1. Provide the Commissioning Authority with a training plan four weeks before the planned training.
2. Provide designated Contracting Officer personnel with comprehensive training in the understanding of the systems and the operation and maintenance of each major piece of commissioned equipment or system.
3. Training starts with classroom sessions, if necessary, followed by hands on training on each piece of equipment, which illustrates the various modes of operation, including startup, shutdown, power failure, etc.
4. During any demonstration, should the system fail to perform in accordance with the requirements of the O&M manual or sequence of operations, the system will be repaired or adjusted as necessary and the demonstration repeated.

5. The appropriate trade or manufacturer's representative provides the instructions on each major piece of equipment. This person may be the start-up technician for the piece of equipment, the installing contractor or manufacturer's representative. Practical building operating expertise as well as in-depth knowledge of modes of operation of the specific piece of equipment are required. More than one party may be required to execute the training.
6. The training sessions follows the outline in the Table of Contents of the operation and maintenance manual and illustrate whenever possible the use of the O&M manuals and record drawings for reference.
7. Training Includes:

    a. Use the printed installation, operation and maintenance instruction material included in the O&M manuals.
    b. Include a review of the written O&M instructions emphasizing safe and proper operating requirements, preventative maintenance, special tools needed and spare parts inventory suggestions.
    c. Discuss warranties and guarantees.
    d. Cover common troubleshooting problems and solutions.
    e. Explain information included in the O&M manuals and the location of plans and manuals in the facility.
    f. Discuss any peculiarities of equipment installation or operation.

8. Fully explain and demonstrate the operation, function and overrides of any local packaged controls.

B. Schedule training after functional testing is complete, unless approved otherwise by the Contracting Officer.

END OF SECTION 28 08 00

SECTION 28 13 00 – ACCESS CONTROL SOFTWARE AND DATABASE MANAGEMENT

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 280010 "Supplemental Requirements for Electronic Safety and Security" for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

C. Section 281400 "Access Control System Hardware" for hardware required to run software specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Software for access control system units.
2. Software for access control system supplementary computer equipment.

1.3 DEFINITIONS

A. RAM: Random-access memory.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Software for access control system units.
2. Software for access control system supplementary computer equipment.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

1. Manufacturers' Published Instructions: Record copy of official installation and testing instructions issued to Installer by manufacturer for the following:

a. Installation and programming instructions for operating system software, including all applicable software add-ons.
b. Manufacturer's recommended setup and testing procedure for operating system software and applicable software add-ons.

c. Troubleshooting instructions for operating system software and applicable software add-ons.
d. Installation and programming instructions for antivirus and security protection software.
e. Manufacturer's recommended setup and testing procedure for antivirus and security protection software.
f. Troubleshooting instructions for antivirus and security protection software.

B. Installer Qualification Data

1. Qualifications per specification 280010 “Supplemental Requirements for Electronic Safety and Security”
2. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.

C. Field Quality Control Reports

D. System Testing Plans

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data

B. Warranty Information

C. Software and Firmware Operational Documentation

D. Field Test and Inspection Reports

PART 2 - PRODUCTS

2.1 SOFTWARE FOR ACCESS CONTROL SYSTEM UNITS

A. Description: This category covers software products intended for installation on access control system units.

B. Performance Criteria:

1. Regulatory Requirements:

a. Listed and labeled in accordance with NFPA 70 and NFPA 72, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.
b. Comply with NFPA 1, NFPA 730, NFPA 731, and ICC IBC.

2. Listing Criteria:

a. Cybersecurity: UL CCN CYBR; including UL 2900-1 and UL 2900-2-3.
b. Access Control System Functionality: UL CCN ALVY; including UL 294.

c. Hardware that Interfaces with Fire Alarm or Mass Notification: UL CCN UOJK; including UL 864.
d. Additional Characteristics:

1) Provides event logging, notification, and escalation.
2) Provides threat-level management and response actions.
3) Provides dashboard view of system health and status.
4) Maintains credential-driven user records and access history.
5) Supports network-connected credential readers and controlled doors.
6) Supports add-on software.
7) Supports video-surveillance system (VSS) and intrusion detection system (IDS) integration.
8) Provides email alerts and text alerts.
9) Software provides a daily, scheduled, full system save point for backup purposes.

e. Options:

1) Operating System: Windows 10.
2) User Interface: Web browser and mobile app.
3) Web Browser: Microsoft Edge (Chromium) or Google Chrome.

f. Additional Characteristics:

1) Access Control System Unit Antivirus and Security Protection Software:

a) Endpoint detection and response.
b) Anti-ransomware.
c) Managed threat response.
d) Exploit prevention.
e) Active adversary mitigations.
f) Email/Text alerts.

2) Visitor registration and Management Database Software.

a) Visitor audit trails and reporting.
b) Badge printing.
c) Unassisted visitor registration and badging.
d) Preconfigured security levels of users.
e) Filters out unsolicited visitors.
f) Supports integrated credential readers.
g) Automated Alerts:
h) Visitor check-in.

3) Package check-in.

a) Watch list alert.
b) Expired badges.
c) Panic message.
d) Current visitor.
e) Maximum visitor.

f) Frequent visitor.
g) Time based.

4) Mobile Credential Validation Database Software:

a) Mobile Validator and Mobile Registration engines.
b) Network-based functionality and database synchronization.
c) Credential audit trail exporting.
d) Contact or contactless verification.
e) Integrated card reader.
f) Biometric finger scanner.

PART 3 - EXECUTION

3.1 INSTALLATION OF ACCESS CONTROL SOFTWARE

A. Comply with manufacturer's published instructions.

B. Programming:

1. Scheduling:

a. Access through controlled entry points, at doors with card readers or other authentication devices/requirements, shall be scheduled to be unlocked during business hours and locked during non-business hours.
b. Specific doors and specific times may need to be coordinated further during programming, contractor shall coordinate with contracting officer.

C. Reference Standards:

1. Installation of Access Control Software: Unless more stringent requirements are specified in Contract Documents or manufacturers' published instructions, comply with requirements in guide information for UL CCN ALVY, UL CCN CYBR, and UL CCN UOJK.
2. Consult Contracting Officer for resolution of conflicting requirements.

D. Interfaces with Other Work:

1. Coordinate installation of new access control software with existing equipment and system conditions.
2. Coordinate with Section 281400 "Access Control System Hardware" for physical and functional interfaces with access control system and for compliance with system-wide performance requirements.
3. Coordinate with Section 281500 "Integrated Access Control System Hardware Devices" for products that require access control software specified in this Section for operation and compliance with system-wide performance requirements.

E. Systems Integration:

1. Integration of access control management system with other systems shall allow user to have a single interface and platform in which control of system notifications, alerts, and programmed system responses across all systems may be configured, controlled, monitored, and updated.
2. Integration with other systems as listed below is required:

a. Video Surveillance System (VSS)

1) Coordinate with Section 282000 “Video Surveillance” for VSS coordination and system requirements.

b. Intrusion Detection System (IDS)

1) Coordinate with Section 283100 “Intrusion Detection” for IDS coordination and system requirements.

3.2 FIELD QUALITY CONTROL OF ACCESS CONTROL SOFTWARE

A. Field tests and inspections must be witnessed by Contracting Officer or Contracting Officer specified party(ies). The authority having jurisdiction (AHJ) shall be included at their discretion, coordinate with Contracting Officer for scheduling and required attendees.

B. Tests and Inspections:

1. Perform manufacturer's recommended tests and inspections.

C. Nonconforming Work:

1. Software will be considered defective if it does not pass tests and inspections.
2. Remove and replace defective software and retest.

D. Collect, assemble, and submit test and inspection reports.

E. Manufacturer Services:

1. It is acceptable to engage factory-authorized service representative to support and/or supervise field tests and inspections. This shall be at no additional cost to Contracting Officer. In addition, it is at the discretion of the Contracting Officer to require the representative to be present for testing and inspection.

3.3 SYSTEM STARTUP

A. Perform startup service.

1. Complete software installation and startup checks in accordance with manufacturer's published instructions.
2. It is at the discretion of the Contracting Officer to require the representative to be present for start up services at no additional cost to Contracting Officer.

3.4 MAINTENANCE

A. Software and Firmware Service Agreement:

1. Technical Support: Beginning at Substantial Completion, verify that software and firmware service agreement includes software and firmware support for a minimum of two years.
2. Upgrade Service: At Substantial Completion, update software and firmware to latest version. Install and program software and firmware upgrades that become available within two years from date of Substantial Completion. Verify that upgrading software includes operating system and new or revised licenses for using software.

    a. Upgrade Notice: No fewer than 30 days to allow Contracting Officer to schedule and access the system and to upgrade computer equipment if necessary.

3. Upgrade Reports: Prepare report after each update, documenting upgrades installed.

END OF SECTION 28 13 00

SECTION 28 14 00 – ACCESS CONTROL SYSTEM HARDWARE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 280010 "Supplemental Requirements for Electronic Safety and Security" for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

C. Section 281300 "Access Control Software and Database Management" for hardware required to run software specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Access control system units.
2. Access control system power supplies and battery chargers.
3. Access control system supplementary computer equipment.

1.3 DEFINITIONS

A. DGP: Data gathering panel.

B. LAN: Local area network.

C. NFC: Near Field Communications.

D. RAID: Redundant array of inexpensive disks; redundant array of independent disks.

E. RAM: Random-access memory.

F. WAN: Wide area network.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Access control system units.
2. Access control system power supplies and battery chargers.
3. Access control system supplementary computer equipment.

B. Shop Drawings:

1. Project general notes.
2. Hardware and equipment locations.
3. Device layout.
4. Block diagram and cable/conduit routing.
5. System communications details.
6. System mounting details.
7. Hardware and equipment installation details.
8. Secondary power calculations.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

B. Installer Qualification Data

1. Qualifications per specification 280010 “Supplemental Requirements for Electronic Safety and Security”
2. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.
3. Manufacturers' Published Instructions: Record copy of official installation and testing instructions issued to Installer by manufacturer for the following:

    a. Installation instructions for access control system servers.
    b. Installation instructions for access control system DGPs.
    c. Installation instructions for access control system input/output interface.
    d. Installation instructions for access control system door controllers.
    e. Installation instructions for access control system workstation.
    f. Manufacturer's recommended testing and inspection procedure for operation of access control system servers, panels, input/output interfaces, door controllers, and workstations.
    g. Installation instructions for access control system power supplies and battery chargers.
    h. Manufacturer's recommended testing and inspection procedure for operation of access control system power supplies and battery chargers.
    i. Installation instructions for access control system printers.
    j. Manufacturer's recommended testing and inspection procedure for operation of access control system printers.

C. Field Quality Control Reports

1.6 CLOSEOUT SUBMITTALS

A. Record Documentation.

B. Operation and Maintenance Data.

C. Warranty Information.

D. Extra Material

1. Credential Reader: One of Each type
2. Special Tools: Furnish to Contracting Officer proprietary equipment, keys, and software required to operate, maintain, repair, adjust, or implement future changes to integrated access control hardware devices, that are packaged with protective covering for storage on-site and identified with labels describing contents.

E. Field Test and Inspection Reports

1.7 WARRANTY FOR ACCESS CONTROL SYSTEM HARDWARE

A. Special Installer Extended Warranty: Installer warrants that fabricated and installed access control system hardware perform in accordance with specified requirements and agrees to repair or replace systems, assemblies, components, or products that fail to perform as specified within extended-warranty period.

1. Extended-Warranty Period: Two years from date of Substantial Completion; full coverage for labor, materials, and equipment.

B. Special Manufacturer Extended Warranty: Manufacturer warrants that access control system hardware perform in accordance with specified requirements and agrees to provide repair or replacement of components or products that fail to perform as specified within extended-warranty period.

1. Extended-Warranty Period: A minimum of Five years from date of Substantial Completion; coverage for labor, materials, and equipment.
2. Follow-On Extended-Warranty Period: Eight years from date of Substantial Completion; full coverage for materials only, free on board destination, freight prepaid.

1.8 WARRANTY FOR BATTERIES

A. Special Manufacturer Extended Warranty for Batteries: Manufacturer warrants that the batteries perform in accordance with specified requirements and agrees to provide repair or replacement of batteries that fail to perform as specified within extended-warranty period.

1. Initial Extended-Warranty Period for Li-ion Batteries: Five years from date of Substantial Completion; full coverage for materials only, free on-board destination, freight prepaid.
2. Follow-On Extended-Warranty Period for Li-ion Batteries: Ten years from date of Substantial Completion; prorated coverage for materials only, free on-board destination, freight prepaid.
3. Extended-Warranty Period for Ni-MH Batteries: Three years from date of Substantial Completion; prorated coverage for materials only, free on-board destination, freight prepaid.
4. Extended-Warranty Period for Ni-Cd Batteries: Five years from date of Substantial Completion; prorated coverage for materials only, free on-board destination, freight prepaid.

5. Warranted Cycle Life for Valve Regulated (Sealed) Lead Calcium or Lead Selenium Batteries: Equal to or greater than that represented in manufacturer's published table, but not less than the following, based on annual average battery temperature of 77 deg F (25 deg C):

    a. For discharge rate not faster than eight hours, discharge duration not longer than eight hours, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than six discharge cycles.
    b. For discharge rate not faster than 30 minutes, discharge duration not longer than 30 minutes, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than 20 discharge cycles.
    c. For discharge rate not faster than 15 minutes, discharge duration not longer than 45 seconds, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than 120 discharge cycles.

6. Warranted Cycle Life for Premium Valve Regulated (Sealed) Lead Calcium or Lead Selenium Batteries: Equal to or greater than that represented in manufacturer's published table, but not less than the following, based on annual average battery temperature of 77 deg F (25 deg C):

    a. For discharge rate not faster than eight hours, discharge duration not longer than eight hours, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than 40 discharge cycles.
    b. For discharge rate not faster than 30 minutes, discharge duration not longer than 30 minutes, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than 125 discharge cycles.
    c. For discharge rate not faster than 15 minutes, discharge duration not longer than 90 seconds, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than 750 discharge cycles.

7. Warranted Cycle Life for Flooded (Wet Cell or Open) Lead Calcium or Lead Selenium Batteries: Equal to or greater than that represented in manufacturer's published table, but not less than the following, based on annual average battery temperature of 77 deg F (25 deg C) and preventive maintenance performed in accordance with manufacturer's published instructions:

    a. For discharge rate not faster than eight hours, discharge duration not longer than eight hours, and voltage at end of discharge not less than 1.75 V, warranted life must be not fewer than 40 discharge cycles.
    b. For discharge rate not faster than one hour, discharge duration not longer than one hour, and voltage at end of discharge not less than 1.75 V, warranted life must be not fewer than 80 discharge cycles.
    c. For discharge rate not faster than 15 minutes, discharge duration not longer than 45 seconds, and voltage at end of discharge not less than 1.67 V, warranted life must be not fewer than 2700 discharge cycles.

PART 2 - PRODUCTS

2.1 ACCESS CONTROL SYSTEM UNITS

A. Description: This category covers head end units, control units, integrated authentication units, and status sensors for access control systems, providing a means of regulating or controlling physical entry into an area, or access to or use of device by electrical, electronic, and/or mechanical means. Access control systems are considered a complete configuration based upon manufacturer's specified system components. Access control systems are intended to comply with applicable life safety access and egress requirements.

B. Performance Criteria:

1. Regulatory Requirements:

a. Listed and labeled in accordance with NFPA 70 and NFPA 72, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.
b. Comply with NFPA 1, NFPA 730, NFPA 731, and ICC IBC.

2. Listing Criteria: UL CCN ALVY; including UL 294.

C. System Compatibility

1. The components listed herein shall be compatible with one another and with the equipment listed in specification 281300 “Access Control Software and Database Management”.
2. The components listed herein shall be in accordance with system requirements defined in specification 280010 “Supplemental Requirements for Electronics Safety and Security”.
3. The components listed herein shall be compatible with all existing software and hardware components that are scheduled for existing to remain or are intended to be reused in conjunction with new components.

D. Access Control System Server:

1. Options:

a. Ethernet port for connection to LAN/WAN.
b. Input Voltage: 120 V(ac).
c. Operating Temperature: 50 to 82 deg F (10 to 28 deg C).
d. Server Type: Windows.
e. Server Model: Refer to schedules
f. Drive Space: 500 GB.
g. Memory: 24 GB RAM.
h. Mounting: EIA/ECA-310 standard 19 inch (482.6 mm) rack mount.

2. Accessories:

a. DIN rails for mounting.

E. Access Control System Door Controllers:

1. Manufacturers: Subject to compliance with design requirements, provide products by one of the following, or an approved equal:

    a. Mercury: LenelS2

2. Additional Characteristics:

    a. Data ports for connection to LAN and downstream controllers.
    b. Integral terminal blocks for connection of inputs/outputs to field devices.
    c. Controller Inputs:

        1) Auxiliary input.
        2) Door contact.
        3) Request to exit.
        4) Cabinet tamper.
        5) Power monitor.

    d. Controller Outputs:

        1) Auxiliary output.
        2) Door strike.
        3) Door operators.
        4) Credential readers.

F. Reader for Card, Fob, or Smartphone:

1. Refer to schedule for more information.
2. Performance Criteria:

    a. Listing Criteria:

        1) UL CCN ALVY; including UL 294.
        2) Plastic Enclosure: UL CCN QMFZ2; including UL 94.

    b. General Characteristics: Visual LED indication of card read.

3. Options:

    a. Operating Temperature: Minus 40 to plus 160 deg F (Minus 40 to plus 71 deg C).
    b. Materials: Long-wearing polymeric plastic.
    c. Colors: Black.
    d. Climate: Indoor and outdoor.

G. Access Control Cards:

1. Performance Criteria:

    a. Reference Standards:

        1) Personal Identity Verification (PIV): FIPS PUB 201-2.

2) Physical: IEC 7810; IEC 10373; IEC 7816 for contact cards; and IEC 14443 for contactless cards.
3) Functional: UL 294; IEC 60839-11-1; and IEC 60839-11-2.
4) Material Durability: INCITS 322.

b. Format: Customizable, including artwork:

1) On Front: Photograph, full name (surname first), affiliation, organization, organization logo, card expiration date, color-coding, two-dimensional bar code.
2) On Back: Physical characteristics of cardholder, employee ID number, card number, magnetic stripe, linear bar code, return address. Include the following message:
3) Stored Data Elements: PIN, employee ID number, two fingerprint templates, electronic facial image, one iris image, digital signature private key and certificate, key management private key and certificate, card authenticate data, and PIV authentication data.

2. Options:

a. Data Format: Wiegand 26 bit
b. Transmit Frequency: 125 kHz and 13.56 MHz.
c. Data Retention: 20 years.
d. Memory Capacity: 32 kB.
e. Operating Temperature: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
f. Materials: PVC laminate.
g. Color: White.
h. Reading Medium: Proximity

H. Request-to-Exit (REX) Motion Sensor:

1. Performance Criteria:

a. Regulatory Requirements: Listed and labeled in accordance with NFPA 70, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.
b. Listing Criteria: UL CCN ALVY; including UL 294.

2. Options:

a. Input Power: 125 V(ac).
b. Operating Temperature: Minus 13 to plus 140 deg F (Minus 25 to plus 60 deg C)
c. Sensing Technology: Passive infrared (PIR)
d. Color: Black
e. Mounting: Wall surface

I. Door Position Switch

1. Manufacturers: Subject to compliance with design requirements, provide products by one of the following, or an approved equal:

   a. Approved Equal

2. Must activate within an inch of separation from the base magnet.
3. Must be capable of 500,000 activations before failure.
4. Must be wired with pry tamper and magnetic tamper lamps.

2.2 ACCESS CONTROL SYSTEM POWER SUPPLIES AND BATTERY CHARGERS

A. Description: This category covers power supplies and battery chargers for access control system units, controllers, input/output interfaces, workstations, printers, and encoders.

B. Performance Criteria:

1. Regulatory Requirements:

   a. Listed and labeled in accordance with NFPA 70 and NFPA 72, by qualified electrical testing laboratory recognized by authorities having jurisdiction, and marked for intended location and application.
   b. Comply with NFPA 1, NFPA 730, NFPA 731, and ICC IBC.

2. Listing Criteria: UL CCN ALVY; including UL 294.

C. Access Control System Power Supply and Battery Charger:

1. Source Limitations: Obtain products from single manufacturer.
2. Additional Characteristics:

   a. Input Voltage: 120 V(ac).
   b. Output Voltage: 12 to 24 V(dc).
   c. Operating Temperature: 32 to 120 deg F (0 to 49 deg C).
   d. Network connection for remote supervision, management, and testing.
   e. Enclosure lock with tamper switch and monitoring.
   f. Built-in battery charger.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with manufacturer's published instructions.

B. Reference Standards:

1. Installation of Access Control System Hardware and Wiring: Unless more stringent requirements are specified in Contract Documents or manufacturers' published instructions, comply with requirements in guide information for UL CCN ALVY.
2. Consult Contracting Officer for resolution of conflicting requirements.

C. Special Techniques:

1. Cable Type: Unshielded
2. Analog Maximum Cable Length: 1000 ft (300 m).
3. Digital Maximum Cable Length: 300 ft (100 m).

D. Interfaces with Other Work:

1. Coordinate installation of new access control hardware with existing equipment and system conditions.
2. Coordinate with Section 281300 "Access Control Software and Database Management" for products that require access control software for operation and compliance with system-wide performance requirements.

E. Systems Integration:

1. Integration of access control management system with other systems shall allow user to have a single interface and platform in which control of system notifications, alerts, and programmed system responses across all systems may be configured, controlled, monitored, and updated.
2. Integration with other systems as listed below is required:

a. Video Surveillance System (VSS)

1) Coordinate with Section 282000 “Video Surveillance” for VSS coordination and system requirements.

b. Intrusion Detection System (IDS)

1) Coordinate with Section 283100 “Intrusion Detection” for IDS coordination and system requirements.

3.2 FIELD QUALITY CONTROL OF ACCESS CONTROL SOFTWARE

A. Field tests and inspections must be witnessed by Contracting Officer or Contracting Officer specified party(ies). The authority having jurisdiction (AHJ) shall be included at their discretion, coordinate with Contracting Officer for scheduling and required attendees.

B. Tests and Inspections:

1. Perform manufacturer's recommended tests and inspections.

C. Nonconforming Work:

1. Software will be considered defective if it does not pass tests and inspections.
2. Remove and replace defective software and retest.

D. Collect, assemble, and submit test and inspection reports.

E. Manufacturer Services:

1. It is acceptable to engage factory-authorized service representative to support and/or supervise field tests and inspections. This shall be at no additional cost to end user or Contracting Officer. In addition, it is at the discretion of the Contracting Officer to require the representative to be present for testing and inspection.

3.3 DEPLOYMENT OF CREDENTIALS

A. Comply with manufacturer's published instructions.

B. Third-Party Issuing of Access Control Credentials:

1. Provision of New Credentials: Contractor must be capable of programming and issuing credentials compatible with end user's access control system.

a. Initial Issuance: Provide spreadsheet or form for collecting data for generation of credentials. Contracting Officer will submit completed form containing data for personnel to be issued credentials. Print and provide initial run, printed and programmed credentials in specified format within four weeks of request.

3.4 SYSTEM STARTUP

A. Perform startup service.

1. Complete installation and startup checks in accordance with manufacturer's published instructions.
2. It is at the discretion of the Contracting Officer to require the representative to be present for start up services at no additional cost to Contracting Officer or End User.

3.5 ADJUSTING

A. Adjust printers and ribbons to function smoothly, and lubricate as recommended in writing by manufacturer.

3.6 PROTECTION

A. After installation, protect access control system hardware from construction activities. Remove and replace items that are contaminated, defaced, damaged, or otherwise caused to be unfit for use prior to acceptance by Contracting Officer.

END OF SECTION 28 14 00

SECTION 28 16 00 – INTRUSION DETECTION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 280010 "Supplemental Requirements for Electronic Safety and Security" for additional abbreviations, definitions, submittals, qualifications, testing agencies, and other Project requirements applicable to Work specified in this Section.

1.2 SUMMARY

A. Section Includes:

1. Enclosures
2. Secure and Access Devices
3. Duress Alarm Switches
4. Master Control Unit
5. Security Fasteners

1.3 DEFINITIONS

A. CCTV: Closed-circuit television.

B. PIR: Passive infrared.

C. RFI: Radio-frequency interference.

D. UPS: Uninterruptible power supply.

E. Control Unit: System component that monitors inputs and controls outputs through various circuits.

F. Master Control Unit: System component that accepts inputs from other control units and may also perform control-unit functions. The unit has limited capacity for the number of protected zones and is installed at an unattended location or at a location where it is not the attendant's primary function to monitor the security system.

G. Monitoring Station: Facility that receives signals and has personnel in attendance at all times to respond to signals. A central station is a monitoring station that is listed.

H. Protected Zone: A protected premises or an area within a protected premises that is provided with means to prevent an unwanted event.

I. Standard Intruder: A person who weighs 100 lb (45 kg) or more and whose height is 60 inches (1525 mm) or more; dressed in a long-sleeved shirt, slacks, and shoes unless environmental conditions at the site require protective clothing.

J. Standard-Intruder Movement: Any movement, such as walking, running, crawling, rolling, or jumping, of a "standard intruder" in a protected zone.

K. Systems Integration: The bringing together of components of several systems containing interacting components to achieve indicated functional operation of combined systems.

L. Zone. A defined area within a protected premises. It is a space or area for which an intrusion must be detected and uniquely identified. The sensor or group of sensors must then be assigned to perform the detection, and any interface equipment between sensors and communication must link to master control unit.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Enclosures
2. Secure and Access Devices
3. Door and Window Switches
4. Microwave-PIR Dual Technology Sensors
5. Duress Alarm Switches
6. Master Control Unit
7. Audible and Visual Alarm Devices

B. Shop Drawings:

1. Functional Block Diagram: Show single-line interconnections between components including interconnections between components specified in this Section and those furnished under other Sections. Indicate methods used to achieve systems integration. Indicate control, signal, and data communication paths and identify programmable logic controllers, networks, and control interface devices and media to be used. Describe characteristics of network and other data communication lines.
2. Site and Floor Plans: Indicate final outlet and device locations, routing of raceways, and cables inside and outside the building. Include room layout for master control-unit console, terminal cabinet, racks, and UPS.
3. System Wiring Diagrams: Include system diagrams unique to Project. Show connections for all devices, components, and auxiliary equipment. Include diagrams for equipment and for system with all terminals and interconnections identified.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

1. Manufacturers' Published Instructions: Record copy of official installation and testing instructions issued to Installer by manufacturer for the following:

a. Installation and programming instructions for operating system software, including all applicable software add-ons.
b. Manufacturer's recommended setup and testing procedure for operating system software and applicable software add-ons.
c. Troubleshooting instructions for operating system software and applicable software add-ons.
d. Installation and programming instructions for antivirus and security protection software.
e. Manufacturer's recommended setup and testing procedure for antivirus and security protection software.
f. Troubleshooting instructions for antivirus and security protection software.

B. Installer Qualification Data

1. Qualifications per specification 280010 “Supplemental Requirements for Electronic Safety and Security”
2. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.
3. Installer Qualifications:

a. An employer of workers, at least one of whom is a Certified Alarm Technician, Level 1.
b. Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.
c. Layout Responsibility: Preparation of Shop Drawings cabling administration Drawings, and field testing program development by a Technician.
d. Installation Supervision: Installation shall be under the direct supervision of Technician who shall be present at all times when Work of this Section is performed at Project site.

4. Intrusion Detection Systems Integrator Qualifications:

a. An experienced intrusion detection equipment supplier who has completed systems integration work for installations similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful in-service performance.

1) At least one of whom is a Certified Systems Integrator.

C. Field Quality Control Reports

D. System Testing Plans

1.6 CLOSEOUT SUBMITTALS

A. Record Documentation

1. Include method of operation and supervision of each component and each type of circuit. Show sequence of operations for manually and automatically initiated system or

equipment inputs. Description must cover this specific Project; manufacturer's standard descriptions for generic systems are unacceptable.

B. Operation and Maintenance Data

C. Warranty Information

D. Software and Firmware Operational Documentation

E. Extra Material

1. Intrusion Detection Devices: Furnish quantity equal to five percent of the number of units of each type installed, but no fewer than one of each type.
2. Fuses: Three of each kind and size.
3. Tool Kit: Provide two sets of tools for use with security fasteners, each packaged in a compartmented kit configured for easy handling and storage.
4. Security Fasteners: Furnish no fewer than 1 box for every 50 boxes or fraction thereof, of each type and size of security fastener installed.

F. Field Test and Inspection Reports

1.7 PROJECT CONDITIONS

A. Environmental Conditions: Capable of withstanding the following environmental conditions without mechanical or electrical damage or degradation of operating capability:

1. Altitude: Sea level to 8000 feet (2440 m).
2. Master Control Unit: Rated for continuous operation in an ambient of 60 to 85 deg F (16 to 29 deg C) and a relative humidity of 20 to 80 percent, noncondensing.
3. Interior, Controlled Environment: System components, except master control unit, installed in temperature-controlled interior environments shall be rated for continuous operation in ambients of 36 to 122 deg F (2 to 50 deg C) dry bulb and 20 to 90 percent relative humidity, noncondensing.
4. Interior, Uncontrolled Environment: System components installed in non-temperature-controlled interior environments shall be rated for continuous operation in ambients of [0 to 122 deg F (minus 18 to plus 50 deg C) dry bulb and 20 to 90 percent relative humidity, noncondensing.
5. Exterior Environment: System components installed in locations exposed to weather shall be rated for continuous operation in ambients of minus 30 to plus 122 deg F (minus 34 to plus 50 deg C) dry bulb and 20 to 90 percent relative humidity, condensing. Comply with UL 294 and UL 639 for outdoor-use equipment. Rate for continuous operation when exposed to rain as specified in NEMA 250, winds up to 85 mph (137 km/h) and snow cover up to 24 inches (610 mm) thick.
6. Hazardous Environment: System components located in areas where fire or explosion hazards may exist because of flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers or flyings shall be rated, listed, and installed according to NFPA 70.

1.8 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer and Installer agree to repair or replace components of intrusion detection devices and equipment that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 FUNCTIONAL DESCRIPTION OF SYSTEM

A. Description: Hard-wired, modular, microprocessor-based controls, intrusion sensors and detection devices, and communication links to perform monitoring, alarm, and control functions.

B. Supervision: System components shall be continuously monitored for normal, alarm, and trouble conditions. Indicate deviations from normal conditions at any location in system. Indication includes identification of device or circuit in which deviation has occurred and whether deviation is an alarm or malfunction.

1. Alarm Signal: Display at master control unit and actuate audible and visual alarm devices.
2. Trouble Condition Signal: Distinct from other signals, indicating that system is not fully functional. Trouble signal shall indicate system problems such as battery failure, open or shorted transmission line conductors, or control-unit failure.

C. System Control: Master control unit shall directly monitor intrusion detection units and connecting wiring.

D. System shall automatically reboot program without error or loss of status or alarm data after any system disturbance.

E. Operator Commands:

1. Help with System Operation: Display all commands available to operator. Help command, followed by a specific command, shall produce a short explanation of the purpose, use, and system reaction to that command.
2. Acknowledge Alarm: To indicate that alarm message has been observed by operator.
3. Place Protected Zone in Access: Disable all intrusion-alarm circuits of a specific protected zone. Tamper circuits may not be disabled by operator.
4. Place Protected Zone in Secure: Activate all intrusion-alarm circuits of a protected zone.
5. Protected Zone Test: Initiate operational test of a specific protected zone.
6. System Test: Initiate system-wide operational test.
7. Print reports.

F. Timed Control at Master Control Unit: Allow automatically timed "secure" and "access" functions of selected protected zones.

G. Automatic Control of Related Systems: Alarm or supervisory signals from certain intrusion detection devices control the following functions in related systems:

1. Switch selected lights.
2. Shift elevator control to a different mode.
3. Open a signal path between certain intercommunication stations.
4. Shift sound system to "listening mode" and open a signal path to certain system speakers.
5. Switch signal to selected monitor from CCTV camera in vicinity of sensor signaling an alarm.

H. Save Record of Events: Use has the ability to reference a record of alarm, supervisory, and trouble events. Sort and report by protected zone, device, and function. When master control unit receives a signal, record a report of alarm, supervisory, or trouble condition. Report type of signal (alarm, supervisory, or trouble), protected zone description, date, and time of occurrence. Differentiate alarm signals from other indications. When system is reset, report reset event with the same information concerning device, location, date, and time. Commands shall initiate the reporting of a list of current alarm, supervisory, and trouble conditions in system or a log of past events.

I. Response Time: A maximum of two seconds between actuation of any alarm and its indication at master control unit.

J. Circuit Supervision: Supervise all signal and data transmission lines, links with other systems, and sensors from master control unit. Indicate circuit and detection device faults with both protected zone and trouble signals, sound a distinctive audible tone, and illuminate an LED. Maximum permissible elapsed time between occurrence of a trouble condition and indication at master control unit is 20 seconds. Initiate an alarm in response to opening, closing, shorting, or grounding of a signal or data transmission line.

K. Programmed Secure-Access Control: System shall be programmable to automatically change status of various combinations of protected zones between secure and access conditions at scheduled times. Status changes may be preset for repetitive, daily, and weekly; specially scheduled operations may be preset up to a year in advance. Manual secure-access control stations shall override programmed settings.

L. Manual Secure-Access Control: Coded entries at manual stations shall change status of associated protected zone between secure and access conditions.

2.2 SYSTEM COMPONENT REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Control Units, Devices, and Communications with Monitoring Station: Listed and labeled by a qualified testing agency for compliance with SIA CP-01.

C. Comply with NFPA 70.

D. Surge Protection: Protect components from voltage surges originating external to equipment housing and entering through power, communication, signal, control, or sensing leads. Include surge protection for external wiring of each conductor entry connection to components.

1. Minimum Protection for Power Lines 120 V and More: Auxiliary panel suppressors complying with requirements in Section 264313 "Surge Protection for Low-Voltage Electrical Power Circuits."
2. Minimum Protection for Communication, Signal, Control, and Low-Voltage Power Lines: Listed and labeled by a qualified testing agency for compliance with NFPA 731.

E. Intrusion Detection Units: Listed and labeled by a qualified testing agency for compliance with UL 639.

F. Interference Protection: Components shall be unaffected by radiated RFI and electrical induction of 15 V/m over a frequency range of 10 to 10,000 MHz and conducted interference signals up to 0.25-V rms injected into power supply lines at 10 to 10,000 MHz.

G. Tamper Protection: Tamper switches on detection devices, control units, annunciators, pull boxes, junction boxes, cabinets, and other system components shall initiate a tamper-alarm signal when unit is opened or partially disassembled and when entering conductors are cut or disconnected. Master control-unit alarm display shall identify tamper alarms and indicate locations.

H. Self-Testing Devices: Automatically test themselves periodically, but not less than once per hour, to verify normal device functioning and alarm initiation capability. Devices transmit test failure to master control unit.

I. Antimasking Devices: Automatically check operation continuously or at intervals of a minute or less, and use signal-processing logic to detect blocking, masking, jamming, tampering, or other operational dysfunction. Devices transmit detection of operational dysfunction to master control unit as an alarm signal.

J. Addressable Devices: Transmitter and receivers shall communicate unique device identification and status reports to master control unit.

2.3 ENCLOSURES

A. Interior Sensors: Enclosures that protect against dust, falling dirt, and dripping noncorrosive liquids.

B. Interior Electronics: NEMA 250, Type 12.

C. Exterior Electronics: NEMA 250, Type 4X.

D. Corrosion Resistant: NEMA 250, Type 4X.

E. Screw Covers: Where enclosures are readily accessible, secure with security fasteners of type appropriate for enclosure.

2.4 SECURE AND ACCESS DEVICES

A. Keypad and Display Module: Arranged for entering and executing commands for system-status changes and for displaying system-status and command-related data.

B. Key-Operated Switch: Change protected zone between secure and access conditions.

2.5 DURESS-ALARM SWITCHES

A. Description: A switch with a shroud over the activating lever that allows an individual to covertly send a duress signal to master control unit, with no visible or audible indication when activated. Switch shall lock in activated position until reset with a key.

1. Minimum Switch Rating: 50,000 operations.
2. Push Button: Finger activated, suitable for mounting on horizontal or vertical surface.

2.6 MASTER CONTROL UNIT

A. Description: Supervise sensors and detection subsystems and their connecting communication links, status control (secure or access) of sensors and detector subsystems, activation of alarms and supervisory and trouble signals, and other indicated functions.

1. System software and programs shall be held in flash electrically erasable programmable read-only memory (EEPROM), retaining the information through failure of primary and secondary power supplies.
2. Include a real-time clock for time annotation of events on the event recorder and printer.
3. Addressable initiation devices that communicate device identity and status.
4. Control circuits for operation of mechanical equipment in response to an alarm.

B. Construction: Desk-mounted console, modular, with separate and independent alarm and supervisory system modules. Arrangements that require removal of field wiring for module replacement are unacceptable.

C. Comply with UL 609

D. Console Controls and Displays: Arranged for interface between human operator at master control unit and addressable system components including annunciation and supervision. Display alarm, supervisory, and component status messages and the programming and control menu.

1. Annunciator and Display: LCD, multiple line(s) of 80 characters, minimum.
2. Keypad: Arranged to permit entry and execution of programming, display, and control commands.
3. Control-Unit Network: Automatic communication of alarm, status changes, commands, and other communications required for system operation. Communication shall return to normal after partial or total network interruption such as power loss or transient event. Total or partial signaling network failures shall identify the failure and record the failure at the annunciator display and at the system printer.

4. Field Device Network: Communicate between the control unit and field devices of the system. Communications shall consist of alarm, network status, and status and control of field-mounted processors. Each field-mounted device shall be interrogated during each interrogation cycle.
5. Operator Controls: Manual switches and push-to-test buttons that do not require a key to operate. Prevent resetting of alarm, supervisory, or trouble signals while alarm or trouble condition persists. Include the following:

    a. Acknowledge alarm.
    b. Silence alarm.
    c. System reset.
    d. LED test.

6. Timing Unit: Solid state, programmable, 365 days.
7. Confirmation: Relays, contactors, and other control devices shall have auxiliary contacts that provide confirmation signals to system for their on or off status. Software shall interpret such signals, display equipment status, and initiate failure signals.
8. Alarm Indication: Audible signal sounds and a plain-language identification of the protected zone and/or addressable detector originating the alarm appears on LED or LCD display at master control unit. Annunciator panel displays a common alarm light and sounds an audible tone.
9. Alarm activation sounds a bell or siren and strobe.

E. Protected Zones: Quantity of alarm and supervisory zones as indicated, with capacity for expanding number of protected zones by a minimum of 25 percent.

F. Power Supply Circuits: Master control units shall provide power for remote power-consuming detection devices. Circuit capacity shall be adequate for at least a 25 percent increase in load.

G. UPS shall be sized to provide a minimum of six hours of master control-unit operation.

H. Cabinet: Lockable, steel enclosure arranged so operations required for testing, normal operation, and maintenance are performed from front of enclosure. If more than a single cabinet is required to form a complete control unit, provide exactly matching modular enclosures. Accommodate all components and allow ample gutter space for field wiring. Identify each enclosure by an engraved, laminated, phenolic-resin nameplate. Lettering on enclosure nameplate shall not be less than 1 inch (25 mm) high. Identify, with permanent labels, individual components and modules within cabinets.

I. Transmission to Monitoring Station: A communications device to automatically transmit alarm, supervisory, and trouble signals to the monitoring station, operating over a standard voice grade telephone leased line. Comply with UL 1635.

J. Printout of Events: On receipt of signal, print alarm, supervisory, and trouble events. Identify zone, device, and function. Include type of signal (alarm, supervisory, or trouble) and date and time of occurrence. Differentiate alarm signals from all other printed indications. Also print system reset event, including same information for device, location, date, and time. Commands initiate the printing of a list of existing alarm, supervisory, and trouble conditions in the system and a historical log of events.

2.7 SECURITY FASTENERS

A. Operable only by tools produced for use on specific type of fastener by fastener manufacturer or other licensed fabricator. Drive system type, head style, material, and protective coating as required for assembly, installation, and strength.

B. Manufacturers: Subject to compliance with design requirements, provide products by one of the following, or an approved equal:

1. Approved Equal

C. Socket Flat Countersunk Head Fasteners:

1. Heat-treated alloy steel, ASTM F835 (ASTM F835M).
2. Stainless steel, ASTM F879 (ASTM F879M), Group 1 CW.

D. Socket Button Head Fasteners:

1. Heat-treated alloy steel, ASTM F835 (ASTM F835M).
2. Stainless steel, ASTM F879 (ASTM F879M), Group 1 CW.

E. Socket Head Cap Fasteners:

1. Heat-treated alloy steel, ASTM A574 (ASTM A574M).
2. Stainless steel, ASTM F837 (ASTM F837M), Group 1 CW.

F. Protective Coatings for Heat-Treated Alloy Steel:

1. Zinc chromate, ASTM F1135, Grade 3 or Grade 4, for exterior applications and interior applications where indicated.
2. Zinc phosphate with oil, ASTM F1137, Grade I, or black oxide unless otherwise indicated.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of intrusion detection.

B. Examine roughing-in for embedded and built-in anchors to verify actual locations of intrusion detection connections before intrusion detection installation.

C. Prepare written report, endorsed by Installer, listing conditions detrimental to performance of intrusion detection.

D. Inspect built-in and cast-in anchor installations, before installing intrusion detection, to verify that anchor installations comply with requirements. Prepare inspection reports.

1. Remove and replace anchors where inspections indicate that they do not comply with requirements. Reinspect after repairs or replacements are made.
2. Perform additional inspections to determine compliance of replaced or additional anchor installations. Prepare inspection reports.

E. For material whose orientation is critical for its performance as a ballistic barrier, verify installation orientation.

F. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 SYSTEM INTEGRATION

A. The park is supported by ALARMSPECIALIST, Inc. at other locations within the park. This facility shall be included with the installation of the system as noted herein.

B. Contractor shall coordinate with ALARMSPECIALIST, Inc. to ensure all components specified are compatible with, and integrate into, the park's overall system and support structure.

C. Integration with Access Control System and Video Surveillance system shall be consistent with integrations requirements at other facilities. This shall be coordinated with ALARMSPECIALIST, Inc.

D. Upon depression/activation of a panic button (duress button), dispatch shall be notified.

3.3 SYSTEM INSTALLATION

A. Comply with UL 681 and NFPA 731.

B. Equipment Mounting: Install master control unit on finished floor with tops of cabinets not more than 72 inches (1830 mm) above the finished floor.

1. Comply with requirements for seismic-restraint devices specified in Section 70548.16 "Seismic Controls for Communications Systems."

C. Connecting to Existing Equipment: Verify that existing perimeter security system is operational before making changes or connections.

1. Connect new equipment to existing control panel in existing part of the building.
2. Connect new equipment to existing monitoring equipment at the Supervising Station.
3. Expand, modify, and supplement existing control/monitoring equipment as necessary to extend existing control/monitoring functions to the new points. New components shall be capable of merging with existing configuration without degrading the performance of either system.

3.4 WIRING INSTALLATION

A. Wiring Method: Install wiring in metal raceways according to Section 270528 "Pathways for Communications Systems." Conceal raceway except in unfinished spaces and as indicated. Minimum conduit size shall be 1/2 inch (13 mm). Control and data transmission wiring shall not share conduit with other building wiring systems.

B. Wiring Method: Install wiring in metal raceways according to Section 270528 "Pathways for Communications Systems," except in accessible indoor ceiling spaces and in interior hollow gypsum board partitions where cable may be used. Conceal raceways and wiring except in unfinished spaces and as indicated. Minimum conduit size shall be 1/2 inch (13 mm). Control and data transmission wiring shall not share conduit with other building wiring systems.

C. Wiring Method: Cable, concealed in accessible ceilings, walls, and floors when possible.

D. Wiring within Enclosures: Bundle, lace, and train conductors to terminal points. Use lacing bars and distribution spools. Separate power-limited and non-power-limited conductors as recommended in writing by manufacturer. Install conductors parallel with or at right angles to sides and back of enclosure. Connect conductors that are terminated, spliced, or interrupted in any enclosure associated with intrusion system to terminal blocks. Mark each terminal according to system's wiring diagrams. Make all connections with approved crimp-on terminal spade lugs, pressure-type terminal blocks, or plug connectors.

E. Wires and Cables:

1. Conductors: Size as recommended in writing by system manufacturer unless otherwise indicated.
2. 120-V Power Wiring: Install according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables" unless otherwise indicated.
3. Control and Signal Transmission Conductors: Install unshielded, twisted-pair cable unless otherwise indicated or if manufacturer recommends shielded cable, according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables."
4. Data and Television Signal Transmission Cables: Install according to Section 260519 "Low-Voltage Electrical Power Conductors and Cables."

F. Splices, Taps, and Terminations: Make connections only on numbered terminal strips in junction, pull, and outlet boxes; terminal cabinets; and equipment enclosures.

G. Install power supplies and other auxiliary components for detection devices at control units unless otherwise indicated or required by manufacturer. Do not install such items near devices they serve.

H. Identify components with engraved, laminated-plastic or metal nameplate for master control unit and each terminal cabinet, mounted with corrosion-resistant screws. Nameplates and label products are specified in Section 270553 "Identification for Communications Systems."

3.5 IDENTIFICATION

A. Identify system components, wiring, cabling, and terminals. Comply with identification requirements in Section 270553 "Identification for Communications Systems."

B. Install instructions frame in a location visible from master control unit.

3.6 GROUNDING

A. Ground the master control unit and associated circuits; comply with IEEE 1100. Install a ground wire from main service ground to master control unit.

B. Ground system components and conductor and cable shields to eliminate shock hazard and to minimize ground loops, common-mode returns, noise pickup, cross talk, and other impairments.

C. Signal Ground Terminal: Locate at main equipment rack or cabinet. Isolate from power system and equipment grounding. Provide 5-ohm ground. Measure, record, and report ground resistance.

D. Install grounding electrodes of type, size, location, and quantity indicated. Comply with installation requirements in Section 270526 "Grounding and Bonding for Communications Systems."

3.7 FIELD QUALITY CONTROL

A. Pretesting: After installation, align, adjust, and balance system and perform complete pretesting to determine compliance of system with requirements in the Contract Documents. Correct deficiencies observed in pretesting. Replace malfunctioning or damaged items with new ones and retest until satisfactory performance and conditions are achieved. Prepare forms for systematic recording of acceptance test results.

1. Report of Pretesting: After pretesting is complete, provide a letter certifying that installation is complete and fully operable; include names and titles of witnesses to preliminary tests.

B. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

C. Tests and Inspections: Comply with provisions in NFPA 731, Ch. 9, "Testing and Inspections."

1. Inspection: Verify that units and controls are properly labeled and interconnecting wires and terminals are identified.
2. Test Methods: Intrusion detection systems and other systems and equipment that are associated with detection and accessory equipment shall be tested according to Table "Test Methods" and Table "Test Methods of Initiating Devices."

D. Documentation: Comply with provisions in NFPA 731, Ch. 4, "Documentation."

E. Tag all equipment, stations, and other components for which tests have been satisfactorily completed.

3.8 ADJUSTING

A. Occupancy Adjustments: When requested within 12 months of date of Substantial Completion, provide on-site assistance in adjusting system to suit actual occupied conditions. Provide up to three visits to Project during other-than-normal occupancy hours for this purpose. Visits for this purpose shall be in addition to any required by warranty.

3.9 DEMONSTRATION

A. Train Contracting Officer's maintenance personnel to adjust, operate, and maintain the intrusion detection system. Comply with documentation provisions in NFPA 731, Ch. 4, "Documentation and User Training."

1. It is acceptable to engage a factory authorized representative to assist in, or lead, the training effort.

END OF SECTION 28 16 00

SECTION 28 20 00 – VIDEO SURVEILLANCE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Section 280010 “Supplemental Requirements for Electronic Safety and Security” for general construction, administration, system requirements.

1.2 SUMMARY

A. Section Includes:

1. Camera
2. Camera Supporting Equipment

1.3 DEFINITIONS

A. AGC: Automatic gain control.

B. BNC: Bayonet Neill-Concelman – type of connector.

C. B/W: Black and white.

D. CCD: Charge-coupled device.

E. FTP: File transfer protocol.

F. IP: Internet protocol.

G. LAN: Local area network.

H. MPEG: Moving picture experts group.

I. NTSC: National Television System Committee.

J. PC: Personal computer.

K. PTZ: Pan-tilt-zoom.

L. RAID: Redundant array of independent disks.

M. TCP: Transmission control protocol – connects hosts on the Internet.

N. UPS: Uninterruptible power supply.

O. WAN: Wide area network.

1.4 ACTION SUBMITTALS

A. Product Data:

1. Each Camera included
2. Camera Supporting Equipment

B. Shop Drawings:

1. Indicate camera locations, mounting equipment, mounting requirements, and coordination with architectural elements.
2. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Functional Block Diagram: Show single-line interconnections between components for signal transmission and control. Show cable types and sizes.
4. Dimensioned plan and elevations of equipment racks, control panels, and consoles. Show access and workspace requirements.
5. Wiring Diagrams: For power, signal, and control wiring.

1.5 INFORMATIONAL SUBMITTALS

A. Coordination Drawings

1. Manufacturers' Published Instructions: Record copy of official installation and testing instructions issued to Installer by manufacturer for the following:

    a. Installation and programming instructions for operating system software, including all applicable software add-ons.
    b. Manufacturer's recommended setup and testing procedure for operating system software and applicable software add-ons.
    c. Troubleshooting instructions for operating system software and applicable software add-ons.
    d. Installation and programming instructions for antivirus and security protection software.
    e. Manufacturer's recommended setup and testing procedure for antivirus and security protection software.
    f. Troubleshooting instructions for antivirus and security protection software.

B. Installer Qualification Data

1. Qualifications per specification 280010 “Supplemental Requirements for Electronic Safety and Security”
2. Manufacturer certification indicating completion of manufacturer specific training and/or accreditation for the installation of products submitted.

C. Field Quality Control Reports

D. System Testing Plans

1.6 CLOSEOUT SUBMITTALS

A. Record Documentation

1. Include an equipment list consisting of every piece of equipment by model number, manufacturer, serial number, location, and date of original installation. Add pretesting record of each piece of equipment, listing name of person testing, date of test, set points of adjustments, name and description of the view of preset positions, description of alarms, and description of unit output responses to an alarm.

B. Operation and Maintenance Data

C. Warranty Information

D. Software and Firmware Operational Documentation

E. Field Test and Inspection Reports

1.7 PROJECT CONDITIONS

A. Environmental Conditions: Capable of withstanding the following environmental conditions without mechanical or electrical damage or degradation of operating capability:

1. Control Station: Rated for continuous operation in ambient temperatures of 60 to 85 deg F (16 to 29 deg C) and a relative humidity of 20 to 80 percent, noncondensing.
2. Interior, Controlled Environment: System components, except central-station control unit, installed in temperature-controlled interior environments shall be rated for continuous operation in ambient temperatures of 36 to 122 deg F (2 to 50 deg C) dry bulb and 20 to 90 percent relative humidity, noncondensing. Use NEMA 250, Type 1 enclosures.
3. Interior, Uncontrolled Environment: System components installed in non-temperature-controlled interior environments shall be rated for continuous operation in ambient temperatures of 0 to 122 deg F (minus 18 to plus 50 deg C) dry bulb and 20 to 90 percent relative humidity, noncondensing. Use NEMA 250, Type 12 enclosures.
4. Exterior Environment: System components installed in locations exposed to weather shall be rated for continuous operation in ambient temperatures of minus 30 to plus 122 deg F (minus 34 to plus 50 deg C) dry bulb and 20 to 90 percent relative humidity, condensing. Rate for continuous operation when exposed to rain as specified in NEMA 250, winds up to 85 mph (137 km/h) and snow cover up to 24 inches (610 mm) thick. Use NEMA 250, Type 4 enclosures.
5. Hazardous Environment: System components located in areas where fire or explosion hazards may exist because of flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers shall be rated, listed, and installed according to NFPA 70.
6. Corrosive Environment: System components subject to corrosive fumes, vapors, and wind-driven salt spray in coastal zones. Use NEMA 250, Type 4X enclosures.

7. Security Environment: Camera housing for use in high-risk areas where surveillance equipment may be subject to physical violence.

1.8 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of cameras, equipment related to camera operation, and control-station equipment that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Three years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 SYSTEM REQUIREMENTS

A. Internet Protocol (IP) based, Power Over Ethernet (POE) enabled, High Definition (HD) cameras shall be installed at locations as indicated to provide visual surveillance and event monitoring in specified areas.

B. Local video storage capabilities and remote viewing capabilities.

C. Surge Protection: Protect components from voltage surges originating external to equipment housing and entering through power, communication, signal, control, or sensing leads. Include surge protection for external wiring of each conductor's entry connection to components.

1. Minimum Protection for Power Connections 120 V and More: Auxiliary panel suppressors complying with requirements in Section 264313 "Surge Protection for Low-Voltage Electrical Power Circuits."
2. Minimum Protection for Communication, Signal, Control, and Low-Voltage Power Connections: Comply with requirements in Section 264313 "Surge Protection for Low-Voltage Electrical Power Circuits" as recommended by manufacturer for type of line being protected.

D. Tamper Protection: Tamper switches on enclosures, control units, pull boxes, junction boxes, cabinets, and other system components shall initiate a tamper-alarm signal when unit is opened or partially disassembled. Control-station, control-unit alarm display shall identify tamper alarms and indicate locations.

2.2 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Video surveillance system shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. Comply with NECA 1.

D. Comply with NFPA 70.

E. Electronic data exchange between video surveillance system with an access-control system shall comply with SIA TVAC.

2.3 VIDEO MANAGEMENT SYSTEM (VMS)

A. The existing video management system and equipment shall be relocated from the Grand Lake entrance facility to Server 116 as indicated in drawings. All components herein shall be integrated into, and be compatible with, the existing system components.

2.4 CAMERAS

A. Camera

1. Camera type defined is not for reference outside of this document and is only meant to differentiate between general camera types specified for the project.
2. Camera selection is defined within equipment schedules in construction drawings. Items and components defined below shall be in coordination with device specified and shall serve as guideline for review of devices submitted other than specified on schedule.
3. Options:

    a. Min and Max Resolution: 2MP – 8MP (1080p – 4k)
    b. Frame Rate: 30 fps
    c. Color: White
    d. IP rating: IP 66
    e. IK rating: IK10
    f. Minimum Horizonal Field of View: 100 degrees
    g. Minimum Vertical Field of View: 30 degrees
    h. Minimum compression standard of H.264, recommended to have H.265.
    i. Operating Temperature Range: -40 to 122 deg F (-40 to 50 deg C)
    j. Sensitivity: Camera shall provide usable images in low-light conditions, delivering an image at a scene illumination of less than 0.5 lux at less than or equal to F1.8.

2.5 CAMERA-SUPPORTING EQUIPMENT

A. Any and all support equipment including, but not limited to, ceiling mounts, wall mounts, and pole mounts, shall be of the same manufacturer as the camera submitted.

B. Minimum Load Rating: Rated for load in excess of the total weight supported times a minimum safety factor of two.

C. Mounting Brackets for Fixed Cameras: Type matched to items supported and mounting conditions. Include manual pan-and-tilt adjustment.

D. Protective Housings for Fixed and Movable Cameras: Steel enclosures with internal camera mounting and connecting provisions that are matched to camera/lens combination and mounting and installing arrangement of camera to be housed.

1. Tamper switch on access cover sounds an alarm signal when unit is opened or partially disassembled. Central-control unit shall identify tamper alarms and indicate location in alarm display. Tamper switches and central-control unit are specified in Section 283100 "Intrusion Detection."
2. Camera Viewing Window: Aligned with camera lens.
3. Alignment Provisions: Camera mounting shall provide for field aiming of camera and permit removal and reinstallation of camera lens without disturbing camera alignment.
4. Built-in, thermostat-activated heater and/or blower units. Units shall be automatically controlled so the environmental limits of the camera equipment are not exceeded.
5. Sun shield shall not interfere with normal airflow around the housing.
6. Mounting bracket and hardware for wall or ceiling mounting of the housing. Bracket shall be of same material as the housing; mounting hardware shall be stainless steel.
7. Finish: Housing and mounting bracket shall be factory finished using manufacturer's standard finishing process suitable for the environment.

E. Dedicated Security Camera Network Switch

1. Refer to schedules for basis of design information.
2. A dedicated network switch shall provide network and POE connection to the cameras as indicated on plan.

2.6 NETWORK VIDEO RECORDERS

A. The existing network video recorder (NVR) and equipment shall be relocated from the Grand Lake entrance facility to Server 116 as indicated in drawings. All components herein shall be integrated into, and be compatible with, the existing system components.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine pathway elements intended for cables. Check raceways and other elements for compliance with space allocations, installation tolerance, hazards to camera installation, and other conditions affecting installation.

B. Examine roughing-in for LAN, WAN, and IP network before device installation.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 WIRING

A. Comply with requirements in Section 270528 "Pathways for Communications Systems."

B. Comply with requirements in Section 271513 "Communications Copper Horizontal Cabling."

C. Surge Protection: When a camera, or any portion of cabling support the camera and/or connected to the camera, is outside the building envelope, surge protection units shall be utilized.

1. Refer to specifications and details on drawings for surge protection unit type and mounting requirements.

3.3 VIDEO SURVEILLANCE SYSTEM INSTALLATION

A. Install cameras and infrared illuminators level and plumb.

B. Installation locations and heights are defined on drawings. Precise field coordination with architectural elements is required prior to rough-in.

1. Contractor shall be aware of ceiling heights, overhangs, trees, and any other obstructions that would obscure the view intended for the camera. Communication with the Contracting Officer General Contractor is required for any such observed issue prior to rough in and installation.

C. Install tamper switches on components indicated to receive tamper switches, arranged to detect unauthorized entry into system-component enclosures and mounted in self-protected, inconspicuous positions.

D. Identify system components, wiring, cabling, and terminals according to Section 270553 "Identification for Communications Systems."

3.4 FIELD QUALITY CONTROL

A. Field tests and inspections must be witnessed by Contracting Officer or Contracting Officer's specified party(ies). The authority having jurisdiction (AHJ) shall be included at their discretion, coordinate with Contracting Officer for scheduling and required attendees.

B. Manufacturer Services:

1. It is acceptable to engage factory-authorized service representative to support and/or supervise field tests and inspections. This shall be at no additional cost to Contracting Officer. In addition, it is at the discretion of the Contracting Officer to require the representative to be present for testing and inspection.

C. Tests and Inspections:

1. Inspection: Verify that units and controls are properly installed, connected, and labeled, and that interconnecting wires and terminals are identified.
2. After all equipment has been migrated to facility, contractor shall validate that connection to all existing equipment is functional and operational prior to integrating new equipment into system.

3. Pretesting: Align and adjust system and pretest components, wiring, and functions to verify that they comply with specified requirements. Conduct tests at varying lighting levels, including day and night scenes as applicable. Prepare video-surveillance equipment for acceptance and operational testing as follows:

    a. Prepare equipment list described in "Informational Submittals" Article.
    b. Verify operation of auto-iris lenses.
    c. Set back-focus of fixed focal length lenses. At focus set to infinity, simulate nighttime lighting conditions by using a dark glass filter of a density that produces a clear image. Adjust until image is in focus with and without the filter.
    d. Set back-focus of zoom lenses. At focus set to infinity, simulate nighttime lighting conditions by using a dark glass filter of a density that produces a clear image. Additionally, set zoom to full wide angle and aim camera at an object 50 to 75 feet (17 to 23 m) away. Adjust until image is in focus from full wide angle to full telephoto, with the filter in place.
    e. Set and name all preset positions; consult Contracting Officer's personnel.
    f. Set sensitivity of motion detection.
    g. Connect and verify responses to alarms.
    h. Verify operation of control-station equipment.

4. Test Schedule: Schedule tests after pretesting has been successfully completed and system has been in normal functional operation for at least 14 days. Provide a minimum of 10 days' notice of test schedule.
5. Operational Tests: Perform operational system tests to verify that system complies with Specifications. Include all modes of system operation. Test equipment for proper operation in all functional modes.

D. Video surveillance system will be considered defective if it does not pass tests and inspections.

E. Collect, assemble, and submit test and inspection reports.

3.5 ADJUSTING

A. Occupancy Adjustments: When requested within 12 months (1 year) of date of Substantial Completion, provide on-site assistance in adjusting system to suit actual occupied conditions. Provide up to two visits to Project during other-than-normal occupancy hours for this purpose. Tasks shall include, but are not limited to, the following:

1. Check cable connections.
2. Check proper operation of cameras and lenses. Verify operation of auto-iris lenses and adjust back-focus as needed.
3. Adjust all preset positions; consult Contracting Officer's personnel.
4. Recommend changes to cameras, lenses, and associated equipment to improve Contracting Officer's use of video surveillance system.
5. Provide a written report of adjustments and recommendations.

3.6 CLEANING

A. Clean installed items using methods and materials recommended in writing by manufacturer.

B. Clean video-surveillance-system components, including camera-housing windows, lenses, and monitor screens.

3.7 DEMONSTRATION

A. Train Contracting Officer's maintenance personnel to adjust, operate, and maintain video-surveillance equipment.

END OF SECTION 28 20 00

SECTION 28 46 00 – FIRE DETECTION AND ALARM

PART 1 - GENERAL

1.1 SUMMARY

A. Work Included:

1. Fire Alarm Control Panel
2. Notification Appliance Circuit Panels
3. Fire Alarm Transmitters
4. Fire Alarm Annunciators
5. Manual Pull Stations
6. Fixed Temperature Heat Detectors
7. Rate-of-Rise and Fixed Temperature Heat Detectors
8. Photoelectric Type Detectors
9. Relay Modules
10. Control Modules
11. Input Modules
12. Fault Isolation Modules
13. Combination Horn/Strobes
14. Strobes
15. Weatherproof/Surface Backboxes
16. Protective Guards
17. Circuit Conductors
18. Surge Protection
19. Batteries
20. Locks and Keys
21. Document Storage Cabinet
22. Instruction Charts
23. Framed Floor Map

B. Scope:

1. Provide a new fire alarm system and preaction sprinkler system control system.
2. Provide a new fire alarm transmitter communication system.

C. In addition, provide design for the following as required in these Contract Documents:

1. Fire Alarm System
2. Fire Alarm Transmitter Communication System

D. In addition, remove existing fire alarm system throughout the entire building.

E. System Design:

1. Design Criteria:

a. Design systems utilizing equipment, appliance, and device layouts depicted in the contract documents and as required by code.

b. Fire Alarm Sequence of Operation: Activation of manual fire alarm box, automatic fire detector, or fire extinguishing system causes system to enter “alarm” mode including the following operations:

1) Local English language annunciation of device location, address and condition and audible and visual alarm signal at control panel and remote annunciators.
2) Manual "acknowledge" function at control panel and remote annunciators to silence audible alarm signal, visual signal remains displayed until initiating alarm is cleared.
3) Transmit “alarm” signal to off-premises equipment, i.e., to local fire department or Contracting Officer's selected vendor. Provide necessary connections to transmitter.
4) Activate fire alarm notification appliances.
5) Activate Emergency Control Functions as required by code.
6) Initiate first interlock for the preaction sprinkler system control sequence upon alarm signal from the clean agent system control panel.
7) Initiate second interlock for the preaction sprinkler system control sequence upon alarm signal from the alarm pressure switch for the preaction sprinkler system.

c. Supervisory Sequence of Operation: Fire sprinkler tamper or supervisory pressure switch activation, causes system to enter "supervisory" mode including the following operations:

1) Local English language annunciation of device location, address and condition and audible and visual supervisory signal at control panel and remote annunciators.
2) Manual "acknowledge" function at control panel and remote annunciators to silence audible supervisory signal, visual signal remains displayed until initiating supervisory is cleared.
3) Transmit "supervisory" signal to off-premises equipment.

d. Trouble Sequence of Operation: System trouble, including single ground or open of supervised circuit, or power or system failure, causes system to enter "trouble" mode including the following operations:

1) Local English language annunciation of device location, address and condition and audible and visual trouble signal at control panel and remote annunciators.
2) Manual "acknowledge" function at control panel and remote annunciators to silence audible trouble signal, visual signal remains displayed until initiating trouble is cleared.
3) Transmit "trouble" signal to off-premises equipment.

2. Design of Fire Alarm Transmitter Communication System: Provide design of the fire alarm transmitter communication system as required by code.

1.2 RELATED SECTIONS

A. Contents of Division 28, Electronic Safety and Division 01, General Requirements apply to this Section.

B. Division 26, Electrical requirements apply to this section.

1.3 REFERENCES AND STANDARDS

A. References and Standards as required by Division 28, Electronic Safety and Division 01, General Requirements.

B. In addition, meet requirements of the following:

1. NFPA 72, National Fire Alarm and Signaling Code, adopted edition.
2. NFPA 70, National Electrical Code, adopted edition.

1.4 SUBMITTALS

A. Submittals as required by Division 28, Electronic Safety and Division 01, General Requirements.

B. Shop Drawings:

1. Submit shop drawings which include documentation required per NFPA 72, Shop Drawings.
2. In addition, provide the following:

    a. Provide system designer NICET certification number or Engineer's signature and seal on shop drawings.
    b. Identification of system designer and evidence of qualification or certification of designer as required by AHJ.

C. Operation and Maintenance Manuals:

1. Provide manuals containing the documentation required in NFPA 72, Completion Documentation.
2. In addition, provide the following:

    a. One year warranty agreement including parts and labor. Warranty period begins upon date of substantial completion.
    b. Instruction chart.

1.5 QUALITY ASSURANCE

A. Quality assurance as required by Division 28, Electronic Safety and Division 01, General Requirements.

B. Meet National Park Service requirements, ordinances and amendments.

C. Meet the following certification requirements:

1. Fire alarm system to be designed by someone meeting or exceeding one or more of the following requirements:

   a. A fire alarm engineer.
   b. A NICET III technician for the type of system to be designed.
   c. As approved by the AHJ.

2. Fire alarm system to be installed by someone meeting or exceeding one or more of the following requirements:

   a. State or municipal certified/licensed fire alarm contractor.
   b. Qualified by the manufacturer.
   c. NICET Level II for the type of system to be installed.
   d. As approved by the AHJ.

1.6 WARRANTY

A. Warranty of materials and workmanship as required by Division 28, Electronic Safety and Division 01, General Requirements.

PART 2 - PRODUCTS

2.1 GENERAL

A. Substitutions:

1. Reference Specification Section 01 67 00, Product Requirements.
2. Submit product data showing features and certification by Contractor that the design will comply with contract documents.

B. Equipment to be supplied by a certified manufacturer representative.

C. Salient Characteristics as outlined in Articles below:

1. Physical
2. Functional
3. Performance

2.2 FIRE ALARM CONTROL PANEL

A. Physical:

1. Provide flush mounted units where installed in finished areas; in unfinished areas, surface mounted units are acceptable, unless otherwise noted.
2. Multiprocessor Based: Configurable as an addressable, point identified system.

a. House the Central Processing Unit (CPU) in fire alarm cabinet with sufficient space to allow maximum system expansion and to enclose alphanumeric display.
b. For control panel display modules, provide keyboard display module with minimum 80-character backlit LCD.

B. Functional:

1. Program Software:

a. Field configuration program provides programmable operating instructions for system. Store resident program in non-volatile memory.
b. Devices meet criterion specified under materials.
c. Verification and display of sensitivity of each addressable smoke detector can be read using the operating software.

2. Control Panel Display Modules:

a. Provide keyboard display module with minimum 80-character backlit LCD. Each alarm/trouble condition appears in English language with description and location of alarm/supervisory/trouble.
b. Alarm/supervisory/trouble may be acknowledged, silenced and system reset from control panel or remote annunciator(s).

3. Auxiliary Relays: Provide sufficient SPDT auxiliary relay contacts for each function in this portion of the Specifications and for equipment interconnections required under electrical and mechanical specifications.
4. Auxiliary Switches: Provide auxiliary equipment control switches with labeled status indicating lights for each switch.
5. System Reset:

a. Key-accessible control function returns system to normal, non-alarm state, if initiating circuits have cleared.
b. Provide reset on both main fire alarm control panel and remote annunciators.

6. Addressing: Provide each initiating device with its own discrete address.

C. Performance:

1. Central Processing Unit (CPU):

a. CPU continuously monitors the communications and data processing cycles of microprocessor. CPU failure generates an audible and visual trouble signal on control panel and remote annunciators.

b. Retain basic life safety software in field programmable non-volatile memory. Provide CPU with minimum capacity of 50 addressable points.
c. Equip CPU with software to provide a control-by-event feature, whereby receipt of an alarm point is programmed to operate control points within system. Provide control-by-event actions for life safety functions in programmable non-volatile memory. CPU software programming for control of systems defined in this Section is installed as part of this Section.

2. System Capabilities:

a. System capable of addressing and operating smoke detectors, manual pull stations, open contact devices and addressable auxiliary control relays on the same communication loop.
b. System capable of displaying sensitivity of each smoke detector, address and condition of fire alarm monitoring points.

3. Power Supply: Provide power supply(s), adequate to serve control panel modules, remote annunciators, addressable devices, notification appliances and other connected devices.
4. Power Requirements:

a. Loss of 120VAC power automatically causes system to transfer to secondary power. Indicate battery power operation by yellow lamp and audible annunciation at control panel and remote annunciator panels.
b. Upon return of 120VAC power, unit recharges batteries to full capacity and maintains battery on float charge. Provide trickle charge adequate capacity to maintain battery fully charged with automatic rate charge.

5. Provide fire alarm panel listed for use in releasing preaction sprinkler system solenoid valves. Coordinate with fire sprinkler system design to ensure compatibility.

2.3 NOTIFICATION APPLIANCE CIRCUIT PANELS

A. Physical: None.

B. Functional:

1. Same manufacturer as control equipment.
2. Minimum six amps of notification power.
3. Three amps per output circuit.
4. Class A and Class B configurable circuits.
5. 24 VDC filtered output voltage.
6. Four power-limited notification outputs.
7. Ground fault detector/indicator.

C. Performance:

1. Built-in synchronization for System Sensor, Gentex, Wheelock, Faraday, and AMSECO.
2. UL and CSFM listed.

2.4 FIRE ALARM TRANSMITTERS

A. Physical:

1. Provide flush mounted units where installed in finished areas.
2. In unfinished areas, surface mounted units are acceptable, unless otherwise noted.

B. Functional:

1. Electrically supervised, capable of transmitting alarm, supervisory and trouble signals over Cellular lines to off-premises receiver.
2. Signal transmitter interfaces fully with receiver station of local fire department or Contracting Officer's selected vendor.
3. For cellular transmitters, provide exterior antenna where required to facilitate communication with supervising station.

C. Performance: None.

2.5 FIRE ALARM ANNUNCIATORS

A. Physical:

1. Backlit alphanumeric LCD, minimum 80 characters long.
2. Locking cover.
3. Test switch.
4. Alarm and trouble buzzer.
5. Buzzer silence switch.
6. Reset switch.
7. Vandal-resistant UV stabilized Lexan (or equal) overlay.
8. Self-contained.
9. Suitable for wet location where located exterior.
10. Flush mounted units where installed in finished areas.
11. Surface mounted units are acceptable in unfinished areas, unless otherwise noted.

B. Functional: Fire alarm annunciation.

C. Performance:

1. Compatible with fire alarm control equipment.
2. Same manufacturer as control equipment.
3. Buzzer silence message.
4. Required modules, control panel, etc. to drive annunciator.

2.6 MANUAL PULL STATIONS

A. Physical:

1. Semi-flush mounted.
2. Red finish
3. Nongrasping operation: maximum pull strength as allowed per ADA and ABA criteria.

4. Semi-flush mounted; flush mounted units where installed in finished areas.
5. Surface mounted units are acceptable in unfinished areas, unless otherwise noted.

B. Functional: Manual fire alarm pull station.

C. Performance:

1. Compatible with fire alarm control equipment.
2. Addressable.
3. Stations do not allow closure without keyed reset.

2.7 FIXED TEMPERATURE HEAT DETECTORS

A. Physical:

1. Provide off-white, low-profile detectors.
2. Flush mounted units where installed in finished areas.
3. Surface mounted units are acceptable in unfinished areas, unless otherwise noted.

B. Functional: Rated 135 degrees F or 190 degrees F as required by space use.

C. Performance:

1. Compatible with fire alarm control equipment.
2. Addressable.

2.8 RATE-OF-RISE AND FIXED TEMPERATURE HEAT DETECTORS

A. Physical:

1. Provide off-white, low-profile detectors.
2. Flush mounted units where installed in finished areas.
3. Surface mounted units are acceptable in unfinished areas, unless otherwise noted.

B. Functional: Responding to 15 degrees F temperature rise per minute and to 135 degrees F fixed temperature as required by space use.

C. Performance:

1. Compatible with fire alarm control equipment.
2. Addressable.

2.9 PHOTOELECTRIC TYPE DETECTORS

A. Physical:

1. LED source, multiple cell.
2. Visual latching operation indicator.
3. Insect screen.

4. Functional test switch.
5. Two-wire operation.
6. Vandal-resistant locking feature.
7. Flush mounted units where installed in finished areas.
8. Surface mounted units are acceptable in unfinished areas, unless otherwise noted.

B. Functional: Smoke detection.

C. Performance:

1. Panel adjustable sensitivity.
2. 360 degree smoke entry.
3. Two-wire operation.
4. Compatible with fire alarm control equipment.
5. Addressable.

2.10 RELAY MODULES

A. Physical: None.

B. Functional: Relay module, powered from control panel.

C. Performance:

1. Addressable.
2. Compatible with fire alarm control equipment.
3. Signaling line circuit interface module that connects to other building systems for control of fire/life safety functions, e.g., air-handler shutdown, fire/smoke damper closure, elevator recall.

2.11 CONTROL MODULES

A. Physical: None.

B. Functional: Control module, powered from control panel.

C. Performance:

1. Addressable.
2. Compatible with fire alarm control equipment.
3. Compatible with preaction sprinkler system solenoid valve.
4. Signaling line circuit interface module that provides notification appliance circuits or system control outputs, including preaction sprinkler system releasing functions.

2.12 INPUT MODULES

A. Physical: None.

B. Functional: Input module, powered from control panel.

C. Performance:

1. Addressable.
2. Compatible with fire alarm control equipment.
3. Signaling line circuit interface module provides initiating device circuits for connection to contact closure initiating devices.

2.13 FAULT ISOLATION MODULES

A. Physical: Provide module with status indicator LED.

B. Functional: Fault isolation module.

C. Performance:

1. Addressable.
2. Compatible with fire alarm control equipment.
3. Signaling line circuit interface modules that provide isolation of wire-to-wire shorts on a signaling line circuit with automatic reconnection upon correction of short circuit.

2.14 COMBINATION HORN/STROBES

A. Physical:

1. Flush wall and ceiling mount.
2. White finish.
3. Insect-proof.

B. Functional: Multi-candela horn/strobes.

C. Performance:

1. Must be compatible with fire alarm control equipment and notification appliance circuit panels.
2. Meet the latest requirements of NFPA 72, ANSI 117.1 and UL 1971.
3. Candela rating as required by NFPA 72.

2.15 STROBES

A. Physical:

1. Flush wall and ceiling mount.
2. White finish.
3. Insect-proof.

B. Functional: Multi-candela strobes.

C. Performance:

1. Must be compatible with fire alarm control equipment and notification appliance circuit panels.
2. Meet the latest requirements of NFPA 72, ANSI 117.1 and UL 1971.
3. Candela rating as required by NFPA 72.

2.16 WEATHERPROOF/SURFACE BACKBOXES

A. Physical: None.

B. Functional: Rated for exterior use.

C. Performance: Compatible with device or appliance.

2.17 PROTECTIVE GUARDS

A. Physical:

1. Wire Guard: Steel construction.
2. Protective Cover: Polycarbonate construction.

B. Functional: Protection of equipment.

C. Performance: None.

2.18 CIRCUIT CONDUCTORS

A. Physical:

1. Copper or optical fiber; color code and label. Type FPL, FPLR and FPLP. Cable type as required by the NEC and the manufacturer.
2. Minimum signaling line circuit and initiating device circuit wire size: AWG18.
3. Minimum notification appliance circuit wire size: AWG14.
4. Fiber optic cable as required by manufacturer.

B. Functional: Circuit conduction.

C. Performance: Same manufacturer as control equipment.

2.19 SURGE PROTECTION

A. 24V Protection:

1. Physical:

a. Hardwired terminals.
b. 30-12 AWG, ABS housing.

2. Functional:

a. Minimum 24v service voltage.
b. MCOV: 33V
c. Clamping Voltage: 39V.
d. Surge Current Rating: 20,000A.
e. Maximum Continuous Current: 5A.
f. UL497B certified.

3. Performance: None.

B. 120V Protection:

1. Physical: Diagnostic LED.
2. Functional:

a. Approved for 20A circuit breakers.
b. MCOV: 150V.
c. Voltage Protection Rating: Minimum 700V.
d. Surge Current Rating: 50,000A.
e. Nominal Discharge Current Rating: 10kA.
f. UL1449 Listed Type 1 SPD.

3. Performance: None.

2.20 BATTERIES

A. Physical:

1. Sealed lead acid.
2. Absorbent Glass Mat (AGM) technology.
3. ABS case.
4. 12 Volt.

B. Functional: Provides battery power.

C. Performance: None.

2.21 LOCKS AND KEYS

A. Physical:

1. Provide same standard lock and key for each key operated switch and lockable panel and cabinet.
2. Provide five keys of each type.

B. Functional: Keyed locks.

C. Performance: None.

2.22 DOCUMENT STORAGE CABINET

A. Physical:

1. Suitable to store as-built drawings, operation and maintenance manual, system data file disk, spare parts, and tools.
2. Constructed from steel with baked enamel finish.
3. Size adequate for storage of full-size as-built drawings, operation and maintenance manual, system data file disk, spare parts, and tools.

B. Functional: Storage of as-built drawings, operation and maintenance manual, system data file disk, spare parts, and tools.

C. Performance: Suitable for mounting.

2.23 INSTRUCTION CHARTS

A. Physical:

1. Frame to be stainless steel or aluminum with polycarbonate or glass cover.
2. Confirm make and model with Contracting Officer prior to ordering.

B. Functional:

1. Printed, framed instruction chart for operators, showing steps to be taken when signal is received (normal, alarm, supervisory and trouble).
2. Easily readable from normal operator's station.

C. Performance: Suitable for mounting.

2.24 FRAMED FLOOR MAP

A. Physical:

1. Graphics: Floor plan of facility.
2. Frame to be stainless steel or aluminum with polycarbonate or glass cover.
3. Confirm make and model with Contracting Officer prior to ordering.

B. Functional: Printed, framed floor plan of facility.

C. Performance: Suitable for mounting.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

A. Provide a complete and operable system compliant with all applicable codes and standards.

B. Install per manufacturer's instructions and recommendations.

C. Obtain Contracting Officer's approval of locations of devices, appliances and annunciators before installation.

D. Circuits:

1. Signaling Line Circuits (SLC): Class B
2. Notification Appliance Circuits (NAC): Class B.

E. Spare Capacity:

1. New Notification Appliance Circuits:

a. Minimum 25 percent spare current capacity.
b. Maximum 10 percent voltage drop.
c. Utilize UL maximum current draw value for notification appliances in calculations.

2. New Signaling Line Circuit: Minimum 25 percent spare device capacity.

F. Power Sources:

1. Primary: Dedicated branch circuits of facility power distribution system.
2. Secondary: Storage batteries.
3. Capacity: Sufficient to operate fire alarm system under normal supervisory condition for 24 hours and operate alarm signals for five minutes at end of standby period.

G. Obtain approval of system design from AHJ prior to installation. Do not begin installation without approval from AHJ and submittal review comments from Contracting Officer.

H. Install in accordance with applicable codes, NFPA 72, NFPA 70 and the Contract Documents.

I. In accordance with manufacturer's instructions, provide wiring, conduit and outlet boxes required for the erection of a complete system as described in these specifications, as shown on Drawings and as required by AHJ.

J. Conceal wiring, conduit, boxes and supports where installed in finished areas.

K. Provide complete raceway system for wiring. Where raceway cannot be concealed, provide surface raceway in finished areas and surface mounted EMT in non-finished areas. Paint exposed raceway to match adjacent surface.

L. Paint all interior raceway to match adjacent surface.

M. Provide cabling and conduits system suitable for wet locations for below grade systems.

N. At junction boxes and termination points, provide identification tags on wires and cables.

O. Route wiring to avoid blocking access to equipment requiring service, access, or adjustment.

P. Existing Components:

1. Existing Fire Alarm System:

    a. Maintain fully operational until new equipment has been tested and accepted.
    b. As new equipment is installed, label it "Not in Service" until new equipment is accepted.
    c. Remove tags from new equipment when put into service and tag existing fire alarm equipment "Not in Service" until removed from building .

2. Disable system only to make switchovers and connections.

    a. Notify Contracting Officer before partially or completely disabling system.
    b. Notify local fire service.
    c. Make notifications at least five working days in advance.
    d. Make temporary connections to maintain service in areas adjacent to work area.

3. Provide fire watch in areas where the system is not functioning if required by the AHJ.
4. Equipment Removal:

    a. Remove existing system after acceptance of new fire alarm system. Restore damaged surfaces.
    b. Package operational fire alarm and detection equipment that has been removed and offer to Contracting Officer prior to disposal.
    c. Remove from site and legally dispose of remainder of existing material.

5. On-Premises Supervising Station: Include, as part of this work, modifications necessary to existing supervising station to accommodate new fire alarm work.

Q. Fire Safety Systems Interfaces:

1. Provide conduit, wiring, boxes and terminations from fire alarm system to monitored components. Provide a separate input module for each switch or relay to be monitored.

    a. Alarm Inputs: Provide connection in accordance with NFPA 72 for the following systems and components:

        1) Fire sprinkler water flow switches.
        2) Fire sprinkler dry-pipe alarm pressure switches.
        3) Fire sprinkler preaction system alarm pressure switches.

4) Special hazard fire suppression system alarm condition.
5) Other alarm inputs.

b. Supervisory Inputs: Provide connection in accordance with NFPA 72 for the following systems and components:

1) Fire sprinkler water control valve tamper switches.
2) Fire sprinkler dry-pipe system low air pressure switches.
3) Fire sprinkler preaction system low and high air pressure switches.
4) Special hazard fire suppression system supervisory condition.
5) Elevator shunt trip power monitoring circuit.
6) Other supervisory inputs.

c. Trouble Inputs: Provide connection in accordance with NFPA 72 for the following systems and components:

1) Other trouble inputs.

2. Fire Safety Functions:

a. Provide power and control conduit, wiring, boxes and terminations to power devices and interface to fire alarm system.
b. Elevators:

1) Provide elevator recall smoke detectors, addressable control relays and connection to elevator equipment per NFPA 72 and as required by the AHJ.
2) Provide elevator shunt trip heat detectors, addressable control relays for shunt trip operation, addressable input module for monitoring shunt trip power and connection to elevator equipment per NFPA 72 and as required by the AHJ.
3) Elevator Lobby and Machine Room Smoke Detectors: Elevator recall for fire fighters' service.
4) Elevator Machine Room Heat Detector: Shut down elevator power prior to Elevator Machine Room sprinkler activation.

R. Inspection and Testing for Completion:

1. System testing and commissioning to be performed by a certified manufacturer representative.
2. Perform inspection and testing in accordance with NFPA 72 and requirements of local authorities, document each inspection and test.
3. Document audibility measurements and verify intelligibility for each space on record drawings.
4. Provide the services of the installer's supervisor or person with equivalent qualifications to supervise inspection and testing, correction and adjustments.
5. Provide tools, software and supplies required to accomplish inspection and testing.
6. Prepare for testing by ensuring that work is complete and correct; perform preliminary tests as required to test system.
7. Correct defective work, adjust for proper operation and retest until entire system complies with Contract Documents.
8. Notify Contracting Officer seven days prior to beginning completion inspections and tests.

9. Notify authorities having jurisdiction and comply with their requirements for scheduling inspections and tests and for observation by their personnel.
10. Diagnostic Period: After successful completion of inspections and tests, operate system in normal mode for at least 14 days without any system or equipment malfunctions.

    a. Record all system operations and malfunctions.
    b. If a malfunction occurs, start diagnostic period over after correction of malfunction.
    c. Replace devices with readings outside of allowed value at time of system check out.
    d. Contracting Officer will provide attendant operator personnel during diagnostic period; schedule training to allow Contracting Officer personnel to perform normal duties.
    e. At end of successful diagnostic period, complete and submit NFPA 72 "Inspection and Testing Form."

S. Designated Personnel Instruction:

1. Provide the following instruction to Contracting Officer designated personnel:

    a. Hands-On Instruction: On-site, using operational system.
    b. Classroom Instruction: Contracting Officer furnished classroom, on-site or at other local facility.

2. Basic Operation: One-hour sessions for attendant personnel, security officers and associated staff; combination of classroom and hands-on:

    a. Initial Training: One session pre-closeout.
    b. Refresher Training: One session post-occupancy.

3. Detailed Operation: Two-hour sessions for engineering and maintenance staff; combination of classroom and hands-on:

    a. Initial Training: One session pre-closeout.
    b. Refresher Training: One session post-occupancy.

4. Furnish the services of instructors and teaching aids; have copies of operation and maintenance data and record drawings available during instruction.
5. Provide means of evaluation of trainees suitable to type of training given; report results to Contracting Officer.

T. Closeout:

1. Closeout Demonstration:

    a. Demonstrate proper operation of functions to Contracting Officer.
    b. Be prepared to conduct any of the required tests.
    c. Have at least one copy of operation and maintenance data, copy of project record drawings, input/output matrix and operator instruction chart(s) available during demonstration.
    d. Have authorized technical representative of control unit manufacturer present during demonstration.

e. Demonstration may be combined with inspection and testing required by AHJ. Notify AHJ in time to schedule demonstration.
f. Repeat demonstration until successful.

2. Substantial Completion of the project cannot be achieved until inspection and testing is successful and:

a. Specified diagnostic period without malfunction has been completed.
b. Approved operating and maintenance data has been delivered.
c. Spare parts, extra materials and tools have been delivered.
d. All aspects of operation have been demonstrated to Contracting Officer.
e. Final acceptance of the fire alarm system has been given by authorities having jurisdiction.
f. Occupancy permit has been granted.
g. Specified pre-closeout instruction is complete.

3. Perform post-occupancy instruction within three months after date of occupancy.

3.2 FIRE ALARM CONTROL PANEL

A. Provide control panels with 120VAC dedicated circuit per NFPA requirements.

B. Do not install cabinets or equipment below the battery cabinet. Do not locate battery and charging system cabinets in ceiling space.

C. Provide instruction charts at each control panel where system operations are performed. Obtain approval from the Contracting Officer prior to mounting.

D. Perform system programming at the fire alarm control panel. Program the system without shutting the system down. Programming is done off line. Provide copy of site-specific program on electronic storage media. Locate in document enclosure.

E. Room Name Labeling: Control panel schedules, programming and labeling for electrical equipment, to use the room names and room numbers that the Contracting Officer adopts at the date of substantial completion of construction. This work is to be done at no added cost to the Contract.

F. Programmable Function Keys: Provide control panel accessible function keys for the notification bypass, fire drill, fire door bypass, elevator control bypass, and supervising station bypass.

G. Programmed control point activation includes selective control of HVAC, fire door release, elevator recall, elevator shunt trip, and other fire safety and auxiliary functions.

H. Provide machine printed labels on switches and indicators.

3.3 NOTIFICATION APPLIANCE CIRCUIT PANELS

A. Provide notification appliance circuit panel power supplies with 120VAC dedicated circuit per NFPA requirements.

B. Do not install cabinets or equipment below the battery cabinet. Do not locate battery and charging system cabinets in ceiling space.

C. Provide power supply(s), adequate to serve modules, remote annunciators, initiating devices, notification appliances and other connected devices or appliances.

D. Provide batteries in locking cabinet manufactured for purpose.

3.4 FIRE ALARM TRANSMITTERS

A. Provide conduit and wiring for connections to the transmitter as required for fire alarm system off site supervision.

B. Verify and provide call sequence and message as directed by Contracting Officer and the AHJ.

3.5 FIRE ALARM ANNUNCIATORS

A. When required by the manufacturer, provide fire alarm annunciator with 120VAC dedicated circuit per NFPA requirements.

B. Provide machine printed labels on switches and indicators.

C. Verify location with AHJ before installation.

3.6 MANUAL PULL STATIONS

A. Provide machine printed address labels on addressable devices.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.7 FIXED TEMPERATURE HEAT DETECTORS

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.8 RATE-OF-RISE AND FIXED TEMPERATURE HEAT DETECTORS

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.9 PHOTOELECTRIC TYPE DETECTORS

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.10 RELAY MODULES

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification. Labels to include description of controlled function.

3.11 CONTROL MODULES

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification. Labels to include description of controlled function.

3.12 INPUT MODULES

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification. Labels to include description of monitored input.

3.13 FAULT ISOLATION MODULES

A. Provide machine printed address labels on addressable devices. Labels to be visible from the floor without magnification.

B. Provide Fault Isolator Modules for signaling line circuit per code requirements and manufacturer instructions.

3.14 COMBINATION HORN/STROBES

A. Provide machine printed labels on notification appliances with appliance circuit number and sequence. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.15 STROBES

A. Provide machine printed labels on notification appliances with appliance circuit number and sequence. Labels to be visible from the floor without magnification.

B. Provide protective guard where device is subject to abuse and where required by AHJ.

3.16 WEATHERPROOF/SURFACE BACKBOXES

A. Provide manufacturer's weatherproof backbox listed for use in areas where the device or appliance is subject to humidity in excess of listed rating.

B. Provide manufacturer surface backboxes where devices cannot be installed recessed.

3.17 PROTECTIVE GUARDS

A. Wire Guard.

B. Protective Cover.

3.18 CIRCUIT CONDUCTORS

A. Provide wiring to meet the requirements of national, state and local electrical codes. Provide color coded wiring as recommended and specified by the fire alarm and detection system manufacturer.

B. Provide Type FPLR cable when in a riser application or FPLP cable when installed in plenums.

3.19 SURGE PROTECTIONS

A. Install in accordance with IEEE C62.41 B3 combination waveform and NFPA 70; except for optical fiber conductors.

B. Provide for alternating current circuits powering fire alarm equipment.

C. Signaling Line Circuits: Provide surge protection at each point where circuit exits or enters a building, rated to protect applicable equipment.

D. Initiating Device Circuits, Notification Appliance Circuits and Communications Circuits: Provide surge protection at each point where circuit exits or enters a building.

3.20 BATTERIES

A. Provide batteries in locking cabinet manufactured for purpose. Provide additional cabinet, if required due to space limitations in control panels.

B. Provide machine printed label with installation date, and date of month and year of battery manufacture (MM/YYYY).

3.21 LOCKS AND KEYS

A. Deliver to Contracting Officer.

3.22 DOCUMENT STORAGE CABINET

A. Provide document storage cabinet adjacent to fire alarm control panel.

3.23 INSTRUCTION CHARTS

A. Install chart adjacent to fire control panel.

3.24 FRAMED FLOOR MAP

A. Provide framed floor plan of facility adjacent to the annunciator panel identifying room names/numbers, device/addresses or fire zone number and description as utilized on the annunciator panel, as required by local AHJ.

B. Check with the local fire department for size and approved mounting location.

END OF SECTION 28 46 00

SECTION 31 10 00 – SITE CLEARING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Protecting existing vegetation to remain.
2. Conserving topsoil.
3. Clearing and grubbing.
4. Disposal of material from clearing, grubbing, thinning and demolition in approved off-site disposal areas.
5. Filling of voids and excavations resulting from the work.

B. Related Sections:

1. Section 01 35 13.22 – Archeological Protection
2. Section 01 57 13 – Temporary Storm Water Pollution Prevention
3. Section 02 41 13 – Selective Demolition
4. Section 31 20 00 – Earth Moving
5. Section 31 23 16 – Topsoil

1.3 MATERIAL OWNERSHIP

A. Except for materials indicated to be stockpiled or to remain on the government's property, cleared materials shall become Contractor's property and shall be removed and disposed of legally except agreed upon felled trees which are to remain onsite. When possible, the Contractor should re-use or sell materials generated by Site Clearing such as timber, firewood, or wood chips.

1.4 SITE CONDITIONS

A. Site conditions existing during the bidding period will be maintained by the government insofar as practical.

B. Actual site condition variations that differ from those of the bidding period and which affect site clearing operations shall be brought to the attention of the Contracting Officer (CO) prior to the commencement of any site work.

1.5 SUBMITTALS

A. The Contractor shall submit the following information to the CO for review before commencing work:

1. Permits for transport and disposal of debris.
2. Demolition procedures and operational sequence.

PART 2 - PRODUCTS

2.1 TREE PROTECTION FENCING

A. Tree protection fencing shall be an orange plastic web fence, 4 feet high minimum.

B. Posts shall be at least six (6) foot long driven a minimum of two (2) feet into the ground. Posts shall be spaced eight (8) feet on center, maximum.

PART 3 - EXECUTION

3.1 PROTECTION

A. The Contractor shall flag the limits of clearing shown on the drawings by accurate field survey with marked stakes or other means acceptable to the CO. Trees to remain shall be clearly identified during this staking process. The CO shall be notified a minimum of five (5) working days prior to scheduled commencement of clearing operations to review the flagged limits. Adjust the clearing limits as directed by the CO.

B. Install erosion-control measures prior to start of any other construction to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust to adjacent properties and roadways in accordance with Section 01 57 13 – Temporary Storm Water Pollution Prevention.

C. Before clearing begins, protect designated trees to remain with tree protection fencing to the approximate diameter of foliage (dripline of the tree) to prevent damage to the trunk, foliage and root system by construction equipment and procedures.

D. Place tree protection fencing as required to protect other plants, adjacent property areas to remain uncleared, monuments, and existing improvements from damage.

E. The Contractor shall repair or replace immediately any damage to existing trees or root systems that are to remain. The Contractor shall employ an arborist licensed in the State of Washington to determine the repair and replacement needs and methods for approval by the CO.

F. Replace damaged shrubs and other vegetation designated to remain with the same size and species.

G. The tree protection fencing shall be maintained for the duration of construction operations. The work shall include immediate replacement of any damaged fence. Fencing shall be removed from the site at the completion of construction operations. The fencing disposal shall be in accordance with local, state, and federal laws and regulations for the disposal of the material.

3.2 UTILITIES

A. Notify all corporations, companies, individuals, or local authorities owning or having jurisdiction over utilities running to, through, or across areas to be affected by site clearing operations.

B. Locate and identify existing utilities that are to remain and protect them from damage.

3.3 CONSERVING TOPSOIL

A. See Section 31 23 16 Topsoil for soil conservation execution.

3.4 CLEARING AND GRUBBING

A. Clearing shall include cutting, removal, and off-site disposal of trees, bushes, shrubs, stumps, fallen timber, brush, refuse, trash, fencing and other incidental materials not required for reuse on the site.

B. The Contractor shall grub the area within the clearing limits to completely remove stumps and root systems to a depth of at least 12-inches, except for those to remain.

C. Depressions, excavations, and voids resulting from the removal of stumps or roots shall be filled with suitable material and compacted as specified under Section 31 20 00 – Earth Moving.

3.5 DISPOSAL

A. Remove all material resulting from site clearing and demolition operations from site except agreed upon felled trees to remain.

B. No burning of any material will be allowed.

END OF SECTION 31 10 00

SECTION 31 20 00 – EARTH MOVING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Preparing subgrades for walks and pavements.
2. Subbase course for walks.
3. Subbase and base course for asphalt and concrete paving.
4. Excavating and backfilling for utility trenches.
5. Porous backfill for walls and foundation drains.

B. Related Sections include the following:

1. Section 01 35 13.22 Archeological Protection
2. Section 01 57 23 – Temporary Strom Water Pollution Prevention
3. Section 31 10 00 – Site Clearing
4. Section 31 25 00 – Erosion and Sediment Control
5. Section 32 12 16 – Asphalt Paving
6. Section 32 13 20 – Site Concrete
7. Section 33 46 00 – Subdrainage

1.3 DEFINITIONS

A. Backfill: Soil material or controlled low-strength material used to fill an excavation.

1. Initial Backfill: Backfill placed beside and over pipe in a trench, including haunches to support sides of pipe.
2. Final Backfill: Backfill placed over initial backfill to fill a trench.

B. Base Course: Course placed between the subgrade and hot-mix asphalt paving or cement concrete paving.

C. Bedding Material: Course placed over the excavated subgrade in a trench before laying pipe.

D. Borrow Soil: Satisfactory soil imported from off-site for use as fill or backfill.

E. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.

1. Authorized Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by CO.
2. Bulk Excavation: Excavation more than 5 feet in width and more than 20 feet in length.
3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction by CO. Unauthorized excavation, as well as remedial work directed by CO, shall be without additional compensation.

F. Fill: Soil materials used to raise existing grades.

G. Rock: Rock material in beds, ledges, unstratified masses, conglomerate deposits, and boulders of rock material 3/4 cu. yd. or more in volume that exceed a standard penetration resistance of 100 blows/2 inches when tested by an independent geotechnical testing agency, according to ASTM D 1586.

H. Structures: Buildings, footings, foundations, retaining walls, curbs, or other man-made stationary features constructed above or below the ground surface.

I. Subgrade: Surface or elevation remaining after completing excavation, or top surface of a fill or backfill immediately below drainage fill, or topsoil materials.

J. Utilities: On-site underground pipes, conduits, ducts, and cables.

K. Embankment: Soil or soil-rock material placed to raise the subgrade or natural grade of the site for embankment construction.

L. Engineered Fill: Clean, crushed, granular material used to create embankments that will support structures. Structural fill material will be material that is considered an all-weather fill material.

M. All Weather Fill Material: Imported material that has minimal fines, is free draining, and can be placed, shaped, and compacted using normal compaction methods during most wet weather with minimal pumping and shoving.

N. Unsuitable Material: Excavated material or material below the natural ground surface in embankment areas or below subgrade elevation in excavated areas, which is unsuitable for its planned use as determined by the CO. Unsuitable material is further defined as material determined to be:

1. Of such unstable nature as to be incapable of being compacted to specified density using ordinary methods at optimum moisture content.
2. Too wet to be properly compacted and circumstances prevent suitable in-place drying prior to incorporation into the Work.
3. Not meeting requirements specified in Part 2 of this Section.
4. Otherwise unsuitable for the planned use.

   The presence of excessive moisture in a material is not, by itself, sufficient cause for determining that the material is unsuitable.

O. Relative Compaction: The ratio, expressed as a percentage, of the in-place dry density of fill materials as compacted in the field to the maximum dry density of the same material.

P. Optimum Moisture Content: The water content at which a soil can be compacted to a maximum dry unit weight by a given compactive effort.

Q. Native Soil: Sand or silt naturally occurring on site that was never imported to the site, is weed free, and supports local native habitats.

R. Earthwork Terminology: Terms used in this Section and not defined herein shall be interpreted in accordance with the definitions given in ASTM D653.

1.4 SUBMITTALS

A. Test Reports: Submit certified test reports of all tests specified to be performed. Test reports shall be signed and sealed by a geotechnical engineer registered in the State of Iowa.

B. Delivery Tickets: Submit a delivery ticket with each load of imported embankment material delivered to the site, stating the type of fill material and the quantity.

C. Product Data: For the following:

1. All aggregates proposed for use.
2. Import fill.

D. Material Test Reports: From a qualified testing agency indicating and interpreting test results for compliance of the following with requirements indicated:

1. Classification according to ASTM D 2487 of each on-site and import soil material proposed for fill and backfill.
2. Laboratory compaction curve according to ASTM D 1557 for each on-site and import soil material proposed for fill and backfill.

E. Pre-excavation Photographs or Videotape: Show existing conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by earthwork operations. Submit before earthwork begins.

F. Contractor is to provide the CO with copies of the documentation for the legal disposal of materials offsite.

1.5 QUALITY ASSURANCE

A. Pre-excavation Conference: Conduct conference at Project site to comply with requirements in Division 01 Section “Project Management & Coordination.”

B. Testing will be performed by an independent testing laboratory selected by the Contractor. All tests will be paid for by the Contractor.

C. Allow access for CO to obtain samples and perform in-place tests. Remove surface materials as required to allow sampling and testing.

D. Tolerances:

1. Construct finished subgrades to plus 0 or minus 1/2-inch of the elevations indicated.
2. Construct finished grade of slopes not steeper than 3:1 plus or minus 1/2 inch and on slopes steeper than 3:1 plus or minus two inches.
3. Maintain the moisture content of fill material as it is being placed within plus or minus 2% of the optimum moisture content of the material as determined by the laboratory tests herein specified.

1.6 PROJECT CONDITIONS

A. Existing Utilities: Do not interrupt utilities serving facilities occupied by Government or others unless permitted in writing by Government and then only after arranging to provide temporary utility services according to requirements indicated.

1. Notify Government not less than two days in advance of proposed utility interruptions.
2. Do not proceed with utility interruptions without Government's written permission.
3. Contact utility-locator service for area where Project is located before excavating.

B. Archeological monitoring will be required during any ground disturbance. Refer to Section 01 35 13.22 Archeological Protection.

C. Demolish and completely remove from site existing underground utilities indicated to be removed. Coordinate with utility companies to shut off services if lines are active.

D. Unfavorable Weather Conditions:

1. Excavating, filling, backfilling, and grading work shall not be performed during weather conditions that might damage or be detrimental to the condition of existing ground, in-progress work, or completed work. When the Work is interrupted by rain, freezing weather, or other conditions deemed unsuitable by the CO, excavating, filling, backfilling, and grading work shall not resume until the site and soil condition (moisture content) are suitable for compaction.
2. Subgrade shall be free from mud, snow, ice, and deleterious material when work is resumed.
3. Soil material that is too wet for compaction shall be left to drain, to be aerated and dried by disking and harrowing or other approved method until the moisture content of the area is uniform and within the specified limits.

E. Erosion Prevention: Protect stockpiles, ditches, embankments, filled, backfilled, and graded areas to prevent erosion until such time as permanent drainage and erosion control measures have been installed per the requirements of Section 01 57 23 – Temporary Storm Water Pollution Prevention.

F. Control of Dust: Employ construction methods and means that keep flying dust to the minimum throughout the entire Contract period. It may include watering and sprinkling, or other approved methods at the construction site of such frequency that will satisfactorily control the dust caused by construction operations.

G. Control of Dirt: Provide for prompt removal from existing roadways of all dirt and other materials that have been spilled, washed, tracked, or otherwise deposited thereon by construction operations whenever the accumulation is sufficient to cause the formation of mud, interfere with drainage, damage pavements, or create a traffic hazard.

H. Indicated locations of existing facilities and systems are approximate. Investigate and determine exact location and nature of utilities, facilities, and systems, and be solely responsible for damages caused by construction activities.

I. Excavation in areas that have or may have underground facilities:

1. Delineate with white paint, flags, or other methods the area to be excavated and contact Underground Location Service to request that facility owners mark the location of underground utilities.
2. Identify temporary survey markings using PINK colored markings in accordance with Underground Service alert requirements.
3. Facility owners will field mark approximate location and number of subsurface installations affected by excavation using the following colors:

    a. Safety RED – Electric power, distribution, and transmission installations, or municipal electric installations.
    b. High Visibility Safety YELLOW – Gas distribution and transmission installations, oil distribution and transmission installations containing or transporting dangerous materials, products, or steam.
    c. Safety Alert ORANGE – Telephone and telegraph installations, police and fire communication installations, and cable television installations.
    d. Safety Precaution BLUE – Water installations and slurry pipelines.
    e. Safety GREEN – Sewer installations.
    f. Do not use above listed colors for Contractor’s temporary markings such as survey points; use U.S.A. designated PINK color.

PART 2 - PRODUCTS

2.1 FILL MATERIALS

A. Material used for fill, backfill, and embankment construction shall be an inert, nonexpansive soil, free from organic matter and other deleterious substances and of such quality that it will compact thoroughly without the presence of voids when watered and rolled.

B. Excavated on-site material shall not be considered suitable for support of paving or other exterior walks and slabs on grade.

2.2 SOIL MATERIALS

A. Embankment: All embankment material shall consist of clean well-graded soil with little or no potential for expansion.

B. Pipe Bedding Material:

1. Aggregate of the following types:

a. Sand and gravelly sand free from organic matter, loam, contaminants, and other deleterious substances.

2. Minimum sand equivalent of 30 as determined by ASTM D2419.
3. Complying with the gradation requirements shown in the following table:

Sieve Size	Percentage Passing
½″	100
No. 4	70–100
No. 16	50–90
No. 50	10–40
No. 200	0–10

C. Base Course: Shall be compacted to at least 95% of the maximum density per ASTM D 1577 and shall be within the percentage passing limits for the sieve sizes shown in the following table:

Sieve Size	Percentage Passing
1-1/2″	100
1/2″	40–80
No. 8	5–25
No. 200	0–6

D. Sand: ASTM C 33; fine aggregate, natural, or manufactured sand.

E. Porous Backfill Material:

1. Porous backfill material shall be crushed stone or processed gravel complying with the grading requirements shown in the following table:

Sieve Size	Percentage Passing
¾″	100
½″	95–100
3/8″	50–100
No. 4	0–50
No. 8	0–8

PART 3 - EXECUTION

3.1 PREPARATION

A. Protect structures, utilities, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.

B. Preparation of subgrade for earthwork operations including removal of vegetation, topsoil, debris, and obstructions, from ground surface is specified in Division 31 Section “Site Clearing.”

C. Provide protective insulating materials to protect subgrades and foundation soils against freezing temperatures or frost.

3.2 TESTING

A. Testing will be performed by an independent testing laboratory selected by the Contractor. All tests will be paid for by the Contractor.

B. Density Tests: Compacted fill, backfill, and embankment shall be tested to verify compliance with specified requirements.

C. Fill or backfill that does not meet the specified requirements shall be removed or conditioned and recompacted until the requirements are satisfied.

D. Any completed earthwork disturbed by subsequent construction operations or adverse weather shall be corrected in a verified manner as directed by the CO.

E. Perform backfill material qualification tests and field density tests for each category of products in Part 2 of this Section. Also provide details of testing on graph format.

3.3 STAKING AND GRADES

A. Layout the work, establish all necessary markers, benchmarks, grading stakes, and other stakes as required, in accordance with the lines and grades of the Plans.

B. Perform field surveys to include the following:

1. Final surveys when excavations, backfills, and embankments are completed.

3.4 UTILITIES

A. Verify on site the location and depth (elevation) of all existing utilities and services before performing any excavation work. Use the one call utility service and other means required to verify locations.

B. Active utility lines encountered, which are not indicated in the Contract Documents, shall be reported immediately to the CO and utility owners involved. The CO and utility authorities shall be permitted free access to determine the measures deemed necessary to repair, relocate, or remove the utility.

3.5 EARTHWORK GENERAL REQUIREMENTS

A. Archeological monitoring will be required during any ground disturbance. Refer to Section 01 35 13.22 Archeological Protection.

B. Erosion Protection: Prevent erosion of the site at all times. Conform to all the requirements as per Section 01 57 23 – Temporary Storm Water Pollution Prevention.

C. Construction Traffic: Disperse travel paths of traffic and construction equipment over entire width of compacted surfaces so as to aid in obtaining uniform compaction. Protect exposed soil layers with high moisture content from excessive wheel loads.

D. On-Site Excavation:

1. Do not excavate or remove any material from the project site which is not within the designated excavation.

E. Disposal of Surplus Material:

1. Excess earth materials, unsuitable materials, and debris shall become the property of the Contractor and shall be removed from the site and disposed of in a legal manner. The Contractor is to provide the CO with copies of the documentation for the legal disposal of materials.
2. Location of disposal site and length of haul shall be the Contractor's responsibility.

F. Maintenance of Excavations, Slopes, and Embankments:

1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.

G. Dust and Dirt Control: Provide satisfactory control or remove dust and/or dirt at the project site and adjacent roadways whenever traffic or buildings that are occupied or in use, are affected.

3.6 ROUGH GRADING AND FILLING

A. Fill or backfill holes, swales, and low points that will not otherwise be removed in the course of the Work, to the indicated grades.

3.7 DEWATERING

A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.

B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.

1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.
2. Install a dewatering system to keep subgrades dry and convey ground water away from excavations. Maintain until dewatering is no longer required

3.8 EXPLOSIVES

A. Explosives: Do not use explosives.

3.9 EXCAVATION

A. The bottoms of excavations shall be level, firm, undisturbed earth, clean and free from loose material, debris, and foreign matter.

B. Excavate to the lines and grades indicated on the Contract Drawings.

C. Support and maintain excavation so that sides are stable and will not move. Where excavations are deeper than 4 feet, the sides of the excavations shall be sloped back at 1.5H:1V or shored for safety. Unshored excavations shall not extend below a plane drawn at 1.5H:1V extending downward from any adjacent existing footings

D. Limits of excavations shall allow for adequate working space for installing forms, and as required for safety of personnel.

E. Except as otherwise indicated, preserve the material below and beyond the lines of excavations. Where excavation is carried below the indicated grade, backfill to the indicated grade as herein specified using materials specified in Section 2.2.

F. Excavation and its restoration, for convenience of the Contractor, shall be at no additional expense to the Government.

G. Place excavated material at sufficient distance from edge of excavation so as not to cause cave-ins or bank slides, but in no case closer than three feet from the edge of excavations.

H. Existing fill soils shall not be considered suitable for fill. Existing fill soils shall be removed and replaced as properly compacted fill.

3.10 EMBANKMENT CONSTRUCTION

A. Fill shall be placed in level lifts not exceeding 8 inches in loose thickness and compacted. The fill shall be compacted to at least 90 percent of the maximum density in accordance with ASTM D 698, except the upper 12 inches. The upper 12 inches of exposed base course and/or subgrade shall be compacted to at least 95% of the maximum density in accordance with ASTM D 698.

B. Each lift shall be brought to within 2 percent of optimum moisture content.

C. Finished fill slopes shall be constructed at an inclination no steeper than 2:1. Fill slopes shall be overbuilt and trimmed back as necessary to expose a well-compacted surface.

D. Remove material that cannot be compacted to the required density within specified tolerances and replace with suitable material at no additional expense to the Government.

3.11 SUBGRADE PREPARTION

A. Notify CO when excavations have reached required subgrade.

B. The subgrade surface shall be free of any water, debris, organic, loose, soft, disturbed, or other deleterious material prior to placement of fill, reinforcing bars, and concrete. If CO determines that unsatisfactory soil is present, continue excavation and replace with compacted backfill or fill material as directed.

C. Perform all cutting, blading, and shaping as required to cut and shape the subgrade to the grades and elevations indicated.

D. Subgrade preparation includes fine grading, reworking as necessary, and preparation of cut, fill, or embankment upon which structure and equipment foundations, pipe, base, and pavement will be placed. Subgrade preparation shall be performed immediately prior to placement of the base course.

E. Maintain proper drainage of the paved areas throughout construction.

F. Subgrade, including trench backfills, shall be compacted to at least 95% of the maximum density obtained by ASTM D 1557 method of compaction.

G. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by CO, without additional compensation.

3.12 STORAGE OF SOIL MATERIALS

A. Stockpile borrow soil materials and excavated satisfactory soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.

1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees.
2. Temporary stockpiles shall be no higher than 3 feet.

3.13 BACKFILL

A. Place and compact backfill in excavations promptly, but not before completing the following:

1. Surveying locations of underground utilities for Record Documents.
2. Testing and inspecting underground utilities.
3. Removing concrete formwork.
4. Removing trash and debris.

B. Place backfill on subgrades free of mud, frost, snow, or ice.

C. Place backfill in horizontal layers not to exceed eight inches of loose material, and compact each layer to specified density before the next layer is placed.

D. Do not backfill on or against structural concrete until the specified 28-day concrete strength has been attained.

E. Do not use compaction equipment and methods that produce excessive horizontal or vertical earth pressures.

3.14 UTILITY TRENCH BACKFILL

A. Place backfill on subgrades free of mud, frost, snow, or ice.

B. Trenches shall be graded uniformly sloped between pipe elevations as shown on the drawings.

C. Place and compact bedding course on trench bottoms and where indicated. A 6-inch layer of bedding material shall be placed beneath utilities. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.

D. Place and compact initial backfill of pipe bedding material to a height of 12 inches over the utility pipe or conduit. Vibrate pipe bedding materials using vibratory equipment.

1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of utility piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.

E. Backfill voids with satisfactory soil while installing and removing shoring and bracing.

F. Backfill and compact over excavations to ninety-five percent relative compaction with bedding material. There shall be no additional payment to the Contractor for over excavations not directed by the CO. Remove unsatisfactory material encountered below the grades shown as directed by the CO and replace with bedding material. Payment for removal and replacements of such unsatisfactory material directed by the CO shall be made in accordance with the provisions of the General Conditions.

G. Install warning tape directly above utilities, 12 inches below finished grade, except 6 inches below subgrade under pavements and slabs.

3.15 SOIL MOISTURE CONTROL

A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 3 percent of optimum moisture content.

1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
2. Remove and replace, or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 3 percent and is too wet to compact to specified dry unit weight.

3.16 COMPACTION OF SOIL BACKFILLS AND FILLS

A. Place backfill and fill soil materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.

B. Place backfill and fill soil materials evenly on all sides of structures to required elevations, and uniformly along the full length of each structure.

C. Fill compaction shall be verified by means of in-place density tests performed per ASTM D6938 (or appropriate alternative when ASTM D6938 is not suitable for the fill material) during fill placement so that compaction may be evaluated as earthwork progresses.

D. Compact soil materials to not less than the following percentages of maximum dry unit weight according to ASTM D 1557:

Fill Application	**Minimum Compaction %**
Retaining wall footing subgrade	95
Concrete slab subgrades	95
Asphalt pavement base and subbase	95
Asphalt pavement subgrade, within 3 feet of finished grade	95
Asphalt pavement subgrade, more than 3 feet below finished grade	90

3.17 GRADING

A. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.

1. Provide a smooth transition between adjacent existing grades and new grades.
2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.

B. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:

1. Unpaved Areas: Plus or minus 1 inch.
2. Walks: Plus or minus 1 inch.
3. Pavements: Plus or minus 1/2 inch.

3.18 BASE COURSES

A. Place base course on subgrades free of mud, frost, snow, or ice.

B. On prepared subgrade, place base course under pavements and walks as follows:

1. Place base course material over subbase course under hot-mix asphalt pavement.
2. Shape subbase and base course to required crown elevations and cross-slope grades.
3. Place base course 6 inches or less in compacted thickness in a single layer.

4. Place base course that exceeds 6 inches in compacted thickness in layers of equal thickness, with no compacted layer more than 6 inches thick or less than 3 inches thick.
5. Compact base course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than 95 percent of maximum dry unit weight according to ASTM D 1557.

3.19 SHORING AND SHEETING

A. Construct and maintain all shoring, sheeting, and slope lay-back necessary to protect the excavation, as need for safety of the employees and as required by applicable State and Federal laws.

B. Where excavations are deeper than 4 feet, the sides of the excavations shall be sloped back at 1.5H:1V or shored for safety. When work area is left open and unattended by Contractor, provide suitable barricades for public safety, regardless of trench depth. Provide suitable exit means in accordance with applicable provisions in OSHA.

C. Do not remove timber or sheeting if it is in a compacted zone. Trim timber or sheeting off at a safe level above that zone.

D. As directed, remove all other sheeting and shoring when no longer needed.

3.20 FIELD QUALITY CONTROL

A. Testing Agency: Contractor shall engage a qualified independent geotechnical engineering testing agency to perform field quality-control testing.

B. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earthwork only after test results for previously completed work comply with requirements.

C. Testing agency will test compaction of soils in place according to ASTM D 1556, ASTM D 2167, ASTM D 2922, and ASTM D 2937, as applicable. Tests will be performed at the following locations and frequencies:

1. Paved Areas: At subgrade and at each compacted fill and backfill layer, at least 1 test for every 2000 sq. ft. or less of paved area or building slab, but in no case fewer than 3 tests.
2. Trench Backfill: At each compacted layer as described in 3.16.A, 1 test for each 150 feet or less of trench length, but no fewer than 2 tests.

D. When testing agency reports that subgrades, fills, or backfills have not achieved degree of compaction specified, scarify, and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.

3.21 PROTECTION

A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing, and erosion. Keep free of trash and debris.

B. Repair and reestablish grades to specified tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.

1. Scarify or remove and replace soil material to depth as directed by CO; reshape and recompact.

C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.

1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to greatest extent possible.

3.22 DISPOSAL OF SURPLUS AND WASTE MATERIALS

A. Disposal: Remove surplus satisfactory soil and waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off Government property.

B. Disposal: Transport surplus satisfactory soil to designated storage areas on Government property. Stockpile or spread soil as directed by CO.

1. Remove waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off Government property.

END OF SECTION 31 20 00

SECTION 31 21 13 – RADON MITIGATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Radon mitigation of the Visitor Center basement and crawlspaces areas

B. Related Sections:

1. Section 07 26 00 – Vapor Retarders

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Radon Control Plan: A radon mitigation plan for controlling radon and exhausting the basement and crawlspace areas in the Visitor Center. The plan shall consist of:

1. Subfloor preparation: Location, and method.
2. Soil-Gas-Retarder: Size, type, quantity, location, installation method.
3. Sump: Size, type, quantity, location, installation method, if necessary.
4. Drain Tile Loop: Size, type, quantity, location, installation method, if necessary.
5. Exhaust Fans: Size, type, quantity, location, installation method.
6. Vent Piping: Size, type, quantity, location, installation method.

C. Qualification Data:

1. Installers: All radon mitigation installers are qualified with successful completion with projects of similar size and scope.

D. Application for Permit, if necessary, completed by the Contractor for submittal to Iowa Health & Human Services.

1.4 INFORMATIONAL SUBMITTALS

A. Product Test Reports: For each product, for tests performed by a qualified testing agency.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

A. Deliver products in good condition.

B. Store products to prevent corrosion, deterioration, and damage.

C. Handle products to prevent damage.

1.6 QUALITY ASSURANCE

A. Accessibility Requirements: Comply with Architectural Barriers Act Accessibility Standards (ABAAS) when installing.

PART 2 - PRODUCTS

2.1 AGGREGATE, IF NECESSARY

A. The aggregate shall consist of a material that will pass through a 2-inch sieve and be retained by a 0.25-inch sieve.

2.2 SOIL-GAS-RETARDER

A. A minimum 6-mil polyethylene or equivalent flexible sheeting material.

B. For placement in crawlspace and unfinished bare soil areas. May be considered for areas where floor is cored open to vent.

2.3 VENT PIPE

A. A minimum 3-inch diameter ABS, PVC or equivalent gas-tight pipe.

2.4 VENT FAN

A. Vent fans, if necessary, for exhausting radon gas through the venting system.

B. Shall be applied in accordance with the manufacturer’s recommendations.

2.5 ACCESSORIES

A. Vent Pipe Connections: A connection or union joint compatible with the vent piping system.

B. Sealants / Caulks: Compatible with the vent piping and penetration areas for sealing air-tight connection, joints, unions, and other penetrations.

C. Junction Box: For electrical connection, if necessary, to power vent fans located in the attic for exhausting the vent system.

D. Shall be applied in accordance with the manufacturer's recommendations.

PART 3 - EXECUTION

3.1 SYSTEM INSTALLATION AND PERFORMANCE

A. Contractor shall design and construct radon mitigation system based on manufacturers and qualified expert's recommendations to provide radon mitigation and ventilation of the Visitor Center.

B. Contractor shall provide system installation plan for CO review and approval prior to system installation.

3.2 PREPARATION

A. Examine areas and conditions, with Installer present, for compliance with requirements and other conditions affecting performance.

1. Proceed with installation only after unsatisfactory conditions have been corrected.

3.3 INSTALLATION, GENERAL

A. Comply with manufacturer's written installation instructions and approved installation plan unless more stringent requirements are indicated. Complete system assembly shall be performed on-site.

3.4 PROTECTION

A. Protect materials and system installation from damage before and after system installation.

END OF SECTION 31 21 13

SECTION 31 23 16 – TOPSOIL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This section includes the following: furnishing, conserving, stockpiling, and placing topsoil on a previously prepared subgrade. Contractor shall use topsoil excavated within the project area prior to importing additional topsoil.

B. This Section includes the following:

1. Section 01 57 23 – Temporary Stormwater Pollution Prevention
2. Section 31 10 00 – Site Clearing
3. Section 31 20 00 – Earth Moving

1.3 QUALITY ASSURANCE

A. Soil-Testing Laboratory Qualifications: An independent or university laboratory, recognized by the State Department of Agriculture, with the experience and capability to conduct the testing indicated and that specializes in types of tests to be performed.

B. Submit soil analysis report for imported topsoil from the State University Agricultural Extension Service or other approved soil testing laboratory. Report shall cover soil textural classification (percentages of sand, silt, and clay), pH, % organic matter, and soluble salts (electric conductivity in millimoles/centimeter), and shall include additive recommendations. Testing will be at the expense of the Contractor.

1.4 DELIVERY, STORAGE AND HANDLING

A. Do not deliver or place topsoil in frozen, wet, or muddy condition.

B. The Contractor shall submit to the CO for approval a complete list of all materials to be used prior to delivery of any materials to the site. Include complete data sources, amount, and quantity. This submittal shall in no way be construed at permitting substitution for specific items described on the plants or in the specification unless approved in writing by the CO. The CO or COR reserves the right to reject materials that are not weed free.

C. Follow all specifications listed in Section 31 10 00 Site Clearing for conserving topsoil. Cover conserved topsoil stockpiles with an approved breathable water repellent fabric. Anchor the fabric around the perimeter to hold it in place.

D. Do not conserve topsoil that is contaminated with seeds from non-native or invasive species.

1.5 SUBMITTALS

A. Sources: Submit name and address of sources for imported soil materials. Sources must be on NPS approved list of Quarries, Borrow Pits and Other Material Sources Outside of NPS Boundaries.

B. Material Test Reports: For imported or manufactured topsoil.

C. Imported Topsoil Sample: 1-gallon volume in sealed plastic bags labeled with source. Each Sample shall be typical of the lot of material to be furnished; provide an accurate representation of color, texture, and organic makeup.

D. Delivery Tickets: Submit delivery tickets to CO on a daily basis upon delivery of these materials:

1. Imported Topsoil

PART 2 - PRODUCTS

2.1 IMPORTED TOPSOIL

A. Topsoil will include natural humus bearing soils or approved soils. If there is a shortage of topsoil remaining from site clearing, notify the CO, who will consult with NPS representative for an acceptable source of additional material located outside the park. If NPS approved borrow pits do not have suitable topsoil, NPS will consider contractor provided options, but will need to inspect the quarry for noxious weeds. Provide notice of at least 1 month before topsoil needs to be applied.

B. All imported topsoil shall be a loam or sandy loam. Topsoil shall be furnished by the Contractor and shall be a natural, friable soil representative of productive soils in the vicinity. It shall be obtained from the top 12″ of well drained areas. Imported topsoil or other imported soil must be inspected by NPS Vegetation Program staff for approval on texture, biologic activity, and absence of exotic plant species prior to delivery.

1. Fertile, friable, loamy soil, reasonably free from subsoil, refuse, roots, heavy or stiff clay, stones larger than 1-inch, coarse sand, noxious seeds, sticks, brush, litter, and other deleterious substances; suitable for the germination of seeds and the support of vegetative growth. The pH value shall be between 7.0 and 8.0.
2. Soil Texture: Soil can be loamy sand, sandy loam, sandy clay loam, or loam. Sand 30-50%, Silt 0-50%, Clay 5-30%.
3. Additives: none
4. % Organic Content: 2.0% minimum.
5. Soluble Salts: Electric conductivity shall be less than 2.0 mmhos/cm

PART 3 - EXECUTION

3.1 EXAMINATION:

A. General: Verify that existing site conditions are as specified and indicated before beginning work under this Section.

1. Grades: Inspect to verify rough grading is within +/- 0.1 foot of grades indicated and specified.
2. Damaged Earth: Inspect to verify that earth rendered unfit to receive planting do to concrete, water, mortar, limewater, or any other contaminant dumped on it has been removed and replaced with clean earth from a source approved by the CO.

B. Surface Grade: Remove weeds, debris, clods, and rocks larger than 8″. Dispose of accumulated debris at direction of the CO.

C. Runoff: Take measures and furnish equipment and labor necessary to control the flow, drainage, and accumulation of water. Ensure that all water will run off grades.

3.2 PREPARATION

A. Remove vegetation prior to stripping of topsoil. Break down grasses, shrubs, sagebrush, and other minor brush less than 3-feet in height and woody plants and trees less than 2-inches in diameter by chipping, mowing, or shredding. Incorporate this material with the conserved topsoil. Ensure material is reduced to less than 2 inch in any dimension prior to incorporation. Remove and dispose off-site, all brush and trees over 3 feet tall or with a diameter over 2 inches.

B. Limits of topsoil stripping shall be verified with the CO prior to removal, and stockpile locations shall be identified so as not to interfere with the other work.

C. Conserved Topsoil: Topsoil refers to the uppermost soil horizon including the top 6 inches which includes organic duff and other materials capable of supporting vegetation.

3.3 PLACING TOPSOIL

A. Scarify compacted subgrade in planting areas and any area to be restored to a minimum depth of 2-inches to bond 1-inch of topsoil minimum after settlement to subsoil. Topsoil shall be free from weeds, sod, clods, and stones larger than 8″, toxic substances, litter, or other deleterious material. Spread evenly and grade to elevations and slopes shown. Hand rake areas inaccessible to machine grading.

3.4 STORING

A. The topsoil, after being stripped, shall be immediately stockpiled for reuse in staging area determined by the CO.

B. Store topsoil in windrow or pile formation with a maximum height of 3 feet protected from wind and water erosion, compaction, and contaminants unless otherwise approved by the CO. Separate topsoil from other excavated material. Do not mix topsoil with subsoil. Stockpiles shall be stabilized with excelsior logs/silt fence or natural fiber (excluding straw) blankets if determined necessary by the CO. Flag or install barriers to keep heavy equipment/vehicles from driving on stockpiled topsoil. Coordinate at least 2 weeks prior to topsoil salvage with CO and NPS to ensure their on-site presence at the initiation of salvage activities. Stockpiles shall be watered down periodically by the Contractor as necessary to mitigate the formation of windblown dust.

C. Stockpile topsoil and surface soil away from edge of excavations without intermixing it with subsoil. Grade and shape stockpiles to drain surface water. Cover to prevent windblown dust.

1. Limit height of topsoil stockpiles to 36″ unless authorized by CO.
2. Do not stockpile topsoil within drip line of remaining trees
3. Stockpile surplus topsoil to allow for respreading a thicker layer of topsoil

3.5 WEED CONTROL

A. All stockpiles shall remain weed free.

3.6 DISPOSAL

A. Move surplus soil to a location within the park as directed by the CO.

B. Remove all other waste material including excess subsoil, unsuitable soil, trash, and debris and legally dispose of them off CO’s property.

END OF SECTION 31 23 16

SECTION 31 25 00 – EROSION & SEDIMENT CONTROL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes:

1. Perimeter sediment control.
2. Site erosion control.

B. Related Sections:

1. Section 01 57 23 – Temporary Storm Water Pollution Prevention
2. Section 31 10 00 – Site Clearing
3. Section 31 20 00 – Earth Moving

1.3 SUBMITTALS

A. Product data for the following:

1. Wattles, including weed-free certification for all materials.

PART 2 - PRODUCTS

2.1 GENERAL

A. To the extent feasible, use wildlife safe materials for erosion control products.

2.2 WATTLES

A. Provide certified weed free logs that consist of drainage filter made of curled aspen wood excelsior and rolled into a cylindrical shape with a consistent width of fibers evenly distributed throughout the cylinder.

B. Wattles shall be a minimum of 9 inches and maximum 12 inches in diameter.

C. Wattles shall be encased in 100% natural fiber biodegradable netting. Photodegradable materials shall not be permitted. Netting opening size shall be ½-inch or greater.

D. Stakes for wattles shall be made of wood from untreated Douglas fir, hemlock, or pine species and shall be 24-inches long.

PART 3 - EXECUTION

3.1 WATTLES

A. Wattles shall be installed in accordance with the manufacturer's recommendations as soon as construction will allow or when designated by the CO.

B. Place sediment logs in a shallow trench, the depth of which shall be 1/3 the diameter of the log. Secure the log every 2 linear feet across the length of the log with a 24-inch wooden 1-inch x 1-inch (or larger stakes) driven a minimum of 14 inches from backfilled grade or 10 inches from the bottom of the wattle into the ground. Do not use wooden stakes that are severely split.

C. When placing more than one log in a continuous section the logs should overlap 2'. Install perpendicular to the contour, in a staggered overlapping pattern so that an unnatural pattern cannot be detected.

D. The Contractor shall exercise care when installing wattles to ensure the method of installation minimizes the disturbance of waterways and prevents sediment or pollution discharge into water bodies.

E. When accumulated sediment reaches a level one-half the height of the wattle, or when the wattle becomes clogged with sediment and no longer allows runoff to flow through, remove and replace the wattle.

END OF SECTION 31 25 00

SECTION 32 13 20 – SITE CONCRETE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes cast-in-place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and special concrete flatwork finish, for the following:

1. Concrete sidewalk.

B. Related Sections:

1. Section 31 20 00 – Earth Moving.

1.3 DEFINITIONS

A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume; subject to compliance with requirements.

B. Broom Finish Concrete Paving: Where shown on the drawings, using a medium bristle concrete finishing brush perpendicular to the direction of travel.

1.4 SUBMITTALS

A. Product Data: For each type of product.

B. Design Mixtures: For each concrete mixture. Submit design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

C. Steel Reinforcement Shop Drawings: Placing drawings that detail fabrication, bending, and placement. Include bar sizes, lengths, material, grade, bar schedules, stirrup spacing, bent bar diagrams, bar arrangement, splices and laps, tie spacing, hoop spacing, and supports for concrete reinforcement.

D. Qualification Data: For manufacturer testing agency.

E. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.

2. Admixtures.
3. Form materials and form-release agents.
4. Curing compounds.
5. Adhesives.
6. Vapor retarders.
7. Joint-filler strips.
8. Repair materials.

F. Joint Plan

G. Material Test Reports: For the following, from a qualified testing agency, indicating compliance with requirements:

1. Aggregates.

H. Field quality-control reports, for accessibility standards.

I. Mockup Panel: A 3-foot by 3-foot square mock-up of the sand finish concrete and color shall be poured and finished at the project site for review and approval prior to proceeding with any sand finish concrete work.

1.5 QUALITY ASSURANCE

A. Testing will be performed by an independent testing laboratory selected by the Contractor. All tests will be paid for by the Contractor.

B. Installer Qualifications: A qualified installer who employs on Project personnel qualified as ACI-certified Flatwork Technician and Finisher and a supervisor who is an ACI-certified Concrete Flatwork Technician.

C. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.

1. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."

D. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified according to ASTM C 1077 and ASTM E 329 for testing indicated.

1. Personnel conducting field tests shall be qualified as ACI Concrete Field-Testing Technician, Grade 1, according to ACI CP-1 or an equivalent certification program.
2. Personnel performing laboratory tests shall be ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician – Grade I. Testing Agency laboratory supervisor shall be an ACI-certified Concrete Laboratory Testing Technician – Grade II.

E. Source Limitations: Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, obtain aggregate from single source, and obtain admixtures from single source from single manufacturer.

F. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:

1. ACI 301, "Specifications for Structural Concrete," Sections 1 through 5.
2. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

G. Concrete Testing Service: Engage a qualified independent testing agency to perform material evaluation tests and to design concrete mixtures.

PART 2 - PRODUCTS

2.1 FORM-FACING MATERIALS

A. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.

B. Chamfer Strips: Wood, metal, PVC, or rubber strips, 3/4 by 3/4 inch, minimum.

C. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces. The form-release agent shall be natural, organic, VOC compliant, and biodegradable.

1. Formulate form-release agent with rust inhibitor for steel form-facing materials.

D. Form Ties: Factory-fabricated, removable, or snap-off metal or glass-fiber-reinforced plastic form ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.

1. Furnish units that will leave no corrodible metal closer than 1 inch to the plane of exposed concrete surface.
2. Furnish ties that, when removed, will leave holes no larger than 1 inch in diameter in concrete surface.

2.2 CONCRETE MATERIALS

A. Cementitious Material

1. Cement shall be from a preapproved source listed on the Iowa Department of Transportation Approved Product List. Use cementitious materials, of the same type, brand, and source, throughout the project.

B. Silica Fume: ASTM C 1240, amorphous silica.

C. Aggregates:

1. Aggregates shall consist of 50% fine aggregate and 50% coarse aggregate.
2. Fine aggregates shall meet the requirements in the tables below:

Quality

Fine Aggregate Quality	Test Limits	Test Method
Shale and Coal	2.0% Maximum	Materials I.M. 344

Gradation

Standard Sieve Size	Percent Passing
3/8”	100
No. 4	90-100
No. 8	70-100
No. 30	10-60
No. 200	0-1.5

When the fine aggregate is sieved through the following numbered sieves – 4, 8, 16, 30, 40, and 100 – no more than 40% shall pass one sieve and be retained on the sieve with the next higher number.

3. Coarse aggregates shall be Class 2 Durability and shall meet the gradation requirements in the table below:

Standard Sieve Size	Percent Passing
1”	100
¾”	90-100
3/8”	20-55
No. 4	0-10
No. 8	0-5
No. 200	0-1.5

4. Coarse aggregates shall meet the quality requirements described below:

Aggregate Quality

Aggregate Quality	Maximum % Allowed	Test Method
Abrasion (Cr. Stone)	50	AASHTO T 96
Abrasion (Gravel)	35 (may be increased by 0.1% for each 1% of particles with at least one fractured face)	AASHTO T 96
Alumina	0.5	Iowa DOT Materials Laboratory Test Method No. 222
A Freeze	6	Iowa DOT Materials Laboratory Test Method No. 221, Method A
Clay Lumps and Friable Particles	0.5	Materials I.M. 368

If the alumina value passes, the A Freeze is not needed. Alumina does not apply to gravel.

Maximum Permissible Amounts of Objectionable Materials

Objectionable Materials	Maximum % Allowed	Test Method
Coal and carbonaceous shale	0.5	Materials I.M. 372
Total of all shale, similar objectionable materials, and coal combined	1.0	Materials I.M. 372
Organic materials, except coal	0.01	Iowa DOT Materials Laboratory Test Method No. 215
Unsound chert particles retained on 3/8" sieve (Nonstructural concrete)	3.0	Materials I.M. 372
Unsound chert particles retained on 3/8" sieve (Structural concrete)	2.0	Materials I.M. 372

D. Water: ASTM C 94/C 94M and potable.

2.3 ADMIXTURES

A. Air-Entraining Admixtures must comply with ASTM C260.

B. Chemical Admixtures must comply with ASTM C494.

2.4 CURING MATERIALS

A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.

B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

C. Water: Potable.

D. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C309, Type 1 or 2, Class A or B.

2.5 RELATED MATERIALS

A. Bonding Agent: ASTM C 1059/C 1059M, Type II, non-redispersible, acrylic emulsion or styrene butadiene.

B. Epoxy Bonding Adhesive: ASTM C 881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class suitable for application temperature and of grade to suit requirements, and as follows:

1. Types I and II, non-load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

C. Joint Sealants: Self-leveling, exterior-rated polyurethane as specified in Section 07 92 00 “Joint Sealants.”

1. Color: Match each concrete color and as approved in mockups and submittals.

D. Color Treatment: Provide color additives that, along with specified concrete materials, result in concrete to match existing sidewalks in the project area.

2.6 STEEL REINFORCEMENT

A. Galvanized Reinforcing Bars: ASTM A 615/A 615M, Grade 60 deformed bars, ASTM A 767/A 767M, Class I zinc coated after fabrication and bending.

2.7 CONCRETE MIXTURES, GENERAL

A. Prepare design mixtures for each type and strength of concrete, proportioned based on laboratory trial mixture or field test data, or both, according to ACI 301.

1. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.
2. Provide concrete with an ultimate strength of 4000 psi.
3. Air entrainment shall be between 6.0% min. and 10.0% max.
4. Slump between ½-inch and 4-inches.
5. Use a locally-obtained sand to match the geology of the park location and meeting ASTM C33 requirements.

B. Limit water-soluble, chloride-ion content in hardened concrete to 0.15 percent by weight of cement.

C. Admixtures: Use admixtures according to manufacturer's written instructions.

1. Use admixture in concrete, as required, for placement and workability.
2. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
3. Use water-reducing admixture in pumped concrete, concrete for heavy-use industrial slabs and parking structure slabs, concrete required to be watertight, and concrete with a water-cementitious materials ratio below 0.50.
4. Use corrosion-inhibiting admixture in concrete mixtures where indicated.

2.8 CONCRETE MIXING

A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M and ASTM C 1116/C 1116M, and furnish batch ticket information.

B. Project-Site Mixing: Measure, batch, and mix concrete materials and concrete according to ASTM C 94/C 94M. Mix concrete materials in appropriate drum-type batch machine mixer.

1. For mixer capacity of 1 cu. yd. or smaller, continue mixing at least 1-1/2 minutes, but not more than 5 minutes after ingredients are in mixer, before any part of batch is released.
2. For mixer capacity larger than 1 cu. yd., increase mixing time by 15 seconds for each additional 1 cu. yd.
3. Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mixture type, mixture time, quantity, and amount of water added. Record approximate location of final deposit in structure.

2.9 DETECTABLE WARNING SURFACE

A. Dome Size: Truncated domes in a detectable warning surface shall have a base diameter of 0.9 inch minimum and 1.4 inches maximum, a top diameter of 50 percent of the base diameter minimum to 65 percent of the base diameter maximum, and a height of 0.2 inch.

B. Dome Spacing: Truncated domes in a detectable warning surface shall have a center-to-center spacing of 1.6 inches minimum and 2.4 inches maximum, and a base-to-base spacing of 0.65 inch minimum, measured between the most adjacent domes on a square grid.

C. Material: detectable warning surface shall be unpainted cast iron.

PART 3 - EXECUTION

3.1 FORMWORK

A. Use either fixed forms or slip forms.

B. Provide straight, full depth, un-warped forms that will resist springing during concrete placement.

C. Firmly stake fixed forms.

D. Construct forms tight enough to prevent loss of concrete mortar.

E. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

F. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

G. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

H. Form liners shall not be used for the final finish.

3.2 REMOVING AND REUSING FORMS

A. General: Formwork that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F for 24 hours after placing concrete. Concrete shall be hard enough to not be damaged by form-removal operations and curing and protection operations need to be maintained.

1. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.

B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.

C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Contracting Officer (CO).

3.3 SHORES AND RESHORES

A. Comply with ACI 318 and ACI 301 for design, installation, and removal of shoring and reshoring.

B. Plan sequence of removal of shores and reshore to avoid damage to concrete. Locate and provide adequate reshoring to support construction without excessive stress or deflection.

3.4 REINFORCEMENT

A. General: When concrete is placed, reinforcement shall be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory provided the minimum nominal dimensions, nominal weight, and the minimum average height of deformation of a hand-wire-brushed test specimen are not less than the applicable ASTM specification requirements.

B. Reinforcement: Place, support, and fasten reinforcement as indicated in the Contract Documents. Do not exceed the placing tolerances specified in ACI 117 before concrete is placed. When necessary to move reinforcement beyond the specified placing tolerances to avoid interference with other reinforcement or embedded items, submit the resulting arrangement of reinforcement for acceptance.

C. Cover: Allowable concrete cover for reinforcement is indicated in the project drawings. Tolerances on concrete cover shall meet the requirements of ACI 117.

D. Tie Wires: After cutting tie wires, turn wires to the inside of section and bend so that concrete placement will not force ends to exposed concrete surfaces.

E. Splicing: Make splices as indicated in the project drawings. Lapped bars shall be placed in contact and securely tied, or spaced transversely apart to permit the embedment of the entire surface of each bar in concrete. Lapped bars shall not be spaced farther apart than 1/5 the required length of lap, or six inches (6”).

F. Mechanical Splices: Mechanical splices for reinforcement not shown on the project drawings shall be submitted for review and accepted prior to use. Mechanical splices shall be in accordance with the recommendations of the manufacturer of the mechanical splicing device.

G. Reinforcement shall not be field bent or straightened except when specifically permitted by the CO.

H. Reinforcement shall not be cut in the field except when specifically permitted by the CO.

I. The following reinforcing steel work will be considered defective and shall be removed and replaced by the Contractor at no additional cost to the Government:

1. Bars with kinks or bends not shown on drawings.
2. Bars injured due to bending or straightening.
3. Bars heated for bending.
4. Reinforcement not placed in accordance with the project drawings and/or specifications.

3.5 JOINTS

A. General: Construct joints true to line as shown in the Contract Documents. All joints shall be tooled.

3.6 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.

B. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301.

1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.

C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.

1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.

D. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.

1. When average high and low temperature is expected to fall below 40 deg F for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

E. Hot-Weather Placement: Comply with ACI 301 and as follows:

1. Maintain concrete temperature below 90 deg F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor’s option.
2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.7 FINISHING CONCRETE

A. Float the surface, but do not over-float, to produce a smooth surface, free from irregularities. Round the edges and joints with finishing tools.

B. Broom finish is achieved by brushing the surface with a stiff bristled broom traverse to the direction of travel.

3.8 CONCRETE PROTECTING AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces to deter moisture loss due to hot, dry, or windy conditions before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.

D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.

E. Cure concrete according to ACI 308.1, by the following method:

1. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.

3.9 JOINT FILLING

A. Prepare, clean, and install joint filler according to manufacturer's written instructions.

B. Remove dirt, debris, saw cuttings, curing compounds, and sealers from joints; leave contact faces of joint clean and dry.

1. Tape sides to minimize staining at Contractor's option to minimize excess filler removal and staining.

C. Install exterior-rated joint sealant filler full depth in saw-cut joints and at least 2 inches deep in formed joints. Overfill joint and trim joint filler flush with top of joint after hardening and remove excess from exposed surfaces. Sealant filler color shall match concrete paving color as determined by CO. Where colors do not match, remove and reinstall until matching at no additional cost to Government.

1. Apply joint sealant using masking tape or other methods approved by the CO to ensure clean, straight sealant lines. Remove tape prior to initial curing.
2. Sealant exhibiting smeared edges, feathering, irregular lines, voids, or contamination of adjacent surfaces remove and replace at no additional cost to the Government.

3.10 FIELD QUALITY CONTROL

A. Testing and Inspecting: Contractor shall engage a qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.

B. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:

1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu. yd., but less than 25 cu. yd., plus one set for each additional 50 cu. yd. or fraction thereof.
2. Testing Frequency: Obtain at least one composite sample for each 100 cu. yd. or fraction thereof of each concrete mixture placed each day.
3. When frequency of testing will provide fewer than five compressive-strength tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.
4. Slump: ASTM C 143/C 143M; one test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture. Perform additional tests when concrete consistency appears to change.
5. Air Content: ASTM C 231, pressure method, for normal-weight concrete; ASTM C 173 volumetric method, for structural lightweight concrete; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
6. Concrete Temperature: ASTM C 1064/C 1064M; one test hourly when air temperature is 40 deg F and below and when 80 deg F and above, and one test for each composite sample.
7. Compression Test Specimens: ASTM C 31/C 31M.

    a. Cast and laboratory cure two sets of two standard cylinder specimens for each composite sample.

8. Compressive-Strength Tests: ASTM C 39/C 39M; test one set of two laboratory-cured specimens at 7 days and one set of two specimens at 28 days.

    a. Test one set of two field-cured specimens at 7 days and one set of two specimens at 28 days.
    b. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.

9. Test results shall be reported in writing to CO, concrete manufacturer, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7- and 28-day tests.
10. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by CO but will not be used as sole basis for approval or rejection of concrete.

11. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by CO. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C 42/C 42M or by other methods as directed by CO.
12. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
13. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

3.11 INSPECTIONS

A. Accessibility Inspection: Accessibility inspection shall be performed by the Contractor in a two-part process to ensure compliance of final installation. The two-part process inspections shall occur pre-installation and post-installation.

1. Pre-installation inspection shall examine slopes, dimensions, base materials, and preparation work prior to concrete placement. Check formwork, embedded items, reinforcing, etc. Pre-installation work shall conform to National Park Service Accessible Route Design Standards.
2. Post-installation inspection shall examine completed concrete placement. Report non-compliance with ABAAS for running and cross slopes, landings, vertical changes in elevation, transitions, imperfections or flaws, improper drainage, slippery surfaces, and dimensional accuracy.

B. Contractor shall utilize the National Park Service CQC Accessibility Inspection Report as basis for recording findings. Any non-compliant conditions shall be documented. Supplement Report with digital images, notation, red-line drawings, sketches and dimensions, product literature, submittal, etc. to explain installed work. Submit completed Report to CO within 24 hours of commencing inspection.

3.12 DETECTABLE WARNING SURFACE

A. The detectable warning surface shall be located as shown in the Plans. Placement of the detectable warning surface shall be in accordance with the manufacturer's recommendation for placement in fresh concrete, before the concrete has reached initial set, or on a hardened cement concrete surface or asphalt pavement surface.

B. Vertical edges of the detectable warning surface shall be flush with the adjoining surface to the extent possible (not more than ¼ inch above the surface of the pavement) after installation.

C. Embossing or stamping the wet concrete to achieve the truncated dome pattern or using a mold into which a catalyst-hardened material is applied shall not be allowed.

END OF SECTION 32 13 20

SECTION 32 33 00 – SITE FURNISHINGS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. New bench
2. New double sided bench
3. New trash receptacle
4. New three-sided kiosk
5. New anchored bike loop
6. Existing bike rack
7. New flagpole

B. Related Sections:

1. Section 10 75 16 – Gound-Set Flagpoles

1.3 SUBMITTALS

A. Product Data: For each type of product indicated.

1.4 PRODUCT DELIVERY, STORAGE AND HANDLING

A. Deliver products in good condition.

B. Store products to prevent corrosion, deterioration, and damage.

C. Handle products to prevent damage.

1.5 QUALITY ASSURANCE

A. Accessibility Requirements: Comply with Architectural Barriers Act Accessibility Standards (ABAAS) when installing.

PART 2 - PRODUCTS

2.1 NEW BENCH

A. New free-standing bench with backrest to be installed by Contractor as indicated on plans.

B. Basis of Design Product and Manufacturer: Dumor Site Furnishings, 520-80TX

1. Performance Characteristics:

a. General: 8’ steel bench with wood grain recycled plastic lumber seat and back
b. Dimensions: 95” L x 19 13/16” D x 33 1/16” H
c. Seat and Back Material: 100% high-density polyethylene

1) Color: Antique Mahogany

d. Support Material:

1) Color: Textured Rust
2) Finish: Baked-on polyester powder

a) All parts are processed through an 8-stage iron phosphorous wash system.
b) Parts are coated with a zinc-rich epoxy primer to an AVERAGE of 4-5 mils.
c) Parts are then finished with a top coat of TGIC-polyester powder to an AVERAGE of 4-5 mils.
d) Powder is cured at the powder manufacturers specifications using combination of infrared and convection heat for approximately 20 minutes.
e) Finished parts shall comply with the following American Standard Test Method (ASTM) for coating and coating method: ASTM-D-523, ASTM-D-3363, ASTM-D-1737, ASTM-D-3359, ASTM-D-2794, ASTM-B-117 and ASTM-D-3451.

2.2 NEW DOUBLE SIDED BENCH

A. New free-standing double-sided bench with half backrest to be installed by Contractor as indicated on plans.

B. Basis of Design Product and Manufacturer: Dumor Site Furnishings, 524TX-B1

1. Performance Characteristics:

a. General: 6’ steel island bench and half backrest with wood grain recycled plastic lumber seat and back
b. Dimensions: 77 1/8” L x 38 3/8” W x 33 3/8” H
c. Seat and Back Material: 100% high-density polyethylene

1) Color: Antique Mahogany

d. Support Material:

1) Color: Textured Rust
2) Finish: Baked-on polyester powder

a) All parts are processed through an 8-stage iron phosphorous wash system.
b) Parts are coated with a zinc-rich epoxy primer to an AVERAGE of 4-5 mils.
c) Parts are then finished with a top coat of TGIC-polyester powder to an AVERAGE of 4-5 mils.
d) Powder is cured at the powder manufacturers specifications using combination of infrared and convection heat for approximately 20 minutes.
e) Finished parts shall comply with the following American Standard Test Method (ASTM) for coating and coating method: ASTM-D-523, ASTM-D-3363, ASTM-D-1737, ASTM-D-3359, ASTM-D-2794, ASTM-B-117 and ASTM-D-3451.

2.3 NEW TRASH RECEPTACLE

A. New trash receptacle to be installed by Contractor as indicated on plans.

B. Basis of Design Product and Manufacturer: Bearsaver, HA Series Double Trash Enclosure, ADA Compliant – HA2-CH, Surface Mounted

1. Performance Characteristics:

a. General: Powder coated, corrosion resistant steel housing and doors with two 40 gallon rigid liners.
b. Dimensions: 44” L x 24” W x 52” H
c. Color: Forest Brown
d. Hardware: Zinc coated hinges, handles, striker plates and hardware. Heavy-duty piano hinge with stainless steel pins.
e. Labels: (1) Trash, (1) Mixed Recycling

2.4 NEW THREE-SIDED KIOSK

A. New three-sided kiosk to be installed by Contractor as indicated on plans.

B. Basis of Design Product and Manufacturer: Fossil Graphics, Upright Post - Kiosk - 86, Surface Mounted

1. Performance Characteristics:

a. General: Triangular kiosk with steel posts to mount three interp panels (by others)
b. Dimensions: 86” H, width and depth determined by interp panel dimensions
c. Hardware: Tamper resistant screws and washers, tamper resistant bolts, angled mounting “L” brackets, and base plates

d. Support Material: Steel 6” x 2” post

1) Color: Textured Rust
2) Finish: Baked-on polyester powder

2.5 NEW ANCHORED BIKE LOOP

A. New anchored bike loop to be installed by Contractor as indicated on plans.

B. Basis of Design Product and Manufacturer: Dumor Site Furnishings, Model 290-00/S-2 Bike Rack

1. Performance Characteristics:

a. General: 2” (2 3/8” OD) ASTM A513 schedule 40 steel tubing and 3/8” thick ASTM A36 steel plate. Stainless steel expansion anchors (1/2” x 3 3/4").
b. Dimensions: 30” long x 6” deep x 36” high
c. Color: Textured Rust
d. Finish: Baked-on polyester powder

1) All parts are processed through an 8-stage iron phosphorous wash system.
2) Parts are coated with a zinc-rich epoxy primer to an AVERAGE of 4-5 mils.
3) Parts are then finished with a top coat of TGIC-polyester powder to an AVERAGE of 4-5 mils.
4) Powder is cured at the powder manufacturers specifications using combination of infrared and convection heat for approximately 20 minutes.
5) Finished parts shall comply with the following American Standard Test Method (ASTM) for coating and coating method: ASTM-D-523, ASTM-D-3363, ASTM-D-1737, ASTM-D-3359, ASTM-D-2794, ASTM-B-117 and ASTM-D-3451.

2.6 EXISTING BIKE RACK

A. Existing free-standing bike rack shall be removed, stored, and re-installed by contractor as indicated on plans.

2.7 NEW FLAGPOLE

A. New flagpole to be installed by Contractor as indicated on plans.

B. See section 10 75 16 for flagpole Basis of Design.

PART 3 - EXECUTION

3.1 PREPARATION

A. Examine areas and conditions, with Installer present, for compliance with requirements for correct and level finished grade, mounting surfaces, installation tolerances, required clear space, and other conditions affecting performance.

1. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION, GENERAL

A. Comply with manufacturer's written installation instructions unless more stringent requirements are indicated. Complete field assembly of site furnishings where required.

B. Unless otherwise indicated, install site furnishings after concrete paving has been completed.

C. Install all site furnishings level, plumb, true, and securely anchored at locations indicated on Drawings.

3.3 INSPECTIONS

A. Performance of accessibility inspection using DSC, CQC Accessibility Inspection Report required within seven days of completion of work and linked to the construction completion schedule.

3.4 CLEANING

A. After completing site furnishing installation, inspect components. Remove spots, dirt, and debris. Repair damaged finishes to match original finish or replace component.

END OF SECTION 32 33 00

SECTION 32 91 13 – SOIL PREPARATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This section describes preparation of soil including application of herbicide and soil amendments.

B. Related Sections:

1. Section 31 20 00 – Earth Moving
2. Section 32 92 00 – Turf and Grasses
3. Section 32 93 00 – Plants

1.3 SUBMITTALS

A. Submit product data of herbicide and fine compost delivered to the site to the CO.

B. Submit to the CO for approval a sample of fine compost.

1.4 PRODUCT DELIVERY

A. Deliver fertilizer to the site in original unopened containers, each bearing the manufacturer's guaranteed analysis.

1.5 PROTECTION

A. Protect utility lines, storm drainage lines, site improvements, and underground irrigation system during execution of work. See Section 015000, Temporary Facilities and Controls.

PART 2 - PRODUCTS

2.1 SOIL AMENDMENTS

A. Fine compost: Well-decomposed, stable, and weed free organic matter sourced locally. The pH shall be between 6.0 and 8.5 when tested. Manufactured inert material (plastic, concrete, ceramics, metal, etc.) shall be less than 0.5 percent on a dry weight or volume basis, whichever

provides for the least amount of foreign material. Minimum organic matter shall be 40 percent dry weight and soluble salt contents shall be less than 6.0 mmhos/cm.

Compost shall meet the following:

	Min.	Max.
Percent passing 2”	100%	
Percent passing 1”	90%	100%
Percent passing ¾”	70%	100%
Percent passing ¼”	40%	75%

2.2 TOPSOIL

A. Clean, fertile, friable, natural soil material free of debris, roots, stones, weeds, and grass.

PART 3 - EXECUTION

3.1 HERBICIDE APPLICATION

A. Apply contact herbicide over all areas of weed or grass growth within landscaped area.

B. Apply in two applications as follows:

1. First application, apply seven days prior to performing earthwork.
2. Second application (to kill new vegetation), apply after grading has been completed and 48 hours prior to planting.

C. Observe manufacturer’s recommended period prior to working in treated areas.

D. Apply at manufacturer’s maximum recommended application rate.

E. Do not apply herbicide to impervious areas unless specifically directed to do so by the CO.

3.2 SOIL PREPARATION

A. Remove stones, mortar, gravel, concrete, asphalt, rubbish, debris, and other materials larger than 1 1/2 inches from planting areas. See Section 329200, Turf and Grasses.

1. If existing soil base is not suitable for use, remove top 8 inches and replace with suitable topsoil.

B. Seeding area finished soil surface shall be 1 inch below edge of pavement or sidewalk elevation prior to hydroseed application.

C. See Section 31 20 00, Earth Moving, if existing soil is not sandy or silty, thus requiring over-excavation and placement of fill material.

D. Amend soil with compost at the rates described below to achieve a total 9 3/4 inch layer of amended soil material. Apply in the following order.

1. Add 1 3/4 inches of fine compost.
2. Rototill, in two directions, to a depth of 9.75 inches, to create a uniform mix of soil and fine compost.

E. Rake and smooth planting area to a tolerance of 1 inch, plus or minus, in 10 feet.

F. See Section 32 93 00, Plants, for application of prepared soil mix in planting pits.

END OF SECTION 32 91 13

SECTION 32 92 00 – TURF AND GRASSES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Seed Mix 2

B. Related Sections:

1. Section 31 10 00 – Site Clearing
2. Section 31 20 00 – Earth Moving
3. Section 31 23 16 – Topsoil

1.3 SUBMITTALS

A. Product Data: Virgin wood fiber mulch

B. Warranty: Sample of special warranty.

1.4 QUALITY ASSURANCE

A. Installer Qualifications: A qualified landscape Installer whose work has resulted in successful establishment of plants.

1. Experience: Three years' experience in successful landscape installation in addition to requirements in Division 01 Section "Quality Requirements."
2. Installer's Field Supervision: Require Installer to maintain an experienced full-time supervisor on Project site when work is in progress.

B. Measurements: Measure according to ANSI Z60.1.

C. Preinstallation Conference: Conduct conference at Project site prior to installation.

1.5 MATERIALS

A. Seed Mix 2: Native Turf Replacement – Mowed Frequently

1. To be furnished by NPS.
2. Performance Characteristics:

a. Seeding Rates: 200 seeds per square foot minimum rate
b. Consists of the following seed mix:

Species	Common Name	% Mix by weight
Achillea millefolium	Yarrow	15
Allium cernum	Wild nodding onion	2
Antennaria neglecta	Field Pussytoes	8
Avena sativa	Annual oats	2
Bouteloua curtipendula	Sideoats grama	5
Carex eburnea	Ivory Sedge	2
Carex sprengelii	Sedge	2
Dalea purpurea	Purple Prairie Clover	2
Danthonia spicata	Poverty oats	30
Elymus virginicus	Wild rye	3
Fragaria virginiana	Wild strawberry	10
Juncus tenuis	Pathrush	5
Lolium multiflorum	Annual rye	3
Prunella vulgaris	Heal-all	13
	Total	**100**

1.6 PROJECT CONDITIONS

A. Field Measurements: Verify actual grade elevations and construction contiguous with new plantings by field measurements before proceeding with planting work.

B. Interruption of Existing Services or Utilities: Do not interrupt services or utilities to facilities occupied by CO or others unless permitted under the following conditions and then only after arranging to provide temporary services or utilities according to requirements indicated:

1. Notify CO no fewer than two days in advance of proposed interruption of each service or utility.
2. Do not proceed with interruption of services or utilities without CO's written permission.

C. Planting Restrictions: Plant between September 15 and November 1 or February 1 and April 15 when soil temperatures are consistently within the 40-49 degrees F range. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.

D. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained. Do not seed on ground that is excessively wet, frozen, or has more than ¼" of snow cover. Apply products during favorable weather conditions according to manufacturer's written instructions and warranty requirements.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Delivery: All seed mix material will be delivered to the Project site by NPS.

B. Bulk Materials:

1. Do not dump or store bulk materials near structures, utilities, walkways, and pavements, or on existing turf areas or plants.
2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.

C. NPS to deliver plants after preparations for hydroseeding have been completed and Contractor is ready to install immediately. If planting is delayed more than six hours after delivery, protect seed mix material from weather and mechanical damage.

1.8 WARRANTY

A. Special Warranty: Contractor agrees to overseed seeding areas that fail in materials, workmanship, or growth within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Germination less than 70%.

2. Warranty Periods from Date of Planting Completion:

a. 12 months from date of planting completion.

3. Include the following remedial actions as a minimum:

a. Immediately overseed failed seeding area following all specified project conditions.
b. Provide extended warranty for period equal to original warranty period, for replaced plant material.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive seeding for compliance with requirements and conditions affecting installation and performance.

1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within a planting area.
2. Do not mix or place soils and soil amendments in frozen, wet, or muddy conditions.
3. Suspend soil spreading, grading, and tilling operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.
4. Uniformly moisten excessively dry soil that is not workable and which is too dusty.

B. Proceed with seeding only after unsatisfactory conditions have been corrected.

C. If contamination by foreign or deleterious material or liquid is present in soil within a planting area, do not remove any soil unless directed by the CO.

3.2 PREPARATION

A. Notify a minimum of two weeks prior to seeding date to allow proper delivery of seed.

B. Protect structures, utilities, sidewalks, pavements, and other facilities and turf areas and existing plants from damage caused by planting operations.

C. Install erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

D. Stake perimeter of seeding locations and adjust locations when requested and obtain Project Representative’s acceptance of layout before planting. Make minor adjustments as required.

E. Roughen embankments and fill areas by track walking.

3.3 INSTALLATION

A. Seeding Seed Mix Method Options

1. Two-Step Hydroseed Method

a. First Application (Seed): Hydroseed Application: In order to ensure that the seed is being applied at the correct rate areas to be seeded shall be marked off/flagged at a minimum of half acre increments prior to seeding. Add 500 lbs/acre of tracer material consisting of virgin wood fiber mulch to the water/seed in the hydraulic equipment to provide visible evidence of uniform application. Adding this 500 lbs/acre during seeding, means that only 1500 lbs/acre will be added during the

second application of just mulch, which would result in a total of 2,400 lbs/acre of mulch applied.

2. Hand Broadcast/Rake/Hydromulch Method (for slopes flatter than 2H:1V)

   a. First Application (Seed): Hand Application: In order to ensure that the seed is being applied at the correct rate areas to be seeded shall be marked off/flagged at a minimum of half acre increments prior to seeding. Seed shall be applied by hand or with hand seeders such as a belly grinder. Seed will be distributed in such a way as to assure all species in the seed mix are evenly distributed across the area, regardless of seed size. Lightly rake the seed into the soil to a depth of ¼". Do not smooth with the backside of the rake, light furrows in the soil are desired.
   b. Second Application (Mulch): 2,400 lbs./acre of long strand virgin wood fiber and;

      1) Apply mulch with a non-toxic, and natural and biodegradable plant-based psyllium/alpha plantago tackifier. Depending on product apply per the manufacturer's specifications, generally at a rate of 80 lbs. per acre on slopes less than 3H:1V and apply at a rate of 100 lbs./acre on slopes greater than 3H:1V.
      2) Apply mulch with a non-toxic, 100% environmentally safe tracer dye material applied per the manufacturer's specification generally at a rate of around 200-300 pounds per acre to provide visible evidence of uniform application.
      3) The hydromulcher must be equipped with a built-in continuous agitation and recirculation system of sufficient operating capacity to produce homogeneous slurry and a discharge system that will apply slurry to the designated areas at a continuous and uniform rate. No fertilizers, mycorrhizae, or other amendments shall be added to the hydroseed slurry.
      4) The Contractor shall spray designated areas with the slurry in a sweeping motion and in an arched stream until a uniform coat is achieved, with no slumping or shadowing, as the material is spread at the required rate. The hydromulch slurry should float down from the arched stream, as opposed to being shot directly at the ground. During hydromulching the existing native vegetation must be protected from damage (including, but not limited to, coating with mulch, damage by direct spray, and dragging hose). Notify the CO/DSC three weeks before hydromulching to ensure that NPS oversight of seeding operations can be coordinated.

3. Hydroseed/Hydromulch Method (for slopes steeper than 2H:1V)

   a. First Application (Seed): Hydroseed Application Hydroseed Application: In order to ensure that the seed is being applied at the correct rate areas to be seeded shall be marked off/flagged at a minimum of half acre increments prior to seeding. Add 500 lbs/acre of tracer material consisting of virgin wood fiber mulch to the water/seed in the hydraulic equipment to provide visible evidence of uniform application. Adding this 500 lbs/acre during seeding, means that only 1,900 lbs/acre will be added during the second application of just mulch, which would result in a total of 2,400 lbs/acre of mulch applied. The hydroseeding shall proceed following the specifications above for the hydromulch application on top of broadcast seed.
   b. Second Application (Mulch): 1,900 lbs./acre of long strand virgin wood fiber and;

1) Apply mulch with a non-toxic, and natural and biodegradable plant-based psyllium/alpha plantago tackifier. Depending on product, apply at manufacture's specification or at a rate of 80 lbs. per acre on slopes less than 3H:1V and apply at a rate of 100 lbs./acre on slopes greater than 3H:1V.
2) Apply mulch with a non-toxic, 100% environmentally safe tracer dye material applied per the manufacturer's specification or at a rate of around 200-300 pounds per acre to provide visible evidence of uniform application.
3) The hydromulching shall proceed following the specifications above for the hydromulch application on top of broadcast seed.

B. Hydroseed to establish a uniform stand of grasses.

C. Seeding rate shall be as recommended by the seed producer.

3.4 PLANT MAINTENANCE

A. Block off seeded areas from visitors until the grasses are well established.

B. Apply treatments as required to keep plant materials, planted areas, and soils free of pests and pathogens or disease. Use Plant Health Care practices including Integrated Pest Management whenever possible to minimize the use of pesticides and reduce hazards. Treatments include physical controls such as mechanical controls such as traps and biological control agents.

3.5 CLEANUP AND PROTECTION

A. During seeding, keep adjacent paving and construction clean and work area in an orderly condition.

B. Protect plants from damage due to landscape operations and operations of other contractors and trades. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

END OF SECTION 32 92 00

SECTION 32 93 00 – PLANTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Plants
2. Double Processed Mulch

B. Related Sections:

1. Section 31 10 00 – Site Clearing
2. Section 31 20 00 – Earth Moving
3. Section 31 23 16 – Topsoil

1.3 UNIT PRICES

A. Work of this Section is affected by unit prices specified in Division 01 Section “Unit Prices.”

1. Unit prices apply to authorized work covered by quantity allowances.
2. Unit prices apply to additions to and deletions from Work as authorized by Change Orders.

1.4 DEFINITIONS

A. Bare-Root Stock: Plants with a well-branched, fibrous-root system developed by transplanting or root pruning, with soil or growing medium removed, and with not less than minimum root spread according to ANSI Z60.1 for type and size of plant required.

B. Container-Grown Stock: Healthy, vigorous, well-rooted plants grown in a container, with a well-established root system reaching sides of container and maintaining a firm ball when removed from container. Container shall be rigid enough to hold ball shape and protect root mass during shipping and be sized according to ANSI Z60.1 for type and size of plant required.

C. Finish Grade: Elevation of finished surface of planting soil.

D. Manufactured Topsoil: Soil produced off-site by homogeneously blending mineral soils or sand with stabilized organic soil amendments to produce topsoil or planting soil.

E. Pesticide: A substance or mixture intended for preventing, destroying, repelling, or mitigating a pest. This includes insecticides, miticides, herbicides, fungicides, rodenticides, and molluscicides. It also includes substances or mixtures intended for use as a plant regulator, defoliant, or desiccant.

F. Pests: Living organisms that occur where they are not desired, or that cause damage to plants, animals, or people. These include insects, mites, grubs, mollusks (snails and slugs), rodents (gophers, moles, and mice), unwanted plants (weeds), fungi, bacteria, and viruses.

G. Planting Area: Areas to be planted.

H. Planting Soil: Standardized topsoil; existing, native surface topsoil; existing, in-place surface soil; imported topsoil; or manufactured topsoil that is modified with soil amendments and perhaps fertilizers to produce a soil mixture best for plant growth.

I. Plant; Plants; Plant Material: These terms refer to vegetation in general, including trees, shrubs, vines, ground covers, ornamental grasses, bulbs, corms, tubers, or herbaceous vegetation.

J. Root Flare: Also called “trunk flare.” The area at the base of the plant’s stem or trunk where the stem or trunk broadens to form roots, the area of transition between the root system and the stem or trunk.

K. Subgrade: Surface or elevation of subsoil remaining after excavation is complete, or the top surface of a fill or backfill before planting soil is placed.

L. Subsoil: All soil beneath the topsoil layer of the soil profile and typified by the lack of organic matter and soil organisms.

M. Surface Soil: Soil that is present at the top layer of the existing soil profile at the Project site. In undisturbed areas, the surface soil is typically topsoil; but in disturbed areas such as urban environments, the surface soil can be subsoil.

1.5 SUBMITTALS

A. Product Data: For each type of product indicated:

1. Double Processed Mulch: manufacturer source information.

B. Samples for Verification: For each of the following:

1. Double Processed Mulch: 1-gallon volume in sealed plastic bags labeled with source. Each Sample shall be typical of the lot of material to be furnished; provide an accurate representation of color, texture, and organic makeup.

C. Qualification Data: For qualified Landscape Installer. Include list of similar projects completed by Installer demonstrating Installer’s capabilities and experience. Include project names, addresses, and year completed, and include names and addresses of COs’ contact persons.

D. Product Certificates: For each type of manufactured product, from manufacturer, and complying with the following:

1. Manufacturer's certified analysis of standard products.
2. Analysis of other materials by a recognized laboratory made according to methods established by the Association of Official Analytical Chemists, where applicable.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Instructions: Recommended procedures to be established by CO for maintenance of plants during a calendar year. Submit before start of required maintenance periods.

B. Warranty: Sample of special warranty.

1.7 QUALITY ASSURANCE

A. Installer Qualifications: A qualified Landscape Installer whose work has resulted in successful establishment of plants.

1. Experience: Three years' experience in successful landscape installation in addition to requirements in Division 01 Section "Quality Requirements."
2. Installer's Field Supervision: Require Installer to maintain an experienced full-time supervisor on Project site when work is in progress.

B. Preinstallation Conference: Conduct conference at Project site prior to installation.

1.8 MATERIALS

A. Pollinator Garden

1. To be furnished by NPS.
2. Consists of the following planting palette:

Species	**Common Name**
Antennaria plantaginifolia	Pussytoes
Aralia nuducaulis	Wild Sarsaparilla
Aralia racemosa	American Spikenard
Asclepias tuberosa	Butterfly milkweed
Ceanothus americanus	New Jersey Tea
Fragaria virginiana	Wild Strawberry
Osmunda claytoniana	Interrupted Fern
Rudbeckia hirta	Black-Eyed Susan
Solidago speciosa	Showy Goldenrod

1.9 PROJECT CONDITIONS

A. Field Measurements: Verify actual grade elevations, service and utility locations, irrigation system components, and dimensions of plantings and construction contiguous with new plantings by field measurements before proceeding with planting work.

B. Interruption of Existing Services or Utilities: Do not interrupt services or utilities to facilities occupied by CO or others unless permitted under the following conditions and then only after arranging to provide temporary services or utilities according to requirements indicated:

1. Notify CO no fewer than two days in advance of proposed interruption of each service or utility.
2. Do not proceed with interruption of services or utilities without CO's written permission.

C. Planting Restrictions: Plant between September 15 and November 1 or February 1 and April 15 when soil temperatures are consistently within the 40-49 degrees F range. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.

D. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained. Do not seed on ground that is excessively wet, frozen, or has more than ¼" of snow cover.

1.10 DELIVERY, STORAGE, AND HANDLING

A. Delivery: All plant material will be delivered to the Project site by NPS.

B. Bulk Materials:

1. Do not dump or store bulk materials near structures, utilities, walkways, and pavements, or on existing turf areas or plants.
2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
3. Accompany each delivery of bulk fertilizers and soil amendments with appropriate certificates.

C. Handle planting stock by root ball.

D. NPS to deliver plants after preparations for planting have been completed and Contractor is ready to install immediately. If planting is delayed more than six hours after delivery, set plants in their appropriate aspect (sun, filtered sun, or shade), protect from weather and mechanical damage, and keep roots moist.

1. Heel-in bare-root stock. Soak roots that are in dry condition in water for two hours. Reject dried-out plants.
2. Set balled stock on ground and cover ball with soil, peat moss, sawdust, or other acceptable material.

3. Do not remove container-grown stock from containers before time of planting.
4. Water root systems of plants stored on-site deeply and thoroughly with a fine-mist spray. Water as often as necessary to maintain root systems in a moist, but not overly-wet condition.

1.11 WARRANTY

A. Special Warranty: Contractor agrees to repair or replace plantings and accessories that fail in materials, workmanship, or growth within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Death and unsatisfactory growth, except for defects resulting from abuse, lack of adequate maintenance, or neglect by CO, or incidents that are beyond Contractor's control.
b. Structural failures including plantings falling or blowing over.
c. Faulty performance of tree stabilization and edgings.
d. Deterioration of metals, metal finishes, and other materials beyond normal weathering in the surrounding area.

2. Warranty Periods from Date of Planting Completion:

a. 12 months from date of planting completion.

3. Include the following remedial actions as a minimum:

a. Immediately remove dead plants and replace unless required to plant in the succeeding planting season.
b. Replace plants that are more than 25 percent dead or in an unhealthy condition at end of warranty period.
c. A limit of one replacement of each plant will be required except for losses or replacements due to failure to comply with requirements.
d. Provide extended warranty for period equal to original warranty period, for replaced plant material.

1.12 MAINTENANCE SERVICE

A. Initial Maintenance Service for Plants: Provide maintenance by skilled employees of Landscape Installer. Maintain as required in Part 3. Begin maintenance immediately after plants are installed and continue until plantings are acceptably healthy and well established but for not less than maintenance period below.

1. Maintenance Period: 12 months from date of planting completion.

B. Continuing Maintenance Proposal: From Installer to CO, in the form of a standard yearly (or other period) maintenance agreement, starting on date initial maintenance service is concluded. State services, obligations, conditions, and terms for agreement period and for future renewal options.

PART 2 - PRODUCTS

2.1 PLANT MATERIAL

A. Plant material to be furnished by NPS.

2.2 PLANTING SOILS

A. Topsoil shall be as defined in Section 31 23 16.

2.3 MULCHES

A. Double Processed Mulch: Premium hardwood mulch, aged to a natural dark brown color (no dye), and processed to 1 – 2 inches derived from the mechanical grinding or shredding of the above-ground portions of trees. It may not contain visible amounts of soil, branches, or leaves. It shall be free of weeds and weed seeds. It may not contain more than 1/2% by weight of manufactured inert material (plastic, concrete, ceramics, metal, etc.). Tub grindings shall not be used as mulch.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive plants for compliance with requirements and conditions affecting installation and performance.

1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within a planting area.
2. Do not mix or place soils and soil amendments in frozen, wet, or muddy conditions.
3. Suspend soil spreading, grading, and tilling operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.
4. Uniformly moisten excessively dry soil that is not workable and which is too dusty.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

C. If contamination by foreign or deleterious material or liquid is present in soil within a planting area, do not remove any soil unless directed by the CO.

3.2 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities and turf areas and existing plants from damage caused by planting operations.

B. Install erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

C. Lay out individual plant locations. Stake locations and adjust locations when requested and obtain CO's acceptance of layout before excavating or planting. Make minor adjustments as required.

3.3 EXCAVATION FOR PLANT MATERIAL

A. No excavation into existing grade will be authorized prior to approval from CO.

B. Topsoil removed from excavations may be used as planting soil.

C. Obstructions: Notify CO if unexpected rock or obstructions detrimental to trees are encountered in excavations.

D. Drainage: Notify CO if subsoil conditions evidence unexpected water seepage or retention in planting pits.

E. Fill excavations with water and allow to percolate away before positioning plantings.

3.4 PLANTING

A. Before planting, verify that root flare is visible at top of root ball according to ANSI Z60.1. If root flare is not visible, remove soil in a level manner from the root ball to where the top-most root emerges from the trunk. After soil removal to expose the root flare, verify that root ball still meets size requirements.

B. Remove stem girdling roots and kinked roots. Remove injured roots by cutting cleanly; do not break.

C. Set plant stock plumb and in center of planting pit or trench with root flare 2 inch above adjacent finish grades.

1. Use topsoil for backfill.
2. After placing some backfill around root ball to stabilize plant, carefully cut and remove burlap and rope if applicable from tops of root balls and from sides, but do not remove from under root balls.
3. Backfill around root ball in layers, tamping to settle soil and eliminate voids and air pockets. When planting pit is approximately one-half filled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed.
4. Continue backfilling process. Water again after placing and tamping final layer of soil.

D. When planting on slopes, set the plant so the root flare on the uphill side is flush with the surrounding soil on the slope; the edge of the root ball on the downhill side will be above the surrounding soil. Apply enough soil to cover the downhill side of the root ball.

E. Spread Double Processed Mulch throughout planting area after all planting material has been installed. Mulch shall be a minimum of 2" deep with all mulch pulled 2-3" away from plant root crowns. Water lightly after application is complete.

3.5 PLANT MAINTENANCE

A. Maintain plantings by pruning, cultivating, watering, weeding, fertilizing, mulching, restoring planting saucers, adjusting and repairing tree-stabilization devices, resetting to proper grades or vertical position, and performing other operations as required to establish healthy, viable plantings.

B. Fill in as necessary soil subsidence that may occur because of settling or other processes. Replace mulch materials damaged or lost in areas of subsidence.

C. Apply treatments as required to keep plant materials, planted areas, and soils free of pests and pathogens or disease. Use Plant Health Care practices including Integrated Pest Management whenever possible to minimize the use of pesticides and reduce hazards. Treatments include physical controls such as hosing off foliage, mechanical controls such as traps, and biological control agents.

3.6 CLEANUP AND PROTECTION

A. During planting, keep adjacent paving and construction clean and work area in an orderly condition.

B. Protect plants from damage due to landscape operations and operations of other contractors and trades. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

C. After installation and before Substantial Completion, remove nursery tags, nursery stakes, tie tape, labels, wire, burlap, and other debris from plant material, planting areas, and Project site.

3.7 DISPOSAL

A. Remove surplus soil and waste material including excess subsoil, unsuitable soil, trash, stakes, ties, and debris and legally dispose of them off CO’s property.

END OF SECTION 32 93 00

SECTION 33 14 16 – SITE WATER UTILITY DISTRIBUTION

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Replacing the existing water service to the Visitor Center building.

1.2 RELATED SECTIONS

A. Section 31 20 00 – Earth Moving

1.3 REFERENCE STANDARDS

A. ASTM D2683 – Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Fittings.

B. ASTM D3261 – Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing.

C. ASTM D3035 – Standard Specification for Polyethylene (PE) Plastic Pipe (DR-PR) Based on Controlled Outside Diameter 2015.

D. ASTM F1055 – Electrofusion Type Polyethylene Fittings for Outside Diameter Controlled Polyethylene Pipe and Tubing.

E. ASTM F2648 – Standard Specification for 2 to 60 inch (50 to 1500 mm) Annular Corrugated Profile Wall Polyethylene (PE) Pipe and Fittings for Land Drainage Applications.

F. AWWA C901 – Polyethylene (PE) Pressure Pipe and Tubing, 3/4 In. (19 mm) Through 3 In. (76 mm), for Water Service.

G. NSF 61 – Drinking Water System Components – Health Effects

1.4 SUBMITTALS

A. See Section 01 31 00 – Project Management & Coordination, for submittal procedures.

B. Product Data: Provide data on pipe materials, pipe fittings, valves and accessories, and water meter box.

C. Manufacturer’s Certificate: Certify that products meet or exceed specified requirements.

D. Submit manufacturer’s recommended fusion procedures for the products.

E. Project Record Documents: Record actual locations of piping mains, valves, connections, thrust restraints, and invert elevations. Identify and describe unexpected variations to subsoil conditions or discovery of uncharted utilities.

F. Disinfection Plan meeting the requirements of Section 3.7.

G. Contractor to maintain all testing and quality control documentation and assurance procedures. Contractor to provide the following documents to the CO:

1. Quality control test reports
2. Fusion reports for each weld as reported by the datalogger
3. Certified laboratory test data for the materials and products to be used in the work shall be submitted to the CO for approval
4. Results of the quality control tests required during the performance of the work shall be submitted to the CO within 48 hours of completion
5. An independent testing / inspection firm hired by the contractor shall provide the following submittals to CO:

    a. A statement attesting that the Contractor's work is in accordance with the requirements of the project documents
    b. Informal daily "pass" or "fail" reports
    c. Formal weekly reports including all test logs and comments to include density and moisture content test logs, indicating location of tests by coordinates and elevation
    d. Upon completion of backfill activities, all density and moisture content test logs and comments compiled and submitted
    e. Sources and test results of all borrowed materials used for backfill.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Deliver and store valves in shipping containers with labeling in place.

B. Protect polyethylene tubing from direct and indirect UV exposure.

C. The manufacturer shall package product in a manner designed to deliver the pipe and fittings to the project neatly, intact and without physical damage. During transportation each pipe shall rest on suitable pads, strips skids, or blocks securely wedged or tied in place.

D. During loading, transportation, and unloading, every precaution should be taken to prevent damage to the pipe. The handling of the pipeline shall be in such a manner that the pipe is not damaged by dragging it over sharp and cutting objects. Cuts or gouges that reduce the wall thickness by more than 10% are not acceptable and must be cut out and discarded.

E. Handle the pipe in accordance with the PPI Handbook of Polyethylene Pipe (2nd Edition), Chapter 2. All pipe and accessories shall be loaded and unloaded by lifting with hoists or by skidding in order to avoid shock or damage. Under no circumstances shall materials be dropped. Pipe handled on skidways shall not be rolled or skidded against pipe on the ground. Slings, hooks or pipe tongs shall be padded and used in such a manner as to prevent damage to the exterior surface or interior of the pipe. All pipe and fittings shall be subjected to visual inspection at time of delivery and before they are lowered into the trench to be laid.

F. Materials, if stored, shall be kept safe from damage and shall not be stacked higher than the limits recommended by the manufacturer. The bottom tiers shall be kept off the ground on timbers, rails, or concrete. Pipe shall not be stored close to heat sources. The contractor shall be responsible for all security, damage and loss of pipe, excluding Acts of God.

G. The interior of the pipe as well as all sealing surfaces of mating components (i.e. flange faces) shall be kept free from dirt or foreign matter at all times. The open ends of all sections of joined and/or installed pipe (not in service) shall be plugged to prevent insects, animals, or foreign material from entering the pipe line or pipe section. The practice of stuffing cloth or paper in the open ends of the pipe will not be permitted. Use waterproof nightcaps to prevent the entrance of any type of natural precipitation into the carrier or containment pipe and will be secured to the pipe in such a manner that the wind cannot blow them loose. Where possible, the pipe shall be raised and supported at a suitable distance from the open end such that the open end will be below the level of the pipe at the point of support.

H. All pipe and fittings shall be subjected to visual inspection at time of delivery and before they are installed or lowered into the trench to be laid. Defective, damaged, or unsound pipe will be rejected. Cuts, punctures, or gouges that penetrate or reduce the wall thickness by 10% or more are not acceptable and must be removed and discarded.

1.6 WARRANTY

A. See Section 01 77 00 – Closeout Procedures, for additional warranty requirements.

PART 2 - PRODUCTS

2.1 WATER PIPE

A. Pipe:

1. Pipe for main line piping shall be polyethylene pipe (not tubing) in accordance with AWWA C906, material designation code of PE4710 and all applicable ASTM Standards.
2. Main line piping shall be SDR 9.
3. Pipe shall be NSF 61 certified for potable water.

B. Fittings

1. Butt fusion fittings shall be made of HDPE material with a minimum material designation code of PE4710 and all applicable ASTM standards. Molded and fabricated fittings shall have a pressure rating equal to the pipe unless otherwise specified on the project documents. All fittings shall meet the requirements of AWWA C901, C906 and all applicable ASTM standards. Markings for molded fittings shall comply with the requirements of ASTM D3261. Fabricated fitting shall be marked in accordance with ASTM F2206. Socket fittings shall meet ASTM D2683. Fabricated fittings shall be manufactured using a data logger to record fusion time, pressure and temperature, and shall be marked with a unique joint identifier that corresponds to the joint report. A graphic representation of the time and pressure data for all fusion joints made producing fittings

shall be maintained for a minimum of five years as part of quality control and will be available upon request of CO.

2. Electrofusion fittings shall be made of HDPE material with a minimum material designation code of PE4710 and meet ASTM F1055. Electrofusion fittings shall have a pressure rating equal to the pipe unless otherwise specified on the project documents. All electrofusion fittings shall be suitable for use as pressure conduits and have nominal burst values of four times the working pressure rating of the fitting. Marking of electrofusion fittings shall comply with the requirements of ASTM F1055. All electrofusion fittings shall be properly stored in compliance with the manufacturer's recommendation.
3. Saddle fusion could be used to fuse branch saddles, tapping tees and other HDPE fittings onto the wall of the main pipe. Saddle fusion shall be done in accordance with ASTM F2620 or PPI TR-41 or the fitting manufacturer's recommendations. Saddle fusion joints shall be made by qualified fusion technicians. Qualification of the fusion technician shall be demonstrated by evidence of fusion training within the past two years on the equipment to be utilized on this project in accordance with ASTM F3190.
4. Flanges and Mechanical Joint Adapters (MJ) shall have a minimum material designation code of PE4710 and meet all applicable AWWA and ASTM standards. Flanged and MJ adapters can be made to ASTM D3261 or machined in compliance with ASTM F2206. Flanges and MJ adapters shall have a pressure rating equal to the pipe unless otherwise specified on the plans. Markings for molded or machined flange adapters or MJ adapters shall be per ASTM D3261. Fabricated (including machined) flange adapters shall be marked per ASTM F2206. Installation of all Flanged adapters shall follow the guidelines of the Plastics Pipe Institute TN-38.
5. Glands, bolts, and gaskets shall be manufactured in accordance with AWWA C153. Bolts and nuts shall be grade 2 or higher.

2.2 BEDDING AND COVER MATERIALS

A. Bedding: As specified in Section 31 20 00.

B. Cover: As specified in Section 31 20 00.

2.3 TRACER WIRE / WARNING TAPE

A. Tracer Wire: #12 AWG copper clad steel insulated wire

B. Warning Tape: Non-detectable, plastic, blue in color, imprinted with "Caution Water Line Below" or "Caution Water Line Buried Below" in large letters.

PART 3 - EXECUTION

3.1 GENERAL

A. Verify that restroom service connection and water main size, location, and invert are as indicated.

B. All HDPE pipe and fittings shall be cut, joined, and installed in accordance with the manufacturer's recommendations. Joining, laying, and pulling of polyethylene pipe shall be accomplished by personnel experienced in working with polyethylene pipe systems.

3.2 PREPARATION

A. Cut pipe ends square, ream pipe and tube ends to full pipe diameter, remove burrs.

B. Remove scale and dirt on inside and outside before assembly.

C. Prepare pipe connections to equipment with flanges or unions.

3.3 FUSION AND JOINING

A. Fusion Joining Requirements:

1. All HDPE pipe shall be joined to itself by the heat fusion process which produces homogeneous, seal, leak tight joints. Tie-ins between sections of HDPE pipe shall be made by butt fusion whenever possible.
2. Butt Fusion: The pipe shall be joined by the butt fusion procedure outlined in ASTM F 2620 or PPI TR-33. A record or certificate of training for the fusion operator must be provided that documents training to the fundamentals of ASTM F 2620.Considerations should be given to and provisions made for adverse weather conditions, such as temperatures below freezing, precipitation, or wind, which is accepted by the CO.
3. Electrofusion: Electrofusion joining shall be done in accordance with the manufacturers recommended procedure. Other sources of electrofusion joining information are ASTM F 1290, PPI TN 34, and PPI Municipal Advisory Board (MAB) Generic Electrofusion Procedure for Field Joining of 12 Inch and Smaller Polyethylene (PE) Pipe. The electrofusion processor must be capable of reading and storing the input parameters and the fusion results for later download to a record file. Qualification of the fusion technician shall be demonstrated by evidence electrofusion training within the past year on the equipment to be utilized for this project.

B. Fusion Operators:

1. The employer of the fusion machine operator is responsible for the fusion joint quality of the fusion weld made by that individual. The employer is responsible for documenting all training and qualification records for that individual, including compliance to any code requirements for fusion/bonder operators.
2. All HDPE fusion equipment operators shall be qualified to the procedure used to perform pipe joining. Fusion equipment operators shall have current, formal training on all fusion equipment employed on the project. Training received more than two years prior to

operation with no evidence of activity within the past 6 months shall not be considered current.

3. For Projects with at least 5,000 feet or with pipe larger than 24 inches, operators or their supervisor must have a current McElroy Fusion Training Certificate for the equipment to be used on the project.

C. Butt Fusion Equipment:

1. For 6″ and larger pipe sizes, the pipe butt fusion machine shall be a hydraulic fusion machine capable of butt fusing HDPE pipe. The carriage must be removable from the chassis for in-ditch use. The machine must be compatible with an electronic data recording device, McElroy Data Logger or equal. Accessories will include all butt fusion inserts for the specified range of pipe sizes, a pyrometer kit for checking the surface temperature of the heater, extension cord of appropriate gauge (25′ minimum), and hydraulic extension hoses (minimum of four). The butt fusion machine will be McElroy, or approved equivalent.

3.4 TRENCHING

A. See Utility Trenching in Section 31 20 00 and plan details for additional requirements.

B. Hand trim excavation for accurate placement of pipe to elevations indicated.

C. Form and place concrete for pipe thrust restraints at each change of pipe direction. Place concrete to permit full access to pipe and pipe accessories. Provide thrust restraint bearing on subsoil.

3.5 INSTALLATION – PIPE

A. Establish elevations of buried piping to ensure not less than 3 feet of cover over the top of the pipe.

B. Verify trench cut is ready to receive work and excavations, dimensions, and elevations are as indicated on layout drawings.

C. Install pipe to allow for expansion and contraction without stressing pipe or joints.

D. Install access fittings to permit disinfection of water system performed.

E. Slope water pipe and position drains at low points.

F. Install trace wire 12 inches below finished grade; coordinate with Section 31 20 00.

3.6 PIPE JOINING

A. High density polyethylene pipe shall be heat fused and pressure tested as per manufacturer’s guidelines before installation. During assembly and prior to installation, pipe must be laid out in such a way as to minimize interference to pedestrian and vehicular traffic.

B. Cuts or gouges that reduce the wall thickness by more than 10% are not acceptable and must be cut out, discarded and the pipe rejoined.

C. Each butt fusion shall be recorded and logged by a datalogger affixed to the fusion machine. Joint data shall be submitted as part of the as-built documentation.

D. Mechanical joining – Polyethylene pipe and fittings may be joined together or to other materials by means of flanged connections or mechanical couplings designed for joining polyethylene pipe or for joining polyethylene pipe to another pipe material. Mechanical couplings shall be fully pressure rated and fully thrust restrained and installed in accordance with manufacturer's recommendations.

3.7 DISINFECTION

A. Disinfection of the water service line shall be in conformance with AWWA C651.

1. New or repaired water potable water systems shall be disinfected prior to use. Prior to utilization of newly constructed or altered potable water piping systems, all affected potable water piping shall be disinfected using procedures prescribed in AWWA C651. The method to be followed shall be that prescribed by the Health Authority or, in case no method is prescribed by it, the following:

    a. The pipe system shall be flushed with clean, potable water until potable water appears at the points of outlet. Minimum flushing velocity shall be 3 feet per second.
    b. Introduce highly chlorinated water to the pipe at a point within five pipe diameters of the pipe's connection to the existing potable system.
    c. Introduce water containing a minimum of 25 mg/L free chlorine until the entire new pipe contains a minimum of 25 mg/L free chlorine.
    d. Retain chlorinated water in the pipe for at least 24 hours and no more than 48 hours.
    e. Flush pipe using potable water until chlorine residual equals that of the existing potable water system.
    f. Dispose of chlorinated water to prevent damage to the environment. Dechlorinate highly chlorinated water from testing before releasing into the ground or sewers. Obtain Jurisdiction approval prior to flushing activities.
    g. Test water mains according to AWWA C651. If the initial disinfection fails to produce satisfactory bacteriological samples, flush the main again and reinitiate the sampling process. If check samples show the presence of coliform organisms, re-chlorinate the main prior to flushing and sampling until satisfactory results are obtained.

3.8 PRESSURE AND LEAKAGE TESTING

A. Conduct hydrostatic pressure testing of installed polyethylene pipe in accordance with ASTM F2164.

B. For HDPE mains, fill the main slowly ensuring fill rate does not exceed capacity of air release devices. Once air has been expelled from the system, gradually raise the pressure to 160 psi. Add makeup water as necessary to maintain this pressure as necessary for four hours. After four-hour period, reduce main pressure to the 150 psi test pressure and monitor for one hour. Do not increase

pressure or add make-up water during this one-hour period. The test is passed and considered acceptable if the main pressure does not drop more than 5% (7.5 psi) during the one-hour period.

C. If any defects or leaks are revealed, they should be corrected, and the pipeline retested after a minimum 24-hour recuperation period between tests. Total testing conducted on a section of pipeline shall not exceed eight hours within a 24-hour period.

3.9 FIELD QUALITY CONTROL

A. See Section 01 40 00 – Quality Requirements, for additional requirements.

B. Perform field inspection and testing in accordance with Section 01 40 00.

C. Testing shall be conducted by Contractor or an independent testing and inspection agency employed by the Contractor.

D. Pressure and leak test water piping.

E. If tests indicate Work does not meet specified requirements, remove Work, replace and retest at no cost to Government.

END OF SECTION 33 14 16

SECTION 33 40 00 – STORM DRAINAGE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Installation of roof drain line and inlet.
2. Installation of trench drain and associated storm drain.

B. Related Sections:

1. Section 01 57 23 – Temporary Stormwater Pollution Prevention
2. Section 31 20 00 – Earth Moving
3. Section 33 46 00 – Subdrainage

1.3 SUBMITTALS

A. Product Data: For inlet, frame, grate/lid, trench drain, and storm drain pipe.

B. Manufacturer's Certification: Products meet or exceed specifications.

PART 2 - PRODUCTS

2.1 PIPE

A. Storm drain pipe shall be lined corrugated polyethylene pipe and shall meet the requirements of AASHTO M 294 Type S or D.

B. Corrugated polyethylene storm sewer pipe manufacturers shall participate in the National Transportation Product Evaluation Program (NTPEP) work plan for HDPE (High Density Polyethylene) Thermoplastic Pipe and be listed on the NTPEP audit website displaying they are NTPEP compliant.

C. Corrugated polyethylene couplings and fittings shall meet the requirements of AASHTO M 294 Type S or D.

D. All joints for corrugated polyethylene storm sewer pipe shall be made with a bell/bell or bell and spigot coupling and shall conform to ASTM D3212 using elastomeric gaskets conforming to ASTM F477. All gaskets shall be factory installed on the pipe in accordance with the manufacturer's recommendations.

2.2 INLETS

A. Inlets shall be precast and shall conform to the dimensions shown in the plans. All precast concrete items shall meet the requirements of AASHTO M199.

B. Inlet reinforcement shall be as set forth in the Plans.

C. The knockout diameter shall not be greater than 20 inches. Knockouts shall have a wall thickness of 2 inches minimum to 2.5 inches maximum. Provide a 1.5″ minimum gap between the knout wall and the outside of the pipe. After the pipe is installed, fill the gap with joint mortar.

D. The precast base section may have a rounded floor. The walls may be sloped at a rate of 1:24 or steeper.

2.3 FRAME AND LID

A. Castings for metal frames for inlets shall be cast steel, gray iron, or ductile iron.

B. Lids shall be solid and shall be cast steel or ductile iron.

C. Lids shall be seated properly to prevent rocking.

2.4 TRENCH DRAIN

A. Channels

1. Trench drain channel shall be manufactured from polyester resin polymer concrete with integrally cast-in stainless steel edge rails. Polymer concrete shall be conform to the following:

Compressive Strength:	98 MPa
Flexural Strength	26 MPa
Tensile Strength	14 MPa
Water Absorption	0.07%
Frost Proof	YES
Coefficient of Expansion/Contraction	2.02x10-5/°C
Water Vapor Transmission	0.0364 g/m^2
Non-Flammable	YES
Manning's Roughness, n	0.011
Resistant to Weathering	YES
Dilute Acid and Alkali Resistant	YES
SF Sealant Groove	YES

2. Channels shall be 4 inches nominal interior width with an overall width of 6 inches.
3. Channel invert shall have a U-profile to allow efficient drainage. Sloped channels shall have a built-in slope of 0.5%.
4. All channels shall be interlocking with a male/female joint.

B. Grates

1. Grates shall be manufactured from ductile iron to ASTM A 536-84 – Grade 65-45-12.
2. Grate shall have an overall width of 6 inches.
3. Grate shall have maximum slot widths of ½".

C. Concrete: Concrete shall have a minimum strength of 4000 psi.

D. Grates and trench drain system shall withstand HS-20 traffic loading.

2.5 ANIMAL GUARD

A. Animal guard shall be 4″ stainless steel mesh.

PART 3 - EXECUTION

3.1 PIPE INSTALLATION

A. Excavation, Trenching, and Backfill

B. Verify trench cut is ready to receive work and excavations, dimensions, and elevations are as indicated on layout drawings. Trenches must be of sufficient width in the pipe zone to permit proper installation and bedding of the pipe and to provide the required compaction of backfill.

C. Hand trim excavation for accurate placement of pipe to elevations indicated.

D. All ledgerock, boulders, and stones shall be removed to provide a minimum of 6 inches clearance under all portions of the pipe.

E. Placement of bedding material shall precede the installation of all pipe. This shall include necessary leveling of the native trench bottom or the top of the foundation material as well as placement and compaction of required bedding material to a uniform grade so that the entire length of pipe will be supported on a uniformly dense unyielding foundation.

F. When, after excavating to the foundation level, the material remaining in the trench bottom is determined to be unsuitable by the CO, excavation shall be continued to such additional depth and width as required by the CO. Unsuitable foundation materials shall be disposed of at an approved site. The trench foundation shall be backfilled to the bottom of the pipe zone with sand bedding, or other suitable material, and compacted to form a uniformly dense, unyielding foundation.

1. Lay pipe to slope gradients noted on layout drawings; with maximum variation from true slope of 1/8 inch in 10 feet.
2. Pipe zone bedding material shall provide uniform support along the entire pipe barrel, without load concentration at joint collars or bells. All adjustment to line and grade shall be made by scraping away or filling in with bedding material under the body of the pipe and not by blocking or wedging. Bedding disturbed by pipe movement, or by removal of shoring movement of a trench shield or box, shall be reconsolidated prior to backfill.
3. Pipe zone bedding shall be placed in loose layers and compacted to 90 percent maximum density. Bedding shall be placed, spread, and compacted before the pipe is installed so that the pipe is uniformly supported along the barrel. Lifts of not more than 6 inches in thickness shall be placed and compacted along the sides of the pipe. Material shall be worked carefully under the pipe haunches and then compacted.
4. If the CO determines that the material existing in the bottom of the trench is satisfactory for bedding the pipe, the bedding material specified is not required, provided the existing material is loosened, regraded, and compacted to form a dense, unyielding base.
5. Before lowering and while suspended, inspect pipe and each fitting for defects. Installation of defective material is not permitted.
6. Where pipe, fittings or joint materials have been soiled by earth in handling, thoroughly clean soiled surfaces by wire brushing and wiping until all traces of earth are removed before joining pipe.
7. Maintain interior of all pipes thoroughly clean. After each line of pipe has been laid, carefully inspect, identify and remove dirt, trash, rags and other foreign matter from interior. Protect completed work.
8. Install pipe, fittings, and accessories at the locations indicated on layout drawings and in accordance with manufacturer's instructions. Seal watertight.
   Trenches shall be backfilled as soon after the pipe laying as possible.

    a. Pipe zone backfill material shall be clean earth or sand, free from clay, frozen lumps, roots, or moisture in excess of that permitting required compaction. Rocks or lumps larger than 3 inches maximum shall not be used for pipe zone backfill.
    b. Pipe zone backfill shall be placed in loose layers and compacted to 90 percent maximum density. Backfill shall be brought up simultaneously on each side of the pipe to the top of the pipe zone. The pipe shall then be covered to the top of the pipe zone and the materials compacted in a manner to avoid damaging or disturbing the completed pipe.
    c. Backfill above the pipe zone shall be accomplished in such a manner that the pipe will not be shifted out of position nor damaged by impact or overloading. If pipe is being placed under existing paved areas, or Roadways, backfill above the pipe zone shall be placed in horizontal layers no more than 6 inches thick and compacted to 95 percent maximum density. If pipe is being placed in non-traffic areas, backfill above the pipe zone shall be placed in horizontal layers no more than 6 inches thick and shall be compacted to 85 percent maximum density.
    d. Protection

        1) Water is not permitted to run or stand in trench while pipe laying is in progress, before the joints are completely set, or before trench has been backfilled.
        2) Protect pipe and bedding cover from damage or displacement until backfilling operation is in progress.

3.2 INLET INSTALLATION

A. The excavation for all catch basins shall be sufficient to leave 1 foot in the clear between their outer surfaces and the earth bank.

B. The cover or grating of a catch basin shall not be grouted to final grade until the final elevation of the ditch in which it is to be placed has been established, and until permission thereafter is given by the CO to grout the cover or grating in place. Covers shall be seated properly to prevent rocking. Leveling and adjustment devices that do not modify the structural integrity of the metal frame, grate or cover, and do not void the originating foundry's compliance to these specifications and warranty are allowed. Leveling and adjusting devices that interfere with the backfilling, backfill density, and grouting will not be allowed. The hardware for leveling and adjusting devices shall be completely removed when specified by the CO.

C. In the event any pipe enters the manhole through the precast concrete units, the Contractor shall make the necessary cut through the manhole wall and steel mesh. The steel shall be cut flush with the face of the concrete and shall be cut in such a manner that it will not loosen the reinforcement in the manhole wall.

D. The ends of all pipes shall be trimmed flush with the inside walls.

E. Rubber gaskets or flexible plastic gaskets may be used in tongue and groove joints of precast units. All other joints and all openings cut through the walls shall be grouted and watertight. Mortar must be composed of cement, sand, and water. The proportion of sand to cement measured by volume must be 2 to 1. Mortar must contain only enough water to allow placing and packing. Sand particles must be no larger than 1/2 the size of the recess or space in which the mortar is to be placed.

F. If gaskets are used, handling of the precast units after the gasket has been affixed shall be done carefully to avoid disturbing or damaging the gasket or contaminating it with foreign material. Care shall be exercised to attain proper alignment before the joints are entirely forced home. During insertion of the tongue or spigot, the units shall be partially supported to minimize unequal lateral pressure on the gasket and to maintain concentricity until the gasket is properly positioned.

G. Backfilling around the Work will not be allowed until the concrete or mortar has thoroughly set.

H. Catch basins shall be watertight.

I. Backfilling

1. Place structure backfill in uniform layers. Bring backfill up uniformly on all sides of structures. Backfill layers must be at most 0.67 foot thick before compacting the thickness must be at most 4 feet when compaction is performed by ponding or jetting.
2. Do not use compaction equipment or methods that may cause excessive displacement or damage structures.
3. Do not place structure backfill the catch basin has been inspected by the CO and authorized for backfilling.
4. Compact structure backfill to a relative compaction of at least 95 percent.

J. Catch basins shall be constructed on a compacted or undisturbed level foundation. If the Contractor elects to use a separate cast-in-place base, the concrete shall be Class 4000.

K. All excavations 4 feet or more in depth shall be shored with structural shoring or shall meet open-pit requirements.

3.3 TRENCH DRAIN INSTALLATION

A. Ensure ground conditions are compacted and suitable for construction.

B. Excavate trenches to sufficient dimensions to permit proper installation of stabilizing concrete and trench drain.

C. Concrete shall be placed to encase trench drain as shown in the Plans.

D. Install trench drain in accordance with manufacturer's instructions. The drain shall be finished 1/8 inch below finish grade while ensuring positive drainage of adjacent areas toward the drain.

E. The existing trench drain shall be cleaned of all debris.

3.4 PROTECTION

A. Protect pipe and bedding from damage or displacement until backfilling operation is in progress.

END OF SECTION 33 40 00

SECTION 33 46 00 – SUBDRAINAGE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Installation of foundation drains and underdrains.

B. Related Sections:

1. Section 01 57 23 – Temporary Stormwater Pollution Prevention
2. Section 31 20 00 – Earth Moving
3. Section 33 40 00 – Storm Drainage

1.3 SUBMITTALS

A. Product Data: For foundation drain pipe and fittings.

B. Manufacturer’s Certification: Products meet or exceed specifications.

PART 2 - PRODUCTS

2.1 PIPE

A. Corrugated polyethylene pipe shall meet the requirements described in 33 40 00 – Storm Drainage.

B. Perforated PE Pipe and Fittings:

1. Foundation drains shall be 4-inch diameter, corrugated, and meet the requirements of ASTM F405 or AASHTO M 252, Type CP.
2. Fittings shall be bell-and-spigot ends for loose joints compatible with the pipe and shall meet the requirements of ASTM F405 or AASHTO M 252.

2.2 GEOTEXTILE FABRICS

A. Geotextile fabrics shall meet the requirements described in 31 20 00 – Earth Moving.

2.3 SOIL MATERIALS

A. Soil materials shall meet the requirements described in 31 20 00 – Earth Moving.

PART 3 - EXECUTION

3.1 EARTHWORK

A. Excavation, Trenching, and Backfill shall be performed as described in 31 20 00 – Earth Moving.

3.2 FOUNDATION DRAIN INSTALLATION

A. Place impervious fill material on subgrade adjacent to bottom of footing after concrete footing forms have been removed. Place and compact impervious fill to dimensions indicated, but not less than 6 inches deep and 12 inches wide.

B. Lay geotextile filter fabric for separation in trench and overlap trench sides.

C. Place supporting layer of drainage course over compacted subgrade and geotextile filter fabric, to compacted depth of not less than 4 inches.

D. Provide minimum pipe slope of 0.50% unless otherwise noted.

E. Add drainage course to width of at least 12 inches on side away from wall and to top of pipe to perform tests.

F. After satisfactory testing, cover drainage piping to width of at least 12 inches on side away from footing and above top of pipe to within 12 inches above top of slab.

G. Install drainage course and wrap top of drainage course with geotextile filter fabric for separation.

H. Place layer of geotextile filter fabric for separation over top of drainage course, overlapping edges at least 4 inches.

I. Place backfill material over compacted drainage course. Place material in loose-depth layers not exceeding 6 inches. Thoroughly compact each layer as described in 31 20 00 – Earth Moving. Final backfill to finish elevations and slope away from building.

3.3 FIELD QUALITY CONTROL

A. Tests and Inspections:

1. After installing drainage course to top of piping, test drain piping with a digital level on the top of the pipe to ensure adequate slope before backfilling.

B. Remove obstructions, replace damaged components, and repeat test until results are satisfactory.

C. Drain piping will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

END OF SECTION 33 46 00

EFFIGY MOUNDS NATIONAL MONUMENT

Redesign Parking and Replace Curbs,
Walks, and Pavement
EFMO 224159

Division 02 through Division 33

Final Construction Documents
PROJECT SPECIFICATIONS

SECTION 00 01 07 – SEALS PAGE

PART 1 - GENERAL

1.1 SUMMARY

A. Professional seals by Design Professionals and others responsible for preparing Final Construction Documents:

1. Civil Engineer

a. Divisions:

1) Division 2
2) Division 31
3) Division 33

b. Sections:

1) 32 12 16
2) 32 13 20
3) 32 17 00
4) 32 31 13

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Emily Maggio

Name: Emily Maggio

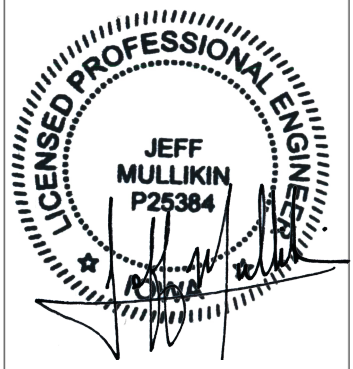
My license renewal date is: 12/31/2027

Divisions and sections covered by this seal are as noted above.

2. Electrical Engineer

a. Division:

1) Division 26



CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Engineer under the laws of the State of Iowa.

Jeff Mullikin

Name:

My license renewal date is: 12/31/2026

Divisions and sections covered by this seal are as noted above.

3. Landscape Architect

   a. Sections:

      1) 31 23 16
      2) 32 91 13
      3) 32 92 00
      4) 32 93 00

CERTIFICATION

I hereby certify that this engineering document was prepared by me or under my direct personal supervision and that I am a duly Licensed Professional Landscape Architect under the laws of the State of Iowa.

Name: Lindsay Martin

My license renewal date is: 06/30/2026

Divisions and sections covered by this seal are as noted above.

END OF SECTION 00 01 07

SECTION 02 41 13 – SELECTIVE SITE DEMOLITION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Sawcutting and removal or recycling of existing asphalt concrete pavement and base course material.
2. Removal or recycling of existing cement concrete walkways.
3. Removal and return of existing site features to the NPS include:

    a. Picnic table
    b. Bench
    c. Trash receptacle
    d. Retaining wall stone

4. Demolition, removal, and disposal of the following items:

    a. Concrete curb
    b. Flagstone pavers
    c. Gates
    d. Signage
    e. Pedestrian lighting
    f. Catch basin
    g. Storm drain
    h. Culvert
    i. Shed

B. Related Sections include the following:

1. Section 01 57 23 – Temporary Pollution Prevention
2. Section 31 10 00 – Site Clearing
3. Section 31 20 00 – Earth Moving
4. Section 31 25 00 – Erosion and Sediment Controls

1.3 DEFINITIONS

A. Remove: Remove and legally dispose of items except those indicated by CO to be salvaged, reused, reinstalled, or to remain on EFMO property.

B. Remove and Salvage: Remove items indicated; clean, service, and otherwise prepare items for reuse; store and protect against damage. Reinstall items in locations indicated in the plans and specifications or as directed by the CO.

C. Existing to Remain: Protect items identified to remain against damage and soiling during demolition. When permitted by the CO, items may be removed to a suitable, protected storage location during demolition and then cleaned and reinstalled in their original locations.

1.4 MATERIALS OWNERSHIP

A. Except for items or materials indicated to be reused, salvaged, or otherwise indicated to remain on the EFMO property, demolished materials shall become the Contractor's property and shall be removed and legally disposed of at the Contractor's expense.

B. Materials for removal and return to the NPS:

1. The following materials within the limits of disturbance shall be removed and returned to the NPS at the project site:

a. Picnic table
b. Bench
c. Trash can
d. Retaining wall stone

C. Materials for removal and disposal:

1. The following materials shall be removed from the site and legally disposed of.

a. Concrete curb
b. Flagstone pavers
c. Gates
d. Signage
e. Pedestrian lighting
f. Catch basin
g. Storm drain
h. Culvert
i. Shed

1.5 QUALITY ASSURANCE

A. Comply with the local noise ordinances regarding hours of operations and allowable noise levels. Monitor vibrations generated by the demolition activities and received by the existing structures in the immediate vicinity of the project.

1.6 SCHEDULING

A. Notify the Contracting Officer (CO) 48 hours prior to beginning demolition work.

B. Contractor is responsible for verifying location of utilities.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.1 UTILITY SERVICES

A. Utility requirements: Locate, identify, disconnect, and seal or cap off indicated utility services where indicated.

1. Arrange to shut off indicated utilities with utility companies.

3.2 PREPARATION

A. Install erosion and sediment control measures prior to demolition.

B. Install site protection fencing prior to demolition.

C. Conduct demolition operations and remove debris to ensure minimum interference with roads, driveways, and other adjacent occupied and used facilities.

1. Do not close or obstruct roads, or other adjacent occupied or used facilities without permission from the CO and authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways.

D. Conduct demolition operations to prevent injury to people and damage to facilities to remain. Ensure safe passage of people around demolition area.

1. Erect temporary protection, such as walks, fences, and railings, where required by CO.

2. Protect existing improvements, appurtenances, and landscaping to remain.
3. Erect protective fencing around drip line of individual trees or around perimeter drip line of groups of trees to remain according to requirements specified in Section 311000 “Site Clearing.”
4. A vertical sawcut shall be made between any existing pavement that is to remain and the portion to be removed. Where sawcutting is required, the sawcut shall be a minimum of 3 inches deep. When the existing pavement is more than 3 inches thick, the portion below 3 inches may be broken after the sawcut is made.

3.3 POLLUTION CONTROLS

A. Use water mist, temporary enclosures, and other suitable methods to limit the spread of dust and dirt. Comply with governing environmental protection regulations.

1. Do not create hazardous or objectionable conditions, such as ice, flooding, and pollution, when using water.

2. The Contractor shall be responsible for ensuring that special precautions are undertaken during sawcutting operations. No concrete (asphalt or cement) or concrete by-products are

to be discharged into any storm drain or surface water. Sawcutting operations will increase the pH of water, therefore filtering is not acceptable.

B. Remove and transport debris in a manner that will prevent spillage on site improvements or public roads.

3.4 DEMOLITION

A. Remove existing asphalt, concrete and base course material as shown in the plans. Approximate existing asphalt depth, including base course material, is 22 inches. Approximate existing concrete walkway depth, including base course material, is 8 inches.

B. Filling Below-Grade Areas: Completely fill below-grade areas and voids resulting from demolition of existing facilities shown on the plans with soil materials according to requirements specified in Section 31 20 00 – Earth Moving.

C. Damages: Promptly repair damage to adjacent facilities caused by demolition operations.

3.5 DISPOSAL AND RECYCLING OF DEMOLISHED MATERIALS

A. Promptly dispose of or recycle demolished materials: If recycling requires additional time to accomplish, provide for orderly storage of materials. Comply with the waste management plan.

B. Transport demolished materials off EFMO property and legally dispose or recycle material.

END OF SECTION 02 41 13

SECTION 26 05 00 – COMMON WORK RESULTS FOR ELECTRICAL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section

B. Applicable codes include the following:

1. International Building Code 2024
2. ASHRAE 90.1-2016
3. International Fire Code 2024
4. National Electric Code 2023
5. NPS Building Standards

1.2 SUMMARY

A. Section Includes:

1. Electrical equipment coordination and installation.
2. Common electrical installation requirements.

1.3 COORDINATION

A. Coordinate arrangement, mounting, and support of electrical equipment:

1. To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.
2. To provide for ease of disconnecting the equipment with minimum interference to other installations.
3. To allow right of way for piping and conduit installed at required slope.
4. To connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.

PART 2 - PRODUCTS (NOT USED)EXECUTION

3.1 COMMON REQUIREMENTS FOR ELECTRICAL INSTALLATION

A. Comply with NECA 1.

B. Measure indicated mounting heights to bottom of unit for suspended items and to center of unit for wall-mounting items, except as otherwise indicated in drawings.

C. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide maximum possible headroom consistent with these requirements.

D. Equipment: Install to facilitate service, maintenance, and repair or replacement of components of both electrical equipment and other nearby installations. Connect in such a way as to facilitate future disconnecting with minimum interference with other items in the vicinity.

E. Right of Way: Give to piping systems installed at a required slope.

F. Exposed raceways/conduit: Where conduit or raceway is routed in an exposed location, the conduit and raceway shall be painted to match adjacent finishes. Coordinate required finishes with Contracting Officer.

END OF SECTION 26 05 00

SECTION 26 05 19 – LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Building wires and cables rated 600 V and less.
2. Connectors, splices, and terminations rated 600 V and less.

1.3 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Field quality-control test reports.

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 CONDUCTORS AND CABLES

A. Salient Characteristics

1. Physical:

a. Conductors shall be new and unused. Wire and cable shall be copper single conductor type with 600 V insulation, unless otherwise noted. Conductor shall be soft annealed Class B, per ASTM B-3 for solid wire and ASTM B-8 for stranded wire. Conductors shall be minimum 98% conductive.

2. Functional:

a. Insulation shall be flame retardant, heat resistant polyvinyl chloride (PVC), ethylene propylene (EP) or polyethylene (PE) with minimum insulation thicknesses per table 310-13 of the NEC. The insulation shall conform to the requirements of UL 83 ICEA S-68-516 for EP, ICEA S-61-402 for PVC and PE.
b. Type THWN or THHN wire and cable shall have an outer nylon jacket conforming to UL-83. Cables shall be manufactured to meet the standards of Insulated Cable Engineer's Association (ICEA).
c. Copper Conductors: Comply with NEMA WC 70.
d. Conductor Insulation: Comply with NEMA WC 70 for Types THWN-2 or XHHW-2.

3. Performance:

a. None.

2.2 CONNECTORS AND SPLICES

A. Salient Characteristics:

1. Physical:

a. Factory-fabricated connectors and splices of size, ampacity rating, material, type, and class for application and service indicated.

2. Functional:

a. None.

3. Performance:

a. None.

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

A. Feeders: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

B. Branch Circuits: Copper. Solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

A. Exposed Feeders: Type THHN-THWN-2, single conductors in raceway.

B. Feeders Concealed in Ceilings, Walls, Partitions, and Crawlspaces: Type THHN-THWN-2, single conductors in raceway.

C. Feeders Concealed in Concrete, below Slabs-on-Grade, and Underground: Type XHHW-2, single conductors in raceway.

D. Exposed Branch Circuits, Including in Crawlspaces: Type THWN-2, single conductors in raceway.

E. Branch Circuits Concealed in Ceilings, Walls, and Partitions: Type THHN-THWN-2, single conductors in raceway. Metal-clad cable, Type MC is not allowed.

F. Branch Circuits Concealed in Concrete, below Slabs-on-Grade, and Underground: Type XHHW-2, single conductors in raceway.

G. Cord Drops and Portable Appliance Connections: Type SO, hard service cord with stainless-steel, wire-mesh, strain relief device at terminations to suit application.

H. Class 1 Control Circuits: Type THHN-THWN-2, in raceway.

I. Class 2 Control Circuits: Type THHN-THWN-2, in raceway.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

A. Conceal cables in finished walls, ceilings, and floors, unless otherwise indicated.

B. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.

C. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.

D. Support cables according to Division 26 Section "Hangers and Supports for Electrical Systems."

E. Identify and color-code conductors and cables according to Division 26 Section "Identification for Electrical Systems."

3.4 CONNECTIONS

A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A and UL 486B.

B. Make splices and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.

C. Delete subparagraph below unless aluminum conductors are specified.

D. Wiring at Outlets: Install conductor at each outlet, with at least 12 inches (300 mm) of slack.

E. Provide crimped ferrules for stranded wires at any screw terminal locations.

3.5 FIELD QUALITY CONTROL

A. Perform tests and inspections and prepare test reports.

B. Tests and Inspections:

1. Perform each visual and mechanical inspection and electrical test stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.

C. Test Reports: Prepare a written report to record the following:

1. Test procedures used.
2. Test results that comply with requirements.
3. Test results that do not comply with requirements and corrective action taken to achieve compliance with requirements.

D. Remove and replace malfunctioning units and retest as specified above.

END OF SECTION 26 05 19

SECTION 26 05 26 – GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUBMITTALS

A. Product Data: For each type of product indicated.

B. Other Informational Submittals: Plans showing dimensioned as-built locations of grounding features specified in Part 3 "Field Quality Control" Article, including the following:

1. Ground rods.
2. Grounding arrangements and connections for separately derived systems.

C. Field quality-control test reports.

1.3 SUMMARY

A. This Section includes methods and materials for grounding systems and equipment.

1.4 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with UL 467 for grounding and bonding materials and equipment.

PART 2 - PRODUCTS

2.1 CONDUCTORS

A. Salient Characteristics:

1. Physical:

a. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.
b. Bare Copper Conductors:

1) Solid Conductors: ASTM B 3.
2) Stranded Conductors: ASTM B 8.
3) Tinned Conductors: ASTM B 33.
4) Sizes and types of conductors in four subparagraphs below are typical examples. 28-kcmil bonding cable in first subparagraph below is slightly larger than No. 6 AWG.
5) Bonding Cable: 28 kcmil, 14 strands of No. 17 AWG conductor, 1/4 inch (6 mm) in diameter.
6) Bonding Conductor: No. 4 or No. 6 AWG, stranded conductor.
7) Bonding Jumper: Copper tape, braided conductors, terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.
8) Tinned Bonding Jumper: Tinned-copper tape, braided conductors, terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.

2. Functional:

a. None.

3. Performance:

a. None.

2.2 CONNECTORS

A. Salient Characteristics:

1. Physical:

a. Listed and labeled by a nationally recognized testing laboratory acceptable to authorities having jurisdiction for applications in which used, and for specific types, sizes, and combinations of conductors and other items connected.

2. Functional:

a. Bolted Connectors for Conductors and Pipes: Copper or copper alloy, bolted pressure-type, with at least two bolts.
b. Pipe Connectors: Clamp type, sized for pipe.
c. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

3. Performance:

a. None.

2.3 GROUNDING ELECTRODES

A. Salient Characteristics:

1. Physical:

a. Ground Rods: Copper-clad; 5/8" in diameter by 96 inches in length.

2. Functional:

a. None.

3. Performance:

a. None

2.4 GROUNDING BUSBARS

A. Salient Characteristics:

1. Physical:

a. Grounding Bus: Predrilled rectangular bars of annealed copper, ¼ by 4 inches in cross section, with 9/32-inch holes spaced 1-1/8 inches apart. Stand-off insulators for mounting shall comply with UL 891 for use in switchboards, 600 V and shall be Lexan or PVC, impulse tested at 5000V.

2. Functional:

a. None.

3. Performance:

a. None.

PART 3 - EXECUTION

3.1 APPLICATIONS

A. Conductors: Install solid conductor for No. 8 AWG and smaller, and stranded conductors for No. 6 AWG and larger, unless otherwise indicated.

3.2 EQUIPMENT GROUNDING

A. Install insulated equipment grounding conductors with all feeders and branch circuits.

B. Install insulated equipment grounding conductors with the following items, in addition to those required by NFPA 70:

1. Feeders and branch circuits.
2. Lighting circuits.
3. Receptacle circuits.

3.3 INSTALLATION

A. Grounding Conductors: Route along shortest and straightest paths possible, unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

B. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance, except where routed through short lengths of conduit.

1. Use exothermic-welded connectors for outdoor locations, but if a disconnect-type connection is required, use a bolted clamp.

C. Grounding and Bonding for Piping:

1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes, using a bolted clamp connector or by bolting a lug-type connector to a pipe flange, using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.
2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
3. Bond each aboveground portion of gas piping system downstream from equipment shutoff valve.

D. Ground Rod: Re-use existing otherwise, drive rods until tops are 2 inches (50 mm) below finished floor or final grade, unless otherwise indicated.

E. Concrete-Encased Grounding Electrode (Ufer Ground): Re-use existing.

3.4 FIELD QUALITY CONTROL

A. Perform the following tests and inspections and prepare test reports:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements listed below.
2. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal, and at individual ground rod. Make tests at ground rods before any conductors are connected.

a. Measure ground resistance not less than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.

b. Perform tests by fall-of-potential method according to IEEE 81.

B. Report measured ground resistances that exceed the following values:

1. Power and Lighting Equipment or System with Capacity 500 kVA and Less; 10 ohms.

END OF SECTION 26 05 26

SECTION 26 05 29 – HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Hangers and supports for electrical equipment and systems.
2. Construction requirements for concrete bases.

1.3 DEFINITIONS

A. EMT: Electrical metallic tubing.

B. IMC: Intermediate metal conduit.

C. RMC: Rigid metal conduit.

1.4 PERFORMANCE REQUIREMENTS

A. Design supports for multiple raceways, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

B. Design supports for multiple raceways capable of supporting combined weight of supported systems and its contents.

C. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.

D. Rated Strength: Adequate in tension, shear, and pullout force to resist maximum loads calculated or imposed for this Project, with a minimum structural safety factor of five times the applied force.

1.5 SUBMITTALS

A. Product Data: For the following:

1. Steel slotted support systems.
2. Nonmetallic slotted support systems.

B. Shop Drawings: Show fabrication and installation details and include calculations for the following:

1. Steel slotted channel systems. Include Product Data for components.
2. Nonmetallic slotted channel systems. Include Product Data for components.
3. Equipment supports.

C. Welding certificates.

1.6 QUALITY ASSURANCE

A. Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code – Steel."

B. Comply with NFPA 70.

1.7 COORDINATION

A. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.

B. Coordinate installation of roof curbs, equipment supports, and roof penetrations. These items are specified in Division 07 Section.

PART 2 - PRODUCTS

2.1 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

A. Steel Slotted Support Systems:

1. Salient Characteristics:

a. Physical:

1) Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
2) Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4.
3) Painted Coatings: Manufacturer's standard painted coating applied according to MFMA-4.
4) Channel Dimensions: Selected for applicable load criteria.

b. Functional:

1) None.

c. Performance:

1) Comply with MFMA-4, factory-fabricated components for field assembly.

B. Nonmetallic Slotted Support Systems:

1. Salient Characteristics:

a. Physical:

1) Structural-grade, factory-formed, glass-fiber-resin channels and angles with 9/16-inch- (14-mm-) diameter holes at a maximum of 8 inches (200 mm) o.c., in at least 1 surface.

b. Functional:

1) Fittings and Accessories: Products of channel and angle manufacturer and designed for use with those items.
2) Fitting and Accessory Materials: Same as channels and angles.

c. Performance:

1) Rated Strength: Selected to suit applicable load criteria.

C. Raceway and Cable Supports:

1. Salient Characteristics:

a. Physical, Functional and Performance:

1) As described in NECA 1 and NECA 101.

D. Conduit and Cable Support Devices:

1. Salient Characteristics:

a. Physical:

1) Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.

b. Functional:

1) None.

c. Performance:

1) None.

E. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:

1. Powder-Actuated Fasteners:

    a. Salient Characteristics:

        1) Physical:

            a) Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.

        2) Functional:

            a) None.

        3) Performance:

            a) None.

2. Mechanical-Expansion Anchors:

    a. Salient Characteristics:

        1) Physical:

            a) Insert-wedge-type, zinc-coated steel, for use in hardened portland cement concrete with tension, shear, and pullout capacities appropriate for supported loads and building materials in which used.

        2) Functional:

            a) None.

        3) Performance:

            a) None.

3. Concrete Inserts:

    a. Salient Characteristics:

        1) Physical:

            a) Steel or malleable-iron, slotted support system units similar to MSS Type 18; complying with MFMA-4 or MSS SP-58.

        2) Functional:

            a) None.

        3) Performance:

            a) None.

4. Clamps for Attachment to Steel Structural Elements:

    a. Salient Characteristics:

        1) Physical:

            a) MSS SP-58, type suitable for attached structural element.

        2) Functional:

            a) None.

        3) Performance:

            a) None.

5. Through Bolts:

    a. Salient Characteristics:

        1) Physical:

            a) Structural type, hex head, and high strength. Comply with ASTM A 325.

        2) Functional:

            a) None.

        3) Performance:

            a) None.

6. Toggle Bolts:

    a. Salient Characteristics:

        1) Physical:

            a) All-steel springhead type.

        2) Functional:

            a) None.

        3) Performance:

            a) None.

6. Hanger Rods:

    a. Salient Characteristics:

1) Physical:

a) Threaded steel.

2) Functional:

a) None.

3) Performance:

a) None.

2.2 FABRICATED METAL EQUIPMENT SUPPORT ASSEMBLIES

A. Description: Welded or bolted, structural-steel shapes, shop or field fabricated to fit dimensions of supported equipment.

B. Materials: Comply with requirements in Division 05 Section "Metal Fabrications" for steel shapes and plates.

PART 3 - EXECUTION

3.1 APPLICATION

A. Comply with NECA 1 and NECA 101 for application of hangers and supports for electrical equipment and systems except if requirements in this Section are stricter.

B. In first paragraph below, support spacings in NECA 1, Table 1, are more detailed and specific and generally stricter than those permitted by NFPA 70 for EMT, IMC, and RMC. Revise to suit Project, and consider retaining reference to NECA table for critical projects.

C. Maximum Support Spacing and Minimum Hanger Rod Size for Raceway: Space supports for EMT, IMC, and RMC as required by NFPA 70. Minimum rod size shall be 1/4 inch (6 mm) in diameter.

D. Retain paragraph below for projects where seismic design requirements do not apply. Consider retaining for light-commercial projects only.

E. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1-1/2-inch (38-mm) and smaller raceways serving branch circuits and communication systems above suspended ceilings and for fastening raceways to trapeze supports.

3.2 SUPPORT INSTALLATION

A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this Article.

B. Raceway Support Methods: In addition to methods described in NECA 1, EMT may be supported by openings through structure members, as permitted in NFPA 70.

C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb (90 kg).

D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:

1. To Wood: Fasten with lag screws or through bolts.
2. To New Concrete: Bolt to concrete inserts.
3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
4. To Existing Concrete: Expansion anchor fasteners.
5. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inches (100 mm) thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inches (100 mm) thick.
6. To Steel: Welded threaded studs complying with AWS D1.1/D1.1M, with lock washers and nuts.
7. To Light Steel: Sheet metal screws.
8. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate.

E. Drill holes for expansion anchors in concrete at locations and to depths that avoid reinforcing bars.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

A. Comply with installation requirements in Division 05 Section "Metal Fabrications" for site-fabricated metal supports.

B. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.

C. Field Welding: Comply with AWS D1.1/D1.1M.

3.4 PAINTING

A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide minimum dry film thickness of 2.0 mils (0.05 mm).

B. Touchup: Comply with requirements in Division 09 painting Sections for cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.

C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

END OF SECTION 26 05 29

SECTION 26 05 33 – RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes raceways, fittings, boxes, enclosures, and cabinets for electrical wiring.

1.3 DEFINITIONS

A. FMC: Flexible metal conduit.

B. IMC: Intermediate metal conduit.

C. LFMC: Liquidtight flexible metal conduit.

D. RNC: Rigid nonmetallic conduit.

E. RSC: Rigid steel conduit

1.4 SUBMITTALS

A. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.

1.5 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

B. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 METAL CONDUIT AND TUBING

A. See Editing Instruction No. 1 in the Evaluations for cautions about naming manufacturers and products. Retain one of first two paragraphs and list of manufacturers below. See Division 01 Section "Product Requirements."

B. Salient Characteristics:

1. Physical:

a. Fittings for Conduit (Including all Types and Flexible and Liquidtight), EMT, and Cable: NEMA FB 1; listed for type and size raceway with which used, and for application and environment in which installed.

1) Fittings for EMT: Steel, set-screw or compression type.
2) Coating for Fittings for PVC-Coated Conduit: Minimum thickness, 0.040 inch (1 mm), with overlapping sleeves protecting threaded joints.

b. Joint Compound for Rigid Steel Conduit or IMC: Listed for use in cable connector assemblies, and compounded for use to lubricate and protect threaded raceway joints from corrosion and enhance their conductivity.

2. Functional:

a. None.

3. Performance:

a. Rigid Steel Conduit: ANSI C80.1.
b. Aluminum Rigid Conduit: ANSI C80.5.
c. IMC: ANSI C80.6.
d. PVC-Coated Steel Conduit: PVC-coated Rigid Steel Conduit.

1) Comply with NEMA RN 1.
2) Coating Thickness: 0.040 inch (1 mm), minimum.

e. EMT: ANSI C80.3.
f. LFMC: Flexible steel conduit with PVC jacket.

2.2 NONMETALLIC CONDUIT AND TUBING

A. Salient Characteristics:

1. Physical:

a. RNC: NEMA TC 2, Type EPC-40-PVC, unless otherwise indicated. Type EPC-80-PVC for areas that will be subjected to vehicular travel.
b. Fittings for RNC: NEMA TC 3; match to conduit or tubing type and material.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 SURFACE RACEWAYS

A. Salient Characteristics:

1. Physical:

a. Surface Metal Raceways: Galvanized steel with snap-on covers. Manufacturer's standard enamel finish in color selected by the Contracting Officer.

2. Functional:

a. None.

3. Performance:

a. None.

2.4 BOXES, ENCLOSURES, AND CABINETS

A. Salient Characteristics:

1. Physical:

a. Metal Floor Boxes: Cast or sheet metal, semi-adjustable, rectangular.
b. Nonmetallic Floor Boxes: Nonadjustable, round.
c. Cast-Metal Access, Pull, and Junction Boxes: Cast aluminum with gasketed cover.
d. Cast-Metal Outlet and Device Boxes: Ferrous alloy, Type FD, with gasketed cover.
e. Hinged-Cover Enclosures: Type 1, with continuous-hinge cover with flush latch, unless otherwise indicated.

1) Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
2) Nonmetallic Enclosures: Plastic.

f. Cabinets: Type 1, galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.

1) Hinged door in front cover with flush latch and concealed hinge.
2) Key latch to match panelboards.
3) Metal barriers to separate wiring of different systems and voltage.
4) Accessory feet where required for freestanding equipment.

2. Functional:

a. None.

3. Performance:

a. Sheet Metal Outlet and Device Boxes: NEMA OS 1.
b. Cast-Metal Outlet and Device Boxes: NEMA FB 1.
c. Nonmetallic Outlet and Device Boxes: NEMA OS 2.
d. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.
e. Cast-Metal Access, Pull, and Junction Boxes: NEMA FB 1
f. Hinged-Cover Enclosures: NEMA 250.
g. Cabinets: NEMA 250.

2.5 HANDHOLES AND BOXES FOR EXTERIOR UNDERGROUND WIRING

A. Salient Characteristics:

1. Physical:

a. Color of Frame and Cover: Gray.
b. Configuration: Units shall be designed for flush burial and have open bottom, unless otherwise indicated.
c. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure.
d. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
e. Cover Legend: Molded lettering, "ELECTRIC." or "TELEPHONE." depending upon the cabling the box is enclosing.
f. Conduit Entrance Provisions: Conduit-terminating fittings shall mate with entering ducts for secure, fixed installation in enclosure wall.
g. Handholes and boxes are to be fiberglass with Polymer-Concrete Fram and Cover. Sheet-molded fiberglass-reinforced polyester-resin enclosure joined to polymer-concrete top ring or frame.

2. Functional:

a. None.

3. Performance:

a. Comply with SCTE 77, TIER 15.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

A. Outdoors: Apply raceway products as specified below, unless otherwise indicated:

1. Exposed Conduit: Rigid steel conduit.
2. Concealed Conduit, Aboveground: EMT.
3. Retain subparagraph below if Project includes small amounts of exterior underground wiring, 600 V and less, and Division 26 Section "Underground Ducts and Raceways for Electrical Systems" is not being used.
4. Underground Conduit: For horizontal underground conduit, utilize RNC, Type EPC-80-PVC, direct buried. Underground bends are to utilize PVC Coated Rigid Steel prefabricated sections. Vertical underground conduit (risers) shall be PVC Coated Rigid Steel.
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.
6. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.

B. Comply with the following indoor applications, unless otherwise indicated:

1. Exposed, Not Subject to Physical Damage: EMT.
2. Exposed, Not Subject to Severe Physical Damage: EMT.
3. Exposed, Subject to Severe Physical Damage: RSC.
4. Concealed in Ceilings and Interior Walls and Partitions: EMT.
5. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.
6. Damp or Wet Locations: IMC.
7. Raceways for Optical Fiber or Communications Cable in Spaces Used for Environmental Air: EMT.
8. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 3R, stainless steel in damp or wet locations.

C. Minimum Raceway Size: 3/4-inch (21-mm) trade size.

D. Raceway Fittings: Compatible with raceways and suitable for use and location.

1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings, unless otherwise indicated.
2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with that material. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer.

E. Install nonferrous conduit or tubing for circuits operating above 60 Hz. Where aluminum raceways are installed for such circuits and pass through concrete, install in nonmetallic sleeve.

F. Do not install aluminum conduits in contact with concrete.

3.2 INSTALLATION

A. Comply with NECA 1 for installation requirements applicable to products specified in Part 2 except where requirements on Drawings or in this Article are stricter.

B. Keep raceways at least 6 inches (150 mm) away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.

C. Complete raceway installation before starting conductor installation.

D. Support raceways as specified in Division 26 Section "Hangers and Supports for Electrical Systems."

E. Install no more than the equivalent of three 90-degree bends in any conduit run except for communications conduits, for which fewer bends are allowed.

F. Conceal conduit and EMT within finished walls, ceilings, and floors, unless otherwise indicated.

G. Raceways Embedded in Slabs:

1. Run conduit larger than 1-inch (27-mm) trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support.
2. RNC, Type EPC-40-PVC, rigid steel conduit, or IMC before rising above the floor.

H. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.

I. Raceway Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors, including conductors smaller than No. 4 AWG.

J. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb (90-kg) tensile strength. Leave at least 12 inches (300 mm) of slack at each end of pull wire.

K. Install raceway sealing fittings at suitable, approved, and accessible locations and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install raceway sealing fittings at the following points:

1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
2. Where otherwise required by NFPA 70.

L. Expansion-Joint Fittings for RNC: Install in each run of aboveground conduit that is located where environmental temperature change may exceed 30 deg F (17 deg C), and that has straight-run length that exceeds 25 feet (7.6 m).

1. Install expansion-joint fittings for each of the following locations, and provide type and quantity of fittings that accommodate temperature change listed for location:

    a. Outdoor Locations Not Exposed to Direct Sunlight: 125 deg F (70 deg C) temperature change.
    b. Outdoor Locations Exposed to Direct Sunlight: 155 deg F (86 deg C) temperature change.
    c. Indoor Spaces: Connected with the Outdoors without Physical Separation: 125 deg F (70 deg C) temperature change.
    d. Attics: 135 deg F (75 deg C) temperature change.

2. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per deg F (0.06 mm per meter of length of straight run per deg C) of temperature change.
3. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at the time of installation.

M. Flexible Conduit Connections: Use maximum of 72 inches (1830 mm) of flexible conduit for recessed and semirecessed lighting fixtures, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.

1. Use LFMC in damp or wet locations subject to severe physical damage.
2. Use LFMC in damp or wet locations not subject to severe physical damage.

N. Set metal floor boxes level and flush with finished floor surface.

O. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

P. Install plastic bushings at conduit terminations prior to pulling wire.

3.3 INSTALLATION OF UNDERGROUND CONDUIT

A. Direct-Buried Conduit:

1. Excavate trench bottom to provide firm and uniform support for conduit. Prepare trench bottom as indicated in drawings for pipe less than 6 inches (150 mm) in nominal diameter.
2. Install backfill as indicated in drawings.
3. After installing conduit, backfill and compact. Start at tie-in point, and work toward end of conduit run, leaving conduit at end of run free to move with expansion and contraction as temperature changes during this process. Firmly hand tamp backfill around conduit to provide maximum supporting strength. After placing controlled backfill to within 12 inches (300 mm) of finished grade, make final conduit connection at end of run and complete backfilling with normal compaction as indicated in drawings.
4. Install manufactured duct elbows for stub-ups at poles and equipment and at building entrances through floor unless otherwise indicated. Encase elbows for stub-up ducts throughout length of elbow.
5. Install manufactured PVC coated rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through floor.

a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches (75 mm) of concrete for a minimum of 12 inches (300 mm) on each side of the coupling.
b. For stub-ups at equipment mounted on outdoor concrete bases and where conduits penetrate building foundations, extend steel conduit horizontally a minimum of 60 inches (1500 mm) from edge of foundation or equipment base. Install insulated grounding bushings on terminations at equipment.

6. Underground Warning Tape: Comply with requirements in Section 260553 "Identification for Electrical Systems."

3.4 INSTALLATION OF UNDERGROUND HANDHOLES AND BOXES

A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.

B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch (12.5-mm) sieve to No. 4 (4.75-mm) sieve and compacted to same density as adjacent undisturbed earth.

C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 1 inch (25 mm) above finished grade.

D. Install handholes and boxes with bottom below the frost line, 30” below grade.

B. Install removable hardware, including pulling eyes, cable stanchions, cable arms, and insulators, as required for installation and support of cables and conductors and as indicated. Select arm lengths to be long enough to provide spare space for future cables, but short enough to preserve adequate working clearances in the enclosure.

C. Field-cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.5 PROTECTION

A. Provide final protection and maintain conditions that ensure coatings, finishes, and cabinets are without damage or deterioration at time of Substantial Completion.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 26 05 33

SECTION 26 05 44 – SLEEVES & SLEEVE SEALS FOR ELECTRICAL RACEWAYS & CABLING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Sleeves for raceway and cable penetration of non-fire-rated construction walls and floors.
2. Sleeve-seal systems.
3. Sleeve-seal fittings.
4. Grout.
5. Silicone sealants.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 SLEEVES

A. Salient Characteristics:

1. Physical:

a. Wall Sleeves:

1) Steel Pipe Sleeves: Type E, Grade B, Schedule 40, zinc coated, plain ends.
2) Cast-Iron Pipe Sleeves: Cast or fabricated "wall pipe," equivalent to ductile-iron pressure pipe, with plain ends and integral water stop unless otherwise indicated.

b. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies: Galvanized-steel sheet; 0.0239-inch (0.6-mm) minimum thickness; round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.
c. PVC-Pipe Sleeves: Schedule 40
d. Molded-PVC Sleeves: With nailing flange for attaching to wooden forms.

e. Molded-PE or -PP Sleeves: Removable, tapered-cup shaped, and smooth outer surface with nailing flange for attaching to wooden forms.
f. Sleeves for Rectangular Openings:

1) Material: Galvanized sheet steel.
2) Minimum Metal Thickness:

a) For sleeve cross-section rectangle perimeter less than 50 inches (1270 mm) and with no side larger than 16 inches (400 mm), thickness shall be 0.052 inch (1.3 mm).
b) For sleeve cross-section rectangle perimeter 50 inches (1270 mm) or more and one or more sides larger than 16 inches (400 mm), thickness shall be 0.138 inch (3.5 mm).

2. Functional:

a. None.

3. Performance:

a. Wall Sleeves:

1) Steel Pipe Sleeves: ASTM A 53/A 53M

b. PVC-Pipe Sleeves: ASTM D 1785

2.2 SLEEVE-SEAL SYSTEMS

A. Salient Characteristics:

1. Physical:

a. Modular sealing device, designed for field assembly, to fill annular space between sleeve and raceway or cable.
b. Sealing Elements: EPDM rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
c. Pressure Plates: Carbon steel
d. Connecting Bolts and Nuts: Carbon steel, with corrosion-resistant coating, of length required to secure pressure plates to sealing elements.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 SLEEVE-SEAL FITTINGS

A. Salient Characteristics:

1. Physical:

a. Manufactured plastic, sleeve-type, waterstop assembly made for embedding in concrete slab or wall. Unit shall have plastic or rubber waterstop collar with center opening to match piping OD.

2. Functional:

a. None.

3. Performance:

a. None.

2.4 GROUT

A. Salient Characteristics:

1. Physical:

a. Nonshrink; recommended for interior and exterior sealing openings in non-fire-rated walls or floors.

2. Functional:

a. Premixed and factory packaged.

3. Performance:

a. ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.
b. Design Mix: 5000-psi (34.5-MPa), 28-day compressive strength.

2.5 SILICONE SEALANTS

A. Salient Characteristics:

1. Physical:

a. Silicone Sealants:

1) Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below.

b. Silicone Foams:

1) Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, nonshrinking foam.

2. Functional:

a. Silicone Sealants:

1) Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces that are not fire rated.

3. Performance:

a. Silicone Sealants:

1) Sealant shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION FOR NON-FIRE-RATED ELECTRICAL PENETRATIONS

A. Comply with NECA 1.

B. Comply with NEMA VE 2 for cable tray and cable penetrations.

C. Sleeves for Conduits Penetrating Above-Grade Non-Fire-Rated Concrete and Masonry-Unit Floors and Walls:

1. Interior Penetrations of Non-Fire-Rated Walls and Floors:

a. Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint. Comply with requirements in Section 079200 "Joint Sealants."
b. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall so no voids remain. Tool exposed surfaces smooth; protect material while curing.

2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
3. Size pipe sleeves to provide 1/4-inch (6.4-mm) annular clear space between sleeve and raceway or cable unless sleeve seal is to be installed or unless seismic criteria require different clearance.
4. Install sleeves for wall penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.

5. Install sleeves for floor penetrations. Extend sleeves installed in floors [2 inches (50 mm)] above finished floor level. Install sleeves during erection of floors.

D. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies:

1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.
2. Seal space outside of sleeves with approved joint compound for gypsum board assemblies.

E. Roof-Penetration Sleeves: Seal penetration of individual raceways and cables with flexible boot-type flashing units applied in coordination with roofing work.

F. Aboveground, Exterior-Wall Penetrations: Seal penetrations using steel pipe sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch (25-mm) annular clear space between pipe and sleeve for installing mechanical sleeve seals.

G. Underground, Exterior-Wall and Floor Penetrations: Install cast-iron pipe sleeves. Size sleeves to allow for 1-inch (25-mm) annular clear space between raceway or cable and sleeve for installing sleeve-seal system.

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at raceway entries into building.

B. Install type and number of sealing elements recommended by manufacturer for raceway or cable material and size. Position raceway or cable in center of sleeve. Assemble mechanical sleeve seals and install in annular space between raceway or cable and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

3.3 SLEEVE-SEAL-FITTING INSTALLATION

A. Install sleeve-seal fittings in new walls and slabs as they are constructed.

B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.

C. Secure nailing flanges to concrete forms.

D. Using grout, seal the space around outside of sleeve-seal fittings.

END OF SECTION 26 05 44

SECTION 26 05 53 – IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Identification for raceway and metal-clad cable.
2. Identification for conductors and communication and control cable.
3. Underground-line warning tape.
4. Warning labels and signs.
5. Instruction signs.
6. Equipment identification labels.
7. Miscellaneous identification products.

1.3 SUBMITTALS

A. Product Data: For each electrical identification product indicated.

B. Identification Schedule: An index of nomenclature of electrical equipment and system components used in identification signs and labels.

1.4 QUALITY ASSURANCE

A. Comply with ANSI A13.1 and ANSI C2.

B. Comply with NFPA 70.

C. Comply with 29 CFR 1910.145.

1.5 COORDINATION

A. Coordinate identification names, abbreviations, colors, and other features with requirements in the Contract Documents, Shop Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual, and with those required by codes, standards, and 29 CFR 1910.145. Use consistent designations throughout Project.

B. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

C. Coordinate installation of identifying devices with location of access panels and doors.

D. Install identifying devices before installing acoustic ceilings and similar concealment.

PART 2 - PRODUCTS

2.1 RACEWAY AND METAL-CLAD CABLE IDENTIFICATION MATERIALS

A. Salient Characteristics:

1. Physical:

a. Color for Printed Legend:

1) Power Circuits: Black letters on an orange field.
2) Legend: Indicate system or service and voltage, if applicable.

b. Self-Adhesive Vinyl Labels: Preprinted, flexible label laminated with a clear, weather- and chemical-resistant coating and matching wraparound adhesive tape for securing ends of legend label.
c. Snap-Around Labels: Slit, pretensioned, flexible, preprinted, color-coded acrylic sleeves, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action.
d. Snap-Around, Color-Coding Bands: Slit, pretensioned, flexible, solid-colored acrylic sleeves, 2 inches (50 mm) long, with diameter sized to suit diameter of raceway or cable it identifies and to stay in place by gripping action.
e. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; 2 inches (50 mm) wide; compounded for outdoor use.

2. Functional:

a. None.

3. Performance:

a. Comply with ANSI A13.1 for minimum size of letters for legend and for minimum length of color field for each raceway and cable size.

2.2 CONDUCTOR AND COMMUNICATION- AND CONTROL-CABLE IDENTIFICATION MATERIALS

A. Salient Characteristics:

1. Physical:

a. Color-Coding Conductor Tape: Colored, self-adhesive vinyl tape not less than 3 mils (0.08 mm) thick by 1 to 2 inches (25 to 50 mm) wide.
b. Marker Tapes: Vinyl or vinyl-cloth, self-adhesive wraparound type, with circuit identification legend machine printed by thermal transfer or equivalent process.

c. Aluminum Wraparound Marker Labels: Cut from 0.014-inch- (0.35-mm-) thick aluminum sheet, with stamped, embossed, or scribed legend, and fitted with tabs and matching slots for permanently securing around wire or cable jacket or around groups of conductors.
d. Metal Tags: Brass or aluminum, 2 by 2 by 0.05 inch (50 by 50 by 1.3 mm), with stamped legend, punched for use with self-locking nylon tie fastener.
e. Write-On Tags: Polyester tag, 0.010 inch (0.25 mm) thick, with corrosion-resistant grommet and polyester or nylon tie for attachment to conductor or cable.

1) Marker for Tags: Permanent, waterproof, black ink marker recommended by tag manufacturer.

2. Functional:

a. None.

3. Performance:

a. None.

2.3 UNDERGROUND-LINE WARNING TAPE

A. Salient Characteristics:

1. Physical:

a. Tape Characteristics:

1) Detectable three-layer laminate, consisting of a printed pigmented polyolefin film, a solid aluminum-foil core, and a clear protective film that allows inspection of the continuity of the conductive core, bright-colored, compounded for direct-burial service.
2) Overall Thickness: 5 mils (0.125 mm).
3) Foil Core Thickness: 0.35 mil (0.00889 mm).
4) Weight: 28 lb/1000 sq. ft. (13.7 kg/100 sq. m).
5) 3-Inch (75-mm) Tensile According to ASTM D 882: 70 lbf (311.3 N), and 4600 psi (31.7 MPa).

b. Color and Printing:

1) Inscriptions for Red-Colored Tapes: ELECTRIC LINE, HIGH VOLTAGE.
2) Inscriptions for Orange-Colored Tapes: TELEPHONE CABLE, CATV CABLE, COMMUNICATIONS CABLE, OPTICAL FIBER CABLE.
3) Inscriptions for Blue-Colored Tapes: CONTROL CIRCUIT CABLE.

2. Functional:

a. Tape:

1) Recommended by manufacturer for the method of installation and suitable to identify and locate underground electrical utility lines.

2) Printing on tape shall be permanent and shall not be damaged by burial operations.
3) Tape material and ink shall be chemically inert, and not subject to degrading when exposed to acids, alkalis, and other destructive substances commonly found in soils.

3. Performance:

a. Color and Printing:

1) Comply with ANSI Z535.1 through ANSI Z535.5.

2.4 WARNING LABELS AND SIGNS

A. Salient Characteristics:

1. Physical:

a. Self-Adhesive Warning Labels: Factory printed, multicolor, pressure-sensitive adhesive labels, configured for display on front cover, door, or other access to equipment, unless otherwise indicated.
b. Baked-Enamel Warning Signs: Preprinted aluminum signs, punched or drilled for fasteners, with colors, legend, and size required for application. 1/4-inch (6.4-mm) grommets in corners for mounting. Nominal size, 7 by 10 inches (180 by 250 mm).
c. Metal-Backed, Butyrate Warning Signs: Weather-resistant, nonfading, preprinted, cellulose-acetate butyrate signs with 0.0396-inch (1-mm) galvanized-steel backing; and with colors, legend, and size required for application. 1/4-inch (6.4-mm) grommets in corners for mounting. Nominal size, 10 by 14 inches (250 by 360 mm).

2. Functional:

a. Warning label and sign shall include, but are not limited to, the following legends:

1) Multiple Power Source Warning: "DANGER – ELECTRICAL SHOCK HAZARD – EQUIPMENT HAS MULTIPLE POWER SOURCES."
2) Workspace Clearance Warning: "WARNING – OSHA REGULATION – AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES (915 MM)."

3. Performance:

a. Comply with NFPA 70 and 29 CFR 1910.145.

2.5 INSTRUCTION SIGNS

A. Salient Characteristics:

1. Physical:

a. Engraved, laminated acrylic or melamine plastic, minimum 1/16 inch (1.6 mm) thick for signs up to 20 sq. in. (129 sq. cm) and 1/8 inch (3.2 mm) thick for larger sizes.

1) Engraved legend with black letters on white face.
2) Punched or drilled for mechanical fasteners.
3) Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2. Functional:

a. None.

3. Performance:

a. None.

2.6 EQUIPMENT IDENTIFICATION LABELS

A. Salient Characteristics:

1. Physical:

a. Adhesive Film Label: Machine printed, in black, by thermal transfer or equivalent process. Minimum letter height shall be 3/8 inch (10 mm).
b. Self-Adhesive, Engraved, Laminated Acrylic or Melamine Label: Adhesive backed, with white letters on a dark-gray background. Minimum letter height shall be 3/8 inch (10 mm).
c. Engraved, Laminated Acrylic or Melamine Label: Punched or drilled for screw mounting. White letters on a dark-gray background. Minimum letter height shall be 3/8 inch (10 mm).
d. Stenciled Legend: In nonfading, waterproof, black ink, or paint. Minimum letter height shall be 1 inch (25 mm).

2. Functional:

a. None.

3. Performance:

a. None.

2.7 MISCELLANEOUS IDENTIFICATION PRODUCTS

A. Salient Characteristics:

1. Physical:

a. Cable Ties: Fungus-inert, self-extinguishing, 1-piece, self-locking, Type 6/6 nylon cable ties.

1) Minimum Width: 3/16 inch (5 mm).
2) Tensile Strength: 50 lb (22.6 kg), minimum.
3) Temperature Range: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
4) Color: Black, except where used for color-coding.

b. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws, or stainless-steel machine screws with nuts and flat and lock washers.

2. Functional:

a. None.

3. Performance:

a. Paint: Paint materials and application requirements are specified in Division 09 painting Sections.

PART 3 - EXECUTION

3.1 APPLICATION

A. Accessible Raceways and Cables of Auxiliary Systems: Identify the following systems with color-coded, self-adhesive vinyl tape applied in bands:

1. Revise list below to suit Project.
2. Fire Alarm System: Red.
3. Fire-Suppression Supervisory and Control System: Red and yellow.
4. Combined Fire Alarm and Security System: Red and blue.
5. Security System: Blue and yellow.
6. Mechanical and Electrical Supervisory System: Green and blue.
7. Telecommunication System: Green and yellow.
8. Control Wiring: Green and red.

B. Power-Circuit Conductor Identification: For secondary conductors No. 1/0 AWG and larger in vaults, pull and junction boxes, manholes, and handholes use color-coding conductor tape. Identify source and circuit number of each set of conductors. For single conductor cables, identify phase in addition to the above.

C. Branch-Circuit Conductor Identification: Where there are conductors for more than three branch circuits in same junction or pull box, use color-coding conductor tape. Identify each ungrounded conductor according to source and circuit number.

D. Conductors to Be Extended in the Future: Attach write-on tags to conductors and list source and circuit number.

E. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, signal, sound, intercommunications, voice, and data connections.

1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.
2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
3. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and Operation and Maintenance Manual.

F. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Comply with 29 CFR 1910.145 and apply self-adhesive warning labels. Identify system voltage with black letters on an orange background. Apply to exterior of door, cover, or other access.

1. Equipment Requiring Workspace Clearance According to NFPA 70: Unless otherwise indicated, apply to door, or cover of equipment but not on flush panelboards and similar equipment in finished spaces.

G. Equipment Identification Labels: On each unit of equipment, install unique designation label that is consistent with wiring diagrams, schedules, and Operation and Maintenance Manual. Apply labels to disconnect switches and protection equipment, central or master units, control panels, control stations, terminal cabinets, and racks of each system. Systems include power, lighting, control, communication, signal, monitoring, and alarm systems unless equipment is provided with its own identification.

1. Labeling Instructions:

a. Indoor Equipment: Adhesive film label. Unless otherwise indicated, provide a single line of text with 1/2-inch- (13-mm-) high letters on 1-1/2-inch- (38-mm-) high label; where 2 lines of text are required, use labels 2 inches (50 mm) high.
b. Outdoor Equipment: Engraved, laminated acrylic or melamine label.
c. Elevated Components: Increase sizes of labels and letters to those appropriate for viewing from the floor.

2. Equipment to Be Labeled:

a. Panelboards, electrical cabinets, and enclosures.
b. Disconnect switches.
c. Enclosed circuit breakers.

H. Arch Flash Hazard warning label: Refer to Specification 26 05 73 section 3.5(B) for labeling requirements associated with the Arc Flash Hazard Analysis study.

3.2 INSTALLATION

A. Verify identity of each item before installing identification products.

B. Location: Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment.

C. Apply identification devices to surfaces that require finish after completing finish work.

D. Self-Adhesive Identification Products: Clean surfaces before application, using materials and methods recommended by manufacturer of identification device.

E. Attach nonadhesive signs and plastic labels with screws and auxiliary hardware appropriate to the location and substrate.

F. System Identification Color Banding for Raceways and Cables: Each color band shall completely encircle cable or conduit. Place adjacent bands of two-color markings in contact, side by side. Locate bands at changes in direction, at penetrations of walls and floors, at 50-foot (15-m) maximum intervals in straight runs, and at 25-foot (7.6-m) maximum intervals in congested areas.

G. Color-Coding for Phase Identification, 600 V and Less: Use the colors listed below for ungrounded branch-circuit conductors.

1. Color shall be factory applied.
2. Colors for 240/120-V Circuits:

    a. Phase A: Black.
    b. Phase B: Red.

3. Field-Applied, Color-Coding Conductor Tape: Apply in half-lapped turns for a minimum distance of 6 inches (150 mm) from terminal points and in boxes where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding. Locate bands to avoid obscuring factory cable markings.

H. Aluminum Wraparound Marker Labels and Metal Tags: Secure tight to surface of conductor or cable at a location with high visibility and accessibility.

I. Painted Identification: Prepare surface and apply paint according to Division 09 painting Sections.

END OF SECTION 26 05 53

SECTION 26 51 00 – LIGHTING FIXTURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Adjust list below to suit Project.
2. Lighting fixtures, lamps, and ballasts.
3. Emergency lighting units.
4. Exit signs.
5. Lighting fixture supports.

B. Related Sections include the following:

1. Division 26 Section "Lighting Control Devices" for automatic control of lighting, including time switches, photoelectric relays, occupancy sensors, and multipole lighting relays and contactors.

1.3 DEFINITIONS

A. CRI: Color-rendering index.

B. LER: Luminaire efficacy rating.

C. Luminaire: Complete lighting fixture, including ballast housing if provided.

1.4 SUBMITTALS

A. Product Data: For each type of lighting fixture, arranged in order of fixture designation. Include data on features, accessories, finishes, and the following:

1. Physical description of lighting fixture including dimensions.
2. Emergency lighting units including battery and charger.
3. Driver.
4. Energy-efficiency data.
5. Life, output, and energy-efficiency data for lamps.
6. Photometric data, in IESNA format, based on laboratory tests of each lighting fixture type, outfitted with lamps, ballasts, and accessories identical to those indicated for the lighting fixture as applied in this Project.

a. For indicated fixtures, photometric data shall be certified by a qualified independent testing agency. Photometric data for remaining fixtures shall be certified by the manufacturer.
b. Photometric data shall be certified by a manufacturer's laboratory with a current accreditation under the National Voluntary Laboratory Accreditation Program (NVLAP) for Energy Efficient Lighting Products.

B. Shop Drawings: Show details of nonstandard or custom lighting fixtures. Indicate dimensions, weights, methods of field assembly, components, features, and accessories.

1. Wiring Diagrams: Power wiring.

C. Qualification Data: For agencies providing photometric data for lighting fixtures.

D. Field quality-control test reports.

E. Operation and Maintenance Data: For lighting equipment and fixtures to include in emergency, operation, and maintenance manuals.

F. Warranties: Special warranties specified in this Section.

1.5 QUALITY ASSURANCE

A. Luminaire Photometric Data Testing Laboratory Qualifications: Provided by an independent agency, with the experience and capability to conduct the testing indicated, that is an NRTL as defined by OSHA in 29 CFR 1910.7.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

C. Comply with NFPA 70.

1.6 COORDINATION

A. Coordinate layout and installation of lighting fixtures and suspension system with other construction that penetrates ceilings or is supported by them, including HVAC equipment, fire-suppression system, and partition assemblies.

PART 2 - PRODUCTS

2.1 LIGHTING FIXTURES AND COMPONENTS, GENERAL REQUIREMENTS

A. Salient Characteristics:

1. Physical:

a. Metal Parts: Free of burrs and sharp corners and edges.

b. Sheet Metal Components: Steel, unless otherwise indicated. Form and support to prevent warping and sagging.
c. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position.
d. Reflecting surfaces shall have minimum reflectance as follows, unless otherwise indicated:

1) White Surfaces: 85 percent.
2) Specular Surfaces: 83 percent.
3) Diffusing Specular Surfaces: 75 percent.
4) Laminated Silver Metallized Film: 90 percent.

e. Plastic Diffusers and Covers:

1) Acrylic Lighting Diffusers: 100 percent virgin acrylic plastic. High resistance to yellowing and other changes due to aging, exposure to heat, and UV radiation.
2) Lens Thickness: At least 0.125 inch (3.175 mm) minimum unless different thickness is indicated.
3) UV stabilized.

f. Glass: Annealed crystal glass, unless otherwise indicated.

2. Functional:

a. Refer to drawing sheets.

3. Performance:

a. Refer to drawing sheets.

2.2 SOLID STATE LIGHTING FIXTURES

A. Housing, where applicable:

1. Steel bonderized or equal rust projected, or aluminum, rigid construction. Minimum gauge thickness shall be as follows:

a. Interior locations: No. 20 gauge steel, no. 16 gauge aluminum.

B. Finish:

1. Baked enamel finish (except when otherwise specified).
2. Concealed interior surfaces (this applies to interior hardware, circuit boards, etc.) matte black.
3. Concealed exterior surfaces: matte black.
4. Visible surfaces: color and texture as specified below for each fixture type or as selected.
5. Exterior fixture finish: refer to Lighting Fixture Schedule.

C. Light Emitting Diode (LED) requirements:

1. Correlated color temperature (CCT) for phosphor-coated white LEDs must have one of the following designated CCT's and fall within the following binning standards:

    a. Outdoor:

        1) 2700K defined as 2725 +/- 145K
        2) Amber

    b. Indoor:

        1) 3500K defined as 3465 +/- 245K
        2) 4000K defined as 3985 +/- 275K

2. Color spatial uniformity shall be limited to variations in chromaticity for different directions within 0.004 from the weighted average point on the CIE 1976 (u',v') diagram.
3. Color maintenance shall be limited to a maximum change in chromaticity of 0.007 on the CIE 1976 (u',v') diagram over the lifetime of the product.
4. Color rendering index.

    a. Color rendering index to be determined using ANSI C78.377-2008 and applicable IESNA standards.
    b. Laboratory tests must be produced using specific module(s)/array(s) and power supply combination that will be used in production.
    c. Manufacturers must provide a test report from a laboratory accredited by NVLAP or one of its MRA signatories.

5. Lumen depreciation.

    a. Lumen depreciation to be measured using IESNA LM-80-08 standard for IES approved method of measuring lumen maintenance of LED light sources.
    b. Phosphor coated white LED module(s)/array(s) shall deliver at least 70% of initial lumens for a minimum of 35,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.
    c. Colored LED module(s)/array(s) shall deliver at least 50% of initial lumens for a minimum of 35,000 hours when installed in-situ and operated at 100% output and the maximum specified operating temperature.

6. Acceptable LED manufacturers:

    a. CREE
    b. Nichia
    c. Osram Opto Semiconductors
    d. Philips Lumileds

D. Luminaire Efficacy:

1. Luminaire efficiency shall be measured using IESNA LM-79-08 standard for electrical and photometric measurements of solid state lighting products.

2. Manufacturer shall provide published luminaire efficacy, which is defined as luminaire light output divided by luminaire input power measured in a 25 degree Celsius environment. Efficacy shall include power supply, thermal, optical, and fixture losses. Minimum efficacy of 80 lumens per watt.

E. Thermal Management:

1. Solid state light fixture shall not exceed LED manufacturer's maximum junction temperature requirements when operated in-situ at fixture manufacturer's maximum ambient operating temperature and 100% light output.
2. Solid state light fixtures shall be thermally protected using one of more of the following thermal management techniques:

    a. Metal core board
    b. Gap pad
    c. Internal monitoring firmware

3. Solid state lighting fixture housing shall be designed to transfer heat from the LED board to the outside environment.

F. Power Supplies/Drivers:

1. Power supply shall have a power factor of 0.90 or greater for primary application.
2. Power supply input current shall have Total Harmonic Distortion (THD) of less than 20%.
3. Power supply shall have a minimum operating temperature of minus 20 degrees Celsius of below when used in luminaires intended for outdoor applications.
4. Power supply output operating frequency to be equal to or greater than 120Hz.
5. Power supply shall operate with sustained input variations of +/- 10% (voltage and frequency) with no damage to the driver.
6. Power supply shall tolerate sustained open circuit and short circuit output conditions without damage and without need for external fuses or trip devices.
7. Power supply output shall be regulated to +/- 5% across published load range.
8. Power supply shall have a Class A sound rating.
9. Power supply outputs shall have current limiting protection.
10. Power supply shall operate LEDs at constant and regulated current levels. LEDs shall not be overdriven beyond the diode manufacturer's specified nominal voltage and current.

G. Solid State Lighting Controls:

1. Control interface to dimmable power supplies shall consist of one of the following:

    a. Line Voltage Dimming. Controls to be rated for magnetic or electronic low voltage transformer operation.
    b. Low voltage (0-10V) control. Controls to be compatible with either current sink or current source operation.

2. Dimmable LED power supplies shall use pulse width modulation (PWM) to regulate power to LEDs

a. Dimmable power supplies shall have 12-bit or greater resolution to obtain flicker-free operation throughout the dimming range.

H. System Installation:

1. All hardwired connections to solid state lighting fixtures shall be reverse polarity protected and provide high voltage protection in the event connections are reversed or shorted during the installation process.
2. All solid state lighting fixtures (100% of each lot) shall undergo a minimum eight-hour burn-in test during manufacturing.
3. Solid state lighting installations shall be UL listed as a low-voltage lighting system including, but not limited to, luminaire, power supply, controller, keypad, and wiring.

I. Warranty: Luminaires, drivers, and controllers for solid state lighting systems shall be covered by a two-year warranty against defects in workmanship or material. Warranty shall include in-warranty service program providing for payment of authorized labor charges incurred in replacement of inoperative in-warranty equipment.

2.3 LIGHTING FIXTURE SUPPORT COMPONENTS

A. Comply with Division 26 Section "Hangers and Supports for Electrical Systems" for channel- and angle-iron supports and nonmetallic channel and angle supports.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Lighting fixtures: Set level, plumb, and square with ceilings and walls. Install lamps in each fixture.

B. Adjust aimable lighting fixtures to provide required light intensities.

END OF SECTION 26 51 00

SECTION 31 10 00 – SITE CLEARING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Protecting existing vegetation to remain.
2. Removing trees.
3. Conserving topsoil.
4. Clearing and grubbing.
5. Disposal of material from clearing, grubbing, thinning and demolition in approved off-site disposal areas.
6. Filling of voids and excavations resulting from the work.

B. Related Sections:

1. Section 01 35 13.22 – Archeological Protection
2. Section 01 57 13 – Temporary Storm Water Pollution Prevention
3. Section 02 41 13 – Selective Demolition
4. Section 31 20 00 – Earth Moving

1.3 MATERIAL OWNERSHIP

A. Except for materials indicated to be stockpiled or to remain on the Owner's property, cleared materials shall become Contractor's property and shall be removed and disposed of legally except agreed upon felled trees which are to remain onsite. When possible, the Contractor should re-use or sell materials generated by Site Clearing such as timber, firewood, or wood chips.

1.4 SITE CONDTIONS

A. Site conditions existing during the bidding period will be maintained by the government insofar as practical.

B. Actual site condition variations that differ from those of the bidding period and which affect site clearing operations shall be brought to the attention of the Contracting Officer (CO) prior to the commencement of any site work.

1.5 SUBMITTALS

A. The Contractor shall submit the following information to the CO for review before commencing work:

1. Permits for transport and disposal of debris.
2. Demolition procedures and operational sequence.

PART 2 - PRODUCTS

2.1 TREE PROTECTION FENCING

A. Tree protection fencing shall be an orange plastic web fence, 4 feet high minimum.

B. Posts shall be at least six (6) foot long driven a minimum of two (2) feet into the ground. Posts shall be spaced eight (8) feet on center, maximum.

PART 3 - EXECUTION

3.1 PROTECTION

A. The Contractor shall flag the limits of clearing shown on the drawings by accurate field survey with marked stakes or other means acceptable to the CO. Trees to remain shall be clearly identified during this staking process. The CO shall be notified a minimum of five (5) working days prior to scheduled commencement of clearing operations to review the flagged limits. Adjust the clearing limits as directed by the CO.

B. Install erosion-control measures prior to start of any other construction to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust to adjacent properties and roadways in accordance with Section 01 57 13 – Temporary Storm Water Pollution Prevention.

C. Before clearing begins, protect designated trees to remain with tree protection fencing to the approximate diameter of foliage (dripline of the tree) to prevent damage to the trunk, foliage and root system by construction equipment and procedures.

D. Place tree protection fencing as required to protect other plants, adjacent property areas to remain uncleared, monuments, and existing improvements from damage.

E. The Contractor shall repair or replace immediately any damage to existing trees or root systems that are to remain. The Contractor shall employ an arborist licensed in the State of Washington to determine the repair and replacement needs and methods for approval by the CO.

F. Replace damaged shrubs and other vegetation designated to remain with the same size and species.

G. The tree protection fencing shall be maintained for the duration of construction operations. The work shall include immediate replacement of any damaged fence. Fencing shall be removed from the site at the completion of construction operations. The fencing disposal shall be in accordance with local, state, and federal laws and regulations for the disposal of the material.

3.2 UTILITIES

A. A.Notify all corporations, companies, individuals, or local authorities owning or having jurisdiction over utilities running to, through, or across areas to be affected by site clearing operations.

B. B.Locate and identify existing utilities that are to remain and protect them from damage.

3.3 CONSERVING TOPSOIL

A. Remove topsoil before excavating any area. Do not strip topsoil in frozen, snow covered, or saturated conditions. Only areas with low invasive plant cover (max of 10-15% cover) and less than 2H:1V slopes will have topsoil salvaged, as determined by the CO. Remove the topsoil in one pass. Use equipment capable of excavating small, isolated pockets of topsoil. Hand salvage topsoil in areas where topsoil cannot be mechanically removed.

3.4 CLEARING AND GRUBBING

A. Clearing shall include cutting, removal, and off-site disposal of trees, bushes, shrubs, stumps, fallen timber, brush, refuse, trash, fencing and other incidental materials not required for reuse on the site.

B. The Contractor shall grub the area within the clearing limits to completely remove stumps and root systems to a depth of at least 12-inches, except for those to remain.

C. Depressions, excavations, and voids resulting from the removal of stumps or roots shall be filled with suitable material and compacted as specified under Section 31 20 00 – Earth Moving.

3.5 DISPOSAL

A. Remove all material resulting from site clearing and demolition operations from site except agreed upon felled trees to remain.

B. No burning of any material will be allowed.

END OF SECTION 31 10 00

SECTION 31 20 00 – EARTH MOVING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Preparing subgrades for walks and pavements.
2. Subbase course for walks.
3. Subbase and base course for asphalt and concrete paving.
4. Excavating and backfilling for utility trenches.
5. Outfall protection

B. Related Sections include the following:

1. Section 01 35 13.22 Archeological Protection
2. Section 01 57 23 – Temporary Strom Water Pollution Prevention
3. Section 31 10 00 – Site Clearing
4. Section 31 25 00 – Erosion and Sediment Control
5. Section 32 12 16 – Asphalt Paving
6. Section 32 13 20 – Site Concrete

1.3 DEFINITIONS

A. Backfill: Soil material or controlled low-strength material used to fill an excavation.

1. Initial Backfill: Backfill placed beside and over pipe in a trench, including haunches to support sides of pipe.
2. Final Backfill: Backfill placed over initial backfill to fill a trench.

B. Base Course: Course placed between the subgrade and hot-mix asphalt paving or cement concrete paving.

C. Bedding Material: Course placed over the excavated subgrade in a trench before laying pipe.

D. Borrow Soil: Satisfactory soil imported from off-site for use as fill or backfill.

E. Excavation: Removal of material encountered above subgrade elevations and to lines and dimensions indicated.

1. Authorized Additional Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions as directed by CO.
2. Bulk Excavation: Excavation more than 5 feet in width and more than 20 feet in length.
3. Unauthorized Excavation: Excavation below subgrade elevations or beyond indicated lines and dimensions without direction by CO. Unauthorized excavation, as well as remedial work directed by CO, shall be without additional compensation.

F. Fill: Soil materials used to raise existing grades.

G. Rock: Rock material in beds, ledges, unstratified masses, conglomerate deposits, and boulders of rock material 3/4 cu. yd. or more in volume that exceed a standard penetration resistance of 100 blows/2 inches when tested by an independent geotechnical testing agency, according to ASTM D 1586.

H. Structures: Buildings, footings, foundations, retaining walls, curbs, or other man-made stationary features constructed above or below the ground surface.

I. Subgrade: Surface or elevation remaining after completing excavation, or top surface of a fill or backfill immediately below drainage fill, or topsoil materials.

J. Utilities: On-site underground pipes, conduits, ducts, and cables.

K. Embankment: Soil or soil-rock material placed to raise the subgrade or natural grade of the site for embankment construction.

L. Engineered Fill: Clean, crushed, granular material used to create embankments that will support structures. Structural fill material will be material that is considered an all-weather fill material.

M. All Weather Fill Material: Imported material that has minimal fines, is free draining, and can be placed, shaped, and compacted using normal compaction methods during most wet weather with minimal pumping and shoving.

N. Unsuitable Material: Excavated material or material below the natural ground surface in embankment areas or below subgrade elevation in excavated areas, which is unsuitable for its planned use as determined by the CO. Unsuitable material is further defined as material determined to be:

1. Of such unstable nature as to be incapable of being compacted to specified density using ordinary methods at optimum moisture content.
2. Too wet to be properly compacted and circumstances prevent suitable in-place drying prior to incorporation into the Work.
3. Not meeting requirements specified in Part 2 of this Section.
4. Otherwise unsuitable for the planned use.

   The presence of excessive moisture in a material is not, by itself, sufficient cause for determining that the material is unsuitable.

O. Relative Compaction: The ratio, expressed as a percentage, of the in-place dry density of fill materials as compacted in the field to the maximum dry density of the same material.

P. Optimum Moisture Content: The water content at which a soil can be compacted to a maximum dry unit weight by a given compactive effort.

Q. Native Soil: Sand or silt naturally occurring on site that was never imported to the site, is weed free, and supports local native habitats.

R. Earthwork Terminology: Terms used in this Section and not defined herein shall be interpreted in accordance with the definitions given in ASTM D653.

1.4 SUBMITTALS

A. Test Reports: Submit certified test reports of all tests specified to be performed. Test reports shall be signed and sealed by a geotechnical engineer registered in the State of Iowa.

B. Delivery Tickets: Submit a delivery ticket with each load of imported embankment material delivered to the site, stating the type of fill material and the quantity.

C. Product Data: For the following:

1. All aggregates proposed for use.
2. Import fill.

D. Material Test Reports: From a qualified testing agency indicating and interpreting test results for compliance of the following with requirements indicated:

1. Classification according to ASTM D 2487 of each on-site and import soil material proposed for fill and backfill.
2. Laboratory compaction curve according to ASTM D 1557 for each on-site and import soil material proposed for fill and backfill.

E. Pre-excavation Photographs or Videotape: Show existing conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by earthwork operations. Submit before earthwork begins.

F. Contractor is to provide the CO with copies of the documentation for the legal disposal of materials offsite.

1.5 QUALITY ASSURANCE

A. Pre-excavation Conference: Conduct conference at Project site to comply with requirements in Division 01 Section “Project Management & Coordination”.

B. Testing will be performed by an independent testing laboratory selected by the Contractor. All tests will be paid for by the Contractor.

C. Allow access for CO to obtain samples and perform in-place tests. Remove surface materials as required to allow sampling and testing.

D. Tolerances:

1. Construct finished subgrades to plus 0 or minus 1/2-inch of the elevations indicated.
2. Construct finished grade of slopes not steeper than 3:1 plus or minus 1/2 inch and on slopes steeper than 3:1 plus or minus two inches.
3. Maintain the moisture content of fill material as it is being placed within plus or minus 2% of the optimum moisture content of the material as determined by the laboratory tests herein specified.

1.6 PROJECT CONDITIONS

A. Existing Utilities: Do not interrupt utilities serving facilities occupied by CO or others unless permitted in writing by CO and then only after arranging to provide temporary utility services according to requirements indicated.

1. Notify Owner not less than two days in advance of proposed utility interruptions.
2. Do not proceed with utility interruptions without CO's written permission.
3. Contact utility-locator service for area where Project is located before excavating.

B. Archeological monitoring will be required during any ground disturbance. Refer to Section 01 35 13.22 Archeological Protection.

C. Demolish and completely remove from site existing underground utilities indicated to be removed. Coordinate with utility companies to shut off services if lines are active.

D. Unfavorable Weather Conditions:

1. Excavating, filling, backfilling, and grading work shall not be performed during weather conditions that might damage or be detrimental to the condition of existing ground, in-progress work, or completed work. When the Work is interrupted by rain, freezing weather, or other conditions deemed unsuitable by the CO, excavating, filling, backfilling, and grading work shall not resume until the site and soil condition (moisture content) are suitable for compaction.
2. Subgrade shall be free from mud, snow, ice, and deleterious material when work is resumed.
3. Soil material that is too wet for compaction shall be left to drain, to be aerated and dried by disking and harrowing or other approved method until the moisture content of the area is uniform and within the specified limits.

E. Erosion Prevention: Protect stockpiles, ditches, embankments, filled, backfilled, and graded areas to prevent erosion until such time as permanent drainage and erosion control measures have been installed per the requirements of Section 01 57 23 – Temporary Storm Water Pollution Prevention.

F. Control of Dust: Employ construction methods and means that keep flying dust to the minimum throughout the entire Contract period. It may include watering and sprinkling, or other approved methods at the construction site of such frequency that will satisfactorily control the dust caused by construction operations.

G. Control of Dirt: Provide for prompt removal from existing roadways of all dirt and other materials that have been spilled, washed, tracked, or otherwise deposited thereon by construction operations whenever the accumulation is sufficient to cause the formation of mud, interfere with drainage, damage pavements, or create a traffic hazard.

H. Indicated locations of existing facilities and systems are approximate. Investigate and determine exact location and nature of utilities, facilities, and systems, and be solely responsible for damages caused by construction activities.

I. Excavation in areas that have or may have underground facilities:

1. Delineate with white paint, flags, or other methods the area to be excavated and contact Underground Location Service to request that facility owners mark the location of underground utilities.
2. Identify temporary survey markings using PINK colored markings in accordance with Underground Service alert requirements.
3. Facility owners will field mark approximate location and number of subsurface installations affected by excavation using the following colors:

    a. Safety RED – Electric power, distribution, and transmission installations, or municipal electric installations.
    b. High Visibility Safety YELLOW – Gas distribution and transmission installations, oil distribution and transmission installations containing or transporting dangerous materials, products, or steam.
    c. Safety Alert ORANGE – Telephone and telegraph installations, police and fire communication installations, and cable television installations.
    d. Safety Precaution BLUE – Water installations and slurry pipelines.
    e. Safety GREEN – Sewer installations.
    f. Do not use above listed colors for Contractor’s temporary markings suck as survey points; use U.S.A. designated PINK color.

PART 2 - PRODUCTS

2.1 FILL MATERIALS

A. Material used for fill, backfill, and embankment construction shall be an inert, nonexpansive soil, free from organic matter and other deleterious substances and of such quality that it will compact thoroughly without the presence of voids when watered and rolled.

B. Excavated on-site material shall not be considered suitable for support of paving or other exterior walks and slabs on grade.

2.2 SOIL MATERIALS

A. Embankment: All embankment material shall consist of clean well-graded soil with little or no potential for expansion.

B. Pipe Bedding Material:

1. Aggregate of the following types:

a. Sand and gravelly sand free from organic matter, loam, contaminants, and other deleterious substances.

2. Minimum sand equivalent of 30 as determined by ASTM D2419.
3. Complying with the gradation requirements shown in the following table:

Sieve Size	Percentage Passing
½"	100
4	70-100
16	50-90
50	10-40
200	0-10

C. Base Course: Shall be compacted to at least 95% of the maximum density per ASTM D 1577 and shall be within the percentage passing limits for the sieve sizes shown in the following table:

Sieve Size	Percentage Passing
1-1/2"	100
1/2"	40-80
No. 8	5-25
No. 200	0-6

D. Sand: ASTM C 33; fine aggregate, natural, or manufactured sand.

E. Outfall protection: Shall be broken limestone, dolomite, quartzite, or granite complying with the following gradation requirements:

1. Nominal 6-inch size.
2. 100% passing the 9-inch screen.
3. 100% retained on the 3-inch screen.

PART 3 - EXECUTION

3.1 PREPARATION

A. Protect structures, utilities, pavements, and other facilities from damage caused by settlement, lateral movement, undermining, washout, and other hazards created by earthwork operations.

B. Preparation of subgrade for earthwork operations including removal of vegetation, topsoil, debris, and obstructions, from ground surface is specified in Division 31 Section "Site Clearing."

C. Provide protective insulating materials to protect subgrades and foundation soils against freezing temperatures or frost.

3.2 TESTING

A. Testing will be performed by an independent testing laboratory selected by the Contractor. All tests will be paid for by the Contractor.

B. Density Tests: Compacted fill, backfill, and embankment shall be tested to verify compliance with specified requirements.

C. Fill or backfill that does not meet the specified requirements shall be removed or conditioned and recompacted until the requirements are satisfied.

D. Any completed earthwork disturbed by subsequent construction operations or adverse weather shall be corrected in a verified manner as directed by the CO.

E. Perform backfill material qualification tests and field density tests for each category of products in Part 2 of this Section. Also provide details of testing on graph format.

3.3 STAKING AND GRADES

A. Layout the work, establish all necessary markers, benchmarks, grading stakes, and other stakes as required, in accordance with the lines and grades of the Plans.

B. Perform field surveys to include the following:

1. Final surveys when excavations, backfills, and embankments are completed.

3.4 UTILITIES

A. Verify on site the location and depth (elevation) of all existing utilities and services before performing any excavation work. Use the one call utility service and other means required to verify locations.

B. Active utility lines encountered, which are not indicated in the Contract Documents, shall be reported immediately to the CO and utility owners involved. The CO and utility authorities shall be permitted free access to determine the measures deemed necessary to repair, relocate, or remove the utility.

3.5 EARTHWORK GENERAL REQUIREMENTS

A. Archeological monitoring will be required during any ground disturbance. Refer to Section 01 35 13.22 Archeological Protection.

B. Erosion Protection: Prevent erosion of the site at all times. Conform to all the requirements as per Section 01 57 23 – Temporary Storm Water Pollution Prevention.

C. Construction Traffic: Disperse travel paths of traffic and construction equipment over entire width of compacted surfaces so as to aid in obtaining uniform compaction. Protect exposed soil layers with high moisture content from excessive wheel loads.

D. On-Site Excavation:

1. Do not excavate or remove any material from the project site which is not within the designated excavation.

E. Disposal of Surplus Material:

1. Excess earth materials, unsuitable materials, and debris shall become the property of the Contractor and shall be removed from the site and disposed of in a legal manner. The Contractor is to provide the CO with copies of the documentation for the legal disposal of materials.
2. Location of disposal site and length of haul shall be the Contractor's responsibility.

F. Maintenance of Excavations, Slopes, and Embankments:

1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.

G. Dust and Dirt Control: Provide satisfactory control or remove dust and/or dirt at the project site and adjacent roadways whenever traffic or buildings that are occupied or in use, are affected.

3.6 ROUGH GRADING AND FILLING

A. Fill or backfill holes, swales, and low points that will not otherwise be removed in the course of the Work, to the indicated grades.

3.7 DEWATERING

A. Prevent surface water and ground water from entering excavations, from ponding on prepared subgrades, and from flooding Project site and surrounding area.

B. Protect subgrades from softening, undermining, washout, and damage by rain or water accumulation.

1. Reroute surface water runoff away from excavated areas. Do not allow water to accumulate in excavations. Do not use excavated trenches as temporary drainage ditches.
2. Install a dewatering system to keep subgrades dry and convey ground water away from excavations. Maintain until dewatering is no longer required

3.8 EXPLOSIVES

A. Explosives: Do not use explosives.

3.9 EXCAVATION

A. The bottoms of excavations shall be level, firm, undisturbed earth, clean and free from loose material, debris, and foreign matter.

B. Excavate to the lines and grades indicated on the Contract Drawings.

C. Support and maintain excavation so that sides are stable and will not move. Where excavations are deeper than 4 feet, the sides of the excavations shall be sloped back at 1.5H:1V or shored for safety. Unshored excavations shall not extend below a plane drawn at 1.5H:1V extending downward from any adjacent existing footings

D. Limits of excavations shall allow for adequate working space for installing forms, and as required for safety of personnel.

E. Except as otherwise indicated, preserve the material below and beyond the lines of excavations. Where excavation is carried below the indicated grade, backfill to the indicated grade as herein specified using materials specified in Section 2.2.

F. Excavation and its restoration, for convenience of the Contractor, shall be at no additional expense to the Governement.

G. Place excavated material at sufficient distance from edge of excavation so as not to cause cave-ins or bank slides, but in no case closer than three feet from the edge of excavations.

H. Existing fill soils shall not be considered suitable for fill. Existing fill soils shall be removed and replaced as properly compacted fill.

3.10 EMBANKMENT CONSTRUCTION

A. Fill shall be placed in level lifts not exceeding 8 inches in loose thickness and compacted. The fill shall be compacted to at least 90 percent of the maximum density in accordance with ASTM D 698, except the upper 12 inches. The upper 12 inches of exposed base course and/or subgrade shall be compacted to at least 95% of the maximum density in accordance with ASTM D 698.

B. Each lift shall be brought to within 2 percent of optimum moisture content.

C. Finished fill slopes shall be constructed at an inclination no steeper than 2:1. Fill slopes shall be overbuilt and trimmed back as necessary to expose a well-compacted surface.

D. Remove material that cannot be compacted to the required density within specified tolerances and replace with suitable material at no additional expense to the Governement.

3.11 SUBGRADE PREPARTION

A. Notify CO when excavations have reached required subgrade.

B. The subgrade surface shall be free of any water, debris, organic, loose, soft, disturbed, or other deleterious material prior to placement of fill, reinforcing bars, and concrete. If CO determines that unsatisfactory soil is present, continue excavation and replace with compacted backfill or fill material as directed.

C. Perform all cutting, blading, and shaping as required to cut and shape the subgrade to the grades and elevations indicated.

D. Subgrade preparation includes fine grading, reworking as necessary, and preparation of cut, fill, or embankment upon which structure and equipment foundations, pipe, base, and pavement will be placed. Subgrade preparation shall be performed immediately prior to placement of the base course.

E. Maintain proper drainage of the paved areas throughout construction.

F. Subgrade, including trench backfills, shall be compacted to at least 95% of the maximum density obtained by ASTM D 1557 method of compaction.

G. Reconstruct subgrades damaged by freezing temperatures, frost, rain, accumulated water, or construction activities, as directed by CO, without additional compensation.

H. The upper 12 inches of soils underlying pavement section shall be compacted to at least 95 percent of the maximum dry density based on the ASTM D-1557 method of compaction. If soft areas are identified during proof-rolling, remove and replace or scarify existing material.

3.12 STORAGE OF SOIL MATERIALS

A. Stockpile borrow soil materials and excavated satisfactory soil materials without intermixing. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.

1. Stockpile soil materials away from edge of excavations. Do not store within drip line of remaining trees.
2. Temporary stockpiles shall be no higher than 3 feet.

3.13 BACKFILL

A. Place and compact backfill in excavations promptly, but not before completing the following:

1. Surveying locations of underground utilities for Record Documents.
2. Testing and inspecting underground utilities.
3. Removing concrete formwork.
4. Removing trash and debris.

B. Place backfill on subgrades free of mud, frost, snow, or ice.

C. Place backfill in horizontal layers not to exceed eight inches of loose material, and compact each layer to specified density before the next layer is placed.

D. Do not backfill on or against structural concrete until the specified 28-day concrete strength has been attained.

E. Do not use compaction equipment and methods that produce excessive horizontal or vertical earth pressures.

3.14 UTILITY TRENCH BACKFILL

A. Place backfill on subgrades free of mud, frost, snow, or ice.

B. Trenches shall be graded uniformly sloped between pipe elevations as shown on the drawings.

C. Place and compact bedding course on trench bottoms and where indicated. A 6-inch layer of bedding material shall be placed beneath utilities. Shape bedding course to provide continuous support for bells, joints, and barrels of pipes and for joints, fittings, and bodies of conduits.

D. Place and compact initial backfill of pipe bedding material to a height of 12 inches over the utility pipe or conduit. Vibrate pipe bedding materials using vibratory equipment.

1. Carefully compact initial backfill under pipe haunches and compact evenly up on both sides and along the full length of utility piping or conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.

E. Backfill voids with satisfactory soil while installing and removing shoring and bracing.

F. Backfill and compact over excavations to ninety-five percent relative compaction with bedding material. There shall be no additional payment to the Contractor for over excavations not directed by the CO. Remove unsatisfactory material encountered below the grades shown as directed by the CO and replace with bedding material. Payment for removal and replacements of such unsatisfactory material directed by the CO shall be made in accordance with the provisions of the General Conditions.

G. Install warning tape directly above utilities, 12 inches below finished grade, except 6 inches below subgrade under pavements and slabs.

3.15 SOIL MOISTURE CONTROL

A. Uniformly moisten or aerate subgrade and each subsequent fill or backfill soil layer before compaction to within 3 percent of optimum moisture content.

1. Do not place backfill or fill soil material on surfaces that are muddy, frozen, or contain frost or ice.
2. Remove and replace, or scarify and air dry otherwise satisfactory soil material that exceeds optimum moisture content by 3 percent and is too wet to compact to specified dry unit weight.

3.16 COMPACTION OF SOIL BACKFILLS AND FILLS

A. Place backfill and fill soil materials in layers not more than 8 inches in loose depth for material compacted by heavy compaction equipment, and not more than 4 inches in loose depth for material compacted by hand-operated tampers.

B. Place backfill and fill soil materials evenly on all sides of structures to required elevations, and uniformly along the full length of each structure.

C. Fill compaction shall be verified by means of in-place density tests performed per ASTM D6938 (or appropriate alternative when ASTM D6938 is not suitable for the fill material) during fill placement so that compaction may be evaluated as earthwork progresses.

D. Compact soil materials to not less than the following percentages of maximum dry unit weight according to ASTM D 1557:

Fill Application	Minimum Compaction %
Retaining wall footing subgrade	95
Concrete slab subgrades	95
Asphalt pavement base and subbase	95
Asphalt pavement subgrade, within 3 feet of finished grade	95
Asphalt pavement subgrade, more than 3 feet below finished grade	90

3.17 GRADING

A. General: Uniformly grade areas to a smooth surface, free of irregular surface changes. Comply with compaction requirements and grade to cross sections, lines, and elevations indicated.

1. Provide a smooth transition between adjacent existing grades and new grades.
2. Cut out soft spots, fill low spots, and trim high spots to comply with required surface tolerances.

B. Site Grading: Slope grades to direct water away from buildings and to prevent ponding. Finish subgrades to required elevations within the following tolerances:

1. Unpaved Areas: Plus or minus 1 inch.
2. Walks: Plus or minus 1 inch.
3. Pavements: Plus or minus 1/2 inch.

3.18 BASE COURSES

A. Place base course on subgrades free of mud, frost, snow, or ice.

B. On prepared subgrade, place base course under pavements and walks as follows:

1. Place base course material over subbase course under hot-mix asphalt pavement.
2. Shape subbase and base course to required crown elevations and cross-slope grades.
3. Place base course 6 inches or less in compacted thickness in a single layer.
4. Place base course that exceeds 6 inches in compacted thickness in layers of equal thickness, with no compacted layer more than 6 inches thick or less than 3 inches thick.
5. Compact base course at optimum moisture content to required grades, lines, cross sections, and thickness to not less than 95 percent of maximum dry unit weight according to ASTM D 1557.

3.19 SHORING AND SHEETING

A. Construct and maintain all shoring, sheeting, and slope lay-back necessary to protect the excavation, as need for safety of the employees and as required by applicable State and Federal laws.

B. Where excavations are deeper than 4 feet, the sides of the excavations shall be sloped back at 1.5H:1V or shored for safety. When work area is left open and unattended by Contractor, provide suitable barricades for public safety, regardless of trench depth. Provide suitable exit means in accordance with applicable provisions in OSHA.

C. Do not remove timber or sheeting if it is in a compacted zone. Trim timber or sheeting off at a safe level above that zone.

D. As directed, remove all other sheeting and shoring when no longer needed.

3.20 FIELD QUALITY CONTROL

A. Testing Agency: Contractor shall engage a qualified independent geotechnical engineering testing agency to perform field quality-control testing.

B. Allow testing agency to inspect and test subgrades and each fill or backfill layer. Proceed with subsequent earthwork only after test results for previously completed work comply with requirements.

C. Testing agency will test compaction of soils in place according to ASTM D 1556, ASTM D 2167, ASTM D 2922, and ASTM D 2937, as applicable. Tests will be performed at the following locations and frequencies:

1. Paved Areas: At subgrade and at each compacted fill and backfill layer, at least 1 test for every 2000 sq. ft. or less of paved area or building slab, but in no case fewer than 3 tests.
2. Trench Backfill: At each compacted layer, 1 test for each 150 feet or less of trench length, but no fewer than 2 tests.

D. When testing agency reports that subgrades, fills, or backfills have not achieved degree of compaction specified, scarify, and moisten or aerate, or remove and replace soil to depth required; recompact and retest until specified compaction is obtained.

3.21 PROTECTION

A. Protecting Graded Areas: Protect newly graded areas from traffic, freezing, and erosion. Keep free of trash and debris.

B. Repair and reestablish grades to specified tolerances where completed or partially completed surfaces become eroded, rutted, settled, or where they lose compaction due to subsequent construction operations or weather conditions.

1. Scarify or remove and replace soil material to depth as directed by CO; reshape and recompact.

C. Where settling occurs before Project correction period elapses, remove finished surfacing, backfill with additional soil material, compact, and reconstruct surfacing.

1. Restore appearance, quality, and condition of finished surfacing to match adjacent work, and eliminate evidence of restoration to greatest extent possible.

3.22 DISPOSAL OF SURPLUS AND WASTE MATERIALS

A. Disposal: Remove surplus satisfactory soil and waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off Government's property.

B. Disposal: Transport surplus satisfactory soil to designated storage areas on Government's property. Stockpile or spread soil as directed by CO.

1. Remove waste material, including unsatisfactory soil, trash, and debris, and legally dispose of it off Government's property.

END OF SECTION 31 20 00

SECTION 31 23 16 – TOPSOIL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This section includes the following: furnishing, conserving, stockpiling, and placing topsoil on a previously prepared subgrade. Contractor shall use topsoil excavated within the project area prior to importing additional topsoil.

B. This Section includes the following:

1. Section 01 57 23 – Temporary Stormwater Pollution Prevention
2. Section 31 10 00 – Site Clearing
3. Section 31 20 00 – Earth Moving

1.3 QUALITY ASSURANCE

A. Soil-Testing Laboratory Qualifications: An independent or university laboratory, recognized by the State Department of Agriculture, with the experience and capability to conduct the testing indicated and that specializes in types of tests to be performed.

B. Submit soil analysis report for imported topsoil from the State University Agricultural Extension Service or other approved soil testing laboratory. Report shall cover soil textural classification (percentages of sand, silt, and clay), pH, % organic matter, and soluble salts (electric conductivity in millimoles/centimeter), and shall include additive recommendations. Testing will be at the expense of the Contractor.

1.4 DELIVERY, STORAGE AND HANDLING

A. Do not deliver or place topsoil in frozen, wet, or muddy condition.

B. The Contractor shall submit to the CO for approval a complete list of all materials to be used prior to delivery of any materials to the site. Include complete data sources, amount, and quantity. This submittal shall in no way be construed at permitting substitution for specific items described on the plants or in the specification unless approved in writing by the CO. The CO or COR reserves the right to reject materials that are not weed free.

C. Follow all specifications listed in Section 31 10 00 Site Clearing for conserving topsoil. Cover conserved topsoil stockpiles with an approved breathable water repellent fabric. Anchor the fabric around the perimeter to hold it in place.

D. Do not conserve topsoil that is contaminated with seeds from non-native or invasive species.

1.5 SUBMITTALS

A. Sources: Submit name and address of sources for imported soil materials. Sources must be on NPS approved list of Quarries, Borrow Pits and Other Material Sources Outside of NPS Boundaries.

B. Material Test Reports: For imported or manufactured topsoil.

C. Imported Topsoil Sample: 1-gallon volume in sealed plastic bags labeled with source. Each Sample shall be typical of the lot of material to be furnished; provide an accurate representation of color, texture, and organic makeup.

D. Delivery Tickets: Submit delivery tickets to CO on a daily basis upon delivery of these materials:

1. Imported Topsoil

PART 2 - PRODUCTS

2.1 IMPORTED TOPSOIL

A. Topsoil will include natural humus bearing soils or approved soils. If there is a shortage of topsoil remaining from site clearing, notify the CO, who will consult with NPS representative for an acceptable source of additional material located outside the park. If NPS approved borrow pits do not have suitable topsoil, NPS will consider contractor provided options, but will need to inspect the quarry for noxious weeds. Provide notice of at least 1 month before topsoil needs to be applied.

B. All imported topsoil shall be a loam or sandy loam. Topsoil shall be furnished by the Contractor and shall be a natural, friable soil representative of productive soils in the vicinity. It shall be obtained from the top 12″ of well drained areas. Imported topsoil or other imported soil must be inspected by NPS Vegetation Program staff for approval on texture, biologic activity, and absence of exotic plant species prior to delivery.

1. Fertile, friable, loamy soil, reasonably free from subsoil, refuse, roots, heavy or stiff clay, stones larger than 1-inch, coarse sand, noxious seeds, sticks, brush, litter, and other deleterious substances; suitable for the germination of seeds and the support of vegetative growth. The pH value shall be between 7.0 and 8.0.
2. Soil Texture: Soil can be loamy sand, sandy loam, sandy clay loam, or loam. Sand 30-50%, Silt 0-50%, Clay 5-30%.
3. Additives: none
4. % Organic Content: 2.0% minimum.
5. Soluble Salts: Electric conductivity shall be less than 2.0 mmhos/cm

PART 3 - EXECUTION

3.1 EXAMINATION:

A. General: Verify that existing site conditions are as specified and indicated before beginning work under this Section.

1. Grades: Inspect to verify rough grading is within +/- 0.1 foot of grades indicated and specified.
2. Damaged Earth: Inspect to verify that earth rendered unfit to receive planting do to concrete, water, mortar, limewater, or any other contaminant dumped on it has been removed and replaced with clean earth from a source approved by the CO.

B. Surface Grade: Remove weeds, debris, clods, and rocks larger than 8″. Dispose of accumulated debris at direction of the CO.

C. Runoff: Take measures and furnish equipment and labor necessary to control the flow, drainage, and accumulation of water. Ensure that all water will run off grades.

3.2 PREPARATION

A. Remove vegetation prior to stripping of topsoil. Break down grasses, shrubs, sagebrush, and other minor brush less than 3-feet in height and woody plants and trees less than 2-inches in diameter by chipping, mowing, or shredding. Incorporate this material with the conserved topsoil. Ensure material is reduced to less than 2 inch in any dimension prior to incorporation. Remove and dispose off-site, all brush and trees over 3 feet tall or with a diameter over 2 inches.

B. Limits of topsoil stripping shall be verified with the CO prior to removal, and stockpile locations shall be identified so as not to interfere with the other work.

C. Conserved Topsoil: Topsoil refers to the uppermost soil horizon including the top 6 inches which includes organic duff and other materials capable of supporting vegetation.

3.3 PLACING TOPSOIL

A. Scarify compacted subgrade in planting areas and any area to be restored to a minimum depth of 2-inches to bond 1-inch of topsoil minimum after settlement to subsoil. Topsoil shall be free from weeds, sod, clods, and stones larger than 8″, toxic substances, litter, or other deleterious material. Spread evenly and grade to elevations and slopes shown. Hand rake areas inaccessible to machine grading.

3.4 STORING

A. The topsoil, after being stripped, shall be immediately stockpiled for reuse in staging area determined by the CO.

B. Store topsoil in windrow or pile formation with a maximum height of 3 feet protected from wind and water erosion, compaction, and contaminants unless otherwise approved by the CO. Separate topsoil from other excavated material. Do not mix topsoil with subsoil. Stockpiles shall be stabilized with excelsior logs/silt fence or natural fiber (excluding straw) blankets if determined necessary by the CO. Flag or install barriers to keep heavy equipment/vehicles from driving on stockpiled topsoil. Coordinate at least 2 weeks prior to topsoil salvage with CO and NPS to ensure their on-site presence at the initiation of salvage activities. Stockpiles shall be watered down periodically by the Contractor as necessary to mitigate the formation of windblown dust.

C. Stockpile topsoil and surface soil away from edge of excavations without intermixing it with subsoil. Grade and shape stockpiles to drain surface water. Cover to prevent windblown dust.

1. Limit height of topsoil stockpiles to 36″ unless authorized by CO.
2. Do not stockpile topsoil within drip line of remaining trees
3. Stockpile surplus topsoil to allow for respreading a thicker layer of topsoil

3.5 WEED CONTROL

A. All stockpiles shall remain weed free.

3.6 DISPOSAL

A. Move surplus soil to a location within the park as directed by the CO.

B. Remove all other waste material including excess subsoil, unsuitable soil, trash, and debris and legally dispose of them off CO’s property.

END OF SECTION 31 23 16

SECTION 31 25 00 – EROSION & SEDIMENT CONTROL

PART 1 - PART GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes:

1. Perimeter sediment control.
2. Site erosion control.

B. Related Sections:

1. Section 01 57 23 – Temporary Storm Water Pollution Prevention
2. Section 31 10 00 – Site Clearing
3. Section 31 20 00 – Earth Moving

1.3 SUBMITTALS

A. Product data for the following:

1. Wattles, including weed-free certification for all materials.
2. Erosion control blanket, including weed free certification for all materials.

PART 2 - PRODUCTS

2.1 GENERAL

A. To the extent feasible, use wildlife safe materials for erosion control products.

2.2 WATTLES

A. Provide certified weed free logs that consist of drainage filter made of curled aspen wood excelsior and rolled into a cylindrical shape with a consistent width of fibers evenly distributed throughout the cylinder.

B. Wattles shall be a minimum of 9 inches and maximum 12 inches in diameter.

C. Wattles shall be encased in 100% natural fiber biodegradable netting. Photodegradable materials shall not be permitted.

D. Stakes for wattles shall be made of wood from untreated Douglas fir, hemlock, or pine species and shall be 24-inches long.

2.3 EROSION CONTROL BLANKETS

A. Erosion control blankets shall be composed of certified weed free processed all-natural fibers (straw is not acceptable) mechanically bound between two all-natural fiber nettings to form a continuous matrix with a minimum dry weight of 0.4 pounds per square yard according to ASTM 6475.

B. Netting shall be a loose weave, non-welded, with movable, floating, jointed netting.

C. U-shaped wire staples shall be a minimum of 12-inches long and 8-gauge thickness to facilitate installation without bending.

D. Ensure materials are protected from weather and/or handling damage.

PART 3 - EXECUTION

3.1 WATTLES

A. Wattles shall be installed in accordance with the manufacturer's recommendations as soon as construction will allow or when designated by the CO.

B. Place sediment logs in a shallow trench, the depth of which shall be 1/3 the diameter of the log. Secure the log every 2 linear feet across the length of the log with a 24-inch wooden 1-inch x 1-inch (or larger stakes) driven a minimum of 14 inches from backfilled grade or 10 inches from the bottom of the wattle into the ground. Do not use wooden stakes that are severely split.

C. When placing more than one log in a continuous section the logs should overlap 2'. Install perpendicular to the contour, in a staggered overlapping pattern so that an unnatural pattern cannot be detected.

D. The Contractor shall exercise care when installing wattles to ensure the method of installation minimizes the disturbance of waterways and prevents sediment or pollution discharge into water bodies.

END OF SECTION 31 25 00

SECTION 32 12 16 – ASPHALT PAVING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes:

1. Hot mix asphalt (HMA) for parking areas, drive aisles, and miscellaneous paving.

B. Related Sections:

1. Section 31 20 00 – Earth Moving
2. Section 32 13 20 – Site Concrete
3. Section 32 17 00 – Paving Specialties

1.3 SUBMITTALS

A. Product Data for the following:

1. Asphalt, aggregate, asphalt binder, asphalt mix design.

B. Asphalt mix design submittal shall be submitted in writing by the Contractor at least 30 days prior to the start of paving operations and shall include as a minimum:

1. Percent passing each sieve size.
2. Percent of asphalt binder.
3. Performance graded asphalt binder certified test results and Material Certificate certifying the PG grade, including source and location.
4. Mixing temperature.
5. Compaction temperature.
6. Percent anti-stripping additive.

C. Results from tests described in Section 3.6.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Job mix formula shall conform to Iowa Department of Transportation Specifications, latest edition, and the requirements described below.

B. HMA shall be asphalt binder grade 58-34S.

C. Aggregate shall be crushed stone, gravel, slag, sand, and filler from an approved source, and shall meet the requirements below:

Combined Aggregate Gradation

Standard Sieve Size	**Percent Passing**
3/4”	100
1/2"	90-100
3/8”	90 max.
No. 8	28-58
No. 200	2-10

D. Prime coat will not be required.

E. Basic asphalt binder content shall be 5.5% for Aggregate Type A, ¾-inch.

F. Tack coat may be SS-1, SS-1H, CSS-1, CSS-1H, CQS-1, or CQS-1H.

PART 3 - EXECUTION

3.1 GENERAL

A. Provide alternate routes around closed or obstructed traffic ways.

B. Where HMA is placed in areas not retained by curb faces, walls, and other vertical surfaces, the closed edges of the asphalt concrete shall be compacted to meet the same density requirements as the other asphalt surfaces.

C. Where a new asphalt surface joins an existing asphalt surface, tack coat shall be applied to the existing surface at the rate of 0.1 gallons per square yard. Fine building sand shall be swept into the joint to function as filler. Joints shall be trimmed to form near vertical surfaces and neat lines prior to asphalt placement.

D. HMA for roadways shall be placed in two lifts.

3.2 PREPARATION

A. Soil subgrades should be proof-rolled with a loaded dump truck or heavy compactor to verify density. Any areas of yielding subgrade disclosed during this proof-rolling operation should be over-excavated and replaced with a suitable structural fill material.

B. The upper 12 inches of soils underlying pavement section shall be compacted to at least 95 percent of the maximum dry density based on the ASTM D-1557 method of compaction.

C. The subgrade and base course material shall be compacted to at least 95 percent of the Modified Proctor maximum dry density (ASTM D-1557).

D. Contractor shall be responsible for protection of the subgrade. If rutting or any other damage occurs to the subgrade or subbase as a result of hauling operations, the subgrade and subbase shall be repaired immediately.

3.3 COMPACTING

A. Thoroughly and uniformly compact the asphalt surface by rolling. In places inaccessible to rollers, use alternate equipment approved by the CO. Do not cause cracking, shoving or undue displacement.

B. Monitor the compaction process with nuclear density gauges calibrate according to the ASTM D2950 calibration section within 6 months before use. Check the standard and reference on each day of use according to the ASTM D2950 standardization and reference check sections. Compact to at least 91.0 percent of the maximum specific gravity (density) determined in 2.1.A/B.

C. Continue compaction until surface marks are eliminated and cracks are sealed.

3.4 TACK COAT

A. Equipment

1. Asphalt distributor shall have the following:

    a. Heater for uniformly heating the asphalt;
    b. Full circulation spray bar adjustable to 15-foot width;
    c. Positive controls including tachometer, pressure gauge, volume measuring device, or calibrated tank to uniformly deposit asphalt over the full width within 0.02 gallons per square yard of the required rate; and
    d. Thermometer for measuring asphalt temperature in the tank.

B. Surface Preparation

1. Clean the existing surface of loose material, dirt, or other deleterious material by approved methods. When the surface is concrete, remove excess joint and crack filler.

C. Weather Limitations

1. Apply asphalt tack only when the following apply:

a. Surface is dry and unfrozen;
b. Ambient air temperature is above 35 °F and rising; and
c. Weather is not foggy and rainy.

D. Tack Coat Application

1. Apply tack coats when the entire surface area on which the coat is to be applied is free of moisture.
2. Place a tack coat to form a continuous, uniform film on the area to be covered. Tack coat may be diluted with water at a 1:1 ratio to improve application.
3. Tack coat shall be spread at an undiluted rate of 0.03-0.05 gallons per square yard.
4. Tack the vertical face of exposed, longitudinal joints as a separate operation at a rate of 0.10-0.15 gallons per square yard. Tack before the adjoining lift is placed. Lightly paint or spray vertical surfaces of all fixtures, curbs, or cold mixture with which the hot mixture will come in contact to facilitate a tight joint with the fresh mixture.
5. Allow tack coat to adequately cure prior to placement of HMA. If tack coat surface becomes dirty from weather or traffic, thoroughly clean and, if necessary, retack.

3.5 JOINTS, TRIMMING EDGES, AND CLEANUP

A. Make transverse joints vertical to the depth of the new pavement at connections to existing pavements and previously placed lifts. Form transverse joints by cutting back the previous run to expose the full-depth course.

B. Apply an asphalt tack coat to the edge of both transverse and longitudinal joints according to 3.4.

C. Place the asphalt concrete mix as continuously as possible. Do not pass rollers over an unprotected edge of freshly laid asphalt concrete mix.

D. Dispose of material trimmed from the edges and other discarded asphalt concrete mix in a manner consistent with state and local regulations. Asphalt material may be considered hazardous waste. Submit copies of the disposal permits.

3.6 FIELD QUALITY CONTROL AND TESTING

A. Placement temperature

1. Sample temperature of first load and as determined by the CO thereafter.
2. Sampling shall occur in hauling vehicle before dumping.

B. Density Testing

1. Take nuclear gauge density readings.

a. Density testing shall be in accordance with AASHTO T 166.

b. Sampling frequency shall be one per 100 tons.
c. Sample in-place after compacting.

C. Monitor the compaction process with nuclear density gauges and compact according to:

1. The lower specification limit is 91.0 percent of the maximum specific gravity (density) determined according to AASHTO T 166 and AASHTO T 209.
2. The percent compaction will be determined using the average maximum specific gravity (AASHTO T 209) from all samples tested each day.

3.7 QUALITY CONTROL / QUALITY ASSURANCE

A. The contractor is responsible for performing Quality Control monitoring and/or testing. Take samples and perform tests as described in Section 3.6.

END OF SECTION 32 12 16

SECTION 32 13 20 – SITE CONCRETE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes cast-in-place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and special concrete flatwork finish, for the following:

1. Concrete sidewalk.
2. Concrete parking areas.
3. Concrete curb and gutter.
4. Concrete valley gutter.

B. Related Sections:

1. Section 31 20 00 – Earth Moving.

1.3 DEFINITIONS

A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume; subject to compliance with requirements.

B. Broom Finish Concrete Paving: Where shown on the drawings, using a medium bristle concrete finishing brush perpendicular to the direction of travel.

1.4 SUBMITTALS

A. Product Data: For each type of product.

B. Design Mixtures: For each concrete mixture. Submit design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.

C. Steel Reinforcement Shop Drawings: Placing drawings that detail fabrication, bending, and placement. Include bar sizes, lengths, material, grade, bar schedules, stirrup spacing, bent bar diagrams, bar arrangement, splices and laps, tie spacing, hoop spacing, and supports for concrete reinforcement.

D. Qualification Data: For manufacturer testing agency.

E. Material Certificates: For each of the following, signed by manufacturers:

1. Cementitious materials.
2. Admixtures.
3. Form materials and form-release agents.
4. Curing compounds.
5. Adhesives.
6. Vapor retarders.
7. Joint-filler strips.
8. Repair materials.

F. Joint Plan

G. Material Test Reports: For the following, from a qualified testing agency, indicating compliance with requirements:

1. Aggregates.

H. Field quality-control reports, for accessibility standards.

I. Mockup Panel: A 3-foot by 3-foot square mock-up of the sand finish concrete and color shall be poured and finished at the project site for review and approval prior to proceeding with any sand finish concrete work.

1.5 QUALITY ASSURANCE

A. Testing will be performed by an independent testing laboratory selected by the Contractor. All tests will be paid for by the Contractor.

B. Installer Qualifications: A qualified installer who employs on Project personnel qualified as ACI-certified Flatwork Technician and Finisher and a supervisor who is an ACI-certified Concrete Flatwork Technician.

C. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.

1. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."

D. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified according to ASTM C 1077 and ASTM E 329 for testing indicated.

1. Personnel conducting field tests shall be qualified as ACI Concrete Field-Testing Technician, Grade 1, according to ACI CP-1 or an equivalent certification program.
2. Personnel performing laboratory tests shall be ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician – Grade I. Testing Agency laboratory supervisor shall be an ACI-certified Concrete Laboratory Testing Technician – Grade II.

E. Source Limitations: Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, obtain aggregate from single source, and obtain admixtures from single source from single manufacturer.

F. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:

1. ACI 301, "Specifications for Structural Concrete," Sections 1 through 5.
2. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

G. Concrete Testing Service: Engage a qualified independent testing agency to perform material evaluation tests and to design concrete mixtures.

PART 2 - PRODUCTS

2.1 FORM-FACING MATERIALS

A. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.

B. Chamfer Strips: Wood, metal, PVC, or rubber strips, 3/4 by 3/4 inch, minimum.

C. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces. The form-release agent shall be natural, organic, VOC compliant, and biodegradable.

1. Formulate form-release agent with rust inhibitor for steel form-facing materials.

D. Form Ties: Factory-fabricated, removable, or snap-off metal or glass-fiber-reinforced plastic form ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.

1. Furnish units that will leave no corrodible metal closer than 1 inch to the plane of exposed concrete surface.
2. Furnish ties that, when removed, will leave holes no larger than 1 inch in diameter in concrete surface.

2.2 CONCRETE MATERIALS

A. Cementitious Material

1. Cement shall be from a preapproved source listed on the Iowa Department of Transportation Approved Product List. Use cementitious materials, of the same type, brand, and source, throughout the project.

B. Silica Fume: ASTM C 1240, amorphous silica.

C. Aggregates:

1. Aggregates shall consist of 50% fine aggregate and 50% coarse aggregate.
2. Fine aggregates shall meet the requirements in the tables below:

Quality

Fine Aggregate Quality	Test Limits	Test Method
Shale and Coal	2.0% Maximum	Materials I.M. 344

Gradation

Standard Sieve Size	Percent Passing
3/8”	100
No. 4	90-100
No. 8	70-100
No. 30	10-60
No. 200	0-1.5

When the fine aggregate is sieved through the following numbered sieves – 4, 8, 16, 30, 40, and 100 – no more than 40% shall pass one sieve and be retained on the sieve with the next higher number.

3. Coarse aggregates shall be Class 2 Durability and shall meet the gradation requirements in the table below:

Standard Sieve Size	Percent Passing
1”	100
¾”	90-100
3/8”	20-55
No. 4	0-10
No. 8	0-5
No. 200	0-1.5

4. Coarse aggregates shall meet the quality requirements described below:

Aggregate Quality

Aggregate Quality	Maximum % Allowed	Test Method
Abrasion (Cr. Stone)	50	AASHTO T 96
Abrasion (Gravel)	35 (may be increased by 0.1% for each 1% of particles with at least one fractured face)	AASHTO T 96
Alumina	0.5	Iowa DOT Materials Laboratory Test Method No. 222
A Freeze	6	Iowa DOT Materials Laboratory Test Method No. 221, Method A
Clay Lumps and Friable Particles	0.5	Materials I.M. 368

If the alumina value passes, the A Freeze is not needed. Alumina does not apply to gravel.

Maximum Permissible Amounts of Objectionable Materials

Objectionable Materials	Maximum % Allowed	Test Method
Coal and carbonaceous shale	0.5	Materials I.M. 372
Total of all shale, similar objectionable materials, and coal combined	1.0	Materials I.M. 372
Organic materials, except coal	0.01	Iowa DOT Materials Laboratory Test Method No. 215
Unsound chert particles retained on 3/8" sieve (Nonstructural concrete)	3.0	Materials I.M. 372
Unsound chert particles retained on 3/8" sieve (Structural concrete)	2.0	Materials I.M. 372

D. Water: ASTM C 94/C 94M and potable.

E. Color: Concrete color to closely match existing concrete as approved in mockups and submittals.

2.3 ADMIXTURES

A. Air-Entraining Admixtures must comply with ASTM C260.

B. Chemical Admixtures must comply with ASTM C494.

2.4 CURING MATERIALS

A. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.

B. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

C. Water: Potable.

D. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C309, Type 1 or 2, Class A or B.

2.5 RELATED MATERIALS

A. Bonding Agent: ASTM C 1059/C 1059M, Type II, non-redispersible, acrylic emulsion or styrene butadiene.

B. Epoxy Bonding Adhesive: ASTM C 881, two-component epoxy resin, capable of humid curing and bonding to damp surfaces, of class suitable for application temperature and of grade to suit requirements, and as follows:

1. Types I and II, non-load bearing, for bonding hardened or freshly mixed concrete to hardened concrete.

C. Joint Sealants: Self-leveling, exterior-rated polyurethane as specified in Section 07 92 00 “Joint Sealants.”

1. Color: Match each concrete color and as approved in mockups and submittals.

D. Color Treatment: Provide color additives that, along with specified concrete materials, result in concrete to match existing sidewalks in the project area.

2.6 STEEL REINFORCEMENT

A. Galvanized Reinforcing Bars: ASTM A 615/A 615M, Grade 60 deformed bars, ASTM A 767/A 767M, Class I zinc coated after fabrication and bending.

2.7 CONCRETE MIXTURES, GENERAL

A. Prepare design mixtures for each type and strength of concrete, proportioned based on laboratory trial mixture or field test data, or both, according to ACI 301.

1. Use a qualified independent testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.
2. Provide concrete with an ultimate strength of 4000 psi.
3. Air entrainment shall be between 6.0% min. and 10.0% max.
4. Slump between ½-inch and 4-inches.
5. Use a locally-obtained sand to match the geology of the park location.

B. Limit water-soluble, chloride-ion content in hardened concrete to 0.15 percent by weight of cement.

C. Admixtures: Use admixtures according to manufacturer’s written instructions.

1. Use admixture in concrete, as required, for placement and workability.
2. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
3. Use water-reducing admixture in pumped concrete, concrete for heavy-use industrial slabs and parking structure slabs, concrete required to be watertight, and concrete with a water-cementitious materials ratio below 0.50.
4. Use corrosion-inhibiting admixture in concrete mixtures where indicated.

2.8 CONCRETE MIXING

A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M and ASTM C 1116/C 1116M, and furnish batch ticket information.

B. Project-Site Mixing: Measure, batch, and mix concrete materials and concrete according to ASTM C 94/C 94M. Mix concrete materials in appropriate drum-type batch machine mixer.

1. For mixer capacity of 1 cu. yd. or smaller, continue mixing at least 1-1/2 minutes, but not more than 5 minutes after ingredients are in mixer, before any part of batch is released.
2. For mixer capacity larger than 1 cu. yd., increase mixing time by 15 seconds for each additional 1 cu. yd.
3. Provide batch ticket for each batch discharged and used in the Work, indicating Project identification name and number, date, mixture type, mixture time, quantity, and amount of water added. Record approximate location of final deposit in structure.

PART 3 - EXECUTION

3.1 FORMWORK

A. Use either fixed forms or slip forms.

B. Provide straight, full depth, un-warped forms that will resist springing during concrete placement.

C. Firmly stake fixed forms.

D. Construct forms tight enough to prevent loss of concrete mortar.

E. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

F. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

G. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

H. Form liners shall not be used for the final finish.

3.2 REMOVING AND REUSING FORMS

A. General: Formwork that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F for 24 hours after placing concrete. Concrete shall be hard enough to not be damaged by form-removal operations and curing and protection operations need to be maintained.

1. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.

B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.

C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Contracting Officer (CO).

3.3 SHORES AND RESHORES

A. Comply with ACI 318 and ACI 301 for design, installation, and removal of shoring and reshoring.

B. Plan sequence of removal of shores and reshore to avoid damage to concrete. Locate and provide adequate reshoring to support construction without excessive stress or deflection.

3.4 REINFORCEMENT

A. General: When concrete is placed, reinforcement shall be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory provided the minimum nominal dimensions, nominal weight, and the minimum average height of deformation of a hand-wire-brushed test specimen are not less than the applicable ASTM specification requirements.

B. Reinforcement: Place, support, and fasten reinforcement as indicated in the Contract Documents. Do not exceed the placing tolerances specified in ACI 117 before concrete is placed. When necessary to move reinforcement beyond the specified placing tolerances to avoid interference with other reinforcement or embedded items, submit the resulting arrangement of reinforcement for acceptance.

C. Cover: Allowable concrete cover for reinforcement is indicated in the project drawings. Tolerances on concrete cover shall meet the requirements of ACI 117.

D. Tie Wires: After cutting tie wires, turn wires to the inside of section and bend so that concrete placement will not force ends to exposed concrete surfaces.

E. Splicing: Make splices as indicated in the project drawings. Lapped bars shall be placed in contact and securely tied, or spaced transversely apart to permit the embedment of the entire surface of each bar in concrete. Lapped bars shall not be spaced farther apart than 1/5 the required length of lap, or six inches (6”).

F. Mechanical Splices: Mechanical splices for reinforcement not shown on the project drawings shall be submitted for review and accepted prior to use. Mechanical splices shall be in accordance with the recommendations of the manufacturer of the mechanical splicing device.

G. Reinforcement shall not be field bent or straightened except when specifically permitted by the CO.

H. Reinforcement shall not be cut in the field except when specifically permitted by the CO.

I. The following reinforcing steel work will be considered defective and shall be removed and replaced by the Contractor at no additional cost to the Government:

1. Bars with kinks or bends not shown on drawings.
2. Bars injured due to bending or straightening.
3. Bars heated for bending.
4. Reinforcement not placed in accordance with the project drawings and/or specifications.

3.5 JOINTS

A. General: Construct joints true to line as shown in the Contract Documents. All joints shall be tooled.

3.6 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.

B. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301.

1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.

C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.

1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.

D. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.

1. When average high and low temperature is expected to fall below 40 deg F for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

E. Hot-Weather Placement: Comply with ACI 301 and as follows:

1. Maintain concrete temperature below 90 deg F at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor’s option.

2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.7 FINISHING CONCRETE

A. Float the surface, but do not over-float, to produce a smooth surface, free from irregularities. Round the edges and joints with finishing tools.

B. Broom finish is achieved by brushing the surface with a stiff bristled broom traverse to the direction of travel.

3.8 CONCRETE PROTECTING AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces to deter moisture loss due to hot, dry, or windy conditions before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.

D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.

E. Cure concrete according to ACI 308.1, by the following method:

1. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape.

3.9 JOINT FILLING

A. Prepare, clean, and install joint filler according to manufacturer's written instructions.

B. Remove dirt, debris, saw cuttings, curing compounds, and sealers from joints; leave contact faces of joint clean and dry.

1. Tape sides to minimize staining at Contractor's option to minimize excess filler removal and staining.

C. Install exterior-rated joint sealant filler full depth in saw-cut joints and at least 2 inches deep in formed joints. Overfill joint and trim joint filler flush with top of joint after hardening and remove excess from exposed surfaces. Sealant filler color shall match concrete paving color as determined by CO. Where colors do not match, remove and reinstall until matching at no additional cost to Government.

1. Apply joint sealant using masking tape or other methods approved by the CO to ensure clean, straight sealant lines. Remove tape prior to initial curing.
2. Sealant exhibiting smeared edges, feathering, irregular lines, voids, or contamination of adjacent surfaces remove and replace at no additional cost to the Government.

3.10 FIELD QUALITY CONTROL

A. Testing and Inspecting: Contractor shall engage a qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.

B. Concrete Tests: Testing of composite samples of fresh concrete obtained according to ASTM C 172 shall be performed according to the following requirements:

1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu. yd., but less than 25 cu. yd., plus one set for each additional 50 cu. yd. or fraction thereof.
2. Testing Frequency: Obtain at least one composite sample for each 100 cu. yd. or fraction thereof of each concrete mixture placed each day.

    a. When frequency of testing will provide fewer than five compressive-strength tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.

3. Slump: ASTM C 143/C 143M; one test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture. Perform additional tests when concrete consistency appears to change.
4. Air Content: ASTM C 231, pressure method, for normal-weight concrete; ASTM C 173 volumetric method, for structural lightweight concrete; one test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
5. Concrete Temperature: ASTM C 1064/C 1064M; one test hourly when air temperature is 40 deg F and below and when 80 deg F and above, and one test for each composite sample.
6. Compression Test Specimens: ASTM C 31/C 31M.

    a. Cast and laboratory cure two sets of two standard cylinder specimens for each composite sample.

7. Compressive-Strength Tests: ASTM C 39/C 39M; test one set of two laboratory-cured specimens at 7 days and one set of two specimens at 28 days.

    a. Test one set of two field-cured specimens at 7 days and one set of two specimens at 28 days.
    b. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.

8. Test results shall be reported in writing to CO, concrete manufacturer, and Contractor within 48 hours of testing. Reports of compressive-strength tests shall contain Project identification name and number, date of concrete placement, name of concrete testing and inspecting agency, location of concrete batch in Work, design compressive strength at 28 days, concrete mixture proportions and materials, compressive breaking strength, and type of break for both 7- and 28-day tests.
9. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by CO but will not be used as sole basis for approval or rejection of concrete.
10. Additional Tests: Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by CO. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C 42/C 42M or by other methods as directed by CO.
11. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
12. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

3.11 INSPECTIONS

A. Accessibility Inspection: Accessibility inspection shall be performed by the Contractor in a two-part process to ensure compliance of final installation. The two-part process inspections shall occur pre-installation and post-installation.

1. Pre-installation inspection shall examine slopes, dimensions, base materials, and preparation work prior to concrete placement. Check formwork, embedded items, reinforcing, etc. Pre-installation work shall conform to National Park Service Accessible Route Design Standards.
2. Post-installation inspection shall examine completed concrete placement. Report non-compliance with ABAAS for running and cross slopes, landings, vertical changes in elevation, transitions, imperfections or flaws, improper drainage, slippery surfaces, and dimensional accuracy.

B. Contractor shall utilize the National Park Service CQC Accessibility Inspection Report as basis for recording findings. Any non-compliant conditions shall be documented. Supplement Report with digital images, notation, red-line drawings, sketches and dimensions, product literature, submittal, etc. to explain installed work. Submit completed Report to CO within 24 hours of commencing inspection.

END OF SECTION 32 13 20

SECTION 32 17 00 – PAVING SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes the following:

1. Pavement surface preparation.
2. Furnishing and installing:

a. New pavement markings.
b. Accessible parking signage.

B. Related Sections include the following:

1. Section 32 12 16 – Asphalt Paving
2. Section 32 13 20 – Site Concrete

1.2 SUBMITTALS

A. Submit material certificate to the Contracting Officer (CO), signed by the material producer and contractor, certifying that materials comply with these specifications.

B. Shop Drawings:

1. Shop Drawings of all signs and supports. Do not order material or begin fabrication until the Contracting Officer's (CO) approval has been obtained.
2. Show sizes and thicknesses of all members, types of materials, methods of construction and assembly, type of surface treatment, complete dimensions, hangers, brackets, anchorage, fasteners, relationship to surrounding work by other trades, shop finishes, sign designs, layouts, and lettering, and other pertinent details of fabrication and installation.

C. Product Data: Pavement markings, signs, and detectable warning surface.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Pavement Markings

1. All pavement markings and accessible parking space symbols shall be waterborne paint and shall be on the Iowa Department of Transportation Approved Product List.
2. Edge lines, stop bars, arrows, crosswalks and lane delineations will be white paint unless otherwise specified. Paint for accessible parking space symbol shall be blue.

3. Pigment content: Percent pigment by weight of the finished product to be from 45.0% to 55.0% by weight for white.
4. Resin solids: Composed of 100% acrylic emulsion polymer.
5. Nonvolatile Vehicle: No less than 42.0% by weight for white paint and no less than 44.0% by weight for yellow paint.
6. Volatile Organic Compounds: Not to exceed 1.25 pounds per gallon excluding water and VOC exempt solvents per ASTM D 3960.
7. Flash Point: Closed cup flash point shall be no less than 100°F per ASTM D56.
8. Density: 12 pounds per gallon minimum per ASTM D 1475. The density of the production batches shall not vary by more than + 0.2 pounds per gallon from the density of the qualification samples.
9. Color:

    a. For white, the color after drying shall be a flat white, free from tint, furnishing good opacity and visibility under both daylight and artificial light.
    b. For yellow, the color after drying shall be within the follow CIE chromaticity limits when measured with an instrument having a 2 degree observer, using a standard C illuminant, and 45/0 or 0/45 geometry:

CIE Data Limits	**Y**	**x**	**y**
Minimum	0.5400	0.432	0.428
Maximum	0.5910	0.501	0.455

    c. For blue, the color after drying shall be within the follow CIE chromaticity limits when measured with an instrument having a 2 degree observer, using a standard illuminant D_{65} and 45/0 or 0/45 geometry:

Chromaticity Coordinates							
1		**2**		**3**		**4**	
x	y	x	y	x	y	x	y
.105	.100	.220	.180	.200	260	.060	.220

    Blue paint shall have a daytime luminance factor of 5% minimum to 14% maximum with CIE 2 degree standard observer and 45/0 or 0/45 geometry and CIE standard illuminant D_{65}.

10. Viscosity: No less than 80 or no greater than 90 Krebs Units at 77°F per ASTM D 562.
11. No-Pick-Up Time: Less than 5 minutes per ASTM D 711.
12. Directional Reflectance: 84.0% minimum for white, 50.0% minimum for yellow.
13. Dry Opacity: Minimum contrast ratio of 0.955.
14. Flexibility: No cracking or flaking shows when tested according to Federal Specification TT-P-1952b.
15. Pigment Particle Size: Grind of no less than 3 on a Hegman Grind Gage when measured according to ASTM D 1210.
16. Reflectorizing spheres:

    a. Apply glass beads to all pavement markings unless otherwise noted. Glass beads shall be dual coated beads (silicone and silane).

b. Glass beads shall not contain more than 200 ppm total of lead, antimony, or arsenic. Use transparent, clear, colorless glass spheres that are:

1) Free from milkiness, dark particles, and excessive air inclusions.
2) Essentially clear from surface scarring or scratching.
3) Free of hard lumps and clusters.
4) Readily dispensed under any conditions suitable for paint striping.
5) Virgin class for 20 sieve and larger glass spheres.

c. Glass beads shall include a minimum of 80% true spheres
d. Glass beads shall have a minimum refractive index of 1.50.
e. Glass beads shall meet AASHTO M247 requirements and the below gradation requirements. Glass beads shall meet either percent passing or percent retained gradation but are not required to meet both.

Sieve Size	% Passing	% Retained
16	100	0
20	85-92	8-15
30	50-67	25-35
40	0-37	30-50
50	0-15	15-35
80		0-10
Pan		0-2

17. Skid-Resistance: The surface, with properly applied and embedded surface beads, must provide a minimum resistance value of 45 BPN when tested according to ASTM E303.

B. Signs

1. General

a. The sign face shall be uniform, flat, smooth, and free from defects, scratches, chips, wrinkles, gel, hard spots, streaks, extrusion marks, and air bubbles.
b. The front, back, and edges of sign panels must not have bends, router chatter marks, burns, sharp edges, loose rivets, delaminated skins, excessive adhesive over-spray, or aluminum marks.
c. Do not store sign panels directly on the ground. Keep sign panels dry at all times and store:

1) In a dry environment.
2) On edge vertically whether indoors or outdoors.
3) In enclose, climate-controlled trailers or containers in areas of high heat and humidity.
4) Indoors whenever the panels will be stored for more than 30 days.

2. Aluminum Sheeting

a. Sheet aluminum signs shall be constructed of an alloy and temper conforming to ASTM B209 and shall be aluminum alloy 5052-H38.
b. The thickness shall be 0.08 inches for signs with the longest side of 35 inches or less, and 0.125 inches for signs with the longest side greater than 36 inches.

c. Aluminum sheeting must be pretreated for corrosion resistance as specified in ASTM B449.
d. Aluminum sheeting shall be protected from exposure to grease, oils, dust, and contaminants.
e. The aluminum sheeting must be free from buckles, warps, dents, cockles, burrs, and other defects resulting from fabrication.
f. The base plate for standard route markers must be die cut.
g. The aluminum sheeting must be attached to the frame with 3/16-inch-diameter rivets. The rivets must be placed at least 1/2 inch from the web channel edges. The rivets must be made of aluminum alloy 5052 and be anodized or treated with a conversion coating to prevent corrosion.

3. Retroreflective Sheeting

a. Retroreflective sheeting shall comply with ASTM D4956, including supplementary requirements.
b. Use Type XI retroreflective sheeting. Type XI retroreflective sheeting must have Class 1, 3, or 4 adhesive backing.
c. Adhesive backing shall be pressure sensitive and fungus resistant.
d. Reflective sheeting shall be applied to in the manner specified by the sheeting manufacturer. The applied sign face shall not have bubbles, wrinkles, or foreign material beneath the reflective sheeting.
e. Splices in the retroreflective sheeting shall overlap by at least 1 inch unless otherwise specified by the manufacturer.
f. The retroreflective sheeting on either side of a splice must not exhibit a color difference under incident and reflected light.

4. Sign covering

a. The covering shall consist of 4 mils minimum thickness black polyethylene sheeting of sufficient size to entirely cover the sign and shall extend over the edges of the sign and fastened to the back.

5. Signposts

a. Signposts shall be nominal 4" x 4" treated Douglas Fir or treated Southern Pine, Construction Grade.
b. Ensure a straight line from the centers of the ends of a spot does not deviate from the longitudinal axis of the post at any point by more than 0.5% of the length of the post.
c. Color: NPS Brown (RAL 6008 or Cardinal T032-BR62)

C. Detectable Warning Surface

1. Dome Size: Truncated domes in a detectable warning surface shall have a base diameter of 0.9 inch minimum and 1.4 inches maximum, a top diameter of 50 percent of the base diameter minimum to 65 percent of the base diameter maximum, and a height of 0.2 inch.
2. Dome Spacing: Truncated domes in a detectable warning surface shall have a center-to-center spacing of 1.6 inches minimum and 2.4 inches maximum, and a base-to-base spacing of 0.65 inch minimum, measured between the most adjacent domes on a square grid.
3. Material: detectable warning surface shall be unpainted cast iron.

PART 3 - EXECUTION

3.1 PREPARATION

A. The Contractor shall clean the pavement of dust, dirt, old pavement markings, concrete curing compounds, and other foreign material which may be detrimental to the adhesion of the pavement marking materials.

B. Examine areas and conditions, with Installer present, for compliance with requirements for correct and level finished grade, mounting surfaces, installation tolerances, required clear space, and other conditions affecting performance.

1. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PAVEMENT MARKING APPLICATION

A. The material shall be applied to the pavement by equipment designed and manufactured specifically for the application of pavement markings.

B. Pavement markings shall be applied in accordance with the layout shown on the Drawings. No paint shall be applied to new bituminous pavement until the top course has cured at least one week, and allow two weeks curing for newly installed bituminous concrete curbing.

C. All parking stalls shall be single stripe and shall be spaced equally. The line indicated on the drawings is on the center line of the stall marking.

D. all parking stall markings shall be straight with sharp corners and clean edges.

E. All markings shall be applied in one coat with brush, spray, or marking machine over clean dry pavement surfaces, when the atmospheric temperature is at or above 45°F, and when the weather is otherwise favorable in the opinion of the CO.

F. Application rate: Apply paint to a minimum wet film thickness of 14 mils for standard markings and 20 mils for symbols.

G. Immediately remove drips, overspray, improper markings, paint, and thermoplastic tracked by traffic with an authorized method.

H. Use only skilled workmen who are experienced and normally employed in the work of installing pavement markings. Supply all the necessary equipment and materials required for the work.

I. The Contractor shall protect the buildings, walks, pavement, curbing, trees, shrubs, mulch, and other site fixtures from over-spray of paint and damage from pavement marking operations.

J. Traffic shall not be permitted on the pavement until the paint is thoroughly dry.

K. All pavement markings shall form a continuous, smooth line or curve consistent with the alignment and radius shown on the plans. Contractor shall use paint application equipment capable of producing uniform line widths and maintaining straight lines or constant-radius curved lines. Prior to paint striping application, contractor shall mark out proposed striping which shall

be reviewed and approved by the CMR prior to installation. Deviations or irregular markings shall be removed and reinstalled at no cost to the Government.

3.3 SIGNAGE

A. Sign Fabrication

1. All fabrication shall be completed before degreasing metal surface and applying reflective sheeting.
2. Sign blanks and panels shall be to size and shape and be free of buckles, warps, dents, and other defects resulting from fabrication.
3. Finished signs shall be smooth and flat and free from blisters, wrinkles, cracks, tears, and delamination.
4. All letters shall be cut clean and sharp.
5. Reflective sheeting and transparent film shall be applied mechanically with the equipment and in a manner specified by the sheeting manufacturer.
6. On extruded aluminum or formed steel sign panels, the sheeting shall be lapped over the longitudinal edges of each panel a minimum distance of 1/4 inch.
7. Sheeting shall be bonded to the panels as recommended by the manufacturer.

B. Protection and Temporary Covers

1. Completed sign panels shall be protected and maintained in good condition, free from dirt, scratches, hand marks or other blemishes. The panels shall be transported in a manner that prevents damage.
2. Subsequent to erection, if required by the CO, exterior signs shall be covered until the actual use is desired. Material used to temporarily cover any sign panel shall effectively conceal the message, be non-injurious to the panel, its finish, its structural integrity, and shall be of sufficient durability for the time period required.

C. Installation

1. Erection of all signs shall be performed by experienced sign erectors. Signs shall be installed true, plumb, and level, located where shown on the Drawings or as required by the CO. No field cutting of any sign work will be allowed. Exercise extreme care in all handling and stacking of signs to avoid chipping.
2. All work shall be rigidly anchored to the supporting construction in accordance with the approved shop drawings.

D. Clean-up

1. Surfaces of signs shall be cleaned as recommended by the sign manufacturer after installation.
2. All defective work shall be removed and replaced with work conforming to the specified requirements.

E. Erection of Posts

1. Drive or embed posts to ensure sign faces and supports are within 3/16 inch of plumb over 3 feet.
2. Do not damage the top of posts during driving.
3. Install steel sign supports and square tubular steel sign supports in accordance with the contract plans.
4. If installing posts in new concrete, the Contractor may form holes before placing the new concrete.

3.4 DETECTABLE WARNING SURFACE

A. The detectable warning surface shall be located as shown in the Plans. Placement of the detectable warning surface shall be in accordance with the manufacturer's recommendation for placement in fresh concrete, before the concrete has reached initial set, or on a hardened cement concrete surface or asphalt pavement surface.

B. Vertical edges of the detectable warning surface shall be flush with the adjoining surface to the extent possible (not more than ¼ inch above the surface of the pavement) after installation.

C. Embossing or stamping the wet concrete to achieve the truncated dome pattern or using a mold into which a catalyst-hardened material is applied shall not be allowed.

END OF SECTION 32 17 00

SECTION 32 31 13 – GATES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Gates: Metal Swing Gate.

B. Related Sections:

1. Section 32 13 20 – Site Concrete

1.3 SUBMITTALS

A. Product Data: For each type of product indicated. Include material descriptions, dimensions of individual components and profiles, and finishes for fences and gates.

1. Gate posts and fittings.
2. Reinforcements and attachments.
3. Accessories: Locks, latches, post brackets.
4. Gates and hardware.

B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.

C. Samples for Initial Selection: For components with factory-applied color finishes.

D. Samples for Verification: Prepared on Samples of size indicated below:

1. In 12-inch lengths for components and on full-sized units for accessories.

E. Product Certificates: For gate, from manufacturer.

F. Field quality-control reports.

G. Operation and Maintenance Data: For the following to include in emergency, operation, and maintenance manuals:

1. Gate hardware.

H. Warranty: Sample of special warranty.

1.4 QUALITY ASSURANCE

A. Preinstallation Conference: Conduct conference at Project site.

1. Review required testing, inspecting, and certifying procedures.

1.5 PROJECT CONDITIONS

A. Field Measurements: Verify layout information for gates shown on Drawings in relation to property survey and existing structures. Verify dimensions by field measurements.

1.6 WARRANTY

A. Special Warranty: Manufacturer's standard form in which Installer agrees to repair or replace components of gates that fail in materials or workmanship within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Faulty operation of gate.
b. Deterioration of metals, metal finishes, and other materials beyond normal weathering.

2. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 METAL ENTRANCE SWING GATE

A. General: Comply with ASTM F 1184 for gate posts and single and double swing gate types.

1. Classification: Swing

a. Gate Leaf Width: As indicated on plans.
b. Gate Height: As indicated on plans.

2. Classification: Swing.

a. Gate Frame Width and Height: As indicated on plans.

B. Material:

1. Gate Posts

a. Steel.

2. Bracing, and Frame

a. Steel.

3. Finish

a. Powder-Coat Finish: Prepare, treat, and coat nongalvanized ferrous metal to comply with resin manufacturer's written instructions and as follows:

1) Prepare uncoated ferrous-metal surfaces to comply with SSPC-SP 6/NACE No. 3, "Commercial Blast Cleaning."
2) Treat prepared metal with iron-phosphate pretreatment, rinse, and seal surfaces.
3) Apply thermosetting polyester or acrylic urethane powder coating with cured-film thickness not less than 1.5 mils (0.04 mm).
4) Color: Match NPS Brown (RAL 6008 or Cardinal T032-BR62)

2.2 FITTINGS

A. General: Comply with ASTM F 626.

B. Gate Frame: steel, as indicated on plans.

C. Fasteners: According to ASTM F 626.

D. Finish:

1. Gate components shall be hot-dipped galvanized after welding.
2. Following galvanization, prep the parts and coat above finish grade components with high build epoxy and silicone alkyd enamel topcoat.
3. Color: "NPS brown."

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas and conditions, with Installer present, for compliance with requirements for a verified survey of property lines and legal boundaries, site clearing, earthwork, pavement work, and other conditions affecting performance of the work.

1. Do not begin installation before final grading is completed unless otherwise permitted by CO.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

A. Stake locations of gates. Do not exceed intervals of 50 feet or line of sight between stakes. Indicate locations of utilities, underground structures, benchmarks, and property monuments.

3.3 INSTALLATION, GENERAL

A. Install gate as shown on plans.

3.4 GATE INSTALLATION

A. Install gates level, plumb, and secure for full opening without interference. Install ground-set items in concrete for anchorage. Adjust hardware for smooth operation and lubricate where necessary.

B. Install gate hardware such that the swing arm gates rotate open in the downstream direction only.

C. Install the gate so that both swing arm gates may be properly secured when in either the open or closed position.

3.5 ADJUSTING

A. Gates: Adjust gates to operate smoothly, easily, and quietly, free of binding, warp, excessive deflection, distortion, nonalignment, misplacement, disruption, or malfunction, throughout entire operational range. Confirm that latches and locks engage accurately and securely without forcing or binding.

B. Lubricate hardware and other moving parts.

3.6 DEMONSTRATION

A. Train government’s personnel to adjust, operate, and maintain gates.

END OF SECTION 32 31 13

SECTION 32 91 13 – SOIL PREPARATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This section describes preparation of soil including application of herbicide and soil amendments.

B. Related Sections:

1. Section 31 20 00 – Earth Moving
2. Section 32 92 00 – Turf and Grasses
3. Section 32 93 00 – Plants

1.3 SUBMITTALS

A. Submit certification of quantities of herbicide and fine compost delivered to the site to the CO.

B. Submit to the CO for approval a sample of fine compost.

1.4 PRODUCT DELIVERY

A. Deliver fertilizer to the site in original unopened containers, each bearing the manufacturer's guaranteed analysis.

1.5 PROTECTION

A. Protect utility lines, storm drainage lines, site improvements, and underground irrigation system during execution of work. See Section 015000, Temporary Facilities and Controls.

PART 2 - PRODUCTS

2.1 SOIL AMENDMENTS

A. Fine compost: Well-decomposed, stable, and weed free organic matter sourced locally. The pH shall be between 6.0 and 8.5 when tested. Manufactured inert material (plastic, concrete, ceramics, metal, etc.) shall be less than 0.5 percent on a dry weight or volume basis, whichever

provides for the least amount of foreign material. Minimum organic matter shall be 40 percent dry weight and soluble salt contents shall be less than 6.0 mmhos/cm.

Compost shall meet the following:

	Min.	Max.
Percent passing 2”	100%	
Percent passing 1”	90%	100%
Percent passing ¾”	70%	100%
Percent passing ¼”	40%	75%

Maximum particle length of 6 inches.

2.2 TOPSOIL

A. Clean, fertile, friable, natural soil material free of debris, roots, stones, weeds, and grass.

PART 3 - EXECUTION

3.1 HERBICIDE APPLICATION

A. Apply contact herbicide over all areas of weed or grass growth within landscaped area.

B. Apply in two applications as follows:

1. First application, apply seven days prior to performing earthwork.
2. Second application (to kill new vegetation), apply after grading has been completed and 48 hours prior to planting.

C. Observe manufacturer’s recommended period prior to working in treated areas.

D. Apply at manufacturer’s maximum recommended application rate.

E. Do not apply herbicide to impervious areas unless specifically directed to do so by the CO.

3.2 SOIL PREPARATION

A. Remove stones, mortar, gravel, concrete, asphalt, rubbish, debris, and other materials larger than 1 1/2 inches from planting areas. See Section 329200, Turf and Grasses.

1. If existing soil base is not suitable for use, remove top 8 inches and replace with suitable topsoil.

B. Seeding area finished soil surface shall be 1 inch below edge of pavement or sidewalk elevation prior to hydroseed application.

C. See Section 312000, Earth Moving, if existing soil is not sandy or silty, thus requiring over-excavation and placement of fill material.

D. Amend soil with compost at the rates described below to achieve a total 9 3/4 inch layer of amended soil material. Apply in the following order.

1. Add 1 3/4 inches of fine compost.
2. Rototill, in two directions, to a depth of 9.75 inches, to create a uniform mix of soil and fine compost.

E. Rake and smooth planting area to a tolerance of 1 inch, plus or minus, in 10 feet.

F. See Section 329300, Plants, for application of prepared soil mix in planting pits.

END OF SECTION 32 91 13

SECTION 32 92 00 – TURF AND GRASSES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Seed Mixes

B. Related Sections:

1. Section 31 10 00 – Site Clearing
2. Section 31 20 00 – Earth Moving
3. Section 31 23 16 – Topsoil

1.3 SUBMITTALS

A. Product Data

1. Seed Mix 1: Include seed source and seed tag.
2. Virgin wood fiber mulch

1.4 QUALITY ASSURANCE

A. Installer Qualifications: A qualified landscape Installer whose work has resulted in successful establishment of plants.

1. Experience: Three years' experience in successful landscape installation in addition to requirements in Division 01 Section "Quality Requirements."
2. Installer's Field Supervision: Require Installer to maintain an experienced full-time supervisor on Project site when work is in progress.

B. Project Representative Provide quality, size, genus, species, and variety of plants indicated, complying with applicable requirements in ANSI Z60.1.

C. Measurements: Measure according to ANSI Z60.1.

D. Preinstallation Conference: Conduct conference at Project site prior to installation.

E. Warranty: If less than 70% germination occurs, overseeding is required.

1.5 MATERIALS

A. Seed Mix 1: Non-Native Turf Replacement

1. Contractor shall furnish only regional genotype plant materials; no other species, cultivars or varieties shall be permitted without CO approval. Regional genotype is defined as the state of Iowa and/or neighboring states within 200 miles of Effigy Mounds National Monument.
2. Performance Characteristics:

    a. Shall have 0% noxious weed presence
    b. Seeding Rates: 200 seeds per square foot minimum rate or recommended rate by the seed producer.
    c. Consists of the following seed mix:

Species	Common Name	% Mix by weight
Fescue rubra	Red Fescue	50
Trifolium repens	White clover	50
	Total	**100**

B. Seed Mix 2: Native Turf Replacement – Mowed Frequently

1. To be furnished by NPS.
2. Performance Characteristics:

    a. Seeding Rates: 200 seeds per square foot minimum rate
    b. Consists of the following seed mix:

Species	Common Name	% Mix by weight
Achillea millefolium	Yarrow	15
Allium cernum	Wild nodding onion	2
Antennaria neglecta	Field Pussytoes	8
Avena sativa	Annual oats	2
Bouteloua curtipendula	Sideoats grama	5
Carex eburnea	Ivory Sedge	2
Carex sprengelii	Sedge	2
Dalea purpurea	Purple Prairie Clover	2
Danthonia spicata	Poverty oats	30
Elymus virginicus	Wild rye	3
Fragaria virginiana	Wild strawberry	10
Juncus tenuis	Pathrush	5
Lolium multiflorum	Annual rye	3
Prunella vulgaris	Heal-all	13
	Total	**100**

C. Seed Mix 3: Native Turf Replacement – Mowed Infrequently

1. To be furnished by NPS.
2. Performance Characteristics:

a. Seeding Rates: 200 seeds per square foot minimum rate and/or 100lbs/acre
b. Consists of the following seed mix:

Species	Common Name	% Mix by Weight
Achillea millefolium	Yarrow	3
Allium canadense	wild garlic	2
Allium cernum	wild nodding onion	2
Amorpha canescens	Leadplant	2
Anemone canadensis	Canada Anenome	2
Anemone virginiana	Tall Thimbleweed	2
Avena sativa	Annual oats	3
Antennaria neglecta	Field Pussytoes	7
Aquilegia canadensis	Columbine	2
Asclepias syriaca	Common milkeweed	1
Bouteloua curtipendula	Sideoats grama	5
Carex eburnea	Ivory Sedge	3
Carex retrorsa	Knotsheath sedge	3
Carex sprengelii	Sedge	3
Coreopsis palmata	Prairie Coreopsis	2
Cryptotaenia canadensis	Canadian Honewort	1
Dalea purpurea	Purple Prairie Clover	3
Danthonia spicata	Poverty oats	6
Elymus hystrix	Bottlebrush Grass	1
Elymus virginicus	Virginia Wild Rye	1
Enemion biternatum	False rue anenome	1
Fragaria virginiana	Wild strawberry	5
Galium boreale	Northern Bedstraw	1
Gentiana alba	White gentian	1
Juncus tenuis	Pathrush	4
Lespedeza capitata	Round-headed Bushclover	1
Lolium multiflorum	Annual rye	3
Lobelia siphilitica	Great blue lobelia	1
Maianthemum racemosum	Solomon's Plume	1
Mitella diphylla	Bishop's Cap	2
Monarda fistulosa	Wild Bergamot	1
Osmorhiza claytoni	Sweet Cicely	1
Polemonium reptans	Jacob's Ladder	3
Polygonatum biflorum	Solomon's Seal	1
Prunella vulgaris	heal-all	4

Species	Common Name	% Mix by Weight
Schizachyrium scoparium	Little Bluestem	10
Solidago nemoralis	Gray Goldenrod	3
Zizia aurea	Common golden alexanders	3
	Total	**100**

D. Native Lawn Extension: Native Turf Replacement – Mowed Frequently

1. To be furnished by NPS.
2. Performance Characteristics:

a. Seeding Rates: 0.10 acres minimum rate 200 seeds per square foot
b. Consists of the following seed mix:

Species	Common Name	% Mix by weight
Achillea millefolium	Yarrow	15
Allium cernum	Wild nodding onion	2
Antennaria neglecta	Field Pussytoes	8
Avena sativa	Annual oats	2
Bouteloua curtipendula	Sideoats grama	5
Carex eburnea	Ivory Sedge	2
Carex sprengelii	Sedge	2
Dalea purpurea	Purple Prairie Clover	2
Danthonia spicata	Poverty oats	30
Elymus virginicus	Wild rye	3
Fragaria virginiana	Wild strawberry	10
Juncus tenuis	Pathrush	5
Lolium multiflorum	Annual rye	3
Prunella vulgaris	Heal-all	13
	Total	**100**

1.6 PROJECT CONDITIONS

A. Field Measurements: Verify actual grade elevations and construction contiguous with new plantings by field measurements before proceeding with planting work.

B. Interruption of Existing Services or Utilities: Do not interrupt services or utilities to facilities occupied by CO or others unless permitted under the following conditions and then only after arranging to provide temporary services or utilities according to requirements indicated:

1. Notify CO no fewer than two days in advance of proposed interruption of each service or utility.
2. Do not proceed with interruption of services or utilities without CO's written permission.

C. Planting Restrictions: Plant between September 15 and November 1 or February 1 and April 15 when soil temperatures are consistently within the 40-49 degrees F range. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.

D. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained. Do not seed on ground that is excessively wet, frozen, or has more than ¼" of snow cover. Apply products during favorable weather conditions according to manufacturer's written instructions and warranty requirements.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Delivery: All seed mix material for Seed Mix 2 and 3 will be delivered to the Project site by NPS.

B. Packaged Materials: Deliver packaged materials in original, unopened containers showing weight, certified analysis, name and address of manufacturer, and indication of conformance with state and federal laws if applicable.

C. Bulk Materials:

1. Do not dump or store bulk materials near structures, utilities, walkways, and pavements, or on existing turf areas or plants.
2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.

D. NPS and Contractor to deliver plants after preparations for hydroseeding have been completed and Contractor is ready to install immediately. If planting is delayed more than six hours after delivery, protect seed mix material from weather and mechanical damage.

1.8 WARRANTY

A. Special Warranty: Contractor agrees to overseed seeding areas that fail in materials, workmanship, or growth within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Germination less than 70%.

2. Warranty Periods from Date of Planting Completion:

a. 12 months from date of planting completion.

3. Include the following remedial actions as a minimum:

a. Immediately overseed failed seeding area following all specified project conditions.
b. Provide extended warranty for period equal to original warranty period, for replaced plant material.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive seeding for compliance with requirements and conditions affecting installation and performance.

1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within a planting area.
2. Do not mix or place soils and soil amendments in frozen, wet, or muddy conditions.
3. Suspend soil spreading, grading, and tilling operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.
4. Uniformly moisten excessively dry soil that is not workable and which is too dusty.

B. Proceed with seeding only after unsatisfactory conditions have been corrected.

C. If contamination by foreign or deleterious material or liquid is present in soil within a planting area, do not remove any soil unless directed by the CO.

3.2 PREPARATION

A. Notify a minimum of two weeks prior to seeding date to allow proper delivery of seed.

B. Protect structures, utilities, sidewalks, pavements, and other facilities and turf areas and existing plants from damage caused by planting operations.

C. Install erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

D. Stake perimeter of seeding locations and adjust locations when requested and obtain Project Representative’s acceptance of layout before planting. Make minor adjustments as required.

E. Roughen embankments and fill areas by track walking.

3.3 INSTALLATION

A. Seeding Seed Mix Method Options

1. Two-Step Hydroseed Method

a. First Application (Seed): Hydroseed Application: In order to ensure that the seed is being applied at the correct rate areas to be seeded shall be marked off/flagged at a minimum of half acre increments prior to seeding. Add 500 lbs/acre of tracer material consisting of virgin wood fiber mulch to the water/seed in the hydraulic

equipment to provide visible evidence of uniform application. Adding this 500 lbs/acre during seeding, means that only 1500 lbs/acre will be added during the second application of just mulch, which would result in a total of 2,400 lbs/acre of mulch applied.

2. Hand Broadcast/Rake/Hydromulch Method (for slopes flatter than 2H:1V)

    a. First Application (Seed): Hand Application: In order to ensure that the seed is being applied at the correct rate areas to be seeded shall be marked off/flagged at a minimum of half acre increments prior to seeding. Seed shall be applied by hand or with hand seeders such as a belly grinder. Seed will be distributed in such a way as to assure all species in the seed mix are evenly distributed across the area, regardless of seed size. Lightly rake the seed into the soil to a depth of ¼". Do not smooth with the backside of the rake, light furrows in the soil are desired.
    b. Second Application (Mulch): 2,400 lbs./acre of long strand virgin wood fiber and;

        1) Apply mulch with a non-toxic, and natural and biodegradable plant-based psyllium/alpha plantago tackifier. Depending on product apply per the manufacturer's specifications, generally at a rate of 80 lbs. per acre on slopes less than 3H:1V and apply at a rate of 100 lbs./acre on slopes greater than 3H:1V.
        2) Apply mulch with a non-toxic, 100% environmentally safe tracer dye material applied per the manufacturer's specification generally at a rate of around 200-300 pounds per acre to provide visible evidence of uniform application.
        3) The hydromulcher must be equipped with a built-in continuous agitation and recirculation system of sufficient operating capacity to produce homogeneous slurry and a discharge system that will apply slurry to the designated areas at a continuous and uniform rate. No fertilizers, mycorrhizae, or other amendments shall be added to the hydroseed slurry.
        4) The Contractor shall spray designated areas with the slurry in a sweeping motion and in an arched stream until a uniform coat is achieved, with no slumping or shadowing, as the material is spread at the required rate. The hydromulch slurry should float down from the arched stream, as opposed to being shot directly at the ground. During hydromulching the existing native vegetation must be protected from damage (including, but not limited to, coating with mulch, damage by direct spray, and dragging hose). Notify the CO/DSC three weeks before hydromulching to ensure that NPS oversight of seeding operations can be coordinated.

3. Hydroseed/Hydromulch Method (for slopes steeper than 2H:1V)

    a. First Application (Seed): Hydroseed Application Hydroseed Application: In order to ensure that the seed is being applied at the correct rate areas to be seeded shall be marked off/flagged at a minimum of half acre increments prior to seeding. Add 500 lbs/acre of tracer material consisting of virgin wood fiber mulch to the water/seed in the hydraulic equipment to provide visible evidence of uniform application. Adding this 500 lbs/acre during seeding, means that only 1,900 lbs/acre will be added during the second application of just mulch, which would result in a total of 2,400 lbs/acre of mulch applied. The hydroseeding shall proceed following the specifications above for the hydromulch application on top of broadcast seed.
    b. Second Application (Mulch): 1,900 lbs./acre of long strand virgin wood fiber and;

1) Apply mulch with a non-toxic, and natural and biodegradable plant-based psyllium/alpha plantago tackifier. Depending on product, apply at manufacture's specification or at a rate of 80 lbs. per acre on slopes less than 3H:1V and apply at a rate of 100 lbs./acre on slopes greater than 3H:1V.
2) Apply mulch with a non-toxic, 100% environmentally safe tracer dye material applied per the manufacturer's specification or at a rate of around 200-300 pounds per acre to provide visible evidence of uniform application.
3) The hydromulching shall proceed following the specifications above for the hydromulch application on top of broadcast seed.

B. Hydroseed to establish a uniform stand of grasses.

C. Seeding rate shall be as recommended by the seed producer.

3.4 PLANT MAINTENANCE

A. Block off seeded areas from visitors until the grasses are well established.

B. Apply treatments as required to keep plant materials, planted areas, and soils free of pests and pathogens or disease. Use Plant Health Care practices including Integrated Pest Management whenever possible to minimize the use of pesticides and reduce hazards. Treatments include physical controls such as mechanical controls such as traps and biological control agents.

3.5 CLEANUP AND PROTECTION

A. During seeding, keep adjacent paving and construction clean and work area in an orderly condition.

B. Protect plants from damage due to landscape operations and operations of other contractors and trades. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

END OF SECTION 32 92 00

SECTION 32 93 00 – PLANTS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Plants
2. Double Processed Mulch

B. Related Sections:

1. Section 31 10 00 – Site Clearing
2. Section 31 20 00 – Earth Moving
3. Section 31 23 16 – Topsoil

1.3 UNIT PRICES

A. Work of this Section is affected by unit prices specified in Division 01 Section “Unit Prices.”

1. Unit prices apply to authorized work covered by quantity allowances.
2. Unit prices apply to additions to and deletions from Work as authorized by Change Orders.

1.4 DEFINITIONS

A. Bare-Root Stock: Plants with a well-branched, fibrous-root system developed by transplanting or root pruning, with soil or growing medium removed, and with not less than minimum root spread according to ANSI Z60.1 for type and size of plant required.

B. Container-Grown Stock: Healthy, vigorous, well-rooted plants grown in a container, with a well-established root system reaching sides of container and maintaining a firm ball when removed from container. Container shall be rigid enough to hold ball shape and protect root mass during shipping and be sized according to ANSI Z60.1 for type and size of plant required.

C. Finish Grade: Elevation of finished surface of planting soil.

D. Manufactured Topsoil: Soil produced off-site by homogeneously blending mineral soils or sand with stabilized organic soil amendments to produce topsoil or planting soil.

E. Pesticide: A substance or mixture intended for preventing, destroying, repelling, or mitigating a pest. This includes insecticides, miticides, herbicides, fungicides, rodenticides, and molluscicides. It also includes substances or mixtures intended for use as a plant regulator, defoliant, or desiccant.

F. Pests: Living organisms that occur where they are not desired, or that cause damage to plants, animals, or people. These include insects, mites, grubs, mollusks (snails and slugs), rodents (gophers, moles, and mice), unwanted plants (weeds), fungi, bacteria, and viruses.

G. Planting Area: Areas to be planted.

H. Planting Soil: Standardized topsoil; existing, native surface topsoil; existing, in-place surface soil; imported topsoil; or manufactured topsoil that is modified with soil amendments and perhaps fertilizers to produce a soil mixture best for plant growth.

I. Plant; Plants; Plant Material: These terms refer to vegetation in general, including trees, shrubs, vines, ground covers, ornamental grasses, bulbs, corms, tubers, or herbaceous vegetation.

J. Root Flare: Also called “trunk flare.” The area at the base of the plant’s stem or trunk where the stem or trunk broadens to form roots, the area of transition between the root system and the stem or trunk.

K. Subgrade: Surface or elevation of subsoil remaining after excavation is complete, or the top surface of a fill or backfill before planting soil is placed.

L. Subsoil: All soil beneath the topsoil layer of the soil profile and typified by the lack of organic matter and soil organisms.

M. Surface Soil: Soil that is present at the top layer of the existing soil profile at the Project site. In undisturbed areas, the surface soil is typically topsoil; but in disturbed areas such as urban environments, the surface soil can be subsoil.

1.5 SUBMITTALS

A. Product Data: For each type of product indicated.

1. Plant Materials: Include quantities, sizes, quality, and sources for plant materials.
2. Double Processed Mulch: Manufacturer source information.

B. Samples for Verification: For each of the following:

1. Double Processed Mulch: 1-gallon volume in sealed plastic bags labeled with source. Each Sample shall be typical of the lot of material to be furnished; provide an accurate representation of color, texture, and organic makeup.

C. Qualification Data: For qualified Landscape Installer. Include list of similar projects completed by Installer demonstrating Installer’s capabilities and experience. Include project names, addresses, and year completed, and include names and addresses of COs’ contact persons.

D. Product Certificates: For each type of manufactured product, from manufacturer, and complying with the following:

1. Manufacturer's certified analysis of standard products.
2. Analysis of other materials by a recognized laboratory made according to methods established by the Association of Official Analytical Chemists, where applicable.

E. Material Test Reports: For imported or manufactured topsoil.

F. Maintenance Instructions: Recommended procedures to be established by CO for maintenance of plants during a calendar year. Submit before start of required maintenance periods.

G. Warranty: Sample of special warranty.

1.6 QUALITY ASSURANCE

A. Installer Qualifications: A qualified Landscape Installer whose work has resulted in successful establishment of plants.

1. Experience: Three years' experience in successful landscape installation in addition to requirements in Division 01 Section "Quality Requirements."
2. Installer's Field Supervision: Require Installer to maintain an experienced full-time supervisor on Project site when work is in progress.

B. Provide quality, size, genus, species, and variety of plants indicated, complying with applicable requirements in ANSI Z60.1 to Contracting Officer (CO).

C. Measurements: Measure according to ANSI Z60.1. Do not prune to obtain required sizes.

1. Trees: Measure with branches and trunks or canes in their normal position. Take height measurements from or near the top of the root flare for field-grown stock and container grown stock. Measure main body of tree for height and spread; do not measure branches or roots tip to tip.

D. Plant Material Observation: CO may observe plant material either at place of growth or at site before planting for compliance with requirements for genus, species, variety, cultivar, size, and quality. CO retains right to observe trees further for size and condition of balls and root systems, pests, disease symptoms, injuries, and latent defects and to reject unsatisfactory or defective material at any time during progress of work. Remove rejected trees immediately from Project site.

1. Notify CO of sources of planting materials seven days in advance of delivery to site.

E. Preinstallation Conference: Conduct conference at Project site prior to installation.

1.7 MATERIALS

A. Tree Plantings

1. Consists of the following planting palette:

Species	Common Name
QUERCUS ALBA	WHITE OAK
QUERCUS BICOLOR	SWAMP WHITE OAK

1.8 PROJECT CONDITIONS

A. Field Measurements: Verify actual grade elevations, service and utility locations, irrigation system components, and dimensions of plantings and construction contiguous with new plantings by field measurements before proceeding with planting work.

B. Interruption of Existing Services or Utilities: Do not interrupt services or utilities to facilities occupied by CO or others unless permitted under the following conditions and then only after arranging to provide temporary services or utilities according to requirements indicated:

1. Notify CO no fewer than two days in advance of proposed interruption of each service or utility.
2. Do not proceed with interruption of services or utilities without CO's written permission.

C. Planting Restrictions: Plant between September 15 and November 1 or February 1 and April 15 when soil temperatures are consistently within the 40-49 degrees F range. Coordinate planting periods with maintenance periods to provide required maintenance from date of Substantial Completion.

D. Weather Limitations: Proceed with planting only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained. Do not seed on ground that is excessively wet, frozen, or has more than ¼" of snow cover.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Packaged Materials: Deliver packaged materials in original, unopened containers showing weight, certified analysis, name and address of manufacturer, and indication of conformance with state and federal laws if applicable.

B. Bulk Materials:

1. Do not dump or store bulk materials near structures, utilities, walkways, and pavements, or on existing turf areas or plants.
2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
3. Accompany each delivery of bulk fertilizers and soil amendments with appropriate certificates.

C. Deliver bare-root stock plants freshly dug. Immediately after digging up bare-root stock, pack root system in wet straw, hay, or other suitable material to keep root system moist until planting.

D. Do not prune trees before delivery. Protect bark, branches, and root systems from sun scald, drying, wind burn, sweating, whipping, and other handling and tying damage. Do not bend or bind-tie trees in such a manner as to destroy their natural shape. Provide protective covering of plants during shipping and delivery. Do not drop plants during delivery and handling.

E. Handle planting stock by root ball.

F. Deliver plants after preparations for planting have been completed and install immediately. If planting is delayed more than six hours after delivery, set plants in their appropriate aspect (sun, filtered sun, or shade), protect from weather and mechanical damage, and keep roots moist.

1. Heel-in bare-root stock. Soak roots that are in dry condition in water for two hours. Reject dried-out plants.
2. Set balled stock on ground and cover ball with soil, peat moss, sawdust, or other acceptable material.
3. Do not remove container-grown stock from containers before time of planting.
4. Water root systems of plants stored on-site deeply and thoroughly with a fine-mist spray. Water as often as necessary to maintain root systems in a moist, but not overly-wet condition.

1.10 WARRANTY

A. Special Warranty: Installer agrees to repair or replace plantings and accessories that fail in materials, workmanship, or growth within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Death and unsatisfactory growth, except for defects resulting from abuse, lack of adequate maintenance, or neglect by CO, or incidents that are beyond Contractor's control.
b. Structural failures including plantings falling or blowing over.
c. Faulty performance of tree stabilization and edgings.
d. Deterioration of metals, metal finishes, and other materials beyond normal weathering in the surrounding area.

2. Warranty Periods from Date of Planting Completion:

a. 12 months from date of planting completion.

3. Include the following remedial actions as a minimum:

a. Immediately remove dead plants and replace unless required to plant in the succeeding planting season.
b. Replace plants that are more than 25 percent dead or in an unhealthy condition at end of warranty period.

c. A limit of one replacement of each plant will be required except for losses or replacements due to failure to comply with requirements.
d. Provide extended warranty for period equal to original warranty period, for replaced plant material.

1.11 MAINTENANCE SERVICE

A. Initial Maintenance Service for Plants: Provide maintenance by skilled employees of Landscape Installer. Maintain as required in Part 3. Begin maintenance immediately after plants are installed and continue until plantings are acceptably healthy and well established but for not less than maintenance period below.

1. Maintenance Period: 12 months from date of planting completion.

B. Continuing Maintenance Proposal: From Installer to CO, in the form of a standard yearly (or other period) maintenance agreement, starting on date initial maintenance service is concluded. State services, obligations, conditions, and terms for agreement period and for future renewal options.

PART 2 - PRODUCTS

2.1 PLANT MATERIAL

A. General: Furnish nursery-grown plants true to genus, species, variety, cultivar, stem form, shearing, and other features indicated in Plant Schedule or Plant Legend shown on Drawings and complying with ANSI Z60.1; and with healthy root systems developed by transplanting or root pruning. Provide well-shaped, fully branched, healthy, vigorous stock, densely foliated when in leaf and free of disease, pests, eggs, larvae, and defects such as knots, sun scald, injuries, abrasions, and disfigurement.

1. Collected Stock: Do not use plants harvested from the wild, from native stands, from an established landscape planting, or not grown in a nursery unless otherwise indicated.
2. Contractor shall furnish only native, regional genotype plant materials; no other species, cultivars or varieties shall be permitted without CO approval. Regional genotype is defined as the state of Iowa and/or neighboring states within 200 miles of Effigy Mounds National Monument.

B. Provide plants of sizes, grades, and ball or container sizes complying with ANSI Z60.1 for types and form of plants required. Plants of a larger size may be used if acceptable to CO, with a proportionate increase in size of roots or balls.

C. Root-Ball Depth: Furnish trees with root balls measured from top of root ball, which shall begin at root flare according to ANSI Z60.1. Root flare shall be visible before planting.

2.2 PLANTING SOILS

A. Topsoil shall be as defined in Section 31 23 16.

2.3 MULCHES

A. Double Processed Mulch: Premium hardwood mulch, aged to a natural dark brown color (no dye), and processed to 1 – 2 inches derived from the mechanical grinding or shredding of the above-ground portions of trees. It may not contain visible amounts of soil, branches, or leaves. It shall be free of weeds and weed seeds. It may not contain more than 1/2% by weight of manufactured inert material (plastic, concrete, ceramics, metal, etc.). Tub grindings shall not be used as mulch.

2.4 MISCELLANEOUS PRODUCTS

A. Burlap: Non-synthetic, biodegradable.

B. Tree Stake: Douglas Fir, size: as indicated on plans, pencil-pointed end, beveled top

C. Tree Stake Tie: Flat woven polypropylene, flexible.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive plants for compliance with requirements and conditions affecting installation and performance.

1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within a planting area.
2. Do not mix or place soils and soil amendments in frozen, wet, or muddy conditions.
3. Suspend soil spreading, grading, and tilling operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.
4. Uniformly moisten excessively dry soil that is not workable and which is too dusty.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

C. If contamination by foreign or deleterious material or liquid is present in soil within a planting area, do not remove any soil unless directed by the CO.

3.2 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities and turf areas and existing plants from damage caused by planting operations.

B. Install erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

C. Lay out individual plant locations. Stake locations and adjust locations when requested and obtain CO's acceptance of layout before excavating or planting. Make minor adjustments as required.

D. Wrap trees with burlap fabric over trunks, branches, stems, twigs, and foliage to protect from wind and other damage during digging, handling, and transportation.

3.3 EXCAVATION FOR TREES

A. No excavation into existing grade will be authorized prior to approval from CO.

B. Topsoil removed from excavations may be used as planting soil.

C. Obstructions: Notify CO if unexpected rock or obstructions detrimental to trees are encountered in excavations.

D. Drainage: Notify CO if subsoil conditions evidence unexpected water seepage or retention in tree planting pits.

E. Fill excavations with water and allow to percolate away before positioning trees.

3.4 TREE PLANTING

A. Before planting, verify that root flare is visible at top of root ball according to ANSI Z60.1. If root flare is not visible, remove soil in a level manner from the root ball to where the top-most root emerges from the trunk. After soil removal to expose the root flare, verify that root ball still meets size requirements.

B. Remove stem girdling roots and kinked roots. Remove injured roots by cutting cleanly; do not break.

C. Set plant stock plumb and in center of planting pit or trench with root flare 2 inch above adjacent finish grades.

1. Use topsoil for backfill.
2. After placing some backfill around root ball to stabilize plant, carefully cut and remove burlap, rope, and wire baskets from tops of root balls and from sides, but do not remove from under root balls. Remove pallets, if any, before setting. Do not use planting stock if root ball is cracked or broken before or during planting operation.
3. Backfill around root ball in layers, tamping to settle soil and eliminate voids and air pockets. When planting pit is approximately one-half filled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed.
4. Continue backfilling process. Water again after placing and tamping final layer of soil.

D. When planting on slopes, set the plant so the root flare on the uphill side is flush with the surrounding soil on the slope; the edge of the root ball on the downhill side will be above the surrounding soil. Apply enough soil to cover the downhill side of the root ball.

E. Drive stakes on the windward side of trees. Ensure proper depth for stability, considering the size and weight of the plant being supported. Use appropriate ties that won't damage bark, allowing for growth while maintaining security.

F. Spread Double Processed Mulch evenly across 6' diameter ring around each tree after all planting material has been installed. Mulch shall be a minimum of 3" deep with all mulch pulled 6" away from tree trunk. Water lightly after application is complete.

3.5 PLANT MAINTENANCE

A. Maintain plantings by pruning, cultivating, watering, weeding, fertilizing, mulching, restoring planting saucers, adjusting and repairing tree-stabilization devices, resetting to proper grades or vertical position, and performing other operations as required to establish healthy, viable plantings.

B. Fill in as necessary soil subsidence that may occur because of settling or other processes. Replace mulch materials damaged or lost in areas of subsidence.

C. Apply treatments as required to keep plant materials, planted areas, and soils free of pests and pathogens or disease. Use Plant Health Care practices including Integrated Pest Management whenever possible to minimize the use of pesticides and reduce hazards. Treatments include physical controls such as hosing off foliage, mechanical controls such as traps, and biological control agents.

D. Check stake stability regularly, especially after severe weather events. As trees grow, adjust ties to prevent girdling or bark damage. Replace any damaged stakes promptly to maintain consistent support. Once trees are established and showing signs of self-sufficiency, remove the stakes to allow for natural strengthening.

3.6 CLEANUP AND PROTECTION

A. During planting, keep adjacent paving and construction clean and work area in an orderly condition.

B. Protect plants from damage due to landscape operations and operations of other contractors and trades. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

C. After installation and before Substantial Completion, remove nursery tags, nursery stakes, tie tape, labels, wire, burlap, and other debris from plant material, planting areas, and Project site.

3.7 DISPOSAL

A. Remove surplus soil and waste material including excess subsoil, unsuitable soil, trash, stakes, ties, and debris and legally dispose of them off CO's property.

END OF SECTION 32 93 00

SECTION 33 40 00 – STORM DRAINAGE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes the following:

1. Installation of catch basins and associated storm drain.

B. Related Sections:

2. Section 01 57 23 – Temporary Stormwater Pollution Prevention
3. Section 31 20 00 – Earth Moving

1.3 SUBMITTALS

A. Product Data: For catch basins, frame, grate/lid, outfall protection material, and storm drainpipe.

B. Manufacturer's Certification: Products meet or exceed specifications.

PART 2 - PRODUCTS

2.1 PIPE

A. Storm drainpipe shall be lined corrugated polyethylene pipe and shall meet the requirements of AASHTO M 294 Type S.

B. Corrugated polyethylene storm sewer pipe manufacturers shall participate in the National Transportation Product Evaluation Program (NTPEP) work plan for HDPE (High Density Polyethylene) Thermoplastic Pipe and be listed on the NTPEP audit website displaying they are NTPEP compliant.

C. Corrugated polyethylene couplings and fittings shall meet the requirements of AASHTO M 294 Type S.

D. All joints for corrugated polyethylene storm sewer pipe shall be made with a bell/bell or bell and spigot coupling and shall conform to ASTM D3212 using elastomeric gaskets conforming to ASTM F477. All gaskets shall be factory installed on the pipe in accordance with the manufacturer's recommendations.

E. ASM D 3350 minimum resin Cell Classification 335420C.

F. Minimum pipe stiffness at 5% deflection complying with ASTM D 2412.

2.2 CATCH BASINS

A. Catch basins shall be precast, Class C concrete and shall conform to the dimensions shown in the plans. All precast concrete items shall meet the requirements of AASHTO M199.

B. Catch basin reinforcement shall be as set forth in the Plans.

C. The knockout diameter shall not be greater than 20 inches. Knockouts shall have a wall thickness of 2 inches minimum to 2.5 inches maximum. Provide a 1.5” minimum gap between the knout wall and the outside of the pipe. After the pipe is installed, fill the gap with joint mortar.

D. The precast base section may have a rounded floor. The walls may be sloped at a rate of 1:24 or steeper.

2.3 FRAME AND GRATE

A. Castings for metal frames for catch basins shall be cast steel, gray iron, or ductile iron.

B. Castings for metal grates for catch basins shall be cast steel or ductile iron.

C. Grates shall be seated properly to prevent rocking.

D. Grates shall be capable of supporting HS-20 loading.

E. Grates shall be ABA-compliant.

PART 3 - EXECUTION

3.1 PIPE INSTALLATION

A. General

1. Clean pipe interior and joints prior to lowering into trench. Keep pipe clean during construction.
2. Begin at the lowest point in the line. Lay groove or bell end pointing upstream.
3. Place pipe with lifting holes at the top of the pipe and fill lift hole with non-shrink grout or manufactured plugs.
4. Assemble joints as specified by the pipe manufacturer.
5. Use a saw to cut ends of pipe flush with inside wall manholes, intakes, and structures. Do not use hammer or other means to break pipe.
6. Provide manholes and catch basins as specified in the Contract Documents.

B. Excavation, Trenching, and Backfill

1. Verify trench cut is ready to receive work and excavations, dimensions, and elevations are as indicated on layout drawings. Trenches must be of sufficient width in the pipe zone to permit proper installation and bedding of the pipe and to provide the required compaction of backfill.
2. Hand trim excavation for accurate placement of pipe to elevations indicated.
3. All ledgerock, boulders, and stones shall be removed to provide a minimum of 6 inches clearance under all portions of the pipe.
4. Placement of bedding material shall precede the installation of all pipe. This shall include necessary leveling of the native trench bottom or the top of the foundation material as well as placement and compaction of required bedding material to a uniform grade so that the entire length of pipe will be supported on a uniformly dense unyielding foundation.
5. When, after excavating to the foundation level, the material remaining in the trench bottom is determined to be unsuitable by the CO, excavation shall be continued to such additional depth and width as required by the CO. Unsuitable foundation materials shall be disposed of at an approved site. The trench foundation shall be backfilled to the bottom of the pipe zone with sand bedding, or other suitable material, and compacted to form a uniformly dense, unyielding foundation.
6. Lay pipe to slope gradients noted on layout drawings; with maximum variation from true slope of 1/8 inch in 10 feet.
7. Pipe zone bedding material shall provide uniform support along the entire pipe barrel, without load concentration at joint collars or bells. All adjustment to line and grade shall be made by scraping away or filling in with bedding material under the body of the pipe and not by blocking or wedging. Bedding disturbed by pipe movement, or by removal of shoring movement of a trench shield or box, shall be reconsolidated prior to backfill.
8. Pipe zone bedding shall be placed in loose layers and compacted to 90 percent maximum density. Bedding shall be placed, spread, and compacted before the pipe is installed so that the pipe is uniformly supported along the barrel. Lifts of not more than 6 inches in thickness shall be placed and compacted along the sides of the pipe. Material shall be worked carefully under the pipe haunches and then compacted.
9. If the CO determines that the material existing in the bottom of the trench is satisfactory for bedding the pipe, the bedding material specified is not required, provided the existing material is loosened, regraded, and compacted to form a dense, unyielding base.
10. Before lowering and while suspended, inspect pipe and each fitting for defects. Installation of defective material is not permitted.
11. Where pipe, fittings or joint materials have been soiled by earth in handling, thoroughly clean soiled surfaces by wire brushing and wiping until all traces of earth are removed before joining pipe.
12. Maintain interior of all pipes thoroughly clean. After each line of pipe has been laid, carefully inspect, identify, and remove dirt, trash, rags, and other foreign matter from interior. Protect completed work.
13. Install pipe, fittings, and accessories at the locations indicated on layout drawings and in accordance with manufacturer’s instructions. Seal watertight.
14. Trenches shall be backfilled as soon after the pipe laying as possible.
15. Pipe zone backfill material shall be clean earth or sand, free from clay, frozen lumps, roots, or moisture in excess of that permitting required compaction. Rocks or lumps larger than 3 inches maximum shall not be used for pipe zone backfill.
16. Pipe zone backfill shall be placed in loose layers and compacted to 90 percent maximum density. Backfill shall be brought up simultaneously on each side of the pipe to the top of

the pipe zone. The pipe shall then be covered to the top of the pipe zone and the materials compacted in a manner to avoid damaging or disturbing the completed pipe.

17. Backfill above the pipe zone shall be accomplished in such a manner that the pipe will not be shifted out of position nor damaged by impact or overloading. If pipe is being placed under existing paved areas, or Roadways, backfill above the pipe zone shall be placed in horizontal layers no more than 6 inches thick and compacted to 95 percent maximum density. If pipe is being placed in non-traffic areas, backfill above the pipe zone shall be placed in horizontal layers no more than 6 inches thick and shall be compacted to 85 percent maximum density.
18. Protection

    a. Water is not permitted to run or stand in trench while pipe laying is in progress, before the joints are completely set, or before trench has been backfilled.
    b. Protect pipe and bedding cover from damage or displacement until backfilling operation is in progress.

3.2 CATCH BASIN INSTALLATION

A. The excavation for all catch basins shall be sufficient to leave 1 foot in the clear between their outer surfaces and the earth bank.

B. The cover or grating of a catch basin shall not be grouted to final grade until the final elevation of the ditch in which it is to be placed has been established, and until permission thereafter is given by the CO to grout the cover or grating in place. Covers shall be seated properly to prevent rocking. Leveling and adjustment devices that do not modify the structural integrity of the metal frame, grate, or cover, and do not void the originating foundry's compliance to these specifications and warranty are allowed. Leveling and adjusting devices that interfere with the backfilling, backfill density, and grouting will not be allowed. The hardware for leveling and adjusting devices shall be completely removed when specified by the CO.

C. Ladder rungs shall be grouted in the precast concrete walls. Rungs shall be uniformly spaced at 12 inches and be vertically aligned.

D. In the event any pipe enters the manhole through the precast concrete units, the Contractor shall make the necessary cut through the manhole wall and steel mesh. The steel shall be cut flush with the face of the concrete and shall be cut in such a manner that it will not loosen the reinforcement in the manhole wall.

E. The ends of all pipes shall be trimmed flush with the inside walls.

F. Rubber gaskets or flexible plastic gaskets may be used in tongue and groove joints of precast units. All other joints and all openings cut through the walls shall be grouted and watertight. Mortar must be composed of cement, sand, and water. The proportion of sand to cement measured by volume must be 2 to 1. Mortar must contain only enough water to allow placing and packing. Sand particles must be no larger than 1/2 the size of the recess or space in which the mortar is to be placed.

G. If gaskets are used, handling of the precast units after the gasket has been affixed shall be done carefully to avoid disturbing or damaging the gasket or contaminating it with foreign material. Care shall be exercised to attain proper alignment before the joints are entirely forced home.

During insertion of the tongue or spigot, the units shall be partially supported to minimize unequal lateral pressure on the gasket and to maintain concentricity until the gasket is properly positioned.

H. Backfilling around the Work will not be allowed until the concrete or mortar has thoroughly set.

I. Catch basins shall be watertight.

J. Backfilling

1. Place structure backfill in uniform layers. Bring backfill up uniformly on all sides of structures. Backfill layers must be at most 0.67 foot thick before compacting the thickness must be at most 4 feet when compaction is performed by ponding or jetting.
2. Do not use compaction equipment or methods that may cause excessive displacement or damage structures.
3. Do not place structure backfill the catch basin has been inspected by the CO and authorized for backfilling.
4. Compact structure backfill to a relative compaction of at least 95 percent.

K. Catch basins shall be constructed on a compacted or undisturbed level foundation. If the Contractor elects to use a separate cast-in-place base, the concrete shall be Class 4000.

L. All excavations 4 feet or more in depth shall be shored with structural shoring or shall meet open-pit requirements.

3.3 PROTECTION

A. Protect pipe and bedding from damage or displacement until backfilling operation is in progress.

END OF SECTION 33 40 00